A History of Modern Philosophy

Volumes One and Two

Harald Høffding

translated from *Den nyere Filosofis Historie*, 2nd edition (1903–04)

1894–1895 / 1903–04

Translator's note. This is the whole of Høffding's history of modern philosophy in English, in the order of the work. The first volume (1903) runs from the philosophy of the Renaissance and the new natural science through the great systems of the seventeenth century and English empiricism to the French Enlightenment and Rousseau; the second (1904) from the German Enlightenment and Kant through the philosophy of Romanticism and positivism to the philosophy of Germany from 1850 to 1880. Each volume opens with its title pages and its table of contents, the printed page numbers kept, and ends with Høffding's notes, as in the original; those to the Tenth Book are given as footnotes. The second volume closes with the index of both volumes, its headwords translated, its order (the Danish alphabet's) and its page references as printed. The numbers in the margin are the pages of the second Danish edition (1903–04); each volume is paged separately. The translation is made from the diplomatic transcription of that edition published in this collection, which keeps the printer's errors as printed; the English gives the intended sense. The Introduction and the Tenth Book were translated first, from the e-book text of the same edition, and then checked against the transcription. The Third and Ninth Books, with their notes, were translated from the transcription with Claude (an AI model) in fifteen parts, working from one brief and one glossary of terms; each part was checked by script against the Danish, printed page by printed page (page numbers, notes, paragraphs, emphasis, Greek), and where the Danish was corrected afterwards the English was mended to follow it. The rest of the work — the First, Second and Fourth to Eighth Books, the remaining notes, the title pages, the tables of contents and the index — was then translated in the same way, in thirty further parts, from the transcription of the whole work, with the glossary extended to the terms of these Books. The Introduction and the Tenth Book, too, were translated with Claude. Until now the work was available in English only in B. E. Meyer's translation of 1900, made from the German edition of the first Danish edition.

A HISTORY OF
MODERN PHILOSOPHY

A HISTORY OF
MODERN PHILOSOPHY
AN ACCOUNT OF
THE HISTORY OF PHILOSOPHY FROM THE CLOSE OF THE RENAISSANCE
TO OUR OWN DAY
BY
HARALD HØFFDING
SECOND, REVISED EDITION
VOLUME ONE
FROM THE CLOSE OF THE RENAISSANCE TO ROUSSEAU
COPENHAGEN
DET NORDISKE FORLAG
ERNST BOJESEN
1903

AARHUS STIFTSBOGTRYKKERI.

[V]

TO THE MEMORY
OF
MY TEACHER AND FRIEND
PROFESSOR HANS BRØCHNER
THIS WORK IS CONSECRATED

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Introduction

It will be natural that, when a new presentation of an important period in the history of philosophy appears, the question should be asked what conception of philosophy the author takes as his ground, and what significance and value he ascribes to the history of philosophy. My answer is that the studies whose provisional conclusion is marked by the publication of this book had precisely the aim of orienting me with respect to what philosophy really is. As we come to know a person through his biography, so we must be able to come to know a science through its history. And it will be all the more natural to try this road, since experience constantly shows that in the domain of philosophy conflicting ways of seeing assert themselves, so that there is no presentation of philosophy to which one could point as giving an exhaustive concept of it. It will then be natural here — as in the domain of religion — to apply the comparative method. The history of philosophy treats the attempts which individual thinkers have made to discuss the ultimate problems of knowledge and of life. The most important result will be to get clear about which problems philosophy occupies itself with, how those problems are posed at different times, and by what the different ways of posing them, and of attempting to solve them, are determined in detail. If such a result can be approximately reached, the study of the history of philosophy will be of no slight importance for continued philosophizing. *[XI]*

For my own part I have been strengthened, by the study of the history of modern philosophy, in the assumption I had arrived at by another road: that philosophical inquiry turns upon four principal problems. These four problems are now to be briefly characterized. A more searching discussion and comparison of them I have given in the treatise *Filosofiske Problemer* (University Programme 1902). *[XII]*

1. *The problem of knowledge* (the *logical* problem). However different the several sciences may be in respect of subject and of method, they all nevertheless work with human thinking. Every time they form a concept, utter a judgment or draw an inference, they presuppose the general forms and principles of thinking. Here, then, there shows

itself the possibility of a peculiar discipline, one which investigates the forms in which thinking moves and the principles it must by its nature take as its ground, whatever the objects it may concern. This discipline, *formal logic*, treats however only one part of the problem of knowledge. Those forms and principles do not lead beyond thinking itself, but only make it possible for thinking to be in agreement with itself, to be consistent. Every time they are applied to given phenomena, which our thinking has not itself constructed but must take as they come, the question arises with what right that application takes place — with what right we assume that not only our thinking, but existence itself, which expresses itself in the given phenomena, is consistent, agrees with itself. Herewith shows itself the possibility of a discipline which investigates the conditions of a knowledge of existence, and the limits of such knowledge. This discipline is *epistemology*.

2. *The problem of existence* (the *cosmological* problem) arises when it is asked how the existence of which we are members must be supposed to be constituted, when we draw the ultimate consequences of all that we know, or can conjecture with the help of the most probable hypotheses. We call this problem the cosmological (from *kosmos*, world, and *logos*, doctrine), because it leads to a discussion of the possibilities that present themselves to thought when it seeks to work up what is given in experience into a general world-view, or when by speculation's bold road it attempts to construct one. The
[XIII] different philosophical systems are attempts in this direction — attempts whose value depends on how significant and how comprehensive are the experiences they take as their ground, and how much consistency and power of combination is displayed in their constructions.

3. *The problem of valuation* (the *ethical-religious* problem) arises through this, that we do not merely stand to existence in an intuiting and understanding relation, but also have our feeling set in motion by it, so that we utter judgments which ascribe value to it or deny it value. Of particular importance are the judgments we utter upon human actions, our own and others'. All such appraisal rests — like all knowledge and understanding — upon certain presuppositions, which it is a matter of exhibiting and determining. This task belongs under the ethical problem. When the valuation concerns not merely human actions and institutions but existence, life as a whole, the religious problem arises, which leads especially to a discussion of the relation between the ethical ideals and actual existence, so that it becomes a combination of the cosmological and the ethical problem.

4. *The problem of consciousness* (the *psychological* problem). For all three of the problems indicated it shows itself that their treatment presupposes an empirical acquaintance with human conscious life. Psychology presents the *actual* development

of human knowledge, which must be known before one can discuss the conditions of knowledge's *validity*. And since the relation between the spiritual and the material is a principal point within the problem of existence, psychology is presupposed in cosmology as well. As regards the problem of valuation, finally, psychology discusses partly the constitution of the very feelings that lead us to make valuations, partly the possibilities which actual conscious life presents for a further development in the direction valuation demands. In consequence of this close relation to the philosophical problems, psychology is itself to be reckoned a part of philosophy; and one will therefore also, conversely, quite naturally come in psychology to touch upon philosophical problems. And even if one were to declare all three of the preceding problems insoluble, or arisen through misunderstanding, the psychological problem — the question of the nature and the laws *[XIV]* of conscious life — would still remain as philosophy's last stronghold.

— If it is asked what factors will, in the nature of the case, come to influence the treatment and the answering of these problems, the first to be named must be the personality of the one who philosophizes. The problems named have this in common, that they lie at the boundary of our knowledge, where the exact methods fail; it cannot then be otherwise than that the inquirer's personality must come to determine his train of thought, without his needing to be conscious of it. The "personal equation" must be of greater importance in philosophy than in other scientific domains. The historical, comparative method therefore becomes of particular importance here; by its help the personal element can most easily be exhibited. That personal element is not always something to be pushed aside; its presence is often a condition of a problem's being able to come to be at all. There are thoughts which can come to be only upon a determinate psychological soil. — It will, in the second place, depend upon which observations are taken as the ground. Here the development of natural science acquires especially great importance for modern philosophy. As will appear, the decisive problems of modern philosophy are determined by the very fact that modern natural science came into being. But to this there come as well — especially as regards the problem of valuation — historical conditions, intellectual movements in other domains. — Finally, the posing of problems and their solution are determined by the consistency with which the starting points once seized are held fast and carried through.

Both the characterization and the criticism of every philosophical attempt will always lead back to these three points. Among them I lay most weight, in the presentation that follows, upon the first two. An inconsistency in a great thinker is often precisely a natural consequence of *several* trains of thought having opened themselves to his genius,

without his having been able to pursue them far enough to discover their mutual conflict. And it may be of the greatest importance that these different trains of thought be brought out. Best of all, of course, is when profundity and consistency are united.

[XV] An attempt at posing or at solving a problem may have a twofold interest. In the first place it may be regarded as a symptom, as the historical expression of intellectual currents. From this side the history of philosophy is a part of general cultural history. In the second place it may be examined with respect to the real, definitive clarity that has been reached by it. This consideration is more specifically philosophical and technical. The two sides from which the philosophical phenomena can be regarded will of course, in the case of different phenomena, stand in very different relations to one another. Sometimes the cultural-historical interest may even stand in inverse relation to the purely philosophical.

If I am briefly to indicate wherein my presentation of the history of modern philosophy differs from its predecessor in our literature — Brøchner's work of 1873–74 — I would name first the particular weight I lay upon the personal factor, upon the relation to empirical science, and upon the cultural-historical significance of the philosophical phenomena. Next I would point to my endeavor to give the posing of problems precedence over their solution. Solutions may die, and the problems live on; otherwise philosophy would not have had so long a life as it has had. Finally it goes without saying that a renewed study of the sources, as well as the use of the abundant literature of recent years, has set many phenomena in a new light. But in spite of all this it has been a dear thought to me, in this work, to be able to continue Brøchner's labor in the field of the history of philosophy; and I can only hope that my book may not be unworthy to be consecrated to the memory of that noble inquirer, whose admiration for the heroes of thought was at once so deep and so understanding.

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First Book

[1]

The Philosophy of the Renaissance.

1. The Renaissance and the Middle Ages

The period in the European life of the spirit that is called the Age of Rebirth is marked partly by a break with the unfree, incoherent and limited view of life of the Middle Ages, partly by a widening of the gaze to new domains and an unfolding of new spiritual powers. In every birth both go together: the tearing loose from the old and the unfolding of the new life. But it is surely unique in the life of the human spirit that both sides stand out so fully at the transition to a new age as here. In other periods of transition, as a rule, either criticism and negation or the new and positive fullness takes on an overwhelming character. It is characteristic of the Renaissance in this respect that it is chiefly the new life that brings with it the criticism of the old. On this rest its health and strength. It laid the foundation of a new, free, human conception of nature and of human life, which it was itself able to express only in broad outline, and which, when worked through in detail, proved to contain problems enough for centuries of thought and research, but which in its great main features has undergone no essential change. Every attempt to orient oneself, with the help of history, with regard to the great theoretical and practical problems must therefore look back to this time, which has the freshness and power of the breakthrough and, in a richness full of presentiment, embraces what more divided research is often able to grasp only one-sidedly. *[2]*

Before we proceed to set forth the philosophizing of the Renaissance as it appears in its most important thinkers, we must cast a glance back at the Middle Ages, from which it was a matter of breaking loose.

It would be wrong to take the Middle Ages for an age of sheer darkness. Not only did there unfold, beneath or outside the official dominion of the Church, a bright folk life,

fresh with nature, which has left its monuments in the new, budding national literatures. But even within the learned world it will be very difficult to fix a definite boundary between the Middle Ages and the age of rebirth. Especially in the Romance countries, and most of all in Italy, the connection with antiquity was really never wholly broken. The justification for speaking of the Renaissance as a special period must then be sought in the fact that there came a time when the knowledge of nature and of human life became so full that it could not be contained within the bounds of the ecclesiastical notions. And even this point in the development will not be easy to fix definitely. But this difficulty does not concern us more closely here, since we shall confine ourselves to setting forth the philosophy that took shape at the close of the Renaissance, that is, in the course of the sixteenth century, and can therefore disregard germs and beginnings, which moreover have a literary-historical rather than a philosophical interest.

Even if one disregards the germs of further development that are found within the spiritual life of the Middle Ages, and keeps to medieval thought in its classical form, one will find that the Middle Ages made their significant contributions to spiritual development and were by no means the desert or the world of darkness for which they are still often given out. They deepened the life of the spirit and sharpened and exercised its powers in a significant way, and in any case they stand behind no other period in the energy with which they used the means of culture at their disposal, although these, owing to the historical conditions, such as they happened to be, were very limited. In later, happier periods, which commanded a rich material, the working-up and inward appropriation of this wealth has hardly taken place with so great a power as that which
[3] the Middle Ages applied to their meager material. Some individual points on which medieval thinking especially offers advantages and has had a preparatory effect shall be indicated here.

Medieval thinking was theological. The theology of a monotheistic religion proceeds from the fundamental thought that there is one first cause of all things. Apart from the great difficulties this thought presents, it has the great and beneficial effect that it accustoms one to look beyond the differences and particulars and prepares for the assumption of a lawful connection in all things. To the unity of the cause the unity of the law must correspond. The Middle Ages educated men to this thought, to which natural man, who, overwhelmed by the multiplicity of phenomena, inclines to polytheism, is not disposed. They thereby mark a preparation for a conception of the world determined by science. For all science endeavors to lead the phenomena back to as few principles as possible, even if it must indeed acknowledge that the thought of one single, highest law

is an unattainable ideal.

For carrying out its ideas in detail, medieval thinking, as already remarked, had only a very poor material at its command. But all the greater was the labor it spent on it. The poverty of realities had to be outweighed by the wealth of formalities. Thinking displayed a formal acuteness, a skill in making distinctions and in setting up arguments, that is altogether unique. One wishes this skill a better application, but a skill it was, and it has been of great importance for the development of the spirit. It trained organs that could come into function in more fruitful domains. It was bound in the long run to lead to a testing criticism of the very presuppositions which for a long time had stood as the fixed framework that men did not dare to examine.

The greatest merit of the Middle Ages lies in their absorption in the inner world of mental life. Classical antiquity stopped at a harmonious relation between the inner and the outer, and took an interest in the inner life only in its relation to the outer life in nature and the state. For medieval faith the eternal destiny of the personality was determined by the events of the inner life. It was a question of life and death whether the mental growth unfolded as it should. No wonder that a fine and deep sense developed *[4]* for what went on within. The self-absorption of the mystics has the same significance for the development of the psychological sense as the distinguishing and arguing of the scholastics for the development of the logical sense. It became clear that the spiritual world is, after all, just as much a reality as the material world, and that it is in it, after all, that man has his true home. The way was paved for a more thorough discussion of the great problem of mind and matter than antiquity had been able to undertake. But first and foremost there was won for human life a domain of experience that was its own in the very narrowest sense, where no outer powers could penetrate. Thereby the possibility of a more deeply reaching spiritual liberation was given.

Yet it was precisely the great misfortune of the Middle Ages that none of these motives, fruitful in themselves, could stir freely and fruitfully. There is in medieval thinking, as in medieval architecture, a craving toward the great and the infinite, and there is at the same time a striving to fit all the elements of the knowledge of the world that men possessed, as bearing stones and pillars, into the great edifice of thought. At the bottom: the world of nature, as Aristotle had set it forth; above it: the world of grace, broken forth in Christ; and highest: the prospect of the eternal world of glory. The ideal was a harmonious gradation of *natura, gratia* and *gloria,* such that the higher domain did not break off but completed the lower. This endeavor is most perfectly expressed in Thomas Aquinas (1227 — 1274), the perfecter of scholasticism, one of the greatest

systematizers who have ever lived. He inspired Dante, was canonized in 1323, was called *doctor angelicus* in the theological schools of the Middle Ages, and still stands as the classical thinker of the Roman Church, Pope Leo XIII having decreed in the year 1879 that his philosophy is to be made the foundation in the Catholic institutions of learning. But despite the grandeur of a doctrinal edifice which for so many still contains the elements of existence and throws light on their connection, it has nevertheless had, from its very beginning, highly questionable defects. Only in an artificial way were the elements,
[5] drawn from such different sides, joined together. In order to unite the natural knowledge which, it was believed, Aristotle had summed up with the supernatural presuppositions of the Church, this knowledge had partly to be reinterpreted, partly to be prevented from working out all its consequences. Aristotle's philosophy was really designed to present existence as a harmonious gradation. The fundamental concepts with which Aristotle operated were taken from the phenomena of organic life. He saw nature as a great process of development, within which the higher stages were related to the lower as form to matter, or as reality to possibility. What on the lower stage was only possible (potential) became real (actual) on the higher stage. Aristotle himself was not able to carry through this significant view. In what direction its consequence lies is clear enough. As an ecclesiastical thinker, Thomas Aquinas had nevertheless to break completely with this consequence, to push back the doctrine of unity (monism) to which it led, and to put a doctrine of duality (dualism) in its place. This shows itself in a characteristic way in his psychology and in his ethics. According to the Aristotelian psychology the soul is the "form" of the body: what is given in the body as mere possibility comes forth in mental life in full activity and reality. But so intimate a relation between soul and body conflicted with the ecclesiastical presuppositions, and although Thomas in words adheres to Aristotle and calls the soul the form of the body, he nevertheless in reality treats the soul as a being wholly different from the body, just as he does not hesitate to assume "forms" without matter — in order to make room for the angels! In ethics a similar dualism shows itself. From the Greeks he takes over a series of natural cardinal virtues: wisdom, justice, courage and self-control; but whereas for the Greeks these made up all virtue, he places above them, as a higher layer, the three "theological" virtues: faith, hope and love, which arise only in a supernatural way. The development is thus broken, and Thomas does not even inquire whether the forms of will denoted by the theological virtues might not very well find their place within the natural virtues, as special forms of these. —
[6] As regards the conception of the world as a whole, the idea of natural development stood in sharp opposition to the dogma of creation and the assumption of miraculous

intervention. On closer inspection dualism shows itself on one point after another. But such closer inspection medieval thinking had to keep away. It *had to* agree with the teaching of the Church. There was to be no thinking, no understanding of the world, that was independent of theology. And since men had once acquired the conviction that the Aristotelian philosophy was particularly well suited to represent natural knowledge in the erection of the scholastic edifice of doctrine, it came to be regarded as heretical to depart from Aristotle, and, be it noted, from Aristotle as he was known and understood in the Middle Ages. In other words, inquiry and thought were arbitrarily brought to a halt, so that the building that had been erected should not be shaken. The Aristotelian philosophy, which for its time marked so extraordinarily great an advance, was fixed as eternally valid. For the doctrine of organic life and of mental life the damage done by this was not the greatest; for in these domains lay Aristotle's real merit, already in his own time. But the development of an exact natural science was hindered, of a science that could explain through what elementary processes and according to what general laws the development of forms that Aristotle assumed took place. The trouble with the Aristotelian philosophy was that it fixed organic growth as the type of everything that happens in nature. It could therefore at most give a description of forms and properties, not a real explanation of their coming into being. The mechanical view of nature was thus excluded in advance. Men cut themselves off from the way of progress — without, however, being able to overcome the difficulties by the means at their command. The principle of authority hindered a freer and wider discussion of the problems and fixed dualism as a permanent result. The principle of authority is, of course, itself a form of dualism, since it presupposes an irreconcilable opposition between human knowledge and the goal that is to be reached. Men were then reduced to living on the meager content they had, and to interpreting this content, by artificial distinctions and argumentations, as the Church wished it. No wonder that a strong hunger for fullness of content was *[7]* bound to arise, and a great enthusiasm for the new wealth that streamed in from all sides in the centuries of the Renaissance!

Even in the domain into which the Middle Ages particularly delved, these defects show themselves. Self-absorption was anxiously religious. It was not allowed to develop into free knowledge of the spiritual world. The dogmatist always stood on guard against the mystic, the surge of whose inner life so often led him beyond what the Church sanctioned as true and right feeling. Just as little as the Church could give the new experience free play, just as little did it dare to let inner experience go its own ways. It saw a danger in the possibility that man, by withdrawing into himself, might be able to

place himself in an immediate relation to what is of highest value and thereby become independent of the Church. It sensed that self-knowledge, just as well as knowledge of nature, offered possibilities of spiritual liberation and opened the way to a quite different conception of the world from the theological one. — Yet it was not only the anxious restriction to religious moods that hampered the development of inner experience. Here too dualism was at work and prevented the recognition of a real lawfulness in the inner domain, as well as of a natural connection between the mental and the bodily. Mental life could not be rightly understood until a freer and more comprehensive view of nature developed. —

The significance of the Renaissance, then, is that it marks the period in which the limits and one-sidednesses of the medieval view of life could be combated with the help of new experiences and points of view. However diverse the phenomena that meet us within the philosophy of the Renaissance, they nevertheless have in common a great enthusiasm for, and confidence in, nature, a striving to carry through a unified conception of existence and to assert the lawfulness and legitimacy of natural human life. The Renaissance begins with this last point, with "the discovery of man" (to use Jacob Burckhardt's expression), and leads later to the development of a new world-picture and
[8] a new investigation of nature.

A. The Discovery of Man

2. Humanism

It was no accident that Italy became the home of the Renaissance and thereby the cradle of modern thought. In Italy the connection with antiquity had been best preserved, and when the literature of antiquity was brought forth again, the Italians could appropriate it in a quite especially independent way, because it was the work of their own antiquity, flesh of their flesh and blood of their blood. Toward Greek literature too, which in the fifteenth century once more became the object of enthusiastic study, the Italians stood in several respects with more understanding than the northern peoples. This whole return to the literature of antiquity, to the study of ancient history, poetry and philosophy, had the great significance for the history of culture that men now became aware that outside the Church and Christianity too there existed a human life of the spirit which followed its own laws and had its own history. The spiritual world was widened, and a comparative consideration of human affairs became possible. In several of the thinkers of the Renaissance we already find the comparative method applied to the understanding

of what is human. Moreover, the works of antiquity could serve as models that guided thought until it could work in a more independent way. Many germs of thought, which in antiquity itself had only just begun their growth, only to be checked by the age of the Middle Ages, so harsh for free spiritual development, could now unfold. After a long slumber thought sought to take up its work where it had laid it down at the close of antiquity.

But that a living understanding and an independent appropriation of the literature of antiquity became possible was due in turn to the historical conditions in Italy. The division into the many small states, constantly tossed about by internal political struggles in which, as a rule, no means were shunned to obtain and secure power, brought with it a dissolution of the medieval order of society. In the Middle Ages man was valued according to his connection with the Church and the corporation. Natural man, with his purely individual life of feeling, was not heeded and not regarded as legitimate. *[9]* The political struggles round about in the cities had now, on the contrary, developed a reckless striving to assert one's personality, a striving that naturally led to fuller results only in the fortunate holders of power. On the other hand, the dominion of power that prevailed in the Italian states was bound to draw many individuals away from public life and, since the need to assert individuality had after all been awakened, to direct them to develop their personality as widely and freely as possible in the relations of private life and in artistic and literary interests. It is Burckhardt's merit (in his work on the culture of the Renaissance) to have shown how the bent toward individualism and the need for purely personal development were bound to arise in Italy in the fourteenth and fifteenth centuries under the influence of the historical conditions. It was not, then, merely the outer discovery of the literature and art of antiquity that conditioned the Italian Renaissance. In that case it would have been a merely learned movement, a predominantly receptive process. It was first and foremost the practical discovery of human nature as a domain — for each individual the domain lying nearest, — in which there was ample room and occasion both for gaining experiences and for undertaking a work of development. What the Middle Ages had made possible only in the form of religious mysticism was now carried further and freed from limits and bonds. Individual mental life is felt as a reality and awakens interest in and for itself, apart from what it is attached to. It was a discovery of no less significance than the discovery of the new continent on the earth and of the new worlds in the heavens. In the poets (Dante, Petrarca) the interest in, and occupation with, one's own self and its events naturally appears first and in the grandest style.

Humanism therefore denotes not merely a literary tendency, a school of philologists, but also a tendency of life, which was characterized by interest in the human, both as an object of consideration and as a basis for action. It is, both in literature and in life, a tendency that presents extraordinarily many differences. The Church for a time
[10] showed itself favorably disposed toward it, and humanistic interest even prevailed for a while on the papal throne. When Pius II had a crusade preached, it was presumably essentially in order to save civilization; in his conviction Rome was now, after the fall of Constantinople, the last refuge of literature, and he had perhaps mourned the fall of the Greek imperial city fully as much from humanistic as from religious motives. Humanism has something indeterminate in its character. It denotes the discovery of the human; but how this human element is conceived, and how its development is sought, still remains uncertain. For a time it looked as if the humane and the traditional might be able to go together in a peaceful way. But the new content soon burst the old leather bottles.

Here we now have the task of showing how, on the basis of the practical interest in the human, there appears in the thinkers of the time a series of attempts to conceive and understand it theoretically as well. It is attempts to emancipate psychology and ethics from theology that we have here to bring forward.

3. Pietro Pomponazzi and Niccolò Machiavelli

The whole of medieval thinking had, as we know, rested on an ancient foundation, since it built on Aristotle, whose ideas it had bent toward what the dogmas demanded. But opposed to this "Latin" interpretation of Aristotle stood partly the old Greek interpreters, who conceived his doctrine more naturalistically, partly the Arabian interpreters, first and foremost Averroës, who conceived his doctrine in a pantheistic spirit. On the question whether Aristotle taught the immortality of the soul the three groups of interpreters clashed. According to the Greek interpreters, whom men in the Middle Ages took no account of, Aristotle taught a natural development of mental life from lower to higher stages, but in such a way that even the highest stage was not independent of the natural conditions. According to the Averroist interpretation the highest form of thinking becomes possible only by man's being made to partake in an eternal reason; but this
[11] partaking lasts only for a time, and it is not the individual souls but the universal reason, in which they momentarily partake in their highest acts of thought, that is immortal. The theological Aristotelians, on the other hand, with Thomas Aquinas at their head, maintained the immortality of the soul as a doctrine grounded on the soul's capacity to know and to will the universal and eternal.

Pietro Pomponazzi discussed this much-disputed question in a remarkable little work *(de immortalitate animi* 1516). Rightly has this work been seen as the introduction to the philosophy of the Renaissance. It deserves this place by the manner in which the problem is treated. It does discuss what Aristotle's real meaning is, but declares at the same time that it will examine the question according to natural reason, independently of all authority. And its endeavor aims at asserting the natural connection within mental life and, at the same time, the possibility of an ethical conduct of life on the ground of nature. Pomponazzi was a celebrated teacher of philosophy, first in Padua, later in Bologna. He was born in Mantua in 1462 and died in Bologna in 1525. Of his personal circumstances little is known. He was an excellent disputant and speaker; his writings, however, do not distinguish themselves in a literary respect.

Against Thomas as well as against Averroës, Pomponazzi lays weight on the lawful and continuous development of nature which Aristotle taught in his Physics, and which permits no leaps, nor the introduction of wholly new principles not prepared by the preceding development. He carries this thought through even further than the master Aristotle himself, who taught that the highest activity of reason ("the active reason") could not be explained from the successive development. How it was then to be conceived and explained, Aristotle had not pronounced, and precisely thereby he had given occasion to the conflicting interpretations. That even the highest thinking cannot be independent of natural conditions, Pomponazzi proves especially by the proposition, expressly set up by Aristotle, that all thinking presupposes representations which must at the outset be given through sense perception. And he maintains that the Aristotelian definition of the soul as the form of the body, or its perfect reality, likewise does not make it possible to *[12]* assume an independent subsistence of the soul.

Will there not, then, arise from this result questionable consequences in an ethical respect, since for natural man the prospects of reward or punishment after death cannot then become motives? Pomponazzi maintains, on the contrary, that these prospects are questionable in an ethical respect, since they hinder man from doing the good for its own sake. The full development of man's nature in the various domains has its satisfaction in itself. And while scientific and artistic development does not fall in equal measure to everyone's lot, there is one development from which no one is excluded: ethical development. And this has its goal and its value in itself. The reward of virtue is virtue itself, and the punishment of the vicious is vice itself. Whether, therefore, man is mortal or not, death is still to be despised, and however things may stand with the state after death, one must still not turn aside from the way of the good. — These utterances, which

are found in the fourteenth chapter of the book, recall on the one hand the Apology of Socrates and point on the other forward to Spinoza's and Kant's conception of the ethical. Several humanists had, it is true, come forward with the idea of an independent ethics; but in Pomponazzi this idea acquires a special significance through the connection with a specifically religious problem in which it appears.

The question of immortality is, according to Pomponazzi, an insoluble problem (problema neutrum); one may add in his spirit: a neutral problem also in the sense that no real ethical values depend on its answer. But with such questions the point is to get it settled how far our knowledge can really reach. Evidently, for Pomponazzi the chief interest of his investigation is bound up with the light it throws on the nature of our knowledge. The problem is only the occasion. He distinguishes sharply between the standpoint of the philosopher and that of the legislator. For the legislator the point is to find motives that can make men upright, and he may perhaps find it right to make use of the faith in immortality as an educative motive; but the philosopher has only to ask after
[13] the truth without letting himself be moved by fear or hope. A similar opposition can, as Pomponazzi teaches, be found within the individual man. With his reason he seeks only to draw correct conclusions from given presuppositions; what results reason is to arrive at therefore does not depend on man's will. With his will man can hold fast to a faith that his reason cannot ground. Pomponazzi thus distinguishes between faith and knowledge, and he can therefore, with regard to the final decision of the problem he treats, submit to the doctrine of the Church. Whereas the scholastic Duns Scotus had declared two centuries before that something can be true for the philosopher which is not true for the theologian, Pomponazzi's meaning would rather be expressed in the proposition that something can be true for the theologian which is not true for the philosopher. It must be remembered, after all, that as a philosopher he ended with a problema neutrum. There is therefore no reason, either, to doubt that this doctrine of faith and knowledge was meant in full earnest, or to take it for irony or for a mere evasion[1]). He gives the impression of a seeking nature, for whom inquiry is a matter of full earnest. He compares the philosopher to Prometheus, because he is constantly gnawed by restless thoughts in his craving to search out the divine secrets. It may very well be that reason was more developed in him than the will that was to bind him to faith, but it may nonetheless have been his earnest when he held fast to the claims of both. Only, a closer development of the connection between will and faith is lacking in him. He confines himself to pointing to the will as the foundation of faith without giving further explanation, although such an explanation would be of great interest. —

Besides the work discussed here, Pomponazzi also wrote on sorcery and sought to give a natural explanation of events that were interpreted as effects of supernatural agency. This work is of interest through its endeavor to assert the principle of natural causes, although the causes he makes use of themselves, in our view, belong under the concept of superstition. For it is especially the influence of the stars that he relies on, an influence which the age, of course, did not regard as supernatural. — In a third work he discusses the question of divine predestination and man's will. He acutely demonstrates *[14]* the conflict between them. As a philosopher he keeps to what stands firm in experience: the reality of the human will; its relation to divine agency he leaves standing as an insoluble problem, here too making use of the distinction between faith and knowledge.

This distinction did not help him, however. In Venice the Inquisition had his work de immortalitate burned, and if he had not had a powerful protector in Pope Leo X's friend Cardinal Bembo, he might perhaps not have escaped sharing the fate of his book. —

At first glance it might seem strange to place Machiavelli beside Pomponazzi. The Florentine statesman seems to have nothing to do with the Bolognese scholastic. And yet the common enemies of these two contemporary thinkers were hardly wrong when they held them to be of the same leaven. It was the ancient faith in human nature that they both, each in his own way, wished to see reborn. And both were strongly influenced by the authors of antiquity. As Pomponazzi sought to continue Aristotle's naturalistic psychology and ethics, so Machiavelli shows himself on important points influenced by the Greek historian Polybius, who in turn points back to older Greek thinkers' conception of the course of development of states. But it was Machiavelli's own observations of the events of his time, in which he himself was involved, that begot in him the need to go back to antiquity. Nicolo Machiavelli was born in Florence in the year 1469 of an old and respected family that had come down in the world. He early entered the diplomatic service of the republican government of his native city and visited as its envoy Pope Julius II, Cesare Borgia, the Emperor Maximilian and King Louis XII. He thereby had ample opportunity to gather observations of men and events. Of fateful significance for the development of his views was the fact that precisely this period in the history of Italy was one in which cunning and intrigue, treachery and cruelty were the means by which political decisions were carried out. He himself, in his Florentine History (in the introduction to the fifth book), made the remark about the history of *[15]* Italy in the fifteenth century that one cannot admire the conduct of the Italian princes of that time for its greatness and nobility of heart, though it does not on that account afford less material for consideration, when one sees how such noble peoples were kept

in check by weak and badly led arms. It was bound to stand for Machiavelli as the result of what he had experienced that cunning and strength were the one thing needful for the statesman. He himself does not show himself to be any superior politician; his greatness is in the world of thought, not in that of action. And yet he grieved deeply when he was forced to exchange the latter world for the former. When the Medici overthrew the free constitution of Florence (1512), Machiavelli was dismissed, and during the enforced private life that he then led for a number of years he composed his celebrated works. Had he borne his political misfortune with greater dignity, his literary works too would perhaps have been kept in a higher style. As it was, he worked out especially that one of his works which has become most famous, the book on the prince *(Il principe)*, with the purpose in view of gaining the favor of the Medici. He partly attained this purpose, but it cannot be denied that a stain was thereby put on his character, and that by having the political power of a single family in view he came to stress individual arbitrariness and recklessness in politics more strongly than he would perhaps otherwise have done. Often it looks in his exposition as if it were in the last instance indifferent what purpose the ruler wished to carry through, and as if he in any case regarded it as his task to give directions for the various purposes one might happen to set oneself. Not only in "The Prince", but also in his real chief work, the *Discorsi,* the discourses on the first ten books of Livy, this shows itself. Whoever would found a republic must go about it thus; whoever would found a monarchy, thus; whoever would found a tyranny, thus. And yet it is certain that there was one purpose for which Machiavelli's heart beat, and which in his innermost being he embraced with his thought in the midst of the diplomatic intrigue as well as in the working-out of his works, namely the unity and greatness of Italy. It is
[16] certain, too, that he always felt himself a republican; this is seen from the advice he gave the Medici concerning the constitution of Florence. And there also stood before his mind an ideal of man, of health, strength and wisdom, which was to tower high above the wretchedness by which he felt himself surrounded, and above the cowedness that the medieval Church had brought over human nature. It was on this point that he himself looked to antiquity and demanded that it be imitated. Although he was himself a poet and lived with the poets, and although he himself had a good deal of the unrestrained man of pleasure in him, the aesthetic Renaissance was not enough for him, any more than the gospel of pleasure was for him the highest. It was the Renaissance of strength and greatness that he demanded. "When I," he says (in the introduction to the first book of the Discorsi), "consider the reverence that antiquity inspires in all, how, e.g., men give a great sum of gold for a fragment of an ancient statue in order to have it constantly

before their eyes, for the adornment of their houses and as a model for artists, while on the other hand I see that the marvelous deeds which history relates to have been performed in the ancient states by their princes, generals, citizens and legislators, and by all those who have labored for the greatness of their fatherland, are coldly admired rather than imitated, nay, that everyone on the contrary seems to avoid everything that can recall them, so that we now have not the least trace of ancient virtue left, then I must both wonder and lament at it." In his highest ideas he is akin to the great spirit that appears in Michel Angelo's finest works. But his tragic fate was that the great purposes so often receded behind the multiplicity of the means. He was so taken up with the thought that it is of no use to have a purpose, be it ever so noble, when one has not powerful means to carry it out, that in the end he forgot the purpose over the means, or omitted to examine whether the means he admired would also in reality and in a lasting way work in the direction of the great purposes that were undoubtedly the highest for him. He became so taken up with the strength of the means that he came to attribute value to it in and for itself, detached from the purpose that was to sanctify it. As a true man of the Renaissance he admires the display of strength in and for itself, indifferent as to the direction in which the strength works. Although he declares himself against the *[17]* purely aesthetic Renaissance, he has nevertheless himself introduced it into a domain where the point is precisely always to have strength determined by the purpose. The contradiction that expresses itself on this point in his writings, and especially in "The Prince", has often led to his being judged most unjustly. And the same contradiction was the fate of his life, if one will not call it his guilt. When the Medici were again driven out, after the taking of Rome by the imperial troops in 1527, and the free constitution was restored, his countrymen would not accept his service, because they regarded him as a turncoat. His last days were thereby embittered. A contemporary who is otherwise not favorable to him said of him after his death: "I believe that his own circumstances pained him; for he truly loved freedom, and that in more than the usual way. But it tormented him that he had involved himself with Pope Clement." He died in 1527.

There runs, like an undercurrent, through Machiavelli's works a constant comparison between ancient and Christian morality. He has (in the "Discorsi") set himself the task "of comparing ancient and modern events"; but even where he does not expressly undertake such a comparison, it can be traced in his conception and manner of exposition. He will not content himself with relating and describing; he will also find the causes of the difference between the old and the new, and this he cannot do except by going back to the great opposition between the ancient and the medieval view of life. Machiavelli

thereby becomes the most consciously pronounced representative of the opposition to the Middle Ages.

In the Middle Ages it stood as the task of the state to help men toward their highest goal, blessedness in the world beyond. The prince was not, indeed, to work for this directly, but still indirectly, especially by providing for the preservation of peace. In this direction goes the conception of the state in Thomas Aquinas. For Machiavelli, on the contrary, the national state has its purpose in itself. Its strength and health within, its power and expansion without, on these everything depends. And it is not an ideal state
[18] that he is thinking of, but definite, actually given states. "Since I," he says (The Prince. ch. 15), "intend to write things that may be useful to those who read me, it seems to me better to keep to the reality of things than to give myself up to empty speculations. There are many who have imagined for themselves republics and principalities such as have never been seen or known. But what purpose do these fantasies serve? There is so great a distance between the way in which men live and the way in which they ought to live that by studying only the latter one learns rather to ruin oneself than to keep oneself up. He who in all things and everywhere would show himself an honorable man cannot avoid perishing among so many who are bad. A prince who wishes to maintain himself in power must therefore learn not always to be good, but to make use of good and evil according to necessity." And in accordance with this it is said in the discourses on Livy: "Wherever a decision has to be taken on which the welfare of the state depends, one must not let oneself be stopped by considerations of justice or injustice, humanity or cruelty, honor or dishonor, but, disregarding everything else, keep to what can save the state and assert its freedom." (Discorsi. III, 41). Whoever cannot in this way free himself from the usual moral notions must live as a private man and not venture forth among those who govern (I, 26).

What is not clear in Machiavelli is whether he regards the justice and sense of honor that in certain cases hinder the necessary political action as real virtues or only as imagined virtues. In some places (as The Prince ch. 8) he speaks of qualities that seem to be virtues and that can bring about the prince's ruin. He does not go further into the great problems that arise here, namely those of the relation between the ethics of the private man and that of the statesman. Is there here an opposition in principle, two kinds of virtue? Or should it not be a misunderstanding when one calls a quality a virtue that hinders the attainment of something which is a good for the whole circle that the valuation concerns? Machiavelli is not free from a certain ambiguity here, which several modern declaimers against "morality" share with him. It looks as if he wishes to get

beyond the opposition between good and evil, and yet it is only a new valuation, thus a *[19]*
new application of the concepts good and evil, that he aims at. This new valuation itself becomes, as has already been shown, uncertain in him because of the unclear relation between power and the purpose that power is to serve. The aesthetic admiration for the reckless display of strength pushes aside the thought of the sanctifying purpose. What he blames men for is that they show energy neither for evil nor for good: of small crimes men are not afraid, but from crimes in the grand style, "whose greatness would blot out their infamy", they shrink back. In the year 1505 Pope Julius II ventured with his cardinals, without an army, into Perugia to depose Baglioni, who held power in this city. Now when Baglioni did not use the opportunity to seize the foolhardy pope and thereby at one stroke free himself from his mortal enemy, seize great riches and give the princes of the Church a salutary lesson for the future, it was certainly not for moral reasons; for Baglioni was a man who did not shrink from incest and from the murder of near relatives. "The conclusion was drawn from this that men do not know how to preserve any dignity in crime, any more than they are able to be perfectly good, but that they shrink back from a crime when it bears the stamp of greatness and loftiness." (Discorsi. I, 27).

What Machiavelli turns against, then, is the petty, the dull and the timid. That was his real complaint against his contemporaries, as is also seen from his Florentine History. When, then, he asked the reason why men had moved so far from the greatness of antiquity, he found this reason in education, which has caused weakening and cowedness, and education in turn is bound up with religion. The ancients loved honor, greatness of mind, bodily strength and health, and the ancient religions bestowed divine esteem on those mortals who had won merit as great generals, heroes or legislators. Their religious ceremonies were magnificent and often connected with bloody sacrifices, which were bound to dispose minds to fierceness. Our religion, on the contrary, places the highest goal in another world and teaches us to despise the honor of this world. It glorifies *[20]*
humility and self-denial and places the quiet, contemplative life above the practical, outward-turned one. If it demands strength of us, it is rather strength to suffer than to act. This morality has made men weak and handed the world over to foolhardy men of violence, who have discovered that most men, in the hope of paradise, are rather disposed to put up with insults than to avenge them. (Discorsi. II, 2). Machiavelli does add that this is a false interpretation of Christianity, due to human cowardice; but it is hardly his meaning thereby to take back his whole comparison between ancient and Christian morality, and it is clear enough to which side his own sympathy goes. Religion

is for him essentially a means in the hand of the legislator. The legislator needs it as a support for the laws. The wise man foresees much that the multitude does not suspect and will not believe, and then the gods must step in and sanction the necessary laws. The history of Lycurgus and Solon bears witness to this, and Savonarola worked so powerfully on his countrymen because it was believed (whether rightly or wrongly) that he stood in immediate intercourse with God. Religion is especially a firm foundation for the maintenance of unity and good manners among the people. Numa's religious institutions were an essential cause of the strength and greatness of the Roman Empire, and every prudent prince must uphold the national religion, even if he personally regards it as an error. (Discorsi. I, 11—12).

Machiavelli has no feeling that in religion he stands face to face with a spiritual power which unfolds according to its own involuntary laws, and which politics therefore cannot always, without further ado, take into its service or have at its disposal. He relies on the whole too much on shrewd calculations, on arbitrary intervention and on the uniformity of circumstances, and overlooks that the greatest things in history often come like a stream that suddenly rushes forth from springs hitherto invisible. He himself, it is true, emphasizes the multiplicity of great and unforeseeable events to which the age had been witness, and which might lead to the belief that "God and fortune" ruled everything. But he does not think that the human will is therefore powerless. Against a
[21] raging flood there is perhaps nothing to be done at the moment; but when the storm is over, we can dig canals and throw up dikes, so that the damage becomes less the next time. Fortune comes to rule especially where no resistance has been prepared. (The Prince. ch. 25). It is, however, a rather external relation that he here sets up between the involuntary and the arbitrary. The statesman who stands wholly outside the great spiritual currents will hardly have such an understanding of them that he can draw canals and build dikes in the right way. — Machiavelli lays great weight on the necessity that institutions and constitutions — religious as well as political — be reborn from time to time by being "led back to their principle." In the course of time there come additions and changes that can corrupt the source of their essential strength. The point then is to remember the beginning, to reflect upon one's origin. The occasion for this can be given by external misfortunes, as when Rome was reborn after the Gallic war, or by definite institutions, such as the tribunes and censors of Rome, or by eminent men whose personalities become models, as when St. Francis and St. Dominic again set the original Christian model before men's eyes and thereby saved the Church from the ruin that would otherwise have been near. (Discorsi. III, 1). This is one of the most brilliant

considerations to be found in Machiavelli. But it shows at the same time the limits of the shrewdness that he believed he could reduce to a system. The greatest event that occurred in his lifetime was precisely such a renewal according to the original principle in the religious domain, but at the same time a renewal that was not a mere repetition. Through the Reformation spiritual development was led into new paths, and no politician who would regard it merely as a means, without having a deeper understanding of its value, has been able to exert any lasting influence on the course of things. That through the "renewal" wholly new forces can come forth, he will not understand. — Machiavelli on the whole had no eye for the causes working in secret. There were many sides of life besides inward religiosity that escaped his attention. The development of the practical interests, of trade and industry, of mechanics and economics, and the way in which these interests and the new social strata that represented them had grown strong enough *[22]* to determine to an essential extent the politics of the new age — this was something to which his predominantly formal conception of politics as a game of intrigue and a struggle for power blinded him. His politics hovered in the air, because they were not tied to the great ideas in which the constantly working forces find expression. Like so many realists he missed reality, because he wished to seize it on the surface of events.

It was natural that Machiavelli, in his striving for an understanding of history, had to keep to the causes he could catch sight of, or thought he caught sight of. So, after all, Pomponazzi in his criticism of superstition had to content himself with astrological explanations. The merit of both is this: to have asserted the principle of natural causes. They were on the way, and they made it possible for their successors to get further along this way. As regards Machiavelli in particular, he stands not only as the founder of scientific politics in modern times, but also as the founder of comparative ethics. What he had indicated in broad outline, it became the task of later times to carry out in detail, avoiding the rocks on which he had run aground.

Shortly after the death of Pomponazzi and Machiavelli there was an end of the free Italian life of the spirit. When Florence lost its freedom for the second time (1530), the turning point had come. Reaction now broke forth mightily, and the Inquisition was organized anew to meet the Protestant movement. Exile or martyrdom now became the lot of those Italians whose religious or scientific views led them into new paths. And yet it was, as we shall see later, precisely in this period that the new world-view and the new method of natural science were developed by Italian thinkers, Italy's great contribution to the spiritual development of modern times. First we follow the conception of man, and have to bring forward some significant phenomena that show us how the fundamental

[23] thoughts of the Renaissance worked outside Italy.

4. Michel de Montaigne and Pierre Charron

The various sides of the Renaissance show themselves in hardly any of the thinkers of the time so clearly and fully as in Montaigne. He has, in the first place, the pronounced individualism which in France in the sixteenth century, no less than in Italy in the preceding centuries, was developed through the political struggles, and in Montaigne's case also through the religious conflicts he witnessed. France, moreover, had its humanism in the sixteenth century. Montaigne was born (1533) of a noble family in southern France, received a careful and learned education, made journeys in Italy and afterwards lived for the most part on his estate, enjoying his freedom and pursuing his studies, carefully guarding himself against being drawn into the political and religious movements of the time, which he looked upon as upon a play. That he held back was certainly not grounded merely in his being led to skepticism by seeing the many parties and the changing views, but chiefly in his wish to preserve his own personality, such as it happened to be. It was this that was the most important object of his study. His *Essais* (whose first two books appeared in 1580, the third book in 1588) are a book personal in a high degree, both in content and in form. "I study myself more than any other object," he says (III, 13); "that is my metaphysics, that is my physics." And the book does indeed give us full information about his interests, studies, habits and manner of life. It is written in a free form, without systematic connection. Thinking is for him a kind of play or pastime; he lets his thoughts go as they will, and gives them in turn to his readers just as unbound and unordered as he himself received them. And he declares at once in the preface that he really writes only about himself (je suis moy mesme la matiere de mon livre), and that, had manners and custom permitted, he would gladly have gone still further in giving particulars about himself. In the third book of the "Essais" he goes further in this respect than in the first two, appealing to the greater freedom that age gives. This self-preoccupation, this giving
[24] oneself up to one's own thoughts and rocking oneself in one's own moods, is a modern trait; it bears witness that the individual is freed from traditional presuppositions and, without troubling himself about any authority, confidently dares to abandon himself to the currents in his own nature and let himself be carried by them.

Another characteristic trait is his great reading, especially in the literature of antiquity. His writings swarm with quotations, and he has a great store of characteristic traits with which to illustrate human opinions, feelings and drives. The comparative method he has applied to the spiritual domain to an even greater extent than Machiavelli. And that

not merely, as has often been thought, in order to draw skeptical consequences from the multiplicity of characters, views and tendencies that he brings forward. He is interested in the individual nuances, and he has at the same time the humanist's joy in the rich material. It is the gaze out into the immeasurable world of spiritual phenomena that captivates him, not merely the mutual difference or conflict that there may be between them.

To be sure, both his individualism and his humanism were bound to make him an opponent of every attempt to force upon men and carry through, dogmatically, one single general doctrine. He attacks both theological and philosophical dogmatism (especially II, 12). If — he says to the orthodox — we were really in earnest about faith in the supernatural, — "if we were really touched by the ray from the Deity," — then our life would have to look quite different from what it does. All relations would be transformed in accordance with the holy demands, and the wars of religion with their ignoble passions could not have split us. It is, then, hardly divine powers, but rather custom and tradition, that lead us, in so far as it is not the passions that determine our faith. In the wars of religion God's cause alone could hardly have raised a single company. And apart from the influences that make themselves felt, it is clear that our representation cannot grasp anything that lies wholly beyond the limits of our being. Our own qualities we can think of as heightened or diminished, but beyond them we cannot get. Moreover, every being is most interested in itself, in its peculiar nature. From this it follows that we *[25]* represent the Deity to ourselves in human shape and think that everything in the world is governed for the good of men. But would not the animals reason in a corresponding way from their nature? He who thinks he knows something about God inevitably drags God down into what is low. Of all human opinions about religion that one seems to be the most probable which acknowledges God as an incomprehensible power, the author of all and the sustainer of all, pure goodness and perfection, graciously receiving the worship that men render it, under whatever form they may conceive it and in whatever way they may show it their reverence.

But because the orthodox assertions must be rejected, there is still no reason to presume upon human reason. Man is at once the most wretched and the proudest of all creatures. Pride is his innate disease. He feels himself highly exalted above the rest of creation, and yet there is (as Montaigne seeks to show through a detailed enumeration of traits that bear witness to the understanding and feeling of animals) not so great a distance between him and the animals as he thinks. He has no reason to set himself apart from the host of beings. With his knowledge things stand badly. The senses are uncertain

and erring. We cannot convince ourselves that they teach us the truth. They present the world to us as their nature and condition bring with them. It is not the external object but the senses' own condition that appears to us in sense perception: les sens ne comprennent pas le subject estangier, ains seulement leurs propres passions. If we were to be able to rely absolutely on sense perception, we should have to have an instrument that could check it — and then in turn a means of checking this instrument, and so on to infinity. Nor does reason lead to any final decision: every reason that is adduced for an assumption itself needs a reason in turn, and so we could go on going back to infinity. — To this is added that both we ourselves and the objects change and alter incessantly; there is nothing fixed and constant. And the wealth of differences is so great that it is
[26] hopeless to attempt to set up general laws or types. No law can exhaust the multiplicity of cases. The more exact the investigation, the more differences are discovered. And in the attempt to bring the differences found under common points of view, one will find that they conflict with one another, so that the comparison cannot lead to any result. — The constant changes and the great differences show themselves also as regards the moral and civil laws. No natural law can be pointed out that is followed by all. Manners change with time and place. What kind of goodness is it that yesterday was in force, but tomorrow no longer, and that becomes a crime when one crosses to the other bank of the river? Can truth be bounded by mountains, so that it becomes a lie on the other side? — Doubt then becomes the last resort. But neither may doubt be fixed as something that definitely holds. We dare not say that we know nothing. Our result is: *what do I know?* —

This whole line of thought in Montaigne has often been taken as skepticism, men seeing in it Montaigne's last word. Pascal already took him so. But this does not reach the ultimate foundation of Montaigne's view of life, the point about which everything finally turns in him, and by which he comes to suggest an entire world-view. Montaigne is, to be sure, too much a causeur to develop his world-view in purely philosophical form. But his last word is not the bewildering diversity of phenomena, not skepticism, nor individualism either. Behind everything in him there rises an infinite background: the idea of nature in its majesty and infinity as that from which the fullness of phenomena derives, and as that whose power branches out in a peculiar way into each individual being[2]).

Montaigne will not merely refute puffed-up knowledge. He even downright praises ignorance, because it gives nature free play and does not, with reflection and art, hinder "our great and mighty mother Nature" from leading us. By ignorance he does not then mean raw and thoughtless emptiness, but the ignorance that is the fruit of an

understanding of the limits of our being. Only by knocking against a door can we convince ourselves that it is shut, not by standing passively before it. Only for "a good head" (une teste bien faiste) is ignorance a good and wholesome pillow. Both in great *[27]* things and in small the concept of nature plays a role in Montaigne. If the talk is, e.g., of illness, he will have nothing to do with the physicians, who only corrupt nature and hinder it in its own right development. Like every other being, every illness has its definite time to develop and to end, and it is in vain to intervene. Let nature rule; it understands the matter better than we. In the order of nature one must acquiesce. Like the harmony of the world, our life is composed of opposites, of different tones, all of which belong to the great connection, illness and death as well as health and life. — If the talk is of education, then likewise the point is to let nature have its free course. The communication of knowledge is to be only a means for the development of feeling and character, and the experience of life is the best school for self-control. — His very skepticism he really takes back outright when he comes to the concept of nature. The only way, he says, in which one can secure oneself against measuring things with a false yardstick is to set before one's eyes "our mother Nature in her whole majesty": she shows us a pervasive and constant multiplicity, within which we discover ourselves, and everything we call great, as a vanishing point[3]). We thereby avoid drawing arbitrary and narrow limits. Thereby tolerance arises. As we have already seen, Montaigne's final result in a religious respect is that the unknown Deity is worshipped by the various nations under just as many different forms. — And by this consideration Montaigne grounds his conservatism. I am weary of the new, he says, under whatever shape it presents itself. Not because the existing laws are always reasonable. The validity of laws does not rest on their being just, but on the very fact that they are laws: c'est le fondement mystique de leur auctorité! Custom is the mistress of the world. Conscience feels itself bound to what it is accustomed to respect, and cannot tear itself loose from it without pain. The wise man must indeed inwardly free his soul in order to judge things without prejudice; but outwardly he ought to follow the existing laws and customs. It is foolish pride to presume oneself able to put something better in its place. Although *[28]* Montaigne does not say it outright, it seems nonetheless to lie in his line of thought that nature makes itself known in what becomes manners and custom, and his conservatism thereby falls under his faith in nature, although, of course, the new — on Montaigne's own view — would have to be just as "natural" as the old.

By the concept of nature, finally, the first trait we emphasized in Montaigne is grounded, his individualism. Everyone, he thinks, by listening attentively to what goes

on within him (s'il s'escoute), will discover in himself a manner of being peculiar to him, a ruling character (forme sienne, forme maistresse, forme universelle), which resists outer influence and keeps away emotions that are incompatible with it. This character is in its innermost ground unchangeable. I can wish myself another "form", loathe and condemn the one I have. But what has its deepest root in my nature cannot really be repented of, since repentance can concern only what it is in our power to change. It is through this ruling form that nature stirs in us, and in each of us in a peculiar way, for which reason everyone ought to be judged by his own measure. Montaigne thus asserts nature in the single individual, just as he asserts it in the great whole. In opposition to the Church's demand for an absolute rebirth he claims that a complete transformation is impossible. He denies the reality of repentance if it is to be such a transformation. Repentance is possible only if there is an inner foundation that can be wounded by the full and clear picture of the evil action that has been committed. While reason remedies the other sorrows and pains, it produces the pain of repentance, which is the heaviest, because it arises within, just as the heat or cold of fever is more penetrating than that which comes from without. In a corresponding way there is felt an inner satisfaction, a noble pride, in doing good and having a good conscience. To build the reward for good actions on the approval of others would be to build on an all too uncertain foundation, especially — Montaigne thinks — in times like the one he lives in, where to be an object
[29] of people's esteem is a bad sign. Everyone must have an inner pattern (un patron au dedans) by which he can judge his actions. Only you yourself can judge your conduct. Others see not so much your nature as your art. Even those nearest to us look differently on our actions from strangers. Through the greater insight into our nature that they have occasion to gain, they perhaps discover that the brilliant outward deeds are only fine jets of water that shoot up from a swampy ground. From the outer no conclusion can be drawn to the inner. (Essais. III, 2).

Montaigne does not explain more closely how that "patron au dedans" develops, and how it is related to the "forme maistresse". Here he leaves a great problem to subsequent thought. But it is clear that, just as he praised ignorance and confident surrender to nature only for him who has a good head, so too in the practical domain it cannot be his meaning that one should without further ado let nature provide. The individual nature develops into its peculiar form only when reflection and will work with it, and the "ruling form" can be preserved from distortion only by watchfulness and labor. Montaigne is aware that he is no hero of the will. If he has any virtue, it is, he thinks, rather the favor of fortune than the work of the will. He is grateful for belonging to a good family (une

race fameuse en preud'hommie) and for having received a good education. He has a natural abhorrence of most vices (particularly cruelty), but he does not vouch for how it would have gone with him if his nature had not been so happy; inner strife and debate he cannot endure. His reason does strive to take the seat of honor in his soul; but it often has enough to do not to let itself be corrupted by the drives, and cannot always reform them. And yet it was his conviction that the highest pleasure was bound up with virtue; the struggle could only be a transition. The lower enjoyments are only momentary and fleeting and easily bring repentance with them. A perfect satisfaction, equally far removed from sensual enjoyment and from a toilsome struggle to hold fast to the commands of reason, is found only where it is the very innermost nature of the soul that expresses itself in good actions. To have developed such a nature in oneself *[30]* that every occasion for inner struggle falls away — that is for Montaigne the highest. He knows that he himself stands only in the third rank, belonging neither to those who fight bravely nor to those who do not need to fight. But he admires the "loftiness of heroic souls". His own weakness does not make him blind to the strength of others. Although he himself creeps on the earth, he can still see their high flight. And neither does he regard it as a small matter that he has kept his moral judgment sound in an age in which virtue seems to be nothing but a "jargon du collège". (I, 19; 36. — II, 11. — III, 13).

There is an ancient character in the way in which Montaigne asserts the concept of nature. But this concept is carried by him beyond the limited form it has in the Greek thinkers. It is made infinite by being brought into so close a connection with the concept of individuality. When Montaigne sets nature in opposition to men's artifices, it is partly because these fix certain single forms as alone legitimate and thereby overlook the fullness of nature, partly because they violate the single peculiar individuality, whose right to develop "sur son propre modele" is the same as that of every other. The two thoughts, that of infinity and that of individuality, had the future before them. And through his skepticism, his great power of observation and his humanist learning Montaigne gained the freedom of mind and the material for impressing the significance of these concepts. His skepticism — which is really an assertion of the right of thought — led him to seek beyond the artificial barriers, and his rich experience and learning taught him what was to be found on the other side of the barrier: new individual forms, in which the one and the same infinite nature expresses itself. —

Montaigne died in 1592. A thinker of his kind could really have no disciple. His thinking is too personal for that. There were combined in him elements that could not recur in the same combination in others. He who stands nearest as his disciple, and

as the one who systematized his ideas, accordingly represents only a single side of his
[31] world of thought, even if it is a very significant side. It was acquaintance with Montaigne that led Pierre Charron to philosophical ideas. He was born in Paris in 1541, was first an advocate, but later went over into the clerical estate and became a celebrated preacher. He published apologetic writings for the Catholic faith against Protestants and freethinkers. But for all that, the most remarkable thing in the book of lasting interest that he left behind, and which appeared at Bordeaux in the year 1600 (three years before his death) under the title *De la sagesse,* is precisely a thoroughgoing defense of the main thought that we have found in Pomponazzi, Machiavelli and Montaigne alike: the thought of human nature as the foundation of ethics and politics. He lays great weight on the ignorance that follows from true self-knowledge. Human nature, on closer investigation, appears to us both in its wretchedness and in its greatness. Natural reason cannot be the measure of all things; one ought therefore, in Charron's opinion, to keep to the old doctrine of the Church without entangling oneself in new doctrines. Like Montaigne, and in an even higher degree than he, Charron thus draws conservative consequences from doubt. But what is remarkable is this: he goes more closely into the relation between religion and morals, which Montaigne had not directly touched upon. Religion and morals belong together, but must not be confused, since each has its own mainspring (ressort). Above all, goodness must not be made dependent on religion, since it thereby becomes something accidental and does not have its source in "the good mainspring of nature". I want, says Charron, a man to be a good man even if there is no paradise and hell. It is to me a terrible saying when someone says that if he were not a Christian, if he did not fear being damned, he would do this or that. I want you to be a good man because nature and reason will it so, — because it is demanded by the general order of things, of which you are only a part, — and because you cannot act against yourself, your being and your purpose; — let come of it what will. If this stands firm, then religion can come afterward. But the reverse relation is pernicious.

[32] So resolute a recognition of the independence of ethics is remarkable in the mouth of a Catholic cleric. It is a psychological riddle how Charron, who not long before had come forward as a zealous Catholic preacher, came to it. But in any case it bears witness that a strong tendency in this direction was stirring in the age.

5. Ludovicus Vives

The great interest that the age of the Renaissance took in the human was bound, naturally, to bring about an eager psychological study of mental phenomena. In all the thinkers

named so far, psychological observation and psychological interest play an essential role, without the psychological inquiry, however, appearing as a distinct endeavor, separated from the farther-reaching philosophical consequences that were drawn from it. But a purely theoretical psychological study was also pursued at this time. And if any single person is to be named as the representative of pure psychology at this time, it must be the Spaniard LUDOVICUS VIVES. He was a quiet scholar, an eager humanist, the friend and collaborator of Erasmus of Rotterdam, but his chief significance lies in the domain of psychology and pedagogy. He was born in Valencia in 1492, studied in Paris, and later lived in Bruges right up to his death (1540). For a time he was tutor to the daughter of the English king Henry VIII, but broke with the English court over the divorce case. Vives was a serious Catholic, but his peace-loving, sound and mild way of looking at things led him, in a candid letter to Pope Hadrian VI (1522), to speak for a general council that should heal the breach in the Church by distinguishing between what is necessary for piety and morals and what can be left to free discussion. Later, too, he worked for tolerance and for a mild and free conception of Christianity. As a pedagogue he has an extraordinarily great significance for his time. Many of the good pedagogical ideas that the Jesuits afterwards carried out are said to derive from Vives, with whom Loyola is said to have been personally acquainted. His occupation with philological criticism and with pedagogical problems led him on to the question of the method of science in general, and he demands that experience be the foundation of all knowledge. As regards psychology, he himself sought to carry out this demand, standing as he does as the founder of modern empirical psychology. Like his whole age, he builds on a great body of material taken from antiquity; but he seeks to confirm it by his own experience, and his work *De anima et vita* (Bruges 1538) is rich in original observations. His description of particular mental phenomena, especially of the emotions, is still instructive, and his book had an extraordinarily great influence on the psychological theories of the sixteenth and seventeenth centuries[4]). *[33]*

Vives's clear empirical standpoint shows itself especially in the section of the first book of De anima that bears the heading "What the Soul Is". After pointing out the difficulties bound up with answering this question, since it is easier for us to say what the soul is not than what it is, he remarks that it is not really of interest to us either to know *what* the soul is, but *how it acts,* and that the commandment to know ourselves concerns not the essence of the soul but its functions. It is established here with great assurance that it is the mental phenomena we have to do with directly, and that empirical psychology can disregard entirely every purely speculative theory of the essence of the

soul. To this standpoint Vives on the whole remains faithful in his purely descriptive procedure. It could not, of course, be otherwise than that much had to stand for him as certain experience that was later to fall before criticism; but Vives nevertheless did much toward letting the psychological standpoint be separated off from the speculative and the theological on the one side and the physical on the other, a separation that Descartes carried out in a clear and decisive way. Descartes owes Vives more than one would think from the particular passages where he mentions him.

As is evident already from the title of Vives's book, he places the concepts of soul and life in close connection with each other. He conceives the soul as the principle of life, the principle of all organic life, not merely as the principle of the life of consciousness.
[34] He endeavors to conceive the lower forms of life as the foundation of the higher. His psychology is a physiological psychology, which lays great weight on describing the phenomena, as far as possible, from the physiological side as well. Here, naturally, he uses the physiology of his time. He shows himself an inquirer emancipated from the Aristotelian school by referring cognition to the brain. There were still Aristotelians who denied that the nerves (as the later Greek anatomists had already shown) issued from the brain and not from the heart. According to Vives, on the other hand, the vital force has its seat in the heart, where the first and the last sign of life is given, and whose hampered or free function finds expression in the emotions. The brain he takes, in agreement with a doctrine handed down from the Stoics and Galen, to be filled with fine air or breath (spiritus tenuissimi ac pellucidi). Its vibrations were supposed to stand in connection with what went on in consciousness; but Vives does not express himself more precisely on this. What the relation is between the soul that acts in the brain and the one that acts in the heart, he likewise leaves undecided. In his view it was only the properly human soul that was created immediately by God; the plant soul and the animal soul (that is, the principle of organic life and of sensation) are produced by the force of matter. The human soul, in thought and aspiration, is not content with the finite and the sensible, but seeks an infinite object; from this its divine origin is derived, for the cause must correspond to the effect. Vives shows himself here a decided spiritualist, as one would also have to expect from his religious presuppositions. On this point too he leaves great problems for the thinking that followed.

His psychology confines itself to setting out the various mental phenomena. Only later inquiry sought the principles of the mechanics that could lead over from one phenomenon to another.

6. The Development of Natural Law

Italian humanism had an intellectually aristocratic stamp. Its great significance was the founding of a free life of the spirit. But it did not much concern itself with how things went with human life in the wider circles. Church, state and social life it left to look *[35]* after themselves; it was essentially intellectual and aesthetic tasks that it busied itself with. Even Machiavelli forms no real exception to this, in spite of his great interest in nationality and politics; for it is, after all, the ruler's display of force that most draws his gaze, and the deeper-lying forces and conditions of social life he does not heed. Humanism fell when political freedom fell in Italy; but that political freedom fell, and that no national political life developed, was owing to this narrow conception of the human in humanism, its fear of going into depth and into breadth. This was also the reason why it left the religious question alone and was content to push it aside. The northern nations, among whom the Renaissance appeared with less brilliance, carried the liberation deeper in and farther out.

It was the great merit of the *Reformation* that it was not content to go around the religious question, but went directly into it and in the religious domain proclaimed the same principle of personality that humanism had pronounced in other domains. The Reformation is the application of the thought of the Renaissance to the religious domain. Not to be understood as if Luther and Zwingli had first appropriated that thought and then applied it. What is great about these personalities is precisely that in their own experience of life they discover that thought anew and cast it in an altogether peculiar form. They assert the immediate personal experience of life itself as the real foundation of religion, and, supported by it, they wage their struggle against the medieval Church and theology. The inner, human forces were freed from the artificial forms. Christianity was here really — to use Machiavelli's expression — led back to the principle from which it had originally issued. Even if *Luther* did not enter on a critical investigation of primitive Christianity, he nevertheless seized on an important point in the notions of the earliest congregation, building as he did on the Pauline doctrine of justification by faith. It was the inward, personal adherence to Christ by which man was raised above all external relations. This thought he developed especially in his writing "*Von der Freiheit* *[36]* *eines Christen-Menschen*" (1520). When personality was thus freed from all external authority in its innermost relation, the one on which its eternal destiny was supposed to depend, it could not but seem natural to carry out a similar liberation in other domains as well. Nor can it be denied that the Lutheran Reformation had an awakening effect in many respects. In his struggle against the Church Luther was also bound to come

to speak for the right of what is naturally human. He asserts the independence of the family and the state over against the Church, and above all he asserts — in natural connection with his doctrine of justification by faith alone — the significance of the conduct and activity of ordinary human life over against Catholicism's undervaluation of it in relation to asceticism. But a positive relation to the natural human interests Luther could not, after all, ground. Toward science he stood — especially in his earlier period — suspicious, by reason of his principle of faith, and in political life he held to the principle of passive obedience. For when he wrote on "the freedom of a Christian", he divided his consideration into two parts: first he shows the inner freedom that the Christian has, then he refers the full unfolding of this freedom to the world beyond and maintains that the Christian, according to his outer man, is precisely "ein dienstbarer Knecht und Jedermann unterthan". In this dualism between the inner and the outer lay the reason why Lutheranism did not come to play the role in the spiritual and political development for which there might in itself have been the possibility, given the great and powerful personality of its founder. The religious and the human were merely placed side by side like Sunday and weekday, without inner, essential connection. And so it went in particular with the relation of faith to science. At Worms Luther appealed to the Bible and to clear, evident reasons. But as to the mutual relation between these two principles, no clarity was reached[5]), except in so far as it can be called clarity that a scholasticism was soon developed in Lutheran theology which was less grand and more narrow-hearted than the medieval one.

Yet at Luther's side there stood a man who with quiet enthusiasm sought to unite
[37] the thoughts of the Reformation and of the general Renaissance. Philip Melanchton, "the Teacher of Germany", represents, in contrast to Luther, a more rational side of Protestantism. He was not only a theologian, but a philologist and philosopher, and when the harsh condemnation of unregenerate man by Luther and the Lutherans grated on his ears, it was presumably not only because it conflicted with his mild conception of the human in general, but especially because it fell upon his beloved classics. At the University of Wittenberg Melanchton lectured not only on theology, but also on natural philosophy and on philosophical sciences such as psychology, logic ("dialectic") and ethics. An elegant presentation and a great knowledge of ancient literature distinguish these lectures. Of greatest interest is his ethics. In natural philosophy he polemicizes (as will be mentioned later) against the new Copernican doctrine, and as a psychologist he cannot measure himself against Vives, whose work is roughly contemporary with his. Of great significance for his ethics was his doctrine of *the natural light,* in which he is

especially influenced by Cicero's rendering of the Stoic philosophy, though in such a way that he also relied on the Apostle Paul's words about the law as written in men's hearts. At the basis of all inferring, counting and calculating, of all acceptance of the first principles of the sciences and of all moral judgment, there lie certain representations implanted by the Deity and innate in everyone (noticiæ nobiscum nascentes, divinitus sparsæ in mentibus nostris), which make it no accident that scientific knowledge and moral valuation never die out in the human race. (This doctrine Melanchton develops partly in his *Liber de anima,* partly in his *Erotemata dialectices).* This natural light was, however, darkened by the Fall, and therefore its essential moral content had to be proclaimed in the ten commandments on Mount Sinai. This assumption, that it is the same content that is proclaimed in the ten commandments and in the natural moral law, was common to the Reformers. It is found in both Luther and Calvin, and it goes far back in the Middle Ages. Through this doctrine, renewed by Melanchton, it came to stand firm that one could arrive by the way of clear reasons at the same result as the Old Testament revelation. But although Melanchton thus gives the natural law, which for him meant both the moral law and the highest principle for all relations of *[38]* right, an independent, natural and rational grounding, it nevertheless followed from his theological presuppositions that the ethical conduct of life possible on this foundation could concern only the outer, civil life, not the innermost stirrings of the soul. He defines moral philosophy as "the knowledge of the commandments concerning all moral actions (honestis actionibus) of which reason sees that they agree with human nature and are necessary in civil life, so that one seeks, as far as possible, to find the origin of these commandments by scientific means". The most inward and deepest element in mental life was thus, in the Reformer's view, not to "agree with human nature," but could only be set in motion by supernatural influence. He does indeed hold (in his *Philsophiæ moralis epitome*) that philosophy can teach that there is one God, that he ought to be worshipped, and that it is God's will that upholds the great difference between good and evil. The first three commandments fall within moral philosophy just as well as the last seven. But the real, inward relation to God, that "in which we have to do immediately with God," falls outside philosophy. Here the error shows itself of setting the content of the ten commandments equal to the natural law. For it was surely hardly Melanchton's meaning to deny to Old Testament piety an immediate and inward relation to God. And on the other hand, that equation bore witness to the theological prejudice that what is most inwardly human could be thought only in religious form. For Melanchton the humane was only the outer, civil sphere. We meet here again, even in this humanist, the

opposition between the inner and the outer. But the concession that had been made, namely that human life in its outer relations could be ordered by laws that could be rationally grounded, was nevertheless of great significance. Civil life, the whole life of the state, could then unfold independently according to its own purely human principles.

And in Melanchton's school this independence was developed still more sharply than had happened with the master himself. The Danish theologian Niels Hemmingsen, who has been called the Teacher of Denmark, just as his teacher Melanchton was called that
[39] of Germany, set out in his writing *De lege naturæ apodictica methodus* (1562) the demand for a strictly scientific development of natural law. As he himself states at the close of the writing, he has deliberately kept all theological considerations away, so that it may become clear "how far reason can reach without the prophetic and apostolic word". He wants by way of inquiry to gain a clear and distinct knowledge of the essence of right. Only that is a true law which does not merely rest on the authority of the prince and the magistrate, but has its "firm and necessary ground", which cannot be sought anywhere else than in the very nature and purpose of the law. In human nature there are given germs of the moral and the just, and besides these a judgment for distinguishing the just from the unjust. These means one must use if one wants a law in the true sense of the word. Hemmingsen then shows that there is in man's being a sensuous and a rational element, and that the animal drive awakened by sense-sensation is to obey reason, which contains the true human law of nature (lex naturæ seu recta ratio). The purpose of human life is the knowledge of the true and the practice of the good. Above family life and political life there rises the spiritual life (vita spiritualis as opposed to vita oeconomica and politica) in the worship of God, in which the highest purpose is to be sought. At this point the natural forces no longer suffice, for no one can know the right way of worshipping God except by divine revelation. But as far as the knowledge of natural law itself is concerned, revelation is not needed. Hemmingsen, like Melanchton, does regard the ten commandments as a brief epitome of natural law, (epitome legis naturæ), but only after having developed the natural law does he examine its agreement with the ten commandments. Here, then, the advance has been made that ethics and the doctrine of right ("natural law" comprises both) are independent of theology and do not rest on supernatural authority.

But with this the movement also comes to a halt within Lutheranism. Individual jurists such as Oldendorp and Winkler did indeed hold fast to the thoughts that had been developed. But it was only in the Reformed countries that the inner and outer conditions
[40] were present for a more thoroughgoing liberation of ethics and the doctrine of right.

In Zwingli and Calvin there was not that anxious separation between the freedom of personality in the inner life and bondage in everything outer. Here the break with the principle of authority is far greater than in Lutheranism. The man made free by virtue of election by grace feels his right to self-determination in all spiritual and worldly things. Zwingli, that clear, powerful nature, in whom the reformer's religious inwardness was joined with the thinker's consistency, with the humanist's love of antiquity and with the republican's political impulse toward freedom, spoke for self-government in both the civil and the ecclesiastical domain. And Calvin continued his work, with less talent and greater narrowness. It was their thoughts that lay at the basis of the great struggles in France, the Netherlands and England in the sixteenth and seventeenth centuries, the struggles in which the civil, religious and scientific freedom of modern times came into being. The modern state and modern science owe their existence to these struggles. What Italian humanism, from softness and intellectual aristocracy, and what Lutheranism, from an anxious impulse toward inwardness and obedience, could not lead to, the disciples of the Swiss reformers accomplished with the weapons of the spirit and of the hand. Here too, indeed, a hard struggle had to be fought out to burst the confessional barriers within which the Reformed Church, just as well as other churches, sought to hold the spiritual movement. But it was of great significance that this Church had a bolder conception of the relation between the inner and the outer and did not set the innermost personal life in sharp opposition to all human interests. Thereby the Reformed countries became the second home of modern philosophy, after its first home, Italy, had been subjugated by the powers of reaction. And thereby modern natural law in particular had its origin in these countries: the conception of right that lies at the basis of all political and social endeavors at reform, right into the nineteenth century.

The wars of religion in the Netherlands and in France, and the publicistic literature to which they gave occasion, became of great significance for the development of the idea of popular freedom and of the independence of the state over against the Church. The struggle to assert religious freedom was closely bound up with the struggle to preserve *[41]* civil freedom. The right of subjects, through their lawful representatives, to resist and in the extreme case to depose a prince who violated the right of the people was asserted with great force. And as its ground the claim was put forward that all governing authority springs from a transfer that is conditional on the fulfillment of definite duties on the part of those who govern. Thus the principle of popular sovereignty was proclaimed, and the idea of an original contract between people and government was laid at the basis of all discussion of constitutional law. The contract theory had already been used in

the medieval strife between Church and state as a weapon against the Church's claims to supremacy. The memory of the origin of the Roman imperial power in the Roman democracy here worked together with consequences that were drawn from the notion of a paradisal state of nature without relations of right or of the state. Now this theory was used on the side of the people against the princes, whereas in the Middle Ages it had been employed in the service of the princes against the Church. A series of Protestant publicists, in the first rank Hubert Languet and François Hotman, built their writings of agitation on it. But it was also taken up on the Catholic side. Over against the power of princes — especially where it was lukewarm or heretical — one found one's account in appealing to the right of the people, which was derived from an original contract. In this spirit a series of Jesuit writers, in part men of great acuteness and energy, expressed themselves. The Church was then to stand as the only society that had arisen from above, whereas the state had come into being from below.

The wars of religion, both the physical and the literary, were naturally bound to lead patriotic men who were concerned for the subsistence of the state, threatened as it was by the struggle of the confessions, to fix their gaze on the conditions of political life that were independent of the disputed religious assumptions. There then formed in France, in opposition to Huguenots and Leaguers, a party of "politicians", whose view was expressed in literature by Jean Bodin in his book on "the State" *(La republique),* which appeared in 1577. This remarkable man, who by his superstition recalls the darkest
[42] side of the Middle Ages, while his political and historical conception points far beyond his time, was a French jurist who for a time stood in the favor of Henry III, but later forfeited it by speaking for the rights of the estates and of the people over against the king. For a time he stood close to the chief of the "politicians", the Duke of Alençon, and was later among the first to join Henry IV, when the latter ended the civil wars by gathering the contending parties under the protection of tolerance. He died, however, two years before the Edict of Nantes appeared. In the history of the doctrine of right his significance rests on the clear and consistent way in which he developed *the concept of sovereignty.* He distinguishes definitely between sovereignty and government. The highest power is not one with sovereignty. For power can be transferred for a time. And power can be divided, whereas sovereignty is indivisible. Just as little as there can be several gods, if God is to be an absolute being, just so little can there be several sovereigns in a state. Bodin distinguishes between the form of the state and the form of government. The difference between the forms of state rests on who has sovereignty. And the marks of sovereignty are legislative power, the right to declare war and peace,

the right of pardon and the right to appoint the highest officials. Sovereign authority can lie with the whole people, with the aristocracy or with the prince. But the form of government can quite well be monarchical although the form of state is democratic; such is the relation where the whole people elects the king.

Bodin felt a need, over against the Protestant and Catholic stirrers-up of unrest, to insist on the concept of absolute sovereignty. Yet he is not for that reason a denier of the right of the people. Sovereign authority is subordinate to morality and natural law; it cannot abolish the personal right of property, and therefore it cannot impose taxes either. Bodin did not see that legislative authority could not in the long run be maintained without the authority to grant taxes, and that when he placed the latter with the people he in reality also transferred sovereignty to them, so that the difference between the various "forms of state" fell away and only the difference between the "forms of government" remained. This consequence was first drawn from the concept of sovereignty by his successors. *[43]*

Bodin's political theory is distinguished by the weight he lays on the family and the smaller societies or corporations. He demands that their independence and peculiar character be preserved and developed, in so far as they are compatible with the common tasks of the state. He has on the whole a sound eye for real human life in its various directions, and he reproaches Machiavelli, of whom he says that he "has never sounded the waters of political science", for his belief in tyrannical subtleties as the most important thing in politics. In a special little writing (on the method of history) he shows the significance of the comparative and historical method for the doctrine of right. "In history," he says, "the best part of universal law is contained, and what is very useful for a correct judgment of the laws is that from them we can come to know the manners of the nations, the foundations of the state, its course of development, forms, revolutions and downfall." He holds that by this way one must get away from the assumption of a golden age: "in comparison with our time the so-called golden age would stand as an iron age!" His belief in progress he builds especially on the development of industry. He strongly emphasizes the inventions and discoveries of modern times, as well as the livelier intercourse between the various nations. Yet he distinguishes between progress in culture and progress in morality; in the latter respect his belief is not so confident. — Of his writing on the philosophy of religion and its significance there will be talk later.

The one who first really founded the theory of natural law in modern times was Johannes Althusius. This man, whom it is GIERKE'S merit to have drawn forth from a most undeserved oblivion, was born in Diedenshausen in 1557, studied in Basel and

probably in Geneva, was for a time professor of law at Herborn and later, through a long series of years, burgomaster of the East Frisian town of Emden, whose rights he zealously defended against the encroachments of the East Frisian counts and knights. He was an ardent Calvinist and followed with enthusiasm the struggle for religious and political freedom waged by the Netherlanders, kin to the East Frisians. In his writings he alludes several times to this struggle. In Friesland the old freedom of the peasants had
[44] survived, while in the rest of Germany it had perished, and Althusius accordingly asserts the rights of the peasant estate alongside those of the nobility and of the burghers. Thus both his religious faith and his studies, both the events of the day in the neighboring country and the social conditions in his own narrower fatherland, combined to nourish in him the view in the philosophy of law and of the state that he developed in his writing on politics *(Politica methodice digesta atque exemplis sacris et profanis illustrata.* 1603). Of his other writings, besides several purely juridical ones, mention should be made of his writing on the virtues that find application in intercourse with other men *(Civilis conversationis libri duo).* He died after a long and active life in the year 1638.

Althusius agrees with Bodin that sovereignty is one and indivisible. There can be only one sovereign in a state, just as there can be only one soul in a body. But he polemicizes against Bodin's assumption of several forms of state. Sovereignty can lie in only one place, and it cannot be transferred or alienated. Princes and aristocrats can exercise government, but never possess sovereignty; it must have its seat in the whole people, which never dies, and to which every ruler is answerable for his administration. It is the welfare and need of the people that the state is to further and help; here lie the cause and the purpose of the state. The individual ruler dies, but the people does not. And the ruler is only the single man, but the people is the many. From the people, therefore, all authority must go out, and to it it constantly returns.

Only the form of government, not the form of state, can differ. It is the people bound together by common conditions of life (corpus symbioticum) that is the source of all authority, and that cannot permanently hand over its power to any magistracy whatever. All who exercise governing authority stand as representatives of the people, and as soon as they overstep the limits that are set to their activity by the natural law contained in the ten commandments and by the welfare of society, they cease to be servants of society and become private men, to whom one owes no obedience in that in which they overstep the limits of their authority. Yet it is not every individual's business to assert the limit of
[45] the power of the magistracy. In every well-ordered state there is, besides the magistrates, yet another kind of "administrators", namely overseers (ephori), whose office it is to

elect the supreme magistrate, to assert the inalienable rights of the people, to remove the magistrate who violates them, but also to defend and support every magistrate within the limit of his authority. Such ephors were known both in the states of antiquity, as senators and tribunes of the people, and in modern states as the estates, electors and councillors. Under one form or another a supervisory authority must always be organized[6]).

A historical proof of his idea of the sovereignty of the people Althusius finds not only in the fact that there is thus in most states an authority which, on behalf of the whole people, is to criticize the conduct of the supreme magistracy, but also in the fact that the people has thrown off the rule of a tyrannical prince. He refers to the Netherlanders' struggle for freedom, and in the preface to the third edition he states that he has dedicated his work to the estates of Friesland because they, like the other Netherlanders, had seen that sovereignty could not be inseparably bound to the person of the prince, but that he could have only the use of it, while the real right of majesty belonged to the people, — and because they had fought for this principle with such great bravery, good sense and endurance.

A philosophical proof he finds by looking to the purpose and the cause of political life. The purpose is, after all, the welfare not of the magistracy but of the whole people. And the cause of men's joining together and forming states is the helpless and needy condition of the individual. The feeling and need of the *individual* persons are the ultimate cause of political life. First, narrower societies are formed, such as the family and the neighborhood. These narrower societies are the nursery of the larger societies, which are formed by the union of several small ones. Then a political, not merely a private, society is obtained. The state is the universal society, which includes all the smaller ones within itself. Althusius here recalls Bodin, who, as we saw, laid such great weight on the significance of the smaller societies within the large. It is the people, represented in the various circles of society and by the ephors, that is the holder of *[46]* sovereignty. On this point these older writers have a great advantage over the teachers of natural law of the eighteenth century, who really knew only the single individual and the state as a whole. It must be added, however, partly that the corporations had by then shown themselves bound up with many abuses, partly that absolute monarchy in France and its imitations elsewhere had done everything to press the intermediate links lying between the individual and the state down to as great an insignificance as possible.

Society in its simplest form, which lies at the basis of all others, has arisen by agreement, which in turn has its motive in human wants. To understand rightly the contract theory thus put forward, one must consider somewhat more closely how Althusius

conceives the agreement that is to lie at the basis of all social life. It is need that leads to it. But Althusius seems not to conceive this need as purely egoistic. Especially where he speaks of the origin of the family and of the community of neighbors, he uses as explanation not only "necessity" but also "natural feeling" (naturalis affectio). He agrees with Aristotle's statement that man is a "social animal" in a far higher degree than bees or other animals living in societies. It remains unclear how that "natural feeling" and this social inclination are related to individual self-preservation. Althusius felt no need of a closer psychological analysis. Yet it would have lain all the nearer to hand for him in that he confines the purely individual compact to the simplest society and lets the other societies grow out of it. The assumption of a natural growth of society obviously stands in a certain opposition to the immediate arbitrariness of the single individual. Even if, according to Althusius, it is the will of the individuals who found society that lets society come into being, it is nevertheless, by his own doctrine, only the first push that they can give; later on they appear not as individuals but as members of corporations, not as independent atoms but as elements in a compound molecule whose motions do not depend on the single atom alone. There are no *pure* individuals; all individuals
[47] are involuntarily socially determined. From the beginning, the contract theory, by its assumption of pure individuals, stands in opposition to the historical conception of the development of social life, which regards individuals as the *products* of society, not merely as its *active forces.* Althusius wanted to unite both points of view. Here too the decisive problem came forward later.

Connected with this is another question: was the original contract that founded society entered into consciously or not? It is clear that only if it was entered into consciously can it be meant as a real historical event, taking place at a definite time and place. Althusius seems not to conceive the matter so. Society, he says, is formed by an *express or tacit* compact (pacto expresso vel tacito). But what lies in a "tacit" compact? The assumption of a contract is, according to Althusius, necessary not only for understanding the origin of society, but also for understanding the transfer of governing authority to the magistrates. There is thus first a social contract, then a contract of rulership, the former between the individuals among themselves, the latter between the whole people and those who are to govern. With regard to this latter contract Althusius now says (and it must hold for the first contract too) that the rights the people has not expressly transferred to the prince it keeps for itself, and that even where the people submits to a magistracy without express conditions, the general demands of equity and justice and the content of the ten commandments must nevertheless be understood as

implied: "for it is not probable that the people, to its own ruin, has given the prince unbridled freedom, and the people has not wished to deprive itself of the capacity for self-preservation or to give one individual a power greater than that of all the others together." In other words: Althusius infers from what must *reasonably* be the people's meaning to the content of the *original* contract, and he therefore presupposes that the people behaved reasonably when the contract came into being. But unfortunately the people behaves reasonably always just as little as the princes do, and the justification for assuming a "tacit" reason on that occasion is therefore unprovable. It is evident, then, that that contract (especially in its "tacit" part) rather expresses Althusius's ideal *[48]* of a society than a historical event. Natural law, and especially its contract theory, is a mythological expression of the ideal demand that must be made on the order and development of society. It places its ideal of the future back in the past. It is perhaps a question whether most of the teachers of natural law really conceived the original contract as a historical fact. Bodin already saw that the idea of the golden age rested on an illusion, and so it goes with every notion of an absolute state of nature. The great significance of natural law rests on its having had, in the idea of the original contract, a form in which the right of the individual over against society and its authorities could be asserted with clarity and emphasis. But it was only Kant who taught that there is a difference between a guiding, regulative idea and something factually given. With the help of this distinction one can see at once the defects of natural law and its great significance. —

Bodin and Althusius continue Machiavelli's work of making political theory a purely human science. Althusius does rely, as the very title of his work shows, on biblical history just as well as on profane history. But it is only purely historical examples that he takes from it. He does not operate with theological arguments. He assumes with Paul that all authority is of God; but God's working is here not immediate but mediate: the forces and drives that lead to the formation of society and the installation of authority stem from God through nature. The authority of the prince and of the ephors stems *immediately* from the people, *mediately* from God. Not to the prince alone, but to the ephors or estates as well, has God committed the care of the state. The theological hypothesis has thus been pushed further into the background. As an ardent Calvinist, however, Althusius sets the furtherance of religion as a chief purpose of the state. He declares himself decidedly against religious freedom, with reference to Deuteronomy 13, 12, which demands the death penalty for him who speaks for another faith than that of the fathers. Religious freedom, he holds, would make faith uncertain and abolish the unity that ought to reign

in the state; he that is not with me is against me. Althusius's theological standpoint also
[49] comes to light in his denying the independence of individual ethics over against theology. What the individual is obliged to do he learns from the ten commandments, in so far as it does not belong to what political theory and the doctrine of right enjoin. What then remains for ethics is only the rules for worthy conduct in word and deed, which Althusius treated in his writing on "cultivated social intercourse" (civilis conversatio). He thus consistently holds fast to the opposition, fixed by Melanchton, between the inner and the outer man, although he cherishes a disquieting interest in outer human life and considers its ordering by purely natural laws possible. He recognizes the natural forces in social life with an energy that would have aroused offense in Melanchton. What the Emden councillor formulated in his "Politics" were ideas that had been fostered during the Dutch war of liberation and that later, after passing through Rousseau's passionate mind and being sharpened there to their extreme formal consequence, came to make up the circle of dogmas that ruled the French National Convention. It was the gospel of political freedom. Melanchton and his school would never have admitted that there was such a gospel. In glaring contrast to this gospel stands Althusius's own rejection of religious freedom, a testimony to the inner discord in the line of thought that had been inherited from the Middle Ages and that even the boldest tendency within ecclesiastical Protestantism could not get beyond.

That it was not Althusius but Hugo Grotius who for the age that followed came to stand as the real founder of natural law in modern times had its reason partly in the radical, democratic standpoint that the author of the "Politica" represented, and which was most different from the one taken by the leading statesmen and teachers of constitutional law of the seventeenth century, partly in the larger and fuller historical grounding that Grotius gave the principles of natural law, and partly, not least, in the fact that Grotius, in his famous work "On Law in War and in Peace" *(De jure belli ac pacis* 1625), took as his point of departure the question of the justification of war and of the law that holds in war. Thereby he tied natural law to the law of nations and started from
[50] a phenomenon that occupied the thoughts of statesmen and touched the interests of all. It was not the inner relation between magistracy and people, but the outer relation between sovereign states, that was the point of departure for his consideration. What put the pen in his hand was the sight of the many wars that were often begun without reasonable ground and that let loose all wild passions, as if men, once war had begun, were transformed into raging beasts. He was then led to inquire how far all right must really vanish here, and this question led him back to the ultimate foundation of right.

Hugo de Groot was born in Delft in 1583. In his student days at Leyden he already won great renown for learning. At sixteen he accompanied Oldenbarneveldt to Paris, where Henry IV is said to have presented him to his court as the prodigy from Holland. He early attained high posts in his fatherland as jurist and statesman. When Maurice of Orange, with the help of the orthodox clergy, overthrew Oldenbarneveldt and had him executed, Grotius was sentenced to imprisonment for life, from which he escaped through his wife's stratagem. He now took up residence in Paris, where for a time he received support from the French government. Here he worked out the work to which his name is attached. Later he became Swedish ambassador in France. He died in Rostock in 1645, on the way home from a visit to Sweden.

Grotius remarks that it is easier to say what wrong is than what right is. Wrong is everything that conflicts with the nature of the social life that is led by rational beings. — It is seen here that Grotius stands nearer to experience and history than his predecessors. For that human social life rests on definite conditions is seen best when these are lacking; there is then a kind of experiment before us. If those conditions were always present, one would not notice them. — It is an involuntary need that leads man to live together with others. Even if we did not directly need others, we would still seek intercourse and fellowship with them. It is true that we cannot do without the help of others; but the real ground of social life does not lie in that. Already in children one can trace an inclination to do good to others and to have pity on them in misfortune. This natural *[51]* impulse (appetitus societatis) is developed and clarified when fellowship with others becomes, by the help of language, a spiritual intercourse, and when man acquires the capacity to lay general principles at the basis of his knowledge and his action, so that he treats similar cases in a similar way. In the care to preserve a social life of this kind natural law has its origin. Human actions are now judged by their agreement with, or their conflict with, what such a social life requires. Natural law thus springs from principles that lie in man himself (ex principiis homini internis), and would hold even if there were no God. Yet God can be called the author of natural law, since he is the author of nature and has thus willed that this law should hold.

The first and most important provision of natural law is that agreements and promises are to be kept. This is contained in the very original will that brings society about. Every obligation has its ground in a preceding agreement or in a given promise. Positive law (jus civile) here springs from natural law. For social life could not subsist if men could not bind themselves to one another. The whole of civil life here rests, then, on an eternal and unchangeable justice, which holds for God as well as for all rational beings. Human

nature, which is the mother of natural law, is thus the grandmother of positive law, since the source of positive law is the necessity, grounded in natural law, of standing by one's word.

But what, then, is it that gives the principle of the keeping of promises and compacts its original force? Here it appears how the contract theory lies at the basis in Grotius as well as in Althusius. The very principle that compacts are to hold is in fact laid down by an original compact! Only this original compact need not have been entered into with full consciousness. It can be tacit, understood, lie in the nature of the circumstances, that is to say: it is only another expression for the will to keep society up, which in turn is
[52] the condition of all social life. Society is regarded as the product of a free association, in which each sets in his will as part of the common will. All particular relations ought to be judged by this original will (ex primæva voluntate). The state is a body that has a voluntary origin (corpus voluntate contractum). From this is derived, among other things, the important provision that the right of the individual can never wholly vanish over against society: it surely cannot have been his will to annihilate himself! He cannot give up the right of self-preservation that he had before he entered society.

As in Althusius, the mythological notion of the state of nature and of its replacement by political life through an act of will appears here. Yet where Grotius, with the help of this notion, derives special relations of right (such as the injustice of lying[7]), private property, prescription, customary law), he readily uses the idea of a *tacit* agreement, an expression of what the beneficial ordering of the relation requires (aut expresse, aut ex negotii natura tacite promittere, silentio convenire), so that it may be doubtful in his case too how far he conceived the social contract as a historical event. But it is remarkable that, although Grotius starts from wrong and the absence of peace as the leading facts, he nevertheless has a very idyllic conception of human life as it would look if it were not ordered by society. He conceives the social need as direct, disinterested, and its relation to self-preservation remains, as in Althusius, unclear. Only Hobbes here brought the stir from which greater clarity about the psychological presuppositions of right could arise.

Grotius parts from Althusius in maintaining that the people, after it has been constituted by the social contract, can transfer sovereignty permanently to a prince or an estate, just as well as the individual can give himself into slavery. He polemicizes directly against Althusius's doctrine of popular sovereignty, although, for what reason is unknown, he does not name him. He insists that the question here is not which form of state is the best — something on which there can always be different opinions, — but the
[53] will that is the source of right (ex voluntate jus metiendum est). And what should stand

in the way of the people's being able to deliver itself wholly into the power of one or of several? Even in a democratic state the minority must, after all, subordinate itself to the majority, and women, children and the poor have no share in the decisions, and so none in sovereignty either. The original will can produce a necessity that binds for all time. It is of no use to say that the state exists for the sake of the whole people, not for the sake of those who govern; for a guardianship exists for the sake of the wards, not for the sake of the guardian, and yet he has authority over them. — By thus giving the contract by which power is transferred to those who govern a different character from the one by which the people is constituted as a people, Grotius becomes the forerunner of Hobbes; it thereby becomes possible for absolute monarchy, too, to rely on the contract theory. His arguments do not, however, shake Althusius's claim that it is hardly probable that the people should have wished to sign away its freedom for all eternity. Althusius's conception obviously agrees best with the contract theory, both in its mythological and in its rational meaning.

It is, as already remarked, war that gives Grotius the occasion for his inquiry into the concept and origin of right. War can, in the first place, take place between individuals among themselves. But on the presupposition of a social condition and a political life, just war between individuals is confined to self-defense, and the power of punishment is in the hands of the government. In the second place, war can be raised by the individual against the state. This is rebellion, and its inadmissibility stands firm for Grotius, who here too polemicizes against Althusius. He holds that those who have granted the representatives of the people a right of rebellion have accommodated themselves too much to the circumstances of a particular time and place. Grotius himself, on the other hand, seems to have taken too little account of his countrymen's struggle against their ruler, a struggle grounded on the "right of nature". In the third place, war can be waged on the part of the state against the individual. Such a war is just when it is directed against one who has himself done wrong. The inquiry into this point thus leads to an inquiry into the nature and justification of punishment. Grotius treats this point in a *[54]* very interesting way. He is a decided opponent of the theory of retribution. Punishment does not have its purpose in itself; only a savage and inhuman mind can find satisfaction in any pain whatever being inflicted, when no further good is thereby attained for the person himself or for others or for the whole of society. In the fourth place, there can be war between states among themselves. This subject is treated at the greatest length. The conditions for the justice of a war and the considerations of humanity that ought to be observed during war are derived from the principles of natural law. The chief principle

is that men do not cease to be men because they become enemies, and that war, which exists only for the sake of peace, must be waged in so honest and humane a way that it does not make a peace impossible. Grotius is fully aware that the law of nations lacks the sanction that the power of the state affords within the single people. Great states seem able to permit themselves anything. But even they need ordered intercourse with other peoples, and the public opinion of humanity cannot be indifferent to them, just as war, too, can be waged with vigor only when the people itself believes in the justice of the cause. By this insistence on the demands of humanity and by pointing out the analogy between the conditions of intercourse among peoples and the conditions of intercourse within the single people, Grotius made an extraordinary impression on the statesmen of the seventeenth and eighteenth centuries. Gustavus Adolphus is said always to have had his book with him in his tent. It is said to have been Grotius's authority that kept Louis XIV from making good his threat of giving no quarter in his war with the Dutch, just as the indignation that the devastation of the Palatinate aroused would hardly have risen with such force in advance of the temper that Grotius's work had awakened. And to this very day the instructions of most states for their officers in wartime are in essential part built on the rules set up by Grotius[8]), which it is not, however, our task here to follow out in detail. The point was only to show that the famous philosopher of law and teacher
[55] of the law of nations has his place among those who discovered the human.

An important question in constitutional law and the law of nations at that time was how far men ought to be punished for their religious opinions, and how far one state might make war on another for the sake of religion. As to the first point Althusius stood at a narrowly confessional standpoint, and the wars of religion bore witness to what the general view of the second point was. Grotius begins by asking what is to be understood by the true religion. It must be the one that is common to all times, and its content he finds to be: the unity and invisibility of God, God as the author of all things and as the one who cares for everything. These assumptions, he holds, can be proved in a natural way, and their general diffusion testifies moreover to ancient traditions. At least the existence of God (whether one then assumes one god or several) and the government of providence are self-evident, generally diffused and morally necessary truths. Those who deny them violate human society and ought to be punished. Another people should be warred against for religious reasons only when its worship of God has an immoral and inhuman character. When it is asked whether one should make war on a people because it will not accept Christianity, Grotius answers that, apart from the quality of the Christianity that is offered it, the rejection of a faith that rests on historical testimony and

arises only through supernatural influence cannot fall under human judgment. There is a wider horizon here than in Althusius, and a natural theology is set up. On this point Grotius shows himself akin to the endeavors that we now pass on to discuss.

7. Natural Religion

It could not be otherwise than that the powerful spiritual movement of the Renaissance, and the widening of the spiritual horizon that took place, must reach into religious life. The Reformation itself is already evidence of this. But the Reformation is at the same time precisely an attempt to set definite limits to the religious movement. Among the most remarkable phenomena of the Renaissance, especially toward its close, belongs on the other hand the endeavor to find a religion laid down in human nature itself and *[56]* independent of external forms and traditions. And just as with natural law, two ideas here go hand in hand: for what really has its ground in human nature must also be found again in the various historical forms under which the human life of the spirit appears. The comparison between the various contending confessions, whose number was, after all, being increased precisely at that time, worked together with the newly awakened belief in the self-sufficient power of human nature. Already within Italian humanism there appear endeavors to develop a circle of religious ideas that was to be independent of the doctrine of the Church. It was above all Plato's doctrine that was taken as the foundation, as it was conceived by the Neoplatonists. The Platonic Academy in Florence became of great influence here, beyond the borders of Italy as well. In the heartfelt faith in a personal God and in the immortality of the soul the individual found a satisfaction of his religious need which at once set him free with respect to so many of the dogmas and ceremonies of the Church and yet made it possible for him to find a deeper symbolic meaning in them. In Germany and the Netherlands this humanist current was joined by the mystical tendency, which already in the Middle Ages had worked in quiet for a more inward and freer religiosity. The Swabian Sebastian Franck and the Dutchman Coornhert came forward in this spirit against the new Protestant dogmatism. They pointed to an inner religious experience that can be lived through under very different confessional forms. For Franck the invisible Christ was one with the natural light, whose spark is found in every human mind and which could therefore be kindled among the heathen too. One can, said Coornhert, follow the law of Christ without knowing his name. These two men are forerunners of the philosophy of religion of a later time, which sought the understanding of positive religion and of its significance not by external criticism of the finished dogmas, but by immersion in the spiritual ground of nature and

in the inner experiences that are the ultimate source and constant abode of all dogmas. They themselves were made to feel confessional fanaticism. The name by which it was
[57] customary to designate those who took an independent stand toward the confessions was attached particularly to Coornhert, whom the Dutch preachers called "the prince of the Libertines"[9]).

Natural religion was developed into still more definite forms in France and England. The long and violent wars of religion in France could not but awaken the need to get beyond the strife of the confessions and open the eye to the human side of each of them. In the year 1577 Henry of Navarre (the later Henry IV) wrote: "Those who straightforwardly follow their conscience belong to my religion, and I belong to the religion that all brave and good people have". Montaigne had at the same time, by his own path, arrived at the notion of a universal religion that could be found in all, no matter how they conceived the Deity. The most remarkable contribution in this direction, however, was made by Jean Bodin, whose ideas in the domain of political theory have already been brought forward. He left behind a dialogue between seven men *(Colloqvium heptaplomeres),* each of whom stands at his own religious standpoint, which they develop each for himself without reserve, but in the greatest friendliness and with the finest understanding of the personal relation to religious notions. A rich Catholic in Venice gathers a circle of friends around his table for conversations on various subjects, among which the question of religion then comes for a time to take first place. Among the participants, besides the host, who is a Catholic, are a Lutheran, a Calvinist, a Jew, a Mohammedan and two men who, each in his own way, profess a universal theism. Remarkable is the objective, dramatic manner in which the various standpoints are brought forward. Only the Lutheran perhaps stands somewhat in the shade; at any rate he does not shine by acuteness. When one day the host had mixed artificial apples among the natural ones that had been set on the table, the Lutheran guest was the only one who confidently bit into one of them without suspecting anything amiss. The host, who himself always holds to the doctrine of the infallible Church but is no fanatic, draws from this the moral that if sight, the keenest of our senses, can thus be mistaken, how then should our mind, bound to the senses, be able to arrive at certain knowledge of
[58] the highest things! — After all the standpoints have come forward, on several points with the display of great learning, the dialogue ends without a definite conclusion. A choir sings the hymn: "Behold, how good and how pleasant it is when brethren dwell together in unity!" Then the participants embrace one another and part. "Thereafter they practiced piety and integrity in admirable concord, in their life together and in

common studies, but never again held any debate on religion, although each maintained *his* religion in holy conduct."

Bodin wrote this work in his 63rd year, that is, in 1593. It was not published, but was spread in copies and was well known among the learned of the seventeenth and eighteenth centuries. It is mentioned not infrequently, and as a rule with the greatest abhorrence. Only a few, such as Leibniz in his later years, declared themselves in favor of its being published. But only in the year 1841 was it published (in a somewhat abridged translation) by Guhrauer, in 1857 complete in Latin by Noack, and it can now serve as a measure of what, a couple of centuries ago, people were fairly well agreed in calling ungodliness.

Since Bodin himself has not directly given us to understand which of the persons expresses his own views, there have been differing opinions on the matter. For a time people were inclined to think that Bodin sympathized with Judaism, whose representative is drawn with particular force and expertise. But if one attends to what is otherwise known of Bodin's religious tendency, and compares it with what is put forward in the course of the conversations, one arrives at another result. There is extant from Bodin a letter of an earlier time to a friend, in which he expresses himself in the spirit of universal theism as we already know it. "Do not let yourself," he says, "be confused by the different opinions about the religions. Only hold fast in your mind that the true religion is nothing other than the turning of a purified soul to the true God." "This is my religion, or rather Christ's." There follows an account of how men would lie in eternal darkness if exalted men did not from time to time come forward as exemplars. To their number belonged, besides the Jewish patriarchs and prophets, the Greek and Roman sages as well. Plato is especially praised as the one who proclaimed the idea of God *[59]* and of the immortal soul. What he intimated Christ fulfilled, and after him chosen men have worked in his spirit. This same idea of a continued revelation through a series of sages, both before and alongside the Mosaic, Christian and Mohammedan religion, now appears also as characteristic of Toralba, one of the two persons in "Heptaplomeres" who maintain universal theism. He argues that the best religion must be the oldest. The first man received his knowledge and his piety from God himself. Faith in one God goes back, according to the testimony of the Bible, to before the Jewish religion; it was the religion of Abel, Enoch and Noah; it was the religion of Job. When men abandoned this natural religion, which together with reason is implanted in men (cum recta ratione mentibus humanis insita), they went astray in intricate labyrinths. The natural law and the natural religion are sufficient; the other religions — of Christians,

Jews, Mohammedans and heathens — are superfluous. For natural religion one needs no special learning or upbringing; one is created for it, one has it given in one's own nature. The reason that is implanted in us discerns and judges what is good and what is bad. But in some chosen ones this religious and moral capacity appears with greater power than in others. Toralba praises (just as Bodin does in that letter) Plato because he — "not without divine help and enlightenment" — attained so high a wisdom concerning divine things. The essence of the Deity is infinite, cannot be comprehended by thought, let alone expressed in words. One comes nearest to it by calling God the eternal being, exalted above everything bodily, infinite wisdom, goodness and might. And in this Greek philosophers meet with Hebrews and Ishmaelites, so that this idea must be assumed to be implanted by nature.

To this is added yet another trait in Toralba that recalls Bodin himself. However far advanced Bodin was in several respects in relation to his time, in one respect he was nevertheless deeply sunk in it. He believed with a convinced heart in witchcraft
[60] and demonic possession, and he even wrote a book against the ungodly men who recommended caution and leniency in the punishment of witches. It was the devil himself, he thought, who brought forth such mildness toward the possessed. In the second book of the dialogue this point is touched upon in a discussion between Toralba and Senamus, who is otherwise the one among the interlocutors who stands nearest to Toralba. Toralba declares that it is a mistake of the more recent physicists that they want to explain everything by natural laws and recognize no intervention of God or of the demons. Senamus replies to this that all natural science is thereby overturned, since it presupposes definite laws of nature. Toralba adduces several demon stories as proof of his view, and here can rejoice in the full assent of the Lutheran. Senamus replies that even if there were demons who could intervene in the corporeal world, it was still not thereby said that wonderful phenomena were to be explained by them; and on the other hand: if one assumes the intervention of demons, why then does one not explain everything in nature by demons? Moreover he urges the necessity of criticism toward the witch stories.

Toralba's standpoint is Bodin's own, as it is known from his personal utterances. And yet he puts into the mouth of Senamus a criticism that Toralba is unable to refute. Senamus is on the whole the critical spirit in the dialogue. To the question who is to be judge in religious disputes, the Lutheran answers: Christ, who is God! Senamus points out to him that the dispute concerns, among other things, precisely the divinity of Christ. The Calvinist refers to the witnesses and the documents. Senamus asks where the

genuine witnesses and testimonies are to be found. The Catholic refers to the Church, — but Senamus asks which of the many churches is the true one. The Lutheran appeals to the Bible, the Mohammedan to the Koran; Toralba rejects this appeal to authorities and holds that the wise must decide the matter; but Senamus then asks who is wise. And when at last the Catholic wants a distinction drawn between what is essential and what is inessential in religion, Senamus shows that the great dispute arises precisely when this boundary is to be drawn.

Senamus takes the same stand toward Toralba's natural religion as toward the heathen, the Mohammedan, the Jewish and the Christian religion. He thinks that they are all *[61]* well-pleasing to God when they are practiced with a sincere mind. He himself gladly visits all kinds of temples, wherever they are found. He will offend no one: him who believes most, the Catholic, who is "religiosissimus", just as little as those whose religion is merely faith in one God. He thus does not set up natural religion as a particular religion in opposition to the positive religions. In Senamus Bodin has let thoughts come forward that contain a corrective to the dogmatism into which, as history shows, natural religion can lapse just as well as the positive religions. Even if his own standpoint coincides with Toralba's, he has nevertheless intimated a more far-reaching consideration than the latter's. It is not impossible that Bodin in his last years had begun to entertain doubts about his own earlier assumptions in the domain of theology and demonology, and that he let these doubts find a voice through Senamus, without having been able to form a new conception on the basis of what he has Senamus say.

The idea of natural religion was brought forward in fully finished form before wider circles by Lord Herbert of Cherbury, in his book *De veritate* (1624). Like Althusius, Grotius, Coornhert and Bodin he was a statesman who took part in the public affairs of his time and had occasion to gather extensive experience and to make comparisons. He belonged to a noble family in Wales; his forefathers were doughty knights, and he himself did not deny their warlike spirit. War and duels are a main subject of his autobiography. He was quick to get his sword out of its sheath, although he asserts that it was always for the sake of others that he drew it. He was born in 1582 or 1583 and received a careful education, which he afterwards constantly continued on his own. It was his wish to form himself into a citizen of the world. After having been married to a rich heiress already in his 16th year, he studied at Oxford and then came to court. Later he was in the camp of Maurice of Orange, traveled in Germany and Italy, and was then for a number of years English ambassador in France. But neither at court, in the camp nor in his diplomatic post did he forget his "beloved studies". His famous book he *[62]*

finished in Paris, where he let Grotius read it. Although Grotius's judgment was highly favorable, he still had misgivings about publishing it, since he knew it would meet much opposition. Once, when he sat irresolute in his room, which was open to the south, he resolved to obtain a supernatural decision of his misgivings. With the book in his hand he threw himself on his knees and implored God, who is not only the author of the outward light but from whom all inward enlightenment comes, for a sign that might urge him to publish; if the sign failed to come, he would suppress the book. At once he heard a faint sound, which came from the clear sky and resembled nothing on earth; and now he confidently let the book go forth[10]). This event does not stand, as has been thought, in any contradiction with the conception Herbert develops in his book. For he assumes that God not only works in human reason but also answers men's prayers through particular manifestations. Like Bodin's Toralba, he assumes miracles. Shortly after the book's publication he was recalled from his post as ambassador, because his brisk conduct did not suit the hesitant policy of James I. The rest of his life he spent partly in London, partly at his castle in Wales, occupied with historical and philosophical studies. During the Civil War his conduct was wavering; but in the end he joined Parliament. He would have liked best, as before, to combine public activity with his studies, and he could not get over his political disappointments. He died in the year 1648.

Herbert makes his way to his doctrine of natural religion by an epistemology that is not without historical interest. — If the truth is to be grasped by us, there must be faculties in us for it. Besides discursive reason and the inner and outer sense there is a natural instinct that leads to the development of certain truths common to all men (notitiæ communes). This Stoic doctrine we have already found in Melanchton and his school. In Herbert it serves as the foundation for the universal truths that the comparison of the various religions led men to assume. But peculiar to him is the way in which he connects
[63] the origin of these common notions with the instincts that lead to the self-preservation of the individual and of the species. For the natural instinct from which they spring is the fundamental force in every man and in the whole world: the instinct of self-preservation in the widest sense, through which divine providence manifests itself. Without conscious reason everything in the world strives after what is best for it. And since the "common notions" serve preservation and the maintenance of peace, the instinct of preservation leads to their being developed and recognized. It is thus the truths most important for the individual and for society that become the object of instinctive recognition. When they are separated from impure admixtures and methodically ordered, they constitute a sum of divine wisdom. Such truths are the following: mutually contradictory assertions cannot

all be true; there is a first cause of everything; nature does nothing in vain; do not do to others what you would not have them do to you. Thus the most important logical, ethical and religious ideas are to be grounded, in the manner indicated, on natural instincts. Herbert gives no more than intimations. The relation between the fundamental truths and the natural instinct remains unclear. There are evidently a good many psychological intermediate links to be understood between the instinct and the fundamental truths. The task is — to use modern expressions — to understand how the struggle for existence can lead to fixing certain representations as self-evidently valid. But these intermediate links Herbert does not seek out, whereas he sometimes makes the natural instincts and the fundamental truths wholly one.

Herbert makes a front on three sides. First, against those who set faith above knowledge, and by faith understand what their sect assumes, while they seek to gain entry for this faith by threats of punishments in another life. With this the assumption of the corruption of nature is usually connected. An ungodly faith, for it conflicts with the providence that manifests itself in the natural instincts! Second, against those who want to prove everything by way of reasoning (by discursus luxuriosus). This scholastic pedantry does not see that natural instinct, just as well as the inner and the outer sense, stands above reason. It manifests itself before the capacity to reason stirs, and works independently of it; it is found in all, works without hesitation and uncertainty, and is *[64]* a necessary condition of self-preservation. Third, he turns against those who derive all knowledge from the senses and take our mind to be originally an unwritten tablet (tabula rasa). Those fundamental notions are precisely the conditions of our being able to have experiences! And it surely cannot be the objects themselves that make us act back upon them! The agreement between men's representations proves the existence of a natural instinct, which besides could also be inferred from the goodness of God, who could not leave man to stand helpless.

Natural religion is now grounded precisely on such an instinctive foundation. In his autobiography Herbert occasionally develops this more closely. There arise in us faculties and drives which cannot find full employment or reach full satisfaction in anything that experience can offer; only the perfect, the eternal and the infinite could bring rest. The Deity itself is their real object. In every sound and normal man there is therefore the same capacity for religion, even though it is developed in the most different ways and perhaps finds no expression in any outward cult. At the foundation of all religion lie five propositions which have validity everywhere: there is a highest, divine being; it ought to be worshipped; the most important part of its worship consists in virtue joined

with piety; faults and crimes are to be atoned for by repentance; there is reward and punishment after this life. Whatever in the individual religions does not conflict with these propositions but serves to confirm them may be accepted. That there are those who deny them is to be explained by the fact that so many false and unworthy notions have arisen in the religions that some are thereby led to reject all religion, without excepting the natural one. The five propositions are in themselves sufficient; one should keep to them and let the disputed questions lie. Through the instincts on which natural religion is grounded, a constant inner revelation takes place to individuals, so that they become independent of what the priest says.

[65] It is natural to apply to Herbert Senamus's remark that it is by no means easy to draw the boundary between the essential and the inessential in religion. It would also be impossible to give a proof that those five points really recur in all religions. Even if they could be shown to be universal, it would by no means follow, moreover, that they are grounded on natural instincts. And even if they had such an instinctive origin, it would — as already touched upon — still remain to be shown *how* they develop. Herbert proceeds too briskly and dogmatically. On his way he passes by a whole series of epistemological, psychological and religious problems. But the point of view he sketched in broad outline was shaped with a vigorous hand and became of great historical significance. It was established that there is a religious consciousness which is independent of all existing church communities and takes a critically testing stand toward them.

8. Religious Speculation (Jakob Böhme)

Lord Herbert busied himself by turns with sword and with book, and from diplomatic negotiations at the court of Paris he went back to his closet to work on his work "on truth". But the need of the age for new thoughts, and for bursting the rigid ecclesiastical forms in which the movement of the Reformation had so early settled down to rest, manifested itself under less brilliant circumstances too. Jakob Böhme let awl and last lie in order to write his "*Morgenröte im Aufgang*". He too had looked about him in the world before he took up the pen. As a journeyman shoemaker he had traveled at his trade; later he settled in Görlitz, and in the quiet workshop thought worked in a race with the hand. He was not a wholly unlearned man; he calls himself a naturalist and philosopher, and had made the acquaintance of astronomical writings; remarkably enough he is a Copernican, inasmuch as he lets the earth and most of the planets move round the sun; he had made the acquaintance of the physician Paracelsus's writings on the philosophy of nature, and probably also of the works of the mystic Valentin Weigel,

spread in copies, in which Paracelsian ideas were worked together with the doctrines of earlier mystics. He was at the same time a pious man who knew his Bible and the *[66]* Lutheran doctrine. But his problems he set himself, and he treated them with a spiritual freedom and strength that still make his writings valuable and that led him beyond the dogmatic barriers. While Bodin and Herbert come to the religious problem more from without, for Böhme it presents itself within the confessional notions themselves; he goes on working with these until the stream of thought breaks through all dams. In his naive but vigorous self-confidence the shoemaker of Görlitz too bears the stamp of the Renaissance, just as the elements of learning that he uses for his edifice of thought point back to the philosophical and scientific movements of the time, which had found their way to him too through many intermediate links and transformations. He uses what is at his disposal to solve the problems that for a time were bringing him to despair. A remarkable example of how the philosophical intellectual tendency can break forth even where inner and outer circumstances seem bound to exclude it! So long and so hard had he struggled with the tormenting thoughts that it stood before him as a revelation, a higher enlightenment, when he caught sight of the possibility of a solution. In the 19th chapter of "Morgenröte" he has described his doubts, and how the light then dawned on him, and it was to him "as though life had been born in the midst of death". He was then 25 years old, and according to the legend it was in an ecstatic state, into which the sight of a shining metal dish put him, that he first saw his fundamental thoughts clearly before him. And they still had to ripen for twelve years before his first work (Morgenröte) was written (1612). That the shoemaker dared to philosophize so boldly aroused great offense in the parish priest, who had got hold of a copy of the book. He denounced Böhme as a heretic from the pulpit and got the authorities to forbid him to write. Only for some years did Böhme observe the prohibition. In the last period of his life he unfolded a fruitful activity as an author. He died in Görlitz in 1624.

The religious problem presents itself to Böhme under two main forms. — He fell into "hard melancholy and sadness" on seeing how small a place man occupies in the outer *[67]* world. It is high up to heaven, where the learned say that God dwells. And globes and elements, which fill space, seem to go their way without any heed being paid to how things stand with man! God's remoteness and man's vanishing place in nature was thus the one stumbling block. But to this was added: the struggle between good and evil, both in the human world and in nature, together with the sight of how the ungodly fare just as well in the world as the pious, and how the barbarous peoples hold the best parts of the earth and have better fortune with them than the believers. That was the other

stumbling block: How is the existence of evil compatible with the existence of God? —

The first problem, the one posed by the fact that the world of matter is so great and the world of spirit so remote or so small, he solves by means of the thought that it is God's force and essence that stirs in everything, in us as well as in the outer masses. God is not separated from nature, but is related to it as the soul to the body. Heaven is not up there above the blue; it is in your own innermost being, where the divine life stirs in you. God is not remote; you live in God, and God in you, and if you are pure and holy, you are God. The same forces stir in you as in God and as in the whole of nature. Fire, air, water, earth — everything is God. Or else you are not God's image. You cannot be any other matter than God himself. When you see the stars, the globes and the deeps, you see God, and in the same God you too live and are. The globes and the elements, to be sure, are not the clear Deity itself; but the force from which they draw their life rules in you too. The inner motions in man's being are of the same nature as those that take place in God. In all processes of nature God is hidden; only in the spirit of man is he known. One understands nothing of God as long as the soul is separated from him, as if it were a separate being.

If one wants to call this line of thought heathen, Böhme answers: "Hear and see, and mark the difference. *I write not heathenishly, but philosophically.* Nor am I a heathen, but
[68] I have the deep and clear knowledge of the one great God, who is all." And he knows how much it has cost him to reach this thought, which has made the world alive for him and taught him that spirit is not remote and alien in it, but one with what stirs innermost in everything. He has tried to push such thoughts aside, but great anguish and temptation have then led him anew to seek them out. (Morgenröte im Aufgang. Ch. 19 and 25). The religious need of inward unity with the Deity, of abolishing God's remoteness, here meets with the philosophical need to find connection in everything, and both are supported by the doctrine, set up by the natural philosopher Paracelsus, that man consists of the same elements as the universe.

But Böhme does not stop at nature as it now is. The question arises for him how this outward, sensible world came into being? why is everything not in such inward and harmonious connection that temptations such as those he had experienced could not arise? Already Paracelsus, the revolutionary chemist and physician, by whom Böhme is evidently strongly influenced, could put him on the way here, since he had taught that the diversity of the elements had arisen by a process of separation and splitting. Böhme follows this track with his usual boldness, and it leads him at the same time to a solution of the second and most serious problem. But the problem of the origin of evil

had been sharpened for him by the fact that he will not have God separated from the world and from the souls, but would conceive their life as a part of God's own life. Evil consists for Böhme precisely in wanting to separate oneself from the whole connection, in wanting to be a whole, although one is only a part. Thereby arise the hard division, the externality, the strife and the torment in the world.

Böhme gives his elucidation of this question in mythological form, using biblical and alchemical representations, often in a baroque, but also often in a magnificent poetic combination. Of this mythological or symbolic form of exposition he is clearly aware. That he is able at all to solve the lofty questions that press upon him he explains by the fact that the same spirit and the same elements that stir in the world, and have stirred there from all eternity, also stir in him. In other words, it is his own inner experiences *[69]* and struggles that he uses to interpret nature and its history and to understand what is innermost moving in everything. And when he depicts a series of processes and catastrophes, a whole world-drama in vivid images, he does not thereby mean that such things took place literally thus, at a definite time and place, — nor that it was something that unfolded historically and successively. He wants to depict what takes place everywhere and at all times, an eternal fundamental relation between the forces of the world, a strife that has not been fought once for all, but is constantly being fought, and in which he feels that he himself stands. His utterances on this are surprisingly clear. "The meaning is not that there is in heaven a particular place or a particular corpus where the fire of the divine life rises . . . but *I speak in a corporeal manner for the sake of the reader's want of understanding* . . . For you cannot name any place, either in heaven or in this world, where the divine birth does not go on." "And although I have here written how everything comes into being, and how everything is formed and fashioned, and how the Deity rises, you must not think that there was first to be quiet and extinction and then the rising to take place. O no, but *I must describe piecemeal for the sake of the reader's want of understanding.*" "The birth of nature goes on to this very day in the same way as in its first beginning." "In the divine forces (Qualitäten) there is neither beginning, middle nor end." "You must know that what I write here is not a story that has been told me by others, but I must myself stand in this battle, and often I get, like other men, a broken leg." — In presenting Böhme's thoughts it is therefore justified to let the mythological form play a subordinate part.

Böhme declares himself decidedly against creation out of nothing. "Many writers have indeed written that heaven and earth were created out of nothing; but I wonder that among so many excellent men there was found none who could describe the right

ground, since after all the same God who now is has been from eternity. Where nothing is, nothing comes to be either; everything must have a root, otherwise nothing grows
[70] forth, and had not the spirits of nature existed from eternity, then no angel, no heaven and no earth would have come into being." It is Böhme's conviction that nothing can be explained from an absolute unity. There must lie in the essence of the Deity a plurality of moments; a difference must assert itself in it which cannot from the first have been arbitrarily produced, but is precisely as a dark source of universal life. In purely logical or psychological form Böhme expresses this (in his "Theosophiske Fragen") by saying that God is not merely a Yes, but that a No must step into opposition to this positive moment, since otherwise there would be no joy and sensation, on the whole nothing that could stand out definitely (no *Erheblichkeit*). Without such an inner original opposition between a positive and a negative moment there would be no will either; pure unity has nothing in itself that it can will. This original element of opposition Böhme calls, in his mythological language, God's wrath and the foundation of hell. There is originally in God wrath just as well as love, a foundation for hell just as well as for blessedness. Where indeed should wrath, all that is sharp, bitter, hard and poisonous in the world, come from, if it did not have its root in God? But originally wrath is only a moment that serves to let love step consciously forth. Life demands such opposition; only because there are different "qualities" in nature is there mobility, growth and drive in everything. The word quality Böhme interprets by a naive etymology, which expresses a brilliant thought, as coming from quallen, Quelle: "Qualität ist die Bewegligkeit, quallen oder treiben eines Dinges." "Qualificirung" is one with "Bewegung". He will not stop at ascribing dead and resting properties to things, but dissolves everything into active forces and tendencies. That heat, e.g., is a "quality" thus means that it shines, warms, dissolves, melts. The plurality of the qualities — i.e. of the fundamental forces — thus stands firm for Böhme, since no motion, no life and no consciousness would be possible without such a difference. But from the first the fundamental forces are in harmonious interplay. The bitterness in the root of nature serves only to let the sweetness come forth all the
[71] more strongly. The bitter quality is in God too, but as "a triumphant source of joy".

But even if this original unity of opposition is recognized, the riddle remains standing. How can there be reconciled with it the experience of a hardness and a bitterness in the world which are by no means merely moments in an inner harmony, and above all of the sharp strife between good and evil in the inner and in the outer world? Here is the crux of all religious speculation. Just as little as it is possible to derive plurality from absolute unity, just as little is it possible to derive disharmony from absolute harmony. No wonder

that Böhme's mythologizing fantasy, at this point, where his feeling is moreover in such strong motion, gets the better of his thinking completely. It will not be possible to give a vivid account of the drama in which the persons of the Trinity, the original "Qualitäten" and "Quellengeister" and the host of archangels appear. The main thing is that one of the elements in the original nature of the Deity, the element of wrath, the No, pushes forward with a tendency to spread over the whole. It was not content to be a moment, but wants to be totality. That was Lucifer, who was blinded by his splendor and his power and his seat in the highest center, and now would triumph over the whole Deity, have a prouder and more splendid "qualification" than God himself. Then strife and torment were born in the world; a fire was kindled that has never since been quenched. The opposites were loosened from the original harmonious connection, hard matter was hurled out into space, and only then did the world come into being as we now know it. Evil thus existed before man; man received the germs of it in his nature. Only now has Böhme come to the point where the story of creation in Genesis begins, a story for which he otherwise has not much respect, since there is a good deal in it that conflicts "with philosophy and sound reason", for which reason he also cannot believe that "the dear man Moses" wrote it. On the whole he follows his own thoughts, trusting in the inner testimony of the spirit. Nor does he let himself be impressed by the learned: "I do not need the formula and art of philosophers, astronomers and theologians, since I have not learned from them, but have another teacher, which is the whole of nature."

[72] Above all Böhme is indignant at those who teach that God has purposely chosen some for blessedness and destined others for damnation. Even if Paul or Peter had written it, he will not believe it. God does not will evil. Nor did he know beforehand that it would go as it went. And yet evil must have its root in God, who is the author of everything! This root is the negative moment, "the foundation of hell", — that in God which is not God, if by God one understands only love (not wrath). And yet it is after all a divine element that breaks loose from the original harmony — it was "God against God", as Böhme (Morgenröte. Ch. 14 § 72) expressly says. Thereby the strife in the world becomes so serious and so violent; there are divine forces on both sides. Hence the anguish and agony that the struggle brings, and from which Böhme saved himself only by fleeing to the thought of the heart of the Deity, which constantly works to force hardness and bitterness back within their bounds.

In his theological speculation Böhme applies the psychological truth that the involuntary precedes and forms the foundation of the voluntary. He finds the origin of evil not in God's voluntary willing, but in the involuntary ground of nature, which

contained different elements. But the great leap from difference to strife, from opposition to division, he overlooked. How pride comes to be in Lucifer he cannot explain. Here he narrates instead of giving grounds. And yet his thought has ventured as far as the thoughts of so inwardly religious a man seldom venture. From the theological side one must consistently regard him as he regards Lucifer. He has set a dangerous "quality", thought, in motion, and once this source has begun to flow it cannot be stopped again. Yet it is not pride but an inward need of understanding that leads him on from step to step. The naive self-confidence of newborn thinking keeps him from wavering and stopping. He expressly regards his task, as we have already heard in a characteristic utterance, as a purely philosophical one. "What is still hidden?" he asks in another place, "Christ's pure doctrine? No, but philosophy." This, then, he wants to bring, and he attaches such great significance to the understanding he has won that he calls his
[73] first book "Dawn Breaking Forth"; the time of fulfillment must now be near! There is in reality dawn in his thinking. He felt the new forces, even if he did not see whither they led.

Two great laws Böhme laid at the foundation of his speculations: the law of opposition as the condition of all motion and of all consciousness, and the law of development as the progressive unfolding of differences. In the law of opposition it is clearly enough psychological experiences that he starts from, as can be seen especially from the characteristic proposition: "Kein Ding ohne Widerwertigkeit mag ihm selbst offenbar werden." In the thought of development as the arising of diversities he is influenced by Paracelsus. The word development he introduced into the German language[11]). — By the consistent use of these laws he carried religious thought to its limit, a limit that no later attempt of a similar kind has been able to move.

There might seem to be a great difference between Böhme's religious speculation and the natural religion maintained by Bodin, Cherbury and Grotius. Böhme was, or thought he was, a positively believing Lutheran. But he saw consistently that if divine light and life work in the whole of nature as a counterweight to hardness and bitterness, access to the light cannot be limited. Everyone can work his way or grow forth through "wrath" to "love". Jews, Turks and heathens, who know nothing of Christ, nevertheless stand on an equal footing with Christians. "Whoever has love in his heart and leads a merciful and meek life and fights against wickedness and presses through God's wrath into the light, he lives with God and is one spirit with God. For God needs no other service." These words Bodin might very well have put into the mouth of his Toralba. Martensen finds them "strange and without foundation". But they have a secure foundation in

Böhme's own fundamental thoughts. Later "speculative" theologians might wish for his steady gait as a thinker. What interests us here is that in the midst of the territory of Lutheran orthodoxy, in an unlearned artisan, we find the same universal tendency in religious respects that is peculiar to the philosophy of the Renaissance, and that is so closely connected with the fact that men again believe in the natural forces and dare to *[74]* use their own thoughts.

B. The New World-View

9. The Aristotelian-Medieval World-Picture

Just as little as man and the world can be separated from each other, just as little can the conception of man be separated from the world-view. There is therefore something artificial in following the change that the conception of man undergoes without regard to the changes in world-view. There is a constant interaction between them. Böhme's religious speculation is thus evidently in part conditioned by the new world-view (especially by Paracelsian and Copernican ideas) and can properly be understood only when one takes it into account too, although what predominates in him is his conception in terms of the psychology of religion. Conversely, in several of the thinkers whom we shall here consider as founders of the new world-view there will be found valuable ideas and points of view in psychological and ethical respects, which in and of themselves could entitle them to a place among those who have discovered man. But it must be permitted to give them their place under the point of view that indicates their decisive significance in the history of thought.

To understand the magnificent change in world-view that is the greatest deed of the Renaissance beside its art and its discovery of man, we must first bring forward the main features of the world-picture that still in the fifteenth century stood unshaken for learned and lay alike. It derived from Aristotelian physics and Ptolemaic astronomy and had been interwoven with biblical representations, a combination that was very easily possible, since the whole world-picture kept to what the senses immediately taught and merely gave it a more exact determination and a systematic execution.

Even for the ordinary conception there is a great difference between the motions *[75]* in the heavens and what happens here on earth. Here there seems to reign a constant change. Phenomena wane and wax, arise and pass away. Motions arise, but they cease again after a shorter or longer time. Otherwise up yonder: the stars go their regular course without any change being noticed, and their path leads them always back along

the same ways in an eternal circular motion without rest. Aristotle laid this opposition between the heavenly and the earthly or sublunary region at the foundation of his world-view. He relied on what observation seemed to teach, while he also referred to the old belief, among Hellenes and barbarians alike, that makes heaven the seat of the eternal gods. The heavenly region is incorruptible; motion in it is eternal and absolutely regular. The sublunary region, on the other hand, is the home of corruptibility, where motion and rest, coming into being and passing away, succeed one another.

The two regions must then also consist of different matter. The heavenly bodies must consist of a matter that does not belong in any definite place and can therefore continue its motion eternally. This matter, "the first body", Aristotle calls the ether. It is what fills the space of heaven, and in which the eternal circular motion takes place. Only circular motion can be eternal, since it always returns into itself and from every point in the circle passes to every other point. In the sublunary region every motion stops at a certain point, namely when the body has reached its "natural place". Here rectilinear motion reigns, which goes either from the center of the world outward and upward, or toward the center of the world inward and downward. That whose natural place is at the center of the world or near it is called heavy; that whose natural place lies upward toward the upper boundary of the sublunary world is called light. The constant changes in the earthly regions now come from the elements' not always being in their natural place. The heavy element is earth, the light one fire; between both lie water and air. Fire, after all, constantly rises aloft, and the stone thrown aloft always falls down again. Each element has a craving for its definite place in the space of the world. The four elements
[76] can pass into one another, but cannot themselves be dissolved into simpler constituents, whereas all bodies are composed of them.

There can be only one world. If one imagined for a moment that there were several, the heavy elements would at last have to gather around one center, and there would then be only one world. The earth is, as observation shows, at the center of the world, and from it we come, going outward and upward, first through the three other sublunary layers and then to the region of the ether, whose matter is the purer the farther it is removed from the earth. The heavenly bodies, first the moon, the sun and the other planets, then the fixed stars, sit fastened in solid but transparent spheres that turn about their axes. This assumption Aristotle took over from the astronomer Eudoxus. That the globes could float freely in the space of heaven the ancients could not conceive. All the spheres move about the earth as their center; the moon, the sun and the other planets each have their own sphere; all these spheres, and thus the whole system of the

world, are enclosed by "the first heaven", the sphere of the fixed stars, which is moved directly by the Deity itself, while lower principles or spirits guide the motions of the lower spheres. The highest is what encloses and limits. The uppermost or outermost part of the world is the most perfect[12]).

The Aristotelian world-scheme offered a framework that not only corresponded to what sensory observation seems to teach, but could also take into itself the new observations made by the astronomers of later antiquity. It turned out, to be sure, that one had to conceive the motions of the planetary spheres as very complicated when one took into account the forward and backward motions of the planets; but this was remedied by conceiving each single planet as attached at once to several spheres, each corresponding to one of its motions. The motions were conceived as taking place not in simple circles, but in smaller circles, epicycles, whose centers moved on larger circles. And this system of epicycles then became, as research advanced, ever more complicated. Ptolemy of Alexandria (in the 2nd century after Christ) gave an account of the system of the world that became the authority in the astronomical domain throughout the Middle Ages, just as Aristotle became so in the philosophical domain. *[77]*

Since the Aristotelian-Ptolemaic system, like the Bible's conception of the world, took its stand on the standpoint of sensory observation, for which the earth stands still while the sun and the stars move, it was well suited to be combined with the religious notions prevailing in the Middle Ages. As Aristotle had remarked, it also agreed, after all, with the old belief in heaven as the seat of the Deity. An essential difference made itself felt, however, between the ancient and the medieval conception of the world. Aristotle by no means placed the earth at the center of the world because he thought that everything else in the world existed for its sake or for that of its inhabitants. On the contrary, for him the highest was what encloses and limits, and the earth was the place of the lowest matter. But for the medieval conception of the world it was precisely the earth and the events on it around which everything turned. Nevertheless it could well use the Aristotelian scheme and the idea of the imperfection of the sublunary world, since it let the heavenly forces descend to intervene in earthly affairs and lead them to the right goal.

The whole mode of representation was — apart from the difficulties that the compound spheres and the epicycles caused — clear and vivid. It amounted to this: that sensible space was absolute space, and that in this space there were absolute places. The globe on which the philosopher or the astronomer found himself he took, just as the practical man constantly does, for the absolute center of the world. And the world that took form around this center in the various spheres had its limit in the outermost sphere, beyond

which there was nothing. It belonged to the ancient view of nature to think everything formed and limited. The unlimited was one with the dark and the unformed, if not with evil. Here too the scheme could rely on sensory intuition and fantasy, which always stops at a certain point without any need to ask what lies beyond it.

[78] A world-picture such as this could be attacked from two sides. First and foremost by bringing forward observations and calculations that conflict with it or at least make another conception possible or probable. But also — and this interests us here not least — by contesting the naive trust itself in the absolute validity of sensible space. If it turns out that every determination of place depends on the observer's position, the absolute difference between the heavenly and the earthly region, and between the natural places within the earthly region, falls away. Along both paths the Aristotelian-medieval world-picture was attacked in the research of the Renaissance. And as will appear, the latter path was entered upon first.

10. Nicolaus Cusanus

Already at the close of antiquity there showed itself within philosophy a need to burst the clearly articulated and limited Aristotelian world-picture. Plotinus, the founder of the Neoplatonic school, expressed a thought which, when carried further, was bound to lead beyond every dogmatic limitation and partitioning of nature. For him, as for Plato and Aristotle, harmonious limitation was not the highest. The highest was something that lay beyond, or led beyond, every barrier. The infinite was thus no longer something negative, but became the expression of a positive, unsurveyable fullness, before which thought and language are at last powerless. The finite forms, beings and regions are then each of them only limited revelations of the infinite. Although Plotinus still stands on the ground of Greek thinking, thought in him nevertheless often threatens to lose itself in mysticism. He became the forefather of the mysticism of the Middle Ages, a tendency which again and again came forward with its protest against the conceptual dogmatism of Scholasticism. In the view of nature this tendency led to an emphasis on the connection of the inner forces in spite of the differences of the outer phenomena, and to a feeling of dissatisfaction with fixed forms and limits in every domain. Mysticism has thereby prepared the way for the doctrine of evolution. It was therefore of great

[79] significance that the Neoplatonic philosophy in particular was cultivated with zeal in the period of the Renaissance. The Academy in Florence has especial merit here.

Nicolaus Chryffts, called Cusanus after his native town, did not first come under the influence of Neoplatonic ideas through these new studies. He was brought up at

Deventer by the "Brethren of the Common Life", who were strongly influenced by German medieval mysticism. He was born in 1401 at Cues, a small town near Trier, but fled from there as a boy because of his father's ill-treatment. A nobleman with whom he took refuge then had him study at Deventer. Later he studied law, mathematics and philosophy at Padua. But the mystical keynote, and the problems which it set in motion in him, never left him. For a time it seemed as if he would pass wholly over into practical life. He was first an advocate, then became a cleric, and took part with zeal in the negotiations on the reform of the Church, at first attaching himself to the Council of Basel, but later seeing the only possibility of a reform of the Church in the strengthening of the papacy. Later still, as Bishop of Brixen, he had a hard struggle to sustain with the sovereign of the land, Archduke Sigismund of Austria. For a time he was even imprisoned. Pius II made him a cardinal and sent him to Germany to visit churches and monasteries. Here too he had hard struggles to sustain through his zeal to reform the manners of the clergy and of the monasteries. In particular he combated superstition in a way that betrays a great spiritual superiority over his age. He censured the use of superstition for ecclesiastical purposes, and the procuring of ecclesiastical revenues by means of relics and bleeding hosts. Witchcraft and sorcery he regarded as remnants of paganism, which were kept alive by the very persecutions with which they were visited. He found that there were most witches where people most believed in witches. Two old women who had the reputation of being witches he examined himself; they were half mad, and "what they believed had really happened to them, the Devil had put into them in dreams". — But amid all this participation in the struggles of his time he was occupied with his problems. The fundamental thought of his most interesting and best-known work *(De docta ignorantia)* came to him, by his own account, as he was sailing back to *[80]* Italy from Constantinople, where he had been on a papal errand. It was followed by a whole series of other writings, in which he partly developed his ideas further, partly modified them. He died at Livorno in the year 1464, during the preparations for the crusade, planned by his friend Pius II, against the enemies of faith, of science and of literature.

Cusanus stands on the border between the Middle Ages and the Renaissance, and between theology and philosophy. His innermost interest is, to be sure, theological. It is the problem of the Trinity that he wants to solve. But he treats it in such a way that he comes to develop epistemological and scientific thoughts of great interest. It is bound up precisely with his mystical tendency that the sharp distinction between the domains of concepts does not exist for him. The obscure and abstruse element which is found in his

writings beside the many excellent ideas arises from this confusion. — It is of no interest for us to go through the whole of Cusanus's philosophy. The point is only to show how he arrived at thoughts that shook the world-picture supported by the testimony of the senses, the inquiry of antiquity and the authority of the Church.

Cusanus conceives cognition as a connecting and assimilating activity. Sense perception apprehends the scattered impressions, but already in this apprehension thought expresses itself, partly in that the separate impressions are put together into a whole, and partly in that there arises a wonder which spurs one to go further. The various sense images are connected by the faculty of representation (phantasia), the various complexes of representations by reason (ratio); but at all these stages thought (intelligentia) is at work, which at last seeks to lead all differences back to absolute unity. But at this point the difficulty arises, for without plurality and difference (alteritas) our thought cannot know. At its summit, then, thought finds precisely its limit. It can only approach absolute unity in that a mystical vision, for which all the radii of existence run together in one center, takes the place of the cognitive process and gives thought rest. The perfection of
[81] thought is thus its cessation.

This conception of cognition (which is developed in the work *De conjecturis* ch. 16) makes clear what Cusanus means by "conscious ignorance" (docta ignorantia), an expression which already occurs in Bonaventura. It is thought's consciousness of its own limit, joined with the mystical presupposition that the best and highest lies beyond this limit, though in continuation of the process that led to the limit. In the very work that bears the name De docta ignorantia the matter is developed thus: Our cognition seeks to reach beyond the oppositions and contradictions which experience presents. It discovers on closer investigation that even the greatest oppositions, even mutually contradictory states, coincide in one, if only the point of unity can be shown. Thus maximum and minimum coincide by means of the concept of magnitude: both are absolute quantities, each in its own direction; both are superlatives, and that on one and the same scale. The concept of magnitude itself embraces them both. But all the intermediate terms on the scale our cognition cannot find. It remains forever stuck in oppositions. It works its way forward from term to term, but cannot reach the perfect unity of the oppositions, any more than one can produce a perfect circle by giving a polygon ever so many sides. Our thought is related to truth, which is the perfect unity of oppositions, just as the polygon is related to the circle.

The infinite is the measure and ideal of the finite. In it the oppositions meet. An infinite curve is the same as an infinite straight line. But every line that we apprehend

is either straight or curved. We cannot reach beyond the finite determinations. And yet that is necessary if there is to be a knowledge of God. God can be thought only by the canceling of all particular determinations: but for us nothing remains when all particular determinations are removed. God as unity in all, as comprehending all, even what is mutually contradictory (omnium rerum complicatio, etiam contradictoriarum), is therefore incomprehensible. No name can be ascribed to him. For every name derives from a distinguishing and separating, by which the one is set against the other. But God has no opposite, any more than the concept of magnitude has any opposite on the whole *[82]* scale from maximum to minimum. In the religious cult conscious ignorance is replaced by holy ignorance (docta ignorantia by sacra ignorantia), negative theology by positive. Definite predicates and names are then ascribed to God; but none of these can then be taken in the proper sense. When God, e.g., is called light, what is meant is not a light that forms an opposite to darkness; in the infinite light even darkness is light, just as the minimum lies on the same scale as the maximum.

The Aristotelian school has failed to recognize this truth, without which no access stands open to a mystical theology. That is why it has not been able to comprehend the doctrine of the Trinity rightly; for it has not seen the necessity of a principle that can connect the oppositions expressed by the Father and the Son: Spiritus sanctus est nexus infinitus! And that is why it has not led to any true knowledge of nature. It stops at the difference between possibility and actuality, between matter and form, between what produces and what is produced, and does not seek a bond that can embrace all these oppositions and express their unity. This bond is motion. Motion is neither merely produced nor merely producing, but both at once, and so it is too with the other oppositions. And this juxtaposition of the connecting principle in the Trinity and in nature is not accidental for Cusanus: the Holy Spirit is the bond of nature, is one with nature as the sum of all that happens through motion. — Cusanus does not distinguish between the coincidence (coincidentia), the comprehension (complicatio) and the connection (connexio) of the oppositions, although these are indeed quite different relations that are denoted by the three expressions. His examples point mostly to a connection of extremes through transitions and intermediate terms, and to a comprehension of them through the continuity thus brought about. But that is something other than a complete canceling of the oppositions. There is a want here of full clarity about the nature of the very concept that denotes his highest point of view. And this is bound up with the lack, mentioned before, of a distinction between the different domains of knowledge. *[83]*

It is especially characteristic of Cusanus that he slides easily from mystical inquiry

into the divine processes over to the processes of nature, and can use the same points of view in both places. That the two processes, the life of the Deity and the life of nature, nevertheless do not coincide is, to be sure, clear to him. The relation between them is that between the closed unity (coincidentia, complicatio, connexus) of the oppositions and their separated appearance in time and space. Nature is an unfolding (explicatio, evolutio) of what was joined in complete unity in the Deity, just as the line is the unfolding of the point (by repeated positing), and just as actuality is the unfolding of possibility. But how this conversion from the complicative to the explicative form takes place, — how the absolute coincidence of the oppositions is replaced by their appearance as contending powers, — how the confusion of multiplicity replaces the harmony of unity, — that is a riddle which our thought cannot solve. Every finite existence is a limitation, and the great riddle is precisely how the infinite can limit itself. In one place Cusanus says that there is no positive cause of multiplicity, since only the unity in things can derive from God. If you appeal to omnipotence to explain how multiplicity comes into being, that is the same as saying that you do not understand it![13])

But even if Cusanus ends at this point in dualism, since a gulf remains between his knowledge of God and his knowledge of nature, he has nevertheless, in the idea of motion as the connecting principle in nature and as that which makes the world a whole, and no less in the idea of the development of the world as a progressive process of complication and explication[14]), expressed fruitful points of view which anticipate the thoughts of later philosophers. They made it possible for him to maintain at once the peculiar character of individual beings (as distinct complications or contractions) and also the continuous connection and transition between them. Whereas for antiquity and the Middle Ages alike the completed actuality was the highest, the significance of possibilities and of development now began to come into its own. And that again was
[84] closely bound up with the change which the concept of infinity had undergone. In place of the finite, firmly limited forms, the emphasis was now laid on infinite variation, on the constant transitions which let the oppositions stand as the extreme cases or stages on a continuous scale.

In religious respects a difficulty arose for Cusanus from his denial, on principle, of the right to apply any concept whatever to the Deity. To be consistent he would then have had to declare all religious representations symbolic or mythological; but this step he could not take as a believing Catholic. In Docta Ignorantia he helps himself by making a certain distinction among the predicates. Though all predicates must be denied of the Deity, the negation of the greatest imperfections is nevertheless to be

"truer" than the negation of the highest predicates. It is thus to be truer to say that God is not a stone than that he is not life or spirit; — and from this it follows in turn that one may with greater right call God life or spirit than stone. It is clear that, when *all* oppositions are to coincide or vanish in the Deity as in the infinite principle, even the greatest difference between predicates that we can make from our standpoint becomes of altogether vanishing significance. In his later writings Cusanus therefore also sought more positive determinations and laid more weight on positive theology in relation to negative, without, however, succeeding in removing the difficulties which he himself had pointed out.

Within the knowledge of nature the fundamental thoughts of Cusanus find several very important applications. In canceling the absolute oppositions through the constant transitions, he is led to discover the relativity in our representations. The epistemological law of relation has in him one of its first representatives. Thus he maintains the relativity of the concept of the atom. *In our thought,* he says, we can always carry the division of things further; but the *real* division always stops at a part which is in fact indivisible for us. Such a magnitude, in fact indivisible because of its smallness, is an atom. This actual stopping is therefore no proof that an absolute minimum has been reached, any more than the actual stopping in the summation of magnitudes is proof of an absolute *[85]* maximum. — Just as an infinite division is impossible in reality, because we always stop at a certain point in the process of division, so an infinite motion is also impossible, because every body, owing to its own nature and that of its surroundings, will always meet with resistance. A perfectly round sphere on a perfectly smooth base would continue forever in the motion into which it had once been brought, if nothing in its state were altered. It would have to be regarded as an atom, and the force imparted to it could not cease to act. A sphere that was not perfectly round could not take up the imparted motion in such a way that it became a natural motion which as such could not cease. This is one of the most remarkable intimations of the law of inertia that occur before Galilei. And it springs from Cusanus's great capacity to think every thing in the definite relations through which it first acquires its full peculiar character. — Of still greater interest is his doctrine, arising from the same way of looking at things, of the relativity of the determination of place and of motion. The world can have neither center nor circumference, for then it would have to stand in relation to something outside it by which it was limited, and it would not be the whole world. Wherever a man may be: on the earth, on the sun or on another star, he will think himself to be at the center. Now put together these various possible points of view, and you will see that it is impossible to ascribe any figure to

the world, any more than to ascribe motion to it. Every point in the world can with equal right be called the center and be said to lie on the periphery. And since the earth does not stand at the absolute center of the world, it cannot be at rest. But we do not notice its motion, since we have no absolutely fixed point to compare it with. It is with us as with one who is on board a ship that drifts with the current, when he does not know that the water is flowing and cannot see the shores: how should he discover that the ship is being moved? — Cusanus thus moves the earth out of the midpoint of the world, because on the whole no absolute midpoint can be given, and with the central position absolute rest must also fall away. On the other hand, he does not teach that the
[86] earth moves round the sun; he cancels the geocentric theory, but does not establish the heliocentric. Those who might think that the earth thereby loses in dignity he consoles with the thought that perfection does not depend on size. The inhabitants of the earth are the more perfect the more they approach the ideal that accords with their nature; but it is not said that this ideal is the same as that which holds for the inhabitants of other globes. Man does not strive after another nature, but only to become more perfect in his own nature. Nor can the perishability on the earth prove its imperfection, for nothing is absolutely perishable. It is only the peculiar form of existence that ceases, in being dissolved into its elements. And such processes of dissolution can take place just as well on other stars as on the earth, since they, no less than the earth, must be assumed to be subject to external influences[15]). By this doctrine a decisive blow was dealt to the Aristotelian-medieval world-picture. The security which the closedness of this picture presupposed was broken. Reflection had lifted the world-sphere from its foundation and set it in a motion that could now no longer stop. At the same time the sharp opposition between heaven and earth was canceled within the picture; the possibility that the same laws and relations hold in both regions had been expressed. An important contribution had thereby been made to the new world-view, and a contribution that sprang from the innermost core of the profound thinker's system. If the ancients did not arrive at the doctrine of the relativity of motion, that was, says Cusanus, because they lacked Docta Ignorantia. They did not see that all our concepts hold only in certain relations and fall away with these. Thus the same thought which according to Cusanus makes a knowledge of God impossible makes it possible to know that the world-picture cannot be so limited and so at rest as had hitherto been supposed.

11. Bernardino Telesio

The philosophy of the profound cardinal is the greatest work of thought that the fifteenth century produced. It prepared the way for a new world-view by dissolving the rigidity into which concepts had fallen under the dominion of Scholasticism. But not only from *[87]* the inner workshop of thought itself, from the world of outer experience too, important preparatory steps were taken. In the sixteenth century experience began to become the watchword. People no longer wanted to read in old books, but in the ever-young book of nature itself. This need expressed itself, as was natural, at first in an unclear but all the more passionate way. People wanted to stamp out of the earth the fullness of content which they missed in books. And they had not critical sense enough to sort out the genuine experiences, nor to keep their own mystical inspirations or favorite ideas away. As representatives of this titanic impulse toward experience and knowledge of nature may be named the German physician and chemist Paracelsus and the Italian physician and mathematician Cardanus, both restless, storm-tossed men, who besides ideas of genius in the domain of their special sciences also expressed philosophical views. They contested the Aristotelian view of nature on several points, especially the doctrine of the four elements, and they maintained the unity of the different parts of the universe as regards matters and forces, in opposition to the Aristotelian-medieval duality of a celestial and a sublunary world. But although they regard themselves as pioneers, especially as regards method, their inquiry is nevertheless far from scientific and methodical. They do not yet know the stricter requirements for the method of the knowledge of nature which were impressed by Leonardo da Vinci, Galilei and Baco[16]). Nor do their thoughts arrange themselves into an easily surveyable whole, as — in spite of all methodical defects — was the case with Bernardino Telesio, whom we shall therefore consider as the representative of the philosophy which, supported by experience, precedes the founding of modern natural science and has yet freed itself from the Aristotelian-medieval conception.

With Telesio we set foot on the soil of southern Italy, which in antiquity nurtured so many thinkers, and which was to win this fame again in the period of the Renaissance. It was the homeland of Telesio, Giordano Bruno and Campanella. These three thinkers stand in the same line; the tendency of their ideas is the same, however characteristic the differences they otherwise present. Telesio influenced the two others. He was born at *[88]* Cosenza near Naples of a noble family, and used his fortunate outward circumstances to undertake comprehensive studies. After studying in Milan he came to Rome, where he suffered ill-treatment when the city was taken by the Imperial troops under the Constable

of Bourbon. Later he went to Padua, and here the break with Aristotle is said to have taken place in him. The foundation was thereby laid for a sharp opposition between the northern Italian and the southern Italian schools. While in Padua and Bologna people went on cultivating Aristotle and more or less blindly swearing by his word, in southern Italy they struck out in new directions and opened new paths, seeking at the same time to build as far as possible on the awakening natural science. After a stay in Rome with Pope Paul IV, who valued him highly and even wanted to make him an archbishop, Telesio returned to southern Italy. He lectured in Naples and founded an academy in Cosenza. In the year 1565 the first part of his chief work "On the Nature of Things" (de rerum natura) appeared; later (1587) appeared the part that treats of man in his psychological and his ethical aspect. Telesio died in the year 1588. In his last years he was the object of sharp attacks from the monks, who thought that everything fell with the Aristotelian concepts, and a few years after his death his works were put on the list of forbidden books.

Telesio sets his endeavor in opposition to that of all his predecessors: they, in their great self-confidence, wanted to fathom nature with their reason, just as if they were God's equals in wisdom; he himself has greater humility and wants to reach only a human wisdom, limited to what the senses teach and what can be inferred from the likeness with things sensed. He wants, then, to rely on sensuous experience, and on that alone. Thereby he thinks he can reach a surer knowledge than his predecessors. Non ratione, sed sensu! is his motto. But he is convinced that sense perception will never lead him into contradiction with itself, nor (as he adds, however, only in a later edition — perhaps taught by experience) into conflict with the Church. Aristotle, on the other
[89] hand, contradicts experience, himself and the Church alike!

The most significant feature of Telesio's philosophy of nature is his endeavor to put the relation between matter and force in place of the relation between matter and form that runs through Aristotle. It was characteristic of the aesthetic view of nature of antiquity that it found the explanation of natural phenomena in the forms in which they appeared; the natural processes were understood through the fully developed results to which they led. The new view of nature goes further back and asks about the acting forces. According to Telesio there are two such forces (principia agentia), an expanding one, which he calls heat, and a contracting one, which he calls cold. Heat and cold are thus for him not matters or mere properties, but forces which express themselves in two different kinds of motion. It would thus not have lain far from Telesio to conceive everything that goes on in nature as motion. These two forces act on matter, which is

never either increased or diminished, but under their influence takes on the most various forms, according to the different relation between contraction and expansion. Besides, it is uniform everywhere, without there being any ground to make a difference between celestial and terrestrial matter.

Next, he also comes into definite opposition to Aristotle by teaching that space is something different from matter, whereby the Aristotelian doctrine of the natural places of the different elements received its death-blow. The different places in space have no different properties. There can even be an absolutely empty space; it is pure imagination when Aristotle believes that nature shudders at the void. Aristotle's doctrine of the elements fell away not only through the falling away of their "natural places", but also through there remaining only the one and imperishable mass (moles), which takes on different shapes because of the different relations between expansion and contraction. If different elements are to be assumed, there ought to be only two: one in which the expanding force acts, another in which the contracting force acts. The first has its center in the sun, the second in the earth. Telesio adheres on this point to the Aristotelian world-picture, except that he conceives the mass as uniform in both and only the forces as different, and assumes a livelier interaction between the celestial and the terrestrial *[90]* than Aristotle. He also expressly opposes the way in which Aristotle lets the celestial spheres be guided by special spirits. Not compelled by an alien principle, and not by craving for something that lies beyond it, but because it is its own nature, the heaven moves round. And likewise the earth keeps still because it is its nature, cold and dark as it is. God does not intervene in nature at particular points; but he has endowed every being with its nature and its way of acting. And there is harmony between the impulses of self-preservation of the particular beings, just as in an organism there is harmony between the particular organs, in that they serve the whole by each acting according to the peculiar character of its nature. There are therefore no special final causes, but it is a satisfaction for every cause to follow its own nature, and thereby harmony arises between the particular beings. In place of an outward adjustment and intervention on God's part, Telesio assumes that the divine wisdom shows itself precisely in this, that what happens according to the necessary laws is in itself serviceable to a purpose. The heaven does not move round for the sake of the earth, but in accordance with its own nature; and yet its motion benefits the earth.

There is in Telesio an evident endeavor toward a more correct and more fruitful setting up of concepts than the Aristotelian philosophy of nature offered. But he gives no methodical, inductive demonstration of the validity and necessity of his fundamental

concepts, and so far denies his own program, according to which he wanted to build merely on sense perception. Against the fundamental concepts themselves objections could be made, and were made. A contemporary philosopher, the Platonist Patrizzi, who on essential points was dependent on Telesio, directed against him, in an interesting discussion partly with Telesio himself, partly with one of his disciples[17]), various telling objections, and remarked among other things that what Telesio called matter could not be apprehended by sense perception, which shows us only the particular, changing qualities, not absolute, passive matter. He asked, moreover, how it was possible to derive the whole
[91] multiplicity of phenomena from the two forces. These objections both struck home, and the latter was especially confirmed by the highly arbitrary and naive explanations that Telesio gives of particular natural phenomena. There was as yet no material for scientific explanations in detail, and on this point Telesio did not show the same critical spirit as in setting up the fundamental concepts valid for the whole knowledge of nature. It was arbitrary too when Telesio called the fundamental forces heat and cold. Among other things it had the consequence that the origin of heat through motion always had to be regarded as secondary or as the awakening of a pre-existing heat, while the origin of motion through heat was to be the original fact, which — as Patrizzi likewise showed — is an arbitrary assertion. When, finally, heat was centralized in the sun and cold in the earth, this was not, according to Telesio, to exclude that the earth produced heat under the influence of the sun; but then — declares Patrizzi — the earth must surely also be able to move, since heat produces motion. Telesio himself did not answer this remark; one of his disciples, on the other hand, stated that there were indeed now those who thought that the earth moves, but they had proved nothing but their own acumen. The time when the Copernican theory could gain decisive influence on the world-view had not yet come. Telesio's philosophy of nature would have had to be radically altered in its details to agree with the new astronomy, which let the earth, the "cold" body, move, and the sun, the "warm" body, stand still! But that does not exclude that Telesio on important points played into the hands of the new world-view. This is best seen from the indignation he aroused among the Aristotelians. Throughout Italy great disputations were held, in which "the forces" contended with "the forms", and in which the struggle at times — as at a disputation in Venice in the year 1573 between Telesians and Paduan students — was on the point of being fought out with bodily weapons. The philosophy of southern Italy and that of northern Italy here stood face to face. —

The mental nature stands, according to Telesio, in an intimate relation to the material, is indeed not really different from it at all. The material forces, "heat" and "cold", must

have the capacity for sensation, for otherwise they could not subsist, since, in order to be *[92]* able to gather themselves to resist each other, each of them must notice the opposite force approaching and feel satisfaction in its own subsistence and activity. Every material thing has, besides, a need of contact with other things, for which reason it follows them when they are removed; this too presupposes sense perception and feeling. Special sense organs therefore need not be ascribed to things; a sense organ is, after all, only a path along which an impression is conducted, and which can be dispensed with in immediate contact. Telesio thus finds, in mythological, animistic fashion, the necessary presupposition of the interaction of bodies in their ensoulment. Of more lasting significance is another argument on which he bases his theory of ensoulment, namely that, if the fundamental forces and the fundamental matter did not possess sensation, its origin in the beings that consist of them would be altogether inexplicable, for what one does not have, one cannot give either. He maintains, then, the impossibility of explaining the origin of consciousness from matter, if one does not assume consciousness as an original capacity of matter. But he thereby at the same time bears witness that the fundamental matter and the two fundamental forces with which he began are by no means sufficient for the explanation of everything in the world. Here, surely, a new force is added! And yet — Telesio would not admit the justice of this criticism. For just as he ascribes consciousness to matter, so he conceives the soul as a material being. If the soul were not material, how then could it be affected by material forces? How should it be able to notice expansion and contraction if it were not itself expanded or contracted? In pleasure the soul is expanded, in pain it is contracted. And since this can happen quickly and easily, it must consist of a very fine matter. This soul-matter (spiritus) has its place in the cavities of the brain, which is proved partly by the nerves springing from the brain, not, as Aristotle believed, from the heart, and by the matter of the nerves resembling that of the brain, not that of the heart, partly by death setting in when the cavities of the brain are filled with too thick a matter, or when the brain is injured in some other way, just as unconsciousness in apoplexy, fainting and sleep can also set in without alterations in the other parts of the body. Already in Vives and Melanchton we have met with this animal spirit, which is *[93]* now the seat of the soul, now, as here in Telesio, the soul itself, a representation that had been handed down from Greek physicians and philosophers. — It is evident, of course, that when Telesio thinks of all matter as endowed with consciousness, he cannot have so much against thinking of consciousness itself as something material. Yet it turns out that he makes a distinction between consciousness itself and what goes on in that "spirit" which moves in nerves and brain. For in pointing out that a uniform motion in the

"spirit" is never connected with sense perception, but that an alteration in the motion of the "spirit", determined by the action of things, must set in if sense perception is to become possible, he remarks: Sense perception (sensus) is therefore an apprehension (perceptio) of the outer impressions and the inner alterations, first and foremost of the inner alterations, since it is only through them that we know the outer influences[18]). When sense perception is thus not simply one with the alterations of motion in the inner soul-matter, but is the "apprehension" of them, the new question arises, of course: how then does this apprehension become possible? By the inserted "perceptio" Telesio has expressed an involuntary admission that the matter is not so simple after all. He thinks he has solved the problem by making the soul material; but the goblin moves house with him and asks how, then, what goes on in the material soul is *apprehended!* —

In an interesting way Telesio seeks to show that all knowledge is sense perception. He denies the difference between sense perception and thinking in such a way that he makes all thinking into sense perception. Once a motion has been aroused in the "spirit", it can later be called forth again, a habit forming itself of coming into this motion, or perhaps the motion itself in part persisting. The knowledge connected with such accustomed or repeated motion is memory. Now when a thing that has earlier been sensed as a whole later presents only some of its properties to our immediate apprehension, we can nevertheless, by virtue of the likeness with what was earlier apprehended, supply what
[94] is missing and intuit the thing as a whole, although it is given only as a fragment. We can represent to ourselves fire with all its properties, although we perhaps only see it shine and do not notice the heat or the consuming power. In such an apprehension as a whole of what is given in fragments consists, according to Telesio, the understanding (intelligere), which he would therefore rather call a memory or an estimation. Even the highest and most perfect knowledge contains no more than the capacity to discover the unknown properties and conditions of things by means of the likeness with a case that is known to us as a whole. The absolutely unknown cannot be known. There must always be a point of contact with something given, and thus a point of attachment for sense and for memory, the latter of which is nothing but prolonged sense perception. To infer is only to know the missing properties in the manner described. Even pure logic and mathematics have their origin in sense perception, since this shows us something similar to what the logical and mathematical fundamental propositions contain. Sense observation gives me immediate examples of the whole being greater than the part, and of my not being able at once to affirm and deny the same thing, just as well as it immediately shows me that snow is both white and cold, and that man is two-legged. Through all degrees of

approximation the simplest sense observations hang together with the ideal scientific principles. There is therefore no ground for splitting our cognitive faculty in two. It is in its essence one and rests throughout on the relation of likeness — or perhaps on several very compound relations of likeness — between the objects apprehended by immediate sense perception, and therefore it is sense perception throughout.

The presupposition of this whole doctrine is that likeness and difference are "sensed" just as well as other properties. "The spirit", says Telesio, "notices the likeness and difference of the things sensed; what has the same effect it notices as one, what has different effects it notices as different things." The question is whether these are quite simple acts of sense. There is in any case this discrepancy between likeness and difference and other properties, that the former — more than the latter — presuppose a process of comparison, so that one must distinguish within sense perception between a passive and *[95]* an active side. In the apprehension of likeness and difference the active side outweighs the passive. It is then only a terminological question whether one will use the word sense perception of both sides, or will say (with Cusanus) that in all sense perception there is a thinking. Yet Telesio himself sometimes came to set the apprehension of likeness in opposition to sense perception. When Patrizzi reproached him with despising reason (ratio), he answered: "I by no means despise reason, that is to say: the knowledge of things which *not sense perception, but the likeness between the things apprehended by sense perception* affords us, and I would never think either that it ought to be despised. But I shall always maintain that one must believe sense perception more than reason." To this Patrizzi in turn answers very rightly: "But of this likeness, then, you do not seem to assert that it can be apprehended by sense perception; by what, then, is it apprehended, if not precisely by reason?" — In the apprehension of likeness, then, something is added that does not lie in the simplest acts of sense. Telesio's attempt to make knowledge absolutely alike at all stages has thus not succeeded, but it was in itself a justified attempt, a counterpart to the attempt made by Cusanus. Both sketches contain fruitful thoughts in psychological and epistemological respects. —

Just as all matter does, the soul too seeks to preserve itself. It strives after many different goods and has many different drives; but when it makes a choice and subordinates all goods to one single good, it always appears clearly that the standard in this choice is self-preservation and its conditions. All goods get their value from their relation to self-preservation, and to the natural activity by which a being is preserved and maintained an immediate feeling of pleasure is attached. No being, and therefore not the "spirit" either, can strive after any other end than self-preservation. It appears especially

that all knowledge is a means to it. Wisdom becomes the sum of all virtue, since right knowledge is of so great importance for being able to find the means of self-assertion. The various particular virtues are the various capacities that self-assertion has need of.
[96] The individual cannot live securely and comfortably except in interaction with others, and in order that this interaction may become intimate and fruitful, individuals must be joined as closely as if they made up a single compound being, so that they work together like the organs in an organism. In this way the social virtues are motivated, whose sum is humaneness (humanitas), and whose possibility rests on the need of social life, intimacy and goodwill. But the highest of all virtues[19]) is nevertheless magnanimity (sublimitas), which regulates the impulse toward honor. Man will not be despised; he cannot bear to be regarded as unfit to do his part, or as impure or as harmful. He then seeks a support in the judgment of others, since his own power does not always bear sufficient witness. Magnanimity now holds fast that only that honor is to be sought which is grounded on the individual's own inner goods, and that these goods themselves are to be preferred to the honor that may be connected with them. Honor is for magnanimity not a necessary, an inner good; what matters is to be worthy of honor; there is a sufficient source of satisfaction in purity and ability themselves. —

But all this, both knowledge and self-preservation, concerns, according to Telesio, only the material soul, the one that develops from the seed (spiritus e semine eductus). But now there shows itself in man, besides this sensuous knowledge and natural self-preservation, also a higher need and sense, which points beyond the present life and its preservation, and which must therefore be explained by God's having implanted another soul, which as an incorporeal form joins itself to the naturally developed one when the body is fully formed. Only thus is it understood that man in his contemplation can forget his sensuous necessities, and, on the whole, that he is not satisfied with the present life. How this "form added from above" (forma superaddita) is related to the natural soul, Telesio does not explain more closely. By this doctrine, which is perhaps due merely to an accommodation to theology, he breaks the integral character of his philosophy. And he overlooks that already within natural ethics he has described a state of character that is raised above merely physical self-preservation. Several psychological
[97] intermediate terms are missing for an understanding of how magnanimity can develop out of the original impulse of self-preservation. The endeavor ought rather to have been directed to filling up these gaps than to taking up supplements that do not fit the whole foundation. —

With all its imperfections Telesio's system nevertheless stands as one of the most

remarkable that the period of the Renaissance produced. Earthbound and sensuous in its foundation, it nevertheless rises boldly toward the heights. It drew consequences and indicated ways of looking at things which could be developed with ample grounding only much later. And yet it also set its own time in strong motion, and gained considerable influence on thinkers like Bruno, Campanella and Bacon.

12. The Copernican World-Picture

The traditional conception of the world had been shaken by Cusanus's criticism of the doctrine of absolute motion and by the polemic of Paracelsus and Telesio against the Aristotelian elements and forms. But the ground was first drawn out from under it by Copernicus's new world-picture. He showed that there was every reason to regard things in the opposite way from that to which people were accustomed. That was what Luther, who had himself turned so many things upside down, complained of: that the Copernicans wanted to stand the world on its head. No wonder that the new world-picture aroused the greatest enthusiasm in some, by reason of the infinite horizon it seemed to open, and the greatest indignation in others, because they seemed to lose the clear and surveyable frame within which everything that one believed one knew about the world and the forces of the world had hitherto fitted. A hard struggle began, which cost noble blood and what is dearer than blood. It was not only a struggle against religious misgivings, but also a struggle against trust in immediate sense perception. Human knowledge was to learn that existence in itself can be constituted quite otherwise than it immediately presents itself to us. It was a lesson in distinguishing between the subjective apprehension and existence itself which became of great philosophical significance. Freedom of mind was then especially needed in this matter, the capacity to *[98]* disregard what the senses immediately seem to teach and what religious ideas seem to demand. Among the champions of the new hypothesis the necessity of a "free mind" is accordingly constantly emphasized. Copernicus's first disciple, Joachim Rheticus, says with regard to his master's relation to the old astronomers: "Whoever would inquire must have a free mind"; Kepler too regarded Copernicus's free mind (animus liber) as an essential condition of his work; and Galilei again used the same expression (ingegno libero) of Kepler[20]). It was a process of liberation that was at stake.

The founder of the new conception of the world stood on the ground of humanism. Nicolaus Copernicus (whose real name is said to have been Coppernick) was born of a well-to-do family at Thorn on 19 February 1473. Poles and Germans have contended over his nationality; probably he descends from a German family that had long lived in

Polish lands. He is not known to have spoken or written Polish. At Cracow, where there was a flourishing university, he studied the humanities, mathematics and astronomy. Through the influence of his uncle, the Bishop of Ermland, he was received as a canon or member of the cathedral chapter at Frauenburg. This brought with it, to be sure, a lower degree of clerical dignity, but no theological training was required, and only very few of the canons were able to perform clerical functions. They lived more as noblemen and humanists than as clerics. It is a wrong picture that has been formed of Copernicus when he has been imagined as a monk in his cell. For the time being the young canon spent ten years in Italy, where at Bologna, Rome and Padua he pursued astronomical, medical and humanistic studies. Of his teacher at Bologna it is related that he is already supposed to have had his doubts about the correctness of the Aristotelian-Ptolemaic system. After his return home from Italy he stayed for some years with the Bishop of Ermland, partly as physician, partly as courtier. In this period, between his thirty-third and thirty-sixth years, the main features of his system took shape in his mind. He says himself that from 1506 he began to write down the scientific exposition and grounding
[99] of his doctrine. After his uncle's death he lived the greater part of his life in the castle of Frauenburg; an exception is formed by some years in which he managed some of the chapter's estates. It was thus not only astronomical studies that occupied him. He was an administrator and a physician, and took part besides in negotiations on reforms of the coinage, and he had not laid his humanistic studies wholly aside. In 1509 he published a Latin translation of Theophylact's Epistles. He was thus active in many directions. He followed, moreover, what went on in the world. As a physician he belonged, to be sure, to the old school and took Avicenna as his basis. But he and several of his colleagues stood freely and sympathetically toward the religious movement. A reform was wished for, and in any case people wanted the struggle to be fought with spiritual weapons. It seems to have been the intellectual tendency of Erasmus that prevailed among the canons of Ermland.

In many years Copernicus let his work rest. He was constantly polishing it, and he felt great misgivings about publishing it. His ideas nevertheless became known far and wide, perhaps especially through a little abstract (commentariolus) which he composed, and which has been found in manuscript in recent years. The decisive push toward publication came from a young, enthusiastic adherent, Joachim Rheticus, who was a teacher at the university of Wittenberg. He traveled (1539) to Frauenburg and stayed for two years with Copernicus in order to study his work. From his hand the learned world received the first detailed report of the new system (Narratio prima de libris

revolutionum. Printed in Danzig 1539—40). He also got the master to overcome his misgivings and to permit the printing.

Copernicus's last years were not good. The Catholic reaction began, and the freer, humanistic spirit that had hitherto prevailed among the canons was now hampered and persecuted by a new bishop, Johannes Dantiscus, formerly a man of the world and an erotic poet, now a fanatical churchman. In spite of his earlier friendship with Copernicus he now seemed to make a special point of vexing him. Copernicus was compelled to break off his connection with various persons close to him, whom the bishop regarded as heretical. He lived his last years in spiritual solitude. In the spring of 1543 he was *[100]* struck by hemorrhage and paralysis and died on 24 May 1543. On the day of his death he received a copy of his work, but by then consciousness had already gone.

It does not belong here to bring forward all the astronomical questions that Copernicus comes to touch upon. What interests the general history of thought are the presuppositions from which the reformer of astronomy starts, and the great features in which his new world-picture appears. By its presuppositions his system bears the spiritual stamp of the time, and by the character of the world-picture it intervened in the progressive development of the time.

Among the presuppositions there are two which are of special interest.

It was the artificial and intricate character of the old world-picture which led Copernicus to think about the possibility of conceiving the celestial phenomena in another way. This whole system of manifoldly connected spheres and epicycles seemed to him to conflict with the simplicity and purposiveness that nature otherwise displays. The sagacity of nature (naturæ sagacitas) otherwise reaches its end by the simplest ways, without circumlocution and with harmonious connection between the acting elements. It prefers to attach many effects to one cause rather than to multiply the number of causes. This principle of the simplicity of nature was for Copernicus and his successors not merely a methodological but a metaphysical principle. They would have been at a loss had they been asked how, then, they knew that nature always acts in the simplest way. It was for them an immediate certainty, a religious faith. They still felt themselves near to the heart of nature. It became an essential task for the philosophy of the following age to discuss the justification of such presuppositions. For the time being people boldly took hold, and it turned out that the presupposition had in this case been happily chosen. It set the task of finding a point of view from which the order of the world could be surveyed and the present confusion removed. In particular, by virtue of the principle of simplicity, it could not but appear absurd that the whole universe should move round the *[101]*

earth, the great mass round the small. Would it not be simpler if the great mass rested and only the small ones moved?

With the principle of simplicity was combined the principle which had already led Cusanus beyond the old system of the world (without, however, leading him to a new one), the principle of relativity. When a motion takes place in space, then — says Copernicus — sense perception cannot immediately tell us what it is that moves. It may be the thing sensed that moves. But it may also be that the one who senses is himself moving. And it may finally be both the thing and the one who senses that move, with different speeds or in different directions. When we sail, the shores, after all, seem to move away from us, although in reality it is we, not they, that move. Let us then suppose that the earth, the place from which we sense what goes on in the universe, moves, and let us then see whether we do not get a simpler and more natural conception of the world out of it than when we assume that it is the objects of sense perception that move![21]) — Copernicus's whole work is a mathematical demonstration that under this presupposition the phenomena would look precisely as sense perception shows them to us.

The new world-picture, then, has the sun as the midpoint of the world and the source of light. Round it move, fastened to concentric spheres, the planets, in whose series the earth now takes its place between Venus and Mars. The earth moreover moves round its own axis. Outermost, the whole universe is enclosed by the fixed sphere of the fixed stars, which is immovable. Whether the world is finite or infinite, Copernicus will decide nothing. But so much stands firm for him, that the diameter of the earth's orbit must be vanishing in relation to the earth's distance from the sphere of the fixed stars, since no such apparent advance and retreat is observed in the fixed stars as in the planets, which are nearer to the earth. This enormous enlargement of the universe, which for Copernicus himself was nevertheless not a bursting of the old frame, was one of the great stumbling blocks of the new theory[22]), and Copernicus felt it himself. He consoles himself that one must after all rather grant it than assume such a swarm of circles as
[102] would be necessary if the earth were assumed to be at the center. And is it not, after all, more reasonable that what is enclosed should move than that the all-enclosing frame itself should be in motion? The sphere of the fixed stars is for Copernicus the absolute place, that in relation to which every place and every motion is determined (communis universorum locus). The heaven is, besides, the most venerable thing in the universe; immobility must therefore belong to it. And should it move, this would have to happen with so violent a speed that there would be far greater reason to fear that it would break

to pieces than to entertain, with Ptolemy, such a fear for the earth's part, if it were to move. That water and air are not left behind in the earth's revolution comes from their making up one whole with the earth itself and therefore sharing in its motion. It is just the same with the objects on the earth. — Thus Copernicus seeks briefly to answer the physical objections that might be made against his theory. Of the cause that makes the planets move round the sun he says nothing. With the ancients he takes circular motion to be the natural motion of bodies. A rectilinear motion occurs only when a part is loosened from its connection with the whole to which it belongs[23]). It is then always a sign that something is not as it ought to be. — Copernicus's theory has a descriptive character. It is an attempt to show how the things in the world must look from a definite position. But the compelling necessity of taking up this position he could not establish. And many difficulties in detail still remained unexplained.

And yet one of the greatest among his successors praised him precisely for the courage with which he developed his view without letting himself be stopped by any difficulty, trusting more to reason than to sense perception. Galilei even finds it understandable that he did not go more at length into the counter-arguments; he paid no more heed to them than the lion pays to the barking of dogs. Without such courage the new view would not have come forward and would not have been tested. Copernicanism stands in the history of thought as a great example of how necessary it is that ideas and hypotheses *[103]* be set up which can show investigation the way. The venture has its right in science too.

Yet Copernicus did not conceive his doctrine as a mere hypothesis. — He did not himself see to the publication of his work. It was entrusted to the preacher Osiander in Nuremberg, who, so that the book should not arouse indignation, put a preface in front of it, in which the new doctrine was presented as a mere hypothesis, which one could, purely mathematically, have the pleasure of trying to take as a basis. The new assumptions, he says, may be just as good as the old, and no one must expect that astronomy can teach anything that is certain. This preface, which the reader could not but believe derived from the author himself, no doubt did much to keep the theory from arousing great attention for the time being. If the author was not in earnest with it, it was, after all, of no great interest. Luther mocked at it in his Table Talk, and Melanchton declared in his lectures on physics (Initia physices) that it was only from a need of something new and of showing one's acumen that people invented these strange things, but that it was nevertheless not seemly (honestum) and could set a harmful example when, merely to exercise one's mind, one engaged in absurd assumptions of this kind, instead of receiving with reverence the truth that God reveals and resting content with

it, especially since the new doctrine obviously conflicts with the statements of the Bible. Melanchton therefore also regarded his young colleague Rheticus as a restless head. Only comparatively few still joined the new doctrine in the course of the sixteenth century. Among the most enthusiastic of these were Giordano Bruno and Kepler, while Galilei still held back for fear of ridicule, and Tycho Brahe let only the planets, but not the earth itself, move round the sun. Bruno already protested in strong words against the idea that the timid preface could derive from Copernicus himself. He intimates that there were those who contented themselves with reading the preface and then dismissed the matter when they saw that it was only a question of mathematical speculations. Bruno asserts that that preface must have been added by "some ignorant and presumptuous ass", who
[104] wanted to make the book palatable to the other asses, and he appeals to Copernicus's own foreword, from which it is seen that the theory was his serious opinion, just as he also, after all, speaks in the book itself not only as a mathematician but also as a physicist. And later Kepler, relying on letters from Osiander himself, gave an express proof that the preface was not genuine[24]).

Even in Italy, where the ecclesiastical reaction was now in full swing, it was some while yet before the new world-picture became so well known that it could arouse a persecution. That it came so far was due perhaps not least to the enthusiastic thinker who not only took up the new world-picture and wove it into his train of thought, but at the same time enlarged it and drew the most important consequences from it with regard to the general world-view, while in return his struggle for the new ideas led to his ruin.

13. Giordano Bruno

a. Biography and Characterization

In Bruno we find the ideas of Cusanus, Telesio and Copernicus bound together into a distinctive whole, the greatest philosophical edifice of thought that the Renaissance erected, and in several respects of a prophetic character. The scientific world-view of the modern age was founded in some of its most essential features. On the other hand, Bruno sacrifices at the altar of his time and shares its superstition and its extravagant fancy, to which his passionate and impatient mind disposed him. Clear methodical thinking was not his affair; only the great lines took clear shape and were followed with an enthusiasm that stood its test. But in particulars he was not clear about himself or his thought; he did not know how far it had moved away from the old, to which he still felt himself bound, however much he could mock and scorn it. He is a figure in whom neither the

intellectual nor the moral physiognomy is drawn purely and clearly. He himself felt the contradictions and the inner struggles within him. They sprang from his nature, but *[105]* they were nourished by the circumstances under which he and his thinking developed.

His fate was decided early. He himself said that when one has buttoned the first button of one's coat wrongly, none of the others can be buttoned rightly. This fits the step that Filippo Bruno, who was born at Nola in southern Italy in the year 1548[25]), took when, at fifteen years of age, he entered a monastery in Naples. His family seems to have been noble, and he had received a good preparatory education; otherwise nothing is known of the years of his childhood, and in particular not what determined him to become a monk. It was perhaps in an exalted mood that he took this resolution. Strong oscillations between boundless enthusiasm and deep dejection, and between mystical rapture and critical inquiry, belonged to the peculiarities of his character. He took a step that cannot be undone, and yet his whole following life was an attempt to take this step back. In the monastery, where he took the name Giordano, it soon showed how little he could bend his spirit under discipline. He was repeatedly accused of heresies. First he made himself suspect by removing all images of saints from his cell and keeping only a crucifix; later, when he had already been ordained a priest, by defending the Arian doctrine. He declared later, during the examination before the Inquisition in Venice, that from his eighteenth year on he had harbored doubts about the dogma of the Trinity. When he noticed that an accusation was being prepared against him, he fled, to escape imprisonment, to Rome, and when the accusation followed hard on his heels there, he fled, after laying aside his monk's habit, from Rome as well. That was at the end of the year 1576. And now began his restless wandering from place to place, amid passionate intellectual labor and no less passionate struggle for the new ideas he had arrived at. There is, however, much that suggests that it was not only the intellectual side of his nature that rebelled against monastic discipline. He was a strongly sensual nature and himself declared that not even the snows of the Caucasus could cool his fire. He had a hard struggle to go through in order to ennoble his drives, so that a "heroic affect" could develop out of the "animal affect," and it was not through asceticism that he could *[106]* strive toward this goal. A series of poems which he later interprets allegorically (in De gl' heroici furori), but which were surely love poems originally, bears witness to this. In a comedy whose material must have been gathered during his stay in his native region, circumstances are depicted and expressions used whose coarseness is striking, even when one knows that the inhabitants of Nola were known for their free speech. Though he entered a monastery early, he did not thereby make himself a stranger to life,

which indeed stirred so strongly in him that it soon drove him out again. Essentially, however, it was intellectual doubts that led him to a break. — After his flight from Rome he wandered about northern Italy for some years. At Noli near Genoa he ran a school for small boys for a time and at the same time taught some young noblemen astronomy. Probably he was already at this time an adherent of Copernicus; he himself dates his acquaintance with him back to his very young years. At first he could regard the new doctrine as nothing but madness, but soon its truth and significance dawned on him, especially when he had learned enough mathematics to follow the grounding in detail. He then felt the need to go further than his teacher, who, with ancient and medieval astronomy, still held to the limitation of the world and the immobility of the eighth sphere. For Bruno the universe is unlimited, since every attempt to posit an absolute limit is unjustified, and it includes within itself an infinity of worlds like ours, each with its own center. The Deity works at every point of this infinite universe as the inner soul that embraces and animates all things. Such were the thoughts that now developed in Bruno. Besides by Copernicus he is, as already remarked, strongly influenced by Cusanus and Telesio. How the influence of these thinkers appears in him will become clear in the exposition of his doctrine in its particular motivation. Here it need only be stressed that this altered conception of the world brought him into sharp opposition to the reigning Aristotelian philosophy, on which the theology of the time also relied, and which stood
[107] and fell with the old world-picture. Since he found no lasting place in northern Italy, though he roamed through it as far as Venice, he crossed the Alps and came to Geneva (1579). Here he seems to have been counted as a member of the Reformed Church, although he himself, during the examination before the Inquisition in Venice, denied having gone over. For his name has been found entered in the register of the university, even in his own hand, and a condition of this was assent to the Calvinist confession of faith. Moreover, records have been found which show that he carried on a violent quarrel with a professor at the academy and with the town's ministers; it is laid especially to his charge that he called these last "pedagogues"; he has also taught heretical doctrine. He is therefore excommunicated and must — in order that the excommunication may be lifted — express repentance for what he has done. This episode evidently presupposes that Bruno at any rate outwardly joined the Reformed Church. He then at once got to feel Calvinism's theocratic rule, and this has surely, together with other experiences from his stay in Protestant lands, laid the ground for the sharp judgment he later passes on Protestantism, especially in its Calvinist form. What was historically to develop out of Calvinism, Bruno did not suspect; he got to feel only its dark sides, and they caused

him to leave Geneva after a few months' stay and go to southern France. At Toulouse he began to lecture at the university to great applause and was appointed professor, a position he held for two years. Here too he lectured on astronomy and on Aristotelian philosophy. It was his calmest time, although here too he came into conflict with the learned, probably on account of the new things he had in mind. It is worth noting, in order to understand Bruno's relation to Catholicism rightly, that already here in Toulouse he took steps to be readmitted to the Catholic Church, steps which he later took again in Paris. He thought he could live in harmony with the Church without returning to the monastery. He hardly regarded his relation to the Church as broken. What was essential in Christianity he thought he could unite with his new ideas, and the interpretation of the dogmatic representations that was the presupposition of this reconciliation he believed the Church could permit. The repellent impression he had received of Protestantism *[108]* nourished the need for union with the Church he had left, even though he thought that it needed a thorough reform. But he was told — both in Toulouse and in Paris — that the first condition was that he should go back to his monastery, and to that he could not agree. His wish was to live in quiet for his studies, without again coming under the monastic rule. — This conception of Bruno's relation to Catholicism is supported by his statements during the examinations before the Inquisition, just as, on the whole, his return to Italy only becomes intelligible on this presupposition.

When the civil war put a stop to Bruno's university activity in Toulouse, he went to Paris (1581), and appeared here as an academic teacher with great applause. He expounded here, besides scholastic subjects, partly the so-called Lullian art, a kind of schematism of thought with which he was much occupied and which he treated in a whole series of writings, partly the art of memory, to which he also ascribed great importance. King Henry III had him summoned in order to obtain information from him about his ideas. By the king's favor and recommendation he came, when in the year 1583 he had gone over to England, into the house of the French ambassador. In the writings he published in London he praises in lofty tones the hospitality and high culture of the ambassador (the Marquis de Castelnau) and his family. He was less satisfied with the English, who struck him as barbarians. Even the learned on the whole formed for him no exception to this. Especially in the first of a series of Italian writings he published in London, *Cena de le ceneri* (The Ash Wednesday Supper), his ill will toward the learned and unlearned rabble found vent. The fact was that here in England he appeared for the first time as a reformer of the world-view and now, in his own expression, learned what it is to cast pearls before swine. Bruno was now not only enthusiastic about the new

ideas that took shape more and more for him, and the time for whose proclamation had now come, but he also had a sense of his own importance, to which he gave vent in a rather bombastic way. It was not only conservative pedants who could feel irritated by
[109] the letter to the vice-chancellor of the university at Oxford with which he announced his appearance there. He presents himself here as a teacher of pure and blameless wisdom, as a philosopher famous at the other European academies, who is a stranger only among barbarians and brutes, an awakener of sleeping souls, a vanquisher of ignorance, whose gaze is directed only at spiritual culture, hated by the stupid and the hypocrites, greeted with jubilation by the honest and the earnest, etc., etc. Much in this bombast must of course be put down to the taste of the time. The University of Oxford opened its doors to him, and for some time he gave lectures there on his psychological and astronomical theories. His ideas of the unity of mental life under the many different forms under which it appears in the world simultaneously and successively, and of the infinity of the universe, since the earth, far from being the absolute center, was only one of the infinitely many globes in constant motion, none of which whatever has any right to stand as absolute center — these ideas might well make the scholastics at Oxford dizzy and give them offense. The lectures had to be broken off. At a public disputation in Oxford too, where by his own account he reduced his opponent, "the coryphaeus of the university," to silence fifteen times, he thought he had received proofs of the ignorance of the specialists. And the same thing happened at the disputation he has described in detail in "Cena de le ceneri." Yet he did not give up the hope of progress for his ideas, but used his stay in the French ambassador's house to work out a series of writings in Italian and in dialogue form, in which he has given the fullest and grandest picture of his views. However restless and however quickly broken off his life was, he nevertheless attained the greatest happiness a thinker can attain: to have his ideas developed as fully and clearly as was possible at the stage at which he stood. In spite of all opposition, his stay in London was therefore a happy time for Bruno. Some distinguished men such as Phillip Sidney and Fulco Greville showed him friendship; he came into contact with aristocratic circles and was even received by Queen Elizabeth, whom he praises in his writings as Amphitrite, the queen of the sea. His severe judgment of the English he does
[110] indeed limit more closely in a later writing. Yet it would surely be too bold to suppose that even within narrower, more select circles any real understanding of his views was formed. Of this not the slightest trace can be shown; of Bruno's stay in England we know anything at all only through his own accounts. Philosophical interest in England went in a different direction from Bruno's, both now and in the following time; the time for the

understanding of Bruno came only later. The ideas related to Bruno's which are found in Shakespeare, especially in Hamlet, are rather to be derived from Montaigne and other contemporary authors, in so far as they are not due to the great poet's own thinking[26]).

It was an astonishing productivity that Bruno developed during his stay in London. Besides a mnemotechnic writing (with the introductory letter to the Oxford chancellor mentioned above) he published here five important Italian dialogues which contain an exposition of his philosophical ideas. In the first, "The Ash Wednesday Supper," the Copernican theory is treated, with the grounding and the extension Bruno gave it. Then followed "On the Cause, the Principle and the One" *(De la causa, principio et uno),* Bruno's chief work, in which his speculative ideas and his ideas in the philosophy of religion are developed. It is continued in the dialogue *On the Infinite Universe and the Worlds,* which sets out more closely his doctrine of the infinite, divine principle that reveals itself in an infinity of worlds and beings. After he had in these writings developed his views from the theoretical side, he gives in "The Expulsion of the Triumphant Beast" *(Spaccio de la bestia trionfante),* to which is joined *the secret doctrine of the Pegasean ass* and "On the Heroic Affects" *(De gl' heroici furori),* his moral philosophy and his practical view of life. These writings are distinguished in many passages by profundity and by an exposition full of mood and poetry. But with these alternate passages where the old scholasticism still sticks its head out, or where imagination passes over into extravagant fancy, or where there is a struggle with language to utter the unutterable. Perhaps this last is what is most characteristic of Bruno. His world-view is essentially motivated by the new way of seeing which Copernicus put forward. But Bruno sees that the consequence leads *[111]* much further, and that in the same moment one removes the earth from its resting place at the center of the universe, one properly loses the right to posit rest and limit anywhere in the world. Together with the endeavor to ground the world-view on given experience there then goes a movement that leads beyond the limits of experience. And as convinced as Bruno was that the highest reveals itself in nature, so convinced was he also that any revelation whatever, or series of revelations, is insufficient to express its fullness and its unity, so that no oppositions hold for it and no word fits it. No thought, no number and no measure suffice here, although nature is determined throughout by thought, number and measure. He is equally convinced of the necessity of the scientific conception and of its insufficiency. In one of his philosophical poems he says: "My thought and my sense teach me that no act of thought, no measurement and no calculation can comprehend a force, a mass and a number that extends beyond every lower limit, every middle, every upper limit."

Con senso, con raggion, con mente scerno,

Ch'atto, misura, et conto non comprende

Quel vigor, mole, et numero, che tende

Oltr' ogn' inferior, mezzo et superno.

The "learned ignorance" (docta ignorantia) that we know from Nicolaus a Cusa is Bruno's last word too. But he stands nearer to the world of experience and has a greater interest in it than the theologian, for whom the inquiry into the nature and limit of thinking was essentially a ladder to mystical absorption in the thought of the Deity. Bruno uses the ideas he owes to Cusanus above all for the grounding and development of the new world-picture. Yet he does not lose himself in the intuition of the external world. He holds the conviction that the Deity works in the innermost part of the world, and that this innermost part is found at every point. The external, childish distinctions between heaven and earth have fallen away, but only (just as later in Böhme) to make room
[112] for the feeling that the highest is everywhere where the sense is open to it. But then the work begins anew in the domain of the inner life. Here too the strong oppositions stir, a surge of waves without limits, a craving toward an infinite goal. Especially in the writing on "the heroic affects" this is expressed in a wealth of poems, allegories, psychological reflections and ethical ideas. Pleasure and pain are bound together, so that the one cannot be chosen without the other; only through danger and ruin does the way go to victory; and the danger is not only outer but inner too; the will must fall in order to be able to rise high enough; in repentance the soul soars upward like the swan from the marsh. There is for Bruno a tragic relation between knowledge and truth, and between the will and its goal. But the highest is for him to feel this disproportion in the constant striving. "Even if the goal desired is not reached, and even if the soul is consumed by the strong craving, it is enough for it that so noble a fire is kindled in it":

Eh bench' il fin bramato non consegua,

E'n tanto studio l'alma si dilegua.

Basta che sia si nobilmente accesa!

In his own nature and in his own fate Bruno had oppositions in full measure. Ideal craving and a passionate need of knowledge on the one side, fantastic self-esteem and sensual ardor on the other: here tasks enough were set, and here there were reefs enough on which the development of character could founder, even apart from the external circumstances. And how complicated and changing were these! A runaway monk who could not find his place in the Protestant lands where he went about, but passed a far severer judgment on Protestantism than on the Church under whose ban he lay! A

thinker who wanted to overthrow all inherited systems for the sake of a new one, which his thought constantly strove to shape, and which could find no understanding among his contemporaries! A southerner who felt himself in northern Europe as in exile among barbarians! —

After two years' stay in England Bruno went back with the French ambassador to Paris, where he made his second vain attempt at a rapprochement with the Catholic Church. At the university he held a public disputation, at which he attacked the Aris- *[113]*
totelian philosophy and defended the new conception of the world. The freedom of thought was asserted here in strong and clear words. The invitation piece for this disputation *(Acrotismus,* probably printed first in Paris 1586, later in Wittenberg 1588) is one of Bruno's clearest writings. It gives in concise form the most important ideas that had been developed in the Italian dialogues. He left Paris soon after the disputation, probably on account of the political disturbances. Thereafter he wandered about among various German universities. At Marburg he was refused the right to teach, which led to a violent scene between him and the rector of the university, who thereupon struck him off the list of students, on which he had already been entered. Probably it was as a Catholic that he was excluded; his writings were hardly known. At Wittenberg, on the other hand, he was allowed to lecture, and here he lived two quiet years. But a new duke came, and he then had to move on. In his farewell oration he praises German science, represented by Cusanus, Paracelsus and Copernicus, and glorifies Luther for his struggle against the power of the Church: like a new Hercules he had fought Cerberus with the triple crown! —

Once more the wandering life began. After short stays in Prague and Helmstädt he went to Frankfurt, where he settled for about a year in order to print a series of writings which, partly in the form of didactic poems, were to give a systematic exposition of his doctrine. Unlike the London writings, these writings are composed in Latin, probably because they were meant for the German learned. This series of works is remarkable, moreover, in that Bruno approaches an atomistic view of nature (in the writing *De triplici Minimo),* and that in his exposition of the new world-picture (in the writing *De Immenso)* he uses Tycho Brahe's investigations of the orbits of comets, which excellently supported the ideas he had developed earlier. But side by side with this there are found in these writings several abstruse and symbolizing developments, which are of interest only in showing what one could resort to at that time for want of scientific data. On this point as on other points Bruno has a certain likeness to Kepler, in whom too great scientific ideas and symbolizing speculations were joined in a remarkable way. *[114]*

Before the printing of these books was finished he had to leave Frankfurt. In the preface to De triplici Minimo, whose publication he did not manage to see through, the publishers say that the author was taken from them by a sudden accident (casu repentino avulsus). Probably he was expelled; documents have been found which establish that the magistracy of Frankfurt had been unfavorably disposed toward him from the very time of his arrival. The expression the publishers use does not suggest a voluntary departure. Yet he had received an invitation to come to Italy, which he would surely have followed in any case. In a sonnet he has compared himself to the moth that flutters toward the light. By this he was thinking of the irresistible but painful need of truth. But a similar need seems to have stirred in him at the thought of his fatherland. A new fatherland he had not been able to find during his long wanderings north of the Alps. But how much danger must not threaten the runaway monk when he returned? Perhaps the circumstance that the invitation came from Venice made him feel safe. A young Venetian nobleman, Giovanni Mocenigo, who had studied one of Bruno's mnemotechnic writings, learned from his bookseller that the author was staying in Frankfurt. He then invited him to Venice in order to be able to receive instruction from him. It seems as if he believed that Bruno also understood secret arts, and that it was really these he wanted to be initiated into. Bruno accepted the invitation. After his sudden departure from Frankfurt he stayed for a time in Zürich, where he gave lectures to some young men, and then he set out on the fateful journey across the Alps (autumn 1591). In Venice he taught Mocenigo and at last had his lodging in his house. But his pupil complained, when some time had passed, that he was not learning all that he wished. And he also came to have scruples of conscience about harboring a heretic. At the bidding of his confessor he then denounced Bruno to the Inquisition and shut him up in his house until he was fetched
[115] to the prison of the Inquisition (May 23, 1592).

In the examination Bruno broke down. He protested against several detached and wanton statements which, on Mocenigo's information, were ascribed to him; he declared that he had always held fast to the faith of the Church in his innermost being, even if as a runaway monk he had been prevented from taking part in the Church's divine service, and even if in his philosophy he taught things that indirectly led to conflict with the dogmas — and at last on his knees he begged pardon for all his errors and expressed the wish for a penance that might open the bosom of the Church to him again. — In order to understand this denial of what he himself had so enthusiastically proclaimed, one must hold fast to the fact that Bruno himself never thought that the bridge between him and the Catholic Church had been broken. His repeated attempts at reconciliation bear

witness to this. And the repellent impression he had received of Protestantism must also have disposed him to reconciliation. While still in Venice he had declared to various persons, who confirmed it to the Inquisition, that he had a writing in hand which he meant to lay before the Pope, and which he hoped would make it possible for him to live in Rome, occupied with literary pursuits. He meant also to lay before the Pope the writings he now acknowledged, and hoped to be able to obtain absolution for what he had transgressed. He perhaps thought that the symbolic truth he could ascribe to the ecclesiastical representations in his philosophy might cause it to be tolerated, especially since, by virtue of "learned ignorance," he granted faith an independent place outside the domain of science. He therefore begins boldly to tell the inquisitors his life and to set out his philosophy for them; he tells them, e.g., that what the Church calls the Holy Spirit is for him the world-soul that holds the universe together, and that while souls "catholicamente parlando" went after death to paradise or purgatory or hell, he taught, "seguendo le raggion filosofiche," that mental life was imperishable and constantly took on new forms, which he found expressed in the old doctrine of the transmigration of souls. He does not properly appeal, like Pomponazzi, to the claim that something can be true in theology which is not true in philosophy. He assumed not a double truth but a double form of truth. The relation between his philosophy and the doctrine of the Church stood for him roughly as the relation between the Copernican theory and sensuous *[116]* intuition. Yet it must be granted that he did not come to full clarity with himself. All the easier is it to understand that he — the man of moods who swung between exaltation and dejection — could break down, and that, in order to save the prospect of a quiet literary life such as he had imagined for himself after the restless years as a wandering scholastic, he — just like Galilei later — made the confession that was wanted, instead of setting his understanding of Christianity against the ecclesiastical-dogmatic one.

Perhaps he would have escaped with this, and then he would not have come to stand in history as the heroic figure we now know. But the Roman Inquisition had received word of his case and demanded his extradition, partly because he was guilty of such grave heresies, partly because by flight he had evaded earlier accusations in Rome and Naples. After some hesitation the Venetian government handed him over, and a new trial now began in Rome. Of the records of this trial only a few fragments are known, and the rest is said, without its being known how, to have disappeared from the papal archive[27]). One does not know, therefore, how it came about that Bruno sat over six years in prison in Rome. It is only seen that the Roman inquisitors, among whom the later Cardinal Bellarmine especially distinguished himself, showed more zeal than the Venetian ones.

Bruno's books were gone through and a list drawn up of eight grave heresies, whose retraction was demanded. Probably, among other things, the assumption that there are infinitely many worlds was laid to his charge[28]). It was heretical because it seemed to conflict with revelation, which cannot have taken place more than once. For the same reason the assumption of antipodes had been found heretical in the Middle Ages. Yet it is seen from the remains of the examinations that are known that the Pope wished to restrict the retraction to be demanded of Bruno to such propositions as the Church had condemned from of old; the propositions directly connected with Copernicanism were then to be left aside. But Bruno stood firm against this precise demand. He denied that
[117] he had made himself guilty of any heresy; it was the inquisitors who had interpreted his opinions in a heretical sense; he had nothing to retract. This attitude of his confirms the conception set out above of his relation to Catholicism. With death before his eyes he held fast to his conception of Christianity as against the Church's, and he was no more willing to abjure his religious than his philosophical views. He composed a defense which he wished laid before the Pope; it was opened, but not read. On February 9, 1600, the sentence was pronounced to Bruno. He was degraded, excommunicated and handed over to the secular power, the governor of Rome, with the usual hypocritical request that he might be punished mildly and without the shedding of blood. To this Bruno answered with a threatening look: "You who pass sentence on me perhaps feel greater fear than I on whom it is passed!" He meant, no doubt, that they felt fear of the truth, while he himself had now overcome the fear of what there may be to suffer in the service of truth. Now he saw clearly what he had not seen in Venice: that it was the free search for truth itself that had to be asserted. Now he did not waver. On February 17, 1600, he was burned on the Campo di fiore and suffered death with steadfastness. He turned away a priest who wanted to hold out a crucifix to him, and died without any outcry. His ashes were scattered to the wind. — But on the spot where he was burned his statue was erected in the year 1889 by contributions from the whole civilized world, and the Italian state is now defraying the cost of a splendid edition of his works. —

Having come to know the life and personality of this greatest philosopher of the Renaissance, we now pass on to set out his philosophy in detail.

b. The Grounding and Extension of the New World-Picture

Giordano Bruno is one of the first thinkers to have a clear consciousness that great thoughts are owed to a long, successive series of experiences. He thinks he has uttered great thoughts, but he is also conscious of what he owes to his predecessors, especially

the great astronomers on whose observations he builds. While in the Renaissance men *[118]*
were still inclined to look back to antiquity as the source of truths, just as the Church looked back to the time when the revelations took place, Bruno maintains that it is those now living who are older than "the ancients," since they have more experiences to build on than these. Eudoxus did not know as much as Hipparchus, and he not as much as Copernicus. And Copernicus he praises not only as the one who continued the work of his predecessors, but above all as the one who had loftiness (magnanimita) and strength enough of mind to raise himself above the prejudices of the multitude and the illusions of the senses and set forth a new world-picture. In the Latin didactic poem "on the immeasurable and the innumerable worlds" he breaks out in a hymn in honor of Copernicus. But he reproaches Copernicus with having stopped too early and not drawn all the consequences of his thought. There was needed, then, an interpreter who could find out all that lay in Copernicus's discovery, and this dignity he ascribes to himself. He has opened our eyes to the infinity of the universe, shown that there could be no absolute limits to it, any more than there could be fixed "spheres" that marked off the different regions of the world from one another — shown that one law and one force rule in the whole universe, so that wherever we go we do not get away from the Deity who rules in it, any more than we need at all to seek it outside ourselves. — The task will now be to find the grounds on which Bruno supports his setting up and carrying further of the Copernican world-picture. They can be traced back to two main considerations, the one of an epistemological, the other of a religious-philosophical kind.

The old world-picture, with the earth in the center and with the sphere of the fixed stars as outermost limit, cannot rightly appeal to the testimony of the senses. If we take up the different sense-images we receive when we move, we see that the horizon is constantly shifted when we change our standpoint. So far is it, then, from sense perception establishing an absolute center of the world and an absolute limit of the world, that it, rightly interpreted, rather establishes the opposite: namely the possibility of constantly changing and extending the limit of our world. And with this testimony of sense perception agrees the capacity of our imagination and our thought constantly to *[119]*
add number to number, magnitude to magnitude, form to form, just as there also stirs in us a drive and craving that never rests content with what has been attained. It would be — Bruno thinks — unthinkable that our imagination and our thought should surpass nature, and that to this constant possibility of new intuition there should not correspond realities in the world. From the subjective impossibility of setting a limit and positing an absolute center he then concludes that there is no limit and no center. In this proof he

relies, as he himself says, on the fundamental condition of our knowledge (la conditione del modo nostro de intendere). It is in good consistency with this consideration — a consequence, however, that he only occasionally hints at — that Bruno remarks in one place that one cannot properly conceive the universe as a totality when it has no limit.

Since the horizon always forms itself anew around the place of the beholder as center, every determination of place must be relative. The universe looks different according as one thinks of it as beheld from the earth, from the moon, from Venus, from the sun, etc. One and the same point will, according to the different points from which it can be seen (respectu diversorum), be center, pole, zenith, nadir. Determinations such as up and down therefore do not mean, as the old world-picture presupposes, anything absolute. Only by fixing definite points of view can one give such expressions a definite meaning. — And with the relativity of place follows the relativity of motion. Motion is apprehended only in relation to a fixed point, and it then depends on where one thinks of this fixed point. One and the same motion will look different seen from the earth and from the sun, and wherever I then imagine myself to be, my standpoint will always appear to me as motionless. The distinction between what is at rest and what is in motion can therefore not attain absolute certainty. The old world-picture presupposes as given what was precisely to be proved: that the earth is the fixed point by which every motion is to be measured. — With the relativity of motion follows the relativity of time. For an absolutely regular motion cannot be demonstrated, and we have no sign that can
[120] establish that all the stars have taken exactly the same position in relation to the earth that they had before, and that their motions are absolutely regular. An absolute measure of time we therefore cannot obtain. Since motion looks different from the different stars, there will, when it is to serve as the measure of time, be just as many times in the universe as there are stars.

No more than the determinations of place can the concepts of heaviness and lightness have absolute meaning. Heaviness was indeed for Aristotle a tendency to seek toward the center of the world, and since earth is the heaviest element, the earth had to be at the center of the world! But the properties heavy and light belong to the parts of each single globe in relation to the whole globe. When what is heavy falls, it is because it seeks back to the place where it belongs, and where it can best abide. The particles of the sun are heavy in relation to it, just as the earth's particles are in relation to it. In relation to the universe as a whole the concepts of heaviness and lightness hold just as little as determinations of time and place and motions. They acquire meaning only in relation to a particular globe or a particular system. — It is the same theory of gravity we have

already met in Copernicus, only that Bruno lays the main weight on the parts' seeking back to their whole from the impulse of self-preservation. Copernicus also relied on the relativity in our determinations; but he stopped halfway. Bruno's merit is to have carried this principle through and shown what consequences it leads to. In Bruno we also find for the first time a decisive answer to one of the most important objections to Copernicus, namely that things which fall down to the earth would not be able to strike it perpendicularly below the place from which they fall, but must strike it some distance to the west. For Bruno shows that a stone thrown down from the top of the mast will fall at the foot of the mast, because from the outset, by impressed force (virtu impressa), it shares in the ship's motion. If, on the other hand, the stone were thrown down from a point that did not itself belong to the ship, it would fall some distance further back. Bruno here strikes the significant line of thought that later led Galilei to the discovery of the law of inertia. *[121]*

In close connection with the principle of relativity stands, in Bruno, the principle that nature is everywhere essentially like itself (indifferenza della natura). From conditions as they are here with us he concludes to the conditions elsewhere in the universe. An experience from his childhood years led him onto this path. From the mountain Cicada near Nola, which lay before him covered with forests and vines, he looked across toward distant Vesuvius, which seemed to him both small, bare and barren. But when he had once wandered there, he saw that the two mountains had exchanged character; now Vesuvius was the large and wooded one, and Cicada the small and naked one. The same principle that leads him to ground Copernicanism and to extend it by asserting the infinity of the universe leads him quite naturally also to assume essentially the same conditions everywhere, where particular experiences do not speak against it. He then thinks of the other globes as like the earth, and of the other systems as like the solar system, so that the fixed stars become suns surrounded by planets. There is no ground for assuming anything other than that the same forces work everywhere. It is therefore unjustified when Copernicus, with the usual conception, lets all the fixed stars be at an equal distance from us, seated in one and the same sphere. It is perhaps only in appearance that they always keep the same distance from us and from one another. Distant ships seem to us motionless, and yet often move with no small speed. Whether this is the case with the fixed stars, only very long-continued observations, which have not yet been begun, will certainly be able to show. But that such observations have not been begun comes precisely from men's having been so convinced that the fixed stars did not change place in relation to us and to one another. — Thus the principle of "the

indifference of nature" too (or, as it is called nowadays, the principle of actuality) shows itself, no less than the principle of relativity, from which it can be derived, fruitful in leading to new investigations. — Bruno has a far clearer eye for the fact that theoretical and subjective considerations need confirmation by way of experience than the usual expositions credit him with. "What could we think if the many observations were not
[122] before us?" he asks. He is by no means merely an enthusiastic visionary of the infinite. By critical intellectual labor he has sought to show on what dogmatic presuppositions the old world-picture rested, and to establish the justification and naturalness of adopting other presuppositions. And the burden of proof here rests, he thinks, rather on the one who asserts the limitation of the universe, since experience shows that the limit always shifts for us wherever we go. Why, too, should the universe stop precisely at the eighth sphere, as even Copernicus thought? Why not assume a ninth and a tenth, and so forth? Because our sense perception has a limit, one is not entitled to conclude that the universe too is limited. — Bruno's great merit is the energy with which he has thought his way into the new world-view and demanded that it be tested in detail. Thereby his doctrine became more than an anticipation of genius. The epistemological foundation on which he builds has its lasting significance. It is not to be denied for all that, that the passionate consistency with which he proceeds often makes him speak with greater certainty than he strictly has a right to. It is no wonder that the momentum with which he comes, and which is necessary to be able to overcome the resistance, carries him beyond the mark.

By the relativity of the determinations of place Bruno had, as we have seen, overthrown the old doctrine of the elements, which characterized them by heaviness and lightness as absolute properties and assigned each of them to its "natural" place in the universe. The distinction between the heavenly and the sublunary world was thereby also abolished, and with it the prejudice that no changes could take place in the heavens. But what lay especially close to Bruno's heart was to overthrow the assumption of the fixed spheres. He shows how this assumption is connected with the assumption of the earth as absolute center. As soon as one thinks one's way into the idea that each globe is in a certain sense a center and can move freely in space, just like the earth, the necessity of assuming the spheres falls away. And why should the globes need external forces in order to be moved? Each of them has, like every being in the world, an inner impulse of
[123] motion that carries them along; every globe and every little world has a source of life and motion in itself, and space is the great ethereal medium in which the all-embracing world-soul works, without any need of particular sphere-spirits to set the individual regions in motion. A confirmation of his conception Bruno found in TYCHO BRAHE'S

investigation of the comets. The Latin didactic poem "on the immeasurable and the innumerable worlds" he perhaps even worked out in order to be able to show how these investigations confirm the assumptions he had developed on other grounds in the Italian dialogue "on the infinite universe and the worlds." He praises here the Danish inquirer as the first among the astronomers of the time (Ticho Danus, nobilissimus atque princeps astronomorum nostri temporis) and as the one who has put an end to the fixed spheres which were to enclose our world in layers[29]). The comets indeed went right through the "spheres" whose crystal masses were to separate the regions of the world from one another! —

So much for the considerations in Bruno that can be summed up under the point of view of an epistemological grounding of the new world-picture. We now pass on to what might be called the religious-philosophical grounding. This is drawn from the idea of the infinity of the Deity, which stands firm for Bruno from the outset, and which he could presumably also presuppose that his readers and opponents shared with him, even if they did not make clear to themselves what lay in this idea. For Bruno it stood as a contradiction that to the infinite cause there should not correspond an infinite effect. If the Deity, which in its original unity embraces all that unfolds itself in the universe, is infinite, the universe, which is the unfolded form of God's being, must be so too. No force limits itself, and the infinite force has nothing that can limit it. If the Deity is conceived as the principle of the good, must it not then be assumed that it will impart all that it is able to? Should it be envious or niggardly? The infinite perfection must manifest itself in infinitely many beings and worlds. It is not justified to ascribe to the Deity a force or a possibility that does not become actuality. This opposition between possibility and *[124]* actuality holds only for finite beings and may not be transferred to the Deity. We should then get two gods, one possible and one actual or acting, in opposition to each other, and that would conflict with God's unity and be blasphemous. — Of this blasphemy Jacob Böhme, as we have already seen, was not afraid. His religious-philosophical speculations recall Bruno's in several respects, just as the new world-view also plays a part in him. But it was the problem of evil, not the problem of the connection of the world, that chiefly occupied Böhme. — Of the religious-philosophical proof on which Bruno relies he is, moreover, not the first originator. It had, as he himself mentions, already been put forward by Pietro Manzoli of Ferrara, who under the name of Palingenius published a Latin didactic poem *(Zodiacus vitæ,* printed in Lyons 1552), in which the infinity of the universe is taught (see the poem's 12th book), with a religious motivation similar to Bruno's, although the world-picture is otherwise conceived in the traditional way with

fixed spheres; outside the eighth sphere Palingenius imagines an incorporeal and infinite world of light. It does not lessen Bruno's originality that he thus makes use of earlier thinkers, here Palingenius, as on other points Cusanus, Copernicus and the old atomists (see LUCRETIUS: *De rerum natura* I, v. 944—1013). Everywhere he brings their thoughts into a larger connection and carries them through with greater consistency and on a fuller basis of experience than they had been able to.

For Bruno it was as if he first breathed freely when the limits of the universe had widened out for him into the infinite, and when the fixed spheres had fallen away. There was no longer any barrier to the flight of the spirit, no "thus far and no further"; and the narrow prison in which the old faith had held the spirits must now open its doors, and the pure air of a new life could be breathed in by him. These thoughts he has expressed in some sonnets prefixed to the dialogue on "the infinite universe." The picture of reality which his thought had labored so enthusiastically and assiduously to shape acquired for
[125] himself a symbolic meaning. The outer infinity became for him a symbol of the inner. Not all symbolism is built up on such firm ground.

c. Fundamental Philosophical Thoughts

Bruno's great achievement as a thinker is the grounding which, starting from the nature of our sense perception and thinking, he gives of the new world-picture, and the extension of it which he at the same time shows to be necessary. But in close connection with this world-view of his stand his general fundamental thoughts, his doctrine of the ultimate principles of existence. On the basis of a thorough study of Bruno's Latin writings and a comparison between them and the Italian writings, Felice Tocco has sought to show that, with regard to Bruno's fundamental philosophical conception, three stages are to be distinguished. At the first stage he stands close to the Neoplatonists, in that he sees in the world as well as in human knowledge an emanation of the Deity. This stage is represented above all by the work De umbris idearum (The Shadows of Ideas). At the second stage he conceives the Deity as the infinite substance which subsists beneath all the change of phenomena, as the unity of all the oppositions of existence. This stage is represented by his chief works, the Italian dialogues published in London. At the third stage he conceives that which lies at the ground of phenomena as an infinity of atoms or monads, without, however, giving up on that account the thought of the substance that stirs in all things. This stage appears in the Latin didactic poems published in Frankfurt, especially De minimo. With this conception of Bruno's philosophical development in the last ten years (1582—1592), during which he was at liberty, I can on the whole agree,

especially when sufficient weight is laid on what Tocco states, that the transitions and differences between these stages were not conscious to Bruno himself. In particulars I believe that several points must be conceived somewhat differently than the indefatigable Italian Bruno scholar conceives them.

Bruno began as a Platonist. In an eternal idea, which is one with the essence of the Deity, all things have their ultimate ground. It is the task of our knowledge to raise *[126]* itself from the confused multiplicity of the sensible to the unity which stirs in all things, even though the highest knowledge we can attain is only a shadow of the divine ideas. But on one, and probably very essential, point Bruno differs from Plato. While Plato by ideas means the general concepts, what is common to the individual phenomena, the universal, Bruno declares expressly that the concepts by which we raise ourselves above the confused multiplicity of the senses are not a graded series of general concepts (universalia logica), but concepts which express the real connection between phenomena, so that instead of an unformed multiplicity of parts one obtains a firmly connected and formed whole, whereby the individual parts become more intelligible than as long as they are held each apart, just as we understand the hand only in its connection with the arm, the foot in its connection with the leg, the eye in its connection with the head. The highest unity, which is set up as the ideal goal of knowledge, is thus not an abstract idea, but the principle of the real, lawful connection which alone gives the individual phenomena subsistence and makes them intelligible to us. Bruno here puts forward a thought which in various forms extends through the whole history of modern philosophy. While ancient philosophy turns its attention above all to the *idea* or the *form,* modern philosophy turns its attention above all to the *law.* The lawful connection of existence is the fundamental fact which it seeks to think through and interpret. Bruno only touches on this thought in the work on "the Shadows of Ideas." He does not go further into it, because his chief interest is mnemotechnical: he recommends linking the representations together in memory in the same way as the corresponding phenomena belong together in reality, since this connection makes both recollection and practical application easier. — It thus becomes not difficult to understand Bruno's transition to the conception which he develops in the Italian dialogues in close connection with the consequences of the new world-picture. What he really seeks, already at his first stage, is the inner principle which lies at the ground of the real connection between the individual things as well as of these themselves. Inward, not outward or upward, is his watchword from the first. *[127]*

In the dialogues "on the Cause, the Principle and Unity" we find the full development of Bruno's fundamental thoughts. The constant endeavor is to grasp the universe as a

whole moved by inner forces, within which all things stand in close connection, and which is itself the unfolding of what the infinite principle, which is the highest thought to which we can attain, contains within itself. When Bruno distinguishes between the cause and the principle, he means by this that the infinite unity can be seen partly in its opposition to what springs from it, — and then Bruno calls it cause, — partly as that which stirs in the multiplicity of phenomena, — and then he calls it the principle or the world-soul. In the former form it is not accessible to our thought, but only an object of faith. The true philosopher and the believing theologian differ from each other in that the latter seeks the Deity outside and above nature, the former in nature. As already appears in the discussion of the world-picture, Bruno lays great weight on the infinite cause having to have an infinite effect. But precisely because the effect is infinite, it cannot be surveyed, and the knowledge which can be had of the Deity, even when it is regarded as principle (immanent cause) or world-soul, must therefore always remain imperfect. That knowledge is possible at all rests on this principle stirring in us just as well as in everything else in the world. The new world-picture proves the illusoriness of seeking the seat of the Deity outside us, and teaches us that it is to be sought innermost within us, "in a more inward relation to us than we stand in to ourselves" (di dentro piu che noi medesmi siamo dentro à noi). The Deity is the soul of our soul, as it is the soul of the whole of nature.

The world-soul is in Bruno the principle which connects and orders all things in nature; through space, which is not empty but an infinite ethereal medium, it produces interaction between the individual things. Just as space does not exclude bodies, but precisely makes the intuitable connection between them possible, so the world-soul, as
[128] the bearer of the world-ideas (mundus intelligibilis. De immenso VIII, 10 cf. De la causa ed. Lag. p. 267) acts not from outside or as something foreign to things, but precisely as the law of their own being. It is an artist who from within develops and produces the forms which natural phenomena assume: not a spirit that sets the world in motion from outside, but the inner principle of motion. It acts in its wholeness at every single place, in every single part. The universe is not as a whole in its individual parts; but in each of these parts the world-soul stirs as a whole: Anima tota in toto et qvalibet totius parte![30]) There stirs an inner impulse of motion, a vital force in every thing, a will, which leads things along the ways that make self-preservation possible for them. It is this inner need that leads the iron to the magnet, — that lets the drop of water as well as the whole earth assume the spherical form, which best holds the parts together. This inner principle of motion, in which the world-soul makes itself known in every single

being in the world, stands in Bruno in opposition to the external moving spirits of the old philosophy of nature, and marks an advance, insofar as it rejects external causes where the inner ones are sufficient. It prepares for the discovery of the law of inertia, which Bruno hints at in his astronomical dialogues. But it is at the same time for him, as is already clear from what has been adduced, a purposive, teleologically acting principle. It leads with inner necessity to preservation and development. Teleology and mechanics are united in it. The various beings and worlds are moved by the inner principle in such a way that their subsistence is secured and the systems of the world are not dissolved before their time. Providence and the acting force in nature (la provida natura) are thus not different, but one and the same. In the Latin didactic poem De immenso (V, 3) Bruno shows more closely how this unity of teleology and mechanics appears in nature. Those combinations of the elements which are not purposive quickly perish: male adorta cito pereunt. Through constant tossing (exagitatio) of the elements there must then at last arise combinations which have subsistence and viability for a longer time. In this way it can be explained that systems of the world can be formed which subsist for centuries. While Bruno in this mechanical explanation joins the atomists of antiquity, who — as *[129]* is seen from Lucretius's didactic poem — had similar ideas, it is nevertheless his firm conviction that a providence is not excluded thereby: in the tendency which drives toward constantly new, at last victorious attempts, he sees the expression of a governing will, he traces a mens paterna cuncta moderans. This is characteristic of Bruno's attempt to unite Platonic idealism with a realist view of nature. At the first stage (in "the Shadows of Ideas") it was the connection of nature in which he found the idea; here he goes a step further and finds the idea in the very struggle to subsist. — Yet Bruno's view of nature, through its emphasis on the inner force, retains a poetic character. On some points he approaches a purely natural-scientific mode of consideration, but his need to find again everywhere the same motives which he knew from himself prevented him from taking the step fully. He has expressed points of view which can be of interest also *after* a mechanical natural science has been founded, but which could not themselves lead to its foundation. Among the points where he approaches the mechanical view of nature belongs especially his statement that motion in space is the principle of all change. It is the thought on which modern natural science rests.

Bruno's philosophy acquires a realist stamp through his sharp polemic against Plato and Aristotle, who set the idea or the form in opposition to matter. Thereby these became fantastic, and matter is degraded to a purely passive and unfruitful principle. In agreement with the older Greek philosophers of nature (perhaps also influenced by

Paracelsus) Bruno states the doctrine of the conservation of matter. From the seed comes the stalk, from the stalk the ear, from the ear bread, from bread chyle, from chyle blood, from the blood semen, from the semen an embryo, from the embryo a man, from the man a corpse, from the corpse earth! There thus takes place a constant metamorphosis of matter. It subsists beneath the change of the "forms." And the forms which it assumes in the individual cases it receives not from outside, but from its own womb. Art produces the forms by joining together or dissolving the matter; nature, on the other hand,
[130] produces its forms by unfolding what is present in it as disposition, by explicating what is complicated in the original principle of nature. Far from being something low and mean, matter is therefore a divine being (cosa divina), the origin and mother of natural things, indeed is one with nature itself. Bruno says that this consideration of the significance and originality of matter for a time even led him toward the view that the forms were something merely external and vanishing in nature. Presumably it was Telesio's philosophy of nature that influenced him then. But he nevertheless saw that there must be an original principle by which the forms find their explanation. Form and matter, actual acting and passive receiving, must originally be united in the being that lies at the ground of all things. The divine activity does not have its matter outside itself, nor does it need to be set going from outside. Possibility and actuality here coincide, just as matter and form, passivity and activity. Bruno distinguishes between two kinds of substance, which he calls now form and matter, now spiritual and material substance, and both have eternal subsistence. The world-soul is both the eternal form, which comprises within itself the individual forms that arise and vanish, and the infinite spirit, which subsists beneath the change of the finite spirits. All that exists is ensouled and formed, in different degrees and kinds, at least in disposition (secondo la sustanza), if not in actual development (secondo l'atto). But both substances are for him in the end nevertheless only one and the same substance; they spring from one root, are to be traced back to one being. No plurality of substances can be assumed in the world, but only *una originale et universale sustanza,* which lies at the ground of all the differences that existence presents. Indeed, he even declares that what produces the differences in things does not belong to the essence of things, but only to their sensible appearance. In the highest unity, which is the goal of knowledge, the difference between spirit and matter is thus to be overcome, as well as the difference between form and matter, acting and suffering, actuality and possibility[31]). In short, all oppositions and differences in existence are in the eternal being joined into unity and agreement (unità et convenienza), without any
[131] change in the one or the other direction being possible, and without anything being

able to set itself in opposition to it, any more than anything can stand in a relation of likeness to it, since there is nothing other than it. In its eternal subsistence the difference between hour, day, year and century falls away, and likewise the differences of the finite beings among themselves, — between ant and man, between man and star, — become vanishing in relation to the infinite being which embraces all things. But for that very reason our thought cannot grasp this infinite unity and fullness either. Only a *negative* concept of it can we form, in that we strive to keep differences and oppositions out. All that we comprehend we shape as unity, but the absolute unity surpasses our capacity. — In this negative theology Bruno shows himself a disciple of Cusanus, just as in his doctrine of the relativity of knowledge. He halts (De la causa. ed. Lagarde. p. 268) at the difficulty of how the Deity can be *both* possibility *and* actuality, when it is to be an absolute unity. Like Cusanus, he endeavors to bring life into the concept of the Deity by ascribing to it possibility and not merely the actuality concluded once and for all, but he takes this determination back by the mystical declaration that possibility and actuality are to be one. Thereby, at the same time, dualism breaks forth in his system; for in the finite world possibility and actuality are separated, and there then arises a strong opposition between this finite world and the eternal substance, where the opposition between possibility and actuality does not exist. If the eternal substance, which in its unity (complicamente et totalmente) comprises the whole content of existence, is perfect, what significance then has the process of unfolding which advances through the surging of the oppositions and the constant transition from possibility to actuality? Only when one admits an original imperfection or incompleteness (that which consists in having the content *only* complicatively, not also explicatively) or — with Jakob Böhme — a fall in the Deity, can the finite development become compatible with the idea of God. It is Böhme's merit to have seen this clearly. Bruno oscillates, without quite noticing it himself, between mystical contemplation of the unity and enthusiastic devotion to *[132]* multiplicity. Yet his thought is most occupied with tracing the eternal unity beneath the constant development and through all oppositions. He refers to a series of confirmations by experience (verificationi) of his doctrine. The constant conversion of matter into new forms is the expression of the infinite unity, which cannot be realized exhaustively in any single form. Only by means of a common, underlying principle could oppositions such as minimum and maximum, dizzying fortune and sudden ruin, birth and death, love and hate, pass over into each other. Through continuous transitions the highest degree of heat and the highest degree of cold stand in connection with each other, which indicates that one single principle lies at the ground (a consideration which Bruno directs against

Telesio). On the whole, he who would understand the greatest secrets of nature must attend to how the oppositions in their minima and maxima pass over into each other. Everywhere in nature the oppositions work together, and on closer consideration there always appears a common ground to which they each lead back. — When one notices the manner in which Bruno thus "verifies" his doctrine of the unity of oppositions, this doctrine loses a good deal of its mystical character. It is the continuity between the oppositions, not their absolute identity, that he proves. Thereby his doctrine acquires a realist character; it calls for seeking all intermediate links and intermediate causes and for finding the highest in that which makes their connection and cooperation possible. This is also the significance of his doctrine of the world-soul, just as it was the same thought that came forward already in the work De umbris idearum. This whole line of thought puts him in an optimistic mood. All perishing, dissolution, malformation and suffering do not touch that which has true subsistence. They belong to the sensible world, where possibility and actuality do not always go together, and where the oppositions come forward in sharp conflict. They vanish when the gaze is turned to what eternally subsists. But they vanish also when one conceives each thing in its individual nature, according to the definite manner in which it fills a place in the great connection, where
[133] even the smallest link has its significance: "All is perfect, because in its own peculiarity (De imm. II, 12: in sua individuitate) it is a being which is not limited by anything else; this is the inner measure of perfection". And even what in certain respects and for some beings is evil is for other beings and in other respects good. For him who turns his gaze to the whole connection, discord therefore dissolves into harmony. Nature is like a choirmaster who has many voices in his choir, but also knows how to make them sound together harmoniously. — Yet it is only when Bruno's thoughts dwell in these high regions that he is able to hold fast this harmony. Where he immerses himself in the actual oppositions of life, he feels them to be so much in conflict that the fundamental mood becomes more complex and acquires a tragic character, or in any case bears the stamp of resignation. From this side he conceives existence in his work "on the heroic affects."

Those statements of Bruno's in which individual peculiarity is set up as the measure of perfection and the significance of the small elements for the great results is emphasized are found in the Frankfurt writings, thus in the last group of works, which, according to Felice Tocco's demonstration, express Bruno's third standpoint. Here he moves even further than in the London writings from the Neoplatonic absorption into a supernatural principle. And whereas in the London writings he emphasized above all the connection

and the interaction, here he emphasizes above all the individual elements between which the interaction takes place. The individual is emphasized in its peculiarity: there are, he asserts e.g., no two things, two cases, two times which are entirely alike. And these elements, of which all things in the end consist, are what is properly substantial. What presents itself to our senses as something continuous consists of small bodies, just as number consists of units. The atom (or minimum or monas) is that which has no part, and which is itself the first part of a phenomenon. Within every domain we must go back to these constituent elements, which are what is not dissolved when the whole phenomenon is dissolved. A composite being cannot be substance. Bruno states definitely that division cannot go on to infinity, and he reproaches not only the physicists but also *[134]* the mathematicians for assuming such an infinite division. However, it is not really his meaning to establish absolute atoms. He conceives atoms of different kinds, and atoms of the one kind can enclose atoms of another kind. It is then clear that the concept of the atom is with him a relative concept. Of the two members of which his definition of the minimum consists:

Est minimum, cujus pars nulla est, prima qvod est pars, he can really hold fast only the second. The atom is the first constituent element we need for the explanation of a sensible phenomenon; but there is no reason to lay it down that it itself has no parts; that we do not take account of them is another matter. Bruno himself accordingly uses the concept of minimum of great wholes too: thus the sun with its whole planetary system is a minimum in relation to the universe. Indeed, even the universe is called a monad. And further: the minimum is not merely a part of matter, but acting force, soul and will. The purpose lies implanted in it — whereby it becomes possible to hold fast the unity of teleology and mechanism. Seen from this side too, Bruno distinguishes between atoms (or minima or monads) of different degrees: the world-soul too, God himself, is called a monad. This determination is necessary: for what is not one, is not. God is the monad of monads, that is, he is the being in all individual beings. This says the same as when God in the Italian writings was called the soul in souls. Bruno has thus not given up his fundamental thought from the earlier stages of one single substance that stirs in all things. Even though he has seen the necessity of the concept of the atom for an exact view of nature, he nevertheless applies it only to material phenomena, and it remains an unsolved problem how the individual mental beings are related to the spiritual substance whose individual forms they are[32]). From his interest in the old doctrine of the transmigration of souls one might conclude that he assumed that the individual souls after death passed over into new bodies. But this fantastic notion he has

not seriously embraced. He stopped at the thought that the spiritual substance, just as
[135] well as the material, subsists beneath constantly changing forms, but the new forms need not be identical with the earlier ones. He lays great weight on the need for continuation of life and for constant development; and this need is not disappointed, he thinks, even if it is an illusion when we believe we know about the form of existence into which we pass[33]). —

Beside many ideas in which the poet gets the upper hand over the thinker, and beside many allegorizing and arbitrary speculations (especially in the work De monade), which we have not brought forward here because they no longer have any interest, Bruno's philosophy presents an attempt, remarkable for his time, to unite an ideal fundamental conception with the scientific world-view. Bruno did not, to be sure, know the exact grounding of this world-view which Galilei and Kepler first brought; but in several of his thoughts he has approached them.

d. Theory of Knowledge

The endeavor that rules in Bruno is to come to know nature. And for him that is to come to know the Deity. An advance in the knowledge of nature is for him like a revelation. Hence his enthusiasm over Copernicus and Tycho Brahe and his zeal to think through the consequences of their discoveries. Thinking, reason, is for him only the means of finding the essence of nature; it must not put itself and its forms in the place of nature. It is his complaint against Plato and Aristotle that they have set up logical differences, oppositions of thought and distinctions as if they were real oppositions in the very essence of nature. It is, as he states several times, the foundation of all "natural and divine knowledge" to distinguish between logical opposition and real unity, so that one does not — as Aristotle does, e.g., with his distinction between form and matter — divide with reason what is undivided according to nature and truth. Reason is to conform to nature, not nature to reason. In Bruno's view it is in particular quite arbitrary to think
[136] that in nature the oppositions are as sharply separated as in our thinking. In nature itself the oppositions do precisely pass over into each other and thereby show themselves closely related, whereas in a logical system they are placed as far from each other as possible.

As regards sensuous knowledge too, Bruno inculcates the critical distinction between what belongs only to our cognitive activity and what belongs to nature. This distinction comes forward especially at his last stage, where the atomistic turn in his way of looking at things gave particular occasion for it. In the work De minimo, where the doctrine

of atoms is set up, it is taught that the minima (atoms or monads), which are what is properly real, are not themselves perceived as such. It is thus only through their united action that sense perception arises. To the differences which our sensations present there need not, therefore, correspond equally great differences between the physical elements. Yet Bruno will not go so far as to assume only one homogeneous matter (communis materia), because it would then be difficult to understand how the differences for sense perception could arise. That light, the fluid and the dry (the atoms) should consist of one and the same matter, he cannot conceive. This is connected with the fact that his atomism holds only for the "dry" or solid matter (materia arida), not for light and for the ethereal fluid which brings about connection between the small bodies. He does not see that the difficulty of understanding the origin of the differences of sensation must in any case present itself as regards the "dry" matter, to which, after all, different qualities of sensation correspond. He approaches the principle of the subjectivity of the sense qualities, but he does not adopt it fully. Only as regards the elements of feeling attached to the sensations does he teach that they have merely subjective and relative significance. Their validity depends on the perceiving subject, whose valuation they express. They hold *ex latere potentiarum,* not *ex latere objectorum.* (De minimo. ed. 1591. p. 59). The same consideration can in part be applied to the moral and aesthetic concepts; yet Bruno holds fast that there is something good in and for itself (simpliciter), which it is the business of thinking to find, in that it leaves the limited human standpoint. The good can, he thinks, be determined in such a way that it does not merely express *[137]* a relation to human nature (ad speciem humanam contractum). The consequence of Bruno's view of nature would seem, however, to have to be this: that just as little as any absolute determination of place — that is, any determination of place without relation — is possible in the universe, just as little is moral or aesthetic valuation possible except in relation to a definite subjective basis. On this point Bruno is not yet a Copernican.

And this is connected with his being unable to give up the presupposition that our knowledge itself does after all express in itself the innermost nature of existence and is a testimony to it. It is, he says, one and the same ladder by which nature in its producing descends to the individual things, and by which we in our knowledge ascend to the universal laws. Nature's producing is an unfolding, an explication; our knowledge is a gathering together, a complication. The two processes correspond entirely to each other. To understand something is to give it an abbreviated, simplified expression. All knowledge is a simplifying, because it seeks unity — and from this, says Bruno, it is clearly seen that the essence of things consists in unity! — It is clear that he here himself

transfers his form of thought to existence itself, only that it is not the form of distinction but the form of unity. This he takes to be more essential than the other. Neither of them, however, can be dispensed with. Here lies a great problem which was taken up only much later.

The highest object of knowledge marks for Bruno at the same time its limit. The absolute unity is inaccessible to our discursive and distinguishing thinking. We cannot form any concept that can express it. It is the object of faith. Only as revealed in nature, as world-soul, is it an object for philosophy. And yet — the world-soul too, God as the soul in nature, the being in all beings, is surely to be raised above all oppositions (between form and matter, actuality and possibility, spirit and matter etc.)! It too must then withdraw from knowledge. — While Nicolaus a Cusa, by whom Bruno is here strongly influenced, swings over on this point into positive theology, and ever more
[138] the older he grows, Bruno halts only for a moment to state that "the deepest and most divine theologians" have taught that God is honored and loved more by silence than by speech, and to praise the superiority of negative theology over the scholastic, — and thereupon he turns back again to nature. The ladder of knowledge which he builds does not reach quite to the top, and Bruno himself is also most at home in the middle of the ladder, where the place of those who strive is.

Positive theology has, according to Bruno, only practical, not theoretical significance. The new astronomical system for which he fought, and which met with such great opposition from the believers in the Bible, he defended (like Galilei and Kepler later) by saying "that the holy books do not treat of proofs and speculations concerning natural things, as if it were science that was at issue, but on the contrary, for the good of our mind and our heart, would guide moral action by their commandments". Therefore, he adds, they had to speak a language which all understand. With this he meant to stop the mouth of some "impatient and strict rabbi". He himself is convinced that the new philosophy which he teaches will support true religion far more than any other philosophy, by teaching that existence is infinite like its cause, and that in spite of all change no absolute perishing takes place. This statement (Cena. p. 169—173) contains the clearest account of Bruno's relation to positive religion that is to be found. In other places (De l'infin p. 318. De gl' her. fur. p. 619) he expresses himself in a more intellectually aristocratic way, as when he says that faith is for the crude people who are to be governed, but thinking for the contemplative natures who know how to rule both themselves and others. It is no wonder that the relation had to present different sides to him, especially during the violent struggle he had to wage against prejudices that supported themselves on religious

notions. — Where he occasionally attacks or alludes to Christianity, it is Protestantism rather than Catholicism that must bear the brunt. And this is surely precisely because he fears that the beneficial practical consequences of positive religion will be lost through Protestantism (especially through Calvinism), which produces a rebellious disposition, strife between those closely related, civil war and endless dogmatic polemic. Every *[139]* pedant, after all, has his catechism, which he is planning to publish, if he has not already published it, and to which he demands that all others conform. What offends Bruno not least is the Protestants' praise of faith in opposition to works. It opens the way to barbarism, he thinks. It leads people to appropriate and consume what their forefathers founded, without increasing it. Hospitals and poorhouses, schools and universities are owed not to the new but to the old Church, and it is unjustified when these decriers of works appropriate them. Instead of reforming, they take away precisely the good from religion. These statements (in Spaccio p. 446 f. 466 f.) have evidently arisen from Bruno's experiences in Geneva, France and England. The new shoots of life in the Protestant spirit he did not catch sight of, and it was probably also difficult, in the midst of the ferment and under the mutual denunciations of heresy, to catch sight of them. His sympathy is evidently on the side of the old Church, and one understands that he felt homeless. As freely as his thought moved in infinite cosmic space, so hard did it find it to settle into the motley human world. He did not see that Protestantism was a Copernicanism in the spiritual domain: that it would make each individual a center of the world. The fanaticism and pedantry of the Protestant theologians made him blind to the process of liberation which had here been begun. Admittedly, also only begun.

In his symbolizing writings (Spaccio and Cabala) Bruno satirizes several dogmatic notions, thus the incarnation and transubstantiation. It was, after all, precisely this that brought him down, however much he asserted that in his philosophy he brought out the viewpoints of true religion. It is interesting to see what a definite difference he makes between the dogmatic notion of Christ and the historical Christ. In "Spaccio," where the constellations are to be reformed, the centaur Chiron also comes under treatment. Momus, the god of mockery, makes a number of witticisms on the subject of his double nature, which is nevertheless to constitute one person, but he is ordered to take his understanding captive like a good boy. Zeus recalls that this Chiron, while he lived on *[140]* earth, was the most righteous of men, that he taught men the art of healing and the art of music and showed them the way upward to the stars. Therefore he is not driven out of heaven, but gets his place at the altar as its only priest. This trait is illuminating with regard to Bruno's relation to Christianity. That it meant nothing in the eyes of Cardinal

Bellarmine goes without saying.

e. Ethical Ideas

Bruno has not given a coherent ethics, although he says (in the preface to Spaccio) that it is his intention to give a moral philosophy "grounded on the inner light". We meet here again "the natural light," which in the sixteenth century was brought forward in its independence. This plan Bruno did not come to carry out. On the other hand, he has given what he calls "preludes" in two symbolizing writings: Spaccio (The Expulsion of the Triumphant Beast) and De gl' heroici furori (On the Heroic Affects).

These two writings, although they came out shortly after each other, stand in a certain mutual opposition. Spaccio is more optimistic, more directed to human life as a whole, particularly to its social life. The work on the heroic affects, on the other hand, leads us into the inner struggle and opposition in the single striving individual; the pain of the constant striving is emphasized, although it is seen precisely as a testimony that one is setting up ideal demands; it is the striving of the individual, not of society and the species, that is depicted. The latter work is an ethics for the chosen few, whose inner life goes into the depths and the heights; the former is for all.

In Spaccio the frame is this: Zeus, by whom is meant not the highest Deity but each one of us, insofar as there are divine forces in him, makes the resolution to reform heaven. He says to the gods: Our scandalous stories are there to be read in the constellations. Let us now return to justice, that is, to ourselves. Let us reform the inner world, and then
[141] the outer will be reformed too. — And this reformation is now depicted in the form that the constellations receive new names, the various virtues taking the place of the more or less ambiguous figures of gods and animals. In discussing which virtues should be represented in heaven, Bruno has occasion to undertake a "revaluation of values" (to use a modern expression). The account of this is very prolix, but has a number of interesting episodes. There is occasion here only to dwell on a couple of characteristic traits.

In the first and highest place in the new order truth is set, for it rules over all things, and it determines the place of each thing; on it all things depend. If one would conceive something that could surpass truth and determine its validity, then this something would be precisely the real truth! Thus nothing can precede truth. It is desired by many, but beheld only by few. It is often combated, but nevertheless needs no defense; it grows precisely the more it is disputed. — This valuation is evidently connected with Bruno's struggle for the new world-view and with his conviction that the new ideas would widen the outlook and ennoble the mind.

In Bruno's world-view the necessary connection of the oppositions and their passing over into each other play a great role. The same law he finds in the domain of mental life, especially in that of feeling. If there were no change, no pleasure could be felt. Every feeling of pleasure consists in a transition, a motion. And pleasure presupposes pain as its background. Hence no pleasure without an admixture of sorrow. This kinship of the oppositions is what makes repentance possible. It gives birth to the need for a higher stage of life than the one on which one has hitherto stood. It is by virtue of this that Zeus resolves to reform himself and the whole world of the gods. Beneath all changes only truth subsists, and it is in its light that the reform is to take place. These are, after all, the two thoughts which Bruno (like Heraclitus in antiquity) constantly holds fast: the oscillation of all things through oppositions, and the eternity of the world-law beneath all change. From this it is understood how both truth and repentance receive a place in his ethics. Repentance is to take the place of the swan in the new order. Like *[142]* the swan it emerges from rivers and marshes and strives through purification to attain shining purity. It consists in a displeasure at the present state, sorrow at having found satisfaction in it, and a need to raise oneself from the lower regions up toward the sun. Although error is its father and injustice its mother, it nevertheless has a divine nature in it; it is like the rose that is plucked among thorns, or like the bright spark that flies out of the dark flint. —

Among those who seek a place in heaven under the new order is also freedom from labor (ocio). It praises the happy childhood of the human race, when it did not need to labor and was free from all cares and from the unrest of endeavor, and when there was neither misfortune nor vice and sin. But to this praise of the golden age Zeus replies that man has hand and thought in order to use them, and that it is his task not merely to follow the promptings of nature, but to produce another nature, a higher order, by the power and freedom of his spirit, without which he cannot maintain his dignity as the god of the earth. In the golden age, with its freedom from labor, men were no more virtuous than the animals now are; indeed, they were perhaps even duller than these. Want and need have produced industry, sharpened thought and led to the discovery of art, and thus, from day to day, under the pressure of necessities, ever new and wonderful inventions are developed out of the depths of the human spirit. Thereby it moves away from the animal and approaches the divine. Injustice and wickedness grow at the same time, it is true. But in the animal state the virtues are lacking as well as the vices; for one must not confuse virtue with the absence of vice. Self-control is present only where there is resistance to overcome; otherwise animal dullness becomes a virtue. Chastity is

no virtue in a cold and dull nature. In northern Europe it is therefore hardly a virtue; but in France, still more in Italy, most of all in Libya. — The result is that freedom from labor can be acknowledged only as the necessary opposite of labor (ocio — negocio). Labor
[143] and rest ought to form a natural rhythm. For a high-born spirit, freedom from labor will be the greatest burden when it does not come as relief from strenuous activity. —

In the work on the heroic affects too, weight is laid on the point that, since existence is composed of great oppositions, the life of feeling in those who are not animally narrow must be of a complex character, and this is made a measure of the development of feeling. The stupid man keeps to the present state without thinking of what goes before and what follows after, of the opposite, which nevertheless lies so near, of the absent, which nevertheless is always a possibility. His joy can therefore be without care, without fear and without repentance. Ignorance is the mother of sensuous bliss and of the animal paradise. He that increases his knowledge increases his pain. With growing knowledge one sees more possibilities, sets oneself a higher goal, — and it then becomes so much the more difficult to reach it. The heroic affect arises when one does not let oneself be held back from craving toward a high goal because pain and danger are bound up with this striving. The butterfly which is attracted by the light does not know that it is death; the heroic man knows it, and seeks the light nevertheless, because he knows that pain and danger are evils only for the limited and sensuous consideration, not seen from the standpoint of eternity (ne l'occhio de l'eternitade). It is even a necessity that with all higher striving a pain must come along, since the goal constantly rises the further one advances. One steps carefree into the boat to begin the voyage — and soon one is out on the infinite sea, which overwhelms sense and thought. The further one reaches, the more one sees that absolute satisfaction is impossible. One discovers that its object is infinite, and thereby strife and unrest come into the finite nature. But one then feels satisfaction in the noble fire being kindled, even if pain is bound up with it. It is a higher form of self-preservation that leads one to continue the ideal striving in spite of the harmony that has been abolished. What binds the will (vinculum voluntatis) is at every stage a love, but this love can concern something that lies high beyond the finite existence of the individual.

[144] — This whole line of thought has interest not only through containing an answer to objections that have often been directed, right up to the most recent time, against the ethics which uses happiness or welfare as its measure, but also through the opposition in which Bruno here stands both to the ancient and to the medieval conception. To be placed in the strife of oppositions and to be out on the ocean of infinite striving is for

Bruno the highest. It brings with it a richness and fullness in the inner life which he cannot imagine that any resting state that could be attained would bring with it. However much the work on the heroic affects reminds one of Plato and Plotinus, to whom he himself refers, it nevertheless characterizes Bruno precisely as a modern thinker. He is here a forerunner of Lessing's and Kant's idea of eternal striving as the highest, just as his position toward the notion of the golden age also recalls the culture-historical conception of a later time. As in his conception of the world, so also in his ethical ideas he works with a wide horizon — and with the notion that it can always become wider.

14. Tommaso Campanella

Like Bruno, Campanella too was a monk who came into contact with the new thoughts of the time and had a receptive mind for them. But he already belongs to the reaction. With the Renaissance philosophers he demands that a new science and a new philosophy must come, because the book of nature is now opened as never before. But he has at the same time another reason for his demand, namely that the old philosophy is heathen. It was impossible, he thought, to unite the Aristotelian philosophy with the faith of the Church, although in the Middle Ages, to be sure, this had been held possible. A new doctrine of nature and a new political theory, a whole new philosophy must be founded, in order that the Church may subsist, and in order that it shall not, in face of the new peoples in the other parts of the world, be ashamed to come with a culture which has not been produced within the Church itself. The bold hopes of the Renaissance thinker are united in Campanella with the believing Catholic's humility toward the Church. He denies that Bruno was burned on account of his scientific assumptions; in his apology *[145]* for Galilei he urges the Church to let research grounded on experience go its free course, since the book of nature, when it is thoroughly studied, will surely agree with Holy Scripture. But when the Church condemns the new astronomy in Galilei's doctrine, he bows. It is probably also a satisfaction to him that he now need not give up the view of nature which he had learned from Telesio and stated in his earlier writings. By his attempt to set up a philosophy other than the scholastic, he nevertheless incurs a bitter enmity, which was probably a contributing cause of his long imprisonment. There is something tragic in his figure, in that his struggle against scholasticism is regarded with displeasure by those in power in Church and state, because they had now learned to fear any change in what had been handed down. All his enthusiasm for the cause of the Church did not help him. Nor was he able in his inner self to triumph over the forces of the past. There are significant thoughts working their way forward in him; but they

are clothed in scholastic and mystical forms; and when he had succeeded in bringing them out, they had already been stated by another road, in clearer form and in better connection with the science of the time, by Descartes.

Campanella was born in Stilo in Calabria in the year 1568. His first name was Giovan, which he exchanged for Tommaso when, at fourteen years of age, he entered the Dominican order. It was, by his own account, not only religious need, but perhaps just as much the hope of being better able to pursue his studies in the monastery, that led him to take this step. By his zeal and his acuteness he soon aroused attention and fear among the monks. He attacked the Aristotelian philosophy with such good arguments that it was thought he had not come by them in a natural way, and his inclination to the "secret sciences" could strengthen this opinion. Telesio's philosophy of nature aroused his enthusiasm, and it was a grief to him that he was not allowed to visit the old thinker; only when Telesio lay on his bier did he get to see him. In essential points he follows the Telesian philosophy, as whose eager defender he came forward. To escape the enmities
[146] he had aroused, he traveled to Rome and from there to Florence and Padua. He did not succeed in obtaining a chair from which he could proclaim his ideas. People were wary of the new doctrine. His manuscripts were even stolen from him, — and he saw them again only when, on the way home from northern Italy, he had to appear before the Inquisition in Rome. He seems, however, to have come off fairly lightly from the trial to which he was here subjected. A harder one awaited him in his homeland. Amid the prevailing ferment of minds and the discontent with the Spanish regime, disturbances broke out in Calabria. Campanella, who had already drawn attention to himself by his philosophical opposition, could not but become an object of suspicion through the socialist ideas which he took up early, and no less through the manner in which, from the signs of nature and of the times, he believed he could predict great upheavals in the year 1600. He was accused of plans hostile to the state and of heresy, was repeatedly subjected to the most terrible torture and held in prison for 27 years. The best time of his life passed in this way. The vigorous man's enthusiasm, however, was not crushed. In prison he wrote poetry and thought, and when he got his underground hole exchanged for a better abode, he gave himself up to eager studies. His manuscripts were seen into print by friends who visited him. At last, however, he became free by being handed over to Rome. The Pope protected him and let him flee to France, where he lived his last years in peace, with a pension from the French government. He died in 1639, two years after Descartes's first epoch-making work had appeared.

Campanella wants a philosophy which proceeds from experience. That is why he

was enthused by Telesio's program and overlooked the imperfect manner in which this program had been fulfilled. That is also why he took a lively interest in Tycho Brahe's and Galilei's observations and was, as already touched on, for a time wavering in his view of the universe. If Galilei is right in his conclusions, he said, we must philosophize in a new way. And since Galilei constantly relies on observations, one can refute him only with other observations. It lies very close to Campanella's heart as a Catholic that the Church should not expose itself to ridicule by condemning the new doctrine. Therefore *[147]* he wrote in prison his *Apologia pro Galileo,* which one of his friends saw into print in Frankfurt in 1622. But the Church would not (as we shall later have occasion to discuss more fully) go the way which Campanella so urgently called on it to take: to let the book of nature and the book of revelation be interpreted each according to its own rules, in the conviction that they would surely agree in the end, especially when one holds fast that the Bible quite naturally keeps to the sensuous and popular conception of the world. When the judgment on Galilei then came, Campanella was so eager a Catholic and at the same time so eager a Telesian that he bowed to it[34]). Only, on the strength of Tycho's and Galilei's investigations, he rejected the fixed spheres. Like Telesio he assumed two world-forces struggling against each other, the expanding (heat) and the contracting (cold), and the first had its source in the sun (as centrum amoris), the second in the earth (as centrum odii). The sun, with the other stars as its helpers, moves around the earth.

Campanella's view of nature has, like Telesio's, an animistic character, in that he thinks that the interaction of things, and especially the mutual attraction of the opposite forces, would be unthinkable if they were not ensouled. In order to be able to act on each other, they must perceive each other. He repeats Telesio's arguments with closer elaboration. Of greatest interest is the consideration that sensation cannot arise from the cooperation of the elements, since the origin of such an entirely new capacity or property would be a creation out of nothing. In antiquity the answer had been given to this (by Lucretius) that nature after all presents so many examples of the product having other properties than the producing elements. But Campanella replies that in all such cases it is for us, too, a production out of nothing, since we do not understand what it is in the elements that contains the possibility of the new property. That sensation arises from material elements is thus just as much a creation out of nothing as that the bodily arises from the incorporeal. That in the elements which contains the possibility of the new property must be essentially akin (ejusdem rationis) to it, without the latter therefore *[148]* needing to pre-exist in the same way as it appears in the result (eodem modo qvo nunc). — Conceived in this way, the idea of the ensoulment of all things has a significance which

is entirely independent of animism, and which comes into its own precisely only when animism gives way to the mechanical explanation of nature. — It is, however, only the sensuous or bodily soul that Campanella will in this way bring into connection with the rest of nature. The higher, spiritual part of the soul he assumes to be created out of nothing, as Telesio assumed.

Campanella's philosophy of nature develops, like Bruno's, into an entire metaphysics, which in turn is brought into connection with religious ideas. All that is appears as power (potestas), knowledge (sapientia) and drive (amor). — To be is first and foremost to be able, to have the capacity to assert oneself. In its highest, infinite degree and form power is found in the Deity; it is limited in every finite being, in which the fullness of being is always restricted in greater or lesser degree by non-being: there is nothing that God is not; with his essence, not by external influence (essentiando, non exterius agendo), he stirs in all things; — but there is much that a finite being is not. The multiplicity of the finite beings shows this relative non-being; they limit each other and to that extent negate each other. — What holds of power holds also of knowledge and of drive. Without knowledge power cannot act. But this knowledge is inherent in power, is one with power. The capacity to perceive what is to be acted against cannot be separated from the very capacity to act. The instinct of self-preservation in animals shows us this clearly; but it holds for everything in nature. Therefore knowledge must be original, not first the product of external influence. The presupposition of all other knowledge is knowledge of oneself. Such an original self-knowledge is found in every thing, but "hidden" (as intellectio abdita), since no opposition and change asserts itself. Campanella attempts to show how self-knowledge is possible, as against Telesio's proposition that all knowledge presupposes a change. This proposition, Campanella thinks, holds only for the knowledge acquired through external experience. In order to know other things, one
[149] must be influenced by them and thereby changed. But in order to know oneself one need not be influenced or changed: for what one is here to know, one *is*, after all, and need not become it. Of itself the individual being has an "original, hidden thought" (notio abdita innata), which is one with its nature. When it is objected: why then do we not at once have clarity about ourselves, but must learn to know our nature from our actions? Campanella answers that this self-knowledge lying in our nature is constantly hindered or checked by influences from outside. It is necessary for self-preservation to learn to know other things and thus to be constantly changed. Thereby the original knowledge is obscured: the soul lapses into self-forgetfulness and ignorance of itself. The lower animals in particular are strangers to themselves and live as in constant madness owing

to the constant influence from outside. But that "hidden" self-knowledge is nevertheless the necessary presupposition for anything being able to be known about other things. That I perceive heat means that I perceive myself as heated. Every knowledge of other things is a particular determination, a modification of my consciousness, and presupposes it. Only in the infinite being can the original self-knowledge unfold freely and clearly, because external influence here falls away. But in the finite beings the doctrine of the original self-knowledge has the significance that all knowledge is built on our immediate sensation. The skeptics may be right with regard to all derived knowledge; but this always presupposes a consciousness of our own immediate states and their modifications, and this consciousness is a reality which cannot be doubted. It goes with our knowing as with our being able: when I can lift a burden, it is because I can lift my arm; and thus I can know something about other things only because I can know my own state as determined by these other things. I can be mistaken in the interpretation of my state, but not in my apprehending it in a definite way. Campanella refers to Augustine, who had already shown that immediate consciousness does not deceive. "For me," said Augustine, "it is the most certain of all things that I am. Even if you deny that I am, and say that I err, you nevertheless thereby precisely admit that I am; for I cannot be mistaken if I am not." *[150]* — It is by this doctrine that Campanella reaches the point which in Descartes became the starting point of the whole of modern philosophy. Campanella's Metaphysics, in which this doctrine was stated, appeared, it is true, only the year after Descartes's *discours sur la methode,* but it was certainly for the greater part written already during his time in prison. Moreover, the thought of immediate consciousness as the foundation of all knowledge is found earlier in several of the thinkers of the Renaissance, in Cusanus, in Montaigne, in Charron and in Sanchez (the last of whom we have not yet had occasion to discuss). — But Campanella goes further back still. His "hidden" knowledge is what might now be called potential consciousness; it is possibility of consciousness rather than actual consciousness, and Campanella also teaches that it is constantly hindered by external experience. Real self-knowledge is, after all, also won, not by withdrawing from all changes (for an unchangeable being we could not conceive as conscious), but by noticing how our being unfolds by means of the changes and through our actions. And this Campanella then also admits when he says that "the manifold changes veil our original being and constantly make us into new beings, whereby the comparison with what is past and the unity in our being, and thus also acquaintance with ourselves, is hindered." Here another kind of self-knowledge is hinted at than that half-mystical one which is to be independent of all change. —

As power and knowledge stir in all things, so too does drive. This shows itself in every thing seeking to preserve itself. The stone will continue to be stone, and when it is hurled into the air, it seeks back to the earth, where it belongs. Plants and animals seek even through propagation a self-preservation that extends beyond the life of the single individual. Indeed, in the need of light and heat to spread themselves over and into all things, Campanella sees the same eternal drive or "love" (amor) at work. And just as power to move oneself is the presupposition of power to move everything else, and
[151] knowledge of oneself the presupposition of knowledge of everything else, so too love of oneself is the presupposition of love of everything else. The special forms which love assumes toward different objects presuppose the "hidden" love (amor abditus) with which every being holds fast and asserts its own being. Happiness for man, as for every being, consists in such self-preservation; if it consisted in something else, self-annihilation would be the consequence. Virtue is the rule which must be followed in order that the highest purpose, full self-preservation, can be attained; what the law is for society, virtue is for the individual man. By three ways the goal can be reached here in the world; by preserving one's own personality, by continuing one's life in one's children and by leaving behind honor and fame. A fourth way leads to eternal life in God, where man gets a share in the infinite being from which his own existence has been separated by the admixture of non-being. All morality and religion Campanella thus sets in close connection with self-preservation. The relation is most intimate in the case of religion: for the unconscious or "hidden" love of oneself, which lies at the ground of all else that we love, is in turn, in its innermost ground, love of the infinite being which stirs in the limited being of all things, and only by participation in it can the highest blessedness be reached. Therefore there is an original "hidden" religion (religio abdita) in all, just as well as a "hidden" knowledge and drive, and really as one with these. The individual, positive religions may err, but not that inner, original religion which is their common presupposition. A revelation is a necessity only because the positive religions (religiones superadditæ) present differences and errors.

In his individual ethics Campanella quite recalls Telesio. Like the latter, he sets magnanimity (sublimitas) as the summit of all virtue, in that it expresses the true nobility, the inner self-preservation, which looks down on external honor and, in its striving to win freedom for itself and others, does not deny itself under the harshest tortures or the longest imprisonment. In his social ethics he goes through the various societies, from the narrowest, family and household, through commune and state to the world-empire,
[152] a universal human society, whose highest leader is the Pope, and whose senate is formed

by the secular princes. Campanella occupied himself eagerly with politics. Perhaps in his youth he even nourished bold plans for the future of his fatherland and regarded himself as called to be a religious-political reformer. Great obscurity still rests over the conspiracy in which he was accused of having taken part. Utterances that occur in his portrayal of the magnanimous man might indicate that he had had great plans which he *would* not confess. But however that may be, later he looked toward a theocratic world-dominion with the Pope at its head, spread over the whole earth, so that Spain (as is developed particularly in the work *On the Spanish Monarchy)* has the task of bringing the peoples of the other parts of the world under the Church, while the "most Christian" King of France (according to the preface to *Universalis philosophia)* has the charge of subduing the heretics in Europe. Campanella's reactionary position comes forward here in his politics, as above in his view of nature. He stands above all in sharp opposition to Machiavelli, whom (especially in *Atheismus triumphatus* 1636) he regards as the representative of the evil principle.

But the remarkable thing is that Campanella here, as in the domain of the view of nature, is extraordinarily strongly influenced by the thoughts which he wants to, or must, combat. As an appendix to his ethics and politics his ideal portrayal of the future, "The City of the Sun" (*Civitas solis)*, has been published. It came into being in prison, while the pressure of the present lay heavily upon him. A century earlier the English chancellor Thomas More, and at about the same time as Campanella, Bacon of Verulam had given such social ideals; the common model for them all Plato gave in his "Republic." Campanella describes an ordering of society in which science, that is, natural science and philosophy, rules, and in which material labor has come into its own. There is no priesthood and no nobility. Those who govern are those who have received the best theoretical and practical education. The inhabitants of the City of the Sun find ridiculous the thought that the craftsmen should not be just as noble a class as other members of society. No private property, no separate dwellings and no private *[153]* family life are permitted, since thereby egoism would be favored and the burning love of fatherland which animates the inhabitants of the City of the Sun would be diminished. The authorities regulate sexual unions according to physiological considerations, so that the state may obtain healthy and well-endowed citizens. It is, after all, they think, the species and not the individual that is to be preserved through propagation! Pride is regarded as the worst vice. Each is set to the activity which best suits his abilities, and from the yield of labor each receives what he needs and deserves. Chance is not to be allowed to rule over breeding, over vocational activity or over distribution. That

would only result in injustice and the distress of many. "There now live 70,000 people in the city of Naples; of them hardly ten or fifteen thousand work, and these are worn out and ruined by excessive, constant and daily labor, while the rest give themselves over to idleness and fall into sloth, avarice, sickliness, lust and love of pleasure. But in the City of the Sun, where services, arts, labors and occupations are distributed to all, each individual need work barely four hours a day; the rest of the time he can use to acquire knowledge in a pleasant way, to discuss, read, tell stories, write, walk or exercise mind and body in an enjoyable fashion."

This picture of the future, which hovered before Campanella while he sat in his prison "in chains and yet free, solitary and yet not alone, silent and yet raising his cry" — as it says in one of his sonnets, stands in a certain opposition both to the philosophical and to the political-religious view which he develops in his other writings. With the strong emphasis on self-assertion it does not agree that the individual in the "City of the Sun" must so entirely renounce finding his own way to his goal. Among other things Campanella forgets that he has expressly regarded propagation as a form of the preservation of the *individual*; here it is to be *only* the preservation of the species that is at stake. Still more remarkable is the opposition between the "City of the Sun" and the theocracy which Campanella in his philosophical and political writings sees as the goal. All hierarchy and all absolutism has here vanished. There is really no room
[154] for positive religion either. The " hidden" religion seems here to be enough, and it is even stated that Christianity really only sanctions what natural thinking leads one to assume, — a sanction which, in so perfect a social order as that of the City of the Sun, no longer seems to be necessary. One can, to be sure, as Sigwart remarks in his excellent treatise on Campanella and his political ideas, find an agreement between the plan of the theocracy and the plan of the scientifically guided state of the future; but it is nevertheless characteristic that Campanella has worked out his ideas in this double form, and in such a way that not the theocracy but the socialist state, where the right of labor and of science stands in the first line, is the final ideal. The powers of his time, the very representatives of the reaction in state and Church, he thus regarded as means and as educative forces which were to lead forward toward the ideal goal, which was diametrically opposed to the existing order. Although Campanella is in so much a man of the reaction, he nevertheless still has the blood of the Renaissance in him, and he looks out beyond the limits of the given social world with the same enthusiasm as Giordano Bruno beyond the limits of the outer, sensible heaven.

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Second Book

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The New Science.

1. The Task

Practice precedes theory, and art precedes science — even if a new practice may later develop out of the theory, a new art out of the science. Humanism developed, as we have seen, out of the political and social conditions in the Italian states, and out of humanism as a tendency in the practical conduct of life there developed in turn a new doctrine of man. In a corresponding way mechanical natural science developed out of the industry that flourished in the Italian cities. In order to procure means of power and splendor, the rulers had to favor handicraft and industry, and the personal strength and strong self-esteem of the citizens of the cities found expression in eager and ingenious activity in the domain of industrial labor and industrial inventions. There was a contest between the cities in this respect. They sought to outdo one another in professional skill and brooded jealously over new inventions and machines. The practical application of the forces of nature increased acquaintance with their mode of operation and was bound to awaken interest in finding their laws. Neither Leonardo's nor Galilei's appearance is intelligible except in connection with Italian industry, any more than Pomponazzi and Machiavelli are intelligible except in connection with the development of the Italian spirit and Italian politics.

To take a single example. Galilei began his famous *Investigations concerning Two New Sciences* by having the principal character in the dialogue, Salviati, say: "The inexhaustible *[156]* activity in your famous arsenal, gentlemen of Venice, seems to me to offer thinkers a large field for speculation, especially in the domain of mechanics, since machines and apparatus are constantly being made there by numerous craftsmen." To this Sagredo replies: "You are quite right, sir, and since I am by nature eager for knowledge, I often

come here, and the experience at the command of those whom, on account of their outstanding mastery, we call 'the first,' has often opened to me the causal connection of wonderful phenomena that were formerly held to be inexplicable and incredible."

The development of the new world-picture also had a preparatory effect in this direction. It led to seeking a connection in nature other than the one that sense perception immediately offered, and was bound to lead to the question by what forces and laws the system of the world that thought constructed on the basis of observation was held together and kept in activity. This question the new world-view did not itself answer, or when it attempted to, it was in a fantastic, poetic, animistic manner. It was indeed a significant advance that Telesio put the concept of force in the place of the Aristotelian form. By this it was established that the explanation could not be sought in the quality of the phenomenon itself, its completed shape, as with an artist's work, which is understood when one discovers what model hovered before him. But the concept of force is still too vague and unfruitful so long as it is not constructed on the basis of insight into the regular connection between phenomena. If one knows according to what law phenomenon B follows phenomenon A, then one also knows what force and how much force must be ascribed to A; for force then means only the conditions that A contains for the occurrence of B. But to discover this, observation and description are not enough. One must go back to the simplest relations and investigate by way of experiment the conditions on which the occurrence of phenomena depends. It is no longer merely a matter of a picture. The picture must be analyzed into its individual features, and their connection
[157] investigated. Only thereby can full certainty of the picture's validity be won. Several of the investigators who stand as founders of the new mechanical natural science therefore also worked further on the development of the new world-picture, just as they in part got the problems for their mechanical investigation from its relations.

Here too it was a thought of antiquity that was taken up. Archimedes, the founder of statics and hydrostatics, had already in the third century before Christ expressed thoughts that contained the germ of a mechanical view of nature. Through the disfavor of the times this germ did not develop until after the lapse of millennia. Archimedes, now that his intellectual tendency had found favorable conditions of life, was among the authors who were eagerly studied, edited and translated in the 16th century. It was, however, in reality a new science that was to be founded. The philosophical interest of this event lies in its significance for the life of the spirit, and this significance is of a many-sided kind. — In the first place, a new method developed here, and with it a new use of the human cognitive faculty, which reached deep into the character

and direction of the development of the spirit. New spiritual needs and habits arose. Experiment and analysis pressed forward ahead of contemplation and construction, if they did not actually displace them. — In the second place, it was especially the material side of existence to which the new method could be applied. The question then had to arise what scope the results found here had — whether they contained a knowledge of the whole of existence, or what their relation was to the mental side of existence. The revolution that took place through the new world-view had already led profound thinkers to see that the riddles of the world do not merely, as the naive conception supposed, lie essentially outside us and beyond us, in the great cosmic relations, but hide in our own inner self and in the very smallest natural phenomena accessible to us. Through the new method and its results this became still more evident. — Finally, it could not be otherwise than that the more exact knowledge of nature, and the ability, following upon insight into the laws of nature, to foresee and in part to control the course of phenomena, must strengthen human self-esteem and carry further what humanism *[158]* had begun in a more aesthetic and theoretical form.

2. Leonardo da Vinci

With this great artist we come back again into the Renaissance. He ought to be named in the history of philosophy because in the scattered thoughts that have been extracted from his posthumous manuscripts there is found the first clear statement of the principle and method of exact natural science. Not only in the visual arts but also in anatomy, engineering and mechanics was this most versatile mind of the Renaissance at home, and he had the need to give himself an account, in philosophical ideas, of the path he took in his research, as well as of the impressions he received of the human life around him. He was born near Florence in the year 1452 and received instruction in the art of painting from Verocchio, who himself, besides the visual arts, practiced weaving, metal casting and the goldsmith's art. He was called by Ludovico Moro to Milan, where he founded an academy of sciences. It has been supposed that a part of his notes were intended to be communicated here. In Milan Leonardo worked not only as painter and sculptor but also as engineer, musician and arranger of court festivities. After Ludovico Moro's fall he worked chiefly in Florence and Rome. His last years he spent in France, where he died in 1519. His wandering life and many-sided activity brought it about that he did not come to work out the works on scientific and philosophical subjects that he had sketched. They would have reached deep into the development of scientific thinking and hastened it

considerably. Thoughts that one has been accustomed to attribute to Galilei or Bacon are already found expressed by Leonardo, but lay buried in his manuscripts, which have only in the most recent times become more closely known. The main thoughts that interest us in this connection aim partly at emphasizing the significance of experience, partly at enjoining that without the application of mathematics no full certainty is reached for the results of our knowledge. Wisdom is the daughter of experience, and therefore it is also a
[159] product of time. He rejects speculations that do not find their confirmation in experience, which is the common mother of all sciences. But he will not stop at mere observation. It is his conviction that to every active element (potentia) in nature are attached effects of a definite kind, which unfold in a definite order. Necessity is the eternal bond, the eternal rule in nature (freno e regola eterna). It is into this that we must penetrate, and — so Leonardo's meaning must presumably be understood — it is by virtue of necessity that mathematical knowledge can be applied to experience: it makes it possible to infer from the given phenomena to others that stand in a relation of condition to them. In mechanics this shows itself most clearly and simply: mechanics is "the paradise of the mathematical sciences." — In the thoughts that thus appear to us through a combination of Leonardo's aphorisms, the fundamental problems of modern epistemology are already contained. Through his practical application of them he became one of the founders of modern mechanics and engineering.

What else is available of Leonardo's ideas bears the stamp of a robust naturalism. He regards the soul partly as the principle of sensation, memory and thinking, partly also as the formative principle of the organism. Insofar as we wish to know anything about the soul beyond this, he refers us to the monks, "those fathers of the people, who by inspiration know all secrets." He expresses (as Montaigne and Bruno later did) the idea of the circulation of matter through the organic and inorganic world, and draws from it a peculiar conclusion, which only after a long lapse of time crops up again in Diderot: "In the matter that dies, life continues to exist without its being noticed; and when that matter is then taken up again into the nutritive organs of living beings, it awakens once more to sensuous and mental life". In agreement with this he finds in man's constant craving and expectation a testimony to what stirs in nature itself, of which man is the model. His hope of attaining a clear knowledge of nature is supported by the thought that nature follows the simplest paths, a thought that on the whole played a great part
[160] in the first times of modern research.

Interesting as these hints are, it is nevertheless not they but the idea of the combination of experience and exact thinking that gives Leonardo a prominent place in the

history of thought. A century was to pass before this idea, through great investigators who had to discover it by their own path, could be carried further.

3. Johannes Kepler

Attention has already been drawn to how the new world-view, in the form of mystical speculation, stirred at the beginning of the 17th century even in Böhme's quiet workshop. His contemporary and, on certain points, kindred spirit Johannes Kepler succeeded in finding a more exact foundation for the new view than it had hitherto been able to rest on. Through a rare union of apparently conflicting faculties and interests he was led, by enthusiastic and faithful work, from lofty, soaring speculations to the founding of an exact science of experience. In common with Bruno, Kepler has the great need of a wide horizon and of expressing the thoughts that move his mind. "I know," he says, "no greater torment than not to be able to bring out what I feel within me — let alone to state the opposite of what I think". He did indeed come to feel opposition on the part of orthodoxy; this, in connection with the slight spiritual sense and great superstition of the time, cast dark shadows into his life. But his love of truth and untiring zeal for work held him up despite all adversities.

Kepler was born in the year 1571 in Weil in Würtemberg. He is a Swabian, like so many of Germany's most profound minds. He received his education at the theological seminary of Tübingen, where humanities, philosophy, mathematics and astronomy were studied to no small extent. Here he was introduced to the Aristotelian philosophy of nature, to which he long held fast. In astronomy he had Maestlin as teacher, who privately doubted the correctness of the Ptolemaic system, although officially he continued to lecture on it. Theology was laid aside when he reluctantly took up a post as teacher *[161]* of mathematics at the gymnasium in Graz. This call became decisive for his — and for science's — future. He now formed the plan of giving a new philosophy of nature that combined Copernicus's system with the old doctrine of psychical beings as movers of the heavenly bodies. He separates himself *(Epitome Astronomiæ Copernicanæ.* Op. ed. Frisch. VI, p. 136 f.) expressly from Bruno by assuming the sphere of the fixed stars as the boundary of the universe, since he holds that on Bruno's theory the fixed stars must be so far removed from one another that we could see only far fewer of them than we actually see. The sphere of the fixed stars encloses a hollow, in whose middle the sun is found, about which the planets, among them the earth, move. Bruno has here seen more rightly than Kepler. But in return the latter was able to subject his ideas —

despite their originally mystical form — to a more exact verification than Bruno had been able to. Kepler's first writing *(Mysterium cosmographicum.* 1597) proceeds from theological and Pythagorean presuppositions. He wants to conceive the universe as an image of the Trinity: to the Father corresponds the center, to the Son the surrounding sphere, and to the Spirit the relation between the two, expressed in the geometrical relations between the different spheres in which the planets move. The divine spirit reveals itself, namely, in the harmonic proportions of magnitude in the universe. Kepler seeks to show that in the different spherical surfaces in which the planets move one can inscribe the five regular solids set up by Pythagoras, that is, the solids that have nothing but equal faces with equal sides and equal angles. The fundamental forms of geometry and the distribution of the globes in space are also supposed to correspond exactly to each other. This is the cosmographic mystery over which Kepler was enthusiastic, and which he carried with him as a guiding idea that partly furthered, partly hampered his subsequent investigations. It expressed his never-failing conviction that definite mathematical relations must be demonstrable in the universe, and guided him therefore always to renewed research; but it caused him lasting trouble through the presupposition,
[162] from which he set out with all antiquity and the Middle Ages, that the heavenly bodies moved in circles, since the circle was the most perfect figure.

Among the men to whom Kepler sent his writing was also Tycho Brahe, whom he greeted as "the prince among the mathematicians of the century." Tycho answered kindly, but declared that, relying on 35 years of observations, he could not go along with Kepler's speculations, however much ingenuity he found in them. In particular he advanced his objections to the Copernican theory. The connection thus begun between the two men led to Kepler — when Tycho shortly afterwards moved to Prague — also taking up his residence in Prague and later, after Tycho's death, being entrusted with his great body of material, about which he had earlier written to his teacher Maestlin: "My opinion of Tycho is that he possesses riches which he — like so many rich men — does not use in the right way". Now he had himself inherited these riches and with them the possibility of a further development and confirmation of his ideas. It was on the basis of Tycho's observations that he found the laws named after him and — what interests us most in this connection — was at the same time led to set up the mechanical view of nature in place of the animistic one he had hitherto embraced.

Kepler lived his later years in Linz, in hard struggle against Protestant and Catholic fanaticism and in painful efforts to procure means for the publication of his works. To his native region in Würtemberg he had to return home for about a year to save his

mother, who was accused of witchcraft and threatened with the stake. He died in 1630 in Regensburg, where he had traveled to claim a sum owed to him by the Imperial Diet. —

Through his idea of the significance of quantitative relations in nature Kepler became one of the founders of exact natural science. He arrived at this idea partly by a theological, partly by a psychological, partly by a natural-philosophical path. — The theological path we know already. In order that the universe might become the beautiful image of the essence of the Deity, it was ordered in definite quantitative relations. Kepler, like Copernicus and Leonardo, had the conviction that nature works according to simple and *[163]* clear rules. Simplicity and ordered regularity (simplicitas atque ordinata regularitas) in a view of nature therefore speak in its favor in his eyes from the outset. The task is to lead everything back to as few and simple principles as possible. — The psychological grounding lies in the fact that the human mind sees through quantitative relations most clearly; it seems to be quite properly created to apprehend these. In qualitative respects the activities in nature appear very different to different subjects (pro habitudine subjecti); full certainty is attained only by keeping to the quantitative side; this side therefore brings the real truth. — Finally, the natural-philosophical grounding lies in matter itself, as we know it from experience: "where there is matter, there is geometry" (ubi materia, ibi geometria). The world shows itself in fact to partake of quantity (mundus participat quantitate).

The question *which* quantitative relations, then, lie at the ground of the universe, Kepler at first (in Mysterium cosmographicum) believed he could answer purely a priori. From this error he was led away by immersing himself in Tycho Brahe's material of experience and in his own observations. It is well known how indefatigably he worked until he had found the relation that agreed entirely with the observations, and that he was thereby led to put the ellipse in place of the circle, idolized since antiquity.

In his doctrine of hypothesis Kepler has given a clear account of his conception of the nature of scientific research. He had occasion in several ways to set up such a doctrine. Tycho had sharply rejected Copernicus's hypothesis as pure fancy (imaginatio) and protested against every method that proceeded a priori (or, as it was also said at that time, ab anteriore). But besides this, both Tycho's and Copernicus's theories were regarded by opponents (e.g. Ursus of Dithmarschen) as purely arbitrary assumptions that aimed only at making fools of people. And Osiander, as we know, had in his day, to avert offense, declared Copernicus's conception of the world to be a hypothesis that was merely to make the mathematical relations intuitive, but not to be held seriously. In *[164]* his unfinished *Apologia Tychonis contra Ursum* (cap. 1) Kepler now discusses the whole

question of the significance of hypotheses, and he comes back to it many years later, in the first book of *Epitome astronomiæ Copernicanæ.* All science — he says — rests on certain presuppositions, and one does not wait to do research until one has found absolutely first presuppositions, any more than one waits to build one's house until one has investigated whether the interior of the earth is solid. Hypothesis means, in the widest sense of the word, what one assumes as certain in a proof. In this sense geometry too rests on hypothesis, and the observation on which one builds in natural science is also a hypothesis. In a narrower sense an astronomical hypothesis is the sum of representations by whose help an investigator demonstrates the ordering of the motions of the heavenly bodies. Such representations can by no means be formed arbitrarily. They are to stand their test by the conclusions drawn from them agreeing with the actual phenomena and not leading to absurdities in physical respects. And they can be dispensed with as little as a physician can avoid forming a representation of the disease instead of stopping at the bare symptoms. Science begins with observations, builds its hypotheses on them, and next seeks the causes that bring about the assumed connection. There is in Kepler still some uncertainty concerning the relation between the first and second members of this threefold division. For when he regards Tycho as the one who has supplied the observations on which astronomy builds, and Copernicus as the one who has supplied the best hypothesis, then the hypothesis has, after all, come before the observations. This can very well be the case, when it is the hypothesis grounded on a preliminary investigation that guides toward the understanding of the facts found and toward the search for new facts. So it went with Kepler himself: it was his Copernican presuppositions and his ideas of the significance of quantity in the world that gave him an eye for what lay in the riches that Tycho himself did not rightly value.

[165] As regards the third member, the demonstration of the causes, Kepler gradually made stricter demands than he had made in his first, animistic period. He began (in Mysterium cosmographicum) by having the planets guided by souls, or else the whole system by the world-soul in the sun. But later he broke with animism. In his epoch-making treatise on Mars (*Astronomia nova s. physica coelestis,* tradita commentariis de motibus stellæ Martis. 1609) he declares that it is physical causes that must be demonstrated. "True" causes *(veræ causæ)* or probable causes now become a frequent demand with him. He approaches the conviction that the causes assumed for natural phenomena must themselves be demonstrable as operating in nature. In the second edition of his animistic youthful writing he adds to the expression "moving souls" (animæ motrices) the following note: "In my treatise on Mars I have shown that there are no such things,"

and remarks that in place of the word soul one ought to put the word force. "Formerly," he continues, "I believed that the force that moves the planets was really a soul . . . . But when I considered that this moving force decreases with distance, ... I concluded that it must be something corporeal." (Opera I. p. 176). — In place of the Mysterium cosmographicum there now came for Kepler the idea of a Physica coelestis, which he still lacked the means, however, to carry through. It was carried a mighty step further by Kepler's great contemporary

4. Galileo Galilei

When Galilei, in a letter some years after Kepler's death, says that "he has always esteemed Kepler as a free (perhaps all too free) and acute mind, but that his own way of doing research is different from Kepler's," he is no doubt thinking in this characterization essentially of Kepler's earlier, animistic theories. He has hardly had clearly before him the turning point that came for Kepler through the investigation that led to the famous laws. The two great investigators work in reality in the same spirit: they seek to combine deduction and induction, mathematics and experience, and they thereby stand as founders of the exact science of experience. Only that Kepler begins more from the a *[166]* priori, deductive side, Galilei more from the experimental, inductive side. In Galilei the really decisive step is taken by which natural science was constituted as an independent science and a great ideal for all research was set up.

Galilei, like Bruno, became a martyr for the new world-view. There is in him an interesting interaction between his astronomical and his physical investigations. His experiments and his study of Archimedes led him to find difficulties in the Aristotelian-Ptolemaic system, and conversely he was led by his zeal to ground the new world-picture to find new physical laws. Only slowly and hesitantly did he come forward as a declared Copernican. He was born in Pisa in 1564 and studied philosophy, physics and mathematics there, just as he was also an enthusiastic cultivator of poetic literature. He preferred Plato and Archimedes to Aristotle, and thereby the foundation of his later views was already laid in his student years. As professor, first in Pisa, later in Padua, he lectured on the old system after he had long been convinced in his heart of the truth of the new. He writes in a letter to Kepler (in August 1597): "It is a misfortune that those who seek after truth, and who do not follow a wrong method, are so rare. Many years ago I came to Copernicus's opinion, and from this standpoint I have found the causes of many natural phenomena that are certainly inexplicable by the usual hypothesis. I have

written down many reasons and refutations, which, however, I have not dared to bring to light, since I am deterred by the very fate of our teacher Copernicus, who, though he has won immortal fame with a few, has with countless others (so many fools are there in the world) become an object of laughter and scorn." Only when he had constructed a telescope for himself and discovered the moons of Jupiter did he declare himself publicly (in the year 1610) for the Copernican system. With that the signal for persecution was given. Although Galilei made one brilliant discovery after another that could support his hypothesis — thus he discovered the sunspots and the phases of Venus — monks and
[167] theologians nevertheless spoke out against him with increasing vehemence. Aristotelian philosophers would not even look through Galilei's telescope themselves, so as not to see the offensive sight of changes in the heavens and thereby lose their faith in the old world-picture. Galilei was right when he said that even if the stars themselves came down to earth and bore witness, his opponents would not let themselves be convinced. He sought in vain to show that his views did not conflict with the Bible. Had he remained in Padua, which stood under the dominion of Venice, he would no doubt have been safe in his person. But in order to have more leisure and a more favorable financial position, he accepted a position in Florence as the Grand Duke's "mathematician," and here he could easily be reached from Rome. The College of the Inquisition now (in the year 1616) took the momentous step of putting Copernicus's work on the list of forbidden books ("until it had been corrected") and of declaring his doctrine heretical. Galilei was summoned before Cardinal Bellarmine and is said — according to what is claimed on the Catholic side — to have received an express injunction not to embrace and spread the condemned doctrine; proof that such an injunction was really given cannot, however, be furnished. Galilei did not, however, give up his research on that account. Through his investigations of the comets he came into conflict with the Jesuits, and the mighty order stood as one man against him. He was moreover working constantly on a complete account of the struggle between the two systems of the world, which he had often announced in earlier writings. If he merely expressed himself hypothetically, there was, he thought, no danger. He thus used the same method that Osiander had used when he published Copernicus's work. In the year 1632 his famous work appeared, which is described on the title page as *"Dialogue, in which in conversations on four successive days the two most important systems of the world are discussed,* the Ptolemaic and the Copernican, in such a way that, without any decision being made (indeterminatamente), the philosophical and natural grounds for the one and for the other of the two parties are set forth". The persons conversing are Salviati and Sagredo, two of Galilei's friends,

and Simplicio, who is the representative of the Aristotelian philosophy. Salviati is the deliberate, critical investigator; he sets forth the grounds, but prefers to draw no definite *[168]* conclusion and seeks to hold back the fiery Sagredo, through whom Galilei gives vent to the freest of his ideas, so as then, if need be, to have Salviati call them back. Yet it is clear enough to every reader in which direction the author's sympathy goes, and in Rome they did not let themselves be fooled. The book was forbidden, and Galilei was summoned to Rome. He seems not to have been actually tortured, but only threatened with the rack, and then on the 22nd of June 1633 had, kneeling, to abjure "the false doctrine that the sun is the center of the world and immovable, the earth not the center, but movable," and to swear "that he would not in future either orally or in writing state anything from which this doctrine could be derived, but on the contrary would denounce to the Inquisition any heretic or anyone suspected of heresy whom he came to know!" — That he swore a false oath there is no doubt. His conviction he did not give up. Instead of burning him bodily, they punished him with the burning pain caused by having to keep silent about one's conviction. Even if Galilei, by his nature, did not feel this pain as strongly as a Kepler would have felt it, it was nevertheless strong enough to embitter his life. To this were added grief and blindness. He lived under constant supervision in the neighborhood of Florence, always occupied with scientific ideas. The life he saved by his false oath was still to be of extraordinary value through the publication of his second main work: *Investigations (Discorsi) concerning Two New Sciences,* which — to escape the censorship — was printed in Holland (1638). This work, which contains the founding of modern physics, had, as has been aptly said, to sneak its way into literature. Galilei died in 1642; he was mentally active right up to his last moment. — We shall now bring forward the most important points of view from which Galilei has significance for the general history of thought[35]).

a. The Method and the Principles

Galilei objects to the old logic that it is excellent for regulating and correcting the train of thought, but is not a means for the discovery of new truths. Such discovery comes *[169]* about when one derives an assumption from certain experiences and then seeks by deductive means to show that the assumption set up agrees with other experiences. The analytic method (metodo risolutivo) and the synthetic method (metodo compositivo) thus mutually supplement each other. If by induction one meant an investigation of all possible cases, inductive inference would be either impossible or useless: impossible when one could not get hold of all the cases, useless when one could manage all the cases.

It is therefore only a matter of investigating the most characteristic cases; from them one then infers to the others. But one must set up a proposition as a conjecture before one can find the real proof of it. Confidence that the proposition is correct contributes much to one's finding the proof of it. On the other hand, what one infers from general principles must be proved in the particular case by experiment. Thus Galilei proved by experiments on the inclined plane his assumption, derived a priori, that the paths traversed with uniformly increasing velocity are to each other as the squares of the times.

But when deduction and induction are thus to work together, a difficulty arises from the fact that in our deductive reasonings we assume the relations to be simpler than they are in reality. The mathematical propositions hold only for ideal figures. It is no defect in real things that their figure never coincides entirely with any of the figures that geometry constructs. The irregular figure that the stone has, it has to perfection. But in the application of abstract reasoning to concrete relations, the manifold and intricate conditions must be taken into account. It is our art of calculation, not the figure of things, that is imperfect. In empty space — we assume — all bodies would fall with the same velocity. Since we cannot produce an absolutely empty space, we investigate how the relations are in denser and in thinner media, and when we see that the velocities approach one another the thinner the media are, we consider the proposition proved. That a body thrown along a horizontal surface will continue to move when all obstacles
[170] are kept out is proved by the fact that the more one manages to exclude the obstacles, the longer the motion is continued.

In order to be able to find points of attachment for deduction in experience, and in order to be able to establish to what degree experiences approach the ideal presuppositions on which deduction rests, Galilei — like Kepler — lays great weight on quantitative relations. Only by the measurement of phenomena does it become possible to decide how far they agree with what deduction requires. Therefore it is Galilei's principle to measure everything that is measurable, and to make measurable what is not.

Galilei's chief method is the one that Stuart Mill much later called the method of proportional changes, and that one could also call the method of limits. It makes it possible to combine abstract thinking with concrete observation. Through approximations the concrete cases come to share in the ideality of the abstract rules. As with Plato the "idea" was the model for real things, so with Galilei the law is the ideal expression of the connection between things.

When the law is thus found by analysis of what is given in experience, it follows of

itself that the causes by which phenomena are explained must themselves be given in experience. Galilei embraces the same principle that Kepler expressed in the demand for "true causes" (veræ causæ). In a very apt way Galilei shows that the appeal to the divine will explains absolutely nothing, precisely because it explains everything equally easily. It is as easy for God to let the universe move about the earth as to let the earth move about the sun. Here the one becomes just as simple as the other. But if we keep to what is moved, there is evidently a great difference in simplicity between the two assumptions; the one requires a far greater expenditure of force than the other. Before the infinite the finite differences vanish, but they remain when one compares two finite things with each other. To such a comparison science must keep. If one appeals to a miracle, the one can be assumed just as well as the other; every criterion is lacking. Besides, one overlooks that all the works of nature and of God are miracles, even if they can be explained in a *[171]* natural way.

Not only the theological but also the animistic explanation of nature is excluded by the method employed. To explain the mutual influence of things, e.g. the attraction of the magnet, by "sympathy" is merely to dispose of a whole host of questions with a fine-sounding word. Thus gravity too is only a name. We do not know what it is that draws the stone down toward the earth, any more than what it is that holds the moon in its orbit around the earth, or what keeps the stone in motion after it has left the throwing hand. Galilei therefore rejects the assumption put forward by Kepler that the moon has an influence on ebb and flood. Here his fear of "occult qualities" leads him astray.

b. The New System of the World

However often Galilei says in his dialogues that he will not settle on any view, his own standpoint is nevertheless clear enough, even if we did not know it from his letters. He admires Copernicus because, despite the testimony of the senses, he managed to gain and hold fast his conviction of the motion of the earth. And he himself, through his astronomical discoveries, brings forward a series of proofs that the universe cannot, as the Aristotelians held, be divided into an unchangeable, heavenly world and a changeable, earthly one: the sunspots and the new fixed stars show that changes also take place in the heavens. And — Galilei remarks — with what right does one really ascribe a higher value to the unchangeable than to the changeable? "I for my part," he has Sagredo say, "consider the earth to be most noble and admirable precisely on account of the many diverse changes and productions that take place on it without ceasing". Simplicio objects to this that while the changes that take place on the earth may be for the good of men,

the heavenly changes would be altogether purposeless. A similar reasoning he applies to
[172] the enormous extension that the universe (as Tycho Brahe had already pointed out) must have on Copernicus's system: of what use is the enormous space between the outermost planet and the sphere of the fixed stars? In his answer Salviati applies to final causes the same consideration that Simplicio applies elsewhere to natural causes: how dare we presume to know the purposes of the Almighty! There is, after all, so much in our own body whose significance we do not know; how then should we be able to know the significance of what is more remote? And how difficult is it not to discover how a thing enters into the whole connection to which it belongs, when one has no possibility of seeing what effect its removal will have? It is to set limits to the divine power to believe that the enormous dimensions of the universe should make its working at all points impossible. The sun shines on the grape and ripens it as if that were the only thing it had to do, and so God and nature too may well provide for every single thing, even if the content of the universe extends to infinity. Galilei's standpoint here recalls Bruno's and Böhme's line of thought in face of the same problem. — Galilei is rather inclined (like Bruno) to assume the infinity of the universe; he doubts the right to speak of a center of the world; and if he does not express himself definitely on this, it is perhaps because Bruno's fate made him wary.

It has rightly been a matter of wonder that Galilei, in his discussion of the systems of the world, has not mentioned Kepler's laws, whose derivation from the law of gravitation in Newton's hands was to become so mighty a support for the new system. Galilei presumably attributed to them only a purely mathematical interest and did not suppose that they could bear witness to the physical connection of the universe.

c. The First Propositions of the Doctrine of Motion

In support of the new world-picture Galilei appeals, like his predecessors, to the principle of simplicity. Nature does nothing in vain, and therefore it always goes the simplest ways and seeks by few and simple causes to attain a multiplicity of effects. This principle
[173] of simplicity, which in astronomy he set against the complicated systems of Ptolemy and Tycho, led him in physics to set up the first laws for the changes of material phenomena. It seemed to him to be the simplest thing that a thing should remain in the state in which it was, when no change occurred. Kepler had already set up the proposition that a body could not of itself pass from rest to motion. Galilei now discovered that a body also cannot of itself change its motion or pass from motion to rest. This is the law that was later (with an expression used by Kepler) called the law of inertia. Galilei does not,

however, set it up as a general proposition. That both the direction and the velocity are preserved he has nowhere stated. In the "Dialogues" it is the preservation of velocity that he chiefly dwells on; he is thinking here chiefly of horizontal motion, and he assumes that circular motion will continue just as well as rectilinear motion. In the "Discorsi" the proposition appears more clearly. A motion, it is said here, can only increase when new force (impeto, momento), is supplied, and only decrease when an obstacle (impedimento) is applied, in both cases, then, only under the influence of external causes. If external causes are kept away (dum externæ causæ tollantur), the motion will be continued with the velocity once attained[36]). That this is a thought experiment Galilei expressly states; but he proves the correctness of the proposition by experiments showing that the more external causes are kept away, the longer the motion continues unchanged. Here the interaction between deduction and induction by means of the limiting cases shows itself, and Galilei here at the same time himself gives an example of his claim that the way of proof is other than the way of discovery. For to appeal to the simplicity of nature is no proof, even if this thought can lead to significant discoveries.

In his investigations of falling motion too Galilei uses the principle of simplicity as his starting point. "To the investigation of naturally accelerated motion I have been led by attentive consideration of nature's usual procedure and ordering in all its operations, since it is wont to employ the simplest and easiest means . . . When therefore I observe *[174]* that a stone falling from the position of rest from a considerable height gradually acquires new increments in velocity, why should I not believe that such increments come about in the very simplest way? And we shall not be able to find any increment simpler than the one that always comes in the same way". — But here too it is only in the finding and setting up, not in the proof proper, that the general principles are used. —

As Galilei begins his investigations of motion, he says: "On a very old subject I bring an entirely new science". He emphasizes in particular that no one has yet investigated the quantitative variations of motion in free fall or of projectile motion. In finding laws for these variations Galilei is conscious of having founded a new science, "into whose deeper secrets it is reserved for more superior minds to penetrate." But already Galilei himself saw clearly that if motion is to be thought of in continuous connection, account must be taken not only of the actual, sensible motion, but also of the force or incitement (energia, momento, impeto) to motion that is present at every moment — whether a body is at rest or in motion (Discorsi III).

The step that was taken by this founding was of extraordinary significance. The later history of natural science has shown that the laws of motion contain the key to all

scientific knowledge of material nature. Galilei himself had a presentiment of this. In the "Dialogues" he states that he has never been able to comprehend how a transformation of substances into one another was possible. When a body presents properties it did not have before, he does not consider it impossible that this happens through a mere change in the arrangement of the parts, without anything being annihilated or anything coming into being. With this it was clearly stated that the understanding of the qualitative changes in nature becomes possible only when these can be conceived as quantitative changes, that is to say here, as motions in space. The great question for the philosophy of the modern age now became how it would take its stand toward the great principle
[175] that was hereby set up, and that triumphantly established its applicability in one domain of nature after another.

d. The Subjectivity of the Sense Qualities

Already in Copernicus and Bruno it appeared that the new conception of the world led to posing the problem of knowledge in a new way, in that, in place of the picture of the world that immediate observation gave, an entirely different one had to be put when thought drew its consequences. In Galilei this appears still more clearly.

When he lays the principle of simplicity at the ground of so many important questions, he distinguishes expressly between the ease and simplicity with which nature works and the often intricate and laborious procedure we must employ to convince ourselves of nature's simplicity. What is difficult for us to understand is very easy for nature to do. There appears here, then, a definite opposition between existence and knowledge. It was Galilei's view that in the clearest knowledge we have, the mathematical, this opposition in a certain way falls away. Here human knowledge shares in the necessity with which the Deity thinks the truths that lie at the ground of the connection of existence — only that what we think successively, laboriously working our way forward from inference to inference, the Deity knows by a simple beholding (di un semplice intuito, as opposed to: con discorsi). To that extent the same opposition between simplicity and multiplicity remains on this point too. And it is only in intensity, in rigor and necessity, that human knowledge at its summit thus becomes one with the divine. In extent it falls infinitely far behind, and the most insignificant thing that goes on in nature is impossible to see through even for the deepest inquiry. Socrates saw clearly this imperfection of our knowledge, and yet he was praised as the wisest of all. — Thus both by its simplicity and
[176] by its infinity existence stands in opposition to our knowledge. Galilei teaches a docta ignorantia just as Cusanus and Bruno do.

The opposition between knowledge and existence comes out no less in the distinction that Galilei, like all Copernicans, enjoins between absolute and relative motion. Only in relation to something unmoved do we apprehend motion. The motion common to several things is — as far as their relations to one another are concerned — as if it were not present. While the ancient conception of the world confidently presupposed visible space as one with absolute space, the thought now had to make its way that absolute space, as the immovable container of all material things, was not an object of sense perception at all, since the apparently resting frame might itself be in motion in relation to a point lying outside it, and so on. Copernicanism, especially under Galilei's acute treatment, trained people generally in the application of the epistemological law of relation. While the Peripatetic philosophy would otherwise keep change out of the heavens, it nevertheless regarded the rising and setting of the sun and the stars, the alternation of day and night, as real changes. Galilei's representative in the Dialogues replies to this: "All this is only change from the earth's point of view. Think the earth away: then there is no sunrise or sunset, no horizon, no meridian, no day and no night!"

When a whole series of changes in the heavens thus lost their absolute character and were explained merely by the standpoint of the beholder, Galilei also found occasion, for earthly phenomena, for a similar transfer of the differences from the objective to the subjective domain. We have already seen that he denied the transformation of substances and would assume only a rearrangement of parts; with that the principle of the subjectivity of the sense qualities was really already given. But Galilei has stated this principle in a still more definite form. In a remarkable section of an astronomical polemic *(Il Saggiatore.* 1623) he has stated that the only properties we must of necessity ascribe to bodies are figure, magnitude, motion or rest. These properties, which he therefore calls the first and the real properties (primi e reali accidenti), we cannot even by the *[177]* strongest exertion of the imagination separate from things. On the other hand, it is due only to a prejudice nourished by the senses when we regard taste, smell, color, heat etc. as absolute properties of things. They are only names we give to things when these call forth certain sensations in us. They have in reality their seat not in things but in the sensing body (nel corpo sensitivo). Take the sensing being away — and all those qualities fall away at the same time[37]). — The parallelism between this consideration and the line of thought employed in the defense of the new world-picture is easily seen. It became of great significance for modern epistemology, as soon as the latter could work its way entirely out of the swaddling clothes of dogmatism.

5. Francis Bacon of Verulam

a. Predecessors

The new science, which was founded under the influence of the experiences and inventions of practical life, was bound to lead to an extension of the inherited logic. But even among those who did not stand especially under the influence that called the new science to life, even in the circle of the humanists, the need of a new logic expressed itself, and ever more vividly, through the fifteenth and sixteenth centuries. From this sprang a series of attempts at reform, programs and announcements, with manifold presentiments of what should be put in place of the scholastic logic, which was really suited only to drawing formal conclusions from presuppositions already given by authority. In the Middle Ages it was, after all, chiefly theology and the doctrine of right that made use of logic, and both started from presuppositions fixed by authority. This whole series of attempts at reform culminates in Francis Bacon of Verulam. This man, who has often been represented as the founder of empirical science, hardly deserves even the name of a Moses who looks into the promised land. There is certainly something prophetic about
[178] him, and he has often expressed, in a brilliant way, thoughts that illuminate the course and conditions of human inquiry, just as he is also clearly conscious of his opposition to scholasticism; but the promised land had — without his having noticed it — been conquered by Vinci, Kepler and Galilei. He declares modestly that he is not himself a warrior, but the herald (buccinator) who rouses men to the battle. But the inquirers who founded modern empirical science did not need the notes of his horn to be fired for the struggle. Bacon does not for that reason lose his significance in the history of philosophy. To a rare degree he took into himself the thoughts and hopes that were bound to stir in the age in which the new science came into being. If he did not help to found it, he was touched by the medium in which it developed, and he prophesied of the way in which such a new science was bound to intervene in human life. He has the consciousness of a fundamental change in lines of thought and interests, far more clearly than any other of the thinkers of the period of transition. He will therefore keep a great name, even if a good deal must be subtracted from the idolization his countrymen have practiced with him, and which neither his works nor his personality deserve. — Before we go on to consider the man and his work more closely, we must dwell a moment on his predecessors in the 16th century.

From the circle of the humanists came PIERRE DE LA RAMÉE (Petrus Ramus). In the middle of the 16th century he waged a violent battle against the Aristotelian logic. To

be sure, he only continued[38]) thereby the endeavors which, already from the preceding century on, had sought to bring logic nearer to practical applicability, especially for rhetoric. And he himself declared that he had had the German humanists and pedagogues Agricola and Sturm as his teachers. But the whole tendency nevertheless finds in him its most gifted and most accomplished representative, and he waged the battle with an energy that contributed mightily to shaking the absolute rule of scholasticism at the universities of Western Europe.

Ramus was born in 1515, the son of a charcoal-burner in north-eastern France. That he was a charcoal-burner's son he often had to hear — after that age's manner of polemic — from his opponents. But he was not ashamed of it. Not because his family was noble *[179]* (though it had come down in the world); but because by his passionate eagerness to learn he had worked his way up to a high scholarly position. He began as servant to a rich student in Paris, and when he had done his day's work, he studied at night. It was the scholastic logic into which he was first introduced; but he felt dissatisfied with it and came with enthusiasm to know Plato's dialogues, which seemed to him to describe the living, actual activity of thought to a far higher degree. In his master's disputation of 1536 he defended the radical proposition that everything Aristotle has said is false. The bitterness of the Aristotelians was great, and it grew with his detailed criticism of the old logic. The university demanded that Ramus's books be suppressed, since he was an enemy of religion and of public peace in imparting to the young a dangerous love of the new. Francis I then prohibited Ramus's books by an edict, and at the same time prohibited Ramus from attacking Aristotle and other ancient authors! Under Henry II, however, Ramus received freedom to teach again and now worked at the Collége de France, where his lectures gathered nearly 2000 listeners. It was a movement such as had not been known since Abailard's days. In 1554 Ramus published his logic *(Institutiones dialecticæ)* in Latin, and the year after in French. The fundamental thought of his doctrine is that one must first investigate how thinking is used by nature, before one sets up rules for it. He therefore refers to the first philosophers, who as yet had no artificial logic. And not only these philosophers, but also the great statesmen, orators and poets, as well as the mathematicians of antiquity, show us in their works the involuntary use of the faculty of thought. Here we find the rules unconsciously applied. Ramus's humanist interest finds satisfaction in the fact that the study of the ancient authors can thus be taken into the service of logic. Like his humanist predecessors, he sets up two principal logical functions: the finding (inventio) of arguments, and the use of these arguments by the judgment (judicium) for the closer grounding and illumination of the

subject. The judgment is therefore often called, in the school language of the following
[180] period, secunda pars Petri. Ramus occupied himself mostly with this latter part and especially worked out the doctrine of the syllogism, and the striking thing appears here, that at bottom he does not depart much from the Aristotelian logic. The purely formal character of Ramus's reform shows itself in the great weight he lays on divisions into two (dichotomies). Without regard to the nature of the subject, the exposition is divided at every point into two members, since (according to what logic teaches) a certain predicate in question either holds or does not. Hereby a new scholasticism was really introduced. — There did, however, stir in him a lively need to return to nature, and he did good by simplifying exposition and by insisting that art always builds on nature. But nature he found in the books of antiquity instead of in ever-living thought. He did not immerse himself in the study of the psychology of thinking, and he could not yet use the procedure of the new science as a model in his description of the method of thinking.

Ramus, who had gone over to Protestantism, was murdered on St. Bartholomew's Night 1572. It is not impossible that the hatred of a fanatical scholastic showed the murderers the way. His corpse was maltreated by the Catholic students.

But his reform of the teaching of the liberal arts made its way abroad. Ramism flourished in Germany, Switzerland and Scotland. In Denmark it was represented by Anders Krag (1558—1600), who after studying in France became professor in Copenhagen. At the University of Cambridge in England, Ramism was combated by the zealous scholastic and mystic everard digby, who was probably Bacon's teacher. But here too it found a zealous defender in William Temple (the Elder), who led it to victory and by his polemical writings against scholasticism laid the foundation for a freer tendency in philosophy, which since that time has always characterized Cambridge in contrast to conservative Oxford. The heated dispute about method between Digby and Temple was bound to arouse great attention and was surely of no small significance for Bacon's development[39]).

While Ramism went on its triumphal march in northern and western Europe, there
[181] stirred at the same time in many a deep need of real knowledge, a thirst for well-grounded acquaintance with actual nature. The writings of Telesio and Campanella, and in part those of Bruno too, bear witness to this. A remarkable example of a deep-going criticism of inherited knowledge, of a clear notion of how much is required in order rightly to know anything, but at the same time of an inability to find a positive realization of this ideal, is offered by Franz Sanchez, the author of the "Treatise on the noble and lofty science that nothing is known" *(Qvod nihil scitur),* which appeared in 1581. Sanchez was

the son of a Spanish physician who had moved to Bordeaux, and he himself worked as a professor of medicine, first in Montpellier, later in Toulouse. Although the title of his chief work might seem to stamp Sanchez as a skeptic, it would surely be unjustified to regard him merely from this side. He has a lively feeling of the imperfection of human nature and especially of human knowledge. But doubt is for him not an end but a means. His skeptical work is only the introduction to a series of works of a special and empirical kind. It was also his intention to work out a separate treatise on method. He praises observation and experiment, in union with judgment, as the right paths to knowledge. His motto is: Go to the things themselves! But he does not have the great hope, which animates his contemporary Bacon, of how far it will be possible to advance along this path. He sees how many riddles even the smallest thing contains, — and also that all things in the world hang closely together, and that the world extends into the infinite: therefore a perfect knowledge of things stands before him as an unattainable ideal. The attempts he himself made in the direction of the philosophy of nature, which recall those of Telesio and of Bacon, do not give him the satisfaction that these two thinkers felt in their similar attempts.

On one single point Sanchez goes deeper than both Bacon and Ramus. He goes back, namely, to the source of all certainty in the human mind itself. No external knowledge, he says, can be as certain as that which I have of my own states and my own actions; this is immediate as that can never become. I am more certain that a thought, a drive or a volition stirs in me than I can be that I see a certain object or a person outside me. *[182]* On the other hand, inner experience falls short of outer experience in distinctness and exactness. — By this line of thought Sanchez stands as a predecessor of Campanella and Descartes. But he was as little able as Campanella to make it fruitful. This was reserved for Descartes.

In Bacon both the demonstration of the defects of knowledge and that of the right method appear far more fully than in any of the humanists and empiricists who before him had sought to reform logic. And the explanation of this has rightly been sought not only in Bacon's peculiar personality, but also in his position in the midst of the luxuriantly pulsing life of England. He has — as has rightly been said — as his predecessors not only philosophers and men of science, but practical investigators of nature, engineers, seamen and adventurers.

b. Bacon's Life and Personality

There has been a long dispute about Bacon's character. His honor as a man, not merely as a thinker, has been alternately destroyed and rescued. This dispute falls away for anyone who attentively reads his *Essays* and his diaries, published some years ago. He has been so sincere in them that we cannot be mistaken about him, especially when we connect with these direct or indirect self-confessions the traits of character that come to light in his philosophical writings.

Francis Bacon was born in 1561. He was a son of Queen Elizabeth's Lord Keeper of the Great Seal, Nicolas Bacon, and a nephew of the first minister, Lord Burleigh. After having studied at Cambridge, where he probably studied scholasticism under Digby, he accompanied an embassy to Paris. The brilliant prospects that opened before him fell away with his father's early death. He was the youngest son and now had to make his own way; his powerful uncle would do nothing for him. A strong desire for power, wealth and honor animated him, but it was nevertheless not the only motive that drove
[183] him forward. In one of his earliest treatises he says: "Is not the pleasure of the feelings greater than the pleasure of the senses, and is not the pleasure of thought greater than that of the feelings? Is not the pleasure of thought the only one of which one is never surfeited, and is it not knowledge that frees the mind from all unrest?" The need of knowledge and the desire for power, wealth and honor did not merely go side by side in him, but he found a justification for the latter motive by subordinating it to the former: he sought power and wealth in order to have means for carrying out the great scientific plans — nothing less than a complete renewal of science (instauratio magna) — over which he brooded. And here a third element in his character must now be noted — one to which his writings above all bear witness — namely his great sanguineness and his no less great self-esteem. He saw a great work before him, and it needed great means. By this he excused to himself the crooked moral paths he took, and even baseness faded for him in the brilliant light in which he saw his striving. In order to carry out his plan he needed leisure and means to collect observations and make experiments. Therefore he threw himself into a political career. But in him, as in Machiavelli, the means gained the upper hand over the end that was to have sanctified it. Had he, instead of laying out his plan on so great a scale, as regards both end and means, attentively followed the works of his great contemporaries Tycho Brahe, Gilbert, Kepler and Galilei, he would have found there enough to think about and a sufficient foundation for working out a program for the future. His sense of the grandiose led him astray and, with the help of his lust for power, entangled him in relations that corrupted his character and plunged him into

misfortune. He lacked a living moral sense to form a regulating counterweight to the other motives. In his "Essays" he says: "The best character consists in having a reputation for openness, but practice in proceeding secretly and, should occasion arise, in being able to dissemble." And in another place: "There are no two more fortunate qualities than to have a little of the fool and not too much of the honest man." Bacon's ambition led him into worldly and political life, and he thereby obtained material for observation *[184]* and for knowledge of men, which he has used in his writings (especially in the "Essays" and in the seventh and eighth books of *De dignitate et augmentis scientiarum).* He praises Machiavelli because he sets forth so openly and honestly what men are accustomed to do, not what they ought to do. Knowledge of this is necessary, in order that the simplicity of the dove should not come to lack the wisdom of the serpent. But here again the means became an end for him. For to get on in the world one must direct one's conduct by men's actual conduct, and he therefore warns against remaining the same when circumstances change; the thing is to let the mind adapt itself to occasion and convenience (ut animus reddatur occasionibus et opportunatibus obseqvens, neqve ullo modo erga res durus aut obnixus). And he urges always having several ends before one, so that one may attain the subsidiary end even if the principal end is not attained. A piece of advice especially dangerous for his own character! —

Bacon first tried to make his way as a petitioner for office. It did not succeed. In Parliament he spoke against a government bill and thereby fell into decided disfavor with the queen. His friendship with Essex (if it can be called friendship) helped him for a time out of difficult pecuniary circumstances. But when Essex was about to fall, Bacon drew back from him. Indeed, during his trial Bacon even offered the queen his services, which were accepted. And when Essex had made his desperate attempt at rebellion, Bacon appeared as a witness against him and after his execution wrote a piece in justification of the government's conduct. A zeal that cannot be explained by concern that justice should take its course, but which is understood when one reads the following utterance in an essay (of followers and friends), printed in 1597, thus a few years before Essex's catastrophe: "There is but little friendship in the world, and least of all between equals. The friendship that exists is between superior and inferior, where the fortune of the one comprehends that of the other." Of this kind was Bacon's "friendship" for Essex, and he took care in good time that his "friend's" fortune should not come, in an unpleasant way, to comprehend his own. Whether men really are accustomed to behave in this way *[185]* is a question by itself. The William Temple mentioned earlier, who had been Essex's secretary, at any rate did not behave so, and therefore also had to go into exile.

In spite of all Bacon's attempts to adapt himself to circumstances, he nevertheless had no success under Queen Elizabeth. Things went better under James I, especially since he knew how to ingratiate himself with the king's various favorites. The dark sides of Bacon's character here had unhappy and far-reaching historical consequences. He was himself of the conviction that the king ought to comply with the justified wishes of the House of Commons, but then use his sovereign power to carry through the codification of the laws and the colonization of Ireland, and, in external affairs, to act energetically by making himself the head of a Protestant league. The attention of the people would thereby be diverted from constitutional questions. But with the unstable and weak king, who was puffed up with a sense of his power, it was impossible to carry through such a course. Herbert of Cherbury, who held a similar conception of foreign policy, had, as is known, to lose his post as ambassador in France a little later for this reason. Bacon adapted himself to the king's whims — whether quite unconsciously, as his biographer Edwin Abbot thinks, is surely a great question. And yet vigorous action on his part could have been of great significance. For himself the adaptation had, for the time being, favorable consequences. He advanced at last to Lord Chancellor, Lord of Verulam and Viscount of St. Albans. But to reach this high goal he had not only to adapt his political views, but also to work in favor of the king's favorites. This caused his abrupt fall (1621). Bacon had declared certain monopolies, which were favorable to Buckingham's relatives, lawful and beneficial. When Parliament grew bitter at this, the king put the blame on his advisers. People then turned against Bacon and struck at him with a charge of bribery. Bacon at once declared himself guilty and was deprived of his dignities by the House of Lords and sentenced to a large fine and to sit in prison as long as the king pleased. Since — apart from his own guilt — he fell as a victim of the conduct of the king and his
[186] favorites, he got off lightly. He was imprisoned only a couple of days and never paid the fine. His last years he lived in quiet, occupied with scientific works. This quiet he could have procured for himself earlier and at a cheaper price. He obtained what he had declared in his Essays to be the most desirable thing: to die in the midst of serious work, since the wound is not felt when one falls in a fierce battle. He died in 1626. — By his friends and his servants he was loved and admired. That belongs, together with his great interest in study and the great faith he had that human culture would be able to develop so richly and gloriously as had never yet been dreamed, to the traits that cast a reconciling light over his character after all the dismal things that are otherwise to be told of him.

During his activity as judge, statesman and courtier he had not forgotten his studies.

The plans for his chief works arose in him early. A writing of 1607 *(Cogitata et Visa)* is the first detailed draft of Bacon's most famous work, *Novum Organum* ("The New Logic"), which appeared in 1620, after Bacon had reworked it twelve times. It investigates the grounds of the imperfection of the sciences, sets forth the obstacles to true knowledge that lie in the very nature of the human mind and in the conditions under which it develops, and then goes on to describe the inductive method. The work is unfinished, and it is laid out on a scale that it was impossible to carry through in Bacon's time. Only two hundred years later did Stuart Mill, in his Logic, do the work that Bacon, with his imperfect material, had dared to undertake. — A writing of 1605 *(Advancement of learning)* is the first draft of *De dignitate et augmentis scientiarum* (On the Worth and Advancement of the Sciences), which appeared in 1623. This work is an encyclopedic survey of the sciences, which contains many apt remarks, especially concerning the gaps that still remained to be filled. — The rest of Bacon's works are for the most part collections of material which now have as good as no interest.

As an author Bacon is vigorous and apt in expression, and often extraordinarily happy in the choice of images. But it is incomprehensible how anyone could have compared *[187]* him with Shakespeare[40]). His imagination is abstractly intuitive and symbolizing, and lacks Shakespeare's glowing energy, individual nuancing and atmospheric inwardness.

c. The Obstacles, Conditions and Method of Knowledge

Science, says Bacon, has been cultivated for only a very short time, and this holds especially of natural science, which is the mother of all science. The Greeks occupied themselves mostly with moral philosophy, the Romans with the doctrine of right, and after the rise of Christianity the most gifted threw themselves upon theology. Only in idle hours and as a handmaid was natural science cultivated, not as a principal task, not with the end in view by which alone it can become truly fruitful: to make human life richer and better. And besides, wrong methods were employed. It was held that the human mind was too exalted to occupy itself with experiments, especially since it was supposed able to spin the truth out of its own interior. People were content with what had been handed down. They had too great a reverence for the past and its great thinkers, whom they called the ancients, though it is in fact precisely we who are the ancients, since we can look back on a far greater experience than they. To this was added a false religious zeal, which did not see that natural science teaches us to know God's power, just as religion teaches us to know God's will. As faith is to restore to us the innocence we lost by the Fall, so science is to restore to us the power over nature that

we then lost. But most of all, men's faint-heartedness and distrust of their own strength have done harm. They have lacked the courage and the hope to set themselves great tasks.

There is reason now to have good hope, precisely because our own faults are to blame for the imperfection under which we suffer. The right method must be capable of being found. It is the one that will not, like the spider, spin everything out of itself, nor, like the ant, merely collect material, but which, like the bee, takes the material into itself and works it up. If the material is collected, and if one is able to keep prejudices and
[188] preconceived opinions away, then the right interpretation of nature will soon be found. The most important step of all that must be taken is then to procure as rich and many-sided a collection of facts as possible. Then the human mind will, by its own involuntary drive, work them up and interpret them. — In this assertion of the involuntary activity of mind, which comes into operation as soon as the material is merely at hand, Bacon is perhaps influenced by Ramus, whom he mentions with recognition, though he objects against him that his method is too simple. The difference between them rests on Bacon's having seen the necessity of a large mass of material of experience, in order that the faculty of invention and the judgment may work in a fruitful way. — The most important error that can be committed is, according to Bacon, that of going forward too quickly, of hastening to the general fundamental propositions instead of ascending slowly to them through many intermediate steps. The human mind needs lead, not wings.

The end is to enrich human life through the inventions that can be made when one knows nature. Knowledge is power; for we have the possibility of producing things when we know their causes. By the help of inventions and mechanical skills men have gradually worked their way out of barbarism and have attained a civilized life. Only along this path can the misery and misfortune that men still suffer under be remedied. The thing is to gain greater dominion over nature; but that is gained only by obeying it. — Yet one would do Bacon an injustice by attributing to him the opinion that science is to be cultivated only for the sake of practical utility. Above the external yield that knowledge can procure he places the very contemplation of things (comtemplatio rerum above fructus inventio). We rejoice in the light because we can work, read and see one another by its help; but the very sight of the light is nevertheless more glorious than its manifold uses.

But there is hope of reaching the great goal only if one comes without prejudices. As with the kingdom of heaven, one can enter the kingdom of man, which is to be founded on science, only by becoming as children. Prejudices and preconceived opinions

must be laid aside. The thing is, after all, to interpret nature, not to anticipate nature: interpretatio, not antecipatio. Bacon therefore seeks to give a comprehensive doctrine *[189]* of the unjustified anticipations. This is his famous doctrine of the illusions of the mind (idola mentis), which must be purged away in order that the mind may become a clean and smooth tablet (tabula abrasa), on which things can write their true nature. Bacon distinguishes four classes of such illusions.

Some illusions are grounded in human nature itself and are therefore common to the whole species (idola tribus). To these illusions of the species belong those which arise because we conceive things according to their relation to us and in likeness to ourselves (ex analogia hominis), and not in relation to and in likeness to existence (ex analogia universi). Our mode of conception, our way of sensing and thinking, surely cannot be the measure of things! In particular we have an inclination to presuppose greater order and regularity in things than is actually found there. The evenness (æqvalitas) of our own mind we presuppose in things too. And what conflicts with assumptions we have once formed, we are inclined to overlook. The negative cases are easily passed over when one draws conclusions from experience. To what acts suddenly and immediately upon us we ascribe the greatest significance, and we are inclined to take what is more remote to be of the same nature. On the other hand, we also carry the unrest and constant craving of our own mind over into nature, so that we will posit no limits to its extension or to the series of causes. Or one settles down content by finding the explanation in a purpose and positing final causes (causæ finales), an explanation that is evidently taken from our own nature, not from that of the universe. Finally, our results are very easily determined by our feelings and drives, by our hope or our fear, and this influence of feeling on knowledge often takes place in an altogether imperceptible way.

Another class of illusions springs from the individual nature of the single person. Bacon calls them (alluding to a passage in Plato) "cave-illusions" (idola specus), since each one's individuality is, as it were, the cave from which he looks at existence, and in which the light of nature is refracted in a peculiar way. These illusions of individuality are determined by original nature, upbringing, company and reading. There are some *[190]* who are most inclined to dwell on the differences of things, others who are most inclined to dwell on their likenesses, — some who most love the old, others who turn with predilection toward the new, — some who seek the elements of things, others who stop at the compound phenomena as they are immediately given.

As the worst of all illusions Bacon regards those that are due to the influence of words on thinking. He calls them "idols of the market-place" (idola fori). Words are formed

according to the needs of practical life and according to the popular understanding (ex captu vulgi), and the boundary lines they fix between things often cannot be used by sharper thinking. Words are formed for something that does not exist at all, and on the other hand words are lacking for things that experience actually offers. Hence much verbal dispute.

While the first three classes of idols have their ground in the nature of man, the last class, "idols of the theater" (idola theatri), is due to the influence of inherited theories. These theories may be very ingenious and sharply thought out and yet miss the goal. He who follows the right path reaches the goal with slight gifts more quickly and surely than the best runner once he is off the right path; indeed, precisely by his skill as a runner the latter will move further and further from the goal. The method of experience that Bacon recommends, in contrast to the brilliant speculations of earlier times, leaves — in his view — no great room for acuteness and strength of mind to exercise themselves. The right method levels the difference between minds. When a circle is to be drawn freehand, differing gifts may become of significance; but when compasses are used, such difference falls away. — This fourth class of idols is hard to keep apart from the second, since Bacon makes reading, tradition and authority contribute to the coming into being of individual peculiarity.

Bacon's doctrine of the idols is a small piece of critical philosophy, an attempt to
[191] distinguish between what belongs merely to the subjective nature of knowledge and what belongs to existence. In Montaigne, Cusanus, Bruno and Galilei we have met beginnings in the same direction. Unfortunately Bacon lacks the grand point of view that comes forward, especially in Galilei, through the Copernican theory, which inculcates the relativity of the standpoint. And Bacon is inclined to regard the ways of viewing that we involuntarily adopt as purely deceptive. Thus it is for him a deception of the senses (fallacia sensuum) when the senses show us things otherwise than science explains them. He has, moreover, not investigated more closely how it is possible to purge the mind so that it becomes a blank tablet. The conditions and forms that lie in our original individuality and in common human nature we surely cannot abolish, even if we can discover them. At this point a contradiction comes out in Bacon. For, as we have seen, it is his conviction that when the material is laid ready, the mind works it up involuntarily according to its own inner capacity (vi propria atqve genuina). But is not something then added here to the material — and by what right is it added? What guarantee is there that the mind's own inner power can lead us to see things no longer according to their relation to ourselves, but according to their relation to the universe? How do we convince

ourselves that we have really reached analogia universi? — Here later epistemology took hold. Bacon is its forerunner by his doctrine of the idols. And especially interesting is the thought that the presupposition from which Leonardo, Copernicus, Bruno and Galilei so confidently started, namely that nature always follows the simplest paths, could have a purely human and subjective origin, so that its justification must be discussed. —

The method that must be followed in working up the material is induction. This method had, to be sure, long been used, but Bacon found in its usual application an essential error, which he brought under the illusions of the species, namely that one keeps to the cases in which a phenomenon has shown itself and regards them as sufficient to ground an insight into its nature. This kind of induction Bacon calls induction by simple enumeration (inductio per enumerationem simplicem). He demands that it be *[192]* supplemented by an investigation of the "negative instances", that is, the cases in which the phenomenon does not occur, although the circumstances are akin to those under which it occurs. And besides he demands a table of degrees, which shows under what circumstances the phenomenon decreases or increases. By this procedure we can get so far that we can form a provisional conception of the nature of the phenomenon, or, as Bacon calls it, its "form". By the form Bacon understands the property that is always present when the phenomenon is present, always absent when it is absent, and which decreases or increases in the same proportion as it does. After having eliminated the properties that do not satisfy these requirements, we retain the form. In investigating the nature of death we must thus disregard whether it takes place by drowning, by burning, by piercing, by apoplexy, by lack of nourishment, etc. With heat we must disregard the special ways in which heat is produced, the composition of the hot substances, etc.

When Bacon is often accused of conceiving induction as a mere collecting of material, it must be noted that he lays great weight on the formation of provisional hypotheses for survey and orientation. The first interpretation is only an attempt. It is the "first vintage", which serves merely as a trial (Nov. Org. II. 20). It is undertaken because truth is more easily won through an error, if only it is clearly thought through, than through a chaotic heaping up of material. It is one of Bacon's most famous aphorisms: citius emergit veritas ex errore qvam ex confusione. As such a provisional hypothesis Bacon sets up, as regards the "form" of heat, the proposition: heat is motion. — And his method of induction is now continued after the setting up of this provisional hypothesis and with it as guide. The thing now is to find, or to produce experimentally, such cases as are especially suited to illuminate and more closely ground the assumption that has been set up. Of such "prerogative instances" Bacon enumerates a whole multitude. Among

other things he mentions (as instantiæ solitariæ) cases that present the property under discussion under circumstances that otherwise have absolutely nothing in common
[193] with all the other cases in which it occurs, or which do not present it, although they otherwise have all other circumstances in common with the cases in which it occurs. These are the methods that Mill later called the method of agreement and the method of difference. As instantiæ viæ he designates cases and experiments that show us the phenomenon "on the way", in its coming into being, during which its nature ("form") will most readily reveal itself. As instantiæ irregulares et deviantes he designates cases in which the phenomenon has, as it were, "gone astray", varies, assumes irregular forms. Through such variation the thing more readily discloses its proper nature. Akin to these are the transitional forms (instantiæ limitaneæ), which present some properties that can be brought under one form, and others that can be brought under another form.

Bacon intimates that a whole series of further operations still belongs to the complete carrying through of the method of induction. But Novum organum ends with the enumeration of the various kinds of prerogative instances. —

The detailed description of the method of induction that Bacon gives was a real advance for its time. It betrays a clear understanding of essential points. What he overlooked, and what puts him in the shade beside the founders of the new science, is the subordinate position he assigns to quantitative determination and to deduction. Exact empirical science becomes possible only when, by exact measurement of the phenomena, one obtains a foundation for mathematical deduction. He does, to be sure, give quantitative determination a place in the inductive process; a special kind of prerogative instances is the instantiæ qvanti, also called doses naturæ. But he did not see their decisive significance. And with this his underestimation of deduction is connected; the most brilliant deductions are the mathematical ones. Even if Bacon did not, as he has been accused of doing, altogether misjudge the significance of deduction (the "first vintage" speaks against that), he nevertheless gives the deductive process an altogether subordinate place in his method of experience. He had no inkling that a deduction gives
[194] the real proof of the correctness of the inductive conclusion.

What Bacon wants to reach by way of induction is a determination of the nature or "form" of things, of their peculiar essence, each by itself. But properly speaking such a determination is not an induction in the modern sense of the word. It states not the dependence of one phenomenon on another, not a law, but a concept. It is (as FRIES has already remarked in his history of philosophy) rather to be called abstraction than induction. And Bacon himself here refers to Plato as his model. There is, he says, a

difference between the idols of the human mind and the ideas of the divine mind. It is these latter that he wants to seek: they are the eternal forms of things. The form is the definition of the thing, the thing itself (ipsissima res). He praises Plato because he saw that the "form" is the true object of science. Plato's error was, partly, that he separated the forms or ideas from the things themselves, whereas they are precisely to be sought in the individual phenomena themselves, and partly that he did not resolve the compound forms or natures into the simplest, purely elementary ones. What must be established is, according to Bacon, precisely the simplest, indissoluble properties (naturæ simplices) of things[41]). Such simple properties are color, heaviness, solubility, expansibility, etc.

Seen from this side, the concept of "form" in Bacon points back to ancient and scholastic philosophy. But he wavers not a little in his conception of this concept. In some places the "form" means not a definition or a determination of the nature of the thing, but the law according to which the activity proceeds in which the phenomenon consists. "The forms", he says in one place, "are fictions (commenta) of the human mind, unless one is willing to call the laws of activity or of motion forms". The form of heat is thus the same as the law of the phenomena of heat. By this he wanted to distinguish his "forms" from those that had hitherto been assumed. This law one must know if one is to be able to produce the phenomenon. Both in theory and in practice it is then the law that must be found. This practical side Bacon no doubt also gets expressed when he speaks of the form as the expression of the essence of things, in that he calls the form the active nature of the thing (natura naturans), the source from which its individual *[195]* properties spring (fons emanationis). Plato too, after all, regarded the ideas as the true causes. But the nuance that — without Bacon himself noticing it — comes out in the concept of form, in that it designates now the essence of the thing, now the law of its production, is nevertheless characteristic of his doctrine of method. He shows himself here to stand at the transition between the old and the new philosophy. He starts — like Plato and the scholastics — from the assumption that there is a certain number of "simple natures", whose combination makes up the various phenomena. But the presupposition that things are subject to definite laws he does not set up or investigate.

A connection between the two meanings of the concept of form arose for Bacon from his asserting so definitely that one could learn to know the essence of things only by investigating the process by which they come into being, as well as the inner connection between the parts of which they consist. But both of these lead us beyond the sensible. For the changes in nature take place by degrees, through small transitions (per minima), so small that we notice them as little as we can sense the minute parts of which

things must be assumed to be composed, and whose kind of composition (schematismus) produces the properties (forms!) that they present to us. The task of inquiry is then to find the hidden process (latens processus) that continuously connects the various stages of development which our leaping and periodic observation apprehends, and the hidden arrangement (latens schematismus) that underlies the sensible properties. Nature is too subtle for our senses. The motion in which the essence of heat consists we do not sense as such. The hidden process escapes us here. It goes similarly with the development of the organism, with the process of nutrition (when one is to have all the stages, right from the taking up of the nutritive matter to its complete working up into flesh and blood), with voluntary motion (when one is to have all the members, right from the first motion in the representation through the inner processes to the bending and motion of the limbs). And the task becomes especially difficult to solve when one must find at
[196] every point what is lost, what remains, and what is added; for Bacon regards it as certain that the total sum of matter is neither increased nor diminished. — This doctrine (Nov. org. I, 10, 50; II, 5—7. 40—41) of the continuous process, and of its individual members as the conversion into a new form of what is given in another form, without anything being lost or gained as regards quantity, belongs to the most brilliant anticipations in Bacon — although he himself, by his method, wanted to exclude anticipation and put interpretation in its place. He set up the principle of continuity, which has later proved so fruitful. Bacon here comes involuntarily to admit that every interpretation must start from certain principles. His whole doctrine of the hidden continuous process and the hidden arrangement he sets forth purely dogmatically, merely illustrating it with a few single examples.

By drawing the consequence of the doctrine of the hidden process one discovers that the two meanings of the concept of form in Bacon are in conflict with each other. According to the first meaning (form = simple nature = property that makes up the essence of the thing) we start from the assumption that the properties which observation establishes also belong to the things themselves. By the second meaning (form = law of the process by which things come into being) it turns out that things in reality consist of minute parts and arise through minute changes, none of which can be observed by itself. With the latter meaning the subjectivity of the sense qualities agrees (since heat in itself or in ordine ad universum is motion), while it conflicts with the first meaning (heat as natura simplex, like color, etc.). Just as Bacon by the first meaning recalls Plato and scholasticism, so by the second meaning he recalls Democritus, without, however, having really been an atomist. It is characteristic that the utterance cited above, that

the "forms" are fictions and illusions of the species unless by form one understands law, occurs precisely in a context where he has just before praised Democritus because of his demand that nature be resolved into its elements rather than that abstractions be made. — It is again here the relation between quality and quantity at which Bacon stops short. Entangled in Platonic and scholastic presuppositions, he did not consistently carry *[197]* through what lies in his brilliant idea of the continuous process by which qualitative differences are resolved into quantitative ones; and therefore he does not recognize the subjectivity of the sense qualities in the resolute way that Galilei does.

d. Theory of Science, Theology and Ethics

The inductive method is to hold for all sciences, for the human sciences (ethics, politics, logic) as well as for the natural sciences. Bacon on the whole strongly emphasizes the unity of science in spite of its division into special subjects. The complete isolation of the single subject will condemn it to unfruitfulness. If one asks where the common foundation of the various sciences is to be found, he answers somewhat differently in his two chief works. In Novum organum natural science is called the great mother of all sciences. But in the work on the advancement of the sciences (De augmentis scientiarum) the honorary name of mother of the sciences is given to the science that contains the fundamental propositions common to the various sciences. This science of the common fundamental propositions Bacon calls prima philosophia. But such a science has not yet been developed, though it would be of great significance, since a system of such common fundamental propositions would contain a testimony to the unity of nature. As such common fundamental propositions Bacon names (Nov. org. I, 80. 127. De augm. III, I): Equals added to unequals produce unequals; things that agree with one and the same third agree with each other; nature reveals herself especially in minima; everything changes, but nothing perishes. These are in part propositions with which Bacon confidently operates in his doctrine of method. Of their origin and grounding he does not speak more closely, but it is of course his opinion that they are themselves a fruit of induction. There undeniably arises thereby a circle in his line of thought, since on the one hand he uses these fundamental propositions as presuppositions of the inductive method, and on the other hand regards them as results of induction, a circle that is unavoidable for all forms of pure empiricism. And Bacon wanted, after all, as we have *[198]* seen, to have the mind made a blank tablet before the work of knowledge begins.

Of the grounding of the first fundamental propositions by induction he speaks not in his doctrine of prima philosophia, but in quite another context, namely where he wants

to draw the boundary between philosophy and theology. Bacon warns strongly against mixing philosophy and theology together; one thereby gets a fantastic philosophy and a heretical theology. Their sources are altogether different. Philosophy starts from sense observation; theology builds on divine inspiration. In science the human mind is affected by sense perception; in faith it is affected by another spirit; therefore faith is nobler than science. And the more absurd and incredible a divine mystery is, the greater the honor one shows God by believing it nevertheless, the more glorious is the victory of faith. Once one has accepted the principles of religion, one can draw logical consequences from them, just as from the first philosophical principles. But the significant difference is that in philosophy the principles themselves are subjected to a closer testing by induction, like all other propositions; but in religion the first principles stand fast by divine authority, as in the game of chess, where the first rules of the game cannot be discussed. Yet Bacon thinks that there is still wanting a closer investigation of how far reason can be applied in the religious domain, as well as of the question how far agreement in religious assumptions is necessary.

Besides the ecclesiastical theology, which Bacon mainly has in view in these determinations, he nevertheless also takes a natural theology to be possible, even if it is not of great extent. It can only refute atheism and show the necessity of assuming a first cause. When one begins to find natural causes (causæ secundæ), one often becomes so captivated by them that one thinks one needs no first cause. But later it is precisely the fixed mutual connection of the causes that shows the existence of a Deity. Here belongs also the point that, even if natural science has no use for final causes, which on the contrary corrupt science by applying a merely human motive to nature, the justification
[199] for assuming final causes does not fall away, and these can very well be united with the continuous process which it is the task of natural science to find. The final cause is like a nun; it is barren in science, but has religious significance.

In his Essays Bacon discusses the relation between atheism and superstition. Here too he says that beginning inquiry leads to atheism, but deeper inquiry to the assumption of a God. The chief causes of atheism, however, he finds in religious controversies, in scandals among the clergy, in a bent for mockery, and in enlightened, peaceful and happy times (for misfortune and adversity bend the mind to religion). Atheism degrades man, since man needs a support for his spiritual being, so that the bodily may not drag him down too far toward the animal, and since man is ennobled by his relation to a higher nature, as the dog is ennobled by its relation to man. But superstition is nevertheless worse than atheism. Better no opinion of God than an opinion unworthy of God; the

former is only unbelief, the latter is insult. Superstition produces immoral qualities far more readily than unbelief does. It is dangerous to the state, since it founds a power in the people that can exceed the power of the government. Then the wise are forced to follow the fools. It is especially in barbarous times that superstition arises.

The duality that appears in Bacon's doctrine of faith and knowledge, and which clearly bears the stamp of being a compromise, consistently casts its reflection into his psychology and his ethics. — The sensible soul of man is, like that of the animal, material. It consists of a fine fiery air (aura ex natura flammea et aërea conflata), which streams out from the brain through the nerves and is nourished by the blood. Bacon here follows Telesio. He misses, however, a closer demonstration of how this fine air can move the dense and hard body. With Telesio he also agrees in ascribing to all things a receptivity to impressions, which he would rather call perception (perceptio) than sensation (sensus). He misses a closer investigation of how the former can pass over into the latter, that is, how the general receptivity to impressions can become sensation, or how unconscious mental life can pass over into conscious mental life. — Besides the sensible and material *[200]* soul Bacon assumes, like Telesio, a spiritual soul implanted by God; but about this only religion, not philosophy, can teach us anything. — The special psychological observation of the phenomena of mental life, which goes its way independently of materialism and spiritualism, Bacon does not know. Here he falls short of Sanchez and Campanella, to say nothing of Descartes. Only in passing does he intimate it, in the proposition that we know nature through direct light, God through a light refracted in the world as a medium, and man, who is his own object, by reflected light (radio reflexo).

As regards ethics, Bacon distinguishes between the doctrine of the exemplars and the doctrine of the means and ways of approaching these exemplars. This latter doctrine (de cultura animi) has, he thinks, been greatly neglected, and for the same reason that explains the imperfection of the sciences generally: people have preferred to revel in notions of the ideal rather than laboriously to inquire how it can be realized. As regards the exemplar, in its highest form it belongs to religion, not to philosophy, since it is supernaturally revealed. The philosophical doctrine of the exemplar has on the whole been well set forth already by the ancient philosophers. Bacon misses, however, investigations of the first sources of the ethical (fontes ipsi rerum moralium). He thus demands a doctrine of the psychological development of the ethical. And he himself gives the main outlines of such a doctrine. In every thing there is found a twofold need: by the one it seeks to assert itself as a whole; by the other it seeks to act as part of a greater whole. The first has the individual good, the second the common good, as its

object. The ancient philosophers wrongly laid the greatest weight on the individual good, in that they set knowledge, contemplation, as the highest, whereas the active life is surely to be set highest. Our task here in life is not to be mere spectators. — Political theory Bacon will not go into more closely. For the work (De augmentis sc.) of which it was to form a part is dedicated to King James I — and before such a master of the art of
[201] government silence was for him the most seemly course! —

Here at last we meet Bacon again as courtier. It was the misfortune of his life that he entered upon that career, just as it was the misfortune in Bruno's life that he became a monk. But in spite of this he was able to develop thoughts that were good expressions of the new paths on which inquiry was about to enter, and which especially heralded the philosophical inquiry that was to contribute mightily to giving his fatherland an eminent place in the history of the modern development of the spirit.

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Third Book

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The Great Systems.

After the time of the new ideas and discoveries there comes a time of attempts to order and systematize, to lead the many thoughts and facts back to simple and fixed fundamental thoughts. These attempts were made with great confidence that the right foundation had been found. In place of analysis, construction was put. This had the great importance for thinking that it could now become fully clear what lay in the points of view which the Renaissance and the new natural science had brought forward. They were established with a dogmatic assurance of which the minds of the preceding time had not thought, because they did not give themselves so clear an account of the individual principles and their range. What had earlier hovered more or less obscurely before the mind was now formulated with express consciousness. And out of the presuppositions so formulated systems were built up, each of which laid claim to take the place of the old scholastic system, which was only now wholly overthrown. From scholasticism, however, was inherited the endeavor to give knowledge an absolute conclusion, to bring thought to rest in a principle which itself should contain no problem. Men wished to build as high as the building that had been torn down had reached. The natural need, which always stirs more or less strongly in the human mind, to bring coherence into all representations that are taken to be valid, expressed itself here so vividly and energetically, and at the same time in so brilliant a manner, as in no other period of the history of modern philosophy. *[203]* This was connected with the fact that tasks had been set which had been set neither before nor later with the same novelty and sharpness. It was the new picture of the world and the new science that had to be worked together with the rest of the spiritual content that stood firm for consciousness. It therefore had to be the problem of existence that came to stand in the first rank. Bruno had treated it on the basis of the new picture of the world. But now the new mechanical explanation of nature had been added, and the great question of the relation between the material and the mental was thereby set

before thinking. The most important of the hypotheses that are possible concerning this relation were put forward by the systems of the seventeenth century with a clarity and energy that give these attempts of thought a lasting value. But other problems weave themselves together with this one. The question of the relation between God and the world even presses forward for a while in front of it. With it, in turn, the question of the unity or multiplicity of existence is connected. And finally it becomes an important problem whether the mechanical explanation of nature makes it possible to ascribe any real significance to the concept of purpose.

The problem of knowledge and the problem of valuation recede before the problem of existence, although they constantly exert their influence as more or less consciously motivating and impelling forces. In Descartes, the first in the series of the great systematizers, there is still, in the presentation as well, a clear endeavor to find the way to constructive thinking through analysis. In Hobbes and Spinoza construction overshadows analysis. In Leibniz the analytic tendency begins again to work its way forward, as a transition to the predominant importance the problem of knowledge and the problem of valuation were to acquire in the eighteenth century.

The conditions of cultural history in the seventeenth century present an analogy to the concluding, systematic tendency that at this time formed the main current in philosophy. There is one fundamental mood that pervades the different domains, politics, church and thinking. It is the time of the absolute sovereignty of the state. The state
[204] has freed itself from ecclesiastical tutelage and now demands absolute submission of individuals and of the smaller circles. The principle of sovereignty is even carried through, in its very sharpest forms, by the two thinkers who mark the high point of construction. The individual now seeks shelter and rest after the storms of the Renaissance and the age of the Reformation. Intellectual contemplation and mysticism draw many to them. A tendency to quietism makes itself felt. But in the philosophical systems these tendencies are satisfied through a thinking elaboration of the new ideas and discoveries. They thereby acquired the significance of experiments for testing the carrying power of important trains of thought.

1. René Descartes

a. Biography and Characterization

This man, the founder of modern philosophy, was born in Touraine on 31 March 1596, of a noble family. The sickly boy early showed unusual gifts, and his father used to

call him "my philosopher" on account of the many questions he had to ask. So that he might receive a careful upbringing, he was sent to the Jesuit college La Flèche, recently founded by Henry IV. He always remembered his Jesuit teachers later with gratitude, and it was a great sorrow to him that the Jesuits came forward as opponents of his philosophy. At La Flèche he learned natural philosophy and philosophy according to the scholastic system, but was most occupied with mathematics, and very early he seems to have busied himself with ideas that led him to his great mathematical discovery, the founding of analytic geometry, the application of algebra to geometry. He has himself described his development in the years of his youth in his epoch-making work *Discours de la méthode,* since it is at the same time the history of the genesis of his philosophy. On leaving school he felt dissatisfied, despite the much he had learned. Many facts he knew, and many fine thoughts had been imparted to him; above all he admired the stringent method of mathematics. But those facts and thoughts stood before him as disconnected fragments, and mathematics was to him only useless cobweb-spinning. He then laid *[205]* his studies on the shelf and threw himself into the worldly life of Paris. His theoretical interest, however, he could not wholly deny: among his papers was found a treatise on the art of fencing that dates from this time. Soon he grew weary of the thoughtless life and suddenly vanished from his friends, withdrawing to a solitary part of Paris in order to study in peace. It now became more and more his ideal to lead a retired life devoted to reflection and study. His motto became: "Happy he who lived in concealment!" (Bene vixit, qvi bene latuit!). His friends found him after the lapse of two years, and he was torn out of his solitude. Then he resolved to come to know "the great book of the world." Perhaps practical life, which puts all thoughts to the test, had taught men truths that learned speculation could not discover! And besides, he wished to try himself under the vicissitudes of life. He then went as a volunteer, at his own expense, into military service under Maurice of Orange, though he used every quieter time for studies, especially mathematical ones. From Holland he went to Germany, where the Thirty Years' War was about to break out. He joined the army the Elector of Bavaria was gathering against the rebellious Bohemians. While he lay in winter quarters (1619—20) at Neuburg on the Danube, a scientific breakthrough took place in him, as he found the general method that later guided him both in his philosophical and in his mathematical studies. In a memorandum from that time, which was found among his papers, he has even given a definite date for the emergence of the decisive thought: — "the 10th of November 1619, when I was filled with enthusiasm and found the foundation of a wonderful science!"[42]) He lived shut up in his room and gave himself up to thoughts that led him to his general

doctrine of method. It occurred to him that, as in general a work done by many in common will be more imperfect than one carried out by a single person, so our imperfect knowledge derives from the many different teachers who each instill their opinions in us, from the influence of the different drives, from the different conflicting judgments we hear from scholars and from practical people. Let us then, in order to remedy this
[206] imperfection, begin afresh, disregard what has been handed down, and slowly erect our building on one single foundation! The right method consists in accepting only what is thought clearly and distinctly, in resolving every difficulty into its individual parts, and in starting from what is simplest and most easily understood, so as then step by step to come to the more composite questions. It was the analytic method that stood before his inner sight in its whole generality. In the mathematical domain this method led him to the idea of a science that was more general than the individual mathematical sciences, since it investigates relations, proportions in general, no matter whether they obtain between figures or numbers or other things. It was a general theory of magnitude or of functions, of which analytic geometry was a special application. — His thoughts worked so eagerly with these ideas that he fell into an exalted state. He had remarkable dreams, and the next day he promised the Holy Virgin to make a pilgrimage to Loreto, so that she might give his thoughts success. (He did not fulfill this vow, however, until a convenient opportunity offered itself). This vow of pilgrimage is a peculiar introduction to modern philosophy — a counterpart to the supernatural voice in which Herbert of Cherbury a few years later heard a sanction of his "natural religion"! — Yet he still thought it was too early to proceed to the working out of his philosophy. After having taken part in the capture of Prague and in a campaign in Hungary, he returned home to France and took over some estates that had fallen to him by inheritance. His family would gladly have seen him married and placed in an official position; but his mind was not set on that. He formed the resolution to devote his life to science, and in order to be able to pursue it in peace, he moved to Holland (1629). Before that he had repeatedly had occasion to develop his philosophical ideas, especially about method, in literary circles in Paris. Two remarkable unfinished treatises, which were first published long after his death, "Rules for the Direction of Thinking" and "The Search for Truth by the Help of the Natural
[207] Light," probably date from this time. They give a significant presentation of the analytic method. In the first period in Holland he (as is seen from Discours de la méthode, parts 3 and 4) occupied himself with the speculative ideas he later presented at length in his "Meditations." He laid the foundation of his theology and his psychology and found a way that, through doubt itself, leads him to the starting point of all coherent philosophical

knowledge.

When Descartes settled in Holland, it was surely not only because he sought peace and because — as he said — he could philosophize better in a cooler climate. A contributing reason was that he expected to find greater freedom there for his inquiry. The reactionary movement that brought Bruno to the stake and Galileo under mental torture had spread to France. When some young investigators in Paris in the year 1624 wished to defend atomism against Aristotelian physics in a public disputation, the theological faculty declared the doctrine that everything consisted of atoms heretical, since it conflicted with the Catholic doctrine of the Eucharist. The disputation was forbidden at the last moment, after nearly a thousand people had assembled to attend it. The authors were arrested and sent away from the city. And on 6 September 1624 the Parlement forbade — on pain of death! — the putting forward of fundamental propositions that conflicted with the old and tried authors, and the holding of disputations not approved by the theological faculty! — This certainly could not but make the atmosphere in Paris less agreeable for a philosopher. In Holland he could expect to find freer air. Besides the purely philosophical speculations mentioned, he occupied himself in his new place of residence with studies in natural science and worked out a treatise that was to be called "The Universe" (le Monde) and to give a presentation of the development and structure of the world according to purely mechanical laws. He supposed that God had created matter as a chaos and thereafter ordered it according to the same laws that he now constantly follows in the preservation of the world. In this way he reconciled his faith in the dogma of creation with the idea of a development according to natural, still demonstrable laws, in the application of which to the genesis of our world-system he stands as the predecessor of Kant and Laplace. But then came the news of Galileo's *[208]* condemnation and of the Inquisition's condemnation of the Copernican doctrine, which Descartes had presupposed in his hypothesis — and he then laid the work aside. He would not, he wrote to his friend Father Mersenne, teach anything that conflicted with the faith of the Church, and besides it was, after all, his motto that one lived best in concealment (bene vixit, qvi bene latuit); he wanted above all to have peace and be free from fear and vexation; and therefore under the present circumstances he would confine himself to studying for himself alone. — Descartes's fine motto thereby received an ugly application, and this whole affair has rightly been called a stain on his character. From his letters it is seen that he is in full agreement with Galileo in his results. And even if one now has every reason to suppose that Descartes was a sincere Catholic, there is nevertheless hardly any question that fear, and perhaps most of all the need of peace,

was the predominant motive for his reticence. Later (in "Principia philosophiæ") he presented his hypothesis of cosmic development — but in disguised form. He threw, as his first biographer says, sand in the eyes of the Inquisition.

Descartes had not yet had anything printed. And yet his philosophical and scientific ideas were known in wider circles, partly in Paris, partly also in Holland. The Cartesian philosophy, as has rightly been said, was taught before it could be studied in print. Several of his disciples brought it forward at the Dutch universities, where it occasioned hard struggles, which also caused its founder much disquiet. Only in the year 1637, at the urgent requests of his friends, did he have four treatises published (*Essays philosophiques.* Leyden 1637), which were to give characteristic examples of his inquiry and its results. In the first treatise (Discours de la méthode), the only one that has direct philosophical significance, he gave here the history of his ideas and the main features of a new epistemology and metaphysics. In the second and third treatises (Dioptrique, Météores) he gave examples of a strictly mechanical explanation of nature, and
[209] in the fourth (Géometrie) he founded analytic geometry. — The complete presentation of his philosophy he gave some years later in his *Meditationes* (1641) and in *Principia Philosophiæ* (1644). He had had the Meditations sent in manuscript to several of the thinkers of the time, e.g. the famous Jansenist Antoine Arnauld, Gassendi, Hobbes, and their objections, with Descartes's replies, were printed together with the work itself, which thereby acquires an interesting dialogical character. The discussion with Gassendi continued for a long time and took on a rather bitter character. While the opposition on the side of Gassendi and Hobbes was purely philosophical and therefore always very instructive, the new philosophy met resistance for quite other reasons on the side of the Jesuits and of orthodox Protestantism. At Utrecht, Gröningen and Leyden hard struggles were fought, since the theologians held to the scholastic philosophy as a bulwark for the faith. At last prohibitions against new opinions were issued. The Dutch, says Descartes in a letter, honor the beard, voice and bearing of the theologians more than their integrity. He considered the Protestant theologians worse than the Catholic ones. He had come under a double fire. The Protestant theologians accused him of skepticism, atheism, the dissolution of universities and of church and state, but besides warned against his philosophy as papism; and the Catholic theologians accused him not only of heretical opinions, e.g. the assumption of the motion of the earth (which he had nevertheless sought to conceal), but also of inclining to Protestantism and attending Protestant services.

The last work of Descartes that appeared in his lifetime was the interesting treatise on the emotions *(Les passions de l'âme),* which appeared in 1649. Its origin is owed

to Princess Elisabeth of the Palatinate (a daughter of the unfortunate Bohemian king Frederick of the Palatinate), with whom he was in lively correspondence. In his letters to her he developed his ethical ideas. With another gifted princess too he came into correspondence, namely Queen Christina of Sweden. At her invitation he traveled to Stockholm in order personally to introduce her to his philosophy. This stay "in the land of bears, ice and rocks" (as he calls it in a letter), as well as the life of the court, cannot have been favorable to his health. The year after his arrival he contracted an illness that *[210]* caused his death (1650).

The main features of Descartes's character appear clearly in his life. A love of study and thinking, which was faithfully preserved through his whole life, and which in particular moments, when new ideas dawned on him, could pass over into affect and enthusiasm, was the most important and finest trait of his nature. His favorable and independent circumstances made it possible for him to carry through the plan of life he had laid down, which aimed above all at preserving peace and equanimity for the sake of his studies. He did not lack courage, as he showed on his travels; but he was too anxious and deferential toward authorities. The least attractive trait in him is his lack of ability to acknowledge the merits of others. He had a great feeling for what was new in his standpoint and firmly denied having learned from his predecessors, maintaining that even the assumptions he had in common with them received in his hands a quite different justification. He names Plato, Aristotle, Epicurus, and among the moderns, among others, Telesio, Campanella, Bruno as authors he has known, but — despite all agreement — has not learned from, since his principles were quite different. Only Kepler does he acknowledge having learned from; in particular, the German investigator's mechanical view of nature, his emphasis on the importance of the concept of quantity and his partial discovery of the law of inertia were of great influence on Descartes's philosophy of nature. But this trait too is connected with his complete absorption in his own thoughts. As a thinker, sharp distinction, the clear leading back to simple points of view, was his chief quality. He has thereby done a deed of lasting significance: definitively put an end to arbitrariness and fantasticality as far as method is concerned. If errors are committed in his thinking, it is in any case easy to see where the error lies. Descartes found an essential difference between geometry and metaphysics in this, that in the former there is no difficulty about the first principles, whereas what matters is to draw consequences from them, while in metaphysics, on the contrary, the great dispute stands precisely about the principles themselves, or about the right determination of the fundamental concepts, since it requires great effort of thought to work oneself free of the prejudices *[211]*

we are inclined to cherish. This he himself gives (Resp. ad sec. obj.) as the reason why in his Meditations he has followed the analytic method. But his ultimate purpose was nevertheless a construction, a synthesis, that could present the whole system of our knowledge in its logical grounding. And he often makes the transition from analysis to construction all too quickly, takes the fundamental concepts as finished before they are. Moreover, it was his conviction that concerning the special relations of nature we can only arrive at hypotheses if we do not wish to remain standing at the individual data of experience. In putting forward such hypotheses he was very bold, and his historical significance rests in part precisely on the enthusiasm and the resistance he thereby aroused.

b. The Method and Presuppositions of Knowledge

Since in all knowledge, whatever object it concerns, we use our understanding, it is, Descartes thinks, of importance to investigate this itself closely. The understanding works alike everywhere; therefore the science of it becomes the universal science, and the true method can therefore be only one. It consists essentially in starting from what is simple and clear, and from that seeking to understand what is composite and obscure. First of all one must by experience (*induction* or enumeration) bring together everything that is needed to throw light on the question to be treated. This material is then ordered in such a way that the simplest, immediately evident relations are laid at the foundation. To find something immediately evident is the first condition for being able to think. Every transition of thought takes place by immediate intuition, *intuition.* It is by intuition that I learn that a triangle is bounded by three lines, that a sphere has only one surface, that when I think, I must exist. By such immediate intuiting the first principles are apprehended. There are propositions that are so simple that we cannot think them without holding them to be true. To these belong: that the same thing cannot both be
[212] and not be, that nothing cannot be the cause of something, and that there cannot be more in the effect than there is in the cause. It is not, however, Descartes's meaning that such propositions immediately are *given.* He expressly maintains that intuition relies on individual cases; but in the individual example a general truth can be found contained, and we then know it immediately as self-evident (simplici mentis intuitu), even if it is not yet formulated as a general proposition. The great question of how general principles can be derived from individual cases, so that intuition can pass over into a fundamental proposition, Descartes does not go into more closely. He only admonishes us to be very cautious in putting forward principles[43]).

From the propositions that are clear in themselves we derive further consequences by *deduction*. It consists in a series of intuitions, in that by a continuous movement of thought one goes from link to link and so connects what was not immediately evident in its connection. If one here comes upon something that cannot be comprehended, one must stop, since it would be fruitless to go further. Either one lacks sufficient experiences, or one stands before questions which, by the nature of the human mind, cannot be answered. If one succeeds in carrying the deduction through, we have led the unknown back to the known, and the first, simple truths have been extended to hold for a larger domain. The items of knowledge accessible to us are mutually connected by a wonderful bond and can be derived from one another by way of inference, so that one will be able to find them if one only advances gradually from the very simplest items of knowledge. — Descartes, as is seen from the way in which deduction is conditioned by intuitions of simple relations of experience, is no opponent of the method of experience, just as he was always eager to gather experiences and make experiments. But he maintains that scientific knowledge is reached only by strict thinking, and that experience has its proper significance in giving occasion for it. He does not, on the other hand, discuss searchingly whether it is possible to reach a complete collection of everything that can throw light on the subject (complete induction or enumeration), and he underestimates besides *[213]* the necessity that the results of deduction be verified. Science arose for him through inferences from data of experience and consisted in logically deduced hypotheses. —

But how far must I now go back to find the very simplest? Where do I find an unshakable foundation for my knowledge? These questions Descartes answers in Discours de la méthode (Chap. 4) and — more fully — in the Meditations. About every content of knowledge, every object of knowledge, we can doubt. All assumptions and sense perceptions may rest on illusion. Let us suppose that an evil being had created us and incessantly instilled errors into us! Even then there would still be one thing we could not doubt: our own thinking, our own consciousness. All doubt, all error is after all a thinking, an activity of consciousness. However many incorrect, non-existent things I represent to myself, the very faculty of representation (vis imaginandi) still constantly proves its reality. In so far as I think (or: have consciousness), I exist: Je pense, donc je suis (Cogito, ergo sum). It is, as already remarked, an immediate intuition that cannot be further grounded, any more than it can be further defined what thinking or existence is. It has an immediate clarity and distinctness that is the model and measure for every other insight. — Descartes declares it a misunderstanding to take that proposition as an inference, but he has undeniably himself given occasion for this misunderstanding by

saying *donc* (ergo)[43 b]).

That simple intuition, which cannot be shaken by any doubt, is not yet any knowledge. How do we get further? How can we ground the objective validity of our knowledge, the reality of what we represent and think? — To answer these questions Descartes uses the causal principle, or rather the proposition that effect and cause must correspond to each other. This proposition belongs, as we have seen, according to Descartes to those that can be discovered by intuition. The natural light teaches us that there cannot be more in the effect than there is in the cause. From this it follows that nothing can arise
[214] from nothing, and that the perfect cannot arise from the imperfect. If we apply this to our representations, then it is clear that some of these can be explained as having arisen from external causes, others as produced by ourselves. But in neither of these ways can the representation of God as the infinite being, the sum of all perfection and reality, be explained. Since I myself am a finite being (of which my doubt and my desire convince me), I cannot have produced such a representation. It cannot have arisen through the combination of individual observed perfections; for then it would not contain the unity and indissolubility that is the distinctive mark of the representation of God. Every external cause is also finite. There remains, then, only that God himself is its author. It is innate in us in the sense that our faculty of thinking, as soon as it awakens to clarity, can by its own power grasp this representation as true. — By an innate representation (idea innata), then, Descartes understands not a representation that is *given* from the first, but one that we have the capacity to develop. The idea of God, says Descartes, is innate in the same sense as the representation of myself is. "The innate representations proceed from the faculty of thinking itself"[44]). — But the word "innate" was fateful; it gave occasion to many misunderstandings.

When the reality of the representation of God has thus (be it noted: by virtue of the causal axiom) been secured, a firm foundation, according to Descartes, has been won for the reality of our knowledge. Everything I know belongs to the infinite reality that this representation expresses, is one of the perfections the concept of God contains, and thereby stands firm for me. Only a finite being can err; the possibility of erring falls away in the infinite being, in whose perfection I share through my clear and distinct representations. Popularly, Descartes expresses this by saying that God, who is the ultimate cause of all our sensations and representations, cannot deceive, since he is a perfect being. But, when one looks more closely, he indicates a less external connection between the representation of God and the reality of our knowledge. Since I can perceive my imperfection only with the help of a more or less distinct representation of perfection,

and since the representation of God is the representation of the highest perfection, which *[215]* comprehends everything, this representation serves me as an ideal by which I can measure and correct my imperfect knowledge[45]). When Descartes draws the concept of God into epistemology, it is not really God in the religious sense he means; the idea of God is for him one with the idea of the continuous and all-embracing connection in existence, in which everything that is to have reality must be able to find its place. Thus it is said in the sixth Meditation: "By nature as a whole (natura generaliter spectata) I mean nothing other than either God himself or else the connection (coordinatio) of created things established by God." What I am to acknowledge as true must be able to find its place in this connection, which, then, could just as well be called nature as God. It is by its connection with everything else that the individual thing acquires reality for me. The criterion for distinguishing between dream and the waking state is that the experiences of waking life I can connect with all my experiences and memories without a break in the connection (perceptionem earum absqve ulla interruptione cum tota reliqva vita connecto). — This is the deeper meaning of Descartes's theological grounding of the validity of knowledge. But interesting as this line of thought is, the theological turn could nevertheless have been avoided altogether, since already the causal principle, which is after all in its turn a presupposition of the reality of the representation of God, is sufficient to bridge the gap between the knowing subject and existence. The causal principle and the other first fundamental propositions Descartes (in Responsiones secundæ) expressly exempts from the knowledge that is first to be grounded by the representation of God. On a closer development of what lies in the causal principle, it would appear that it leads precisely to the assumption of a connection within which the individual phenomena have their definite place. But the time for a closer discussion of the causal relation and the causal principle came only a century after Descartes's time. *[216]*

c. Theological Speculation

Descartes begins analytically and critically, but very quickly leaps over into theological and spiritualistic speculation. In his "Je pense, donc je suis" he finds the starting point for both; for the present we keep to his theological ideas, which are of interest to us in that they provide a stricter development and grounding of the content of "natural religion."

Descartes denies that the existence of God can be proved from the world given in experience. The series of causes could after all go on to infinity, and we must not confuse the impotence of our understanding to follow such an infinite series with the necessity of assuming a first cause. Moreover, the concept of God is presupposed as known

beforehand when I am to prove the right to call such a first cause God. His proof is more direct. It consists, as we have already seen, in the impossibility that the representation of an infinite being could be formed by a finite being. This representation is indeed even a presupposition of the representation of myself as a finite, limited being; for the finite presupposes a limitation of an infinite; the infinite is the positive, by whose limitation the finite comes into being. The concept of God is then formed by removing the limitation (thus by the negation of a negation found to hand). With this last determination Descartes comes considerably closer to the view that the concept of God is formed as we form all our ideals: by extension and by the removal of barriers (besides by the combination of features from different contexts). Gassendi pointed out against Descartes that we in fact form ideal representations by extension and combination; but Descartes answered that this very capacity to extend the representation of the relatively perfect given in experience presupposes that we have our origin from God. Instead of following Gassendi to the study of the psychology of the formation of ideals (a subject that even to this day has been very little treated), Descartes keeps to the *finished* representation of God: since it concerns something infinite, it supposedly cannot have been produced by a finite
[217] being. This inference itself would after all only be correct if it were the representation itself, and not merely its object, that had to be infinite. —

Yet Descartes has still another proof, the ontological. While there he went from the representation of God back to the cause of this representation, here he seeks to show that when we clearly and distinctly think the concept of an infinite and all-perfect being, we must also assume the existence of such a being: for by denying it existence we would deny it a perfection and so contradict ourselves. Against this reasoning (which is found in the fifth Meditation) Gassendi already objected that existence is certainly the condition for something's being able to have perfections or imperfections, but is not in itself any perfection. Descartes held to the point that we do after all ascribe existence to things in the same way as we ascribe properties to them. —

Descartes is clear that the concept of God whose validity he thinks he has proved by the two proofs cited is not the popular one. Most people do not regard God as an exalted and infinite being, the sole author of everything, but think by the word God of a finite being that can be worshipped by men. It is no wonder that there are those who deny the existence of such a God. His own higher concept of God Descartes develops further by determining God as the absolute substance. That God is a substance means that he is a being which is able to exist by itself (per se), needing no other thing in order to exist. Descartes does indeed also use the concept of substance of finite beings

(thus of soul and matter, which we shall go into more closely in what follows); but he declares that it cannot be used in one and the same sense (univoce) of the infinite being and of the finite beings. Since the finite beings all depend on the infinite, they can only improperly be called substances. Especially in his letters Descartes carries the strict concept of substance through. "One cannot," he writes, "prove the existence of God without regarding him as the most perfect being of all; but he would not be so if anything happened in the world that did not proceed from him. Natural philosophy already teaches that not the least thought can come up in the human mind without *[218]* God's willing it and having willed from eternity that it should come up." This is also one of the reasons why Descartes rejects atomism: an atom would be something that subsisted by itself; but only one being, the highest of all, can be independent of every other being! — In his writings Descartes expresses himself more vaguely, since a wider and more popular definition of the concept of substance slips in beside the strict one. According to the laxer usage, substance means the same as thing or being. Substance is that which has certain properties. From the properties we infer a substance. In this sense (as bearer of properties) a finite being can very well be a substance. It is clear that it must lead to unclarity when the two meanings of the word are not kept sharply apart. But Descartes conceives soul and matter as substances also in the sense of absolutely independent beings. And just as he thus applies the strict concept of substance to finite beings, so too he conversely applies the loose concept of substance to the infinite being, since the content of this concept is after all all possible reality and perfection — thus all possible properties! It was therefore a criticism that cut deep into Descartes's system when Hobbes and Gassendi objected against him that we have no positive concept at all of substance as bearer of properties, since we know only properties (accidents, attributes — or, as we would now say, phenomena). Descartes conceded that we do not apprehend substance immediately, but infer it from the properties; but he maintains that we can think it, even if we cannot form any image of it[46]). Here too lay a problem that came to play a great role in the subsequent philosophy. —

It was with the help of the causal principle that Descartes arrived at the assumption of God's existence. But this same proposition might seem to lead beyond the concept of God: for must not God then also have a cause? From this difficulty Descartes seeks to free himself by distinguishing between that which has its cause outside itself and that which has its cause in itself. God must be, in relation to himself, what the cause is *[219]* in relation to the effect, thus be the cause of himself (causa sui). As the infinite being, God has an inexhaustible power in himself, so that he needs nothing other than himself

in order to exist. In this way the concepts of substance and cause of itself are closely connected. Antoine Arnauld, the great Jansenist theologian, who later became an ardent Cartesian, objected (with a line of thought that already occurs in Thomas Aqvinas) that the concept of cause of itself contained a self-contradiction and in any case could not be applied to God. For, he said, no one can give what he does not have, and for anyone to be able to produce himself, he would have to exist already beforehand. And if God is cause, he is also effect; the cause precedes the effect in time; but in God there can be no distinguishing between past and future or between possibility and actuality. Descartes answers that when we prove the existence of God by means of the causal principle, we must also apply this principle to God himself, which is possible only by conceiving him as the cause of himself. But he adds that the expression cause is used of God only by analogy, on account of the imperfection of human knowledge, and that one may well call God the cause of himself, but not the effect of himself, since thereby something in him would come to stand lower than something else! The relation of time has, moreover, no validity at all for God, since it is due only to human apprehension, is a modus cogitandi. — There seems then undeniably not to remain much of the concept of cause in its theological application. The Cartesian philosophy of religion could not but give rise to a need to discuss the justification and significance of the application of our fundamental concepts in theology.

The proposition that the relation of time has no validity in God must particularly be remembered when Descartes teaches that the eternal truths (such as, e.g., that a thing cannot both be and not be, — that the sum of the angles in a triangle is equal to two right angles) depend on the absolute will of God. God's reason and his willing are not distinguished. God has not first perceived the eternal truths and then laid them down.
[220] His willing is eternal and unchangeable like his essence. Here too the great difference shows itself between Descartes's concept of God and the current one. There is really not a single psychological expression that can be transferred to God. But this consequence Descartes did not yet draw.

One difficulty remains, even if one grants Descartes to be right in all his assumptions. Through our doubt we perceived our imperfection, and its concept presupposed the concept of the perfect. Thus we rose up to the concept of God. But after we have risen to this height, imperfection nevertheless continues to subsist — and how is it at all possible that imperfection could have come into being, when it is an ideal and perfect being that lies at the foundation of everything? Descartes answers that imperfection necessarily clings to everything finite, and everything created is finite. It is, besides, only a negation

that brings about imperfection; and this negation makes the world more perfect than it would otherwise be, since the multiplicity in the world thereby becomes greater. Even if the eye is the most beautiful thing in the body, the other parts of the body still have no reason to complain, since the body could not consist of nothing but eyes. Descartes indicates in these remarks a theodicy similar to the one Leibniz later presented at length.

d. Philosophy of Nature

By virtue of the criterion of truth Descartes had found, he concludes that our sense-sensations must derive from something that is different from our consciousness. But this something need not — according to Descartes — by any means be so constituted as the senses present it to us. The significance of sense perception is first and foremost practical; it is to inform us of what is beneficial or harmful to us. And even if we regard it as the task of sense perception to help us to a knowledge of things, it is not necessary that our sensations *resemble* things, provided only that they correspond to them as the word corresponds to the thought. If we look to the way in which sense perception takes place, we find that all impressions on the senses are contacts; it is thus only the surface of things that we come into relation with. If we wish to think how things are constituted *[221]* apart from the way in which they affect us, then there remain only extension, divisibility and mobility as properties we cannot — even in our imagination — think away. They are the simplest and clearest representations we can have of matter, and it turns out that by keeping to them — thus to the purely geometrical properties — we can reach a simple and clear understanding of everything that goes on in the material world. It is the phenomenon of motion that contains the explanation of all other phenomena. On the other hand, nothing is explained by assuming "forms" or qualities in matter. The qualities belong to our consciousness; but what really goes on in the material world is only motions outside our organism or within it. In science we must think of the material world as it would be if no human being perceived it. — It is the principle of simplicity that played so great a role in the struggle for the new picture of the world, and that is now used to ground the leading back of all the properties of matter to extension and motion. A consequence of this leading back is that the qualities are relegated to the subjective domain. Since Descartes can hardly have known Galileo's "Saggiatore," he has probably come to the principle of the subjectivity of the sense qualities by his own path, through the attempt to think through the conditions for a purely objective and strictly necessary explanation of material processes. From notes in Descartes's hand from the years 1619—20 it is seen that he did not yet at that time adhere to the

mechanical explanation of nature. The transition to it must have taken place in him in the years 1620—29, in connection with his studies in natural science, especially in the domain of optics and physiology. Besides Kepler's influence, there has probably also been at work here the influence of a circle of younger French investigators who adhered to atomistic views and were especially influenced by Sebastian Basso, the real renewer of the atomistic doctrine in modern times; Descartes himself names Basso among his predecessors. It was two young men of this circle who wished to hold the
[222] disputation in Paris mentioned earlier and thereby gave occasion to the terrorist decree against new opinions. Since Sebastian Basso in turn seems to have been influenced by Giordano Bruno's writings, a historical connection appears here between Descartes and the philosophy of the Renaissance. — Harvey's discovery of the circulation of the blood too has (as we shall see later) intervened strongly in Descartes's line of thought.

In the attempt to explain all material phenomena by means of extension, divisibility and mobility as the fundamental properties of matter, Descartes proceeds deductively. He goes from cause to effect. He is fully conscious that by this path only a hypothesis can be reached, since there is always the possibility that the given phenomena could have been produced by other causes than those we start from. Here, then, experience and experiment must come in as confirming, and Descartes declares that for this so great a quantity of experiments is needed, and such costly ones, that they far exceed his means and can be brought about only by the cooperation of many. Descartes, then, has in no less a degree than Bacon an eye for the significance of experience, but he lays greater weight than the latter on the guiding principles and hypotheses. From Galileo he likewise differs in his interest in principles and deductions. He took the deductive procedure to be the properly scientific one and lacked Galileo's ability to throw light on great and comprehensive relations through investigations of particulars — and this although he himself, in his doctrine of method, sets the intuition bound up with individual observation as the starting point of thinking. But his task was a different one from Galileo's. His mission was to bring to full clarity and consciousness that the time was past when an explanation of nature could be reached by assuming mystical forces or supernatural intervention. He set up the ideal for all natural science: the derivation of phenomena from their causes with the same transparent necessity with which a mathematical conclusion follows from the presuppositions laid down. The principle of simplicity, to which he appeals, is really at the same time the principle of transparency. In the ideal thus set up he was not mistaken. But he ascribed to the deductive application
[223] of general principles, and to the hypotheses grounded on it, greater significance than

they have in reality, and his philosophy of nature thereby acquired a dogmatic character, which, especially in the hands of the Cartesians, stood out markedly and provocatively. In many a mechanical scholasticism took the place of the scholasticism of substantial forms and qualities. —

From the fundamental properties of matter it follows that there can be no absolute atoms, since no limit can be set to divisibility — that there can be no empty space, for space means extension, and extension presupposes an extended being, thus matter — that the material world is infinite, for it is impossible for us to set limits to extension, and where there is extension, there is also matter. Further, it follows that when material changes occur, these must all be explained according to the laws of motion, which thus become the highest laws of nature. These laws too Descartes seeks to derive by deduction, and the foundation for this deduction he finds in the concept of God, which in his philosophy of nature comes to play a role analogous to the one it plays in his epistemology. From the unchangeableness of God, which is part of his perfection, Descartes derives that the sum of motion produced at the creation is preserved during the preservation of the world (which according to Descartes is a constantly continued creation). Motion can be distributed differently among the different parts of the world, but no motion is lost and none begins anew. — There lies in the theological grounding Descartes gives of the old proposition of the constancy of motion an involuntary admission that the fundamental properties of matter from which he starts in his philosophy of nature are too simple and abstract. For it turns out that it is really the divine force constantly at work in the world that is constant; the constancy of motion is derived from the constancy of this force. Had Descartes seen this clearly, he would also have found a more correct proposition than the one he sets up. For it is not motion but force, energy, that is constant. With his proposition Descartes gets into difficulty at every transition from motion to rest or the reverse. But even in its imperfect form his proposition is an interesting predecessor of the modern principle of the conservation of energy. — Special laws of motion too Descartes *[224]* derives from God's unchangeableness. So first and foremost the law of inertia, which in him for the first time is consciously and systematically set up. It would, Descartes thinks, conflict with the divine unchangeableness if any thing that can be regarded as simple and undivided could lose the state it is in without an external cause. Thereby the possibility is rejected that such a thing could of itself (sua sponte) and without external influence pass over from motion to rest or the reverse. Descartes probably arrived at the principle of inertia under the influence of Kepler, who set up one half of it, but independently of Galileo. Already in the treatise "Le Monde," which was worked out before Galileo's

Dialogues appeared, he had in fact set it up. — In setting up the principle of inertia Descartes makes a reservation, leaving open the possibility that souls and angels could act upon matter. This reservation is made in the interest of his spiritualistic psychology, which we shall consider later.

Descartes's concept of matter bears the stamp of simplicity and clarity and is indeed also grounded by the principle of simplicity. It is a natural outcome of the tendency that also found expression in Kepler and Galileo and sought to lead all qualities back to quantitative relations, whereby exact natural science became possible. But just as he is inclined to overlook the hypothetical character of his deductive results, so he is also inclined to regard his definitions, set up by abstraction and by virtue of the principle of simplicity, as exhaustive. He thinks he knows the essence of matter completely by the determinations extension, divisibility and mobility, without, however, being able to give any guarantee for it. When he speaks of matter as *the extended substance,* he does not notice that it is an abstraction he has made into a being subsisting by itself. He has not proved the justification for regarding the geometrical properties of things as absolute. For it does not necessarily follow from the fact that these properties are the most important for the knowledge of nature. Properties such as extension, divisibility and mobility may very well be due to the action of things upon us (like colors, taste, smell,
[225] etc.), without their therefore *resembling* the things themselves. And besides there is — as already shown — a determination that is still more fundamental than the three properties Descartes keeps to, namely force, energy. — It became the business of subsequent inquiry to discuss the questions indicated here. Descartes's philosophy of nature signified for the time being, in all its one-sidedness, an immense advance in clarity. We shall now briefly consider some of the most important applications he makes of the general principles he has set up. It is always the fundamental thoughts we keep to; in detail his explanations are often very unfortunate and arbitrary. Spinning hypotheses was his fault; his greatness was the happy grasp of general points of view.

When everything in material nature is to be explained from extension, divisibility and mobility as the only properties of matter, it follows of itself that final causes cannot be applied. But Descartes gives a theological rather than an epistemological grounding for the exclusion of the teleological explanation of nature. Since God is an infinite being, there must be much in his working that we cannot grasp; it would be presumptuous to wish to search out his purposes. There is, after all, in the unbounded world infinitely much that does not affect us at all; what meaning can one then attach to its having been created for our sake? The only purpose that can be thought for everything that happens

must be God himself. It is thus because we dare not set limits to the essence of the world and of God that the teleological view is rejected.

If we are now to understand nature in detail, then according to Descartes we must start from the simplest, most easily seen-through phenomena, from the processes we constantly see going on before our eyes, and by means of them explain what goes on in the small and in the hidden, and what has gone on in the past. The principle of simplicity thus passes over, for Descartes, into the principle of actuality, which requires the distant and unknown to be explained by the near and known. Best of all, we must take as our foundation insight into the construction and working of our machines, since we have the most distinct representations of these. Therefore, says Descartes, I describe the world as *[226]* if it were a machine (terram totumqve hunc mundum *instar machinæ* descripsi). The present state of the universe we explain to ourselves by supposing that the parts of matter from the very first (for matter and motion were created at the same time) were in whirling motion about certain centers. In these centers the smaller parts had to gather, which arose during the whirling motion through the friction of the larger parts against one another. Thus arose the various larger and smaller globes. Some of these globes gradually lost their independence and were swept along by the vortices that go on about larger globes. So it has gone with the earth. Descartes thinks that according to his theory he can call the earth immovable, since it does not change place within the medium that whirls it along with itself around the sun, and since, besides, outside the part of the universe we can observe, there may be stars in relation to which the earth is immovable! By such a subtle application of the relativity of motion Descartes seeks — as it turned out, in vain — to escape standing as a Copernican heretic. Gravity is explained as a consequence of the pressure of the vortex currents on the parts in space, whereby they are pressed toward the globes or in any case held in their vicinity.

Descartes is the first in modern times who has attempted to give a mechanical theory of the evolution of the world-system. He does indeed make a polite reservation toward theology, conceding that the world was created perfect and finished; he only wishes to show the *possibility* that the present state of the world can have developed according to natural laws from a more imperfect initial state. If one asks how this initial state is then to be thought, he answers that it will not matter much whether we think of it in the one way or the other: for by virtue of the natural laws matter will run through all the states it is capable of assuming, so that after the exhaustion of more or fewer imperfect states the present state will in any case be reached. This declaration (Princ. phil. III, 47), which contains the program for all mechanical theory of evolution, is of greater interest

[227] than Descartes's attempt to depict the cosmogonic process in detail. — A chief defect of his hypothesis was that it made no attempt to explain the connection that Kepler's laws had established. These laws continued to stand as an indifferent fact until Newton's brilliant thought discovered the physical connection from which they could be derived, and which could be derived from them. —

Just as Descartes seeks to explain the connection and development of the world according to the general laws of nature, so he also seeks to explain the organism and organic life according to purely mechanical laws. Not only astronomy, but also physiology was in his view to be a purely mechanical science. Just as in his cosmogonic hypothesis he abstracts from theology, so he abstracts in his organic mechanics from psychology and thinks of the human body as formed of material parts and working according to the laws of heat and motion, without any soul (whether "vegetative" or "sensitive" or "rational") intervening. This view, which is a consequence of Descartes's general principles of the philosophy of nature (since the organism, as a material being, must fall under the general laws of matter), received for him a confirmation from experience in William Harvey's discovery of the circulation of the blood (1628). Harvey stands in the first rank among the founders of modern natural science; he is for physiology what Galileo was for physics. He gave the mystical forces their deathblow in the domain of physiology by showing that the motion of the blood is not due to its own force or that of the soul, but is determined by the contraction of the heart, which presses it out into the body. The general laws of motion thus hold in the organism as outside it. Descartes was one of the first eminent men to adhere to Harvey's doctrine, and his adherence, which was declared in Discours de la méthode (chap. 5), was a powerful support for the new doctrine, which had so much resistance to overcome, since it stood in sharp conflict with the old view of organic life[47]). Descartes depicts in various writings (especially in Traité de l'homme) how the human body can be conceived as a mere machine. Here too his general way of looking at things has brought clarity about, even if his special explanations were
[228] unfortunate. One of the greatest anatomists of the time immediately following, Niels Steensen, acknowledged that the method Descartes introduced serves to show the insufficiency of what had earlier been taught about the significance of the various organs, and to put the questions about it in a clearer way. Descartes extended the mechanical view that Harvey had carried through for the circulation of the blood to the domain of the physiology of the nerves. In the nerves he assumes, in agreement with ancient physiology, currents of "animal spirits" (esprits animaux), which are formed by the fine parts of the blood, after being heated in the heart, being carried up to the brain and filling

its cavities, while the rest of the bloodstream continues its way through the vessels. From the brain they stream out through the nerves to the muscles. These currents can now, according to Descartes, be set in motion by impressions that do not become conscious to us; this happens in involuntary movements, as when in falling we put out our hands, or when we continue our walk without thinking of it. Such involuntary movements can take place purely mechanically, even against our will (mente invita et *tanqvam in machina).* Descartes expresses what goes on here by saying that the "animal spirits" are reflected (esprits réfléchis), and he gives a clear description and explanation of what is now called reflex movement. In animals there is, he thinks, no reason to assume anything other than that all functions and actions go on in the involuntary and machine-like way. There is no reason to ascribe a soul to them. When the lamb flees at the sight of the wolf, it is because the rays of light that from the wolf's body strike the lamb's eye, through the "reflected" currents of "animal spirits," set its muscles in motion. And when the swallows come in spring, it is to be explained in analogy with a clock's striking its strokes at a definite time. One sees, after all, that animals cannot adapt themselves to new and more complicated conditions, any more than a machine can. In any case: when we can explain the behavior of animals purely mechanically, there is no reason for assuming an animal soul. And when man can make wonderful automata, should not nature be able to produce still more wonderful ones? — The assumption that animals are mere machines recommends itself to Descartes also for the reason that, in his view, *[229]* one would otherwise have to ascribe immortality to them — and should an oyster or a sponge be immortal? It is otherwise with man, in whom, on account of the consciousness that makes itself known in each of us, we must assume a soul, a thinking substance in interaction with the material substance, with the capacity to intervene in directing the motions of the "animal spirits." Immediately, according to Descartes, the soul stands in connection only with a single part of the brain, the pineal gland (glandula pinealis), which drew Descartes's attention because it is not, like so many other parts of the brain, a double organ, and because it seemed to him to be placed about in the middle of the brain and above the passage through which the "animal spirits" from the anterior cavities of the brain are in connection with the posterior ones. The "animal spirits" strike against the pineal gland and thereby affect the soul, awaken sensation, feeling and drive in it, and the soul answers in turn — likewise by striking (!) against the pineal gland and thereby directing the animal spirits in a certain direction. Often (as in self-control) thrust and counter-thrust go in opposite directions, and it then depends on who can strike hardest![48])

Descartes has the merit of having described and analyzed reflex movement and thereby thrown clear light on the involuntary activity in us. He has also the merit of having maintained that mental activity is bound to the brain, and immediately only to it. And finally he has — despite the imperfect anatomical presuppositions at his disposal — the merit of having presented the spiritualistic psychology with great vividness and in its full consequence. His psychological theory we shall now consider somewhat more closely.

e. Psychology

Only in man does Descartes assume a soul, and he conceives it as a being distinct from the body, a substance of its own, the bearer of the mental properties, just as the body is the bearer of the material properties. Since, as we have seen, he carries his mechanical
[230] view of nature through into the domain of organic life as well, he must assume that in the organism, as little as in the rest of material nature, can any motion arise that is not a continuation of earlier motion, or any motion vanish that is not continued in a following motion: for the principle of the constancy of motion is indeed a chief proposition of his philosophy of nature. But when he now lets the soul "push" the pineal gland in order to set the "animal spirits" in motion, and in turn itself receive "pushes" through the action of the "animal spirits" on the pineal gland, then motion must indeed arise or vanish at this point without compensation! Even if one could attach any meaning to the soul's pushing or being pushed, Descartes's spiritualistic psychology nevertheless comes here into open contradiction with his philosophy of nature. In reality he thinks of the soul as a material being when he lets it push and be pushed; this is what always happens to spiritualism when it is thought through with full clarity. Descartes's gifted correspondent, Princess Elisabeth of the Palatinate, remarked quite rightly that it was easier to conceive the soul as material than to understand how, if it is not material, it can move matter and be moved by it. In one single place Descartes seems to want to restrict the soul's action to bringing about a change of direction in the "animal spirits", which in themselves could just as well move in the one direction as in the other; but even this mere directing would then conflict with the principle of inertia and could only be set up as an arbitrary exception to it.

When Descartes in general sets up the soul as a peculiar being, distinct from the body, it is because consciousness and extension are such wholly different properties that he can conceive them only as borne by different beings. Je pense, donc je suis! was his starting point, and by thinking he meant, as he expressly explains, consciousness in general

(omnia, qvæ nobis consciis in nobis fiunt). He could therefore, he remarks, just as well say: I feel, therefore I am, or: I will, therefore I am. It is the fundamental fact of consciousness that he wants to bring out. And he has deserved well of psychology by having set up consciousness (mens) as the starting point, whereas the concept of soul (anima) is — *[231]* and especially was — an ambiguous and mystical concept. He thereby paved the way for modern empirical psychology. It is not impossible that in this he was influenced by Vives, who distinguished so clearly between mental phenomena as what is really given and the essence of the soul in and for itself; Vives's psychological work is cited in several places by Descartes. But Descartes thinks that in thinking or consciousness he has a perfectly clear and exhaustive concept of the essence of the soul, just as in extension he thinks he has a perfectly clear and exhaustive concept of the essence of matter. Now if, says Descartes, I can think the concept of the soul clearly and completely without needing to presuppose the concept of matter, and can think the concept of matter clearly and completely without needing to think of the concept of the soul, is this not proof that we have before us here two wholly different beings or substances, independent of one another? (It is — as Descartes himself expressly remarks in one place — not the individual body that is substance in the strict sense, but matter considered as a whole: corpus generaliter sumptum). — We have earlier mentioned the double meaning in which the concept of substance appears in Descartes. It does not merely denote a bearer of properties, a something that "has" the peculiarities that experience presents. Even in this vaguer meaning of the word Descartes infers too much when he makes the "something" of thinking different from the "something" of extension; it could indeed be the same "something" that had both properties! But in the stricter meaning substance is a wholly self-subsistent and independent being. It lies, says Descartes, in the nature of substances that they *mutually exclude one another.* And this he assumes of soul and matter, because he can think of each of them by itself alone without needing to think the other party along with it. Man, then, is to consist of two beings that mutually exclude one another! Thereby the interaction between them becomes incomprehensible. When the one substance affects the other, they cannot be wholly independent of one another; something then takes place in the one that cannot be understood without my thinking of the other. The relation becomes incomprehensible, and Descartes himself admits it. *[232]* Those who never philosophize, he says in a letter, have no doubt that soul and body interact and make up one being. But philosophically it is impossible to think at once the difference and the connection between soul and body. No inference and no comparison, he says in another place, can teach us how consciousness, which is incorporeal, can set

the body in motion; but a sure and evident experience teaches it to us daily; it is one of those things that must be understood through themselves, and that are obscured when we compare them with other things. — Descartes speaks here as though he had seen the soul push the pineal gland and receive a push in return! The knot of the problem is precisely that immediate experience cannot decide the matter, but that inference and comparison are our only way to an intelligible hypothesis free of contradiction.

It is a common assumption that medieval dualism recurs in Descartes's theory of soul and body. But what really led Descartes to his theory is his endeavor to form clear and distinct concepts. From the independence of the concepts of one another he then infers the independence of the corresponding beings of one another. It is the clear and sharp characterization of the difference between mental and material phenomena that is Descartes's great merit; but it is also this that leads him astray. As a dogmatist he trusts in the completeness of his concepts and in the agreement of existence with his concepts. The psychological dualism that we have met in Telesio, Bacon and Campanella Descartes carries through in extreme form — and he has thereby contributed much toward leading beyond it. He has formulated the popular metaphysics of soul and body so sharply that its difficulties and contradictions stand out clearly. — Gassendi and Arnauld rightly objected against him that, because consciousness and extension are the chief properties of the soul and of the body, it does not follow that their essence is thereby exhausted. They have thereby struck precisely the decisive point in Descartes's theory. —

The consequence of Descartes's sharp distinguishing between soul and body is felt in
[233] his special psychology. He investigates carefully which mental activities and phenomena derive from the soul itself, and which are due to the body's influence on the soul.

Sense perception is due to the body; it arises through the "animal spirits" acting on the soul by way of the pineal gland. Yet Descartes emphasizes that popularly more is counted as belonging to sense perception than should be. For a distinction must be made between the motion in the brain produced by the impression (which can also be found in animals), — the action of this brain-motion on the soul, whereby the sensation (e.g. of color) arises, — and thirdly the judgments that are involuntarily passed on external things on the occasion of the motions of the bodily organs. Such an involuntary judgment underlies the conception of things as external and as situated at determinate places in space. It is the motions of the eye and of the head that, through the corresponding motions in the "animal spirits", occasion the soul to pass such a judgment. This third moment in sense perception is due to the soul itself.

Likewise there are two kinds of memory: one of material things, which rests on after-

effects or traces of the impressions in the brain, and one of mental things, which rests on lasting traces in consciousness itself. — Thinking proper (intellectio) and representation (imaginatio) are distinguished from one another in that in thinking proper the soul acts alone, but in representation it makes use of bodily images. Representation, like sense perception and material memory, belongs to the soul only insofar as the soul is united with the body; the soul in its purity (anima pura) can be thought without representation and sense perception. — The difference between drive and will likewise rests on the drive's being called forth by the body, while the will belongs to the soul itself. Descartes ascribes to us an immediate consciousness of the causal freedom of the will, although he admits that the indifference of the will is rather an imperfection in our knowledge than a perfection in our will. The will has a greater scope than the intellect; we can choose and hold fast to something that we do not understand. The will is unlimited, the intellect limited: by this Descartes explains the possibility of error. — The emotions (passions) are due to the body's influence on the soul, but the inner feelings (émotions intérieures) arise *[234]* in the soul in consequence of its own thoughts and judgments. The inner, purely mental feeling is indeed, so long as the soul is united with the body, accompanied by an emotion, a sensuous feeling: but this is then due to the soul's action on the brain, inasmuch as, e.g. in intellectual love, the mood of the soul affects the pineal gland. In emotion joy precedes love, in inner feeling the reverse. Although by this dualistic conception Descartes really blocks his own way to a coherent history of the development of the feelings, he has nevertheless deserved well of the psychology of the feelings by his attempt to trace the various feelings back to certain elementary forms from which they arise by combination. As elementary feelings he assumes wonder, love, hatred, desire, joy and sadness. Of these and their combinations he gives a purely descriptive and analytical account, constantly making use of his physiological theory. In explaining the physiological conditions of the emotions he lays the chief weight on the interaction between brain and heart. Only in the case of admiration, which he conceives as a purely intellectual feeling, does he not think that any influence of the brain on the heart takes place. He emphasizes the connection of the emotions with the instincts: they dispose the soul to will what is useful and to shun what is harmful; even without the soul's intervention, the motion in the "animal spirits" that calls forth the emotion leads to the performance of purposive actions. But the purposiveness that shows itself here is not absolute; therefore experience and reason must intervene to correct it. Of importance for the development of the feelings is the connection between a representation in the soul and a certain state of the "animal spirits"; only through such a representation, not quite directly, can the will intervene

determiningly in the bodily motion. The struggle between reason and sensuousness is carried on, according to Descartes, not within the soul itself, but is a struggle between the soul or the will on the one side and the body or the "animal spirits" on the other, and
[235] the struggle is over which can push the pineal gland the harder! (Passions de l'âme. I, 47). — Here too we find the spiritualistic theory carried through by Descartes with all desirable consistency.

f. Ethics

When Descartes set up universal doubt as the starting point of his philosophy, he provisionally excepted the moral domain, although he saw that his philosophizing must have consequences for it too. So that his conduct of life should not be left without guidance while he analyzed everything theoretically, he set up (in the Discours de la méthode, chap. 3) the following provisional rules for his conduct: 1. To keep to the religion and manners of his fatherland. 2. To carry out consistently what he has once resolved. 3. To strive to conquer himself rather than fortune, to subordinate his wishes to the order of the world, and to remember that our own thoughts are the only thing that is wholly in our power.

Later, when he had developed and published his philosophical views, he was wary of venturing into the moral domain. The theologians, he says, who were so angry with him on account of his physics, would certainly not leave him in peace if he wrote on ethics. Apart from some statements in the work Passions de l'âme, it is in his letters to Princess Elisabeth, to Queen Christina and to Chanut (French envoy in Sweden) that we find his ethical ideas developed. They are only a further unfolding of the provisional rules of the Discours de la méthode; they betray the influence of Stoic philosophy, particularly of Seneca.

What means, asks Descartes, can philosophy teach us for attaining the highest happiness, which vulgar souls vainly expect from the course of things, but which we can only get from ourselves? Three means present themselves: clear knowledge of what is right, firm willing of it, and the suppression of every wish and every want that is directed at what is not in our power. Only through the intervention of the will does knowledge come to rule in us, so that the motion of the "animal spirits" follows it. Whereas the
[236] sensuous feeling or emotion confuses our thought and makes it overvalue external goods, it now becomes possible to hold fast to the inner enjoyments of the soul, which are eternal like the soul itself, because they are built on the firm ground of the knowledge of truth. The most important knowledge is the knowledge of God as the author of all

things, as the all-perfect being of whom our mind is an emanation, and the knowledge of the soul as a being distinct from the body. Next comes the knowledge of the infinity of the world, which roots out the belief that everything should exist for the sake of the earth and the earth for our sake. Knowledge teaches us to regard ourselves as parts of a whole (family, society, state) and to prefer the interest of the whole to our own interests. But above all it is a matter of distinguishing between what is in our power and what is not. Only the use of our will is wholly in our power. In the feeling of this consists *magnanimity* (générosité), which is the key to all virtues. The magnanimous man feels at once his own power to accomplish great things and his own limitation. Instead of the notion of a capricious fortune, which is a mere phantasm, it is a matter of holding fast to the conviction that everything rests on the eternal and unchangeable will of Providence. Thereby the passions are purified, and room is made for the spiritual love (amor intellectualis) of God, the sweetest and most beneficial of all the feelings we can have in this life (although it must be left to the theologians to decide whether it, such as the natural man can have it, is sufficient to bestow eternal blessedness on man). By sufficient exertion of heart and thought this feeling can also become the strongest of all. — As in Descartes's epistemology and in his philosophy of nature, so in his ethics too thinking culminates in the idea of the infinite being as the ultimate foundation. A mixture of rationalism and mysticism is characteristic of Descartes, as of the seventeenth century as a whole. This side of Descartes's philosophy was particularly developed by his immediate successors. *[237]*

2. Cartesianism

The Cartesian philosophy could not but exercise great influence on the age. By its demand for methodical doubt and analysis, for disregarding all inherited presuppositions, it could not but awaken free inquiry and lead to independent thinking. On this point its most important significance doubtless lies. It has thereby exercised influence on many more than those who became Cartesians in the narrower sense of the word. By its demand for a purely mechanical explanation of nature it intervened as a reforming force in natural science, especially in physics and medicine, even though several of its special hypotheses soon proved unfortunate. It stated the principle of mechanism with an energy that has been of lasting significance. By its spiritualistic psychology and its theological speculations it appealed to the current conception, for which the resolute and absolute distinction between soul and body was very clear and appealing, and which

also found in the Cartesian proofs a new foundation for agreement between theology and science, in place of the one that was falling with the scholastic philosophy. In the last part of the century we accordingly see the greatest minds of the French Church, Bossuet and Fénélon, as zealous Cartesians. In eminent Jansenists, such as Arnauld and Nicole, it likewise found adherents, although the greatest figure of Jansenism, Pascal, here took up a special position, which will be characterized later. Under Arnauld's direction an excellent account of logic was worked out in this circle *(La logique ou l'art de penser.* 1662), which is still of interest and value. In the learned Order of the Oratory too Cartesianism found great support. In wider circles it was spread by popular lectures in Paris and the provinces. From Madame de Sevigné's letters one sees what interest it aroused in the high aristocracy, among women too. It gained no small spread in Holland through the appearance of several younger university teachers. But both in France and in Holland it also met with violent opposition. The Jesuits and the orthodox Protestant clergy sought in every way to crush it. The new philosophy ascribed too much to free human reason
[238] and gave doubt a dangerous significance. It taught the subjectivity of the sense qualities, which was not compatible with the dogma of the Eucharist (especially in its Catholic form), and besides it assumed the motion of the earth and the infinity of the universe. The cause of theology seemed to be bound up with the old scholastic doctrine. At several Dutch universities Cartesianism was forbidden; in any case the Cartesians were kept away from theological chairs and from offices in the ministry. In Rome Descartes's works were put in 1663 on the list of forbidden books, and in France repeated royal decrees forbade the teaching of Cartesianism at the universities. All this did not prevent it from becoming the ruling philosophy, whose dominion was first broken by the tendency proceeding from Locke and Newton, a tendency which it had, moreover, itself prepared.

Among the more independent Cartesians the difficulties and contradictions in Descartes's doctrine could not, in the nature of the case, but lead to attempts to develop the master's thoughts further. In the dogmatic part of the Cartesian system there were especially two points that contained stings. The one was the relation between the absolute substance, which played so great a role in Descartes's theory of knowledge, doctrine of nature and ethics, and the finite substances, the finite world as a whole. How can an absolute being have a finite world outside itself, and what reality can such a finite world have? This problem could not but press forward particularly in an age that had a bent for mysticism and for the recognition of absolute powers; it is indeed the age of absolutism in politics, and it became so by a natural analogy in philosophy too. Soon the Cartesian philosophy glides over into mysticism. The other point was the relation between soul and body

as substances that mutually excluded one another and yet were to stand in interaction with one another. The master himself had indeed even declared that this could only be understood so long as one did not philosophize. There were some of the disciples who went on philosophizing on this point, and who found here an incomprehensibility that for them recurred everywhere in experience where different things were to act on one another. By this road Cartesianism was led on to restricting knowledge to establishing the succession and external connection of phenomena, while the inner connection be- *[239]* tween them was declared incomprehensible. Thereby the treatment of the problem of causality by the later epistemology was prepared. — Both points work together in the so-called *occasionalism,* yet in such a way that the mystical element, the tendency to gather all reality and all causality in the absolute substance, is the predominant one. We consider in all brevity the most important representatives of this tendency.

Louis De La Forge, a French physician, stated clearly (in his *Tractatus de mente humana,* 1669, in French as early as 1661) that just as difficult as it is to see how the mind moves the body, just so difficult is it to comprehend how one body can move another body. Perhaps, however (cf. *L. Stein* in Archiv für Gesch. d. Philos. I. p. 56), another Cartesian before him, Géraud De Cordemoy, a Parisian advocate who later became tutor to the Dauphin, had stated this thought; in that case he must be regarded as the real founder of occasionalism. And while de la Forge declares that the relation between the soul and the body is intelligible only on the assumption that God has originally put them in connection with one another, Cordemoy goes further and teaches that it is just as impossible for the soul to get new representations as for the body to get new motions, if God does not intervene. The finite beings, souls as well as bodies, are only occasions for God to intervene. Thus, e.g., our will is the occasion for God to move our arm. How the tendency to concentrate all activity in an absolute being cooperates in the development of occasionalism is seen from the following statement of the Cartesian Sylvain Régis (in his *Cours de philosophie)*:"In order to produce actions properly so called one must act of oneself and by oneself, i.e. by one's own power, and it is certain that only God can act thus. From this it follows that only God is a truly efficient cause."

These thoughts came to consistent development in the Netherlandish Cartesian Arnold Geulincx (the name is pronounced Gölinks). He was born in Antwerp in 1623 of Catholic parents, studied at Louvain and became professor there. As a Cartesian he was in ill favor, and they contrived to drive him from his post. He then went to Leyden, *[240]* where he went over to Protestantism. In Leyden too Cartesianism was struggling with the old scholasticism, and Geulincx met with much opposition. After a series of straitened

years, in which he worked as a teacher of philosophy, he died of the plague in 1669, shortly after having obtained a professorship at the university. The most important of Geulincx's works that appeared in his lifetime is the *Ethica* (1665) (which, however, appeared complete only later, in 1675). The occasionalist doctrine appears partly as the background of his ethics, partly — more fully — in the Metaphysics published after his death (1695). — His collected works have recently (1891—93) been published by *Land*, who has also written an interesting monograph on him. *(Arnold Geulincx und seine Philosophie.* Haag 1895). In the following characterization we shall keep especially to the thoughts that are of significance for Geulincx's ethical standpoint.

It is a peculiar mixture of rationalism and mysticism that characterizes Geulincx's philosophy. And at the same time it shows itself clearly in him how the problem of causality grows out of the Cartesian doctrine of substance.

Geulincx sets his ethics in sharp opposition to the prevailing, Aristotelian ethics, which built on man's striving after happiness and laid great weight on practice and unconscious habit. The ethical consists, according to Geulincx, in obedience to the law of reason that God has implanted in us. Not in obedience to God's will; for this is fulfilled whatever we may do, and besides: everything, be it ever so exalted and holy, is subject to the discussion of reason. Ethics presupposes the knowledge of nature. For man must come to know his position in the world in order to be able to act rightly. Self-examination (inspectio sui) is the foundation of ethics. It teaches me that only my thought and my will belong to me, whereas my body is a part of the material world. And in the material world I have produced nothing and can produce nothing. For by what right dare I assert that I produce something, when I do not know at all how it comes to be? And besides: my acting cannot reach beyond the limit of my being, cannot flow out to other things. My
[241] will has significance only for my own soul. Just as little can things act on one another or on me. How can the acting of a thing, which is its own inner state, acquire significance for other things? It does not itself know, after all, how the change outside it is produced; its inner state does not, after all, extend wholly over into the other things! — If you think that you yourself move your limbs, although you do not know how it happens, you might just as well think that it is you who composed the Iliad, or who makes the sun go across the sky, and the child in the cradle could with the same right believe that its will directly sets the cradle going when the mother fulfills its wish to be rocked. If my willing and acting are to have significance beyond my own being, the Deity, the only true and real cause, must intervene. The same one who put motion into matter and gave it laws also formed my will and united it with the material body in such a way

that will and motion come to correspond to one another, just as two clocks that go alike and strike at the same time, not because the one acts on the other, but because they are produced by the same one. Better than this image, which had already been used by Cordemoy and has become classic in the discussion of the relation between soul and body, is an image that Geulincx uses in his *Annotata majora in principia Cartesii.* When I can procure food and clothes for money, it is not the metal's own natural power that brings this about, but its value rests on human institution (ex hominum instituto, qvo valor iste ei consignatus est); in a similar way it is not the material motion itself that is able to produce sensations and representations in me, but this happens by virtue of divine institution (nulla vi sua, sed instituto qvodam decretoqve divino). — So strongly does Geulincx emphasize the Deity as the *only acting* being that *Land* is probably right that the expression occasionalism does not quite fit his conception, since what *occasions* does after all to some degree *cooperate.* He stands in any case nearer to the standpoint of Spinoza and Leibniz than the other "occasionalists".

By the sharp emphasis on the difference and self-containedness of the individual beings the problem of causality, which concerns the possibility of a transition and *[242]* connection between them, clearly emerges. And since the divine will is conceived as the unifying power that embraces all the individual beings and acts in them, it is also intimated that the problem becomes wholly insoluble if a unity in existence is not assumed behind all differences, a continuity that extends through them all and beyond them all. Since, finally, it is said that not motion itself, but an element that is put into it or attached to it, is able to produce the mental states, a deeper solution of the problem of soul and body than the Cartesian one is intimated. Admittedly all this appears in Geulincx in a theological or mythological form that prevents full clarity and consistency. But his philosophy offers great interest through its energetic emphasis on the point that the acting of a thing and its being cannot be separated.

Geulincx himself draws from his doctrine the ethical consequence that we stand as spectators before the world, or rather before God (for the world cannot itself produce a picture of the world in me; only God can do that, and as a spectator of what takes place in the world I am therefore a constant miracle: ego ipse spectator mundi maximum sum et juge miraculum!). We are absolutely dependent on the will of the Deity. The supreme principle in ethics then becomes that where I can accomplish nothing, I shall also will nothing: ubi nihil vales, ibi nihil velis! The chief virtue in Geulincx's ethics therefore becomes humility. A characteristic contrast to the magnanimity of Descartes and of Telesio and Campanella! The vigorous self-assertion of the Renaissance had

given place to its opposite. I have, says Geulincx, no right to anything, not even to myself. It cannot then be my task to further my happiness, but only to do my duty. Yet happiness is perhaps furthered in just this way in the best manner; for, thinks Geulincx, our misfortunes really come from our all too eager striving after happiness. — Into the details of this ethics of thoroughgoing resignation we cannot go either. In many of Geulincx's ethical remarks a fine and heartfelt mood comes to light. —

[243] Occasionalism gets its extreme form in Nicolas Malebranche. The mystical side of Cartesianism decidedly presses forward into the foreground here. Malebranche, the son of a high official in Paris, entered the Oratory in the year 1660, at the age of 22. His weak health had already, while he was a child, nourished in him the desire for a quiet, contemplative life, and the last part of his life he spent for the most part in his cell. By chance he came one day upon one of Descartes's writings (Traité de l'homme), and the clear development of thought interested and inspired him to such a degree that for emotion he could scarcely finish reading it. From that moment his life was dedicated to philosophy. His chief work is the *Recherche de la vérité,* of which the first volume appeared in 1674. Of his other works, particular mention must be made of the *Entretiens sur la Métaphysique* (1687), which many regarded as his finest work. As an author Malebranche is clear, elegant and witty, but also somewhat verbose. The religious and the philosophical interest go together in Malebranche, and he thought he had found the way to a complete reconciliation between them. He did not see that his thinking led him to results similar to those that Spinoza, whom he abhorred (le misérable Spinoza, as he calls him), had already reached earlier. Just as little as Spinoza does Malebranche seem to have known the earlier occasionalists (Cordemoy and Geulincx), a proof that a natural tendency of development was stirring here. Malebranche kept his freshness of mind into his high old age and died in 1715.

Malebranche begins in his Recherche de la vérité by investigating the causes of errors and the means of freeing oneself from them. A chief source of error is the belief that the senses, which are given us for practical purposes, should be able to teach us to know the essence of things, and that we rightly transfer the sensuous qualities that we sense to the things themselves. Material things are in themselves only extended, not colored, hard, soft etc. The imagination leads us astray, since it stands under the influence of sense perception and is also dependent on our feelings. Malebranche undertakes an interesting investigation of the psychological and physiological nature of the feelings
[244] and here states views (particularly on the role that the influence of the nervous system on the blood vessels plays in emotion) that to this very day have attracted attention. —

The errors both of sense perception and of the imagination derive from the action of the body. But the understanding too, which after all belongs to the soul in itself, must be subjected to a critique. The chief question is, whence our representations (ideas) come. Only our own mental states and activities do we apprehend immediately; other things we know only through the representations or ideas we have of them. Now these ideas can be produced neither by the things nor by ourselves: for no finite being can be an absolute cause. To be a cause is something divine, and it is therefore paganism to ascribe causality to finite things. There can be only one cause, the Deity; the natural causes are only occasional causes (causes occasionelles — hence the name of this whole tendency of philosophy). When a ball is set in motion because another pushes it, this other ball is not the cause. There is no necessary connection between the motions of the two balls[49]). The collision of the balls is only the *occasion* for the author of all material motion to carry out the decree of his will by imparting to the pushed ball a part of the motion of the pushing one. The moving force of bodies is only his will, which upholds the bodies. And it is likewise with the human mind: without God's action it can neither sense nor will. One cannot help oneself by assuming that God has given bodies and souls the capacity to act, imparted to them a part of his power. For God cannot do that: God cannot create gods! Just as the true religion teaches that there is only one God, so the true philosophy teaches that there is only one cause.

Our knowledge, then, can only be explained by our being dependent on God with regard to all our thoughts. God stands in an immediate relation to every mind; he is "le lieu des esprits". Through this connection with God, the author of all things, our mind beholds the created things: the representations of them must indeed be in God, since he could not otherwise have created them. Strictly speaking, all the particular representations we have are only limitations of the representation of God, the infinite *[245]* being. Every act of thought is a closer determination of an indeterminate representation of what is, which never lets go of us. In a similar way all our striving is at bottom a striving toward God; only in union with him is true happiness, and it is he who awakens the need of happiness in us; but we often stop at a finite, limited good and regard it as the highest good — just as we regard the finite beings as true causes. Malebranche's ethics thus forms a complete analogy to his theory of knowledge.

A difficulty in Malebranche's doctrine that we behold everything in God is this: why it is then really necessary to assume an actual material world. Our knowledge of nature, according to Malebranche, finds its full explanation in God's action on us, and the actual material world is not needed at all: God's representation of it (the intelligible extension,

as Malebranche calls it) is enough. According to one report Berkeley, on a visit to Malebranche shortly before the latter's death, is said to have made this objection to him. The same objection also came out in a correspondence that Malebranche's former pupil, the mathematician Mairan, carried on with him. Mairan puts to Malebranche the alternative: either material extension does not exist (since, according to Malebranche, it is not needed to explain our knowledge of it), or else material extension, as Spinoza taught, belongs to the very essence of God (it then being not merely the intelligible extension, not merely the representation of extension, but actual extension itself that we behold in God). The old mystical philosopher was ill at ease at these objections, and the correspondence with Mairan closes with his appealing to religious faith as the ultimate foundation of all certainty. —

While Cartesianism in the series of thinkers we have discussed here led over into a theological idealism, in another circle of thinkers it helped to produce new forms of skepticism. It contained several elements that could lead in this direction. In the first place the principle of methodical doubt, the whole critical tendency with which
[246] Descartes's philosophy began. Not everyone could pass over from doubt and analysis to dogmatic settlement with the same speed as the founder of Cartesianism. In the second place the very dogmatism of Descartes could not but challenge criticism, especially as the dogmatic results themselves raised new problems, or old problems in a new form. In the third place Descartes himself maintained an opposition between positive religion and scientific knowledge, an opposition that, when more sharply stressed, could lead to denying any decisive value to the philosophical results Descartes might have reached, since they did not, after all, lead to the highest truth. These motives worked in varying degree and form on a series of men at the end of the seventeenth century who, for all their differences, have this in common, that at least for a time they stood under the influence of the Cartesian philosophy and later went over to a more or less pronounced skepticism in philosophical respects. Only a quite brief characterization of them can be given here.

Joseph Glanvil (1636— 1680) worked his way in his youth out of Puritanism and the scholastic philosophy that reigned in Oxford during his time as a student. He became a great admirer of Bacon and Descartes, especially of the latter, and cherished great hopes for the future of the science of experience. In his chief work *(Scepsis scientifica* 1665), which he dedicated to the Royal Society, he spoke up for unprejudiced inquiry; he counted himself a skeptic only in the original meaning of this word, in which it denotes one who seeks. He joins Descartes entirely in the demand for a mechanical explanation of

nature (the mechanical hypothesis) as opposed to the scholastic doctrine of qualities and forms. Nature is for him the great machine (the great automaton). But he stresses more strongly than Descartes that all our knowledge of nature is hypothetical; he traces out with great zeal all the grounds that indicate that our knowledge will always be afflicted with imperfection; and he shows how precisely the advance of experience brings new phenomena that may necessitate a change in the hypotheses we formed earlier. Like the occasionalists, but independently of them, he emphasizes very strongly the difficulties *[247]* of conceiving an interaction of soul and body, and like Geulincx and Malebranche, but before them, he comes to see the difficulties that lie in any causal relation whatever. "No cause", he says (Scepsis scientif. cap. 23), "do we know by immediate intuition, but only through its effects. When we infer that one thing is the cause of another, we build only on its constantly going together with it; for *the causal relation itself cannot be sensed* (causality it self is unsensible). But an inference from concomitance to causal relation is not compelling." Glanvil here anticipates Hume. He lays particular weight on such cases in which cause and effect are highly unlike one another, and where we therefore can have no clear and necessary insight into the relation between them. When, for all his critical and empirical sense, he defended the belief in witches, it was because he held it dogmatic to deny witchcraft, since one could not know a priori what causal relations took place in the world. His skepticism thus led him not to put full trust in the "mechanical hypothesis". He is here, like the kindred spirits who are to be named in what immediately follows, evidence that the new thinking had not succeeded in driving out the medieval notions.

High above all the men in this group, and high up into the sphere of eminent minds as a whole, towers the figure of Blaise Pascal. Pascal too (1623—1662) was influenced by Descartes and always kept some Cartesian views in his circle of ideas. But his violently and deeply stirred inner life led him to turn his back on philosophy. And yet there is always a certain kinship between Pascal's religious thinking and Descartes's philosophizing: both proclaim conviction free of authority. In his *Fragment d'un traité du vide* Pascal speaks out against the power of authority in the scientific domain, and in the treatise *De l'esprit géometrique* he gives clear indications of a doctrine of method, with express acknowledgment of Descartes's Discours de la méthode. Later, after his religious breakthrough, he fights on the side of the Jansenists against the validity of papal authority in the face of facts. From Rome he appeals to heaven. And on the occasion *[248]* of Galileo's condemnation he utters (in the 18th Provincial Letter) the often repeated saying that if the earth really goes round, no decree will be able to prevent it; it will go

round with all men, whether they believe it or not. Like Bruno and Descartes he assumes the infinity of the universe and uses this thought in his religious agitation to make the soul dizzy with the feeling of being a vanishing element. He agrees with Descartes in his conception of soul and body. The body is to him only a machine, an automaton. The sharp opposition between thought and extension he uses to impress upon us man's spiritual worth: as a material being man is admittedly to be reckoned as nothing over against the enormous masses; he is only a reed — but a *thinking* reed! This is a genuinely Cartesian thought. But even though this thought could fill Pascal with enthusiasm, it did not for him contain the highest. Above the world of matter and of mind there stood for him the supernatural world of love, which comes into being for man through God's immediate revelation. Religion is the heart's immediate experience of God (dieu sensible au coeur). What philosophical proof can lead to is at best an abstract God, perhaps the God of truth, but not the God of love, the one true God. Deism and atheism come for Pascal to much the same thing. What he yearns for is to have his heart filled, and no inquiry could help him to that. But besides this direct need to attach his feeling to an infinite object, another psychological motive is also at work in Pascal, which conditions the decisive difference between him and a figure such as Malebranche. It is the anguish with which he feels himself existing in a world without limits, without stability, a world concerning which knowledge can arrive at no firm results. Skeptics and dogmatists contend; neither of the two parties can be proved right — and yet one of them must be right; the very contention between them shows that the skeptics are at bottom right. It is no use wanting to ground the first principles of knowledge on human nature: for where is human nature to be found? It changes constantly, or else it is ruled by habit.
[249] In practical respects man is driven on by restlessness and vanity, or he follows custom and usage. Man is full of oppositions and contradictions. Doubt and uncertainty reign within him. "He is to be judge of all things and is yet a wretched worm; he is to be the bearer of truth and is yet a sewer of uncertainty and error; he is at once the glory of the universe and its disgrace!" — In the face of all these contradictions Pascal knows only one rescue: to take refuge in the historically revealed Christianity. There his own loving and trembling soul found rest. In order to lead others there he had devised an account that rested on the biblical history and was to show how all oppositions and contradictions in the nature and condition of man fell away through the revelation given in the biblical writings. It is certainly a piece of good fortune for Pascal's memory that the positive part of his apology for Christianity was not carried out, so that his thoughts *(Pensées)* on human nature and its condition, which were only to make up the

preparatory part of the apology, can now act by their own power. It is the strong feeling of the weight and value of life that animates Pascal. In opposition to the rationalistic tendency, which in Descartes's philosophy received such good support, Pascal appeals to the living personality, to the heart, to conscience as the ultimate decisive motive in the view of life. A century before Rousseau he raises the question of the value of scientific knowledge for personal life. In the face of his need of love and his anguish of soul the answer becomes that "the whole of philosophy is not worth an hour's trouble!" And he had the plan of writing against those who go too deeply into the sciences (qui approfondissent trop les Sciences). That he himself found the solution in the Christian, particularly the Catholic, more particularly the Gallican, and most particularly of all the Jansenist doctrine, he would not regard as the result of habit or tradition; he would not see that here he still left a place open for the skeptical arguments that he himself had brought into the field. And when he was asked what one was then to do — what one was to resort to in order to bring the doubting thoughts to rest, he answered as a good Cartesian: We are automaton as well as we are mind. Both must be brought to believe, *[250]* the latter by conviction, the former by habit. And when conviction will not come, then begin with habit: act as if you believed; take holy water and have masses said; it will make a beast of you and make you believe (cela vous fera croire et vous abêtira)! That is cutting the knot — with the help of the philosophy that he at once uses and combats. —

Pierre Daniel Huet, Bishop of Avranches (b. 1630, d. 1721), likewise wished to abolish free inquiry in favor of religious faith, which in him appears more pronouncedly than in Pascal as the faith of the Church. (Cf. *Barach: Huet als Philosoph.* Wien 1862). He also still holds to essential parts of the scholastic philosophy. For a time he had been a Cartesian, and he felt great enthusiasm for the clear and simple method of the new philosophy. But he later came to doubt whether there could be any rational criterion of truth at all: we must then again have a criterion that this is the right criterion of truth — and so on to infinity! Descartes urges us to begin with the simple and from it to pass over to the composite; but there is nothing so easy and simple that it could stand as indubitable! Descartes thinks that we cannot doubt thought itself; why not? We know thought, after all, only in the same way as everything else: by directing our attention to it — that is, by a new thought. An absolute beginning is therefore not reached. Descartes's "Cogito, ergo sum" is an inference whose legitimacy can be called in doubt, and his proof of the existence of God falls away when one sees that existence is not a property. Only the authority of tradition and of the Church could furnish a foundation for knowledge, and Huet, who was a learned humanist, offended at the Cartesians' disdain for the ancients,

therefore seeks to show that the religions and philosophical systems of antiquity are due to a tradition stemming from the Jewish people.

Pierre Bayle (1647—1706) had come to know Cartesianism during his student years in Geneva, and in him it now displaced the scholastic philosophy into which he had been introduced at a Jesuit college. He admired especially the Cartesian demand for
[251] clear and distinct concepts and for leading complicated questions back to simple and immediately evident principles. To several of the doctrines of Cartesianism he later (during his university work at Sedan and in Rotterdam) took up a critical attitude. He was quite as much a scholar and man of letters as a philosopher; he took an interest in literary phenomena and speculative opinions in their motley multiplicity, and the need of clear and distinct concepts led him to sharp pointing up of the difference between standpoints, to decided emphasis on the sting of problems, and to the unmasking of illusory solutions. In his masterly characterization of Bayle Ludwig Feuerbach shows that Bayle was by no means a systematic doubter, but that it was the difficulty and multiplicity of the individual subjects that gave birth to doubt in him. Philosophy was for him essentially criticism, its significance more negative than positive. Its results decidedly conflict with those of theology, which Bayle shows in the case of a single dogma (that of the Fall) by setting up a series of theological and philosophical fundamental propositions that prove to be wholly incompatible with one another. By this setting up (in his *Réponse aux questions d'un provincial)* Bayle opened the most important debate on the problem of evil since Jakob Böhme's "Aurora". Already in his *Dictionnaire historique et critique* he had maintained (especially in the article on the Manichaeans) that the assumption of two principles in the world, a good and an evil one, would be very difficult to refute if it were defended by unprejudiced and acute thinkers. If Augustine had not left Manichaeism, he would, thought Bayle, have been able to make something of it that could have confounded the orthodox. By the way of reason this doctrine could not, in his view, be refuted. According to Bayle a choice must be made between reason and faith, and faith shows itself all the more glorious the more contrary to reason the dogma is that is to be believed. Bayle himself had certainly made his choice and was sincere in his sharp distinction. But this choice has not prevented him from acknowledging a humanity independent of all dogmatism. He lays (especially in *Pensées diverses à l'occasion de la comète)* great
[252] weight on the natural instincts that keep the individual and the species alive without conscious reasoning. And as regards in general the conduct of men in public and private life, he maintains that it is determined far more by their natural temperament and their character than by their belief or unbelief. He can therefore (which aroused great offense)

very well conceive of a state of atheists. Ludvig Holberg later remarked, precisely on the occasion of this line of thought of Bayle's, that it made "the whole disputation" on whether freethinkers could live morally "in vain" (Ep. 210). This line of thought stands in natural connection with Bayle's zealous struggle for toleration, and no less with his maintaining the independence of ethics over against theology. The Cartesian method is applied by him here in an interesting way: just as the proposition that the whole is greater than the part is so clear and distinct that no revelation can be accepted if it conflicts with it, — so too the simplest ethical judgments, by which we distinguish between good and evil, are so clear that no revelation can abolish them, but the validity and value of every revelation must conversely be tested with them as the standard. If one rejects the natural light here, one has no standard at all. On this point a possibility comes out in Bayle that would make the conflict between faith and knowledge lose its significance: namely this, that the natural instincts and the immediate sense for the true and the good were sufficient to carry life! It became the task of the eighteenth century to develop this possibility, which the Renaissance had already seized in its discovery of man, but which the victorious reaction had not allowed to get beyond the narrow circle of the thinkers. The thought also still needed a fuller development. —

3. Pierre Gassendi

formed a counterweight to Descartes, especially in the French world; yet more by his method than by his results. He wished to be a philosopher of experience and to plead the cause of sense perception against Descartes's intuitive and deductive method. While Descartes, with greater speed than his own method allowed, rose to the highest ideas and *[253]* tended to ascribe to them an immediate grounding in the human mind, Gassendi maintained the empirical origin of our concepts. The very polemic *(Disqvisitio metaphysica.* Amsterodami 1644) that he published against Descartes is among the most interesting documents in the debate between the empirical and the deductive school. When Gassendi nevertheless ends in spiritualistic results, this is due to his clerical position, which made it necessary for him to combine theology and empirical philosophy. He was born near Digne in the south of France in 1592, became a priest in 1617 and later professor of philosophy at Aix, a position he later exchanged for a professorship of mathematics in Paris, in which he worked until his death in 1655. In his youth Gassendi published a sharp critique of the scholastic philosophy; after that the severe decree of 1624 against the new doctrines and the condemnation of Galileo caused him to hold back. He occupied

himself with physical and astronomical studies, but did not venture to declare himself for the Copernican doctrine. In his later writings he developed the atomistic doctrine partly in the form of a renewal of the Epicurean doctrine. He defends the personality and philosophy of Epicurus against the current attacks. These works became of great significance, since the atomistic theory, which Bruno and Basso had prepared, was excellently suited for use by modern natural science. Gassendi was an admirer of Galileo, and the fundamental propositions that he sets up for the knowledge of nature betray Galileo's influence *(Opera omnia.* Lugd. 1658). Starting from the premise that nothing can become nothing or arise from nothing, and that something must therefore subsist through all changes, he shows that if change is to be explicable at all, the something that subsists through all changes, i.e. matter, must be a multiplicity of solid and indivisible elements, i.e. atoms. Mathematically, division can be continued to infinity, but physically divisibility has a determinate limit. All changes in nature are to be explained by the motions of the atoms: all causes are causes of motion. Between the atoms there is an
[254] empty space; otherwise their motion would not be possible. Space, like time, is a thing of its own kind, to be called neither a being nor a property. Only by contact can motion arise and cease; therefore all causes must be material. The soul moves the body only insofar as it (as vegetative and sentient soul) is itself material. Gassendi differs from Descartes by maintaining that solidity, not merely extension and mobility, must be ascribed to matter, and by setting a limit to physical divisibility. He teaches moreover (under the evident influence of Galileo) that it is not motion itself, but the impulse of motion (impetus), that is constant; the impulse of motion does not fall away because motion passes over into rest. These improvements of the new mechanical view of nature mark a significant advance. On the other hand Gassendi recalls the predecessors of Descartes and Galileo, especially Telesio, by assuming sensation in all atoms and using this assumption, as well as the notion of a world-soul, in his philosophy of nature. A similar half-heartedness appears in his attempt to combine spiritualistic notions with his atomistic and mechanical doctrines: the atoms are created by God; and to the proposition that all causes are material God and the supernaturally implanted soul or part of the soul (anima rationalis or intellectus) form an exception. This compromise was even more inconsistent than the Cartesian one, since the soul itself was split into two parts, a material and an immaterial. But it brought the advantage that the learned, and especially the theological, world grew used to not regarding atomism as an absolutely godless doctrine. Natural science could then make use of the atomistic hypothesis undisturbed, as its needs required.

4. Thomas Hobbes

a. Biography and Characterization

When in the year 1588 the Spanish Armada was about to put to sea in order to attack England, the terrifying rumors of war caused the wife of the clergyman in Malmesbury *[255]* to give birth prematurely to a little son. She bore, says the latter later in a versified autobiography, twins, namely me and fear. He derived from this his timorous, peace-loving temperament, which nevertheless — as his life shows — did not exclude energetic thinking and a delight in literary feuds. The boy, Thomas Hobbes, received his first instruction in his native town and at the age of 15 came to Oxford, where he learned the scholastic logic and physics, which interested him very little. He left the university early and became tutor to a young nobleman of the family of Cavendish. To this family he remained attached all his life. His position gave occasion to repeated journeys round Europe. He gathered experience of the world and cultivated literature with zeal, especially the classical historians and poets. He followed the events in his native country with an attentive eye, and a translation of Thucydides which he published in 1629 was not prepared without regard to the political storm clouds that already in the first years of the reign of Charles I threatened on the horizon. For a time, however, his interests were drawn in an entirely different direction. By chance he came upon Euclid's geometry and thereby discovered that there existed a rigorous deductive science. He had not suspected this until then, for mathematics did not yet belong to the regular subjects of instruction in England, and was indeed even regarded as devilry. He now obtained a model for his thinking. A mathematician by profession he did not, to be sure, become; his later attempts in that direction turned out most unfortunately and involved him in long literary feuds in which he decidedly drew the shortest straw. But the foundation of his philosophical tendency had now been laid. The great weight he lays on deductive thinking places him close to Descartes, while it places him in decided opposition to Bacon, with whom he had stood in a personal relation, having in Bacon's last years been of assistance to him as secretary and in the translation of his books into Latin. From Bacon he received no decisive philosophical impulses, even if one can perhaps show on individual points that, after his own peculiar philosophizing had been set going by impulses from other quarters, he took up Baconian ideas and wove them into his system. *[256]* The problem that set his thinking going he is said, according to what his first biographer relates, to have received in a conversation that probably took place in a scientific circle in Paris, and in which the question was raised in passing what sensation is. As this

question was not answered, Hobbes set himself to think about it. It then dawned on him that if material things and all their parts were always at rest or in uniform motion, all difference (discrimen) between things, and consequently all sensation, would fall away. From this he then concluded that difference of motion (diversitas motuum) is the cause of everything. Now he thought, by his own account, of nothing but motion day and night, waking and sleeping. It had dawned on him that the deductive method, which he had not long before come to know in its most perfect form, was excellently suited to application on the basis of the fundamental proposition that everything is motion. Probably he also at the same time (c. 1630) arrived by his own path at the conviction of the subjectivity of the sense qualities[50]). That Galilei had already in 1623 stated this principle, Hobbes did not know. Only after it had dawned on him does he seem to have become aware of Galilei's writings, and later on a journey to Italy he came to know the man of whom he himself later (in the dedication of De Corpore) said that "he first opened to us the first gate to the whole of physics, namely the nature of motion." Thereby his already begun line of thought received a more definite form and a firmer basis. Galilei is said also to have drawn his attention to the possibility of treating ethics deductively in analogy with physics. If this account is correct, then Hobbes's whole system had thereby found its provisional conclusion for him. For a long time yet, however, only in his mind and thought and in the confidential exchange of ideas that took place within a circle of men whom the monk Mersenne, Descartes's and Gassendi's friend, gathered about him in Paris, and to which Hobbes also belonged. "From the moment," says Hobbes in his autobiography, "that I had communicated my ideas to Mersenne, and he in turn
[257] had made others acquainted with them, I too was counted among the philosophers. His cell was better than all schools." It was also Mersenne who later sent Descartes's Meditationes round to various thinkers, among others Gassendi and Hobbes, and thereby called forth one of the most remarkable philosophical discussions of the 17th century. Hobbes's critique of Cartesianism is an interesting document for the illumination of his own philosophy. Hobbes, to whom the late Lord Cavendish had secured economic independence, now intended, when in the year 1637 he came back to England after several years' absence, to work up his ideas into a threefold system: Corpus — Homo — Civis, a doctrine of the laws of matter, of man and of society, as far as possible by the deductive method on the foundation of the general laws of motion. But the execution of this great plan was prevented for a long time by the outbreak of the English Civil War. His temperament made him look with anxiety on the political strife growing in passion. It seemed to him to shake the foundation of all ordered social life. He feared that the

elementary drives would stir with unbound and unconditioned violence. His experience of the world had taught him how much the impulse of self-preservation and egoism can work not only for good but also for evil, and that a strong authority is needed to keep the stream of these forces within its proper bed. He was ultra-conservative precisely because in his theory he was ultra-radical and went back to the purely elementary presuppositions. But surely his moving in the circles of the high nobility also contributed much to determining his political sympathies. He therefore misjudged the significance of the English Revolution, although the grounds he gave for his condemnation were purely naturalistic and thereby diametrically opposed to the grounds that held the adherents of the Stuarts together. For Hobbes the principle of authority was a derived principle; for them it was an absolute, divine, supernatural principle: this was the great opposition that came out as soon as Hobbes worked out his political theory. And it is the grounds given, not the results, in Hobbes's political theory that have given it its lasting and in a certain sense unique significance in modern ethics and political theory. In the year 1640 he composed a work that contained his psychological and ethical-political theory *[258]* in its main features. This work, which circulated in copies and was later published in two parts, Human Nature and De Corpore politico, has recently (1888) been published by TØNNIES from the oldest manuscripts and with the original title *Elements of Law.* It is one of his freshest and most instructive works and ought to form the foundation of every study of Hobbes. It was composed with the troubled circumstances of the time in view and as a word of warning. Hobbes believed that he would have been exposed to personal danger from the side of the opposition if the King had not dissolved Parliament. When the unrest increased, and when the new Parliament began to proceed severely against the King's men, he fled (at the end of 1640) over to France in order there to be able to continue his studies in peace. But the grave circumstances of the time gave him no peace to turn his thoughts away from political problems, and he then worked up the last part of "Elements of Law" into an independent work *(De cive),* which appeared in 1642 in a small, in 1647 in a larger edition. It was intended to be the third part of the whole system. From the earlier treatise it differs by stressing far more strongly the opposition between the state of nature and life in the state, and by insisting on the necessity of the sole authority of the power of the state in religious matters. Hobbes was an opponent on principle of all hierarchy, the Protestant as well as the Catholic. This made his stay in a Catholic country and in a circle of exiled clerical royalists insecure. There came a formal breach when he had once more laid his studies in natural philosophy aside in order to write a political work. This was the famous work *"Leviathan* or the

Matter, Form and Power of a Commonwealth ecclesiastical and civil", which appeared in London in 1651. It was named after the mighty sea monster mentioned in the Book of Job, of which it is said: "Upon earth there is none like him; he is made without fear." With this mighty one Hobbes compares the absolute power of the state. He here carries through the doctrine of sovereignty founded by Bodin and Althusius to its extreme consequences — but on a naturalistic basis and in an antihierarchical spirit. That it was not especially the absoluteness of the royal power but of the power of the state that he
[259] asserted was interpreted as if with this work he were angling for Cromwell's favor. And the episcopal royalists were indignant at the statement (in a concluding section which was left out in the later Latin edition of 1670) that just as Elizabeth had overthrown the Catholic hierarchy, the Presbyterians had overthrown the Episcopalians, only to be overthrown themselves by the Independents — "and so we are brought back to the independence (independency) of the first Christians." The consequence was that Hobbes lost his position as teacher of mathematics to the young King Charles II and was denied access to the court. In England conditions were now calmer; there was again a firm and ordered power of the state, an exerciser of sovereignty, and Hobbes thought that it was now safest for him to return home. After a toilsome journey in wintertime he came home to England at the end of 1651, and he could now enjoy the benefit of complete freedom of the press to work in peace on the publication of his works, which he had never lost sight of. In 1655 appeared *De corpore,* which comprises logic, the doctrine of fundamental concepts (philosophia prima), the doctrine of motions and magnitudes, and the doctrine of physical phenomena. In 1658 appeared *De homine,* which for the most part is an optics (to illuminate the nature of the sense of sight) and otherwise gives only brief outlines of the psychology of language and of the feelings. This second part of the system can measure itself neither in content nor in form with the first part of Elements of Law or with the opening chapters of Leviathan. The third part of the system was the earlier published work De cive, and the whole system was thus now complete. Then came the Restoration, which Hobbes greeted with joy. He came into favor with his former pupil Charles II, who often conversed with him. He lived with the Cavendish family as an old student, much occupied with mathematical and theological polemics. His antihierarchism was made into atheism, and a Hobbist counted as the same as what was later called a freethinker. Although he ascribed to the power of the state the sole right to determine what should be taught, he himself interpreted the Bible in a critical and rationalistic spirit. For him the original faith of Christianity was only this, that Jesus was the Messiah, not any speculative doctrine whatsoever. The eternity of the

punishments of hell he rejected. He assumed no incorporeal spirits. But for the most part *[260]* he referred to the incomprehensibility of the doctrines of faith and advised accepting them without thinking about them, just as one does best to swallow bitter pills whole without chewing them. Hobbes preserved his physical and mental energy into his high old age; at 87 years of age he translated the whole of Homer into iambic meter. He died at 91 years of age (1679).

Hobbes is an acute and energetic thinker. Thinking, he says in the preface to De corpore, will be esteemed the highest enjoyment if one but once experiences the joy that is caused by the meeting between our mind and the glorious universe (pulcherrimi mundi cum anima congressus). He made the most thoroughly considered attempt in modern times to make the knowledge of natural science the foundation of all our knowledge of existence. It is the most profound materialistic system that has been set up in modern times. But Hobbes's writings, which are distinguished by their vigorous and clear exposition, contain besides many interesting remarks in logical and psychological respects. He may be called the founder of English psychology, and thereby of the properly English school in philosophy. Above all, however, it was his ethical and political points of view that became of far-reaching significance. By his bold, though one-sided naturalism he set people's thoughts in motion, partly by provoking them. He broke with scholasticism in the domain of the human sciences, just as Copernicus had broken with it in the domain of astronomy, Galilei in that of physics and Harvey in that of physiology. By the side of these men Hobbes, with justified self-esteem, places himself as founder of the science of society: this (he says in the preface to De corpore) is no older than his book De cive. By his naturalistic grounding of ethics and politics he roused a movement which has aptly been compared with the one Darwin roused in the 19th century.

b. First Presuppositions

Science stands for Hobbes in opposition partly to sensation and memory, partly to theology. It does not, like sensation and memory, stop at the individual phenomena, *[261]* but seeks to know them from their causes; now it goes from cause to effect, now from effect to cause. From this it follows that science can only concern what has an origin; thereby theology is excluded, the doctrine of the eternal God, who has not come into being. — So at least I define science, says Hobbes. Perhaps someone wishes another definition; that is his affair. Definition is a free and arbitrary matter, a mere explanation of what we mean by a name, and the use of names is arbitrary. Science begins with the definition of the underlying concepts. One arrives at these concepts by analysis

(resolutio) of what is given to sensation (a sensibus ad inventionem principiorum). The validity of the principles, the fundamental concepts, itself cannot be the object of proof, only of definition. They are known in themselves — otherwise they are not principles. Since they are fixed by arbitrary naming, they are produced by ourselves. It is art or construction, not science, that fixes the principles: here we ourselves produce the truth: ratiocinationis prima principia, nempe definitiones, *vera esse facimus nosmet ipsi* per consensionem circa rerum appellationes (De corpore. cap. 25, 1., cf. cap. 3, 8—9).

This sharp emphasis on the purely arbitrary in the setting up of the first presuppositions Hobbes makes properly only in the work De corpore; in Elements of Law it does not appear so clearly. It is due to the ever more predominant deductive method: when all true science is deduction, the presuppositions of deduction, which cannot themselves be deduced, must come to stand as purely arbitrary. But then it cannot be understood what Hobbes means by saying that the principles must be "known by themselves" (per se nota), if this is to mean more than that one is conscious of what is meant by the words employed. And this immediate consciousness itself is then a presupposition which properly ought to be set up by itself. It is a kind of intuition or, as Hobbes himself in one place[51]) calls it, a reflection. Besides, Hobbes declares, as we know, that we arrive
[262] at the principles by resolution or analysis (resolutio, analysis) of the given; it is then not altogether arbitrary what principles we set up, and there is more in the setting up than mere naming. Indeed, Hobbes says expressly that analysis is an inference from the given to the principles or definitions (analysis est ratiocinatio a supposito ad principia, id est, ad definitiones. De corp. cap. 20, 6): thus the first principles of inference are themselves set up by inference, in so far as they are set up by analysis. It turns out, then, that the situation in setting up the principles is more complicated than the pure theory of arbitrariness assumes.

Hobbes has also in another way limited the arbitrariness in setting up the principles, or in naming. One cannot, for instance, give one and the same thing two mutually opposed names. Of this principle Hobbes declares (De corp. cap. 2, 8) that it is the principle of all inference, and so of all science. But it is surely not itself set up arbitrarily; it precisely restricts arbitrariness: if I have given a thing a name, I cannot add to it *any whatever* other name. The same holds everywhere we construct: only the first step can be purely arbitrary; the following ones are determined by the first. — Apart from the wavering and one-sidedness that shows itself here in Hobbes, it is his great merit to have brought out so sharply the active and constructive element in knowledge. Neither in Descartes nor in Bacon did this come sufficiently forward, and the decisive point where

one turns from induction and analysis to deductive thinking therefore did not receive sufficient illumination with them. —

When we have now arrived by the way of analysis at the first principles, the task is to derive the phenomena from them. As a general principle Hobbes sets up that there is a single cause of all the properties that the phenomena present, namely *motion.* This assertion, he thinks, needs no proof; it will be granted by anyone who thinks it through without prejudice (De corp. cap. 6, 5). To this proposition, that all change is motion, a series of other propositions attaches itself, which Hobbes sets up on occasion without examining their closer relation to one another or to the one first set up. Thus the general causal principle (which is set up in the work on freedom and necessity), the principle *[263]* of inertia, the proposition that the cause of motion must always lie in the motion of a contiguous body, and the proposition of the conservation of matter. In a single place, however, Hobbes proves the proposition that all change is motion by means of the causal principle and the principle of inertia, and for these two propositions too he sets up proofs on occasion, which, however, on closer examination turn out to presuppose precisely what was to be proved[52]).

From the proposition that all change is motion, and the propositions more or less closely connected with it, Hobbes now wishes to build up his system by the deductive way. After logic has first examined the method, and philosophia prima the most important fundamental concepts, geometry discusses the mathematical laws of motion, mechanics the effects of the motion of one body on another body, physics the effects of the motions that take place in the small parts of bodies, the doctrine of man and the state (what would now be called the human sciences) the motions that take place in men's minds and determine their conduct toward one another. If the system could be developed purely deductively, we should in these various disciplines merely have progressive special applications of the general laws of motion, starting from the first self-created presuppositions, through the series of steps Corpus — Homo — Civis. And this is evidently the ideal that stands before Hobbes. But just as we have seen that the doctrine of the self-creating activity in the fixing of the presuppositions is, by his own account, subject to considerable restrictions, so it turns out also that at three places in the system there is a break in the strict deductive connection, since the relations are too complicated for a single, progressive method to manage them. These three points lie at the transition from geometry to mechanics, from mechanics to physics, and from physics to psychology.

In the first place, the system would form a strict unity if mechanics could without more ado be derived from pure geometry, without wholly new presuppositions being

taken up. But in mechanics the cause, or the concept of force, comes forward, while
[264] geometry presupposes only extension and the general logical principles. That Hobbes lays no weight on the difference between geometry and mechanics is connected with his not distinguishing between ground and cause. Yet he admits the difference in so far as he concludes the geometrical section of De corpore with the remark: "In this section we have considered motion and magnitude in themselves, and in an abstract manner. The next section deals with the phenomena of nature, i.e. with the motion and magnitude of real bodies." (De corp. XXIV, 9).

Although Hobbes lets science go partly from effects to their causes, partly from causes to their effects, he is nevertheless clear that only when one goes from cause to effect does knowledge have strict necessity. When we begin with a given phenomenon and regard it as effect, the knowledge of the cause can always be only hypothetical: what one can find out can only be the manner in which the phenomenon *could* have been produced; but whether it really *has been* produced in this manner cannot be decided purely deductively. When we start from the given and seek its real causes, we do not ourselves produce the principles, but must try to find the principles which have been employed by the Author of nature in producing it. Herein physics differs from geometry and mechanics (De corp. cap. 25, 1. De homine, cap. 10, 5). — We must therefore, at the transition from mechanics to physics, halt our deduction in order to examine what phenomena are given, and form our hypotheses by analysis of them. The question is whether something analogous does not already hold for the presuppositions of logic, of "the first philosophy," of geometry and of mechanics, so that these too are not apodictic but of a hypothetical character. This seems to follow from the fact that analysis is necessary in order to find them[53]).

The third point where the successive advance in the system gives place to more complicated relations is at the beginning of the human sciences. For, says Hobbes, the causes of the mental motions can be found not only deductively, that is, from the propositions of mechanics and physics; here everyone has, by self-observation (per uniuscujusqve proprium animum examinantis experientiam), access to knowledge of
[265] the principles on which political theory builds: the human feelings and drives and their tendencies. One can therefore come to political theory not only by the long series of deductions from geometry through mechanics and physics, but also directly, inductively, from the facts of self-observation, which are set up as new starting points for deduction. (De corp. cap. 6, 7). It is the independence of empirical psychology over against the science of material nature that Hobbes here acknowledges. He had all the more occasion to call attention to this point since he had himself, as we know, published psychological

and political-philosophical writings (Elements of Law, De cive and Leviathan) before the work which in the system comes before the human sciences (De corpore). He ought here to have gone a step further. For what could be grasped deductively from physics can in any case only be the physiological processes that are connected with the mental phenomena. As phenomena of consciousness the mental phenomena can be known only by the inductive way of self-observation, and the break in the deductive process is therefore greater than Hobbes sees.

The last two points are, moreover, on closer examination connected. For the reason why at the transition from mechanics to physics we must break off deduction is properly this, that physical phenomena appear to us as qualities, but these as such are due to the sentient subject. What is given in physical phenomena is thus subjective phenomena, whose objective cause is to be sought, which cause must necessarily — on Hobbes's presuppositions — be motion. (De corp. cap. 25, 3). It is thus the mental phenomena that at both points break the unity of the method and of the system.

c. The Object of Science

In agreement with his first fundamental proposition, that all change was motion, Hobbes declares that the object of science is the corporeal. By body (corpus) is understood everything that is independent of our thinking and fills a part of space. As independent of us it is called substance. Body and substance are one and the same. An incorporeal substance *[266]* is for Hobbes a contradiction. But substances and bodies we do not perceive immediately, only by inference. It is only the properties (the accidents) that we observe; of substance as such we have properly no representation, although we infer that there is something that underlies the properties. The essential properties of body are extension and motion. All other properties are only subjective phenomena in the sentient consciousness. Space and time too are, according to Hobbes, subjective, since he understands by space and time such representations as the extension and motion of bodies call forth in us, and which we could retain even if all bodies disappeared.

One might interpret this to mean that Hobbes renounces knowledge of the essence of things and keeps only to their properties, in so far as these can be apprehended with the help of sensation. And Hobbes has even a very lively understanding of the fact that the very circumstance that something can appear to us and be apprehended by us is for the philosopher the most remarkable of all facts. "Of all phenomena, the very fact that something appears to us as phenomenon (id ipsum *τὸ φαίνεσθαι)* is the most wonderful of all, so that, since the phenomena are principles for the knowledge of everything else,

sensation is the principle for the knowledge of the principles themselves, and all science is derived from it, while its own causes can only be found by beginning with it." (De corp. cap. 25, 1). Here the sentient consciousness is set as the first starting point for all knowledge. It is a statement that recalls Descartes's "Cogito, ergo sum," and which might seem to conflict with the assumption that Hobbes's system is materialistic. It is also only in so far as Hobbes proceeds strictly deductively from his original postulates or definitions that he can be called a materialist. His materialism falls away every time the strict deductive procedure, which would derive everything from the one proposition that change is motion, is broken. That Hobbes wants such a derivation is certain, though it is also certain that he cannot achieve it, and that he sees that he cannot achieve it. Whether
[267] one will call him a materialist or not therefore easily becomes a dispute about words. Edv. Larsen understands (in his monograph on Hobbes, Copenhagen 1891. p. 186) the matter thus: Hobbes is a materialist only in his doctrine of method, because our knowledge in his opinion could only give an account of motion, but he does not wish to give a materialistic metaphysics. Hereby, I believe, a distinction between method and system is put into Hobbes which he does not know. He is too much of a dogmatist to distinguish in this way. He trusts that things are such as they must be according to our definitions. He is a materialist in the same sense as Descartes is a spiritualist. Both become entangled in contradictions through their assumptions; but of these contradictions they were not conscious.

We therefore also see Hobbes, just after he has brought out with great emphasis what problem lies in the very fact that something becomes a fact for us, pass with great composure to asking whence, then, sensation, the principle of all knowledge, comes, and answer this question in consistency with his first fundamental proposition: since all change is motion, sensation too, which is due to a change, must be motion. Sensation, he says, is *nothing but* a motion in the small parts of the sentient body. (De corp. cap. 25, 2, cf. Leviathan, cap. 6). And what he here says of sensation he says, in the objections to Descartes, of consciousness as a whole (mens *nihil aliud erit* præterqvam motus in partibus qvibusdam corporis organici). Thus for Hobbes psychology properly belongs entirely under the doctrine of motion. And even if he distinguishes between motion itself and its appearance (apparition) to us, as when he insists on the subjectivity of the sense qualities, he nevertheless maintains that *what really exists*, when we have sensations or feelings, is only motion. The feeling of pleasure is, e.g., nothing but a certain motion in the heart (pleasure is nothing *really* but motion about the heart, Elements of Law, VII. 1). Appearance, apprehension, feeling, consciousness become for Hobbes, where he

speaks strictly deductively, only a seeming (nothing really, El. of Law, I. 2, 5). — But a seeming, we must add, that presents a great problem: for motion, according to Hobbes, *[268]* can produce only motion, and it then stands as an inexplicable marvel that at certain points, besides motion, there comes into being the seeming, the "apparition", which we call consciousness. It is Hobbes's great significance that he has so clearly established the limitation of the materialistic hypothesis. His energetic deductive thinking has here led him to the decisive point. No later materialistic attempt has been carried through with the clarity and freedom from prejudice of his. He has thereby given a contribution of lasting value to the elucidation of the place of mental phenomena in existence. What is cast in one piece always carries its reward with it. —

As regards Hobbes's special doctrine of nature, it need only be remarked that he no more than Descartes assumes an empty space, but conceives the smallest solid particles as moving in an ethereal fluid. Motion vanishes for him no more than for Gassendi in the transition to rest. He conceives space and time as composed of small parts, each smaller than any part experience can show us (minus qvam datur); even in such vanishingly small parts of space and time there is motion, but only as tendency to motion (conatus, endeavour). And this tendency to motion does not vanish because an equally great opposite tendency to motion holds it in check; it then becomes tension (nixus). By the help of these concepts, set up (De corpore, 15, 2; 22, 1), it becomes possible for Hobbes, who here (like Gassendi with his corresponding ideas) stands under the influence of Galilei, to maintain the continuity of motion in a more perfect manner than Descartes. And he has (as is seen from a treatise first printed in the appendix to Tönnies's edition of Elements of Law) applied these thoughts in a polemic against Descartes (El. of Law p. 215).

d. The Limits of Knowledge. Faith and Knowledge

Every representation we can form is, according to Hobbes, finite and limited. A knowledge of the infinite is therefore excluded. When one hears the words "eternal and infinite," one must be prepared to hear absurdities. "Infinite" can only be used in a negative sense, *[269]* of that for which we can set no limits. This word signifies nothing in the thing itself, but a lack of power (impotentia) in our mind. We express by it our own limitation, not a positive property of any being. — For Descartes, conversely, the infinite was the positive concept: the finite arose by limitation of the infinite. These are two different views of life that express themselves in this opposition.

Just as little as there can be a knowledge of the infinite, just as little can there be a

knowledge of the world as a whole. Only a few questions can be raised about the whole of the world — and none of them can be answered. Science cannot say anything about the world's magnitude, duration or origin, not even whether it has an origin at all. The knowledge of the first things, the first fruits of wisdom (primitiæ sapientiæ), is reserved for the theologians, just as in Israel the first fruits of the harvest were offered to the priests. So Hobbes expresses himself in De corpore (cap. 26, 1). He further argues here that we cannot close the causal series in such a way that we could know for certain that it could not be continued further back. Even if we assumed a first cause, this cannot be something unmoved, for nothing can be set in motion by something that is not itself in motion; what we stop at must then be something eternally moved. But there is in itself no contradiction whatever in assuming that the world has no beginning. Therefore, says Hobbes, I cannot praise those who boast that they have proved by natural reasons that the world has an origin. — In a remarkable opposition to this stands the manner in which Hobbes expresses himself in his political-philosophical writings, (El. of Law I, 11, 2; De cive, cap. 14—16; Leviathan, cap. 11—12). He here acknowledges, to be sure, the incomprehensibility of God, but he finds no difficulty in inferring back from the world to God: by continually going back, we must necessarily come to an eternal cause which in turn has no cause! Yet it is not only theoretical but especially practical reasons that lead to natural religion, since man, on account of his weakness and dependence, renders
[270] God obedience. — This opposition is all the more striking as it cannot be explained by the political-philosophical writings being older than De corpore. For in the objections to Descartes, which were written at about the same time as De cive, it is expressly maintained that the creation of the world cannot be proved. The matter can hardly be explained in any other way than that Hobbes in his political-philosophical writings, from practical motives, attributed greater probative force to the grounds of reason than he strictly could. — But he always maintains that we can form no representation of God. It goes with us here as with the blind man who warms himself at the fire: of the cause of the warmth he can form no representation. We dare not transfer our properties and states to God — neither pain nor need nor understanding nor will; for they all presuppose a limited being in order to have any meaning. Only negative, superlative and indefinite expressions are permissible, and they are then used to express our obedience and admiration, not what God in himself is. The word incorporeal too, applied to God, is only an honorific predicate. For since everything that is, is body, God must also be so, which also agrees with the opinions of the old Fathers of the Church. — Natural religion is not sufficient, for reason is imperfect and the affects are strong. A revelation

is then necessary. But the discussion of it belongs in political theory. For since it goes beyond human comprehension, it can be grounded only on authority, and authority a private man can obtain only by working miracles; since now the time of miracles is past, the state ought to decide all religious questions. Religion is not philosophy but a law; it demands not discussion but obedience.

e. Psychology

Hobbes opens the series of eminent investigators in the domain of psychology who are the pride of English philosophy. With his usual vigorous grasp and his eye for the great, elementary lines he has given valuable contributions to the understanding of mental life. It was properly the endeavor to obtain an empirical foundation for ethics and political theory that led him into psychology; but this gains independent interest in his exposition. *[271]*
— We first bring forward Hobbes's physiological theory, in so far as it concerns the conception of the life of consciousness. — What goes on in the external world is, as we have seen, only motion. When this motion propagates itself inward through sense organs and nerves to the brain and from this to the heart, it meets here with a certain resistance and reaction, since the inner organs are themselves constantly in a certain motion. It is on account of this reaction, which is like a striving outward, that we place the object of sense-sensation outside us. How the relation more precisely stands between these inner processes (action and reaction) and sense-sensation, Hobbes, as we have already seen, does not go into. — As is seen, Hobbes stands somewhat closer to ancient and scholastic physiology than Descartes, since he (though only in De corpore, cap. 25, 4, not in Elements of Law, I, 2, 8) regards the heart as the place where sense-sensation arises. This assumption was probably called forth by the effects on feeling attached to sense-sensation, in which the heart plays an essential role. In the heart is the source of life, and according to the manner in which the impression from without alters the vital motion, which is one with the motion of the blood, namely either facilitates or impedes it, there arises pleasure or pain respectively. Thereby is called forth in turn an involuntary striving to hold fast the motion that brings pleasure with it, or to check the motion that brings pain with it. Already in the movements of the fetus this original need (conatus primus) makes itself known.

What really takes place in the world is, as we have seen, according to Hobbes only motion — motion outside us and in us. The subjectivity of the sense qualities is therefore in Hobbes a consequence of his metaphysics of motion. But not only by the deductive but also by the inductive way he seeks to establish the subjectivity of the qualities. He

points to the fact that we see images of things in places where they in fact are not, as in reflection, in dreams and in sense illusions, — that different persons can apprehend the properties of things differently, — that we sometimes see things double, — that by a
[272] blow on the eye or pressure on the optic nerve light arises, which nevertheless in fact is not an external light. From this he concludes that the content of our sense-sensations is always only "appearance" of motions in brain and nerves, nothing real.

Sensation arises when there is a reaction from the interior of the organism against the impression (a reactione cordis versus exteriora). But, says Hobbes, one cannot turn the proposition round and teach that everything that reacts has sensation. Then all bodies would be able to have sensation! Even if a glimmer of sensation always arose in such a reaction (phantasma), it would at once vanish if there were no capacity *to hold it fast and recall it,* and it would not be sensation proper (sensio). To sensation proper belongs a judgment about the objects, built on comparison of the sense impressions. (Nam per sensionem vulgo intelligimus aliqvam de objectis per phantasmata judicationem, phantasmata scilicet comparando et distingvendo, De corp. XXV, 5). Sensation therefore presupposes memory and comparison. The individual sensations must be capable of being distinguished from one another. It is therefore a necessary condition of sensation that the sense impressions *vary.* Under entirely uniform influence all sensation falls away. Just as little as I have any sensation of the bones in my arms, although they are surrounded by very sensitive membranes, just as little would a man be able to have sensation if he stared motionless at one and the same unchanging thing. To sense the same thing constantly and not to sense at all come to the same. — This was, as we have seen, the point with which Hobbes's whole philosophizing began: that without change there would be no sensation. Through the proposition that all change is motion he then passed over into his philosophy of motion, from which (as he certainly does not himself notice) he has difficulty in coming back to sensation. No wonder that he so strongly stresses that the very fact that something is apprehended by us (or can be phenomenon for us) is the most remarkable of all phenomena; about it his whole philosophy turns, since it constructs partly the causes of sensation, partly the effects of sensation. —

[273] Just as several different sense impressions obscure one another, so the memory images, which are nothing but sense-sensations weakened by the removal of the object, will be further weakened by the new sense impressions that come. In dreams, where such new impressions do not come, the memory images come forward as the expression of something present, often together with features and properties that are taken from other representations. — The memory images arise according to certain laws. They follow the

same order as the original sense perceptions. The motions in the brain are connected in such a way that when the first is repeated, the others follow after on account of cohesion, just as water on a smooth table is drawn after the finger that leads a part of it in a certain direction. Thus the representation of the apostle Andrew can bring us to think of the apostle Peter, his name can bring us to think of a stone, etc. This is the law that is now called the law of contiguity. But Hobbes also lays great weight on what might be called the law of interest. The course of representations is regulated not only by the connection between the original sense-sensations, but also by the influence of feeling and drive. A certain order and connection comes into our representations from the fact that we always have a purpose we seek to reach, and which makes us seek for means to reach it. It is the constant regard to our purposes (freqvens ad finem respectio) that brings system into our thoughts. In dreams (besides the external impressions) this consciousness of the purpose is lacking, and thereby the capriciousness of dream representations is better understood. — For the origin of language, the involuntary association between an object that sets feeling in motion and a vocal outburst or an action in which the feeling finds vent is for Hobbes not a sufficient explanation. Language arises, as already mentioned, according to Hobbes by arbitrary naming. Man sets a mark in order to be able to recognize a thing again, as when seafarers set a mark on a rock they pass by, in order to be able to recognize the place again. Language is a mark (nota) for the individual before it is a sign of communication (signum) for him over against others. And only when signs of communication have thus been introduced — does a society become possible. — There lay in Hobbes's own doctrine of association conditions for a more natural and more *[274]* correct psychology of language than this theory, which ascribes too much to conscious reasoning and arbitrariness. — Arbitrary naming plays, as we have seen, a great role in his epistemology. It makes possible the construction of the first fundamental propositions or definitions. And when such construction has once taken place, all thinking, according to Hobbes, proceeds as an addition or subtraction of names. He seeks to make all thinking a kind of algebra of names. In a judgment, e.g., it is expressed that one name denotes the same thing as another name. Since Hobbes has not seen the significance of the relation of similarity in the involuntary association of representations, he consistently stops at this wholly external conception of thinking, which at the same time suits so well his deductive tendency. —

Pleasure and pain are, as we have seen, closely bound up with the facilitation or checking of the vital process. The impulse of self-preservation is the fundamental need in man. Pleasure and pain arise from its momentary satisfaction. When experiences

have been made and there is a capacity to form representations of the future, the more composite feelings arise. These can, according to Hobbes, be traced back to the feeling of power or of impotence. When I feel pleasure in something that in the future may cause pleasure, this presupposes that I have power to procure it for myself. This power may be my own mental or bodily strength, or that of my friends, on whose assistance I rely, or an authority by whom I feel myself protected; perhaps even God the Almighty alone disposes of the power that makes it possible for me to feel pleasure in the representation of the future. The feeling of pain arises, on the other hand, from the representation of impotence to procure future goods or avert future evils. All men have a constant and restless craving for power, which only death brings to an end. It is not always that one expects greater joy than that which one possesses; but one wishes constantly to secure better what one has. The special feelings arise during the struggle or race between men. There arises triumph in those who come before others, humility in those who fall behind, hope when one is in breath, despair when one is tired, anger at a sudden obstacle, the
[275] feeling of pride when a strong obstacle is broken through, weeping when one suddenly falls, laughter when one suddenly sees others fall, pity when we see one whom we wish well fall behind, indignation when we see one whom we wish ill get ahead, love when we help another during the race, the feeling of happiness when we constantly get ahead of those before us, the feeling of unhappiness when we constantly fall behind, — and only with death does the race end! As long as we live, we set ourselves purposes and feel craving. All feeling of pleasure is bound up with a craving toward a goal. If we conceive a highest goal, and that it has been reached, then not only all craving but also all feeling would fall away. Unhindered progress to ever more distant goals would be the greatest of all goods. The enjoyment itself of what has been reached is a craving: the soul of the one who enjoys moves through the parts of what is enjoyed. For life is motion, and if this cannot proceed straight forward, it becomes circular motion.

In his closer explanation of pity and love Hobbes traces these feelings back to the impulse of self-preservation and the feeling of power. Pity arises from our coming, at the sight of others' misfortune, to think of a possible misfortune for ourselves; love consists in the feeling of one's own power that one has in being able to help others. The joy of knowledge, too, is a kind of feeling of power, as well as the joy of wealth and rank; only science gives very little power, since so few understand it. Yet Hobbes also assumes an immediate joy of knowledge, which is an inconsistency. — Cruelty, he explains, is contempt for another's pain and sorrow; it springs from the assurance of being oneself in safety; on the other hand, Hobbes takes it to be impossible that anyone, without

something being gained for him thereby, directly rejoices in the suffering of others. Hobbes's explanation of all feelings from the impulse of self-preservation becomes artificial and one-sided in detail. But he nevertheless always has firm and natural ground under his feet in regarding the feelings in their appearance and motivation during the struggle to subsist and to get ahead. There is not the refinement in him that there is in his contemporary La Rochefoucauld, who in his "Maxims," in which he set down his "experiences" of the Fronde and of salon life, constantly operates with the trick of putting a conscious ulterior motive into every feeling. Hobbes has far more sense for *[276]* the involuntary. He would hardly have been able to write down an assertion such as the one La Rochefoucauld has put forward, that one weeps only either to get a name for tenderness of heart, or to be pitied, or to avoid the shame of not weeping! —

As regards the psychology of the will, there are found in Hobbes good remarks on the relation between drive, deliberation and willing, directed against the scholastic psychology, which makes "the will" a peculiar, abstract faculty, without connection with the more elementary expressions of activity. Besides, Hobbes has given a brilliant plea in defense of determinism in the little work "Of Freedom and Necessity," which was called forth by a discussion with an English bishop. The bishop held that if the will were not free of causes, the laws would be unjust when they forbid actions; deliberation would be useless, admonition, praise, blame and punishment in vain, and books, weapons, instruments and medicines of no use! Hobbes answered: the law acts as a deterrent, and so as a motive; punishment is not meant to torment the criminal for what cannot be otherwise, but to make him better and to deter others; without deliberation the action often turns out badly, does not become what it otherwise can become, and for the same reason instruments and medicines etc. often cannot be dispensed with; praise and blame are conditioned by the goodness or corruptness of the action and have nothing to do with necessity or non-necessity.

f. Ethics and Political Theory

With the same energy with which Hobbes founds his whole conception of existence on the laws of mechanical motion, he founds his whole conception of ethical and social life on the impulse of self-preservation. If one asks about the relation between Hobbes's ethics and political theory on the one side and the rest of his system on the other, Hobbes has already conceded that one can arrive at ethics and political theory by two roads, partly through deductions from the doctrine of bodies, partly through immediate psychological *[277]* experience. But the deductive road he has not attempted at all, which is connected

with the fact that his ethico-political writings were composed before his writings on natural philosophy. He makes no attempt to derive the impulse of self-preservation from mechanics; he does not see self-assertion as a general tendency in all that exists, as Telesio and Bruno did before him, and Spinoza later. He contents himself with establishing it as a fact. There is, however, yet a third road by which, according to Hobbes, ethics and political theory could be introduced: by arbitrary definition, arbitrary fixing of the first principles. For in ethics and political theory we are dealing with our own life; its valuation and ordering rest on our will. Therefore, according to Hobbes, ethics and political theory can be constructive sciences like geometry and mechanics. For all moral and political laws presuppose a contract among men to lead a social life on certain conditions. From this contract the particular moral and political rules can be deduced. As with the general scientific presuppositions, however, regard for experience plays an essential role here too in the setting up of the principles; this setting up is not unmotivated. On the contrary, it is not the least interesting part of Hobbes's ethico-political theory that sets out to show how men come to construct a social life for themselves.

It is experience of bitter need and danger that leads them to it. It is wrong to suppose (with Aristotle and Grotius) that man has an original need of and capacity for social life. On the contrary. Since each individual's need and feeling of power lead him to take possession of as much as possible of what nature offers, he comes into collision with the others, who have a like need and feeling of power. Fear will perhaps even lead them to try to forestall one another in using violence, so as not to be taken by surprise themselves. That this is how matters stand is shown by the accounts of our barbarian forefathers and of savages; and it is seen in the precautions for safety that we all take, and in the relations of states to one another. The state of nature, i.e. the condition of man as it is apart from the life of the state, is a *war of all against all* (bellum omnium
[278] contra omnes). Here no other rule holds than the individual's need and power. Where power is slight, cunning takes its place. — This does not mean that all men are evil by nature. It is not human nature itself that is evil; but ruinous actions can spring from it. And even if there were fewer evil men than good, the good would still have to take precautions against the evil.

There would be no way out of this dismal condition if reason did not have its root in human nature just as much as passion does. Right reason leads men to seek better means for their preservation than they have so long as they fight each for himself. They discover that the evil is common, and that there are common means to employ against it. Thereby *the first and most fundamental moral law of nature* comes into being: peace is

to be sought, and if it cannot be attained, aid for war is to be sought! But the condition for this is that the individual give up the unconditional right that he has in the state of nature. It is the impulse of self-preservation that dictates this surrender. From this there further follows: keeping one's word, gratitude, obligingness, forbearance and equity are to be practiced, but arrogance and encroachment avoided; what cannot be divided is to be used in common; mutual strife is to be settled by arbitration; each individual is to keep his mental faculties unimpaired (for otherwise right reason and its commands cannot come to prevail). All these consequences can be summed up in the old saying: Do not do to others what you would not have done to yourself. — Should anyone think that these commands of reason could have no power against the passions, Hobbes concedes that fear and hope, anger and ambition, greed and vanity can indeed prevent their being acknowledged; but — he adds — there is no one who does not at times have a calm mind (sedatus animus), and then he will acknowledge them in their necessary connection with the impulse of self-preservation. All duties and virtues follow from the one proposition that peace is to be procured; they are means to peace. Therefore the natural moral law is also the will of the Deity; reason is implanted by God, and Christ, the apostles and the prophets proclaim the same commands that can be derived from right reason. *[279]*

It is not impossible that Hobbes was influenced in this line of thought by Herbert of Cherbury, who, as we know, saw the significance of the natural law in its preventing mutual destruction, and saw in the right setting out of the "common notions" a means to peace. It is known from Hobbes's biography (Vitæ Hobbianæ Auctarium) that Hobbes stood personally close to Lord Herbert. —

When Hobbes appeals to right reason, he is fully aware that he is not assuming, with Plato and the Scholastics, a special mystical faculty that could furnish a common standard in all decisions about what is good and evil. He expressly denies (Elements of Law, II, 10, 8) that any such ready-made recta ratio is to be found in the world: Right reason is not existent. What stands firm is only the end after which all living beings strive, and when reason awakens to clarity, it will also find the necessary means of reaching this end. Hobbes gives reason its basis and content in self-preservation and its means, and is fully clear that reason works only when the mind is calm (De cive III, 26: sedato animo). In constructing the presuppositions of ethics and political theory he takes such a calm state of mind as his basis. And it is experience that has taught him that only through the union of the impulse of self-preservation with the composure that makes deliberation possible can there be any question of ethical laws for human actions. An idealization of the given is thus necessary, for passionate excitement, shortsightedness

and inconsistency must all be kept away if reason is to be "right." The ideal and consistent demands that peace requires must, moreover, be upheld by a constant will. It is not merely a matter of outwardly observing the commands, but of inward adherence to them for their own sake. The natural moral law binds the individual before the tribunal of his own conscience (in foro conscientiæ). — Hobbes has undeniably not noticed to how high a degree he is idealizing here. If the impulse of self-preservation in its elementary form is always to be the only motive, then such a conscience as Hobbes here conceives,
[280] a feeling or will that holds fast to the necessary conditions of peace for their *own* sake, becomes impossible. It goes with him here as when he assumes a desire for knowledge as an immediate feeling of pleasure; this too cannot be directly derived from the impulse of self-preservation. Hobbes lacks the idea of a displacement of motives, by which what first stands as a means can pass over into becoming a purpose. Such a metamorphosis is quite conceivable on the basis of the impulse of self-preservation: it might well be that one uses the means best when one does not use them *as* means!

It is of little use, now, that the individual acknowledges the natural law before the tribunal of his conscience. An authority must be brought into being before which all bow. The individual must hand over his natural right to such an authority on the presupposition that all others also hand over theirs to the same authority, and this authority, the power of the state, then concentrates in itself all the right that in the state of nature was distributed among the many. This contract, by which the life of the state replaces the state of nature, contains at once an agreement of the individuals among themselves and a common submission to an authority (whether this be a single person or an assembly of several persons). In this Hobbes's doctrine differs from that of Althusius, who distinguished between the social contract and the constituting of the authority; for Hobbes the two coincide. But Hobbes, no more than the earlier teachers of natural law, assumes that this contract is entered into with full consciousness. It can be understood or supposed (a supposed covenant, pactum subauditum), and need not be express (pactum expressum). Examples of understood contracts are found in the relation between the victor and the vanquished whose life he has spared: had he not trusted in his obedience, he would not have spared his life. Likewise a tacit compact is entered into between the infant and the one who brings it up and protects it, as a rule the mother. That the contract has taken place Hobbes *infers* from the nature of the relation. When the contract is understood or tacit, it comes close to being a mere point of view, used for illumination
[281] and valuation. But it is then characteristic of Hobbes that he uses precisely this point of view. The contract presupposes (where it is express) the individuals standing over

against one another as independent beings, each with his impulse of self-preservation, and society is now to be the product of their interaction. By this point of view it becomes possible for Hobbes to deduce the life of the state from individual self-preservation. He imagines men sprung up out of the earth like mushrooms, each in his own place, and afterwards joining together. This is a thought experiment that is necessary for the constructive method. But Hobbes does not inquire expressly how experience is related to this thought, as indeed in general the relation between experience and the arbitrary presuppositions of constructive thinking remains unclear with him.

The contract theory is meant to show that it is in every individual's well-understood interest to obey. The authority that is to be obeyed must be unconditional. The power of the state must be as strong as possible, so that each individual over against it is utterly insignificant. It must have the authority to punish, the right to wage war and make peace, disposal over all property, and the decision as to which opinions and doctrines may be proclaimed. To absolute authority on the one side corresponds absolute obedience on the other. Only what would be worse than death — e.g. to kill oneself or one's nearest — the power of the state cannot command. When this enormous power is abused, there is a breach of the moral law of nature, by which the holders of the power of the state are also bound before the tribunal of their conscience, — but not a breach of the laws of the state, which cannot bind the very authority whose will upholds them. If one finds these hard conditions, Hobbes answers that with the power to protect there follows unavoidably the power to oppress. Human life simply cannot be free of inconveniences. The freedom of the state of nature is bound up with insecurity and strife, the constraint of the life of the state with security and peace. It is impossible to limit the power of the state. For power can be limited only by power, and to limit power is therefore the same as to divide it. If one restricts the sovereign's power, then sovereignty in reality lies with the one or those who can punish or dismiss the so-called sovereign. This holds whether one *[282]* wishes to distinguish between ecclesiastical and secular power, or between executive and appropriating power. — Hobbes regards monarchy as the best form of government. Democracy is in reality an aristocracy of orators. Monarchy has the advantage that there is only one who can commit abuses, but above all that party strife is avoided and secrets are more easily kept. It is (by virtue of the original contract) in reality the people that governs; its will is carried out through the monarch, to whom in the original contract everyone's natural right has been transferred.

In all religious and moral questions it is for the sovereign to decide. The sovereign is to determine how God is to be worshiped: otherwise what is worship for one would be

blasphemy for another, and a constant source of strife and dissolution would be present. Just as little can individuals decide what is good and right in practice; that would be a constant seed of rebellion. It is clear, too, that only when the definitions are common can there be certainty that the conclusions will be common. When Hobbes calls ethics and politics constructive sciences because "we ourselves" fix the first principles, then in practice this "we ourselves" becomes one with the sovereign. Here the strangeness of Hobbes's whole standpoint comes clearly to light. His free, naturalistic inquiry has led him to see the necessity of an absolute authority whose reason must be acknowledged as recta ratio. But he then nevertheless himself gives a grounding of this authority, grounds its recta ratio by his own recta ratio, and in this grounding makes use of the freedom of thought and of expression whose extirpation he demands. The fruits of knowledge that he himself has plucked he uses to impress on us the necessity of a prohibition against eating of the tree of knowledge. He used the freedom of the press under the rule of the Independents to publish the first edition of Leviathan (1651), and later, under the Restoration, he took refuge in the Dutch freedom of the press to bring out the second edition of Leviathan (1670); Charles II had expressly refused him permission for a new edition of the heretical work. Like so many of the best men of the seventeenth
[283] century, Hobbes unites in himself the Renaissance's vigorous self-confidence and feeling of freedom with the conviction of the absoluteness of the great sustaining powers. In his doctrine he marks the strong reaction against the Renaissance and the Reformation. The rebellious opinion concerning freedom of conscience and concerning the right of individual knowledge in the moral and political domain he derived partly from the study of the Roman and Greek authors (especially Aristotle) at the universities, partly from the free access to the reading and interpretation of the Bible that came with the Reformation. Against this he set up the strictest caesaropapism: Christ's kingdom begins only with his coming again to judgment; until then God speaks only through his vice-gods: the holders of the power of the state. He declares himself against Catholics, Anglicans and Puritans alike. If the pope is to be the shepherd of the faithful, he becomes their real sovereign, and the state is dissolved. If each individual believer is to be considered a partaker of the divine wisdom, there will be a strife of all against all. If the bishops are to have any authority beyond what follows from royal appointment (the Anglicans think that they have "something more, they know not what, of divine right"), there will also be strife between church and state — and an ecclesiastical tyranny besides. The Protestant clergy thought that the papacy had been overthrown so that they could step into its place. But — asks Hobbes — what then have we gained by deliverance from the pope's

tyranny, if these insignificant men (these petty men) are to replace him? Especially since they are people whose mental qualities are not favorable to a peaceful social life (qvorum ingenium paci et societati aptum non est)! Hobbes thus begins with polemic against free discussion, but ends with polemic against the small authorities who had pushed themselves forward in the place of the great one. It is really they whom he means to strike. And in this relation lies the explanation of his strange position.

Hobbes shares the tendency to seek a center, the concentration and the resignation that are characteristic of the seventeenth century, and that in philosophy find expression not only in his political theory but appear in analogous features in the mysticism of Fenelon and Malebranche and in Spinoza's pantheism. And he has the great merit in political theory of having established that in a state there can be only one center of *[284]* gravity. But he mistakes the center of gravity: all forces other than gravity vanish for him, and consistently, then, all masses gather at the center. Power and obedience become all in all. The spontaneous content of life, manners and custom, public opinion, the quiet and hidden ways by which even the holders of power are dependent on what moves among the many, and on the other hand the whole unfolding of life which it is the task of power to protect and to subordinate itself to as its serving foundation — for all this Hobbes has no eye. His eye is held captive by that one point. And his thinking is in the lapidary style; the finer nuances escape him. All the greater is his sense for the great, elementary conditions of life of human society. He belongs to the same side of the English mind from which Malthus and Darwin, both of whom can be said to continue Hobbes's thought, have sprung.

Nevertheless there are, as already touched on, hints in him that lead out to a wider horizon.

The duty of the ruler is summed up in the proposition that the welfare of the people is the highest law. For the state is not instituted for the sake of the rulers, but for the sake of the citizens. The despotism for which Hobbes speaks is of the same kind as that which appears in the enlightened, reforming statesmen of the eighteenth century (Frederick II, Joseph II). He is not only an ardent opponent of hierarchy in all its forms, but also of every kind of class rule, such as may be found even in "free" countries. By the absolute center of gravity that he establishes he means to abolish the small centers of gravity, so that all the elements can come to move in natural orbits. And the absolute authority itself is grounded by way of reason. Hobbes makes use of free discussion. He hoped that his "Leviathan" would "fall into the hands of a prince who would consider it himself, without the help of any interested or envious interpreter, and then use his sovereign authority to

protect the public teaching of the doctrine it contains, so that the theoretical truth might be converted into practical utility"[54]). But he hopes not only for the enlightenment of
[285] princes, but also for that of the people. Against the hierarchy he relies not only on the jealousy of the power of the state, but also on the advancing intellectual culture and criticism of the people: Paulatim eruditur vulgus! (De homine, XV, 13). Self-contradiction between the propositions of the doctrine, or between the doctrine and the life of its preachers, will sooner or later be discovered. A wide horizon opens here, wider even than the one that stood before the heroes of the Enlightenment in the eighteenth century; for Voltaire and Diderot expressly disclaimed the thought of an enlightenment of the vulgus. And this wide horizon does not, properly speaking, come about through an inconsistency, when one goes to the bottom of Hobbes's presuppositions. For by the "contract" through which individuals give up their self-will, this will is not really meant to vanish, but only to be taken up into a higher connection. What is lacking in Hobbes is the means of showing how the isolated will can undergo a metamorphosis by being thus worked into a larger whole. But by his first and his last word he points beyond the imperfection of his own doctrine, an imperfection that is outweighed by the natural force and the acuteness in the foundation and construction of the doctrine.

g. Opposing Tendencies in English Philosophy

The history of English philosophy in the seventeenth century is not exhausted with Bacon, Herbert of Cherbury and Thomas Hobbes and with the tendency they represent. Besides this tendency, which takes its foundation in what is given in external or internal nature, and works on this foundation by means of the inductive or the deductive method, there is also another tendency, which historically points back to Neoplatonism, and which assumes a foundation raised above all experience for the highest ideas, especially for the ethical ideas. To this belongs Robert Greville, Lord Brooke, who in his thirty-fifth year fell at Lichfield (1643), fighting for the cause of Parliament. In his work on "The Nature of Truth," which appeared two years before his death, and for which he perhaps received the impulse from Lord Herbert's work, he protested against knowledge's being
[286] merely a human faculty, empty in itself, that would first have to go looking for its content: only light is receptive to light, and knowledge is a ray of the divine nature itself. True knowledge raises us above sense perception — of which the Copernican system is an example. All plurality, all relations of space and time are mere appearance; thinking finds rest only in the knowledge of the one cause of all things. The knowledge of the unity of all things in God frees us from envy, shows me the happiness of others as my

own, disperses the darkness of the soul and heals all pain. Experience, on the other hand, can show us no real causes; it can only show us that the one phenomenon precedes the other[55]).

The Platonic tendency to which Lord Brooke belonged found a home about the middle of the century at the University of Cambridge, while Oxford continued to hold to Scholasticism. A year after Leviathan had appeared, there appeared (1652) Nathaniel Culverwel's "Discourse of the Light of Nature." Culverwel had died in 1651; he was Master of Arts and Fellow at Cambridge. His writings are Platonizing like Brooke's, but not so mystical, and he lays greater weight on psychological experience. But he assumes eternal and universally valid truths in both the practical and the theoretical domain. These truths are one with God's essence and are upheld in the world by God's will. Thus for him, as for Brooke, philosophy and theology merge into one, whereas Bacon and Hobbes drew a sharp boundary between them[56]). In a more systematic form the same tendency, whose distinguishing mark is a close union of Platonic philosophy and Christian theology, appears in Ralph Cudworth († 1688) and Henry More († 1687), both attached to the University of Cambridge. Whereas for Bacon and Hobbes it was a principal matter that truth must work its way forward in the world, come into being in man's consciousness through strenuous labor, these men returned to the Platonic doctrine of the eternal truth, which does not come into being, and to which man's consciousness need only open itself in order to have a share in it. This tendency turned both against Puritanism, which was hostile to philosophical thinking, and against Hobbes (and, in More's case, also against Spinoza), who set philosophy in opposition to theology. It is characteristic that Henry More was led into philosophical studies through *[287]* doubt about the truth of Calvinism; he had been brought up strictly Calvinist, and the fact that he was able to work his way out of the doctrine that had been impressed on him by the energy of his own thinking was to him a testimony to the independence of thought over against everything external, an independence that he then asserted not only against the narrow Puritan theology but also against the realist tendency of Bacon and Hobbes. More was in touch with Descartes and took up several of his ideas, only not the sharp distinction between mind and matter; for him the mental was extended, though not an object of touch. In his work *De anima* (Rotterdam 1678. cap. 1) he defines a mind as a substance that can indeed be penetrated but not divided (substantia penetrabilis et indiscerpibilis), whereas body is a substance that can be divided but not penetrated (substantia impenetrabilis et discerpibilis). What is characteristic of body is not that it is extended, but that it is impenetrable. Everything that exists has extension in length,

breadth and depth, the mind as well as the body. *(De anima.* cap. 8: Responsiones ad Hobbesiana). In this way he mediates between Hobbes and Descartes and thinks he has refuted Hobbes's objection to spiritualism, that we cannot form any concept of an immaterial substance. Extension is to be common to both kinds of substance, and More goes even a step further, in that he, like several of the philosophers of the Renaissance, regards space as a divine attribute: that God has extension follows from his omnipresence. *(De anima.* cap. 2, § 4—5). The difference between God and the individual soul is only that God is unlimited, the soul limited. In this bold mysticism More is influenced by the Kabbalistic doctrine. His mystical conception of space recurs in Newton, who on the whole was influenced by More. — From Descartes More differs not only in assuming that all substance is extended, but also in taking activity (in the case of mind original activity, in the case of matter communicated activity) into the concept of substance. *(De anima.*
[288] cap. 1 § 3). Hereby he prepares (like Gassendi and Hobbes) Leibniz's philosophy of nature. — We cannot go further into this tendency here; it is more a learned mysticism than new philosophical thinking. It has had great significance for the spiritual development of England, but has not brought new thoughts into the history of philosophy.

A greater interest attaches in this respect to an author who met Hobbes on his own ground, the inquiry into the psychological foundation of ethics and social life. Richard Cumberland (b. 1632, studied at Cambridge, died as Bishop of Petersborough 1718) published in the year 1672 a work remarkable in the history of ethics: "Philosophical Inquiry into the Natural Laws" *(De legibus naturæ disqvisitio philosophica),* which is expressly directed against Hobbes. He declares that he means to continue the work of Hugo Grotius by another road than the one the famous Netherlander followed. Instead of inquiring how the moral laws of nature make themselves known in experience and history, he means to inquire into their origin and their causes. In this inquiry he means to start from experience and to reason his way forward from the nearer to the more remote causes. He is sure that by this road he will at last come back to a divine source of those laws; but he will not use the easy method of the Platonists, which lets the innate ideas bear immediate witness to this lofty origin: "I have," he says, "not been so fortunate as to arrive at the knowledge of the natural laws in so convenient a way." And in any case he will not build his ethics on such a hypothesis, although he acknowledges the sound and good influence of "our Platonists" in moral respects. Against Hobbes, on the other hand, he points to the spontaneous tendencies in human nature, whereas Hobbes had taken the arbitrary adaptation of wills as his basis. From birth onward there takes place an unfolding of the human faculties and powers, a spontaneous use

of the organs, which is bound up with an immediate satisfaction. Sense impressions are met by a spontaneous inclination to receive them, to turn toward them. In this way the first experiences are had and the first mental operations performed: remembering, comparing and counting. In this immediate need (impetus), which is not produced by *[289]* the impressions, but only released by them, the natural and self-evident truths of reason (naturalia rationis dictata) have their ground. And such an immediate need also underlies the first striving to produce goods and shun evils — and that without the individual's own goods and evils being at once separated from the common ones: on the contrary, it is spontaneously acknowledged that the good that is common to all rational beings stands above that which falls to a single being only, and that the single being attains its peculiar happiness precisely by striving after what is good for the whole to which it belongs. Clear consciousness and arbitrary willing can only develop further what the spontaneous tendencies already lead to; art can work on the foundation of nature, but cannot take its place. — While Cumberland thus stresses the original natural instinct, in opposition to Hobbes's construction of society out of clearly conscious, independent individuals, he maintains on the other hand that the natural is not only what can be traced in the first manifestations of life, but also what appears as ripe fruit when growth and development are complete. Human nature is therefore to be sought not only in the elementary drives, but just as much in what the highest spiritual development displays. Even if man, then, at first confines his interest to himself, it is not on that account unnatural for him to attach his interest to what brings happiness to all. The state of nature is not given up because there develops a magnanimity (generositas) for which the good is what works toward the preservation and perfecting of the whole human species, nay perhaps even of the whole universe, so far as we have a concept of it. — What Cumberland urges, then, is partly the possibility of a development from the elementary drives to magnanimity, without the external and mechanical notion of contract needing to form the middle term, partly the harmony between individual and common goods. In both respects he put forward thoughts that were more fully grounded and developed within English philosophy in the eighteenth century, and that already, in Cumberland's own time, were quietly taking shape in Spinoza's Ethics and Leibniz's philosophy of law. *[290]* Cumberland's exposition is still unclear and imperfect, but he has pointed to decisive points. Even if his work (to judge by its title page and preface) was called forth chiefly as a plea against Hobbes, it nevertheless has an independent significance in the history of ethics.

5. Benedict Spinoza

a. Biography and Characterization

Spinoza is the central thinker of the seventeenth century. All the threads of thought run together in him: mysticism and naturalism, theoretical and practical interest, which in the century's other thinkers stand in greater or lesser opposition to one another, and — where they appear in the same personality — awaken inner discord, are here sought to be carried through consistently, and the reconciliation between them is found precisely in their being carried through consistently. Whereas so many know no other way of connecting the threads of thought than to cut each of them off separately and then tie them together in an external fashion, the greatness of Spinoza's thinking is that he will set no arbitrary limits, but trusts in the harmony of the trains of thought in their innermost essence. Through the quiet labor of thought to which he devoted his life, he sought to shape a work that could take the new world-view and the new science into itself and let them unfold all their consequences, and yet maintain the reality and independence of the mental in existence, — a work in which the multiplicity and individuality of phenomena could be fully acknowledged, and yet the unity, the all-embracing order of the world, be seen as the power that in its innermost bears all and works in all. And this work was for him not a purely theoretical labor; it was to him the satisfaction of a personal need of clarity and self-understanding, and it helped him to the cheerful fundamental mood that is won by working one's way through the unrest and darkness of existence to the understanding of eternal laws. His chief work (*Ethica*) has thereby become a work of art, not merely a purely speculative work. It has aptly been
[291] compared, in its five books, with a drama in five acts. And this comparison can be carried through in detail. In the first book the most general propositions are given, which unroll before our eyes a great, infinite background for human life. In the second book there is added to these a series of propositions borrowed from physics, by which the strictly mechanical view of nature is laid down, and the book then passes on to an inquiry into human knowledge. A provisional conclusion is offered by the end of the second book — and, on a not unfounded conjecture, the first two books originally formed a whole by themselves. There appears a prospect of attaining what Spinoza had set as his goal: the knowledge of the connection in which our mind stands with the whole of nature. In this knowledge he saw (in an autobiographical fragment) the only good that was closely bound up with his personality, and of which no external accident could rob him. But so swift a conclusion of the drama is prevented by a cloud appearing over the horizon: it

is not only thoughtlessness and false combinations of representations that hinder true knowledge; it is first and foremost the feelings and passions by which we are tossed about. Here a new and serious resistance is to be overcome — and in order to be overcome it must first and foremost be known. In the third book, then, Spinoza gives a masterly natural history of the feelings. He demonstrates the connection of the feelings with the impulse of self-preservation, and shows how they are transformed under the influence of representations. His chief endeavor is to find a fixed causal connection in the mental as well as in the material domain. He wants to find the law of the swell of waves by which we are tossed about. In this very law he finds deliverance. In the fourth book the light begins to break through again. No passion, it is true, can be directly extirpated; but it can be checked by another passion that is stronger. Once this has been seen, the task is to make the knowledge of the good, that is to say, of the conditions of the preservation of our being, a power in the mind, and this comes about by this knowledge itself awakening pleasure in us, both through the end it shows us and through its being itself a mental activity, an expression of our power. This knowledge will unite us with individuals akin to us in essence, since it turns out that we are subject to common conditions, and that *[292]* only common striving leads each of us to the goal. With this the drama approaches its conclusion a second time. But there still remains a question: that of the individual's complete working-through of his personality by uniting the scientific knowledge, whose possibility the first two books showed, with the practical course of development, the discipline through the struggle for existence, that the third and fourth books depicted. In the fifth book it is shown how the clear understanding of our passions in their natural connection raises us above them, and unites with all our other knowledge of nature into an immediate beholding of our being as one single one of the forms in which the eternal Deity, working in the world of mind as in that of matter, unfolds its being. As we thus see ourselves — and all things — "under the aspect of eternity," all the unrest and obscurity of time and finitude vanish, and with the mental freedom whose development the fourth book depicted there is now united the inward feeling of being one with the eternal and infinite being. — Thus within one frame Spinoza finds room for philosophy of religion, physics, theory of knowledge, psychology and ethics, just as the carrying through of realism by the strict application of the law of causality does not hinder, but for him precisely makes possible, the mystical unity with the Deity. If we add that this whole great and manifold connection of thought is set out in the form of propositions and proofs after the mathematical pattern (more geometrico), we have the picture of a work unique of its kind. No wonder that it was not understood in its own time; nay, that

the time of understanding began only a hundred years after its appearance, so far as the idealist side of its content is concerned, and came only in our own day, so far as the realist side of its content is concerned. For the history of philosophy the task with regard to a work such as this is first and foremost to inquire how it came into being, and on what historical presuppositions it rests. It stands before us like a cut crystal; if possible, we should discover something of the process of crystallization. Here Spinoza's personality and personal circumstances and his philosophical development may perhaps be able to
[293] help us. There can hardly be any doubt that the manifold content that is here gathered into so peculiar a frame is held together first and foremost by the personality of the author. After that, an analysis of the system itself will show us how the presuppositions and elements that the author has taken into himself are used by him, and how far he has reached his goal. —

Baruch Spinoza was born on 24 November 1632 in Amsterdam, the city in which his parents, Spanish Jews who had fled from the Inquisition, had taken refuge. The gifted boy made his first studies at the Jewish academy in Amsterdam, where he was introduced to the Talmud and to medieval Jewish philosophy. Thereby the foundation was laid for an essential side of his thinking, the endeavor to hold fast to and carry through the thought of the Deity as the one infinite being, as this thought comes forward in the higher popular religions, especially in Judaism. In the course of his studies, Maimonides (a Jewish philosopher of the twelfth century) and Leo Hebraeus (a Jewish mystic of the time of the Renaissance) seem in particular to have been of significance for him. Their books were in his library. The oriental or mystical tendency always lay at the basis of his thought and gave him the quality of being cast in one piece, the determined direction toward one point, that distinguishes his thinking. Early on he harbored doubts concerning the Mosaic theology, and he therefore became an object of suspicion on the part of the Jewish theologians, which in turn led to his withdrawing from the synagogue. He felt the need of a wider intellectual horizon and began to study the humanities and natural science. In Latin he was taught by the physician van Ende, who had a name as a freethinker. The most important source for Spinoza's biography, the pastor Colerus, who — in spite of his great abhorrence of Spinoza's doctrine — collected information about his life with great love of truth, thinks that in this "school of Satan" Spinoza no doubt learned other things than Latin. According to Colerus's account, Spinoza is said to have been in love with van Ende's gifted daughter, who helped her father with the teaching, but to have been rejected for a fellow pupil who knew better how to make his
[294] way as a suitor. There is, however, some difficulty in reconciling this account with what

is known of the ages of the two persons. — Colerus says that Spinoza at this time left theology and gave himself wholly to "physics." It is probable that among the authors he studied in order to become acquainted with the newer view of nature, Giordano Bruno was one of the most important. Bruno, it is true, is never named by Spinoza, and his books were not in Spinoza's library, but the latter's first work (especially a little dialogue that is incorporated in the work *On God, Man and his Happiness*) clearly recalls Bruno. Here he found a philosophical conception that made it possible for him to unite what was for him the essential element in the religious ideas with a scientific understanding of nature. Bruno's doctrine of the infinity and divinity of nature made it possible for him to unite the concept of God and the concept of nature in an inward way. This influence of the philosophy of the Renaissance on Spinoza has become probable only since the discovery (some thirty years ago) of Spinoza's first work. Previously one held to Colerus's remark that Spinoza, after he had given up theology for "physics," was long in two minds about which teacher he should follow, but that then Descartes's works had fallen into his hands and had appealed to him by their endeavor to build everything on clear and distinct grounds. It was thought that Spinoza began in philosophy as a Cartesian and only gradually turned critically against Descartes. Against this stands the fact that the very first work we know from Spinoza's hand criticizes Cartesianism on some of the most decisive points. Spinoza was never a Cartesian, but he learned much from Descartes and used several of his concepts, and in part followed him in his terminology. But in this period of transition, too, before his own system took more definite shape for him, he must have studied the more recent Scholastic authors, who at that time — before the philosophy of Bacon and Descartes had made its way — dominated philosophical teaching in Germany and the Netherlands. Many expressions and propositions in Spinoza's later as well as in his earlier works point to such a Scholastic influence, just as Spinoza himself in several places refers to the Scholastics. It appears — when one takes all this together — *[295]* that they are very comprehensive and diverse circles of thought, and no small literature, that Spinoza traversed in the course of his philosophical development. Closed and cast in one piece as his chief work stands before us, it nevertheless has its roots branching out to many sides within the philosophical tradition[57]). This does not diminish his originality; it is the mark of genius to be able to take in and work over as no one else can. The peculiarity and originality of the building do not suffer from the stones' having been fetched from many places.

Under the influence of all these motives of thought, which streamed in upon him from many sides, Spinoza in his views moved further and further away from the synagogue.

His open apostasy was foreseen, and in order to avoid it attempts were made to hold him back by offering him a pension. But neither this nor a fanatic's attempt to assassinate him kept him from following the way of his thought. He was then solemnly expelled by the Jewish clergy (1655). They even succeeded — since the Protestant congregation too regarded him as a dangerous man — in having him banished from Amsterdam for a time. He took up provisional residence with a friend in the country near Amsterdam. He supported himself by grinding optical lenses, a skill he owed to his rabbinical upbringing, since every future rabbi had to learn a trade. His friends in the city fetched his lenses and sold them for him. It seems that already at this time a circle of younger men had gathered around him and his thoughts. At this time the work on God and man (the "short treatise") and perhaps the first draft of the later "theological-political treatise" came into being. Spinoza turned with indignation from the fanaticism of the positive churches, and with ever greater inwardness and self-understanding found the task of his life in the independent working out of a circle of thoughts that could spread clarity over the nature of man and his position in existence. Experience had (as he says at the beginning of the unfinished treatise "on the Improvement of the Intellect") taught him that neither riches,
[296] sensual enjoyment nor honor could be a true good for man, but that the only thing that could fill the mind with ever new satisfaction was the constant striving after knowledge, by which the soul is bound to what endures while all else changes. Spinoza's thinking had — however speculative and abstract the forms it took, and however remote from ordinary human life it might seem to lie — a decidedly personal and practical motive. Full mental clarity was to him a necessity of life.

In the year 1661 he moved to Rhynsburg, a small town near Leyden. Here he began the working out of his famous chief work. It is mentioned in letters to his friends (especially in the letters to Oldenburg and to Vries). Copies of the worked-out part of the work existed early, and Spinoza's young friends in Amsterdam, mostly medical men, studied the book together and in cases of doubt turned to the master for explanation. His life was not so solitary as it has often been depicted. His correspondence, of which we have had fuller knowledge only since the find of manuscripts that was made some forty years ago, and in which the "short treatise" was also discovered, shows him in touch with a not inconsiderable circle of men of different station and intellectual tendency. The study of this correspondence is of considerable interest for the understanding of Spinoza and his thoughts as well as of the spiritual condition of the time; in my popular account of Spinoza I have treated this subject in a separate chapter, "Spinoza og hans Korrespondenter," to which I must refer here. In the nearby university town of Leyden

he had several friends. Thus he came here into a confidential relation with the Danish naturalist Niels Stensen, who was studying in Leyden at this time. Later, when Stensen had gone over to Catholicism, he addressed to Spinoza in an open letter (Epistola ad novæ philosophiæ reformatorem), in which he calls him his former very intimate friend (admodum familiaris), an appeal to follow him into the bosom of the one saving Church. Niels Stensen's biographer, the keeper of the national archives A. D. Jørgensen, explains Spinoza's silence in the face of his former friend's appeal in the following way: "The restless spirit of love of Christianity drove Steno to an attempt to make his friend a *[297]* partaker of its happiness; philosophy's infinite indifference to the individual personality with its desires and its cares let Spinoza see that this man was irretrievably lost to the knowledge of the truth, and that the right answer to his address was therefore silence." I believe that the excellent historian (who was the first to bring to light the interesting relation between Steno and Spinoza) here does both Spinoza and philosophy a great injustice. I have great reverence for the figure of Steno, for his religious as well as for his scientific personality. But I nevertheless believe that in all propaganda, however noble the name it calls itself by, other motives are at work besides the spirit of love (if this is always present at all); and in any case this spirit of love did not prevent Steno and many others from looking forward to a blessedness from which, they believed they knew, very many would be excluded. It is in any case unjust to deny Spinoza the spirit of love because he will not force his views on others. His loving faith was directed to man, and he found what is human behind very different forms of faith. An unwilling witness, the pastor Colerus, relates of him how sympathetic he was toward others, and how he comforted and cheered the members of his household when they were visited by sorrow or sickness. When his landlady in the Hague asked him whether she could be saved in the religion she had, he answered her that her religion was good and that she should not seek any other; she would surely be saved by it if she led a pious and quiet life. He sympathized most with the freer tendencies within Protestantism. He had a different view of the significance of Protestantism from that of his kindred spirit Bruno, and quite naturally so, since he lived in the freest country in Europe, which admittedly did not mean much at that time. When later a young man, Albert de Burgh, who had lived in Spinoza's household and who had gone over to Catholicism, repeated Steno's attempt at conversion, Spinoza answered him in a letter in which he maintains the validity of free knowledge and sets justice and love as what is essential in all true religion. It must not be forgotten that what not least offended Steno in Spinoza was that the latter had spoken *[298]* up for freedom of religion. To work for freedom of religion was for Steno to do harm to

one's soul! Should there not lie in Spinoza's conviction that a true faith cannot be forced on anyone either by spiritual or by bodily violence, and in his caution about shaking the faith of others when only there was seriousness in it, a greater and truer interest in the individual personality than the one that finds outlet in passionate propaganda?[58]) —

After a couple of years' stay in Rhynsburg Spinoza moved to Voorburg near the Hague (1663) and later (1670) to the Hague itself. Here too he had many friends, in part men in high positions, such as the brothers de Witt. His way of life was frugal, and he showed great unselfishness toward relatives and friends in money matters. A bright and cheerful mood marked his way of taking life. "Why," he asks in a passage of his Ethics, "should one rather still hunger and thirst than drive away melancholy? In my conviction no deity, and in general no one who is not envious, can take pleasure in my impotence and my misfortune, or count tears, sobbing and fear as merit; but on the contrary, the greater the joy we feel, the greater the perfection to which we pass, that is to say, the more we partake of the divine nature. Therefore the wise man will refresh and strengthen himself with food and drink, as well as with fragrance, flowers, music, tasteful dress, bodily exercise and plays." In spite of his mystical vein Spinoza was no ascetic. In his life as in his philosophy an energetic seeking to be absorbed in one thought went hand in hand with a sense for the manifold reality whose inner law, in his conception, this thought precisely expresses. Both his life and his philosophy are distorted if one does not hold fast to the different currents that actually moved in him, even if he had perhaps not (as he believed) found the redeeming word that could let all these currents come into their full right and into reconciliation with one another.

The first work Spinoza published was an exposition of the Cartesian philosophy (1663), originally intended for a young man (the Albert de Burgh already mentioned)
[299] whom he did not yet want to initiate into his own philosophy. It is characteristic that he thus regards Cartesianism as an introduction to his own doctrine. On the "Ethics" he worked and polished constantly. He was very cautious about communicating it to others; before he allowed his disciples to show the manuscript to strangers, he gathered careful information about their character and circumstances. Thus he refused his friend Tschirnhausen permission to show the Ethics to Leibniz; only after a longer personal conversation with Leibniz did he himself show him his chief work. According to a hypothesis put forward by TØNNIES[59]), the first two books of the Ethics were worked out by themselves, whereas the realist psychology of the third book, which betrays the influence of Hobbes, who also strongly influenced Spinoza in political theory, seems to be due to a later working out. This explains several discrepancies between the conception

in the second and in the third book. In relation to the "short treatise" the "Ethics" marks the advance that Spinoza no longer conceives the relation between mind and matter as a causal relation, but as a relation of identity. Spinoza made several attempts at publishing this chief work of his, but gave it up again when even the rumor of its publication aroused great agitation. He was in ill repute after the publication of his theological-political treatise (1670), although he had not yet developed his real system there. The book defended freedom of religion as a human right, with particular reference to the fact that the diversity of personalities must also bring with it diversity in faith. It further undertook a purely historical inquiry into the biblical writings (especially those of the Old Testament, whose late origin is maintained) and a psychological characterization of their authors. A definite distinction was made between religion, whose purpose is practical, in that it aims at bending the minds of men to obedience and morality, and science, which has a purely theoretical purpose; and on the basis of this distinction the complete freedom of science was maintained. Finally there was polemic against the assumption of miracles, since the laws of nature are the laws of God's own essence. — This work, which contained so many thoughts of the future, was regarded as the epitome of all ungodliness. The experiences Spinoza had in this way led him not only to hold back *[300]* his "Ethics," but also to decline an invitation to take up a professorship at Heidelberg; he feared that he would — in spite of all promises — not have sufficient freedom of teaching. — A disease of the chest that was hereditary in his family, and that had been consuming him for a long course of years, brought on his death (21 February 1677). About his death, as about his life, wild rumors were spread, which the truth-loving pastor Colerus refuted after careful inquiry — although he himself is astonished that a freethinker could live so beautifully and die so calmly, and although he is offended that Spinoza's barber, in a bill submitted to the estate, calls his deceased customer "the late blessed Mr. Spinoza!" —

In Spinoza's personal development we thus see the different elements of his system come forward from different sides. It will now be the task, through an analysis of the finished system, to see how he has united these elements, and how far the union has really succeeded.

b. Theory of Knowledge

If in setting out Spinoza's system one wished to follow his own "Ethics," one would have to begin with certain definitions and fundamental propositions and from them in turn derive a series of propositions. The question would then remain unanswered how Spinoza came to his definitions and fundamental propositions, which contain the whole

system within them, — and yet it is really this question that is of the greatest interest historically. If one begins with the definitions and fundamental propositions on bare ground, as Spinoza himself does in his chief work, the most natural supposition is that these first principles are either set up quite arbitrarily (as Hobbes thought) or else are the object of an immediate intuition. Neither was Spinoza's meaning. By following the fashion of the time, which demanded a strictly deductive exposition, he has therefore given occasion to misunderstandings of his system. From the unfortunately unfinished treatise "on the Improvement of the Intellect" it is seen that he gave serious thought to
[301] scientific method and had in mind to write a kind of inductive logic. In a certain sense it can be said that the principles on which Spinoza builds the deductive exposition in the "Ethics" came about for him by reflection on, and construction of, the presuppositions on which our empirical knowledge of existence rests.

If by experience one understands merely an apprehension of things as they present themselves to us spontaneously and by chance, it gives us no true knowledge. It stops at the given as it appears at a definite time and place, and an assumption then settles fast in consciousness merely because no contrary facts have so far shown themselves, without there being any guarantee that such are not to be found. This kind of experience Spinoza calls experientia vaga; it corresponds to Bacon's inductio per enumerationem simplicem. True knowledge must be a knowledge of reason, which does not stop at the single given phenomenon, but sees it as the consequence of a general law, as a single expression of a general order of things that extends far beyond it. Knowledge of reason considers the phenomena given in experience in their definite connection with one another, and finds in this connection the explanation of each single phenomenon. Knowledge of reason (ratio) starts from the given. Spinoza urges the necessity "that we always derive our concepts from real beings, proceeding as far as possible along the series of causes." First, then, an account of the facts must be procured, and after that the fundamental concepts and fundamental propositions must be derived from them. In his "theological-political treatise" Spinoza draws an analogy between the interpretation of a book (the Bible) and the explanation of nature. First it must be established what the given content is. Next one must make sure of certain general relations and laws that hold good at every point. In the case of physical nature these are the laws of motion; in the case of mental nature Spinoza would most likely name the laws of the association of representations. It is, in a word, the idea of the lawfulness that reigns in all things, which reveals itself in the single changing phenomena and makes each one of them explicable to us. In this lawfulness
[302] there is revealed for Spinoza the true, eternal essence of things. Not to the advantage of

clarity, he distinguishes between "the fixed and eternal things, on which the changeable single things depend in so inward and essential a way that without them they can neither be nor be conceived," and "the laws that are written in those fixed and eternal things as in law books." What he calls the fixed and eternal things can be nothing other than the order of nature itself in the material and in the mental domain, that is, the very fact that definite laws hold for the phenomena. These laws reveal themselves equally completely in nature as a whole and in its single parts. It is to them that one must go back if one wishes to reach a completion of knowledge. The series of phenomena is infinite, inexhaustible; there we could go on passing from link to link. But true knowledge concerns what connects the links, and what is therefore present everywhere, — those fixed and eternal things, which thought is able to grasp, while the series of phenomena extends to infinity, without completion. — Spinoza distinguishes expressly between the real connection that those fixed and eternal things (entia realia) express, and the mere general concepts (abstracta et universalia), which have no objective reality.

The truth and validity of our knowledge rest, according to Spinoza, on no external criterion, not on agreement with anything outside it. It is the clarity and distinctness that go with perfect consistency that give the assurance of truth. Error always rests on something limited and isolated having been made into a whole. Error vanishes when one simply proceeds with strict consistency from the erroneous presuppositions in which one is entangled. Intellectio fictionem terminat. In strict logical consistency, then, our thinking has a norm of truth that it can apply everywhere, even where it is in error. Truth, says Spinoza, illuminates both itself and error (veritas norma sui et falsi est), just as light makes known both itself and darkness. The starting point of all our inquiry naturally becomes, then, the investigation of the rule of truth that is given in the very nature of our intellect. This is the foundation on which we build in our inquiry after *[303]* what is eternal and necessary in things, after what underlies all things (res omnium prima).

Spinoza here hints that it is only by means of our subjective reason that we can find "the eternal things," "the first of all things," in nature. Had he pursued this hint further, the problem of knowledge in all its sharpness would have presented itself to him: by what right do we assume that existence itself follows the norm that holds for the relations of our thoughts to one another? But this problem does not arise for him. It stands fast for him that our rational presuppositions are also those of things, nay, that what is eternal in things corresponds to them; an objective order of the world corresponds to the subjective order of thought. There is the same relation between the objective objects of our thoughts

(which Spinoza denotes by the Scholastic expression essentiæ formales) as between our thoughts among themselves. To what is for us the first thought, from which all other thoughts are derived, there corresponds what is in existence the first thing, the origin and source of all else. When Spinoza has the laws that explain the single phenomena be "written in the fixed and eternal things as in their law books," he means thereby (we could say from our standpoint) to express that the connection that makes things intelligible is just as much a reality as the single detached "things" each by itself, nay, that it far rather than they deserves the name of reality, since it is eternal and all-embracing. And here too we must see the origin and meaning of what he in his deductive exposition calls substance. For by substance he understands that which exists in itself and is conceived through itself, and through which all else exists and is conceived. Spinoza's substance, the fundamental concept of his whole philosophy, is thus really causality represented as a thing or a being. For causality, the causal relation, is according to Spinoza understood through itself and is what makes everything else intelligible to us. And since the causal relation is the presupposition of a knowledge of real existence, which makes itself known to us through experience, we see that Spinoza's whole philosophy can be said to be a construction of the presuppositions that hold for all scientific experience, in such a
[304] way that these presuppositions are made one with the innermost essence of things. But causality, or the causal relation, which Spinoza thus thinks of as a real power on which the single things depend and in which they exist, corresponds in turn for him to the necessity with which we derive conclusions from premises, or in other words: Spinoza does not distinguish between ground and cause; the causal relation is for him not a relation in time, where the cause goes before and the effect follows after, but an eternal relation of ground; the relation between "the fixed and eternal things" (or "the first of all things") and the single phenomena is not that the former were first and then the latter arose as their effects. For true knowledge, according to Spinoza, the relation in time drops away. Knowledge of reason is, according to Spinoza, an apprehension of things "under the aspect of eternity" (sub specie æterni): it keeps to that in things in which the eternal necessity makes itself known in them, and sees them only as examples of it. The causal principle itself Spinoza regards as a necessary presupposition. In the Ethics he sets up as a fundamental proposition that a thing must have its cause either in itself or in something else. Now since substance, that which exists through itself and is conceived through itself, cannot have its cause in something else, it must have its cause in itself and thus exist with necessity. If one denies this proposition, one must also deny the proposition that there must necessarily be a cause of every thing (either in it or outside

it). And that it is a proposition of reason is evident also from the fact that for two things to be related as effect and cause means that the concept of the one thing can be derived from that of the other; the knowledge of the effect is only the knowledge of a particular property of the cause; there is something common in their concepts. Only when in this way the relation in time, and in general *the difference between cause and effect, drops away,* so that the effect is seen as the logical consequence of the cause (which would admittedly be the ideal completion of knowledge), can substance or "the eternal things" be called the cause of phenomena. And only on the presupposition of this confusion of ground and cause does it become intelligible how Spinoza could hope to build up a purely rational knowledge of existence, or, as one can also express it, to know existence *[305]* as being rational in its innermost essence. The counterpart to Spinoza is formed here by the conception that Brooke and Glanvil (as well as the occasionalists) had already hinted at and that Hume later carried through: that cause and effect are *wholly different* things, an assumption that must lead to doubt about the intelligibility of existence. —

Knowledge of reason (ratio) does indeed lead us to see what is eternal and necessary in things, but it is nevertheless, according to Spinoza, not the highest form of knowledge. In it the universal law and the single phenomena still stand in opposition to one another, and thinking here is still discursive, and wins its result by comparison and inference. The highest form of knowledge, according to Spinoza, is the intuitive, where the single phenomenon is seen immediately as if shone through by the general order of things, as when I see immediately that two lines are parallel to each other when each of them is parallel to one and the same third, or as when in the knowledge I have of something I at the same time see immediately what belongs to knowing something. The difference between the universal and the individual has here dropped away entirely; their unity is seen at one glance. It is Spinoza's highest aim to understand as much as possible in this immediate, intuitive way; he declares, however, that it is very little that he has so far been able to understand in this way. In the last book of the "Ethics" he seeks to understand the single soul in its inward unity with the eternal being: an intuition that forms the conclusion of all his theoretical and practical striving. The chief condition for rising to this intuition is that one tear oneself loose from external, accidental experience and find the necessary connection with the help of knowledge of reason. — Yet it is not easy to hold fast to a sharp difference between knowledge of reason and intuition; for both show us things "under the aspect of eternity," and moreover: in his description of intuitive knowledge Spinoza says that it *"proceeds* from an adequate concept of the essence of certain divine attributes to an adequate knowledge of the essence of things":

[306] here, then, the discursive has not entirely dropped away. — What Spinoza aims at is nothing less than the highest goal of all knowledge: to unite individuality and continuity, the single and the whole connection, in as inward a way as possible. He succeeds in this only by postulating an intuition that can recall now the artistic apprehension, now mystical beholding, according as the weight is laid on the individual or on the universal moment. —

It is a great pity that Spinoza did not get his theory of knowledge finished. But the fragment we have nevertheless shows that the fundamental definitions and axioms do not come in him as if shot out of a cannon, but are constructed on the basis of a reflection on the presuppositions of all understanding of existence. What is dogmatic in Spinoza appears in the confidence with which he sets up these presuppositions and treats them as realities, not in his having altogether overlooked the existence of such presuppositions.

c. The Fundamental Concepts of the System

α. When Spinoza begins his deductive exposition by defining what he understands by cause of itself and by substance, and by laying down as a fundamental proposition that what does not have its cause in something else must have it in itself, he has really thereby only postulated the possibility of a conclusive knowledge of existence. Existence could not be known if the causal principle did not hold, and causal knowledge could not be completed if there were not something which had its cause in itself and did not again point the inquiring thought beyond itself, as every single phenomenon does. When Spinoza defines substance as "that which subsists in itself and is understood through itself, that is to say, whose concept does not need the concept of any other thing from which it should be formed" (per substantiam intelligo id, qvod in se est et per se concipitur, hoc est id, cujus conceptus non indiget conceptu alterius rei, a qvo formari debeat), he has nevertheless, in his view, not thereby given a purely subjective determination of
[307] a concept, but formulated an objective fact, a reality. For him there is no doubt that substance exists. Its existence is given; but when we reflect on what substance is, we see that its existence is also necessary, since there is nothing that can exclude it, inasmuch as it has its cause in itself. Likewise it follows from the essence of substance that it cannot be produced or pass away; that it cannot be divided; that it cannot be limited; that there cannot be more than one substance. Everything else to which we ascribe existence can then only be properties (attributa) of the one substance or single phenomena (modi, ways) in which it reveals itself. That there appear in existence different properties (such as consciousness and extension) which we cannot reduce to a common form does not

justify (as Descartes thought) the assumption of different substances: they must be conceived as different properties or attributes of one and the same substance. Just as little does the fact that experience shows us a multiplicity of finite things or phenomena, separated in time and space, justify us in assuming a multiplicity of substances. The existence of finite things or phenomena is conditioned through and through, and true subsistence (substantiality) can be ascribed only to the universal order of things, the comprehensive being which bears them all.

β. With the concept of substance, the concept of God and the concept of nature coincide for Spinoza. Through the concept of substance he points back to Cartesianism and scholasticism, through the concept of nature also to Bruno and the Renaissance, and through the concept of God to the religious ideas which first of all set his thinking in motion. God is called substance as the infinite being which reveals itself under an infinity of attributes. Everything that is said of substance must then hold of God too, so that everything that is, is in God and can neither be nor be conceived without God. Both matter (extension) and mind (consciousness) are then divine attributes and do not denote beings outside God, and just as little are the single things or phenomena (modi) outside God. Just as little as God has anything outside himself, just so little are there differences or oppositions in God's essence; here there is no difference between possibility and actuality, no difference of time. Everything that follows from his essence follows with *[308]* eternal necessity; God's freedom consists precisely in this necessity determined by his essence. When God is called the cause of things, it must be well noted that cause and effect are here not two different things, but the effect a revelation of the essence of the cause. God's work is not different from himself, as the work of a human master is. God is the cause of things in the same sense in which he is his own cause; he is an indwelling (immanent), not a transcending (transscendent) cause; his work remains in him, and he in it; he cannot go beyond the compass of his essence. A distinction must indeed be drawn between God and the sum of all the single phenomena (modi): the phenomena make up the produced nature (natura naturata); God is the producing nature (natura naturans). But an external separation does not exist. According to Spinoza it is a purely *abstract* way of regarding things when we regard the single phenomena isolated from "the whole order of nature," that is, from substance, the producing nature. We thereby throw away the condition of understanding them. To see existence *abstractly and superficially,* as it appears to the senses and the imagination, through the isolated appearance of the phenomena, is for Spinoza the opposite of seeing it *as substance* and with the aid of the intellect. Every single phenomenon is only a limited form of the one infinite substance,

arising through a negation, a canceling of all the other forms under which it appears. Every particular determination is a negation[60]).

γ. Substance, God or nature, appears under different properties or attributes. "By attribute I understand," says Spinoza, "that which the intellect perceives of substance as making up (constituting) its essence." Of the infinitely many attributes which the infinite being that we call God or nature has, we know only two, mind and matter. The essence of God or nature appears to us only under these two forms, as working partly in the world of material phenomena, partly in that of mental phenomena. Spinoza's relation to
[309] experience appears clearly here. Not only is it a matter of experience that we know only two attributes, but the very definition of the concept of attribute rests on experience, namely on the fact (which must have dawned on Spinoza in the interval between the "short treatise" and the definitive working out of the "Ethics") that the mental and the material are each understood according to its own laws. The difference between them is then irreducible; we cannot derive the mental from the material or the material from the mental, but within the mental world by itself and within the material world by itself we can demonstrate connection and causal relation. The fact that the mental and the material are irreducible differences in existence is what Spinoza expresses by calling them attributes and by defining the attribute as that which expresses the essence of substance: for substance is, after all, that which subsists in itself and is understood through itself. One must therefore not leap over from one attribute to another. "So long," says Spinoza, "as things are regarded as mental phenomena (modi cogitandi), we must (debemus) explain the order of nature, or the causal connection, through the attribute of mind alone; and so long as we regard them as material phenomena (modi extensionis), the whole order of nature must (debet) be explained through the attribute of extension alone." There is thus only one substance, one causal connection, but it can be regarded in two ways. It has hardly been Spinoza's meaning, however, that the attributes should be only points of view which *we* apply to substance[61]). He says expressly, after all, that an attribute is what the intellect perceives as *constituting* the essence of substance, and a doubt whether what our intellect perceives as constituting also really constitutes cannot be shown in Spinoza. When he says, then, that we *must* explain the mental by the mental and the material by the material, this is a necessity which, like all necessity for him, is an expression of the eternal essence of existence.

If it is asked what it really comes from that one cannot explain the mental side of existence by the material, or conversely, the answer lies contained in the ideal of causal explanation which Spinoza has set up, and according to which cause and effect must

be homogeneous (4th and 5th axioms of the first book of the Ethics). In a letter (Ep. 4) *[310]* Spinoza says still more clearly: "When two things have nothing in common with each other, the one cannot be the cause of the other; for since there would be nothing in the effect which was also in the cause, everything that was in the effect would have arisen out of nothing." When this fundamental proposition is consistently held fast, one has the key to Spinoza's whole system. For from it follows the doctrine of God or substance as the *indwelling* cause of all phenomena (modi), and so the abolition of the theological dualism of God and world. And from it follows further the impossibility that the cause can be material and the effect mental, or conversely, and so the doctrine of mind and matter as expressions of one and the same existence, one and the same causal connection, and with that the abolition of the spiritualist dualism of soul and body. Just as it was the *general* epistemological consideration of the conditions for understanding what is given in experience that led Spinoza to set up the concept of substance (as the concept of hypostatized causality), so it was the *special* consideration of the possibility of understanding the mental and the material that led to the doctrine of attributes. This doctrine is essentially conditioned by the conviction of the possibility of conceiving the material side of existence as a continuous whole, and of the impossibility of deriving the mental side of existence from it. Spinoza makes a stand against Descartes and against Hobbes alike. His theory of attributes is an attempt to solve the difficulties from which Cartesianism and Hobbesianism (spiritualism and materialism) suffer. He has set a new, original hypothesis over against spiritualism and materialism.

Now Spinoza has not strictly held fast to this meaning of the doctrine of attributes. With the problem of the relation between mind and matter he has in several places confused the problem of the relation between knowledge and its object. It is easy to see that these are two different problems; for both mind and matter (existence from its mental as well as from its bodily side) are objects of knowledge, and the problem of knowledge comes up with regard to all the sides from which existence can be seen. But since Spinoza, *[311]* as a dogmatic philosopher, confidently relied on the agreement between our reason and the essence of things, he had to assume — as already set forth — that to every relation of thought there corresponded a relation in existence. And this *epistemological* doctrine of identity he then confused with the *psychophysical* doctrine of identity which his doctrine of attributes expresses. When he lays down the proposition: "The order and connection of representations is the same as the order and connection of things" (Eth. II, 7), this proposition is ambiguous, since it can be understood both epistemologically and psychophysically. Spinoza himself understood it rather epistemologically. This is

seen from his example: to my concept of the circle corresponds the circle existing in reality (Eth. II, 7 Schol). Psychologically it would have had to run: to my representation of the circle corresponds the state my brain is in when I have this representation. But when Spinoza, in immediate connection with this, maintains that mental phenomena are to be explained by mental phenomena, material phenomena by material ones, it is an entirely different problem that he has to do with. In the third book of the Ethics, where the point of view is psychological, not epistemological, he accordingly applies the doctrine of attributes purely psychophysically. — This confusion is naturally connected with the fact that the problem of knowledge in Spinoza does not appear as an entirely independent problem. Only through a critical revision of our knowledge — with regard to its capacity to know the mental as well as the material side of existence — does the difference between the two problems come clearly to light[62]).

Just as, according to Spinoza, there is an inward relation (an indwelling relation, a relation of immanence) between substance (God or nature) and the single phenomena, so for him the great fundamental forms and fundamental properties that existence presents must also be substance's own attributes. It was a bold consequence of this when Spinoza declared matter or extension a divine attribute. Matter, on the usual theological conception, is only something derived, a product of mind. Just as little as Bruno and
[312] Böhme could Spinoza rest content with this conception, which conflicts above all with his fundamental thought of the indwelling causal relation. With regard to Spinoza's doctrine of matter as a divine attribute it must be well noted that by matter he here means what is substantial in material phenomena, that in them which subsists through all changes and in spite of all divisions, and which can neither suffer, arise nor pass away. The material side of existence is, just as much as the mental side, a revelation of the Deity or of substance; to its "infinite power of thought" (potentia infinita cogitandi) corresponds the infinite material extension, of which the single phenomena are parts. What lies at the ground of existence, from whichever side it is regarded, is the causal connection and the unity to which the causal connection bears witness.

δ. The single phenomena (modi) are, as will be remembered from what has gone before, the particular ways in which substance comes forth within the different attributes. Since the attributes hold for substance, they hold also for the single modi: every single phenomenon can therefore be regarded from infinitely many sides, just as well as the whole of existence, of which it is a member or part, only that we know, after all, no more than two of these sides. The definition of modus is: a determination of substance, or: that which has its subsistence in something else, through which it is

also understood. As affectio substantiæ a modus is understood only through substance and has its subsistence in it — just as, one might say, the single wave subsists only as part of the sea and is understood only by being seen as part of the sea. In every single phenomenon there stirs an impulse of self-preservation which is a part of the divine power working in all things. But in each single modus this power stirs only in a limited way; one cannot, then, understand the way in which the single modus acts or suffers without seeing it in its interaction with all other modi. The state of the single modus is therefore understood only through something other than itself. Here an external (transscendent) causal relation comes into play. But however externally the single phenomena stand over against one another, they are nevertheless, all together and each by itself, only determinations (affectiones, modificationes, determinationes) of the one infinite substance. In the produced nature (natura naturata) the producing *[313]* nature (natura naturans) is at work at every point.

Within natura naturata Spinoza distinguishes between necessary and infinite modi, which follow immediately from God's eternal nature, and finite modi, which need other modi for their coming into being. Within the attribute of extension he names (Ep. 64) as infinite modi motion and rest; here he seems in particular to be thinking of the conservation of motion, or of the law that through all the changes of matter the same relation between motion and rest is preserved, — the law which makes it clear that every single body is only a part of the whole universe. Within the attribute of thought he names the "infinite intellect" (intellectus infinitus), which is an immediate effect of the infinite divine power of thought, and which — since it is placed in analogy with the constancy of motion — must be taken to mean the mental energy that constantly remains the same in the world, however much the single mental phenomena may change. (Perhaps it is chiefly the conservation of motion and "the infinite intellect" that Spinoza, in the treatise on the improvement of the intellect, has in mind with the expression "fixed and eternal things.")

ε. For a closer characterization of the system set forth here, a series of problems and difficulties to which the system leads shall only be indicated further. — God or substance is spoken of as acting, although the form of time is to hold only for the modi, natura naturata, not for substance, natura naturans. This is connected with the confusion of ground and cause mentioned above. — There subsist two causal relations: one among the modi mutually, an external, transscendent one, and one between substance and the modi, an immanent one. How are the two to be reconciled? — The unity of substance and the multiplicity of the attributes and modi are not brought into harmony; no answer is given

to how the one substance can appear under different points of view, or how an infinity of effects can follow from the one substance. — And when for Spinoza the concepts substance, nature and God cover one another, he introduces with the designation God a
[314] determination of value which is grounded only by his defining perfection as reality; the question whether this definition can be held fast is not examined more closely, and at a later point in his exposition he even sets up — as we shall see — a concept of perfection according to which not all reality is perfect. — Thus even the boldest speculative system leaves unanswered questions enough behind it. What is great in Spinoza is his resolute carrying through of the thought that existence must be rational; from this he concludes that its essence must be identity, absolute unity. Consistently, he would have had, instead of several systematic fundamental concepts, only a single one: substance. To this concept one may perhaps ascend — but once one has reached it, one can in any case not get down again. — These objections to Spinoza's system do not, however, strike his fundamental endeavor, the fundamental direction of his thinking, but are made possible for the most part by the dogmatic procedure he employs.

d. Philosophy of Religion

It will be of interest to stop for a moment to see in what relation Spinoza's philosophy stands to the current religious views. When he named his highest concept (substance) God, he had himself at once given occasion for such a comparison. Already in his correspondence we see this point discussed. On a particular occasion he writes (Ep. 73): "I hold an opinion of God and nature far different from the one which the modern Christians are accustomed to defend. For I maintain that God is (as one says) the indwelling, not the external, cause of all things. That is to say, I assert with Paul that in God all things are and are moved . . . But when some think that the 'theological-political treatise' aims to show that God and nature, where by nature one understands a mass or corporeal matter, are one and the same, they are entirely mistaken." — From the usual theological conception Spinoza's philosophy separates itself especially on two points: in not conceiving God and the world as two different beings, and in not ascribing human
[315] properties to God. In the first period one made short work by calling his conception atheism and materialism. Henry More, e.g., says in a writing in which he sets out to refute Spinoza's "gross atheism" (crassissimus atheismus) that God and matter are one in his system. Therefore, says More, stones, dirt, lead, dung are God, since these things are, after all, matter! So Spinoza is then a "sordid and befouled atheist" (sordidus et lutulentus atheus)[63]). A small sample of the polemics of the seventeenth century! — People were

at a loss with the new system on account of its peculiarity; it could not easily be fitted into the current rubrics of heresy, and in its sharply cut forms it nevertheless provoked offense on several points. Later the expression pantheism came up (first, it seems, in Toland). Here one had a word that hit the mark better. And yet there is no full agreement about its use. According to some it denotes a view that conceives God's relation to the world as immanent; according to others what is essential to pantheism is that God is not thought of as a personal being[64]). By the first of these marks there is no doubt that Spinoza's system is pantheistic; by the second it is less clear how one is to designate it.

In one of his letters (Ep. 54) Spinoza says that, so as not to confuse the divine nature with the human, he does not ascribe to God human properties such as will, intellect, attention, hearing, etc. The more precise grounding of this is found in several places in the "Ethics." — It is clear that the applicability of the psychological determinations must fall away with the falling away of the relation of time, and time holds for Spinoza only for natura naturata, not for natura naturans. With the relation of time all change falls away too; God can pass neither to greater nor to lesser perfection and can therefore feel neither joy nor sorrow, love nor hate. Further, God as substance has nothing outside himself, no external object toward which feeling or will could direct itself. Our consciousness, our intellect as well as our feeling and our will, is determined by such external objects, and when they fall away, consciousness too falls away for us. — If one wants to use words such as intellect and will of God, one must make it clear to oneself that their meaning *[316]* must undergo such a change that at bottom it is only the word that is common to the two uses, much as one uses the word dog both of the constellation the Dog (Sirius) and of the barking animal on earth. The whole impulse to transfer human properties has therefore no objective validity. If triangles could think, they would conceive God as triangular; if circles could think, they would conceive God as round.

On the other hand Spinoza recognizes the mental as an independent side of existence. God is substantia cogitans as well as substantia extensa; cogitatio (thinking, mind) is one of God's attributes, and so expresses God's essence! Intellect and will belong only to the produced nature, not to the producing; but there must be a ground in the latter for their being able to arise, and if the "infinite intellect" has eternal subsistence through all the change of mental phenomena, this too must have its ground in the essence of the Deity. One might perhaps express Spinoza's conception thus: God is mind, but not personality, because only a limited being can be a personality. (Spinoza mentions the expression personality in a single place only, but declares it obscure.) It must be noted, however, that Spinoza, by declaring cogitatio a divine attribute and by speaking (as will

be mentioned later) of God's infinite love of himself (of which man's love of God is a part), himself undertakes just such a sublimation of the psychological properties as he denies on other points. Against cogitatio as a divine attribute objections could be raised similar to those which Spinoza himself brings against the current conception. — Those to whom it is absolutely a matter of bringing a philosophical system under one "ism" or another will to that extent be at a loss with Spinoza, especially since every thinking
[317] theism must undertake sublimations of the same kind as Spinoza's.

e. Philosophy of Nature and Psychology

By setting up extension as one of the attributes of existence and by demanding (by virtue of the very definition of the attribute) that in the explanation of a phenomenon one is to keep to the attribute under which one regards it, Spinoza has already established a purely mechanical explanation of nature as the only scientific one. As falling under the attribute of extension, all bodily phenomena are to be explained only by bodily causes, by causes that act from other points of space than the one at which they themselves are. It is therefore very easy for Spinoza to give a proof of the principle of inertia: this really lies already in the setting up of extension as an attribute. Just as dogmatically as Descartes and Hobbes, Spinoza sets up the fundamental propositions of mechanics as eternal truths — although he must concede that he cannot derive motion from extension itself[1]. There can be no differences of essence between bodies; these differ only in respect of motion and rest, speed or slowness. The laws of motion are eternal laws of nature and of reason. — In the doctrine of attributes is contained also the rejection of every assumption of purposes in nature. Since the one attribute cannot determine the other, no material phenomenon as such can have its ground in a thought, which the teleological
[318] explanation presupposes. In a longer note to the first book of the "Ethics" Spinoza seeks to show that the teleological conception springs from man's practical need and wish, from his impulse of self-preservation in connection with his original ignorance of the real causes; he assumes that everything is arranged for the good of men and that the gods work for this; where no reasonable purpose can be found, they content themselves with finding the explanation in God's will, this refuge of ignorance (asylum ignorantiæ),

[1] *Spinoza* makes this concession in the last letter we have from his hand (of 15 July 1676), an answer to the philosopher and mathematician Tschirnhausen, who had asked with great acuteness how one could derive from the general concept of extension the figure and motion of bodily things. Tschirnhausen pointed out that this difficulty does not exist in Descartes, who assumed that God created matter in motion; but Spinoza, for whom extension is a divine attribute, then lacks the concept of force which in Descartes was conceived in theological form. Spinoza declares the definition of matter as extension insufficient and announces a closer discussion of the question, which he did not manage to give.

which is always at their service. If mathematics had not pointed out a rational way to knowledge, the teleological conception would forever have hindered the knowledge of nature. The teleological conception, however, not only hinders the knowledge of nature but also presupposes unworthy representations of the Deity: if God acts according to purposes, he is, after all, not perfect, but becomes so only when he has reached his purposes! Men make bold to apply moral and aesthetic concepts to nature, find order and confusion, beauty and ugliness, good and evil in it, supposing that it is designed to appeal to their feeling. They do not see that the perfection of nature is one with its might, and that this might is infinite, so that it is able to produce all possible degrees and forms of existence, those which to us stand lowest just as well as those which to us stand highest.

In a series of lemmas in the second book of the Ethics Spinoza indicates how he conceives a mechanical view of nature carried through. — According to the law of inertia every body is determined to motion or rest by another body, this by a third, and so on. The way in which a body is determined depends partly on its own nature, partly on the nature of the body that acts on it. — When bodies of the same or different size are so held together by other bodies that they touch one another, or when they communicate their motions to one another in a certain ratio, they are said to be mutually united and all together form one body or one individual, which is distinguished from other individuals by the definite union between the single bodies of which it consists. Such an individual continues to be the same even if some parts are taken in and others given off, and even if the size of the parts and the direction of the motions change, provided only that the *[319]* whole form of the composition continues to be the same. Several such individuals can be united into an individual of the second order, and several individuals of the second order into an individual of the third order, and so on, until at last the whole of nature stands before us as one single great individual, whose single parts vary infinitely without the whole individual undergoing any change.

In all its stages and degrees nature is animated. This too follows from the doctrine of attributes: since each attribute expresses the whole substance, to every form under the one attribute there must correspond a certain form under the other attribute. To the different stages of individuality in material nature, then, there correspond different stages of mental life in mental nature. Just as the human body consists of many composite individuals of different constitution, so the thought which makes up the nature of the human mind is also composed of many different thoughts. The mind and the body are, after all, one and the same substance, or rather one and the same part (modus) of

substance, and what under the attribute of extension appears as more or less composite forms of motion appears under the attribute of thought as more or less composite forms of thought. When I have a drive or make a decision, there takes place simultaneously under the attribute of extension a bodily determination according to the general laws of motion and rest, and this is to be understood in such a way that at bottom it is one and the same thing that happens. By this assumption, as Spinoza shows, one is spared from maintaining an influence of the soul on the body or conversely, something of which no one can form a clear concept. To say that the soul sets the body in motion is at bottom to say that one does not know how this motion comes about. This assumption has the further advantage that one does not set arbitrary limits to the activity of material nature. No one, after all, has yet found the limit of what can be effected by the laws of material nature itself, so that one has no right to invoke the intervention of the soul because the
[320] body seems to act in a marvelous way. Nor does the appeal to the "freedom" of the will help. This so-called "freedom" (real freedom, according to Spinoza, is that which consists in following the necessity of one's own nature) is an illusion which arises from one's not knowing the causes of one's own acting. The chatterer thinks he acts freely when he lets his tongue run, just as the stone, if it could think, would think that it fell to the ground of its "free" will. Besides, the decision is, after all, dependent on memory, and of this we are not always masters! —

Although Spinoza rejects every explanation of mental phenomena by material causes, he nevertheless uses — which is also fully justified on his presuppositions — the connection between the material and the mental to throw light on mental phenomena where these cannot be understood immediately. Every sense-sensation corresponds to a bodily state. Now this state has indeed its cause in the influence of other bodies, but this influence is modified by the constitution of our own body: from which it follows that the sense-sensation corresponds not only to the nature of the external bodies but first and foremost to the nature of our own body. Nevertheless we involuntarily regard every sense-sensation as the expression of something real, until other sensations or representations arise in us which exclude the assumed reality. According to Spinoza, then, it is through the mutual struggle of sensations and representations that our conception of reality is cleared and determined. — Sensations call up representations of what resembles, or has occurred together with, the object of the sensation[65]). The laws of the connection of representations are for Spinoza mental laws of nature, the counterpart of the laws of motion within the attribute of matter. Spinoza distinguishes sharply between the association of representations and thinking proper. This difference has been discussed in

the section on his theory of knowledge. Sense-sensation and the association of representations express man's state as a member of the series of phenomena, of the infinite natura naturata. The higher stages of knowledge, rational knowledge and intuitive knowledge, are reached by going back to that which stirs in everything as common law and essence, that is, to the infinite substance, and seeing everything taken up into it, or seeing it *[321]* sub specie æterni. — Hobbes's psychology has here the advantage over Spinoza's that it shows the possibility of how thinking proper, by concentration on a purpose or a task, can develop out of the involuntary association of representations. Spinoza still, in scholastic fashion, posits thinking proper as an entirely peculiar activity, different in essence from ordinary representing. All the more remarkable, then, that he nevertheless confuses the epistemological and the psychophysical points of view in the doctrine of attributes — although on the other hand this confusion would hardly have occurred if he had seen that the laws of the association of representations are in reality not suspended in thinking proper. —

The treatment Spinoza gives of the psychology of the feelings belongs to the finest he has accomplished, and that has been accomplished in psychology at all. In the first place he has the merit of maintaining the necessity and the justification of a purely natural-historical investigation of the human feelings and passions. He will (as an inquirer) neither mock them, laugh at them nor weep over them, but seek to understand them, and will therefore regard them with the same equanimity as if it were material phenomena of nature or geometrical figures that he had before him. Next, he has brought forward the most important points of view that come into application in the explanation of the feelings.

As regards the position and significance of feeling in mental life, Spinoza held different views at the different stages of his philosophical development. In his first work, "the short treatise" (II, 16, 5), his conception is entirely intellectualistic. It is knowledge, the representations, that determine all the phenomena of mental life. The will is merely a consequence of the representation, an affirming or rejecting. And here, moreover, the representations themselves are still determined by the influence of the objects, since in this writing he assumes an influence of the one attribute on the other. Already in his writing on the Cartesian philosophy he indicates that the representations are themselves expressions of our activity, so that one cannot distinguish between the arising of the representation and its affirmation or rejection. And with the definitive doctrine of attributes this indication is carried further, but in such a way that intellect and will *[322]* are now made completely one. To think is an activity of the mind, and all activity of

the mind is thinking: so, roughly, one might express Spinoza's conception here; it is formulated in the corollary to the 49th proposition of the 2nd book of the "Ethics": Voluntas et intellectus unum et idem sunt. But in the 3rd book of the "Ethics," where the natural history of the feelings is given, an entirely new conception of the will appears, since the will is here determined as conscious impulse of self-preservation (Eth. III, 9 Schol). It is now so far from being one with, let alone a consequence of, knowledge, that knowledge is precisely dependent on the will: "we do not seek, will, desire and wish for anything because we think that it is good, but conversely, we think that something is good because we seek, will, desire and wish for it." Here, then, need and will are one, and the representations dependent on them. In his treatise on the history of Spinoza's development, in which Tönnies has thrown light on this point in a brilliant way, he conjectures with great probability that it is Hobbes's influence that has produced this last change in Spinoza's conception. Hobbes puts drive and will in close connection with each other and with the impulse of self-preservation, and he also stresses the influence which the thought of a purpose has on the connection of representations. In Spinoza's work as it lies before us there is an evident contradiction between the definition of will II, 49 Coroll. and the definition given III, 9 Schol., so that it is highly probable that two drafts have been worked together without all traces of different conceptions having been removed.

From the impulse of self-preservation as the expression of the nature of each single being Spinoza now proceeds in his exposition of the psychology of the feelings in the 3rd book of the "Ethics." Even if he thereby changed something in his psychology, he changed nothing in his general philosophy. In every single being the infinite nature, the Deity itself, is at work. The striving of each single being to preserve itself is then only a part of the infinite divine working in the whole of existence. Already in the "short treatise" (I, 5) he saw (like Bruno) in this impulse of self-preservation, living in all beings, the revelation of divine providence. And this need is one with the very nature
[323] of things, takes on a different character according to the nature of the single beings: to exist is to strive to preserve oneself. This striving is called drive when it is bound up with consciousness: by drive, then, is understood the very essence of man, insofar as it is thought of as determined to activity by reason of a change in its state. When this drive is regarded in particular from the mental side, it is called will. Pleasure arises when this striving is furthered, pain when it is disappointed. Be it noted: it is the very transition to a more favorable ("more perfect") state that is bound up with pleasure, and the very transition to a less favorable ("imperfect") state that is bound up with pain. The states

themselves, as at rest, are not bound up with feelings. Nor, however, is any express comparison made between the states: the feeling corresponds immediately to the very change of the states.

Just as Spinoza here states what we now call the biological significance of feeling, its connection with the advance or decline of life, and just as he also has a clear eye for the significance of the relation of contrast for the arising of the feelings ("the law of relation in the domain of the feelings"), so he has also put the third main point in the psychology of the feelings, the significance of the representations and the combinations of representations for the development of the feelings into special kinds, in an excellent light. — From drive, pleasure and pain he derives, with the help of the laws of the combination of representations, the special kinds of feeling. We love what causes pleasure, but hate what causes pain. We love further what furthers what we love, just as we hate what harms what we love. And when we represent to ourselves a being that resembles ourselves as having a certain feeling, we involuntarily get the same feeling ourselves. From this compassion can arise — but also envy and wounded ambition, namely when we wish to possess alone what the other enjoys, or when we wish the other to be disposed entirely as we are disposed. — According to these general principles Spinoza now carries out his consideration in detail. He has founded the theory of evolution in the domain of the feelings, by showing how our feeling, under the influence of the representations and by mutual composition and mixture, can take on forms that are highly different from the *[324]* elementary initial forms. It is the conditions of growth of feeling that he has formulated. — How far it is possible to keep exclusively to the one starting point he has seized on, the individual impulse of self-preservation, is a question that cannot be discussed here.

f. Ethics and Political Theory

It is characteristic of Spinoza that there is a practical motive for his thinking, and that his thinking nevertheless rises high above the world of finitude and experience in which alone practical motives and practical striving are possible. Spinoza rejects the validity of every valuation of existence; existence follows its eternal laws, and our ethical judgments, which rest on a comparison between the more or the less perfect in relation to the ideal of human life that we form for ourselves, cannot find application to the eternal essence of nature itself, for which time, measure and number do not hold. Spinoza insists on the subjectivity of the ethical qualities as energetically as the subjectivity of the sense qualities had been insisted on from Galilei's time onward. Insofar as Spinoza holds fast to his conception of existence "under the aspect of eternity," he can have no real ethics,

and needs none either. Of ethics there can be question only in a world where the relation of time holds, where there is becoming and development, change and opposition, and where models can be formed and purposes set up. As Spinoza has aptly expressed it: the judgments of good and evil rest on comparison — but in the eternal unity with itself of the infinite, divine nature all comparison falls away, because all difference and opposition has fallen away. The investigation of Spinoza's theory of knowledge showed us that all *riddles* fall away under "the aspect of eternity"; we now see that all *tasks* fall away too.

But just as Spinoza distinguishes between natura naturans and natura naturata, he also distinguishes between the purely theoretical standpoint and the practical standpoint.
[325] The practical standpoint holds only within natura naturata. Spinoza sets up an ethics with the same right as that with which he maintains the significance of the modi of the world of experience, the finite world. There is no greater inconsistency in his setting up an ethics than there is in his setting up, at all, other concepts than the concept of absolute substance.

The possibility of an ethics is closely connected in Spinoza with his psychological doctrine of the feelings. The foundation of ethics is the impulse of self-preservation which lies at the ground of the manifold feelings and passions. For out of the impulse of self-preservation there can develop a need not to be tossed about by all the feelings and passions which external causes call up in us. As a part of nature, as a modus among other modi, man is dependent and is tossed about like the waves of the sea without knowing goal or direction. We naturally strive, then, to assert our being, after full, inward activity. The fundamental ethical virtue is *strength of mind* (fortitudo), an inward might which makes man free and independent. Virtue is the very essence or nature of man, insofar as he has the might to produce effects that can be understood through the very laws of man's nature. For man to be able to attain this character, he must make it clear to himself that no feeling and no drive can be checked in us except by a stronger feeling or drive. (This proposition was already maintained by BACO: De Augm. Scient. VII, 3). Spinoza builds his proof of this proposition on the law of inertia, which, by virtue of the identity of mind and matter, has its counterpart within the mental side of existence. Ethical development becomes possible through the fact that feelings can be called up which gradually displace the feelings by which they themselves were originally called up. When, by comparing different forms and stages of human conduct of life, we form for ourselves an ideal of human nature (idea hominis, tanqvam naturæ humanæ exemplar), we call everything that brings us nearer to this ideal good, everything that hinders us from approaching it evil, and there arises a drive to approach the ideal as much as

possible. The knowledge of good and evil is — since it concerns what is truly useful and harmful to us — nothing other than conscious joy at approaching the ideal or conscious *[326]* sorrow at moving away from it, and this knowledge can, precisely because it is one with a feeling, become a might in the soul. Man now soon sees that he cannot attain full strength of mind and freedom of mind by his own help. The whole of outward and inward life must be ordered and made secure. Above all it is a matter of developing a true knowledge; this too is reached only by united strength, just as, on the other hand, there is nothing that better unites men than the striving after such goods as can be common to all. The passions that bring strife among men always concern something which the individual alone can possess. The more each individual attains freedom and strength of mind, the less strife will there be among men, and the more will they act in common, because they all seek what is truly useful to human nature. Nothing is so useful to men as men. Best of all would it be if all could become as one soul and one body, seeking the common utility. Those who strive after their own true utility therefore wish nothing for themselves which they do not wish for other men, and thereby they become just, faithful and honorable in their conduct. The strength (fortitudo) in which virtue consists therefore appears not only as *courage* (animositas), by which man immediately asserts his personality, but also as *magnanimity* (generositas), in that man strives to help others and to bind them to himself in friendship. — By distinguishing between courage and magnanimity Spinoza makes an advance in relation to Hobbes and the thinkers of the Renaissance, in that he gets an eye for the different elements and possibilities, of which some can lead more in an individual, others more in a social direction. Courage is a predominantly individual virtue, magnanimity a social virtue. But Spinoza is hindered, already by his starting point in the pure impulse of self-preservation, from entering on a closer investigation of the relation between self-assertion and devotion.

Characteristic of Spinoza is the valuation which, after setting up the model of human development, he makes of the different feelings. The feeling of pleasure is in itself good, the feeling of pain in itself evil. Therefore hatred, fear, contempt, pity, remorse *[327]* and humility are evil — insofar as they are not necessary intermediate links for the arising of feelings of pleasure. This is grounded by the fact that the feeling of pleasure corresponds to an advance in perfection, the feeling of pain to a decline. He who is led by true knowledge, the spiritually free man, has his gaze directed toward the positive development of life in himself and in others. He thinks of nothing less than of death; his wisdom is a meditation on life (meditatio vitæ), not on death. He strives, as far as he can, to outweigh the hatred, anger and contempt of others by love or magnanimity. He

who fights hatred with hatred lives a wretched life; but when hatred is conquered by love, the conquered yield with joy, not because their strength diminishes but because it grows. There is no way in which a man can better display his strength of mind than by developing others to live according to their own free knowledge[66]).

While Spinoza's ethics leads in this way from the impulse of self-preservation as its foundation out to practical action and the founding of society, there also stirs in him another tendency, which from the same starting point leads in a mystical and individualistic direction. From his earlier psychological standpoint there still constantly presses forward the presupposition that man's real essence consists in pure knowledge: the more this can develop, the more does man's real nature come forth, the more is he truly self-active, and the true impulse of self-preservation therefore leads to the development of knowledge. It was, after all, also the knowledge of the connection in which our mind stands with the whole of nature that stood for Spinoza as the only good which could not be taken from us by external causes. Therefore Spinoza can, almost in the same breath (cf. Eth. IV, 22 Coroll. with 26 Dem.), set up the two propositions that the striving after self-preservation is the first and only foundation of virtue, and that the striving after knowledge (conatus intelligendi) is the first and only foundation of virtue. Alongside the realistic psychology, according to which Spinoza conceives feelings and passions as positive powers in our being, there runs another conception, according to
[328] which feeling is defined as imperfect or confused representation (idea inadæqvata sive confusa, see the concluding explanation in the 3rd book of the "Ethics"). According to this other conception feeling falls away with the full and clear development of knowledge. The more we understand ourselves and our states as determined by the infinite nature, by the Deity which stirs in us as well as in everything that acts on us, the more do we see ourselves not as single, isolated, powerless beings, but as taken up into and as one with the Deity. We feel pleasure at this thought, since it is the fruit of our mind's highest activity, and pleasure at the thought of the being that is the cause of the joy of knowledge we feel. Thereby there arises a spiritual love of God (amor intellectualis dei), which gives us the highest peace. God does not stand here as an external object; our own innermost power is a part of the infinite power that stirs in everything, and our love of God is one with the infinite love with which God loves himself: God's love of men and men's love of God are one and the same. We see ourselves and everything under the aspect of eternity. — Spinoza the mystic comes into conflict with Spinoza the psychologist; for the intellectual love (the human and the divine) is to be raised above becoming and opposition, but Spinoza has himself, after all, explained feeling

as corresponding to a *transition* — either an advance or a decline! Only by explaining feeling as unclear thinking and abolishing the law of relation can Spinoza hold fast to the mystical conclusion of the ethics.

In the fifth book of the "Ethics," where this mysticism prevails, is found also Spinoza's doctrine of immortality. In that intellectual love of God, grounded on the understanding of our unity with the absolute being, our true essence expresses itself: this essence, then, is not dependent on time, but is a part of the infinite intellect (intellectus infinitus), which continues to subsist although the single mental phenomena arise and pass away, just as motion subsists through all material changes. As an immediate form of the divine working we are immortal. But this immortality — which is not a *continued* subsistence but an *eternal* subsistence, since the relation of time has no significance at all here — seems able to belong only to those in whom the highest knowledge, and with it the full *[329]* and clear self-activity, is developed[67]). Whether this subsistence as an eternal element of thought (modus cogitandi æternus) is personal or not cannot be seen from Spinoza's exposition; his notion of immortality is in any case very different from the popular one. And at the close he remarks that the assumption of the eternity of our mind is not a condition of ethics, pointing out that he himself has developed his ethics without using this assumption. Blessedness is not the reward of virtue but is one with virtue itself. For only when we can feel joy in our mind's highest working can we master our passions. —

The deep and sound thoughts in Spinoza's ethics would have come to a fuller unfolding if he had carried out his psychological theory in detail. As it is, intellectualism gains the upper hand in him, especially when he is to depict the highest forms of spiritual life. His ethics thereby becomes too much an ethics for philosophers. The significance of immediate devotion and of artistic intuition does not come into its own, although in his "intellectual love," grounded on intuition, there are hints in these directions. But no one will be able, with an unbiased sense, to immerse himself in Spinoza's wonderful work without confirming Goethe's words about the boundless unselfishness that shines forth from every proposition, about the lofty resignation that once and for all bows to the great laws of existence instead of scraping through life with the help of petty consolations, and about the air of peace that runs through the whole book. —

But peace, according to Spinoza, must be won through strife. It is won by the individual through the struggle with the passions, in which a new, higher passion is developed, and it is won by men in union through the savagery of the state of nature, where everyone takes as much right as he has might, being replaced by the life of the state through the establishment of an authority that can subdue the conflicting drives

and bring harmony between the impulses of self-preservation of the individuals. In his political theory, as in his psychology, Spinoza is decidedly influenced by Hobbes. Only
[330] he does not have Hobbes's blind and one-sided fear of the multitude. He is consistent enough to hold fast that human nature is the same in all, and that safeguards are needed just as much against those who rule as against those who are to be ruled. One can just as little presuppose the best motives in the rulers as in the multitude. And if it is thought that the multitude has no understanding of public affairs, Spinoza answers that this comes from its being kept in ignorance of them. Spinoza is one of the first democratic philosophers of law in modern times; but since his "political treatise" was not completed, we do not know his doctrine of democracy in detail. It is characteristic that while Hobbes maintains that a democracy is in fact always an aristocracy — of orators, Spinoza maintains that a monarchy is always an aristocracy — of officials. In reality, of course, the political functions must always be exercised by a few — but it depends on where one finds the ultimate source of authority. According to Spinoza it depends on the state's being built on the united might of the many, in that they are united into one will, just as the strength of the individual also grows by being united with that of the others. But this spiritual unity, which is the condition of the unity of might, can be present only where the state strives after what sound reason points out as useful to all. One can put up with much for the sake of peace; but if slavery, barbarism and isolation are to be called peace, then peace is the most wretched of all things for men. The state is not to make men into beasts or machines, but precisely to bring it about that their mental and bodily activity can unfold freely, and that they can use their free reason.

Only on the free adherence of individuals does the might of the state rest, and the individual therefore does not give up his natural right on entering the state, but has it precisely secured and developed. The state would act against its own purpose by suppressing the individuals' freedom of thought and of speech; it would first and foremost make the noblest and best men in the land its opponents.

As Spinoza maintains the freedom of expression, so he also maintains religious freedom. It is the main subject of his "theological-political treatise." He wants to show
[331] here that religion has nothing to do either with the state or with science, and that the state, by its nature and its purpose, must let the inner spiritual life develop with full freedom. The state can have power only over outward action, not over disposition. It can lay down an outward cult, and its dominion in this respect is to be preferred to a hierarchy. But even when it uses its might in this way, it does best to let religious worship consist merely in the exercise of love and justice. The mental nature of men

(ingenium) is so various that what awakens reverence in the one causes laughter in the other. It is therefore best to let everyone keep his free conviction and right to interpret the religious documents according to his own intellectual tendency (ex suo ingenio). — From science religion separates itself by its practical character. It aims at piety, not at truth. The prophets were not men who were especially highly gifted in intellectual respects. Imagination was stronger in them than understanding. Their task was to impart to the people, through images and symbols, representations that could bring it to live in that obedience to God which shows itself in love of one's neighbor. The multitude, which cannot follow the inferences of reason, must have historical models for guidance. Only thereby does it come to obey not out of mere fear but out of its own will and devotion. What theoretical opinions men have of God — whether they think he is fire or spirit or light or thought — is a matter of indifference, so long as they have in him a model that can lead them on the right way. — While according to Spinoza the prophets do not see the truth in its pure form, but only in images, the divine wisdom was revealed immediately in Christ, the only one who has had the capacity to apprehend something which is not contained in, and cannot be derived from, the first grounds of our knowledge. In a letter (Ep. 73) Spinoza declares that he does not understand what the Church means by the doctrine that God has taken on human nature; it seems to him as absurd as that the circle should take on the nature of the square. But he assumes that God's eternal wisdom, which reveals itself in all things and especially in the human mind, has revealed itself most of all in Christ. — Spinoza demands that the biblical writings be subjected to a purely historical investigation. The Bible is to be explained by itself, *[332]* just as nature is explained by itself. Only when one has established what the Bible really teaches can one raise the question whether this teaching is true or not. Just as Spinoza has maintained the independence of the investigation of nature and of ethics, so he here maintains the independence of the historical method. His historical eye shows itself in the demand that in the interpretation of the writings one is to take account of what one knows of the individualities of the authors. One is also to investigate the origin of the writings. The investigation Spinoza undertakes leads him to assume that the five books of Moses were composed in the time of the Exile. Even if more recent research does not always go so far, it nevertheless has a forerunner in Spinoza.

It was through his philosophy of religion that Spinoza first had an influence. Before Lessing and Goethe, and after them the speculative philosophers, pleaded his cause at the end of the eighteenth century, his views had an influence of no small extent in the Netherlands, both in such a way that individuals quietly found nourishment for their

life of thought and feeling in his ideas, and in such a way that religious sects were even formed, now of a more mystical, now of a more rationalist character, whose conception and application of Christianity derived, through more or fewer intermediate links and modifications, from the writings of the solitary philosopher. Right up to the end of the eighteenth century Dutch orthodoxy had frequent occasion to denounce Spinozist thoughts as heresy; his books were repeatedly forbidden and confiscated. In its more mystical form religious Spinozism united with the influence of Jakob Böhme's writings, which from the end of the seventeenth century were eagerly read in the Netherlands. — Remarkably enough, the three elements which were bound together into a peculiar unity in Spinoza's thinking have later each in its own time won enthusiastic adherence: first the religious, in the Dutch sects, then the idealist, in the German Renaissance at the end of the eighteenth century, and finally the realist, among philosophers and natural scientists in our own day. And even thinkers who officially would not acknowledge him
[333] were nevertheless deeply and strongly influenced by him. This holds in particular of the man who has often been set up, by himself and by others, as Spinoza's antipode: Leibniz.

6. Gottfried Wilhelm Leibniz

a. Biography and Characterization

In Descartes, Hobbes and Spinoza we find three great systems developed which, for all their differences, have this in common, that they assert a purely mechanical connection of nature as regards the material side of existence. In these systems the consequences had been drawn from the principles whose establishment had made the rise of exact natural science possible. They were attempts to show how existence looks when these principles hold absolutely for all material phenomena (whether one then holds that there are other phenomena than the material ones, or not). To many of the old school these new principles and systems were the epitome of arbitrariness and ungodliness. A reaction makes itself felt in various forms at the end of the seventeenth century. In philosophical respects the reaction of greatest interest is the one which will not give up what had really been won in scientific respects, and which, not in an external way, by compromising or by patching incompatible elements together, but in an internal way, through an investigation of the presuppositions that lay at the basis of the mechanical view of nature itself, sought to lead beyond it and to reconcile it with the ancient and medieval conception on which men had turned their backs with such great energy. In this sense Leibniz signifies a reaction against those three great thinkers. But perhaps never

has a reaction been carried through in so profound a way, even if Leibniz, consciously and unconsciously, accommodated himself in his manner of expression to conservative views far more than his standpoint entitled him to. What from the first offended Leibniz in the modern ideas, for all his admiration for them and for those who bore them, was that they threatened the aesthetic and the religious contemplation of nature. The legitimacy of the concept of purpose was rejected in the world-view, and the world seemed to become *[334]* a mere machine. The aim of his life was to bring about a reconciliation between ideas from opposite regions of the world of the spirit, and in working to reach this aim he developed a wealth of thoughts and points of view, of discoveries and hints, which is perhaps unique in the history of thought. In almost every scientific field he intervened creatively and stimulatingly, and there is an inner agreement between, a common type for, his thoughts in the widely different fields in which he worked.

Gottfried Wilhelm Leibniz was born in Leipzig on 21 June 1646. His father, who was a jurist and professor of moral philosophy, died early, and the boy, who was often left to himself, was very fond of spending his time in his father's library, where he read all the books he could manage, at first chiefly novels and historical works, later scholastic works. Despite all the obstacles of foreign languages and of the abstruse character of the content, he nevertheless sought by his own efforts to understand what he got hold of. Those around him, he relates in an autobiographical sketch, feared first that he would become a poet, and next that he would become a scholastic: "they did not know that my mind could not be filled exclusively by one kind of thing". The struggle between the mechanical and the teleological view of nature began early in his mind. He relates that as a quite young man he wandered about in a wood near Leipzig and pondered whether he should hold to the scholastic philosophy or join the new doctrines. The need of an all-round education and of acquainting himself with the exact sciences led him away from Leipzig. On the occasion of a slight he gave up taking the doctor's degree in his native city. He took the degree in law at Altorf and then went to Mainz, where, despite his youth, he was taken into the Elector's service. Plans for the improvement of the science of law and of the law-books occupied him greatly in these years alongside his philosophical and scientific studies. He wanted to root out the scholastic barbarism in jurisprudence, just as at the same time he became more and more convinced of the decisive victory of modern natural science over the scholastic doctrine of nature. There was even, by his own later *[335]* statement (in the Nouveaux Essais), a time when he inclined very strongly to "the side of the Spinozists". But the ruling thought in him was nevertheless this: to maintain the assumption that existence is determined by purposes without giving up mechanical

natural science. With zeal he defends the new scientific conception against the men of the older time, his teacher Jakob Thomasius and the famous jurist and physician Herman Conring. But it was his endeavor to show that the mechanical connection of nature, far from excluding, actually presupposes a divine thought of providence. In a small treatise *(Confessio naturæ contra Atheistas)* and in his letters to Jakob Thomasius from the years 1668—69 this double endeavor appears clearly. Leibniz is here still a dualist; there is still for him a gulf between the two principles he wants to connect. In studying Kepler, Galilei and Descartes he feels, by his own account, "as if transported into another world"; but the old world is to be connected with the new. A journey to Paris on a mission for his government gave him the opportunity for new studies. Alongside his practical business and alongside the remarkable proposal he laid before Louis XIV for a French conquest of Egypt (in order to divert the lust for conquest from Germany), he immersed himself in mathematical studies in Paris under the guidance of Huyghens and was led to the invention of the differential calculus, at the same time as he zealously studied the Cartesian philosophy. His standpoint around the year 1670 bears the stamp of the influence of Hobbes's writings. In a treatise on natural philosophy *(Hypothesis physica nova.* 1671) he asserts, in a manner similar to Hobbes and Gassendi, the continuity of motion by means of the thought of tendencies of motion (conatus) in the smallest moments and the smallest parts. This thought stands in connection with the discovery in the following years of the significance of infinitely small quantities in mathematics. From this time on he combats Descartes's doctrine of the conservation of motion and begins to put the conservation of force in its place. He finds the difference between mind and matter in this: in the material domain a tendency which does not pass over into motion
[336] can exist only in the single moment, but in mental life the tendency can, by means of memory and thinking, reach beyond the moment. He has expressed this pregnantly in the proposition that a body is a momentary mind or a mind without memory. In a letter from the year 1671 we read: "As the activity of bodies consists in motion, so the activity of souls consists in striving (conatus) or, so to speak, in a minimal or punctual motion, just as the soul properly is only in a point in space, while the body occupies a place". We have in these hints what one might call Leibniz's second philosophy. He seeks to go beyond dualism and asserts continuity by putting mental striving into the differentials of matter, but is still not clear about the relation between the material and the mental. An obscure point of union is found in the concept of striving; beyond this indefinite hint Leibniz does not here get. Only this stands firm for him, that the solution can be found only by going back from what is given in the phenomena to its elements

and presuppositions.

Just as Leibniz worked in the fields of various sciences, it was also his endeavor to come into personal contact with eminent men of various intellectual tendencies. In Paris he associated with Huyghens, to whose mathematical and scientific ideas Leibniz owes much: through him he was surely led to the principle of the conservation of force, which plays so great a part in his thinking. Even earlier he had addressed himself to Hobbes in writing, though no correspondence seems to have come about. During his stay in Paris his attention was directed to Spinoza, whose "Theological-Political Treatise" had appeared a few years before, while the "Ethics" was as yet accessible only to a narrow circle. The mathematician and philosopher Tschirnhausen, who belonged to this circle and was one of Spinoza's most acute correspondents, was together with Leibniz in Paris and gave him a survey of Spinoza's views. Through a common friend Tschirnhausen informed Spinoza that he had met in Paris "a man well versed in the various sciences and free from the usual theological prejudices, by the name of Leibniz, with whom he had formed a friendship, since he was a man who, like himself, worked at developing his *[337]* mind and regarded this as the most important thing". Tschirnhausen considers his new friend worthy to know the "Ethics" and asks the master for permission to show it to him, reminding him that Leibniz had already earlier written Spinoza a letter concerning the "Theological-Political Treatise". Spinoza answers: "I believe I know this Leibniz through his letters. But I do not know why he, who is a court councillor in Frankfurt (i.e. Mainz), has gone to France. So far as I have been able to see from his letters, he is a man of a free mind and well versed in every science. Yet I believe it would be unwise to entrust my writings to him so quickly. I should first like to know what he is doing in France, and to hear what my friend Tschirnhausen thinks of him when he has associated with him longer and come to know his character better". Spinoza evidently did not have real confidence in Leibniz, whose many-sidedness and pliancy were certainly not always favorable to his intellectual character and in any case gave his spiritual life and his activity a stamp that was entirely opposite to Spinoza's withdrawn and concentrated existence as a thinker. Spinoza's mistrust was justified. For however near Leibniz stands to Spinoza in his final results, and when he is consistent, he nevertheless in his later writings seizes every opportunity to distance himself from him, and that often in expressions one would not have expected of a man who was "free from the usual theological prejudices". At this time, however, he was very eager to come into contact with Spinoza. After a four years' stay in Paris (1672—76), where he — a proof of how different minds he knew how to value and learn from, — associated with the great Jansenist theologian and zealous

Cartesian Antoine Arnauld, he went by way of England to Holland, where he stayed for two months, during which he repeatedly visited Spinoza and discussed philosophical subjects with him. Only through the most recent literary finds[68]) has this personal relation between the two great thinkers been discovered, a relation which became so intimate that Spinoza at last overcame his misgivings and showed him the "Ethics". Leibniz himself does mention in a single place (in the "Theodicy") his visit to Spinoza,
[338] but gives the impression that the conversation turned on the events of the day! It was not commendable to have been so near to Spinoza. One sees accordingly that several of the letters in the edition of Spinoza's works which his friends saw to immediately after his death have been abridged, and the names of the writers of the letters often omitted, and Leibniz, who had looked forward to the publication with great longing, was very dismayed to find one of his own letters (an entirely innocent one, to be sure) printed with his name given. That Leibniz had written Spinoza several letters, no one had previously suspected. — Spinoza's line of thought exerted a strong influence on Leibniz, especially in leading him definitively beyond the dualistic theory. One finds Spinoza's thought again in him when at this time he objects against the Cartesians that the difference between mind and matter did not exclude their being properties of the same being. We have marginal notes and excerpts of Leibniz's dating from his first reading of Spinoza's "Ethics". There are three points on which he at once raises objections, and all three are characteristic of his line of thought. The first is the fifth proposition of the first book, which asserts that there can be only one substance; the second is the exclusion of intellect and will from "the producing nature" (Eth. I, 31); the third is the appendix to the first book with the famous polemic against final causes. — As little as Leibniz ever was a Cartesian or a Hobbist, so little was he ever a Spinozist. But he received mighty impulses from his predecessors. As regards Spinoza it must especially be noted that Leibniz embraces the identity hypothesis put forward by Spinoza and carries it further. Hardly any greater misunderstanding of Leibniz could be conceived than to conclude, from his assertion of the plurality of substances, that he disputed the identity hypothesis. His endeavor aimed precisely at uniting both assumptions; how far he succeeded is a question in itself. —

At the end of the year 1676 Leibniz took up a position with the Duke of Hanover (Braunschweig-Lüneburg) as court councillor and librarian. A mass of administrative,
[339] historical and political work laid claim to him, and his restless zeal to keep up in every field brought with it an enormous correspondence. In the library in Hanover are kept the correspondences he carried on with over a thousand different persons. But just

as it was the main thought of his philosophy that existence consists of a multitude of individual beings, each peculiar in itself, in mutual harmony and continuity, so too there is a connection between his endeavors in the various fields. He had an eye for small relations, for the vanishing differences which nevertheless, by their joining together and totalization, can prove to be realities. His mathematics, his physics and his philosophy show us this trait. But no less weight did he lay on continuity, the inner connection between all things. It was the connection between past and present that interested him in his historical studies, and it was a similar interest that led him to procure information about China through the missionaries of the Jesuit order. He had a sense for the practically useful as well as for science and Church. He saw to the development of the mines in the duchy, worked for the founding of academies, proposed that the monasteries be transformed into scientific institutions, negotiated with Bossuet about a union between the Catholic and Protestant Churches and later with Protestant theologians about a union between the Lutheran and the Reformed Churches. To Peter the Great, whom he highly admired and with whom he had several meetings, he put proposals for reforms and for measures to promote philology and natural science. As a philosopher of language he had so sound an eye and asserted with such energy the significance of experience for the study of language that Max Müller has been able to say of him that, had he been understood and supported by the learned men of his time, the science of language would have been founded a century earlier as an inductive science. In the field of medicine too he put forward significant ideas. — No wonder that he can declare in a letter: "How extraordinarily distracted I am cannot be told. I make researches in the archives, bring out old papers and collect unprinted documents. I receive and answer letters in great number. But I have so much that is new in mathematics, so many thoughts in philosophy, *[340]* so many other literary observations which I do not like to let be lost, that I often do not know what I should do first." Nor, accordingly, did Leibniz succeed in collecting himself for a work typical of his personality and his line of thought. He was not a "man of one book" like Spinoza. His ideas he expressed in an infinity of letters and small treatises, which are not yet all printed. The most important philosophical works brought to light so far are found in Gerhardt's edition (7 vols.); but much still lies in the library in Hanover, and a new collected edition of his works is being planned, arranged by the international association of the scientific societies of the various countries. An incredible capacity for work and mobility of thought this man had. And just as existence in his system is mirrored in countless ways in the individual souls, so too he had a need to make his system intuitive from the very different standpoints with which he came into contact. It

was during his correspondence with the Jansenist Arnauld that he developed his system to full consciousness in himself. But he had nothing against giving it a scholastic form in order to make it acceptable to his friends among the Jesuits. He cast it in a more popular form when he philosophized with the Electress of Hanover or with the Queen of Prussia, or when he was to give Prince Eugene an exposition of his doctrine (the later so-called "Monadologie").

But we have not yet followed the development of Leibniz's philosophical views to their conclusion, to the point where he could say that he believed he had found "a new expression for the inside of things" (une nouvelle face de l'intérieur des choses). We have seen how he held fast to the legitimacy of mechanical natural science, but wanted to reconcile the teleological way of looking at things with it, first in a more dualistic manner, later (following Hobbes and Spinoza) in a more monistic way. It was his conviction that "the moderns had carried the reform too far" by making everything into machines, but it is no less his conviction that hitherto men had only "prostituted themselves" by trying
[341] to refute the strict mechanical connection even in the material processes that proved to be most closely bound up with mental phenomena.

In a letter from the year 1697 Leibniz wrote: "Most of my opinions have been matured at last after a deliberation of twenty years; for I began at a very young age to think . . . Yet I have changed and changed again my views according to the new knowledge I acquired, and only about twelve years ago did I feel satisfied." This statement points to the year 1685. In that year Leibniz worked out a small treatise *(Petit discours de métaphysique),* which at the beginning of the following year he sent to Arnauld. It states that existence consists of individual substances; in each of these the world is expressed from one of the infinitely many possible sides from which it can be seen in the divine thought; each one is a divine thought — realized by constant "emanation". Extension and motion are only phenomena, just as much as the sense qualities. Only force, not motion, persists; and force is one with substance. This treatise had been prepared by an earlier one (of 1680), in which it is first developed that the innermost nature of motion — motion, to which the modern systems reduced everything in the material nature! — was not yet sufficiently cleared up (intima motus nondum patent), and then it is asserted that to subsist (to exist, to be substance) and to act are one and the same, and that all force or striving is an acting[69]). The proposition that every substance acts, and that everything that acts is substance (omnem substantiam agere, at omne agens substantiam appellari), is perhaps the most important in the whole Leibnizian philosophy. By it an end was made of the mysticism or mythology that conceives the essence of things as a resting

essence, an unchangeable subsistence in itself. Leibniz himself found in it the possibility of his union of teleology and mechanism, in that the forces which make up the innermost nature of individual beings and phenomena, and are that which persists through all changes, are assumed to flow out from the divine being and to act in the direction of valuable purposes. The divine purposes are then fulfilled in nature without a breach of the lawful connection, but precisely through it. The advance in the "Discours" of *[342]* 1685 over the treatise of 1680 consists chiefly in this, that the *individual* character of the substances is asserted, and that they are conceived in analogy with the souls of men and animals. Each individual being is a little world, develops according to its inner laws, by its own force, a peculiar radiation of the divine force, and this development takes place, on account of the common origin, in harmony with the development of all other beings. — With this Leibniz's system was complete. It first came before the public in a treatise which was printed in the Journal des Savants in 1695 *(Système nouveau de la nature et de la communication des substances).* And the name monad he used only later (1696). He hardly borrowed the name from Bruno; on the whole no direct influence of Bruno on Leibniz can be shown historically, however much kinship there is in several of their ideas. Rather (according to what has been conjectured in the most recent time) he took the name from the mystical chemist van Helmont the Younger, with whom, just shortly before it is first found in him, he had come into personal contact. —

Leibniz fulfilled his own definition of substance as that which constantly acts. Even after his system was formed, he went on working to shape its fundamental thoughts to the greatest possible clarity. His extensive correspondence and his wide personal acquaintance gave him ample opportunity for this. Thus during a stay in Vienna he worked out (1714) for Prince Eugene an exposition of the doctrine of monads, which after his death was printed under the title *Monadologie.* At the same time he followed attentively what was stirring in his time. With two eminent thinkers he entered into a discussion, which he conducted with great chivalry. John Locke's attempt to derive all knowledge from experience he criticized in his *Nouveaux Essais* in a way that is brilliant and still to this day instructive, asserting in particular the significance of the inner dispositions, of the conditions lying in the knowing beings' own nature. As Locke died before Leibniz had his work finished, he would not publish it, and so it came not to exert the influence on the philosophical discussion that it would surely otherwise have *[343]* had. Only in 1765 did it appear. The other contemporary against whose views Leibniz felt called upon to take up his pen was Pierre Bayle. Bayle had given a critique of the attempts of various theologians and philosophers to demonstrate a harmony between

the assumption of an almighty divine government and the moral and physical evil in the world, and had asserted that philosophically the assumption of two opposite principles, a good and an evil one, could be defended far more easily given the world's actual constitution. Against this Leibniz sought in his *Théodicée,* which appeared in 1710, to show the agreement between theology and philosophy and to ground the claim that the actual world is the best possible world. It is Leibniz's weakest work, diffuse and loose in form, vague and in a bad sense popular in its line of thought. Nowhere, moreover, does he accommodate himself more to notions to which, by strict consistency, he could give no place in his system. It was unfortunate that it was above all this work through which Leibniz was known in the eighteenth century. The "Nouveaux Essais" lay forgotten in the library in Hanover, and the brilliant and inspired small treatises in journals in which he had expressed his real fundamental thoughts were accessible only to few.

There can be no doubt that Leibniz meant his attempt to unite and reconcile religion and philosophy honestly. It was a thought which from his youth on runs through his whole life. But a properly religious nature he was not. This is shown by the ease with which he could disregard confessional oppositions. It was natural rather than positive religion that lay close to his heart. And it was his sense for the individual and characteristic that made it easy and at the same time interesting for him to familiarize himself with the various ecclesiastical systems, just as it was his faith in the harmony among all individual peculiarities and endeavors that led him to the wish for a reconciliation between the contending Churches, a wish which was not shared by men like Bossuet and Spener, who could see no legitimate peculiarity in confessions other than their
[344] own. Arnauld urged him rather to be converted to the true faith than to speculate about the connection of things. Characteristic of Leibniz is a story he liked to tell about two brothers, one of whom had become a Catholic and then sought to convince the other, while the latter just as eagerly sought to win him back for Protestantism; the result was that they both mutually convinced each other, — and the end of the whole thing was that God, on account of their great zeal, had mercy on them both! What interested Leibniz was the inner force and striving in the monads, not merely the particular way in which force manifests itself in the individual monad, and which is often regarded by the latter as the only right one. Leibniz's faith concerned the universal harmony, which for him was not first a result of the future, but was already present in minds, if only they could become conscious of it. In every book Leibniz read — and he read many books — he found something that was interesting and instructive to him, and since he knew how differently things can look from different points of view, he was inclined to

defend and excuse even where he could not approve. With people of all stations in life he knew how to talk about their interests. His recognition of the monads extended to the lower forms. He avoided killing animals, and if he had used an insect for microscopic examination, he put it carefully back on the leaf from which he had taken it. Mild and cheerful, not subject to strong emotions, constantly occupied with thinking and study — so he went through the world. — His last years were lonely. Because of his many labors he had, as he himself said, not had time to marry — before it was too late. Whereas in an earlier time he had been in favor at the Hanoverian court, and had advanced to "Geheime-Justizienrath", he seems in later years — after the death of his princely patrons — to have been in a kind of disgrace. The clergy and the people suspected his orthodoxy because he did not go to church; his name was interpreted in Low German as Lövenix, i.e. Glaube-nichts. When he died (1716), his bier was followed by only a few. — His motto was: "As often as an hour is wasted, a part of life is lost"; it was set on his coffin together *[345]* with a symbol he had been fond of: a spiral with the superscription: Inclinata resurget (it bends only in order to rise).—

b. The Doctrine of Monads

Now that we have gained a survey of the historical development of Leibniz's philosophy, the task is to give a systematic exposition of it. The writings which must chiefly be taken as the basis here are *Discours de métaphysique* (1685), *Système nouveau* (1695), *Lettre à Basnage* (1698), *De ipsa natura* (1698) and *Monadologie* (1714). But a mass of letters and smaller treatises must be compared with these.

It is by an analysis of the fundamental propositions of natural science that Leibniz grounds his philosophy. He acknowledges these fundamental propositions as unconditionally as Hobbes and Spinoza, but he finds them to be derived from presuppositions lying deeper. Physics has its ground, for him, in metaphysics. The first and foremost thing, then, is to see how, according to Leibniz, the transition from physics to metaphysics is to be made.

When the new science aimed at reducing everything in nature to motion, Leibniz entirely approves of this endeavor. But as a philosopher he asks: what reality has motion itself, and what cause has it? — All motion is relative. Whether something is assumed to be moved will depend on the standpoint from which it is seen. The real reality is not motion itself, but the force which is its cause, and which persists even if the motion ceases. The Cartesians, who taught that the sum of motion in the world is the same, but also that motion passes over into rest and conversely, are forced to assume constant

breaches of continuity, since both the transition from motion to rest and the converse became inexplicable. Only by means of the *concept of force* (or tendency, conatus) can the connection be maintained. There is then only a relative difference between motion and rest: they are different phenomenal forms of force. It is force that persists. There is no
[346] absolute rest. Not in the smallest moment one may think of is a moved body in one and the same place in space. If one asks what force then is, Leibniz answers: *that in the present state which brings with it a change in the future* (ce qu'il y a dans l'état présent, qui porte avec soi un changement pour l'avenir) (Lettre à Basnage). This is what is real in existence. Leibniz also arrives at this concept through a critique of the concept of the atom as the atomism of the time conceived it. The assumption of absolutely hard atoms would conflict with the conservation of force and with continuity, since in every collision force would be lost. The atoms must therefore (as Huyghens already taught) be elastic — and must accordingly consist of parts; absolute atoms are consequently a self-contradiction. There is, when we examine the matter thoroughly, activity everywhere. "The atoms are," says Leibniz in a letter, "only the effect of a weakness in our imagination, which likes to rest and therefore hurries to come to an end with the divisions and analyses". (Letter to Hartsoeker 1710). In his polemic against the Newtonian Clarke, Leibniz distinguishes between a body's losing motion, considered as a whole, in a collision with another body, and force's being lost: the motion which the body considered as a whole (in relation to other bodies) loses passes over to the parts, which by the impact are set in an inner vibration; the forces are not annihilated, but distributed among the small parts. The same thing happens here as when one changes large money into small coin. (Cinquième écrit de Mr. Leibniz § 99). The difference between the solid and the fluid is relative, just as motion is relative. From this point of view too only force has a real reality.

That force is the real reality Leibniz expresses, in the language of the time, by saying that force is substance and that all substance is force. But the concept of force is in turn closely connected with the *concept of law:* force is, after all, only that in a state which brings it about that a change occurs in the future. The presupposition of every application of the concept of force is then a lawful connection between the changing states. The principle of sufficient reason, which says that there is a ground for everything,
[347] is therefore the real fundamental proposition of Leibniz's philosophy, although he sets it up as an express fundamental proposition only in his later writings (the Theodicy and the Monadology). And with the concept of law another main concept of Leibniz's is in turn connected: *the concept of individuality.* For individuality means for him the law according to which changes in a being's states succeed one another. Each individual has, so to

speak, its formula. And just as in a mathematical series one can infer the following terms when one has found the law for the succession of the first terms, so it is also with the inner changes of an individual being. In the law for their succession consists individuality itself: *la loi du changement fait l'individualite de chaque substance particulière* (Lettre à Basnage). In the "Discours de métaphysique", where the concept of the individual substance is introduced for the first time, the relation between the substance and its changing states is compared with the relation between subject and predicate in a logical judgment: just as in a logical judgment it must be possible to find the predicate contained in the subject as soon as one knows the subject completely, so he who knows the nature of the individual will be able to derive from it all its changing states. There is continuity between these, since the following one always has its ground in the preceding one. Each of these substances is therefore a little world for itself, develops from its own inner drive and has its own inner life. Only in a miraculous way can such an individual substance arise or pass away. And each substance has its own inner law, different from others.

If it is asked why the real, the persisting (the substance) must be individual, Leibniz answers: because only *unities,* individual beings, can exist. He reproached the Cartesians not only for ascribing absolute reality to motion, but also for ascribing absolute reality to extension and making it substance. Already in his notes on reading Spinoza's "Ethics" he writes that he is not sure that bodies are substances; it is otherwise with souls. This remark indicates the direction in which his thinking went. In his later works the following line of thought recurs constantly under manifoldly varied turns of phrase: extension presupposes something which is extended, that is, repeated several times. Extension *[348]* means multiplicity, composition, aggregate. But the nature of the composite depends on what composes it (the components). Not the aggregate, but the ones of which it consists, have reality. If there were not absolutely simple substances, there would therefore be no reality at all. The assumption of the reality of extended matter brings with it, moreover, the difficulty that its divisibility, by virtue of the law of continuity, goes on to infinity. The true unities therefore cannot be material, extended; if, on the other hand, no such absolute unities existed, matter would be a mere seeming without any real significance. Only the absolutely simple and indivisible can be substance, and it cannot be material.

Before we ask what, then, it is, one more important determination must be brought forward. Leibniz started from the fundamental propositions of natural science. The most important of these was for him the principle of the conservation of force[2]. It was for him

[2]As already remarked, this law derives from Huyghens, who in his doctrine of the pendulum further developed hints which Galilei had given, and which we have also found in his disciples Gassendi and Hobbes. (See E. MACH: Die Geschichte und die Wurzel des Satzes von der Erhaltung der Arbeit. Prag

the foundation of the laws of nature (fundamentum naturæ legum). This very foundation of the laws of nature can, according to Leibniz (see especially De ipsa natura, Erdmann's edition p. 155), be understood only by a teleological argument: it is necessary that the same quantity of force be preserved in the world; but this necessity depends on the divine wisdom and order reigning in the world; the first principle or "sufficient reason"
[349] of mechanism is a final cause, *a teleological principle.* In a small treatise directed against the physicist Papin *(De legibus naturæ,* Dutens' edition, III, p. 255) Leibniz carries this out as follows: cause and effect must always correspond to each other, be equivalent. If there were more in the effect than in the cause, we should get a perpetuum mobile; if there were less in the effect than in the cause, the same cause could never be brought about again, nature would constantly go backward, perfection would be diminished. Both would conflict with the divine wisdom and immutability. When the equivalence of cause and effect obtains in nature, to the exclusion of the two other possibilities, then this, according to Leibniz, cannot be grounded purely mathematically or mechanically, but must be explained teleologically.

The law according to which the individual beings, which are the true realities, unfold their changing states is for Leibniz a law according to which they advance to more perfect states, not merely to new states. The relation between cause and effect is thus here identical with the relation between means and purpose. In its innermost ground existence is so disposed that the purely mechanical laws act in the direction of what the laws of justice and of love demand. In the law inherent in its own being (lex insita) both kinds of laws are united in the case of every individual. — Leibniz is, as one sees, so occupied with the law of the conservation of force that he overlooks that, teleologically seen, what matters is not that the same quantity of force is preserved, but that it is preserved in a form favorable and useful for the subsistence of life and for continued development. This is, moreover, a consideration which has come clearly to light only through the investigations of the most recent time. — But although Leibniz thus grounds the mechanical connection of nature teleologically, it is nevertheless not his intention to reintroduce final causes and to permit an explanation of natural phenomena by appeal to inner forces or to purposiveness. In particulars one must proceed strictly mechanically; it is only mechanism as a whole, or rather the first principles of mechanism themselves, that prove to become intelligible only through teleological consideration. The mechanical

1872. p. 10 f.). Leibniz maintained the principle of the conservation of force against the Newtonians, who denied it. See his interesting polemic against Clarke (Opera philos. ed. Erdmann. p. 746 ff.). On the relation between Descartes, Newton and Leibniz with regard to this principle see Rosenberger: Isaac Newton u. seine physikalischen Principien. Leipzig 1895. p. 409—412.

connection of nature is the fact from which Leibniz's whole philosophy proceeds, just as much as Hobbes's and Spinoza's. — *[350]*

The technical name for the absolute individual beings which for Leibniz make up the true reality is *monads.* It is a Greek word which means unities, and which thus designates the beings only from the formal side. If we now ask what these individual unities, with their inner states unfolded according to a definite law, really are, Leibniz answers that we must conceive them in analogy with our own souls. In the "Discours" this is not stated definitely. But in the following correspondence with Arnauld we read: "The substantial unity presupposes a complete (accompli), indivisible and naturally indestructible being, since its concept encompasses everything that is to happen to it. Such a thing cannot be found in figure or motion, . . . but only in a soul or a substantial form *in the likeness of (à l'exemple de)* what is called an I". (Letter to Arnauld, Nov. 1686). The objective analysis is thus here supplemented by a subjective analysis. In ourselves we find, by way of self-observation, inner states (sensations, feelings, thoughts) which change, and a need or drive which leads us to pass over from one state to another. By means of this subjective observation we must — by way of analogy — make the nature of the monads intelligible to ourselves. Our soul is only a single monad, but we have no reason to believe that we stand alone in nature. On the contrary. By virtue of the law of continuity, which excludes leaps in nature, we must assume that there are infinitely many degrees of the kind of existence we know from ourselves: c'est partout et toujours la même chose, aux degrés de perfection près. The law of analogy bids us everywhere to embrace the principle: "Just as with us" *(tout comme ici)*[70]*). The law of analogy,* according to which we conceive all other existence as differing only in degree from our own, is for Leibniz only a special form of the law of continuity and springs, like it, from the principle of sufficient reason. If one does not assume such a general ensoulment of lower and higher degrees, it becomes entirely unintelligible how sensation and consciousness can arise. Just as the origin of motion became inexplicable for the Cartesians because they did not go back to force, so the origin of consciousness was also a riddle for them, because they took no account of the obscurer degrees and forms of mental life, the unconscious mental life, the need and craving which first awaken to the clarity of consciousness at higher stages of life. In ourselves we constantly observe, after all, an alternation of clearly conscious, obscure and unconscious states. In analogy with this we must conceive monads of all degrees with respect to clarity and obscurity. There are slumbering, dreaming and more or less waking monads. The uniformity of nature leads to this assumption. The part of matter which makes up the human body cannot, after all, alone be endowed with the *[351]*

capacity to sense and to will. Something analogous (whether one wants to call it soul or not) must be found at lower stages. In a letter (of May 1704) Leibniz calls this application of analogy "mon grand principe des choses naturelles".

The way by which all mythologies and earlier speculative systems — without being conscious of it — had come to the assumption of the mental as ruling in existence, this way Leibniz goes in full consciousness. He points clearly to the point where all metaphysical idealism springs up. And it is again the presupposition of the complete intelligibility of existence that guides him here. From this presupposition follow the continuity of all existences and the analogy of all beings with our own being.

But as Leibniz has come to this point in the development of his thought, he stands at a turning point. He builds his philosophy on an analysis of material phenomena, guided by the endeavor to find what is real and persisting in them. Now that he has arrived at the assumption that, properly speaking, only representing beings exist, with differing obscurity and clarity in their representations, matter becomes for him a *mere* phenomenon. It is a seeming for the senses when bodies seem to exist. The senses do not deceive us, however, any more than they deceive us when we believe that the sun goes round the earth. The phenomena are real insofar as there is a regular connection
[352] between them, a law according to which they succeed one another. It is this law that is the mark of what is reality for us. But it does not force us to assume that anything exists other than monads and their representations. — At this point there would be the possibility of an entirely new line of thought: a testing of whether, from the purely subjective standpoint, from the assumption that the world exists only in the form of representation, one would arrive at the same result as by the objective way Leibniz has gone so far. This new way Leibniz did not take, and when in his last years he heard of Berkeley's attempt in this direction, it struck him as a mere paradox (see his letter to des Bosses of 15 March 1715). Leibniz himself evidently did not become clear about the great difference between being a phenomenon *of* something and being a phenomenon *for* someone. Both relations can lie in the concept of phenomenon; Leibniz begins with the first relation and ends in the second. And it is only after he has set up the concept of substance as force and as individual unity that he — in accordance with the law of analogy — finds its essence to consist in representing activity. Precisely because this last assumption is the end point of his train of thought, it is understandable that he does not draw all the consequences of it[71]). Leibniz continues to treat material phenomena as the expression of an objective order of things. By virtue of the law of continuity everything in the material world stands in reciprocal action; a change in one part of the world

extends to all parts. And from this it is concluded that it is the *whole* universe which is represented by each monad in a more obscure or clearer way. Every monad represents the whole universe, just as every single point in the physical universe feels everything that happens in the whole universe. The very assumption that there is a multiplicity of monads shows that Leibniz did not go the purely subjective way, for then he would have had to do with only one monad and its picture of the world. The assumption of other consciousnesses than my own already presupposes, after all, the application of the inference from analogy on which Leibniz builds his metaphysical idealism. —

An empirical confirmation of his hypothesis that there is force, life and soul behind all apparently dead mass he found in the discovery of the small organisms which Swammer- *[353]*
dam, Leeuwenhook and other naturalists found with the help of the microscope. There opened up here for him a world of infinitely small beings in the domain of organic life, just as he himself had helped to open up the world of infinitely small quantities, and just as in his philosophy he had asserted that behind the apparently homogeneous mass the true reality was found, consisting of individual beings with a nature akin to our own. Arising and passing away, birth and death were for him only phenomena, expressions of processes of contraction or unfolding, of obscuration or clarification in monads. The absolute leaps and differences fell away (no less than the homogeneous mass) the more clearly he became aware of the infinite individual nuances of life. With visible satisfaction Leibniz in all his later expositions refers to the works of the great contemporary naturalists just named.

With his concept of the monad Leibniz apparently enters into a sharp opposition to Spinoza. For the latter there was only one substance, and all individual beings stood to it as the waves to the sea. For Leibniz the true reality is a multiplicity of individual forces, each acting according to its inherent law, so that all its effects unfold from its own original character. Leibniz himself often sharply emphasized this opposition, most clearly in a letter to Bourguet (Dec. 1714): "I do not know how one can extract Spinozism from my doctrine. It is precisely by the monads that Spinozism is dissolved. For there are as many true substances, . . . as there are monads, whereas according to Spinoza there is only one single substance. He would be right if there were no monads; then everything outside God would be vanishing and be dissolved into simple modifications". But it is the same fundamental concept of substance from which both Leibniz and Spinoza proceed. All the determinations and states of the substance are to follow from its own inner nature, not to be due to influence from outside. Leibniz asserts this just as definitely in the case of each of his monads as Spinoza asserts it in the case of his one and only

[354] substance. The difficulty of this absolute concept of substance became for Spinoza this: how the multiplicity of the attributes and of individual beings could be explained from the assumption of a single substance. The difficulty of the same concept became for Leibniz this: how there can be agreement between the states of the different monads when each of them is determined only from within, and none of them has holes, windows or doors (to use Leibniz's own images) by which something from outside can come into them. And the difficulty became for Leibniz, besides, that of reconciling the doctrine of monads with his theological presuppositions.

Leibniz regards both these difficulties as solved by one thought. That there is agreement between the states of the different monads he takes as a fact: there is a universal harmony or concord (accord) between all beings in the world. Clearly enough, as already touched on, he starts here from the observation of the general physical reciprocal action in the world and from the empirical (and yet grounded on an inference from analogy) assumption that the observations and representations of the individuals agree in such a way that a common picture of the world can be constructed in its broad outlines. But since the real reality is a multiplicity of entirely self-subsistent and independent beings which (as Leibniz says in a letter) are to be regarded as little gods, since they have life in themselves, there must — he concludes — be a common origin of that harmony. "God alone", we read in the "Discours" (§ 32), "brings about the connection or coherence between the substances, and it is through him that the phenomena of the one meet with and agree with those of the other, so that there is reality in our representations". God has from the first created the individual monads in such a way that in the one regard is taken of the other; the agreement between them is due to the common origin. They arise from God by an outflowing or radiation (émanation, fulguration), as individualizations of the one divine force, just as each of our thoughts is a radiation of the force of our mind. God sees the universe from infinitely many different points of view, and each of these different points of view he realizes by creation as an individual substance. The
[355] states of the individual monads thus correspond to one another as the different views one gets of a town from different vantage points agree with one another (Discours § 14). The spontaneity in the monads and the conformity between the monads thus both find their explanation in the fact that it is the divine force and thought that make up their innermost essence. The harmony between them is present from the first, is "pre-established". Each of them is a peculiar concentration of the universe which stood before God's creating thought, is a universe en miniature, and the universe consists only of such concentrations. Leibniz is a metaphysical Copernicus: he shows how the world

can be seen from far more points of view than one is accustomed to think, and that the one point of view in itself has the same legitimacy and validity as the other. The world is for him not something that lies outside or between the individuals, but consists precisely of and in the individual beings.

Leibniz thus puts the whole problem back into the thought of the creating God. He passes over from philosophy to theology. But that does not free him from the difficulties: for how does there arise in the divine thought or imagination a world that can be seen from infinitely many points of view? And these possible points of view do not stand in the divine imagination as resting images; but just as Leibniz constantly asserts that all possibility is a little actuality, an incipient reality, so he assumes here too an original need or craving for existence (inclinatio ad existendum, exigentia existendi): the possibilities seek to press forward to full reality, which is granted only to those that fit together best into a harmonious world-whole. It is a mythological prelude which decides the struggle for existence before actual existence begins, so that the latter can then be a harmonious unfolding of inner germs in each being. In the exposition of the Theodicy we shall return to this point. It has been to the decided detriment of the clarity and significance of Leibniz's philosophizing that theological points of view play so great a part in him from first to last. And it does not help at all toward the solution of the problems to clothe them in a theological form. A certain feeling of devotion makes one rather stop asking questions and raising objections; but such a stopping is no solution. *[356]* It is easily seen that the problem, in the form in which it comes to present itself to Leibniz, is insoluble. If the monads are absolute substances, they cannot be produced. Only the appeal to a supernatural creation covers over the evident self-contradiction. Just as Leibniz, by virtue of his concept of substance, excludes real reciprocal action between the monads, so he would consistently have had to exclude their coming into being out of a substance different from them. Here Malebranche said the redeeming word: God cannot create gods! — Leibniz's difference from Spinoza — if one disregards that evident self-contradiction — in reality falls away, since the creation of the monads is not concluded at any point of time once for all, but constantly takes place: the radiation or outflowing of the monads goes on uninterruptedly (Discours § 14. 32. — Monadologie § 47: Dieu seul est l'unité primitive ou la substance simple originaire, dont toutes les Monades créées ou dérivatives sont des productions, et naissent, pour ainsi dire, par des fulgurations continuelles de la divinité de moment à moment). Leibniz lays weight on its being the source, the force to act, that thus uninterruptedly wells out from the Deity into the monads; but Spinoza too, after all, explains the striving which leads each

individual being to assert itself as a single manifestation of the power that acts in all individual beings. Consistently carried through, Leibniz's philosophy here differs from Spinoza's only in that the individual beings are drawn more into the foreground and their common origin stands as an obscure background (which is lit up more strongly only when Leibniz seeks connection with the theologians), whereas in Spinoza the individual beings are seen more as offshoots or ramifications of the infinite being which is his first and his last thought. The difference between the two systems can be reduced to the same difference as there is, according to Leibniz, between two views of the universe, produced by a turning of it. For both, existence is a pyramid, which Spinoza sees from the apex, Leibniz from the base[72]). — There falls, as already emphasized, a shadow both
[357] on Leibniz's intelligence and on his character when, despite this state of the case, he so often distances himself from Spinoza in more or less heresy-hunting expressions. —

Leibniz's theory of the relation between soul and body is a simple consequence of his doctrine of monads. What we call our body is a group of monads, each of which develops for itself according to inner laws and by inner force, just as our soul does. From the state of one monad one can, by virtue of the universal harmony, infer the states of other monads; it is this that is popularly expressed by saying that one being acts on another. The monad from whose state one can, by such an inference, derive those of the others is said to act. When the inner development of the individual monad is hindered, this is due to the obscurity, the chaotic state of the elements in its own interior. But since the nature of every monad, that is, the degree of obscurity or clarity and the force to pass over to new states with which it is endowed, depends from the first on a regard to the other monads together with which it is set into existence, there correspond to the inner hindrances definite states in other monads, which are therefore popularly called their causes. When Leibniz wants to express himself quite popularly, he uses the image of two clocks adjusted to each other once for all to designate the relation between soul and body. Popularly and accommodatingly he also calls the body a composite substance. But this manner of expression is too crude for his real conception. Matter is for him, after all, only a phenomenon, a world of external multiplicity whose states express for the senses the inner changes in the monads, which are the real reality. One thing "expresses" or "represents" another, in Leibniz's usage, when there is a constant and regular relation between what can be asserted of the two things (Letter to Arnauld, Oct. 1687). What is concentrated, united into unity, in the interior of the monads appears to sense perception as an extended multiplicity. In a letter of 4 Nov. 1696 (to the Electress of Hanover) Leibniz writes: "Souls are unities, bodies multiplicities . . . But although

the unities are indivisible, they nevertheless represent the multiplicities, much as all *[358]* lines from the periphery meet in the center. *In this union (réunion) consists the wonderful nature of consciousness (sentiment);* and it is also this that makes every soul like a world for itself". — The stringency with which Leibniz, in agreement with Hobbes and Spinoza, asserts the mechanical connection of nature makes it impossible for him to keep to the usual dualistic conception. The identity hypothesis stands, on his presuppositions and according to his whole system, as the only possibility. The difficulties to which this hypothesis is subject Leibniz has made still greater by the very absolute substantiality he ascribes to the monads. From the stamp of unity which consciousness has, he concludes with dogmatic haste that the conscious being must be an absolute, indestructible unity which can receive nothing from outside. Had he, instead of merely conceiving the bodily in analogy with the mental, *also* conceived the mental in analogy with the bodily, he would have seen the necessity of assuming that the individual consciousnesses, despite their wonderful stamp of unity, stand in reciprocal action with the rest of existence by virtue of the law of continuity on which he himself lays so great weight, but is able to apply only within each individual monad, not to the relation between them. Development comes for Leibniz to mean only unfolding. The individual organism, for him, really does not come into being at all; it exists from the first; its birth and growth consist only in its being freed from its envelope and in its dimensions being enlarged. It was the so-called encasement hypothesis (the one germ being thought to lie inside the other up through the series of generations) which Leibniz embraced like many naturalists of his time. And so in the mental domain too development became not a taking up of new content, but only a clarification of the content which was given from the first in a chaotic and obscure form. — Concentration cannot come into being according to Leibniz; new centers, new monads cannot form. Here he cuts the knot. The problem of individuality perhaps contains an insoluble knot for human thinking; but to cut it is not to solve it.

Yet this very cutting is — just as much as the distance he felt between himself and *[359]* Spinoza — only an outcome of the energy with which Leibniz thought through the concept of individuality. It appears in him with a brilliance and clarity as in no other thinker. And the peculiar turn he gave the concept of substance by applying it to the individual elements of existence (the monads) is characteristic of the transition from the seventeenth to the eighteenth century. In place of mystical resignation and submission to the absolute powers there now enters the free, peculiar striving of the individual for clarity, for the unfolding of its own being. It is a revolution in the world of thought — a prelude to the revolution in the external world. But one presupposition Leibniz's system

shares with the other great systems: the presupposition of the complete rationality of existence. It is the principle of sufficient reason that guides him in his theological speculation and in his doctrine of monads as in his mathematical and scientific works. This presupposition too passed as an inheritance to the new century, whose typical initiator he was.

c. Psychology and Theory of Knowledge

From the significance that the analogy with the life of consciousness has for Leibniz as the foundation of his metaphysical idealism, it follows that his real point of departure must be the same as Descartes's: immediate self-observation, inner experience. It gives us the first term of the analogy. Leibniz is also clear that here lies a point of departure for his thinking, and he has devoted great attention to it. It is the "Nouveaux Essais" that is the chief work here. The polemical side of this work could only be properly assessed once Locke's theory of knowledge, against which it is directed, had first been set out. But in this context the weight lies on Leibniz's own psychological and epistemological views. The work mentioned gains significance as a verification of Leibniz's general philosophical ideas, since in the introduction he declares that here he will disregard his
[360] doctrine of monads and take his stand on purely empirical ground.

The chief question that presents itself is whether the soul is empty from the outset, like an unwritten tablet (tabula rasa). Long before, in the "Discours" (§ 27), Leibniz had expressed himself on this question. Even then he found that this notion of an unwritten tablet was false. It is due to inexact observation. One overlooks the small, obscure stirrings in the soul and keeps only to what is clearly conscious, which arises only later, when outer experiences have long been at work. The Cartesians made themselves guilty of this error just as much as the empiricists. The less difference and opposition there is between our sensations, the less any single thing in us stands out in relation to the rest of the content, the more obscure, in a word, the life of consciousness is, the more easily one overlooks it. More exact observation will be wary of setting limits. There are all possible transitions between obscurity and clearness. The obscure changes in us that do not come to real consciousness Leibniz calls perceptions. Already in them a multiplicity is united in a single state. At this stage stand the lowest beings (in Leibniz's system: the monads at the lowest stage). The expression consciousness (sentiment)[3] Leibniz first uses where the sensation (the perception) is more distinct and is accompanied by

[3]Sentiment in Leibniz should be translated by consciousness. By conscience Leibniz understands something more than necessarily lies in the word consciousness, namely connaissance *reflexive,* thus rather "self-consciousness".

memory. The highest stage of mental life presents express attention (apperception) or reflection, which is directed toward the obscure sensations. (These three degrees are best described in the little treatise *Principes de la nature et de la grâce* § 4, cf. *Monadologie* § 14 ff.). But at every stage it is an activity, a spontaneity, that unfolds itself. We are active, even in the most obscure states. Just as there is no tablet that does not by its constitution have an influence on what is written on it, so our nature is preformed from the outset. In an involuntary, instinctive way we apply principles of which we can only later become conscious. What is contradictory we reject, even if we have never heard *[361]* tell of the principle of contradiction. There are both practical and theoretical instincts. The impulse of self-preservation and the need to help others work involuntarily. There is on the whole far more in us than we think of. Here, as in material nature, it is the small magnitudes that underlie what comes forth distinctly and to the senses. No more than motion can consciousness arise from nothing or all at once. Corresponding to the conservation of force (or properly — by virtue of the identity hypothesis — as one with it) Leibniz assumes the uninterrupted subsistence of mental life through states of consciousness and of unconsciousness, through clearness and obscurity. The apparent disappearance of mental life is only a conversion into a more obscure, more elementary form. The question whether there are innate ideas Leibniz thus answers to the effect that there are innate capacities, dispositions, which unfold as soon as experience gives occasion for it, and which underlie all theoretical and practical activity. As an example of the obscure elements we easily overlook, he points to what may be called sensations of life, the as a rule faint sensations that correspond to the organic functions, and that have an influence on the state we are in at every moment. Our outer senses are indeed more distinct; but since the qualities of sense are utterly different from the motions that correspond to them within matter, Leibniz assumes that our sense-sensations are only apparently simple, but are in reality just as compound as the corresponding motions. What comes forth clearly for consciousness is for him always something compound that presupposes more obscure elements, just as, when we hear the roar of the sea, we really gather the impressions from thousands of waves into one sensation. If we pay attention to it, we shall find that we are never without sensation. Our soul is constantly active. Constantly we receive small sensations, and constantly they release our activity. We are never altogether indifferent; the assumption of indifference is only an expression of our inattentiveness to the small differences. Involuntarily we turn to the right or to the left, even where there are no outer grounds for the choice. Our nature is always working in us to produce greater satisfaction (à se mettre mieux à son aise). A certain *[362]*

unrest always makes itself felt, a lack of full satisfaction, which drives us forward, a spur which as a rule we notice only when from unrest it turns into pain. There are always small hindrances to overcome, and demands are always made for exertion of strength. No feeling of pleasure is possible without such unrest, without minimal hindrances; indeed, the hidden spur maintains itself in joy itself and heightens it, so that a constant progress becomes possible. Only through these small sensations (petites perceptions) does the connection between the different periods of life of one and the same individual become clear; for the clearly conscious states often stand as contradictory. And as they at once explain the difference and the connection between the states of one and the same individual, so they also explain both the difference and the connection between different individuals. Even if one will call the soul an unwritten tablet from the outset, one must nevertheless grant that no two tablets are alike in constitution. If one looks at the small nuances and differences, one will find in the whole of nature no two altogether alike beings. There are not two leaves that are alike, let alone two human beings. Here too the assumption of indistinguishability is due only to our ignorance. But in the obscure sensations of which the individual does not become conscious, the rest of existence makes itself known in him and determines him, without his knowing it. Through them he is connected with the rest of existence. Even in sleep this connection is not suspended. There is only a difference of degree between sleep and waking: the turning away of attention is a partial sleep.

It is a great merit that Leibniz has earned in psychology by drawing attention to the significance of the vanishingly small elements. He has opened a new world for self-observation, admittedly a world where it feels the want of an instrument that could correspond to the microscope in the domain of outer observation. And he has indicated points of view by which it becomes possible to maintain the continuity of the mental side of existence to a far higher degree than the usual conception assumes, and indeed at bottom also than Leibniz himself assumed when he let the monads be without windows
[363] and thereby shut them off from unconscious influences from the rest of existence. The elements of existence (monads) differ from one another only by the degrees of clearness or obscurity of their inner states, and likewise the states of the same individual differ only by such differences of degree. It is a herald of the fact that we are entering the century of the Enlightenment. All the conditions are assumed to be present — only the light must be brought: on this optimistic faith Leibniz and his posterity built. Yet it must be noted in a purely psychological respect that Leibniz points toward a deeper view of mental life than the one rationalism and the age of the Enlightenment took as

their foundation. In the first place it is for Leibniz an *infinite* content that is given in obscure form, and full clarification of it is thus impossible for a finite being. Secondly, he assumes, as an independent element beside the sensations or representations, the need and the tendency (appétit, tendance) to pass over to new sensations. Striving or willing is to that extent for him what is innermost in man, and it is a striving that cannot cease. Toward the passions (passions) he takes a somewhat wavering stand. At times he regards them as obscure thoughts (pensées confuses); and on this view they must then fall away with sufficient clarification; — at times he declares that they are different both from representations and from feelings of pleasure and pain, but are tendencies that are aroused by representations and accompanied by feelings of pain; on this view they are thus closely connected with the never resting striving or willing[73]). By carrying the latter point of view through, Leibniz would have arrived at a more realistic conception of mental development than the one that sees it exhausted by the concept of enlightenment.
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Self-observation gives us, as we saw, the first facts from which we proceed in our thinking, *the first truths of fact. The first necessary or a priori truths* we have in identical judgments, that is to say, those in which subject and predicate can be shown to be one and the same concept. Common to the first truths of fact and to the first a priori truths is the immediate relation: in the one case between the understanding and its object, in the other between the subject and the predicate of the judgment. This immediate relation is *[364]* in both places proof that nothing more certain can be reached. Leibniz came early to this distinction between a priori truths and truths of fact. As early as 1678 (in a letter to Conring of 3 Jan.) he makes this distinction and demands that all a priori fundamental propositions be traced back to identical judgments. He therefore divides the axioms into identical and a priori. The significance of identical judgments rests on this, that when other judgments can be traced back to such, their validity is thereby given. Leibniz thus demands that proof be given for all fundamental propositions one lays down, in so far as they are not identical. He sympathizes with Locke's criticism of "the innate ideas," in so far as Locke thereby wished to combat indolence and prejudice. But Locke ought to have distinguished between necessary truths and truths of fact. Even if a truth is innate in the sense in which Leibniz uses this word, a proof of it must nevertheless, on his view, be given; "innate" truths must be *learned* like other truths. The clarification may be a hard labor, and full clearness is found only in identical judgments. Leibniz has great merit in logic for having laid down the principle of identity, whereas the Aristotelian and scholastic logic kept to the principle of contradiction. He sketched a logic in which

every judgment is formulated as a relation of identity, and has thereby anticipated a method that English logicians (Boole and Jevons) have taken up in our day. But these sketches were not published before 1840, so they have not exercised any influence. — Just as the principle of identity is our criterion of truth for pure thinking, so the lawful connection that the principle of sufficient reason demands is the criterion of truth in the domain of experience. And just as Leibniz is the first to lay down the principle of identity as the fundamental logical principle, so he is also the first to lay down the principle of causality as a special principle for actually given relations. He thereby brings forward the principle that bears all real knowledge, and since he at the same time distinguishes it sharply from the purely logical principles, he makes it natural to discuss its nature more
[365] closely. It is not to be a formal logical principle; nor is it to be drawn from experience; but it is nevertheless to be a principle of reason and to hold for all actual relations (and that both in the supernatural and in the natural world!): how is this possible? That was the problem Leibniz left to his successors. He himself was content to analyze it out of the confusion, partly with general logical, partly with special physical principles, in which it had occurred among his predecessors. Indeed, he does not even distinguish clearly between ground and cause, which is connected with the fact that the principle is at once to be a principle of reason and to hold for actual relations, — and in the last instance with the fact that the laws of facts are seen as testimony to the divine reason that is the origin of the world[74]).

His whole life long Leibniz occupied himself with plans for a kind of alphabet of thought (Alphabetum cogitationum humanarum), which was to be formed by a thoroughgoing analysis of our knowledge, so that one arrived at the fundamental concepts that condition it. By combination of these fundamental concepts he then thought one would be able to find new truths. And by a suitable system of signs one would at the same time be able to obtain a universal language, something corresponding, for all concepts, to what the mathematical language of signs is for magnitudes. It was thus to be at once a logic, an encyclopedia and a grammar, at once a means of gathering what had been learned and of making new discoveries. As the sharp distinction between a priori truths and truths of fact dawns on him, he nevertheless restricts his "Ars combinatoria" or "Characteristica universalis" to the a priori domain. The plan was, despite many attempts, never carried out, essentially, no doubt, because he gradually saw that such a general language of signs really presupposed that our knowledge was already complete. But the plan is characteristic of dogmatic philosophy.

d. Theodicy

It was Leibniz's task as a thinker to lead beyond the mechanical view of nature without breaking it. The way he took was to try to show that the forces at work in the mechanism *[366]* are determined by purpose, so that the whole graded series of mechanical causes and effects, seen from within, is a graded series of means and ends. What is innermost in the world is at every point striving, development, progress. In this thought, which besides bringing harmony between mechanism and teleology also brought harmony between the states of the different monads and between soul and body, he thought he also had a means of reconciling religion (both the positive and the "natural") with reason[4]. It was his conviction that his philosophy satisfied the demands the strictest orthodoxy could make. But the religious problem was just then sharpened by a contemporary writer for whom Leibniz cherished the greatest esteem. Pierre Bayle maintained, as we have mentioned in its place, a disharmony between religion and reason and found especially the doctrines of positive religion contrary to reason. He relied here especially on the problem of evil and found the Manichaean doctrine of two principles in the world, one evil and one good, more in agreement with experience than the orthodox doctrine, to which he nevertheless bowed in the obedience of faith. Leibniz's gifted disciple, the Queen of Prussia, was strongly influenced by Bayle's misgivings and asked Leibniz to refute them. Leibniz thought himself well prepared for it. From his first sketches on he had had an eye for the problem of a theodicy, i.e. the problem of how the assumption of the world's origin in an almighty, all-good and all-wise being can be reconciled with experience of the physical and the moral evil in the world. Early on he had planned *[367]* a work on this; Bayle's "Reponse à un Provincial" and the Queen's request led him to work out and publish his *Essais de Théodicée sur la bonté de Dieu, la liberté de l'homme et l'origine du mal,* which appeared in Amsterdam in 1710.

Against Bayle's assertion that the doctrines of religion are contrary to reason and yet must be believed, Leibniz sets the distinction between something's *being above* reason and its *being contrary to* reason. This distinction is connected with his division of truths into eternal truths (which are grounded in the principle of identity) and truths of fact (which belong under the principle of sufficient reason). Against the first kind of truths

[4] In a recently published statement by Leibniz he places his conception of the relation between *God and the world* in a characteristic way together with his conception of the relation between *soul and body* and between *teleology and mechanism:* "The general mechanical laws are expressions of the will of the Deity (decreta voluntatis divinæ), and the special mechanical laws in each body are expressions of will or forms of the soul in its striving after perfection . . . Everything in nature can be proved by final causes (causæ finales) just as well as by efficient causes (causæ efficientes)." (Bodemann: Die Leibniz-Handschriften der kgl. öffentl. Bibliothek zu Hannover. 1895. p. 89).

of reason nothing that is to be believed can be contrary. The existence of God, e.g., is possible only if the content of the concept of God, the properties ascribed to God, are not self-contradictory. But an assumption can be raised above what experience teaches, since the rational necessity we find, with the help of the principle of sufficient reason, in the connection of phenomena is always conditioned: the very fundamental propositions of natural science, according to Leibniz, find their explanation only in a thought of providence, a teleological principle. It is the very principle of the truths of fact, the principle of sufficient reason, that leads us beyond experience, since the harmony between the monads and the lawful connection in all that happens finds its explanation only in an absolute being that has produced the world by a rational choice. Each single thing, each single event by itself is contingent: we then obtain no completion of the series of grounds or causes, no full satisfaction of the principle of sufficient reason, unless we go outside the series (extra seriem, see Opera ed. Erdm. p. 83) to a cause that is the cause of itself. That the divine choice by which the given world came into being was a rational choice is inferred from the fact that several worlds were possible besides this actual world: this one can then only have come into being because it was the best. But the best world does not mean a world that has no dark sides and defects; it means merely
[368] a world in which the defects are vanishing in relation to the perfections. Every possible world that was to consist of finite beings would have to present imperfections, since finite nature brings limitation with it and therefore cannot fully take the divine nature into itself. From limitation (metaphysical evil) spring suffering (physical evil) and sin (moral evil). Evil thus has in a certain way its origin in God: but not in God's will, which always wills the good; rather in God's intellect, in which the possible pictures of the world dwell from eternity. God has created actuality, not the possibilities. The source of evil is the imperfection that clings to every possible world of limited beings; but by the divine choice of creation this imperfection is reduced to the least possible and made to serve as means and as foil for so much the greater perfection. — In a somewhat more mythological form and carried out in livelier colors, Leibniz's conception becomes really the same as Jakob Böhme's. And both approach Bayle, since they (as Leibniz himself remarks), like him, assume two principles, only that they think they can bring them to agree within one and the same nature of God, whereas Bayle found it necessary to assume two gods. Leibniz and Böhme let the possibilities press forward and struggle within the divine nature, a prelude to the struggle in actual existence.

Bayle had maintained that the Manichaean doctrine had its strength in the testimony of experience to misfortune and sin in the world, while an a priori consideration would

rather maintain the unity of the ground of the world. Leibniz must then follow him over into the domain of experience. Frequently, to be sure, he confines himself to saying: this world must be the best possible; otherwise God would not have chosen it! He overlooks the objection that could be made (and that Schopenhauer, Leibniz's antipode, put forward much later), that even if the chosen world was the best possible, it was not thereby said that it was good enough to be actualized! The need to exist, the craving for actualization, which Leibniz ascribes to the eternal possibilities (he can, after all, everywhere think of possibilities only as minimal actualities), might well be of evil! *[369]* Leibniz must then try to show that the actual world deserves to exist, since for a thorough and comprehensive consideration there remains a surplus of light and perfection. — He demands that one must not look merely at a single part of the world, but think of the whole. A sound that by itself would sound ill can make an excellent effect in a musical ensemble. Cover a painting except for a small spot: it will then seem a meaningless and unclear juxtaposition of colors; but in the whole painting it helps to determine the impression. A bitter ingredient in a dish serves to heighten the flavor, while by itself alone it would be unpalatable. — But how far, then, is the consideration to be extended? Existence, Leibniz answers, is infinite. The thought of providence, the purpose that underlies existence and conditions its stepping forth into actuality, we therefore cannot reach. Only a small, vanishing part of the world do we know — and it is perhaps the one where the most evils are found! Human purposes and ideals must not be made the only thing or what is decisive in the last instance. God pursues purposes that comprise the whole of existence, in which there are surely many other beings than human beings[75]). — And should this great, infinite work then be altered or taken back for the sake of our misfortunes and sins? All the more since everything hangs together in such a way that if, e.g., Tarquinius's deed of rape or Judas's betrayal were taken away, the whole historical development would become a different one. God has placed the single beings in the world; but of their actions they are themselves the cause; the source, not the stream, he has created. And although he knew what would stream from this source, he nevertheless created it, because the world in which it occurred was after all better than any other. Without Judas's betrayal the Savior's atoning death falls away: all the more reason, then, to say with the old church hymn: O felix culpa! — When the evil in the world is emphasized so strongly by many, it is because it draws attention to itself, and because habituation weakens the sense for the good we possess. Hindrances and resistance there must be, so that our activity may constantly be aroused anew and we may not become dulled. Even the blessed and the angels must meet resistance if they are not to become *[370]*

"stupid." Although existence in itself is continuous and harmonious, it is nevertheless purposive that apparent leaps should appear, breaks in the connection, disharmony: it exercises thought and increases the beauty of the whole. — Leibniz even undertakes to justify the dogma of eternal damnation: in existence as a whole he assumes such a fullness of light and perfection that even if the majority of human beings were eternally tormented, their pain would nevertheless be vanishing in relation to that infinite sum of happiness!! — Whether existence regarded as a whole advances in perfection, or whether the perfection it possesses merely varies in form through changing states, is a question Leibniz will not decide. But he is in no doubt that the fundamental mood of one who looks at existence with a widened and understanding gaze will be joy in the beauty and perfection of things and in the constant development, a joy that is widely different from Stoic patience in the face of necessity. —

Leibniz comes, without noticing it, to grant Bayle far more right than he wishes. Apart from the fact that in a certain sense he too assumes two principles, he must in fact concede to Bayle that when the consideration is confined to *experience,* it is very difficult to maintain teleological optimism. Leibniz's appeal to the infinity of existence and the limitation of our knowledge is really an abandonment of the proof: for whence does he know anything of how things stand in other regions of the universe? It is a faith he appeals to; but thereby he declares the question insoluble by way of reason. In a letter contemporary with the Theodicy (to Bourguet, printed in Gerhardt III, p. 550 f), he accordingly also says: "We can survey only a small part of the chain of things, and it is this part that presents the most evils and is therefore well suited to exercise our faith and our love of God." — Nor does the appeal from the single to the whole untie the knot: for precisely in Leibniz's own system, where there is life and soul at every single point in existence, it becomes impossible to ground the sacrifice of the single for the whole. The suffering in the single monad does not fall away because it increases the
[371] beauty of the picture of the world that the monads together form. The single monad will rightly be able to complain of the role assigned to it, and say that it would rather not be than be in hell, even if this hell could not be dispensed with as a kind of bass in the great orchestra of existence. Leibniz's own sense for the infinitely small, for individual differences, ought consistently here to have led him away from teleological optimism.

e. Philosophy of Law

Leibniz's ethical ideas and his ideas in the philosophy of law are closely connected with his general philosophy. His doctrine of monads culminates in the idea of existence as a

"City of God," ordered according to the fundamental law of justice, according to which the forces that underlie all mechanism act. The teleological point of view is common to his conception of the world and his ethics.

The foundation of all ethics and philosophy of law (both are comprised for him in the concept of natural law) he finds in the immediate, instinctive need of happiness. Each single feeling of pleasure corresponds to a progress, a "perfection," which either consists in the force being increased, or in a greater inner harmony being attained in the content the force comprises. "Perfection," he says (in a little treatise *Von der Glückseligkeit,* one of the few German works we have from his hand), "shows itself in the force to act, as indeed every being consists in a certain force, and the greater the force, the higher and freer is the being. Further: in all force there shows itself — and the more so, the greater it is — a multiplicity springing from a unity and contained in a unity (Viel aus Einem und in Einem), since there is one thing that determines a multiplicity outside itself and forms it beforehand in itself. Now unity in multiplicity is nothing other than agreement." Leibniz means by this to express that the feeling of pleasure is conditioned by the fullness of and the harmony between the forces that sustain life, whether one is conscious of these conditions or not. Happiness consists in a constant feeling of pleasure. The condition for the production of happiness is wisdom: the clarification of the understanding and *[372]* the exercise of the will. But involuntary striving already leads in this direction. Leibniz assumes, like Grotius, the "incomparable Grotius," as he calls him, an involuntary need to act not merely for one's own but also for the happiness of others. Love is joy in the happiness of others, in that one has made it one's own; even disinterested love is possible only through the happiness of others as it were shining back upon ourselves. As the chief ethical virtue Leibniz, following Aristotle, lays down justice, which he defines as the love of the wise man (caritas sapientis): love sets the end, and wisdom knows how to find the right means and the right distribution. Justice consists in giving each his own, in distributing goods in the right way, both in the ordering of public relations and in the imparting of what the individual has at his disposal. Justice in this sense (as legislative and distributive justice) aims at benefiting all and is different from strict right (jus strictum), whose purpose is the preservation of peace in society. (See the treatise *De rationibus juris et justitiæ.* 1693).

There is something prophetic in Leibniz's philosophy of law in spite of his partial use of ancient and scholastic elements. By setting universal utility or happiness as the end of law and morality he is a forerunner of so-called utilitarianism. His ideas recall Cumberland's, and he was himself gladly surprised to find a kindred line of thought in

his younger contemporary Shaftesbury.

Although ethics and the philosophy of law are built in Leibniz on a purely psychological foundation, they nevertheless cannot, in his view, do without a theological sanction. He agrees with Grotius that morality and law are not to be grounded on the arbitrary commands of the Deity. But like all the highest principles and laws, the laws of morality and of law too spring in the end from the divine thought and are upheld by the divine will. In disinterested love, to be sure, the striving after one's own happiness becomes one with the striving after the happiness of others. But it is found only in chosen natures (les âmes bien nées). One cannot, without presupposing divine reward and punishment in an
[373] other life, prove that the moral good *and* what is most useful to the individual himself *always* coincide. Therefore natural religion is necessary as a guarantee of morality. And it gives at the same time the highest point of view for human morality and doctrine of right, since the consideration is widened to the divine law that reigns in the whole of existence.

In Leibniz's philosophy of law, as in his whole system, the fundamental thought of harmony between individual beings comes forth. It was this thought that bore his inquiry in all domains, and that he bequeathed to the new century, for which his system is in so many respects typical.

7. Christian Wolff

What Leibniz had expressed in letters and treatises that were accessible and intelligible only to a few, Christian Wolff developed broadly, flatly and systematically in voluminous works. He has an extraordinarily great significance for having introduced rational thinking into wider circles and made "the principle of sufficient reason" valid in all domains. Whereas Leibniz wrote predominantly in French or Latin (although for a time he thought of writing his "Theodicy" in German, precisely on account of that language's philosophical "virginity"), Wolff used the German language and in part created the German philosophical terminology. A popular Wolffianism became an essential component of the German Enlightenment in the eighteenth century. Through Wolff a whole series of thoughts from the great systems of the seventeenth century found their way out into the educated public. Another circle of thoughts came from England through the movement awakened by John Locke, which the following main division will set out.

Christian Wolff, who was born in Breslau in 1679, studied mathematics and phi-

losophy besides theology and became professor of mathematics at Halle in 1706, but lectured chiefly on philosophy. He was strongly influenced by the writings of Descartes and Leibniz, but through Tschirnhausen also by Spinoza, whose theory of knowledge Tschirnhausen (without naming Spinoza's name) had carried out in broader form in his *[374]* *Medicina mentis.* Although Wolff himself did not like to be regarded as Leibniz's disciple, and although he modified Leibniz's thoughts in various ways, it was nevertheless with full right that the philosophy he introduced into the German universities, and which reigned there right up to Kant's time, was called the Leibniz-Wolffian philosophy. To be sure, the innermost kernel of Leibniz's line of thought, his metaphysical idealism and the doctrine of monads, did not lend itself to such a popular systematization as Wolff gave. But the tendency to unite natural science and theology by conceiving the world as a great mechanism that is designed to serve divine purposes and brought by "pre-established harmony" into agreement with mental life is pervasive in Wolff[76]) as in Leibniz. The world is conceived as a law-determined whole of individual beings with God as the ultimate ground. And the logical bearer of the whole system is the principle of sufficient reason.

It is interesting to see how Wolff is not merely the popularizer of dogmatic philosophy but also becomes its completer. It was a significant step in Leibniz that the principle of identity and the principle of sufficient reason were laid down as two different principles, the one holding for all truths of reason, the other for all truths of fact. Wolff now tries to derive the latter principle from the former. He wants, then, to make it self-evident that there is a ground or a cause for everything that happens, and he found it a defect in Leibniz that he had given no proof of "the principle of sufficient reason," which nevertheless plays so great a role in his as well as in the other great systems. Wolff's proof (which is found in *Vernünftige Gedanken von Gott, der Welt und der Seele des Menschen, auch allen Dingen überhaupt,* den Liebhabern der Wahrheit mitgetheilet von Christian Wolffen. 1719. § 30) is as follows: "Where there is no ground present, there is also nothing from which one can comprehend why something exists, and so it must arise from nothing. Now since it is impossible that something can arise from nothing, everything that exists must have its sufficient ground why it exists." — It is easy to see that this proof moves in a circle; for that something cannot arise from nothing means, after all, the same as that *[375]* everything has its ground or its cause, and that was precisely what was to be proved. But by this line of thought Wolff got dogmatism set completely into a system. All certainty was traced back to the principle of contradiction, and philosophy became a system of propositions that could all be grounded purely logically. The principle of contradiction

contains, as must indeed be noted so as not to do Wolff an injustice, the guarantee for all empirical assumptions as well as for all inferences. When I observe something, I cannot at the same moment not observe it; this demand every observation must satisfy. Wolff by no means wants to make all science formal logic. But he wants to show that rational principles reign everywhere. It was rationalism that made its entry.

Wolff applies it not merely to the conception of the world but also to theology. Just as from the principle of sufficient reason he derives propositions of natural science such as the principle of inertia and the law of continuity (nature makes no leap, because the transition from one state to another would then be incomprehensible), so he also wants to derive from this principle not merely the existence of God (the world must have its ground, which has its ground in itself), but also the conditions to which every positive revelation must be subject. There must — Wolff thinks — be definite marks by which revelation can be distinguished from empty imagination and false pretense. Revelation cannot be contrary to God's perfection and cannot contain contradictions. Although it is a miracle and to that extent is contrary to "contingent" (factual) truths, it must not be contrary to the necessary truths: God can make the sun stand still, but cannot change the relation between the diameter and the circumference of the circle. And Wolff maintains that a world in which miracles are rare is more perfect than a world in which they are frequent; miracles require only might, but the natural order requires in addition a wisdom that thinks of the whole, not merely of the single.

Wolff himself became a victim of "the principle of sufficient reason." The pietist
[376] theologians in Halle saw a danger in the rationalism he proclaimed, and when in a university oration he had praised the Chinese philosopher Confucius for his pure morality and declared it to agree with his own, the storm broke loose. With King Friedrich Wilhelm I people are said to have made use of Wolff's application of the principle of sufficient reason to human actions, representing to the soldier-loving king that when one of his grenadiers deserted, one could not, according to Wolff's doctrine, hold him responsible. According to what Wolff relates in his autobiography, the king is said to have been shown a passage in the writings of one of his theological opponents where this consequence of his determinism was drawn. There then came (1723) a royal cabinet order according to which Wolff, on account of his ungodly doctrines, was to be dismissed and within 48 hours be out of the King's realms and lands; if not, he would be hanged. Wolff went to Marburg, where he continued his activity until the new King of Prussia, Friedrich II, who was himself among his admirers, called him back to Halle, where he worked until his death (1754).

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Fourth Book

[377]

The English Philosophy of Experience.

The great systems sprang from the confidence that there was sufficient material and sufficient clarity of thought to erect edifices that could take the place of the medieval world-view, which had been overthrown by the inquiry of the Renaissance and by the coming into being of the new science. In a certain sense this confidence was not unfounded either; through the new discoveries, methods and fundamental propositions a decisive light had been cast on the direction in which human thinking had to move with regard to some of the most important problems, and it is the special significance of the seventeenth century that it worked out with energy and consistency the most important hypotheses concerning the relation between the mental and the material in existence. But the very fact that several hypotheses were possible was bound to arouse the attention of the critical thinker. And to this was added that those who with such great confidence and genius erected those edifices of thought had indeed discussed the nature and working of thinking, but only as an introduction to their real work. With dogmatic haste they hurried from the examination of thinking to finding the solution of the riddles of existence. The significance of the classical English school in the history of philosophy is marked by the fact that it makes the investigation of the development of human knowledge and of the forms and presuppositions at its disposal into an independent problem. John Locke and his successors secured the independence of the problem of knowledge over against the *[378]* problem of existence, which in the great systems had entirely overshadowed it. They pushed epistemology forward ahead of metaphysics. If (following Kant's usage) one understands by dogmatism a tendency which, without sufficient examination of the conditions and limits of knowledge, uses our concepts to fathom the essence of things, while the critical philosophy examines the cognitive faculty itself before it begins to speculate about existence, then the critical philosophy decidedly begins with John Locke.

But behind this purely philosophical opposition between dogmatic and critical philosophy lies a more comprehensive historical opposition. It is not only the philosophical systems that are criticized: the critical examination turns against every authority, every established power. The concept of substance in philosophy had, as we have seen, its analogue in absolute authority in the life of the state. Now comes the age of emancipation, together with that of criticism. The process of which Leibniz's monads are symbols, the individual's becoming independent, appears with sober clarity in Locke's great settling of the accounts of thought with authorities and traditions in the domains of science, education, the state and the church. Leibniz and Wolff asserted the principle of sufficient reason and practiced applying it. John Locke and his successors subjected all principles — in the end even the principle of sufficient reason itself — to a thorough investigation.

Together with the energetic treatment of the problem of knowledge goes a no less energetic discussion of the problem of valuation. In the English school ethics comes to a more independent position than it could obtain within the great systems, whose real interest led the gaze far out beyond human life. And through the method of experience employed in this school, the foundation of philosophical ethics now receives a more
[379] thorough illumination.

1. John Locke

a. Biography and Characterization

The first critical philosopher was born in the same year as the greatest of the dogmatists. John Locke was born on 29 August 1632 in the neighborhood of Bristol. His father was an attorney, but took part as a cavalry captain in the Civil War on the side of Parliament. His independent and liberal cast of mind passed by inheritance to the son. The upbringing the future philosopher received from his father was of such a kind that he could draw on memories of it in working out his epoch-making treatise on education. In particular the feature *(Thoughts on Education* § 40) that the relation of obedience necessary in childhood is gradually to be replaced by a free relation of friendship is taken from the upbringing the author himself had received. It was more by the effect of contrast that the experiences he had at school and at the university gained significance for his philosophy. For six years he went to school in Westminster, where he learned the ancient languages by a strictly grammatical method and was plagued with learning by rote and with composing Latin dissertations on subjects he did not understand. Natural science was not taught — only in summer a little geography after supper. His warnings against learning by rote

and against beginning the teaching of languages with grammar are grounded in what he had thus himself experienced at school. When he came to Oxford (1652), Puritanism and scholasticism reigned there. Locke felt dissatisfied with both. His philosophical awakening he received (by his own later statement) through the study of Descartes's writings. Acquaintance with these was a great encouragement to him, since he had put down his small progress in scholastic philosophy to his lack of talent in the philosophical direction. Gassendi and Hobbes too he studied, and they had great influence on his intellectual tendency.

Great tolerance was shown in Oxford at that time. Not only the university's chancellor John Owen, but the mighty Protector Oliver Cromwell asserted freedom of thought for all Protestants. This time set its lasting marks on Locke's thinking throughout his whole *[380]* life. With the Restoration the episcopal church came to power again. Locke had at first thought of becoming a clergyman, but his free, "broad-church" (latitudinarian) conception of Christianity now made it impossible. In a treatise of 1667 (which was later expanded into his famous *Letter on Toleration)* he denies to anyone whatsoever the right to force speculative opinions and particular forms of worship on others. And in a treatise headed *Error* (which, like the preceding one, is printed in *Fox Bourne's* excellent "Life of Locke". London 1876) he says: "He who inquires, and who after honest inquiry accepts an error instead of a truth, has done his duty better than he who accepts the profession of the truth (the truth itself he does not accept) without having inquired whether it is right or not". He lays the chief weight on the moral side of religion and demands as few dogmas and ceremonies as possible. The episcopal church with its thirty-nine articles and numerous ceremonies was then closed to him.

He decided to become a physician and studied chemistry and medicine. Thereby was laid the foundation of a friendship with the famous chemist Robert Boyle, as well as with the no less famous physician Sydenham. Locke's endeavors as a philosopher form an analogy to those of his two friends. — Boyle (who was six years older than Locke) asserted the experimental method in chemistry over against alchemists and medical chemists, who set themselves other purposes than the purely scientific one. He was the first to make clear what chemical analysis aims at: namely, to find the elements of compound substances, that is to say, the demonstrable constituents that cannot be further dissolved. He predicted that far more elements would be discovered than were then imagined, and disputed the uncompounded character of many substances that had hitherto been regarded as elements. Boyle also emphasized the peculiarity of chemical compounds that the product has other properties than the elements. — Sydenham

asserted the method of experience in medicine and paid homage to the same principles
[381] that Locke expressed in a small treatise *On Medicine,* where the necessity is impressed of building on observations, not on axioms. Locke often accompanied Sydenham on his visits to the sick.

Yet medicine did not become Locke's life's work either. He made the acquaintance of the Earl of Shaftesbury, the well-known politician under Charles II, and was soon attached to his family as friend, secretary, physician and tutor of two generations. Locke's political views were (as can be seen from short treatises of his younger days) liberal, like his religious views. He stood with zeal and conviction on the side of the Whig party and became entangled in its fate. Shaftesbury's fall in 1672 drew Locke's with it. He had to give up the posts Shaftesbury's confidence had procured him. He now spent some years traveling in France. His travel notes have historical interest and testify to his good power of observation and his attention to all sides of life. When later his noble friend had become entangled in a conspiracy and had to flee to Holland, Locke did not consider himself safe in England either and likewise went (1683) to Holland, where by degrees a good many exiled Whigs gathered. For a time he had to live in hiding, since the English government demanded his extradition. In these years he worked on his writings, especially on *Epistola de tolerantia* (1685), which appeared in English some years later, and on his chief work "on the human understanding". At the same time he was, so far as one can see, occupied with the preparations for the Revolution. He stood in close connection with the Prince of Orange and his consort, and returned to England at the beginning of 1689 together with the Princess.

Locke forgot neither philosophy for politics nor the reverse. At the beginning of 1690 his chief work *Essay on Human Understanding* appeared, one of the most remarkable and most influential works in the history of philosophy. The first germ of it lies far back in Locke's development. He relates in the preface that the occasion of the work's coming into being was a discussion he had with some of his friends. When they could not solve the problems they had set themselves, it occurred to Locke that it would
[382] perhaps first have been necessary "to examine our own powers and see what objects our understanding was fitted to deal with, and what not". In a copy of Locke's Essay which is in the British Museum, Locke's friend James Tyrrell has written: "I remember that I myself was one of those who were present when the principles of morality and revealed religion were discussed". It was thus a discussion of ethical and religious subjects that led to the closer investigation of knowledge. This first beginning was made in the winter of 1670—71. Later Locke worked on his work during his stay in France and during

his exile in Holland. In 1687 the work was finished. An abstract of it was printed the following year in the Dutch journal "Bibliothèque universelle", and in 1690 it appeared complete in English. It consists of four books: the first criticizes the doctrine of innate ideas and principles; the second shows that all ideas derive from experience, and resolves the complex ideas into their simple constituents, so that agreement may thereby more readily become possible; the third investigates the influence of language on thinking, combats the scholastic philosophy of words and impresses that our concepts of species may not without further ado be regarded as valid for nature; the fourth distinguishes between the different kinds of knowledge and determines the limits of knowledge. The epistemology proper is thus contained in the fourth book. It and the second book (on the origin of ideas from experience) seem to have been written first; the first (the critique of innate ideas) and the third (on language) were added later.

Under King William Locke had no small influence. He stood close to the king and to several of the leading men. He worked for freedom of the press, tolerance, sensible coinage and trade legislation and for a better poor-relief, and held various offices. His *Two Treatises on Government,* which appeared in 1690, avow, besides having a theoretical aim, that they are also a defense of the Revolution that had led to the new order of things. Of his other writings there must be named (besides *Thoughts on Education* 1692) the book on "the reasonable nature of Christianity, presented according to Scripture" *(The Reasonableness of Christianity as delivered in Scripture).* In his conception of Christianity *[383]* Locke lays the weight on the fact that the faith of the oldest church was comprised in the proposition that Jesus is the Messiah. Christianity is for him the gospel of love; it would not torment men with incomprehensible dogmas (of the Trinity, the Atonement, eternal damnation), but extend the law of nature and of reason by showing how men can attain eternal blessedness. He himself believed he was a believing Christian, and his letters as well as his life testify to religious inwardness. He was a zealous reader of the Bible and in his last years worked out a commentary on the Epistles to the Corinthians. But he felt most attracted by the Christian tendencies that had least dogmatics and hierarchy. During his stay in Holland he lived for a time with a Quaker, with whom he formed a faithful friendship. Later, in London, he accompanied King William when the latter visited the Quakers' meetings in disguise in order to get to know this sect. To an English Quaker woman Locke wrote that, just as it was women who first saw the risen Savior, so perhaps it was also women who were now to proclaim the resurrection of the spirit of love. He became the object of violent attacks on the part of the theologians, and since the source of his theology was found in his philosophy, this too was sharply attacked,

especially by Stillingfleet, Bishop of Worcester, whom Locke answered repeatedly in very lengthy polemical writings. It was the last single combat between the scholastic and the modern philosophy. What sharpened the ill will toward Locke's theological standpoint was that it lay so near to that of the deists that a writing such as John Toland's "Christianity not mysterious", which appeared in 1696 and was publicly burned the year after in Dublin, seemed only to draw an obvious consequence from it. Since Locke's writings were also read by the student youth, the heads of the colleges in Oxford resolved that Locke's Essay might not be used at the university. Locke remarked, when he heard of this, that because there are those who put on blinkers or turn their heads away, not everyone will put up with doing without his eyes; and in this history proved him right.

In his last years Locke, who was never married, lived in the house of Mr. Masham
[384] at Oates near London. Lady Masham was a daughter of the Cambridge philosopher Cudworth and a highly gifted lady. After several years of suffering from asthma he died there in the year 1704. A gentle disposition, great love for his friends, sincere search for truth and firm faith in the significance of personal and political freedom, these are the traits of character that stand out especially in Locke, as we know him from his works and letters. Characteristic of him is what he wrote, the year before his death, to a young friend, the later deistic writer Anthony Collins: "To love truth for truth's own sake is the most important part of human perfection in this world and the seed-plot of all other virtues."

b. The Origin of Ideas

Since Locke means to call human knowledge to account, his first task is to investigate where the ideas (ideas) with which it operates derive from. By idea (idea) he means everything we can think of. This task would fall away if there were innate ideas in the literal sense, and it is against this assumption, then, that he first turns his criticism. It has been held that the idea of God was such an innate idea; likewise the most fundamental logical and moral principles were supposed to be so. Such propositions as that every thing is what it is, and that we are to do to others what we would that they should do to us, are assumed to be originally implanted in human consciousness. But in that case — Locke holds — they would have to be the very first ideas to appear to consciousness. That this is not so one can easily convince oneself by investigating children, idiots, savages and unenlightened people. The consciousness of such beings contains only single, definite, concrete ideas and observations, no general principles. And experience shows us individuals and peoples without an idea of God and without the moral ideas

proper. — If one would infer innateness from the fact that such ideas and principles are understood and assented to as soon as they are made distinct to consciousness, that is an unwarranted inference, for what has to be made distinct and proved is not innate, but *[385]* must be learned. There must of course be an original capacity (capacity) to learn, and one may rightly speak of natural laws or fundamental propositions, inasmuch as there are assumptions at which man arrives by the natural use of experience and of his power of thought. Among these Locke counts, besides the logical and mathematical truths, also the most important religious and moral assumptions. But a *natural* law — he insists — is by no means the same as an *innate* law. The doctrine of innate ideas springs in great part from indolence; one would be spared the investigation of the development of ideas.

This polemic seems to be directed especially against the scholastic philosophers, the Cambridge Platonists and Herbert of Cherbury; this last is the only representative of "innate ideas" who is named by name. Descartes is hardly aimed at in the first instance, just as the closer explanation Descartes gives of the unfortunate expression "innate" means that he is not hit by Locke's criticism. — Locke himself used an expression that has been misunderstood, namely the old image of consciousness as an unwritten tablet (tabula rasa). It was not his meaning thereby (as has often been believed) to deny original powers in the soul prior to experience. — It is an unfortunate side of philosophical catchwords that they easily convey too crude notions. Even with the said correction of Locke's doctrine, Leibniz's criticism is nevertheless to the point, since Locke partly overlooks the significance of the obscure, more or less unconscious elements, partly the involuntary, spontaneous way in which original tendencies assert themselves. The very moment of activity that Locke still recognized in the formation of ideas was reduced by some of his successors and in the end abolished. —

Locke's own answer to the task is that all ideas, by which is meant all content of consciousness, stem from experience, partly outer experience (sensation), partly inner experience (reflection). Outer experience arises when an impression or a motion produced in some part or other of the body awakens a sensation (perception) in the soul. Inner experience arises when the soul also receives impressions of the activity that is unfolded *[386]* (e.g. in memory and comparison) in working up the ideas derived from outer experience. By "reflection" we thus apprehend our own states and activities, by "sensation" the influences of other things. In every such immediate apprehension or reception of ideas consciousness behaves as good as (for the most part) passively. It is however only the simplest ideas that come into being through such immediate and passive reception.

The simple ideas that arise through outer experience need no more resemble the

properties of the things that call them forth than the word resembles the idea. Only the so-called primary qualities, solidity, extension, figure, mobility, cannot be separated from the outer things themselves; the secondary qualities, such as color, taste, smell etc., correspond only to the power that things, by means of the primary properties, have to produce certain ideas in us. — The expressions primary and secondary qualities Locke seems to have taken over from Robert Boyle; the doctrine itself, which has often been ascribed to Locke as its first originator, stems, as we have seen, from Galilei, Hobbes and Descartes. Locke does not go further into the problem that lies in there being here so great a difference between the outer cause and the inner effect. —

The simple ideas are the material of consciousness, which it works up in different ways. The activity of consciousness shows itself partly in the formation of *complex* ideas by combination of simple ideas, partly in the formation of *ideas of relation* by bringing simple ideas into a certain mutual connection, partly in the formation of *abstract ideas* by separating simple ideas from others with which in reality they occur together. All ideas that are not due to immediate impressions have, however exalted and significant they may be, come into being through the compounding, uniting and abstracting activity of consciousness and on the foundation of immediate observations. — Of these three kinds of derived ideas it is the first two that have most interest. They are formed, moreover,
[387] with the co-operation of abstraction.

1) To *complex ideas* belong *properties* (modes) such as space and time. In forming the idea of space we rely on the senses of sight and touch. At the same time we here employ abstraction, distinguishing between space and solidity, a distinction that is as clear as that between the bushel and the grain. The idea of time we form by means of the inner sense, which shows us a succession of ideas. And when it appears that our power of representing space and time to ourselves as enlarged remains the same however many additions we have made, the idea of infinity arises. — To the concepts of properties belong also the ideas of power and motion as well as all ideas of compound colors, forms etc., and in the domain of the inner sense the ideas of sense perception, memory, thinking, attention etc.

While the concepts of properties present no difficulty, it is otherwise with the *idea of thing, being or substance.* It too is formed by compounding. Our idea of a thing or a substance is the idea of the properties or powers we ascribe to it. But the remarkable thing is that we nevertheless represent the thing or the substance itself to ourselves as something different from the properties and powers, namely as their bearer or foundation (support). From experience stems everything we ascribe to substance. This holds even of

the idea of God, which is an idea of substance that is formed by enlarging and heightening our ideas of mental properties, which are taken from the inner sense.

2) As an example of *ideas of relation* (ideas of relation) may be named the idea of cause and effect. This idea we form on the foundation of the observation that properties and things arise, and that their arising is due to the influence of other properties and things. Other examples are the ideas of relations of time and space and of identity and difference. The moral ideas too belong to the ideas of relation, since they are formed from the simple ideas of our actions compared with the idea of a law. — While by compounding we combine several ideas into one idea, ideas of relation are formed by two ideas (simple or complex) being placed together so that they can be apprehended *[388]* at once without becoming one. Locke does not investigate whether the idea formed by compounding does not have another nature than the simpler ideas of which it is formed, in analogy with what Boyle had emphasized in chemical compounding. Nor does he investigate whether ideas ever occur except in some relation or other, and whether the idea of relation is produced by mere juxtaposition; the relation, after all, does not lie in any of the elements each taken by itself.

3) By *abstraction* we separate an idea from other ideas with which in reality it occurs together. Thus we form the idea of white by disregarding the other properties it occurs together with in chalk, snow or milk, and the idea of space in and for itself by distinguishing between space and content, just as we distinguish between the bushel and the grain.

c. The Validity of Knowledge

After the coming into being of ideas has been shown, the question of their validity arises. Now there is for Locke no doubt that the simple ideas have real validity. They stem from reality and must therefore agree with reality. The secondary qualities, to be sure, do not resemble the things that call forth the ideas of them, but they correspond to the things because they are the latter's constant effects. — As regards the derived ideas, their validity cannot be established in this way, since they are after all formed by compounding, juxtaposition and extraction, thus by our activity. That they contain no self-contradiction is not enough to establish their real significance. They are not copies of things, but their significance consists in being models or patterns (archetypes, patterns), which consciousness uses when it orders and names things (thus in mathematics and moral philosophy). In mathematics and moral philosophy we take account only of the content of things, of the ideas formed by us by free choice, not of the connection things *[389]*

have in nature (Essay IV, 4, 5). Yet this holds only of the ideas of properties and relations (modes and relations). The concept of substance, the idea of an unknown bearer of the properties, can only have validity when there is in reality such a combination of properties as it indicates. The idea of a centaur is a false, the idea of God a true idea of substance. The concept of substance is not itself a model, but its model must be outside us if it itself is to have validity. But what it is that underlies the properties of things we do not know. We know the material substance as little as the mental, and the Cartesians therefore have no right to assert that matter cannot think.

Knowledge is the perception of the mutual agreement or disagreement of ideas. In its simplest form it is intuition, immediate intuiting. On this point Locke decidedly approaches Descartes, as when he says that we have an intuitive knowledge of our own existence; doubt itself says so: for if I know that I doubt, I have as certain a perception of the existence of the doubting thing as of the thought I call doubt. Besides our own existence we apprehend by such immediate intuiting the simplest fundamental relations between our ideas. A proof (demonstration) arises when a series of intuitive cognitions is joined together. Each step in the demonstration is an immediate intuiting. These two kinds of knowledge are the only strictly reliable ones. All other knowledge is only assumption, more or less probable conviction; to this belongs the sensuous knowledge of the things outside us.

Demonstrative knowledge concerns not only mathematical propositions, but also the existence of God — and this is the only case in which we can, by way of proof, know real existence besides our own. Locke's proof of the existence of God is the usual one: from the world (or from our own existence) one infers God as its cause. But it acquires a special interest from the fact that it builds on the causal principle, which is declared to
[390] be an intuitive truth: we have an immediate certainty that what does not exist cannot produce reality (nonentity cannot produce any real being)! (Essay IV, 10, 3). There must then be an eternal being. And it cannot be matter, Locke continues, for matter cannot produce understanding. — Only apparently is there a contradiction between this last proposition and the one cited earlier about the possibility that matter can think. For in the earlier proposition he asserts that we do not know the innermost essence of matter; here he proceeds from matter as we know it[76]).

Just as there is in Locke's doctrine of the origin of ideas a sharp opposition between the passivity in the reception of simple ideas and the activity in the formation of derived ideas, so there appears in his doctrine of the validity of knowledge a no less sharp opposition between sensuous knowledge, which only attains to probability, and intuitive and

demonstrative knowledge, which brings full certainty and necessity with it. Remarkably enough, this opposition appears in setting up the proposition on whose validity Locke's proof of the connection of knowledge with reality rests, the causal principle. The very concept of a causal relation we form by sensuous experience: "That which produces a simple or complex idea we denote by the general name cause, and that which is produced, by the name effect." (Essay II, 26, 4). The validity of the causal principle is established by intuition (Essay IV, 10, 3), and the causal principle itself is used confidently both in inferring from the simple ideas to the things that have produced them, and in the proof of the existence of God. The idea of God is — like the idea of cause — taken from experience, though with the co-operation of the combining and enlarging (idealizing) activity of the mind; the existence of God is proved by inference with the help of the causal principle. Locke's standpoint as empiricist as regards the origin of ideas, and as rationalist as regards their application, appears clearly here. By the boldness with which Locke sets up and uses the causal principle he still belongs to dogmatism. But by the demand for psychological explanation and epistemological grounding he has led onto *[391]* the path of the critical philosophy. Locke's transitional position is, however, seen most clearly in the concept of substance. It was the self-evident fundamental concept in the great systems, the last, absolute resting point of thought. Only Hobbes shook it, and here, as on several points, he is Locke's predecessor. Locke treats it with a certain irony, calls substance an "I know not what" and compares the belief in it with the Indians' belief in the elephant as necessary to bear the earth; — and yet he does not reject it. He explains it as a self-made concept, just as the mathematical and moral-philosophical concepts are; and yet he ascribes to it an outer model and foundation.

d. Philosophy of Religion

Locke's religion did not stand in opposition to his philosophy. He did not need as many surreptitious moves as Leibniz in order to unite them. Since he believed he had given a philosophical proof of the existence of God, he believed he had at the same time asserted the truth of natural religion by way of reason, just as he thereby also obtained the theological presuppositions without which, in his conception, no ethics was possible. His ethics is a theological doctrine of happiness: from the natural need of happiness springs the law that states the conditions under which our own happiness can be attained in connection with that of others; but this law, found by the natural way of reason, which is summed up in the old saying that we are to do to others what we would that they should do to us, must, in order to obtain authority, be regarded as sprung from and upheld by

the will of God.

Although Locke held that tolerance in the state ought to be extended to all who professed natural religion, his own personal standpoint was nevertheless that of faith in revelation. In the spring of 1695 he writes to Limborch: "This winter I have carefully considered wherein the Christian faith consists. I have drawn from Holy Scripture itself, but kept the opinions of sects and systems away." The results of his investigations he set
[392] down in his writing on "the reasonable nature of Christianity". Revelation is for him an extension of natural religion, which can be grounded by reason, and on the other hand reason is the constant control of faith in revelation. In his chief work (Essay IV, 18, 2) he defines faith as assent to a proposition that is not set up by inference of reason, but is built on trust in him who puts it forward as coming from God and communicated in an extraordinary way. Whether something really is revelation, reason must decide, and faith can never convince us of anything that conflicts with reason: for the knowledge we have that revelation stems from God can never be as certain as the knowledge we ground on the very agreement or disagreement of our ideas. Revelation is nevertheless necessary. For although nature testifies sufficiently to God, men have not rightly used their reason. From indolence, sensuality and fear they came under the dominion of superstitious priests, since the few who followed reason could gain no influence on the great multitude. The greater part of men lack time and ability to follow the proofs of reason. Christ was then sent to enlighten, strengthen and help. Faith in him as Lord and Master has the promise of an eternal life. Even the most unlearned, even those whose life passes in heavy toil, can understand Christ's example and teaching as it appears in the Gospels. To the doctrinal developments of the Epistles Locke attaches less weight. The dogma of the Trinity he can find neither in the Gospels nor in the Apostles' Creed.

Beside Christian Wolff (who on other points too, e.g. in the unfortunate grounding of the causal principle, recalls Locke) Locke stands as the founder of religious rationalism. His writings and especially his religious standpoint influenced the greatest minds of the eighteenth century; Voltaire and Frederick II stand here in the first rank. Most remarkable, however, is to see how religious rationalism breaks forth as a democratic movement, — as indeed, for that matter, every new religious current does. In combating the theology of the church, Locke pays special regard to the needs of the unlearned
[393] and of those who labor hard: he demands a Christianity that can be accessible and comprehensible to them. But Locke's standpoint was nevertheless still dogmatizing: he ascribes to "natural" religion so great a certainty as regards proof, and so great a necessity as the foundation of ethics, that he even excludes those who reject it from the

benefits of religious freedom! Not only Catholicism and Protestantism, but rationalism too has its intolerance.

e. Philosophy of Law and of the State

Locke's writing on "civil government" is an occasional piece. He says in the preface that his intention is "to establish the throne of our great restorer King William, to make good his title by the consent of the people, which is the only title for a lawful government, and which he possesses in far higher degree than any other prince in Christendom, and finally to justify before the world the English people, whose love of their lawful and natural rights and firm resolution to uphold them saved the nation when it stood on the brink of slavery and ruin." In the first part of the writing Locke refutes the doctrine set up by the royalist Filmer of kingship as a divinely instituted, patriarchal institution. Political power, Locke asserts, is different from the father's power over his children, the master's power over his journeymen and the lord's power over his slaves. It is the power to give laws, to execute these laws and to defend society against outer violence, and all this only for the sake of the common good. Such a power can only be established by free agreement. Such an agreement is entered into tacitly even now, when the state of nature has ceased, as every time the son takes over the father's place in society. This agreement amounts essentially to this, that the will of the majority is to be law, because it is the only way in which society can come to act as a whole. The state of nature is not, indeed, as Hobbes held, a state of war; but it nevertheless brings with it inconveniences that can only be remedied by fixed laws, impartial judges and executive power of the state. The state of nature is already a state of freedom, but freedom is better upheld in society. It is then very far from the case that the natural rights are given up on entering into society. *[394]* The right of property, e.g., is a natural right, grounded in man's working of the soil or of other material. God has given the world to the industrious, not to the lazy, and all economic value springs from labor. The state can therefore as little abolish the right of property as ground it, but has only to secure and uphold it. As it is with the right of property, so it is also with personal freedom; slavery conflicts with nature and ought therefore not to be maintained by the state.

In Locke's political theory his strong emphasis on the legislative power is especially of significance. By the introduction of a legislative authority no arbitrary power is established; for one leaves the state of nature precisely in order to avoid arbitrariness. What is aimed at is the greatest possible regard for the general good in all decisions. Only with the approval of the majority ought taxes to be levied: otherwise the right of property

is abolished, for I do not own what another can take against my will! — Locke demands that the legislative power be separated from the executive and the judicial[77]); but the legislative power is nevertheless the highest: with its establishment the constitution of the state is given, and moreover he who gives the laws stands above him who executes them (Of Govern. II, 141. 143).

The highest power nevertheless remains with the people, and it comes into use when the executive power comes into conflict with the legislative. No power on earth except the people can decide such a conflict. By virtue of the last, inalienable right, the right of self-preservation, the people appeals to heaven and carries out its will. This is not rebellion: the rebellion is committed by those who break the laws. Nor does it lead to the dissolution of the state, for in the first place the presupposition is that the majority of the people feels the inconveniences that are to be remedied, and in the second place the people does not so easily get out of its old forms as is believed. —

Locke's free spirit of inquiry, united with his warm and practical participation in the great events of his people's history in his time, led him to formulate the great fundamental
[395] propositions of popular freedom in a way that became of decisive significance not only for later doctrine of law and the state, but also for the history of the peoples in the following centuries. Montesquieu and Alexander Hamilton are his disciples, Rousseau's doctrine of popular sovereignty can rest on him, and the North American and the French Revolutions were illustrations of what Locke calls the appeal to heaven. The constitutional life of the modern states is essentially built on Locke's fundamental propositions. The limitation of this whole political philosophy coincides with the limitation of the constitutional question. Behind it rises the social question. By deriving the right of property from labor Locke had, without knowing it, posed this problem, which only in a much later time was to come to the fore.

2. Discussions in Moral Philosophy and the Philosophy of Religion

a. The Doctrine of the Moral Sense

Through the Renaissance and the seventeenth century self-preservation can be traced, under different forms, as the foundation on which the conduct of life is thought to be built. With quite few exceptions all attempts to ground an ethics proceed from the individual, its craving and need, its subsistence and its welfare. It makes here in

principle no difference whether one thinks only of the individual's subsistence in this world, or whether one (as e.g. Locke) also takes its subsistence in another world into consideration. Over against this way of considering things, which in Hobbes found its sharpest and most consistent defender, there was set up within philosophical ethics especially the appeal to reason as that which is to regulate men's conduct. Spinoza's profound attempt to show what metamorphoses or displacements the impulse of self-preservation can undergo was little heeded. It is SHAFTESBURY's merit to have brought out the significance of immediate feeling, determined by instinct, for ethical valuations. It is a salutary and fruitful opposition to the rising rationalism that Shaftesbury initiates; and as so often, where a refreshing current of thought breaks forth in modern times, it is *[396]* ancient influences that determine his ideas. The ancient harmony and self-limitation and the ancient trust in nature are found again in Shaftesbury, to be sure combined with a certain amount of modern sentimentality. Anthony Ashley Cooper Shaftesbury, born in London in 1671, was a grandson of Locke's friend. Locke assisted as physician at his birth and was later his teacher. The greatest influence came from the careful instruction in the classical languages. He had a woman teacher who spoke Latin and Greek with fluency, so that he learned to speak these languages as his mother tongue. Sound Greek thoughts were stamped early on the boy's mind. His further development he received on travels in Italy and France, which brought him knowledge of men and gave him occasion to cultivate his artistic interests. For some years he was a member of the House of Commons. King William valued him highly and wished to give him a high office, but he declined; his health was weak, and his mind was set on a quiet, literary life. He died comparatively young in Naples in 1713.

Shaftesbury's works, of which he himself collected most in three volumes under the title *Characteristicks of Men, Manners, Opinions, Times* (London 1711), are not written in the form of calm, systematic development of concepts; they give outbursts and reflections, often in the form of letters or dialogues, often with great poetic swing and mood, but also often with sentimental rhetoric instead of reasons. He is, both in a good and in a less happy sense, the first philosopher of feeling. The significance of immediate feeling is asserted in opposition to reasoning reason, calculating egoism and outer sense perception. He asserts the connection between the beautiful and the good, taking up again the ancient conception of virtue as a harmony between the drives in the single human being and among men in their mutual relations. Finally he asserts the independence of ethics from religion, while at the same time it is his conviction that ethical feeling, through faith in the harmonizing Deity working in all things, is perfected as religious feeling.

[397] While Shaftesbury regarded his teacher Locke with great respect and directs his sharpest polemic against Hobbes, he nevertheless declares (in his *Letters to a young Man at the University),* that Locke by his criticism of innate ideas has overthrown the foundation of morality. He grants that there is a meaning of the word "innate" in which "innate ideas" are absurd. He himself uses it in the same meaning as natural, in agreement with nature, instinctive, in opposition to what is due to art, culture and discipline. (Cf. also *The Moralists,* III, 2). The question is surely not, he says, of the moment at which one body comes out of another, or of the point of time at which our ideas are formed, but of whether the human constitution is such that certain ideas arise naturally in the course of development. And one cannot assert that we get the ideas of love and justice only from experience and the catechism. Then there would also have to be a catechism that teaches the bird to fly and to build its nest and teaches man and woman to find each other! — It is, says Shaftesbury, who here agrees with Grotius, Cumberland and Leibniz, an instinct that binds the individual to the species, an instinct that is just as natural as the instincts of propagation and of care for offspring. Man cannot and never could subsist outside society. Wrongly is the state of nature set in opposition to the state of society. On closer investigation of the development of the human species one finds a whole series of different states of nature, without in any of them social life and the drives that bear it falling away entirely. — By this consideration (which is found in The Moralists II, 4) Shaftesbury places himself in opposition to the contract theory of natural law, which let society arise through the association of independent individuals. He goes back to the obscure standpoint of instinct, where individual and society do not yet stand in opposition to each other, and he sees at the same time that the state of nature and the state of culture are relative concepts. It is one of the most important germs of thought to be found in him. But the spiritual atmosphere of the eighteenth century was not favorable to its further development.

[398] Although Shaftesbury lays great weight on instinct, he does not overlook the significance of thinking. It makes a reflection on our inner states possible, and thereby these become objects of feeling and judgment. It is Locke's inner experience to which he here appeals. Through reflection on what stirs involuntarily in us, peculiar feelings arise: of esteem or contempt, of admiration for the noble and right, of aversion to the ignoble and perverse in ways of thinking and acting, feelings that are akin to aesthetic pleasure and displeasure, but are distinguished from these by their active character, prompting to action. Such a feeling Shaftesbury calls a "reflected feeling" (reflex affection) or "moral sense" (moral sense). As springing from natural instincts it is itself natural, original

(Inquiry concerning Virtue and Merit).

There is, to be sure, a "cool philosophy" which teaches that there is no natural faith or justice, no real virtue, because self-love and striving for power are the only forces at work. Shaftesbury thinks that this doctrine perhaps springs from an aversion to letting oneself be led by nature to serve purposes that lie beyond oneself. He himself teaches that all beings strive after happiness. But, he says, it makes a great difference whether one finds happiness in striving after common purposes, or whether one narrows one's interest to one's own advantage, perhaps even to one's own self-preservation. There is no absolute opposition between the selfish and the sympathetic feelings: for on the one hand love and friendship bring self-satisfaction, since by a kind of reflection one comes to share in the happiness one prepares for others, and on the other hand our conditions of life are so closely bound up with those of others that we cease to provide for ourselves when we cease to provide for the common goods. The point is to bring harmony among the different promptings that stir in the mind. That man is the architect of his own happiness who acquires for himself an inner foundation of order, peace and concord. Happiness rests on what is inward, not on what is outward. The harmony and beauty of the feelings condition the forms and manners of true social life: what satisfies the demands of social life is also what brings harmony into the individual's inner being. And therefore — writes Shaftesbury in his "Letters to a young man at the University" — seek the beautiful in everything, even in the smallest things! — He intimates the ideas of *[399]* a modern evolutionist aesthetics when he says (in his *Miscellaneaus Reflections)*: "The forms that are ugly produce at the same time discomfort and disease, and the forms and proportions that are beautiful yield advantage by making fit to act and to be of use."

The sense of order and harmony in which, according to Shaftesbury, moral feeling consists is directed not only to human society, but turns to the whole universe and thereby becomes religious reverence. The order of nature stands as admirable. Misfortunes and evils exist only for the limited consideration: our finite thought must often take for imperfect what would prove to be a perfection if we could see it from the standpoint of the whole. The universe meets no outer resistance. It follows its own inner harmonious order, which has its ground in the thought of God. Ethics thus leads over into religion, and Shaftesbury speaks up for "a noble theism, which conceives God as the all-loving and protecting one and therefore as ideal model." On the other hand, it would be to begin at the wrong end if one would (with Locke) build ethics on religion. Disinterestedness would thereby fall away. Virtue is its own reward. What reward could better suit it? —

With Francis Hutcheson (1694—1747), a Scot who was born in the north of Ireland

and, after having directed a private academy in Dublin, worked as professor of moral philosophy in Glasgow, Shaftesbury's ideas received a more systematic form and were at the same time spread in wider circles. His *Inquiry into the Ideas af Beauty and Virtue* (1725) gives an investigation of aesthetic and ethical feeling which contains many interesting observations and gives a closer determination of several points that in Shaftesbury had been left more indeterminate. After his death his *System of Moral Philosophy* was published (1755). It is his endeavor to ground ethics on observation of real human nature. He finds there, besides the selfish instincts, also an immediate need to help and gladden
[400] others. And he finds on the other hand an equally immediate feeling of joy and approval in the face of actions that spring from such a need to help and gladden. Reason is only the power of finding means to given purposes; it enters the service of sympathy and cannot be dispensed with, since that immediate feeling otherwise works blindly and shortsightedly; but it is not reason alone that leads to the valuation of human actions. Just as little as the moral sense can do without reason, just as little can it do without experience, which alone can teach us the effects or tendencies of actions. The moral sense works only when it is confronted with observations of human actions and their effects. When Hutcheson observes how the moral sense works on the foundation of such observations, he finds that, with equal degrees of happiness that can be expected to result, the action is valued highest whose effects in happiness extend to the greatest number of persons, — yet so that the dignity or moral significance of the persons can make up for the number, — and that when the number of persons who benefit by the actions' effects is the same, the action is valued highest that causes the highest degree of happiness. He sums up these observations in the principle that has so often been repeated since: that action is best which causes the highest happiness for the greatest number (the greatest happiness for the greatest number) (Inquiry, 2. ed.p. 177.). — Among the many interesting investigations to be found in Hutcheson's writings, there is especially to be singled out his discussion of how the moral judgment on an action is determined by the relation between the feeling of humanity (benevolence) that is its motive, and the power and skill (ability) found in the agent. He assumes that the action's moral goodness stands in direct proportion to the feeling of humanity to which the action testifies, and in inverse proportion to the power the agent is presupposed to possess. (Inquiry. 2. ed. p. 194 f).

The moral sense cannot be explained by experience, for it manifests itself quite immediately and instinctively, — nor by education, custom and association of ideas, for in that way — Hutcheson holds (System, I, p. 32. 57) — no entirely new senses or ideas

can arise. He grants that there is to that extent something mysterious in the moral sense. *[401]* He assumes it to be originally implanted by God and sees a proof of the Creator's wisdom in the fact that the moral sense approves only actions that serve the welfare of others, or that serve one's own welfare in a way that agrees with the welfare of others. But it works even in those who do not believe in any God. The ethical is for Hutcheson as independent of theological ideas as of egoistic ulterior motives, although he grants that where the moral sense is weakened, the command of authority may be the only means that can take up the fight with the affects. — Not always, moreover, in Hutcheson's conception, does the moral sense work as an immediate instinct. There can develop out of it a feeling of obligation, inasmuch as — even where the immediate prompting does not lead to doing or forbearing — a consciousness can assert itself that by not making a particular decision one would come into conflict with what love of humanity demands, and thereby suffer harm to one's inner peace (serenity), perhaps also bring upon oneself other inner or outer inconveniences. The feeling of duty is thus, according to Hutcheson, a derived and a vicarious feeling. —

In particular ethical questions Hutcheson applies the principle of welfare as a standard. There are found in his System of Moral Philosophy a multitude of acute, humane and liberal remarks on particular moral, social and political matters. — An excellent biography and characterization of Hutcheson and his significance has been given by W. R. Scott in his book *Francis Hutcheson* (1900).

Joseph Butler, who after having studied at a Dissenting academy later went over to the established church and died as a bishop in 1752, an energetic thinker and able observer, gave Shaftesbury's doctrine of the moral sense a still more theological form than Hutcheson had done. Connected with this was the fact that Butler had an especially sharp eye for the dark sides of life and felt himself very far from Shaftesbury's enthusiastic optimism.

His conception of ethics he developed in his religious discourses *(Sermons* 1726). He does indeed, like Shaftesbury and Hutsheson, set the ethical in close connection with *[402]* the single human being's immediate relation to the species, but he stresses still more strongly than they the quite immediate inner compulsion that makes itself known in the moral sense, which he would rather call conscience. We are too shortsighted to find out what in the particular case will bring happiness. Over happiness the Lord of the world disposes. But we are so constituted that, quite apart from the outcome, we condemn falsehood, violence and injustice and approve love and benevolence. For the principle of welfare set up by Hutcheson there is no use here; Butler keeps merely to the inner,

subjective side of the ethical. In Shaftesbury he blames especially that the authority of conscience, the inner moral feeling, over against all other elements and promptings in the soul is not brought out in his doctrine. Conscience is of a higher nature than all other powers. It is destined to rule over the world. By a fine psychological analysis Butler shows how such an immediate satisfaction as moral feeling gives is different from the feeling of pleasure attained by egoistic calculation: in the immediate instincts we do not consciously make ourselves our purpose; on the other hand, egoism proper presupposes experience of the feelings of pleasure won by immediate devotion and the conscious setting up of them as purposes; but this reflection and calculation can then precisely hinder the attainment of full satisfaction.

Although according to Butler there is an immediate satisfaction in following the commands of conscience, he nevertheless has so sharp an eye for the oppositions in human nature that he cannot rest content with Shaftesbury's joy in the harmony of our own nature. If it is not specially guaranteed to us that in the end we shall quite personally have our need of happiness satisfied by the good and the right, this will not hold us fast: "Though virtue or moral perfection consists in feeling for and striving after what is in and for itself good and right, yet we shall nevertheless, *when the enthusiasm is gone* (when we sit down in a cool hour), be unable to justify either this or any other pursuit to ourselves before we are convinced that it will serve our happiness or at least
[403] not conflict with it" (Sermon XI). The "cool" hours Shaftesbury esteemed as little as the "cool" philosophy. For Butler it is in the last resort only the prospect religion gives of another life that can help us over such hours. Yet he knows also another way by which ethics leads over into religion. The need to feel love and admiration that lies in our nature finds in God the highest possible object. Religion widens our feeling and alters its direction, but creates no entirely new kind of feeling; our nature cannot be transformed. So for Butler the opposition between self-interest and admiring devotion withdraws from the ethical relation into the religious one, without receiving any definitive solution.

Butler's opposition to Shaftesbury is the opposition of the pessimist to the optimist. The eighteenth century began decidedly optimistically. Nature and human life were seen in a bright light, and great hopes of happiness and progress were cherished. Leibniz, Locke and Shaftesbury expressed this optimism under different forms. By Shaftesbury and the freethinkers natural religion, which built on the consideration of the connection and harmony of nature, was praised in opposition to the obscure mysteries and inhuman dogmas of revealed religion. Both personal mood and theological conviction led Butler to oppose this opposition. In his remarkable writing *Analogy of Religion, Natural and*

Revealed, to the Constitution and Course of Nature (1736) he sought to show that the most important objections urged against revealed religion also strike the faith of natural religion in a providential government in nature. Observation of nature cannot establish that it has a wise, just and good cause. When one is offended at the doctrine of election and damnation, one must not forget that countless germs in nature perish without unfolding, and that it is only few men who come to full moral development in this life! When one is offended at the dogma of the Atonement, one must not forget that in nature the innocent in fact suffer for the guilty! If one takes offense at Christianity, one must also take offense at nature. The mystery is at bottom equally great in both. — What helps Butler himself over this difficulty — as regards both nature and revelation — is the *[404]* thought that both in the natural and in the supernatural world we are only parts of an immeasurable whole which we cannot survey from our place.

By this consideration Butler strikes the smooth, veiling optimism that overlooks the real conditions. There is just as much a thoughtless cult of nature as a thoughtless faith in revelation, and the dogmatism of natural religion can become no less questionable than that of positive religion. But Butler's proposition can be turned round and then comes to declare that the same contradictions are found in Christianity as in nature: the question then becomes what solution of the riddle of existence has been reached in revelation, when it becomes only a kind of doubling or reflection of nature and its riddles? The spear with which Butler wounded his opponents wounds himself.

On the other hand, the author of the "Fable of the Bees", Bernard De Mandeville (born in Holland of a French family, lived as a physician in London and died in 1733), seems to have avoided wounding himself. The Fable of the Bees is a poem that appeared in 1708 and was sold as a broadsheet in the streets of London. It depicts a society of bees in a most flourishing and mighty condition. All worked zealously to satisfy the common necessities. Restlessness, discontent, voluptuousness, vanity, fraud prevailed — but all of them helped toward the common good. Precisely the viciousness in the single parts of the state made it as a whole a paradise, just as the single discords in a piece of music contribute to the harmony of the whole. Even the poor lived better than the rich had formerly lived. But then there were some who began to cry: Away with corruption! Good God, if only we were honest! And the cry won assent. The most zealous cheats were also the most zealous to join in the cry for honesty. The gods heard the cry. Hypocrisy vanished. Luxury ceased. Wars of conquest fell away. Priestly rule and officialdom were curtailed. The poor were provided for by the public. People contented themselves with the products of their homeland and needed no import of costly goods. Seafaring

therefore ceased. The population declined — and in the end the whole swarm withdrew
[405] into a hollow tree. Contentment and honesty had been attained — but the splendor and the power were gone. — The moral is clear enough: the contentment and morality of the individual cannot coexist with the culture of society.

In a later edition *(The Fable of the Bees, or Private Vices Publick Benefits.* 6th. ed. London 1732) Mandeville added notes, partly in dialogue form, in which he sharply develops the difference between his own views and Shaftesbury's. — Two systems, he says, could not be more contradictory than Lord Shaftesbury's and mine. It is man's egoistic interests, his need of food and drink, his ambition, envy, love of enjoyment, laziness or impatience, that lead to labor, culture and social life. Contentment and virtue, on the other hand, lead one to be satisfied with what one has. An original social impulse there is none. Without hypocrisy society cannot subsist: only think what the consequence would be if we were to give honest expression to all our ideas and moods! The task of the true statesman is this: to make society mighty by getting men's egoistic interests to work together for the general good; but it would be foolish to root out the necessary evils. The current virtues were invented by ambitious politicians, who could more easily secure their dominion and power when they could get men to become self-sacrificing and obedient; only thereby did it become possible to rule over great masses. But with increasing civilization the connection between the individual's vice and constant discontent and the good of the whole shows itself. Philanthropy, which would abolish poverty, has a harmful effect. In a special treatise *(On Charity and Charity Schools)* Mandeville insists how dangerous it is to give the poor more enlightenment than befits their station: how then will one get the simplest and least attractive kinds of work done? — One must make one's choice between nature and culture, between morality and social progress. The dirt in the streets of London cannot vanish without the magnificent traffic vanishing too. Evils and goods are inseparably bound up with
[406] each other in the course of development.

Mandeville thus dissolves the harmony of Shaftesbury and Leibniz. He makes the most forceful plea that comes from the pessimistic side in the eighteenth century and has real merit in the treatment of the problem of culture. In order to come to terms with the theologians, he declares that since the author of nature is incomprehensible to us, we have no right to call it cruel. Indeed, he even believes he is working in the interest of faith in revelation and of Christian morality by establishing the vanity of the world and the insufficiency of human reason as well as of pagan virtue — in opposition to Shaftesbury, who favored deism and set pagan virtue above Christian (Fable of the Bee's.

6th. ed. II, p. 431 f.). This theological finale seems rather stuck on — but it had its effect. The clergy came out far more sharply against Shaftesbury than against Mandeville.

An interesting monograph on Mandeville is available by P. Sakmann *(Bernard de Mandeville und die Bienenfabel-Controverse.* Freiburg in B. 1897).

b. Freethinkers

Under a great many different forms and in a great many different directions, thought had, ever since the dawn of the Renaissance, tried its powers on the great problems of life. It was no wonder, then, that in wider circles too the conviction could arise that a view of life and a conduct of life needed no authority as a support, but that the insight into the laws of nature and of human life that man can procure for himself by the powers of his own mind can suffice as a guiding star. Early on, therefore, there appeared, in Italy from the fifteenth century onward, in the more northern countries in the following centuries, men for whom natural religion was enough, or who perhaps did not even need it in its usual form. From the sixteenth century onward the name *Deism* became common for those who did not accept a revelation in the Church's sense of the word; later it was stressed as the special mark of this tendency that it denied the miraculous intervention of the Deity in the course of things. Often no great distinction was made between deists and atheists. Every new conception in the philosophy of religion people were for the *[407]* time being inclined to call atheism. From the beginning of the eighteenth century the word *Freethinker* appears in English literature. It denotes thinking freed from authority. The word can — like the word deist or atheist — denote the most diverse views; none of these expressions therefore has any philosophical value. But it is a sign of the times that the so-called freethinkers become more numerous and more self-conscious from the beginning of the new century. (Cf. FR. EILSCHOW: *Philosophiske Breve.* Copenhagen 1748. p. 119 ff.). The deistic or freethinking literature is for the most part of very slight value in a philosophical respect, which does not diminish its great significance in a cultural-historical respect, as a symptom of the spread of scientific and philosophical ideas and in general of what is stirring in wider circles. But to follow such symptoms in detail is the business of the history of literature more than of the history of philosophy. As regards the English deists, Leslie Stephen *(History of the English Thought in the 18th Century,* Chap. 3) remarks, no doubt rightly, that on the whole they were inferior to their opponents, while the reverse was the case in France. The explanation is presumably to be sought partly in the fact that the greater resistance the Catholic Church could offer provoked the attackers and set greater forces in motion, partly in the fact that in France

it was not a purely abstract dogmatic discussion that was carried on, but the social and political opposition was closely bound up with the religious and theoretical. For both reasons there arose an enthusiasm that English deism lacked.

Only a single man among those usually called the English deists has made contributions of value to the history of thought. That man is John Toland. He was born in 1670 in the north of Ireland and was brought up in the Catholic religion; that is all that is known of his childhood. In the preface to *Christianity not Mysterious* (1696) he says: "After I had been brought up from the very cradle in the grossest superstition and idolatry, it has pleased God to make my own reason, and men who used their reason, the instruments of my conversion." He found no difference between the authority of the Pope
[408] and the authority that the Protestant clergy laid claim to. For him it was now a matter, just as at the beginning of Christianity, of choosing between the clear teaching of Christ and the intricate doctrines of the scribes. What Toland took to be the clear teaching of Christ he has set out in the work named. Locke's friend, the physicist Molyneux in Dublin, at first speaks of Toland with respect in his letters to Locke; he calls him (letter of 6 April 1697) "an honest freethinker and a learned man" (a candid free-thinker and a good scholar). (This is, as far as I know, the first time the word freethinker occurs). But later he was repelled by Toland's vanity and his inclination to make propaganda in coffee-houses and other public places. No less offended were the clergy and the magistrates. Such zealous preaching was directed against him that a pious man (according to what Molyneux relates) would no longer go to church, because more was said there about a certain Toland than about the Lord Jesus. The book was burned, and an Irish member of parliament wanted to have the author burned as well.

In the period that followed, Toland occupied himself with works of literary history and of politics. He came forward as a defender of the Protestant succession and thereby came into favor at the court in Hanover, where he met Leibniz. With Leibniz's pupil, too, the gifted Queen Sophia Charlotte of Prussia (a Hanoverian princess), Toland came into high favor. It interested the Queen to hear discussions between the Berlin theologians, Leibniz and Toland. In the preface to his *Letters to Serena* (1704) Toland praised woman's capacity for intellectual work on the strength of the experiences he had had here. Serena is Sophia Charlotte. — Through his activity as a publicist Toland came into such good circumstances that he could buy himself a house in the country. He could now satisfy his feeling for nature and live "free from care as well as from ambition, always with a book in his hand or in his head". Later things went downhill for him again, and his last years were weighed down by poverty and illness. He died in a village near London (1722).

We come to know Toland's relation to Christianity from the work mentioned above, *[409]* in which he seeks to prove "that there is nothing in the Gospel either *contrary to* or *above* reason, and that the Christian doctrine cannot rightly be called a mystery." By reason he understands the mental faculty that leads us to certainty by comparing the doubtful and obscure with what we clearly know. This faculty we must make use of in religion too; for only by it, not by the authority of the Church, can we convince ourselves of the divinity of Scripture. The dogma of creation or the stories of miracles do not conflict with reason. What we do not see through completely is not on that account at once a mystery. In the New Testament the word mystery — Toland maintains — is never used of what is in and of itself incomprehensible to us, but of what had hitherto been hidden, but had been revealed and can therefore now be grasped. Why, moreover, should God require that we believe what we cannot grasp? Our diligence is not increased by it, and there is reason enough anyway to apply ourselves to humility! It was only priests and philosophers who had made Christianity a mystery. —

The later deists, Collins, Tindal, Morgan etc., went further than Toland, in that they explained away or denied the miracles. The dispute was conducted, on their side and on their opponents' alike, with great vehemence, but in an unscientific manner. Both parties lacked historical sense. What the orthodox explained as arbitrary miracles the deists explained as equally arbitrary inventions of cunning priests. —

Toland's most important work is his "Letters to Serena". It is directed against the Cartesian school's concept of matter, which was still held fast — though with misgivings — by Spinoza. Toland maintains that motion must be regarded as just as essential and original a property of matter as extension and solidity. If one conceives matter as passive and at rest, one must assume a supernatural cause that sets it going. But on closer investigation we find motion everywhere in nature; it is only to the senses that there seem at particular points to be rest and standstill; to the conception of thought there is an inner activity (internal Energy, Autokinesis) in everything. Every motion must be explained from another motion, and this holds also of the motions that, like those *[410]* of animals, seem to arise of themselves without a cause. Consciousness does not mark any break in the succession of motions. God must be assumed to have created matter as active. The order and harmony in the universe testify that he guides the motions of matter; without such an assumption the origin of organic life in particular will be inexplicable.

This line of thought of Toland's came to have an influence on French materialism around the middle of the century. It was not entirely new; Huyghens and Leibniz had

already corrected the Cartesian philosophy of nature in a similar direction and in a far more profound way. But Toland carried the doctrine of the eternity of motion out into far wider circles.

Toland developed his concluding ideas in *Pantheisticum* (1720). He depicts here, in a fictitious account, how there are, here and there throughout Europe, circles of unprejudiced men who gather for frugal social meetings, at which they discuss various important questions with calm and good sense. They are called pantheists (a word that seems to derive from Toland himself), because they hold that God is one with the force that works, creates and orders in the whole universe, and can therefore be distinguished from the world only conceptually (ratione). They are entirely at odds with Epicurus and are not materialists. All individual beings, each after its kind, they take to have sprung from motion and thought, thought, which is the force and harmony of the universe. They have a liturgy of their own, and they practice an unconditional tolerance. Their endeavors concern the welfare of the state and of humanity without regard to parties. Their doctrine they keep hidden, as an esoteric doctrine (accessible only to the initiated); for there are erring and obstinate people, whom one must handle as nurses handle their stammering little ones. — Toland wrote this work in a mood similar to the one from which the Freemasons' movement, which Hettner has called the inner mission of English deism, sprang at about the same time. The new ways of looking at things need cultivation in narrower circles, and in part new symbols as well. Only, the age of criticism and enlightenment was not fitted for the founding of such societies or for
[411] the creation of viable symbols. Free individualism and critical examination had (and have) a long work still to do, and neither societies nor symbols are created by conscious arbitrariness.

3. Newton and His Significance for Philosophy

Just as at the transition from the Renaissance to the seventeenth century we had to insert an examination of the founding of modern natural science, without which the subsequent philosophical development could not be understood, so here too, at the beginning of the eighteenth century, we must bring forward a great figure from the history of natural science, perhaps the greatest it presents. In Newton not only is the ideal of inquiry that the new science had set up brought to completion, so that his work stands as a model of scientific method as a whole; but his discoveries and results in natural science themselves gain great influence on the subsequent philosophical development, not only

in his own country but also in France and Germany. And finally Newton himself has his own peculiar philosophy, which is of interest for the history of philosophy both as the expression of a great natural scientist's world-view and by the influence it exerted.

Isaac Newton was born at Woolsthorpe near Nottingham on 25 December 1642. Already as a child he had an interest in mechanics and constructed machines, mills and clocks. He was at first destined to be a farmer, but when, as a shepherd boy, he let cows and sheep look after themselves while he became absorbed in his reflections, he got his wish to be allowed to study. At the University of Cambridge he soon went so far by studying on his own that he surpassed his teachers. Very early he got the ideas for his great scientific discoveries in the domains of mathematics, optics and astronomy alike. Like several others of the greatest minds in the history of science, he kept ideas conceived early through long periods of silence and quiet work, until he could give them full execution and demonstrability. The idea of the differential calculus he had, *[412]* it seems, before Leibniz, although the latter published his method first; together with the proof of the composition of light and of gravity as a universal law, it makes up his great discoveries, on which it is not our task here to dwell more closely. His chief works are *Principia philosophiæ naturalis mathematica (1687)* and *Opticks* (1704). After he had been professor at Cambridge for a number of years, he lived in his later years in London, where he held a high position at the Mint. He died at a great age in 1727.

The great philosophical significance of Newton's discovery of the law of gravitation was first and foremost that it was now established that the physical laws that hold at the surface of the earth hold also in the universe, as far as we are able to survey it. Newton's fundamental thought, which according to the legend dawned on him in a happy moment, was to extend what we see taking place on the earth beyond its domain and to test whether the inferences we can draw in this way agree with the observations. Gravity is greater in the valleys than on the mountain heights; but there is no reason why it should cease because we thought of ourselves on the moon or on one of the planets. Might not the motion of the moon be conceived as a motion of falling, so that in the direction of its motion it constantly deviates from the line it would follow by virtue of the law of inertia just as much as the law of falling bodies requires according to the distance at which the moon is from the earth? What guided his thought in this conjecture he has formulated in his chief work as a special rule of inquiry (regula philosophandi): "such properties as cannot be intensified or diminished, and which belong to all bodies of which one can have experience, ought to be taken as properties of all bodies whatever … For one ought surely not lightly to invent dreams that conflict with the connection

of experiments (tenor experimentorum), and one ought not to abandon the analogy of nature, since nature is wont to be simple and always to agree with itself … It is by virtue of this rule that we learn that all bodies gravitate mutually toward one another." — What Newton has set up in this rule of inquiry is the principle on which the whole of modern
[413] science and philosophy rest. We have met it in Copernicus, Bruno, Galilei and Leibniz under various forms and in various applications. In Newton it finds its most scientific application.

It was a magnificent widening of view. It now became clear that the fixed, law-governed order of nature reigns not only here on earth but in the whole universe. The mechanical view of nature founded by Kepler, Galilei and Descartes thereby became both more certain and more comprehensive. The world now came to stand as a great machine. People had, no doubt, surmised that it was one; but only now had they found what held it together. And only now was a scientific proof given of the Copernican hypothesis. If the heavenly bodies follow the law of gravity, the larger globe must necessarily stand at a center in relation to the smaller ones, and a small globe like the earth could not be the absolute center. — A great confidence in the method of natural science, a self-esteem on behalf of the human mind for having solved such a task, was bound naturally to develop even in those who could not see at first hand how the task had been solved.

The very method Newton employed was quite as important as his results. The ingenious idea that, by virtue of the principle of actuality, extends what has been established for narrower circles of experience to wider circles, is only the first step. The next step is the strict deduction of the consequences that the idea set up contains. And the third step is the proof that what thus follows consistently from the idea agrees with experience. An unsurpassed cooperation of deduction and induction makes Newton's chief work a model for all scientific inquiry. First he derived the consequences of the assumption that all the planets move according to the same laws as falling bodies at the surface of the earth, and next he showed that these consequences were also actually confirmed by experience. From this he then concluded at last that it is the same force that acts in both places. He goes from the phenomena to the law, and from the law to the force (like Leibniz, see above).

This force he calls attraction, adding expressly that by this designation he does not
[414] wish to establish anything about the nature of the force, but merely to express that it acts so as to bring one body nearer to a larger body. At bottom he does not assume any force acting at a distance. For both in the "Principia" (2, 11) and in the "Opticks" (Query 18—24) he declares that the centripetal force should rather be called impulse (impulsus) instead

of attraction (attractio), since the most reasonable hypothesis would be the assumption of an ethereal matter that pervades everything and is thinner in the neighborhood of the globes than farther away; by this assumption perhaps not only gravity but also light and heat could be explained. But his chief standpoint is nevertheless the one stated in the Scholium generale (at the end of the third Book of the Principia): "The ground of the properties of gravity I have not yet been able to derive from the phenomena, and I do not occupy myself with inventing hypotheses". Pemberton *(A View of Sir Newton's Philosophy,* London 1728. p. 407) relates that Newton complained to him of having been misunderstood, as if by the word attraction he had meant to give an explanation, although it was only his intention to draw attention to a force in nature whose cause and mode of action it must be the business of further inquiry to seek. Yet he (and still more his disciples) was inclined to regard it as an immediate effect of God. — Newton's doctrine does not stand in the irreconcilable opposition to Cartesian physics that the Cartesians, Leibniz and also many of Newton's friends supposed. At bottom he agrees with Descartes in his general points of view, although he corrects his special assumptions by his more perfect method. Quite unjustly was he accused of wanting to reintroduce the "occult qualities" of scholasticism. He has satisfied his own demand: to set up only demonstrable causes (veræ causæ).

But Newton does not enter the history of philosophy only through his results and his method. There appears in him as well, partly as the background of his doctrine in natural science, partly as its consequence, a peculiar world-view, which it will be of interest to bring forward.

There is a remarkable connection between Newton's religious conception and his mathematical doctrine of nature, a connection that is brought about by his doctrine *[415]* of space. (The chief passages are the Scholium after the Definitions in the "Principia" and Query 31 in the "Opticks"). Wrongly, says Newton, does the popular conception (vulgus) assume that the sensible times, spaces, places and motions are the true ones. It determines them by their relation to sensible things. But it is by no means certain that there is any body whatever that is at absolute rest, so that we could use it as a starting point for the determination of place and for distinguishing between real and apparent motion. For there to be real motion (such as, e.g., the principle of inertia presupposes), there must be an absolute space and an absolute time, which are not determined by relation to anything external whatever (sine relatione ad externum qvodvis). There must be absolute, immovable places in order that absolute determination of place can take place; but such places sensation cannot show us. The absolute places (loca primaria)

are places both for themselves and for all other things. True space and true time are mathematical space and time, which are not objects of sensation. — Strangely enough, we find in the great natural scientist the dogmatic tendency to leap from the phenomenal and relative to the absolute. He postulates a space in itself (a kind of locus sui), just as Descartes, Spinoza and Leibniz postulate a cause of itself (causa sui). He does not let the mathematical way of considering things be merely a point of view that can guide us in calculating the relations of phenomena, but sets it up as the *true* one in opposition to the sensible or vulgar one, which does not get beyond the relative. A mathematical abstraction is made into the true reality. In practice (in rebus humanis) we may keep to sensible space and forget that sensation can show us no absolute places; but as thinkers (in rebus philosophicis) we must disregard the senses!

Some wavering can, to be sure, be shown in Newton's statements on this point[79]), but the assumption of absolute, ideal space as a reality is bound up with what is innermost in Newton's world-view. Space is for him not an empty form, but God's organ, through
[416] which he acts in the world as omnipresent and at the same time immediately perceives the states of things. It is a "boundless and uniform sensorium". — Extension, then, is for Newton (as for Henry More) not the decisive distinguishing mark of material things. Only that is material which besides extension also has solidity. In his conception of space Newton stands, on the side of philosophy of nature, closest to Gassendi by his own account, while as regards its application in the philosophy of religion he is influenced by Henry More. — From Gassendi he also took over the atomic theory, which thereby came to play an important role in the natural science of the following period, especially in chemistry (since the founder of chemical atomism, John Dalton, had his concept of the atom from Newton). —

As for the existence of God itself, Newton proved it from the purposive and harmonious arrangement of the world, and he thereby struck up a theme that was varied endlessly throughout the whole eighteenth century. — Whence comes it that nature does nothing to no purpose, but always follows the simplest ways? Whence comes all the order and beauty we see in the world? The motion of the planets about the sun in orbits that are concentric with the sun and lie practically in the same plane, — in short, the whole wonderful framework of our solar system (elegantissima compages) cannot be explained according to mechanical laws, and cannot have developed in a natural way. Only in a supernatural way can the masses, distances, velocities and densities of the various globes have been fitted to one another, and only by a supernatural force can it be explained that the planets move in circles instead of following gravity and plunging

into the sun. And to this is added the remarkable structure, organs and instincts of animals! — (Besides in the Scholium generale in the "Principia" and Queries 28 and 31 in the "Opticks", Newton discusses this subject in his letters to Bentley. Cf. BREWSTER: *Life of Newton,* II. p. 125 ff., and ROSENBERGER: *Isaac Newton,* p. 267—273).

Yet the world-machine was not quite perfect in Newton's opinion. The attempts to prove God's existence from the purposiveness of nature easily come to work against themselves: for if nature is created perfect, it can surely go its way by the forces that are *[417]* laid down in it, and God's continued activity is unnecessary. This awkwardness does not appear in Newton's theology, since he assumes that through the mutual influence of comets and planets irregularities have arisen in the system that would make a divine regulation necessary. — This point was sharply attacked by LEIBNIZ, who mockingly compared Newton's system of the world to a clock that from time to time has to go to the clockmaker. Newton's disciple CLARKE defended his doctrine by maintaining that it surely would not do to make God superfluous. So: one first wants to prove God's existence by the ingenuity of the mechanism of nature — but then becomes afraid that, if the ingenuity is too great, God's existence will be refuted by it, — and this although the ingenuity would surely, properly speaking, have to be infinitely great in order to furnish a proof of the existence of a God! — This part of Newton's thinking is untenable. Fortunately, the great natural scientist's own spirit and method led his successors beyond his unfortunate philosophy. KANT and LAPLACE showed the possibility that the solar system may have developed in a natural way into its present form, and the great French mathematicians, especially Lagrange and Laplace, established that the irregularities were periodic and compensated themselves. There lay in Newton's doctrine and method a deeper philosophy than he himself got out of it. Kant and Laplace were better Newtonians than Clarke. — How dangerous an appeal to the supernatural can become in a scientific respect can be seen from a cavalier remark of Newton's disciple MACLAURIN, who has given an excellent exposition of his doctrine. Apropos of the upheavals that seem to have taken place on the earth in earlier times, he says: "Since God has formed the universe in dependence on himself, so that it must be renewed from time to time by him, it does not seem to be a very important question whether those great changes have been produced by the action of external circumstances or by the same forces that formed things at the first". All doctrine of evolution is thereby declared superfluous, a whole series of scientific tasks is struck out. Nothing lay further from the spirit in which Newton's real doctrine was thought out. *[418]*

4. George Berkeley

a. Biography and Characterization

We put off for a while following the ideas of Locke, Newton and the deists on their wanderings through Europe, and, having shown the source of these ideas, we turn to a thinker who, partly continuing, partly opposing, takes them up for a new philosophical working-through, which leads him far away from the line of thought in which popular consciousness moves and alone can move. What is peculiar to George Berkeley, one of the finest and clearest minds the history of philosophy knows, is, however, that he pleads the cause of immediate, practical consciousness against scientific abstractions and speculations, while at the same time by his own results he comes to stand in the sharpest opposition to this consciousness and to offend it in the sharpest way by his paradoxes. In other respects too Berkeley unites in his way of thinking opposites that might seem irreconcilable. Sharp criticism and childlike religious faith, the love of inquiry and glowing missionary zeal have perhaps never been joined in so intimate and peculiar a way as in Berkeley.

George Berkeley was born at Dysert in County Kilkenny in Ireland on 12 March 1685. He belonged to an English family that seems to have come to Ireland immediately after the Restoration and was a collateral branch of the noble family of Berkeley. George Berkeley's friend Swift is said to have presented him to the Earl of Berkeley with the words: "My lord, here is a young man of your family. I can assure your lordship that it is a far greater honor for you to be related to him than for him to be related to you". This much is certain, that the young man early showed brilliant abilities. He studied in Dublin, where the writings of Boyle, Newton and Locke were at that time taken as the basis at the University. Very early the fundamental thought of his later philosophy dawned on him. In his notebook *(Common-place Book)* from his young years, which Fraser has published in his excellent biography of Berkeley *(Life and Letters of George*
[419] *Berkeley,* Oxford 1871) and later in the first volume of *The Works of George Berkeley* (Oxford 1901), we see his ideas in their successive unfolding. It was early his conviction that when science and philosophy are freed from meaningless abstractions and foggy words, the conflict between knowledge and faith will also be done away with. He works to lead men back to immediate experience and intuition. He wants to put an end to the scholasticism that still remains, and to the one that has stepped into the old one's place. His central thought is the proof that in the literal sense of the word there are no abstract ideas. With this thought his ingenious discoveries as well as his personal outlook on life

are very closely bound up.

Berkeley's first significant work was *New Theory of Vision* (1709); he showed here how the sense of sight and the sense of touch work together in our perception of space, and that space "in itself" was an abstraction to which no immediate sensation corresponds. The following year his chief philosophical work appeared, *Principles of Knowledge,* in which he first criticizes the old theory of abstraction and then seeks to establish that the concept of matter is an invalid abstraction, and that we have no other immediate objects of knowledge than our own sense-sensations. How from here he makes the transition directly to his theological notions will be shown later. A few years after these works appeared he went to London, moved in literary circles and began to combat the freethinkers in articles in Steele's periodical "The Guardian". His polemic has a childlike and naive character. So, for example, when (in *Remarks on "Collins' Discourse of Free-thinking")* he wonders that the freethinkers do not seek in sensual enjoyments a substitute for supernatural blessedness, but for the most part live a retired life in their studies! The freethinkers — he thinks — proclaim publicly that they have less reason for virtue than others. One knows the strength of passion but poorly if one thinks that the beauty of virtue is a sufficient counterweight in the hour of temptation! — After publishing a brilliant popular exposition of his philosophical ideas in the form of dialogue *(Dialogues between Hylas and Philonous,* 1713), he spent several years traveling in France *[420]* and Italy. His travel diary shows us his open eye and his great feeling for nature, history and social conditions. When he came home, he took up a clerical office in Ireland, but did not have peace for long. He despaired of being able to carry out his plans of leading the men of old Europe back to nature and simplicity in thought and life. Europe seemed to him decrepit. From the new virgin land on the other side of the Atlantic he expected a new golden age, for faith, science, art and life alike. With this was joined zeal for spreading Christianity among the heathen and a romantic feeling for free nature. In a poem ("On the Plan of Transplanting Arts and Sciences to America") he has expressed these thoughts. He formed the plan of founding a college in the Bermudas for the training of missionaries for America. When after long exertions he had obtained a promise of state support for this undertaking, he settled on Rhode-Island to direct the preparations for the establishment of the college. But it turned into a great disappointment. The promise was not kept, and after three years' stay in America he had to return (1731). He never got over the shattering of this great hope. His view of life was darker from this time on. — Memories of the beautiful natural surroundings in which he lived on Rhode-Island are found again in the rich scenery of the dialogue *Alciphron* (1732), in

which he resumed his polemic against the freethinkers, besides seeking anew to give a clear and popular exposition of his philosophical ideas. In this work he expresses an idea, similar to the one his friend Butler developed at length a few years later in his "Analogy", of the difficulties common to natural and revealed religion. He directs the polemic against two types of freethinkers, those of Mandeville's kind as well as those of Shaftesbury's kind, and he triumphs when he can show how they fall out with each other, Mandeville's representative declaring that natural religion is a half-measure when it is not supported by the supernatural. — Berkeley's dialogues still have no slight interest,
[421] both through the philosophical and theological problems that are discussed in them and through the beautiful language and the on the whole successful form of dialogue.

As Bishop of Cloyne in the south of Ireland (from 1734), Berkeley showed himself equally zealous as a pastor, philanthropist and patriot. He was loved and respected not only by the Protestants but also by the Catholics (who made up almost five-sixths of the population of his diocese). He spoke up for the admission of Catholics to the universities. In his *Querist* (1735—37) he set a multitude of social and Irish-patriotic questions in motion. Even medicine did not lie outside his interest. In "tar-water" he believed he had found a universal remedy against various diseases, which he sought to explain more fully in one of his last works *(Siris* 1744), which is remarkable as well for the mystical-Platonizing direction his thinking takes here. The last works from his hand were treatises on fossils and on earthquakes. For this subjectivist and idealist nothing in the world of reality was indifferent and alien. He spent his last years in Oxford, where he died in 1753.

By the chief purpose Berkeley set himself in his philosophy, his position in the history of thought forms an analogy to Leibniz's. Both want to carry thought beyond the purely mechanical view of nature without, however, denying it. They undertake an examination of the concepts with which Descartes, Spinoza, Newton and, still to a great extent, Locke confidently operate as realities. They take the world-picture constructed by natural science and philosophy not as an absolute reality but as a phenomenon. But whereas Leibniz dissolves the mechanical world-picture chiefly by the objective road, by leading motion back to force and making force one with the monad's subjective impulse toward new states, Berkeley goes chiefly by the subjective road, seeking to show psychologically how our ideas of space and matter — on which, after all, the whole mathematical-physical knowledge of the world rests — come into being. Berkeley continues Locke's investigation of the origin of ideas and discusses the question what we can really claim to know beyond the reality that is given in our own sensations and

ideas. He pursues his consistency with a ruthlessness he would perhaps hardly have *[422]*
possessed, for all his love of truth, if his living religious faith had not stood ready to fill in at once what his philosophical criticism cut away. In the rejection of absolute space and absolute matter to which his philosophy led him, he felt himself nearer to his God, indeed felt himself in immediate contact with him. His intimate and primitive religious feeling could not be content with the clockmaker theology, the external relation between God and the world, that Locke, Newton and most of the theologians and freethinkers of the time professed. Yet it is not as a religious thinker but as a psychologist and critical philosopher that he has his significance. In our exposition we follow him only to the door that leads him into theology. For him his philosophy was itself a crusade aimed at winning back the Holy Land. But it went with him as it went with the crusaders: he came to work for other tasks and purposes than those he had set himself.

b. On Space and Abstract Ideas

In his "Theory of Vision", one of the most ingenious psychological works ever written, Berkeley investigates the psychological nature of the perception of space. He shows that distance and size are not perceived immediately from the first. The perception of distance and size arises in that visual sensations are combined with sensations of the movement or straining of the eyes (sensations Berkeley calls sensations of touch, whereas nowadays they are called sensations of movement) as well as with memories and associations (or, as Berkeley calls them, suggestions) of sensations of touch that we have previously had in immediate relation to the object. The idea of touch that the visual sensation calls up fuses with it to such a degree that afterwards we immediately "see" the object before us at a certain distance and with a certain size. The practical significance of the visual sensation lies in its giving warning of the sensations of touch we would get at closer quarters. But the connection between the two kinds of sensation is due to experience *[423]*
and practice. We "see" distance and size in the same way as we "see" shame or remorse in a man's look. Sight and the sense of touch have nothing in common; only custom joins them in a common result. But to this result — space — no real single sensation corresponds. Space corresponds only to the practiced subjective association between the two kinds of ideas of sense. In and of itself it is an empty word.

One sees how Berkeley takes Locke's demand in earnest: to test the validity of ideas by investigating their origin. Space no longer hovers as an incomprehensible general something above all the senses; its source is led back to a process of association. This is what we know about it. Berkeley himself, however, conceives it also as a divine language

of signs, in which we are trained by way of association.

In his investigation of abstract ideas (in the Introduction to *Principles of Knowledge)* his polemic is directed against Locke as the last important representative of the old theory of abstraction, which ascribed to us a faculty of abstraction in the sense of a faculty of forming ideas that contain only marks common to several objects. From three ideas of a green, a red and a yellow thing we should, e.g., be able to "abstract" the idea of a color in general (neither green, red nor yellow). That ideas of this kind exist in consciousness Berkeley denies. We can of course divide an idea, e.g. represent to ourselves a part of an object without the others. But we cannot form new ideas whose content would be what is "common" to several qualities. We have words that denote this common element, but not ideas of the common element. This does not exclude our being able to form general ideas that hold for a whole group of phenomena; but this comes about in that an idea that is in itself concrete, individual (particular), is used to represent (represent or stand for) all other concrete ideas of the same kind. We think in examples!

In this investigation too Berkeley has made a significant contribution to psychology.
[424] Leaving it to psychology to show what is still lacking in Berkeley for a complete theory of general ideas, we pass on to see what philosophical consequences Berkeley draws from his psychological investigations.

c. Epistemological Consequences

Whereas — says Berkeley (in the *Principles)* — as regards the objects of the inner sense, no one thinks of ascribing existence to them when they are not perceived, it is held that the objects of the outer sense exist whether they are perceived or not. But this is an unprovable and unreasonable assumption. Objects of knowledge, by the nature of the case, exist only in so far as they are known; their being consists in being perceived (their esse is percipi). If one looks more closely at what underlies that assumption, one finds the old theory of abstraction again. For surely it is the most violent abstraction of all to represent to oneself the object without everything that makes it an object for us. We should then have to disregard all our sensations! It is held, to be sure, that it is only the secondary qualities (color, taste, smell etc.) that are to be disregarded, not the primary ones (extension, solidity). By matter is meant something that has only the primary qualities, something that has extension and figure and can move. But it is overlooked that the primary qualities are not perceived without the secondary ones, and that they themselves, just as much as these, exist for us only by means of sensations. If extension, e.g., is to be assumed to exist outside consciousness, one must ask: is it extension as

sight shows it to us, or as touch shows it to us? Perhaps as both together show it to us — but then we have already seen that the idea of it is due only to an association. There cannot exist an extended and moving something that is neither large nor small, far nor near, and moves neither fast nor slowly! Even if there were solid, figured and moving substances outside consciousness, whence would we know it? How can one appeal to the testimony of the senses for something that is not sensed at all? *[425]*

But is not the difference between reality and imagination thereby done away with? This Berkeley most emphatically denies. Real nature, he says, I do not deny; I deny only abstract matter. Everything that is seen, felt, heard or in general perceived by the senses I take to be real, but not matter, this unknown something (somewhat — if indeed it may be termed somewhat) without sensible properties and inaccessible to all sensation and comprehension. The difference between reality and imagination rests on the difference between sense-sensation and memory or fantasy. Sense-sensations make a stronger impression on us, are more distinct, and appear in a definite order that we cannot break. We have as well the consciousness that we have not produced them ourselves. On this rests my concept of reality, says Berkeley; let others then see whether they can find more in the concept of reality they have! — Nor is the validity of natural science shaken by that overthrow of the concept of matter. Natural science, after all, seeks to explain phenomena by other phenomena, by causes that can themselves be sensed, not by assuming mystical substances. To explain the phenomena is merely to show that under such and such circumstances we get such and such sensations. What natural science establishes is a definite and law-governed relation in the occurrence of our sensations, so that the one sensation can stand as a sign of the other. We thereby become able to infer forward and backward. By careful observation of the phenomena accessible to us we discover the general laws of nature, or we set up certain formulas for the phenomena of motion. Natural science has to do neither with force nor with matter, only with phenomena. — Every proof of a law among the phenomena and every inference from such a law rests in the end on the fundamental presupposition that the author of the phenomena always acts uniformly and in observance of general rules — a presupposition that, according to Berkeley (Principles § 107 in fine), cannot, to be sure, be proved.

What, then, is the author of the phenomena or sensations, in so far as I do not produce them myself? While Berkeley holds the causal principle to be unprovable in so far as it concerns the relation of the phenomena among themselves, he has no doubt *[426]* of its validity when the question becomes one of the origin of our passively received

sensations. There must be an activity that comes into force where we ourselves are passive. Now our own power of calling up and altering ideas is the only example of activity we have. In this power Berkeley finds our real essence expressed: the soul is the will (the soul is the will), he says (Commonplace-book ed. 1871 p. 428). "As far as I can see, the words will, soul and spirit do not denote different ideas" (Principles § 27). All inner activity Berkeley calls willing, thinking too: "What I think of, whatever it may be, I call idea (idea). Thought itself, or thinking, is no idea, but an act, that is to say willing, or — in contrast to the effects — will" (Commonplace-book p. 460). Its being is not to be perceived but to perceive (its esse is percipere). Will is the only form of activity we know. About the knowledge we have of our willing, and so of the real essence of our spirit, Berkeley is not at one with himself. Only this stands firm for him, that our knowledge of our own active nature cannot be an idea (idea), since every idea is passive and therefore presents nothing active to us. "The will is purus actus, or rather pure spirit, which cannot be imagined, sensed or grasped" (Commonplace-book p. 462). Not an idea but a concept (notion) is what we have of ourselves. But in one place (Principles § 27) he teaches that we know spirit, or that which acts, through its effects (that is, the changes of ideas); in another place (Dialogues between Hylas and Philonous, 3rd Dialogue) he ascribes to us an immediate knowledge of our spirit and its ideas, in that we perceive them by self-observation (by reflection). But however we may perceive it: for forming the thought of the source of the phenomena we have only the *analogy* with our own spirit to hold to. We think of God in analogy with our spirit, only with its powers raised infinitely. The real ideas now become those he implants in us, and we now understand as well how objects can exist even if we do not perceive them: they exist potentially,
[427] as possibilities, in God as their constant cause. (Cf. Commonplace-book p. 489). The divine will makes itself known to us in the order and connection of our sensations, and divine providence in the purposiveness this order presents. If nature is to be a being distinct from God, it is a heathen chimera. The usual conception has God first create matter and then matter act upon us. Why this detour? Why not have God produce our sensations immediately? The more clearly we think, the more we shall find ourselves in immediate relation to God. There are no secondary causes. In God we are, live and move, says Berkeley, and he thinks he has more right to say it than Spinoza. "Every sensation that I receive from without according to the general laws of nature, i.e. independently of my will, proves the existence of a God, i.e. of an unextended, incorporeal spirit". (Commonplace-book p. 463). — To this mystical-pantheistic conception Berkeley, as is seen from *Siris,* gave himself up more in his later years; but we cannot go further

here into the more detailed form he gives his thoughts in this direction. We merely draw attention to a point of likeness between Berkeley and Leibniz at the transition from their philosophy of experience to their speculative philosophy: they both make use of the principle of analogy, which is the principle of all metaphysical idealism. In the dialogue Alciphron (IV § 21) Berkeley maintains — against those who deny that we have any concept of God, — that we know God by way of analogy, relying on such human properties as are real perfections.

Berkeley's merit is above all this, that with an energy not weakened by the paradoxical results to which his thinking led him, he held fast to the question: whence do we know that things are more than our sensations and ideas? By what right do we leap from consciousness, which alone is immediately given, over to things, which are never immediately given? — As regards Berkeley's own answer to this question, it is seen from the foregoing exposition that he by no means makes existence a mere series of sensations. In the first place he distinguishes between the sensations and spirit; only the being of the *[428]* former consists in being perceived (esse = percipi); the being of the latter is to perceive (esse = percipere), that is, to act. What is innermost in existence is for him willing and acting. In the second place he acknowledges the principle of causality and uses it to solve the question raised. But whereas the usual conception (popular metaphysics) starts from the assumption that the cause of our sensations must resemble them, Berkeley starts from the assumption that the cause of our passive states must be thought in analogy with what acts in ourselves. — He is an idealist, but not a subjectivist. But his idealism at once takes on a theological form. His theological notions lie ready, poised to spring, while he philosophizes, and as soon as he has drawn the knot tight and shown an open question, they press forward to fill in what is lacking. Thinking thereby halts too early. The theological notions themselves are not tested. The whole conception of the world becomes anthropomorphic. The connection of nature becomes only a connection in the divine will, which holds the phenomena together, and the phenomena become only a collection of arbitrary signs. The principle of causality itself is not examined more closely, and yet it carries the whole solution of the problem. On this last point Berkeley's nearest successor took hold, continuing and opposing, just as Berkeley himself had done toward Locke.

5. David Hume

a. Biography and Characterization

English philosophy has this great significance in the history of the development of the life of the human spirit, that by its method of experience it demands that account be rendered not only for the edifices of thought that speculative philosophers erect, but also for the unconscious and unproven presuppositions on which the popular conception of the world and the special sciences build. Thus Locke had demanded an account of where our ideas in general come from, and Berkeley had shown what problem, when
[429] one takes this account in earnest, lies in the idea of space and in the idea of the material world. This critical investigation of knowledge reaches its high point within the English school of the eighteenth century in the philosophy of David Hume. He undertakes an investigation of the two concepts on which the whole earlier philosophy rested, and which Locke and Berkeley had not yet attacked in earnest: the concept of substance and the concept of cause, the two concepts that were the binding cement in all speculative, scientific and popular edifices of thought. On the causal principle, or the principle of sufficient reason, rested Leibniz's great system of harmony, Newton's mechanics of the world and the popular conception of a world of things subject to certain laws. All started from the reasonableness of existence, its rationality, presupposed with greater or less consciousness that existence contained in itself something corresponding to our reason. This presupposition Hume tests. And he is the first who has in earnest undertaken such a test and has descended into the depths in which the forces that hold the inner and the outer world together for us spring up, depths that lead away from the regions in which speculative philosophers, special inquirers and common sense move. Hume himself felt, and expressed in a characteristic way, the strange state of solitude and dissolution in which the thinker finds himself who perseveringly pursues the problem of knowledge down into these depths, and the opposition there is between the strictly theoretical and the instinctive, practical and popular conception of the world. Only the intellectual passion that awoke ever anew, together with the hope of winning fame by going the way consistently to its end, made it possible for him, by his own statement (in the concluding section of the first Book of his chief work), to complete his work.

In Hume's character intellectual zeal and ambition were united with good nature, benevolence and a mild view of weaknesses and prejudices, but at the same time also with a certain love of ease, which did not want to be disturbed by polemic in the cultivation of literary interests. He was born on 26 April 1711 on the estate of Ninewells in the south

of Scotland, the second son of a landowner. In his autobiography he says: "Very early I *[430]* was seized by a passion for literature, which has been the ruling passion of my life and a great source of enjoyment to me." His family wanted him to be a lawyer, but he "felt an insurmountable aversion to everything but philosophy and learning". His ideal was a carefree existence in which he could satisfy his inclinations for inquiry and cultivate the company of a few chosen friends; but he wanted as well his literary activity to make his name honored. While still quite young he believed himself on the track of new thoughts: a new "scene of thought" opened up for him. But an attack of hypochondria (which he describes in a letter to a physician, printed in BURTON: *Life and Correspondance of David Hume.* Edinburgh 1846 I p. 30 ff.) halted his meditations for a time. Presumably he here noticed at once the strange opposition between the world of reflection and the world of practical daily life which he later described in his chief work. He resolved to lay his studies aside and become a merchant. But practical life could not hold him. He sought out solitary places to stay in France and wrote there his chief work: "Treatise on Human Nature, an Attempt to Apply the Method of Experience to the Mental Domain" *(Treatise on Human Nature etc.).* It came out in London 1739—1740 and consisted of three parts, of which the first deals with knowledge, the second with the feelings, the third with the foundation of morality. It carries the treatment of these various questions considerable steps further, and still stands in the front rank among the classic works of philosophy. But for the time being it had no success. "It fell", he says himself, "dead-born from the press, and did not even attain the honor of exciting a murmur among the zealots". Hume's literary ambition, which seduced him into declaring his excellent brainchild dead-born, had a fateful effect. The fame he did not win by it he sought to win by a series of smaller treatises (Essays), partly of philosophical, partly of economic and political content; for a time he went away from philosophy altogether to cultivate history; indeed, in the end he even disowned his significant work of youth entirely, *[431]* declaring — so as not to be molested by his theological critics (who, then, had after all begun to "murmur") — that he acknowledged only the exposition of his philosophical doctrine he had given in the "Essays". But brilliant as many of these smaller works are, they could not gain so great a significance in philosophical discussion as the chief work could have gained if Hume had used the literary fame he later won to bring life into the "dead-born" child, and had not disavowed it so that it should not cause him disturbance. As regards the problem of knowledge in particular, it was above all through the abridged and muted exposition in *Enquiry concerning Human Understanding* (1749) that Hume's philosophizing gained influence on the further development of thought,

while the radical exposition in the "Treatise", which cuts the very bond between our thoughts, and in general between the elements of our being, lay forgotten for a long time. — That the motive of Hume's disavowal of his work of youth is the one given here is seen from *Letters of David Hume to William Strahan* (Oxford 1888. p. 289 f.). There is no question, as has sometimes been thought, of Hume's really having changed his views on main points. Yet it is psychologically understandable that the highly strung intellectual state in which Hume wrote his work of youth could not last. After having thought with the learned, and better than the learned had thought hitherto, he felt a need to talk with the unlearned. After giving a popular exposition of his philosophical and economic ideas in his "Essays", he threw himself into history. "As you know", he wrote to a friend, "there is no post of honor on the English Parnassus that can more truly be said to stand vacant than that of history". The place he had obtained — after strong opposition on the part of the orthodox — as librarian of the Advocates' Library in Edinburgh gave him good opportunity for learned studies. His History of England made him still more popular than his Essays had done. As a historian he has the merit of
[432] being the first who sought to make history more than a history of wars, in that he took account of social conditions, manners, literature and art. His historical work began to appear two years before Voltaire's famous "Essai sur les moeurs". While he was liberal in his philosophical views, in his judgment of historical personalities he started from royalist and Tory points of view. — Yet he did not put philosophy on the shelf altogether. Studies in the philosophy of religion especially occupied him in his later years. To this his *Natural History of Religion* (1757) and his *Dialogues on Natural Religion* bear witness, the latter work of which he held back for reasons of caution, so that it appeared only a couple of years after his death.

Hume was not only a philosopher and historian. He had a need to take part in practical life. As secretary of an embassy he made (1748) a great journey through Holland, Germany, Austria and northern Italy. And later he exchanged his post as librarian in Edinburgh for a post as secretary to Lord Hertford, when the latter, after the Peace of Paris in 1763, went to France as English ambassador. Hume was by then already famous and was received in the most brilliant way at court and in literary circles. He was, like Franklin later, in fashion, perhaps precisely because of his plain, little elegant manner. When after three years' stay he traveled back to England, he took Jean Jacques Rousseau with him, to procure a refuge for the man, now driven out of both Switzerland and France. Hume's fine conduct toward Rousseau was repaid by the latter with insane suspicion, and after a spectacular break Rousseau fled back to France again, where the

storm had now abated. — After Hume had held the post of Under-Secretary of State for Scotland for a year, he took up residence in Edinburgh and now led a quiet learned life with chosen friends, until, after a lengthy illness that did not rob him of his equanimity and cheerfulness, he died on 25 August 1776.

b. Epistemological Radicalism

All the sciences, and not least the human sciences, stand in a definite relation to human nature. With this proposition Hume begins his investigation of knowledge, and from it he *[433]* draws the conclusion that by the study of human nature one will discover the foundation of all human science. But this study ought to be undertaken by means of the method of experience, which has already been applied with success in natural science, and whose application to the study of human nature Bacon, Locke and Shaftesbury have already begun. As his *fundamental proposition* Hume now sets up, appealing to experience, the proposition that all our ideas (ideas) spring from sensations (impressions); ideas can never be original. In testing the validity of an idea we have then first and foremost to ask of which sensation it is an after-effect. Dispute and doubt arise when the ideas are unclear and indefinite. Since it now turns out that the sensations from which the ideas derive are on the whole clearer and more distinct than the ideas themselves, disagreement and uncertainty will be brought to an end when one seeks out the corresponding sensations; — if such cannot be shown, the idea must be invalid (Enquiry Sect. 2). In the origin of the sensations Hume sees a problem insoluble for our knowledge, but a problem whose solution is also not necessary for the treatment of his task (cf. Treatise I. 3, 5, where he leaves it open whether it is the external things that call forth the sensations, or God who creates them, or the soul itself that unconsciously forms them; — in other places, as II, 1, 1, he follows the usual mode of expression). — By means of that fundamental proposition (which Berkeley had applied only to the ideas of space and matter) Hume then tests a series of important ideas. — The idea of substance or essence must be declared invalid, since we have no corresponding sensation. We immediately sense only single qualities, more or less firmly connected with one another, but no "essence." — The mathematical ideas of time and space are formed by idealization. Experience shows us only imperfect equality in the magnitude of times and spaces; every standard of measurement that we have is imperfect; — but after experience has given us occasion to compare different *[434]* degrees of equality and different standards, we form the idea of perfect equality and of a perfect standard (a perfect line, e.g.), since the imagination, once it has been set in motion, continues its course even if experience cannot keep up. Since geometry concerns

such ideal objects, it cannot, in its application to real objects, lead to complete exactness. — Just as little as the concept of substance and the mathematical concepts does the idea of existence correspond to any sensation. To think of a thing and to think of it as existing are not two different things. Our idea of the object remains a mere idea even if we think the object as existing, and we do not add a new quality to a thing by conceiving it as existing.

In testing the validity of our knowledge as a whole, Hume distinguishes between the knowledge that consists only in clarifying the relation of our ideas to one another (the formal sciences: logic and mathematics),[5] and the knowledge that leads us beyond the given sensations and convinces us of the existence of something that is not given. This latter kind of knowledge rests on the assumption of the validity of the causal relation, — and here Hume then performs his great philosophical deed by posing in a decisive way
[435] the *problem of causality,* the problem on whose solution all appraisal of the significance of the real sciences depends. At the root of all inquiry lies a need to find "what holds the world together in its inmost core"; in every single problem of the real sciences lies the same fundamental problem, and Hume was the first to see it in its whole scope. It must be firmly kept in mind that Hume never at any time doubts that we must of course apply the causal principle without interruption in theory and practice; he asks only whether this application can be *grounded,* — and it is here that he finds only a negative answer. He discovers that what makes us assume causal relations is the same as what makes us assume something as existing even if it is not given to sensation. One and the same investigation casts — in accordance with Hume's psychological method — light on the two concepts of cause and belief (belief).

One cannot, in grounding the validity of the causal relation, appeal to immediate certainty or intuition; for such we have only concerning simple relations of resemblance and quantity. Nor to logical demonstration; for all our ideas can be separated by us from one another. The motion of the one billiard ball is an event wholly different from the motion of the other billiard ball. There is no object that it is not easy for us to think as existing at one moment and as not existing at the next. There is then no self-contradiction

[5]With regard to mathematical knowledge there is a wavering in Hume. In the Treatise he lays strong emphasis on the fact that we can conceive no absolutely regular line or figure, and he therefore declares that geometry cannot be a strict science, whereas algebra and arithmetic can attain complete exactness, since we can always convince ourselves that two numbers contain an equal number of units (Treatise I, 3, 1). But he nevertheless acknowledges that there is both a geometrical and an arithmetical necessity, which springs from an act of the understanding by which we compare two ideas (I, 3, 14), — and that in geometrical relations too we have a standard, although it is owed to a "fiction" (I, 2, 4). And in the later epistemological work he puts geometry and algebra on an equal footing: "Even if there never existed any circle or triangle in nature, the truths that Euclid has demonstrated would still always retain their certainty and evidence (Enquiry III, 1).

whatever in something's beginning without a cause. The proofs that earlier philosophers (Hobbes, Locke, Clarke — to whom Wolff could be added) have attempted are only sham proofs. It is also clear, of course, that if the causal principle were grounded on reason (either on intuition or on demonstration), we should come into conflict with the "fundamental proposition": for then reason would possess the capacity to produce wholly new ideas that did not derive from experience! — But is the validity of the causal relation not grounded by experience? — To this the answer is that experience only shows us the one event following upon the other, but not the inner necessity in their mutual relation, which is what is meant by the word causal relation. This holds whether by experience we mean inner or outer experience. Hume refutes in particular the assumption (which Berkeley had embraced) that in the immediate consciousness of our willing we should *[436]* have an apprehension of power or activity. A single experience cannot give us the idea of cause, and several experiences give us only a relation of succession. When we have very often seen one phenomenon followed by another, we involuntarily expect this other when the first occurs again; but that is a custom, and therein lies no proof of our right to infer from the past to the future.

In his explanation of the tendency we have to go beyond a given sensation and to assume or expect something that is not given, Hume first takes as his basis the experience that every strongly moved state or mood (disposition, feeling) into which consciousness comes has a tendency to maintain itself and to spread out over the new ideas that emerge. When consciousness is heightened and enlivened by an object, every act of consciousness will be livelier than otherwise as long as the enlivenment lasts. It was indeed this same law that explained the idealization of the mathematical ideas. — In the second place he lays weight on the fact that experience shows us a propensity in our ideas to call one another forth. Every idea has an associating tendency; a gentle power (a gentle force) leads from the one over to others. As various grounds of such association Hume names resemblance, contiguity in time and space, causal relation. There manifests itself here an attraction in the inner world which is as significant for that world as physical attraction is for the outer world. But it is at the same time no less mysterious in its nature: its causes must lie in qualities of human nature that we do not know. The uniting bond between our ideas is just as incomprehensible as the uniting bond between external objects; and it is known only through experience. Indeed, Hume finds (in the Appendix to the first book of the Treatise) the associative connection not merely incomprehensible but even self-contradictory: how can there be a connecting principle *(uniting principle,* Tr. I, 3, 4; *principle of connection,* Tr. I App.), when all our sensations and ideas are

[437] distinct, independent existences? Of this difficulty Hume declares that it is too hard for his understanding. — But the two facts (which we may call expansion and association) stand firm. And by their help it is explained how a lively sensation, which itself — precisely because of its liveliness — stands before us as the expression of something real, can communicate its liveliness, and with it also its stamp of reality, to the ideas that it calls forth by association. What we call belief (belief) is nothing other than the lively stamp that ideas can acquire in this way. Belief is not owed to any new, peculiar impression, but signifies only the peculiar manner in which an idea comes to stand for feeling (feeling or sentiment). We cannot refrain from passing over to the idea associated with the lively sensation, and cannot refrain from holding it fast and seeing it in a stronger light than otherwise. — And since now (in the third place) consciousness (as the ideas of the sense qualities, of time and space, and of substance show) has a propensity to regard its own inner states as external, objective phenomena, we understand how we come to assume causal relations between the things in the world, although the necessity in question here is found, of course, only in ourselves, and is a compulsion which for psychological reasons manifests itself in us to pass over from one sensation or idea to another. The necessity is subjective, just as the sense qualities, space and time are. What we call reason (reason, Treatise I, 3, 16 in fine, cf. Enquiry IX: experimental reasoning) is only an obscure instinct in us, which arises through the repetition of experiences in a definite order. The very arising of this instinct, the very influence of custom, is just as incomprehensible as all connection between the elements of our consciousness in general is. But since the mental activity by which we infer from the same cause to the same effect is of extraordinary practical importance, nature has not been willing to entrust it to uncertain thinking (reason = subtile reasoning Treatise I, 4, 1 and 2), but has let it be owed to the sure, though obscure, instinct, which will not be shaken by any critical examination whatever: nature is stronger than reflection (nature is too strong for
[438] principle! Enquiry XII, 3). Whereas dogmatic philosophy in its various forms believed that by the help of pure reason it could build trains of thought that led far beyond sense perception, as a consequence of Hume's investigations the imagination (imagination) comes to stand as the only faculty that can lead to the assumption of something that is not an object of present sense perception (Treatise I, 4, 7). On the imagination, then, rests the assumption of an external world independent of consciousness. We do not experience the uninterrupted existence of objects, only a certain constancy and coherence — but this is enough for our imagination, which of course continues to move in the direction in which it has once been led, to conceive as great a uniformity and coherence as it is at all

able to. The intervals between our sensations we fill up by feigning constant beings. — By quite the same principle the idea of the ego or the self is explained. The invalidity of the concept of substance has already been shown. Our sensations and ideas can very well subsist without leaning on or inhering in such a substance. If one wishes to ground one's assumption of a soul-substance, one must show a sensation on which it is grounded, and that one cannot do, since all our sensations change. If one takes the word substance in a vaguer sense, of that which has separate subsistence, why then should our sensations and ideas, each of which can subsist separately, not be substances? — Hume warns theologizing metaphysicians in particular against the assumption of a soul-substance in which the single sensations and ideas subsist as particular modifications, since one would thereby give Spinozism a dangerous support: it was in just this way, after all, that Spinoza conceived the relation between God and the things in the world! — But even if we disregard the idea of a soul-substance and merely ask how matters stand with the idea of the self as the expression of our personal identity, we are entangled in great difficulties. No sensation or idea is unchangeable and constant. We always find a special inner state given, never "ourselves" as a whole. The identity that we ascribe to ourselves is of the same nature as that which we ascribe to external objects: it is owed only to the ease with which we pass over from one sensation or idea to another, an ease that *[439]* makes us overlook the enigma in the fact that one element can lead over at all to another, different from it! —

Hume is a poser of problems of the first rank. He is able, as few have been able, to think a concept or a relation through in such a way that the hidden difficulties come to light. By his energetic labor of thought he succeeds in drawing forth the fundamental concept that bears all the practical thinking of men, all their exact science, all their speculation and all their religion. He has shown that there are problems enough for philosophy even after the time of the great systems is past. But to see his significance rightly, one must study him in his chief work (Treatise). In his Essays[80]) the investigation is essentially limited to the concept of cause, and the radical manner in which Hume pursues "the uniting principle" under all its forms (in mathematical knowledge, in the concept of the self), as well as the method he follows, in particular the thoroughgoing application of the law of expansion, one would not suspect from the abridged and subdued exposition. — The uniting principle, whose incomprehensibility he shows, he nevertheless acknowledges as operative: partly in assuming that we have ideas not only of single things but also of their relations (time and space, resemblance and difference, e.g.), partly by assuming the association of ideas as a fact, partly, finally, by the weight he lays on the imagination's

tendency to spread and extend itself, a tendency which is, of course, a tendency in consciousness to preserve its identity with itself in spite of all differences. All this certainly stands in a wonderful contrast to the absolute, "substantial" difference between the single sensations and ideas which is the presupposition from which Hume starts, — a presupposition that marks the limit of his thinking. One could with greater right find the difference between the elements of our consciousness incomprehensible: for to
[440] comprehend is to connect, to find connection!

c. Ethics

The second part of Hume's chief work treats of the feelings. It is of great significance for the history of psychology; but we can treat it here as an introduction to the ethics, which forms the third part of the work, and which Hume later rendered in shorter form in his *Enquiry concerning the Principles of Morals* (1751). — In his conception of the psychological nature of the feelings he has Spinoza as his predecessor and was probably influenced by him. His exposition is richer and fuller than Spinoza's. The conception of both rests on the significance that the ideas and their mutual connection have for the development of feeling. Hume shows how one feeling connects itself with another feeling by means of the connection between the ideas of their objects. His statements waver on the question whether a direct association between feelings takes place: for the most part he does ascribe to the feelings — as to the ideas — a tendency to reproduce other feelings (II, 1, 4, cf. *Dissertation of the Passions* in Essays), but in other places (II, 2, 8) he teaches that the connection of feelings does not take place when there is no connection of ideas. At the root of this wavering there probably lies the recognition that feeling does not play a wholly passive role in these processes, something for which Hume, by his doctrine of expansion, which is of such far-reaching significance in his theory of knowledge, has shown that he has a clear eye. — As a psychologist he is significant also for strongly emphasizing that only feeling, not mere reason, can give rise to acts of will. He points out that sensations and ideas are easier to establish than drives and tendencies when these are not violent. Therefore the first germs of the will are easily overlooked. And just as the will arises only from feeling, not from reason alone, so too it is only feeling that can check feeling and prevent action (II, 3, 3). Here too he agrees with Spinoza in a psychological proposition that is often overlooked but is of great scope. Feeling is an original and immediate state, but reason manifests itself in reflection and comparison
[441] and therefore gains influence on feeling only in so far as it can test the ideas connected with the feeling. — An imperfection in Hume's psychology is that he does not bring

out the connection of feeling with original instincts[81]). Shaftesbury saw more clearly here than he. Hume is inclined to regard the instincts as merely derived (as when in his theory of knowledge he explains the causal instinct by custom). Here his conception is limited by his empiricism. A related trait shows itself in his writing of history, for it has been objected against it that he takes no account of the racial peculiarities of the peoples whose history he depicts. —

The psychological investigation of the relation of feeling to reason gains immediate significance for Hume's ethics by containing the answer to the question whether morality is grounded on reason or on feeling. Reason merely establishes relations or facts. But a moral judgment arises only when a feeling is set in motion by the idea of an action, after all the relations and facts that concern this action have been brought forward. Only because our feeling is set in motion do we call something good or evil. The moral qualities (good and evil) therefore have validity only from the standpoint of feeling beings — just as the sense qualities have validity only from the standpoint of sentient beings[82]). But this does not rob these qualities of their significance: in practice we apply moral valuations just as confidently as the sensible qualities, although none of them expresses objective and eternal relations. (See, besides Treatise III, 1, 1, especially the essay *The Sceptic* in "Essays"). Hume (who on this point again agrees with Spinoza) stands here in sharp opposition to the moral philosophers who would derive morality from reason alone and regarded the moral principles as eternal truths, a standpoint that even after Locke's time was maintained by Clarke and Price (just as Cudworth had maintained it against Hobbes). In this connection Hume makes the following remark (Treatise III, 1, 1 in fine): "In every ethical system that I have hitherto met with, I have always remarked that the author for a time reasons in the usual way, asserting the existence of a God or bringing forward observations on human affairs; — but suddenly I am surprised to find *[442]* that instead of the usual words *is* and *is not* I meet with propositions in which *ought* or *ought not* forms the connection. It is an imperceptible transition; but it is of the very greatest significance. For this *ought* or *ought not* expresses a new relation, which ought to have been explained, and it ought to have been shown how this new relation can be derived from other relations that are altogether different from it. But authors are not usually so careful." Hume finds in this transition, and in the impossibility of explaining it by the establishing and comparing reason alone, a sign that it is feeling that is at work here (Treatise III, 1, 2).

The next question is what feeling it is, then, from which morality springs. In all actions and qualities that are the object of moral approval one common trait is found

again: that they are, mediately or immediately, to the good and utility of the agent himself or of others. This trait wrings from us — independently of all upbringing and authority — approval, esteem, perhaps admiration. And since we approve actions that do not benefit ourselves, the feeling that lies at the root of the approval cannot be of a selfish nature. He who pronounces moral judgments leaves his private standpoint and takes up a standpoint that is common to him and others. If we were to value merely according to selfish interests, we could arrive at no common valuation. Even if the recognition of justice as a virtue were originally owed to each single individual's need of peace and security, the interest in the general order of law can nevertheless be explained only by sympathy with that which preserves human life as a whole; the motive that originally makes something a virtue need not be the one that can later lie at the root of the valuation. It is, then, sympathy or fellow-feeling (fellow feling) that is the real foundation of morality. Even the valuation of the virtues that bring advantage only to the agent himself (such as prudence in one's own affairs) is best understood from sympathy. — Sympathy itself Hume explains in accordance with the "fundamental proposition" of
[443] his theory of knowledge: it arises in that the sight or the lively idea of the expressions or the causes of others' joy or pain calls forth in us a lively feeling of joy or pain, that is, in that a mere idea (idea) passes over into *sensation* (impression) by virtue of its connection with a lively sensation.

Hume's ethics bears the stamp of a clear head and a warm heart. One must not seek in it light on deeply penetrating ethical crises and oppositions. He does not even take up Hutcheson's interesting attempt to derive the feeling of duty from sympathy. The inner and outer resistance that the working out of an ethical character may meet he does not stop to consider. But he has a full and clear conviction of the foundation of morality in human nature, and he indicates the fruitful distinction between the first motive for the development of a quality of character or an institution and the motive from which the later valuation springs. If he has often been called a skeptic, this designation at any rate does not fit him as an ethicist. In a special investigation *(A Dialogue,* in Essays) he discusses the objections that could be raised from the mutually conflicting moral ideas, manners and customs of different peoples and times. One could, he says, just as well find a difficulty in the fact that the Rhine flows to the north and the Rhone to the south! Both rivers flow in opposite directions because of one and the same law, the law of gravity, owing to the different slope of the ground. Because men come to different results under different circumstances, they may very well start from the same principle. Everything that men have called good or evil has been something that was regarded as directly or

indirectly useful or harmful. The differences, then, do not shake the principle. —

As an economist Hume is Adam Smith's most important predecessor. In his economic essays he asserts the importance of awakening the acquisitive drive. By compulsion no one is brought to work ably. New wants must arise in order that the desire for advancement may come into being. If no other wants show themselves than those that can be satisfied by the meagerest yield of the soil, the soil will not be cultivated as well *[444]* as possible. Agriculture flourishes only in interaction with commerce and industry. And thereby is also formed the middle class, which is the surest and best support of public freedom. — Into the problem of the relation of the acquisitive drive to sympathy Hume enters as little as his friend and successor Adam Smith. They seem to have assumed a harmony between the effects of the two motives.

d. Philosophy of Religion

Hume's most important merit with regard to the religious problem rests on his clear distinction between the question of the possibility of grounding religion by the way of reason and the question of the actual origin of religion in human nature, a distinction that has analogies both in his theory of knowledge and in his ethics. The first question is discussed in his "Dialogues on Natural Religion," the second in "Natural History of Religion."

As regards the latter, the historical question, Hume seeks to show that it is not a need of understanding that leads to faith in divine beings. On the contrary, what for the thinker is a great stumbling block, the misfortunes and disorders in nature, is for the common man precisely a motive to believe. Faith is called forth by feelings that arise in the course of life: fear and hope, suspense and uncertainty, anxiety before the enigmatic. With this cooperates the general tendency to conceive all other beings in likeness to oneself. History seems to show that polytheism is the original religion. And this agrees with the natural course of development: consciousness rises gradually from lower to higher stages, and as, in the strong agitation into which it is put, partly by fear, partly by enthusiasm, it exalts its object for the imagination and thinks it ever more perfect and nobler, it at last arrives at the idea of a single, infinite and incomprehensible God. There takes place here, in the domain of religious feeling, a process of idealization similar to the one that according to Hume lies at the root of the formation of the mathematical principles and of the principle of causality. The transition from polytheism to monotheism cannot be explained by merely intellectual motives; but by the way of feeling men have reached *[445]* the same result to which rational consideration leads: that there can be only one God.

Now when this exalted stage has been reached, a reaction easily takes place, as history shows. A need is felt of intermediate beings between the one, infinite, purely spiritual God and the world. Between these two opposite poles — the exalted, non-intuitable and the limited, intuitable conception of God — a constant oscillation takes place. — On closer investigation Hume finds conflicting tendencies and qualities in the nature of religion. Sublimity and coarseness, faith and doubt, purity and immorality, grandeur and horror are mixed together in a remarkable manner. There is so much that is enigmatic in religion, and so much conflict between the different religions, that Hume prefers to withdraw into the calm, though obscure, regions of philosophy. — In his essay "on miracles" Hume discusses in particular the question of the supernatural. His main thought here is that no testimony is sufficient to ground a miracle unless the falsehood of the testimony itself would be a greater miracle than that which it concerns. But he maintains that there is no miracle that has been able to be supported by such testimony. —

As regards the question of the truth of religion, there is difficulty in finding out Hume's real standpoint, since he has discussed this question in dialogue form. He puts forward three different standpoints. Demea represents a mystical orthodoxy, which builds partly on a priori grounds of reason, partly on postulates of feeling. Cleanthes represents a rationalist deism, which relies especially on the purposiveness in nature. Philo appears partly as a skeptic, partly as a naturalist. That Demea does not represent Hume's own standpoint there can be no doubt. Cleanthes he has, by his own account, made the hero of the dialogues; but it is evident, and he also says so in letters (Burton I p. 332. Letters to Strahan p. 330), that Philo's considerations interest him himself the most, although he lets him give up his standpoint at the end. Philo's standpoint undoubtedly comes closest to Hume's own. The main thoughts that Philo urges are the following. —

[446] Who could blame us if, faced with such great and difficult questions, we declared that we knew nothing? We go here, after all, far beyond the domain of experience, and system stands against system! Against Cleanthes's inference from the order and purposiveness in nature to the existence of God, Philo makes a series of objections. 1) Why seek the cause of the order and purposiveness outside the world? There could, after all, be forces at work in the world itself that bring order and harmony with them — perhaps through many upheavals and provisional adaptations. Nothing, after all, can subsist entirely without order and connection. And just as skill develops through attempts and trials, different systems of the world could have succeeded one another until the present system could arise as the one that offered the most favorable conditions for continued subsistence. 2) Experience shows us mind and thought as finite, limited phenomena.

With what right does one explain the whole of the world from a part of it? Thought itself — and so also the plan of the world that is ascribed to God — needs explanation just as much as everything else in the world, and when we stop at it as the last cause, is that not because of the satisfaction of finding our own nature again behind things, just as we are inclined to see our own shapes in the clouds? 3) Is there any special problem to pose concerning the world as a whole? When we have explained the coming into being of the single parts or phenomena, it is no special problem how all these parts together have come into being. It is only an arbitrary act of consciousness that makes us conceive them as one collected whole. 4) From the world as experience shows it, with all its imperfection, pain and want, one can at any rate not infer a *perfect* cause. In one of his Essays *(Of a Particular Providence and of a Future State)* Hume poses the following dilemma: if there is justice in the world, there is no reason to seek another world, and if there is no justice in the world, this world cannot be assumed to have been created by God. — Philo, however, will not deny the existence of God. He only warns against conceiving God in too close an analogy with man. He declares that the difference between the theist and the atheist, the dogmatist and the skeptic is in the end only a *[447]* verbal difference: the theist after all admits that the divine mind is extremely different from the human, the atheist that the principle which rules in the world nevertheless has a certain analogy with the human mind; the dogmatist admits that there are great difficulties connected with his assumption, and the skeptic that in spite of the difficulties we cannot remain in mere doubt. What matters is that no prejudices weaken the natural love of mankind and the feeling of justice. —

Hume's treatment of the religious problem marks a great advance. His dialogues stand as a lasting monument beside Kant's critique of theology in "Kritik der reinen Vernunft." An answer to these two works has not really been given. As regards the psychological side of religion, there is surely need of a greater depth and fullness than Hume's sober mind could give; with respect to the historical understanding of religion, too, his material was insufficient. But in both respects he has indicated thoughts that later inquiry has carried further. His sound method of experience gives his investigations lasting value. He is the most important predecessor of modern positivism.

6. Hume's Successors and Critics in England

Hume marks the high point of the development of philosophical thinking in England in the eighteenth century. From the points of view that Bacon and Hobbes already, but

above all Locke, had urged, the last consequences that could be drawn had been reached; the line of thought once struck up had been carried to its end through a labor in which one eminent thinker took hold where the other let go. Only from wholly new points of view, which did not lie within the horizon of the English school but had to come — and did come — from another side, could the development be carried further. This does not exclude that even after Hume's time, before philosophical work in England comes to a
[448] long pause that lasts into our century, there appears a series of significant works, partly supplementing and continuing Hume's work, partly opposing and reacting against it.

a. Hume's work in the domain of ethics and economics was continued in an independent manner by his friend Adam Smith, who was born at Kirkaldy in Scotland in 1723 and studied at Glasgow, where he heard Hutcheson with admiration, and later at Oxford. To judge from the manner in which he later speaks of the English universities in his famous economic work, the conditions at Oxford did not appeal to him. According to what Lord Brougham relates (*Lives of Philosophers of the Time of George III* p. 179), he is said to have experienced the narrow-mindedness of the university authorities himself: his copy of Hume's "Treatise," which he was studying, was confiscated, and he received a reprimand for reading such a work! A trait characteristic of the manner in which Hume and his works were regarded in the chief seat of English learning, where, moreover, Hobbes and Locke had earlier been treated in a similar manner. For a number of years (from 1751) Adam Smith worked as professor at Glasgow. His course on moral philosophy consisted of four parts: natural theology, ethics, natural law (after Montesquieu's historical method) and economics. Only the second and the fourth part of this course did he succeed in working out into independent works. As regards the natural theology, we know only what can be inferred from his writings. In religious respects he was more conservative than Hume; perhaps he stood to Hume as Cleanthes to Philo (in Hume's Dialogues). Of the third part of Smith's course we can form a notion from a set of lecture notes which was found a few years ago, and which, besides the doctrine of right, contains an exposition of economics (the first draft of the Wealth of Nations). (See *Lectures on Justice, Police, Revenue and Arms.* Delivered in the University of Glasgow by *Adam Smith.* Reported by a student 1763. Edited by Edwin Cannon. Oxford 1896). He lays weight here on showing the development of public and private law under the influence of economic and industrial progress. His ethical and economic works too bear witness to his zeal to
[449] use the study of real, historical conditions to illuminate questions of principle, and it is one-sided when Buckle, in his otherwise instructive characterization of Adam Smith in his English history of civilization, describes his method as purely deductive. Interesting

essays from his hand on the origin of language and on the history of astronomy also bear witness to his historical sense. — A short time after the publication of *Theory of Moral Sentiments* (1759) he withdrew from his university work and thereafter stayed some years in France, where in particular his acquaintance with Quesnay, Turgot and Necker became of significance for his economic studies. After his return home he lived for ten years in his native town Kirkaldy and here worked out the work that above all has made him famous, since the beginning of scientific economics is usually dated from it: *Inquiry into the Nature and Causes of the Wealth of Nations* (1776). (Danish translation by DRÆBY under the title: Undersøgelse om Nationalvelstands Natur og Aarsag. Copenhagen 1779—80). In his last years Smith held a post in the customs service at Edinburgh, where he died in 1790.

The two works on which Adam Smith's significance as a thinker rests stand in a peculiar contrast to each other, since in his economic work he takes the acquisitive drive, in his ethical work sympathy, as his basis. BUCKLE has held that it was Smith's intention to set up two tendencies of human nature, each in its abstract form, in order thereby to make a clear, deductive line of thought possible. Against this stands the fact that the contrast between the two works is indeed characteristic but not absolute, and that Smith's method is not purely deductive. It remains an unsolved question how Smith himself conceived the relation between his two works; a psychological and historical investigation of the relation between the two tendencies in human nature he has not given. But he has often been done an injustice by forgetting that he is the author not only of the work on national wealth but also of the work on the moral feelings.

As an ethicist Smith lays great weight on the fact that the moral feeling arises only when man lives in society with others. There then manifests itself in him a natural *[450]* instinct to imitate the looks and conduct of others, to put himself in their place, to feel and suffer with them. It is an involuntary propensity which the most selfish man cannot entirely subdue. This instinctive sympathy acquires its more definite character through the arising of ideas of the causes and the effects of the mood and action of others. We approve the feelings of others when we are conscious that under the same circumstances we would have feelings of the same kind and strength. The feeling must stand in a certain proportion to the cause that has called it forth if we are to be able to sympathize with it. But when the other's feeling becomes the motive for actions that bring about effects for a third person, we put ourselves in his place too, and we can then sympathize with the action only when its effect stands in a suitable proportion to the original cause of the motive. In the case of benevolence and love of mankind we sympathize with both

giver and receiver. In the case of gratitude and revenge, sympathy with the agent ceases when the effect stands out of proportion to the original cause. In opposition to Hume, Smith maintains that regard to the utility of the qualities of character that are approved is not the original ground of approval. It is a factor that comes in later and lends support; but it is an "after thought," a thought that comes after the involuntary judgment which springs from one's putting oneself in the place of the agent and of the one affected.

Our first ethical judgments concern others, whose conduct we follow as impartial spectators. But we soon learn that others also observe and judge our conduct, and we then learn to judge it ourselves — provisionally from the standpoint of the others. Only by living together with others do we thus learn to hold up a constant mirror to our own actions. We divide ourselves into two persons, an agent and a spectator. And this inner spectator does not now go on being a mere representative of the outer censors: we quite naturally ascribe to him a greater knowledge of us than other men can have, and involuntarily equip him as an inner judge, to whose superior wisdom and justice
[451] we appeal from the short-sighted and unjust judgments of those around us, just as, on the other hand, remorse can arise when our will is in conflict with what the inner, impartial spectator can approve, even if no man knows of what stirs within us. — The internalization and idealization of the moral feeling that Smith here depicts he could have explained better if he had made use of Hume's doctrine of the imagination's tendency to hold fast and to heighten its objects. But Smith's exposition, although some intermediate links are missing, is of extraordinary interest and marks a significant step toward the understanding of the development of the moral feeling. He rejects the assumption of a peculiar, original moral sense that would be complete once and for all, just as he rejects the explanation of all morality from egoism or from pure reason. The great significance of reason for moral development he does not deny; but the single moral experiences from which the general moral propositions of reason can be derived we come, in his view, to make by means of that involuntary and instinctive sympathizing with the state and conduct of others. Only in generalization, not in the original observations of good and evil, does reason cooperate. On this point he agrees with Hutcheson, whom he praises as the one who first saw clearly here. — In the last part of the "Theory of Moral Sentiments" Smith gives an excellent criticism of earlier theories, which is still of great interest.

What has given occasion to assume not merely a certain contrast but even a contradiction between Smith's two chief works is the great weight that he lays, in the work on national wealth, on the individual's acquisitive drive being allowed to move freely. The source of all wealth is thrift and industriousness, and these develop only where

the acquisitive drive is not cowed. "The drive that moves us to thrift," says Smith (Om Nationalvelstand. Danish trans. I, p. 469), "is a desire to better our condition, a desire which is indeed in general calm and without passion, but which nevertheless follows us from the cradle to the grave." How the individual can satisfy this drive in the best way, he himself finds out best of all. The state is not to intervene here with commands *[452]* or prohibitions. Let everyone buy where it is cheapest, whether at home or abroad. Supply and demand will regulate everything in the best way. Under the influence of the relation between supply and demand, the division of labor, which is the condition of all development of culture, has come into being, since both the individual himself and the others found their account in his limiting his labor to what, according to his abilities, he could do best. This shows itself both in small things and in great. What is prudence in every private household can surely not be folly in a great kingdom! (Om Nationalvelstand, II, p. 42—44). The task of the government is only to protect against external violence, to maintain internal peace, and to establish and maintain such public works and institutions as private persons cannot find it to their advantage to provide. By this last determination (Om Nationalvelstand, II, p. 333) Smith thus does acknowledge that there are tasks that cannot be accomplished through the interaction between the single individuals, each led by the acquisitive drive. In general it should be well noted that Smith himself, in composing his economic work, stands at the standpoint of the "impartial spectator." The ethical standpoint is so far from being pushed aside that it is precisely what lies at the basis. When he praises the system of liberty, it is because it is "generous and liberal" (Om Nationalvelstand, II, p. 320). When he demands free movement for the individual according to his own calculation of what brings advantage, it is because he believes that the individual can "be led by an invisible hand to promote an end that he has not set himself." "By seeking his own advantage he often promotes that of society in a more effective way than if he really sets out to promote it" (II, 42). — Let the word "often" be noted. People have (as KNIES: Die politische Oeconomie vom geschichtlichen Standpuncte. Neue Aufl. Braunschweig 1883 p. 226 has shown) quoted this statement of Smith's with the omission of "often," and thereby his doctrine has come to stand as a doctrine of the absolute harmony of egoistic interests, which makes all sympathy and ethical action superfluous, indeed harmful. Thereby there would certainly *[453]* arise an insuperable contradiction between Smith's two works.

In order to understand Smith's significance as an economist rightly, one must remember what a burden lay upon the whole of civic life, even in the most advanced countries. He pleads the cause of the industrious and thrifty citizens against ignorant, extravagant

and arrogant governments, relying on the experience that England, in spite of war and extravagance, has advanced in prosperity owing to the good husbandry of private persons and their uninterrupted efforts to better their own condition (Om Nationalvelstand I, p. 476). He demands recognition for the labor that is done in obscurity and in small things, and which is nevertheless what produces the main sum of the nation's strength. And he rejoices at the rise of wages in the immediately preceding period, because the laborers make up the greatest part of society. — Smith's point of view, then, is of an ethical-social nature in his economic work too, even if he lays greater weight on production than on distribution, and even if he assumes too trustfully that the points of view of private economy can without further ado be transferred to those of national economy. In his lectures he treated, under the common name of doctrine of right (jurisprudence), justice, economy (police), public revenue and preparations for war (Lectures p. 3). The third and the fourth part of his course belonged closely together. —

b. When David Hartley is named among Hume's successors, it is not because he was particularly influenced by Hume, but because he has illuminated from a new side questions with which Hume occupied himself. He himself declares that he received the decisive influences from Newton and Locke. He was born near York in 1705. His study was at first theology, but he gave it up, since he could not subscribe to the 39 Articles, and went over to medicine. Early on he conceived the plan of writing a work in which he would work his religious, philosophical and physiological ideas together. The main thought of this work he got from an attempt by a less well-known author, Gay, to explain all higher feelings as having arisen by association of the lower ones. Gay asserted that
[454] Hutcheson, with the concept of "sympathy," had introduced an occult cause (vis occulta), and then attempted to explain compassion as having arisen by the association of ideas. Hartley's work *(Observations on Man, his Frame, his Duty and his Expectations)* appeared in 1749. Hartley worked as a physician and is described as a noble and humane man († 1757). — His significance in the history of psychology rests on his attempt to explain all higher mental phenomena by the association of quite simple sensations and ideas. All thinking and feeling, even the most ideal and exalted, is supposed to have developed in this way, and the power that our will can have over ideas, feelings and actions is likewise owed to association. The laws of association are, according to Hartley, the highest mental laws of nature. Hartley does indeed define association as a connection between ideas that are given simultaneously or in immediate succession (Observations. London 1792. I, p. 66); but he nevertheless also describes the phenomenon that is now called association by resemblance (I, p. 291 f.), just as he also speaks of an immediate

relation of coincidence (coincidence) between ideas, such as appears in the intuitive certainty that is the foundation of all other certainty. — On the physiological side there corresponds to association a connection between different vibrations in the minute parts of the brain, such vibrations uniting by repetition into one vibration. On the mutual relation between ideas and vibrations Hartley will set up no theory; in their ultimate source they are closely connected, but he will join neither spiritualism nor materialism. — By virtue of the law of association mental life develops step by step from lower to higher forms. Thus complex ideas are formed that can make up so complete a unity that the single ideas from which they have arisen are no longer noticed, just as a compound matter can have other qualities than its elements. Further, activities that were at first carried out with full consciousness will by repetition become unconscious, or become what Hartley calls secondarily automatic activities. Finally, the life and strength with which certain ideas appear can later pass over to other ideas that are connected with them by association. It is on these three derived laws that all mental development rests, *[455]* in particular the process by which the primitive, sensuous feelings and expressions of will give place to more ideal expressions. In this way egoism can be worn down and abolished, and a disinterested and universal compassion can develop. On the other hand, envy and cruelty can also develop according to these same laws.

In manner of exposition and in conception Hartley recalls Spinoza. He has carried association psychology a significant step further by setting up the theory of psychical combinations with other qualities than those of the elements, and by bringing out the phenomenon of the displacement of motives. And like Spinoza, Hartley is convinced that by virtue of such psychological laws a transition between the greatest mental opposites is possible: self-preservation can pass over into mystical self-forgetfulness through association and displacement of motives. The love of God arises, it is true, says Hartley, partly from egoistic grounds; but since God is the cause of all things, infinitely many associations will gather in the idea of him, and this idea can thereby become so predominant that the ideas of everything else, ourselves included, vanish in relation to it. — A peculiar union of mysticism and realism is found in Hartley as well as in Spinoza; but he lacks the latter's superior clarity.

Hartley's theory was not much heeded until the natural scientist and theologian Joseph Priestley (1733—1804), famous for his discovery of oxygen, for his struggle against the doctrine of the Trinity and for his enthusiasm for the French Revolution, came forward toward the end of the century as its adherent and popularized it. Priestley himself describes his general standpoint as materialism, but he assumes — following a

theory set up by the Jesuit Boscovich — that the essence of matter consists in force, attractive or repulsive force, and that the atoms are to be conceived as points of force, since this is in reality all that we know of them. Solidity is only a sensible quality, which does not express the essence of matter but only an effect of it on the senses. If this is
[456] how matters stand, there is (as Priestley develops in his *Disquisitions on Matter and Spirit.* London 1777) no reason to assume two different substances; it may very well be the same substance to which both the physical and the psychical forces belong. Priestley is convinced that this conception agrees far better with the original Christian conception than the spiritualistic one, which was taken over from heathen philosophy.

Another eminent natural scientist also took up Hartley's theory, the physician Erasmus Darwin (1731—1802), grandfather of Charles Darwin, a man who had talents as a natural scientist, as a poet and as a philosopher. In his chief work *Zoonomia or the Laws of Organic Life* (London 1794 ff., German trans. by J. D. Brandis. Hannover 1795 ff.) he explains the origin of the instincts by experience and association under the influence of the impulse of self-preservation and of adaptation to circumstances. He goes further than Hartley by laying weight (see especially the remarkable 39th chapter of the 1st part) on the fact that the qualities acquired in this way can be inherited; thereby the psychological doctrine of association is developed into a biological doctrine of evolution, which has a good deal in common with the hypothesis that Lamarck set up some years later concerning the evolution of species, and which, like that hypothesis, is a forerunner of the great hypothesis to which the name Darwin was to be attached.

c. Of quite another kind is the position that Thomas Reid and his successors took up toward the philosophy of Locke and Hume. In opposition to the persevering analysis that was the strength and greatness of the classical English school, but that threatened to undermine the foundation of the secure, popular conception of the world no less than that of the speculative and the religious one, appeal is made here to the judgments of naive consciousness and of sound human understanding, and philosophy is given the task of giving an account of what lies in these judgments and of systematizing their content. Tired of analysis and wary of it, one turns toward the immediate and demands its right. It is a reaction which, in various forms, makes itself felt in the course of the century, as herald of the thoroughgoing reaction against the whole tendency of the
[457] eighteenth century that came at the beginning of the new century, — a reaction that in many respects was fruitful and justified in asserting the right of historical facts, but that at the same time was often highly uncritical when it came to determining what in the single case was given as a real fact. The Scottish school (so the tendency founded by Reid

was called, which can already be traced in Hutcheson and Smith, but which with Reid became dominant at the Scottish universities) has deserved well of the human sciences by its sound, descriptive psychology; but it suffers from the fault of confusing description with explanation and of putting theoretical points of view into the description. —Thomas Reid (1710—1796), the most important man of this tendency, was first a clergyman, later professor at Aberdeen and at Glasgow. In his most considerable work *(Inquiry into the Human Mind on the Principles of Common Sense.* 1764) he relates that he had been an adherent of Locke's and Berkeley's philosophy, but that Hume's "Treatise" had shown him that it leads to dangerous consequences: to the overthrow of all science, all religion, all virtue, all sound human understanding (common sense)! He therefore subjected the whole tendency to a criticism and discovered that it conflicted with experience; it is also on the ground of experience that he wants to combat it. He wants to follow the method of Bacon and Newton. — What Reid presented in shorter form in his "Inquiry" he later carried out, with application to the special psychological domains, in his larger works: *Essays on the Intellectual Powers of Mind* (1785) and *Essays on the Active Powers of Mind* (1788).

At the root of all our knowledge there lie, according to Reid, certain instinctive presuppositions that no doubt can shake. They are the same that irresistibly guide the assumptions and conduct of all men in the ordinary circumstances of life. They are the principles of sound human understanding (principles of common sense), which are older than all philosophy and therefore have greater authority than it. They are components of our original constitution, which stems from the hand of God.

To these original principles belong first and foremost the belief in a material external world and in the existence of the soul. Every sensation that I receive awakens by a natural *[458]* prompting (natural suggestion) the assumption of an external object of the sensation, as well as of a self that has the sensation. Between sense perception and memory on the one hand and imagination on the other there is not merely a difference of degree but a qualitative difference, which cannot be explained further. The belief in what is perceived or remembered is a simple act of consciousness that cannot be described further, any more than it can be explained by association. It is certainly a remarkable trait of our nature that we face here: by a kind of natural magic (by a natural kind of magic) the sensation calls forth for our consciousness something that we have never experienced, and which we nevertheless at once apprehend and believe in! — Related to perception by the senses (perception, which besides sensation also includes that natural suggestion) is the causal instinct, a natural but inexplicable propensity to believe that the connections

between phenomena that we have observed in the past will also continue in the future. On this principle all science is built, as well as all superstition; it may be called the inductive principle. All sound human understanding acknowledges it, and we would put in the madhouse anyone who did not acknowledge it. — In the practical domain we have the moral sense as an example of an original principle. "Just as we," says Reid (Active Powers, p. 238), "by the external senses obtain not only the original ideas of the various qualities of bodies, but also the original judgments that this body has this quality, that one another quality, so by our moral faculty we obtain both the original ideas of good and evil in action, of worth and unworthiness, and the original judgments that this conduct is good, that one evil, this character worthy, that one unworthy." The testimony of nature makes itself known through the moral sense in a manner similar to that in which it does through the external senses. The first principles, in the theoretical as in the practical domain, have an intuitive certainty (intuitive evidence) that cannot be resisted when the mind is mature, calm and unprejudiced. —

[459] Reid and the Scottish school mark a reaction against Hume similar to the one that the Cambridge school in its time marked against Hobbes. It is characteristic that this reaction now itself appeals, not to eternal truths, but to the testimony of experience. It believes, to be sure, and this is its great misunderstanding, that it can read eternal truths straight off from the simplest observations, and it assumes as many original instincts as there are enigmatic phenomena. A very easy manner of explanation! Moreover, it extends intuition far beyond what Descartes, Locke and Leibniz had called by that name: the simple apprehension of likeness or difference. The most complicated acts of consciousness are declared intuitive and are thereby regarded as justified. This tendency was bound to lead to a halt in all inquiry and analysis. And it does not really touch Hume at all, since he had himself declared that in practice he embraced "the principles of sound human understanding," but only found extraordinarily great difficulties in grounding them. Reid's justification against Hume lay in the fact that Hume overlooked the original and constant connection between the sensations, which is just as much a fact as the single sensations each by itself. But he cut himself off from a scientific refutation of Hume by his appeal to Common Sense, which was an abandonment of all philosophy.

After Hume there came in general a long ebb in the history of English philosophy, which is quite understandable. As already remarked, the line of thought that had been entered upon in the English school was now carried through; the line had run out. And on the other hand, partly the political, partly the religious interest now pressed forward into the front rank: politics again laid claim to eminent powers, after people had long

lived on the after-effects of the Revolution of 1688, and the religious feeling, which latitudinarianism and deism had damped down, now flared up in Methodism, a strong and deep popular reaction against the preceding rationalist period. But these conditions cannot be illuminated more closely here. An excellent characterization of them is given by Leslie Stephen in his work: *History of English Thought in the 18th Century.*

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Fifth Book

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The Philosophy of the French Enlightenment and Rousseau.

General Characterization

Through David Hume philosophical thinking had been brought to a point where further progress was possible only by the adoption of entirely new points of view. The answer to the question Hume posed in the "Treatise" of 1739 came only with Kant's "Kritik der reinen Vernunft" of 1781. The interval lying between these two great works is significant in the history of thought above all because philosophical thinking was made into a weapon of battle. The peaceful labor of thought was interrupted when the call to arms was sounded, the call to a spiritual battle against the traditions and institutions of the past, which the impatient human spirit now called to account. In the domain of material nature the new science had led to a magnificent result, magnificent not only in its wide horizon but also in its exact grounding. In the domain of mind one had, in the principle set up by the English school, that the validity of every idea must be proved by its origin in experience, a means of testing all ideas for which respect was demanded. And in the social and political domain one had, in the fundamental propositions of natural law that had been formulated on the basis of the struggles of the sixteenth and seventeenth centuries, a standard which one boldly set up against the principle of authority that had hitherto upheld the social order. Had one then finished the investigation, so that one now had the right to proceed to practical application? One *[461]* certainly had no other right than that which life's own mighty need gives. The solitary thinkers would hardly have concluded; but their time was over for the present. Some years before Hume went over to France in order to work out his chief work there in greater solitude, two young Frenchmen had visited England, not, however, in order to ripen their thoughts in solitude, but on the contrary to open mind and senses to the

English life of the spirit and to take into themselves the living movement that the healthy English soil had fostered. Voltaire and Montesquieu brought home from England, as from an entirely foreign world, entirely new notions of philosophy, religion, church and state. It was a traveler's acquaintance that they made with English thoughts and conditions, and their purpose was not to keep what they had learned to themselves in order to develop it further, but to let it go out into the world in as free, light and simple a form as possible so as to awaken, provoke and transform. And for this purpose one used not only the method and results of English inquiry: the whole development of thought from the Renaissance through the great systems to the philosophy of experience supplied means for waging the battle. Behind the violent and radical criticism that was now applied to everything handed down lay a dogmatism that believed itself essentially complete as regards conception of the world and view of life. One had to stand firm oneself in order to be able to overthrow what existed. When one wants to settle accounts all at once, as in an instant, with everything that history has produced in the spiritual and the social domain, one must have great confidence in one's own standpoint. But so it goes everywhere in history: when the day of judgment is to be held, there is no time to wait until all the premises are in order. Christianity and the Church did not wait to destroy ancient culture until they could give a full grounding of their superiority. Ancient culture was shattered. Spiritual currents are ruthless. When a spiritual need has long been dammed up, there comes a point where it breaks through, however much of value it may destroy on its way. That the Church had not really overcome ancient culture
[462] became clear in the Renaissance. And that the philosophical and political revolution at the end of the eighteenth century did not really overcome the old order of things became clear in the Romantic reaction at the beginning of the nineteenth century. In reality one had not finished philosophizing in the chambers when one began to philosophize, first in the salons and then in the streets and lanes. But the philosophical incompleteness of the weapons of thought with which one took the field takes nothing away from the significance of the historical mission of the men who opened the battle. Where life is at stake, one must use the weapons one has. And although one does not always get the impression of intellectual superiority from the French philosophers of the middle of the eighteenth century, indeed, although by their eager striving to simplify and popularize they often came to make the great small and to profane what was truly exalted, there nevertheless lay, even behind their dogmatism, their shortsightedness and their pettiness, a flaming faith in light, in progress, in humanity, for whose sake many sins may be forgiven them. What Voltaire exclaims in his memorial words on a young, early departed,

noble thinker (Vauvenargues): "Whence did you get that mighty flight in this century of pettiness?" — this can serve as the motto of the whole movement. Particularly when one treats the French philosophy of the eighteenth century not in a history of literature and culture but in a history of philosophy, the strong subjective undercurrent must not be forgotten; what the surface current presents may often seem petty and insignificant. Yet — there was one in whose deeply stirred mind the subjective undercurrent became all but sole master, so that he gained the power to plead, in a way it had never been done before, the cause of immediate feeling, of individual mood, the cause of nature against culture and reflection, the cause of the great, simple, human values of life against every kind of aristocracy and refinement. In Jean Jacques Rousseau a new element of life, a new side of the life of the spirit, came forward and set itself in opposition both to the new critical wisdom and to the old order of things. Deeper down than any relation of authority and any critical reflection reaches, he pointed out living human feeling as that whose assent *[463]* no tradition and no thinking can do without, and as that which is the foundation of all value of life. The significance of feeling had never before been asserted in this simplicity and purity, however often mystical tendencies had pleaded its cause. Not so much by his ideas and theories as by his very personality (despite all its frailties) does Rousseau mark a great turning point in philosophical respect: with him the problem of valuation becomes decisive, whereas the problem of existence and the problem of knowledge had hitherto all but dominated everything. And in turning both against the old order of things and against the new criticism, he proves that there is something that neither of them allows to come into its own. Of the philosophers of the eighteenth century he is the one who most decidedly points toward the future and has its seeds within him. He is, after all, also the thinker of his time who takes the greatest interest in the child.

That the radical consequences — both in their negative and in their positive form — were drawn precisely in France and not in England itself, where the decisive principles had come forward, had its good historical reason. Taine has shown in a brilliant way (in "L'ancien régime") how the French language and the French spirit had been polished at court, in the salons and in classical art into organs for simple and clear ideas, for a thinking that wants to discuss the greatest problems with the least possible preparation, and that does not strive to extend its domain but only wants to work through what has been won. Yet the classical spirit and the polished language would not be a sufficient explanation. They explain only the form, not the content and the direction. The real cause of the practical consequences of English philosophy being drawn in France and not in England is to be sought in this, that the old order of things on French soil stood far

more rigid and stubborn against the new than in England, and that at the same time it was far more hollow and rotten than on English soil. The English social order had absorbed the new forces and was able to use them as awakeners to continued development. In
[464] France the old social order was so consistently carried through that it either had to crush all criticism or be crushed by it.

1. Voltaire and Montesquieu

In his *Lettres sur les Anglais,* which were written 1728—1730 and appeared in Amsterdam in 1735 (after they had already appeared in 1732 in an English translation in London), Voltaire (b. 1694, d. 1778) set before his countrymen a new natural science, a new philosophy and a new social order. He set Locke and Newton against Descartes, Socinians, Quakers and other Dissenters against Catholics, parliamentary constitution against absolute monarchy. Even where he did not directly allude to a comparison but meant to remain purely descriptive, the meaning was clear enough, and no one mistook it; the book was burned. Many Frenchmen, moreover, regarded Voltaire (as later Montesquieu) as a bad patriot because he admired the English character. But those letters mark a turning point in the history of culture: the grafting of English thoughts onto the Continent. Later Voltaire published a good popular exposition of Newton's physical theories (*Eléments de la philosophie de Newton,* mis à la portée de tout le monde. 1738), which contributed much to winning the new natural philosophy diffusion and victory. Voltaire has significance not as an independent thinker but as the great abridger and popularizer. The title of one of his most important philosophical writings: *Dictionnaire philosophique portatif* (1764) is characteristic in this respect. The views are ready-made; it is only a matter of shaping them so that they can easily be carried about. He praises Locke because he has given us the history of the soul, whereas so many "raisonneurs" had previously given us its romance, and because he derives everything that is in the understanding from sensation. But he is not disinclined to go further and regard all memory and thinking as merely continued and altered sensation; when Condillac (who will be discussed later) had set up this doctrine, Voltaire subscribed to what "the great philosopher" had said (Art. Sensation
[465] in Dict. phil.). But if everything comes from the senses, what can we then know of the eternal and infinite and of the nature of the soul? As good as nothing. Peculiar to Voltaire — in contrast both to Condillac and to Locke — is the skeptical result he extracts from the proposition that everything in our consciousness comes from sensation. Officially, to be sure, he assures us, with an irony there is no mistaking, that revelation has taught

us about the soul as a spiritual substance. But his conviction of "the limits of the human mind" (see Art. "Bornes de l'esprit humain" in Dict. phil.) finds expression in the words with which he ends the article on the soul in his philosophical Dictionnaire: "O man, God has given you understanding to guide you to what is right, not to penetrate into the essence of created things!"

That everything comes from sensation, Voltaire nevertheless has no doubt. But sensation itself? Locke had said, to Voltaire's great delight, that there was surely nothing to prevent God from having given matter the capacity to sense. That the communication of this capacity could, according to Locke, take place only by a *miraculous* act, Voltaire did not notice; he seizes on the remark and uses it as a weapon against the spiritualists. Here, however, there is also expressed an endeavor in his system to trace everything back to two principles: God and matter. He then had no use for minds as intermediate links. But what, then, is matter? Well, that we really know just as little as we know what soul is. Yet Voltaire does not doubt its existence, and he even ascribes to it an eternal existence: "No fundamental proposition," he says (Art. "Matière" in Dict. phil.), "is more generally recognized than this, that *nothing comes from nothing*. The opposite is surely inconceivable." "Perhaps my reason is led astray by this old and generally accepted idea [that nothing comes from nothing]; but it tells me that matter must be eternal, since it exists; if it existed yesterday, it has existed before. I see no probability that it should have begun to come into being, nor any cause why it should have come into being at one moment rather than at another." (Le philosophe ignorant. 1766. p. 32). — And he seeks to show that religion does not suffer from this assumption of an eternal matter that is ordered in a purposive manner by the Deity. "To be sure, we now *[466]* happily know by means of faith that God drew matter forth out of nothing, but" etc. The assumption of an eternal matter also presents various difficulties, it is true, and one must not flatter oneself that one can overcome them; philosophy cannot give an explanation of everything. But the belief in an eternal matter ordered by God presents no moral misgivings: we have the same obligations whether we have an ordered or a created chaos under our feet! — Voltaire maintains the slight significance of speculative and dogmatic decisions in practical and moral respect. It is perhaps on this point that he has his greatest significance in philosophical respect, and here too his skepticism proves to be different from mere blasé indifference. Of the practical earnestness Voltaire could show when it mattered, his appearance as defender of the oppressed and the innocently condemned bears sufficient witness. He showed in deed that even if it was irony or malice when he so often referred to what revelation taught, his whole will was in it when

he set practice in opposition to speculation. Toward the belief in immortality his attitude seems to have been that it was necessary for upholding morality in humanity. (Le bien commun de tous les hommes demande qu'on croye l'âme immortelle. Lettres sur les Anglais. XIII). For the belief in God he nevertheless sought (although here too he relied on "le bien commun")[83]) to give a grounding, relying (like Newton) on the purposive order in nature. He defends the assumption of final causes (causes finales); only one must keep to the effects that are invariably the same at all times and everywhere, not to the special and derived effects. Voltaire emphasizes sharply and with teasing relish the dark sides of the world, and he combats the optimism which, with its gaze turned on the great laws that govern the whole, overlooks or disdains the sufferings and misfortunes that this general order of things may bring to the individual beings. The destruction of Lisbon by earthquake gave him, as is well known, special occasion for mockery of the
[467] best of all possible worlds. He strikes Leibniz's optimistic system right in the heart when he has his Candide (as he was being flogged at an auto-da-fé that was held as a festival of atonement after the destruction of Lisbon!) say to himself: "If this is the best world, what then are the others like?" Voltaire was indignant that Leibniz, Shaftesbury and, after them, Pope in verse declared that all was good, and that one could not demand that God should change his eternal laws for the sake of so wretched a being as man. "One must surely admit," he says (Art. "Tout est bien" in Dict. phil.), "that this miserable being has the right to cry out in all humility and — all the while it cries out — to seek to understand why those eternal laws have not come into being for the sake of each single individual!" For Voltaire himself the answer to this last question was given in his doctrine of God and matter, which was perhaps motivated precisely by reflection on the evils that the lawful order of nature brings to individual beings. He believes in a good God, but the suffering in the world is to him a testimony that this God is not almighty but constantly has a resistance to overcome, and this resistance he finds in eternal matter. The moral side of religion was to him the essential thing. "After our holy religion," he says, beginning with his usual official bow (Art. "Religion" in Dict. phil.), "which is quite certainly the only good religion, which, pray, is then the least bad? Should it not be the simplest? Should it not be the one that teaches much morality and very few dogmas? the one that strives to make men just without making them absurd?" Everything in religion that goes beyond the worship of a highest being he declares to be superstition. The thought of simplicity, which the thinkers of the Renaissance applied to the understanding of the natural world, is now applied by the thinkers of the Enlightenment to the supernatural. Over against positive religion Voltaire's criticism keeps to the fully finished dogmas;

about their origin, about their springing from the human need of feeling, about the symbolic value they might have as expressions of dearly bought mental experiences, he does not ask. He amuses himself over the capriciously arranged islands in the ocean of religion and does not consider that what thus appears scattered, contradictory and *[468]* irregular on the surface may have its subterranean connection and perhaps be the effect of volcanic forces in the depths. The unconscious or half-conscious, the involuntary and spontaneous, the obscure movement of feeling he did not understand. He saw everything as in clear daylight. Dawn and twilight were for him signs of ambiguity, insanity, stupidity, absurdity or knavery. Against such things he operates with the alternative that one must be either mad or a knave (fou — fripon) to believe them. What produces superstition is the madness and stupidity of fanaticism, which can rise to insolence (insolente imbécillité); but then the knaves come and make use of the follies for their own ends[84]). An elementary psychology of religion that undeniably satisfies the principle of simplicity and is in a high degree "portable"!

Voltaire was convinced that the moment had come when accounts could be settled with all prejudices. It was not his meaning that all prejudices were bad: there are those that are approved by reason (Art. "Préjugés" in Dict. phil.). But of the sufficiency of reason, of what was then called reason, he had no doubt. He rejoiced (in his letters to d'Alembert) that "the age of reason" had come. On every point he saw the power of the Church give way, in the consciousness of the educated as in the state. To be sure — there was a limit to the spread of enlightenment: "the rabble," "the cobblers and the servant girls" could not have a share in it. But Voltaire consoles himself: "It is not a question of preventing our lackeys from going to Mass or to sermon; it is a question of tearing the fathers of families from the tyranny of the impostors and of spreading the spirit of tolerance"[85]). He had no inkling that the lackeys too would soon begin to philosophize; he had, after all, himself made philosophy "portable." Beyond the dualism between "decent people" (honnêtes gens) and "the rabble" (canaille) he did not see. Here again he kept to the salient oppositions without being able to find the connection, often hidden, to be sure. Yet it must not be forgotten that the gulf between the different parts of the people was, just in his time, as great and yawning as we no longer know it, and *[469]* that Voltaire saw the solution in the care of an enlightened absolutism for the great multitude. —

Montesquieu (1689—1755) had essential qualities that Voltaire lacked, especially that of thinking in such a way that things were seen in their definite connection and not torn loose for the sake of the point. He rises high above the usual line of thought of

the eighteenth century by the idea that underlies his chief work *(Esprit des lois* 1748): that institutions and laws are not arbitrary productions but presuppose definite natural conditions in order to subsist and to act. He emphasizes the connection of the laws with climate, manners and way of life, with religion and with the whole national character. "The laws must," he says (Esprit des lois, I, 3), "be so peculiar to the people for which they are written that it is a pure accident when the laws of one people can suit another." The more recent historical school has in Montesquieu one of its most important predecessors. — But it is nevertheless his endeavor to reach general ideas through the study of historical conditions. On several points he is able, precisely by carrying through his historical method, to prove the sophistry in the defense that had been made of traditional abuses, as in the investigation of personal and political freedom. And his admiration for antiquity and for England leads him to give ideal descriptions that became so many condemnations of the existing conditions in France. As after his return home he transformed his garden into an English park, so his political ideas too acquired an English coloring. And here it should be well noted that in his famous description of the English constitution (Esprit des lois, XI, 6) he builds rather on his study of Locke's book "Of Civil Government" than on more deeply searching investigations of English constitutional life itself in its connection with the history and manners of the nation. Locke's treatise was, after all, a theoretical defense of the Revolution; it was not to describe what historically existed, but to ground the right to develop what historically existed further in a definite direction. Montesquieu now goes even further than Locke, in that he develops, far more sharply than the latter, the theory of the threefold division of power into legislative, judicial
[470] and executive. He did not see that this threefold division in fact fitted only tolerably a definite stage of English history, precisely the time just after the Revolution. Earlier the king had had a greater share in the legislative power, and later Parliament has acquired an ever greater share of the executive power[86]). And yet Montesquieu regarded this system as primordially Germanic: "this beautiful system was invented in the woods." He knew it only as fully finished and as an external form of the people's political life; the whole foundation of self-government in the smaller circles that supports the English parliamentary constitution he did not see. Just as Voltaire took the results of Locke and Newton without taking along the spirit, the investigator's mind, out of which they had grown, so Montesquieu took the form of the English constitution and systematized it without taking along the firm subsoil that in reality supported it. Seen from this side he is akin to the French philosophy of the Enlightenment and belongs in its ranks. One notices in him too "the classical spirit." Albert Sorel says aptly of him: "He does not follow

governments in their historical development . . . He presents them as fixed, complete, definitive . . . No chronology and no perspective; everything lies in the same plane. It is the unity of time, place and action, transferred from the theater to legislation." — It was not, however, Montesquieu's meaning that English forms could be transferred to France without further ado. His thoughts went rather in the direction of a revival of the old French monarchy with an enlightened official class and especially with an independent judiciary. But the spirit of his time put tendencies into his work that conflicted with his own historical method; and especially in his readers this spirit was so powerful that his work acquired a revolutionary significance he had not intended.

2. Condillac and Helvétius

The epistemological and psychological foundation of the French philosophy of the Enlightenment was worked out by Condillac (1715 — 1780), a peaceful thinker who, *[471]* after early becoming a clergyman and after having directed the education of an Italian prince, spent his last years in an abbey whose revenues he enjoyed. In his chief work *(Traité des sensations,* 1754) he carries through, with clarity and breadth and more by a deductive than by an inductive path, the proposition that everything in our consciousness, not merely — as Locke had taught — all content, but also all activities and forms, was merely the transformation of simple passive sensations (sensations transformées). The single sense-sensation appears as a purely passive change in the soul. When this change is so strong that other sensations are excluded, attention arises. When we have two sensations simultaneously, we are said to compare and judge. Memory is only the after-effect of a sensation. Abstracting is the separating out of a sensation from other sensations. The condition of all comparison, judging, remembering and abstraction is attention, which is thus the fundamental phenomenon in knowledge. With its help we are able to conceive distinctly and separately what at first appears in a chaotic conception of the whole. On such separate and thereby distinct conception of what first appears in chaotic mixture — thus on analysis — rests all knowledge. When with the help of language we form signs for each of the elements that analysis presents, we can, by combining these signs and by substituting for one another signs that mean the same, develop a logical calculus, a scientific language of complete clarity and exactness. Only with the help of a language of signs can analysis be completed; but analysis in turn reacts on the language of signs, and all science is really a system of signs whose meanings and mutual relations are completely clear. (Cf. Condillac's work: *La langue des calculs.*

Introduction). Languages are analytical methods, since man involuntarily analyzes, led by his need to make gestures or break out in cries in the face of what arouses his attention. Later, then, conversely, science, exact analysis, becomes the same as a perfect language.

[472] Condillac's theory of knowledge is an attempt to derive everything from experience in such a way that experience itself is conceived as completely passive, but also in such a way that the possibility of a science on the ground of experience is nevertheless shown. By its clarity and simplicity his doctrine won great adherence, and it displaced Cartesianism in the official institutions of learning. It became the philosophy that was taught at the end of the century, during the Revolution and the Empire, until new tendencies asserted themselves. In "Traité des sensations" Condillac employs the mode of exposition of imagining a statue whose senses awaken to activity one at a time; there arises one sensation at a time, and it is investigated how the life of consciousness successively develops and, under the influence of the impressions, acquires its various faculties (attention, comparison, memory etc.), which thus are not original dispositions in nature. Characteristic especially is his attempt to describe attention as a purely passive state and his description of comparison as simultaneous attention to two impressions. — The successive arising of the various faculties takes place under the influence of man's drives and wants. Drive and want arise through comparison of feelings of pleasure and pain, which are a special kind of sensation. Instead of equipping us with a multitude of original faculties, the Author of nature has given us the feeling of pleasure and pain to arouse attention and thereby set analysis going, first in its simpler, then in its higher forms.

Although Condillac's doctrine, seen from one essential side, stands in sharp opposition to the Cartesian, it has points of contact with it not only in the weight laid on the analytical method but also in the spiritualistic point of view that ultimately underlies it. For Condillac maintains decidedly that sensation is different from motion, and that the latter can therefore only be the occasional cause of the arising of sensation in a soul different from the body. To be sure, we know (concedes Condillac, like Locke) the substance of the soul just as little as that of the body. But the capacity to compare (according to Condillac: to have two sensations at once) nevertheless presupposes a *simple* *[473]* substance as the bearer of the sensations. (See especially *De l'art de raisonner* I, 3 and *Discours préliminaire du cours d'étude,* Art. 4). Moreover, Condillac's doctrine concerns only our sensations; these are only the "statue's" own states; if the first sensation is the smell of a rose, then the soul is only the smell of a rose and nothing else. Extension, which according to Condillac derives only from the sense of touch, originally not at all

from the sense of sight, is only the most constant element in our sensations, that around which the others gather; but has one therefore the right to regard it as the expression of absolute reality? Extension is a sensation that is aroused in us, as are other sensations, by something whose essence we do not know. (Traité des sensations, I, 1; II, 11; IV, 5). Condillac betrays the influence of Berkeley ("Barclai," as he calls him), in any case of his "Theory of Vision," whereas it is doubtful whether he knew his chief writings; his doctrine of abstraction does not suggest it.

We have already heard Voltaire refer to Condillac as the great philosopher. As can happen in wartime, the results of the peaceful thinker were used as weapons of war. His doctrine of the transformation of sensation was taken up and his spiritualism left aside. In direct cooperation with the free tendency and with growing sympathy for it, on the other hand, Helvétius (1715 — 1771) wrote his works *(De l'esprit,* 1758. — *De l'homme,* 1773 — 74). The book De l'esprit was treated as one of the most impious books that have ever been written. It was condemned by the Archbishop of Paris, by the Pope and by the Parlement of Paris, and Helvétius, who had expected literary fame from the book, had to stay abroad for a time, among other places for a while with Frederick the Great, who set great store by his unselfish character and after his death wrote to d'Alembert: "With infinite pain I have learned of his passing. His character seemed to me admirable. Perhaps one might have wished that he had consulted his heart rather than his mind." Helvétius was a liberal-minded, mild-thinking and philanthropic man, who used the wealth he had won as a farmer-general in the service of literature and of charity. With great sorrow and indignation he followed (as one sees especially from the posthumous *[474]* work De l'homme) the degradation and inner dissolution of his fatherland. His writings are borne by the conviction of the unhappy consequences of individuals being excluded from active participation in public life. Characters then do not get sufficiently great motives and objects for action. Both literature and morality suffer from it. For both talent and virtue, "mind" (esprit) and "rectitude" (probité), are conditioned and developed by the form of government and by the education that is in turn determined by the form of government. This is the great, serious chief thought that lies behind Helvétius's works. He continues Condillac's theory of the development of all faculties through experience and external influence. Even love of ourselves is acquired: for we cannot feel love without first having felt pleasure and pain. Susceptibility to pleasure and pain is thus the only gift that nature originally gave us. It arouses and sharpens attention, and it determines our action. What comes to fix our attention depends on education — this word taken in the very widest sense, so that it designates everything in our surroundings

and circumstances that has an influence on our development, down to the smallest trifles. Since Helvétius takes the word education in this wide sense, he can maintain that two human beings never receive quite the same education (De l'esprit, III, 1). Endowment he assumes to be the same in all — but the conditions for its development are different. All differences of character Helvétius derives from education, and since this in turn is bound up with public conditions and the form of government, one sees how his theory motivates his sorrow and indignation at the internal conditions of France.

The corruption of manners consists not so much in the excesses to which individuals give themselves up as in the general split between individual and social interest. It is hypocrisy when the moralists attack the private vices of individual persons instead of directing their attack against public abuses and against oppression from above. Morality, the art of legislation and pedagogy are not three different endeavors but rest on the same principle, namely the principle of public utility, the utility for the greatest possible
[475] number of human beings who are united into a state. (De l'esprit, II, 17). What is recognized as good and as virtue (probité) depends on interest. Most people recognize as good only what agrees with their own limited interest. Only comparatively few possess — especially under bad social and political conditions — the enlightened pride, the spiritual nobility, to be able to pass their moral judgments from consideration of public utility without being led astray by the narrow interests of themselves or of their nearest surroundings. Such nobility of soul by no means presupposes the giving up of love of oneself. Such a giving up is an impossibility. But it presupposes that one's own interest is inseparably bound up with the public interest, especially through the need of power and honor. "No one," says Helvétius (De l'homme, V, 1), "has ever contributed to the public good to his own detriment. The citizen who heroically risks his life to cover himself with honor, to deserve public esteem and to free his fatherland from slavery, follows the feeling that for him is bound up with the greatest pleasure.... The good man thus obeys only a noble interest." — It remains unclear in Helvétius whether the need of power and honor does not become of a somewhat different nature when it is inseparably tied to consideration for "bien public" than when it stands isolated and therefore seeks satisfaction by any means whatever. The expression nobility of soul (noblesse d'âme) seems to presuppose that the feeling must after all undergo a certain metamorphosis when — through an excellent education — one has succeeded in connecting the representation of justice so closely with the representation of power and happiness that they fuse completely into one single representation (Note 33 to De l'homme, Sect. 4). What is lacking in Helvétius (as in Condillac) is a closer investigation

of the way in which the feelings are altered under the influence of representations and combinations of representations, investigations that Spinoza, Hume and Hartley had begun.

A chief proposition in Helvétius is that we can recognize only ideas and traits of character that agree with our own ideas and feelings. Mind (esprit — in the sense of capacity for new combinations of thought) we recognize only where the new combi- *[476]* nations of thought we encounter have a certain analogy with our own thoughts and agree with our own interests. And what holds for the recognition of mind holds also for the recognition of rectitude (probité). Without our own interest being set in motion, we recognize nothing. But — says Helvétius — there are two kinds of interest: "There are men who are animated by a noble and enlightened pride, and who keep their mind in a suspended state (état de suspension) that gives new truths free entry. To their number belong some philosophical minds and some young people who have not yet formed opinions so fixed that they blush at changing them . . . . There are other men — and they are the majority — who are animated by a less noble vanity. These can recognize only such ideas as agree with their own, and that justify the high opinion they all have of the ability of their minds" (De l'esprit, II, 3. Cf. De l'homme, IV, 6: "Genius has as its protector and eulogist youth and a few enlightened and upright men"). — Yet besides ignoble vanity Helvétius also names indolence as a hindrance to the recognition of new ideas and new virtues. Both hindrances can, in his view, only be overcome by a sufficiently strong feeling tearing man out of his torpor and his limitation. — Both of Helvétius's works (whose tendency has often been misunderstood) investigate the conditions for the arising of great minds and significant traits of character and for their recognition once they have arisen, and the fundamental condition for him is the close connection of the individual life with the public one. His theories of the original uniformity of individuals, of self-interest as the fundamental thing and of the omnipotence of education all serve to impress one and the same fundamental thought.

Helvétius's religious standpoint is the deistic, with strong emphasis on the unknowability of the Deity. He polemicizes against theological morality because it merely attacks private vices and not the source of evil in public conditions, and against the priesthood because its interest cannot go together with that of the whole people. There runs, in general, a polemical undercurrent through his books — even stronger in De l'homme *[477]* than in De l'esprit, which Diderot nevertheless was not wrong to call a club-blow against all prejudices.

3. La Mettrie, Diderot and Holbach

Already in Voltaire, natural science and its results play an essential role alongside the philosophy of experience. But the influence of natural science comes forward first and foremost in a group of French philosophers who, it is true, avail themselves of Locke's doctrine that everything in consciousness comes from sensation, but whose chief thoughts nevertheless concern material nature. Their philosophy is to be a simple consequence of natural science. They take up again Hobbes's materialistic tendency, but in such a way that they attempt to proceed inductively, whereas Hobbes proceeded predominantly deductively. Their appearance is a testimony that the method and results of natural science demanded, with increasing energy, influence on the view of life and the world-view. And the more impossible it was for them to explain to themselves or to justify the ecclesiastical dogmas and the existing institutions on the basis of the fundamental propositions of natural science, which stood for them as eternal truths, the more these dogmas and institutions had to appear to them as purely arbitrary, as sprung from stupidity, fanaticism or fraud. Voltaire's dilemma, madness or knavery, was here carried out to its ultimate consequences and was hurled forth with a defiance and a passionateness that Voltaire, who does not leave the atmosphere of the court and the salons, had not known. And yet these writers too are clear that their ideas cannot find entry with the great multitude.

The revolutionary tendency in these writers, their conviction that one must reform heaven before one can reform earth, does not prevent their works from having philosophical value. They combat the spiritualistic doctrine, not, it is true, with arguments of principle going beyond what Hobbes and Spinoza had provided, but with a greater
[478] wealth of particular experiences. They rely not merely, like the thinkers of the seventeenth century, on the general principles of mechanical natural science, but also on the results that the science of organic life had produced in the last century. Particularly to be named here is La Mettrie (1709—1751), the real founder of French materialism in the eighteenth century. He was an army physician, but lost his post partly because of his attacks on the prevailing medicine, partly because of his *Histoire naturelle de l'âme* (1745), where he first expressed his views. For a time he now stayed in Holland; but from here he had to flee because of the offense his *L'homme machine* (1748) aroused. He now found a refuge with Frederick the Great, whose reader he became and with whom he seemed to have stood on a very intimate footing. He died suddenly in 1751; evil tongues (especially Voltaire's) said that his death was due to his gourmandise, but there seems to have been a case of poisoning. La Mettrie's character appeals by its boldness and a

certain amiable lightheartedness. But he could also be impudent instead of bold, and frivolous instead of lighthearted, and his philosophical reputation has had to suffer from these unfortunate sides of his character, which are undeniably noticeable in his writings too. In recent times especially Albert Lange in his "Geschichte des Materialismus" and Dubois Reymond in his ingenious lecture on La Mettrie have paved the way for a juster judgment.

La Mettrie wants to build on experience and observation and to apply the comparative method. As a natural scientist he is a pupil of the famous physician Boerhaave, who applied the fundamental propositions of the Cartesian natural philosophy to the study of organic life. The title of La Mettrie's well-known work "L'homme machine" points back to Descartes and could very well have come from the older Cartesian school, if only the soul-substance that now and then touches the pineal gland is pushed out. By the path of the comparative method La Mettrie seeks to show that there is only a difference of degree between animal and man, and that there is therefore no ground for assuming that in man an entirely new kind of substance is added. In order not to misunderstand La Mettrie one *[479]* must note that he has not only written an "L'homme machine" but also a work with the title: *Les animaux plus que machines.* His task is, by bringing out the kinship in structure and mode of operation between the human and the animal organism, to prove that what acts in them is not different in essence. And he goes still further. He emphasizes not only the similarities between man and animal but also the similarities between man and plant (*L'homme plante).* And from the whole graded series that the study of the structure and functions of living beings shows us, he then draws the conclusion that if there is mental life above, it cannot be wholly lacking below. The capacity for sensation he extends to everything living, indeed to everything material: everything in the universe is full of souls, and he refers here (Les animaux plus que machines. Oeuvres philos. Berlin 1755. II, p. 82) to Leibniz's slumbering monads. — This graded series does not, for La Mettrie, consist of forms finished once and for all. He assumes — following partly the naturalist Maillet, whom Voltaire mocked, partly Epicurus and Lucretius — a development from lower to higher forms. As it seems *(Système d' Épicure* § 13. 32—33, 39), he assumed eternal organic germs from which, in interaction with the external conditions, the various forms of life have emerged with progressive perfection. The driving force is need, the wants. A higher mental life arises only when wants other than the purely vegetative appear. Plants need no real soul. The transitional forms between plants and animals have all the more intelligence the more they are obliged to move in order to find their food. And man stands highest because he has the most wants (L'homme plante). — It is

an interesting anticipation of the idea of the struggle for existence that meets us here. And when La Mettrie sets up the proposition: "Beings without wants are also without mind," it is not impossible that he thereby gave Helvétius the motive for one of his chief thoughts.

This whole side of La Mettrie's doctrine set forth so far has its interest quite apart
[480] from the materialistic views that he draws from it. The necessity of the materialistic consequence lies for him in the fact that we see nothing around us but matter, under constantly changing forms. He admits, to be sure, that we do not know the real essence of matter. But we know its properties, namely: extension, motion and sensation. That sensation is a property of matter he derives from experience, which shows us that certain organic states are always accompanied by sensation, and he appeals here particularly to comparative investigation, which shows us mental life varying with organization. If the soul is not material, i.e. extended, how then can it be explained that enthusiasm makes us warm, and that conversely the heat of fever affects the representations? (L'homme machine. Oeuvres, III, p. 75 f.). All our thoughts must be material modifications. And since a great multitude of thoughts is in fact contained in our brain, these must be extraordinarily small in order to be able to "lodge" there *(Traité de l'âme,* p. 103: De la petitesse des idées)! — La Mettrie regards the assumption of a soul-substance as a useless and self-contradictory hypothesis; he wants to content himself with what anatomy and physiology teach us — and he believes that he does so. He maintains that the question of the truth of materialism must be decided purely theoretically. The loftiness of our mind does not depend on whether we can use of it the meaningless word incorporeality, but on its strength, its range, its clarity. It need not then blush at having been born in the mire. La Mettrie has rightly seen here that materialism is compatible with practical idealism. His own moral philosophy, insofar as he has one, is to be sure far from having this stamp. The standard of justice is the welfare of society, he maintains; but this standard he really uses only to distinguish between debauchery (débauche), the feeling of pleasure that brings harm to society, and enjoyment (volupté), the feeling of pleasure that does not harm others. When he has made this distinction, he keeps to enjoyment and gives detailed instructions concerning it *(L'art de jouir),* instructions in which sentimentality
[481] and sensuality work together in a way that arouses disgust.

Denis Diderot (1713—1784) marks at once the high point of philosophical thinking and of revolutionary spirit in the French literature of the eighteenth century. He was a warm-hearted man with a great capacity for entering into subjects and circumstances of the most diverse kind, a French Leibniz, who was able to find life everywhere and

had an eye for peculiarities, but also a need to find the connection in the whole. The poet in him often strove with the thinker; many of his thoughts express only the mood of the moment with crude one-sidedness, without his having sought to find the thread that could connect them with other thoughts which it was nevertheless also his will to maintain. A cynical delight in the coarse and the vulgar could alternate in him with the most exalted ideas and moods. And in his works sentimental effusions stand side by side with brilliant developments of thought. His literary fate has been strange. By his contemporaries he was known only as a writer of comedies, as the author of some writings on the philosophy of nature in a deistic direction and as the editor of the great Encyclopedia, in which consideration for the censorship forced on him the role of a smuggler when he wanted to bring out his own ideas. The work on the Encyclopedia took most of his strength and time. With incredible endurance he carried this giant work through despite all opposition. His many-sided interest showed itself here in a magnificent way, since he wrote not only a series of philosophical articles but also articles on crafts and factories, which rested on thorough study of his own. This work marks a turning point in the history of culture by bringing enlightenment and knowledge out into wide circles. But Diderot did not bring out his real views here. They must be sought in his correspondence (especially in the letters to Mademoiselle Voland, which depict the radical salons of Paris in a way both interesting and delightful) and in his dialogues (especially *Conversation with d'Alembert* and *d'Alembert's Dream),* which were written in 1769 but only appeared in 1830 (under the title: Mémoires, correspondances et ouvrages inédits). Only through these works has one come to know the real Diderot. We keep here to his purely philosophical ideas, all the more since our literature received a few years ago a fine and detailed account of Diderot (Knud Ipsen: *Diderot, hans Liv og hans Gerning*. Copenhagen 1891). *[482]*

In his first writings Diderot still uses, in the manner of popular theology, the representation of God to explain what science seemed unable to explain. But more and more it dawns on him that such an explanation is something quite other than what science understands by explanation. In his work *Interprétation de la nature* (1754) he maintains that the true method is an interplay of observation and thinking, induction and deduction, in which one goes from experience through reason back to experience. At the same time he here takes up again, in a more realistic spirit, questions he had already touched on earlier, namely the great question of how the purposive phenomena of nature are to be explained. Instead of appealing to a cause outside nature, he states the possibility that in the chaos of matter there have been from eternity elements with the

capacity for life and consciousness, and that these elements have gradually been brought together and, through many stages of development, have become animals and men, the combinations and formations that were not viable being successively eliminated. Like La Mettrie, Diderot seems to have come to such ideas partly through reading Lucretius, partly by carrying through Leibniz's doctrine of continuity in the graded series of the monads and connecting it with the investigations of contemporary naturalists into the small organisms. Even if it would be wrong to present Diderot as a mere disciple of Leibniz[87]), the influence of the founder of the doctrine of monads is nevertheless clear in him. He may well have got the idea of motion as an original property of matter from Toland, and the conception of the state of rest as a state of force or tension (nisus) (in the little treatise *Principes philosophiques sur la matière et le mouvement)* may well derive from others than Leibniz. But when Diderot, in his brilliant dialogue "Conversation between d'Alembert and Diderot", not only, like La Mettrie and several of the naturalists of the time, conceives the capacity for sensation, sensibility, as a general and essential
[483] property of matter, but, in working out this thought more closely, introduces a distinction between potential and actual sensibility (sensibilité inerte — sensibilité active) in analogy with the distinction between "dead" and "living" force, one will hardly be able to avoid finding an influence of one of Leibniz's most significant thoughts. For Diderot life and mind are eternal in nature. They are never mere products or results of purely mechanical processes. Their germs are present from the first, and the only question is whether the conditions of development are present. The transition from the potential to the actual form takes place, after all, every time the organism converts food into blood and nerves! However much weight is to be laid on the external conditions, the internal, original conditions are nevertheless for Diderot the essential thing. To suppose that by adding one, two or three dead particles to a dead particle one should be able to form a living system out of them, that is surely a very great absurdity! A rearrangement of particles should be able to produce consciousness! No, what has life and consciousness has always had them and will always have them. Why should not the whole of nature be of the same character? The difference between the lower and the higher levels is then only this: what exists on the higher levels in concentrated form is on the lower levels spread out in a multiplicity of elements. (Lettre à Mlle Voland, 15 octobre 1759). It is this thought that is developed more fully in "Conversation between d'Alembert and Diderot" and in "d'Alembert's Dream", two dialogues that both in form and in content belong among the classic works of philosophy.

d'Alembert makes an objection that is not answered. Even if one ascribes original

sensibility (in potential form) to the particles of matter, how can a consciousness then arise through the joining together of such particles, a consciousness that is not to have its seat in any of these particles but to correspond to the connection among them all? "Listen, philosopher," says d'Alembert, "I do see the aggregate, the tissue of small sensitive beings, but a living being, a whole, a system, a self with consciousness of its unity! That I do not see, no, that I do not see!" — It is thus held fast that here — in the arising *[484]* of consciousness — there lies a problem that is not solved by assuming sensibility as a general property of matter. Diderot thus acknowledges that the monad cannot be explained by mechanical means: it is impossible to derive the unity that characterizes consciousness by way of mechanical composition. — Diderot does not hold fast to the clear insight that thus comes to light in him by flashes. Memory and comparison follow for him (as for Condillac) as a matter of course from the single sensations. Nevertheless that objection shows how keenly Diderot has thought the question through. He does not go by the beaten paths of materialism.

The difficulty of explaining the origin of conscious individuality recedes into the background for Diderot all the more easily because all finite individuals, by reason of the inner connection among them, make up one great whole: "Do you not grant that everything hangs together in nature, and that it is impossible that there can be a gap in the chain? What do you mean, then, by your individuals? There are no individuals, none at all. There is only one single great individual: the whole!" Thus he goes from the one limit of our knowledge to the other, from individuality to totality. He stops for a moment in wonder at what he sees at these limits; but his thought is too much in oscillation for the problems to have time to settle.

A similar oscillation is felt in Diderot's ethical ideas. He takes a critical stand toward Helvétius's attempt to explain all feelings as springing from the striving after one's own advantage, and holds that there is in human nature a foundation for the recognition of the right and the good, a foundation that can never wholly disappear in any human being. With great enthusiasm he defends this assertion in Holbach's salon, and in his earlier writings (especially in an adaptation of one of Shaftesbury's books) he assumed a special moral sense. From his letters it is clearly seen that he has come away from this last assumption. What is called a moral instinct and shows itself in our involuntary actions or judgments of actions is in reality the result of an infinite number of small experiences that begin with the very beginning of life. A multitude of different motives make themselves *[485]* felt here, but none of them needs to come to our consciousness at the moment we judge or act. Everything in us stems from experiences (tout est expérimental en nous); but we

need not become conscious of these experiences (Lettres à Mlle Voland, 2. Sept. 1762 and 4. Oct. 1767). Among these unconscious or forgotten motives there are also some that concern ourselves, especially the need of honor and of an immortal posthumous fame (on the significance of this last motive Diderot carries on a lengthy correspondence with the sculptor Falconet). There are in Diderot hints toward an investigation of the history of the development of moral feeling, and the interesting thing is that his great enthusiasm for "the motives of great and noble spirits" is not weakened by seeing these motives as fruits of development on the ground of experience. Yet he cannot carry this conception through, especially not when he stands face to face with the precepts of current social ethics. It is only in the individual that he finds a natural development, not in society. Existing society offered rules and institutions that seemed to him so unreasonable, especially when he compared them with what was told of the life of savage peoples, that he could only imagine them as having arisen through superstition and lust for power. Between the man of nature and civilized man he could find no natural intermediate links. How could a morality contrary to nature have arisen except through the cunning and arbitrariness of the rulers? "Examine all political, civil and religious institutions closely: unless I am much mistaken, you will find that the human race has for centuries been bowed under the yoke laid upon it by a band of scoundrels. Beware of all who would introduce order! To order is always to make oneself master of others by putting obstacles in their way". *(Supplément au voyage de Bourgainville).* The age's lack of capacity to find historical connection has hardly ever been expressed more clearly. Even Diderot, with his great capacity for living and feeling his way into different conditions, could find no other explanation than arbitrariness and lust for power. The distance between the craving and longing of individuals and what society offered, between the
[486] forces they felt in themselves and the narrow frame the existing order gave them for using these forces, was too great for understanding to be possible. One could hope for everything from the future, but one could find nothing in the past — except villainies. — Religious faith in particular stood for Diderot merely as a source of pernicious effects. It is more than jest when he says in a letter that he would rather lay the blame on gods than on men. Belief in a God is — he maintains — no harmless thing; it begets evil in two respects. In the first place it inevitably brings with it a cult, and ceremonies and theological dogmas soon take the place of natural morality and distort its precepts. In the second place the great amount of suffering in the world conflicts with the idea of a good God; one is then forced into every possible absurdity and self-contradiction in order to gloss over this fact. Reason suffers as morality suffers (Lettres à Mlle Voland, 20.

Oct. 1760 and 6. Oct. 1765).

In Holbach's circle Diderot often discussed such questions. From his letters we get a vivid picture of this circle. Holbach (1723 — 1789) was a German baron who had settled early in Paris, where he gathered about him some of the most radical writers. He occupied himself with chemistry and (probably under the influence of Diderot) also came to philosophical studies. There is hardly any doubt that what Diderot said during his frequent visits to Holbach forms the foundation of the most important of the latter's works, the work that has been called the Bible of materialism: *Système de la nature,* which appeared pseudonymously in 1770. It is also stated in the preface that the author has had help from several friends. Besides Diderot, the mathematician Lagrange, who was tutor in Holbach's house, is thought to have collaborated. What Holbach himself contributed is essentially the systematization. For in the book materialism is set out as a system, a task that neither La Mettrie nor Diderot, who were spirits partly more searching, partly more playful, had attempted. Of new thoughts it really offers none. Its significance lies in the energy and indignation with which every spiritualist and dualist assumption is pursued as harmful both in a practical and in a theoretical respect. *[487]*

Only then — Holbach argues — could it be necessary to assume spiritual causes, a God over against the world, a soul over against the body, if material nature were dead, passive, incapable of moving itself. But when there is now no rest in nature, when motion is a fundamental property of matter, what need is there then of spiritual causes? And besides, they explain nothing, but the invoking of them is only an expression of our ignorance: one inserts the soul or God where one cannot find the natural cause. It is in reality a remnant of the savage's explanation of the phenomena of nature by the intervention of spirits. Every time science has attempted a natural explanation, theology has fought for the supernatural explanation. Of the mental we know nothing but that it is a property attached to the brain. If it is asked how the brain has been able to acquire this property, the answer runs: "It is the result of an arrangement or combination peculiar to the living being, by which a lifeless and senseless matter ceases to be lifeless and acquires the capacity to sense by being taken up into the living being. So it goes with milk, bread and wine when they are taken up into the human organism". (Système de la nature, I, p. 105). The comparison we know from Diderot; but Holbach, so far as can be seen, does not embrace with Diderot the assumption of sensibility as a fundamental property of matter; he does mention it, but himself keeps to the conception that sensibility arises through the combination of elements each of which by itself does not possess that capacity. There is on this point a difference between Holbach and Diderot that carries further-reaching

consequences. For Holbach declares thinking (consciousness) itself to be a motion, not indeed a motion of masses but a molecular motion of the same kind as those that underlie fermentation, nutrition and growth, motions that are not visible but which we infer from their effects. (I, p. 15). There is much that is enigmatic about this motion, as about other
[488] similar ones; but this enigma does not disappear by assuming a spiritual substance. "Let us rest content with the knowledge that the soul is moved and altered by the material causes that act upon it. That gives us the right to conclude that all its activities and faculties prove that it is material". (I, p. 118). All science is then physics. Ethics itself is only applied physics. In place of the imaginary virtues and duties that were derived from man's relation to beings outside nature there now comes a derivation of virtues and duties within man's own nature. From the laws of nature it follows which paths we must take in order to attain our purposes. In this way the concept of duty springs from the concept of nature; for duty means precisely the path we must necessarily take in order to reach our goal (II, p. 229). Each individual seeks his happiness; but reason, "which is nothing but natural science applied to the conduct of man within society" (II. p. 186), teaches him that he cannot become happy if he will separate his happiness from that of others, — such, once and for all, is the will of destiny (ordre du destin), — and that on the other hand he who makes men happy cannot himself be unhappy. In the last chapter of the work, which contains "an outline of the natural law" (Abrégé du code de la nature), these thoughts are developed in an enthusiastic language in which it has been thought that Diderot's spirit can be recognized.

The assumption of a twofold kind of beings, spiritual and material, is according to Holbach the source of the greatest evils for mankind. From the first, men doubtless came involuntarily into the belief in spirits; they did not know the causes of things and therefore quite naturally conceived them as personal beings. But it was the priests' lust for power that first made the belief in spirits into a system. And in particular these priests saw well what power the mysterious has over men. Therefore they led them from belief in visible gods (the sun and other objects of nature) to belief in invisible gods. Spiritualism is theoretically so groundless that there can be no doubt "that this system is the fruit of a very profound and very interested policy on the part of the theologians" (I, 97): there was now, after all, an invisible part of man that could be rewarded or punished in another world! The theologians are the true manufacturers of gods (fabricateurs de la
[489] divinité). "When we will go back to the beginning, we shall always find that ignorance and fear have created the gods; fancy, enthusiasm and deceit have adorned or distorted them, weakness worships them, credulity keeps them alive, custom respects them, and

tyranny upholds them in order to profit by the blindness of men". In this utterance (II, p. 200) Holbach sums up his philosophy of religion. He does not distinguish between natural and positive religion. Once one engages with the representation of gods, a cult becomes necessary; thereby the priests gain power, natural morality is distorted, and persecution sets in. At the same time brooding over how evil can be reconciled with the existence of God will lead to sophistical speculations. — Holbach's reasoning here quite recalls Diderot's, who probably had an influence on the composition of this section. — In religion as in society one had historical facts for which one could find no natural basis and to which one at the same time felt oneself in irreconcilable opposition. One then resorted to the theory of arbitrariness, which seemed so simple. It is indeed in reality so simple that it recalls the mythological theories of the childhood of the human race, which the System of Nature meant precisely to root out. With his cunning priests Holbach introduces a new kind of mythical beings.

He touches on a great problem: what value have religious representations when they can no longer, as in the childhood of the human race, be used to fill the gaps in our knowledge? But an answer to this question the philosophy of the Enlightenment, because of its imperfect understanding of the forces of religious life, could not give. If the world of the gods were really merely pasted on or foisted upon humanity from without, if it had not sprung from the forces and needs of the human spirit itself, then there exists no real religious problem. A new generation and a new time were needed to take this question up. — Only Rousseau, whose solitude in his own age was brought about by his feeling its sting before anyone else, tries his strength on it.

But before we go on to portray this remarkable man, it must still be remarked of the Système de la nature that a certain doubleness makes itself felt in the concept of nature *[490]* that it takes as its foundation. On the one hand nature is defined as a being that exists in itself, is its own cause, as an eternal substance, — thus with reminiscences of Spinoza. But on the other hand (and yet often in the same breath, e.g. II, p. 189) it is granted that we know only the causes that act in experience, not the first causes. Here the concept of nature is grounded on experience, there on construction in thought. But what validity has it in either of these forms? How far can experience lead us? What justification have constructions in thought? None of these questions are raised. The philosophy of the Enlightenment does not doubt its own enlightenment; it exists, after all, to enlighten the others! — And yet it was on a discussion of such questions that the future of philosophy depended. But the spiritual atmosphere in France was not calm enough to undertake it. In a distant German university town, on the other hand, it was already in full swing

when the Système de la nature appeared.

4. Jean-Jacques Rousseau

Rousseau's great achievement is to have brought to light the deep foundation from which judgment could be passed between the philosophy of the Enlightenment and the existing order. He sets up nature and immediate feeling as the foundation of all valuation and demands of all culture and reflection that would go beyond this foundation that they prove their justification by developing and perfecting it; if they hinder and stunt it, they are of evil. For the first time in modern times he raises *the problem of culture*. From the time of the Renaissance the work of culture had gone boldly forward in every domain: now a voice was heard asserting that there was something wrong with the whole movement. What gave occasion to Rousseau's first philosophical writing was a prize question set by a French provincial academy, which called for an answer to the question "whether the renewal of the sciences and the arts had contributed to the refinement of manners". The question struck like lightning into Rousseau's soul. It was
[491] to him, he says, as if he saw a wholly new world, and as if he himself had become a new man! The new world was the world of personality, of living feeling, of the inner life. He saw at once the disproportion between this world on the one side and the existing order together with its critics on the other. It is indifferent whether it was his thought from the start to answer the question with No, or whether Diderot's daughter is right when she relates that it was her father who advised him to answer in the negative. The negative answer is only a formal paradox, which Rousseau more closely limits and explains in his correspondence and in his later writings. But the violent collision between the feeling that in the end is what values everything and the whole brilliant intellectual and aesthetic development, that could come only in Rousseau; it was prepared by his personality and by the course of his life. Jean Jacques Rousseau was born in Geneva on 28 June 1712 of a fairly well-to-do citizen family. His upbringing was of a kind that favored fancy and indefinite feeling; there developed early in him what he himself calls a romantic spirit (esprit romanesque), which led him to crave beyond the present and to find his greatest enjoyments in the world of possibilities. After a vagabond youth, which had the same romantic character that his fancy had already assumed, but which also for a time brought him into circumstances that were degrading and that had no favorable influence on his way of thinking, he went to Paris, where he made his way as a teacher, as a private secretary, and later chiefly as a copyist of music. His copying was for him what the

grinding of lenses was for Spinoza. He had a success with an opera and took the field for Italian music as against French. Nothing suggested that he would win a reputation in the world of thought and come to inaugurate wholly new tendencies in the spiritual life. He had earlier had occasion to acquire a certain general education, and had also occupied himself with philosophical reading. As a contributor to the great Encyclopedia he came into the circle of the "philosophers" and was regarded as one of them. Through his answer to the prize question mentioned he was led into the career of an author. His name *[492]* became known, and as he went on treating the problem that had so suddenly dawned on him, he came more and more into a polemical position toward both the two great contending powers of the time, toward those who attacked as well as toward those who defended the existing order. A new prize question ("On the Origin of Inequality among Men") gave him occasion for a detailed eulogy of the state of nature at the expense of civilization. As he had before denied that the renewal of the arts and sciences had been a good, so he now denied that the exchange of the state of nature for social and civilized life had been a good. The two prize essays mark the negative side of Rousseau's appearance. A new period in his life began when he could withdraw into rural solitude. His encyclopedist friends did not understand him. Something was fermenting in him for which they, with all their criticism and wit, had no sense. His feeling for nature and his enthusiasm for nature, which sustained him in the last part of his life amid great bodily sufferings and amid real and imagined persecutions, were closely bound up with his inclination to live in immediate feeling. His inner life was so overflowing that he could find words only with difficulty for what was stirring in him. At any rate he could seldom find the words at the right moment; therefore he felt ill at ease in the salons. Against the pointed, the refined, the articulated, in which the culture of the mind had ended, he set the plain, the broad, the indefinite. He took an interest in the chaotic beginning of life, in the elements that had not yet formed any distinctly shaped world. In his moods of freedom in the woods at St. Germain or Montmorency he thought he was living again the life of primitive man, before culture had abolished the happy immediacy. The simple and elementary, the great and simple relations of life, were for him venerable and full of sources of joy. He understood what was stirring in the rearguard of the army of civilization, among those whose enlightenment the Encyclopedists gave up as hopeless and whom Voltaire called "la canaille". He found in immediate feeling what can be common even though the life of representations is ever so differently developed. *[493]* His own representations were the children of his feeling. In his "Confessions" he says of himself that he felt before he thought. And in his highest moments, moments such

as he had them on his solitary walks in nature, there was in his mind such a swelling and fullness, such a resonance of countless indefinite representations, such a craving beyond all bounds, that no representation, no image could express what he felt. The independence of feeling he experienced not only through such states, in which it was as good as sole ruler, but also through its influence on the representations. His expectation of the future was dark, his memory of the past bright; the content was determined by the nature of the mood. He has thus learned from his own experience the psychological truth that feeling is just as original and independent a side of mental life as knowledge, and that it by no means merely behaves passively and receptively toward knowledge. The dark side of Rousseau's character shows itself partly as sentimentality, partly as a suspiciousness that rose to madness. A sinister example of how feeling can drive representations forward and form an intricate system of them in order to find expression and explanation is offered by one of Rousseau's last writings: *Rousseau juge de Jean Jacques,* in which he describes the systematic persecution on the part of his former friends of which he believes himself the object. The same systematic constructing that is here undertaken on the basis of a morbid feeling is found also in his famous works: his philosophizing is in reality, as he himself says of his religious faith, only an interpretation of his feeling (exposition du sentiment).

In three works Rousseau has developed the conception of life that he sets in opposition to the refinement and corruption of culture. In *La nouvelle Héloïse* (1761) he portrayed strong and deep love, the beauty and dignity of marriage and family life, the nobility of resignation, the inwardness of religious faith, the glory of nature — subjects that had become foreign to the time, and that drew from him tones that sounded new in the ears of the time and marked a great turning point in poetic literature. In *Émile* (1762) he set
[494] forth a system of education that does not subdue nature but makes itself the servant of nature's development, and he finds occasion besides to develop his religious views. In *Contrat social* (1762), finally, he sets forth his doctrine of the social order that ought to take the place of the tyranny under which humanity groaned.

With these works Rousseau regarded his task as ended. But he was not allowed to cultivate his life in nature in peace. "Émile" was burned in Paris, and a decree of arrest was issued against Rousseau. Now his unhappy years began. He fled to Switzerland; but not even there did he find rest; in Geneva and Bern the governments would not have him, and in Neufchâtel he suffered ill-treatment from the population on account of his religious opinions. He then accepted Hume's offer of an asylum in England, but in his morbid state soon conceived suspicion of his friends there and fled back again to France,

where he wandered about from place to place until a sudden attack of illness ended his life (1778).

To find out Rousseau's real position on the problem of culture, one must not stop at his paradoxical utterances in the two prize essays, but must connect them with various utterances in letters and with the content of his later writings. There then emerges, I believe, a surprisingly clear and significant line of thought, which betrays a deep psychological understanding of the conditions of man's spiritual life.

The state of nature, out of which man is torn by culture, he by no means portrays in ideal colors. He does indeed reject Hobbes's portrayal of the state of nature as a war of all against all; he holds that war presupposes partly needs that man does not have in the state of nature, partly also a greater intercourse among men than was possible; and he holds besides that Hobbes has overlooked pity, which is an involuntary and original human feeling. But the state of nature is for him a purely instinctive state, whose advantage over the civilized state rests on there being a relation of equilibrium between needs and the capacity to satisfy them. Man is guided by the impulse of self-preservation (amour de soi) and easily finds means of fulfilling it. Emotion, fancy and reflection play no part. It is only *[495]* life in common and culture that awaken the capacity for comparison and reflection. Now the state of equilibrium is abolished. The impulse of self-preservation is satisfied at the expense of others and thereby becomes egoism (amour propre)[88]). Representations arise of goods that cannot be attained, and the disproportion between needs and capacities thereby comes into force. One now begins to reflect on the worth and meaning of life instead of following instinct. Weariness of life and suicide, which were not known in the state of nature, now become frequent. Anxiety about the future and about death replaces secure tranquillity. Doubt, that unbearable state, takes the place of the happy thoughtlessness of the state of nature.

Yet according to Rousseau's conception these dissolving and unhappy effects do not set in at once. When he is to name which stage in the development of humanity is the happiest, he does not point to the most primitive state but to a state of incipient social life and civilization, which lies "midway between the carefreeness of the primitive state and the heated bustle of our self-love". This time was the world's real youth, which ought not to have been left behind. Here reflection and refinement have not yet exerted their dissolving influence, and on the other hand instinct has begun to give way to thoughts and feelings[89]). The most dangerous stage is the third; here the dissolving forces spread. It is the stage of corrupt and over-refined culture.

When Rousseau was asked whether it was then his opinion that one should return to

that state which lies so near the primitive, his answer is that it is impossible, as impossible as becoming a child again[90]). One cannot abolish error by going back to the state of ignorance. Once instinct has been replaced by reflection, there is nothing else to do but to put a true and natural knowledge in place of the false. "When men are corrupt, it is better that they be learned than ignorant", says Rousseau in a letter (to Scheib, 15 July 1756), and with this he in reality gives the Enlightenment a recognition, for it implies
[496] that intellectual culture does not increase corruption but rather diminishes it. He wants a culture that does not divide and weaken feeling and strength, a social life that does not make us turned outward and dependent. Only in his own inner self, he says, does man find peace. Only he has lived much who has felt life. But culture had brought doubt, slackness, outwardness and restlessness. And it had also, through the division of labor, brought bondage: social misfortune arose when the one saw that he could use the other's labor and used the superfluous store he could gather to maintain him. All these evils had to be remedied through a new ordering of life for the individual and for society.

There were three domains in which Rousseau's ideas of "the new world" moved: those of education, of religion and of the state. In "Émile" and in "Contrat social" he has expressed these ideas. They roused resistance from the adherents of the old and scorn from those of the new. His books were burned by the authorities, Voltaire called him an arch-madman (archifou), Diderot "the great sophist". Quite naturally: for neither of the parties had gone so deep to the bottom of the problem as Rousseau, none had felt its innermost sting as he had. The adherents of the existing order relied on a supernatural sanction, and the Encyclopedists believed that they themselves had brought forth the true enlightenment and made it accessible to those to whom it was worth the trouble to give it. Why, then, these paradoxes and these plans for the future?

In his *theory of education* Rousseau maintains that nature must be allowed its free course, and that the educator's art must consist in removing obstacles and procuring the most favorable conditions possible for capacities and drives to unfold according to their own nature. One must not try to force culture on from without, either by way of authority or by way of enlightenment. Childhood has its purpose in itself, as every other period of life does, and is not to be treated as mere preparation. One does not know the child, for one has hitherto regarded it only from the standpoint of adults. It has a natural right to free self-development. Therefore do not swaddle it, let it have its own mother's milk, do not spoil it, let it follow its instinct of self-preservation and make its
[497] own experiences, do not preach to it, do not teach it what it can learn for itself, do not awaken wishes and needs in it before it can have them satisfied! — The longer one can

put off mental development, the better. What matters is that the mind be full-grown, so that it shall not languish by using its powers too early. If possible, bring the child to its twelfth year without its being able to tell its right hand from its left: the more quickly will the eyes of reason open in it when the time for that has come! Let it grow up without prejudices, without habits and without knowledge! Otherwise one does not come to know the child's own peculiar nature either. The germ of character must have time to show itself, so that one can then take it in the right way. And need and drive must have time to arise, so that one shall not awaken aversion by forcing on it what cannot be appreciated. — Education, in other words, is to be negative, not positive. "Positive I call", says Rousseau (in *Lettre à M. de Beaumont,* the magnificent defense of "Émile"), "the education that aims at forming the mind before its time and at giving the child knowledge of the duties of the man. Negative I call the education that aims at perfecting the organs, the instruments of our knowledge, before it gives us knowledge itself, and that prepares for reason by exercising the senses. Negative education is not inactive; far from it; it does not give virtues, but it forestalls vices; it does not teach truth, but it guards against error; it prepares the child for everything that can lead it to the true when it becomes able to understand the true, and to the good when it becomes able to love the good". This concept of negative education expresses very strikingly Rousseau's real fundamental thought. Culture is to be the unfolding of nature, not a shell, a heavy form forced on from without. Only that culture is genuine which is nature itself at a higher stage of development. Therefore he urges caution in "introducing" culture. Rousseau has a deeper eye for the peculiar character of the life of the spirit than the adherents both of authority and of "enlightenment". Only what has developed naturally and what has been produced by its own activity is genuine for him. Only through a development of culture of a genuine kind can one leave the secure and carefree time of instinct and yet *[498]* preserve the harmony between need and capacity, between thought and feeling, between the outer and the inner. Like so many mystics he has a fine sense for the conditions of spiritual growth. The very framing of a concept such as that of negative education betrays this sense and has a Socratic character. Rousseau nevertheless does not escape being the child of his time. He does not have full confidence in the involuntary use of the powers, for he sets in motion a shrewd educator and a whole system of intrigues and arrangements so that Émile shall not come under any influences other than those that supposedly suit his age. Systematic construction is characteristic of Rousseau in every domain. His living feeling and enthusiasm lead him to see what no one else could see; but it stops at the great grasp: when the thought is to be carried through in detail, it

becomes evident how far he stands from the real world, and instead of letting himself be guided by real experiences on the particular points, he gives free play to his constructive understanding. Even at Émile's falling in love and wedding "le gouverneur" stands in the background and arranges everything — and not always in a negative and indirect way either. —

It accords with the fundamental thought of Rousseau's pedagogy that the child's *religious conception* must not be imparted to it from without but is to be drawn forth from it under the influence of the needs of the heart. In the famous section "The Profession of Faith of the Savoyard Vicar", which is inserted in "Émile", Rousseau gives us his religion. There is nothing new in his religious ideas. It is the representations of "natural religion" or "deism" in which he finds the expression of his feeling. He will not bring any proof of the validity of these representations, but sets them up as dogmas. Matter cannot have its motion from itself, but the first cause of motion must be a personal will. In opposition to Condillac's doctrine the essential difference between sensation and thought is maintained, and in opposition to materialism the difference between mind and matter as two substances. There are several interesting remarks in Rousseau's development of this point, and it is of particular historical interest that a countryman of Rousseau,
[499] Charles Bonnet, at about the same time in his psychological writings (of which the most important is *Essai analytique sur les facultés de l'âme.* Copenhague 1760) criticized Condillac's attempt to trace everything in mental life back to passive sensation, though it cannot be shown that Rousseau knew these writings. In nature there is revealed a striving after purposes, a unity in the whole plan, that bears witness to the working of a personal God. But Rousseau differs here from the usual "watchmaker philosophy" of the time by maintaining that this harmony and purposiveness in nature is not the reason why he believes in God. The relation is precisely the reverse. His faith springs from an immediate need of feeling. Doubt is to him an unbearable state; when his reason wavers, his faith settles the matter on its own account. And only when inner feeling has thus led him to faith can he find in nature indications that point in the same direction as his dogmas. Although "the Savoyard Vicar" constantly seeks to show what is natural and reasonable in his assertions, he nevertheless declares again and again that he will not philosophize or lecture but only set forth what he feels, and he asks his hearer to find confirmation of his assertions in his own feeling. The practical standpoint of feeling is decisive. On the nature of God he will not brood, except in so far as he is compelled to it "by the feeling of the relations in which God stands to him"[91]). But it is above all in a famous letter to Voltaire (18 August 1756), on the occasion of the latter's poem on the

destruction of Lisbon, that Rousseau decidedly maintains the inner, subjective source of faith: he does not believe in God because everything in the world is good, but he finds something good in everything because he believes in God. When in this letter he defends optimism against Voltaire's criticism, he does indeed rely on the fact that in the great, elementary relations of life there are sources of joy and satisfaction that the refined man of the world does not know; he stresses in particular "the delightful feeling of existing" (le doux sentiment de l'existence indépendamment de toute autre sensation). He assumes besides that the evils for the individual may be necessary for the great connection of the world, and in the belief in immortality he has a last consolation. But the real foundation of the optimism that he, in spite of his sickliness and his wretched *[500]* circumstances, maintains against Voltaire, living in honor, wealth and splendor, is his being inwardly and personally gripped by religious feeling; it is this that enables him to cry out to Voltaire: "You enjoy, I hope!" With respect to content the religion of Rousseau and that of Voltaire were not different; "natural religion" was common to them. But what a difference in inwardness and mood! Rousseau has in an epoch-making way moved the religious problem from the domain of the outward consideration and explanation of the world back to inner, personal feeling and to the way in which this is attuned to life. What Pascal was hindered by his Catholic dogmatism from carrying through, Rousseau came far nearer to in spite of his deistic dogmatism. For a dogmatist he is, by the quickness with which he lands in Cartesian spiritualism and in deistic theology. Here, as so often in philosophy, it is not the result but the motivation that is the new and the lastingly valuable thing. And the motivation is closely bound up with the relation of dependence, so decisive for Rousseau's personality and cast of mind, in which representation stands to feeling and need. The "Savoyard Vicar's" last counsel is: "Keep your soul always in such a state that you constantly *wish,* that there is a God; then you will never come to doubt it!"

That Rousseau's dogmas take the definite form they take is not, however, due merely to the way in which he interprets his feeling. He has subjected the representations to a criticism. In a letter (to M***, 15 January 1769) he says that he has thought through the various systems and decided for the one that seemed to him to have the fewest difficulties. Here, then, thinking is granted a reacting influence on the postulates of feeling. When, e.g., the experience of the evils in the world did not shake his faith in a good God, it was because, by virtue of the doctrine of the two substances (mind and matter), he assumed that matter set bounds to the carrying out of the divine purposes. He did not assume a creation, but only an ordering of an already existing, perhaps

[501] eternal matter. God is the good orderer and guide, but not almighty (see the letter just cited and his Lettre à M. de Beaumont). Here too, then, Rousseau comes to the same result as Voltaire, although the two men stand with so very different a mood before the problem of evil in the world. Rousseau's religion is really joy and enthusiasm that a power can be traced in the world which works for the good. He does not believe in the physical effects of prayer. His prayer is an outburst of enthusiasm in which joy in nature at its height[92]) passes over into a hymn of praise to "the great Being" that works everywhere but for which thought can find no concept; at its height it becomes an ecstasy in which no word can express the deep and strong mood, because the mind craves beyond all limits (Third Letter to Malesherbes). The independence of feeling over against knowledge that Rousseau maintains shows itself here in the fact that feeling at its height can find no adequate expression in representations. From here there is only one step to the knowledge that all religious representations, all dogmas, are symbols. This step Rousseau's deistic dogmatism hindered him from taking.

His natural religion, which he believes to be independent of all tradition, he sets in opposition to the positive religions. He is convinced that if one had kept to the voice of the heart there would have been no other religions than this one. But he was convinced that in his religion he had room for the real kernel of Christianity. To the Archbishop of Paris he writes (Lettre à M. de Beaumont): "Monseigneur, I am a Christian, and a Christian at heart, according to the teaching of the Gospel. I am a Christian, not as a disciple of the priests, but as a disciple of Jesus Christ. My Master speculated very little about dogmas but urged duties greatly: he did not prescribe so much articles of faith as good works; he required faith only in what was necessary in order to be good!" The mysteries of Christianity he left alone; they did not concern the decisive thing, which was the moral kernel. When weight was laid on what goes beyond this kernel and on
[502] the definite dogmatic formulations, one ended in unfairness, falsehood, hypocrisy and tyranny. All this dogmatic learning God is supposed to have enjoined — and on top of that to have set the punishments of hell for those who could not become so learned! Only through books, written by men and guaranteed by men, can one come to know these positive dogmas. The Gospel is the most sublime of all books — but it is still a book. Not on the leaves of a book, but in the heart of man has God written his law (Letter to Vernes, 25 March 1758). —

Rousseau had in his early youth let himself be led into going over to Catholicism. Later he returned to Protestantism and took part in its worship, reserving for himself the Protestant layman's right to free examination of all dogmas by the measure of

conscience and the heart. He was naturally branded as a heretic by both churches; by the Encyclopedists he was regarded as a hypocrite or as a muddled head who would probably end as a Capuchin. He himself called himself the only man of the time who believed in God. —

The third domain in which Rousseau believed he descried a new world was *the social and political.* Here too it becomes evident that he did not give up the hope that the stunting and oppression of nature could be replaced by a development that lets nature come into its own. In his "Contrat social" (I, 8) he even speaks very strongly of the progress made by man's giving up the state of nature and entering social life by a (tacit) contract: "While in the social state he must do without several goods that he had from nature, he gains in return others that are very great: his capacities are exercised and developed; his representations are enlarged; his feelings are ennobled; his whole soul rises to such a point that, if the abuses that attend the new conditions did not often degrade him to such a degree that he sank below the level from which he had set out, he would have to bless without ceasing the happy moment that tore him forever from that level and made him a rational being and a man instead of a dull and stupid animal."

In his theory of the philosophy of law Rousseau forms the conclusion of a whole series of attempts that begin in the struggles of the Reformation and aim at constructing *[503]* the state on the basis of free individual transfer. Rousseau did not invent the idea of the social contract, but he uses it in order to be able to present society as having arisen in accordance with nature: in the state of nature men lived (he thought) isolated; the only transition to the life of the state that could make the state natural was therefore free adherence and transfer. Through it the people comes into being as a people, the individual transferring his absolute authority over himself to the people as a whole. The sovereignty of the people is absolute and inalienable. He transfers Hobbes's absolutist concept of sovereignty to the people as a whole. He agrees with Hobbes that sovereignty can lie in one place only, but instead of assuming with Hobbes that through the social contract a submission to the governing power immediately takes place, he proclaims, like Althusius, whom his expressions also recall[93]), the principle of the sovereignty of the people, and assumes that "the general will" takes up into itself the will of all individuals. Like Althusius he distinguishes between form of state and form of government and assumes only one form of state, but various forms of government. The best form of government is an elective aristocracy. But he rejects the division of powers and denies that sovereignty can be exercised through representatives. The highest authority is and remains with the people itself, which must assemble from time to time to give laws. As

soon as the people is assembled, all governmental authority ceases; all magistracy is suspended. The legislative power — as one with the people itself — is sovereign; the executive is its servant. The legislative power is the heart, the executive is the brain: the brain can be paralyzed and the man can still live, if the heart is intact.

The condition for the exercise of the sovereignty of the people is, according to Rousseau, that the whole people can be assembled. Therefore the state must not be too large; Rousseau's ideal was — besides his own native state of Geneva — the small states of antiquity. The only form under which the sovereignty of the people can be preserved in a large state is — says Rousseau — the federal constitution. Rousseau had worked out a treatise on such a constitution — but this treatise was handed over by him to a friend,
[504] who destroyed it at the beginning of the Revolution so that it should not do harm! It was, as Rousseau remarks, "a wholly new subject, whose principles had first to be set up". Just before the French Revolution this subject was treated by Alexander Hamilton, the founder of the North American constitution of the Union, in "The Federalist". The French revolutionaries, who otherwise took many slogans from "Contrat social", paid no heed either to the hint mentioned of how the goods of a large and of a small state, of centralization and of decentralization, can be united, or to the practical attempt that was made in this direction across the ocean[94]). They kept only to that side of Rousseau's doctrine by which the individual lets his will be wholly absorbed in the general will, which proclaims itself through the decisions of the majority and has full disposal over the life and property, education and religious practice of all citizens. As Hobbes had the individual abdicate in favor of the sovereign governing power, Rousseau had the individual abdicate in favor of sovereign democracy.

There is a definite opposition between Rousseau's treatise on the origin of inequality (of 1755) and his "Contrat social" (of 1762), — but an opposition of which he himself was surely aware. On closer consideration it is clear that in the later account he meant to correct and supplement the earlier. In his "Discours" he shows how economic inequality comes with the division of labor, and how only later on the formation of the state by agreement sanctions and fixes the difference between rich and poor. "The laws", he says (Discours, p. 95), "gave the weak new fetters and the rich new forces; they definitively destroyed natural freedom, fixed forever the law of property and inequality, made a cunning usurpation into an irrevocable right, and for the future brought the whole human race under the yoke of labor, bondage and misery". Here, then, the original contract is conceived as a sanction of inequality. In "Contrat social", on the other hand, it is said (I, 9) that "far from destroying natural freedom, the original contract puts a

moral and lawful equality in place of the physical inequality that nature had been able *[505]*
to produce among men, and while they may be unequal in strength and in mind, they now become equal by agreement and by right". In a note Rousseau adds that it is only under bad governments that this equality is merely apparent; he grants that in fact the social state is beneficial to men only when they all have something and when no one has too much. There can hardly be any doubt that by these words Rousseau means to show how the new account can be reconciled with the earlier. He is indeed right, too, that law and right can be used in both ways: both to sanction existing relations of inequality and to level them out. In the "Discours" he keeps to the first side of the matter, which was indeed the one that most struck the eye in the old society; in "Contrat social", which is to portray an ideal and therefore looks away from actual conditions[95]), he keeps to the other side and urges what ought to happen: precisely because the course of things (already — as the "Discours" showed — in the state of nature) always tends to abolish equality, precisely for that reason — it is said in "Contrat social" (II, 11) — the power of legislation must always strive to maintain it, must see to it that no one becomes rich enough to be able to buy another, and no one poor enough to let himself be bought. That there is nothing fantastic in this consideration the whole subsequent time has shown, it having been more and more recognized as an essential task of the power of the state to even out social oppositions. Rousseau was not a socialist, but had a clear eye for the influence of social inequalities and for the drawbacks that go with the system of private property. More than the earlier writers on natural law he has searched beyond the formal, political principles and found the social question behind all constitutional questions. Only to one thing was he, with his whole time, blind: the significance of free associations. Once the state has come into being, it is to regulate everything. Rousseau here comes into conflict with himself. For in "Émile" he lays weight precisely on individual development, on the unfolding of the individual's natural peculiarity: but how can that become possible under the tyranny of the "general will"? Here lies a great problem, and it cannot surprise *[506]*
us that Rousseau has not seen it: he moved, after all, in the midst of the arbitrariness of the old regime, where neither the general nor the individual will could come forward; he wanted to emancipate them both — and he could, after all, quite calmly leave it to the future to reconcile them. He had set his time problems enough. —

In the history of philosophy Rousseau occupies a position that presents a certain analogy with Leibniz's. They both bring forward the problem of valuation, not being content with the mechanical explanation of nature and the mechanical ordering of society, but asking about the significance of nature and of society for human feeling,

the source of all valuation. And in close connection with this there stand in both of them the emphasis on the right of individuality and the sense for the inner infinity of every individual. But the carrying through of these thoughts was undertaken in very different ways by the two so very different spirits. Leibniz sought to keep the scientific view of nature by conceiving the laws it found as expressions of the harmony among the self-unfolding of countless individual beings. The gradual transition from obscurity to clarity was the fundamental law of all development in this harmonious world-picture. Poor Jean Jacques did not find it so easy to find harmony, either in himself or outside himself. In his ecstatic moments he believed in it, he saw it with his mind's eye. But his monad had an obscure depth in it that could not be cleared up but was precisely darkened more and more, and neither in the world nor in society could he get everything to come together into an intellectual harmony. His feeling was too strong to be explained as confused thinking. Therefore he maintained the right of the irrational. By the way of personal experience of life he had ended in a result similar to the one Hume had come to by the way of epistemology: Hume had indeed shown that all our knowledge rested on unprovable presuppositions. There were then, seen both from the practical and from the theoretical side, problems enough to take up for him who was to succeed in carrying philosophy further.

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Notes to the First Volume

[1]) P. 13. Thus, besides older writers, also G. *Spieker*: *Leben und Lehre des Petrus Pomponatius.* *[507]*
München 1868, p. 8 f. See, on the other hand, F. *Fiorentino*: *Pietro Pomponazzi.* Firenze 1868, p. 30. Fiorentino, however, thinks he finds in Pomponazzi's later writings less warmth in the submission to faith, an effect of the continually pursued criticism. — A fuller account of Pomponazzi's psychology is found in his lectures on Aristotle's book on the soul, edited by *Luigi Ferri. (La psicologia di Pietro Pomponazzi secondo un manoscritto inedito.* Roma 1876). — *Roberto* Ardigò regards (see his address on Pomponazzi 1869, reprinted in vol. 1 of his Opere filosofiche 1882) Pomponazzi as a forerunner of modern science, particularly of positivism, through his emphasis on natural causes and on natural, continuous development.

[2]) P. 26. In my essay *Montaignes Betydning i Etikens Historie* (Det nittende Aarhundrede, 1876) I have already developed this conception of Montaigne. — *Starcke*, in his book *Skepticismen som Led i de aandelige Bevægelser siden Reformationen* (Copenhagen 1890), has presented Montaigne essentially from the negative, skeptical side. As I have tried to show in my essay *Tvivlens Historie i nyere Tid* (reprinted in "Mindre Arbejder"), Starcke in his meritorious book surely lays too much weight on the skeptical notions in a series of writers of the sixteenth and seventeenth centuries. — *Dilthey (Auffassung und Analyse des Menschen im 15. und 16. Jahrhundert.* — Archiv für die Geschichte der Philosophie. 1891. p. 647 ff.) stresses, as I do, that Montaigne's skepticism served only as the means of clearing the way for faith in nature.

[3]) P. 27. Qui se presente comme dans un tableau cette grande image de nostre mère nature en son entière majesté; qui lit en son visage une si generale et constante variété; qui se remarque là dedans, et non soy, mais tout un royaume, comme un traict d'une poincte très-delicate, celuy-là seul estime les choses selon leur juste valeur. *Essais* I, 25.

[4]) P. 33. At about the same time as Vives's work there appeared another psychological work *[508]*
which also had great influence on the views of the following age, although it has neither the fundamental significance nor the lasting interest of the former, namely *Melanchton's Liber de anima.* Vitebergæ 1540. Melanchton keeps far closer to Aristotle than Vives; his account is far more theologizing, and he cannot measure himself against the Spanish psychologist in the art of observation and description.

[5]) P. 36. On Luther's shifting positions with regard to philosophy see F. *Bahlow*: *Luthers Stellung zur Philosophie.* Berlin 1891.

[6]) P. 46. O. *Giercke (Johannes Althusius und die Entwickelung der naturrechtlichen Staatstheorien.* Breslau 1880, p. 58) has shown that this doctrine of the ephors derives from the Huguenots, indeed from Calvin himself, who does conceive of the magistracy as a divine ordinance, but adds that where there are authorities to defend the right of the people against the greed and arbitrariness of princes, such as the ephors in Sparta, the tribunes in Rome and the estates in the modern kingdoms, there they have both the duty and the right to oppose the will of the princes when it is unlawful. Cf. also *Lobstein*: *Die Ethik Calvins,* p. 117 f.

[7]) P. 52. *Grotius* grounds the wrongfulness of the lie on a tacit agreement which coincides with the origin of language (De jure belli ac pacis, III, 1, 11). P. C. *Kierkegaard's* theory (in his dissertation *De vi atqve turpitudine mendacii),* which grounds the reprehensibility of the lie on the necessity of preserving trust in the meaning of words, recalls Grotius's doctrine when this is stripped of the form of the mythologizing contract theory.

[8]) P. 54. Cf. *Henry Maine*: International Law. London 1888. p. 23 f.

[9]) P. 57. On the tendency toward a universal theism within the Italian Renaissance cf. *Burckhardt*: *Die Cultur der Renaissance in Italien.* 4. Aufl. II, p. 236 f., 300 ff. — On Franck and Coornhert cf. the interesting characterizations in *Dilthey* in Archiv für Gesch. d. Phil. 1892, p. 389—400; 487—493.

[10]) P. 62. *Autobiography of Edward Lord Herbert of Cherbury.* Edited by Sidney L. Lee. London 1892, p. 248 f.

[11]) P. 73. Böhme's expression is "Auswickelung." Only later (in Tetens in the eighteenth century) does the word "Entwickelung" appear. Cf. *Eucken*: *Geschichte der philosophischen Terminologie.* Leipzig 1870, p. 127.

[12]) P. 76. Aristotle's world-picture is set forth in his work *On the Universe (περὶ οὐρανοῦ).* See especially I, 3. 8, 9; II, 1, 4, 6.

[13]) P. 83. Cf. *De docta ignorantia* II, 2—3. I do not go further into the way in which Cusanus in later writings *(Possest* and *De apice theoriæ)* modifies his ideas, especially by stressing the activity of the highest principle and by conceiving this principle as the unity of possibility and actuality, later especially as power (posse). Cf. on this F. *Fiorentino*: *Il risorgimento filosofico nel*
[509] *Quattrocento.* Napoli 1885, p. 136 ff. — *Axel Herrlin (Studier i Nicolaus af Cues' Filosofi, med särskilt afseende på dens historiska betydelse.* Lund 1892, p. 22) disputes that the work "Possest" marks a different position from the one taken in "De docta ignorantia." But it seems to me that Fiorentino's view is proved by the fact that Cusanus in "Docta ignorantia" (I, 26) expressly declares that the infinite neither produces, is produced, nor proceeds (Infinitas neqve generans, neqve genita, neqve procedens), whereas in "Possest" he lays very great weight on its activity. I grant that this difference must not be stressed too strongly, since Cusanus in all his writings operates with the relation complicatio — explicatio, and so far cannot at any stage disregard acting and proceeding.

[14]) P. 83. *Richard Falckenberg (Grundzüge der Philosophie des Nicolaus Cusanus.* Breslau 1880, p. 54) has pointed out that the relation complicatio — explicatio comes to mean something quite

different when it is applied to the relation of the Deity to the world (where it denotes a transition from something higher to something lower) and when it is applied within the world (where it denotes a transition from lower stages of nature to higher).

[15]) P. 86. When *Ludwig Lange* (in his interesting essay *Die geschichtliche Entwickelung des Bewegungsbegriffes.* — Philosophische Studien. Herausg. von W. Wundt III, p. 350) remarks that for Cusanus, as the forerunner of Copernicus, the thought of the relativity of place and of motion must have lain particularly near at hand, the relation is precisely the reverse: it was the thought of the relativity of place and of motion that made him the forerunner of Copernicus.

[16]) P. 87. Concerning *Paracelsus* (whose real name was Theophrastus Bombastus von Hohenheim) I must refer to Chr. *Sigwart's* excellent characterization (Kleine Schriften, I). Paracelsus was born at Einsiedeln in Switzerland in 1493 and died, after a very restless life, at Salzburg in 1541. On his significance as a chemist and physician see *Ernst* v. *Meyer*: *Geschichte der Chemie.* Leipzig 1889, p. 57—61. *Julius Petersen*: *Hovedmomenter af den medicinske Lægekunsts historiske Udvikling.* Copenhagen 1876, p. 21—24. — *Cardanus* (b. at Pavia 1501, d. at Rome 1576) is more interesting for his autobiography, which is a rarity for its frankness and is one of the most peculiar documents of the Renaissance, than for his writings, of which the most important *(De subtilitate)* is a chaotic mixture of profundity, fantasy and superstition. He ascribes to himself great powers of observation *(De vita propria.* ch. 23) both in science and in the practice of life; indeed, among his scientific merits, which are recounted in a special chapter of the autobiography (De vita propria. Kap. 44: Qvæ in diversis disciplinis digna adinveni), he counts first and foremost, in the domain of natural science, that he has "brought the consideration of natural things back to a definite art and method, which no one before him had attempted." History has not been able to let him keep this honor.

[17]) P. 90. *Fiorentino* has brought to light the manuscripts that contain Patrizzi's remarks on *[510]* Telesio's book and Telesio's replies to them. See the appendices to vol. 2 of his work *Bernardino Telesio. Studi storici su l'idea della natura nel risorgimento italiano.* Firenze 1872.

[18]) P. 93. *De rerum natura,* VII, 2 (cf. V, 7). — In one single place Telesio has given a definition of sensus without inserting perceptio as an intermediate link, explaining sentire, VIII, 21, by: a rerum viribus exile qvid pati. But here an ambiguity lies hidden in the word pati.

[19]) P. 96. There might seem to be a contradiction in setting up wisdom as the universal virtue (virtus universalis, IX, 6) and magnanimity as the total virtue (virtus tota, IX, 22), since the two virtues surely do not coincide. Telesio's meaning is presumably that wisdom is a condition that cooperates in every single virtue (since the task is always to find means of self-preservation), but that magnanimity (which is called the summit of all the virtues, omnium virtutum veluti apex) concludes the series of the virtues as the trait of character in which they are all comprised as in their most perfect form. — I do not go further into Telesio's polemic, springing from misunderstanding, against Aristotle's doctrine of virtue as habit. That a naturalistic thought should lie at the root of this polemic (Fiorentino: Bernardino Telesio I, p. 316 f.) I am unable to

see.

[20]) P. 98. *Joachim Rheticus*: *Narratio prima.* (In the Thorn edition of Copernicus's work *De revolutionibus orbium coelestium* 1873: p. 490). — *Kepler's* pronouncement on Copernicus is quoted by *Reuschle*: *Kepler und die Astronomie.* Frankfurt 1871, p. 119. — *Galilei* on Kepler: *Opere.* Firenze 1842 ff. VII, p. 55.

[21]) P. 101. Copernicus states the principle of simplicity in his first draft *(Commentariolus,* see *Prowe*: *Nicolaus Coppernicus.* Berlin 1883, I, 2, p. 286 ff.), as well as in his great work De *revolutionibus* (I, 10). The principle of relativity: De revol. I, 5, 8.

[22]) P. 101. This was the chief difficulty that kept *Tycho Brahe* from going over wholly to Copernicus. See his letter to Kepler, December 1599. (*Kepleri Opera omnia,* ed. Frisch I, p. 57). — *Galilei* too found here the greatest difficulty in the Copernican system, and in his *Dialogues on the Systems of the World* works to clear it away.

[23]) P. 102. This explanation, which passed from Copernicus to Bruno and Galilei, and which Copernicus himself probably has from Plutarch (in whom it is said in one place that the earth draws heavy things to itself, not because it is a center, but because it is a whole), is — as Galilei's German translator *Emil Strauss (G.s Dialoge.* Uebers. von E. Strauss. Leipzig 1891, p. 499) remarks — "an ancient surrogate for a theory of gravitation."

[24]) P. 104. *Giordano Bruno*: *Cena de le ceneri.* III. (Opere italiane. ed. Lagarde. Gottinga 1888, I, p. 150—152). — *Kepler*: O*pera.* ed. Frisch, I, p. 245 ff.

[511] [25]) P. 105. The chief source for Bruno's biography is the Inquisition's interrogations from his trial in Venice, which were first published by *Domenico Berti* in his biography of Bruno (Turin 1868). Later Berti published fragments of the interrogations in Rome. Chr. *Sigwart* has collected testimonies on Bruno's appearance at German universities (Tübingen Easter program 1880 and Kleine Schriften I). *Dufour (Giordano Bruno à Genève.* Genève 1884) has published documents on Bruno's stay in Geneva. — An article in the Quarterly Review (October 1902), *Giordano Bruno in England*, contains some interesting remarks on the very few traces of Bruno's stay in England that can be shown, apart from his own statements. Thus one of the persons Bruno mentions in the *Cena, Florio*, was a client of the Earl of Southampton (as was Shakespeare). This Florio translated Montaigne's essays into English. — In the poet Spenser the author of the article thinks he finds traces of Bruno's *Spaccio.*

[26]) P. 110. *Berti (Giordano Bruno da Nola.* Nuova edizione. Torino 1889, p. 196 ff.) concurs in the influence of Bruno on Shakespeare, assumed by some German writers, through common acquaintances or through books. Shakespeare came to London only a year after Bruno had left it. As *Robert Beyersdorf (Giordano Bruno und Shakespeare.* Oldenburg 1889) has shown, an influence is nevertheless highly improbable. In particular it seems to me decisive that no trace is found in Shakespeare of the new astronomical ideas whose prophet Bruno was. Here, surely, there would have been something for a Shakespearean imagination! — Beyersdorf shows very finely how differently Bruno and Shakespeare conceive the idea of the constant change of natural

phenomena and their passing over into one another. For the poet there is here a testimony to the nothingness and wretchedness of human existence, whereas for the thinker it is the unity and eternity of nature that reveals itself in its glory in the midst of perishability. — It must, however, be remarked here, as will be shown more fully later, that Bruno also conceives the alternation of opposites from its bitter and painful side. This comes out especially in the work *De gl' heroici furori,* and in this Bruno — despite all his Neoplatonism — is different from Plato.

[27]) P. 116. A friend of Berti's used the revolutionary period of 1849 to take copies of Bruno's trial in Rome, but did not finish. As for the rest of the records of the trial, it was assumed until quite recently that it must be found in the archive of the Inquisition in the Vatican. But to an inquiry from a German Bruno scholar, Dr. *Güttler*, there came from the Vatican the definite declaration that neither the records nor the confiscated manuscripts were in the archive, and that it was not known where they were. See on this Archiv für Gesch. d. Phil. VI, p. 344 f.

[28]) P. 116. *Schoppe*, a German scholar who had gone over to Catholicism and was an eyewitness of Bruno's condemnation and death, names, in his description glowing with hatred of heresy, among Bruno's "horrible and utterly absurd doctrines" the assumption of innumerable worlds. — When the Inquisition, on the Pope's order, put it to Bruno that the eight propositions extracted *[512]* from his writings had already been regarded as heretical by the most ancient Church, it is quite possible that the assumption of several worlds may have been counted among these old heresies. Later, of course, Galilei had to abjure the doctrine of the motion of the earth as conflicting with Holy Scripture. — Among the Protestants too the assumption of several worlds was heretical. *Melanchton* polemizes against it as an ungodly doctrine, since Christ cannot have died and risen several times. *(Initia doctrinæ physicæ.* Corp. Ref. XIII, p. 220 ff.) — *Campanella*, on the other hand, seeks in his *Apologia pro Galilæo* (Francofurti 1620, p. 9, 51) to establish that this doctrine is not heretical. — Cf. on the probable 8 counts of the indictment: F. *Tocco*: *Giordano Bruno.* Firenze 1886, p. 86.

[29]) P. 123. In the work *De l'infinito universo e mundi* (ed. Lagarde p. 343 f., 363) Bruno had already rejected the solid spheres on physical and epistemological grounds. In *De immenso,* I, 5 he finds this view confirmed by Tycho Brahe's investigations. *Tycho* himself drew the same consequences from his doctrine as Bruno. He says thus in a letter to Kepler of April 1598 (Kepleri *Opera* ed. Frisch. 1858, I, p. 44): "I hold that all reality of the spheres, however one may conceive them, must be excluded from the heavens, which I have learned from all the comets that have appeared, ever since the new star of 1572, and which are in reality celestial phenomena." (On the old view the comets were due to exhalations from the earth and moved only in the sublunary world.) — As F. *Tocco (Opera latine di G. Bruno esposte e confrontate con le italiane.* Firenze 1889, p. 318 f.) remarks, Bruno probably did not know Tycho's works at first hand, since he does not use the excellent argument against the solid spheres that the "new stars" afforded, and that Galilei later constantly employed. (Cf. *Tycho Brahe's* work *De nova stella* (the edition of 1901) (fol. D. 2): "That this star has changed both its size and its color in a distinct manner is not a sufficient proof

that it belongs in the elemental region, below the sphere of the moon . . . . For if it were possible that a new body *came into being* in the ether itself, which I have established above concerning this star on undeceiving grounds against the opinions and dogmas of all investigators, — then it will be regarded as far less impossible and unreasonable that this new star undergoes a *change* in size and color.") Nor does Bruno anywhere mention Tycho's peculiar system of the world.

[30]) P. 128. An interesting anticipation of Goethe's famous words: "*Ihm* ziemt's die Welt im Innern zu bewegen," is found in *De immenso,*V, 12: "God or reason does not stand outside and make the world revolve; for more worthy of him must be the inner principle of motion, which is the own nature, the own soul, that everything has which lives in his bosom and in his body." (*Dignius* enim illi debet esse internum principium motus etc.). — For this inner principle, which
[513] underlies every being's impulse of self-preservation, Bruno, in the last work we have from his hand, the *Summa terminorum metaphysicorum,* dictated during his stay in Zürich in 1591, uses the expression voluntas naturalis. — Cf. *Cena,* Lagarde's ed., p. 163 f., 183 f. *De l'infinito,* p. 370 ff. De la causa 242, 315, 320. —

[31]) P. 130. On the question of the relation between the mental and the material, Bruno expresses himself somewhat differently in his different writings. — In the *Cena,* p. 189, he will not inquire whether mental substance is transformed into material substance and conversely, or not. In *De la causa,* p. 235 ff. (cf. *Spaccio,* p. 409) it is maintained that mental substance, just as well as material substance, persists through all changes. At the same time *De la causa,* p. 261 and 280 ff., maintains the absolute unity of substance, the difference between mental and material substance being due only to a differentiation (explication). Finally, he denies in *De gl' heroici furori,* p. 721, expressly the conversion of the material into the mental and conversely. — Not only the opposition between mind and matter, but also the opposition between freedom and necessity is, according to Bruno, to fall away in the highest principle, which he develops in a series of interesting propositions, *De immenso,* I, 11—12.

[32]) P. 134. Bruno's atomism or doctrine of monads is found in the didactic poem *De triplici minimo,* which appeared at Frankfurt in 1591. It occupies, as Lasswitz (*Geschichte der Atomistik.* Hamburg und Leipzig 1889, I, p. 359— 399) has shown, an interesting place in the history of the development of modern atomism. Yet Lasswitz — so it seems to me — has not noticed the contradiction in Bruno between his polemic against division to infinity and his doctrine of the relativity of the concept of the atom. — *Felice Tocco*, who at first (in *Opere latine di G. Bruno)* had assumed that Bruno also applied atomism to the mental domain, has later (in his work *Opere inedite di G. Bruno.* Napoli 1891) corrected this assumption. He now finds an inconsistency in Bruno, because the latter does not assume mental atoms just as well as material ones. It is, however, a great question whether the concept of the atom is at all suited to be applied to the mental side of existence, and it was perhaps a right instinct that kept Bruno from making this application, although, had he strictly held fast to the relativity of the concept of the atom, it would not have involved all the consequences that at first glance might seem to lie in it. — When

Tocco (Opere latine di G. Bruno. p. 353), apropos of the expression monas monadum (of God), remarks that one cannot see whether God is thereby distinguished from the other monads or thought of as their substance, Bruno himself has clearly explained his meaning when he says: "God is the monad of monads, that is to say, the essential in all beings" (*Deus est monadum monas, nempe entium entitas. De minimo.* ed. 1591, p. 17). — The universe is called a monad in *De gl' heroici furori,* p. 724. — This use of the word monad is understood only by holding fast to the concept of relativity.

[33]) P. 135. Of the transmigration of souls Bruno says (*Spaccio,* p. 410) that if one cannot accept it, it is nevertheless worth thinking about. In the interrogation in Venice too (see *Berti*: *G. Bruno, 2,* ed. p. 402—408) he expresses himself with a certain indefiniteness on this matter. Cf. on his treatment of the doctrine of the transmigration of souls: F. *Tocco*: *Giordano Bruno,* p. 71—75. — He expresses himself most thoroughly on the relation between the world-spirit and the individual souls in the *Cabala,* p. 585—588. *[514]*

[34]) P. 147. In his *Apologia pro Galilæo.* Francofurti 1622, p. 54, *Campanella* says that he might perhaps, through the more recent investigations, be compelled to assume several suns in the universe, so that it would not, as he had hitherto assumed, have two centers, the sun and the earth, but many. Yet he suspends his judgment. — In his *Realis philosophia* (1623) p. 10 he is likewise doubtful. — Later, it is true, he inserted in his *Universalis philosophia seu Metaphysica* (Parisiis 1638), III, p. 71 some remarks which developed how the matter would have to be conceived if Galilei were right (posito qvod vera sint Galilei dogmata), but then added: "Now judgment has been given in Rome against Galilei, and with that what I had earlier [before those remarks] written is established." — A peculiar kind of verification!

[35]) P. 168. The first account of the history of philosophy in which Galilei has been given the place that is his due on account of his clear consciousness of the method of thinking is J. F. *Fries*: *Die Geschichte der Philosophie dargestellt nach den Fortschritten ihrer wissenschaftlichen Entwickelung*. Zweiter Band. Halle 1840. Later *Prantl (Galilei und Kepler als Logiker.* Münchener Sitzungsberichte. 1875) has treated his doctrine of method and shown how it eclipses Bacon's, and *Tönnies (Ueber die Philosophie des Hobbes.* Vierteljahrsschr. für wiss. Philos. III) and *Natorp (Descartes' Erkenntnisstheorie.* Marburg 1882, ch. 6) have established Galilei's significant position at the entrance to modern philosophy, in contrast to the traditional beginning with Bacon and Descartes.

[36]) P. 173. *Galilei*: *Opere.* Firenze 1842 ff., XIII, p. 200 f. — The German translation of the Discorsi (in "Klassiker der Naturwissenschaft" No. 24, p. 57) has an inaccuracy here, through which the simplification of the circumstances that Galilei has in mind drops out. Galilei's words "dum externæ causæ tollantur" (provided only that external causes are kept away) are rendered, namely, by "während äussere Ursachen hinzukommen." — Galilei's thought experiment appears most clearly in another place in the *Discorsi* (Op. XIII, p. 221): Mobile qvoddam super planum horizontale projectum mente concipio, omni secluso impedimento. (*I* conceive of a movable

body thrown along a horizontal plane, in such a way that every hindrance is excluded.) — On the history of the law of inertia cf. *Wohlwill*: *Die Entdeckung des Beharrungsgesetzes.* (Zeitschrift für Völkerpsychologie. XIV—XV).

[37]) P. 177. It is Natorp's merit (in his work on Descartes's epistemology) to have been the first to draw attention to this section of *Il Saggiatore. (Opere de Galilei.* Firenze 1842 ff., IV, p.
[515] 332—338). — The new doctrine was dangerous, since the denial of the reality of the sense qualities was thought to conflict with the Catholic doctrine of the Eucharist, which held that the host kept its sensible properties after its substantial transformation. Later Descartes and Leibniz too had much trouble on account of this question.

[38]) P. 178. As *Prantl* has shown *(Ueber Petrus Ramus.* Münchener Sitzungsberichte 1878).

[39]) P. 180. Cf. *Freudenthal*: *Beiträge zur Geschichte der englischen Philosophie.* (Archiv für Gesch. der Philos. IV—V.) — Croom *Robertson* in Academy. 1892, p. 110.

[40]) P. 187. As a curiosity it may be mentioned that, just as there are those who think that not Shakespeare but someone else, perhaps Bacon, wrote the "Shakespearean" tragedies, so a critical head in the most recent times has found out that Bacon is not the author of the Novum Organum, and has stopped for a moment at the possibility that Shakespeare might have written it, without, however, daring to hold fast to this possibility. See on this Archiv für Gesch. d. Philos. I, p. 111.

[41]) P. 194. On Bacon's relation to Plato see *Nov. org.* I, 105—124; II, 7—13. — In *De augmentis scientiarum,* III, 4 it is said: "It is evident that Plato, that lofty mind, who surveyed everything as from a mountain top, perceived in his doctrine of ideas that the forms are the proper object of science."

[42]) P. 205. *Discours de la méthode* (1637), 2. partie. — *La vie de Monsieur Descartes.* [Par A. Baillet]. Paris 1691, I, p. 51—71. — *Millet*: *Histoire de Descartes avant* 1637. Paris 1867, p. 100 ff.

[43]) P. 212. In the account given here I follow especially Descartes's earliest sketch of a doctrine of method in *Règles pour la direction de l'esprit,* a work which, when carefully studied, removes many current misunderstandings of Descartes. Cf. besides the third Meditation and the Reply to the second set of objections, in which Descartes, at the request of others, attempts a systematic exposition of his ideas. — To the question of how general propositions are formed from the single intuitions, Descartes really gives only this answer: "It is precisely the nature of our mind to form general propositions from the knowledge of the single." *(Resp. ed. sec. obj.).*

[43 b]) P. 213. Whereas Descartes in his doctrine of method *(Règles* etc.) begins by speaking of the clear and distinct intuition of simple relations (intuition) and names the consciousness of ourselves as thinking among the examples of such intuition, he later begins (especially at the beginning of the 3rd Meditation) with the consciousness of ourselves as thinking beings and derives from the simplicity and immediate clearness of this thought the significance of intuition. — The line of thought that underlies cogito ergo sum occurs already in *Augustinus (De vera religione* cap. 49), indeed already in ancient Indian philosophy.

[516] [44]) P. 214. At the end of the 3rd Meditation it is said: "The possibility then alone remains

that the representation of God is innate in me in the same way as the representation of myself is innate in me." This is for Descartes more than a comparison. For I cannot — he thinks — become conscious of myself as a limited and imperfect being without an unconscious use of the representation of a being raised above all limitation (3rd Medit.). — To Hobbes's objection: "There is no innate representation; for what is innate must always be present," — Descartes answers *(Resp. tertiæ* X): "When we say that a representation is innate, we do not mean that it always appears before us; for in that sense there would be no innate representation at all; but only that we have in ourselves the capacity to develop it." — In *Notæ in programma quoddam* Descartes says: "Since I noticed that there are thoughts in me which do not proceed from external objects or from the determination of my will, but merely from my faculty of thinking, I called such *representations*, which are the *forms* of those thoughts, innate, in order to distinguish them from others, which come from outside or are produced by myself. In the same sense we say that magnanimity, or that certain diseases such as gout or stone, are innate in certain families, by which we do not mean that the children suffer from these diseases in their mother's womb, but that they are born with a disposition or capacity to contract these diseases." — The part of Descartes's theological line of thought that infers from the representation of God as the representation of an infinite being to an infinite being as the cause of this representation is due perhaps to an influence of *Vives* (De *anima.* 1538 p. 127 f.), by whom Descartes was also influenced on several points in his psychology.

[45]) P. 215. Cf. *Natorp's* excellent account *(Descartes' Erkenntnisstheorie,* p. 55. ff., 76 f.). Natorp has shown that the use of the concept of God in Descartes's epistemology is not of so external a kind as it appears from the usual accounts. Yet Natorp does not sufficiently emphasize what I believe one must lay the greatest weight on: namely, that this whole epistemological use of the concept of God was altogether unnecessary, since Descartes after all uses the causal principle to establish the reality of the concept of God; when the causal principle is presupposed valid, no theological guarantee is needed for the validity of knowledge.

[46]) P. 218. The strict concept of substance is found in Descartes *Medit. III, Princ. phil.* I § 51. The lax concept of substance is found: *Resp. sec.* and *Princ. phil.* I § 52. — Descartes himself *(Princ. phil.* I, 51) draws attention to the double meaning of the word substance.

[47]) P. 227. On Harvey's discovery and its prehistory see P. Hedenius *Om upptäcktan af blodomloppet.* (Upsala Universitets Årsskrift 1892). See especially on the relation between Descartes and Harvey p. 216 f., 277. Descartes assumed that the heating of the blood in the heart caused it to be pressed out into the body; he had thus not appropriated Harvey's thought of the heart as a contracting muscle. — Harvey too, to be sure, held fast to the old doctrine of the heating of the blood in the heart, but did not apply it in this way. It was only *Niels Steensen* who *[517]* definitely showed that the heart is a muscle, and that there was no foundation for the significance as the very source of life and the seat of the soul that had been ascribed to it from ancient times. (Cf. *Jul. Petersen*: *Bartholinerne.* Copenhagen 1898. p. 84 f.).

[48]) P. 229. Descartes's localization of the soul in the pineal gland was altogether arbitrary and was criticized by the anatomists of the time. *Thomas* Bartholin *(Anatomia.* Lugd. Batav. 1651, p. 356 f.) objects partly that the nerves do not end in the pineal gland, partly that it is so small that the representations would be mixed together [!] in it, partly, finally, that it is placed where the "excrements of the brain" accumulate! In the brilliant lecture that *Niels* Steensen gave in Paris, and which is printed in *Winslöv's Exposition anatomique* (4th part), he remarks that since one does not know what fills the cavities of the brain, the doctrine of the animal spirits may be just as good as that of the excrements of the brain. Against Descartes, whose general method he credits with great merit, he objects that the pineal gland does not stand in connection with any canal, and that it does not float freely, but hangs together with the rest of the brain. — What significance the pineal gland has is not yet known. (*Edinger*: *Vorlesungen über den Bau der nervösen Centralorgane.* 6. Aufl. p. 138). On Descartes's significance for the physiology of the nerves cf. Huxley: *Om den Hypothese, at Dyrene ere Automater.* (Tidsskrift for populære Fremstillinger af Naturvidenskaben 1874). — A remarkable utterance of Descartes's is found in a letter to Mersenne (February 1649), namely that when he denies the animal soul, he denies merely *thinking* (cogitatio), not *life and sense perception*: vitam nulli animali denego, utpote qvam in solo cordis calore consistere statuo; nec denego etiam *sensum qvatenus ab organo corporeo dependet.* This does not agree with other statements, especially not with *Resp. ad sextas obj.* § 9, where 3 degrees of sense perception are distinguished: the first is merely a motion in the brain called forth by the impression (cerebri motus), and this is common to animals and human beings; the second is the immediate sensation of pain, pleasure, thirst, hunger, color, tone, taste, smell, warmth etc., which arises in consciousness (in mente) as the effect of that motion; the third is the judgments that, on the occasion of that motion, we pass on external things. The sense perception he speaks of in the letter to Mersenne must be the first of these degrees; but of it, it is not enough to say that it *depends* on the motions of the brain: it is nothing other than these and cannot be called sensation. What he calls thinking in the letter must correspond to the third degree. There is thus an unclarity with regard to the second degree; it is precisely here that the transition takes place from motion (quantitative processes) to consciousness (where different qualities of sense come to correspond to the different quantities).

[49]) P. 244. By his denial of necessary connection between phenomena Malebranche is a forerunner of Hume. Perhaps Hume even got motives for his theory of causation from Malebranche,
[518] whom he (as can be seen from his *Treatise,* I, 3, 14; 4, 5) studied eagerly. *Mario Novaro* has in his work *Die Philosophie des Nicolaus Malebranche* (Berlin 1893, p. 44—50) drawn an instructive parallel between the theories of causation of Malebranche and of Hume. — Geulincx's philosophy too would present parallels to Hume; but Hume hardly knew it.

[50]) P. 256. *Vita Thomæ Hobbes.* [Autore Rd. B. i.e. Richard Blackbourne]. 1681, p. 14 f. Cf. Telesio's doctrine of sense perception (above p 93). — In his *Vita carmine expressa,* p. 119 Hobbes describes how, after long brooding over the nature of things, he found out that there was only

one single real thing [motion] in the world, although it was "falsified" (sic) in many ways. — F. Tönnies, whose energetic research in most recent times has thrown so much light on Hobbes and his philosophy (see his *Anmerkungen über die Philosophie des Hobbes.* Vierteljahrsschrift für wissenschaftliche Philosophie. III—V), has found a manuscript that contains a treatise by Hobbes which, by its content, must have been composed between the moment when the significance of the deductive method and then of motion dawned on him and the moment when the subjectivity of the qualities of sense dawned on him. This treatise, which is of interest with regard to the historical development of the principle of the subjectivity of the qualities of sense, Tönnies has printed in his edition of Hobbes's *Elements of Law* (Oxford 1888). — Hobbes maintains that as early as 1630 [thus before he came to know Galilei] he had held the conviction of the subjectivity of the qualities of sense. Cf. *Tönnies's* monograph on Hobbes *(Hobbes' Leben und Lehre.* Stuttgart 1896. p. 15).

[51]) P. 261. Cf. Hobbes's treatise *Of Liberty and Necessity.* 3. ed. London 1685, p. 314, where it is said that as regards the definitions of spontaneity, deliberation, will, inclination, drive, freedom, "no other proof can be given than every man's own experience when he reflects on himself (by reflection on himself) and remembers what he himself means when he says that an action is spontaneous" etc.

[52]) P. 262. In *De Corpore,* IX, 9 Hobbes derives the proposition that all change (qualitative change too) is motion from the causal principle and the principle of inertia. — His "proof" of the causal principle (Of Lib. and Necess., p. 315) is as follows: "If something had no cause, it could just as well begin to come into being at the one moment as at the other; — there must then be a cause why it comes into being precisely at this moment!" That here precisely what was to be proved is presupposed, Hume has already shown. — An analogy to this is formed by his proof of the principle of inertia (De corpore, VIII, 10): if a body at rest were surrounded by empty space, there would be no reason why it should move more in the one direction than in the other, and it would thus not move at all unless an external cause were added. Here it is obviously presupposed that there are no "inner" causes. — Possibly this proof of the principle of inertia stems from Galilei. See his "Dialoger over Verdenssystemerne." 1st Day. (Strauss's German translation p. 21).

[53]) P. 264. Hobbes therefore really has no reason to distinguish between definition and *[519]* hypothesis. He says (De Corp., dedic.) that in his logic, his Philosophia prima and his geometrical doctrine of motion he proceeds from definitions, but in his physics from "not unreasonable hypotheses." But those definitions too are hypotheses. — Hobbes requires, for the full understanding of a phenomenon, the demonstration of a continuous progression from cause to effect. This continua progressio (De corpore, IX, 6) recalls Bacon's latens processus continuatus (Nov. org. II, 5—7). I do not consider it improbable that Bacon's doctrine of the continuous process may have come before Hobbes's mind when he had been led by another way to the idea that all change was motion. He made the principle of his deduction what in Bacon was to be the last result of numerous inductions. — Likewise it is not improbable that the name Philosophia prima for the

second part of De corpore is due to Bacon, who (De augm. III, 1) designates by this name the science of the general fundamental propositions that hold for all domains.

[54]) P. 284. A. D. *Jørgensen* gives, in my view, too lurid a picture of Hobbes's political theory, especially in his comparison between Hobbes and Bishop Vandal. *(Peter Schumacher Griffenfeldt,* I, København 1893, p. 218 f.). The moral limitation of the prince's physical right is found no less in Hobbes than in the Lutheran theologian, and in Hobbes moreover with a grounding that makes absolute sovereignty a means, not an end. Cf. e.g. *De cive* XIII, 3: "The state is instituted for the sake of the citizens" (Civitas civium causa instituta est). — In the older edition of *Leviathan* (ed. 1651, p. 193) the hope of influencing princes is expressed somewhat more strongly than in the later one (ed. 1670, p. 172).

[55]) P. 286. *Freudenthal* has (Archiv für Gesch. de Philos. VI, p. 191 ff., 380 ff.) brought forward Lord Brooke's work and shown the interest that attaches to it.

[56]) P. 286. On Culverwel see M. M. *Curtis*: *An Outline of Locke's Ethical Philosophy.* Leipzig 1890, p. 9—18.

[57]) P. 293. *Colerus (La vie de B. de Spinoza,* tirée des écrits de ce fameux philosophe et du témoignage de plusieurs personnes dignes de foi, qui l'ont connu particulièrement. A la Haye 1706, p. 14) regards Spinoza merely as Descartes's disciple in philosophy. *Leibniz (Opera philos.* ed. Erdmann, p. 139) says of him that he has only cultivated some seeds that Descartes had sown. This conception was long dominant. *Kuno Fischer* and *Falckenberg* hold fast to it in their works on the history of modern philosophy. — *Joël (Beiträge zur Geschichte der Philosophie.* Breslau 1876), in opposition to this, emphasized the great and lasting effect of Spinoza's studies of Jewish theology and philosophy of religion. — Chr. *Sigwart (Spinozas neuentdeckter Tractat.* Gotha 1868) and *Avenarius (Ueber die beiden ersten Phasen des Spinozischen Pantheismus.* Leipzig 1868) pointed — on the basis of the youthful treatise of Spinoza published by van *Vloten (Ad Benedicti*
[520] de Spinoza opera quæ supersunt omnia supplementum, Amstelodami 1862): *Kort Traktat om Gud, Mennesket og dets Lykke* — to the probability that Spinoza had been influenced by Giordano Bruno. — Finally *Freudenthal (Spinoza und die Scholastik.* In the Festschrift for Zeller 1887) has shown to how high a degree Spinoza has used scholastic concepts and proofs (especially in *Cogitata metaphysica,* which forms an appendix to Spinoza's exposition of the Cartesian philosophy; but also in the Ethica). And the same scholar has later collected all the older biographies of Spinoza and in general all documents that can throw light on his life and his writings, in his book *Die Lebensgeschichte Spinoza's in Quellenschriften, Urkunden und nichtamtlichen Nachrichten.* Leipzig 1899. — In my work *Spinozas Liv og Lære* (København 1877) I have given a popular account of Spinoza and his philosophy. Later, repeated studies have led me on single points to a different conception than in that work. I take the opportunity to state that the picture used as title vignette in the work mentioned, and which was taken from van Vloten's "Supplementum," has since been shown to represent — not Spinoza, but one of his friends, the philosopher and mathematician Tschirnhausen. A fine and, by all that can be ascertained, genuine portrait of Spinoza is found in

van Vloten and Land's new edition of Spinoza's works.

[58]) P. 298. I must also oppose A. D. *Jørgensen's* reflections (*Nic. Steensen,* p. 57) on Spinoza's relation to experience; but on this point the section on Spinoza's theory of knowledge will contain what is needed.

[59]) P. 299. *Studie zur Entwickelungsgeschichte des Spinoza.* (Vierteljahrsschrift für wiss. Philos. VII) p. 161 ff., 334.

[60]) P. 308. The first book of the "Ethics" develops the concept of substance and the concept of God in a series of definitions, axioms and propositions. — On the connection between the concept of substance, the concept of God and the concept of nature cf. Avenarius: "Ueber die beiden ersten Phasen des Spinozischen Pantheismus," where it is aptly shown that all three concepts, carried through consistently, meet in one point in Spinoza. Yet Spinoza hardly kept the three currents of thought so sharply apart from one another as they are kept in Avenarius's account. — On the relation between substance and modes one must note, besides the first book of the "Ethics," Ep. 12 (in v. Vloten and Land's edition), where a distinction is drawn between the abstract and the substantial conception, and Ep. 50, where it is declared that every determination is a negation. — The expressions natura naturans and natura naturata occur already in Renaissance philosophers, mystics (Master Echardt) and scholastics. According to *Siebeck* (Archiv für Gesch. d. Philos. III, p. 370 ff.) they are said to have their origin in the Greek *φύσις* (as *φύον* and as *φυόμενον*) and to have arisen through the translation into Latin of Greek philosophical writings, probably in the thirteenth century. — As regards the exposition in the Ethics, the geometrical-deductive form is — as already remarked — only a shell. In reality it is a series of analyses that Spinoza undertakes, *[521]* and by God or substance he understands the comprehensive reality to which all phenomena belong, and within which thought can therefore point them out. Cf. on this *Harold Joachim*: *A study of the Ethics of Spinoza.* Oxford 1901. p. 116 f. — Cf. *Descartes's* declaration (in Medit. VI) that by nature in general (natura generaliter spectata) he understands either God himself or the connection of things established by God (coordinatio rerum).

[61]) P. 309. Cf., against this conception of Spinoza's concept of attribute maintained by the elder *Erdmann*: H. *Bröchner*: *Benedict Spinoza.* København 1857, p. 47 ff. — *Mourly Vold (Spinozas Erkendelsesteori.* Kristiania 1888, p. 242) wrongly names Bröchner among those who interpret the attributes "as predicates necessary for our understanding, but accidental for God." Bröchner polemicizes precisely against this conception.

[62]) P. 311. Already in my work "Spinozas Liv og Lære" (1877) p. 100 I have pointed out this confusion in Spinoza. It has also been brought forward by *Windelband*: Geschichte der neueren Philosophie, I (1878), p. 211 and by *Tönnies*: Studie zur Entwickelungsgesch. des Spinoza. (Vierteljahrssch. für wiss. Philos. 1883) p. 176 f.—*Mourly Vold* (Spinozas Erkendelsesteori, p. 202) denies the confusion in Spinoza and regards the doctrine of attributes as springing essentially from the endeavor to ground a "correspondence between idea and object." But when he concedes that the doctrine of attributes in Spinoza also has a psychophysical significance, he must consistently

concede a confusion in Spinoza. For epistemology and psychophysics are two different things, which is already seen from the fact that the "objects" are not merely of a physical but also of a psychical kind; the problem of knowledge concerns, after all, both the knowledge of the psychical and of the physical. — The confusion in Spinoza is seen clearly when one compares Eth.II, 7, Schol., where the relation between the concept of the circle and the existing circle is used to illustrate the relation between the attributes, with II, 17, Schol., where a distinction is drawn between the representation of Peter (idea Petri) that corresponds to Peter's own body as its soul (mens Petri), and the representation of Peter (idea Petri) that Paul has. The first idea Petri has a psychophysical, the second an epistemological significance. Spinoza's highly unfortunate usage of designating the soul (mens) as idea corporis has favored the confusion. This usage stems from Aristotle by way of Maimonides (see *Hamelin*: *Sur une des origines du Spinozisme.* L'année philosophique. 1900. p. 24 f.), to whom Spinoza himself expressly refers as regards the epistemological doctrine of identity, see *Eth.* II, 7 Schol. — In German speculative philosophy (in Schelling and Hegel) the confusion flourishes.

[63]) P. 315. H. Morus: *Demonstrationis duarum propositionum, qvæ præcipuæ apud Spinozium* [sic] *Atheismi sunt columnæ, brevis solidaqve confutatio.* (Opera philosophica. Londini 1679, vol. I, p. 619). — On the different conceptions of Spinoza's concept of God there is an instructive treatise
[522] by J. H. Löwe *(Ueber Spinozas Gottesbegriff und dessen Schicksale.* Appendix to Löwe's book: Die Philosophie Fichtes. Stuttgart 1862), which, however, does not mention More's characteristic interpretation.

[64]) P. 315. Cf. on this question *Ed. Zeller's* remarks in Vierteljahrsschrift für wissenschaftliche Philosophie, I, p. 285 f.

[65]) P. 320. In Eth. II, 18 only the law now called the law of contiguity is mentioned. But the law of similarity is presupposed in Eth. III, 27 and is named in Tract. theol. polit. cap. 4 as an example of a mental law of nature.

[66]) P. 327. A fuller account of Spinoza's ethics than I can give here is found in my work "Spinozas Liv og Lære," p. 120—146.

[67]) P. 329. When it is said in Eth. V.38, Schol. and 40, Coroll. that it is the *intellectus,* not the *imaginatio,* that is eternal, it seems that only those individuals who rise above the stage of imagination (sensuous representation) are organs of the divine activity of thought, which can no more disappear than motion can. A statement in the "short treatise" (2nd part, preface, no. 15) also points this way, to the effect that the soul can become eternal when through knowledge and love it unites itself with the absolute substance. — The thought we meet here in Spinoza often comes forth in the history of thought. I cannot go into it more closely here, but must confine myself to some literary references. Plato's Republic, VII, p. 534 C (Hejse's trans. of the Republic, 2nd part, p. 188). — *Chrysippos* (Diog. Laërt. VII, 157). — Maimonides (see Joël: Zur Genesis der Lehre des Spinoza, p. 66 ff. and Hamelin in L'année philos. 1900. p. 26 f.). — *Salomon Maimon*: Lebensgeschichte. Von ihm selbst geschrieben. Berlin 1792. II, p. 178 f. — *Goethe*: Gespräche

mit Eckermann. 4 Febr. and 1 Sept. 1829. — *Wilhelm* von *Humboldt* (see Haym: W. v. Humboldt, p. 637 f.). — J. G. *Fichte*: Die Thatsachen des Bewusstseins, Stuttgart und Tübingen 1817, p. 197 f. — C. H. *Weisse*: Die philosophische Geheimlehre von der Unsterblichkeit des menschlichen Individuums. Dresden 1834. — *Louis* Lambert (see Ravaisson: La philosophie en France au 19. siècle, p. 223). — *Ernest Renan*: Feuilles détachées. p. 392 f.

[68]) P. 337. See on this Ludwig *Stein*: *Leibniz und Spinoza.* Ein Beitrag zur Entwickelungsgeschichte der Leibnizischen Philosophie. Berlin 1890. (Especially ch. 4).

[69]) P. 341. The treatise of 1680 has no heading. *Erdmann*, who first brought it forward from the library in Hannover, gave it (in his edition of Leibniz's *Opera philosophica,* 1840) the title: De vera methodo philosophiæ et theologiæ and assigned it to 1690—91. That it must be of earlier origin is shown by both internal and external criteria, especially the comparison with "Petit discours de métaphysiqve," which was worked out in 1685 and sent to Arnauld the following year. Cf. on this *Selver*: *Der Entwickelungsgang der Leibniz'schen Monadenlehre bis 1695.* (Philosophische Studien herausg. von W. Wundt, III), p. 433. *Ludwig Stein*: *Leibniz und Spinoza,* p. 117 ff. — Stein, in his excellent account of the history of Leibniz's development, lays too great weight on the influence the study of Plato had on Leibniz. It is admittedly very characteristic that Leibniz, in the years *[523]* so important for his philosophical maturing, occupies himself so much with Plato and, among other things, quotes the well-known passage in the Phaedo on final causes no less than twelve times. But it is nevertheless symptom rather than cause. The conviction of the legitimacy of final causes stood firm for Leibniz from the outset; it sprang from his religious presuppositions, and all through his youth he had sought means of bringing it into harmony with his scientific theories. Plato could bring him nothing new here. — In particular I find it an unnatural explanation when Stein derives Leibniz's proposition of the identity of substance and force from Plato's doctrine (expressed in a single passage) of the "ideas" as active forces. Leibniz had, after all, been occupied with the concepts of force and striving ever since 1670, and it is from his mathematical-physical studies that he carries these concepts over into his philosophy and uses them to bring the concept of substance to life. — When several writers designate the advance Leibniz made in the year 1680 as a *substantialization of force,* I believe that it would be more correct to reverse the relation and speak of a *conversion of substance into force.* It was in reality a dissolution of Descartes's and Spinoza's concept of substance that was brought about by the treatise of 1680.

[70]) P. 350. Leibniz's "Tout comme ici" has the same source as Holberg's "Ligesaa hos os," namely the comedy Harlequin empereur de la lune (Gherardi's Théatre italien, vol. I). See Nouveaux essais, IV, 16, 12 (Erdmann, p. 391 f.): On diroit quasi qve c'est dans l'Empire de la lune de Harléquin tout comme ici. — Besides this passage one must note with regard to the law of analogy: letter to Arnauld, Nov. 1686 (Gerhardt's edition, II, p. 75 f.). De ipsa natura (Erdmann, p. 157). Nouv. Ess. I, 1 (Erdmann p. 205) and III, 6, 14 (Erdmann, p. 312). Statements in letters (Gerhardt, II, p. 270; IV, p. 343).

[71]) P. 352. *Eduard Dillmann (Neue Darstellung der Leibnizschen Monadenlehre.* Leipzig 1891)

seems to me, in his account of the doctrine of monads, which is in many respects instructive and interesting, altogether to overlook the great significance that the law of analogy has for the subjective completion of the doctrine of monads. Dillmann regards the faculty of representation as given from the outset in the concept of the monad and takes the concept of phenomenon in Leibniz merely in a subjective sense (phenomenon for). This conception is already refuted by the fact, conceded by Dillmann himself, that Leibniz without further grounding assumes that the monad represents the universe from the standpoint of its body, and that he supports the assumption that the monad represents the whole universe by the fact that every body, by virtue of the law of continuity, is affected by every change in any part whatever of the world. (Cf. Dillmann, p. 301 ff., 342 ff.). — On Dillmann's hypothesis it would be impossible to understand why Leibniz takes so dismissive a stand toward Berkeley's ideas as soon as he hears of them.

[524] [72]) P. 356. On the different ways in which the relation between Spinoza and Leibniz has been conceived in the course of time, see the first chapter of L. *Stein's* "Leibniz und Spinoza." — E. *Dillmann* finds a fundamental agreement between the two systems, but he thinks at the same time that Leibniz's doctrine of the relation between God and the world is unassailable, *and* he lays no weight on the passages where Leibniz distances himself from Spinoza. (Neue Darstellung, p. 472 ff.). Dillmann lets himself be led by his great and justified admiration for Leibniz into playing the apologist for obvious self-contradictions and evident accommodations. — An instructive investigation of the relation between the two systems has been undertaken by *Robert Latta* in his treatise *On the relation between the philosophy of Spinoza and that of Leibniz* (Mind. 1899).

[73]) P. 363. On the difference between perception and appétit see Monadologie § 15. — Passions as pensées confuses: *Réplique aux reflexions de Bayle* (Erdmann p. 188 a), as tendances ou des modifications de tendances: *Nouv. Ess.* II, 20, 9 (Erdmann, p. 249 a).

[74]) P. 365. Cf. a treatise (without heading) which is printed in Gerhardt's edition, VII, p. 300 ff., where it is first said that all factual data must have their ground, since the creative will that manifests itself in them is not arbitrary, but acts according to grounds (ratio), and then: "It is thus established that all truths, even the most contingent, have an a priori justification (probatio) or a ground why they are rather than are not. And this is precisely what the common saying maintains, that nothing happens without a cause (causa), or (seu) that nothing exists without a ground (ratio)."

[75]) P. 369. The law of the effect of contrast Leibniz already uses in a letter from the beginning of the seventies (Gerhardt's edition, I, p. 61) to defend optimism. Later he develops it especially (as *lex lætitiæ)* in *De rerum originatione* (Erdmann, p. 149 f.). This aesthetic-psychological argument is of ancient origin; it is found already in *Plotinos*, and is repeated (remarkably enough) by *Augustinus*. Modern, on the other hand, is the emphasis on the infinity of existence. Leibniz remarks that Augustine does not yet know la grandeur de la cité de Dieu *(Théodicée* § 19). Cf. *Causa dei asserta §* 57—58.

[76]) P. 374. In the preface to the first edition of *Vernünftige Gedanken etc.* Wolff says: "I had

indeed at the outset resolved to leave the question of the connection between soul and body wholly undecided; but since, by the reasons adduced in chapter 2, contrary to my expectation, I was quite naturally led to Herr v. Leibniz's pre-established harmony, I have retained it and placed this profound discovery in such a light as it has not yet received." — In the preface to the 2nd edition he expresses himself more precisely on the relation of his doctrine to Leibniz's. Leibniz, he says, combined idealism [the unbroken connection of the mental phenomena] with materialism [the unbroken connection of the material phenomena], without examining each of them more closely by itself. That Wolff now thinks he has done, and his result is that "the dualists can unite the theory of the idealists and of the materialists." — This is in reality a correct *[525]* determination of the difference between Leibniz and Wolff on this point: Leibniz embraces the identity hypothesis (though with a spiritualist modification), and Wolff embraces dualism (though with the assumption, taken from the identity hypothesis, of the unbroken connection of the mental and of the material phenomena each by itself). — Very many of the objections that are usually advanced against the identity hypothesis hold for the Wolffian theory, to which the name "duplicism," which does not fit the real identity hypothesis at all, fits very well.

[77]) P. 390. Locke's friend Molyneux explained the apparent contradiction between the two propositions (of which the one is found Essay, IV, 3, 10, the other Essay, IV, 10, 5) by saying that the first passage states that God could by supernatural means endow matter with the faculty of thought, whereas the second passage speaks of matter without such supernatural influence (letter to Locke, 22 Dec. 1692). Locke declares this interpretation to be quite correct (letter of 20 January 1693). *(The Works of John Locke,* 9. ed, London 1794, IX, p. 293—303.)

[78]) P. 394. Locke does not really set up this triad formally. Where he speaks of the need to leave the state of nature *(Civil Government,* II § 124—126) he names legislative power, impartial judges and power to execute the laws as the three things that are needed. Later (§ 136) he insists on the necessity that the same power that gives the laws should not itself judge, since that would open the way to arbitrariness. But when he (ch. 12—13) sets up three powers of the state and examines the relation between them, this triad consists of the legislative, the executive and the federative power. By this last Locke means an authority that represents the state outwardly, towards other societies, in relation to which it finds itself in the state of nature. Now since Locke states that the executive and the federative power had best be united in the same hands (§ 147—148), the difference between them really falls away, and since in another connection he has laid great weight on the distinction between legislative and judicial power, the triad does to that extent come out clearly in him. — In the doctrine of the division of the power of the state Locke has, moreover, Buchanan, Hooker and Sidney as predecessors. Cf. O. *Gierke*: Joh. Althusius, p. 157, 163 f.

[79]) P. 415. In the Scholium after the Definitions there is talk of "the true, absolute and mathematical space" (and likewise of time). "People in general (vulgus)," it is said, "conceive quantities, on the other hand, only in relation to sensible things" (ex relatione ad sensibilia). —

But at the end of Definitio 8 the "true, physical" mode of consideration is set in opposition to the *mathematical:* vere et physice — as opposed to mathematice tantum! — To this belongs also the fact that Newton took the center of the world to be at rest (Principia, III, 10, cf. 12 Coroll.), although he grounded the assumption of absolute space precisely on the fact that no body could be shown to be absolutely at rest.

[526] [80]) P. 439. *Ed. Grimm (Zur Geschichte des Erkenntnissproblems,* Leipzig 1890, p. 571—586) makes an interesting comparison between Hume's two works. — I cannot agree with Grimm that Hume's disowning of his youthful work should derive from his really having come to another standpoint. The motive was surely the same as the one that made his Dialogues on natural religion a posthumous work: he wanted to be left in peace by the orthodox. Cf. *Letters of David Hume to Strahan,* Oxford 1888, p. 289 f., 303—330.

[81]) P. 141. Only in a single place (Treatise, II, 3, 9) does Hume state that feelings or affects, besides arising through experience of goods and evils, can spring from "a natural impulse or an instinct" (natural impulse or instinct), but he then at once declares this instinct to be "altogether inexplicable." As examples of such feelings he names the desire for revenge, the wish for happiness for our friends, hunger and other bodily drives.

[82]) P. 441. Cf. an interesting statement in a letter to Hutcheson: "I wish from my heart that I could avoid drawing the conclusion that, since morality, according to your view, is determined merely by feeling, it concerns only human nature and human life. This has often been urged against you, and the consequences of it are very significant." *Burton: Life and Correspondence of David Hume,* Edinburgh 1846, I, p. 119.

[83]) P. 466. Il est absolument nécessaire pour les princes et pour les peuples, que l'idée d'un étre supréme créateur, gouverneur, rémunérateur et vengeur soit profondément gravée dans les esprits. Art. Athéisme in *Dict. phil.).* At the end of this article it is said that there are fewer atheists now than before, because the true philosophers now acknowledge final causes. "The priest proclaims God to children, and Newton proves him to the learned." — In one of Voltaire's poems (to the author of the book Les trois imposteurs) occurs the well-known line: Si Dieu n'existait pas, il faudrait l'inventer.

[84]) P. 468. In *Lettres sur les Anglais,* XIII it is said (on the occasion of Socrates's belief in his daimonion): Il y a des gens à la vérité qui prétendent qu'un homme qui se vantait d'avoir un génie familier, était indubitablement un fou ou un fripon, mais ces gens-là sont trop difficiles. — In *Essai sur les moeurs et l'esprit des nations* (Tome II, Chap. 9) it is said that Christianity must surely be divine, puisque dix-cept siècles de friponneries et d'imbécilités n'ont pu le détruire." Shortly before, the expression insolente imbécilité was used of a teller of legends. — Art. Fanatisme in Dict. phil.: Ce sont d'ordinaire les fripons qui conduisent les fanatiques. — It was not only religious and ecclesiastical figures that Voltaire's mockery failed to do justice to. But one sees that even old Socrates had to suffer for it. Elsewhere it falls upon Spinoza and upon the naturalist
[527] *Maillet*, whose anticipations of the hypothesis of evolution Voltaire derided. All by virtue of the

principle that what he did not understand was stupidity or fraud.

[85]) P. 468. Cf. D. *Fr. Strauss*: *Voltaire,* 3. Aufl., p. 330 ff.

[86]) P. 470. *Gneist (Das Selfgovernment in England,* 3. Aufl., p. 944) remarks: "The spokesman of the new doctrines [namely the new political doctrines in France] had before his eyes not the English constitution but Blackstone's 'Institutions,' in which the historical course of development of the whole and the intermediate structure of self-government are lacking. Precisely this made the description intelligible to French eyes." A mistake has crept in here, since Blackstone's work did not appear until 1765, whereas "Esprit des lois" appeared in 1748. The relation is precisely the reverse of what Gneist states: it is Montesquieu's theory that influences Blackstone and thereby also gains considerable influence on the political development in England. Cf. on this F. C. *Montague's* introduction to his edition of Bentham's "Fragment on Government." (Oxford 1891, p. 68). Montague also makes the apt remark that Montesquieu, at the sight of the continental governments' system of oppression, was doubtless led to stress the division of power more strongly than accorded with the English constitution and the English theories. — In *The Federalist* No. 47—51 the founders of the constitution of the North American Union, *Madison* and *Hamilton*, examine Montesquieu's doctrine of the division of power in an interesting way and discuss in particular how it is to be arranged when account is taken of the fact that the legislative authority has a constant tendency to appropriate the executive power.

[87]) P. 482. F. *Papillon*: *Histoire de la philosophie moderne dans ses rapports avec le développement des sciences de la nature,* Paris 1876, II, p. 194 makes Diderot's philosophy proceed directly from Leibniz. Against this see *Knud Ipsen*: *Diderot.* p. 206 f. and 210 (where it is doubtless rightly urged that the collection of excerpts and notes found in his works under the title "Éléments de physiologie" proves in how high a degree Diderot is influenced directly by his studies in natural science).

[88]) P. 495. Rousseau develops the difference between amour de soi and amour propre in *Discours* sur *l'inégalité* (note no. 12), and later in the writing *Rousseau juge de Jean Jacques* (London 1780, p. 20 f.). Cf. further on this distinction my book *Jean Jacques Rousseau og hans Filosofi* (Copenhagen 1896) p. 99. This book gives a fuller account of Rousseau's life and thought than there was room for here in the general history of philosophy.

[89]) P. 495. Discours sur l'inégalité, Amsterdam 1755, p. 80. *Lettre à M. de Beaumont* (Petits chefs d'oeuvre. Paris 1859, p. 304 f.). — Whereas this happy condition is depicted here as following directly upon the most primitive, animal condition, Rousseau elsewhere (e.g. in the letter to the Maréchale de Luxembourg, 20 January 1763) depicts a more advanced culture which still bears the stamp of simplicity, but which is corrupted by the introduction of the manners and wants of a *[528]* great civilized country. Such, in Rousseau's view, was the culture of Switzerland in his time, and he therefore fights eagerly against introducing French customs (such as comedy). This question brought Rousseau into sharp conflict with Voltaire and gave occasion to his *Lettre à d'Alembert.*

[90]) P. 495. Cf. letter to Mlle. D. M., 7 May 1764: On ne quitte pas sa tête comme son bonnet,

et l'on ne revient pas plus à la simplicité qu'à l'enfance; l'esprit une fois en effervescence y reste toujours, et quiconqve a pensé, pensera toute sa vie. C'est-là le plus grand malheur de l'état de réflexion etc.

[91]) P. 499. Souvenez-vous toujours que je n'enseigne point mon sentiment, je l'expose. *Émile,* IV (éd. Paris 1851, p. 326). Je ne voulais pas philosopher avec vous, mais vous aider à consulter votre coeur (p. 344). Je ne raisonnerai jamais sur la nature de Dieu, que je n'y sois forcé par le sentiment de ses rapports avec moi (p. 327).

[92]) P. 501. Cf., besides the third letter to Malesherbes, also *Confessions,* XII (éd. Paris 1864, p. 608): "I find no worthier homage to the Deity than the mute admiration which the contemplation of its works calls forth, and which is not expressed in definite acts . . . . In my room I pray more seldom and more drily; but at the sight of a beautiful landscape I feel myself moved, without being able to say why. I have read that a certain bishop, visiting his diocese, found an old woman whose whole prayer was only to say O! and said to her: Good mother, always pray thus; your prayer is worth more than ours. My prayer is of this better kind."

[93]) P. 503. *Gierke*: *Althusius,* p. 201. *Rousseau*: *Lettres écrites de la montagne* (6th letter).

[94]) P. 504. *Contrat social,* III, 15. — Cf. *John Morley*: *Rousseau,* London 1891, II, p. 166—168, and my essay on Hamilton in "Mindre Arbejder."

[95]) P. 505. Cf. *Lettres écrites de la montagne.* Lettre VI (éd. Amsterdam 1764, p. 219): Je ne sors pas de la thése générale . . . Je ne suis pas le seul qui discutant par abstraction des questions de politique ait pu les traiter avec quelque hardiesse.

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The Philosophy of the German Enlightenment and Lessing.

1. Characterization of the Thought of the Enlightenment

In the middle of the eighteenth century the Wolffian philosophy was the dominant one in Germany. It took the place of the renewal of Aristotelian and scholastic philosophy for which Melanchton had laid the foundation, and acquired great significance for German intellectual development, since by its clarity and sobriety it was well suited to popularization. In this respect an important work had already begun before Wolff, the philosopher of law Christian Thomasius laying great weight on breaking through the barriers, hitherto so fixed, between the learned and laymen. Among other things he began to lecture in German and to publish philosophical works in that language, something that aroused much offense, so that the board of censors even sent back one of his works with the message that "one could not censor any writing in which philosophical matters are taught in the German language." Thomasius on the whole worked much, both orally and in writing, to promote a free and enlightened treatment of moral and social conditions in wider circles. In him there was, beside the critical and practical tendency of his nature, a mystical-religious tendency, which explains how he could for a time work together with the Pietists, not merely because the orthodox were common enemies. The Pietist tendency, which developed in Germany from the end of the seventeenth century, worked in the service of spiritual liberation by leading *[2]* back to the inner experiences of mental life from external, authority-bound literalism. The traditional dogmatics and the external ecclesiastical order were pushed aside for the subjective inwardness of the single individual. In the religious domain work was

thus being done for the emancipation of the individual, whereas the state-churchdom of the seventeenth century had compelled individuals to merge in common religious forms. Now, with personal inwardness, individual differences too could come forward; they began to arouse more interest than the confessional differences, and soon value and interest were ascribed to them, even if they did not agree with the scheme of the penitential struggle that Pietism had set up from the first. The universal priesthood was asserted; the right of laymen had of course to be acknowledged, since the great things were accomplished just as well — if not better — in lay souls as in orthodox priestly souls that had all the dogmatic rubrics in order. Pietism led back to the natural, the practical, the useful, in leading away from literalism and dogmatic scholasticism. In this it was akin to the other tendencies of the time — even to the Wolffian philosophy, although for a time it appeared as its bitter enemy. Wolff and his disciples wanted to popularize philosophy, as Pietism wanted to popularize religion. And the two tendencies were united in good accord after Wolff had victoriously asserted his position. A whole series of eminent philosophers of Wolff's school belonged to the Pietist tendency and worked at one and the same time in the service of rational enlightenment and of religious interiorization.

But quite other influences too than those that stemmed from Germany's own philosophical and religious movements made themselves felt in the origin of the so-called Enlightenment. The English philosophy of experience became the object of eager study. Christian Thomasius is already, in his conception of the philosophy of law, strongly influenced by Locke, and in the younger Wolffians Locke's influence is refracted with Wolff's, so that their philosophy finally becomes a combination of the thoroughgoing rationalism that had been learned from Wolff and the philosophy of experience that
[3] Locke had founded. The secret was that Wolff's rationalism could really only lead to the construction of a series of rubrics, and that, the more complete this system of rubrics gradually became, the more there was bound to arise a hunger and thirst for a content of experience that could fill it. There showed itself here a similar effect of contrast to the one that had set Pietism up against orthodoxy in the religious domain. In the philosophical domain this effect of contrast appears in the great interest that empirical psychology attracts. Not speculative metaphysics, but psychology grounded on experience came more and more to be regarded as the fundamental science. The science of mental life, as observation teaches us to know it, became the foundation on which the aesthetic, moral and religious problems were worked out. This became an important turning point in the history of psychology, since it gained greater independence and drew nearer to natural

science. Wolff had still ascribed greater worth to speculative ("rational") psychology than to empirical. This now became otherwise. One sought to take experience as the foundation and only later to examine how far one could get by starting from it. The most eminent of the representatives of the philosophy of the Enlightenment, Johann Nicolas Tetens, expresses this in a very clear way (in the preface to his *Philosophische Versuche über die menschliche Natur und ihre Entwickelung*. Leipzig 1777. I. p. XIII): "One ought not to begin the investigation of the soul with metaphysical analyses, but only end with them. The psychological analysis must come first. Once it has been brought about, the metaphysical analysis, which consists only in the investigation of a few fundamental faculties and modes of action, can very quickly be brought as far as it can go at all. If, on the other hand, that empirical knowledge of the fundamental faculties is lacking, it is fruitless to try to explain them from an organization that is to such a degree unknown to us. To this is added, moreover, that however far one advances in metaphysical psychology, the correctness of its propositions must nevertheless always be tested by means of observations." Kant had already earlier expressed himself in a similar direction *(Nachricht von der Einrichtung seiner Vorlesungen in dem Winterhalbjahre von 1765—66).* He asserts the necessity in philosophy of proceeding analytically, not constructively, and what matters is then first and foremost a foundation of experience. He therefore begins *[4]* his philosophical course with empirical psychology, in which nothing can yet be learned about the nature of the soul, indeed it cannot even be decided whether a soul exists!

It is this dominant interest in empirical psychology that, not least, gives the German philosophy of the Enlightenment its peculiar stamp. Within this common stamp manifold nuances appear, which, however, we cannot go into more closely here. In the first volume of his *Geschichte der neueren deutschen Psychologie* (Berlin 1894) Max Dessoir has given a detailed account of the way in which the psychological problems were treated by the investigators of this time. Here we can dwell only on what has significance for the general history of philosophy[1]).

The psychology of the period of the Enlightenment is grounded first and foremost on Leibniz's thought of the difference between obscurity and clarity as the fundamental difference in mental life, and of representations as the elements of mental life. For the deeper-lying motives and hints in Leibniz's psychology there was no eye. It was the real period of splendor of rationalism. Enlightenment of the understanding was the watchword, and everything in mental life that was not at once transparent was conceived as a chaos of obscure representations. A practical consequence of this intellectualist psychology was that people looked forward to the future with great confidence: only

light, and all will be well! A turning point came when it was recognized that mental life has other elements than the intellectual. English psychology from Shaftesbury and Hutcheson onward had clearly recognized this. Rousseau, whose influence on the German development was extraordinarily great, had with enthusiasm pleaded the cause of feeling and inveighed against the overvaluation of intelligence. And finally Pietism had already worked in this direction. The period of rationalism was succeeded by that of sentimentalism. The word "sentimental" dates from the eighteenth century; it seems to have been formed by the English novelist Sterne and was (at Lessing's suggestion) rendered in German by "empfindsam." It was in the endeavor to develop an aesthetic theory that the peculiarity of feeling as an independent side of the life of consciousness
[5] was first asserted. J. G. SULZER (in essays in the publications of the Berlin Academy from the years 1751—52) and MOSES MENDELSSOHN *(Briefe über die Empfindungen* 1755) became aware, in their investigations of aesthetic feeling, that here there was an immediate and positive side of mental life which did not come into its own by being depicted as a chaos of obscure representations that, on account of our imperfection, were hindered from forming themselves into full clarity; according to Leibniz's and Wolff's psychology feeling was only obscure, inhibited representation. The Wolffian A. G. Baumgarten (*Aesthetica* 1750) had been the first to use the word aesthetics in the modern sense: the doctrine of the beautiful; but, true to his system, he conceived aesthetics as the doctrine that set up rules for obscure representing, thus for the lower part of the cognitive faculty, just as logic gives rules for clear representing, thus for the higher part of the cognitive faculty. The great interest in the inner events of mental life and in poetry and art that was awakened in Germany in the middle of the century was bound to give birth to the need of a new psychology, one that gave the life of feeling a positive and independent position in the mental world. Mendelssohn shows very rightly (Brief IV. Philos. Schriften. Verbesserte Auflage. Berlin 1771. I. p. 22) that if Baumgarten's conception were right, the feeling for the beautiful would have to fall away with the advance of enlightenment, and the higher beings, who are in possession of greater mental clarity, would then be able with justice to complain of "the wretched advantage that closes to them the source of joy with which the lower beings are so richly provided." Characteristic, that precisely so pronounced a representative of the philosophy of the Enlightenment should come to make this reflection! It is in the very interest of enlightenment that he demands that the distinction between obscurity and clarity must not be the only one to be made in mental life. The obscure — he shows — has in and of itself nothing to do with pleasure and pain. He finds the condition of the feeling

of pleasure in an agreement between the manifold that makes itself felt in the soul (or, in the case of sensual pleasure, in the nervous system); but the feeling of pleasure itself is a positive force in our soul, not an inhibited state. In a later work (*Morgenstunden* 1786) Mendelssohn describes feeling as a faculty of valuation (Billigungsvermögen) and has *[6]* thereby pointed aptly to the important function that this side of the life of consciousness performs[2]). — In his work *Ueber die Deutlichkeit der Grundsätze der natürlichen Theologie und Moral* (1762) Kant lays great weight on the difference between cognition and feeling that Sulzer and Mendelssohn had impressed upon us. "Only in our day," he says (IV, 2), "has one begun to see that the faculty of perceiving the true is cognition, the faculty of feeling the good is feeling, and that they must not be confused with one another." Kant is in this not only influenced by his German predecessors, but by Hutcheson and Hume, and most of all by Rousseau. — But the decisive step was taken by TETENS with his distinction between sensation (Empfindung) and feeling (Gefühl), which had hitherto been used synonymously. "Feelings," he says (Philos. Versuche. I. p. 168), "mean, in contrast to sensations, merely that a change or an impression in us and upon us is felt, without our knowing, through this impression, the object that has brought it about. Sensation, on the other hand, points to an object which, by means of the sensible impression, we notice and, as it were, find present in us." Yet the two elements stand so near one another that Tetens — not to the advantage of clarity — will use the word feeling as a common designation for them both. He opposes Leibniz and Wolff, who called all the elements of mental life representations, and wants this word (in accordance with Hume's usage) restricted to denoting reproductions. The faculty of forming and combining representations he calls understanding, and his psychology then leads him to a tripartition: feeling, understanding and will, such that the last two faculties denote the life of consciousness from the active side, the first from the passive side. From Tetens this psychological tripartition came to Kant, except that the latter carried out the distinction between sensation and feeling more consistently. — As to the significance that can reasonably be ascribed to such psychological distinctions, people at that time were not clear. They were inclined to construct a special mental faculty for each new psychological distinction they found, and psychological research knew only by way of exception higher tasks than mere description and classification. The important foundation for an explanatory psychology, which had been created by the doctrine of *[7]* association of Spinoza and the English investigators, was as good as not used. For all that, the psychology of the period of the Enlightenment has had great significance in sharpening the sense for self-observation, and it marks an important and lasting advance

by its recognition of feeling as an independent side of the life of consciousness.

Whereas the strong movement of feeling that was awakened in the middle of the century became in France a womb of revolutionary thoughts, in Germany, where both the sense for and the need of occupying oneself with public affairs were far smaller, it found an outlet partly through psychological study, partly through artistic production in the new tendency inaugurated by Herder and Goethe. It was the intellectual and the aesthetic interests that stood in the front line. It is therefore characteristic that the empirical psychology of the time lays so little weight on the will, the active side of the life of consciousness. Everything turns on enlightenment and on feeling. In deeper minds, however, a want is felt here too. Friedrich Heinrich Jacobi, whose sentimentalism bears the stamp of the time, although — as will be described in a later connection — as a philosopher he stands in sharp opposition to the men of the Enlightenment, has in his first works spoken of the unhappiness of a life of feeling that is cut off from the natural connection with action. After an intimation that modern political conditions gave no possibility for great feelings and characters to have effects outward, it is said in Jacobi's dialogue *Der Kunstgarten* (1779): "Alas, the soul is little helped by feelings and thoughts that have not proceeded from action and do not again aim at action . . . Our most glorious knowledge serves us in the end only for idle contemplation, our most exalted feelings only for solitary, barren enjoyment." It was the time of beautiful souls and noble hearts; people believed themselves capable of the highest; there was fermenting and roaring; there was "Sturm und Drang"; the immediate need of the heart was set against all reason, all rule and all convention, the original force and simplicity of nature, which people thought they had found again, against the forms of culture and society; chaos became the highest, and the capacity to produce new forms created from within stirred
[8] really only in the poetic domain. This period of ferment, in which feeling determined representations, whereas the philosophy of the Enlightenment had from the first asserted the reverse relation as the only possible one, was like a confirmation by experience of the new psychology's doctrine of the independence of feeling. It was the breakthrough of Romanticism in the midst of the camp of the Enlightenment. Through many successive forms it dominated literature from now on and gave rebirth to human imagination after the long intellectual labor. And, as will appear, this whole artistic renaissance worked back upon the philosophical development and gave it, at an important point of its history, a fateful direction. —

The psychological interest of the philosophy of the Enlightenment thus led it to its limit. But in the classical time of the Enlightenment and in its typical representatives

this limitation was not felt. People thought themselves sufficiently equipped with reason to be able in any case to be sure of going the right way, even if they did not think they had gone the way to its end. An enlightened man grants his descendants too that they will advance in enlightenment, and from the necessity of this he does not draw the consequence that in himself there must still be found a good deal of obscurity. In any case — so it was reasoned — what is needed for blessedness our reason can perceive. But to this belonged, it was thought, first and foremost faith in God and immortality; otherwise one could not feel secure. In an infinity of expositions people took up and carried out *the "natural religion,"* which English and French writers had taught. But the polemical relation in which natural religion was placed in France to positive religion came out only in single cases in Germany. Protestant theology was more elastic than Catholic. The Church and the theological faculties were for a large part filled with Wolffians, who did not acknowledge anything as revelation unless it satisfied the demands of reason, but who at the same time thought, from the conviction of their hearts too, that they could establish that the content of the biblical writings agreed with reason throughout. This *ecclesiastical rationalism* thus laid weight on a natural interpretation of the miraculous content of the Bible, because it started from the assumption that Christianity must agree with the religion of reason, indeed really was nothing but a historical proclamation of *[9]* it. Thus the reason that had been gathered from the experience and thinking of the preceding age could be spread through the ecclesiastical organs over the whole people. The Church became an organ of the Enlightenment with all its good and weak sides. An important work of education was done in this way. Yet it was not everyone in whom natural and positive religion could be reconciled in so easy a way. In this time of individualism, when so many different motives were refracted in men's consciousness, the religious development of individuals had to take the most different directions. A remarkable example of a course of development from orthodoxy through Pietism to rationalism and from there to a position that may recall the one Lessing, Herder and Schleiermacher later took has been described by Joh. Chr. Edelmann (1698 — 1767) in his autobiography (edited by Klose, Berlin 1849). The literalism and externality of the orthodox drove him over into Pietism, and he came into intercourse with sectarians and prophets. But their hatred of reason and their lust for domination angered him, and in his anguish he then found out that at the beginning of the Gospel of John one should not translate "The *Word* was God," but "*Reason* (Logos) was God!" If reason is God, religion can contain nothing contrary to reason: with this the transition had been made from Pietism to rationalism. But from the usual rationalism he parted again when he read

Spinoza's proposition that God is the *immanent, inner* cause of things, not their *external* cause. His faith now became that God is the eternal being in all things, inwardly one with every being. All that is true and good in the world is God. The world is eternal. Christ was a man who led men nearer to God by redeeming them from the false notions of God and teaching them mutual love as the highest. The priests brought him down, because they thought he wanted to overthrow their power, although he (like later Edelmann and all his true followers) referred the rabble to the priests. The Last Day breaks forth in every man who awakens from the slumber of error. — Edelmann thought that with the help of historical criticism and symbolic interpretation the true religiousness could
[10] be found in the biblical writings. For Hermann Samuel Reimarus (1694—1768), one of the most zealous champions of "the truths of natural religion" against materialists and other ungodly men, the matter stood otherwise. Just as convinced as he was that the purposiveness in nature (particularly the instincts of animals, on which he wrote a special work) points to the origin of the world from a wise and good God, and that in another life better and more lasting provision is made for human blessedness than could be made in the present life, — just as convinced was he that the content of the biblical writings conflicted with the reason and morality that natural religion enjoins. But this last conviction he thought he must keep to himself. In all quietness he had sought to clarify his thoughts on the biblical history and the biblical writings, and had worked out a work which he called *Apologie für die vernünftigen Verehrer Gottes,* and in which he subjected the biblical literature to a sharp and dismembering criticism from a historical, a scientific and a moral point of view alike. Touching is the reason he gives for not publishing this work: he fears that orthodox fanatics would rob him of the love of his wife and children or else stir up a persecution that might fall upon them. "The reverend gentlemen," he says, "must believe that an honest man causes his mind no small torment when he must dissemble his whole life through." No one suspected what the Hamburg professor, who had so finely conducted proofs both of God and of immortality, had lying in his desk. Lessing, who through Reimarus's family had obtained access to the manuscript after the author's death, published fragments of it, declaring it to be a writing found in the library at Wolfenbüttel. Later David Strauss (*H.S. Reimarus und seine Schutzschrift.* Leipzig 1862) has given an abstract of the whole work. What interests us here is especially Reimarus's general position. Natural religion is enough for him; a revelation is therefore superfluous. But it is moreover physically and morally impossible. God cannot by miracles break into his own work, and he cannot, by revelations that are not imparted to all, and that not all can even come to know, favor some above others.

Most of all, for Reimarus, the doctrine of the eternity of the punishments of hell conflicted *[11]* with the true notions of God, and it was this point that had first aroused his offense. Since, moreover, many details in the biblical narratives aroused his misgivings, it is a purely negative result that Reimarus gets out of his criticism. The only explanation he could give of the origin of these narratives was that they were due to deceit on the part of the Jewish priests and the apostles. The German professor's thorough research thus led him to the same result to which a more light-footed reasoning had led Voltaire. If it is not revelation, it is fraud: that is a dilemma on which from this time many orthodox and freethinkers agreed from the bottom of their hearts, and to which the religious debate is still often brought back, as if since the time of Reimarus and Voltaire no new points of view in the philosophy of religion had come forward.

More peaceably the relation between natural and positive religion took shape for Moses Mendelssohn (1729 —1786), who gave the most popular expositions of the doctrine of natural religion. He was a Jew from Dessau who had followed his Talmud teacher to Berlin and — like Spinoza in his time — had worked his way into Western European literature, led by the need of an intellectual culture going further than Hebrew literature could provide. This was bound up with all the greater difficulty since at that time Jews were forbidden to learn the German language. The young man's energy overcame all obstacles; he learned Latin and German, and Locke and Wolff became his favorite authors. His task was to state in elegant German (of which he became master enough) what Wolff and his disciples had developed in many-volumed books in a long-winded and pedantic exposition. He worked his way up to an admired position in German literature, became the friend of Lessing and Kant, and stands as a type of the popular philosophy of the day. He did not separate himself from his fellow believers, but fought with zeal to procure them a better social position. In his interesting work *Jerusalem oder Ueber religiöse Macht und Judenthum* (1783) he sought to show how the right fundamental propositions for ordering the relation between state and church must lead to a freer position for his fellow believers in civil respects. It was his view that *[12]* Judaism had no dogmas beyond the content of the religion of reason, but that the Jewish religion consisted in a law valid for the Jewish people. And since he now thought he could furnish proof of the immortality of the soul (*Phädon* 1767) and of the existence of a personal God (*Morgenstunden* 1786), there was for him the most beautiful agreement between philosophy and religion. His proof of immortality he grounded partly on the fact that the capacity to think cannot be explained as a product of material composition, and that the soul, since it is thus immaterial, must be imperishable, — partly on the

fact that a being destined for perfection will not be able to be stopped on its way. The existence of God he proves partly by the ontological proof taken from Descartes, partly from the purposiveness of nature. The last-named work (Morgenstunden) came out only after Kant's "Kritik der reinen Vernunft" had heralded a quite new period in the history of philosophy. Mendelssohn declares in his preface that he knows well that the school to which he belongs, the one "that perhaps ruled too high-handedly in the first half of the century," is no longer in repute. Newer tendencies have come forward, which his ill health has kept him from becoming more closely acquainted with; but he expresses the hope that "the all-crushing Kant," whose profundity he admires, will "build up in the same spirit in which he has torn down." He had had greater confidence on his philosophy's behalf when, nineteen years earlier, he published the most celebrated of his writings, the one in which he tried to let Socrates speak as he would speak if he appeared in our days. "After so many barbarous centuries," it is said here (in the "Anhang"), "in which human reason has had to slave for superstition and tyranny, philosophy has at last lived to see better days. All parts of human knowledge have made considerable advances through fortunate observation of nature. Our very soul we have by this way learned to know better. By a closer observation of its effects and states several data have been established from which, by a tested method, more important consequences too can be drawn. The most important truths of natural religion have, through this
[13] improvement of philosophy, attained an evidence that eclipses all the knowledge of the ancients." One understands from this pronouncement why he called Kant, after his "critique of all speculative theology" had appeared, "the all-crushing!"

There remain to be mentioned some interesting attempts that were made, within the philosophy of the Enlightenment, at treating *the problem of knowledge.* The Wolffian philosophy was indeed the dominant one; it was its fundamental thoughts that people sought to work into the general consciousness. But greater weight was laid on experience and the necessity of taking in material, and people sought in various ways to combine Locke's philosophy with Wolff's. From this arose an *eclecticism* which does not quite coincide with the popular philosophy proper, which kept nearer to Wolff. The chief seat of eclecticism was the University of Göttingen, and its chief representatives were Feder and Meiners, whereas the popular philosophy had its chief seat in Berlin, from which Mendelssohn's writings went out, and where Nicolai's *Allgemeine deutsche Bibliothek* and Biester's *Berliner Monatsschrift* spread the thoughts of the Enlightenment among the educated public. Only a few thinkers felt the problem that arose from having to grant that both Locke and Wolff were right, having to admit the necessity both of taking in

material and of working up this material according to reason's own laws. These thinkers belong among Kant's nearest predecessors in the domain of epistemology. The Leipzig professor C. A. Crusius *(Entwurf der nothwendigen Vernunftwahrheiten.* Leipzig 1745) showed that the difference between sense perception and thinking does not coincide with the difference between obscure and clear representing: sense perception can surely be altogether clear and distinct! This demonstration forms an analogue to Sulzer's and Mendelssohn's demonstration that the feeling of pleasure and pain is something other and more than obscure representing. It naturally led Crusius to lay greater weight on experience than Wolff consistently could. He distinguished sharply between the reasons that lead us to perceive something and the causes by which things in reality come into being (grounds of knowledge and real grounds), a distinction Wolff and the whole dogmatic philosophy had blurred. He also refutes Wolff's attempt to derive the principle of causality from the fundamental proposition of contradiction. And as the problem of the relation between thought and reality was thus sharpened, Crusius also *[14]* had to dispute the validity of the ontological proof of God's existence, which of course amounted to this, that if God could be thought, he must also exist. — Still nearer to the decisive problem came Joh. Heinrich Lambert, an investigator known also as a natural scientist and mathematician. In his work *Neues Organon* (Leipzig 1764) he enjoins the necessity of beginning with experience and applying the analytic method. We cannot begin with construction, since what matters first and foremost is to determine rightly the single concepts with which we are to operate. An anatomy of concepts must first be carried out in Locke's manner. But it is not enough to find the simple concepts that are contained in our experience; we must also find in how many different ways they can be combined with one another. Thereby a transition is made from analysis to construction; from the mere setting up of concepts one passes over to discovering the fundamental propositions and postulates that lie in them. This transition Lambert regards as quite easy. If one only begins with the simplest concepts and advances without skipping any possible combination, one will be able to carry out the construction with full certainty. Lambert is, however, clear that what can be reached in this way can only be forms, not any real content; and he raises in a letter (3 Febr. 1766) to Kant, whose thoughts at this time were moving in the same direction as Lambert's, the question: "whether and how far knowledge of the form of our knowledge leads to knowledge of its content?" To that extent he harbors doubt about the justification of philosophical constructions, and comes to touch the decisive turning point without, however, going further into the real problem. But Kant honored his predecessor so highly that he would have dedicated

the "Kritik der reinen Vernunft" to Lambert, had the latter not died before the work's publication. Lambert still belongs to the secure dogmatism of the philosophy of the Enlightenment, since he did not see what had already dawned on Kant early, that the transition from analytic to constructive method presupposes conditions that entail a
[15] limitation in principle of our knowledge (cf. Kant's letter to Lambert 31 Dec. 1765). —

Closest of all to Kant's fundamental thought came the South Schleswiger Joh. Nic. Tetens (who, after having worked as professor of philosophy and mathematics at Bützow and Kiel, held a series of high administrative posts in Copenhagen, where he died in 1807); but he too had himself been set going by the work of Kant in which the critical philosopher's fundamental thought first came forward (the Dissertation of 1770). Tetens's "Versuche über die menschliche Natur" is, in both psychological and epistemological respects, the most important philosophical work in Germany in the period immediately preceding Kant's chief work. In the psychological respect we have already shown its significance. In his doctrine of knowledge Tetens takes as his foundation a thought the holding fast of which would perhaps have brought greater clarity into Kant's philosophy: namely, that all conscious apprehension is apprehension of a relation. By every act of attention we lift what we observe out from its surroundings: "The word: see! expresses at least this much, that the object I observe is a particular thing by itself." Observation is a distinguishing, an "Auskennen." (I. p. 273). The next step is conscious comparison. Here too, by the very act of thought, a relation (of likeness or difference) is fixed between things. A special class of relations we have in space and time, which (as Kant had already taught in the "Dissertation") are forms under which our knowledge orders the material taken up in sensations. "As Herr Kant has remarked, we do not form the concept of time by abstraction from the representation of the objects we sense, but our own acts of sense, which have their succession and duration even if no object is sensed, furnish the material for the abstraction of the concept of time." (I. p. 398, cf. 277). The last kind of relation is found in the relation of dependence (the causal relation), and here Tetens seeks in an unclear way to mediate between Hume and Wolff. Without experience, he says (I. p. 320), I would certainly not suppose that the occurrence of one phenomenon would be followed by a certain definite other phenomenon; but — he adds — this supposition I state in a judgment which my understanding cannot but pass, and which is therefore more than custom or association of ideas, is a true thought, although it goes before
[16] experience. With this, of course, Hume's problem is not answered; Tetens has not shown the possibility that judgments our understanding passes with logical necessity can have validity for real events. When Kant read Tetens's "Versuche," which he valued highly,

and which (according to what Hamann relates in a letter) was always to be found on his table, he had already found the thought that could carry epistemology further on this point.

2. Gotthold Ephraim Lessing

It has appeared in the foregoing that the so-called Enlightenment at several points points beyond itself. It has, no doubt, a certain inclination to think itself complete and to look down pharisaically on the darkness of the past; but it is at the same time an age that is rich in possibilities, and for Germany it became a time of transition to a magnificent period of poetic and philosophical flowering. In the most eminent mind of this period the feeling of belonging to a period of transition expresses itself with particular vividness. Although LESSING stood in a relation of personal friendship with Mendelssohn and Nicolai, a friendship in which the latter were by no means merely the receiving party, he nevertheless did not find himself as satisfied with the Enlightenment as they. He felt a stranger in his time. Unsatisfied by the given forms of the life of the spirit, he laid, like Socrates, the weight on the subjective, personal side of the endeavor after truth. The hunt, he said, is better than the catch. By this stress on the personal feeling of craving and striving he is the child of his time; but it also made it possible for him to understand other times and positions better than the men of the Enlightenment and of sentimentalism were otherwise able to. The difference between times and positions comes out most, of course, when one keeps to the final results, whereas in the inner striving, the subjective forces that have given birth to the results, greater kinship may be found. Lessing's historical eye is connected with his favorite idea of eternal striving. And that his eye was to so high a degree directed toward the future, with hope of what it would bring, is connected still more clearly with this idea. *[17]*

This is not the place to go into Lessing's significance as a poet and as an aesthetic writer, although it might well be of interest to trace the fundamental character of the great seeker in all the different domains in which he worked. Nor do we go more closely into his biography here, since it does not contain data that are of significance for understanding him as a philosopher of religion. He was born at Kamenz in Lusatia in 1729, studied at Leipzig, lived later in Breslau, in Berlin and in Hamburg, occupied with poetic and aesthetic works, until in 1770 he became librarian at Wolfenbüttel. From his youth he had busied himself with philosophical studies and made drafts of philosophical essays. But only in his Wolfenbüttel period did this side of his interests come into the

foreground, especially during the polemic in which he became involved as the publisher of fragments of Reimarus's "Apology" (the Wolfenbüttel Fragments). Here he waged a literary battle against narrow-minded orthodoxy which is unique for the superiority in the art of exposition, learning and wealth of ideas that he displayed. He died in 1781.

Lessing was fully aware that the most important capacity in him was the capacity for criticism. He lacked creative capacity. But he possessed two things of value that distinguished him above most of his contemporaries. He had a living thirst for genuine, original spiritual content, in religious, philosophical and aesthetic respects alike. And he had historical sense and great capacity to appreciate what the past had brought forth of original spiritual content.

His historical sense is perhaps what sets him in sharpest opposition to his contemporaries. If neither orthodoxy, Pietism nor rationalism satisfied him or stood before him as forms in which religious life could develop in the future, it was because he had a clear eye for the historical character of all positive religion. Instead of finding the most important form of appearance of Christianity in the biblical writings, over whose interpretation men contended, he pointed back to the life and the whole religious development and tradition from which these writings have proceeded, and which make their
[18] origin explicable to us. It was not mere tactics when Lessing thus (during his defense of the publication of the Wolfenbüttel Fragments) maintained that Christianity did not stand and fall with the Bible. He asserted that Christianity was older than the Bible, and that it was not on book learning, not on reports of "spirit and power," but on ever living, present "spirit and power" that the future of Christianity depended. *(Ueber den Beweis des Geistes und der Kraft).* He comes out here against the overvaluation of book learning and against the theological tutelage that had become general in the Protestant Church as a consequence of Luther's zealous reference to the Bible as the rule of doctrine. He pointed out that if Christianity was to be preserved, it must have other supports than those to which orthodoxy had hitherto clung. He had in his time greeted the appearance of the Herrnhuters with joy, since in this movement he saw a turning away from orthodox externality, an interiorization that set life above the letter. Christianity, he maintains against his orthodox opponent Goeze, is essentially a matter of the heart, of feeling, and the criticism of the historical and philosophical proofs does not touch those who believe in simplicity.

Yet Lessing, not only in his letters but also in his writings, has stated clearly enough that he did not himself remain standing at the consideration of the Christian religion as the historically revealed highest truth indicated here. In the *Duplik,* in the *Antigoeze,* in

Nathan der Weise, in the *Erziehung des Menschengeschlechts* and in the *Gespräche über die Freimaurer* he develops a whole theory in the philosophy of religion, which is not only highly remarkable for its time, but which is still to this day well suited to awaken reflection. — Lessing's peculiarity was that he furnished friends and opponents alike with new arguments. By virtue of his great love of truth and his clear eye he was often able to find a force on his opponents' side which they themselves did not suspect. He has therefore sometimes been claimed by the Christian side (while at other times he has been accused of dishonesty by the same side!). The world (even the so-called Christian part of it) is simply not used to one's helping one's opponents. But this care to bring out everything that could clear up the problems has nevertheless not prevented Lessing *[19]* from developing his own ideas in their full extent.

The starting point for the exposition of these ideas is best taken from a famous pronouncement (in the "Duplik"): "Not the truth of which a man is in possession, but the sincere effort he has spent to get at the truth, constitutes the worth of man. For not by the possession of truth, but by the search for it, do man's powers expand, and in this alone consists his ever growing perfection. Possession makes one calm, lazy, proud . . . . If God held all truth enclosed in his right hand, and in his left the one, ever living striving after truth, though with the addition that I should always and eternally err, and if he said to me: choose! — then I would humbly throw myself upon his left hand and say: Father, give me this! Pure truth is indeed for thee alone!" — These words were directed both at the orthodox and at the philosophers of the Enlightenment, and both needed to hear them. There might, however, seem to be something Tantalean in the eternal seeking, especially since Lessing is willing to accept the condition that he shall always err. But one must notice that Lessing speaks hypothetically. He *conceives* of a choice between eternal seeking and error on the one side and mere possession of the truth on the other. But he is evidently conceiving of a case that does not occur. This is seen not only from the linguistic form, but also from the grounding of the assertion that the worth of man does not depend on the possession of truth, but on "the sincere effort" he spends to acquire it. This worth is conditioned, namely, by man's powers being *expanded* by the search. Such an expansion would, after all, be impossible if absolutely no result were reached, only constant disappointment. Then paralysis and contraction would be the consequence, not expansion. In and of itself eternal striving without any result whatever is a self-contradiction. It is another matter that every result attained is only provisional and becomes the starting point for new striving. And this is, as the following will show, precisely Lessing's meaning.

[20] His definite relation to positive religion Lessing had already stated in his essay "on the proof of the spirit and of power," and he comes back to it more than once in his later writings. Even if the historical foundation of Christianity were ever so certain, historical truths would nevertheless — he declares — be unable to prove anything here. How can a knowledge of the eternal connection of things be grounded on special, historical facts? How can it be demanded that, on account of certain events 1800 years ago, I should transform all my representations? History is one thing, philosophy another. "This is the ugly, broad ditch which I cannot get across, however often and however earnestly I have attempted the leap!" Lessing holds to this, that however it may stand with the origin of Christianity, it has left its effects in the development. In a fragment headed *Die Religion Christi* he states that while it is difficult to decide whether Christ was more than a man, it is certain that he was a man. Let us then keep to the religion of Christ, to the religion that Christ himself had as a man; the question of the Christian religion, of the truth of the Church's doctrine, must then be left in abeyance.

The religion Lessing professes is not grounded on single supernatural events, but on the great inner connection in nature and history. The various positive religions are for him links in this connection, since he regards them as stages in the spiritual development of humanity. Instead of mocking or being angered at the positive religions, he would rather (as he says in the preface to "Erziehung des Menschengeschlechts") see in them "the course in which the human mind could alone develop in each single place and shall develop further." What education is for the single man, revelation is for the human race. By revelation the race is led forward from lower to higher stages[3]). The people of Israel learned, through the relation of obedience to its God, who stood before it as the mightiest of all gods, to accustom itself to the notion of the unity of the Deity, long before a rational concept of this unity was possible. And through promises of reward and punishment it was trained to do good and shun evil. Only later, through
[21] acquaintance with other, more advanced peoples (Chaldeans, Persians and Greeks), could more ideal religious notions develop in it. The strict discipline it had to go through was necessary in order that teachers for the whole of humanity could come forth from its midst. Christ was the first who enjoined purity of heart with a view not to temporal reward and punishment but to another life. But both the New and the Old Testament are only the elementary books of the human race, at which it cannot remain standing, although it ought not to leave them before their content has been sufficiently drilled in, and although it may be beneficial that the pupil takes his elementary book for the highest science. The advanced pupil who impatiently turns the last page of the book

must beware of awakening premature unrest in his weaker fellow pupils. But on the other hand it can also be harmful to hold the pupil too long to the elementary book; one thereby easily makes him hair-splitting and superstitious. The truths of revelation must be transformed into truths of reason if the human race is really to appropriate them, and Lessing tries, by a symbolic interpretation of the dogmas of the Trinity and of the Atonement, to show what truths may lie hidden in the dogmatic formulas. — Education must reach its goal. The time must come when man does not need the conviction of another life as a motive for his action, but does the good for its own sake. Then the new, eternal gospel, the third age, of which the enthusiasts of the Middle Ages already spoke, will have come! The coming of this future we must patiently await. Through small steps and by roundabout ways development goes forward. In "Gespräche über die Freimaurer" it is said: "The Freemason waits calmly for the rising of the sun and lets the candles burn as long as they will and can. He will not put out the candles too early, lest he should later be obliged to light the stumps again." — For Lessing the significance of religion was in the end decidedly ethical. It is an education to will the good for the good's own sake. The most important thing in Christianity was for him (as he stated in the little dialogue *Das Testament Johannis)* the heartfelt exhortation to love. In the "Freemason Dialogues" (perhaps the most beautiful and clearest expression of Lessing's religious and also of *[22]* his ethical and social views) he conceives of the Freemasons, of whom he here gives an idealized account, as a free brotherhood that wants to abolish the barriers between men that religion, nationality and rank have hitherto set up.

Shortly after Lessing's death it aroused great attention when JACOBI, in letters to Mendelssohn *(Briefe über die Lehre des Spinoza.* 1785), asserted that Lessing had declared himself an adherent of Spinoza. Since Spinoza still, even for the spokesmen of the Enlightenment, stood as the epitome of all ungodliness, it was no wonder that the assertion aroused not only attention but also grief and offense in several of Lessing's friends, especially Mendelssohn.

Jacobi (a kind of modern Herbert of Cherbury) had traveled to Wolfenbüttel to visit Lessing and get help from him in refuting Spinoza's doctrine. On this occasion he showed Lessing Goethe's poem "Prometheus," of which he had a copy. He handed him the poem with the words: "You have given so much offense, so you shall for once be offended yourself!" When Lessing had read it, he said: "I have not been offended. All this I have long known at first hand . . . . The point of view from which the poem is written is the same as my own. . . . . The orthodox concepts of the Deity are no longer valid for me; I cannot use them. ‘*Ἓν καὶ Πᾶν!* (i.e., One and All!) I can do no other. That is what this

poem too amounts to, and I must confess that it pleases me greatly." Jacobi: "Then you are quite in agreement with Spinoza?" Lessing: "If I am to name myself after anyone, I know of no other." Jacobi admits in the course of the conversation that he cannot refute Spinoza and prove his faith in a personal God distinct from the world, but that he helps himself out of the fix by a leap, a salto mortale from knowledge over into faith. After a rather wordy exposition by Jacobi, in which the latter seeks to shine with his (not in all respects thorough) knowledge of Spinoza, Lessing says: "Your salto mortale does not
[23] seem so bad to me, and I understand how a man with his head in the right place can in this way leap out head first in order to get from the spot. Take me along, if it can be done." Jacobi: "If you will only step onto the elastic spot, it goes of itself." Lessing: "Even that would require a leap which I dare not entrust to my old legs and my heavy head."

This is the main course of the famous conversation (6 July 1780). But there came out in its course some utterances of Lessing on single points that are of particular interest. — Lessing declares that he cannot accept a personal God standing outside the world. He is inclined, on the other hand, to think of God as the soul of the world. In agreement with this is the fact that in several philosophical fragments dating from his younger years he developed the thought that there cannot be anything outside God, since God, by thus having something outside himself, would become limited and finite. He asserts an inner, immanent relation between God and the world and will not think of God in analogy with a human personality. With the idea of a personal Deity in unchangeable enjoyment of its supreme perfection he could not reconcile himself. He connected with it, as he said to Jacobi, "such a notion of infinite tedium that he became anxious and ill at ease over it." It was natural that a thinker for whom constant striving was the highest could not put up with a God finished once and for all. This is, moreover, one of the points which show that Lessing's adherence to Spinoza must not be taken all too strictly. For from Spinoza's concept of God becoming and development were excluded. — Lessing next speaks out against "the human prejudice of regarding thought as the first and foremost and deriving everything from it, whereas everything, thought included, depends on higher principles. Extension, motion and thought are evidently grounded in a higher force, which is far from exhausted by them." Here Lessing polemizes against the theology rampant particularly in the Enlightenment, which builds its proof of God on the purposiveness of nature. The pronouncement agrees entirely with Spinoza, in whom
[24] thought as well as extension are attributes of substance, which besides them has infinitely many other attributes. During the following conversations, however, Lessing referred also to Hume's "Dialogues," in which a related line of thought occurs. — Finally Lessing

asks for a purely natural explanation of everything. A leap out into the supernatural he will not engage in. On this point too he points back to Spinoza, of whom he had earlier said in a letter to Mendelssohn: "Spinoza is the first who was led by his system to the possibility that all the changes of the body could spring solely from its own mechanical forces." (It would, however, be more correct to say that it was the thought of this possibility that led Spinoza to his system.)

Mendelssohn wanted to explain the whole conversation as a piece of mental gymnastics on Lessing's part. This is, however, an impossible explanation, even if Lessing may have thought that his sentimental friend could profit from such gymnastics, and even if it would be wrong to ascribe to Lessing all of Spinoza's opinions. Lessing did not attach himself to any definite system, but a circle of thoughts can be shown which were the leading ones in him, in any case in his private chamber. It was not merely indignation that Spinoza had so long been treated "like a dead dog" that led Lessing to plead his cause. On the whole a growing interest in the decried thinker can be traced. Jacobi saw in him the most consistent form of pure philosophy. Enthusiasts like Edelmann and pious orthodox like Jacobi's young friend Thomas Wizenmann felt sympathy with Spinoza in so far as they could not reconcile their religious feeling with the thought of a God distinct from the world. Goethe's artistic-naturalistic faith and Herder's religious-naturalistic faith also attached themselves with enthusiasm to the "One and All" that Lessing had proclaimed[4]). There expressed itself, from various positions, a need of a deeper world-view than the philosophy of the Enlightenment and the popular theology could give. But this need could only be satisfied by the philosophical way when a thorough investigation of the nature and range of human knowledge had first *[25]* been undertaken. An investigation of this kind was needed not only for the sake of the problem of knowledge itself, but also for the sake of the religious problem. The year after that conversation between Lessing and Jacobi, which threw so sharp a light on the religious position of the foremost men of the time, had taken place, and the same year in which Lessing died, Kant's *Kritik der reinen Vernunft* appeared.

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Seventh Book

[26]

Immanuel Kant and the Critical Philosophy.

1. Characterization and Biography

Every great work of the spirit has its fate. Since it offers a wealth of motives, it may at different times be quite different sides and qualities that make it an object of appropriation and admiration, and the thought or the chief motive that guided the work's author need not be the one that determines its lasting significance. In the time that has passed since "Kritik der reinen Vernunft" appeared, a new edition has come out about every tenth year, and to this very day the work is studied to an extent and with a thoroughness like no other philosophical work. On its contemporaries the effect came above all from the overthrow of the proofs of the religious assumptions, together with the energetic assertion of the sublimity of the moral law and the demonstration of its intimate connection with the spiritual nature of man. For a narrower circle it was the treatment of the problem of knowledge that stood in the first line. But here again weight could be laid on different sides. One could stress the assertion of the capacity of reason for knowledge independently of experience, or the limitation of the validity of this knowledge to the world of experience. One could lay the weight on the idealist side of the system, according to which reason determines what reality is; and one could lay the weight on the realist side, according to which the activity of reason acquires real *[27]* significance only when it receives its material from a source to which it cannot itself penetrate. Finally, the chief interest may turn either on the psychological analysis of the nature and working of consciousness and of knowledge, or on the consequences drawn from it concerning the conditions and limitation of knowledge. All these different motives have determined the influence of the system on the time that followed, so that now one, now another has predominated. It is the task of historical inquiry to show the

mutual relation of these different motives in Kant's philosophy and within his course of development. Perhaps it will then be possible to lead them all back to one fundamental motive, which unites in itself the practical and the theoretical interest, the speculative and the empirical tendency, the psychological and the epistemological task, and which Kant himself has designated as *the self-knowledge of reason in scientific form* (Prolegomena § 35).

In Kant's view, thinking about existence begins *dogmatically,* i.e. with an involuntary, often naive confidence in its own power and in the validity of its presuppositions. It therefore believes that it can solve every task, penetrate to the innermost essence of existence. It is the time of the great systems. Later there comes a time when it turns out that these edifices of thought do not reach to heaven, and that the builders are not agreed among themselves about the plan. It is the time of doubt, of *skepticism.* People mock the vain attempts and the mutual contradictions, and console themselves, now with resignation, now with blasé indifference, over the apparently wholly negative result. It is a natural reaction against blind dogmatism. — Kant turns against both tendencies. He holds that there is a task which both the dogmatist and the skeptic neglect: to examine the nature of our mind itself, of our knowledge, in order to discover what forms and powers we have at our disposal for understanding existence, and how far these forms and powers can carry us. The very fundamental form of all our knowledge Kant then found in the unity which all knowledge seeks to bring about. At all its steps knowledge is a gathering, a combining of what is scattered. The simplest sense-image comes about through a
[28] working together of many different impressions — and the same stamp of unity is found in the greatest system that thought can reach in its highest flight. — To this extent the dogmatic systems spring from a real, human need. One wants to embrace everything in one unified image. They are attempts to realize what knowledge at all its steps tends toward. But Kant then goes on to show that knowledge is bound to experience, and that this search for unity, when it ventures beyond the limits of experience, entangles itself in contradictions and in any case can furnish no proof of the validity of its results. It goes with thought (to use one of Kant's many apt images) as it would go with the dove if, because it is so easy to fly in the air, it were to believe that it would be still easier to fly in empty space. It is precisely the resistance that bears it up — but it also, to be sure, sets limits to the motion. — The skeptics do the dogmatic systems an injustice by merely considering them in their fully finished form. One ought to go back to their origin, to examine what human power and need it is that has led to them, and to what conditions the use of this power and the satisfaction of this need are subject. — Through

such self-knowledge Kant will free the human mind from its own earlier works, which easily become obstructing barriers for it. But at the same time, with a clearer insight into conditions and limits, he will work on with the same power from which the works of the past sprang, and he thereby prepares the way for an understanding of these works to which mere negation and criticism could not lead. *Kant has thereby set the whole of the human sciences their program.* He himself, it is true, worked only at the solution of special philosophical problems. But the point of view he adopted here has a general significance — for the study of religion, art, literature and language as well as for the study of institutions and forms of society. The relations are analogous everywhere, and the task is essentially the same: through the study of the powers that produced the work of the past, to set them free to do, with clearer understanding, the deed of the future. What matters is the demonstration and preservation of continuity, while the metamorphosis of the powers goes on. In its deepest ground it was this striving that was astir in Kant's great and profound labor of thought, and with it his central position in *[29]* the history of modern thought is given, even after one has once and for all abandoned a great many of his own favorite doctrines. —

Kant's life offers nothing sensational, nothing interesting or piquant. It was a quiet, sensible life devoted to research. He lived as a plain burgher, with no small touch of pedantry and philistinism. But out of the deep inner ground of this unassuming and inconspicuous outward life great thoughts grew up that illuminate human knowledge and human life. And the feeling for the great and sublime — which is the kind of aesthetic feeling that Kant understood and described best — perhaps thrives best in narrow circumstances, if only one can catch sight of the sky. We unfortunately lack material for a portrayal of Kant's inner, personal development. Those of his biographers who stood nearest to him, and who immediately after his death gave portrayals of his life and character (Borowski, Jachmann and Wasianski), give chiefly outward details and open to us no insight into the inner mental motives that made themselves felt during his development. His surviving letters date mostly from his later years. We are therefore obliged to construct his spiritual development for ourselves out of his writings. But before we attempt to solve this task, we must dwell a little on the main points of his life.

Immanuel Kant was born in Königsberg on 22 April 1724. His father was a saddler of Scottish descent; the name was properly Cant, but the philosopher changed the C to K, so that it should not be pronounced as S. Both parents were capable and sound natures, whose pietist piety let them bear the adversities of life with gentleness. Kant's mother in particular had influence on him. Her pietism did not make her blind to the

beauty and glory of nature, to which she sought to open her son's eyes. Kant often spoke later with enthusiasm of his parents, and the tendency toward interiorization and toward the subjective side of life which was the distinguishing mark of pietism set its stamp on his later development. The pietism of Königsberg was of a particularly gentle
[30] and humane kind. Yet Kant also felt the shadow sides of pietism in the formalism and the spiritual constraint it brought with it, especially in the school, where fixed hours of prayer and compulsory customs bred hypocrisy and unnaturalness. These experiences too came to have influence on Kant's later attitude to religion. In the school the ancient languages were chiefly cultivated, and all his life Kant was a sound Latinist, on whom a Latin quotation fitting the occasion did not fail of its effect. From his sixteenth to his twenty-second year he studied the philosophy of Wolff and the physics of Newton at the University of Königsberg. His teacher in these subjects was Martin Knutzen, one of the most independent of the Wolffians. Kant was indeed enrolled as a theological student, but attended only a few theological lectures. His circumstances were straitened. He made his way by private teaching and later, from his twenty-second year on, took posts as tutor with various noble families in East Prussia. Although he did not himself think he had been a capable teacher, he nevertheless imbued several of his pupils with his deep feeling for freedom and human dignity; for it is hardly an accident that several of Kant's former pupils took the lead in the abolition of serfdom. (At a later point in his life Kant is said to have declared: "My bowels turn over within me when I think of the serfdom in our country".) The position of tutor in the great houses brought Kant an experience of the world that his retired life would otherwise not have been able to procure him, and made him the polished man of society that, on the testimony of many, he could be. But these years were also those in which he began to gather the wealth of thought and knowledge that he displayed as soon as he first appeared as a university teacher and author. At the age of 31 he returned (1755) to Königsberg. In the same year he appeared both as lecturer and as author; he had published only a single small treatise (on a question of physics) before. His lectures were partly popular, for larger circles, on physical geography and empirical psychology, partly strictly philosophical. An enthusiastic portrayal of Kant as a teacher of philosophy in his younger days has been given by Herder, who from 1762 was his hearer for some years (in the second volume of *Briefe zur Beförderung der*
[31] *Humanität).* He himself has spoken of the principles of his teaching in a prospectus from this period *(Nachricht von der Einrichtung seiner Vorlesungen in dem Winterhalbenjahre von 1765—1766).* He lays the weight, first, on a good foundation of experience, so that the young students should not begin to speculate too early, and secondly, on teaching

his hearers not a finished philosophy but *to philosophize.* And this for the reason that a finished philosophy does not exist. Already in the first period of his development, which may be reckoned from 1755 to 1769, he has decidedly broken with dogmatism. He is in philosophical motion. Characteristic of this period is the independent criticism he directs against his teachers Newton and Wolff. In one of his most brilliant works, *Allgemeine Naturgeschichte und Theorie des Himmels* (1755), he criticizes Newton's denial that the present arrangement of the solar system could be explained by mechanical laws of nature, and puts forward the famous hypothesis that our present system of heavenly bodies has developed out of a rotating primordial nebula. In the very fixed connection between all the elements of the universe, to which the laws of nature bear witness, he found a proof that the whole universe has its ultimate ground in an absolute, all-embracing being. In this way he here united his scientific and his religious conception, while he already now rejected the current proofs of the existence of God. In his work *Einzig möglicher Beweisgrund einer Demonstration des Daseins Gottes* (1763) he has developed this standpoint. His confidence in Wolff's dogmatic constructions he had lost. It became his conviction that philosophy must proceed by the way of analysis in order to arrive at certain and clear concepts. In a series of works from the years 1762—63 he has discussed this question. It is perhaps most probable to place at this point the influence that Kant, by his own statements, received partly from Hume, partly from Rousseau. As I shall try to show later, the sharp emphasis on analysis as the philosophical method was bound quite naturally to bring him face to face with the problem of causality as Hume had posed it. For this problem turned precisely on how we can ground the assumption of a necessary relation between cause and effect when these are two different things, so that the effect *[32]* cannot be found by analysis of the cause. Rousseau's strong emphasis on the right of feeling and on the difference between knowledge and faith also produced a revolution in Kant's line of thought. Hitherto, as a good philosopher of the Enlightenment, he had placed man's essence and nobility in the understanding alone; now he learned to find a deeper foundation, which could be common to the learned and the unlearned, and in which the simplest laborer could stand on an equal footing with the greatest thinker. And Rousseau's appeal to immediate feeling and faith was bound to become all the more significant for Kant, as he was, after all, precisely engaged in undermining the proofs on which the assumption of the doctrines of natural religion had hitherto been grounded. On the whole Kant was naturally bound to come, like Rousseau, to ascribe to intellectual development a more *indirect* significance for the life of the spirit than he had at first done. His own critical labor of thought had here led him in the same direction as Rousseau, and

when the latter's *Émile* appeared, Kant was prepared to appreciate it. No wonder that on the day he had got the book he gave up his customary walk at the fixed hour, to the great astonishment of the people in the street. — In his writings from this time he lays great weight on psychology as the foundation of philosophy and particularly of ethics, and appeals to Shaftesbury, Hutcheson and Hume as his predecessors in this respect. Toward speculative constructions and systems he is skeptical. In *Träume eines Geistersehers erläutert durch Träume der Metaphysik* (1766), one of Kant's wittiest writings, he shows how easy it is in a certain sense to construct far-reaching spiritualist explanations, but at the same time how unfinished and ungrounded the concepts are with which one operates. His conclusion is: how much there is that we do not know — but how much there is, too, that we do not need to know! The Socratic ignorance, the "philosophy of ignorance" or the "negative philosophy", was at this time (as is seen from his notes) a favorite thought of his. With his fondness for Latin watchwords he would have used
[33] Cusanus's docta ignorantia, had he known this expression.

Until his 46th year Kant lived as a Privatdocent; the only public appointment he held was a poorly paid post as under-librarian. The straitened times of the Seven Years' War helped to prevent his obtaining a fixed position as a university teacher. A professorship of "poetry" that was offered him he declined. In 1770 he became professor of philosophy and in the same year published a Latin treatise *De mundi sensibilis atqve intelligibilis forma et principiis* (on the form and principles of the world of sense and the world of thought; in what follows we shall for brevity's sake refer to it as "the Dissertation"), in which the fundamental thoughts of his definitive philosophy come forward for the first time. In several statements found in his letters and notes he has pointed to the year 1769 as the year in which his fundamental view came into being. "The year 1769", he says thus in one place, "gave me much light". And in his last years he would properly acknowledge only those writings which were later than 1769[5]). The thought that then dawned on him was that *such forms and presuppositions as are conditions for anything at all becoming an object of our knowledge must have validity for all experience.* This thought he applied in the Dissertation to space and time as the forms of our sensuous intuition. The second period in Kant's philosophical development (1769 —1781) is marked by the emergence of this thought and its application to all scientific knowledge. But it testifies to the greatness of the problem and to the honesty of Kant's labor of thought that he took 11 years to extend this application from sense intuition to knowledge of the understanding. In the Dissertation it is only sense intuition that is bound to subjective forms and presuppositions. The world of sense is merely phenomenal; but with the help

of the understanding we can raise ourselves above it and know things in themselves. The world of thought is real, the world of sense phenomenal. From letters and notes of the seventies it is seen how much labor it cost before the conviction was grounded that the understanding has its forms and presuppositions just as much as sense intuition, and that therefore no scientific knowledge can carry us beyond the phenomenal world of experience. When Kant discovered that knowledge of the understanding, just as much as sense intuition, consists in a synthesis, a synthesizing, combining activity of the mind, *[34]* he was at his goal. But the result of these many years of brooding Kant then, by his own statement, wrote down in great haste, in the course of 4—5 months. He was therefore unable to lay much weight on the form of exposition. Probably he also used earlier drafts, composed at different times, without always examining carefully whether they agreed entirely. As a result *Kritik der reinen Vernunft* (1781) has become a difficult book to read, not only on account of its content but on account of its exposition. It is hardly credible, as one of Kant's biographers relates, that Kant had his good friend the merchant Green test every sentence in the book before it was written down. Had a practical man checked the exposition, it would surely have become clearer. There is a great deal of formality and no little scholasticism in it. Kant himself said of his book in a note: "When one leafs about in it, it seems to be the most pedantic thing of all; and yet it is quite properly designed to abolish all pedantry". Besides contradictions and scholastic pedantries there are also needless repetitions, which may confuse or weary. In spite of all this we nevertheless stand here before an immortal masterpiece of philosophy, a work that marks a milestone on the long wandering of human thought. By penetrating into the innermost nature of knowledge Kant succeeds in finding the conditions on which it rests and the limits beyond which it cannot lead. He shows us thought at once in its greatness and in its limitation. His faith in reason is not slackened by this limitation: for the barriers spring from the very nature of reason and are known according to reason's own laws. Later research has subjected Kant's determinations of the limits to a revision and has held, now that he drew the limits too narrowly, now that he drew them too widely; it is in any case certain that they do not lie precisely where Kant found them; but that does not diminish his greatness as the one who has had the keenest eye for the form and working of knowledge and for the significance of the problems of knowledge for the spiritual life as a whole. — Two years after the appearance of the chief work he gave (in *Prolegomena)* a shorter and more accessible exposition of its line of thought. *[35]*

After the publication of Kritik der reinen Vernunft Kant set himself the task of developing his ethical ideas. Already in the Dissertation he had abandoned the psychological

grounding of ethics that he had earlier embraced in agreement with the Englishmen. In the moral law, as it immediately makes itself known each time we feel ourselves obligated, he found reason at work in the practical domain. In a fresh and clear exposition he has developed his conception of the ethical in *Grundlegung zur Metaphysik der Sitten* (1785), but the chief work is *Kritik der practischen Vernunft* (1788). — Kant needed one more chief work to complete his critical philosophy, which was to embrace all sides of life. In *Kritik der Urtheilskraft* (1790) he discussed the problems that arise from the beauty and purposiveness of nature, and seeks to solve them by analogy with the way in which he had solved the problem of knowledge and the ethical problem. In this work he gives profound hints of a possible connection between the different domains which his critical inquiries had earlier kept sharply apart from one another. —

Kant lived his whole life in East Prussia. But he followed with great interest everything that went on in the world, both in the physical world and in the human world. Books of travel were his favorite reading, and physical geography played, as already mentioned, a great part in his teaching. He kept up with the development of the natural sciences, and not least he followed the political events with eager attention. He was enthusiastic about the North American and the French Revolution, and saw in the general interest with which people throughout Europe followed these events a sign that humanity was advancing in moral respects. News in natural science or news in politics was as a rule the subject of his conversation with his table companions. After the publication of his chief works Kant became a famous man. In the best men of the age he awakened enthusiasm for his labor of thought and for his ideal view of life and its tasks. People made pilgrimages from far away to remote Königsberg to get a sight of him, and a number
[36] turned to him for advice in moral matters. He was in favor with Frederick the Great's powerful minister Zedlitz, who followed his lectures through students' notebooks that were sent to him from Königsberg; to him Kant dedicated Kritik der reinen Vernunft. Frederick the Great himself cared nothing for German literature and philosophy; in his younger years he had been an ardent Wolffian; later he held to Bayle and Voltaire. But Kant was glad to live in the age of Frederick, under a prince who at once conducted the government with a strong hand and allowed thought to move freely. Kant did not consider his time an enlightened time, but a time of enlightenment — a time that is advancing in enlightenment. (See his little treatise of 1784: *Was ist Aufklärung?).*

But there came a time when in high places Kant came to be looked at with other eyes. The great king's successor Friederich Wilhelm II, a dissolute man and a weak character, who was given to spiritism and mysticism and was at the same time frightened

by the powerful movement in France, introduced a rule of the clergy and a sharpened censorship. Zedlitz was dismissed: Wöllner, a bigoted and narrow-minded man, was put at the head of the church and of education, and a board of censors consisting of three theologians was set up. Attention was directed in particular at Kant, the most important representative of free inquiry, and attempts were even made to bring it about that Kant should be forbidden to write! That, however, did not succeed. But the storm broke when in 1794 Kant published his work in the philosophy of religion, *Religion innerhalb der Grenzen der blossen Vernunft,* in which from his standpoint he sought to show the significance of Christianity as a historical form for ethical ideals, while he distanced himself decidedly from the orthodox conception. Kant had been so cautious as to have the theological faculty in Königsberg asked whether the content of the work fell under the theological censorship, and when the answer was in the negative, he had obtained the permission of the philosophical faculty of Jena for its printing[6]). But that did not prevent a royal letter, procured by Wöllner, from being addressed to him, in which the most exalted displeasure was expressed at his inquiries concerning religion, and further unpleasantness was threatened if he continued in the same direction. It was an unworthy treatment that the old thinker (he was now in his 70th year) here had to suffer. He had *[37]* always spoken with reverence of Christianity and its founder and had only used the right of inquiry to test the relation of Christianity to human nature and reason. One sees from Kant's papers that he conscientiously considered what he should now do. He wrote on a slip of paper: "Recantation would be base, but silence in a case like this is the duty of a subject". In his reply he then sought frankly to show that he had not been guilty of any offense, but declared at the same time that "as His Majesty's faithful subject" he would refrain from all authorship in the direction of the philosophy of religion. There lay a reservation in this: the promise, he held, applied only in this king's time — as long as he was *this* king's subject. A few years later Friederich Wilhelm II died, and his successor followed a more liberal policy. Kant now took up his authorship in the philosophy of religion again and published (in the preface to *Streit der Facultäten* 1798) the history of the conflict. This history is a revolting example of how narrow-minded fanaticism often tries to stop the free life of the spirit. Fortunately it cannot be stopped, and the ridiculous therefore unites with the revolting. Those who laid hands on Kant's venerable figure have gained a place in history beside those who laid hands on Galilei and forced him to say that the earth stands still. The spirit stands still as little as the earth. —

Kant's last years were a slow and sad dissolution of the power of thought that had worked so tirelessly. Memory and the capacity for combination fell away; obsessive

representations, especially jingles and melodies from childhood, forced themselves forward; uncanny dreams at night and restless agitation by day tormented the old man. It has been conjectured that he suffered from a disease of the brain. In the good intervals he sat at his desk to work on a last, concluding work, of which some fragments are preserved. It bears the stamp of decrepitude. Beside a few flashes of genius it is full of repetitions from earlier writings. After a long wasting away Kant died on 12 February
[38] 1804.

2. Course of Philosophical Development

It is probably grounded not only in the history of philosophy but also in experience from his own development when Kant says in one place that the first step in philosophy is always dogmatic. And yet Kant does not appear in any of the writings we have from his hand as a full-blooded dogmatist. From the first he has a lively feeling of the scientific unfinishedness of the great systems of speculative philosophy, and already in a treatise of 1758 he speaks with irony of "the gentlemen who are accustomed to throw away as chaff all thoughts that have not passed through the treadmill of the Wolffian or some other famous system". Characteristic of *the first period of his philosophical authorship* (1755—1769) is that he is dissatisfied with the existing doctrinal edifices of philosophy and seeks means to erect a new and more thorough, if not so splendid, building. It is during this work that it dawns on him that a speculative system must be preceded by an examination of the concepts with which we operate, and that very great problems are contained in these concepts. Metaphysics thereby becomes for him a doctrine of the limits of knowledge, and he begins to speak of a critique of reason itself. When many years later (in the preface to *Prolegomena* 1783) he declared that it was David Hume's criticism that *woke him from his dogmatic slumber and gave his inquiries in the domain of speculative philosophy a quite different direction,* this awakening must probably be placed in the years 1762—63, although it is a testimony to the independence and continuity of Kant's philosophical thinking that it is impossible to point to any point in his development where it would be absolutely necessary to assume a definite influence from other authors in order to understand the later development. — Since, as regards the proof of the conception of Kant's course of development set forth in what follows, reference is made to the treatise *Kontinuiteten i Kants filosofiske Udviklingsgang* (Det klg. danske Videnskabernes Selskabs Skrifter. 6. Række), we shall here dwell only on some
[39] main points.

The positive and lasting value of those writings of Kant that precede 1769, the year in which, by his own declaration, his definitive philosophy came into being, rests chiefly on two trains of thought.

a) As already briefly touched on, Kant in his "Allg. Naturgesch." attempts to carry Newton's demand for a demonstrable cause (vera causa) for all phenomena much further than the great natural scientist, to whom Kant himself looked up, had believed possible. He thereby arrived at his famous hypothesis of the origin of the solar system. A striving appears here in Kant that he never gave up. Even in Kritik der reinen Vernunft (in the doctrine of method) he declares that the wildest physical hypotheses are to be preferred to an appeal to the supernatural. This zeal to build further on the natural causal series and to avoid a break in it was connected with the fact that at the beginning of the first period Kant's philosophy of nature and philosophy of religion stand in intimate contact with each other. It was, he held, a false presupposition from which one started when one assumed that nature, left to itself, would produce only disorder and chaos. Not by chance, but according to its own laws, nature produces order and purposiveness. The very mechanical order of nature that embraces all phenomena, and according to whose laws the individual elements work together, bears witness to a ground of unity for existence, to an infinite power that stirs in the individual elements. The individual atoms are (as Kant develops in an interesting treatise of 1756: *Monadologia physica)* points of force, not small extended parts, and the fact that they work together in a law-governed way proves that there exists no original and absolute separation between them. If each element in the world had its own separate nature, it would be a matter of chance whether they fitted together in such a way that a world-whole could come out of them. Their interaction would be impossible if they were not all dependent on a common ground. In this ground of unity the mechanical order of nature and the purposiveness of nature at once find their explanation. Far from harboring fear of the demonstration of physical causes by natural science, one should precisely rejoice that through it the great primordial fact, *[40]* interaction, the connection of nature, comes forward more clearly.

In the causal connection as a fundamental fact, then, Kant here finds the foundation on which he theoretically builds his religious conception. There is a remarkable kinship between his line of thought and Spinoza's, although he quite certainly knew the latter only through expositions that did not let it come into its own. — The current proofs of the existence of God, on the other hand, he already now rejects, especially the "physico-theological" one, which would infer from the purposiveness of nature the necessity of assuming an intervention by a power standing outside nature.

b) But at the same time as Kant found in the natural causal connection itself the "only possible ground of proof" for a religious conception of the world, he was making inquiries into the nature of our thinking. He found that it consists in comparing and analyzing. Every judgment rests on a comparison between a property and a thing: either the property fits the thing, or it does not fit. We therefore constantly operate according to the principles of identity and of contradiction; and can pass over from one concept to another only when the latter can be shown in its identity with the first concept, that is, when I can find the second concept by analysis of the first. Philosophy cannot, like mathematics, begin by constructing and to that extent creating its concepts. It must find them by analysis of experience — but how then can it ground a conviction that the analysis has been carried far enough, so that there are no more marks to discover? The imperfection of earlier philosophy Kant derives precisely from its operating with unfinished concepts, since it had hastily passed over from analysis to construction, trusting that one must be able to construct in philosophy just as well as in mathematics. Kant gives two extraordinarily significant examples of such unfinished or "surreptitious" concepts. The one is the concept of mind, with which Descartes, Leibniz and Wolff had operated in their spiritualist psychology. A complete concept of the mental being that
[41] could justify ascribing to it an independent existence, separated from matter, we do not have. We have an empirical concept of mental phenomena, but empirical psychology cannot say whether a soul-substance exists or not. (The work *Träume eines Geistersehers* is a humorous demonstration of how easy it is to build up a whole spiritualist system if one simply assumes the concept of a soul-substance as given. Swedenborg's writings are used as the example.) The other example is the concept of cause. If thinking is analysis, only such relations are intelligible in which the second term can be derived from the first. But can one, by analyzing one phenomenon, discover the necessity that another phenomenon should occur? And that, after all, is what the concept of cause designates! There is no contradiction in stopping at the first phenomenon. But how then can the concept of cause be a valid concept? — Kant has here (first in the remarkable treatise *Versuch den Begriff der negativen Grössen in die Weltweisheit einzuführen* 1762) come by his own way, but most probably guided by Hume's critical remarks, to the problem of causality, and he lets it stand for the present as insoluble. But a very strong collision must now take place in Kant's consciousness between the two trains of thought, of which the one, in the interest both of natural science and of religion, asserts the causal connection as the great primordial fact, and the other asserts that this primordial fact itself is incomprehensible! Already in "Allg. Naturgeschichte" and "Beweisgrund" he had been at the problem of

causality, in that he had inferred a ground of unity for everything, since the interaction between the elements would otherwise be incomprehensible. What has happened in the treatise on negative magnitudes is a *transposition of the problem from the objective or metaphysical to the subjective or epistemological form.* This transposition was naturally called forth by the sharpened conviction that analysis must precede construction.

Since Kant, then, by analysis of a concept as important for real knowledge as the concept of cause, had come up against a problem he could not solve, it is no wonder that his whole mood at the close of the year 1762—63, so rich in labor of thought (from which five significant treatises date), was decidedly skeptical. With irony he turns against *[42]* "the thorough philosophers, whose number is constantly increasing", and asks them to solve for him the simple questions at which he has come to a stop. Out of this mood sprang, a few years later, "Träume eines Geistersehers". There is no period in Kant's life to which the expression "awakening from dogmatic slumber" fits so well as to this one. He himself later defined dogmatism as "the presumption of wanting to make one's way merely with pure knowledge from concepts, according to principles which reason has long had in use, *without examining the manner in which, and the right by which, it has arrived at them"* (Kritik der reinen Vernunft. 2. Aufl. p. XXXV). When Kant defined dogmatism in this way, he cannot possibly have meant that when he wrote the treatise on negative magnitudes and "Träume" he was still lying in dogmatic slumber. In any case he then did himself an injustice. — There are, however, eminent Kant scholars who do not think that "the awakening from the dogmatic slumber" can be placed so early. Against them the monograph mentioned above attempts a detailed account, in that it takes the disagreement among Kant scholars on this point in particular as evidence for "the continuity in Kant's philosophical course of development". —

The second period in Kant's course of development (1769—1781), the one that ends with his chief work, is taken up by the labor of finding the possibility of a transition from analysis to construction. The year 1769, which he himself has designated as a turning point, led him onto the thoughts that are developed in the Dissertation the following year. A statement in one of Kant's notes indicates clearly what the turning point consisted in: *"I found that many of the propositions which we regard as objective are in reality subjective, i.e. contain the conditions under which alone we can perceive or comprehend the objects".* The discovery he made here he has himself compared to Copernicus's. Just as it is owing to our standpoint on the earth that the heavens are seen to move round us, so it is owing to the constitution of our sense intuition that we apprehend things in space and time. What Newton called absolute space and absolute time are only schemata or forms which

[43] we construct when we notice the way in which we apprehend things. The laws of space and time are the laws of our sense intuition. But for that very reason everything that experience can show us must also fall under these laws (for otherwise it could not be intuited), and it is then understood how applied mathematics can pronounce a priori laws for phenomena. But since in this way everything that is intuited is apprehended by us according to the forms of our faculty of intuition, it is not things in themselves (noumena) but only phenomena that we apprehend by the help of sense perception.

The Copernican principle, that the knowledge of existence is determined by the nature and modes of working of the knowing subject, Kant in the Dissertation applies only to sense intuition. As regards the understanding, he does not yet doubt, any more than before (in "Beweisgrund"), that with the requisite development it will be able to attain knowledge of things in themselves. But the nature of the understanding is not examined more closely. Kant was nevertheless right later in holding that his definitive philosophy had come into being with the Dissertation; for the principle — what for brevity's sake may be called the Copernican principle — had been set up, and it was now only a matter of applying it throughout. It was in itself no wonder that Kant shrank before admitting the impossibility of any scientific knowledge of things in themselves. The old opposition, stemming from Plato, between noumena and phaenomena, between existence as it is in itself and is known by thought, and existence as it appears to sense perception, he had now apparently found a new grounding for. And the sharp difference between intuition and understanding also seemed to speak for their reach having to be different.

Yet it is seen from Kant's letters that shortly after the publication of the Dissertation he felt a great difficulty in its results. How can concepts of the understanding, which we form by the activity of our thought, have validity for things that are altogether independent of us? Since these concepts (as e.g. cause, substance, possibility, actuality and necessity) are formed by us, they cannot after all be mere products of things; and
[44] in any case, if they were mere products of experience, they could not be used to set up propositions that were to have validity without being grounded on experience. Kant — as is seen from letters and notes — proceeded in his treatment of this problem in such a way that he examined the fundamental concepts (categories) with which we operate in our attempts to know existence, and sought to reduce them to as small a number as possible and at the same time to assure himself how many there are. The long time between the Dissertation and "Kritik der reinen Vernunft" was spent on such attempts. It can be seen that he tried different arrangements and divisions. His attention was directed in

particular to such concepts as express a relation. He became clear that a relation can come about not only through comparison but also through combination. Thus the concepts substance, cause and interaction presuppose different combinations: combination of thing and property, of cause and effect, of two causes. And when he thus found that we operate with concepts that express our striving to combine the phenomena in different ways, in different forms, he had gained a point of view from which the knowledge of the understanding, in spite of its difference from sense intuition, nevertheless proved to be homogeneous with it. For when in the Dissertation he had sought to show that space and time are our forms of intuition, this was grounded on their being forms under which our faculty of intuition *orders and synthesizes* the given. They are grounded on a "gathering and ordering power" *(vis animi,* omnes sensationes secundum stabilem et naturæ insitam legem *coordinans), and* they are themselves *schemata coordinandi* (Diss. § 15 D. E.). The understanding too now proves to be a combining, synthesizing faculty; by it we seek to combine the phenomena in certain mutual relations (e.g. as cause and effect). *The concept of synthesis,* which in the Dissertation was applied only to sense intuition, thus now proved to fit the understanding as well. And now the inference had to be drawn in a similar way as concerning time and space: only when we can combine the phenomena in the way that our categories, which express the forms of our understanding, indicate, only then can we understand! Synthetic unity is the condition of all understanding as well as of all sense intuition. By the help of the categories we *[45]* can therefore anticipate experience. The Copernican principle is carried through — but the impossibility of knowing noumena, things in themselves, then becomes at the same time the unavoidable consequence.

The combining principle that had been Hume's stumbling block now proved to Kant to be the fundamental principle of all knowledge, from sense perception up to the highest insight of the understanding. The applicability of this principle draws the limit between what is intelligible and unintelligible for us. But it was not enough for Kant to find, through analysis of the fundamental concepts, the general form or type of the activity of the understanding. He wanted a guarantee that *all* the fundamental concepts had been found. He wanted to set up a complete table of categories a priori. This he believed he could achieve by taking the doctrine of judgments as his basis. Every judgment is a combination of concepts and to that extent a synthesis (this is how he now conceives the judgment, whereas in his treatises of 1762 he had held that the predicate in the judgment must be able to be found by analysis of the subject). As many kinds of judgments as there are, so many special kinds of synthesis, or so many categories, must there be —

he concludes. The earlier logic's doctrine of judgments, however, he must transform somewhat before he can use it. He then finds four classes of judgments, each with three kinds, and so he gets twelve categories. Far more significant than this attempt at systematization, both in epistemological and in psychological respects, is Kant's general thought of synthesis as the fundamental form of the activity of consciousness. He had here found a concept that led both beyond the atomistic psychology that underlay empiricism and beyond the spiritualist psychology on which most idealist systems had hitherto built. Against empiricism, which would regard the unity of the mind as a mere result of the manifold impressions, he asserts the activity of unity as the fundamental character of mental life, which cannot be explained by outward influence alone; against spiritualism, which does have an eye for this fundamental character but would dogmatically lead it back to a mystical substance behind consciousness, he asserts that our knowledge cannot carry us further back than to the fundamental form
[46] and fundamental law of mental life as it appears in experience. At the same time he has thereby gone beyond the psychology of the Enlightenment, which kept to what is clearly conscious and seen through by the understanding. Synthesis is the constant presupposition of consciousness, but need not itself come forward as an object for consciousness. It can work blindly, instinctively, as a hidden art in our innermost nature. — Kant's theory of knowledge was now complete, and he proceeded to work it out. We therefore pass on to set it forth as it stands in his chief work.

3. Epistemology (Kritik der reinen Vernunft)

In setting forth the content of this work we do not follow the work's own division, but an order that lets the great lines of the line of thought to which the work owes its origin come forward more clearly. A natural system is thereby put in place of the more artificial one that Kant himself followed.

a. Subjective Deduction (Psychological Analysis)

The critical philosophy differs from the dogmatic by examining the faculty of knowledge itself and deciding, according to the result of this examination, which tasks it can solve and which lie beyond its reach. But we know the faculty of knowledge only from its activity in experience. The thing to do, then, is to examine experience in order to see whether it might not be composed partly of elements that could be owing to the faculty of knowledge itself, partly of elements owing to the way in which it is roused to activity

from without. What is owing to the faculty itself Kant calls *form;* what is owing to influence from without he calls *matter.* Nowhere has Kant methodically carried out the analysis of experience by which he grounds the distinction between form and matter that bears his whole epistemology. But on closer study of the Dissertation and "Kritik der reinen Vernunft" one finds (as is shown in detail in the monograph on "Kontinuiteten *[47]* i Kants filosofiske Udviklingsgang" p. 26—28) that the forms are discovered when one notices what is constant and universal in our knowledge, while the matter is what can change and vary. Space and time are forms of our intuition: for whatever kind the sensations may be, and however much they change, the forms of space and time in which their content appears to us are the same; a space or a time does not change, however it may be filled. If I abstract from all sense qualities, something is still left: extension and succession. Further: whatever phenomena we may have given, we understand them only when we can combine them as members in definite relations, especially in the way that the concept of magnitude and the concept of cause indicate. It is by a judgment, a combining activity, that we call one phenomenon greater or smaller than the other, or cause or effect in relation to the other. The same form of combination (quantity or causality) can be applied even if the content is quite different. — Common to all the forms, according to what we already saw above, is that they are the expression of a synthesis.

But Kant will not only prove that there are forms with which our knowledge operates, but also decide how many and which these forms are. He holds that, since we have to do here with what lies nearest of all to us, our own knowledge, we must also be able to arrive at a complete and certain insight into all its forms. This is evidently a dogmatic assumption. For the analytic method that Kant employs can give no guarantee of completeness; full certainty can never be attained that one has found all the forms. Nor can we be sure that we have found the fundamental forms. The forms are the constant elements in experience: from the constancy it is inferred that it is the faculty of knowledge itself that is at work. But that is a hypothesis, and can never become anything else. In the preface to the first edition of "Kritik der reinen Vernunft" Kant remarks that "the subjective deduction" (i.e. the demonstration of the forms of knowledge by means of analysis) might look like a hypothesis, since it does after all aim at finding the cause of a given effect (namely of what is constant in experience); he holds, however, that this is *[48]* not so, and promises to prove it on another occasion. This promise, however, he has not kept. His wish for a definitive result has here made him a dogmatist. He could not resign himself to the only way out that is given here: to operate with the foundation found

by analysis as with a hypothesis, leaving it to future inquiry to procure a still better foundation. Kant was caught in the illusion that "critique of reason" was something that could be done once and for all, and did not see that every attempt at critical philosophy must start from presuppositions that are to a certain degree dogmatic, and which it is the task of the critique of reason of the future to treat. — If we now look at the system of forms that Kant sets up, it consists of three groups.

α) *Forms of intuition.* Space is the form of all intuition in the domain of outer experience, time of all intuition in the domain of inner experience. They are forms of *intuition* (not forms of the *understanding*): for in space and time the parts are enclosed by the whole, and make up the whole; one cannot represent to oneself the individual part of space or of time except within the whole of space or the whole of time, and there is only one space and one time. Space and time are *forms,* because every single experience presupposes them. They are the necessary presuppositions that every experience must satisfy. — As a gathering, coordinating activity *(synopsis,* a viewing-together, Kant calls it) intuition in space and time has an active character in relation to mere sensation (Empfindung), which gives us the individual qualities. But Kant compares it above all with the higher degree of activity that lies in the activity of the understanding, and designates it, in relation to that, as receptivity. It does not yet give knowledge proper.

β) Forms of the understanding. Intuition without concept is blind (just as concept without intuition is empty). In intuition a multiplicity is immediately combined into a whole. But knowledge arises only through this combination being performed with consciousness, with an express turning of attention, so that this act of attention *(apperception*
[49] Kant calls it) becomes the common form or unity for the whole content. The unity of consciousness and its identity with itself is the condition for the given content being able to be synthesized. If, e.g., a line is to be known, it must be drawn, i.e. its individual parts must be combined in a definite way, held together by an activity of the mind that successively grasps the individual parts, does not let go of the earlier ones when it passes on to the following, and at last gathers them into a whole. The unity in this operation is the unity of consciousness, and only by means of it does the line come into being as an object for us. When I see the water freeze, I apprehend two states, fluidity and solidity, in a relation of time, but at the same time in such a way that the condition for the occurrence of the second lies in the first. Only by the application of the concept of cause do I apprehend the transition from fluidity to solidity as an objective event. — The synthesis performed in examples such as these Kant refers to the imagination, which forms a middle term between intuition and understanding. When we become

conscious of the forms according to which the imagination works here, we have found the *categories,* the special kinds of synthesis in the domain of the understanding.

In setting up the system of categories Kant starts, as already touched on, from his division of judgments into four classes. Corresponding to these he gets four groups of categories, and corresponding to these in turn four groups of fundamental propositions. To take a single example: to the hypothetical judgments corresponds the concept of cause. For a causal relation between two phenomena means that they are related to each other as the conditioning judgment to the conditioned: *when* the one phenomenon is given, the other follows unavoidably. And the causal principle then states that such a relation exists between every phenomenon and a certain definite other phenomenon.

The twelve categories, which it is superfluous to enumerate here, Kant himself moreover leads back to two divisions: the mathematical and the dynamical categories, *the concept of magnitude* and *the concept of cause* can in reality be said to represent all the categories and to be the two chief forms of the synthesis in which every knowledge consists. — Such a reduction is all the more necessary as the sharp difference between different classes of logical judgments that Kant assumed is altogether untenable[7]), so *[50]* that one cannot by the doctrine of judgments arrive at a system of different categories.

γ) Ideas of reason. By reason in the wider sense Kant understands our faculty of knowledge as a whole. By *reason in the narrower sense* he understands our faculty of knowledge in its tendency toward the *unconditioned* carrying through of synthesis. While intuition forms the chaos of sensations into sense-images in space and time, and while the understanding orders these sense-images in such connection as the concept of magnitude and the concept of cause indicate, reason in the narrower sense contains the demand for a completion of the series of members, for the formation of absolute totalities — thus an absolute beginning and limit of space and time, absolute maxima and minima, absolute completion of the causal series in a first cause[8]). It is a continuation and completion of the coordinating and synthesizing activity that intuition and understanding already exercise. It is synthesis in its highest form. Concepts that designate such an absolute completion Kant calls *ideas.* Whereas the word idea had come in the literature of the seventeenth and eighteenth centuries to mean the same as representation, Kant goes back to Plato's usage, according to which idea designates an object of thought that on account of its absolute character cannot appear in experience. "Plato", he says, "remarked very well that our reason has a natural need to raise itself to cognitions that go much further than that any object which experience can exhibit can ever be congruent with them."

Just as with the forms of intuition and the categories, Kant seeks with the ideas too to prove that there is a certain definite number of them. There are — he holds — three such ideas: the idea of the soul, the idea of the world, the idea of God. For we seek a completed knowledge of inner experience, a completed knowledge of outer experience and a completed knowledge of the origin of everything that is. That these ideas are not invented but proceed from the very nature of reason Kant will prove by their corresponding to the three forms of syllogism that it was customary to distinguish in
[51] logic (the categorical, the hypothetical and the disjunctive). This derivation is, however, very forced, far more forced than the derivation of the categories from the kinds of judgments[9]). It falls away already through the fact that the syllogism requires no other function of thought than the judgment, something Kant himself had earlier shown (in a treatise of 1762 "on the false subtlety of the syllogistic figures"). Kant's unhappy love of systematization has made his work more intricate than it needed to become. But in fact he is right that the ideas of the soul, the world and God have sprung from the involuntary need of consciousness for completion, to get the chain of thought hung up on a firm hook, to form an absolute system as a counter-image of the synthesis that is the fundamental form of thinking.

By analysis of our knowledge as it develops and works in experience, Kant believes he has found all the forms of its working. But with the demonstration of the forms, the demonstration of the justification of their use is not given. Originally no doubt is harbored about this validity. There is a natural tendency to ascribe absolute reality to forms of intuition, categories and ideas alike. Reason acts before it tests. It wants to have its building finished as soon as possible, and only afterwards does it examine whether the foundation too was laid deep and firm enough. The first step of reason is dogmatic. When it has learned wisdom by harm, it becomes skeptical. But the third step, which presupposes mature and manly judgment, is the critical testing of the faculties and reach of reason. For this the subjective deduction, which consists in psychological analysis, is only the preparation. It shows what forms or laws our knowledge follows according to its nature. The next question, which the objective deduction is to answer, is this: to what conditions and limits is the use of those forms subject? —

As was already emphasized in the general characterization of Kant, it is one of his chief merits that he has demonstrated what might be called *the law of the three stages,* a law that holds not only for the development of knowledge but in analogous forms fits the development of the life of the spirit as a whole. Kant has here indicated a point
[52] of view that plays a great part in later thinkers (Fichte, Hegel, St. Simon, Comte). It

points to development not proceeding in so straight a line as people in the period of the Enlightenment were inclined to assume.

b. Objective Deduction

If space and time are forms in which we intuit everything, then no phenomena can occur in experience that are not in space and time and do not obey the laws of space and time. Applied mathematics thereby becomes valid; although it rests on *pure (independent of experience) reason,* it nevertheless has *validity for all possible experiences,* because it formulates only what follows from the general laws of space and time. But on the other hand these laws then also hold *only* for things as they appear to us, for things as *phenomena* (Erscheinungen). For we have no right to make the conditions of our apprehension into the conditions of the very essence and subsistence of things, nor have we any right to regard our forms of intuition as the only possible ones. Only from a human standpoint can we speak of space and time. The mathematical laws are pure laws of reason, which hold for all the experience human beings can have, — and only with that experience and the phenomena it can show us does our knowledge have to do.

The case is similar with the categories: since they are the forms of our understanding, everything that we are to be able to understand must fall under them, must satisfy the conditions they set. To experience belongs not merely that something is intuited (in space or time or both), but also that the various phenomena that are intuited are connected in a determinate way. Experience in the strict sense requires not merely applied mathematics but also applied logic. All phenomena must, if we are to understand them, fall under the concept of magnitude and the concept of cause. The concept of magnitude excludes gaps and leaps (non datur hiatus, non datur saltus): every increase or decrease in extension or in degree must take place continuously. The concept of cause excludes chance and absolute necessity (non datur casus, non datur fatum): every phenomenon must, if the causal principle is to hold, stand in a relation of conditionedness *[53]* to another phenomenon, this again to a third, and so on. All the various fundamental propositions that correspond to the categories can, as Kant in a single place[10]) suggests (unfortunately without developing this thought further), be reduced to a single one: *the fundamental proposition of continuity.* The causal principle is only a particular form of this general fundamental proposition, which demands a continuous connection. That the fulfillment of this demand is a condition of experience, Kant shows in the following way. If there is to be a difference between *experience* and imagination, experience must require a law-determined succession of the phenomena, such that I cannot arbitrarily

exchange the members (as I can in my imaginings, dreams and fantasies), and such that no break in continuity occurs (whereas in the world of dreams I can make leaps as great as you please). The law of continuity (which, after all, comprises the law of the continuity of extension and of degrees as well as the law that all phenomena stand in causal relation) is valid for all phenomena, because it formulates the conditions under which we can have real experience at all (as distinct from imagination). It is the possibility of experience that we formulate in fundamental propositions, which then hold a priori for all possible experiences. — Whereas Wolff vainly attempted to derive the causal principle from the fundamental proposition of contradiction, that is, to give a purely logical proof of it, Kant has here attempted to give an epistemological (or, as he calls it with a barbarous scholastic expression, transcendental) proof of it, by demonstrating its validity as a condition under which experiences can be made. — But here too (as with space and time) the limitation follows along with the grounding. *Only* as a condition of experience does the law of continuity (the law of causality included under it) have validity. Just as geometry would be a subjective figment of the brain if space were not a condition of all outer experience, so the law of causality, and the law of continuity in general, would be only a purely subjective maxim if it did not state the condition of real experience. Our fundamental propositions anticipate the form of experience — but they do not hold beyond the domain of experience, do not hold for anything that cannot come
[54] forth in experience. Experience is, as Kant expresses it, an empirical synthesis which gives all other synthesis reality. We know only phenomena (Erscheinungen), not things as they are in themselves, but only as they are apprehended through the forms of our understanding, which (like the forms of intuition) have significance only from a human standpoint.

As regards the ideas, an objective deduction (a "transcendental" proof) is not possible. For since they denote something unconditioned, while everything that can appear in experience is conditioned and limited, the foundation for a proof that we had, in the case of the forms of intuition and the categories, in the possibility of experience is lacking here. The ideas have their subjective origin in reason's need of unity, but no object given in experience can be congruent with them. An absolute totality, such as the ideas of reason under various forms demand, experience cannot exhibit. We can go on advancing in space, in time and in the series of degrees and conditions — but whether absolute completion is possible cannot be established. — On the ideas, therefore, no science can be built, as it can on the forms of intuition and the categories.

It is a question whether there is the sharp difference between the three classes of

forms that Kant assumes. Continuity, causality, time and space too have, in the shape in which Kant sets them up, an ideal perfection with which no experience is congruent. Continuity is an idea to which only approximations are found in experience. What Kant calls forms are in reality abstractions and ideals that we, in accordance with the nature of our knowledge, set up and use as standards and rules for our inquiry. The fundamental propositions therefore become hypotheses, not proven truths. Experience in the strict meaning in which Kant takes this word in his "transcendental" proof is itself an ideal, and Hume, whose doubt about the causal principle Kant wanted to overcome, denied, after all, precisely that there was experience in the meaning in which Kant takes the word. Kant has thus not solved Hume's problem (which presumably cannot be solved either); but his merit is to have brought forward a side of the problem, or rather of our knowledge, which Hume had left aside, and he has thereby carried epistemology a *[55]* mighty step forward. It was his enthusiasm for the Copernican principle that led him to ascribe to his line of thought a greater force of proof than it had. Instead of stopping at the significance of the forms as types, exemplars and anticipations, he wanted to give a compelling proof of their real applicability, and in this he did not succeed. (Cf. more on this: "Kontinuiteten i Kants filosofiske Udviklingsgang" p. 31—33).

c. Phenomena and the Thing in Itself

Kant's result from the investigation of pure reason was that our thinking does indeed command forms and principles that do not spring from experience, but that these nevertheless have no valid application beyond the domain of experience. Their origin is not empirical, but only an empirical use can be made of them. And since we have experience only by means of the application of our forms of intuition and categories, everything we know is only phenomena, not things in themselves. But then the questions naturally arise: what, then, is this "Ding an sich", this "Noumenon", this "Intelligible", this "transcendental object" (under these various names the thing in itself occurs in Kant)? — and with what right is it assumed at all that there is any such thing?

In putting these questions we must note that the problem of the thing in itself, which came so strongly into the foreground during the first discussion of Kant's philosophy, was not really Kant's main problem. His task was an investigation of what faculty we possess for knowledge without building on mere observations. And he found that it followed from the conditions of this knowledge that it could concern only phenomena. His problem was the relation of reason to experience, not the first origin of the content of knowledge. Nevertheless it was natural that this problem, and the consequences

that seemed to follow from Kant's philosophy with regard to it, should attract special attention.

[56] To the question *what* the thing in itself is, Kant answers that he does not know, and that he does not need to know either, since it can never be met with in experience; everything that becomes an object of experience is phenomenon. We do not even know whether the thing in itself lies in ourselves or outside us. The concept noumenon or thing in itself is merely a limiting concept, a purely negative concept, at which we arrive by investigating the conditions of our knowledge, which are at the same time its limits. But it nevertheless shines through Kant's manner of expression in various places that he is inclined to think of the unknowable in idealist form, perhaps most nearly as Leibniz's world of monads (in which connection it is also of interest that Leibniz himself sometimes called the world of monads the intelligible world as opposed to the world of sense). Kant did, to be sure, firmly dispute the scientific justification of the use Leibniz had made of analogy to construct his metaphysical idealism; but in the background of his consciousness this view nevertheless seems to have maintained itself[11]). And, as will appear later, he ascribes practical justification to analogy when he is to show the possibility of a faith where knowledge is not possible. It is only so long as he speaks from a purely theoretical point of view that he declares the Ding an sich a purely negative concept.

Kant never doubted the existence of things in themselves. With the usual conception he assumes an absolute reality, and his critique aims only at grounding and limiting the knowledge we can have of them. But in the course of his investigations he comes (though always in passing) to state *reasons for the assumption* that something other and more than phenomena exists. Three such reasons appear in Kant. — In the first place, it would be an unreasonable and unjustified assumption that *our* mode of knowledge (in time and space and according to the categories of the understanding) should be the only possible one. We would thereby, such is roughly Kant's meaning, make ourselves guilty of the same error that was committed before Copernicus by taking the earth's standpoint for the absolute standpoint. The concept of the thing in itself thus marks that all our knowledge is conditioned by our nature, — that this nature of ours is a contributing
[57] element in the way in which the objects of experience appear to us. — In the second place: only as regards the form does knowledge have its ground in our nature; the matter, the content, is given to us (in the sensations); toward it we behave receptively. But it must surely have a cause! In a reply to one of his critics Kant says: "The objects, as things in themselves, *give* the matter for empirical intuitions; they contain the ground

by which sensibility is determined to act in its peculiar way"[12]). — The third reason Kant finds, as we shall see later, in the contradictions in which reason entangles itself by going beyond the limits of experience, contradictions that in his opinion can be resolved only by means of the distinction between phenomenon and thing in itself.

The difficulty that Kant's first critics found in his assumption of things in themselves was connected with the second reason. Into the first and the third they did not enter more closely. In his work *David Hume über den Glauben oder Idealismus und Realismus* (1787) JACOBI sought to show that Kant must consistently deny all existence outside our representations, that is, profess a pure idealism [i.e. subjectivism]. The assumption of a thing in itself as the cause of our sensations conflicts with Kant's doctrine of the concept of cause as the expression of a form of the understanding whose application is valid only *within* the domain of experience. A similar objection was made some years later by G. E. SCHULZE in his anonymous book *Ænesidemus* (1792). This objection cannot be refuted from Kant's standpoint[13]). It rests on Kant's own significant fundamental thought, that the limitation of knowledge and its grounding are closely connected.

Kant would have been consistent if he had merely maintained that the matter of knowledge cannot be derived from its form, the varying and special elements not from the constant and general frames. The possibility of experience ceases when no sensations arise. But how sensations arise is a question with which the critical philosophy (epistemology) has nothing to do. Hume had seen this clearly when he (Treatise I, 3, 5) declared the first origin of sensation inexplicable and added: "And this question is of no significance either for the task we have here set ourselves." Kant was, with regard to this *[58]* question, placed in quite the same way as Hume.

On closer investigation it will appear (as shown in "Kontinuiteten i Kants filosofiske Udviklingsgang", p. 33—35) that Kant sees in the Ding an sich not merely the cause of the matter of knowledge but also of the determinate forms in which we apprehend and order it. It is indeed also clear that the question of origin must be raised with regard to the form just as much as with regard to the matter. Especially as regards the apprehension of space, there are in Kant manifold indications that the ground why we apprehend things in precisely this determinate way (in three dimensions, etc.) must lie in the Ding an sich[14]). But he refers in general "all extent and *connection"* in our observations to "the transcendental object" (Kritik der reinen Vernunft. 1. Aufl. p. 494; cf. also p. 653 f.; 660 f.) — Not merely in the matter of our knowledge, then, but also in its forms we have an effect of the Ding an sich. And since the forms will be able to hold only so long as the Ding an sich acts in a constant way, it is clear that the knowledge we can build on the

mere forms can only be hypothetical, — while on the other hand it becomes more than merely phenomenal, since it teaches us how the Ding an sich acts. As soon as this point is brought forward and unfolded in all its consequences, Kant's whole system is therefore altered. Both the apriorism and the phenomenalism are limited, and the inconsistency falls away. For all these three sides of Kant's philosophy are connected with his absolute distinction between matter and form, a distinction that even on purely psychological grounds raises serious misgivings.

Yet — even when the Ding an sich is regarded as the cause of both form and matter in our knowledge — very little can be known about it: "Gar nichts bestimmtes", says Kant (Prolegomena § 32 and 57). It stands as a great X, which cannot be determined scientifically. And yet it is a determination of it that has been attempted in all religions and in all speculative systems. Kant here has the great merit of having determined *the philosophical place of religious and metaphysical speculations.* The conflict between
[59] positive and natural religion, between spiritualism and materialism, between monism and pluralism, etc., concerns precisely this X. Never before had the boundary between science and speculation been staked out in so clear a way.

d. Critique of Speculative Philosophy

As regards the ideas, an objective deduction proved impossible. Nevertheless the attempt has been made to lay the ideas of the soul, of the world and of God at the foundation of sciences that rose high above experience. The result Kant had reached through the general examination of the cognitive value of the ideas he finds confirmed by a special critique of these supposed sciences, — of speculative psychology, speculative cosmology and speculative theology.

α. Critique of Speculative Psychology. — It rests on a fallacy when it has been believed possible to set up a doctrine of the soul as a being separate from the body. One infers from the unity that is the general form of the activity of consciousness to the assumption of a simple substance behind consciousness. From the synthesis one infers the substance. But this is unjustified. The mere form of activity says nothing about the nature of the being that lies at its foundation. Consciousness is not something simple, not a single representation, but a form common to all representations. Psychology is a purely empirical science, which cannot teach us anything about a being or about properties (e.g. simplicity) that do not themselves come forth in experience. Therefore it is also unjustified (as Kant develops especially in the first edition of Kritik der reinen Vernunft) to make the difference between material and mental phenomena into a difference between

two kinds of substances or beings; that would be to confuse a difference in our mode of apprehension with a difference in the things themselves. What lies at the foundation of the outer phenomena might, after all, be the same as what lies at the foundation of the inner phenomena! The dualism would thereby fall away, and with it all the difficulties that have been found in the interaction between soul and body[15]).

β. Critique of Speculative Cosmology. — The concept of the world or of nature *[60]* as a whole presupposes that the concept of magnitude and the concept of cause are extended beyond experience, which always shows us only limited spaces and times and uncompleted causal series. A series of contradictions (antinomies) thereby arises, since mutually conflicting propositions can be proved: a sign that we have ventured beyond the limits of our knowledge! — Kant sets up four such antinomies (corresponding to the four classes of categories and of logical judgments); but they can be reduced to three, of which the first two concern the concept of magnitude, applied to space, time and matter, the third the concept of cause.

1) *Thesis:* The world must have a beginning in time and limits in space; for an infinite series cannot be thought as given. *Antithesis:* The world can have no beginning and no limits, for otherwise an empty time would have to precede its beginning and an empty space be outside it, and neither the beginning nor the limit would then be comprehensible, since one point of time or one place is not different from another in empty time and empty space.

2) *Thesis:* In dividing matter one must at last arrive at something absolutely simple and indivisible (atoms or monads). *Antithesis:* Everything we can apprehend and represent is divisible, and the absolutely simple and indivisible is an idea that can never be supported by experience.

3) *Thesis:* In going back from effect to cause one must at last arrive at a cause that is not in turn an effect; for otherwise one would never have found the complete cause of anything! There must then be an absolute cause at the beginning of the world's events, and perhaps also other absolute causes within the course of the world (namely in beings with "free" will). *Antithesis:* An absolutely first cause is incomprehensible, since no law connects it with its effect. There must surely be something that makes the absolute cause begin to act at precisely this particular moment! —

With regard to all three questions, then, we end according to Kant with problematic judgments. There is a disproportion between our ideas and experience: is it the former that are too large for the latter, or the latter that is too small for the former? We cannot *[61]* lay the blame on experience, since it is precisely the possibility of experience that makes

it possible to draw the boundary between category and idea. The blame must therefore lie in the ideas or in the way in which we apply them. — According to Kant it is two different interests that manifest themselves in the theses and in the antitheses. The theses express, he thinks, the speculative standpoint of dogmatism, and they satisfy at the same time the practical interest in the completion of the trains of thought, just as they are popularly intelligible. The antitheses, on the other hand, appeal to empiricism and to the purely scientific interest, whereas they contain questionable practical consequences.

Toward the first two antinomies Kant takes a different position from that toward the third. In the former — he thinks — both theses and antitheses are false: the world is neither finite nor infinite, matter neither absolutely divisible nor absolutely indivisible! The problems here fall away entirely when we distinguish between our apprehension and the Ding an sich. Our apprehension is a successive synthesis, which goes from member to member; here continuation is always possible and completion impossible; every limit can be overstepped by thought. But what thus holds for our apprehension — that new tasks constantly present themselves to it —, that we have no right to transfer to things in themselves. The idea of the world as a whole has for us only the significance of constantly carrying our inquiry further and preventing us from premature completion. — It is easily seen that although Kant means to reject both antitheses and theses, he in reality grants the antitheses to be right, as he has indeed himself declared them to have sprung from the properly scientific interest. For what the antitheses maintain is infinity in the sense of a constant possibility of continuation, not given, completed infinity, which is a contradiction[16]). — As regards the third antinomy, Kant thinks that both thesis and antithesis can be correct, when the former is understood of the Ding an sich, the latter of phenomena. In experience the causal series is continued constantly, and no exceptions take place: if a phenomenon were exempted from the law of causality, it could not, after all, be distinguished from an illusion! But this constant conditionedness
[62] does not hold for the thing in itself. Man as a being of experience, according to his phenomenal or empirical character, is subject to the law of causality; but as Ding an sich, as homo noumenon, according to his intelligible character, he can be regarded as free. The intelligible character does not appear as phenomenon, but can be regarded as the cause of the whole series of actions in which the empirical character unfolds itself. — Kant does not hereby mean to establish the reality of free will, but merely to establish that if for practical reasons we believe in the freedom of the will, such a belief will be compatible with the lawfulness of the empirical character. He has not, however, succeeded. For how can the intelligible character be the cause of the empirical one, when

the relation of time is not to find application to it? And when the empirical character is seen as an effect of the intelligible one, an empirical (phenomenal) explanation of the origin of character is thereby excluded. Finally, the intelligible character itself is not "freely" chosen; Kant's remarkable doctrine therefore becomes not a "doctrine of freedom" but a fatalism of character: the intelligible character is unalterable — and it determines the whole series of a man's actions!

γ. Critique of Speculative Theology. — In the idea of God Kant finds an expression of reason's need of complete completion. Such a completion would be possible if thought could ground the assumption of a being necessary in itself as the source of everything that has reality. It is thus an ideal of knowledge that asserts itself here, and that it lies in the nature of human reason to have such an ideal Kant finds confirmed by the fact that in science one constantly works at leading all the forces in nature back to a single fundamental force, and by the fact that in the popular religions a constant tendency toward monotheism asserts itself. This ideal of knowledge has its great significance. But it is another matter whether it is justified to make it into an objective being (to hypostatize and realize it), indeed even to make it into a personal being (to personify it).

1) The real proof of God's existence, the proof whose validity is presupposed in all the other proofs that have been attempted, would be the *ontological* one, which derives God's existence from the very concept we can form of him. It alone would lead to a complete *[63]* completion of thinking, by leading to a concept whose object would have to exist if one thinks it at all. We would then end in the thought of something that has its ground in itself. — But such a proof cannot be given. From the concept of a thing the existence of the thing can never be inferred. The concept can be entirely exact and complete, and yet it will be a question by itself whether what is thus thought conceptually exists. For existence is not a property like other properties: existence means merely that the thing, as we think it in its concept (with all its properties), now also really exists. The assumption of existence means the position of what is thought in the concept and adds no new content. A hundred real thalers contain, as far as the concept is concerned, no more than a hundred possible thalers. There is no contradiction whatever in thinking something according to its whole concept and yet thinking that it does not exist. — Of existence we convince ourselves by the fact that the thing belongs within the whole connection of our experience, — that there is a lawful connection between it and other things. Outside the connection of experience we can give no proof of existence. Freedom from self-contradiction is not enough for ascribing existence to the object of a concept.

2) While the ontological proof starts from thought and seeks to get from it to existence

without using experience, the other proofs start from experience. The *cosmological* proof infers from the fact that something exists (e.g. I myself) to an absolutely necessary cause of this. — This proof applies the causal principle beyond all possible experience, and it assumes at the same time a completion of the causal series in an absolutely necessary being, a being that has its ground in itself. The causal principle in itself leads only from member to member in a constantly continued relation of conditionedness, thus always only to finite causes, which must always again be regarded as effects, and are thus only conditionally necessary; by assuming an absolutely necessary being, the cosmological proof presupposes the validity of the ontological proof, since it presupposes that there can be a being that has its ground in itself, i.e. whose existence can be derived from its
[64] concept. Such a completion is impossible for us: we cannot, even if we think of the Deity in the most exalted way, refrain from asking whence it is. There lies here a true abyss for human reason — an abyss that can be thought filled only in an illusory way.

3) While the cosmological proof starts from the general experience that something exists, the *physico-theological* proof starts from the order and purposiveness that nature presents. — But whence is it known that this order and purposiveness is a matter of chance, — that it does not follow from the laws of nature according to which elements and forces act? (Cf. the criticism of the physico-theological line of thought that Kant had already given in his youthful writings). And even if one disregards this, the proof leads only to the assumption of an orderer and master builder who has his material given, not to the assumption of a creator. If one wants to arrive at the assumption of an absolute being, one must use the cosmological and the ontological line of thought, which have already been examined.

Neither by the way of pure thinking nor by that of experience, then, can a proof of the existence of an absolute being be given, and the result obtained earlier, that no scientific knowledge can be founded on the ideas, has thus received its confirmation in the case of all three ideas. But the ideas do not thereby lose their significance. Just as they have sprung from the very structure of our mind, so they act as ideals and principles for our inquiry, although they cannot be used for the knowledge of real existences. The idea of an absolute totality leads us to demand and seek ever greater unity and connection in our knowledge, keeps us from stopping too early. It becomes a guiding thought for us that everything in the world stands in mutual, lawful connection, *as if* it had sprung from a ground of unity. The idea of the soul has an analogous significance for research in the domain of inner experience, and the idea of the world in the domain of outer experience. The ideas have regulative, not constitutive significance. They

guide and regulate our advance within the domain of experience — but by overstepping (transcending) this domain they lose their significance, just as much as the categories do. A "transcendental" knowledge there can be, according to Kant; it consists in the knowledge of the presuppositions and principles on which our empirical knowledge rests; *[65]* but a transcendent knowledge, a knowledge that oversteps the domain of experience, there cannot be. (Kant, however, sometimes uses the word "transcendental" in the same meaning as "transcendent"). — But should we — Kant remarks — arrive by another way than that of scientific knowledge at the conviction of God's existence, then the idea of an absolute being becomes of very great significance, since it helps us to keep all sensuous and anthropomorphic representations away from the thought of the Deity.

The result of the whole critique of speculative philosophy is that it wanted to build a tower that could reach to heaven, but that there turns out to be material only for building a house. About high towers and high-flying metaphysicians there is — says Kant — usually much wind; that is nothing for me; my place is the fruitful lowland of experience! — Kant's speculative successors, like HENRIK IBSEN's Master Builder Solness, thought that building houses as dwellings for human beings was too slight a thing, and therefore, like him, took to building castles in the air. For Kant it was no slight work to labor in "the fruitful lowland of experience", if only one labored with a wide horizon. And this he expresses in his doctrine of the ideas, which enjoins that the highest thing is a never-resting striving that confidently adds stone to stone, even though it knows that it always remains high up to heaven. The thought expressed by Lessing of eternal striving as man's true condition Kant has given a firm and deep foundation in his doctrine of the fundamental form, mode of action and limits of the human mind. But this thought has for Kant not merely theoretical significance. It receives its full and proper significance in the practical domain, as will appear in the consideration of Kant's ethics.

e. Philosophy of Nature

If we look back on Kant's epistemology, it is clear that, as it took shape in "Kritik der reinen Vernunft", it acquired a more constructive character, or, as one can also express it, a more rationalist stamp, than it had in the years when the "awakening" probably took *[66]* place, and when analysis and empiricism gave it its stamp. The fundamental propositions that condition experience Kant now thought, after all, that he could prove a priori. As was suggested above, this proof did not succeed. But he thought himself so certain on this point that he did not content himself with giving a deductive proof of the causal principle, but also thought he could give such a proof of the assumption of certain

original forces of nature and laws of nature. In his *Metaphysische Anfangsgründe der Naturwissenschaft* (1786) he first defines matter as the movable in space and then seeks to show that its essence consists in an interplay of a repulsive and an attractive force. By the repulsive force space is filled; it is therefore the first condition for matter to be; but if it acted alone, *matter would be dispersed in infinite space;* there must then be an opposite force, which brings it about that the repulsion does not go on to infinity, but that the parts of matter are held together; — if, conversely, attraction acted alone, *all matter would be gathered in one point.* Both forces therefore belong to the essence of matter. — Kant here takes the step that Newton did not dare to take: to declare attraction a fundamental force. And in thinking that he can explain the essence and subsistence of matter (the movable that we intuit in space) as the result of the interaction of these two fundamental forces, he rejects atomism, which he himself had professed thirty years earlier (in his "Monadologia physica"). By his attempt to derive matter from forces and by his application of a constructive method in the philosophy of nature he paved the way for the subsequent Romantic-speculative philosophy of nature. He moved further and further away from the scientific mode of view that had intervened so mightily in his philosophical development. Yet Kant himself — in spite of his dynamic constructions — held fast to the strict scientific mode of explanation of all material phenomena. From the concept of matter as the movable in space he derives that all causes of material changes must be external, must be sought in space outside the point that alters its state of motion. With this fundamental proposition natural science stands and falls. All "
[67] hylozoism" (the assumption of inner forces or inner life in the material points) would be the death of natural science. — Of this derivation of the law of inertia there holds what was already suggested in connection with Kant's proof of the causal principle: that the demonstration of an idea as a guiding thought for our inquiry is not the same as the proof of an objectively valid law. But the zeal with which Kant enjoins the law of inertia shows that he is not minded to abate the strict demands on the explanation of the phenomena of nature.

The work on which Kant labored in his very last years was to show, by way of construction, a transition from the general fundamental propositions that the "Metaphysische Anfangsgründe" had set up, to physics. This was an insoluble task, and it was not merely the thinker's enfeeblement in old age that prevented him from completing his work. But the attempt testifies to how strong the constructive and rationalist tendency had gradually become in him.

4. Ethics (Kritik der praktischen Vernunft)

Kant's ethics is best known in its last form, in which it not only made a deep impression on its age but has also in later times stood as the expression of a typical ethical standpoint. What is at once characteristic and meritorious in Kant's ethics must be sought, in the first place, in its finding the law of man's right action in his own innermost being, seeing it as man's own law, which acts practically in him if only he can truly become absorbed in himself and become clear about himself. In the second place, in its having emancipated the ethical from mere theory, from metaphysics and theology, and found an independent foundation for it in man's practical nature. And in the third place: although the moral law is found in man's own being, it nevertheless has a strict and earnest character, points beyond man's individual limitation, lets him see himself as a citizen in a great kingdom, and maintains the sublimity of duty above every regard for self, every sensuous inclination, every immediate impulse toward happiness. Kant sets his face against theology by maintaining the independence of ethics from religion, against the *[68]* enlightenment of the understanding by maintaining man's practical nature, the will, as what is innermost in him, and against the idylls of the current doctrine of felicity by letting the trumpet of unconditional tasks and demands sound. A feeling of freedom and sublimity is the fundamental feeling in which this ethics lives.

Before we enter upon an account of it in detail, the account given earlier of Kant's philosophical course of development must be supplemented by an investigation of the process by which the Kantian ethics came into being.

Manifold elements have worked together here. As already suggested in the biography, Kant's pietistic upbringing has left its effects in the strict earnestness of his ethics. This remark was already made by his contemporaries. Schiller writes to Goethe (22 Dec. 1798): "There is still always something about Kant that makes one, as with Luther, think of a monk who has indeed left his monastery but has not quite been able to obliterate the traces of it". Nevertheless, far too much weight has often (e.g. by Schopenhauer) been laid on this point. When we see how Kant's development actually proceeded, we find that the matter is not so simple. Those after-effects from childhood have in many ways been modified by the intervention of other elements, and on the other hand there were other sides of Kant's nature that could lead him in the direction of ethical rigorism. — Before Kant's definitive ethics, two stages can be distinguished in his course of development as an ethical thinker. (Cf. the fuller argument in "Kontinuiteten i Kants filosofiske Udviklingsgang". ch. III).

a. First Stage (1762–66)

Very decisive was the influence of *Rousseau.* It probably took place at roughly the same time as the influence of Hume, that is, in the fertile year 1762—63. In the ethical domain, as in the theoretical, dogmatism wanted to regulate everything according to its "eternal truths". In opposition to this, Rousseau demanded that one should learn to know human
[69] nature, and this one could do only if one let it unfold freely, so that all theorizing and regulating received only an indirect, negative significance. Imagined wants, mental as well as material, were to be abolished. Not in the clarity of the understanding alone, but first and foremost in feeling, in the inwardness and depth of the heart, lies man's worth. Kant learned from Rousseau a new valuation of man. Earlier he had — as he himself has stated — along with his age set enlightenment as the highest thing and despised the rabble, who know nothing; Rousseau taught him *to honor men.* The idea of man's dignity as a personal being, — an idea that did not leave Kant at any stage of his later development and shows us its continuity, — Kant owes to the influence of Rousseau. — Yet at this time the study of the English moral philosophers (Shaftesbury, Hutcheson and Hume) also worked upon Kant. To their writings he referred his hearers at this time, and their thoughts he wanted to develop further. The distinction between knowledge and feeling, which he carries through in psychological and ethical respects, is due to this French and English influence; perhaps, however, also to influence from Sulzer and Mendelssohn, his German predecessors on this point.

Characteristic of this stage of Kant's ethics (according to his hints — for nothing more is extant from this period) is that all ethical judgments are led back to a foundation of feeling. It is an immediate feeling that dictates them. And the ethical feeling Kant describes particularly (in the small, interesting work Beobachtungen über das Gefühl des Schönen und Erhabenen. 1764) as a feeling of *the beauty and dignity of human nature.* By this psychological foundation this ethics differs decidedly from Kant's definitive ethics, in which he disowns psychology, and he does not want merely to rely on immediate psychological observation, but embarks on a comparative psychology, investigating how the feelings of the beautiful and the sublime, to which the ethical feeling belongs, vary among different races, among different temperaments and between the two sexes. In a fragment from this time he says: "In the first principles of ethical theory one must
[70] take account of the different moral feeling of men, according to difference in sex, age, upbringing and government, races and climates". And in "Nachricht von der Einrichtung seiner Vorlesungen 1765—66" he announces that in his lectures on ethics he will apply a method which is "a fine invention of our time", and consists in first considering

historically and philosophically what happens, before showing what ought to happen. The point, he continues, is to find man's lasting nature in order to be able to see what perfection belongs to him in the state of crude simplicity, and what in that of wise simplicity, and when he wants to raise himself above both.

Yet Kant does not assume, not even at this first stage of his ethics, that immediate feeling is enough. "True virtue can be built only on *fundamental propositions,* which are the nobler and more sublime the more comprehensive they are. These fundamental propositions are not speculative rules, but the consciousness of a feeling that lives in every human breast, . . . the feeling of the beauty and dignity of human nature *(Beobachtungen über das Gefühl des Schönen und Erhabenen.* Königsberg 1764. p. 23). Here, then, there appears the trait characteristic of Kant's ethics at all stages: the sublimity of the guiding ethical principle above man's lower being. Kant remarks that there is a certain kinship between the ethical feeling described and melancholy: for "a soul whose strength is limited feels a peculiar shudder when, filled with a great resolve, it sees the dangers it has to pass through, and has before its eyes the hard but great victory of self-conquest". (Op. cit. p. 28). For the sanguine temperament, therefore, this feeling does not lie within reach. The sanguine man's ethical feeling has the stamp of beauty, not of sublimity, is without fundamental propositions and dependent on momentary impressions (p. 34). — Kant's assertion of the necessity of general fundamental propositions stands in a certain contradiction with his comparative method, which admits the possibility of very considerable variations in the ethical feeling. Perhaps this contradiction was one of the motives that led Kant beyond this first stage. The last work in which this psychologizing ethics still prevails is "Träume eines Geistersehers" (1766). *[71]*

b. The Second Stage (1770–80)

The turning point that came for Kant's philosophy at the end of the sixties led, as shown earlier, to a sharp separation between sensibility and reason, matter and form, and to an emphasis on reason and form as the foundation of what is universally valid in our knowledge. It is possible that Kant made the sharp distinction between reason and the sensuous in our being, to which from this time on he also reckons feeling, earlier in the ethical domain than in the theoretical. The transition probably took place in the following way. When Kant already in the "Beobachtungen" laid such great weight on fundamental propositions, it was because they denote the constant and active elements in our nature as opposed to what is changing and passive. But so long as the fundamental propositions were derived from feeling, were only our consciousness of feeling, the

foundation seemed to be uncertain and empirical. With the prospect that opened up, after the discovery of the Copernican principle, of a knowledge of reason independent of experience, an endeavor to find for ethics too a foundation of reason independent of experience naturally had to go hand in hand. In the "Dissertation" (1770) he expressly distances himself from Shaftesbury's school — reproaching it with Epicureanism! Moral philosophy is now to be a pure science of reason. The moral concepts are obtained not by experience but by pure reason itself. — How Kant's ethical conception took shape in the interval between the "Dissertation" and "Kritik der reinen Vernunft" we know only through some fragments among Kant's posthumous papers. From a note of the seventies it is seen that he laid weight on a "practical idealism", an "idealism of reason, of wisdom", which consists in happiness being sought not outside us but in ourselves. This thought occurs more fully worked out in a fragment published by R. REICKE *(Lose Blätter aus Kants Nachlass.* I. p. 9—16), which (as I have sought to show in "Kontinuiteten i Kants filosofiske Udviklingsgang". p. 43) very likely dates from the time immediately before the publication of "Kritik der reinen Vernunft". Just as the ultimate presupposition of
[72] all knowledge of reason is the inner activity of thought (apperception), which makes coherent apprehension and understanding possible, so too it is self-activity, the capacity to produce our happiness ourselves, that is the firm ground of happiness. The matter of happiness is sensuous, but its form is intellectual: for we preserve our freedom and independence only when our willing is always in agreement with itself. Morality is freedom under a universal law that expresses our agreement with ourselves. It is thus the cause of happiness, and yet it does not have happiness as its purpose. Neither morality nor happiness is dependent on external results, on passive feelings or on the commands of authorities. — The standpoint expressed here thus finds the highest thing in the self-activity by which the individual is the creator of his own happiness. By its individualist and eudaemonist character it differs both from the earlier standpoint and from the later one. But it marks an essential step in the direction of laying the weight on the formal side of the ethical. Kant had here reached a main point of his ethics: the inner relation between action and law, which in his opinion could be present only when the law was purely formal; for only then can the law be independent of experience. The distinction between form and matter, between reason and experience, so important in Kant's philosophy, he has here applied for the first time to the ethical.

c. Third Stage (from 1785)

In the transition to the definitive form that Kant's ethics assumed, three different motives can be seen to have been at work: partly his psychological and culture-historical conception, partly the conviction, gained through "Kritik der reinen Vernunft", that the principles of pure reason are universal, valid for all rational beings, and partly, finally, immediate observation and analysis of ethical feeling as it expresses itself in practice.

α. Kant had early been led to a *developmental-historical conception.* In his famous youthful work "Allgemeine Naturgeschichte" he had put forward a hypothesis on the development of the solar system; in later treatises he had discussed the prehistory of the earth and the origin of the human races. His constant interest in physical geography *[73]* and in anthropology, subjects on which he gave popular lectures, could not but keep his attention alive to the natural history of man, even while his thought sought the ultimate grounds of our knowledge and of our ethical judgments. Two of the most important thoughts of the doctrine of evolution are already found in him: the principle of actuality, according to which the past must be explained by causes we know from present experience, and the principle of the summation of small effects into great and distant results. In the thought of development he found the solution of *Rousseau's problem* concerning the relation of culture to individual happiness. As we have seen, he was strongly occupied with Rousseau at once, when the latter's significant writings appeared (at the beginning of the sixties). At the beginning of the eighties, after the publication of his main work, he again came to concern himself with Rousseau's problem. It had become clearer and clearer to him that individual happiness could not be the measure of the value of historical development. On the whole there developed in him a dark view of human nature, as experience and history teach us to know it. In his notes and in his later published *Anthropologie*, as well as throughout his ethical writings and his writings in the philosophy of religion, we find this empirical pessimism[17]); the beauty of human nature, which he had emphasized in "Beobachtungen", fades for him. We are a lazy, cowardly and false race. Folly combined with a trait of wickedness (Thorheit mit einem Liniamente von Bosheit) is the stamp of our race. Kant quotes in his "Anthropologie" with approval Frederick the Great's words to the optimistic Sulzer: "My dear Sulzer, you do not know what an accursed race we belong to". One cannot without strong repugnance watch the doings of men on the great stage of the world. Far greater than the distress that nature inflicts on men is that which they inflict on one another. The solution of the problem Kant finds in this, that development — unconsciously for the individuals — leads toward a purpose of nature for the species, a goal that is reached precisely through

strife, wickedness and misfortune, and a goal that individuals perhaps, if they became conscious of it, would not care about. In a remarkable little work *Idee zu einer allgemeinen*
[74] *Geschichte in weltbürgerlicher Absicht* (1784) Kant develops this thought, and takes it up again, with special regard to Rousseau, a few years later in the treatise *Muthmaasslicher Anfang des Menschengeschlechts* (1786). The lots of the individual and of the species are different. As an animal being of instinct, the single individual can reach full development; but man's peculiar capacities as a rational being come to development only in the course of the species' history. The development of reason requires long series of generations, for art is long, and life short. What one generation has gained, the next can begin with, and in this way a progress becomes possible in history, but a progress which — so long as the goal has not been reached — brings greater suffering for the individual than the instinctive life of the state of nature. Kant emphasizes in particular that sexual maturity comes to man long before he is mature as a being of culture, and that culture increases the inequality among men. If individual happiness were the goal, man would have had to remain in the instinctive life of Paradise; after he has broken with instinct — a break that in the old story is depicted as a fall — and has entered upon the way of reason, there is nothing else for it but to go the way to its end, so that self-won insight may take the place of instinct in the regulation of life. In place of immediate happiness there then comes worthiness of happiness through free self-activity as the goal for the single individual. But the attainment of this goal presupposes the formation of a free civil society, indeed, in the last instance, of a cosmopolitan society[18]). But a society in which the freedom of the individual, by being bound by law, stands as compatible with the freedom of all others, develops only because distress drives toward it. A constant strife is carried on among men in their life together. They need one another, but their ambition and avarice constantly bring them into strife — individuals among themselves, and states among themselves. This strife is necessary. For otherwise the species would perish from indolence, and otherwise the faculties would not be developed. The trees in the forest seek to rob one another of light and air, but thereby compel one another to seek light and
[75] air in higher regions, and thereby attain in turn a beautiful, tall and straight growth. All culture and art, all social order is the fruit of antisocial tendencies, whose mutual conflict compels to discipline and to develop the seeds of nature to perfect proficiency. But on the way to perfect culture are found the greatest sufferings. The time of transition is painful, and Rousseau was not altogether wrong when he preferred the state of nature to half-civilization.

It is not Kant's meaning that a merely external harmony between the egoistic interests

should be the ideal and distant goal toward which history leads. The highest culture includes morality; for only the good that springs from a morally good disposition is more than illusion and glittering misery. Morality thus stands here for Kant as the keystone of the education of humanity.

Now we find that Kant, the year after the publication of his "Idee einer allgemeinen Weltgeschichte", for the first time declares (in *Grundlegung zur Metaphysik der Sitten.* 1785) that the moral law, as it expresses itself in the clearly conscious conscience, aims precisely at this, that one should act so that the rule one follows can become universal law, and that every human being be treated as an end, not merely as a means. We see, then, that what Kant on his definitive ethical standpoint presents as the content of the moral law or of duty is *an anticipation of the goal of historical development* — just as the a priori propositions in the theoretical domain, by means of the concepts of magnitude and of cause, *anticipated* the course of experience. The ideal, drawn from human experience and human need, of a free society of human personalities — an ideal that again and again, under many different forms, pushes forward in the history of ethics and of religion — denotes for Kant at once the *final goal* of history, progressing through egoism, antagonism, distress and sorrow, and the *content* of the moral law that makes itself known in the individual's inner self. Kant's ethics is the formulation of a historical ideal, and is not so purely a priori as he himself supposed. As in all formation of ideals, experience here too played its part[19]) — The exalted, absolute, unconditional character that the moral law acquired for Kant in opposition to the individual's sensuous nature *[76]* and limited capacity is naturally explained by the fact that it is the goal of the species that in the moral law makes itself known in the consciousness of the single individual. Therein lies the marvelous in the moral law, as in all instincts of the species. The great opposition that for Kant historically existed between the goals of the individual and of the species led him to assume a no less sharp opposition between the moral law and all empirically given elements in human nature. Psychology seemed unable to fill this gap. Only by a leap does Kant now pass from psychology over into ethics. While in "Idee" he had still regarded morality as the highest step of cultural development, it later (Kritik der Urtheilskraft. § 84) stood for him as an impossibility that any degree of natural development could satisfy the unconditional demand of the moral law. He kicks away behind him the historical ladder by which he had reached his definitive ethics.

β. The influence of Kant's epistemological investigations (in "Kritik der reinen Vernunft") makes itself felt very strongly in the development of his ethics into its definitive form. Already the year after the publication of "Kritik der reinen Vernunft" Kant was

working on an exposition of the principles of ethics (as is seen from a letter from Hamann to Hartknoch of 11 January 1782). Sharply as he wishes to distinguish between theory and practice, they were nonetheless closely connected for him. A main thought in "Kritik der reinen Vernunft" was, after all, that a world of experience, a nature, comes into being for us only on the presupposition of objective, universally valid laws. Kant has now expressly conceived the moral law by analogy with the laws of nature: "The validity of the will as universal law for possible actions offers an analogy with the connection of existing things according to universal laws, which is the formal side of nature", — and therefore the moral law, the unconditional demand (the categorical imperative), can be expressed thus: "You shall act as if the fundamental proposition on which your action rests were, through your will, to be made a universal law of nature". (Grundlegung zur Metaphysik der Sitten. 3. Aufl. p. 52. 81). Just as by the law of causality we enter
[77] into the great real world, so the moral law thus opens to us access to an ideal world. There the question is whether my representations are to have objective validity; here the question is whether my actions are to have objective validity. The objective rests everywhere on law. Already in the fragment that acquainted us with Kant's ethics at its second stage, inner consistency, agreement with ourselves in all our willing, was stressed as the decisive thing. This way of consideration is now extended so that the purely individualistic falls away: the law of our action is to be the law for a world of spiritual beings. I come into contradiction with myself when I judge myself otherwise than I judge others!

But now experience can never ground a universal law; the possibility of experience rests, conversely, on the validity of the law. This holds practically as well as theoretically. Human nature as given in experience, which psychology and history teach us to know, can therefore give us no foundation for the moral law. Its origin is independent of all experience, cannot be explained at all empirically or theoretically. Besides, it is the whole of human nature given in experience that is to be subjected to the moral law and regulated by it: so it cannot itself be the source of the law. Experience is always conditioned, but the law is unconditioned. Reason must take its standpoint outside phenomena, if it is to be able to subject everything to its unconditional law. This law does not thereby cease to be the law of man's own innermost being: but it springs from that in man which cannot appear in experience, from the intelligible in man, not from man as phenomenon, from man as Ding an sich, as noumenon, not from man as a sensuous being. Already in "Kritik der reinen Vernunft" (1. Aufl. p. 553) it was declared that "reason itself is no phenomenon and is subject to no sensuous conditions", and the

whole doctrine of the intelligible character shows that the peculiar and fateful turn that Kant's ethics underwent already stood firm in principle for Kant when he wrote his main work. In *Kritik der praktischen Vernunft* (§ 5) it is said that the mere form of the law has nothing whatever to do with phenomena, but makes man independent of everything in *[78]* the world of phenomena. Man himself is thus a citizen of two worlds: as a member of the intelligible world he gives the law to himself as a member of the phenomenal world: he is at once lawgiver and subject. He need not go beyond himself (only beyond his sensuous, phenomenal being) to find the unconditioned. A "You shall!" exists for man only because, besides belonging to the intelligible world, he also belongs to the world of sense.

But thereby both the moral law itself and the possibility of its being applied in the world of experience become altogether inexplicable. It becomes just as impossible to explain "how reason can become practical" as to explain how the intelligible world, things in themselves, are connected with the phenomenal world. No mortal can see how the intelligible world underlies the world of sense! —

One sees the ascent in Kant's ethical idealism. In his zeal to assert the loftiness of the moral law and the dignity of personality he raises them beyond the world of experience, so that it becomes a riddle how they can have anything to do with it. He places the foundation of the ethical beyond the limit of knowledge and lets it share the fate of the disputed limiting concept in which his theoretical thinking found its conclusion. Through his ever more sharply pointed opposition to empiricism he approaches mysticism[20]), and he does indeed expressly declare that mysticism is less questionable in a moral respect than empiricism (Kritik der praktischen Vernunft I, 1, 2 at the end). Kant's rationalism was on the way to passing over into mysticism, just as in its time mysticism (in the form of pietism) had contributed to the development of rationalism. — There is here a decided inconsistency in his philosophy. Since the forms, according to Kant, are gained by analysis of experience, they cannot be separated from phenomena and placed in complete opposition to them. The forms are not intelligible, they are gained by abstraction and analysis and cannot be regarded as belonging to another world than the one in which they were found, even if here they have the great significance of forming the foundation of all objective validity. This must hold for the formal moral law, as well as for the forms *[79]* of intuition and the categories. It is Kant's ethical interest that leads him to assign the moral law a place in the intelligible world (or rather to regard it as the entrance to this world), while he does not think of assigning a similar dignity to the law of causality, which is analogous to it. — As so often, ethical idealism has by its overstraining worked

against itself. One does ethics no service by founding it on something that lies beyond all experience. It is, after all, precisely in the world of experience that the ethical is to live and work. — What is significant in Kant's fundamental ethical conception could very well have been preserved even if he had not turned his back on psychology. The opposition between the ethical ideal and other elements in our nature, which he has so energetically emphasized, is in reality taken from psychological experience, and it is only a mythological expression that Kant gives this experience by making it one with the opposition between the intelligible and the phenomenal world.

γ. In his ethical writings *(Grundlegung* 1785 and *Kritik der praktischen Vernunft* 1788) Kant arrives at the statement of the principle of ethics through *analysis of the ordinary, practically given moral consciousness.* He sets as the task of philosophical ethics to find and to think through the principle that practical human reason involuntarily applies. He presents first 1) the ethical phenomena or data, then finds out 2) the law to which these bear witness, and finally 3) the force that works according to this law.

1) Already the ordinary moral consciousness is clear that the ethical value of an action does not rest on what is attained or accomplished in the external world. Not the external effects, but the inner will is what matters: nothing is good but the good will! In the acting personality itself, not outside it, must the good be found! Only that action is therefore good which springs from duty, or from respect for the moral law. Neither custom and usage nor the experiences of the past nor exemplars — were they even the most exalted — make an action good. Every custom and every exemplar, every ideal
[80] given in experience must, after all, itself first be tested and valued. Even the Holy One of whom the Gospel tells must be compared with our ideal before we can acknowledge him. Neither a theological nor a psychological grounding of ethics is therefore possible. Duty springs neither from authority nor from experience. All feeling is empirical, sensuous, egoistic; even the so-called moral sense and sympathy are at bottom only forms of the impulse toward happiness. The decisive determination in the ethical is *self-legislation* (autonomy). From it follows the inwardness of the law; by being subject to the law of our own being we are independent of experience and authority, and are at the same time active, self-active.

Experience now shows that there are elements in man's nature which only reluctantly bow to this inner law that is contained in practical reason. It is precisely through this that we come to call it a law, and that its demands come to stand as unconditional commands, categorical imperatives. There is talk of an ought, which is nonetheless one with our innermost willing: but this innermost willing meets resistance in ourselves. We have

an inclination not to be autonomous, not to follow the innermost law of our own will. Thereby the ethical task assumes the character of *duty,* moral necessitation, and the moral law comes to stand as exalted above all empirically given elements in our being.

All moral consciousness offers, according to Kant, these two features to analysis: the inwardness of the law and its exaltedness, — its unity with man's being and its sharp opposition to man's phenomenon. Even in the worst scoundrel, Kant asserts, these features can be traced. While in his youthful work (Beobachtungen 1764) he admitted even very considerable variations in moral feeling, he now, in a rather dogmatizing manner, makes human nature more uniform than it is — but this, too, becomes possible for him only by going beyond experience.

Kant lays very great weight on the oppositions in human nature. Precisely extreme cases, he thinks, show us best of all what different elements and forces human nature encloses. The examples on which he bases his analysis are almost all taken from cases of *[81]* conflict. The ethical appears the more clearly in its peculiarity, the stronger the opposing forces are, and the more other auxiliary motives than the purely ethical are kept out. In "the doctrine of method of pure practical reason", which forms the concluding section of "Kritik der praktischen Vernunft", he therefore recommends cases of conflict, which are for ethical analysis what the method of reagents is for chemical analysis[21]). But Kant does not hold merely to this, that the ethical *stands out most clearly* in cases of conflict between the ethical demand on the one side, sensuous and egoistic interest on the other side. He declares even that only in such states is the good will *actually present:* "The state in which I, when certain of my purposes collide with the moral law of duty, am conscious of preferring the latter, is not merely a state that is better than others, but *the only state good in itself".* Schiller caricatured this rigoristic tendency in Kant's ethics in a well-known epigram by drawing the conclusion that one does one's duty only when one does it with loathing. That was not Kant's meaning. As he himself bursts out in a dithyramb to duty, the exalted power in man's inner self, so too it was his meaning that duty should be practiced with cheerful boldness and enthusiasm. He was only somewhat wary of the duties that came all too easily; he would see in them a sign that one had not found one's proper duty. And he demanded that in the relation of duty abstraction should be made from all self-interest and all immediate allurement. The relation of duty is not to abolish the independence of the individual, but only to limit it.

2) But what, then, is the *content* of the law that, according to Kant, makes itself known in every man's consciousness (whether he is able to formulate it or not)? — Since the law (on account of its inwardness and exaltedness) is to be independent of all experience, the

answer to this question seems to be difficult. Kant answers consistently: the moral law is purely formal; it states merely the form the will must have in order to be good. And the form is this, that the principle I follow in my action must be capable of being laid down as the foundation of a universal legislation, i.e. must be able to hold for all other
[82] rational beings in the same situation as I am in. The one who wills is to apply a universal measure to himself, to regard his action as if by it he were about to create a new nature. By means of this rule it will, Kant thinks, be easy in the individual cases to find out one's duty, far easier than to attain happiness. That I may not keep goods entrusted to me, that I may not lie, even if discovery is impossible, now follows of itself: for if this right became universal, all trust between men would fall away. The individual thus regards himself in his valuation as one among many, and he limits his willing according to what the conditions of the life together of the many require. Kant expressly distinguishes his moral principle from the old rule that we must not do to others what we would not have them do to us, a rule that can be interpreted purely egoistically. But there lies in his principle more than the merely formal, mere logical consistency: there lies also the presupposition that there are other interests than my own private ones, — that there are other personal centers in the world than myself. Should that be something that can be known by the a priori way of pure reason? That I am a member of a society whose welfare is my law, I know, after all, only through experience. And as we have seen, Kant's ethics did in reality arise under the influence of his culture-historical conception. He himself comes to admit that the moral law is empirically and historically determined, not purely formal, when he says *(Tugendlehre* § 27): "Legislating reason includes in its idea of humanity in general the whole species". That is, after all, an idea that has its history, and a slow and laborious history! — And Kant has also felt the impossibility of getting anywhere with a purely formal principle, since all action presupposes positive purposes and tasks. But if the categorical imperative, the unconditional in the relation of duty, is not to be violated, the purposes that are set up must be such as have an absolute value: otherwise it cannot be made a duty to have them! The value of my personality — such, roughly, is Kant's line of thought — rests on my capacity to follow the universal law in my inner self; what purpose, then, can it be my unconditional duty to have? Only the one,
[83] comes the answer: to respect the value of other personalities, which likewise is grounded on *their* capacity for self-legislation. While Kant's *first ethical formula* was this: "You shall act so that the principle of your acting can serve as a universal rule!" — his *second formula* becomes this: "You shall act so that you always treat humanity, in yourself as well as in every other, as an end, never merely as a means!" (Grundlegung. 3. Aufl. p. 66.

— Kritik der prakt. Vernunft I, 1, 3. — Tugendlehre. Einleitung § III—IV). In his opinion the second formula can be derived from the first; but this is an impossibility if the first is taken as purely formal. Both formulas presuppose that we actually feel ourselves as members of a kingdom of personal beings. And it is rather the first formula that springs from the second: as surely as the law exists for the sake of personalities, not the reverse! Apart from what is artificial in the derivation, it is a great and significant principle that Kant has stated. It is the principle of personality in its noblest form, a thought that will live when Kant's imperfect and unnatural grounding has long been forgotten, a thought of great ethical value both as against the principle of authority, when this would be more than an educative principle, and as against the external doctrine of utility and happiness, which holds to the shells and forgets the kernel. —

Kant himself, however, now maintained that his moral principle was purely formal, an expression of pure reason. But then the question arises how this intelligible law can determine our will as it works in experience? How can the law become a *motive?* — In his zeal to assert the inwardness and exaltedness of the law Kant had pushed psychology aside. He would not ground the ethical on feeling, since feeling is a passive state, dependent on experiences. But he cannot deny that action becomes impossible if feeling does not drive the work. He then seeks to show that the very thought of the universal moral law awakens in us a feeling of respect and reverence. Respect is, according to him, a feeling that cannot be explained by experience; it is neither pleasure nor pain, but pure interest, awakened by the loftiness of the law in our inner self, an effect that the thought of this law produces on the mind. Here, then, is a motive that can set our empirical willing in motion. But it is, to be sure, a motive that is just as great a *[84]* riddle as the relation between the intelligible and the phenomenal world in general, and, what is still more questionable: a motive whose possibility conflicts with Kant's own conception of the causal principle as valid for all phenomena, inner and outer; respect, after all, is not supposed to be explicable by phenomenal causes. Here again Kant's ethical idealism has led him astray; but here again it is very possible to separate the significant thought from the garb Kant gives it.

3) From the phenomenon, the moral consciousness of practical man, Kant went to the law, and from the law he goes to the force. The law is the expression of a *consciousness of freedom.* Law and freedom are in a certain sense one; if one would distinguish between them, we infer from the law back to freedom (Kritik der prakt. Vern. § 6). The law proceeds from the will itself (Grundlegung. 3. Aufl. p. 104 ff. — Rechtslehre. Einleitung § IV). For the law is, after all, the fruit of self-legislation, the expression of our autonomy as

rational beings. By freedom Kant understands *first and foremost* originality, the capacity to act according to inner principles and forces, independence of the given and of the external. — But Kant does not stop at this definition of freedom. Freedom involves for him also *the capacity to make an absolute beginning,* to initiate an absolutely new causal series. This capacity is for him a necessary consequence of originality and independence. But then "freedom" cannot be found at all in the world of phenomena, which, as Kant has shown in "Kritik der reinen Vernunft", offers no absolute beginnings. If freedom is to be the capacity for absolute production of the motives[22]) that determine the will working in experience (something already contained in the doctrine of respect), then it can be found only in the intelligible world (if one can act and begin there — since time, after all, does not hold!). Kant does not investigate whether independence and the capacity for absolute beginning absolutely belong together. New centers can, after all, form in the world, which, once they are formed, act independently of external influence, according to an inner law. Kant makes each single will a god, instead of letting it acquire
[85] its divinity through development and labor.

And besides the two definitions of "freedom" (as independence and as capacity for absolute beginning) there comes, in a few places (Kritik der prakt. Vernunft. Kehrbach's edition, p. 116 ff. — Religion innerhalb der Grenzen der reinen Vernunft, passim), a third: *the capacity to do just as well the one as the other of two opposite things.* That Kant came to set up such a monstrosity of a concept finds its explanation in the sharp opposition in which he has set man's ideal (intelligible) being to man as he appears in experience. As greatly as he thinks of the former, so meanly does he think of the latter. There is a violent tension between the two poles, and it could not but remain unexplained how man, whose innermost being is good (freedom in the first sense is one with the good), could give room in himself to such motives as experience attests. Kant explains it by assuming an equally inexplicable capacity[23]). — Here again Kant is more in the right in the facts he points to than in his theory of them. Human nature includes the greatest oppositions within itself, seeds of good and seeds of evil. There is a capacity to form ideals of exaltedness and purity, and there is perversity, contrariness and filth. These oppositions must hang together in human nature; they must have developed according to definite laws and under definite conditions. Kant, who otherwise knew well enough the natural investigator's standpoint toward human affairs, here made every solution impossible for himself by his unhappy dualism of ethics and psychology. —

d. Special Ethics

In two books that at many points bear the stamp of the weakness of old age, *Rechtslehre and Tugendlehre,* both published in 1797, Kant has set forth his special ethics. The imperfection of these expositions is connected above all with the systematizing that had gained the upper hand in the old thinker; but at the same time, naturally, the one-sidedness of Kant's whole ethical conception expresses itself in many particular questions. Above all signs of decrepitude and one-sidedness there nonetheless tower a *[86]* series of noble and manly thoughts, Kant's great legacy to the race.

The two books stand in a certain opposition to each other, in that the doctrine of right holds to legality, the external agreement between my action and the law that upholds the freedom of others as well as my own, while the doctrine of virtue also demands morality: that the action is to spring from a disposition, an inner will to acknowledge the law. The moral law (the categorical imperative) demands that my will be limited by such conditions as the freedom of others requires. But the observance of this limitation can come about from fear or from self-interest, not merely from moral motives. The development toward the ideal state in which no war takes place can proceed under the influence of many different motives, which are not all properly moral. The opposition between legality and morality is not absolute: for at the foundation of the establishment of legality, at the foundation of the demand that there shall be laws, lies the categorical imperative: *"Es soll kein Krieg sein,* weder der, welcher zwischen Mir und Dir im Naturzustande, noch zwischen uns als Staaten; denn das ist nicht die Art, wie jedermann sein Recht suchen soll". (Rechtslehre. Beschluss). This confirms the assumption maintained above, that Kant's moral law is an anticipation of the result of historical development. The legal order stands — although it requires only legality — as a morally postulated means. "Reason obliges us", says Kant (Rechtslehre § 49), "by a categorical imperative to strive after a state in which the constitution of society stands in as great agreement with principles of right as possible". Kant thus admits that the realization of the highest moral purpose can also be prepared by the influence of other than purely moral motives. And the very external securing of freedom that the legal order makes possible is a condition for the inner moral disposition to be able to develop (Lose Blätter aus Kants Nachlass. p. 528). In his foundational ethical writings ("Grundlegung" and "Kritik der praktischen Vernunft") Kant states a sharper opposition between legality and morality than he can maintain where the investigation goes into particulars. *[87]*

Kant defines right as the sum of the conditions under which the will of the one can be united with that of the other according to a universal law of freedom. There is only a

single innate right, which is attached to every human personality, namely: *freedom, i.e. independence of the compelling arbitrariness of every other, insofar as it can coexist with the freedom of every other according to a universal law.* The right of property is thus grounded not merely in this, that I have taken possession of a thing before anyone else takes possession of it (prior occupatio), but first in this, that I acknowledge the equal right of others to what they first have taken possession of. — If anywhere, it is clear here that Kant's ethical principles constantly presuppose the existence of a society as historically given. The society itself Kant nonetheless conceives as having arisen by a contract that limits the freedom of the individual by regard for the equal freedom of others. But he does not regard this contract as a historical event. He uses the thought of it as a guiding idea, a principle of reason for the valuation and development of social relations. The historical origin of society he regards as a thing indifferent in this connection. By this standpoint he marks a decided advance over the mythologizing of the earlier teachers of natural law. Only he did not raise the question how the principle of reason itself comes into being in men's consciousness; here lies what is dogmatic in him.

Peculiar to Kant's doctrine of right is the strict carrying through of the individual's rights of man, supported above all by a sharp distinction between person (a being with self-legislation and accountability) and thing. A person must never be treated as a mere means! From this proposition Kant derives not only personal freedom, but also freedom of discussion and the right to a share in legislation. Legislation must not decide anything about the people that the people could not will to decide about itself, — e.g. introduce an unalterable system of dogma that abolishes the right to advance in enlightenment, or introduce hereditary nobility. — With enthusiasm Kant had hailed the North American War of Independence and the French Revolution, and even in his doctrine of right he
[88] has not forgotten this enthusiasm. He looked to the republic as the constitution of the future; but the republican spirit was nonetheless more important to him than the external constitution, and a monarchy could in his opinion very well be governed in a republican spirit. Paternal government he abhorred. The sovereign has no right to make anyone happy against his will. Right has its origin in freedom, not in happiness, and it is for freedom, not for happiness, that the sovereign is to provide. Therefore punishment must not be meted out as a means to a good for society or for the criminal himself, but solely and only for the reason that the criminal has willed his action. Man would be treated as a means, as a thing, if punishment were grounded in any other way than as retribution. Retribution is a categorical imperative. Punishment must not be exercised in order to bring advantage, but is to be exercised though it cause ever so much inconvenience. If

a people were about to emigrate, it ought first to execute the murderers! Fiat justitia, pereat mundus — that means in Kant's translation: "Justice shall reign, even if all the rogues in the world perish by it!" —

In his "doctrine of virtue" (individual ethics) Kant lays the weight above all on the bearing of character that corresponds to his general conception of the ethical. Virtue consists for him in strength of soul (fortitudo moralis), in the force and dignity that follow from the consciousness of having the law for one's acting in one's own inner self and of being bound through this law to a great whole.

The purposes that Kant's individual ethics sets are: 1) one's own perfection and 2) the happiness of others. Not one's own happiness: for one strives after it involuntarily and so eagerly that one regards it as the duty of others also to provide for it. Nor the perfection of others: for only they themselves can provide for it; perfection consists precisely in making oneself one's purpose according to one's own concepts of duty. (As if happiness were so easy to win, and as if one did not need help in one's own striving after the perfect!) — 1) Under the endeavor for one's own perfection belongs the development of all lower and higher faculties, all culture. The point is to make the animal into man: "What is characteristic of man in opposition to the animal is the capacity to set himself *[89]* purposes. Now if humanity in our own person is the purpose, then there is bound up with this the duty to make oneself deserving of humanity by culture" — first and foremost "moral culture", that is to say the capacity to let one's actions be determined by the inner law. But everything that makes man an active and useful member of society Kant derives from human dignity: to be useless and superfluous is to degrade humanity in one's person. A very characteristic section is: "Von der Kriecherei". As a person man is of priceless value. Humble before the law in his inner self, he is therefore to assert himself before all others, even were he to come to stand before a seraph! Humility before others is not a duty, but can be Pharisaism or even baseness, if one would win favor or advantage by it. Therefore become not the slaves of men! Let not your right be trodden under foot! Accept not benefits you can do without! Show steadfastness and shun unworthy wailing under sufferings! Kneel to no one, for the ideal is within you, and what can appear outside you can only be an idol! Deep bows and many outward signs of courtesy are expressions of men's propensity to servility. And when one makes oneself a worm, one cannot complain of being trodden under foot! — 2) With the second purpose that Kant sets up in individual ethics, the happiness of others, there arises for him — without his being clearly conscious of it — an extraordinarily difficult problem: is one in this to be guided by the notions of happiness of others or by one's own? It is a

problem that leads into the innermost depths of personal life, especially when one, like Kant, has so sharp a conception of personal dignity, determined by self-legislation. Kant is very wavering here. Now he says that others have no right to demand of me what I do not think will make them happy, and speaks of the sour merit of promoting the true welfare of others although they do not acknowledge it as such; now he says that I cannot do good to others according to *my* concepts of happiness, and that it cannot be called beneficence when I deprive another of the freedom to become happy in *his own* way. —

[90] The duties toward others Kant divides into duties of love and duties of respect, according as there is to reign a relation of attraction and approach or a relation of distance. The duty of respect follows directly from the first propositions of Kant's ethics. He stands in a more difficult position toward love, partly on account of the sharply cut independence he gives each single personality, partly on account of the pessimistic conception he has of actual human nature. Love must be limited by the other's right, for the other renounces his human dignity when he does not himself have control over himself and his condition. Besides, love is an immediate feeling and need that cannot be regulated by the commands of reason. And as men actually are, it cannot be demanded that one should feel delight in them. It is our duty to will them well and to do them good, and this duty does not lapse with the sad experience that our race, when one comes to know it more closely, is found exceedingly little lovable. The love of benevolence, not of delight (amor benevolentiæ, non complacentiæ), is what can be a duty. A harmony between approach and distance, between love and respect, Kant finds in the relation of friendship: true friendship is indeed "a rare bird, to be likened to a black swan"; but black swans really exist. — One must, on the other hand, excuse Kant, as a bachelor, for having so imperfect an understanding of marriage. He does not speak of it in the doctrine of virtue, but in the doctrine of right. It is a contractual relation by which two persons of different sex bind themselves to enter into sexual relations only with each other, not with others. The sexual inclination Kant speaks of only as a purely isolated need in human nature, sees it only from the purely sensuous side and has no eye for its nuances and possible

[91] connection with some of the most ideal feelings.

5. Philosophy of Religion (Kritik der praktischen Vernunft and Religion innerhalb der Grenzen der blossen Vernunft)

a. Morality and Religion

Kant's ethics is autonomous — is (or wishes to be) independent of all other presuppositions than those that lie in man's innermost being, in the activity that is his innermost nature, independent of physics and hyperphysics, of psychology and theology. It must then be very difficult for Kant to demonstrate a natural transition from morality to religion. For how can dependence be derived from pure independence and self-activity? And yet it was early (from as far back as 1766, see "Träume eines Geistersehers") Kant's conviction that religion could only be grounded on morality. The explanation lies in this, that the unconditional task that is set man in his innermost self is to be solved by a finite, a limited being. The moral law acquaints us with our inner freedom, and thereby with our independence of the whole world of experience. But it is, after all, precisely to be carried through in the world of experience, to permeate our personality completely. There then arises a *need of reason,* a moral want, which demands that the conditions be fulfilled without which the complete realization of the ideal task is impossible. The first condition is continued existence, for only by a progress going on to infinity can the will come into perfect agreement with the moral law. Personal immortality then becomes a postulate of faith. The second condition is harmony between moral striving and the natural impulse toward happiness. By experience it cannot be shown that virtue always leads to happiness or happiness to virtue; where it happens, it is accidental. And yet the highest good must include not merely virtue (the greatest good), but also happiness. There must in the last resort be an agreement between the world of nature and that of freedom. There must then be postulated a power that can bring harmony between the two worlds, so that morality does not stand as a stranger in the world. Therefore the one who is permeated by the moral demand is led to faith in the existence of a God.

Practical reason thus leads to convictions that concern something that lies beyond the *[92]* limits that theoretical reason pointed out. The X (Ding an sich) that theoretical reason had to leave undecided we gain access to when we follow the need that practical reason calls forth in us to postulate a perfect realization of the moral ideals. Kant enumerates three postulates: the freedom of the will (which has already been discussed), the immortality of the soul and the existence of God.

The need from which the postulates spring is not of a sensuous or egoistic kind, but is a consequence of the presence and working of the moral law in man's mind, thus of what is universal and universally valid in his being. It is a pure need of reason (ein Bedürfniss der reinen Vernunft). But just as strongly as Kant stresses the difference of this need from any wish whatever, just as strongly does he stress its purely subjective character. The faith (or rather: the hope) to which it leads is not itself a duty, although it is called forth by the law of duty. There can — says Kant — be no duty to believe what cannot be known. (Kritik der praktischen Vernunft. Kehrbach p. 151. 172 f.). Kant thinks: when man feels the law in his inner self, he will also *unavoidably* feel the need to assume the conditions of its being carried through. And in reliance on this unavoidability he also comes, in a single connection, to declare it a duty toward oneself to have religion, since otherwise one cannot keep one's moral conviction up. (Tugendlehre. Beschluss. — Lose Blätter p. 513). Indeed, the utterance even escapes him that the "right-minded" may well waver in faith at times, but never give it up. (Kritik d. pr. Vern. Kehrb. p. 175).

The question is whether the need of reason from which the postulates spring is really universal, — whether it is necessarily found in all individuals who acknowledge an unconditional moral law and have an infinite ideal in their inner selves. Kant has not seen that this question cannot be decided a priori, but must be answered by means of psychological experience. On his own presuppositions he would not be able to object anything against the one who, without wavering in his moral conviction, but also without
[93] any "postulates" whatever, could look the eternal disharmony between ideal and reality in the face. Indeed, he has even in a single place conceived this possibility. The place is of special interest, since it states clearly what Kant wished to attain by his postulates. "We can", says Kant (Kritik der Urtheilskraft § 87), "conceive a righteous man (such as Spinoza) who holds himself convinced that there is no God and no future life either: how will he judge of the purpose that is set him by the moral law, which he honors in deed? From his obedience to this law he demands no advantage for himself, neither in this nor in another world; unselfishly he will solely and only bring about the good toward which that holy law directs all his forces. But his striving is limited, and from the side of nature he may well now and then expect an accidental concurrence, but not any lawful and regular harmony with the *purpose* he feels himself obliged and impelled to realize. Deceit, violence, envy will always be rife around him, although he himself is honest, peaceable and benevolent. The righteous whom he meets are, by nature, which does not heed how worthy they are of happiness, subjected to want and lack, sickness and untimely death, just as well as the other animals on the earth. And they will go on being so until a vast

grave swallows them all together and hurls them, who could believe themselves the final end of creation, back into the abyss of chaotic and purposeless matter from which they were drawn forth". Kant now thinks that one will not be able to hold fast to the *purpose* that the moral law leads us to set up without the religious postulates. But his ethics would, after all, precisely abstract from all purposes and demand a willing for the sake of the inner law itself — whatever might come of it! Only by an inconsistency can he set up the concept of *the highest good* (union of happiness and virtue) as an *ethical* purpose; for the first proposition of his ethics (Grundlegung p. 1) runs: "Nothing can be conceived in the world or outside the world that could without condition be called good, except *a good will.*" By his postulates (especially the second and third) he in fact takes back the independence of ethics.

The postulates come forth when a need is felt to think the ethical together with *[94]* the whole of the rest of our world-view. It is indeed a need that does not stir in all, and ethically regarded need not stir either. But Kant's great merit with respect to the philosophy of religion is nonetheless precisely this: in a clearer and more deep-going way than had hitherto been done, to lead the religious problem back to a personal need that is determined by the relation to ethical ideals. The position of the religious problem at different times and in different persons will rest on the presence of this need and on the peculiar kind and manner in which it appears. Here there will, as experience shows, be countless individual differences, which Kant's dogmatizing led him to overlook. But with as sure a hand as that with which, in a theoretical respect, he determined the philosophical place of the religious assumptions, he has here found a corresponding determination of place as regards the practical and psychological side of these assumptions.

b. The Religious Postulates in Their Relation to Kant's Epistemology and to "Natural Religion"

By the postulates we come beyond the limits of knowledge. But how is that possible, since the critique of reason has, after all, established that our forms of knowledge simply have these limits? We have, after all, no further sets of forms of knowledge to change to! Kant here lays weight on this, that the categories (it is here, as almost always, the concept of cause that is chiefly in question) do not spring from experience, but from the faculty of thought itself. He finds therein the possibility of using the categories beyond the limit of experience or of phenomena to *think* what the postulates contain, even if we cannot *know* it. For knowledge a co-operation of intuition and thinking would be required. But

intuition is here excluded. The human cognitive faculty is characterized by the difference between intuition and thinking, and by this, that all intuition is sensuous (in time and space). A supersensuous intuition would be fanaticism, would lead to lighting a "magic lantern of figments of the brain". — Thus we conceive God as cause of the world, our
[95] striving as continued and our ideal willing as a first beginning, although these concepts cannot be supplied with intuitive data.

Even such a use of the forms of thought does not, however, agree with the critique of reason, according to which concepts without intuition are empty (just as intuitions without concepts are blind). And — even if one grants such a use — it is in reality not the usual representations of God, freedom and immortality, not what was understood by these words in "natural religion", that Kant can express by his pure concepts. With intuition, after all, *the relation of time must also fall away:* but how can there then be talk of *continued* existence (zukünftiges Leben), of beginning, of creating? And if all data of experience are to be kept away, everything must be removed from the representation of God that is owed to human psychology. We know only an understanding that works its way forward to truth through successive and discursive apprehension and discussion, and only a will that sets itself purposes and thereupon seeks the means of reaching them. That does not fit the unconditioned and infinite being, and it falls away with the relation of time. And if we ask Kant what, then, is left of our psychological concepts when we undertake so radical a change in them that they are to be applicable to God, he answers clearly and candidly (although this answer is constantly overlooked in the popular expositions of his doctrine): only *the bare word* is left! "Name but one property", he says, e.g. (Kritik d. pr. Vern. Kehrb. p. 165), "of which one could not show that, when everything anthropomorphic is separated out, only the bare word is left, without our being able to connect the least concept with it so as thereby to extend our theoretical knowledge." Similar utterances are found in all Kant's main writings[24]). More clearly Spinoza could not have expressed himself. — If it is asked how Kant can have attributed value to the religious representations when they are theoretically altogether empty, the answer is that according to Kant all religious representations are *symbolic.* We must help ourselves with analogies and symbols where the scientific use of our concepts stops. The representations of a personal God and personal immortality, too, then become only
[96] symbols for something that thought cannot express in adequate form. It is no higher knowledge that is gained by passing over from knowledge to faith. The "need of reason" here in reality takes poetry into its service. The postulates are projections out into the unknowable of images that have formed in the world of experience. — There is, when

one thinks through what lies herein, a very great difference between Kant's religious standpoint and the religious standpoint that "natural religion" formulated. Kant, like Lessing, had an esoteric and an exoteric doctrine[25]), even if he has not made the difference between them altogether clear to himself, nor has seen all the consequences that lie in the assumption of the symbolic character of the religious concepts, e.g. this one, that symbols cannot be forced on anyone, but must be the object of free, individual choice, so that my "need of reason" might perhaps find more appealing symbols than those chosen by Kant. Kant has at the same time somewhat too hastily placed everything he felt a need to believe in over into the Ding an sich, without investigating whether the different postulates might not be mutually contradictory (e.g. God's omnipotence and man's freedom — cf. Religion innerhalb der Grenzen der blossen Vernunft. 2. Aufl. p. 215 f. — Lose Blätter p. 544). He has used the limit of knowledge, as the woman in childbed in Holberg uses the screen, to hide the various elements that he wants to be rid of in science, but wants to keep for faith, without thinking that the strife will begin anew — behind the screen! For this demand we do make of our symbols, that they be without self-contradiction.

The symbolic character of the religious representations is closely connected with the fact that religious faith springs from a personal need. We seek forms in which we can represent existence to ourselves as a kingdom in which what for us is of the highest value comes fully into its own. On this point, too, Kant's philosophy of religion — with all its imperfections — marks a great advance over orthodoxy and the theology of the Enlightenment.

c. Positive Religion

As it is with the representations of natural religion, so it must also be with the repre- *[97]*
sentations of positive religion. This consequence Kant drew in the remarkable work *Religion innerhalb der Grenzen der blossen Vernunft* (1793). Instead of asking whether the representations of positive religion can stand before the tribunal of natural science and of historical criticism, Kant asks about their significance for human life in an ethical respect, asks what they contain that can nourish and support life. And this is and remains, after all, always the main question, for on its answer depends the value that can be attributed to these representations (however the verdict of physics and of history may turn out). That Kant keeps to Christianity as the positive religion lying nearest at hand is natural. But it is remarkable to see that it is the dogmas of sin and atonement that he especially discusses, precisely the dogmas for which the period of the Enlightenment had least

sense. He finds deep ethical truths contained in these dogmas, in that what the Bible relates as external, historical events can be interpreted as inner spiritual relations in man's mind, as the expression of contending oppositions in the human will.

In the doctrine of sin and the fall Kant finds expressed a truth confirmed by experience: that, in opposition to the inner willing whose law is the moral law, there expresses itself a propensity to set the claims of sensuousness above those of reason. It is law standing against law [as in Böhme God against God]; for it is not sensuousness itself that is evil. Evil is the will that reverses the right relation, according to which reason rules over sensuousness, and formulates the opposite relation as its rule. The radical evil in man is this perversion of the relation between the motives. Philosophical ethics takes it as a fact, as something given in experience: the Bible presents the origin of radical evil as a historical event, in that it lets man's good nature precede his evil nature in time and the state of innocence be broken off by a fall. By introducing a tempter it admits at the same time that an absolute explanation cannot be given.

But beside radical evil there is always found in man a seed of the good, which can be developed especially through admiration of exemplars. The capacity to respect the good,
[98] even if one cannot practice it oneself, never altogether falls away. This capacity, from which the moral consciousness develops, is the ideal in our nature, the inner heaven in us in opposition to radical evil as the inner hell. There is given in our nature an incomprehensible exemplar, an indwelling ideal, which in the Bible is presented as the Son of God, who descends to earth and takes on human form. The God-man is the idea of human nature in its perfection. Here again what for the philosopher is an eternal relation is presented as a historical event: the relation of the good, of the ideal, to human nature as a force that stands in strong opposition to other tendencies in this nature. The struggle between Satan and Christ is a struggle that is fought in the innermost depths of human nature. In this struggle the God in us must suffer for what our evil will has done. The new man who is to come into being suffers for the sins of the old. To this there corresponds in our experience repentance, which presupposes a new direction of the will that takes over the suffering bound up with working one's way out of the old will. This is the ethical content of the dogma of atonement. The coming into being of the ideal man in us cannot indeed become a completed process; only an approximation is possible; but from the highest point of view the successive approximation is intuited and valued as a whole, the relation of time being left out of consideration.

Kant is clearly conscious that this symbolic interpretation is altogether different from historical interpretation. The latter he leaves to learned research. He asserts that only

through such symbolic-ethical interpretation do the religious books have lasting ethical worth, and that this is in reality the way one goes and has always gone when one has wished to make religion practically fruitful. And such an interpretation is all the more justified, as man's moral capacities have involuntarily exerted their influence already in the development of the religious representations and given the revelations themselves their stamp. By a purely external, historical way we can never come to know the ideal and the divine. The highest exemplar always lies in our own spirit. Only through the God in us do we know the God outside us; and that God in us is the only one before whom all knees bow! (Lose Blätter p. 218). This God in us is the interpreter of everything *[99]* that appears with the claim to be revelation. If Abraham had held fast to it, he would not have wished to lay hands on his son; if the judges of heretics had held fast to it, they would not have condemned anyone for the sake of his dogmas. The moral law in us is more certain than any faith whatever. (Religion innerhalb etc. p. 289 f.).

In Kant's opinion such wonderful things take place in man's inner self that one has no need to assume external events to explain them — especially since every such explanation is illusory. The mysteriousness of the events that the biblical narratives tell of he left undecided; but he "honored the external garb for a doctrine that in the end is grounded on testimonies that are indelibly preserved in the soul and need no miracles." (Religion innerhalb etc. p. 117).

Only this demand does he make, that the historical and dogmatic ("statutory") element in religion is more and more to be subordinated to the inner and ethical element. Historical faith has only provisional and symbolic significance. The visible church is to draw nearer and nearer to the true church, the invisible, in which each individual stands in an inner and immediate relation to the highest truth. Then the degrading difference between priest and layman will also cease; it is conditioned by the necessity of historical learning in order to understand positive religion. The kingdom of God is not a kingdom of priests. After the Reformation there has not been any unconditional progress: whether one forbids laymen to read the Bible, or one says: Search the Bible by all means, but you must not find anything else therein than what we have found, — it comes to the same thing! And then tell us rather at once what you have found: then we shall spare ourselves our reading! (Lose Bl. p. 402). Still more burdensome for the conscientious than the external observances that Catholicism demands is the dogmatic confession of faith. If Christianity consists in that, then the word does not hold: "My burden is light!" —

Kant himself came to feel how strong an opposition there was between his inter-

pretation of Christianity and that of the church. Nevertheless he had the confident
[100] assurance that religious development would go in the direction he had caught sight of. In this he may yet prove right, although the religious oppositions have been sharpened in the time that has passed since his work on the philosophy of religion appeared. For the philosophy of religion itself this work was epoch-making, and it keeps its value for all times by its struggle for the inner against the outer and by its assertion of the significance of the inner events of personal life. In the psychology of religion and in the historical conception of religion it leaves much to be desired. But it was the task of the new century to supplement his thoughts in this respect.

6. Speculative Ideas on an Aesthetic and Biological Basis (Kritik der Urtheilskraft)

a. The Two Worlds and Their Possible Unity

In his *Kritik der Urtheilskraft* (1790) Kant develops thoughts that point beyond the limits within which his philosophy otherwise keeps itself. Instead of the appeal to a moral faith, which in the earlier writings formed the conclusion of his conception, his thinking here ventures on bolder prospects of a great connection within which a unity of the oppositions he had hitherto held apart from one another might be possible.

Kant had operated to a high degree with sharp distinctions. They had been necessary to him for reaching his results. Only by distinguishing between phenomena and things in themselves had he been able to carry through his proof of the real validity of the knowledge of reason, and only by distinguishing between the world of experience and the intelligible world had he thought he could unite the originality and self-legislation of the will with the natural determination of the empirical character. The two worlds, the phenomenal (that of nature, of experience) on the one side, the intelligible (that of freedom, of purposes) on the other, stand over against each other as though they were wholly different and foreign to each other: this is the result of Kant's epistemology and
[101] of his ethics, and his philosophy of religion has remedied it only in a rather external way, through postulates. But in his great critical deliberateness he turns once more against these results and asks whether it is really justified to regard those oppositions as absolute and irreconcilable. — Man himself is, after all, both a natural being and an intelligible being, both phenomenon and thing in itself. Here, then, is a point of unity for the two worlds! Man lives and acts in nature, but is to follow, and can follow, the

law of freedom, and through human development in the world of experience the goal fixed by the ideal law is to be striven for. So the two worlds cannot fall wholly outside each other: there must be a ground of unity for nature and the moral world. — And since Kant has constantly operated with the distinction between our conception and the very essence of things, was there not reason to ask whether the very assumption of those oppositions might not be connected with the nature of our knowing mind? It is not, after all, said that what stands as oppositions for us (with our discursive thinking, which must distinguish and analyze in order to know) is therefore so in the innermost ground of things! Just as Kant had already intimated in "Kritik der reinen Vernunft" that what lies at the ground of the material in our knowledge might be the same as what determines the forms in which we order it, — and that what lies at the ground of material phenomena might be the same as what lies at the ground of mental phenomena, — so now too he finally discusses the possibility that the foundation of the world of nature and of the world of freedom might be one and the same. He does as the master builder does who takes down the scaffolding when the building is finished: the distinctions Kant has used during his inquiry as means of climbing upward he calls back when he is at the conclusion of the work. From the very distinction between knowledge and existence he draws the conclusion that this distinction — which is, after all, made by our knowledge itself — cannot be absolute. But in this he relies on definite facts which lead him to bring forward problems he had not hitherto treated, facts which show that nature, according to its own laws, works in the direction of what our knowing and valuing mind desires *[102]* and demands. Should it then be foreign to the ideal?

b. Aesthetic Consideration

An aesthetic judgment is one by which we call a phenomenon beautiful or sublime. The question then arises whether such judgments can have universal validity, can be the expression of more than a purely individual satisfaction.

The beautiful is distinguished from the agreeable and the good by not resting on the real existence of an object, but only on the image, the conception or representation, that we have of it. Aesthetic satisfaction is disinterested and free. It arises through a free play of the cognitive faculty, which is brought about when an image makes our imagination and our understanding work together in a harmonious way, the particulars of the image being easily and naturally combined into a whole that is immediately understood. Neither the purely material impression nor the pure concept plays a part here: mere material gives no whole, and the pure concept gives an abstract rule, which appears as constraint

over against the material. In the English art of gardening[26]), in musical compositions, in the visual arts, the single elements work together immediately into a conception of a whole that satisfies an essential spiritual need in us. Kant lays very great weight on the entirely immediate in the impression of beauty and underrates the more remote representations that the image can awaken. Only "free" beauty (determined by what Fechner has called the direct factor) is properly beauty; "adherent" beauty (determined by the associative factor) does not rightly bear the name of beauty, since it presupposes definite representations. A flower, an arabesque, a musical fantasia afford free beauty; the beauty of a human being is "adherent," because it presupposes a representation of what it is to be a human being.

Judgments about beauty (i.e. about free beauty) are subjective, insofar as they spring from a feeling that the image awakens in us. But they may nonetheless be universally
[103] valid, since the feeling is here determined by something that is common to all men: the relation between the cognitive powers, the need of a harmonious relation between the faculty of intuition and the faculty of comprehension. Aesthetic judgments (judgments of taste) cannot be proved, but they can be called forth by giving occasion for immediate intuition. The universal validity is exemplary, i.e. it is reached by means of examples, not by rules. Therefore aesthetic criticism is an art, not a science.

The sublime is likewise due to a disinterested satisfaction, but here the relation is more complex. The great, the immeasurable, the infinite in extension and in force overwhelms our faculty of apprehension and crushes our sense of self. The mind is thereby brought to let go of everything finite and sensuous and to feel in itself a power that is raised above every limitation: the power to think the infinite ideas and to frame the unconditioned law. A higher kind of self-assertion is thus awakened over against the mighty in nature, after a preliminary inhibition. What is properly sublime, then, is not the outer phenomena: they are only occasions for the great in ourselves to make itself known. — Here too universally valid judgments can be passed, since judgments about the sublime are grounded in a feeling that must be capable of being called forth in every sufficiently developed human being.

In the phenomena we call beautiful and sublime, nature now works *according to its own laws* to call forth in us a satisfaction independent of all egoistic interest. This is a significant fact, especially since aesthetic feeling, by its disinterestedness, is akin to ethical feeling!

Kant does not stop at investigating the valuation of the beautiful. He also investigates the production of the beautiful in art. This production goes on — like the valuation —

without the guidance of abstract concepts, predominantly involuntarily, and yet in such a way that what is produced becomes an object of general recognition and comes to stand as an exemplar. Art is the work of genius. Genius is "exemplary" originality: a disposition by which nature gives art its rules. Genius does not work according to rules or ideas, and yet rules can be derived and ideas found in its works[27]). The fact of genius thus shows — like the fact of the aesthetic judgment, — that the world of nature and the *[104]* world of freedom are not absolutely separated, but must have a common foundation. These facts too we must — such is Kant's meaning — take into account before we form our concluding conception of existence.

c. Biological Consideration

Not only by producing beautiful and sublime phenomena and by working through men of genius does nature show its agreement with the laws of our mind. In organisms it presents beings of such a constitution that their single parts are understood only as means or condition for the subsistence of the whole being. There is here the same intimate connection between particulars and whole as in the works of genius. In the organic world nature works as genius, in a way that differs both from the mechanical production of a whole by the cooperation of the parts and from the conscious joining of parts into a whole according to a definite plan. Nature's organizing activity has no proper analogy with any of the kinds of causality we know (Kr. d. Urth. § 65). We can make it intelligible to ourselves only when we regard organisms *as if* they were produced under the guidance of the thought of a purpose (whether this is now thought of as conscious or unconscious). But this is only a regulative principle; we may not deny the possibility that the organic species and forms arose through a natural process of development according to mechanical laws. The analogy that exists among the organic forms might point to a common origin, so that nature has reached, step by step, from the lowest forms to the highest. One could imagine aquatic animals first becoming marsh animals and later land animals, so that the forms suited to the circumstances replaced the less purposive ones. (There are — says Kant in Kr. d. Urth. § 80 — probably few even among the most acute naturalists through whose heads such a hypothesis, admittedly a daring adventure of reason, has not passed!). —

The whole opposition between mechanism and teleology, between arising through the blind cooperation of the parts and arising through planned joining, is perhaps due *[105]* only to the nature of our knowledge. Our understanding proceeds discursively, goes from the parts to the whole and sees the whole as their product; if it is to think of the

constitution of the parts as determined by the whole, it can do so only by thinking the representation of the whole as the subjective cause of the formation and combination of the parts. For us, therefore, mechanism and teleology form a great opposition. But in the ground of nature, unknown to us, mechanical and teleological connection might well be united in one principle — only our reason is not able to frame this principle!

With this thought, the most profound to be found in Kant, he takes up again a line of thought that had occupied him much in his youth: namely, that the same thing which makes causal relations between things possible also makes purposiveness and harmony in nature possible. "The continuity in Kant's philosophical course of development" shows itself most clearly of all at this point. His speculative successors would begin precisely where he ended. For Kant himself it was a last prospect, a concluding hypothesis, which was of great value for inquiry, but could not be dogmatically set up as a starting point. His marvelous ability to work at small things with great things in view does not fail him here either. The account of his philosophy cannot end in a more fitting way than with this characteristic trait of his personality as a thinker.

7. The Opponents of the Critical Philosophy

A work like Kant's could not be understood at once. It discussed so many problems, and discussed them in so interwoven a connection, — it presented, both by what it established and by what it denied, so peculiar a standpoint, that it is no wonder its contemporaries could not pass a judgment that bore witness to a full penetration into its peculiarity. Not all the judgments passed on it have interest for the general history of philosophy. The misunderstandings displayed on the part of the philosophy of the Enlightenment
[106] and the Wolffian school bear witness only to the difficulty that one has, from an older, ingrown, self-satisfied standpoint, in entering into a new line of thought. Not everyone had such self-knowledge and modesty — and at the same time so fine a faith in truth — as Mendelssohn displayed in the preface to his "Morgenstunden." — Of considerable interest, on the other hand, is the opposition that Kant's philosophy met with in a group of men who, in various forms, maintained the significance of immediate certainty and of historical tradition, and in general the significance of the mind's collected, concentrated activity, over against Kant's analysis and criticism, which at so many points led him to sharp separations between elements that in reality occur in indissoluble connection. In great part these men do Kant an injustice, in that they overlook Kant's own attempts to join again what he had separated only for the sake of inquiry and clarity. And in part their

objections are directed not merely against Kant's philosophy, but against all philosophy, indeed against all inquiry whatever. But they nonetheless form a salutary counterweight to Kant's analysis by pointing to the living, concrete connection of things, and their opposition relies, moreover, in principle on the thinker whom Kant had especially wished to continue and to overcome, the same to whom he owed his awakening: Hume. On experience, especially on feeling and historical experience, they relied in their struggle against the philosophy of the Enlightenment, to which, in their eyes, Kant was still to be reckoned. They carried Kant's opposition to the Enlightenment so much further that they came into conflict with him as well. Finally, they intimate points of view and ideas that worked in wide circles, far beyond the philosophical domain, and that became of great significance for the awakening of a new poetic and historical conception, a deeper understanding of life as a whole.

a. Foremost in this group stands Kant's fellow townsman and friend Johann Georg Hamann, the Magus of the North, as he has been called. It was through religious experience that he learned to know the power of faith and the great and conflicting forces of life. A religious breakthrough, which came to him during a stay in London, where he had had occasion to experience the weak sides of his character, became decisive for his whole life. There had been a mighty stirring in this deep, fermenting nature, *[107]* which he himself (in a letter to Kant of July 27, 1759) characterized by calling himself "a man to whom the sickness of his passions gives a strength to think and feel that a healthy man does not possess." He felt himself to be like a man in a deep pit who at midday sees stars that those who go about in the daylight cannot see. An intimate feeling of the mystery of life and of the contradictions that existence presents to the finite understanding, as soon as it does not stop at the surface, is Hamann's distinguishing mark. He is fond of enigmatic hints and paradoxical propositions, and his writings swarm with allusions to his chance reading, so that they are often unintelligible without commentary. Baroque fancies alternate with profound ideas and with mighty pathos, which makes intelligible the impression he made on some of the greatest minds of the time. He stood not only for the pious, who sighed over the Enlightenment and unbelief, as the prophet who opened again the springs of the old faith, but he stood for the whole younger generation as the one whose flight of spirit went higher than the reason of the time could follow, and as the one in whom thought, imagination and feeling worked together in a passionate concentration that came forward with tremendous force against the prevailing rationalism and sentimentalism. But precisely because his burning thirst could be satisfied only by a content of life that streamed with collected power, he was a

sworn enemy of all analysis. Already in his first writing *(Sokratische Denkwürdigkeiten.* 1759) he declares it presumptuous to want to carry analysis through to the last elements: that would, after all, be to want to grasp the invisible essence of the Deity itself! Nature and history perhaps contain riddles that only a power wholly other than our reason can solve! Therefore Socratic ignorance is praised. Both our own existence and the existence of things outside us we must *believe* in; no other way to the assumption of them is given. Faith is not a work of reason and therefore cannot be overthrown by reason either. It does not build on grounds, any more than taste and seeing do. Where ignorance stops, the divine genius helps us, which for Socrates was of greater significance than all the
[108] wisdom of the world. — The ground of religion, he declares later (*Zweifel und Einfälle.* 1770), lies in our whole existence and is more comprehensive than the domain in which our knowledge moves. Knowledge is the most abstract form of our existence. — Only through passion do abstractions get hands, feet or wings. Life is to be taken as a whole. Philosophers separate what nature has joined together. To express this concentrated content of life rightly, Hamann makes use of Bruno's principle of the coincidence of opposites (coincidentia oppositorum), which he declares to be worth more than the whole of Kant's critique. Reason (he writes to Kant as early as 1759) — as Hume has shown — is given to you, not that you should become wise by its help, but that you should know your folly and ignorance, just as the law was given to the Jews to convince them of sin. Through experience, tradition and language we get all that we can know.

With great interest Hamann followed the writings of Kant ("the little Magister," later "our Plato") and was naturally occupied above all with "Kritik der reinen Vernunft." Immediately after the work appeared he made a draft of a criticism, which he nevertheless held back — partly because he noticed that "his poor head, over against Kant's, was as clay against iron," — partly so as not to offend Kant, to whom he stood in a debt of gratitude. Only after Hamann's death (1788) was his *Metakritik über den Purismum der reinen Vernunft* printed. He saw in Kant's philosophy a failed attempt to make reason independent of all tradition, faith and experience, and he polemicizes especially against Kant's separation between matter and form and between sensation and understanding. Why this violent, unwarranted, willful separation of what nature has joined together? There is, after all, a constant circulation! Intuitions rise without interruption into reason, and concepts descend into sensation! — In a letter written while he was reading "Kritik der reinen Vernunft" in proof sheets, he calls Kant the Prussian Hume, but declares that he prefers the English one: "Hume is still always my man, because he has at least ennobled the principle of faith and taken it up into his system. Our countryman keeps

chewing the cud on his attack on causality without thinking of the principle of faith. That does not seem to me honest." (Letter to Herder of May 10, 1781.) And already in *[109]* an earlier writing *(Des Ritters von Rosencreuz letzte Willensmeynung über den göttlichen und menschlichen Ursprung der Sprache.* 1772) he had said: "Do you not know, then, philosophers, that there is no physical bond between cause and effect, means and end, but a spiritual and ideal one: namely, the bond of the collier's faith?" And express reference is made here to Hume. —

It was not Hamann's way to write great works and give long developments of thought. But his oracular sayings inspired Herder and Jacobi, who continued his work, in that they, like him, appealing to Hume, set faith against reason. Yet they were more children of the time than "the Magus of the North," who has been called "the most believing among the believers" (yet also "the freest, because he was the deepest"). Hamann was a believing Lutheran. Herder and Jacobi stood far nearer to the Enlightenment, especially to its sentimental form, although they fought the analyzing and grounding reason just as their master did.

b. Hamann lived as a poor warehouse superintendent in Königsberg (b. 1730, d. 1788). His view of life was established even before Kant was awakened from his dogmatic slumber. Johann Gotfried Herder (b. 1744, d. 1803) was as a young man Kant's hearer and was strongly influenced by the standpoint on which Kant stood in the sixties. But in Königsberg he also met Hamann, and the influence of the latter became the properly decisive one. With a great sense for the original, the popular, what unfolds involuntarily, he worked to bring a fresher and more vigorous spirit into literature. Goethe learned (as he relates in "Aus meinem Leben") from Herder that the art of poetry was a gift to the people, to the whole world, not a private inheritance for a few refined, cultivated men. In the spirit of Rousseau, Herder maintained the right of the natural and the human; but he had far more historical sense than Rousseau, and in his view of nature he is influenced by Goethe, with whom, as superintendent in Weimar, he stood in lively interaction until divergent views on art and politics drew them apart from each other. In his chief work *(Ideen zur Philosophie der Geschichte der Menschheit.* 1784—91) he collides with his former *[110]* teacher Kant. Herder cannot accept that the goal is to lie only in the species, not in the single individuals. Each single individual is destined for such happiness and such development as is possible at the given stage, but in order that this goal may be reached there must be interaction among the individuals, and there must be a handing down of the acquired means of culture from generation to generation. It is this connection between individuals and generations that makes there be a humanity — and a philosophy

of history. Already in unconscious nature ideal forces are at work, which form and organize according to a definite type. Leibniz's doctrine of monads is transformed in Herder into a doctrine of organic forces, which work throughout nature in different degrees and at different stages, in analogy with the force that works in our thought. "The force that thinks and works in me is by its nature as eternal a force as that which holds suns and stars together . . . All existence is of the same kind, one indivisible concept." (Ideen. I. p. 7—8). The thought that expressed Kant's last presentiment is thus here laid at the foundation, and by means of "the great analogy of nature" a connection is found in everything, in the universe as in history, a connection that also teaches us to find the bond between science and religion. When Herder constantly uses the word nature in his book, it is so as not to name God's name so often: "Gott ist Alles in seinen Werken!"

In this great connection of receiving and giving man too stands. His reason can by no means go its own ways. The very word for reason, Fornuft, comes from fornemme, to perceive, and points to our getting our thoughts through tradition, language and outer influence. Reason is a product, not something innate. Religion is the first form of spiritual culture. Before the first abstract thoughts could be formed, there went a religious feeling of invisible forces in nature. By this feeling man raises himself above the animal. The disposition to humanity is older than reason. But it is developed only by education, under the influence of exemplars. Man cannot develop everything out
[111] of himself. And precisely because every single man becomes a man only by education, there is an education of humanity. Every single stage of this development must be not merely a means to reach the next, but also an end. All the means of the Deity are ends, and all its ends are means to higher ends. Thus: "What every man is and can be, that must be the end of the human race. And what is that? Humanity and happiness in this place, in this degree, as just this definite, peculiar link in the chain of culture that stretches through the whole species." (Ideen. II. p. 118). — By this line of thought Herder corrects the violent opposition that existed for Kant's philosophy of history between individual and species, and that had far-reaching consequences for his ethics. But Herder contents himself with an enthusiastic declaration and does not enter more closely into the great problems that the idea set up contains when it is to be applied in particulars.

Herder's fundamental philosophical view, so far as it comes out clearly through his lyrical and florid language, is borne by the thought that already played so great a part in Kant's first writings: that the lawful connection of things itself presupposes a ground of unity in existence. And this thought in turn led Herder naturally to sympathy with Spinoza. He saw in Spinoza the most consistent philosopher and found, through repeated

study of his work, nourishment for his need to get beyond all dualism between God and nature, between mind and matter. A God existing outside the world conflicted for him with the concept of God, with the concept of the world and with the concept of space, and personality could not be a property of an infinite being. He therefore decidedly distanced himself from Jacobi's horror of Spinoza. He had a sense for the mystical side of Spinoza, while Jacobi saw in him only the abstract rationalism (and neither of them saw the realistic side). He even wrote to Jacobi: "If you make the most inward, highest concept, which gathers all in one, into a mere name, then it is you who are the atheist, not Spinoza!" It was Herder's hatred of all abstraction that led him beyond the distinction between God and the world. He parted from Jacobi for the same reason as he parted from Kant, and for the same reason as Hamann (who, for the rest, could not understand his *[112]* friends' negative and positive interest in Spinoza) took an interest in Bruno's principle of coincidence. In his writing *Gott* (Gotha 1787) Herder took the side of Spinoza against Jacobi. His acquaintance with Spinoza was not so thorough in particulars as Jacobi's; he reinterprets Spinoza's view of nature in particular by putting into it his "organic forces," in general his modified Leibnizianism; but his writing did much to awaken interest in what is deepest in Spinoza's philosophy. He himself was now regarded as an apostate by the pious band that gathered about Hamann. His religious standpoint had always been different from that on which the Magus of the North, whom he so highly admired, stood. They were akin in the weight they laid on the historical, the traditional, what has arisen involuntarily, in opposition to the time's predilection for the clearly conscious and the arbitrary. Herder was the one who first in earnest shook the current assumption that religious representations were due to arbitrary fabrication and fraud on the part of priests and princes. His sense for folk poetry also taught him to see the original and nature-strong spirit of the old religious books. The need for nourishment for all faculties at once asserted itself here; he wanted a philosophy "for the whole man," and this he found in the religious works that stem from a time when the human mental faculties had not yet begun to work separated from one another. It was a need that was more poetic than philosophical — and at the same time more poetic than religious. For Herder it was always difficult to hold poetry, philosophy and religion apart from one another; in this he is the most important forerunner of Romanticism. For the understanding of religion he worked mightily by maintaining its immediate and involuntary origin in the human spirit and by demanding that the religious writings be read and understood in their own spirit. His own interpretation was (without his always being clearly conscious of it) symbolic and ethical. His standpoint is different both from that of orthodoxy and from

that of rationalism (in any case the current rationalism): in a certain sense a continuation and further carrying out of Lessing's standpoint. Precisely because he saw divine forces
[113] at work everywhere in nature, he felt no need to set revelation in the narrower sense in sharp opposition to the general revelation that takes place through all the life of nature and of man. In the 17th book of "Ideen" and in the writing *Von Religion, Lehrmeinungen und Gebräuchen* (Leipzig 1798) he has developed his symbolic-ethical conception of Christianity. He found an "enormous anti-Christianity" in the prevailing church. Christ was for Herder the spiritual savior of the species, who wished to form God-men who — under whatever laws they may live — promote the welfare of others according to pure fundamental propositions, and themselves, in patience, reign as kings in the kingdom of truth and goodness. Christ's discourses are stamped with the purest humanity. But out of his words speculative dogmas have been constructed, and out of his symbolic actions magical processes have been made. (So wrote a bishop at the end of the last century!) Yet Herder did not doubt that the pure religion of Christ would conquer.

In his criticism of Kant's doctrine Herder gave Hamann's "Metakritik" a broader carrying through, without real originality. His significance is bound up with the positive fullness that he partly brought, partly pointed to. And when the diffuse writings *(Metakritik,* 1799, and *Kalligone,* 1800) in which he criticized Kant's philosophy appeared, the critical philosophy had already been developed further by Kant's successors and led into new paths, in part, to be sure, under the influence of the motives that from the first had determined Hamann's and Herder's position toward it. —

c. Friedrich Heinrich Jacobi (1743—1819), a spirited man of the world and seeker after truth, one of the most characteristic representatives of the "age of genius," contributed in his way a great deal to the peculiar direction that German philosophy took after Kant. With great acuteness he had, as already touched on, pointed out the great difficulties in Kant's doctrine of the Ding an sich, and maintained that this doctrine must go if a consistent system was to be reached. He was right in his criticism, but his intervention did not concern the main problem Kant had posed himself. The doctrine of the Ding an sich was only a consequence of Kant's philosophy, not its main point. Instead
[114] of going back to Kant's *original* task and examining his method in order to find where the error lay, one sought — in great part under the influence of Jacobi's criticism — to correct the error in the *finished* system. Jacobi himself felt unsatisfied with Kant's philosophy for a reason similar to Hamann's and Herder's: because immediate faith did not come into its right. And only faith can grasp the true. All science aims at constructing, at producing its objects by an inner acting, at dissolving everything into pure activity. We

understand a thing only by constructing it. Everything we comprehend we represent to ourselves as a link in a series, in a complete connection, in which all differences are canceled. The most consistent of all systems is subjectivism, which lets one's own ego be the first link in the chain and derives the other links from it. Therefore Jacobi saw in the turn Fichte gave to the critical philosophy the completion of speculation. His open letter to Fichte (*Jacobi an Fichte.* 1799) gives perhaps the best insight into his philosophical conception. Earlier he had maintained that Spinoza was the only consistent philosopher; now he set Fichte higher, because the ego, consciousness itself, must after all necessarily be the first link in the chain of knowledge. But common to all consistent philosophy is that it must deny the original, the unconditioned, that which has its ground and value in itself, the qualitative differences and peculiarities. All philosophy aims at grounding and deriving, at reducing quality to quantity: the immediate, the free and the original it therefore cannot comprehend. But for that very reason — Jacobi thinks — it also shows itself the more clearly, the more consistent philosophy becomes, that it cannot lead to the truth, — and it is for this reason that he urges and goads on to full consistency, for only then does it become manifest that by this way the truth is not reached. His letters on Spinoza impress this upon the philosophy of the Enlightenment; his dialogues on idealism and realism show it against Kant; his open letter to Fichte declares it most clearly and distinctly against this "Messiah of speculation." And once more he had occasion to come forward with his old thoughts against a new system, namely when Schelling thought he had founded a new philosophy of religion by way of the philosophy of nature. (Jacobi's writing: *Von den göttlichen Dingen.* 1811). Although Jacobi was not himself a consistent thinker, he had a great sense for the consistency of systems and was able with *[115]* energy to bring the great lines of a train of thought to light.

That there is a reality different from consciousness, of this — Jacobi thinks — no proof can convince us, since every proof moves within consciousness. Immediate observation is a miracle, which we must assume if we want to have truth: only through faith do things come to be for us. Only through faith does God, the original source of everything, the fount of all value in existence, come to be for us. God cannot be known, only believed. A God who could be known would not be God. Indeed, it is even in the interest of science that there be no God: for God's existence breaks off the series and makes connection impossible! God reveals himself immediately in our inner being (in the outer the true God cannot reveal himself), just as things reveal themselves immediately to our senses. We perceive immediately that there is something that is greater and better than we ourselves: we find God by finding ourselves in God. And finally freedom, in

the sense of the mind's ability to intervene in the world of matter, can only be believed, not comprehended. Freedom does not, indeed, consist in the unreasonable ability to determine oneself without grounds; but since it is absolute self-activity, it is inaccessible to science.

The opposition between science and faith that Jacobi maintained was connected with the strong movement of feeling that dominated the time after Rousseau's appearance. He himself gave expression to this tendency in poetic works. It is the right of feeling that he maintains over against the objective grounds of the understanding, indifferent to all individual emotion. In the moral domain too he pleads the cause of individual feeling. "The beautiful soul," which unfolds itself out of inner germs without regulating itself by general principles, resting securely and rocking itself in its moods, is to him the highest, although he is not blind to the dangers to which it is exposed. He pleads the cause of the exceptions over against the strict, universal law that Kant had set forth in his categorical imperative, which seems to demand the same conduct of all in all cases. Kant's moral
[116] law designates only the formal consistency in action, not the heart out of which one is to act. The law exists for man's sake, not the other way round. The book of life must be written before an index is made to it. A moral system is nothing but such an index made afterwards! And then it wants to regulate life, to set up universally valid rules without respect for individualities and exceptions! Jacobi declares his opposition to this in the open letter to Fichte in the often-quoted words: "Yes, I am the godless atheist who will lie as Desdemona lied in the hour of her death, will lie and deceive like Pylades, who gave himself out to be Orestes, will murder like Timoleon" etc.[28]).

Jacobi, together with Hamann and Herder, has the merit of having held fast the significance of that in life which cannot be converted into universally valid knowledge. He fought for the right of immediacy, of reality and of individuality, and thereby furnished important correctives to the direction on which philosophy was about to enter, and on which he himself wished to goad it on. But it was an illusion when he thought that the objects of his faith were given by immediate revelation. In the first place, he takes the word faith in two very different senses, partly of the involuntary trust in sense observations, partly of the religious faith in what cannot be intuited. In the second place, the content of his religious faith is a good old acquaintance: the Cartesian spiritualism, which now, instead of the proofs the dogmatic school had thought it could furnish, is proclaimed with Jacobi, as with Rousseau, as an object of faith. But would the different individuals, the different "beautiful souls," really find the same content of faith in their inner being? From feeling no definite content can be derived, only the general need to

hold fast the validity of what is of highest value. By fixing definite dogmas (even if only the dogmas of "natural" religion) as the only form of satisfying this need, Jacobi violates the inwardness and individualism of feeling for which he otherwise makes himself the interpreter. And it is really only thereby that the sharp opposition between his faith and his knowledge arises: his faith comes to suffer under all the difficulties of the Cartesian spiritualism. It was in reality two kinds of philosophy that he had, in his faith and in *[117]* his knowledge, and it was then no wonder that he could complain that his head and his heart conflicted with each other: he had taken it into his head that his heart could beat only in an artificial way, and that this was the true nature.

8. The Further Development of the Critical Philosophy

The same need that led Kant's first opponents to raise a protest of feeling against his work led his first independent disciples to try to bring the many branching investigations and the numerous distinctions back to a few simple principles — preferably to a single principle. It was an endeavor that was bound to arise naturally in thinkers whose enthusiasm for what was new and great in Kant was united with a clear eye for his one-sidednesses. And to this was added that, on closer discussion of Kant's doctrine, problems arose that could not yet present themselves with full clarity to the founder himself, and on whose treatment it depended what direction philosophy was to take after the great turning point it had lived through. Three men in particular are to be named here, Reinhold and Maimon as regards the theoretical problems, Schiller as regards the aesthetic and ethical problems.

a. Karl Leonhard Reinhold was born in Vienna in 1758 and as quite a young man became a novice in the Jesuit order; when this order was abolished, to the great sorrow of himself and his comrades (which is described in a letter printed in the biography published by his son), he entered a Barnabite college. It was, however, zeal for study more than religious interest that guided him in this step. He became teacher of philosophy at the college and was strongly influenced by the rationalistic way of thinking that prevailed under Joseph II. At last the opposition between his way of thinking and his position as a monastic brother became so strong that at twenty-five years of age he fled from the monastery. He now became a contributor to Wieland's journal "Merkur" and wrote there, among other things, his *Briefe über die Kantische Philosophie* (1786), a popular *[118]* account through which Kant's doctrine first became properly known in wider circles.

When Reinhold had become professor at Jena in 1787, the university there became a chief seat of the philosophical movement. Almost all the philosophical tendencies that were developed in succession during the attempts to continue Kant's inquiry arose in Jena. (Besides Reinhold himself: Fichte, Schelling, Hegel, Fries, Herbart). Reinhold's chief work is his *Versuch einer neuen Theorie des menschlichen Vorstellungsvermögens* (Prag und Jena 1789), which aimed at deriving the philosophy set up by Kant from a single principle. Reinhold did not, however, remain at the standpoint he had asserted in this work, and to which his significance in the history of philosophy is bound. His great receptivity to the thoughts of others and his disinterested love of truth made him change his standpoint repeatedly when he thought he found significant progress in the solution of the problems in other philosophical writers of the time (not always the most important). Already in 1793 he left Jena and went to Kiel, where he worked until his death in 1823. For Danish literature he is of interest through his friendship with Baggesen (of whom he has given a delightful characterization in his letters to Erhard).

Kant had not, according to Reinhold, gone back to the last presuppositions — or rather to the last presupposition. It was Reinhold's conviction that philosophy could become true science only when it derived all its doctrines from a single principle. Kant had, after all, not only distinguished sharply between sensation and understanding and between theoretical knowledge and practical faith, but he had also applied a double method: partly the analysis of the forms of knowledge (the subjective deduction), which was essentially a psychological investigation, partly the construction of the presuppositions of experience (the objective deduction). These different methods and those distinctions stood as an imperfection for Reinhold: he demanded a single starting point, and from this starting point a single way. By this demand the critical philosophy was led onto the way of speculation. The demand rests on a confusion of the search for unity, which
[119] shows itself in all inquiry, with an absolute principle: the unity that is sought always presupposes a given multiplicity that is to be ordered, and is not in and for itself sufficient to express the ideal of knowledge. It conflicts, moreover, with the nature of our thinking, since every inference consists in the combination of several premises. But in their enthusiasm the thinkers now set themselves an infinite ideal and believed themselves in possession of powers to reach it. — The principle that Reinhold lays at the foundation he calls *the principle of consciousness.* All knowledge is representation (while not all representation is knowledge), and the principle of consciousness now states that every representation stands in relation partly to a subject, partly to an object, so that it is partly to be distinguished from them, partly to be combined with them. Consciousness

itself consists in such a referring of the representation to the subject and the object. Through a series of investigations Reinhold now wishes to show in detail that all the forms and fundamental propositions set up by Kant come forth through a closer carrying through of this first principle, as more special ways in which representations are set in relation (Beziehung) to subject and object. The element in the representation by which it is referred to the subject is its form; the element by which it is referred to the object is its matter. And since matter cannot be produced by form, the object not by the subject, the assumption of a thing in itself becomes necessary: there must be something in the representation that is not produced by consciousness itself, but by a cause different from it. But since Reinhold finds the essence of consciousness in a referring (beziehende) activity, the representation of the thing in itself must necessarily become a self-contradictory representation, since it is, after all, to concern something that cannot be referred to the subject or apprehended by the subject's forming activity. Reinhold accordingly declares: "Of the reality of the thing in itself only a contradictory representation, a mere illusion, is possible." And he declares the connection between the mode of working peculiar to our activity and the thing in itself to belong to "the questions that no one will raise who understands their meaning and knows the limits of what can be represented." (Versuch. p. 456. 460). But by this the problem of the thing in itself acquired a far sharper and more provoking form than in Kant. Reinhold, after all, stresses the unity and activity in consciousness far more strongly than Kant, and the *[120]* enigma of the eternal barrier, the self-contradiction in there having to be something that at once *had to* be thought and *could not* be thought, came forward all the more strongly thereby. — It is especially against Reinhold's doctrine that G. E. Schulze directed his "Aenesidemus." It came forward more and more clearly that either one must make greater concessions to Hume than Kant had made, or one must boldly continue on the way Reinhold had entered upon, and make the unity and activity in our knowledge absolute, whereby the thing in itself would then have to fall away entirely. Reinhold's wavering in his later period came from his being unable to decide for either of the two possibilities.

b. With superior acuteness another of Kant's disciples, Salomon Maimon, took part in the discussion of the questions the master's doctrine led people to raise. He was able at once to keep himself free from the dogmatizing into which so many Kantians fell, and from the titanic speculation to which the demand for the unity of the principle led over. The course of his life had contributed to nourishing his critical independence. He was a Lithuanian Jew of a poor family (born 1754), was trained as a rabbi and early distinguished himself by acuteness and learning. But his thirst for knowledge led him

beyond the Talmud and Jewish theology. In his autobiography, which is a work of the highest interest in psychological and cultural-historical respects, he describes under what wretched conditions he had to work his way forward to scientific knowledge. It was a Hebrew astronomy that he found by chance that taught him that there was another science than the Talmudic. Laboriously he taught himself the elements of the German language, and in order to be able to share in the knowledge of the West he fled from his family (he had been married already at eleven years of age!) and begged his way to Berlin, where he soon awakened Mendelssohn's attention and interest. He now studied Wolff, Spinoza and Locke. Since the imperfect aids that had stood at his disposal had compelled him to supply what was lacking by his own acuteness, he had an ability to
[121] find out what was essential in a line of thought and to draw the consequences from it, which let him stand with independent criticism over against all the views he came into contact with. Thus he stood also over against Kant's writings. From the remarks he wrote down during his study of "Kritik der reinen Vernunft" there came forth a work *(Versuch über die Transscendentalphilosophie.* Berlin 1790), which gave Kant, who saw it in manuscript, occasion to declare that not only had none of his opponents understood him as well as Maimon, but that in general only few possessed in the same degree as he the acuteness necessary for such investigations. In a series of writings, of which only *Versuch einer neuen Logik oder Theorie des Denkens* (Berlin 1794) shall be named here, Maimon developed his epistemological views further. Theoretical thinking was everything for him, — both the highest perfection and the highest happiness. In this he followed the older thinkers of his people (Maimonides, Spinoza). After a long, wandering and insecure existence he lived his last years with a Prussian landowner who took an interest in him. He died in the year 1800.

In Maimon we already find the criticism of Kant's philosophy that will probably become the definitive one. — Since, he says, we can discover the forms of knowledge only by way of experience, we can prove neither their necessity nor their completeness. We can obtain no guarantee that we have brought forward *all* the forms. And the very distinction between matter and form can only be relative: there can be no pure matter, and just as little a pure form, — only approximations to them. The mere sensation is an idea in Kant's sense: in spite of all possible degrees of approximation to a sensation in which form plays no part, this extreme is nevertheless never reached. There are here two limiting points or ideas: on the one side the single element in consciousness, on the other side the perfect synthesis. Our knowledge moves in the interval. It is a grasping, not an encompassing. Every time we try to form a total representation, an all-encompassing

thought, it turns out to be a representation of something limited, something finite. — Against Reinhold Maimon maintains the impossibility of setting up a single highest *[122]* principle. The "principle of consciousness" set up by Reinhold expresses only what is common to all fundamental propositions, and the special fundamental propositions cannot be derived from it. Consciousness in general is an entirely indeterminate concept. — From Kant Maimon separates himself above all by denying the fact on which Kant builds his "objective deduction": that we have *experience* (as something different from subjective representing). Even if Kant is right that our thinking has at its disposal a system of categories, he is nevertheless wrong in holding that we can *in fact* apply these categories to the given. For the given presents only relations of time, not *necessary* transitions. Only in the purely mathematical domain do we have an objectively valid knowledge of reason, a "real thinking." In empirical judgments the consciousness of the subject, as a matter of mere fact, precedes the consciousness of the predicate; the latter cannot be derived from the former. It is otherwise in mathematics, and only there. Hume, then, is not refuted by Kant, and he cannot be refuted at all. When Hume said that the concept of cause was taken from experience, he understood by experience not, as Kant did, a necessary order in the sequence of phenomena, but merely constant observation, which produces custom and expectation in us. The causal principle expresses a postulate, an idea that we seek to apply to phenomena, but whose applicability can be shown only approximately. While Maimon thus takes an intermediate position between Hume and Kant in his conception of the causal principle, he takes up Kant's interesting intimations of the connection between the concept of cause and the concept of continuity. There asserts itself, he shows, a striving in our knowledge to reduce the opposition between phenomena to the least possible. To seek out the cause of a phenomenon is the same as to seek out its continuous coming into being, or to fill out the gaps between our observations. What does one understand in natural science by the word cause but a phenomenon's development and resolution, such that continuity is shown between it and the preceding [and following]? Only by this continuity do observations become experiences. The word experience therefore designates for Maimon not a relation of *[123]* necessity, but only a factual relation of continuity between the observed phenomena. — As regards the much-debated problem of the thing in itself, it falls away entirely for Maimon in a certain sense. The thing in itself was, after all, to be the cause of the "matter" in our knowledge; but there is no pure matter according to Maimon; only in infinite approximation do we come to matter, as to the determination of $\sqrt{2}$: hence the point in our line of thought at which the question of a thing in itself can be posed lies

infinitely far away! That something is *given* to us, or that our cognitive faculty *is affected,* means only that something arises in our consciousness that cannot be derived a priori according to the general laws of consciousness and of knowledge. The given is that in the representation of which we are conscious of no activity. — No faculty whatever contains the ground of its special applications. But a thing, an object, can be given only in and for consciousness. An object that is not an object for anyone is an impossible thought, to be compared with an imaginary number (whereas the given, the point where we are absolutely passive, was to be compared with an irrational number). And even if one will ask about the cause of the given, this question cannot be answered. For it cannot be decided whether the given springs from something different from us or from our own cognitive faculty: we know only what appears to our consciousness, and the cognitive faculty, apart from its functions, is just as much a "thing in itself" as the unknown cause of the "matter" of knowledge assumed by Kant and Reinhold. Both the concept of an absolute subject and the concept of an absolute object are only ideas, limiting concepts, which cannot be drawn into our knowledge[29]). It is the task of our understanding to *grasp* phenomena, i.e. to understand them through their mutual connection, through the *law* of their relations. But the imagination, which in turn is spurred on by our craving to reach the highest, constantly seeks to extend our representations beyond all the limits of experience and wants *to encompass in one image* all the multiplicity that is subject to law. Everything Kant calls ideas, concepts of an unconditioned in various forms, Maimon derives not from reason but from the imagination. The drive to totality in our knowledge
[124] has its ground in the drive toward the highest perfection; therefore it is especially in the ethical and religious domain that the drive to form ideas shows itself. But as soon as we represent such totalities *as objects,* we limit them and thus come to inhibit that drive. For every object must be found within the continuous connection of knowledge. The striving for totality is a perfection; but the representation of totality as an object is an imperfection. It is not this representation, but that striving, that has religious and ethical value. —

Maimon calls himself a skeptic, not only over against the dogmatic philosophy, but also over against the critical philosophy. It was the dogmatism of the Kantians that provoked him to use this designation. In reality he has intimated an epistemological view that was well fitted to lead beyond the difficulties that had arisen during the discussion of Kant's works. Both as regards the relation between thinking and experience (Kant's point of dispute with Hume) and as regards the relation between knowledge and faith (Kant's point of dispute with Jacobi) he had indicated points of view that lent themselves

to scientific carrying out. Maimon is the one among Kant's disciples who has best continued his work (even if the old master liked his attempts to continue his work as little as those of Reinhold and other "hypercritical friends")[30]). But the Romantic ferment of the time hindered a continued development of the critical philosophy in the spirit by which Maimon had been animated. The Romantic need of unity, the drive to revel in the Absolute, to unite thinking with artistic vision, was too strong for Maimon's critical and skeptical deliberateness to be able to hold interest. Its speculative philosophy and its historians see Maimon's significance in his being an intermediate link between Kant and the pure speculation initiated by Fichte. But he has an independent and lasting significance and stands for something far more than one of the "superseded standpoints" over which triumphant speculation strode onward. —

c. There is a certain analogy in course of life and in thinking between Reinhold, Maimon and Schiller, in that all three in their youth had to free themselves by flight from crushing circumstances in order to be able to reach the free development of the *[125]* spirit they craved, and in that their task as thinkers was the same under different forms: standing on the ground of the critical philosophy, to bring about unity and harmony where Kant had put distinctions and oppositions in the first line. Friedrich Schiller (1759—1805), the great poet, whose biography it does not fall to the history of philosophy to give, had throughout his whole life an intense philosophical interest, even if by his own statement he felt his powers properly only in art, and even if he finally used his philosophical theory to prove that the artist alone is the true man, a proof that for him consistently marked the transition from his philosophizing period to the magnificent poetic production of the last decade of his life. It was not first the study of Kant that awakened his philosophical interest. While he studied at the Karlsschule in Stuttgart, his scientific interest was occupied by medicine and philosophy, and speeches and treatises from this time bear witness that thoughts were already taking root in him then which came to influence his later views. Of special interest is *Versuch über den Zusammenhang der thierischen Natur des Menschen mit seiner geistigen* (1780), where it is developed that the pleasure and pain bound up with the organic functions are not only of significance for self-preservation, but also contribute to awakening and attuning or inhibiting and depressing the mental powers, just as, conversely, mental pleasure and pain react upon the organic condition. The ethical ideas Schiller declares at this time betray the influence of Rousseau and of the English moral philosophers of the eighteenth century. After his flight from Stuttgart (the Duke — offended by "Die Räuber" — had forbidden him to publish anything but purely medical writings) he immersed himself in the study of

ancient poetic works, and from this time on Greek humanity in art and life stood for him as the great ideal that had shown itself in history in a happy period, but had been pushed aside during the later development of culture. In a peculiar way this ideal combined with the criticism of the state of culture to which Rousseau had led him. In place of the
[126] vague notion of an unbound life of nature there now came the view of the harmonious unfolding of life determined from within. In art he now saw a power of life that without constraint, through involuntary harmonizing of drives and powers, raises human life above the animal stage, just as in its symbols it leads to grasping the truth long before the abstract thinker can reach it. And not only the beginning of the higher life of the spirit, but also its summit, is marked by art. Therefore Schiller says to the artists (in the poem *Die Künstler.* 1789):

Mit euch, des Frühlings erster Pflanze,

Begann die seelenbildende Natur;

Mit euch, dem freud'gen Erntekranze,

Schlisst die vollendende Natur. In art he saw man's proper distinguishing mark:

Die Kunst, o Mensch, hast du allein.

When Schiller wrote this poem, he already knew Kant's essays on the philosophy of history, which had brought the problem of culture so strongly forward. He first came to a more intimate immersion in Kant's writings in Jena, when he had become Reinhold's colleague. The great thoughts found a well-prepared soil. Schiller, who had had to struggle with outer obstacles as well as with conflicting drives and doubting thoughts, knew how to value the master's struggle to find the ground of truth and his maintaining of the ideal's elevation above the immediate promptings of nature. But his artist's nature and his enthusiasm for the Greeks did not allow him to give up the demand for the harmonious unfolding of human nature as the highest. Thereby the problem was posed to him whose solution was his contribution to the development of philosophy. It is characteristic that from his artistic standpoint he makes the same demand that Hamann made from his religious standpoint: that "one must not separate what nature has joined." "Even in the purest utterances of the divine part of human nature," it says in the treatise *Ueber Anmuth und Würde* (1793), "the sensuous part must not be pushed aside, and the triumph of the one must not be grounded on the suppression of the other. Man's moral
[127] way of thinking is first secure when it springs forth from the whole human being as the united effect of both principles, that is, when it has become nature to him." Therefore Schiller demands beauty and grace of moral action. Grace consists for him in a harmony in the involuntary movements that accompany a freely willed action, and that themselves

point to a moral disposition in the mind. There must be nothing forced, painful or crude in the action if it is to be perfect. This demand, says Schiller, does not, to be sure, accord in its wording with Kant's ethics. But — he goes on — the great thinker became the Draco of his time, because it was not receptive to a Solon, and this led him to overlook that the children of the house had not, after all, deserved that he should care only for the slaves. Is the disinterested feeling in the noblest breast to be made suspect because impure inclinations often usurp the name of virtue? And is perfected humanity not found in the beautiful soul, which, led by immediate feeling, performs the most painful duties and makes the most heroic sacrifices with the ease of instinct? In the beautiful soul it is not the single actions, but the whole character, that is moral. The beautiful soul has no other merit than that it exists. Sensibility and reason, duty and inclination are in harmony, and gracefulness is the outer expression of this. Yet Schiller restricts the demand for grace to the actions that lie *within* the limits of human nature. Tasks may be set for man that lead him out to the very limit of human capacity, where duty and inclination do not harmonize: here strained struggle is the only possibility, and here the demand is to act with dignity, to express the disharmony in such a way that the nobler power is nevertheless seen in its superiority. — A certain wavering asserts itself in Schiller's line of thought. Harmony had to be the highest for him — but he doubts that it can be reached in all cases. Which stands highest, grace or dignity: to this question he does not venture to give any definite answer. This wavering is connected with his not having come (any more than Kant) to the significance of differences of individuality in ethics. He speaks only of the limits of human nature, not of the limits of the single individuality, and he confidently assumes that the limits lie in the same place for all *[128]* individualities. He says expressly: "Our moral judgment brings every individual under the measure of the species, and no other barriers than the barriers of humanity itself are credited to man."

In close connection with the problem of culture Schiller brings forward his ideas in *Briefe über die ästhetische Erziehung des Menschen* (1795). It is the fundamental thought of "Die Künstler" that is developed here in detail. The course of culture has led to the dissolution of the Greek harmony between spirit and nature, reflection and imagination, universality and individuality. It is the division of labor that causes the misfortunes under which we suffer. The conflict of the powers leads to the progress of culture and is in so far a gain for the species; but the single individuals become one-sided and crippled. State and church, laws and manners, labor and enjoyment have become separated, and each individual is only a fragment of a man. This can be remedied only by the completion

of culture. A pure command of reason cannot decide the matter, for drives are overcome only by drives. It is, moreover, a sign of imperfect culture when the moral character can be maintained only by sacrificing the natural one, and when unity and connection can triumph only by violating fullness and multiplicity. Only through aesthetic education is the solution of the problem possible. Take hold of men in their idle hours, in their pleasures, surround them with great spiritual forms, with symbols of the good, and let seeming overcome reality without constraint, let art conquer nature. What matters is to unite the involuntary fullness of natural life with the independence and freedom of moral life, surrender to changing states with the unity of personality, the material drive with the formal drive. In play this task is solved. The powers there work according to their natural laws and yet without being bound to material demands; they work from within, but without self-constraint. Producing and receiving go together as one. Man is raised above the influence of sensuous nature, and yet sensuous nature works according to its own law. He determines himself with freedom; he is self-active; he is guided by
[129] the formal drive; — and yet sensibility and matter are not violated. The condition is that there be a certain superabundance of power that can be applied to the free play of the functions. Where this has been reached lies the proper beginning of human life. Man is only properly man when he plays. In the free play of the powers human nature expresses itself many-sidedly and as a totality, with the possibility of entering on special directions, a possibility that is, however, not realized as long as the aesthetic state lasts. "All other exercises," says Schiller in the 22nd letter, "give the mind a special skill, but thereby also set a limit to it; only the aesthetic exercise leads to the unlimited. Every other state into which we can come points us back to a preceding one and needs a following one for its dissolution; only the aesthetic state is a whole in itself, since it unites in itself all the conditions of its arising and its continuance. Here alone do we feel ourselves as if torn out of time, and our humanity expresses itself with a purity and wholeness (Integrität), as though it had as yet suffered no impairment through the action of outer powers." — The *aesthetic state* that Schiller describes is, then, one in which all human powers work freely and harmoniously, without being set in motion by outer needs, and without any single one being predominant. In it he saw the completion of culture, not merely a means of subduing crudeness and softening disharmony. —

There has been dispute over whether Schiller in "Briefe über ästhetische Erziehung" stands on the same standpoint as in "Anmuth und Würde," or whether he does not in the former set the aesthetic state as the highest, while in the latter, as regards the limiting cases, he still reserved a place for dignity beside grace. As we have seen, he

really already proclaims grace as the highest in "Anmuth und Würde"; but still more decidedly it comes forward as the fundamental thought in "the aesthetic education" that the highest is not reached where there is not harmony and totality[31]). There is a ring of the Renaissance in Schiller's line of thought. Next after Kant's pointing to the state of law built on the principle of freedom, no answer so weighty has been given to the problem of culture posed by Rousseau as in Schiller's setting up of the free and total play of the powers as the highest — not merely as an end, but also as a means that can lead *[130]* onward to perfect culture. He is, however, clear that there is a long way to this goal, and he thought, at any rate as regards himself, that he worked best for it through art. "Follow my advice," he wrote to a friend (letter to Erhard, May 26, 1794), "and for the present let poor, unworthy and immature humanity look after itself. Remain in the bright and quiet world of ideas and leave it to time to carry the ideas into practical life!" He himself, after concluding his philosophical period (1789—95), worked at the presentation of ideas in artistic form. The speculative philosophy that now began to develop he looked upon with aversion, while even in the last year of his life he recalled with enthusiasm the great circle of thought into which Kant had led him: "The speculative philosophy," he writes to Wilhelm von Humboldt (April 2, 1805), "has driven me away with its hollow formulas, if it ever otherwise occupied me. In this barren domain I have found no living spring and no nourishment for myself. But the deep fundamental thoughts of the philosophy of the ideal are an eternal treasure, and for their sake alone one must count oneself happy to have lived in this time." — A more beautiful epilogue could not be imagined for the philosophy whose founding and first fortunes we have been occupied with here.

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Eighth Book

[131]

The Philosophy of Romanticism.

A. The Philosophy of Romanticism as an Idealist Doctrine of Development

The spirit in which Kant's philosophy was carried further in Germany itself was already determined by the objections of his first opponents and by his first disciples' attempts at correction. People missed in him a comprehensive view, totality; the living coherence of the spirit was violated by his analysis and distinguishing, and they wanted ideas that could hold the whole content of the life of the spirit — preferably all at once! They appealed partly to religious faith, partly to artistic intuiting and activity, as organs that let the fullness of life come into its own quite otherwise than critical philosophizing did. And even on purely scientific grounds, it was held, the critical philosophy was insufficient so long as all principles were not derived from a single, absolute principle. What had kept Kant from such a formal completion of his philosophy was the assumption that knowledge always presupposes something besides itself, something that can never be drawn in but always withdraws: Ding an sich. But this assumption led Kant's philosophy into a self-contradiction and was bound up, besides, with his untenable distinction between the matter and the form of knowledge. Let this assumption fall, then — and nothing will prevent the consistent carrying through of philosophy on the basis of Kant's own deepest thought of the nature of spirit as synthesis! What in Kant had been the ultimate presupposition, the fundamental hypothesis, was now drawn forth as the *[132]* principle of beginning, from which a great systematic construction was to be undertaken. It was a thought that filled young thinkers with enthusiasm and made them overlook that it was in reality the philosophers' stone they were seeking. All externality, isolation

and dividedness in the spiritual life would indeed vanish when unity was thus shown in everything, when all forms of life came to stand as steps and phases through which one and the same infinite life expressed itself! Not only the conscious life of the individual human being, but also the life of the species in history and — by way of analogy — the life of nature would thereby be seen in a wholly new light! And not merely a single side of the human life of the spirit would find its explanation; if the guiding principle was grasped deeply enough, it must lead beyond the distinction between science, religion and art and abolish all mental disharmonies.

Such an ideal of knowledge may rightly be called romantic. It stands exalted and remote, awakens longing and enthusiasm, and works more by this exaltedness than by the prospect of clear and sober realization. That the ideal of knowledge could be formed in this way for a whole generation is not explained by the preceding development of philosophy alone, but is understood only through motives that lay in the whole spirit, need and tendency of the age. A purely philosophical motive lay, to be sure, in the conviction awakened by Kant of the originality and activity of the nature of spirit. It was then in itself a justified thought experiment to try whether this originality and activity might not be absolute, without limits and without conditions. But the audacity with which this thought experiment was undertaken is understood only through other currents of the age. The close of the eighteenth century was a time of strong agitation. — The French Revolution stirred up men's minds. It overturned what had hitherto stood firm, and sought to raise a whole social edifice anew by way of construction. Even where its practical ideals and consequences were not accepted, it nevertheless gave a sense for absolute standpoints and for the ruthless carrying through of principles once laid down.
[133] It is more than an analogy when the speculative movement in Germany has so often been compared with the contemporaneous revolutionary movement in France[32]). It is a movement of feeling that finds vent in two different ways, according to the different inner and outer conditions. — At the same time Germany had its poetic golden age. In the poetry of GOETHE and SCHILLER the artistic power had brought forth immortal images as the expression of the craving and striving of life. Should the human spirit be more powerless in the world of thought than in that of art? Should thinking not be able, in its own domain, to reach a unity and harmony analogous to that which art yields in its own? Indeed, should poetry and science not be one in their innermost nature, expressions of one and the same creative power? The most characteristic representative of romantic poetry, NOVALIS, sought in his unfinished work "Heinrich von Ofterdingen" to show poetry as the innermost in everything. He regretted that poetry has a special

name and that the poets form a guild of their own. Poetry is, according to him, nothing other than the peculiar mode of action of the human spirit: does not every human being, in every moment of his life, compose and strive? The distinction between poetry and philosophy is, according to Novalis, superficial and harmful; philosophy is only the theory of poetry, and is to show us that poetry is one and all. The task was to find "the secret word" that solves all riddles in the inner and the outer world. (Novalis' Schriften. 4. Aufl. I. p. 201. 286 ff. — II. p. 240 ff.). The difference between the romantic philosopher and the romantic poet is here this: while the poet regards philosophy as the interpreter of the poetry of life that is found in everything, the philosopher will rather conceive poetry as an intuitive, figurative form of the thought that stirs in everything. — As the third in the alliance, religion came forward. There was stirring in the age — as the appearance of Hamann, Herder and Jacobi already bore witness — a need of a deeper religiousness than the flat and dry rationalism. Nor could the sentimental religion of feeling satisfy. If it is now the business of thinking to show us the unity in everything, can the philosophical striving then be essentially different from the religious need, which also, after all, seeks out beyond the oppositions of life and their unrest? And must not a reconciliation of religion and science be reached, if only one conceives the nature *[134]* of reason thoroughly and consistently enough? Not by taking reason captive, but by going back to the innermost nature of the spirit, which lies at the ground of religion as well as of science, will the philosophy of Romanticism solve the religious problem. Thus in his work *Ueber die Religion, Reden an die Gebildeten unter ihren Verächtern* (1799) SCHLEIERMACHER sought to demonstrate the necessity of religion for the depth and harmony of the life of the spirit. — In the development of natural science at the close of the century, discoveries and ideas came forward that could lead to laying greater weight on the unity in nature than the immediately preceding period had done. The discovery of galvanism, the founding of the new chemistry (after Lavoisier) and of comparative anatomy must in many ways give a new view of nature. And Goethe, whose poetic works exercised so great an influence on those spiritually stirred in the age, was led precisely by his studies in natural science to a presentiment of an inward unity in everything, a presentiment that his studies of Spinoza had nourished in him and that had already exercised influence on Herder. The Goethe-Herderian view of nature furnished the foundation and the direction for the speculation of Romanticism. In particular, Goethe's theory of colors with its polemic against the mechanical explanation of nature, and his "Metamorphose der Pflanzen" with its fundamental thought of an inner unfolding through the various organic forms, gained a significant and fateful influence on the

speculative philosophy of nature. And in his "Faust" Goethe had given passionate vent to the boundless craving for a knowledge that penetrates into the innermost of existence. Schelling was the first who rightly valued Goethe's great poem, which then existed only as a fragment: "To the contradiction that springs from unsatisfied desire for knowledge of things the poet has attached his images in the poem that is the Germans' own poem, and he has opened an ever fresh source of enthusiasm, which by itself alone was able to rejuvenate the science of this age and to spread over it the breath of a new life. Whoever would penetrate into the sanctuary of nature, let him nourish himself on these tones from a higher world and drink in already in his youth the power that goes out in dense
[135] rays of light from this work and moves the world's innermost!" (SCHELLING: *Ueber das akademische Studium.* Vorlesung XI). — Finally, Spinoza's system, which had recently been drawn forth from the darkness of neglect and was acknowledged by friends and foes alike as the ideal of philosophy, exercised an overwhelming power both by its thoughts and by its strict form of unity. — There was thus no lack of motives that could lead men to try to raise bold systematic edifices.

The method became predominantly deductive and constructive. Confirmation by experience was held to be unnecessary — in the confidence that "the inner type of all things" revealed itself in thought just as well as in the whole of existence. (Cf. Schelling, loc. cit.). The material used in the systematizing had been procured by the labor of the preceding age. What the philosophy of Romanticism really wants is to create a systematic form for everything that had come forth in Kant's epistemology and ethics, in Lessing's and Schiller's aesthetic ideas, in Goethe's poetry, in Herder's historical thoughts and in Hamann's fermenting religious spirit. Systems as such are not productive, but presuppose the productive powers. If we now look at the romantic systems in philosophy, we must say that just as little as romantic poetry succeeded in obscuring the great figures of Goethe and Schiller, which on the contrary shine forth again after the romantic period with undiminished splendor, just so little did romantic philosophy succeed in giving the spiritual content that that series of great spiritual heroes had brought forth such a new form that these heroes should now have significance only as forerunners. Whoever seeks real nourishment will often prefer to go back behind the systematizers to the original sources. The speculative period in Germany at the beginning of our century can, moreover, by no means measure itself with the period of the great systems in the seventeenth century in originality and productive power, in rigor and in significance for the treatment of problems. This does not exclude that the eminent men who stand as representatives of the philosophy of Romanticism, in their

eager striving to form the content of the life of the spirit into a coherent whole, have thrown light on several problems and uttered significant ideas whose time will not be over with that of the systems. In literary respects most of these thinkers have harmed *[136]* their continued influence by the use of a scholastic terminology, a kind of jargon, that makes their writings hard of access to those who have not themselves learned to think in this language.

1. Johann Gottlieb Fichte

a. Biography and Characterization

It is a chief peculiarity of German thought, right from the mysticism of the Middle Ages on, that it asserts the independence, inwardness and validity of the spiritual life and lays this assertion at the foundation of its world-view. What is inner and original in us stands as a light in which we — consciously or unconsciously — see everything in heaven and on earth. The greatest deed of the German people, the Reformation, was a struggle for free, inward conviction against ecclesiastical authority. The critical philosophy founded by Kant was a continuation of the Reformation, being a return, leading still further, to the innermost sources of all knowledge and valuation. Kant's first disciple, Joh. Gottl. Fichte, possessed not only the capacity for self-absorption, the mystical vein that leads down into the depths of the inner life, but also the unbending strength of will and the strong self-feeling without which the conviction of the eternal right and supremacy of the inner over the outer cannot be held fast and carried through. He was the son of a Saxon peasant, born at Rammenau in Upper Lusatia on 19 May 1762. His vigorous character, whose shadow sides were obstinacy and pride, he inherited from his mother, as indeed the temperament of eminent men so often comes from the mother's side. As a child he helped at the loom or tended the flock of geese. He distinguished himself by the attention with which he followed the sermons and by the sure memory with which he could reproduce them. Here his sense for spiritual things found its first nourishment, and thereby at the same time the occasion for his further education was given. A landowner from the neighborhood came too late to church one Sunday and was then referred to *[137]* the goose-boy to learn what the pastor had said. Astonished at the boy's abilities, he resolved to take charge of him and help him to study. Fichte's last years at school were a time of strong agitation: Lessing was just then waging his theological struggle, and his polemical writings made an unforgettable impression on the young student. From 1780 on he studied theology, philology and philosophy at Jena and Leipzig. His circumstances were straitened. His benefactor had died, and from home he could get little or no help. He

had a hard struggle to endure against his mother, who wanted at any price that he should be a pastor, while his own plans aimed at a free, many-sided development of his abilities. He was on the point of despairing from want when in the year 1788 he got a place as private tutor in Zürich. During his stay here his plans ripened, insofar as the need to influence his age through thought and word became vividly conscious to him. But the direction of this activity and the means by which he was to act still stood dark to him. It is characteristic of his personality that the will thus goes before the thought. All his life long he is driven as by an inner power to work in the domain of ideas, but he is always able only imperfectly to convert this inner need into the form of thought, although all his life long he works restlessly at new and more perfect expositions of his system. And his philosophy itself rests precisely on the fundamental thought, which he more than any other thinker has brought forward and impressed, that our innermost being is a willing, an acting, and that all representing and thinking is conditioned by this practical capacity, the inmost root of our ego. In a letter from his youth he declares: "I will not merely think, I will act . . . I have only one passion, only one need, only one complete feeling of myself, this: to act outward. The more I act, the happier I feel." — A great good for him, both for the clarifying of his purposes and for the polishing of the unbridled harshness of his character, was that here in Zürich he came to know the excellent woman who later became his wife, Johanna Rahn, Klopstock's niece. Their correspondence
[138] shows how well she understood all sides of his nature. She stood faithfully at his side in good and evil days. — The tutoring in Zürich did not last long, especially because Fichte found that the parents needed education no less than the children, and therefore set up a weekly censure for the parents which drew their attention to the pedagogical faults they had committed in the course of the week! — He set out wandering again, to educate himself further and to find a sphere of activity. During a stay in Leipzig (1790) he studied Kant's works for the first time, and with that his life got its decisive turning point. "I live," he writes in a letter, "in a new world since I have read Kant's 'Kritik der praktischen Vernunft'." He now found definite problems set for his thinking and definite principles for the practical influence on the age that he longed to exert. Kant himself he came to know the year after in Königsberg. To introduce himself to the old master, he worked out a writing *(Versuch einer Kritik aller Offenbarung),* which he laid before him, an application of Kant's doctrine to the philosophy of religion, which Kant himself had not yet treated. The writing won not only Kant's recognition, but since it appeared (through a misunderstanding) anonymously, it was taken by many to be a work by Kant himself. With this Fichte had broken through; the way to literary and scientific

activity now lay open to him. He now accepted his betrothed's offer to set up house with the help of her fortune. In the following years he published writings on the French Revolution and on the freedom of the press, in which he asserted the justification of the efforts for freedom against the reaction that the terrorism of the revolutionary period had begun to produce. Fichte treats practical problems before he immerses himself in the purely speculative problems[32 b]). To these last he was led over naturally when in 1794 he was called as professor to Jena (after Reinhold). Here he unfolded an energetic and brilliant activity as a teacher and developed his peculiar system, which was first set forth in *Grundlage der gesammten Wissenschaftslehre* (1794), but was later reworked again and again, partly so that the system might become more perfect, partly so that it might become more accessible. In neither of these two respects did Fichte ever satisfy himself. The best access to a knowledge of his philosophical view is given by a later *[139]* treatise, *Erste Einleitung in die Wissenschaftslehre* (1797), with whose help a preliminary characterization of Fichte's standpoint will be attempted here.

There can, according to Fichte, be two philosophical systems, each consistent in itself. The one he calls idealism, the other dogmatism. What philosophy is to explain is experience. Our experience contains thoughts of things. Now one can either (with dogmatism) derive the thought from the thing, or (with idealism) the thing from the thought. Which of the two possibilities one chooses depends on what sort of man one is. A philosophical system is not a dead piece of furniture that one can have or part with at pleasure; it springs from the innermost soul of man. The choice will depend on whether the feeling of independence and activity or the feeling of dependence and passivity has the upper hand in us. Yet — Fichte stresses — on closer investigation it will appear that idealism is already better placed than dogmatism on purely theoretical grounds. For from the mere thing, from bare being, one will never be able to derive a thought, a consciousness of the thing and of being. It is the inconsistency of dogmatism that it is a doctrine, a system of thought, and yet cannot establish the possibility of there being thoughts at all. Dogmatism (to which belong materialism, spiritualism, Spinozism) is therefore an impossible philosophy. Idealism, on the other hand, starts from thinking itself as what is original, and from it will be able to derive — not, to be sure, the things themselves, which indeed are never given either, but experience, our definite representations of things. The task of the Wissenschaftslehre is now precisely to show how our representations of things spring from the activity of thought, in that this activity by its nature comes to set itself definite limits. The Wissenschaftslehre starts from the presupposition that there can be nothing in the ego that is not the fruit of the

ego's own activity. While Kant went back from the given multiplicity in the content of consciousness to the unity that embraces it, Fichte, conversely, will start from the original activity of the ego and derive its special forms from it. If his work succeeds, he
[140] will at last have constructed actual empirical consciousness. He holds that philosophy to be false which does not agree with experience. — Of the method Fichte used in this investigation we shall speak in the special exposition of the Wissenschaftslehre. Here only the spirit and direction of Fichte's teaching were to be indicated.

His ethics and doctrine of right Fichte set forth in *Naturrecht* (1796) and *Sittenlehre* (1798), the latter of which is probably the most significant work he produced.

It lay in Fichte's energetic and vehement character as well as in the given circumstances that his activity at Jena did not unfold without frictions and serious struggles, which at last led to a catastrophe. Besides disputes with philosophical colleagues who could not follow him in his reform of the Kantian philosophy, he also came into conflict with the theologians, who took offense that on Sundays he gave lectures to the students on practical ethics, and with the students, because he sought to reform the rough life of the Burschen that had flourished from of old. This last conflict even led to Fichte's being assaulted in his house and leaving Jena for a time. But the most serious and most significant struggle was the *Atheism Controversy* (1799). For this to be understood, Fichte's standpoint in the philosophy of religion must be characterized. It proves to be closely connected with his opposition between idealism and dogmatism. It might indeed seem as if in Fichte's philosophy there could be no room at all for any principle beyond human thought, beyond the human ego; for Fichte wanted, after all, to derive everything from thought, from the ego. But already in the first exposition of the Wissenschaftslehre this frequent misunderstanding was rejected. Fichte does indeed lay down the proposition that everything that is in consciousness must be a product of the ego; but he shows at the same time that there is a very great deal in our consciousness that we are not conscious of having produced. The ego we know in experience is always limited, is placed in a system of limits, has objects (non-egos) over against it that it has not itself produced. There must then be at work in consciousness a more comprehensive principle than the finite (empirical) ego; only in this principle, which Fichte calls the pure or the infinite ego, can
[141] the ground be sought of the world of objects or limits in which our finite ego is placed. The ultimate ground of the limitation by which the finite egos and their limits come to be cannot now be found by the theoretical way; for what can limit an infinite activity? If, on the other hand, our conscience is right that the highest thing is labor and striving, then we understand why there is a finite world: for without resistance no labor, without

means no purposes. The actual world is a material for our duty. And the assurance that the resistance can progressively be overcome, that we can always do our duty and work our way forward toward full spiritual freedom as our highest goal, is grounded on the idea of an order of things that makes conscientious action possible and gives it its full effect, an order of things that does not indeed come forth in sensuous experience, but of which I myself feel myself a member every time I act from purely ideal motives. The essential thing in religion is now that man builds on such a moral world-order, on this divine that is exalted above everything perishable, and regards each of his duties as sprung from this order and as contributing to its development. For it is not an ordering that is finished once for all; it is constantly in development. And just as little is it an accidental ordering, which would in turn presuppose an ordering being outside it. That man personifies his experiences of this world-order into a particular being is harmless, if only it serves to make the thought of it living for his consciousness. But when he thinks of God as a potentate on whose favor future feelings of pleasure depend, then it is an idol he worships, and then he is rightly to be called an atheist. Indeed, every attempt to comprise the infinite principle (the pure ego, the moral world-order) in the form of the concept founders; to comprehend is to limit. As soon as something is comprehended, it ceases to be God, and every so-called concept of God is necessarily the concept of an idol! —

These views Fichte developed in his treatise *Ueber den Grund unseres Glaubens an eine göttliche Weltregierung* (1798) and, after charges had been raised against him, partly in sharpened form in the masterly polemical writings *Appellation an das Publikum* and *Gerichtliche Verantwortungsschrift* (1799). — The course of the controversy was this. After an anonymous pamphlet had raised an outcry on the occasion of Fichte's treatise, the Electoral Saxon government had it confiscated and complained to the government at Weimar that atheistic doctrine was being taught at the University of Jena, which was also attended by Electoral Saxon subjects. Fichte answered this with the polemical writings just named. At Weimar they would have preferred to hush the matter up. They did not want to offend the Electoral Saxon government, nor did they want to stop Fichte's activity. They wished only that Fichte would keep quiet. But that was against his nature, and he addressed a defiant letter to a member of the government at Weimar. All the privy councilors, Goethe among them, were now embittered, and Fichte was dismissed in a sharp form. It was a flagrant encroachment on the freedom of teaching that was thereby committed, since the government in its letter arrogated to itself a judgment on Fichte's doctrine. Despite repeated steps on the students' part, the government held to

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its decision, and Fichte had to leave Jena.

For a number of years he now lived as a private man in Berlin, occupied with new expositions of his system and with popular lectures. It has been thought that a difference of principle in standpoint can be found between the writings he published from this time and the Jena writings. A difference there certainly is; but it can be called one of principle only if one conceives the "Wissenschaftslehre" in its first form as an attempt to derive everything from the individual, finite, empirical ego. As we have seen, such a conception is incorrect. Already in Fichte's first writings he does not really teach a pure idealism by his own definition, for he does assume something that acts in us without our becoming conscious of it. When he calls it an ego, it is the analogy with conscious activity that he uses. Only by the extension of the ego from the empirical to the infinite ego, which becomes possible through this analogy, can Fichte hold fast to his idealism and conceive everything actual as phenomenon or symbol of thought. There is a certain mystical vein in Fichte already in his earlier writings; he felt himself different from the older mystics
[143] only in this, that they were inclined to represent the highest to themselves as something given, as something that could be attained once for all, while for Fichte it was an infinite that constantly demanded new labor. *(Sittenlehre* § 12). — In his later writings he now teaches that the thought (the infinite ego) of which everything is phenomenon is itself only phenomenon of an absolute reality, an infinite power, a life, a light that comes to our knowledge only in broken rays. He thus stresses more than before the mystical (or dogmatic) element lying beyond all conscious thought, which he had already had an eye for in his first exposition of the Wissenschaftslehre. With this there follows a change in feeling, — insofar as it should not, conversely, be the change of feeling that has called forth the altered conception. While Fichte in his first writings stresses rather the restless striving and acting, the constant activity, so that his religion too, which consists in faith in a moral world-order, is essentially "die Religion des freudigen Rechtthuns," — he later lays more weight on devotion to the life that stirs in us and wells up in us without our willing and reason being able to produce it. Characteristic in this respect is the writing *Anweisung zum seeligen Leben* (1806), while Fichte's general philosophical and psychological views, as they took shape in his later years, are most accessible in the posthumous writing *Die Thatsachen des Bevusstseins.* — With the changes named there also follows naturally a change of front in Fichte's polemic. Hitherto he had attacked orthodoxy and dogmatism. Now he attacked the idea-forsaken "Enlightenment," the empty and negative tendency of the understanding that was content with its meager concepts but lacked all deeper spiritual sense. And this polemic gains a peculiar interest

from its giving him occasion to set forth his conception of the historical development of the spiritual life. This is in the remarkable writing *Die Grundzüge des gegenwärtigen Zeitalters* (1806). He distinguishes here five periods in the development. First the *instinct of reason* rules, in that a group of individuals arises (the normal people) among whom life is ordered in noble and perfect forms without either science or art being necessary. When the normal people, in consequence of natural events, are scattered among the animal peoples of nature, the struggle begins between culture and rawness. And the second *[144]* period arises when what the instinct of reason leads to is set up by eminent individuals as law and norm for the others. This is the period of *authority*. Against authority rises the impulse toward freedom and the need to comprehend. It overthrows authority — but with it also the reason that had involuntarily given authority its content and value. Only individual willfulness and sensuality remain in the period of *empty freedom*. With contempt one turns from the darkness of the past, secure in the concepts won, which one makes the measure of everything without having any notion of what it takes to comprehend anything. The character of this time is "Auf- und Ausklärung." It is plainly enough his own contemporaries that Fichte means to characterize hereby. As the mark of the fourth period he gives *the science of reason,* true and many-sided understanding, with full consciousness of the greatness of the task. But mere knowledge is not enough; the goal is reached only when there forms an *art of reason,* an art that aims at ordering all human relations according to the science of reason. — As Fichte himself points out, the third period stands in decided opposition to all the others. It has a negative stamp, while positive powers (instinct and authority, or science and art) rule in all the others. It is a period of transition — and Fichte thus really assumes three stages (like Kant and Schiller), a positive, a negative and a new positive one, so that the new positive powers work their way forward through the dissolution. Dissatisfied with modern times, especially with the critical and rationalist tendency of the eighteenth century, he awaits a new age with a deeper need and better sources for the understanding of life.

The great events of the time soon gave Fichte occasion to speak still more urgent words to his age and his people. After the battle of Jena Berlin was occupied by the French, and Fichte went to Königsberg, where for a time he taught at the university. When this city too was occupied by the enemy, he stayed for a time in Copenhagen, where he came to stand close to the brothers Ørsted. After the conclusion of the peace he returned to Berlin, which for several years still had a French garrison. It was Prussia's *[145]* time of rebirth, and Fichte belonged to the circle of excellent men who by thought, word and deed worked for national restoration. In the winter of 1807—8 he delivered in Berlin

his famous *Reden an die deutsche Nation.* The struggle with arms is over for the present; now what matters is a struggle in the domain of characters and ideas. Of its outer life the nation is no longer master. But the education of the new generation is still in its power. This it must use to bring forth a generation that has the capacity to set itself great purposes and to sacrifice itself for them. What matters is to awaken the whole people, not merely a single class, and what matters first and foremost is to make an independent development of character possible. The starting point must be sought in an original need and drive that is found in every human being; for education cannot bring anything into a human being for which no germ whatever is found in him. Every human being needs esteem, to be esteemed by others at least, if he does not yet feel the need to esteem himself. At its height this need gives birth to the capacity for the free and independent formation of exemplars for one's action. Only such freely and independently formed exemplars can set a living feeling in motion; what is taken in from outside does not act so. The German people in particular will be receptive to such an education by virtue of its sense and zeal for spiritual freedom and inner independence in faith and thought. It is, after all, the people from which Luther's Reformation, Kant's philosophy and Pestalozzi's pedagogy have come forth. — It was Pestalozzi's fundamental thoughts that Fichte here carried through on a larger scale. During his stay in Switzerland he had come to know the great educator[33]).

When the University of Berlin was founded (1810), Fichte became professor there. At the outbreak of the War of Liberation he had plans (as already in 1806) of going along as an orator. It did not happen. But he nevertheless became a victim of the war, being infected by a hospital fever that his wife had contracted in nursing the wounded, and he
[146] died on 27 January 1814.

b. The Wissenschaftslehre

It is the "Wissenschaftslehre" in its original form that constitutes Fichte's most important contribution to the philosophical discussion, and to it we shall here keep. —

The task is to find the first fundamental proposition of all knowledge. If we turn our attention to our ordinary consciousness, we find that we can think both of ourselves (the ego) and of something that is not ourselves (the non-ego). But even when we think of the non-ego, this happens only by virtue of a mental activity, an activity of the ego. The non-ego exists for us only through an activity of the ego, through a "positing." We feel ourselves perhaps bound or compelled to represent to ourselves definite things to which we ascribe reality: but this binding or compelling presupposes something that is bound

or compelled. Limitation presupposes something that is limited. The presupposition of all knowledge is the free, infinite activity of mind which in each single representation appears determinate or bound, but which is not bound absolutely to any single one. In our immediate consciousness this original activity does not appear; there we find only its single products. Of our willing and acting we never become immediately conscious. Limits and results we notice, not that which is limited and which brings forth the results. The first fundamental proposition therefore does not really concern a fact, but an act (not a "Thatsache," but a "Thathandlung"), as Descartes indicated by his Cogito, ergo sum and Kant by his synthesis. To discover this original activity (the pure ego) reflection and abstraction are needed, the capacity to hold fast the thought of an undivided and unlimited activity for which the opposition between subject and object, action and result, does not hold. A kind of higher intuition is required, an intellectual intuition, to grasp what does not appear to immediate apprehension and yet cannot be formed into a concept either, since every concept presupposes opposition. Mental energy and independence are needed to discover this innermost active kernel of our being. Most people would sooner be able to regard themselves as a piece of lava in the moon than as pure mental activity. The assumption of a peculiar soul distinct from the body, as a *[147]* "being" or a "substance" by itself, such as spiritualist metaphysics sets it up, is therefore wholly rejected by the "Wissenschaftslehre." (In a later writing Fichte declares this in a very plain way: "The existence of a soul is absolutely denied. This whole concept is rejected as a bad fiction. And this is not something inessential, but it is a very essential criterion of our system. With the presupposition of such a soul one can neither get into this system nor remain in it." *Die Thatsachen des Bewusstseins.* Stuttgart und Tübingen 1817. p. 105 f.).[34])

But that first fundamental proposition of a mental activity as the source of everything in consciousness does not explain that something other than the ego itself is posited in consciousness. A second fundamental proposition must therefore be laid down, which cannot be derived from the first. If the first is formulated thus: *The ego posits itself,* the second comes to run: *The ego posits a non-ego.* This second fundamental proposition is found, like the first, by reflecting on what is given in consciousness. Now there arises a necessity of connecting the two propositions with each other, of bringing about a synthesis by the connection of thesis and antithesis, and this becomes possible by laying down as the third fundamental proposition: *The ego posits a limited ego in opposition to a limited non-ego.* By this synthesis we get back to immediate consciousness, which can be explained neither by assuming that the ego is posited absolutely, nor by assuming

that the non-ego is posited absolutely, but only by the assumption of a mutual limitation, a reciprocal action of ego and non-ego. — The presupposition of this limitation and reciprocal action is always the unity of consciousness. When the ego and the non-ego are placed in opposition to each other, this very opposing presupposes that it is the same consciousness that posits both members of the opposition.

The method Fichte employs may be called the antithetical. First a proposition is laid down that brings out an essential moment of the matter; then a proposition that brings out an opposite moment (which is also contained in the matter, but cannot be derived
[148] from the first); and both are then sought to be united. The necessity of seeking a union of the opposed propositions lies in the first fundamental proposition itself, that everything in consciousness is due to an indivisible mental activity; for from this it follows that the relation of opposition cannot be fundamental. However limited our consciousness (the finite, empirical ego) may be, the pure ego nevertheless acts in it as the ultimate source not only of all its acting, but also of all its suffering.

We do not follow Fichte in his continued use of this method, by which every synthesis always appears again as thesis, in order by combination with a new antithesis to lead to a new synthesis. The method has its great value, but in the particulars became arbitrary and unfruitful in Fichte's hands, because its application demands a more detailed analysis of experience, here, that is, a fuller psychology, than was at his disposal. What he wants to achieve is a derivation of the forms and fundamental propositions that in Kant were still laid down purely schematically. The principle of identity he finds contained in the first fundamental proposition: the proposition A = A holds originally of the ego in its pure, self-identical, constant activity. The form of time comes forth when various acts of the ego (positing of ego, positing of non-ego) are to be possible in such a way that they are dependent on one another in a definite order. The form of space comes forth when various determinations in the non-ego are to be posited each by itself and yet without interruption and without being dependent on one another in a definite order of time. The causal principle is contained in the third fundamental proposition, which demands mutual limitation, i.e. reciprocal action of ego and non-ego. When the various determinations of the non-ego are also conceived as mutually dependent on one another (as in natural science), the concept of action has been involuntarily transferred from the ego, whose nature it expresses, to the non-ego. — Only the general forms of empirical consciousness, not the special, individual content of experience, can and will Fichte deduce[35]).

In the whole world-picture that forms itself for ordinary consciousness, we have then,

according to Fichte, the product of an involuntary activity unconscious to ourselves. As we project the sense qualities, so we project time, space, logical identity, causality. From this it does not follow that our world-picture is an illusion. The concept of illusion *[149]* can be applied only as the opposite of reality, but this opposition falls away here, or rather: the true reality is in the activity which, without our noticing it, brings forth the world-picture in our consciousness, and this activity is a *necessary, lawful* activity. Only philosophical reflection, which must be well distinguished from ordinary, practical consciousness, discovers that deep-lying power at work in all finite egos, and thereby understands at the same time how the same world-picture forms itself for all egos. This is Fichte's solution of the problem of knowledge. —

Both in order to understand Fichte's method and his result, one must hold firmly that the second fundamental proposition cannot be derived from the first. In this his antithetical method differs from Hegel's so-called dialectical method. And from this there follows too a limitation of purely theoretical philosophy, on which Fichte himself lays the very greatest weight, and which is in the highest degree characteristic of his system. For the question then arises: why does the ego (the pure ego) produce or posit a non-ego within itself? why is the pure, identical activity broken? why does the straight line of the original activity become a curve? Fichte answers that to this question no answer can be given theoretically. To explain that breaking (Anstoss) or bending one would have to assume a force besides the absolute force, and that is a self-contradiction. Only from our moral consciousness, which sets striving and labor, struggle to attain great purposes, as the highest, do we win an understanding. For labor presupposes limits, struggle presupposes resistance. That a world of non-egos, of objects, is set before us has, then, the ethical significance that labor and struggle become possible. Nature is the material for our duty. Object means a standing-against, and a standing-against means resistance. Without a system of limits no ethics. Our highest goal is the freedom and independence that is won by combating all hindrances to our ego. — The practical philosophy thus contains the key to the theoretical.

Both the practical and the theoretical philosophy presuppose the first fundamental *[150]* proposition. The weight that Fichte, in his setting up of this fundamental proposition, lays on the original activity as the first element in our being becomes of special significance for ethics. There is an original need of action for action's own sake, apart from outer occasions and tasks. Both in conscience and in natural impulse this need makes itself known. Kant's categorical imperative, which he himself could not explain, becomes intelligible through this original need of activity at any price, and no less is the unruliness

of the sensuous drives understood by it. The single things or objects in nature stand for us from the first only as means or limits for the satisfaction of our lower or higher drives. Our conception of the world is from the first practical. "The whole system of our representations," says Fichte in the "Wissenschaftslehre" (2. Aufl. p. 288), "depends on our need and our will."

c. Ethics

When the original craving and activity is attached to definite objects, according to the peculiar character of each single individual, it appears as natural impulse and aims at enjoyment. Through enjoyment I become dependent on the objects. But by virtue of the infinite in that craving and activity, which is the expression of my independence over against everything given, all objects, there can develop in me a consciousness, a reflection, that shows me other possibilities besides what is immediately given. On this my freedom rests. It is one and the same primal impulse that expresses itself both in natural impulse and in the reflection and impulse toward freedom that can release me from every sensuous relation of dependence. Natural impulse and the impulse toward freedom need not stand in conflict with each other either: for the satisfaction of the single natural want can take place in such a way that it becomes precisely a means to greater freedom and independence. Thereby the ethical law is given: *each single action shall lie in a series that leads me to complete spiritual freedom.* Thereby the infinite ego is realized in the world of experience. The limits posited are overcome. It is an infinite goal, but a
[151] constant approximation is possible, in that every purpose attained is made a new starting point. — When there is harmony between the two forms in which the primal impulse expresses itself (natural impulse and the impulse toward freedom), we have a feeling of self-esteem, a feeling of pleasure that is very different from sensuous enjoyment and would preferably not be called pleasure at all. In the opposite case we have a feeling of self-contempt. The capacity for having such feelings is called conscience. Only the action that springs from conscience is moral. To act on authority is unconscionable. The first commandment is: *You shall act according to the conviction of your duty!* And this conviction is won by comparing the action contemplated not merely with the conception I have of it at the moment, but with every conception I could think of getting of it, thus by trying whether I could own the action in all eternity. — Moral evil springs from inertia, dependence, — inertia toward reflection, toward raising oneself above what is momentarily given, reluctance to go beyond the present state, a tendency to stay in the given track. Inertia then leads in turn to cowardice and falsehood, slavery being

preferred to toil, and seeming being used as a cover for what one will not defend in open struggle. — But how is it possible that the impulse toward freedom can be awakened where it does not stir? Man cannot, after all, release his own power! The answer to this lies in the doctrine of an instinct of reason that Fichte later developed in "Grundzüge des gegenwärtigen Zeitalters," but which is already found in the "Sittenlehre." In some individuals the original spiritual need can be so strong that in a way inexplicable to themselves and others it raises them above what is sensuously given. They have a kind of "Genie zur Tugend." They then act as awakening exemplars for the others. At this point lies, for Fichte, the explanation of the rise of positive religions. Quite naturally, such individuals endowed with deeper spiritual power, and likewise those who are influenced by their personality, come to regard what has developed in them and goes out from them as a miracle. Out of their empirical ego it certainly cannot be explained. The assumption of such a miracle has its significance when it serves as a means of awakening the slumbering attention and energy. It is thus of great and decisive significance that *[152]* man lives together with other human beings. Only among human beings is he a human being. And in reality the many individuals have only a single goal: the realization of the idea of the ego. My personality is not, for me, the highest, ethically regarded. But it is the only means by which *I* can work for the highest. My goal is reached also when others act ethically. When each follows his conviction, all will work to develop the highest and most inward independence and to realize reason in a society of free beings. What in religious language is called the communion of saints means the presentation of the pure ego in the whole of rational beings. Over against this infinite goal the individual is a mere means and instrument and of vanishing significance. What matters for the individual is to annihilate his individuality — not in mystical absorption, but in active doing for the eternal goal. — In his later writings Fichte expresses himself still more severely about this vanishing significance of the individual personality, in strange contradiction with the weight he otherwise lays on individuality[36]). At last, then, individuality becomes for him what ought not to be, a limitation that is to be abolished, a negation that is taken back. The pure ego crushes the empirical one. It is a nemesis that here overtakes Fichte. He has sought an absolute foundation for his ethics by putting it in inner connection with his speculation, but precisely thereby comes at last to abolish his ethics. Yet on his way, before he plunges into this speculative abyss, he has found ethical thoughts of lasting value. —

Of Fichte's special ethics his conception of church and state is to be mentioned. — The church he conceives as an association of individuals for the awakening and strengthening

of ethical conviction. The common foundation for this is symbols, figurative forms in which the highest thoughts can influence all, the uneducated as well as the educated. Only through such symbols does general and mutual spiritual influence become possible. For as regards the highest thoughts themselves in their pure, scientific form, it will
[153] be impossible to bring about general understanding and agreement. The symbol is generally accessible through its figurative form and at the same time indeterminate in its interpretation. On this its value rests. Even if the ethical teacher of the people does not ascribe literal truth to the symbol, he will nevertheless be able to use it when he does so with enthusiasm and with a living wish to act upon the spiritual life of others[37]). But the spirit of Protestantism demands that all symbols be developed into clearer and more perfect forms. Every symbol (the representations of a personal God and of a personal immortality too, not merely the representations of the positive religions) is only a makeshift symbol. What furnishes the measure for the symbols is the degree to which they are able to make the thought of a higher, ethical world-order plain and living. — Just as every ecclesiastical symbol is a makeshift symbol, so every given state is only a makeshift state. The state has to do only with the outer being of man, and according to Fichte a sharp distinction is to be made between the doctrine of right and ethics. The doctrine of right rests on the principle that whoever (no matter for what reason) will live together with others must restrict his freedom out of regard for the freedom of others. The relation of right rests on the mutual recognition of this. The state's only task is to maintain the relation of right, and where necessary it compels the individual to recognize the freedom of others (as regards body, property and self-preservation). But Fichte does not find the state imperfect merely so long as this outer security is not brought about. He maintains (and in this he plainly goes beyond the concept of a mere state of law or police state) that the state can demand that property be recognized by all only when it strives to bring it about that all come to possess property. Everyone ought to be able to live by his labor. There ought to be neither needy nor idlers in the state. Charity is only a wretched and ambiguous expedient. Already in his "Naturrecht" and his "Sittenlehre" he hints at thoughts in the direction of what has since been called socialism. In *Geschlossener Handelsstaat* (1800) he wants world trade handed over to the state, so that the inner economic development within each single nation can proceed more independently and naturally. He does not fear that the intercourse between the
[154] nations will suffer by this; it is science, not trade, that carries this intercourse, insofar as it has real value. Fichte is, as has rightly been said (see GUSTAV SCHMOLLER's study of Fichte in "Jahrbücher for Nationalökonomie und Statistik." 1865), the first German social

writer. His political theory, despite his sharp distinction between right and morality, nevertheless proves in the end to be borne by ethical points of view. In his expositions of the doctrine of right and of the state from his later years he lays weight on this, that the state, which is to educate from compulsion and through compulsion to freedom, cannot reach this goal without working to procure for all individuals property, leisure and the means of higher culture. —

2. Friedrich Wilhelm Joseph Schelling

a. The Period of the Philosophy of Nature

Schelling is the classic philosopher of Romanticism. Fichte still stands close to the critical philosophy and distinguishes carefully between consciousness and its content; he can be called a Romantic on account of the craving for the boundless that in him replaces Kant's thought of continual progress. But in Schelling there is the genuinely Romantic need to revel in a content that is taken in more by intuiting and symbolism than by testing thought. Thus he revels first in nature and art, later in religion. With this need to revel it was no wonder that, although he began as Fichte's disciple and collaborator, he soon found Fichte's ego too meager. His intellectual tendency went toward great, symbolic intuitions in which the oppositions of things could at once bring each other out and be seen in their connection. Beholding was for him the highest, and when in his speculation he laid weight on oppositions, it was in order to give his thought-images fullness and relief. Fichte, on the other hand, brought out the oppositions because they were the condition of work, struggle and development. Fichte's philosophy was really ethics through and through; Schelling did not concern himself with ethics at all.

Schelling was born at Leonberg in Würtemberg on January 27, 1775. Already at the *[155]* age of fifteen he came as a student to Tübingen, and here, like the rest of the student youth, he was strongly taken up with the French Revolution; he was supposed to be the author of a German translation of the Marseillaise, which led Duke Karl, known also from the story of Schiller's youth, to proceed harshly against the young student. His first studies were theological; he was interested in mythology and the historical interpretation of the Bible. Soon, however, he threw himself into philosophical studies, immersing himself in Kant, Fichte and Spinoza. In various treatises from the years 1794 and 95 he developed Fichte's "Wissenschaftslehre" further. A stay in Leipzig as tutor to two young noblemen gave him the opportunity for studies in natural science, and here he wrote his first work in the philosophy of nature: *Ideen zur einer Philosophien der Natur* (1797). From the first he conceived his philosophy of nature as a supplement to the "Wissenschaftslehre"; later

it led him to set himself in opposition to it, and a very sharp polemical relation thereby arose between the two former friends. From 1798 he worked as professor at Jena, whither the influence of Goethe, Schiller and Fichte had called him. Here in Jena the Romantic school was really constituted. The brothers Schlegel with their gifted wives, Novalis, Tieck, Steffens and others met here with Schelling, and in this "republic of despots" there was an overflowing of poetic, religious, philosophical and scientific ideas, now in conflict, now in accord. Friedrich Schlegel writes to Schleiermacher: "Hier gehts ziemlich bunt und störend durcheinander — Religion und Holberg, Galvanismus und Poesie". And Dorothea Schlegel gives a similar description: "Es geht so recht kunterbunt her mit Witz und Philosophie und Kunstgesprächen und Herunterreissen"[38]). For the time being it was at this period the enthusiasm for nature and art that was strongest in Schelling; the religious enthusiasm of Novalis and Tieck even awakened in him, by way of opposition, an antireligious enthusiasm, which found expression in an "Epicurean confession of faith" in poetic form, which was to have been printed in the Schlegels' "Athenäum" but was held back on Goethe's advice. Schelling later had part of it printed in his journal for
[156] speculative physics; it is now available in full in "Aus Schellings Leben". Leipsig 1869. I. p. 282 ff. The following lines characterize Schelling's philosophy of nature — also in the sense that it looks better in verse than in prose.

Darum, ist eine Religion die rechte,
Müsst sie in Stein und Moosgeflechte,
In Blumen, Metallen und allen Dingen
So zu Luft und Licht sich dringen,
In allen Höhen und Tiefen
Sich offenbaren in Hieroglyfen . . . .
Hinauf zu des Gedankens Jugendkraft,
Wodurch Natur verjüngt sich wiederschafft,
Ist *eine* Kraft, ein Pulsschlag nur, *ein* Leben
Ein Wechselspiel von Hemmen und von Streben.

Besides "Ideen", there must be named as expositions of the "philosophy of nature" *Erster Entwurf eines Systems der Naturphilosophie* (1799) and *Einleitung zu seinem Entwurf eines Systems der Naturphilosophie, oder: Ueber den Begriff der spekulativen Physik* (1799). In *System des transscendentalen Idealismus* (1800) he presents "the philosophy of nature in its connection with the Wissenschaftslehre" and proclaims artistic intuition as the only organ that makes possible the apprehension of spirit and nature, subject and object, in their inner unity, the only standpoint where the oppositions of life, particularly

the opposition between theory and practice, fall away. In the year 1801 he began an abstract systematic exposition of his doctrine *(Darstellung meines Systems* in vol. 2 of "Zeitschrift für spekulative Physik"), but did not complete it. In *Methode des akademischen Studiums* (1803), on the other hand, we have in a freer form an encyclopedic exposition of Schelling's philosophical ideas as they took shape at the close of his first period. A new turn, and soon after a remarkable halt in his production, set in with his departure from Jena (1803). But before we follow him into the second great period of his thought, that of the philosophy of religion, we must give an exposition and characterization of the "philosophy of nature". —

It was from the first in the name of idealism itself that Schelling demanded a more *[157]* positive recognition of nature than Fichte's system, for which nature was only a limit or a means, could afford. The very secrets of the spiritual world can be solved only when we learn to understand nature, so that it is no longer an alien power to us. In Fichte nature is only object; but the object can be understood only if in its essence it is of like kind with the subject. Nature is comprehensible only if it bears the stamp of spirit. And if the forces that work in spirit can already be found in nature, then it can be understood how spirit develops out of nature. Nature then stands as the Odyssey of spirit, as its striving to get back to itself and its inwardness from the form of externality in which it is placed in nature. The way from nature to spirit must be just as possible as the way from spirit to nature (in "Kritik der reinen Vernunft" and in "Wissenschaftslehre").

Modern natural science had sought to make everything in nature into motion, to explain everything by the interaction of material parts. If this explanation exhausts the essence of nature, then either everything ideal must be denied, or else one must assume that it comes into nature from outside. But to explain what is purposive in nature by the influence of a divine understanding, that is, says Schelling (Ideen. 2. Ausg. p. 63), not to philosophize but to indulge in pious considerations. And besides, the chief philosophical problem remains: namely *how we know* the whole system of causes and effects that makes up nature, — how this system *comes to be for us.* The natural scientist lives in nature as an immediately given reality; but the philosopher of nature asks how it can be given to us. (Cf. Ideen. 2. Ausg. p. 4; 27 f.). The same problem appears as the question of the origin of the faculty of sense (sensibility) in nature. How can the organism become its own object? (Erster Entwurf. p. 174)[39]).

No one will be able to deny that a real problem lies here, and that it is clearly posed, both from the standpoint of idealist philosophy and from that of real science. There must — if the spiritual side of existence is to be comprehensible — be other forces at

[158] work and other properties to be found in nature than those that mechanical natural science knows and has expressed in its laws. Schelling's solution, now, is that the same duality of a force going out into the infinite and a limiting force, which Fichte has demonstrated in consciousness, must be found in the whole of nature. Since the life of consciousness rests on opposition (duplicity), the whole of nature, out of which the life of consciousness develops, must also present opposed forces, only in lower potencies, or, to recall Leibniz (see: System des transscendentalen Idealismus. p. 190), all forces in nature are representing forces of different degree. Matter is slumbering spirit, spirit in equilibrium, and spirit is matter in development. But Schelling does not stop at such a general principle or postulate. He believes he can demonstrate in detail the steps by which nature rises to spirit. He starts from the spiritual forces and thinks of them as at work in nature in lower "potencies". In a treatise *Ueber den wahren Begriff der Naturphilosophie* (Sämmtl. Werke. I, 4. p. 84 f.) he has clearly stated his method: "To see the objective in its first coming-to-be is possible only by *depotentiating* the object of all philosophizing, which in the higher potency is = ego, and then constructing from the beginning with this object reduced to the first potency." One is thus to infer from the higher degrees to the lower. Schelling does not investigate whether this is possible (since one and the same object can of course have other forms of state in its lower degrees than in its higher). His "construction" is therefore in reality also a poetic-symbolic interpretation, by which the forces and forms of nature are conceived as stepwise approximations to the life of consciousness. Like all metaphysical idealism he builds on analogy, but he has such great confidence in it that on its basis he believes he can build up a "speculative physics", which is not merely to show us the results of natural science in a new light, but even in the end to take the place of natural science. Nature is understood only "from within", by being seen as visible spirit. By adopting a purely external and empirical point of view Boyle and Newton have corrupted physics, just as Bacon corrupted philosophy! Mathematical astronomy does not, after all, teach us to know the real essence of the
[159] planetary motions! The innermost essence of natural phenomena is understood, namely, according to Schelling, not by explaining the one with the help of the other, but by their all springing from a common ground of which they are, each in its own way, symbolic expressions; it is in this that the unity of nature consists. Thus the three dimensions of space express the inner relations within the Absolute, namely unity, opposition and the combination of both; the circular motion of the heavenly bodies is a symbol of the All, which spreads out in all directions but always returns to itself; in the causal relation there appears, in an externally different way, what in the Absolute is in infinite unity,

etc. (Schelling's dialogue *Bruno* 1802). The empirical natural scientist does not see the symbolic significance of natural phenomena; he explains the external symbol and thinks he has thereby explained everything. *(Methode des akad. Studiums.* Vorlesung XI: Ueber die Naturwissenschaft im Allgemeinen).

The Romantic element in Schelling's philosophy of nature lies not in the problem as he originally poses it to himself, but in the boldness with which he puts his symbolic interpretation of nature in place of science's laborious attempt to explain nature by the mutual connection of the individual phenomena. He overlooks that the unity of nature can be established scientifically only through this connection, which he scorns. On this point lies the decisive difference between Schelling's philosophy of nature and such attempts of thought as those of Spinoza and Leibniz, which precisely took the real connection as their foundation. Not even as confirmation by experience does Schelling think he needs the empirical: since speculative construction presents "the inner type of all things, which by virtue of the common origin must be one", it is sufficient to itself and can penetrate into regions from which experience is excluded by insurmountable limits!

Let us look a little in detail at Schelling's attempt to demonstrate spirit in nature. — The various phenomena and forces are to be shown to be ascending potencies. And at the same time on every step, within every potency, there is to be shown a relation of opposition (or of duplicity, or of polarity), as a lower degree of the relation of opposition that appears in consciousness between subject and object (ego and non-ego). The concepts *potency* and *polarity* are the fundamental concepts of Schelling's philosophy of *[160]* nature. With their help he constructs the scheme in which the various natural phenomena get their place. He arranges them so that there is (in every case symbolically) an ascent to higher steps (potencies) and on each step a new opposition (polarity) in analogy with the oppositions on the lower steps. The qualitative differences here give way to differences in degree and in form. It is interesting to see that Schelling's Romantic attempt to give a speculative natural philosophy, just as much as exact natural science, resolves all qualitative differences into quantitative ones. Schelling has himself drawn attention to this likeness between the "philosophy of nature" and atomism, and on account of it wants to call his doctrine dynamic atomism. The difference is that while mechanical atomism explains everything by a relation between material parts, the "philosophy of nature" explains everything by a relation between forces. But in the clarity and rigor with which the general thought is applied, Schelling stands far behind atomism. The general scheme he has constructed for himself with the help of the concepts of potency and polarity is filled in in a highly arbitrary manner, often by a mere play with words,

so that the name "philosophy of nature" has rightly acquired an ill sound in scientific ears. Yet it is of historical interest to get a survey of Schelling's symbolism of nature.

In the absolute principle *(Urgrunden),* which lies at the ground of everything, there is absolute unity between subject and object. At no point in existence is this unity abolished; but now the one, now the other pole can have the preponderance: on this quantitatively different relation between the poles rests the difference between the various potencies. In *nature* the objective, in *spirit* the subjective pole has the preponderance. If the Absolute can be represented by A = B (where A means subject, B object), nature can be represented by A = B+, spirit by A+ = B (where the plus sign above denotes the predominant pole). Within nature three potencies are distinguished. The first appears in the elementary, attracting and repelling forces that hold the structure of the world together; *gravity* is the
[161] characteristic expression of this first potency. In *light,* as well as in magnetism, electricity and chemical process — the second potency —, there shows itself a combination of what has been isolated by the separating forces, which is already like an intimation of what the process of knowledge is in the spiritual domain. In *organic life* — the third potency — there appears a world in miniature, a system of processes that are both ends and means for one another mutually, and in whose capacity for sensation (sensibility) spirit in nature at last breaks through its limits. The potencies of *spirit* appear in the three activities: *cognition, action* and *art.* In artistic intuition Schelling found the highest form of the life of the spirit when he wrote his "System des transscendentalen Idealismus". "Art is at once philosophy's only true and eternal organ and its document, which ever anew attests what philosophy is unable to present in the external, namely the unconscious in acting and producing and its original identity with the conscious. Art is the highest for the philosopher, because it as it were opens to him the holy of holies, where what is separated in nature and history, and what in life and action as well as in thinking is eternally split apart, burns as in one flame, in eternal, original union. The intuition which the philosopher forms of nature in an artificial way is for art the original and natural one." (System. p. 475). — It is consistent that a philosopher who builds his system with the help of poetic symbolism ends by proclaiming art the highest. For him, as for Novalis, everything is really poetry; the process of nature is an unconscious poetry that breaks forth with consciousness in and for man.

It was not Schelling's meaning that nature was literally the prehistory of spirit. His philosophy of nature differs from the modern doctrine of evolution not only in disdaining the mechanical connection but also — which for that matter is bound up with it — in denying a *factual* transition from one "potency" to the other. The one step does not arise

out of the other, but the producing activity itself of nature or of the Absolute, which appears in experience not itself but only in its products, comes to full unfolding only through a series of forms. Each one of these forms springs from the infinite producing *[162]* itself, not from other forms. There is in the Absolute an eternal unity of subject and object, which does not need to develop through the potencies of nature and spirit, which are only its different reflections in the medium of experience. This Schellingian standpoint, which was in essentials also Goethe's, was dominant among many eminent natural scientists, who saw the kinship of the forms of nature but could not reconcile themselves to the thought of a real descent. For them the kinship consisted in the interior of nature, in the creative imagination of the Deity, or however they expressed it, not between the actual, existing groups of natural beings themselves. — In essentials such a conception was also held by Schelling's friend and disciple, *Henrik Steffens;* although he lays more stress on the individual and the historical, he did not assume a *real* development in nature by which the later forms should have come into being from the earlier. He expressly declares himself against "a gradual transition from one form to the other", although he assumes "one formative force, which goes connecting through all shapes". *(Indledning til filosofiske Forelæsninger.* Copenhagen 1803. p. 81 f.). — On the other hand *Oken,* and the Danish investigators *Olufsen* and *F. C. Sibbern* assumed a *real* development, a development of the forms in time.

b. The Problem of the Philosophy of Religion

There came about, as already mentioned, a break and soon after a halt at a certain point of Schelling's development. He had come forward as a producer very early; his first philosophical work is from his nineteenth year. The period of Romanticism offers not a few examples of such early-ripe and early-arrested productive power. In Schelling the break and the halt were connected partly with the fact that the path he had struck out on could not lead him further than he had come once he had given utterance to his bold ideas, partly with the fact that a new problem appeared before him, and partly, finally, also with purely personal circumstances. *[163]*

The new problem that appeared before him was the religious one. As we have seen, Novalis's religious enthusiasm called forth an opposite mood in Schelling. In the Jena period he was too taken up with nature and art for the more practical problems to arise in him. In a letter from the beginning of the year 1806 he says: "During my retirement in Jena I was less reminded of life, but constantly and powerfully of nature, to which almost all my inquiry was confined. Since then I have learned to see that religion, public faith,

the life of the state is the point around which everything moves." (Aus Schellings Leben. II. p. 78). It was one of his disciples, *Eschenmayer,* who brought him to occupy himself with the religious problem. Eschenmayer maintained in an interesting little work *(Die Philosophie in ihrem Uebergang zur Nichtphilosophie,* Erlangen 1803) that religion was a sphere that lay higher than philosophy. For even if philosophy knows how to overcome all oppositions and to demonstrate the Absolute as the highest unity, there remains the question how then the individual oppositions, the various potencies, arise, and that is answered only in faith in a creating God. Philosophy is unable to ground the existence of a finite world split into oppositions, a world of differences and potencies, different from the absolute unity in which speculation sees the highest. Schelling acknowledged that differentiation contained a great problem, but he held fast that this problem must be capable of solution within philosophy. Philosophy must be extended; but there was no reason to conceive religion and philosophy as two wholly different things. He contends that it is indeed impossible to derive difference from identity, plurality from unity: but precisely for that reason he has, after all, maintained in his philosophy that opposition (difference, plurality) is a principle just as original as unity. A *living* unity is one that has the oppositions not outside itself but within itself, however strong a tension there may be between them. In the work *Philosophie und Religion* (1804), which he wrote after having moved from Jena to Würzburg, he acknowledges that in nature and history forces are stirring that point to a tension so strong that it cannot be derived from the
[164] very idea of the absolute unity, a tension that is to be brought into harmony through the various steps of nature and history. Earlier he had called nature the Odyssey of spirit; now he uses this expression of history as well. The untamed and irrational in nature and history bears witness to a falling away from the eternal idea, to a disharmony that has come in and is to be reconciled. But he maintains against Eschenmayer, as a few years later in a polemic against Fichte, that such opposition and strife is necessary in order that life and harmony may come to be. Without opposition no life. These thoughts were nourished in Schelling, especially after he had settled in Munich, by the study of the old mystics, first and foremost Jacob Böhme, who at this time in general enjoyed a renaissance; St. Martin in France and Franz Baader in Germany renewed his line of thought. In the most significant and brilliant work of Schelling's period of the philosophy of religion: *Philosophische Untersuchungen über das Wesen der menschlichen Freiheit und die damit zusammenhängenden Gegenstände* (1809), Schelling seeks to show that only if one assumes, as he has done in his philosophy, an original opposition in the Absolute, in the essence of the Deity, a dark, irrational ground that is clarified and harmonized by the

very development of the life of the divine being, only then is one justified in conceiving God as a personal being. His philosophy of nature, with its potencies ascending step by step, he here gives a religious significance. Personality does not develop except through opposition to and on the basis of a natural foundation, through strife with resisting forces. Finite beings have this opposition, this basis, *outside themselves.* If the infinite being is to be personality, it must have the relation of opposition *within itself:* there must be in God something that in itself is not God but can become so. Theism, says Schelling in a polemic against Jacobi, can be built up on the ground of naturalism; but from a pure theism, from rationalism's "enlightened God" or from the natureless God of ordinary theology, it is impossible to ground the existence of nature. *Given* oppositions can be overcome — and in this life consists; but from a pure unity given oppositions cannot be explained. In the original difference Schelling sees — still in Böhme's footsteps — the germ of evil. By following the deep need for egoism, for detached self-assertion, *[165]* man follows a will that goes back to the first ground of things. All evil consists in a striving back to the chaos out of which the order of nature has proceeded. And by this it is understood that nature always shows us a residue that is impenetrable to the understanding and cannot be brought under definite laws; it is the old chaos, which has not wholly vanished. But without chaos, without strife and dissolution, no real unity; without discord love does not become manifest. Had God been to avert evil, he would have had to abolish his own personality: in order that evil should not be, God himself would have to not be! —

Schelling's remarkable work, a mixture of profundity and fantasizing, has philosophical interest in that it concedes something irrational, that is, something impenetrable to the thinking that Schelling knew and knew how to practice, and to that extent establishes that speculative philosophy cannot reach its goal but must concede to the critical philosophy that knowledge has limits. To that extent it heralds a reaction against speculation and has a certain realistic character. It is of interest besides in that it opens a new discussion of the concept of God and investigates the conditions under which an absolute and infinite being can be conceived as personality. In accordance with Böhme's proposition: without opposition no consciousness, an opposition *within* God's essence is then postulated as the condition for his being conceivable as personality. On this thought is built *philosophical theism,* as it was later developed, on the basis of Schelling's resumption of Böhme's ideas, by eminent thinkers such as C. H. WEISSE (1801—1866) (in his work *Das philosophische Problem der Gegenwart.* 1842, and later in his *Philosophische Dogmatik.* 1855—62) and HERMANN LOTZE. — The difficulty for philosophical theism

consists in this, that when the opposition whose overcoming is to be the condition of God's personality is an inner one, lies in God's own essence, it cannot be quite in earnest. The struggle then becomes only a game, a divine jest. The personal life that we know,
[166] and of which alone we can form a concept, must struggle with *external* limits; a personal life whose inner forces are not constantly partly nourished, partly checked by external circumstances is unthinkable for us. As regards Schelling in particular, it was not even his meaning that a *successive* process should take place in the Deity; the strife is not to precede the peace; there is to be in God nothing first and nothing last, but an eternal circulation. By this the whole only becomes really incomprehensible; every attempt to think such an eternal circulation ends in dizziness. Already Arnauld, in his criticism of Descartes's doctrine of God as cause of himself (see the first volume of the present work p. 237), demonstrated the self-contradiction therein. Weisse, accordingly, also saw clearly that there could be meaning in the whole thought only if one assumed a real, historical development of God's essence in time. But that conflicts with the concept of God as the absolute being, and the difficulty for philosophical theism is thus not made any less[40]). —

We have seen the break in Schelling's philosophical producing. Soon after comes the halt. At the end of the treatise of 1809 he promises an exposition of the philosophy of spirit on the foundation he had now gained. But despite various starts and announcements, even rather noisy announcements, the fulfillment of this promise failed to come. His productivity was checked. Perhaps it was connected with the loss of his brilliant wife that a hypochondriac mood took hold of him. But in any case Hegel's appearance and victorious advance in the philosophical domain also contributed much to holding Schelling back. In quiet he worked out his philosophy of religion in lectures at Erlangen and Munich, and what came from him publicly was almost nothing but bitter sallies against the victorious rival in Berlin. When, after Hegel's death, the so-called left wing of his school, with Strauss and Feuerbach at its head, drew radical consequences from his doctrine in religious respects, Schelling let himself be called to Berlin (1841) by the Romantic King Friedrich Wilhelm IV as a kind of philosophical rescuer. Now the publication of the thoughts that had occupied him so long was expected. But his appearance in Berlin was a fiasco. His later doctrine came out in print only after
[167] his death, which occurred in 1854. It consists of a *Philosophie der Mythologie* and a *Philosophie der Offenbarung* and aims to show how that Odyssey of spirit, that progressive reconciliation and harmonizing of the restless and conflicting forces, takes place through the development of religious consciousness. One and the same process extends through

the mythologies into Christianity and through the development of Christianity to free religion. In detail this exposition is of no interest[41]); Schelling deals quite as arbitrarily with the facts of the science of religion as in his youth with the facts of natural science, and his ideas now in his later days lack the titanic boldness and Romantic mood that distinguished so many of the thoughts of his youth.

3. Georg Wilhelm Friedrich Hegel

a. Characterization and Biography

Goethe, in a conversation with Eckermann (Feb. 17, 1831), characterized the relation between the first and the second part of Faust thus: "The first part is almost entirely subjective; everything proceeds from a more entangled, more passionate individual. In the second part there is almost nothing subjective; there appears here a higher, broader, brighter, more passionless world". German philosophy in its development from Kant on through the various systems offers a parallel to the difference Goethe found between the two parts of his great work. It begins in Kant and Fichte, and is continued in part in Schelling, with subjective inquiry and craving, with a sharp distinction between ideal and reality. And it closes in Hegel's system with a reconciliation with reality through immersion in its various regions, what was there divined and sought as a distant ideal being now supposed to have been found as the very kernel of things. What Goethe wanted to express in the second part of Faust, the reconciliation with reality won through historical experience and faithful work, was the same as what Hegel wanted to express in his system in opposition to the critical philosophy and to Romanticism. And the parallel fits also in another respect, in that transitions and details are often no less difficult to *[168]* understand in Hegel's system than in the second part of Faust.

What first strikes the reader of Hegel's writings is the abstract character of his developments and the many technical expressions with which he operates. It is this that especially deters present-day readers. But for one who is able to penetrate this shell there is another trait of character that comes most to stand in the foreground, and that must stand in the foreground when Hegel's physiognomy as a thinker is to be rendered with historical justice. It is the great interest in the content of the life of the spirit in all forms and at all stages. Hegel was, despite his speculative method and his abstract form of exposition, nevertheless a realistic nature. He had a lively need to immerse himself in the objective powers of life. His thinking sought to be a re-thinking, a translation of the fullness of life into the form of thought. In order that the rich content might become the property of thought, it had to be formed into a circle of thoughts in which the individual

links hung together just as intimately as the individual moments of existence do, so that no single one of these can be touched without the whole circle trembling. With the help of his dialectical method Hegel believed he could present such a circle of thought. It has gone so tragically for him here that precisely this method, which for him was the way to reach the highest in the domain of thought, has prevented his thoughts from gaining a form and a grounding that could give them the lasting significance they deserve.

Hegel was born in Stuttgart on August 27, 1770. He studied theology in Tübingen together with Schelling, five years his junior. Besides theology he cultivated natural science and philosophy; he was especially taken up with Rousseau and Kant. — 1793—96 he lived in Bern as a private tutor. For the understanding of Hegel's religious development and of his position on one of the deepest problems of his later philosophy, the position of the individual toward the whole, the notes from the Tübingen and Swiss periods, first brought to light by ROSENKRANTZ *(Hegels Leben.* Berlin 1844) and HAYM *(Hegel und*
[169] *seine Zeit.* Berlin 1857), are of the greatest value. Hegel's consideration of the person of Christ is decidedly rationalist, in the spirit of the eighteenth century; what is new and peculiar lies in his consideration of Christianity and its position in world history. Hegel's historical consideration led him beyond the age of the Enlightenment and places him close to Lessing; but the result is nevertheless very different from Lessing's. Hegel's ideal in this period was classical antiquity, the sunlit utopia that has found its most beautiful expression in Schiller's poem "Die Götter Griechenlands". In enthusiasm for this ideal he met with his friend the poet Hölderlin, and he himself sought (in the poem "Eleusis") to give his feeling poetic expression. But the foundation of his conception was a consideration of scientific value. His study of history had taught him to call those times happy in which men gathered in unity around great common tasks, and in which the individual really was a part of the whole, did not come forward with subjective criticism toward it, but was animated by it and absorbed in it. Such a time was classical antiquity, and over against it Christianity really stood for Hegel as a phenomenon of dissolution. Like S. Kierkegaard later, Hegel lays weight on Christianity's being the concern of the individual, and precisely from this point of view he passes judgment on it. When the universal and common vanished because of the corruption of the times and the destruction of civil freedom, the individual was asserted at the expense of the whole, and through this Christianity won its power. Values vanished from life in order to be transferred to the life after death. In his consideration of religion Hegel stands here at the same stage as Feuerbach later, but his special criticism of historical Christianity as the religion of individuals, and that means here in Hegel of the weak and corrupted,

has really found its counterpart only in Nietzsche[41 b]). The thought that the universal and common is what is right, the tendency to assert the whole at the expense of the parts, which later became to so high a degree the sustaining element in Hegel's ethics and in his consideration of life and history, found its first peculiar expression in his enthusiasm for antiquity. Vanished antiquity was the ideal; it lay bright and great; at the subjective, critical, rationalist tendency of the present Hegel feels the same disgust as Fichte in his "Grundzüge des gegenwärtigen Zeitalters"; where Fichte looked with hope *[170]* toward the future, Hegel looked with longing back toward the past, but the dark view of the present is common to all the great Romantics. In his criticism of the Christianity of the present Hegel stood close to Schelling in this period; he followed the latter's production with admiration, and declared himself almost a pupil. His maturing into an independent thinker took place during his stay in Frankfurt, where in 1796—1800 he had a position as private tutor. Here his philosophy came into being. In a letter to Schelling (Nov. 2, 1800) he writes: "The ideal of my youth had at the same time to transform itself in me into a system", — an utterance that is characteristic of his philosophy. To be sure, the ideal lay in the past, in classical antiquity. But when he now proclaims that the ideal must be translated into a system, it means that he wants to break with the Romantic longing and with Romanticism's capricious relation to the reality of the present. The danger and difficulty of such a transition lies in this, that while the ideal means distance, lack of closure and an open horizon, the system is to give a coherent and conclusive understanding. In the translation it then easily happens either that the ideal accommodates itself too much to reality, or that reality is reinterpreted according to the ideal. And Hegel's system came to suffer from both defects. The reality that he shows to be the revelation of an ideal content he has himself all too often given an idealistic reinterpretation. His whole system therefore stands on the ground of Romanticism. But Hegel nevertheless had a deep sense of the real, especially of historical reality, and his enthusiasm for the classical times did not weaken his interest in the present. He made penetrating studies of contemporary conditions; thus during his stay in Switzerland he studied the financial constitution of the canton of Bern down to the smallest details.

From 1801 on Hegel worked as Docent in Jena. The University of Jena was, of course, the hearth of the philosophical movement from the time when Reinhold came forward here as champion of Kant's philosophy; here Schiller, Fichte and Schelling worked; here in these same years Herbart and Fries also studied. For a time Hegel and Schelling worked together. They published a journal in which Hegel polemicized against the "philosophy of reflection", which was unable to rise to the idea of the absolute unity of subject and *[171]*

object and to see everything in its light. This criticism applied especially to Kant, Fichte and Jacobi. But the ways parted on account of the two men's different intellectual tendencies. Schelling's strength was intuition, the quick grasp, the great intimation; but patient execution in detail did not suit him. And with the rapid acquisition of a new standpoint there followed — all through his youth — a rapid abandonment of it as well. Hegel developed slowly, but from the moment when his system stood before him in its main outlines (1800), it did not change, and his whole remaining life was absorbed in the further execution of the method he considered the right one. As he had broken with the "philosophy of reflection" on account of its dualistic doctrine of subject and reality, so he broke with Schelling on account of his formlessness in thinking, his lack of a fixed method. The break was proclaimed in the preface to *Phänomenologie des Geistes* (1807). In Schelling's philosophy, it is said here, the Absolute appears as the night in which all cats are gray, since it is depicted as the unity or identity of all differences. Schelling does indeed acknowledge the oppositions, since he of course operates with the scheme of polarity; but that is precisely a mere scheme, which does not let real development come into its own. Schelling is like a painter who has only two colors on his palette and thinks he can render everything with them. What matters is to show how the various elements of existence pass over into one another with inner necessity. Only through this intimate connection between everything in the world is it made clear that the Absolute is not dead being, resting unity, but process, life, spirit. It is the dialectical method that Hegel here demands; it really contains within it, as will be shown later, his whole system. In "Phänomenologie des Geistes" he wants to give an introduction to the system by showing how ordinary consciousness develops into speculative consciousness, rises through various stages or forms to the knowledge that truth is neither dead being (substance) nor mere subjectivity, but a living unity of both, — a knowledge that the
[172] individual consciousness will now be able to reach all the more easily since "the world-spirit has had the patience to go through all these forms in the long extension of time and to take upon itself the enormous work of world history, in which at every stage it worked out all the content that was possible there". (Phänomenologie. Vorrede). The course of development that this work, unique of its kind, depicts is then at once that of the individual and that of the species; it is at once a psychology and a history of culture that is given, — and in the exposition the two are so entangled in each other that one often does not know which of them one has before one. The historical examples that one glimpses through the abstract exposition Hegel uses only as types, which will repeat themselves again and again when human consciousness works under the same

conditions. Consciousness is always driven forward by its own inner law, since it must take on a new form or rise to a new stage whenever its theoretical and practical results do not correspond to its presuppositions. The fault always proves in the end to lie in consciousness itself, and therefore it is compelled to change its presuppositions or its standard, until — when the very standpoint of pure science is reached — the opposition between presupposition and result, concept and object, is abolished. —

The battle of Jena and its consequences brought about Hegel's emigration to South Germany. He lived to such a degree in speculative thoughts that this catastrophe did not set his patriotic feeling in motion. On the day before the battle he had even, with interest and speculative curiosity, seen "the Emperor, this world-soul" ride out to reconnoiter, and the night before the battle he had finished working out his "Phänomenologie" (see his letter to Schelling of May 1, 1807). (The preface, however, was written only some time later). He felt himself a spectator at a world-historical play and was now merely concerned to find a quiet place. For a time he worked as an editor in Bamberg. After that he became rector of the gymnasium in Nuremberg (1808—16). Here he published his chief work *Wissenschaft der Logik* (1812—16), an exposition of all the fundamental scientific concepts, developed according to the dialectical method. Naturally the wish had to arise in him to work at a university again. 1816—18 he was professor in Heidelberg, where he published his *Encyclopädie der philosophischen Wissenschaften im Grundrisse* (1817), *[173]* and for the rest of his life he worked at the University of Berlin, where he published his *Grundlinien der Philosophie des Rechts* (1821). In the Berlin period he stood at the height of his influence, partly through his magnificent systematics, which seemed to provide a wide and living frame for the whole spiritual content of the age, and which therefore soon gathered a large school around him, partly through his conservatism in the philosophy of right and of religion, which was convenient to those in power. It was the time of broken political promises and of ecclesiastical reaction in Germany, and yet Hegel taught the ideal truth of the existing order of the state and of the ruling religion! However unattractive the forms this tendency in Hegel took, — especially in his sallies against oppositionally minded thinkers who came to suffer under the displeasure of those in power, — there is nevertheless no doubt that his conservatism sprang from his idealistic conception. It was his conviction that the idea was not so powerless as to be unable to penetrate reality, and not only nature but also the state: "Of nature it is granted that philosophy is to know it as it is, that the philosophers' stone lies hidden *in nature itself,* that it is in itself rational, and that knowledge is to grasp this *actual* reason present in nature . . . But the ethical world, the state, is not to be so fortunate

that reason has gained power and might in it, asserts itself in it and dwells in it. The spiritual universe is, on the contrary, to be abandoned to chance and arbitrariness, is to be *God-forsaken!*" (Phil. des Rechts. Vorrede). By this conviction of his one understands his indignation toward those who believed they had more reason than that which had historically developed in the state. On this point Hegel stands close to the so-called historical school, which regards the legal order as a historical work raised above all individual reflection and willing. He even declared that philosophy always comes too late to teach how the world ought to be. Thought is the last product of the world-process. When reflection awakens, it is a sign that a historical form of life is concluded. "The owl of Minerva begins its flight only when dusk has fallen!" — Hegel's conservatism ought
[174] really to have led him to lay far greater weight on the method of experience than he does. For when thought always comes afterward and cannot anticipate the new, how is it then possible to find truth by the way of pure thinking, of the dialectical method?

Hegel died of cholera on November 14, 1831. A few years later his disciples published his collected works, partly from lecture manuscripts (philosophy of religion, aesthetics, history of philosophy etc.).

b. The Dialectical Method

What Hegel calls the dialectical is for him 1) a property of all our thoughts which makes the individual thought necessarily lead over to others, but also 2) a property of things which makes the individual thing necessarily belong together with others. Thereby, for him, the way by which thought reaches truth can at the same time immediately express the innermost life of existence itself. In our thinking existence, existence thinks itself in us.

1) Since every concept is limited, it leads, when consistently thought through, over to its opposite, its negation. Every concept (and especially the fundamental concepts, the categories) comes forward with the claim to express the whole of reality, the whole content of experience; but precisely this claim leads to the demonstration of its limit, thus to the negative. (It goes, then, with the individual concepts as it goes, according to the "Phenomenology of Spirit", with the various stages of human consciousness, which by their own consistency lead beyond themselves. There is a dialectic of concepts, just as there is a dialectic of standpoints). But through the negation a new position emerges: for what is negated is only a determinate, limited content, not all content whatever. The negation thus means that a new concept comes into force. And since this new concept is determined by its relation to and its memory of the earlier concept, it becomes richer

than the latter. The new concept that is formed includes the old one within it, taken up into a larger connection. The negation is then a *sublation* also in the sense that the *[175]* negated concept is lifted up into a higher unity. A unity of opposites comes into being, which comprises both the original concept and its opposite. "In this way now", says Hegel in the introduction to his Logic, "the system of concepts is to form itself and be completed by unceasing, pure progress, taking up nothing from outside". — The Hegelian system therefore unfolds through triads, each concept that is set up being negated, and thereupon the unity of opposites, the higher unity that holds both the position and the negation, being set up, — so that this in turn may be subjected to the same process. If we thus begin with the most abstract thought of all, the concept of being, this thought turns over into the thought of nothing, because pure, differenceless being without content and determinacy is the same as non-being. If both thoughts are combined, the concept of becoming is obtained; for to become is both to be and not to be, since it is the transition from one state to another. But the concept of becoming then leads in turn to the concept of properties that come to be or are dissolved, etc.

2) But this forward-driving dialectic is now the expression of the self-development of *existence*. In our thinking the pulse of existence itself beats, and with the same rhythm as everywhere. Every finite phenomenon points, by virtue of its limitation, beyond itself, is only a moment in the whole great connection. In asserting this objective dialectic Hegel has, as is seen from his application of it in detail, two experiences especially in view. First, the passing of opposites into one another precisely by virtue of the relation of opposition, as in the psychological effect of contrast, in the alternation of life and death, light and darkness (excessive light abolishes sight, as does pure darkness), in the social domain the passing of strict right into crying wrong (summum jus, summa injuria) etc. At the high point of development dissolution begins: that was the law by which Hegel wanted to construct nature and history for himself. But second — and this is the most essential point of view — he is thinking of how the results of the earlier stages determine the development at the later ones. The child's innocence is sublated by the unrest of doubt, of reflection and of passion; but through this purgatory a firm, *[176]* harmonious character develops, in which the immediacy of childhood returns in a higher form. The seed-corn must be dissolved for the plant to come into being; but in the plant what is essential in the seed-corn is preserved. What Hegel wanted to express by his doctrine of the dialectic as world-process is, then, a conviction of the conservation of force and the subsistence of values in existence. In his mystical language: the memory of the world-spirit preserves everything; perishing means merely a sublation of external

existence, not the annihilation of what is essential. Never, surely, has so magnificent an attempt been made to carry through the conservation of force and the subsistence of values in the spiritual domain. When Hegel stood at the turning point at which the ideal was translated for him into a system, it meant that he no longer doubted the subsistence of values in existence. As long as ideal and reality are still sharply distinguished, so long does it remain undecided whether values might not vanish in existence (so that the time of the "happy times" would be over once and for all), even if the elementary forces were not lost but maintained themselves under new forms. —

Important ideas thus lie concealed behind the dialectical method. But first and foremost it must be noted that speculative philosophy here for the first time set up an independent and peculiar method by which pure reason could develop according to its own inner law. Hegel's Logic is the first resolute answer to Kant's "Kritik der reinen Vernunft". Had Hegel been right, then the task that Kant regarded as insoluble would have been solved: to ground knowledge of existence by thought's own way. Since negation is a purely logical operation, which is always in our power, we would, if Hegel were right, always be able to spin the thread of thought further on our own, since from any concept whatever new positive concepts could be conjured forth. But unfortunately that is not how matters stand. It is indeed always in our power to negate, and again to negate the negation, — but the negation of the negation always leads us back to the original position, just as $-(-2)$ is 2 and not a new number. Only if by negating non A I could get (not A again, but) B, only then could the thread of thought a priori be
[177] spun on into the infinite. It is therefore also only apparently that Hegel brings forth the system of our fundamental scientific concepts by a purely dialectical way. In reality the triad (position — negation — higher unity) is only a scheme into which he more or less arbitrarily brings the content of experience into it. It is also a correct remark, which one of Hegel's English disciples has made, that "he was far more anxious to show how the higher category is *intimated* (is suggested) by the lower, than to prove that the lower is *self-contradictory* without the higher; and when this is not proved, the dialectic loses its compelling necessity."[41 c]).

Hegel's dialectical method is a continuation of Fichte's antithetical method. But Fichte saw clearly that the antithesis (as a positive proposition) could not be derived from the thesis. The various propositions that were set up against one another were for him only different attempts to express the given; each proposition means a new incision of thought, and the synthesis then combines what they have each accomplished separately. The new position is thus always won from experience. But according to Hegel

the antithesis should always be derivable from the thesis. — Like Fichte's antithetical method, Schelling's method of potentiation or depotentiation was also a forerunner of the dialectical method. Hegel here marks the conclusion of Romanticism's endeavor toward a purely idealistic construction of the world-view. And it went with him historically as it should according to his own dialectical law, in that thought, once it had properly reflected on what this thoroughgoing speculative idealism was really driving at, sought its way over to wholly different starting points and methods, and Hegel himself became at most a moment in a "higher unity".

c. The System

Hegel's system falls, in accordance with the fundamental law of the dialectic, into three parts. The beginning is made with *logic,* which for Hegel is not merely a doctrine of the forms of our thinking, but an exposition of the eternal thoughts that lie at the ground of existence. He himself recalls the idea of the Logos as the world-creating and world- *[178]*
ordering principle. Both in the world of nature and in that of spirit there reign universal thoughts and laws, which logic presents in their pure, abstract form, by showing how the one concept develops out of the other. — The second part of the system is the *philosophy of nature,* which presents the thought-content of existence not in logical abstraction but in the form of "externality", in space and time. The transition from logic to the philosophy of nature is to take place with dialectical necessity. It is one of the most difficult points for Hegel's method. In reality the transition naturally takes place through the abstraction on which logic builds being sublated by the thinker's returning to the experiences from which he has extracted the abstract concepts he operated with in the logic. — The philosophy of nature has rightly been called the partie honteuse of the Hegelian system. Hegel operates with the concepts of natural science with still greater arbitrariness than Schelling. Already Schelling had put Kepler (in his "Mysterium cosmographicum") and Goethe (in his theory of colors) above Newton; Hegel goes further along the same path. It was a Romantic attempt to displace the mechanical view of nature (an attempt in which no account was taken of the decisive change that had taken place in Kepler's views, see the first part of this work). Hegel aims, like Schelling, to arrange the concepts of natural forces and natural forms so that nature can be seen step by step to work its way up from mere externality to the inwardness of spirit. The most important steps are: mechanics, physics, organics. But just as little as in Schelling do these steps mean a real development. Hegel declares expressly (Encyclopädie § 249): "Nature is to be regarded as a system of steps, of which the one necessarily proceeds from the other . . . but not

so that the one should be produced *in a natural way* (natürlich) out of the other, but in the inner idea that constitutes the ground of nature. *Metamorphosis* belongs only to the concept as such; only its change is development . . . Such nebulous and at bottom sensuous representations as the so-called *origin* of plants and animals from water, and then the *origin* of the more developed animal organizations from the lower etc., thinking
[179] consideration must decline." This utterance is characteristic of the *idealist* doctrine of development, for which the strict mechanical derivation of one natural form from others is externality and sensuousness, a sin against the pure self-movement of the "idea". — The third part of the system is the *philosophy of spirit.* The transition takes place by the form of externality in which the idea appears in nature being sublated again. Inwardness, independence of time and space, take the place of nature's material divisibility and extension. Although here too the demonstration of the dialectical necessity is hard going, the emergence of mental life nevertheless contains no difficulty of principle for Hegel, since he of course conceives the idea, the spiritual principle, as what is innermost, what really subsists in nature. Hegel's philosophy is really philosophy of spirit from first to last, an attempt to make the science of spirit the absolute science, just as materialism is an attempt to make the science of matter the absolute science. When one of Hegel's disciples (the elder ERDMANN) has characterized his system as panlogism, on account of the pervasive endeavor to form all the content of existence into abstract categories and to conceive the movement of thought as a world-law, that is to lay too great weight on the form of Hegel's system. It is surely more correct when another of his disciples (ROSENKRANZ) has characterized the system as philosophy of spirit. The dialectical method really aims to show that everything hangs together as intimately as the thoughts in the mind, forms one great totality, as they do, and that the right expression for the essence of existence is therefore: everything is spirit, and spirit is everything. Kant's "synthesis" has here become a world-principle. — The philosophy of spirit naturally falls, like the philosophy of nature, into three parts. First *subjective spirit* is treated (in the series of steps: soul, consciousness, reason), the spiritual life in individual subjects, — thus the content of what one would now call psychology. Then comes *objective spirit,* the spiritual life as it expresses itself in historically appearing forms of society and institutions. Here belong right, morality (Moralität) and ethical life in family, civil society
[180] and state (Sittlichkeit as distinct from Moralität). The higher unity of subjective and objective spirit is *absolute spirit,* the totality of spiritual life in existence, "spirit in its congregation", where every difference between the individual and what fills and sustains the individual has fallen away. As forms of absolute spirit are named: art, religion and

(speculative) philosophy.

This highest step itself, the culmination of the dialectical movement, must nevertheless be a step on which *human* spiritual life, the only one we know, stands. But Hegel keeps this in a certain ambiguity. Although art, religion and philosophy are human endeavors, he nevertheless conceives them essentially as forms of life of the *world-spirit*. A naive admission that these forms of life are nevertheless realized, and, so far as we know, only realized, at a definite place in the universe, Hegel does make in his philosophy of nature, when he (Encyclopädie § 290 cf. 280) seeks to show that the planets are more perfect globes than the suns, and that the earth in turn is the most perfect planet! Most perfect must here, after all, mean: best suited to the development of the life of the spirit. But what then is the rest of the universe for? Old Böhme's heavy doubts (see the first part) might then arise here again! Despite all his dialectic Hegel has not been able to rise above the geocentric and anthropocentric (earthbound and man-bound) standpoint that we are, once and for all, compelled to take. No dialectic can teach us to jump over our own shadow. —

Of particular interest (both for the history of culture and philosophically) are two sections of Hegel's philosophy of spirit: his doctrine of right and the state and his doctrine of religion.

d. Philosophy of Right

Hegel's youthful ideal had been the ancient state, which stood as a divine power, enveloping and absorbing all individuals, in contrast to the modern individualistic conception, for which the state existed by a contract between individual egoisms. In his *Philosophie des Rechts* (a more detailed development of what in the system is called the objective spirit) he accordingly later also sets ethical life proper (Sittlichkeit), as it appears in the *[181]* life of the family, civil society and the state, in opposition partly to right (das abstrakte Recht) as the expression of the individual will, partly to "morality" as the expression of the subjective conscience, which in its isolation from the objective powers of society turns into sheer arbitrariness, into evil! Only within society can right and morality thrive; they are branches of a whole, not wholes by themselves. In the ethical communities the good has its subsistence as in a world animated by it. What is decisive here is not individual consciousness and arbitrariness. There is in the ethical world something that goes beyond the consciousness of the single individual. In this sense Antigone in Sophocles proclaims that no one knows whence the laws come, that they are eternal. The lives of the single individuals are regulated by ethical powers which do indeed find

their points of attachment in them, but are not dependent on them. Indeed, Hegel even declares (§ 145): "Whether the individual is, is indifferent to objective ethical life, which alone is the abiding, the power by which the life of individuals is governed". Yet the individual first finds its proper sphere in it, is free only by living in it, and the relation between the single individual and society is then so intimate that faith and trust are not the right words, since they still presuppose a certain relation of difference. It is a new and higher nature that has taken the place of the natural will in "Sitte". Where the ethical substance (the spirit of the family, of society, of the state) thus prevails, the individual's particular duties spring from it with necessity and are not difficult to find.

Of the ethical communities the state, again, is the most important. It unites in itself the essence of the family and of civil society as the higher unity of both. It is the full actuality of the ethical idea. Here spirit has realized itself in a far more perfect way than in nature, where it is still asleep. It is "God's march in the world" that the state exists; it is to be honored as an earthly deity. — Just as the single individual's duties follow immediately from his station in society, without his needing to work them out by speculation, so the constitution of the state, too, proceeds from its essence. The constitution does indeed develop in time, historically, but it cannot be made; it is not
[182] "ein Gemachtes". As empirical proof that constitutions cannot be made, Hegel points to the constitution that had been worked out by speculation and that people tried in vain to introduce during the French Revolution, and to the constitution, rational in and for itself, that Napoleon tried to force upon Spain. It is the great merit of Hegel's philosophy of right that it so decidedly stresses the connection of constitutional life with the whole historical character of the state, a character pointing beyond every single individual's wishes and thoughts. He thereby stands close to the historical school and to positivism. Auguste Comte's first independent work (Systéme de la politique positive. 1822) accordingly also met (as is seen from a letter from Comte to a friend in Berlin, 10 Dec. 1824) with a very good reception from Hegel, just as Comte felt an interest in what he heard of Hegel's lectures on the philosophy of history, although Hegel struck him as "encore trop métaphysique" and seemed to him to let "spirit" figure rather too much: Je n'aime point du tout son *esprit,* auquel il fait jouer un rôle si singulier. (See *Littré:* Auguste Comte et la philosophie positive. 2. éd. p. 157).

Hegel did not, however, wish simply to renew the ancient political theory, whose one-sidedness he admitted. The modern state is to be the organization of freedom. Civil society, the family and the single individuals are to have their particularities and interests satisfied in the state. What the state demands as duty is to be the individual's

proper right. The great end of the state should not and cannot be attained without the adherence of the narrower communities and of the individuals *(nicht ohne* das eigene Wissen und Wollen der Besonderheit, die ihr Recht behalten muss. § 260). But in his more detailed working out Hegel nevertheless makes the objective element wholly dominant. Even that modest "nicht ohne" is not respected — as we have indeed already heard that ethical life as objective substance is indifferent to the being or non-being of the single individuals! For a thinker who sees in the state, mark well not the idea or ideal of the state but the historically given state, the earthly deity, substantial reason, the ideals, criticisms and reasonings of single individuals must stand as subjective opinions and wishes, a self-importance, a "Besserwissen", that fails to recognize the deep truth in what historically exists. In accordance with the principle: "Es sollen die Wissenden *[183]* regieren, *οἱ ἄριστοι,* nicht die Ignoranz und die Eitelkeit des Besserwissens", Hegel finds the proper representation of the state in the bureaucracy: "Die Regierung ruht in der Beamtenwelt". In Germany, particularly in Prussia, he saw this principle realized. It was the most satisfying sight that met him when, in his lectures on the philosophy of history (in the twenties), he concluded with a survey of the state of Europe. Thus far had "God's march in the world" come. The speculative reactionary did not see that the foot of the Deity was already lifted for new, mighty strides, which for the time being burst every *system* — without, however, being able to obstruct the prospect of the world of *ideals.*

In the preface to his philosophy of right Hegel makes a spiteful attack on the Kantian FRIES in Jena, who in 1817 had taken part in the German students' festival at the Wartburg, so scandalous to the reactionaries, and had declared there that "among a people in which a genuine public spirit prevails, the conduct of every public affair will receive its life from below, out of the midst of the people, as living communities, held together by the bond of brotherhood, dedicate themselves to every work aimed at the education and welfare of the people". So, exclaims Hegel, the ethical world is to be handed over to the subjective contingency of opinions and arbitrary wills, and the more than thousand-year labor of reason is to be pushed aside for personal feeling! In his zeal to "convert the ideal into a system" Hegel becomes unjust to those who could not find the existing system ideal. History has proved Hegel wrong; it has shown that only through the free adherence of the people does the state gain strength. To consolidate the German Empire founded by war, Bismarck at once introduced universal suffrage.

e. The Philosophy of Religion

With respect to religion, too, Hegel's demand is this: immersion in what has developed historically, not subjective reasoning or individual feeling, — but then also the promise that thought will find its own essence again in the work of history, which is indeed
[184] the world-spirit's own work! He sets his face both against rationalism and against orthodox theology. Orthodoxy clings to the literal expressions of the dogmas and does not see that with increasing culture, reflection and enlightenment the time of immediate, unprejudiced religion is past. Rationalism, on the other hand, makes the concept of God empty and finite, places God outside the world, the infinite outside the finite, and perhaps despairs altogether of a knowledge of God. And when one (like Schleiermacher) tries to remedy this by an appeal to feeling, one does not thereby get beyond one's own subjectivity. Mere feeling cannot establish its own justification; its value depends on its content and object. Feeling cannot be the decisive thing: man has it in common with the animal — and if it depended on feeling alone, the dog would be the best Christian!

The task of the philosophy of religion is now to lead beyond this opposition between literalism on the one side and rationalism and subjective feeling on the other. Philosophy certainly cannot produce religion. But it can know the religion that is, and examine the relation between religion and the rest of man's view of life. And on closer consideration it turns out that the more philosophy attains its full development, the more need, interest and content it has in common with religion.

Speculative philosophy seeks, after all, to know the unity of existence through all oppositions, to understand it as an infinite spirit that contains and comprehends everything. But this same thing is what is expressed in the religious dogmas: they too seek, after all, — in the highest forms of religion — to express that everything in the end springs from an infinite spirit! The difference between religion and philosophy is only that in religion the content is apprehended in the form of *imagination*, so that what for the philosopher are eternal fundamental relations, the expression of an eternal truth valid for all times, is intuited as a historical event and in the form of the image. What is to be apprehended philosophically as moments in one and the same concept appears to the religious imagination as independent opposites, set over against each other in an external way, so that it is only in the religious *cultus* that a concrete unity of the divine
[185] and the human is realized, as the single individual becomes one with the highest spiritual. ("Der Cultus ist, sich diesen høchsten, absoluten Genuss zu geben — da ist Gefühl darin; da bin ich mit meiner besondern Persönlichkeit dabei". *Philos. der Religion.* 2. Aufl. I. p. 215 f.). — The external, figurative, historical form falls away when philosophy converts

the content of religion into the form of thought. The difference between religion and philosophy consists only in this difference of form.

Hegel seeks to show how the various religions can be arranged in a series of stages that leads dialectically from the most elementary forms to the highest religion, the one in which the concept of religion has come to its full development, since in it the Deity is in earnest apprehended as spirit. This religion is Christianity. Humanity, says Hegel, has not needed to wait for philosophy in order to receive the consciousness of the truth. In religion the truth is there — in the form of imagination. Philosophy does not seek to overturn religion, only to convert the truth it contains into the form of thought. What religion sums up in a concrete image, philosophy develops analytically in abstract thoughts; the relation is as that between the clenched and the outstretched hand. — How Hegel conceives this is seen from some examples. — In the notion of creation God and the world are set in opposition to each other; the latter is seen as the product of the former. For thinking this opposition cannot hold: for if the infinite had the finite outside itself, it would of course be limited by it — and so not be infinite. The truth in the dogma of creation is that the finite does not exist in isolation, independently, but always points beyond itself; the essence of the finite is to have a limit, but precisely through the limitation, the negation, it hangs together with the whole content of existence, is taken up into and determined by the infinite. — In the dogma of the atonement this relation between the infinite and the finite is expressed in a still more striking way. God himself is dead, as it says in an old hymn: thus with his infinity undiminished God enters into the finite world, negates himself (in the incarnation) and cancels this negation again (through the suffering, the death and the resurrection). In the great images of dogmatic language it is thus established that finitude and suffering themselves are a divine moment, *[186]* so far from excluding the connection with the highest. Thereby existence has received its explanation for us. What makes up the pain and the anguish in finite existence is precisely what shows it as a member in an infinite connection.

In his speculative zeal Hegel overlooked that for the believer in revelation precisely the decisive thing is that the dogma is more than an image. Every positive religion must assume that at certain points the difference between symbol and reality falls away — that is the condition for the very concept of revelation having validity. In the closing words of his *Vorlesungen über die Philosophie der Religion* he accordingly also states that the disharmony between faith and reflection does indeed fall away for the philosopher when he immerses himself in the essence of the religions and finds in it again the highest ideas of thinking; but the philosophers are, after all, only a small part of humanity — and he

then finds himself forced to close with the words: "How the age is to find its way out of its disharmony must be left to the age itself, and it is not immediately the business of philosophy". — Moreover, Hegel overlooks that a change of form may very well mean an opposition in principle. Hegel interprets the dogma of creation and the dogma of the atonement as symbolic expressions of the connection between the elements of existence, of the infinite life-process that pervades everything and from which we are not loosed even when we feel most bitterly the limitation and pain of finitude. His own vigorous and manly way of taking life shows itself here. But even if this interpretation were the right one, — even if it really were such an experience of life that had found symbolic expression in those dogmas, it is nevertheless a quite different view of life from Hegel's that emerges when creation means a supernatural act, and when the atonement means a historical event by which the supernatural God suffered and died as a man. With the change of form, which Hegel presents as wholly innocent, there in reality follows a transition from a dualistic world-view to a naturalistic or monistic one. This soon came to light in the debate on the philosophy of religion within Hegel's school. He himself
[187] appears here as the Romantic who seeks back to the old forms of the life of the spirit and yet transforms them according to his need. And it is precisely the abiding truth in Hegel's philosophy of religion that the religious representations of the past can have continued value for us only when we can in one way or another find in them an expression of our own experiences and our own thoughts. That naturally does not lead to any historical interpretation; for the nourishing value of the representations may depend on many displacements; but if such a finding again is not possible at all, then the representations of the past are only learned curiosities. — It was a great ideal that stood before Hegel: to show that in history no value had been lost, — even if it was an ideal that his own system was unable to carry out. And in particular his psychology, which made thought the only element of consciousness, was not suited to serve as a foundation for this inquiry. In this respect his rival, whom we shall now describe, decidedly has the advantage over him.

4. Friedrich Ernst Daniel Schleiermacher

a. Characterization and Biography

The demand that Reinhold had set up in his time, that philosophy should be derived from one single principle, found its most resolute execution in Hegel, and the Jena school — so one could call the whole series Reinhold — Fichte — Schelling — Hegel, because it was at the University of Jena that their thoughts were first developed — could seem to form a dialectical process that in its advance gave birth to one system after another

with inner (perhaps even "world-historical") necessity. To demonstrate this process of development over and over again was the Hegelians' favorite undertaking: in this way, after all, their master's ideas were derived historically — by his own method. As we have seen, quite other moving forces than the inner dialectic also played a part. And in any case the Hegelians were embarrassed by a man who, in their master's period of glory, stood at his side with great spiritual authority, and who already long before had *[188]* influenced the thinking and striving people of the time by the weight of his distinctive personality, without its being possible to fit him into that series of stages, toward which he stood precisely in a position of opposition. Schleiermacher's position in the history of philosophy is characterized by his keeping the spirit of the critical philosophy alive within the sphere of the Romantic philosophy, a combination made possible by his Socratic personality, which possessed in a rare degree the capacity to combine full and intimate devotion with clear reflection. He had in his personality what Hegel thought he had in his system, the union of opposites in living unity.

Friedrich Schleiermacher was born on 21 November 1768, of a family of clergymen. His grandfather had been strongly moved by the radical kind of pietism in which subjective feeling led to opposition to the established church, and he had staked his own and his family's existence on it. The father was a practical nature; his rationalist tendencies were pushed back by the need to act practically on people. Enthusiastic about the religious life of the Herrnhuters, he placed his son, when the latter was 15 years old, in their educational institution at Niesky, from which he later passed on to the seminary at Barby. Schleiermacher himself, all his life through, ascribed a decisive significance for his spiritual life to his youth among the Herrnhuters. His nature had indeed already expressed itself before; in a vindication of his views (1801) he says: "My way of thinking has no other ground than my distinctive character, my innate mysticism, my culture proceeding from within." It was a chief trait in him that he set his personal stamp on everything he took in from without. But the Herrnhut religiosity, with its intimate life of feeling, its retirement, its endeavor that the religious content should be lived through in the innermost depths of the single soul, while at the same time life together with like-minded souls was zealously cultivated, gave him a stamp he never lost. Later too he felt himself a "Herrnhuter of a higher order". Higher — for Herrnhutism soon became too narrow for him. He needed a fuller life with people than the cloistered seclusion allowed, and above all there stirred in him an intellectual need and a doubt that led him *[189]* beyond the anxious theology taught by his Herrnhut teachers. After a hard clash with his father he had his wish fulfilled of going to study at Halle. Here he came to know

the theology and philosophy of the Enlightenment, and soon also Kant's writings. It is of great importance for understanding Schleiermacher's development to hold fast (what DILTHEY, in his excellent but unfortunately unfinished *Leben Schleiermachers* (1870), has shown on the basis of Schleiermacher's youthful notes) that he had found his decisive standpoint before he came under the influence of the Romantics and before he came to make a study of Spinoza. At the University of Halle, and later during his stay with a noble family in the country and as a clergyman in a provincial town, he had been led through solitary study and reflection to his distinctive religious standpoint, which brought about for him a harmony between his critical intelligence and his living feeling. This standpoint was the result of the conflict between Herrnhut religiosity and critical philosophy. He never gave up the conviction that in feeling man's innermost life is lived, and that feeling — and feeling alone — brings it into immediate relation to the highest. But he learned from the critical philosophy the definite conditions and limits to which human knowledge is subject. Still more sharply than Kant himself he maintains that all representations that go beyond experience have only symbolic value. It was not only the representations of Christian theology but also the favorite dogmas of the theology of the Enlightenment, of a personal God and personal immortality, that thus now took on another meaning for him. What was distinctive of Schleiermacher — and what makes him one of the most significant figures in the history of the philosophy of religion — was now that what criticism dissolved and denied validity as objective truth did not need for him to lose its religious value, if it could stand as the symbolic expression of an experience that man has in his innermost life of feeling. And these experiences of feeling, the inner moods that can never be fully expressed in words, were for him the properly religious. In the purgatory of the critical philosophy he let
[190] the finite and external in his faith be consumed and kept a kernel that now became so much the more valuable for having stood its test. But he did not take over Kant's philosophy as it stood. On the contrary, he took a very critical stand toward it, so critical that he has even been accused of injustice toward the great master whose disciple he really remained. His criticism concerned in particular the external way in which Kant combined morality and religion. Schleiermacher showed (as is seen from the *Denkmale der inneren Entwickelung Schleiermachers),* published for the first time by Dilthey in the appendix to the biography) that from purely ethical motives no inference can be made to anything that lies beyond knowledge. He thus completes Kant's endeavor to free ethics from dogmatic assumptions by overturning Kant's own moral theology. Only after his views had thus taken shape through his own criticism came the study of Jacobi

and Spinoza and the connection with the Romantic school. Here he now saw a fully formed world-view and an endeavor to gather the whole fullness of life into a form of unity that rose above every external barrier. But in face of these new movements he kept his wonderful capacity to combine devotion with criticism, a capacity that quite naturally hangs together with the spiritual impulse of self-preservation, whose satisfaction requires both, both the taking in of what has nourishing value and the rejection of the opposite. For this need Schleiermacher had an open eye. The assertion of the significance of the single individuality already comes forward in his youthful notes and becomes a chief thought of his all his life through. This emphasis on individuality made him critical of Spinoza's and Schelling's systems, however deeply he sympathized with their striving for a unified conception. Moreover, he found the limits of knowledge overstepped in these systems, and he did not find the significance of immediate feeling recognized. But the great movement of thought and imagination of Romanticism gave him fullness and breadth of view and opened to him spheres for which he had hitherto had no eye. The friendship with Friedrich Schlegel became a turning point for him here. Romanticism's remarkable mixture of individualism and mysticism met him here *[191]* in a pronounced form[42]). However different the two friends were, their life together was nevertheless fruitful owing to Schleiermacher's great capacity to understand another individuality and to convert what could be learned from it into the forms of his own life. Schlegel too, restlessly oscillating as he was, felt polemically disposed toward the attempts to close philosophy in a system; only with him it was rather his arbitrariness, with Schleiermacher besides individualism above all the critical sense, that was decisive here. Schleiermacher found his own tendencies in Schlegel in a chaotic and restless form. And through Schlegel he came into touch with the rest of the Romantics' circle, which gave him rich occasion to use his talent for separating kernel from shell in such a way that what connects kernel and shell is nevertheless not overlooked. — One fruit of the two friends' collaboration was the "Fragments" in the brothers Schlegel's "Athenäum". —

The first work in which Schleiermacher expressed himself at length was *Ueber die Religion. Reden an die Gebildeten unter ihren Verächtern* (1799). Here he sharply separates religion — in a way that will be considered more closely later — from knowledge and morality, and determines immediate intuition and feeling as its organ, but on the other hand seeks to show that intellectual, moral and aesthetic culture only becomes complete by leading back to the immediate living in the feeling of the infinite (the universe or the world-spirit) as that which encompasses and bears all individualities and all finite existence. Spinoza is praised as the one who was pervaded by the lofty world-spirit,

so that the infinite was his beginning and end and the universe his only and eternal love. In a later edition there is added to this an enthusiastic eulogy of Novalis as the one who, in the domain of art, strove in a direction similar to that of Spinoza in the domain of thinking. While these speeches describe religious feeling as that which lets man feel himself, with his whole individuality, one with the eternal and infinite, the next work, *Monologen* (1800), takes the matter up from the standpoint of individuality
[192] itself, maintaining the significance of personal independence and distinctiveness. As we have seen, the conviction of the positive significance of individuality had arisen early in Schleiermacher. It marks his opposition not only to Spinoza and Schelling, but also to Kant and Fichte, who assumed a universal moral law valid for all. It stood firm for him that "every human being is to express humanity in *his* own way, with a distinctive mixture of its elements, so that it may reveal itself under all possible forms, and so that in the fullness of space and time all the differences it holds in its bosom may be developed". Humanity is not to be a uniform mass. (See especially the 2nd Monologue). — In these two works Schleiermacher expressed, in a more indefinite and rhetorical form, the fundamental thoughts that it became the task of his life to carry out and maintain in theory and practice. He was a decidedly practical nature. When he laid weight on individual development, it was not in order to favor isolation and the retired cultivation of one's own self. In his view, energetic assertion of one's own personality, "self-presentation", was the only way in which one can influence others. For one cannot intervene quite directly in the personal life of others. From this side, then, individualism became for him a means. No more than Socrates was he prevented by the need for self-knowledge from influencing others. His work as a clergyman had sprung from the need to unite the two. It consisted for him in an endeavor to lead individuals to reflection on themselves and on the deepest foundation of their life, to immediate living in the eternal and infinite, to liberation from everything finite and sensuous. The dogmas were for him only symbols that could be used as auxiliary representations in this process of interiorization and liberation. Such a conception of the clergyman's work was possible only in an age that, partly in the form of rationalism, partly in the form of Romanticism, took a free attitude toward the dogmas of the church, and even then it gave strong offense, just as it also aroused scandal that Schleiermacher moved in circles that were anything but churchly. In clerical circles he seems not to have felt at ease, and in another age he would perhaps hardly have taken up the clerical calling. In a letter
[193] from the year 1802 we read: "Preaching is now the only means of exercising personal influence on the common way of thinking of the mass of men". Later, during Prussia's

unhappy period and during the reaction after the War of Liberation, he showed his strength and his courage by pleading the cause of national independence and personal freedom through sermons and editorial work and in the church struggles.

It was during his stay in Berlin as chaplain at the Charité hospital (1796—1802) that Schleiermacher came to know life in larger and variously marked circles, entered the atmosphere of the Romantic school and published his first writings. After he had then been a clergyman in a provincial town for some years, he came to Halle as professor of theology (1804), where he worked not only through theological but also through philosophical lectures. Studies of Greek philosophy, especially of Plato (whose dialogues he translated), had widened his horizon and confirmed him in his standpoint. Schelling's philosophy he had greeted with joy. He thought that when idealism recognized the distinctive character of the life of nature, it would also come to recognize the religious life as something different from the mere life of thought and will, so that a "higher realism" could emerge. In Schelling's system he saw only formalism, since the unity of subject and object, of thinking and being, was for him only an abstraction; only living feeling could reach beyond these oppositions. Moreover, he did not find individual existence sufficiently brought out. In Halle he now worked together with HENRIK STEFFENS, and in the modification this man gave to the Schellingian doctrine he found his own views expressed. In Steffens's writings on the philosophy of nature, which rested far more than Schelling's on independent studies in natural science, it appears as a fundamental thought that in nature, from the lowest stages to the highest and through the progressive development of the globe and of organic life, a pervasive *individualizing tendency* shows itself, and in such a way that the more individualized a natural form is, the more it bears the stamp of infinity, i.e. the fuller the content and the greater the oppositions it holds. A proposition such as this: "The stage that most encloses the infinity of nature within itself is the most individual" (Steffens: *Beiträge zur innern Naturgeschichte der Erde.* Freiberg *[194]*
1801, p. 273), could not but appeal to the author of the "Monologues" and of "Speeches on Religion"[43]). A later work, in which Steffens developed this thought in connection with his general ideas about knowledge *(Grundzüge der philosophischen Naturvissenschaft.* Berlin 1806), was the philosophical exposition with which Schleiermacher is said to have agreed most, and on which he himself perhaps (according to a remark of Steffens's, p. XXII) had some influence[44]). — The battle of Jena put an end to Schleiermacher's work in Halle. He now went to Berlin, where by sermons and in other ways he sought to strengthen courage and national feeling. He occupies an eminent place in the series of men to whom Germany owes its national rebirth. When the University of Berlin was

founded, he became professor of theology there, but also lectured on philosophy and its history. At the same time he worked as a preacher on large circles. As a political liberal he was in bad odor in the period of reaction. It went so far that for a time he could not write by post at all, since his letters were opened. The government was also against his conduct in church affairs. Schleiermacher, who himself belonged to the Reformed Church but by his whole standpoint could not attach great weight to confessional differences, was in favor of the king's plan for a union of the two Protestant confessions; but he strongly disapproved of the intention to introduce the union by compulsion.

It could not be otherwise than that the positive content of religion, the more Schleiermacher entered into theological and ecclesiastical work, should come to play a greater part in his expositions. In the later editions of the "Speeches" and in his chief work *Der christliche Glaube* (1821—22) this comes out clearly. But one will not be able to show any difference in principle between the standpoint in the first edition of the "Speeches" and in the work that is probably the greatest Protestant theology has produced since the time of the Reformation. He himself had the feeling of being the same. When he had conceived
[195] the plan of his doctrine of faith, he wrote to a friend of his youth (1818): "A dogmatics, which I have at last overcome myself to write, will show you that I am still quite the same as in the 'Speeches on Religion'." And a few years later (1822), on the occasion of the third edition of the "Speeches", he repeats this remark. He felt himself in very sharp opposition to the pietism and literalism that had grown up around him. It was not in such a way that he had imagined his youthful thought of a renewal of religious life would be fulfilled. At first he had wanted to gain recognition for religion as an essential and sustaining side of the life of the spirit. Later his thought had been a renewal of Protestantism. He looked forward to the founding of the University of Berlin because he hoped to be able to found there "a theological school that could develop Protestantism, as it now is, further and bring it new life" (letter to Brinckmann, 17 Dec. 1809). But the fundamental thoughts on which this school was to be built were in reality those he had expressed in "Reden über die Religion". That he himself did not think he had changed his views is certainly no proof; but a closer inquiry will show that the relation between the youthful writings and the concluding theological work is this: what was first indicated in broad outline and in rhetorical form later received its more definite development, after Schleiermacher's theory of knowledge and ethics had developed (in the way that is seen in the works *Dialektik* and *Philosophische Sittenlehre)*, which appeared after his death), and after he had, through further theological studies, gained a more thorough acquaintance with the historical forms in which religious feeling had found expression within Christianity.

The view that all dogmas had arisen through reflection on immediate experiences of feeling he never gave up. On the other hand, even when he set about working out his doctrine of faith, he was not quite sure where the boundary lay between the properly religious (what he calls in a letter the immanent dogma, by which are probably meant the representations inseparable from religious feeling at its height) and the mythological (what he calls the transcendent or mythical dogma). (Letter to Blanc, 23 March 1818).
What gives Schleiermacher's philosophy of religion its great and abiding interest is the *[196]*
clear way in which he conceives the relation between feeling and representation in the religious domain. The consequence of his conception would certainly have led him much farther away from the ecclesiastical conception than he himself thought he had come. He did indeed see clearly that he felt himself more at one with others than they with him; in a letter from one of his last years (to Reichel, 3 April 1832), after touching on the opposition and misunderstanding of which he is the object "from the exalted on both sides", he utters the beautiful words: "I am learning at least for myself, in quiet, to be at one with many who think themselves very far removed from me, and in that lies a peculiar, life-refreshing power". What is Romantic in Schleiermacher lies precisely in his overlooking the great difference it makes whether one regards the dogmas as symbols of human feelings or as authentic proclamations of the eternal truth. He took a great step toward a deeper understanding of religious phenomena; but at the same time he gave the religious a quite different place and significance from the one ecclesiastical tradition ascribes to it. His whole Socratic personality was needed for him to be able, with the conception he had, to work in the service of the church. Early on he had to defend himself against the charge of dishonesty and improper accommodation. He then pointed with pride to the fact that his abilities could well make another occupation possible for him, were it not precisely his desire and his calling to use them in the work of a clergyman, and with regard to the relation between his religion and his philosophy he declared: "I am convinced that I really have the religion I am to proclaim, even if I should have a quite different philosophy from most of my hearers. Nor is there in me any unworthy cleverness or reservatio mentalis, but I give the words precisely the meaning that a human being gives them *while he is absorbed in religious contemplation* (indem er in der religiösen Betrachtung begriffen ist), only not another meaning besides". (Aus
Schleiermachers Leben. III. p. 284). This statement is characteristic in that it relegates *[197]*
the difference between religious representations to the philosophical, i.e. the theoretical domain, and in that he maintains that the religious representations, when feeling is at its height, in fact take on another meaning than in calmer moments[45]). It is at the height

of feeling that it creates its forms for itself or seizes those that are handed down; then it shows itself as the dogma-creating power, and it is in this, the real hour of the birth of religion, that Schleiermacher wished to regard it; there the lines converge, and there he also found the point of contact between his line and many others that in their further course even went in the opposite direction. —

Schleiermacher's life in Berlin, vexed by opposition from without and from above, was beautified by the happiest home life. After a short illness, during which he kept his full clarity and spiritual freedom to the last, he died on 12 February 1834.

b. Dialectic and Ethics

We shall be able to form a notion of the harmony that existed for Schleiermacher between religious feeling and scientific thinking if we first examine how far, by the way of thinking, he arrives at results that allow religious feeling to move freely, and then how far, by starting from religious feeling, he arrives at results that allow thinking to move freely.

In his theory of knowledge he seeks to mediate between Kant and Schelling. It was clear to him that Schelling overstepped the limits of knowledge, and in his *Dialektik* he accordingly sought to give a new determination of those limits. By dialectic he understood the doctrine of the principles of the art of philosophizing. It forms the preparation for philosophy as a systematic science. It is the task of philosophy to find the inner connection of all knowledge, and dialectic now discusses the very conditions of a knowledge. These fall into two parts: for it belongs to knowledge that each single thought hangs together with other thoughts, and that to the single thought there corresponds a
[198] real being. These two parts cannot be separated. If knowledge is to be valid, there must correspond to the relation between our lower and higher concepts lower and higher kinds of existence, so that, just as the higher concepts contain the ground of the lower, the higher existence contains the ground or force that reveals itself in the lower existence in a multiplicity of phenomena. And in a similar way there must correspond to the connection between concepts expressed in judgments a real connection between the things that are, a causal relation. By this assertion of an agreement between knowledge and being Schleiermacher separates himself from Kant, whose critical scruples he passes lightly over, borne by the Romantic need to immerse himself in existence. On the other hand he recalls Kant by his determination of limits. For the identity of thinking and being is indeed for him a presupposition of all knowledge; but it is a presupposition that cannot itself become a knowledge — although for the other Romantics it was the very highest knowledge. Schelling's attempt to set up a doctrine of absolute identity

led, for Schleiermacher, merely to the setting up of more or less felicitous schemata. He here takes up Kant's doctrine of the ideas, defining idea as a problematic concept, a concept that designates a limit for thought which thought never reaches, let alone gets beyond. In all our knowledge there are two elements, one due to the organic function, sense perception and experience, another due to the intellectual function, which appears as construction and speculation. Neither of the two elements can be wholly lacking. Empiricism is right that the being of the single things is not wholly absorbed in the concept; but it is not right if it asserts that the being of the single things is the whole of being. Speculation is right that all particulars must have their ultimate ground in the source of all being and knowledge; but it hovers between knowledge and poetry when it believes it can derive things from this ultimate source by the way of construction. The idea of God as the unity of thinking and being is the presupposition that, consciously or unconsciously, lies at the basis of all knowledge; but it can only be constructed as a formal schema. With it is combined the idea of the world as the totality of the multiplicity of all *[199]* that is. As the idea of God is the formal point of departure (terminus a qvo), so the idea of the world is the real conclusion (terminus ad qvem) toward which our knowledge moves without ever reaching it. For no more than the point of departure can the end point be converted into real knowledge. As every knowledge we have contains construction and empirical material in varying proportions to each other, so every knowledge we have also lies at a certain point between the point of departure and the end point. All our knowledge is in this respect provisional. It is the task of scientific criticism to compare our actual knowledge with the ideal of knowledge; it is in the theoretical domain what conscience is in the practical domain. An absolute systematic completion is impossible, and in particular it lies in the stated relation between speculation and empirical material that philosophical systems set themselves an impossible task when they seek to derive the finite from the infinite and to give the inner physics of the infinite being. Only in religious feeling is the unity of opposites lived; science is unable to grasp it, either in its meaning as principle or in its meaning as totality. The images in which that feeling finds expression nevertheless fall under scientific criticism. Only such images are admissible as express both the difference between the idea of God and the idea of the world and the inseparability of both ideas. The two ideas are correlata: no God without the world, and no world without God. All images that do not conflict with this rule the philosophical art can admit. But just as it emphasizes against speculative philosophy that the latter gets no further than giving schemata, so it maintains against the religious representation that this gets no further than giving images. The expression person, e.g., can, used of

God, only be an image; likewise the expression force, and whatever others one might wish to use. Atheism is for the most part only the rejection of the validity of images and anthropomorphisms.

Just as the unity of thinking and being is the presupposition of all knowledge, so the unity of willing and being is the presupposition of all action. For as knowledge
[200] would be impossible if there were no point of unity for thinking and being, so action would become impossible if the will were absolutely alien and isolated in the world. The external world must be receptive to our intervention and able to take into itself the ideal stamp of our will. At the basis of all conscience lies the idea of a unity of willing and being, an idea that cannot be different from the idea of a unity of thinking and being, but, no more than that idea, can be formed into a scientific concept.

By this consideration dialectic leads over into *ethics.* Its place in the whole system of science Schleiermacher indicates in the following way. The presupposition and final end of all knowledge (or action) was, after all, the unity of thinking (or willing) and being. At every stage of existence we know, one of these two elements predominates. The science of the part of existence where being predominates over thinking and willing, nature over reason, is called physics, which is again divided into natural history (empirical physics) and natural science (rational physics). The science of the part of existence where thinking and willing predominate, or work toward predominating, over being, reason over nature, is called ethics, which is again divided into two parts, history (empirical ethics) and moral doctrine (rational ethics). But all these oppositions are relative. No nature without reason and willing! Nature is "a diminished ethics"; it shows us the will at a series of stages — in inorganic form, in plant life and animal life and — at the summit — in man. Without this unity of nature and reason, ethics proper would not become possible. It is a further and higher development of something that already makes itself known in nature. (Cf. Dialektik § 213—214). — It was on this point that Schleiermacher found in Steffens's philosophy of nature a necessary supplement to his doctrine. He could not, like Kant and Fichte, set ethics in sharp opposition to nature. He demanded (already in his *Kritik der bisherigen Sittenlehre,* 1803) that ethics be brought into connection with science as a whole, and in this respect he set up Spinoza and Plato as exemplars. He therefore emphasizes with great force that no absolute beginning of ethical development can be assumed. (Philos. Sittenlehre § 103. 124. 325). It is always a given foundation, a
[201] relative harmony, that can serve as point of departure and of attachment. And in this way ethics enters into relation not only with nature but also with history, since every ethical development begins at a certain point of the development of the species. The

single human being has in his innate organs results of the exercise of the preceding generations (Philos. Sittenlehre § 147—148). — Although Schleiermacher's doctrine of development is reached predominantly by the way of construction and has an idealist stamp, he nevertheless here indicates a point of view that has been much discussed in modern biology. Only he thinks that such organization of the nature of the species can be spoken of first in the case of man. The animal species remain standing at the same point. Development, then, takes place in nature, but not in the sense of a real transition from form to form.

The relation in which ethical development stands to nature is a relation of reciprocal action. The ethical process consists partly in an organizing, shaping and forming activity, partly in a symbolizing, expressing and designating activity. To the first kind of activity belongs man's striving to make himself master of nature. The single individual takes possession of and works a part of nature and thereby makes it his property, and men together form a legal community that secures intercourse. To the symbolizing activity belong all the ways in which man gains expression and form for his spiritual life and so puts something into the world that has value only as a sign of inner life. In individual form such a striving expresses itself in religion's need to create poetic forms for feeling, since feeling is in itself incommunicable (unübertragbar). Art conditions the possibility of bringing individual distinctiveness to intuition and thereby bringing about spiritual intercourse with others. Art is for religion what language is for science[46]). Therefore every individual must in a certain sense be an artist. The universal form of symbolizing is science, which expresses what is common, identical and therefore communicable in the life of consciousness. — Thus the whole development of culture falls under ethical development, and ethics is in a radical way freed from the subjective and formal mode of consideration, which prevailed to so high a degree in Kant and Fichte. In his "Monologues" *[202]* Schleiermacher was still impatient that material culture (what he later called organizing activity) played so great a part in life, and he was inclined to see the ethical only in the endeavor to express one's distinctive personality (in symbolizing). But in his maturer conception he seeks to unite the two. A symbolizing striving that does not recognize the significance of natural points of departure he now calls (Philos. Sittenlehre § 209) the cynical one-sidedness; an organizing activity without a sense for symbolizing he calls the economic one-sidedness.

One misses in Schleiermacher's ethics a deeper entering into the subjective oppositions and conflicts. He describes the ethical process without stopping at the crises. And yet it is a chief merit of his ethics that it so clearly sets forth the significance of

individuality. He does indeed grant Kant and Fichte that reason is one and the same in all; — but he adds that the nature of the single individual is not exhausted by this common element. When the species in fact splits into a multiplicity of individuals, each one can gain ethical value only when he expresses the common human nature in a quite definite and distinctive way. In an ethical respect there will therefore be something in the single individual's action that cannot be transferred to or hold for that of others. If he has not been present in his action with his whole and full distinctiveness, then his action is imperfect, then he has not been wholly active. Because of this individualizing of the manner of action there are points where each one must be his own judge. But because he is his own judge, he is not his own teacher[47]). In the concept of personality it lies not only that the single individual sets himself apart from others, but also that he has others beside him. Already in the "Monologues" Schleiermacher had set up the proposition: "Keine Bildung ohne Liebe". The clearer the sense the single individual has for his own distinctiveness, the greater the sense he must also gain for that of others. It is imagination that is the organ here. Conversely, one also becomes conscious of
[203] one's own distinctiveness only within the species. Later Schleiermacher also emphasizes the division of labor, which assigns the single individual a definite calling according to his nature. In this lies a warning against regarding one's individuality as fully finished so long as it has developed only in isolation or in narrow circles. — In his personal experience Schleiermacher had learned to know the value of individuality. He was a virtuoso at discovering and associating with different personalities. He found, as he says in a letter (to Henriette Herz, 17 Dec. 1803), no human being insignificant "who has something distinctive and who presents human nature from one side or another". And it was a precept of duty in his ethics that everyone *should* be distinctive and act in his own distinctive way. In this way, after all, the individualizing tendency in nature is also carried further. —

In a series of academic treatises (which, besides the "Kritik der bisherigen Sittenlehre", were all that Schleiermacher himself published on ethics) he treated the fundamental ethical concepts of duty, virtue and the good, and showed that they do not designate different parts of the content of ethics, but different sides from which one and the same content can be regarded. The concept of the good, which for him is one with the concept of the full realization of spirit or reason in nature (through organizing and symbolizing), ought to be laid at the foundation, since duty and virtue receive their real grounding only through it. —

c. Faith and Knowledge

Dialectic and ethics were to show how far thought and will can reach when they develop according to their own laws. As the third chief form of the human life of the spirit Schleiermacher sets religion, whose source and seat is in feeling, and his endeavor is now to show that it can develop according to its own laws, independently of the other forms and without interfering in their independent development, — and yet in such a way that only in religious feeling are full harmony and reconciliation won.

In his "Speeches on Religion to the Cultured among Those Who Despise It", Schleier- *[204]*
macher opposes in particular two religious standpoints: the one from which religion is conceived as a doctrine (revealed or grounded on mere reason), and the one from which religion is conceived as a means for morality. It consists, he maintains, in the *immediate* consciousness that everything finite is in and through the infinite, everything temporal in and through the eternal. While knowledge goes from thought to thought, from phenomenon to phenomenon, and while the will is directed toward definite tasks, feeling rests immediately in itself, raised above all opposition. (Feeling is for Schleiermacher one with "immediate self-consciousness", from which he distinguishes "objective self-consciousness", which is a kind of self-recognition. Der christliche Glaube § 3). Knowledge and action depend on special talents; in them man acts in a one-sided way and in face of objects that are different from himself. But in feeling there is the fine, infinite movement in which the single individual's full individuality can unfold, and in which individuality is nevertheless precisely irradiated by the infinite. In a description of religious feeling it is a matter of seizing the moment in which the inner life stirs whole and undivided, before it appears in thought and image or in will and action. It is the point where the universal and the individual meet, before the difference between subject and object has yet come into force. In this immediate feeling the individual stands as dependent, not dependent on anything finite whatever, toward which a reaction might be possible, but absolutely dependent. And this absolute feeling of dependence becomes a consciousness of God as soon as reflection awakens and seeks an expression for that on which we are dependent with our whole passive and active being: for "by the expression God is meant precisely the whence *(das Woher)* of our receptive and self-active existence". (Der christliche Glaube § 4,4).

It has been thought that a difference in principle between the "Speeches" and the "Doctrine of Faith" is to be found in this: that in the former religion is depicted more as an immediate feeling of unity than as a feeling of dependence, the latter expression first occurring in the "Doctrine of Faith", — and in this: that religion, especially in the first

[205] edition of the "Speeches", is described as conditioned by the intuition of the universe or the world-all, whereas the "Doctrine of Faith" makes a definite distinction between God and the world. The standpoint of the first work has even been called monistic, that of the second dualistic[48]). But already in the "Speeches" the religious relation is described as a relation of dependence, insofar as one goes back to the point where consciousness in the single moments first comes into being, the point where as yet no difference asserts itself. Here, at the root of consciousness, the universal and the individual meet: but the individual, which is only coming into being, must evidently be dependent. And when in the "Speeches" (especially the 1st edition) expressions such as universe or world-all are mostly used, Schleiermacher himself has repeatedly emphasized that what makes the world a whole is precisely God (Philos. Sittenlehre § 287 and note 2 to the second speech in the 3rd edition). When in the "Dialectic" and the "Doctrine of Faith" he distinguishes between God and the world, he means by the world the totality, by God the unity, and he affirms, as we have seen, a close connection between these two concepts. — His ideas underwent an unfolding, but his feeling of having remained the same did not deceive him.

Both in the "Speeches" and in the "Doctrine of Faith" it is definitely maintained that no concepts and propositions belong to the proper essence of religion, but that they are all derived, due to reflection on the immediate states of feeling. As we have already seen, the word God designates the origin (das Woher) of the peculiar feeling of dependence. It is by virtue of a need for expression and communication that words and images are sought for the feeling that is in itself unutterable. If these images are taken as literal truth, mythology results. It is the task of the doctrine of faith to convert the figurative expressions of feeling into proper expressions, or at any rate to give the figurative expressions a definite limitation. No proposition in the doctrine of faith may be derived from another proposition, but every proposition is to be drawn immediately from religious experience and only afterwards compared with other propositions. (Chr. Gl. § 17). The only proof there can be any question of for such propositions is, then, that others
[206] come to have the same experiences as the expositor. All dogmas that cannot in this way be traced back to immediate experiences of feeling Schleiermacher therefore rejects, or at any rate he declares them to be symbols that are not necessary expressions of religiosity. So the representations of the personality of God, of personal immortality, of creation, of the first human beings, of the origin of sin, etc. He denies that religious experience can lead to assuming that the connection of nature is broken. There would in that case be a relation of opposition between the consciousness of God and the consciousness of

the connection of nature; but there can — Schleiermacher asserts — never arise from religious interest a need to conceive a fact in such a way that its conditionedness by the connection of nature would be canceled by its dependence on God, since these two relations coincide. *Miracle* therefore becomes only the religious name for an event, for what has aroused religious attention and to which a special value therefore attaches. Revelation does not mean a doctrine but a fact of religious significance that cannot be explained from the historical connection. Christians assume such a revelation in Christ, because only by assuming him as a sinless archetype who appeared historically can they explain to themselves their consciousness of redemption, i.e. the experience that what hindered their consciousness of God and by this hindering caused them suffering has now been cleared away. Religious feeling is in itself a feeling of blessedness; but since it constantly occurs in consciousness together with other feelings, which are determined by man's nature as a finite and sensuous being, and since these feelings can partly accord with, partly hinder religious feeling, there arises an opposition between religious pleasure and religious pain. And the canceling of religious pain is something the Christian can make intelligible to himself only if there has appeared in Christ an exemplar whose pure consciousness of God propagates its redeeming and reconciling power through the congregation to the single individuals who are set before the image of his personality. There is the same relation between the consciousness of redemption and faith in Christ as the pure exemplar as between the feeling of dependence and the representation of God. In both places the inference is from effect to cause. *[207]*

The great question becomes whether the justification of such an inference can be shown. From a pure state of feeling no inference can be made, without the help of other experiences, to the cause that has aroused it, — and such other experiences Schleiermacher must consistently keep away. That feeling, because of its need for diffusion and intensification (so excellently described in his time by Hume), leads to the formation of representations of ideal beings is a psychological fact; but this process leads, as Schleiermacher himself acknowledges elsewhere, only to symbols. Feeling's need for symbolism is confused by Schleiermacher with its significance as a fact pointing to the presence of a quite definite cause. As a philosopher he keeps to symbolizing; but he could become an ecclesiastical theologian only because he confused symbolizing with a causal explanation. And this confusion then brought him into open conflict with his own philosophy. His theory of knowledge declared the concept of an absolute being impossible to carry through; but his theology forced him to set up such a concept. His philosophy of nature declared that no absolute beginning could be conceived; but his

Christology forced him to assume an absolute point of beginning in the appearance of Christ, since through him something was supposed to have entered history that could not be explained by the preceding historical development.

Schleiermacher's philosophy of religion is — like Hegel's — an attempt at restoration. Characteristic of it is that it defends religion by showing its significance for the life of the spirit as an interiorizing, infinitizing and harmonizing power, and does not, conversely, judge the life of the spirit by its relation to religion as the highest norm. A standard is thus presupposed that lies beyond that of religion itself. Schleiermacher closes his exposition of the Christian doctrine of faith with a statement that recalls the closing words of Spinoza's Ethics: not a single one of the propositions he has set up in his work loses its meaning, he asserts, even if there is no continued subsistence of personality after death (§ 158). It is thus the life we know and live in the world of experience that is
[208] to be deepened and developed, and from its needs the highest standard is drawn. And whereas the positive religions sought not only to create peace and harmony in the mind, but also represented knowledge and morality, religion according to Schleiermacher has nothing at all directly to do with these domains.

That it was an essential change he made in the concept of religion, he was clearly aware. Neither Protestantism nor Christianity at all had for him reached its final form. A constant development had to take place in the religious domain too, and it was the task of free theology to guide this development. Neither the doctrine of faith nor Christian ethics was for him given once and for all; the writings of the New Testament contain only the conception of the first generation. This holds especially in the case of ethics, that it must constantly be altered and developed — as regards both motives and results: "There is no exposition of ethics that could continue to hold for all the ages of the Christian church, but every exposition has full value only for a certain period." *(Christliche Sitte.* p. 69. cf. p. 94 f.). — This grand conception the church has not adopted. Had Schleiermacher lived in our day, it would have been far more difficult for him "to feel himself at one with many who think themselves far removed from him."

Schleiermacher's philosophy of religion is distinguished from Hegel's by its more correct psychology. It is a main point that religion is referred to feeling, not to mere thinking. Even if the dogmas are dissolved into symbolism, something nevertheless remains: the experiences of feeling that led to the production and acceptance of the dogmas. The problem is whether they fall away or subsist or are converted into other forms when the objective truth of the dogma is rejected. Schleiermacher himself assumed a subsistence, neither a falling away nor a conversion. In that he was wrong. But his

standpoint was whole and harmonious. He knew an undulation between feeling and knowledge, but he did not understand Jacobi's complaints at being a Christian with his heart and a heathen with his understanding. And by his fine understanding of the nature and conditions of personal life he stands as one of the most eminent spirits within the circle of Romanticism. *[209]*

B. The Philosophy of Romanticism as a Pessimistic View of Life

Arthur Schopenhauer

a. Biography and Characterization

Schopenhauer's philosophy arose on the same spiritual soil as the idealist doctrine of development, and yet in sharp opposition to it. The spiritual kinship shows itself in the bold attempt to solve the problem of existence by a purely subjective path; the opposition appears partly in Schopenhauer's critical conception of knowledge, partly in his sharp emphasis on what is disharmonious and irrational in existence. Schopenhauer, like Schleiermacher, is a critical philosopher; he puts the problem of knowledge before the problem of existence, and he lays far greater weight on experience and immediate intuition than Fichte, Schelling and Hegel. Like Schleiermacher, he came to his philosophy while working to appropriate Kant's thoughts and carry them further. He regarded himself as Kant's true heir and denied that anything of significance had happened in philosophy between Kant and himself, although his debt to Fichte and Schelling was no doubt greater than he would admit. But while Schleiermacher embraces an idealist optimism and believes, just as Hegel does, in the development of reason through nature and history, Schopenhauer takes an altogether unique place in the whole of European thought by breaking with the fundamental presupposition of the harmony of existence on which Western theology and philosophy had hitherto always rested, more or less decidedly, and by — appealing to the experience of the misery of life — setting up the proposition that the innermost kernel of existence is a blind, unbridled, never resting and never satisfied need. Thus not only the problem of knowledge but also the problem of valuation is posed here in an entirely different way, and answered in an entirely different direction, than in the preceding group of thinkers, whom he therefore also regarded as his antipodes. But in his solution of the problem of valuation, no less than in *[210]* his solution of the problem of existence, Schopenhauer stands decidedly on the ground

of Romanticism. He continues its opposition to rationalism and "enlightenment"; his philosophy is a thoroughgoing doctrine of the limitation and impotence of reason. He sympathizes with Romanticism's turning back to the Oriental spirit, which is pessimistic in contrast to Western optimism. Not only with Romanticism but also with orthodoxy did he sympathize, since he found the greatest significance of Christianity in its pessimism. The optimistic elements in Christianity he ascribed to the after-effect of Judaism. In a remarkable fragment (*Neue Paralipomena* § 446) (written 1852 at the earliest) he expresses his sympathy with Romanticism and orthodoxy in opposition to humanism and materialism. Since his philosophy is thus closely bound up with his experience of life and his view of life, his expositions too — apart from the epistemological investigations — acquire a personal and living character, a strength and a brilliance, which give them a literary value that the works of his speculative antipodes almost entirely lack. The ability to discuss philosophical questions in clear and vivid form, which distinguished the French and the English school, Schopenhauer — whose culture was also in large part English and French — has at his command in a high degree, and in him it is joined with one of the most peculiarly marked personalities known to modern literature.

Arthur Schopenhauer was born in the then still free city of Danzig on 22 February 1788. His father was a rich merchant, an ardent champion of his city's freedom, who, when Danzig became Prussian, moved with his family and his business to Hamburg in spite of all brilliant offers. The son was to be brought up as a man of the world: "My son" — said the father — "shall read in the great book of the world". Arthur was therefore educated in France and England and spent part of his early youth traveling with his parents round Europe. Then he was placed in a merchant's office in Hamburg, since his father would not yield to his wish to go to the university. Here in Hamburg, by his own
[211] account, his pessimistic view of life was founded. "In my seventeenth year, without any learned schooling, I was seized by the misery of life, just as Buddha was in his youth, when he saw sickness, old age, pain and death. The truth which spoke loudly and clearly out of the world overcame the Jewish dogmas which had also been impressed upon me". (Neue Paralipomena § 656). It was certainly not merely what the young man saw around him that determined his result here (as presumably also with Buddha or with the originator of the Buddha legend). His temperament, his whole personality expressed itself in it. Schopenhauer must be called "burdened" from birth. There was insanity both in his father's and in his mother's family. His father was an energetic but harsh and irritable nature, who suffered from unmotivated attacks of anxiety. He is said to have taken his own life in a fit of derangement. Arthur Schopenhauer inherited his father's

temperament. Melancholy, anxiety and suspicion tormented him. Already as a boy he brooded over the wretchedness of men. For when he was not harboring suspicion toward others, or when his vehemence and unbridled self-esteem did not overwhelm him, his susceptibility to suffering led him to feel pity for others. On a journey in the south of France he could not share his mother's delight in the beautiful landscape, because he could not help thinking of the wretched life led in the dilapidated huts they passed. Already here he was exercised in the thought he later expressed in his chief work in answer to the optimists' appeal to the beauty of nature: "Zu *sehen* sind diese Dinge freilich schön; aber sie zu *sein* ist ganz etwas Anderes". But there were also other elements in him that kept him from seeing life in a bright light. Strong sensual drives stirred in him, which left him no peace and which he could not master, but which again and again "drew him from the heights of heaven to the dust of the earth". In youthful poems from the Hamburg period he has depicted the inner struggle between his striving to lead a life of thought with his gaze on the great and lofty, and the constant pull and snare of voluptuousness. When in his philosophy of life he found the solution of the riddle of the world in the belief that an unbridled, never satisfied need stirs in everything, he built here too on inner experience. His choleric mood, which bred bitterness and vehemence, *[212]* his strong self-esteem and ambition, which chafed at the lack of recognition, and his sensuality all gave him material at first hand. And what was Tantalean in the need he felt lay not only in its unbridledness but especially in the opposition in which it stood to the other need that stirred in him, the need to lead a life of intuiting and thinking. He was a contemplative nature, and in this respect had inherited the temperament and gifts of his mother, Johanna Schopenhauer, the novelist well known in her day. All three of the parts of which, according to Plato, the human soul consists — thought, the sense of honor and self-esteem, the sensual drive — were thus marked here in extreme form and were bound to come into conflict with one another. When in his solution of the problem of existence he would conceive the world by analogy with his own microcosm, it was clear that no harmonious picture could come of it. In a fragment from the year 1814 he speaks against the assumption of a unity in man's nature: "Inner discord is his nature, as long as he lives". And such an inner disharmony he then found in the whole of existence.

After the father's death the mother moved to Weimar, where she was soon received into the circles of Goethe and Wieland. After much resistance the son got his way and was allowed to study. With zeal and brilliant results he threw himself into classical literature, natural science and philosophy. His resolve was now fixed that he would devote his life to thinking. As he said to Wieland: "Life is a precarious business; I have

resolved to spend mine in thinking about it". His stay with his mother in Weimar came to have great significance for him through his acquaintance with Goethe. The young pessimist and the great optimist met in an interest in the theory of colors. They worked together for a time, but to Goethe's great dissatisfaction Schopenhauer — who otherwise, like the other Romantic philosophers, took Goethe's side against Newton — parted from him by giving a physiological explanation of the color qualities, whereas Goethe believed he had given a physical one. Goethe nevertheless followed Schopenhauer's
[213] activity as an author with interest, and felt especially attracted by the weight he laid on immediate intuiting in contrast to speculation and reflection. From the old master the young man, full of self-esteem, who was already complaining bitterly of the world and of men, received this advice in his album:

Willst du dich deines Werthes freuen,

So musst der Welt du Werth verleih'n.

Schopenhauer pursued his academic studies in Göttingen and Berlin. There G. E. Schulze (the author of "Ænesidemus"), here Fichte was his teacher in philosophy. But it was Plato and Kant who became his real masters. His critical sense gave him an interest in Kant's epistemological investigations, whose significance he found misjudged by Kant's successors. But what gave Plato and Kant a special significance for him was that in Plato's opposition between the clear world of ideas and the obscure world of sense, and in Kant's opposition between the law-governed world of phenomena and the "thing in itself" lying beyond all concepts and laws, he found expression for the dualism between thought and will, contemplation and ardor, which his personal experience had impressed upon him. — Besides Plato and Kant, the study of the sacred writings of the Hindus (in Anquetil du Perron's Latin translation) also played its part in the development of his views. Already the Hindus had stopped at the problem that was for Schopenhauer the chief philosophical problem: the problem of physical and moral evil. He set Buddhism and primitive Christianity so high because these religions were faith in a redeemer more than in a creator. The wonder which according to Plato is the beginning of philosophy has — says Schopenhauer — the character of dismay and sorrow. And the problem becomes so sharp because evil, although it is that which ought not to be, must nevertheless have its germ in the very kernel of the world, since it cannot, after all, arise out of nothing. Here lies "the punctum pruriens of metaphysics, which puts humanity in an unrest that can be quieted neither by skepticism nor by criticism."

[214] It was, however, the epistemological grounding of his system that he worked out first. True to his fundamental proposition of standing toward life as a thinking spectator, he

withdrew in the year 1813 — while the youth of Germany were streaming to the colors to free the fatherland, and everything was filled with war fever and the clash of arms — to secluded Rudolstadt, and in this idyllic retreat wrote his doctoral dissertation *Ueber die vierfache Wurzel des Satzes vom zureichenden Grunde* (1813). He means to show here that all our representations stand in a law-governed connection, which appears in four forms: 1) as the relation between ground and consequent, 2) as the relation between cause and effect, 3) as relations of space and time, 4) as the relation between motive and action. After once more insisting on the important difference between ground and cause, the work aims to show that the principle of causality works involuntarily and immediately in every sense perception, since by a construction of which we are not conscious we intuit the cause of our sense-sensations as external objects in space. Schopenhauer means thereby to correct Kant's sharp distinction between intuition and understanding. More than he would admit, he is here influenced by Fichte's Wissenschaftslehre.

As regards the limits of knowledge, Schopenhauer comes to the same result as Kant. Since by the nature of our mind we bring all our representations into such a connection as the principle of sufficient reason demands, and since we know nothing that is withdrawn from this great law of connection and relativity, we do not know the absolute essence of things, the thing in itself. But he now parts from Kant by asserting that there is nevertheless a way to penetrate to the kernel of existence, even if it cannot be done by the way of the knowledge of the understanding. We can penetrate into the fortress by an underground passage. For Kant has overlooked (or at any rate only hinted at it in his doctrine of practical reason and of the intelligible character) that the kernel of existence must be in us, in our own inner being. We carry the "Ding an sich" within us. The craving and striving that express themselves in our pleasure and pain, our fear and our hope, in all feelings and in all willing, are a revelation of the kernel of existence and give us a key to the understanding of the whole of nature. When we keep to the *[215]* standpoint of the knowledge of the understanding, the world is only phenomenon, only representation. But by using the analogy with our own need and willing we discover that the essence of the world is will — in many forms and degrees. — These are the fundamental thoughts of Schopenhauer's chief work *Die Welt als Wille und Vorstellung* (1819).

It was during a four-year stay in Dresden that this work took shape in him. In character it has a certain kinship with Spinoza's "Ethics", in so far as it comprises a whole series of different problems, in mutual connection, within one frame. It gives, in progressive development, theory of knowledge, world-view, aesthetics and ethics.

And the different ideas, according to a remark from 1813 (Neue Paralipomena § 630), had grown together imperceptibly, so that he would not be able to say which of the different parts of his system had formed itself first. It grew up in him like the child in the mother's womb. Philosophy as total intuition or world-view was in his conception an art rather than a science. By virtue of the principle of sufficient reason science goes from proposition to proposition, from phenomenon to phenomenon, from point to point in space and in time. But the philosophical art forms a total intuition which no longer contains the answer to a Why, any more than it makes a new Why possible, but answers the last, decisive question: *What* is the world? This total intuition forms itself in the philosopher in the moments when it becomes possible for him to hold himself in a purely objective, intuiting attitude and to grasp the great, typical features of life. The task then is to translate this total intuition into concepts, and by this endeavor philosophy is distinguished from the fine arts, which remain at intuition and therefore give only fragments and examples, no rule and no whole. This conception of philosophy[49]) is developed in notes of Schopenhauer's which date from the years 1811—1818 and are put together as §§ 1—29 of the "Neue Paralipomena" edited by Griesebach. (Cf. also "Welt als Wille und Vorstellung". Part 2. Ch. 7, 34 and 36). We shall no doubt hit
[216] Schopenhauer's thought if we compare philosophy as an art with historiography, which likewise lies on the border between science and art. And this comparison can then be carried further, since philosophy as an art, according to Schopenhauer, presupposes the critical philosophy, just as historiography presupposes historical criticism. When Schopenhauer thinks that no philosopher before him had held this conception, he is mistaken. Metaphysical idealism in all its forms — especially in Leibniz, Herder and Schelling — is built on a total intuition which illuminates the innermost of the world by means of the analogy with what stirs in man's own inner being. On such an inference from analogy rests Schopenhauer's artistic beholding of existence too. Schopenhauer's answer to the question *What:* "The innermost essence of the world is will!" — is not new either; Kant, Fichte and Schelling had already solved the riddle of the world in this direction, even if Schopenhauer has expressed the thought with greater energy.

Schopenhauer's work can be compared with a drama in four acts (just as Spinoza's "Ethics" was a drama in five acts). The *first book* treats of the world as phenomenon, subject to the principle of sufficient reason. It contains Schopenhauer's epistemology, which had already been laid down in the "Vierfache Wurzel". From the world as mere representation we go back to the will as the innermost essence of the world; therein lies the solution of the riddle of existence. In the *second book* the different grades and

forms of the will in nature are described more closely, and the will to life is depicted as the blind need for existence that stirs in everything, working its way up from grade to grade, taking knowledge into its service, and at last awakening to consciousness of its whole wretchedness. The question arises whether a deliverance from this unblessed, restless craving is not then possible. The *third book* then points to art: in the aesthetic contemplation of nature and of life it is as if the wheel of time stood still, the will brought to rest. But along this path deliverance succeeds only in single moments. If the goal is to be reached completely, it can — as the *fourth book* shows — only be done by wholly abolishing one's will to live, in pity or in asceticism. Existence is a tragedy. In Schopenhauer the drama does not get the bright, reconciling close it has in Spinoza. — *[217]*

After completing his work Schopenhauer made a journey to Italy and stayed for a longer time particularly in Venice. From brooding over the riddles of existence he here once more plunged into the life of the world. On a walk with his beloved he met Byron, who was also staying in Venice at this time. Both the philosophical and the poetic pessimist knew how to accept the goods the wicked world offered. From all that had been able to hold him in Italy he nevertheless tore himself loose and returned home to Germany to work as a lecturer in philosophy. He qualified as a lecturer in Berlin, on which occasion he had a clash with Hegel. His activity as a lecturer did not succeed. This way of working for philosophy did not suit him. Besides, out of defiance he set his lectures at the same hours as those in which Hegel gave his most frequented course. After making a further journey he chose (1831) Frankfurt as his place of residence. With great business ability he had saved his fortune, which was threatened by a bankruptcy, and from now on he led a solitary, quiet life, occupied with study and writing. His pessimism was strengthened by the slight attention that fell to the lot of his work. In his indignation he could imagine no other cause for it than a conspiracy of the envious professors of philosophy. It was the disciples of Schelling and Hegel who now occupied most of the chairs in Germany, and Schopenhauer, for whom Fichte, Schelling and Hegel were only "the three great windbags", could not but feel still greater contempt for the epigones. With justice he was indignant at the mixing of theology and philosophy that flourished. On the other hand he had only few words of recognition for the critical work in the domain of the philosophy of religion and the history of religion that proceeded from the school of Schleiermacher and of Hegel. In the natural sciences he believed he found new confirmations from experience of his doctrine of the will to life; he collected them in his work *Der Wille in der Natur* (1836), in which he himself believes he has given the most thorough and clearest exposition of his cosmological doctrine. It deals with the

subject treated in the second book of the chief work. His ethical views he developed in *Die beiden Grundprobleme der Ethik* (1841). — From the forties on, but especially from
[218] the beginning of the fifties, his writings began to arouse more attention. Several men of letters attached themselves to him and worked to spread the knowledge of his books. There appeared (1844) a new edition of the chief work with the addition of a second part, which, in sections corresponding to those of the first part, gives further developments of its content. Later he also published *Parerga und Paralipomena* (1851), two volumes of shorter, popular essays which illuminate various of his ideas. — It was the pessimistic mood, after the hopes of 1848 had been shattered and reaction had set in in every domain, that bred receptiveness to Schopenhauer's view of life. To this was added the dissolution of the Hegelian school, which in turn awakened a sense of and a need for the critical philosophy, which was after all so essential a part of Schopenhauer's system, just as he now also found greater support in his admiration for Kant, who for many had stood as a standpoint long since overcome. His brilliant exposition won him, once he began to become known, a large circle of readers. Greedily he received the incense now offered to him, and with strictness he held his disciples to procuring him every laudatory review that appeared. Not a drop of his late fame would he lose. His old age became a happy time for him, and in spite of his pessimism he wished for a long life. He himself had always followed the direction toward redemption from the ardor of the will given in the third book of his chief work more than that given in the fourth book. An ascetic he was not, although he admired the ascetics and stood with great emotion before the pictures of St. Francis and Rancé, these men who had wholly overcome the world within themselves. From the ardor of the sensual drives old age had now freed him, to his great relief. His friend and biographer GWINNER relates: "On this theme the old man overflowed with exalted thoughts and deeply moving feelings". *(Schopenhauers Leben.* Leipzig 1878. p.
[219] 526 f.). A stroke of the lungs suddenly put an end to his life on 21 September 1860.

b. The World as Phenomenon for Science

In the exposition of Schopenhauer's philosophy we follow the four books of the chief work, and the first point is then the world as representation or as phenomenon. Only sensation is immediately given, and it corresponds only to the changes of our body. A conception of the world as external comes about only through the understanding, which cannot be separated from sense perception, at once referring the sensation to an external cause, which is conceived as acting in time and as removed from our body in space. This act of the understanding does not come to our consciousness but takes

place involuntarily and unconsciously. At one stroke space, time and causality are released, these forms that lie preformed in our cognitive faculty. Only through causality in immediate connection with time and space does external observation become possible. This theory Schopenhauer sets in opposition to Kant's, according to which the concept of cause first comes into play when the sense images have been formed by means of the forms of space and time. But he agrees with Kant that the causal principle cannot be grounded on experience, since sense observation itself, after all, first becomes possible through its involuntary application. And from the causal principle there follow, according to Schopenhauer, in turn the law of inertia and the law of the conservation of matter as necessary consequences. — The theory of the principle of causality as at work in sense intuition itself Schopenhauer hardly developed except under the influence of Fichte, whose lectures "on the facts of consciousness" he heard in Berlin. A comparison between Fichte's and Schopenhauer's doctrine on this point establishes such an agreement that Schopenhauer, had his ideas been used in this way by another author, would surely have complained of brazen plagiarism and found in it a new testimony to human baseness. The theory mentioned has played a part in modern physiology of the senses, since HELMHOLTZ takes it as the basis of his work on the sensations of tone[50]). It is not conclusive, however; for it must be asked whether that capacity to project and to localize has not come about by development, and whether experience and the association of representations have not had an influence on this development. Schopenhauer denies this. "Physiologically," he says (Vierfache Wurzel, § 21), "the intellect [i.e. the cognitive *[220]* faculty, whose forms are time, space and causality] is a function of the brain, which the brain learns from experience just as little as the stomach learns to digest or the liver to secrete bile."

But when thus the whole way in which the world exists for us is due to forms of our cognitive faculty, the whole world is and remains for us only representation, or rather a series of representations held together by the principle of sufficient reason. This does not mean that it is a lie or a seeming; its empirical reality is not shaken thereby: our experience, after all, comes about precisely through the use of those forms. The consequence, on the other hand, seems to be materialism, since every phenomenon is to find its explanation through another according to the law of inertia and the law of the conservation of matter, which, after all, according to Schopenhauer are supposed to follow immediately from the law of causality. And it is precisely Schopenhauer's opinion that the goal and ideal of natural science must be a complete materialism. He even declares knowledge itself to be a product of the brain, and repeats (in his lecture manuscripts) a

well-known forceful saying of the French author CABANIS in the following statement: "Just as the stomach digests, the liver secretes bile, the kidneys urine, etc., so the brain secretes representations." Yet according to Schopenhauer materialism is right only so long as we are speaking of the world as phenomenon or representation. Materialism founders not only on the fact that the series of material causes goes on to infinity, and that it cannot explain the different forces of nature, but first and foremost on the fact that the whole materialistic world-picture is only our representation, not a thing in itself. The center of gravity of existence thereby falls back into the subject, whose states are what is immediately given. All matter exists only for a knowing being and in its representation.

We seem, then, to be moving here in a circle: matter produces a representation, and matter is itself only an object of representation! This difficulty falls away when it is held fast that time, space and causality — the principle of reason in its different
[221] forms — do not hold for the thing in itself. The world as representation (to which matter itself also belongs) is only the outer side of existence. As soon as we ask *what* it is that comes forth before us in the infinite series of phenomena, ordered according to the principle of reason, the principle of reason itself can of course not help us. Were we merely representing, knowing beings, the question could not be answered. Only when inner experience is brought into connection with outer experience can the answer be reached. The will which is the innermost in man must also be the innermost in the world. Only from man can the world be understood. Our inner being must have its root in what is not phenomenon but thing in itself. The will is Mephistopheles, who comes forth as "des Pudels Kern". If one does not accept this, nature in all its causal series becomes unintelligible; but the veil falls when we take what stirs in us as will to be one with what stirs at the different grades of natural causality. (The clearest statement of this is in "Der Wille in der Natur", at the end of the section on physical astronomy). — With this explanation Schopenhauer does not think he oversteps the limits of knowledge. There does arise, to be sure, the difficulty that our life of will unfolds before us in the form of time, and that the single acts of will are subject to the law of motivation (the fourth of the forms in which the principle of sufficient reason appears). How, then, can the will, which is itself phenomenon and is known only through representation, be made one with the thing in itself? This difficulty seems (as KUNO FISCHER *Arthur Schopenhauer.* Heidelberg 1893. p. 239 remarks) to have dawned on Schopenhauer only in working out the second volume of the chief work, which appeared 25 years after the first. He has to admit that the will too can only be phenomenon; but, he says, it is the phenomenon that is one with our own subject, that which stands in the most intimate

and closest relation to us, where we therefore have the thing in itself most immediately before us and in the very lightest veiling. It is the primal phenomenon (an expression Schopenhauer borrows from Goethe's theory of colors) by means of which we make all other phenomena intelligible to ourselves. If anyone should ask what, in turn, the will is in itself, no answer can be given to that. (Welt als Wille und Vorstellung. II. Ch. 18, 26 and 41). — It is clear that by this limitation of his solution of the riddle Schopenhauer in *[222]* reality admits that he has not solved the riddle. For a primal phenomenon is and remains, after all, a phenomenon, even if it is the one that lies nearest to us. The fundamental presupposition that precisely what lies nearest to us should be the kernel of existence Schopenhauer has not examined. On examination it would turn out that the whole problem of knowledge was really posed anew[51]). This point is of no slight significance for Schopenhauer's philosophy; for only if the will is absolutely one with the thing in itself has he the right to regard it as groundless, as raised above the law of grounding; if it is phenomenon, albeit primal phenomenon, it must in this respect share the fate of all other phenomena.

Even if the will is regarded as phenomenon, there is a question Schopenhauer has passed by, namely the one which Hume and Fichte, each from his side, raised: how far we can immediately observe ourselves as willing. Schopenhauer proclaims an immediate observation of the will, just as he proclaims that what this observation shows us is the innermost nerve of existence. His psychology is romantic, like his cosmology. And the psychological difficulty becomes all the greater since knowledge and will, in his conception, are supposed to be absolutely (toto genere) different. The will in itself is groundless, whereas knowledge everywhere works according to the law of grounding; the will is eternal and unchangeable, whereas everything we know, and knowledge itself, comes into being, develops and changes. Moreover, the will is the master of knowledge. It is the will that guides the course of our representations without our noticing it. And knowledge is from the first only a means for the will. In order that the individual may be able to satisfy the will to life, it must know its relations to other things; all our knowledge is, after all, only a sum of such relations. No wonder that knowledge cannot open to us the access to the Absolute! With full right the mysticism of all ages, especially Christian mysticism, has maintained the limitation of the natural light! — By regarding knowledge as an instrument of the will Schopenhauer becomes a forerunner of the modern doctrine of evolution, of which his very expression *will to life* reminds us. But Schopenhauer's psychological concept of will is very elementary. By *[223]* will he understands need, striving and craving (the Greek *θέλημα*), not the capacity for

deliberation and decision (the Greek *βούλησις*); he expressly wants (Neue Paralipomena § 149) the concept restricted to what is common to the animal and man. As he limits the concept on this side, he extends it on another side, in that he describes all feelings and emotions as manifestations of will and in consequence rejects the setting up of feeling as a distinct side of conscious life. Thus not only all striving and wishing, but also pleasure and pain, hope and fear, love and hate are manifestations of will. It is the constant, blind impulse of self-preservation, the will to life, that stirs in all these forms, that spurs on or checks the development of knowledge, and that gives consciousness unity and connection; on its identity, not on the identity of consciousness, the identity of the personality rests. — Schopenhauer's personal experiences have here led him to insist on differences which Romantic philosophy — especially the Hegelian system — was inclined to blur. From his own inner being he knew the sharp opposition between thought and need; his intellectual and aesthetic gifts led him in one direction, his sensuality, anxiety and irritability so often in quite other directions, and in these elementary forces he then saw outbursts of the obscure power that drives us and everything onward. It was a significant plea against intellectualism that was thereby entered, although it set up an unnatural cleavage as the normal state. This cleavage between knowledge and will is necessary to Schopenhauer's pessimism: for it is the blind, irrational will that explains to him that the world is — as it is.

c. The World as Will

What thus appears to our self-consciousness, our inner experience, as willing, appears, according to Schopenhauer, to our outer experience as our material body. To him this is immediately evident. There is no causal relation between the will and the body;
[224] they are one and the same thing, which is given to knowledge as body and to our self-consciousness as will. The difference arises from the different modes of apprehension, the outer and the inner. Muscular activity is thus not the effect of the will but its sensible phenomenon. For the will is identical not only with the brain but with the whole body, and not only with the force that moves the muscle but also with that which forms the muscle out of the blood. The different organs and functions correspond to the different drives. — Schopenhauer thereby gives the concept of will a new extension, beyond the domain of conscious life. It becomes for him one with everything that is called force of nature. The different forces of nature are only particular forms of a willing at work in the whole of nature. Matter is the visible form of the will. The difference between blind force of nature and deliberate action is only a difference of degree and concerns only

the phenomena, not the essence that reveals itself in them. When Schopenhauer thus leads the concept of force back to the concept of will, instead of — as is usual — going the opposite way, this is in accordance with his fundamental proposition that what is mediately known is to be led back to what is immediately known. All force of nature is conceived by analogy with what we know in ourselves as willing. What really goes on in impact, in attraction, in the turning of the magnetic needle, in the chemical process, in organic growth, becomes clear to us only when we conceive it as different degrees and forms of willing. In "Der Wille in der Natur" Schopenhauer seeks to show this in detail.

Schopenhauer's adherence to the identity hypothesis appears, like so many of his views, without closer grounding. It is not impossible that here — as in his whole doctrine of the will — he got his motives from Fichte, who in his lectures on "the facts of consciousness", at which he had Schopenhauer as a hearer, explained that the body is the outer, material form under which the ego must appear in order to be able to struggle with the material barriers, since matter can be displaced from its space only by another matter. In Fichte, then, a grounding is attempted; in Schopenhauer there is merely a proclamation. And besides, Schopenhauer commits a very great inconsistency in letting *[225]* the brain, without more ado, produce the representations. On this point, then, he speaks altogether materialistically and without feeling any sting, whereas otherwise he has scornful words enough for the materialists. His standpoint is this: the representation of the world is a product of matter (in the brain); but matter itself (the brain included) is only a phenomenon of the will, existing for sensible representation, while the will is the absolute reality. The unpsychological cleavage between representation and will shows itself here in its most glaring form. The Romantic artist's view of existence, which by Schopenhauer's own account led him to his system, has let him pass lightly over glaring contradictions. —

Schopenhauer's philosophy of nature recalls Schelling's. Just as Schelling seeks to show an ascending series of "potencies" in nature, through which matter rises to spirit, so Schopenhauer distinguishes a series of grades through which the will rises from the purely elementary forms to the clearly conscious ones. Lowest stands purely mechanical interaction, where cause and effect are not of different nature and where the relation between them can be made immediately intuitive. With the more special forces of nature (heat, electricity, etc.) the relation becomes less transparent because of the dissimilarity between cause and effect. And the causal relation becomes still more mysterious in the organic domain, where the cause appears as a rousing impulse (Reiz), since there is far more in the effect than in the cause. Finally, in conscious beings, the cause becomes

motive — but here, as it happens, self-observation reveals to us the inner nature of the causal relation, so that we discover that it is a willing. The will strives for the highest possible objectivation, that is, phenomenal, objective appearance. This striving is one with the very need to exist. Hence the infinite wealth of forms and grades in nature! Every grade attained is a barrier beyond which there is craving. The unity and kinship in nature are understood from its being one and the same will that stirs in everything; the diversity and multiplicity, from the infinite impulse to exist being unable to come to a
[226] stop at any grade or to be contained in any form. And the one form stands in the way of the other; hence the struggle in nature, especially in the domain of plant and animal life. The restlessness of the will shows itself also in the motions of the globes through cosmic space without rest and without goal. But in the world of living beings the inner discord in the world-will appears most clearly (die dem Willen wesentliche Entzweiung. Welt als Wille und Vorstellung, I, § 27). Everything presses forward to existence, if possible to organic existence, and then in turn to as high a form of life as possible, and in this throng struggle and mutual destruction arise. Men and animals devour one another and devour the plants, which in turn devour air, water and other substances. Everywhere "Gedränge und Gewirre", and therefore chase, anxiety and suffering! — Through pictures of the life of nature, which he takes from his rich reading, Schopenhauer illustrates this more closely.

In this struggle for self-preservation consciousness, among other things, then arises. It is from the first only a means of self-preservation, since it brings the advantage that motion can be released before the impression (der Reiz) comes, the motive anticipating it. At one stroke the world then comes into being as representation. But through the representations, its products, the will to life continues to work. When we represent life to ourselves as a good and therefore strive to preserve it and develop it, this is due to an influence, unconscious to us, of the world-will on our representations. It deludes us with goods and constantly awakens new expectations, so as to be able to cling to existence by new paths. We are ourselves one with this will to life; therefore we must live, and because we must live we think that life is good. We are driven forward from behind, while we believe we steer by our freely chosen purposes. This holds not only for the self-preservation of the individual but also for the preservation of the species through propagation. The individual feels here the strongest need, and in its satisfaction the highest pleasure — and yet he is thereby only a means for the will's striving after continued subsistence through the species. Even in the individual choice in the relation of the sexes he is led, without knowing it, toward the individual who together with him

can bring the best offspring into the world. At the bottom, everywhere, lies the blind, *[227]* universal need for existence.

While the optimist is blindly taken in by this obscure deceiver, the pessimistic thinker sees through the illusion and discovers that life is a business that does not pay. For the proof of this Schopenhauer refers to experience, which sufficiently establishes the suffering and nothingness of life. But a proof from experience would not be exhaustive. Therefore (see Welt als Wille und Vorstellung, I, § 58—59 and II, Ch. 28 and 46) the a priori consideration, which rests on the very nature of feeling or of will, becomes what is really decisive. Only the feelings of pain are positive: in them stirs the constant need and ardor that keeps life up and drives it forward. Whenever a passing damping of this inner fire is attained, the need being satisfied, the feeling of pleasure arises; its nature is negative, only the sign of a want filled. By an illusion it nevertheless stands before us as a positive state[52]). On the whole, then, the feelings of pain are also the strongest. We notice pain, but not painlessness, sorrow, but not freedom from sorrow, fear, but not security. Well-being is an entirely negative state. Health, youth, freedom — the greatest goods — we do not notice until they are gone. And while habit dulls enjoyment, it creates the possibility of new suffering if what we are used to ceases. — The wretchedness that thus lies at the bottom the great multitude does not notice; the geniuses discover it more easily, because the mental powers stir most strongly in them and their wishes are more lively; resistance and disappointment are then felt all the more strongly. — This whole result agrees very well with the assumption set up earlier, that the law of grounding does not hold for the world-will. The world-will is, both practically and theoretically, a problem, an irrational principle, something which cannot and is not to be comprehended. —

As one sees, Schopenhauer gets something quite different out of his philosophy of nature than Schelling does. Through the different grades of nature the tension grows stronger and stronger, until at last it breaks out into the light of consciousness and thereby gets its sharpest form. There is a more realistic character about his view of nature than about Schelling's[53]). This shows itself also in the distinction he draws *[228]* between the etiology of nature, the demonstration of the efficient forces and causes, and the philosophy of nature, the interpretation of the absolute essence that manifests itself in these forces. Had he only held fast to this distinction! But it does not keep him from sallies against the mechanical view of nature, which, to be consistent, he ought to have let unfold independently, so that he could then afterward attempt his metaphysical interpretation. And closely connected with this is the fact that he, as little as the other

Romantic philosophers, assumes a real development in nature, proceeding step by step in time. He does hint at the influence of need and use on the development of the organs, but he regards it as a mistake on the part of LAMARCK when the latter assumes a historical course of development from lower to higher species. Schopenhauer regards the different forms and grades of nature as expressions of the world-will, radiating from it, but not in real connection with one another. And yet with his "will to life" and by his great emphasis on the significance of throng and struggle in nature he stands as a forerunner of the doctrine of evolution in the form Darwin later gave it. He has an eye for the efficient cause Darwin appeals to, although he does not grant that this cause can produce the differences of species in nature. These stand for him as Plato's ideas, as eternal forms for the outbursts of the unbridled will; an origin they have not.

Even more than by his realistic tendency he comes into opposition to Schelling by his pessimistic interpretation. His personal conception of life here decidedly comes to color his cosmology, and his keen power of observation and deep indignation give his expositions, in so far as they move in the domain of human life, a highly marked character. It is through his philosophy of life that he has above all influenced wider circles.

d. Redemption through Artistic Contemplation

The knot is now tied; the question is whether it can be untied. It is characteristic of
[229] Schopenhauer's directions toward a solution that they address themselves exclusively to the individual. History is for him a play of accidents, like the frost flowers on the windowpane or the figures in a kaleidoscope, and he assumes no progressive development through the race by which evil could be eliminated. The will is the same at all grades — however different knowledge is! It is a great psychological riddle how this can be possible. Knowledge arises as a means in the service of the will; but however good this means may be, it gains no reacting influence on the will itself. On the other hand, in some cases it is supposed to be possible for knowledge to tear itself wholly loose from the service of the will, whereby man's individuality is then abolished and he is wholly absorbed in disinterested contemplation. Thus when one "loses oneself" in the beholding of a work of art. This upheaval and liberation, where the will vanishes and pure beholding reigns, can be explained only by a sudden breakthrough of the faculty of intuition. One now stands in a purely contemplative relation to the world — and one can do so only when one forgets that one belongs to it. With the will, suffering too falls away. Only through art does this become possible. Science goes on constantly

further in the series of grounds, and the will always strives restlessly further. But art is everywhere at the goal; it presents things in their eternal rest, "under the aspect of eternity". Schopenhauer especially praises Netherlandish art for the still, will-less state of mind that expresses itself in it, and that was necessary for so objective a beholding to become possible. But the highest art for him is music, which immediately presents the will, the world-will in its rising and falling, in its elementary and in its complicated forms, and gives us its secret history, its checks, struggles and torments. It is in reality we ourselves who are the taut, struck and quivering string! — But a strong exertion is required to hold fast the artistic relation to existence, and only few have energy enough for it. The will, with its never resting need and misery, always presses on. The geniuses are in a particularly high degree endowed with the capacity to enjoy in the image what one flees in naked reality. *[230]*

The opposition between art and life demonstrated by Kant and Schiller is driven by Schopenhauer to the uttermost extreme. He overlooks the sympathetic absorption in the object, which in turn presupposes that we ascribe a value to it. The value of art will in the end fall away if there is no value of life at all. With a variation on Goethe's album verse one could say: "If you would ascribe worth to art, you must ascribe worth to life!" — It is clear, further, that Schopenhauer necessarily needs a will for the very artistic liberation from "the will"; for — as he rightly says — exertion is required to hold fast the artistic contemplation. For the energy with which ideal purposes are upheld Schopenhauer has no expression, since he denies the possibility that the elementary "will to life" can undergo a metamorphosis. In return he must then posit one of the abrupt transitions in which his system is so rich.

e. Practical Redemption

Even in the geniuses the artistic redemption from life is only momentary and never complete. A perfect calming, a decisive quietive, art does not afford, only a passing consolation. The world-will, which stirs so mightily in us all, as if each of us were the whole, drives us at the same time to commit encroachments on one another. The deterrent authority of the state does indeed damp wrongdoing, but the true overcoming of egoism first comes forth when we bethink ourselves that it is one and the same being that stirs in us all, so that the tormentor in his innermost ground is one with the tormented, although in his illusion he thought himself an entirely different being from his victim. True remorse of conscience and true virtue arise from a presentiment of the illusion of individuation. Love of mankind in particular presupposes the knowledge of

the unity of all. Since every joy, according to Schopenhauer's psychology, presupposes the removal of a pain, love can aim only at relief and therefore appears as pity. In pity Schopenhauer sees "the ethical primal phenomenon", and he declares it inexplicable
[231] unless one presupposes the unity of all in the last ground. From it the highest virtues, justice and love, develop.

But the absolute quietive is attained only by those who wholly deny the will to life in themselves through complete resignation. Only the great ascetics and saints reach so high that the will to live no longer works in them. Asceticism, then, is not mortification in order to attain blessedness in another world, but is the involuntary consequence of the impulse of self-preservation, the need to continue one's own existence and that of the species, having fallen away. In Buddhism and primitive Christianity examples of this are found. For him who has seen through the torment of existence and the illusion of individuation there are no more motives, only quietives. Thereby one glides over into Nirvana, a state that stands as a nothing for those whose highest reality is the world of sense. Schopenhauer expresses himself on this in the following way (Welt als Wille und Vorstellung. II. Ch. 48): "At its highest point my doctrine takes on a negative character, and so ends with a negation. For here it can speak only of what is negated or given up; what is gained and grasped instead it is compelled to describe as nothing, and it can only add the consolation that it is only a relative, not an absolute nothing. For when something is nothing of all that we know, then it is certainly nothing at all for us. But from this it does not follow that it is absolutely nothing, namely that from every possible standpoint and in every possible sense it should be nothing. It follows only that we are limited to a purely negative knowledge of it, and this may very well be due to the limitation of our standpoint." (Cf. with this Schopenhauers Briefe, herausg. von Griesebach. p. 210 f; 359). — The philosopher sees in the involuntary dying away the highest form of redemption. But just as little as the saint needs to be a philosopher, just as little does the philosopher need to be holy. The philosopher has only to set forth his conception of the world in clear concepts, and he then finds the ascetic's ecstasy to be the highest ideal, an ideal before which he bows in reverence, even if for his own person he goes other ways. —

Here again — as with artistic redemption — Schopenhauer fails to give us an explanation of where the energy, the *will*, comes from that can tear us loose from "the will to
[232] life". Even if it is a negative, an inhibiting will, it is nevertheless a will that is required, and in the ascetic far more clearly than in the genius. Besides, this break with the will cannot, after all, take place without a break with the will's forms of manifestation, which

were supposed, after all, to be subject to the law of grounding! Already by his declaration that pity is a psychologically inexplicable phenomenon he sins against his own theory of knowledge. At several points (besides those mentioned, also in his explanation of spiritualistic phenomena, which occupied him much in his later years) he lets, in complete conflict with the spirit of the critical philosophy, "the thing in itself" break through the series of phenomena. Quite apart from this, there appears for Schopenhauer the great difficulty that presents itself in every attempt to derive everything from one principle, namely how opposition and discord within the one world-will (die dem Willen wesentliche Entzweiung) is possible, and further, how it is possible to explain the diversities of phenomenal existence from the one principle that stirs in everything. It was this problem that Böhme wrestled with in his religious speculations; it was this that Tschirnhausen put to Spinoza and Eschenmayer to Schelling. One of Schopenhauer's disciples, Frauenstädt, made a similar objection to his master. Schopenhauer's answer (in letters from the autumn of 1853) really maintains only that the diversities *must* have their ground in the thing in itself, but does not say how[54]).

Schopenhauer's radical pessimism is a phenomenon of great interest for the history of culture. On the way in which he seeks to ground it, not much weight is to be laid. Neither his proofs from experience nor his attempt at psychological deduction hold good when they are to ground absolute pessimism. Yet it would not be justified to conceive the whole as a mere outburst of his peculiar temperament. Every significant individuality is a point of vantage for the human race, from which it catches sight of possibilities and of sides of existence that would otherwise not be brought out. And the energy and freedom of spirit with which Schopenhauer lays bare the disharmonies and the dark sides in nature and culture have brought it about that the problem of valuation has entered an *[233]* entirely new phase. Veiling of facts and blunting of problems will now on this point be more difficult than before. That has its great theoretical and practical significance. And this gain is not shaken by Schopenhauer's truly Romantic attempt to make his feeling of life the feeling of the whole of existence. While his antipodes within the Romantic doctrine of development always strive, after all, to hold back their personalities, so that the content of thought might develop according to its own laws, Schopenhauer's philosophy has a more individualistic character, also in that he makes arbitrary leaps of thought where a transition is to be formed between ideas each of which stands firm for him in his individual philosophy of life, although he cannot show any objective, really grounded connection between them. Fortunately, by his own epistemology, which takes up Kant's investigations again, he has himself put forward correctives. He stands, like

Schleiermacher, as an interesting example of how critical philosophy and a markedly individual conception of life are united. By their very personalities these men illuminate the great problem of the relation between thought and life, even if their own solution should not be the definitive one.

C. An Undercurrent of Critical Philosophy in the Period of Romanticism

Schleiermacher and Schopenhauer stand as eminent testimonies that Kant's thoughts had not come into their full right through the constructions of Fichte, Schelling and Hegel. Kant was no superseded standpoint; in the midst of the apparent sole dominion of Romantic speculation he always had a circle of faithful and understanding followers. There was always an undercurrent which, though it was not able to break through the speculative overcurrent before it got help from natural science and historical criticism, is nevertheless of great importance, partly for philosophical inquiry itself through its results,
[234] partly for the history of philosophy as a testimony that the continuity in the history of thought was not broken off by the speculative philosophers' adventures of thought. First and foremost we must mention here a whole series of men who did not come forward with the claim to be philosophers by profession, but whose view of life had been determined by Kant's philosophy, and who faithfully preserved this view of life during their activity in larger or smaller circles. In several regions of Germany associations were formed of men of the most various age and station for the common study of Kant, and there were many individuals, in part among the most distinguished men of the time, who were able to preserve the kernel of the master's thought while disregarding the inessential in the forms in which he had clothed it. It is of particular interest to see that in the circle of the men to whom Germany owes its spiritual and political rebirth after the defeats at the beginning of the nineteenth century there are not a few of Kant's disciples. Besides Schleiermacher and Fichte we must mention here Wilhelm Von Humboldt, who organized the educational system in Prussia, and Theodor Von Schön, who carried through the abolition of serfdom. "In contrast with the raw reality we see the statesmen of that time approach the problems of agrarian legislation, of freedom of trade and of the constitution of the towns with the idealism of the Königsberg school" (Gneist.) As regards Denmark, we may mention here the great jurist A. S. Ørsted, who in his youth was an ardent Kantian, and who in his old age declared *(Af mit Livs og min Tids Historie.* Copenhagen 1851. I. p. 35) that the influence Kant's ethics had had on his mind had

never at any time been lost. Ørsted says in this connection: "When one of the statesmen who had had a part in leading Prussia's rebirth, and who had always remained true to the spirit from which it went forth (Minister of State von Schön), received on the occasion of his official jubilee a beautiful and warm tribute from his fellow citizens, he testified that if he had accomplished anything useful for his fatherland, he owed it to the way of thinking which his great teacher Kant had implanted in him, and that he must therefore lead the thanks that had been brought to him back to the source of which he was only a little brook. Without otherwise being able to make any comparison, I am seized by a similar feeling when I think about what Kant has been to me and to many *[235]* others in my fatherland." Yet it was not only Kant's influence as an ethicist that held its ground despite the Romantic overcurrent. The strict demand he had made of grounding, and the critical understanding he had gained of the limits of knowledge, made a great many of his disciples regard the speculative turn that philosophy took with his most prominent successors with indignation and contempt. In an exceptionally clear and interesting way this is seen in the ardent Kantian, the physician Joh. Benj. Erhard, whose autobiography and correspondence Varnhagen Von Ense has published. In a letter of May 19, 1794, he says (with reference to Fichte): "The philosophy that *proceeds from* one fundamental proposition and dares to derive everything from it is and remains always a sophistical trick; but the philosophy that *ascends to* the highest fundamental proposition and presents everything else in perfect harmony with it, without deriving it from it, is the true one . . . Kant's philosophy does not reign at all among his disciples; for they want to make reason *constitutive* . . . I have already once written to Reinhold about it and proved to him that a theory of representations is impossible; only an analysis of them is possible." All our judgments, Erhard next shows, are formed by analysis. This holds in the empirical sciences, and it holds also in philosophy, where "we attain all our knowledge by *analysis of the concepts formed by us without reflection;* if one does not attend to this, one constructs for oneself a system of the original concepts, instead of knowing the actual system." The misgivings Erhard expresses here were shared by several of his friends, thus also, as already suggested, by Schiller. The famous jurist Anselm Feuerbach likewise says in a letter that "he who has drawn nourishment from Kant's spirit and knows that play with empty concepts is not philosophy" cannot share the enthusiasm with which the new philosophy was received by most. *(Biographischer Nachlass.* 2. Ausg. I. p. 51). And Wilhelm von Humboldt felt, precisely at the time when Hegel stood at the height of his fame, the need to give a character portrait of Kant permeated by deep admiration, in which Kant is praised for having, by his critical work, given philosophical analysis its *[236]*

proper foundation, for having united "a perhaps unsurpassed dialectic" with "a sense for the truth that cannot be reached by this way", and for having taught philosophizing rather than philosophy, for having awakened to seeking rather than communicated results. *(Ueber Schiller und den Gang seiner Geistesentwickelung.* — Introduction to *Briefwechsel zwischen Schiller und Wilhelm v. Humboldt.* Stuttgart und Tübingen. 1830. p. 45—49).

At the same time as Erhard expressed his misgivings about the constructive turn in philosophy, two of Fichte's hearers in Jena worked out critiques of the "Wissenschaftslehre" which became for them introductions to long labor of thought. Fries and Herbart both set themselves the task of working further in Kant's spirit, since they could not acknowledge as the right one the way by which Fichte wanted to carry Kant's philosophy further. Each in his own way they seek to give a further elaboration to the psychological foundation on which Kant's philosophy in reality — though he was loath to admit it — rested. Thereby they gain great importance for the history of psychology. They maintain the right and importance of empirical psychology over against the speculative constructions. Their sober intellectual tendency and their able insight into natural science place them in several respects as representatives of a line of thought that one does not usually date back to the period of Romanticism. In line with them there later came Beneke, whose merits also lie mainly in the domain of psychology. If these three investigators were able only to maintain and continue an undercurrent, not to break through the overcurrent, it came from the subdued, reserved quality of their nature. Their domain was the empirical and criticism, not the world of striking ideas and brilliant views. But their quiet work has borne its fruit.

a. Jacob Friedrich Fries

Like Schleiermacher, Fries came out of the Moravian Brethren and bore marks of it in his spiritual life, partly in the weight he lays on feeling as the foundation of faith and
[237] presentiment, partly — by a natural effect of contrast — in the criticism he directs at the influence feeling can have in making knowledge dogmatic and in paralyzing the will. And like Schleiermacher he worked his way by independent thinking out of the confessional limitation that reigned at the Herrnhut school in Niesky, where he came in the year 1792, 19 years old. According to what he himself has stated (in notes used by Henke: *Jacob Friedrich Fries.* Aus seinem handschriftlichen Nachlasse dargestellt. Leipzig 1867), it was personal experience and psychological study that brought him away from positive faith. In vain he had exerted himself and struggled to force in himself the devout moods that were demanded in the Herrnhuters' many quiet hours;

he now saw that it was his own natural feeling and imagination that he had wanted to drive up to an artificial height. He also came to have ethical misgivings about the doctrine of atonement. But religious life did not thereby lose its importance for him. Like Schleiermacher, he found a symbolic value in the religious representations, and despite all dogmatic negation he felt spiritually akin to the Moravian Brethren, with several of whose members he kept up a warm friendship all his life. His philosophical development was determined by his being introduced to Kant's philosophy already at the Herrnhut college, where it was presented to the pupils especially in the form Reinhold had given it. Only in secret could Fries read Kant's own works, and here it particularly interested him to see the way by which Kant had come to his results. The psychological analysis on which Kant laid so much weight in his early writings, and which in his "Kritik der reinen Vernunft" is called the subjective deduction, but which he pushed more and more aside, became in Fries's eyes the most important thing. But Fries already now missed in Kant a thorough treatment of the psychological foundation of epistemology, and he set himself as his life's task to bring it about. After leaving the Herrnhut college he studied in Leipzig, where the psychologist Plattner gained a great and lasting influence on his views, and then in Jena, where Fichte was just then at the height of his power and his fame. Fichte's Wissenschaftslehre did not satisfy him at all. He found *[238]* his first demands on scientific method unfulfilled by this attempt to set up one single highest principle and to proceed from it. First psychological description, then analysis and abstraction — and only then, if possible, a construction, which can nevertheless never become anything but hypothetical — these were the fundamental propositions Fries had already gained, partly by his own thinking, partly by his ardent study of mathematics and natural science, which he had begun in Niesky and continued in Jena. Even before he came to Jena, he had worked out a series of essays in which he showed the importance of empirical psychology for the philosophical problems. The objections he had to make against speculative philosophy, and which Fichte's lectures aroused in him, he wrote down, and on the basis of them he later worked out his polemic: *Reinhold, Fichte und Schelling* (1803, published again 1824 under the title *Polemische Schriften.* I), a book which first made his name known, and which still has great interest through its characterization of philosophical method. For some years he lived as a private tutor in Switzerland and continued his philosophical and scientific studies. Then he worked as a lecturer in Jena, in opposition to the speculative philosophy that reigned in this hearth of Romanticism. As professor in Heidelberg he published *Wissen, Glaube und Ahndung* (1805), a popular presentation of his epistemological and religious-philosophical views,

and then his chief work *Neue Kritik der Vernunft* (1806—7), in which he took up Kant's line of thought and sought to correct it and carry it further. Besides Kant he was also influenced by Jacobi, whose emphasis on immediate consciousness and on feeling agreed with his conviction of the fundamental importance of psychological experience. But while Kant wanted to prove too much and thereby led astray into a new dogmatism, Jacobi wanted to prove too little and thereby really remained standing outside philosophy. In a polemic (against Schelling), *Von deutscher Philosophie Art und Kunst* (1812), Fries has developed more closely his relation to Kant, Jacobi and the philosophy of Romanticism. Throughout the last part of his life (1816—43) Fries worked as professor in Jena. Because
[239] of his liberal political views, which he had developed in novels and pamphlets, it was a special joy to him to come back to Karl August's duchy, since this little country was the only one where the promises given by the princes after the Wars of Liberation, of the introduction of constitutions of estates, had been fulfilled. As Kant's disciple Fries laid great weight in his ethics on the feeling of personal dignity and insisted that a vigorous personal life could unfold only in a public life permeated by the ideas of honor and justice; "it is", he says in his Ethics published in 1818 (p. 377), "the disgrace of peoples to slave for the private purposes of single men or single estates, and the true demand for civil freedom amounts to this, that only the public purposes are valid in the state". In this spirit he sought to influence the academic youth, who now after the war had returned to their studies; he wished thereby also to work against the coarseness by which the earlier student life had been marked. Energetically he warned against secret societies, whereas he sympathized with the formation of a general German students' union, by which the bond knit by the common struggle against the conqueror of the world could be preserved. His participation in the Wartburg festival of 1817, where speeches were held in this spirit, and where, in imitation of Luther's burning of the papal bull, a number of (really or supposedly) reactionary books were burned, raised a storm against him which Karl August, who was favorably disposed toward him, sought in vain to avert. From the Prussian and Austrian side his dismissal was demanded, and when Sand, one of Fries's hearers, had murdered the writer Kotzebue (who stood as a symbol of reaction and foreign influence), it became necessary to give way, yet in such a way that Fries gave up only the professorship of philosophy and instead took over a professorship of physics. Already in Heidelberg he had combined teaching in physics and in philosophy, a proof of the thoroughness of his scientific training. He has published a mathematical philosophy of nature, a physics and a popular astronomy, works that won him the recognition of Gauss and Alexander von Humboldt. This side of

Fries's activity is significant in that he maintains the strict mechanical view of nature, which the Romantic philosophy of nature had believed it could push aside as an accident. *[240]* It was Fries's conviction, which he has expressed among other places in his *Geschichte der Philosophie, dargestellt nach den Fortschritten ihrer wissenschaftlichen Entwickelung* (1837 —40), a work outstanding for its time, that it is the strict scientific conception of the phenomena of outer and inner experience that first really gives the philosophical problems of modern times their sting. From this side he sympathizes with Spinoza, whose identity hypothesis he embraces, while he opposes Spinoza as a constructive philosopher. In his *Psychische Anthropologie* (1820—21) he builds on self-observation and physiology and gives interesting contributions both to the doctrine of sense perception, to the doctrine of the association of representations (the psychical mechanism) and to the doctrine of the unity and activity of mental life at all stages. With all its defects this work is a forerunner of the psychology of the most recent times. —

Fries always regarded himself as a Kantian. Two years before his death he still writes to his friend and disciple the theologian De Wette: "As regards the belief in Kant's victory in the future, I think thus: when clear and firm science is once again demanded in the philosophical domain, then we shall be proved right. But when that will happen, I do not know". (Henke p. 268). But it was not his intention to approve Kant's philosophy as it stood. He missed a psychological foundation for the self-knowledge in which, according to Kant, the critical philosophy was to consist. Self-observation is to show us in what forms our knowledge (reason in the widest sense, working both in sense perception and in thinking) involuntarily works. From psychological experience the fundamental concepts that express these forms are then extracted by abstraction (not induction). But the result we can reach in this way can only be probable; there is no absolute guarantee that we have really found the right fundamental concepts. Only that knowledge can be apodictic which builds on the forms of the self-activity of the knowing mind; but this self-activity is never unconditioned, since it must always be awakened by impressions it does not itself produce; and even where we (as in logic and mathematics) have found pure *[241]* forms of thought by abstraction, the knowledge that can be built on them is only purely formal. Nevertheless Fries assumes that a complete system of fundamental concepts (categories and ideas) can be set up, by whose help we can survey the whole domain of the knowledge of nature. He thinks that Kant succeeded in the complete enumeration of the fundamental concepts, and in his own exposition he therefore also follows Kant's system. There is a problem here that Fries has not seen sharply. He admits that all psychological experience is *fragmentary:* it is only by reflection, by "re-consciousness",

an attention directed toward our own involuntary mental activity, that we discover the forms we use immediately and involuntarily, and this reflection takes place successively and occasionally and is never completed. But nevertheless we are thereby to discover the *constant* and *unchangeable* way in which our reason works. (See especially "Neue Kritik", I. p. 198—200). Rightly Fries says that "the whole secret of philosophy lies hidden here." This secret he has not unveiled to the degree he himself thinks. Both Kant's systematics and Jakobi's faith have here exercised too great an influence on him and prevented him from carrying the analysis sufficiently far. — Add to this, moreover, that Fries speaks mostly of the psychological analysis that shows the subjective condition of knowledge, not so much of the logical analysis by which we go back from what is objectively given to the conditions of its being understood (Kant's objective deduction). —

Another chief objection to Kant on Fries's part is that Kant nourished the illusion of being able to give a proof of the objective validity of knowledge. Fries agrees with Maimon that Kant has only shown that we in fact have and use certain categories (qvæstio facti), but not that we have the right to use them (qvæstio juris). The objective validity of our knowledge cannot be proved. We cannot compare knowledge and existence, but only mediate, reflected knowledge with immediate knowledge. Truth does not mean the agreement of knowledge with its object, but the agreement of mediate knowledge with immediate knowledge; every grounding therefore becomes in the end subjective. (Neue
[242] Kritik. I. p. 288 — 295. Polem. Schr. p. 124, 351—354). Kant pointed to this but did not hold fast to it, which in turn has led his speculative successors to confuse the relation between the subject of knowledge and its object (a relation that does not fall outside knowledge itself) with a causal relation, and to revel in mystical ideas of the unity of thinking and being.

Against speculative philosophy Fries maintains not only that we can set up no conclusive *constitutive* principles, but only regulative ones. He states besides that the real philosophical work in any case lies in the regressive, analytical method, which on the basis of what is given leads to the discovery of the fundamental concepts that condition understanding. Speculation leads easily to intellectual laziness; only criticism necessarily makes one diligent. Idealism and dogmatism are not — as Fichte thought — the greatest opposites. Idealism can be just as dogmatic as materialism, when it does not examine its own presuppositions. Only in their results, not in method, do they stand opposed to each other; but of the greatest importance is the opposition that concerns method, the real art of philosophizing, and here it is criticism that forms the opposite of all dogmatism, whether this appears in idealist or in materialist form (Polem. Schr. p. 257). The system

is not so important as the method. The system has importance by bringing order and clarity into our knowledge; but whoever hopes to attain more by its help, whoever wants to extend his knowledge by the system, deceives himself (Neue Kritik § 70).

The limit of our knowledge is marked for Fries by the fact that in science we can form no completed series, can reach no concluded whole. Everything in material nature is subject to the laws of physics, everything in mental nature to those of psychology, and by way of analogy we can imagine that the same relation between a material outer and a mental inner is found everywhere. But even by this way we do not get beyond the finite — and there is on the whole no scientific way from the finite to the infinite and eternal. Only by faith can the eternal be grasped. Faith arises when the limits to which our knowing is always subject are thought of as removed — hence by a negation. Every phenomenon, *[243]* every object of knowing is limited, and the faith that an eternal being lies at the ground of the world of phenomena can therefore arise only by way of negation. Positive concepts of the eternal we cannot form. Every positive representation of it (as in the usual notions of God and immortality) is figurative, and if one wants to shape it into knowledge, one gets mythology. Likewise it is impossible to derive the finite (the phenomena) from the eternal; the speculative attempts at this lead only to philosophical novels, which everyone tells in his own way. There is only one truth; therefore it is one and the same reality which we regard in science as the finite world of phenomena and in faith as upheld by an eternal principle, — just as it is the same phenomenal world which we regard in physics from the outer, in psychology from the inner side. Like Kant, Fries distinguishes the ideas, as representations of absolute totality, from the concepts, whose object is always limited and relative. But he parts from Kant and the Romantics by maintaining that the ideas arise by a negative way. For faith, however, more is required than merely the negation of the limits; faith is a conviction grounded on interest, on the feeling of value. And even if the content of faith cannot be represented except in a symbolic way, certain phenomena can nevertheless be interpreted as revelations of the eternal that is inaccessible to thought. In the beautiful and the sublime in nature the presentiment of the eternal as the true essence of things has its foundation, and the highest beauty and sublimity is that which a human being's personality can offer. For Fries there is a close kinship between aesthetic and religious feeling, far more than in Schleiermacher, of whom his philosophy of religion, however independently it is developed, reminds us in a high degree. *(Handbuch der Religionsphilosophie und philosophischen Aesthetik.* Heidelberg 1832). Since Fries's theoretical philosophy was far more definitely and fully developed than Schleiermacher's, he also keeps himself freer than the latter from any real

or apparent adaptation to church doctrine. His symbolism is far freer and less dogmatic. In Christianity he criticized especially its favoring of passive and humble feelings and
[244] its doctrine of atonement, which conflicts with ethics. In Kant's idea of man's personal dignity he found the clear and complete development of a thought which the Greek and the Christian teaching had not yet been able to carry through, a thought which can reach its realization only through the public life in the state; for only there is man's personal cultivation completed. Through his calling the individual is bound in his inner development to society, and in this Fries finds the justification for regarding ethics, with Aristotle, as a part of political theory, although the fundamental thought of ethics stands higher than all political theory. —

Scorned by Hegel, and still scorned by those who nourish a Romantic admiration for the philosophy of Romanticism[55]), the sober-minded investigator has nevertheless developed, in his theory of knowledge, in his psychology and in his ethics alike, thoughts that constantly prove their validity and value.

b. Johann Friedrich Herbart

This thinker — the most eminent of the men of the critical undercurrent — called himself a Kantian of 1828[56]). This implies that he at once places himself on the foundation for philosophizing given by Kant and thinks he has carried thinking a step further. His admiration for Kant did not exclude a sharp criticism, which went still further than Fries's. With Fries he has in common what the latter called the regressive or analytical method. Already as Fichte's hearer in Jena he became clear that one cannot derive everything from one principle. A principle that included everything in itself could only be the conclusion of thinking, not its beginning. How will one get the derivation going from one single principle? How can one thought pass over into another thought? Already here one notices the tenacious adherence to the principle of identity, to the proposition that every thing is what it is, which is the characteristic mark of Herbart's thinking. All becoming and change contain a contradiction, since identity is thereby constantly canceled. The Romantic systems with their evolutions from an absolute principle were therefore for
[245] Herbart an uninterrupted series of contradictions, constant sins against the fundamental law of thought. These misgivings Herbart laid before his teacher Fichte, who could not but acknowledge the acuteness of his young hearer, although it went against his own favorite concept of the ego, which Herbart found altogether self-contradictory, because the ego was to be at once unity and plurality, and because it was to become and unfold itself.

It was against great resistance from those nearest to him that Herbart, who was born in Oldenburg on May 4, 1776, followed his interest in philosophical thinking, with which the interest in theoretical and practical pedagogy went hand in hand. After his student years in Jena he lived — like several other German philosophers — some years as a private tutor with a patrician family in Switzerland, and here the real foundation of his philosophy came into being. He agreed with Kant that experience shows us only phenomena. But while Kant carefully wants to hold fast to the opposition between phenomenon and thing in itself, Herbart argues (in a sketch of a "Wissenslehre" worked out in Bern) that representation must nevertheless in the end always point to something that is represented, and that is not itself in turn a representation of something else, but is different from all representing. What is different from all representing (which Herbart later calls the reals) we must think in such a way that the contradictions contained in the concepts of experience (first and foremost in the concept of the ego, from which Herbart took his point of departure) fall away. With this the program was given with which Herbart in the year 1802 began his activity as a lecturer in Göttingen. — His course of development, which offers no small philosophical interest, is described in an essay by Robert Zimmermann: *Perioden in Herbarts philosophischem Geistesgang* (Sitzungsberichte der philos. hist. Klasse der kaiserl. Akademie der Wissenschaften. 83. Bd. Wien 1876). Herbart's later life offers no striking features. The quiet, conservative man was absorbed in his studies and lectures and in his pedagogical endeavors. After some years of activity in Göttingen he occupied Kant's chair in Königsberg for a long series of years, but his last years he spent again in Göttingen, where he died in 1841. — Already in two significant writings of 1808 *(Hauptpunkte der Metaphysik and Allgemeine praktische Philosophie)* Herbart had stated his views. His investigations concern partly metaphysics (which for him means both epistemology and the treatment of the problem of existence), partly psychology, partly ethics. In his *Einleitung in die Philosophie* (1813) he gave an excellent propaedeutic, which has lasting value for all who want to learn to philosophize without at once swearing by a system. Although the fundamental thoughts of Herbart's own philosophy naturally come forward at several points, this book is carried throughout by the spirit of the critical philosophy, the very spirit of seeking and testing thought. *Lehrbuch zur Psychologie* (1816) carries further what "Hauptpunkte der Metaphysik" (§ 13) had already suggested as "elements of a future psychology". Even before Herbart published this textbook, his great psychological work *(Psychologie als Wissenschaft, neu begründet auf Erfarung, Metaphysik und Mathematik)* had been worked out, although it appeared only in 1824—25. Finally he presented in *Allgemeine*

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Metaphysik nebst den Anfängen der philosophischen Naturlehre (1828—29) his treatment of the problem of knowledge and the problem of existence.

α. From experience one proceeds, and one must proceed. But, says Herbart, experience does not immediately give us any knowledge. It is not knowledge, but is first to become so by being thought through. The compulsion experience exercises over us is an insurmountable point of departure, beyond which no line of thought is possible. But if we cannot go back, we can go forward. And we must go further, since the sensations we receive do not lie in us as a loose aggregate, but order themselves in forms and series that present tasks for thinking. Until these tasks are solved, thinking gets no rest. It is perhaps a labor of patience that is required here, when the demands of strict logic are to be reconciled with what experience offers. But here one is nevertheless always on firm ground, since one builds on experiences and does not construct castles in the air. The age has been incredibly spoiled by those who simply laid down one single principle and one single method, and this has led to the corruption of the sciences and has awakened
[247] in many an aversion to philosophy. Only by regular investigation can this aversion disappear.

In every sensation something certain and definite is given, which must be taken as it is. An *absolute position* is given. This is not to be understood as if sensation were a copy or immediately gave us a knowledge of things. Things in themselves we do not know — that is a proposition dogmatism will never be able to overthrow. But we know that they are; by sensation itself they are posited. And even if we want to call what our sensations contain seeming, this very seeming becomes unthinkable if one does not assume a being. The proposition must therefore be set up: *So viel Schein, so viel Hindeutung auf Sein.* (Hauptpunkte. p. 20). Every single sensation points to a single thing that is, a single position. It is, says Herbart, a chief proposition in all metaphysics, first set up by the old Eleatics, that *what is is absolutely simple.* As soon as it is thought with inner oppositions, a task arises for thinking, since identity is canceled. But this does not exclude that there can be several things that are, since each single thing that is is the object of absolute position. The relation between the different things that are concerns only thinking, which compares and combines, not the things that are themselves, each by itself.

Now when experience shows us becoming and change, we have here a seeming at which we cannot stop. The task is to find the being that lies at the ground. The problem had first dawned on Herbart through Fichte's ego, which was to be in constant self-activity. Later he saw that the same problem lay in all change, and he therefore

also later put Fichte side by side with the old Heraclitus; the former's assertion: "The ego posits itself," and Heraclitus's: "All flows", are both grounded on experience, but at neither of them can one stop. Herbart adds yet another problem to these, the so-called problem of inherence, which arises because one and the same thing is to have several properties. Under this problem too the concept of the ego falls, insofar as the ego is to include an inner multiplicity in itself. *[248]*

The necessity of assuming a *multiplicity* of things that are (reals) springs from the attempt to explain the experiences of changes and of things with several properties; — to explain means for Herbart to remove the contradictions. Every thing is what it is. When it now appears in experience with another property than before, this cannot be understood by the thing itself alone, but requires the assumption of one or more other reals, to which our conception relates it. The difference with which the real A now appears to me in comparison with its earlier appearance can be explained by the fact that I now think it together with B. A has not changed, but I have put it together with B (in itself equally unchangeable). Thus A keeps its unchangeability, although experience shows it to us changed. The causal principle expresses, according to Herbart, precisely this, that a thing is not changed by acting; the causal relation is thus a timeless relation! Activity (which is indeed a change) appears in experience only because we set the thing in relation to another thing. And when a thing appears with several properties, this likewise is only because we compare it with other things. Put together with B, A looks different from when it is put together with C or D or E etc. (One and the same thing is different in the dark and in the light, for the eye and for the ear etc.). — The method Herbart uses to remove the contradictions of the concepts of experience he himself calls the method of relations *(Methode der Beziehungen).* Not in the single real itself, but in the relation in which we set it to other reals, lies the ground of what experience shows us. But this relation is inessential, accidental for the real itself. It is a matter of complete indifference to A whether we compare it with B or not. Our setting in relation and comparing is an accidental point of view *(zufällige Ansicht).*

Although Herbart, against Kant, maintains the necessity of inferring from the phenomena to the things in themselves (from "Schein" to "Sein"), he nevertheless really comes himself to the result that there is a very great opposition between the phenomena and the reals. In "Psychologie als Wissenschaft" (§ 149, Anm. 2) Herbart states consis- *[249]* tently that to appear as phenomenon (das Scheinen) is no essential property of what is, but that every true explanation of the sensible world must lead to presenting being a phenomenon (das Scheinen) as purely accidental to being. Being and phenomenon

then have wholly different natures. The true reality does not become, is not altered, is not increased or diminished; it is subject to the strict law of identity. It is what it is and needs no development. There is a multiplicity of reals, and from this it is understood that we, who cannot help putting together and comparing, experience changes and compound properties. But each single real is in itself (as absolute position) independent of all others. — The peculiar character of the yield of Herbart's attempt to correct the concepts of experience appears in two propositions: 1) Im Reiche des Seins giebt es gar keine Ereignisse (Allg. Metaphysik § 235); 2) Jedes Kontinuum ist von der Realität ausgeschlossen (Allg. Metaphysik § 209).

Now Herbart obviously removes the supposed contradictions from the reals only by laying them over into our representing and thinking. If nothing is to happen in the world of the reals, things must go on all the more lively in the world of representations and thoughts, where indeed, among other things, those contradictions arise, are treated and are resolved. The inner unrest that is felt before the problems are solved Herbart has himself felt and described. So these great inner events should then be only seeming! Since Herbart grounds his psychology not only on experience but also on metaphysics, this consequence cannot be kept away. It would nevertheless be a misunderstanding to regard Herbart as an extreme realist. Of the real nature of the reals we can, according to him, know nothing, hence not whether they are material or mental either. Undeniably the materialist notion of absolute, unchangeable atoms agrees better with Herbart's reals than the notion of mental beings would do; and Herbart has himself stated (against metaphysical idealism) that the experiences from inner nature have no prerogative over the experiences from outer nature in determining the notions of the nature of the reals.
[250] Nevertheless he cannot help using analogies from our mental phenomena. So when he calls the real's preservation of its identity with itself despite the relation to other reals *self-preservation.* The concept of self-preservation is justified only when there is a real resistance to overcome in order that identity may be preserved, and when a work is performed, when, therefore, something *happens.* Self-preservation is acting (and so far change), not merely being. It is the self-preservation of the reals that comes forward especially in the beings to which we ascribe force and life, and which in our inner experience comes to our consciousness as sensation. And Herbart admits that the only example of self-preservation accessible to us is our own sensations (Allg. Metaphys. § 329). It then becomes clear, and is also admitted by his most acute disciple and interpreter (Drobisch: *Ueber die Fortbildung der Philosophie durch Herbart.* 1876. p. 20), that Herbart in his way — like metaphysical idealism — thinks the reals by analogy with our own

inner states[56 b]). But two contradictions remain standing in his system: between the two propositions that from seeming one must infer being and that seeming is accidental to what is — and between the two propositions that what is is unchangeable and that what is strives after self-preservation.

β. The conflict that has shown itself here between Herbart's metaphysics and psychology he pushes aside by virtue of the presupposition that metaphysics, which is to test all concepts of experience, must also form the foundation of psychology. Psychology is to build not only on experience but also on metaphysics, and, as we have already seen, the problem of the ego is merely a particular form of the problems of inherence and of change and is solved when these are solved. The soul is a real like other reals; its sensations and representations are its expressions of self-preservation. A sensation arises in the soul when it has to preserve its nature over against another real. And since the real that lies at the ground of the mental phenomena is now assumed to be another than the one that lies at the ground of the material phenomena, Herbart lands in a spiritualism that differs from the usual (Cartesian) one in that the beings between *[251]* which the interaction takes place are not of different kinds[57]).

The necessity of assuming a soul-real lies in the fact that our representations always stand in mutual connection and interaction. Partly they fuse together (by assimilation), namely when they are closely akin, partly they combine into groups (complexions), namely when they are mutually different in kind (like color and tone), partly they inhibit one another, namely when they are of the same kind without being able to fuse. The fact that they cannot remain unaffected by one another, or ununited and unconnected, shows that they are expressions of self-preservation of one and the same representing being. They always have a tendency to make up one single activity, insofar as they do not inhibit one another. By assimilation and complexion a total force arises, which expresses what we call our ego (which is thus a result and not a principle), and which comes to have a decisive influence on what can be taken up in us in the future. Only what can fuse with (be apperceived by, Herbart calls it) the dominant groups of representations can win mental subsistence. The apperceiving groups of representations make up the character of the personality.

Herbart's psychology has its great importance by proceeding from the single elements (sensations and representations) as the foundation of all mental phenomena. He thereby strikes into the way first trodden by Hume and Hartley, which a few years after the appearance of Herbart's great psychology was followed again in England by James Mill. Herbart is one of the most eminent representatives of the psychological tendency

that regards the multiplicity of elements as the foundation in mental life, and the unity of consciousness as merely the product of their interaction. What the English call association Herbart calls assimilation (association by similarity) and complexion (association by contiguity). It is a lasting merit he has earned by putting the simple elements in place of the faculties of the soul still embraced by Kant, and by demanding that the mental phenomena be explained by the interaction of these elements according to definite laws. Through the works that have proceeded from him and from the many able psychologists
[252] of his school (Drobisch, Waitz, Zimmermann, Volkmann, Nahlowsky, Steinthal, Lazarus), psychology has made great progress. — Yet the spiritualist assumption of a soul-substance stands in strange contradiction with this emphasis on the multiplicity in mental life as the fundamental thing. If Herbart, instead of grounding psychology on metaphysics, had grounded it exclusively on experience, he would have found in the fact that there is no incoherent multiplicity in consciousness an essential contribution to the very characterization of the life of consciousness, and he would have set himself the task of showing how this characteristic trait (consciousness as synthesizing activity) manifests itself in detail in the psychological laws. As it is, he ends in a psychological atomism and comes into conflict with experience by ascribing to each single representation a tendency to eternal subsistence[58]). — And the elements from whose cooperation he wants to explain everything in consciousness are only elements of cognition. Feeling and willing are to be merely products of relations between representations: feeling arises when a representation is pressed by counteracting tendencies, so that it cannot move freely, and drive when a representation works its way up against obstacles and thereby more and more affects other representations, partly by awakening them, partly by pushing them back. Psychological observation does not confirm this doctrine of Herbart's, that feeling and willing are always derivative in relation to cognition.

It was surely Herbart's endeavor to make psychology an exact science that guided him, both when he proceeded from the multiplicity of elements as foundation and the unity of consciousness as mere product, and when he conceived feeling and willing as mere resultants of opposing representations. As the title of his chief psychological work shows, he wants to ground psychology not only on experience and metaphysics, but also on mathematics. A possibility of this he finds in the fact that representations increase and decrease in clearness, or, as he expresses it, rise and sink, and that this rising and sinking is conditioned by the mutual relations of the representations. Here it is their mutual
[253] inhibition that is decisive. Herbart's mathematical psychology aims at finding definite laws for the mutual inhibition of representations. Now a difficulty lies here for him in the

fact that the forces with which the single representations seek to maintain themselves in consciousness cannot, like physical forces, be measured by a motion in space; it is indeed only a figurative expression when one speaks of the rising and sinking of representations. We lack a measure in mental mechanics. For lack of such a measure Herbart proceeds from the fundamental proposition that the sum of inhibition in consciousness at each moment is as small as possible, since all representations strive to maintain themselves. It then remains to determine mathematically how the inhibition (obscuring) must be distributed among the various representations that are given simultaneously or are striving upward, if it is to be the least possible. — Despite what is interesting in this line of thought, Herbart has not succeeded in showing an agreement between the results he could thus derive by mathematical means and mental observations, any more than the fundamental proposition he built on can be said to be valid. Its validity rests on the assumption of representations as independent forces. But if the nature of the life of consciousness is to be synthesizing activity, the single element has no absolutely independent energy: whether it can subsist in consciousness despite its opposition to other elements will depend not on itself alone, but on the total energy that consciousness can command for its synthesizing activity. The more energetic the synthesis is, the more mutually opposed representations can be embraced without inhibiting one another. And there is the further point that representations are not the only elements. In a state of feeling a representation that is favored by a dominant feeling can go on being far more uninhibited than it should be according to its relation to other representations. — Herbart's school has, in fact, also let his mathematical psychology drop; at any rate it has not carried it further. But even if Herbart erred in his attempt, the attempt itself is a testimony to his energetic searching and to his firm conviction that lawfulness in the *[254]* domain of mental nature is no less than in the domain of material nature[59]).

γ. According to Herbart there is no scientific principle that can combine in itself the explanation of reality and the demonstration of value. The science of valuation (which Herbart calls aesthetics in the widest sense) must therefore be kept separate from the science of the reality of things. In this distinction between theory and practice, too, Herbart is a Kantian. Here too he opposes the philosophy of Romanticism, whose principle of unity was to contain at once the explanation of reality and the standard for valuation. For himself the difference between these two things becomes all the more necessary, since theoretical science, in his conception, ends with the assumption of relationless reals, while valuing judgments concern precisely not realities but relations between realities. When we call something beautiful or ugly, praiseworthy or disgraceful,

it is the relation between the properties of things, or between the different tendencies in a single man, or between the wills of different men toward one another, that we have in view. — The specifically ethical judgments differ from aesthetic judgments in general in that they concern something which can not merely be possessed as a thing of value, but which conditions the unconditioned value of the person himself.

Just as important as it is to subject the concepts of experience to a testing criticism, just so important is it to test the concepts of valuation, especially the ethical ones. We often pass valuing judgments without noticing it, and therefore false accessory notions can easily creep in; motives other than the purely ethical can be stirring and be mixed up with these. The task then is to bring out the simple fundamental relations that valuation concerns, in order to discover by which practical ideas we are guided. A practical idea is an exemplar that involuntarily hovers before us in our judgment when we clearly judge the harmonious or disharmonious relation between a man's conviction and his action, or between the striving of several men in relation to one another. There can be disharmony between what a man considers right and the actual direction of his will; such
[255] a relation we disapprove of as conflicting with *the idea of inner freedom.* Or the energy with which he strives to satisfy his conviction may be too slight and the conviction itself too feeble; such a relation we disapprove of by virtue of *the idea of perfection.* In a similar way *the ideas of right, of equity and of benevolence* manifest themselves in our judgments concerning the relation between different personal wills toward one another. But these judgments are only certain and universally valid when the relations come forward pure and clear, and when extraneous interests are kept away. Both an individual and a social development must take place before man comes to the standpoint where the practical ideas determine valuation. It is a matter of fixing sufficiently strong groups of representations around them, so that they can become a power in the soul. That Kant could pronounce his categorical imperative was a testimony to what development has taken place. Yet he is wrong in holding that the practical ideas must always announce themselves through a violent constraint. He who is able to form the practical ideas and keep them awake in himself will also experience how mild their dominion can be.

In his ethics Herbart is influenced by the English ethicists of the eighteenth century, perhaps especially by Adam Smith, whose "impartial spectator" makes precisely such valuations as Herbart desires. But Herbart wants to keep away any closer psychological explanation of the ethical judgments, just as he also wants to disregard every practical purpose and foundation for valuation. He wants human relations of will to be valued as one values pieces of music. His ethics has an aesthetic stamp, and too little is noticeable

in him of the fact that ethical judgments, where they are genuine and primitive, spring from a will and are grounded on the thought of a practical purpose that is striven for. Nevertheless Herbart's ethics has interest through its clear description and analysis of the most important relations of will, as well as through its endeavor to maintain the independence of ethics from theology. This last has all the more interest since Herbart, in religious respects no less than in political, was a conservative man. In his faith he held to the Protestant Church, but he insists, despite his theologizing philosophy of *[256]* religion, that the concept of God is always thought by means of psychological and ethical determinations that are taken from man. It then depends on the ethical ideas being developed in their purity and independence, in order that the concept of God may become the right one. Herbart's theoretical philosophy offers no points of attachment for theology[60]), and his philosophy of religion therefore stands in a rather loose connection with his other views. Only one notices a consequence of his psychology when he insists on the importance of the form and content of the religious ideas making them able to fill men's minds, to form great and strong (apperceiving) groups of representations in them.

c. Friedrich Eduard Beneke

In a letter to Herbart (May 22, 1824) Beneke writes: "Independently of each other we have arrived at the conviction that psychology needs a thoroughgoing reform, if it is to solve the task that is set it." About the task that is set psychology the two investigators were not entirely agreed. They both set it the task of showing that what Kant — and after him Fries — called the forms were results of mental processes, not originally ready-made modes of action. Even more energetically than Herbart, Beneke conceives them as mental products, whose genesis through law-determined processes of development it is the business of psychology to show. But while Herbart wants to ground psychology not only on experience but also on "metaphysics", which here means on an investigation of the nature of existence that precedes psychological investigation, Beneke maintains that psychology is the fundamental philosophical science, since the concepts of all the other philosophical sciences are mental products, and psychology itself he wants to treat merely as an empirical science. It is indeed the task of philosophy to lead to a world-view; but its nearest object is nevertheless the life of consciousness itself — partly because this has most interest for us, partly because we know this side of existence immediately, while the material side is only known through consciousness, partly, finally, because — *[257]* before we proceed to the highest problems — we must test the powers of the human mind. In thus making empirical psychology the foundation of all philosophy, Beneke

reminds us of the English school, and he also acknowledges himself to be Locke's disciple. Only in Germany, he says, do they not know that psychology, and with it the whole of philosophy, builds on experience; that is why they are there so far behind in philosophy in relation to other peoples, who have let themselves be taught by Bacon, Locke and Hume. Even Kant does not hold fast to the method of experience, but denies it as soon as he passes from critique to positive development. His speculative successors have, despite all their brilliance, only harmed philosophy. They gained a great following because they profited from the enthusiasm Kant had awakened. But this whole speculative period is to be regarded only as a phenomenon of cultural history; not refutation, but historical explanation is what is needed over against these bold systems.

When Beneke expressed himself in this way about philosophy and about the philosophy of the time (in the work *Die Philosophie in ihrem Verhältnisse zur Erfahrung, zur Spekulation und zum Leben.* 1833), he had fought his first battles and published his most important works. In his student days he had been influenced especially by Fries (through the latter's friend and disciple De Wette) and by Schleiermacher. Jacobi too had great influence on him through his insistence on the importance of immediate observation; but Jacobi's dread of analysis and proof repelled him. At 22 years of age he conceived the plan of working toward a reform of philosophy, which was suffering under the wild speculations, and in the same year (1820) he habilitated at the University of Berlin. In a couple of smaller works he developed his conception of the task and method of psychology, and he gave well-attended lectures as Privatdozent. But after the publication of his *Physik der Sitten* (1822), a counterpart to Kant's "Metaphysik der Sitten", his name — although he published an able defense — was struck from the list of lecturers, without his succeeding, despite repeated attempts, in obtaining information about the reasons for
[258] this step. Only the minister, who was a patron of Hegel, declared to him that a philosophy which did not derive everything from the Absolute did not deserve its name! Perhaps it had been inferred from the title that Beneke taught materialism, although by "physics" he meant here only an attempt at a natural, empirical grounding. The book itself is of no small interest for the history of ethics, since it speaks for a psychological grounding of the ethical standpoint and seeks to show how the ethical judgments develop through reflection on the way in which feeling is set in motion over against human actions, one's own and others'. Beneke treats the ethical judgments in quite a similar way to that in which Schleiermacher treats the propositions of faith, and there is hardly any doubt that Beneke is here influenced by Schleiermacher's lectures. Moreover he maintains — in agreement with Jacobi and in opposition to Kant's absolute and universal morality — the

importance of the individual presuppositions in determining what is right and duty in the single case.

Beneke's ethics, however, shows an interesting development. In "Physik der Sitten" he rejects every real content in ethics, every objective valuation, since such a valuation — he thinks — conflicts with the principle that every ethical valuation is to be traced back to the motives. The ethical criterion is determined only subjectively, as the self-agreement of the individual: he who follows his conscience ("Werthgebung") acts morally. Beneke stands here in a similar position to the one Kant held at the second stage of his ethics; only he deviates by maintaining that the individual's peculiar "Werthgebung" can in turn be subjected to a valuation. About 1830 Beneke teaches, under the influence of Bentham, whom we shall come to discuss only later, that the subjective consideration not only does not exclude, but precisely presupposes, an objective principle, a real content. As in Bentham, this becomes the principle of welfare, to which he gives an interesting underpinning in his chief ethical work *Grundlinien der Sittenlehre. Versuch eines natürlichen System derselbers* (1837—40). Five points of view for the objective valuation he sets up here: the character of the human faculties that are furthered or inhibited; the degree in which the single faculty is developed; the multiplicity of mental results; the duration of these; the unmixed progress that is won through them (Sittenlehre *[259]* I. p. 132). It is Beneke's great merit to have first transplanted the objective principle of English ethics onto German soil. While his predecessors from Kant on had wanted to raise ethical valuation above every regard for the effects of actions, whereby ethics acquired a predominantly subjective character, Beneke holds fast to the way in which actions intervene in the weal and woe of living beings as determining for ethical valuation, even if this valuation differs from the juridical one by aiming directly at the disposition from which the actions proceed[61]). —

When his teaching in Berlin had been stopped, he worked for some years in Göttingen, and from here he published his most significant work with the unpretentious title *Psykologische Skizzen* (1825 —27). A place was nevertheless opened to him again at the University of Berlin, and in straitened circumstances, without the reigning Hegelianism allowing him to advance to professor ordinarius, he unfolded a rich activity as lecturer and author in psychological, pedagogical and ethical directions. His pedagogical writings in particular found entry into large circles. — On the first of March 1854 he drowned; probably, broken down by long illness and by despondency, he took his own life in a fit of derangement. —

While Herbart seeks to understand the life of consciousness in its various forms

and at its various stages as the mechanical product of a multiplicity of single elements (through their fusions, complexions and inhibitions), Beneke's psychological theory has a more biological character. He conceives the development of the life of consciousness as a growth from given germs or dispositions, which he calls the *primary faculties* (Urvermögen); by these are meant the faculties of sense perception and of motion; — according to Herbart, on the other hand, the soul is a tabula rasa before other reals, by entering into relation to it, awaken its self-preservation. The primary faculties are bound up with need and striving. The impressions from without that can bring them to full unfolding they seek involuntarily. And under the influence of outer experiences new faculties are constantly formed, since the earlier influences do not entirely disappear but leave traces
[260] or dispositions that act as co-determinants in the later influences. Therefore a constant interaction takes place between the conscious and the unconscious. By the concept of faculty Beneke wants to express precisely only the unconscious inner conditions that, from the first and later at every stage, work together with the outer experiences. We are indeed not able to separate distinctly what stems from the outer experiences from what is due to the inner conditions; but there is no doubt about the constant cooperation of the inner and the outer. Beneke subjects the whole relation between the conscious and the unconscious to an interesting investigation. Of other psychological phenomena that are discussed in his rich chief work (whose most important content he later presented in shorter form in his *Lehrbuch der Psychologie als Naturwissenschaft.* 1833), we may mention the importance of the relation of contrast for the feelings, and the tendency of mental elements to spread their stamp over the whole mental state (Beneke calls this process equalization; it is perhaps better called expansion). When so many elements and laws determine the development of the life of consciousness, it is no wonder that the higher stages can be so different from the lower that they do not seem explicable from them. But just as little as one can see in the germ of a cherry tree any hint of the cherries, which first appear as fruits on the developed tree, just as little can one find the higher forms preformed in the lower forms of mental life. The forms of higher mental life are neither innate, nor have they come into the soul from without, but they arise during the development of the soul according to its peculiar laws. — Beneke's psychology would deserve a monographic treatment that investigated its relation both to the older psychology (especially Hume and Tetens) and to that of the most recent times.

Beneke himself, however, is not entirely clear about the biological character of his psychology, and he does not always hold fast to it. In his great zeal to plead the cause of psychological inquiry over against speculative philosophy, he praises inner experience,

not only at the expense of abstract thinking but also at the expense of outer experience, maintaining that it is far clearer and more exact than the latter, which is to show itself especially in the fact that here we can follow the single elements in their cooperation. But this seems to presuppose that the mental products can be wholly seen through in *[261]* their arising from the elements, which conflicts with the difference in quality between elements and products that Beneke otherwise so rightly emphasizes. "Let us not forget", he says (Skizzen II, p. 329 f.), "that the joining together itself is also something, and that, even if the result contains in itself no more than the sum of its elements, it is nevertheless — precisely as sum and as an organically joined whole — not only different from each single one of its elements, but also from a mere heaping up of them". In every such organic joining (e.g. in the arising of a new faculty) there must always be something that is not transparent to self-observation, just as everywhere in nature the arising of new qualities poses the greatest problems. And in the domain of outer nature one is able far more clearly than in the domain of inner nature to show the conditions under which new qualities arise.

When Beneke has thus come to overestimate the perfection of psychology as a science (or at least its prospect of becoming perfect), it is surely also because he ascribes to it great importance as the fundamental philosophical science. It is from the psychological consideration that we orient ourselves in existence. The principles of the philosophy of nature are taken from psychology: since in our inner experience we have access to knowing a part of existence *as it is in itself,* we naturally conceive the part of existence that we know only as outer, objective being (material nature) *by analogy with* ourselves. The forms and laws of the mental world we therefore transfer by way of analogy to material nature. The philosophy of nature which Beneke thus adopts differs from Schelling's by being conscious of its hypothetical character and by assuming only an analogy between the mental and the bodily, not, as Schelling would, an identity of them. And while Schelling believed he could put this idealist interpretation, which builds on analogy, in place of the mechanical explanation of nature, which he regarded as an error, Beneke holds fast to the justification and necessity of a thoroughgoing material explanation of all material phenomena of nature. Thus he acknowledges the attempts to give a purely *[262]* physiological-anatomical explanation of the phenomena of insanity, however much he insists on the importance of the psychical symptoms. In this respect he joins Spinoza's conception (see especially his book *Das Verhältniss von Seele und Leib.* 1826. p. 219 f.; 243 f.), although other statements point rather to a conception akin to Herbart's, in that they amount to saying that the difficulties of the interaction between the two dissimilar

elements fall away when one thinks of it as taking place between the soul and the being analogous to the soul that lies at the ground of the material elements. Beneke did not attain to clarity about the choice that is to be made here[62]). — Just as the inference by analogy in the philosophy of nature leads us to conceive all material beings as members of a series descending from our own station in existence, so it leads us in the philosophy of religion to form representations of higher beings in likeness to ourselves. All religion and all science of religion, however refined and spiritualized this may be, is nothing but anthropomorphism. And since the distance between God and man is probably far greater than it is between man and worm, — since moreover in the philosophy of religion we lack the support that the philosophy of nature has in mechanical natural science, — the religious representations are objects of faith, and the essential task of the philosophy of religion becomes partly to investigate the psychological development of the representations religion uses, partly to fathom the spiritual need itself that seeks its satisfaction in religion. In the following statement Beneke's conception of the philosophy of religion as applied psychology appears clearly and interestingly: "No longer will the philosophy of religion subdue and cut down the historically conceived religious dogmas according to a preconceived, narrow norm; but in seeking to reconstruct all the forms in which the supersensible reveals itself and has revealed itself to the human mind, and to derive and interpret them on the basis of the most deep-going self-knowledge, it brings them — at once acknowledging them all and subjecting them to a sharp test — into the
[263] organic connection in which they have gradually developed from the first beginning of human culture right up to our days. Here again, then, it is psychology that has the task of unraveling and illuminating the multiplicity, at first sight confusing and wrapped in darkness". (Die Philosophie etc. p. 27).

Both as "psychologist, philosopher of nature (cosmologist), ethicist and philosopher of religion"[63]) the investigator, misjudged in his time, stands as a forerunner of thoughts that in the following generation found a more fruitful soil than he had had before him in Germany's Romantic-speculative age.

D. Transition from Romantic Speculation to Positivism or to Positive Faith

a. Criticism of Hegel and Dissolution of the Hegelian School

The philosophical situation in Germany when Hegel died (1831) was this: a speculative school, founded by eminent thinkers, had won a decided supremacy, while the opposition from the side of the critical philosophy and of specialized research stirred only in smaller circles. It therefore became an event of great significance that within speculative philosophy itself there arose an opposition to the conclusion which the thought of Romanticism had found in Hegel's system. There were thinkers who would not give up speculation, but who did not see its consummation in Hegel; they set themselves the task of keeping Hegel's fundamental thoughts, but of letting him, with his whole system, be absorbed in a higher unity, that is, of treating him as he had treated the preceding philosophers. And the supplement which the Hegelian system was supposed to need was fetched partly from the sphere of experience, partly from that of positive faith. Thus Schelling, who had held himself back as long as Hegel lived, now announced a new philosophy which was to bring speculation into a harmonious relation both to empirical knowledge and to religion. He was called by Friedrich Wilhelm IV to Berlin to counteract *[264]* Hegelianism, and in his lectures here he set forth that by a purely rational way only a knowledge of the general possibilities and laws could be reached, while the knowledge of the real, which is always single and individual, demands an act of will springing from the personal need which cannot rest content with possibilities and general laws. The transition from that rational philosophy to this knowledge grounded on faith and will Schelling called a transition from negative to positive philosophy. Only positive philosophy can maintain the personality of God, which was the chief problem with which Schelling had been occupied ever since 1809, and which now became the problem about which interest gathered in the first decades after Hegel's death. In Hegel's philosophy of religion, after all, a reconciliation between thought and faith was, in his opinion, to have been reached. But soon after his death it became a point of dispute whether Hegel's philosophy could be said to lead to the assumption of a personal God and a personal immortality[64]). Schelling and the thinkers who stood near him, especially C. H. WEISSE (1801— 1866) and The Younger Fichte (1797—1879), maintained that Hegel's system was pantheism, but undertook by way of thought to build up a theism which was not to deny pantheism, but to take it up into itself as into a higher power. They sought to show that all fundamental thoughts gather at last in the idea of personality, and that

this idea must stand as the expression of the highest reality, just as they (by a further working-out of what Schelling had stated in his treatise on freedom in the year 1809) sought to establish the possibility of ascribing the character of personality to an infinite being. Yet at bottom they all come to admit that theism cannot be grounded scientifically. Schelling's transition from negative to positive philosophy was in reality a transition from thought to faith. The younger Fichte, in whom the tendentious aim is stronger than the thinker's interest, returns to the popular theology's God, finished once for all. Weisse, at once the most acute and the most profound of these thinkers, goes back to Schleiermacher, whose philosophy of religion he carries further in an interesting way by
[265] maintaining the close connection of religious feeling with ethical purposes and tasks. In his bold religious thinking he resolutely draws the consequence that a God who is to be a personality must also be subject to development in time, and he insists on the real causal connection as the bearer and frame of all ideal content in existence. He professes (in agreement with Lamarck) a theistic evolutionism. An existence relatively independent over against God becomes possible only through the elements of the Deity themselves being in an infinite process of elevation. From the eternally becoming nature in God springs the material process of nature, after the divine will has received its outward expression in the fundamental laws of mechanics (especially in the law of inertia), which make possible a richer individualization of the divine content of nature. Through the material development of nature there goes a constant struggle between the organic and the purely mechanical, a struggle which is the condition for the rise of mental life. — But the program of speculative theism, which calls for a scientific grounding of the assumption of a personal God, is in fact pushed aside in Weisse too for an appeal to personal faith as the only power that can here lead to the goal.

In a direction similar to that of speculative theism went the views in the philosophy of religion developed by Karl Christian Friedrich Krause (1781—1832), which — together with his views in the philosophy of law — were set out in fairly accessible form in his work *Urbild der Menschheit* (1811). Krause, however, does not care for the expression personality and will not use it of God. He designates his conception as *panentheism,* because it maintains that God has the world in himself, is not absorbed in the world. Krause's philosophy, developed in a long series of writings, which possesses the obscurity of mysticism but also its noble qualities, gained particular significance for the philosophy of law, since he stresses the idea of humanity as an organic whole and as the image of the divine archetype, and since he regards law as the form of this organic whole's development of life. This standpoint made possible a conception at once idealistic

and reformatory, which formed a characteristic contrast to Hegel's conservative doctrine of right and has been of no slight significance for the development of free and humane ideas. Through the works of his disciples, especially Heinrich Ahrens's, Krause's views *[266]* gained influence on wider circles, also outside Germany (especially in Belgium and in Spain). In the philosophy of religion, on the other hand, the ideas of Schelling and Hegel and their consequences became of the greatest significance. —

Within Hegel's own school the sharpening of the problem of the philosophy of religion gave occasion to a split. Some Hegelians held that the master's philosophy, rightly understood, agreed with the current faith and with the doctrine of the Church. Other Hegelians declared that, consistently followed, it must come into irreconcilable conflict with these. Strauss, who himself belonged to the latter, compared this opposition with that between Right and Left in parliamentary assemblies, and it soon became customary to speak of the Hegelian Right and the Hegelian Left. The adherents of the latter were also called the Young Hegelians. The most eminent Right Hegelians were Göschel, Rosenkranz and J. E. Erdmann. The Hegelian Left, which unquestionably gained the greatest significance for the development of thought, is represented by David Friedrich Strauss and Ludwig Feuerbach in the domain of the philosophy of religion, by Arnold Ruge and later by Karl Marx and Ferdinand Lassalle in the domain of the philosophy of law and of society.

Among all these thinkers Feuerbach stands in the front rank. The transition from Romanticism to positivism, from speculative thought to a view of life and world-view built on scientific experience, takes place in an extraordinarily characteristic way in the course of development of this energetic man. In this course of development a whole criticism of the philosophy of Romanticism is contained, and it stands at the same time as typical of many a personal development of life from the faith of childhood to the conviction of manhood grounded on thought and experience. It was Strauss whose *Leben Jesu* (1835) posed the religious problem anew in all its sharpness; but Feuerbach gave one of the most important contributions to its elucidation in the following decades.

It was not exclusively the religious problem, however, with which the philosophical work of the thirties and forties was occupied. An epistemological discussion of great value for its time, not least through its criticism of the speculative method, was undertaken *[267]* by Adolph Trendelenburg in his *Logische Untersuchungen* (1840). Trendelenburg lays great weight on the philosophical problems arising on the ground of experience. It is reflection on what empirical knowledge yields that leads to philosophy. The problem of knowledge Trendelenburg seeks to solve by pointing out motion as a determination

common to thinking and being: the motion of thought in intuition and construction has its parallel in the motion to which all material phenomena can be traced back. And when the motion of will is determined by the thought of a purpose, Trendelenburg finds an analogy to this too in material nature, since he holds it impossible to understand organic phenomena without granting the concept of purpose validity in the knowledge of nature; of the insufficiency of a purely mechanical view of nature he is vividly convinced, however far he stands from speculative philosophy of nature. He emphasizes the finitude and limitation of our knowledge and holds, with Kant, that it is impossible to give our fundamental concepts such an extension that they could be applied to the Absolute. The power of our thought is to behold the subdued rays, broken in a play of colors; but it does not therefore deny the pure light from which they come. As men on whom Trendelenburg's philosophy has exerted significant influence may be named Søren Kierkegaard and Albrecht Ritschl. — Besides this epistemological work Trendelenburg has published a work in the philosophy of law of great value *(Naturrecht auf dem Grunde der Ethik.* 1860)[65]). And finally he has worked as an eminent investigator in the domain of the history of philosophy. Here his endeavors met those of Hegel's disciples, to whom a series of great, in part monumental works on various periods of the history of philosophy are owed. Eduard Zeller and J. E. Erdmann, later Kuno Fischer are especially to be named here. During the struggle of the systems a natural need was felt for historical and comparative orientation, and the history of philosophy as a special discipline really
[268] first came into being in this period.

b. David Friedrich Strauss and the Religious Problem

Already as a young student Strauss (who was born in 1808 in Ludwigsburg and studied theology and philosophy in Tübingen) had raised the question whether the historical components of the Bible, especially of the Gospels, belonged to what for Hegel was the eternal thought-content of religion, or whether they were merely to be reckoned to the form of representation in which this content appears to the popular consciousness. In his *Leben Jesu* (1835) he later sought to establish that in the evangelical history there is present neither history nor conscious invention (the dilemma with which the orthodox theologians are wont to operate), but myth, that is to say unconscious poetry, whose motives were partly the religious ideas that animated the time and the people (especially the messianic idea), partly the mighty impression which the founder of Christianity had made on his disciples. What the Gospels give us is therefore — as Strauss later expressed it — not "the Jesus of history", but "the Christ of faith". As a purely historical person Jesus

is really quite unknown to us; only "the Christ of faith" is a being with definite features. In a concluding treatise to "Leben Jesu" it is set forth that the idea of the God-man cannot find application to a single individual, but that only the whole human race, appearing in an infinity of individuals, can through its constant suffering, striving and working realize the divine. What the doctrine of the Church states about Christ can keep its validity when it is transferred to humanity. In a later work *(Die christliche Glaubenslehre in ihrer geschichtlichen Entwickelung und im Kampfe mit der modernen Wissenschaft dargestellt,* 1840—41, translated into Danish by H. Brøchner) Strauss showed that the reconciliation between Christianity and philosophy which Hegel thought he had reached became possible only by Christianity's being conceived as monism (doctrine of unity), while it is nevertheless in reality a decided dualism. In Christianity, after all, the unity of the divine and the human is realized in only one individual — all other individuals fall outside! And in that one individual the unity of God and man becomes possible only in the form of suffering and through the intervention of supernatural powers! From *[269]* several sides, then, it is precisely in Christianity that the opposition between the two worlds, the divine and the human, appears. Indeed, primitive Christianity even gives up letting the divine world permeate the human and lives in the expectation that the former will in the near future burst the latter in a supernatural way! — The consequence for Strauss becomes a sharp opposition between believers and non-believers and an energetic protest against all attempts at reconciliation.

Strauss's merit with regard to the religious problem is that of having asserted the concept of myth or unconscious poetry against the current dilemma: history or conscious invention. This concept expresses unequivocally what was designated by Kant, Herder and their successors by the expression symbol. It has the advantage that it sets the definite task of showing why precisely these images were formed; it has a more historical tone than the expression symbol. Strauss did not go more closely into the question which mental powers and drives lead to the great images and symbols which through the popular religions have become the property of the species — what it is that sets the whole ideal-forming process going. He later acknowledged that Feuerbach had given a weighty answer to this question.

It had been Strauss's thought to found a free, critical theology. But his heretical views made it impossible for him to obtain any theological teaching post. He lived therefore as a private man, for a long time occupied with literary and biographical studies. Thus he wrote interesting books on Ulrich v. Hutten, Voltaire and others. Toward the end of his life he returned to thinking in the philosophy of religion, in that in the work *Der alte und*

der neue Glaube (1872) he wished to show how the religious problem stood — not only, as in his earlier works, when the consequences were drawn from speculative philosophy, — but also when the consequences were drawn from modern natural science. He had here set himself a task for which he was not equal. He was more scholar than thinker, and more artist than scholar. The book was written with brilliant clarity of style, but
[270] lacked depth and inwardness in its conception of the religious and consistency in the standpoint which it sought to maintain against materialism. In the aesthetic view of life, especially in music, he found a surrogate for the religious cult, a testimony to how little he had thought through all sides of the religious problem. After a hard and painful illness, which he bore with the greatest gentleness and strength of character, he died in the year 1874.

c. Ludwig Feuerbach's Psychology of Religion and Ethics

This energetic and richly endowed thinker is a fine example of how the personal kernel can go on being essentially the same although it appears in the most different forms. It was one and the same interest that moved him at all his changing standpoints. From his earliest youth he felt a need for full and positive human life, for "soulful satisfaction", for occupying himself with the questions whose answering decides the value that life has for man, and whose answering is therefore also for man a matter of conscience. It was — according to what he himself stated in his later years in a self-characterization — this need that first led him to theology and then to philosophy, and within philosophy again from speculative philosophy to a standpoint which roughly coincides with what the French philosopher Auguste Comte calls positivism. Thought was for him always the servant of life. He regarded himself as a positive thinker whose criticism and polemic (which in the eyes of his contemporaries were what was most prominent in him) were only means to separate the husks from the kernel in religion. He wished to penetrate to the living human need that lies behind the religious representations, and therefore he directed so sharp a criticism against these. So far was the essence of man from being absorbed for him in thought, that he says of himself (when he was urged to write his autobiography): "It is impossible for me to express what moves me most deeply; how much more impossible must it then be for me to write it and to give up my inner self to
[271] the indiscreet public." And in another place he says: "How far behind our very being do our thoughts, by which we seek to bring our being to consciousness, nevertheless remain." This incommensurability between thought and personality must be firmly kept in mind if one would understand this ever fermenting and seeking, "dissatisfied" spirit[66]).

Ludwig Feuerbach was born in Landshut in 1804 of a highly gifted family. His father was the famous jurist Anselm v. Feuerbach, and several of his brothers have distinguished themselves in art and science. He studied theology in Heidelberg, but soon went over to philosophy, which he studied under Hegel in Berlin. The discord in his inner self which he could not get resolved in theology was ended by Hegel's philosophy, and he regarded Hegel, who had made him a philosopher, as his second father. For some years he worked as a lecturer in Erlangen, but then withdrew. Partly he seems to have lacked gifts for oral presentation, partly he was opposed from the theological side, since it did not remain hidden that he was the author of the radical work *Gedanken über Tod und Unsterblichkeit* (1830). He then withdrew to the country and took up his abode at Brucksberg, a factory in which he had a share as his wife's dowry, and from here he unfolded a rich activity as an author. In a series of excellent works he treated subjects from the history of modern philosophy. Of these especially to be noted is his *Pierre Bayle,* in which the opposition between theology and philosophy is strongly stressed, and the independence of ethics from theology is maintained. Already here his definitive standpoint in the philosophy of religion is clearly seen, though it first comes forward clearly and definitely in his chief work *Wesen des Christenthums* (1841). While Strauss, like Hegel, kept essentially to the content of the dogmas, Feuerbach goes back to the source of the dogmas in human feelings and drives, in fear and hope, in longing and desire. His task is not the impossible one of formulating the content of the dogmas in scientific concepts, but of understanding their origin psychologically. From the official documents of religion he goes back to the spiritual and inward life which has found expression in them. And as critical as he is toward the dogmas, so sympathetic is he toward the source from which they come. On this rests the great difference there is between his relation to religion and Voltaire's. *[272]* Feuerbach goes much further in criticism and negation than the French freethinker — but nevertheless stands with far more understanding and appreciation before what is inwardly moving in religion than Voltaire with his dilemma: madness or knavery! The studies and experiences of the intervening time had not been wasted. There is a great and characteristic difference between the freethinkers of the eighteenth and of the nineteenth century (when one keeps to the freethinkers who think). It was Feuerbach's endeavor in the philosophy of religion (as it was at about the same time Lyell's in geology) to understand the formations of the past by the help of the forces that are still at work at the present day. It is the so-called principle of actuality that underlies his well-known proposition that all theology is psychology. — In grandeur of plan, in depth and energy none of Feuerbach's other works can be compared with "Wesen des Christenthums",

although both he himself and his friends have placed his *Theogonie*, published in 1857, above it.

The break with speculative philosophy which had been accomplished for Feuerbach as far as the conception of religion is concerned had consequences for his whole philosophical conception. In *Grundsätze der Philosophie der Zukunft* (1843) he gives a program for a new philosophy, in which — in remarkable agreement with utterances in Schelling's "positive philosophy" — it is maintained that only the individual is real, and that this real is unutterable, impenetrable to thought, and therefore to be grasped only with passion. Reality is not a theoretical concern, but "eine Frage auf Tod und Leben". The object of modern philosophy is not what lies beyond experience, but man himself, with nature as his foundation. — Later on he developed in the work *Gottheit, Freiheit und Unsterblichkeit* (1866) his views on the spiritual and the material; this work stands in his production roughly as Strauss's "old and new faith" stands within his. In his last years, which were unhappy through want of livelihood and illness, he was occupied with an
[273] ethics; interesting fragments of this work were published after his death (1872) by Grün.

Already in his "Pierre Bayle" Feuerbach had demanded an "analytic-genetic" philosophy, and a somewhat more detailed program he had given in his "Grundsätze". But beyond a general demand in this direction he got neither here nor in his later works. He never entered upon epistemological investigations, and even where he appears as an epistemological realist he retains something dogmatic and mystical in his line of thought. And he had no inkling that the demand he made was already being fulfilled by the works of Comte and Mill. When he arrived at positivism, it had long since been founded in the French and English schools. The case is similar with his ethics. There is here something tragic in Feuerbach's position within the philosophical development of our century. He lived in a period of transition, and he came to suffer from it. He had used so much strength in working his way out of speculation that he later lacked energy, drive and time for a positive and scientific working-out of his new standpoint. But on the other hand he stands as one of the most pronounced characters, one of the most truth-loving men of the nineteenth century. And he is one of the most important champions of a humane view of life. Humanism was the name he himself held most fitting for his tendency. — We shall now dwell more in detail first on his philosophy of religion, next on his general philosophical standpoint, finally on his ethics.

α. Schleiermacher had determined religious feeling as the feeling of dependence, but had left open whence this feeling got its objects. Feuerbach now sought to show that feeling itself *produces* its objects, so that these do not merely express it, but spring from

it. In place of the mere feeling of dependence he puts the trust of the heart and the living wish, and finds the peculiarity of faith in this, that man's wishes are in it freed from the fetters of reason and of nature. What is the object of man's innermost craving and longing stands in faith as something objectively real, as the Absolute. The opposition between wish and reality is annulled. While therefore Schleiermacher sought to show that every conflict between faith and knowledge rested on misunderstanding or on this, that neither faith nor knowledge had reached its consummation, according to Feuerbach *[274]* the most peculiar religious phenomena arise precisely through the inward craving for the fulfillment of the heart's wishes bursting the barriers set by reason, and at their culmination the religious phenomena therefore acquire an antirational character.

With energy and profundity Feuerbach has gone back to the psychological foundation of religion, men's wishes, need and hope. The wish is, he says (in "Theogonie"), the religious primal phenomenon, the theogonic principle. This assertion is grounded in "Wesen des Christenthums" by a consideration which has general philosophical interest.

Man cannot reach beyond his own being; all conceptions and thoughts bear his own stamp. The objects to which he relates himself therefore teach us his being, since they are only occasions for him to unfold his being. Only by virtue of man's own nature does an object gain power over him. And from the first man has no ground to set barriers to his being. He gives himself over confidently to all representations and ascribes unlimited validity to them. In particular it lies in the nature of feeling that it tends to make its object infinite, to regard it as real. Doubt can first arise when man comes to know his barriers, and when the understanding begins to distinguish between the subjective and the objective, a distinction that is not known at the standpoint of faith. For faith is precisely faith in the absolute reality of subjectivity[67]). — Now it is not everything subjective that becomes the object of religious faith. Religion arises by virtue of a discrimination, a valuation: what man regards as divine is not the indifferent, but that to which he ascribes the highest value. Every man who has a highest purpose has a God. What man praises and extols is God for him; what he blames and rejects is the ungodly. God is the book in which man has inscribed his highest feelings and thoughts. Man's heaven is a collection of flowers formed by choice within the flora of this world. This holds of the heaven of civilized man as well as of that of natural man, only that the former, on account of his culture, does not expose himself in such a way as *[275]* the latter. From man's God and heaven one can therefore come to know his own longing and craving. In the divine attributes we have the attributes which man at the given stage values most highly. That God is personality means: personal life is the highest. That

God is love means: there is nothing that stands above the loving mind. In the Christian religion God suffers: that is to say: suffering for others is divine. And in a similar way, in order to understand religion, we must everywhere make subject what in it is predicate, and conversely. In Christianity the principle of religion comes forward in its whole depth and fullness. The mind, the life of feeling, here acquires an inwardness and a strength, but also an infinity, which paganism did not know; suffering was felt more strongly, and love was therefore also felt all the more inwardly. The levity of the Olympian gods was wrecked on the distress of the heart, but the God of the Christians is a tear of love shed in the deepest concealment over human misery. — The world of wishes and ideals man regards in religion as an independent world beyond, the real world, in sharp opposition to the present world, the world of finitude, of suffering and of struggle.

But, one might say, how can religion be derived from the infinity of the mind, from the omnipotence of feeling, since in religion man feels himself precisely finite, imperfect and sinful! — Feuerbach explains this feeling as a kind of contrast effect: man, who has involuntarily put everything over into his God, now feels himself, in contrast to this, poor and wretched. He sees his own being in God and must then indeed feel empty and poor when he sees himself in opposition to God. But in God his being is also well preserved. Man can in God enjoy his being in far richer fullness than by keeping to his limited actual nature. — Another objection runs thus: "In theology, after all, a distinction has long since been made between God's being and the attributes we ascribe to him, and Schleiermacher in particular developed this distinction!" To this Feuerbach answers that
[276] if one annuls the divine attributes, one annuls the divine being. That distinction is a fruit of skepticism and unbelief. Genuine faith does not make it. And what does one keep back when one takes all attributes away? The history of religion shows that as long as one cares to know something about God, one also knows something. In its classical time religion does not know that distinction. What Feuerbach is concerned with on the whole is to hold the religious phenomena up before consciousness in their peculiar, primitive, involuntary character. Only with religion in its original form will he have to do, not with the religiosity modified by criticism and more or less scientific culture. Within Christianity he thus insists on the difference between original and modern Christianity. Christianity too has had its classical times, and only these interest Feuerbach, not "the modern world's dissolute, characterless, comfortable, belletristic, coquettish, epicurean Christianity." But not even the theologians — he holds — now know what Christianity is. The point has been broken off the original Christian opposition between this world and the world beyond. Protestantism signifies in reality an entirely new view of life, different

from that of primitive Christianity. Its morality is purely human; only its faith leads back to original Christianity. The miracles it relegates to the past, and the Last Day to an indeterminable future — and yet faith in the imminent end of the world belongs to what is innermost in the Christian faith and cannot be separated from the rest of the content of faith; for this world must pass away if the world of infinite wishes is to come. The secret of modern Christianity is therefore, according to Feuerbach, that the theogonic wish, the primal phenomenon, is no longer present. —

This is Feuerbach's theory in the philosophy of religion. If we now ask what value he ascribes to religion, it is, as already touched on, very far from the case that he takes up a polemical attitude toward it where it is original and genuine. Religion is — in its classical period — the only way in which man can become conscious of his nature and of the tasks it sets, — the only way to self-understanding and self-absorption. And it gives *[277]* man a great horizon, enlarges his sensuous consciousness by letting his own being stand before him as something beyond, exalted and infinite. But on the other hand religion brings with it questionable effects. When the believer has everything in his God, and when the supernatural powers are for him the true reality, he cannot feel any need of family life, science, art and the life of the state. The drive toward culture cannot come into being in him. Religion and culture have the same purpose, and the more man hopes to reach by the one of these ways, the less will he seek by the other; there will be an inverse relation between them. It is moreover no altogether harmless projection that takes place when man sees his ideals as attributes of an absolute being which stands as different from himself. The more the difference between God and man is insisted on, the more it will come to this, that the attributes (goodness, justice etc.) are taken in an entirely different meaning when they are used of God and when they are used of man. Man must then take his own conscience and his own reason captive in order to obey the divine will, even if this commands something that conflicts with what is humanly called goodness and justice. Here lies the source of fanaticism, the source of all the dismal phenomena that the history of religion presents. The two sides that religion presents to valuation Feuerbach finds clearly expressed in Christianity's doctrine of faith and love. Love annuls all barriers, makes all one in spite of everything that may separate them; but faith sets barriers between them again, awakens hatred of heretics and lovelessness. Love is only a predicate of God, and the divine subject has other demands to make than those of love; one cannot infer from what love demands to what God demands. Herein lies for Feuerbach the necessity of going away from the standpoint of religion. He protests, however, against his own standpoint being purely negative. He acknowledges the value

of the divine attributes, but precisely therefore he will not have them attached to a divine being as a subject different from them. The true atheist is he for whom these attributes
[278] are nothing, not he for whom their subject is nothing. The attributes first come into their own by being separated from the presupposed subject. This holds especially of love, the disposition in which the unity of the human race expresses itself. It existed before Christianity. Christ was the image under which the unity of the species dawned on the popular consciousness. And even where faith in Christ has vanished, the true being of Christ continues to subsist wherever love reigns. — According to Feuerbach, therefore, no real values are lost by giving up religious faith. What is given up is only the projection. We no longer go over the river to fetch water, because we have discovered that the water we fetched in reality came from the river. —

Without being able to enter into any more detailed criticism of Feuerbach's philosophy of religion, I must yet point out that in his doctrine of feeling or the wish as productive (the theogonic wish) he does not sufficiently attend to the reciprocal relation between feeling and the other sides of the life of consciousness. In and of itself feeling can produce nothing — and yet "the wish out of itself and only out of itself" (Theogonie p. 57) is to bring forth the representation of God. Feeling can only act selectively, intensifyingly and idealizingly on representations which are already present and which man has either from tradition or from his own experience. The wish itself is determined by the experiences man believes he has had concerning the relation between his ideals and the actual course of things. The ideals he projects as divine attributes, and when faith is strong it sees in the bearer of these attributes a victor over all the resistance of reality. The special forms which the religious representations take are determined by representations which have arisen beforehand, by the most different ways. — As regards Feuerbach's valuation of religion, he is perhaps somewhat too optimistic in his assumption that nothing is lost by giving up religion. Even if religion does not contain, and cannot contain, anything but man's being, it might well be that this content worked far more strongly in the
[279] form religion gives it than it otherwise can. In any case there might be natures that were accessible to stronger influence only when it comes in this definite form. This great problem still remains unsolved. — Finally, Feuerbach has overlooked that there can be no question of a strict proof of his proposition that all theology is psychology. The philosophy of religion can only show the possibility of explaining all religious representations as psychological products; but that they are in reality nothing other and nothing more cannot be proved. From the content of faith agreeing with human wishes nothing can in and of itself be inferred for or against. There might perhaps be

one who could accept Feuerbach's method and yet reject his result, holding fast, in passionate assurance, to the reality of that toward which the boundless craving of the mind is directed. That was at least one form under which the religious dispute could be continued after Feuerbach's contribution; perhaps more will appear.

β. If all theology is psychology, all philosophy must of course be so too, and Feuerbach had thus come by way of the philosophy of religion to the standpoint on which Fries and Beneke had long since placed themselves. In his "Grundsätze der Philosophie der Zukunft" he proclaims man — nature, man's basis, included — as the only object of philosophy, anthropology therefore — physiology included — as the universal science. The little witty work is full of striking propositions, but has something oracular in its form that robs it of the value it would have been able to gain through a full working-through of its thoughts. It proclaims the right of the individual fact and of sense perception, but does not go more closely into the problems that thereby arise about the relation between experience and science, between sense perception and thinking, between psychology and epistemology. Feuerbach here stood at his limit.

There is a similar obscurity in his relation to materialism in his later writings. Feuerbach studied physiology with zeal and wrote an enthusiastic review of Moleschott's "Lehre der Nahrungsmittel" (1850), in which he comes out with the following utterance: "The doctrine of foodstuffs is of great ethical and political significance. Foodstuffs become blood, blood becomes brain and heart, thoughts and feelings. Human fare is *[280]* the foundation of human culture and feeling. If you would improve the people, then give them better food instead of declamations against sin. Man is what he eats. (Der Mensch ist, was er isst)." And he adds that when the common people get better food (peas instead of potatoes), a coming revolution will have better prospects of succeeding. This utterance (which is found reprinted in *Nachlass.* II. p. 90) has often been used from the theological side to show how deep Feuerbach had sunk. It shows his ability to impress forcefully and paradoxically what he wants to bring out — but at the same time certainly also obscurity on decisive points, namely concerning the relation between "man" and his "basis". However much Feuerbach in this and in later utterances seems to approach materialism, he nevertheless did not himself find the expression materialism descriptive of his view. (See Nachlass. II. p. 307). In his opinion man cannot be regarded as a mere product of matter. One must start from man, not merely see him as a result, as materialism does. For "life, sensation, thinking is something absolutely original, of genius, uncopiable, irreplaceable, inalienable." The task then is to find an Archimedean point between materialism and spiritualism, so that man is regarded both as a material

and as a spiritual being. *(Werke,* X, p. 162 f.). — What holds Feuerbach back from drawing the consequence to which his materialistic-sounding utterances might lead is the same principle with which he works in his philosophy of religion, the principle of subjectivity one might call it, the principle that the ultimate ground of knowledge can be found only in man, not outside him. This principle he applies first against theology and speculation, later against materialism; but there with far greater clarity than here.

γ. Feuerbach's ethical conception underwent several changes. Early he maintained the independence of ethics from religion and theology, and it was then the ethics of Kant and Fichte that he first referred to (thus in "Pierre Bayle"). In "Wesen des Christenthums" he referred, as already mentioned, to the love of humanity as the feeling in which
[281] the unity of the species made itself known in and for the single individual. Later he stressed the impulse toward happiness as the foundation of ethics in a way that led me in my book *Filosofien i Tyskland efter Hegel* (1872) p. 84 f. to designate his ethics as "the morality of egoism". This, however, has proved, through the moral-philosophical fragments of Feuerbach published by Karl Grün in the year 1874 (in "Nachlass" I.), to rest on a misunderstanding. For Feuerbach the happiness of the individual is not the purpose of morality, but its presupposition. For only he who knows from his own experience what it is to suffer want and wrong can feel pity for others. Pity and love of humanity presuppose that the agent himself has an impulse toward happiness — something Feuerbach asserts especially against Schopenhauer, who makes pity the foundation of morality although he rejects the impulse toward happiness. Pity consists in making others' impulse toward happiness one's own. Ethics acknowledges no isolated impulse toward happiness, no happiness of one's own without the happiness of others. As two human beings belong to the physical origin of man, so two human beings belong also to the origin of morality. Through the relation between the two sexes nature has solved the problem of the transition from the egoistic impulse toward happiness to the acknowledgment of duties toward others. The relation of the sexes forms the foundation of morality by putting a two-sided or many-sided impulse toward happiness in place of the isolated one. As the existence of the single individual is bound to that of others, there arise feelings of belonging together and of community; one's own impulse toward happiness is limited, and there is now only a question of duties toward ourselves in the sense of indirect duties toward others. — Conscience is not a special faculty, implanted once for all, from which a whole ethics could be derived a priori. Conscience is only another word for disposition, heart, pity, humanity. It arises first as bad conscience, after the act. Bad conscience is pity joined with the sting of being conscious that one is the

cause of the sufferer's misfortune. It presupposes that I can feel the other's impulse toward happiness, so that in my own inner self I feel the violation of this impulse and have the accusing image of the sufferer in my soul.

Although Feuerbach expressed his ethical views only in sketches and fragments, they are of no slight interest, partly for showing in what direction the thoughts of this sharp *[282]* critic and untiringly working spirit went concerning such important questions, partly for the agreement between them and views that had already been developed beforehand by the real founder of modern positivism, Auguste Comte. — Within the history of German thought Feuerbach stands as the one who most energetically accomplished the transition from Romantic speculation to critical self-understanding, and led back anew to the discussion of the first presuppositions of all our knowledge and all our valuation.

d. Karl Marx's Speculative Socialism

The greatest theorist of modern socialism is a disciple of Hegel and of Feuerbach. Through Feuerbach KARL MARX (1818—83) was led to a radicalism that acquired a definite direction when in France he became acquainted with St. Simon's ideas. Feuerbach now became too individualistic for him, but Hegel's dialectical method gave him a frame for the picture of the mighty course of social development through history. From the philosophy of Romanticism he kept the presupposition that strict science consists in deriving everything from one principle, and in his endeavors to carry socialism over from the world of utopias onto the firm ground of science he built on the thought that all social order and all spiritual culture (or, as he calls it, all ideology) is determined through and through by economic production and its conditions. This is what has been called the materialist conception of history. In his chief work *Das Kapital,* whose first volume appeared in 1867, Marx wishes to show that economic production of necessity runs through a series of stages ("Entwickelungsphasen"), none of which can be skipped, although the duration of the development can be shortened when the necessary insight is present. Modern capitalism, which by many (especially by "bourgeois" political economy) is held to be the only possible order of production, has arisen under definite historical conditions whose time will one day be past. It reaches its culmination when all wealth is gathered in a few hands, while at the same time the want of the propertyless workers has risen to the utmost. Then comes the reversal with dialectical necessity: the expropriators *[283]* are expropriated! They had "negated" the small owners; now they are themselves "negated" by the organized proletarians, who through universal suffrage have got the power of the state into their hands. By this great revolution, which the Marxists await

as the earliest Christians awaited the Day of Judgment, the social problem is definitively solved. All means of production now become the common property of society.

Thus the constructive method of Romanticism was used to shape a social gospel in the midst of the age of realism. And while contemporary natural science more and more put the doctrine of slow and quiet development in place of an older time's theory of catastrophes, the assumption of a development through the collision of opposites became, through Marxism, dominant in wide circles. Moreover the "materialist conception of history" challenged the criticism of cultural history by its theory that all ideal culture was only a reflex of economic conditions. And although Marx thought he had given a purely objective theory of the future social development, independent of all subjective, ethical principles, his line of thought, on closer consideration, rests in the end on an ethical postulate, namely on the fundamental proposition, reminiscent of Kant's ethics, that the worker is not to be a mere means for production, but that the relation is to be precisely the reverse. (Cf. Das Kapital. I. 2. Aufl. p. 645). — On all these points such great difficulties have arisen that the socialist movement will be compelled to seek a new theoretical foundation if it would avoid building on untenable dogmas. But Marx's giant work keeps its value in the history of thought through the mastery with which an immense material of social experiences is ordered in a grandly planned systematic form.

e. Philosophy in the North

Only at quite few points in the general history of philosophy is there occasion to dwell on thinkers from the Scandinavian North. The philosophical movement has here consisted predominantly in an independent appropriation of philosophical ideas that were
[284] developed in the larger countries, an appropriation which has indeed been of significance for the course of the life of the spirit in the North, but has not led to results that intervened decisively in the general development of thought. The critical and especially the Romantic philosophy thus won disciples in the North too. If something peculiar is to be named in the way in which the philosophy of Romanticism was taken up here, it must be that the practical, personal interest, the interest in the ethical significance of ideas, makes itself strongly felt. And here there appears again a characteristic difference between the development in Sweden and in Denmark.

In Sweden there already comes forward in Thorild, at the beginning of the nineteenth century, the thought which for Swedish philosophy at its culmination came to express the absolute reality: the idea of existence as a living, harmonious whole. Under the influence of Kant, Fichte and Schelling a series of thinkers, among whom especially the energetic

Benjamin Höijer must be named, developed this idea further, until Christopher Jacob Boström (born in Piteå 1797, died in Upsala 1866) systematically completed the "rational idealism". The world which sense perception and experience show us cannot, according to Boström, be the true one, because it presents external oppositions in space, and because it develops in time. When German speculation sought to show the essence of the Absolute as engaged in constant development, this was, according to Boström, a sign that it had not raised itself sufficiently above experience. Even Hegel is an empiricist when he lets the idea unfold itself through the positing and annulling of opposites! Only in the thought of an eternal personality whose ideas are in turn (finite) personalities in mutual harmony does thought, for Boström and the Swedish national school founded by him, find its conclusion. Not only against the idealist doctrine of development of German philosophy, but also against the Christian theology's doctrine of creation and atonement does Boström polemicize in his zeal to hold fast the true reality in its pure, absolute perfection. It was his fundamental proposition that the higher explains the lower, the perfect the imperfect. Rational idealism is an ethical idealism: since the highest for us is a harmonious society of independent personalities, the idea of such a society is used to give the answer to what the "thing in itself" is that underlies the sensuous world of *[285]* phenomena. Boström's philosophy is a Platonism with the modification that it conceives the ideas as concrete personalities, not as abstractions. Idealism has here become a speculative philosophy of personality[68]).

While Swedish thought asserted the principle of personality in the ideal world, Danish thought aimed rather at asserting it in the world of experience. Both Treschow and F. C. Sibbern met German speculation with a sound realism in their conception of the world, and with a sense for psychological experience which kept wild speculations away. When Hegel's philosophy too had made its entry into Denmark, where aestheticians and theologians especially attached themselves to it, Sibbern wrote an able and pertinent criticism of it (1838). "All my philosophy", Sibbern once said, "aims at studying life and reality". Abstract and speculative construction could be nothing for him. Philosophy must start from a given foundation, which it is a matter of analyzing. It is not only the general ideas and laws that make existence intelligible to us: we must also know the factually given starting points from which the processes of development in existence take place. It is a chief thought in Sibbern's world-view (especially developed in his *Spekulativ Kosmologi* 1846) that all development takes place sporadically, from different, often apparently mutually conflicting starting points. Therefore his conception of development could not become as idealistic as Hegel's. While Boström found Hegel too empirical

because he assumed a development, Sibbern's objection to Hegel is that he does not assume a *real, historical* development. And to this doctrine of development as sporadic he gave an extraordinarily interesting application both in his theory of knowledge and in his philosophy of religion. Since the knower himself is always only one of many sporadic elements in existence, he will not be able to survey it in its totality; on the other hand no one other than he will be able to experience how the universal life stirs at this particular point. Thereby Sibbern came to maintain the limits of knowledge against
[286] speculation *(Om Filosofiens Begreb.* 1843. § 21) and to maintain the significance of the single subjectivity against ecclesiastical dogmatism (university programs of 1846 and 1847). — In his psychological works he has given contributions of lasting significance especially in the doctrine of the feelings. His book *Om Forholdet mellem Sjæl og Legeme* (1849), a work significant for its time, shows how he sought to bring his psychological theories into agreement with physiological experiences. He declares himself against the dualistic conception; it is one and the same life that appears as the life of consciousness and as material life[69]).

In a more radical way Søren Kierkegaard came forward against Romantic speculation and against its claim to have reconciled the oppositions of life and of existence. His chief thought was that the different possible views of life stand so sharply against one another that a choice must be made — hence the watchword *Either—Or,* — and a choice by which the single personality must decide out of itself — hence the watchword *the single individual.* He himself designated his thinking as qualitative dialectic, by which he meant to designate the opposite of the Romantic speculation's doctrine of continuous development through inner, necessary transitions. This doctrine was in Kierkegaard's eyes fantastication, but a fantastication to which he had felt himself drawn. Brilliantly gifted as thinker and poet, he had especially a wonderful ability to let reflections unfold, and he endows the possibilities of thought with a power of vividness and a wealth of mood that is unique in our literature. He was supported in this by an art of language that made it possible for him to shape words so that all the small waves on the great sea of mood became visible and all nuances of thought came into their own. Often he plays with language; it is as the warrior who plays with his weapons. But passionately as he loved the life in thought and mood and the cultivation of the art of language, there was nevertheless a still stronger need that led him beyond the world of possibilities to a life in reality and earnestness. A bottomless melancholy made him feel the insufficiency of the intellectual and aesthetic life, while on the other hand this melancholy led him —
[287] through the need for diversion — to the great aesthetic-philosophical production that

fills his first great period as an author (1843—46). He was a Romantic in gifts and mood, and precisely this meant that he learned from his own experience how hard the way can be from the world of reflection and fantasy through the narrow pass of decision (of the Either —Or) to the striving of the single individual in the world of reality.

Qualitative dialectic appears in Kierkegaard's epistemology in the sharp opposition between thought and existence. Even where thought can find a connection, it is not said that this can be held fast in the practice of life. As long as one lives, so long are we in becoming and thus face the uncertain, since there is no guarantee that the future will be like the past. We know backward, but live forward. A purely objective decision is impossible here. Therefore a system of thought that was to comprise existence is impossible. Only an abstract, purely logical system is possible. Such a system can perhaps, as an abbreviation, contain the experience made hitherto in its broad outlines; but life constantly leads to new possibilities and new choices. Besides, there are also at the same time such great differences and oppositions present that the thought is not to be found that could comprise them all in a "higher unity". And in any case the single, individual existence will not be able to come into its full right in such a system. When for Kierkegaard religious truth could appear only in the form of the paradox, this is a natural consequence of his theory of knowledge; for the paradox expresses the relation between an existing spirit and the eternal truth. He insists expressly that when he says that the object of faith is the paradox, it is no concession; the highest, the eternal can be had only through a subjective choice without objective grounding, indeed in conflict with what objective grounding leads one to assume. Outside subjective faith itself there is no criterion of truth: subjectivity is the truth.

In Kierkegaard's ethics qualitative dialectic appears partly in his conception of choice, the decision of the will, partly in his doctrine of the stages. He denies decidedly that there is any analogy between spiritual development and organic development. In the *[288]* spiritual domain no successive unfolding takes place that could make the transition from deliberation to resolution, or from one view of life (one "stage") to another, intelligible. Continuity is broken at every such transition. — As regards choice, psychology is supposed to be able to show only possibilities and approximations, motives and preparations. The choice itself comes with a jerk, a leap, whereby something entirely new (a new quality) is posited. Only in the world of possibilities is there continuity; in the world of reality decision always takes place through a break in continuity. But — one might ask — can this jerk or this leap then not be made an object of psychological observation? Kierkegaard's answer is not clear. He declares that the leap lies between two moments,

between two states, of which the one is the last state in the world of possibilities, the other the first state in the world of reality. From this it would seem to follow that it cannot be observed. But from this it would in turn follow that it must take place unconsciously, — and the possibility that there could be an unconscious continuity behind the oppositions appearing in consciousness could not be rejected. It is therefore by a fiat that Kierkegaard maintains the concept of the leap. — In his doctrine of the different views of life (or, as he calls them with a less happy expression, "stages") he is able to hold fast the necessity of the leap only by conceiving the views of life (the three chief forms: aesthetic, ethical and religious view of life) in their clearly and firmly marked, fully finished, indeed even extreme form. The depiction of the aesthetic stage in particular presents figures of whom it must be said that they have run themselves aground. It is therefore no wonder that the possibilities that can be shown psychologically fall short. Experience shows, after all, that where a development takes place it happens while the forms still have a certain indeterminacy; from one entirely finished form to another, development is mostly impossible. Even if one should appeal to the law of contrast effect, this can find application only where some tension still remains, where life has not congealed. —
[289] Already by this strong stress on the leap in every decision Kierkegaard prevents himself from obtaining a real content, definite real tasks, for his ethics. The decision, after all, is to rest not at all on the value of the content for man and the power which the content, through this value, can gain over man. And Kierkegaard's ethics becomes still more formal as the concept "the single individual" unfolds its consequences for him. The more the single individual is isolated from the species, the less can it obtain definite, real purposes and tasks. In the end, then, Kierkegaard pushes the ethical stage out entirely, and only the two possibilities remain: to live aesthetically or to live religiously. What did not fall under one of these two possibilities was philistinism for him. The choice became: either enjoyment or suffering, either the giving up of personal life or out into the tension and pain which the relation to the eternal truth necessarily brings for a temporal being! The ability to span strong oppositions and to bear the suffering that follows from it became more and more for Kierkegaard the criterion of the height and value of a view of life. It is a standard directly opposed to the natural need and direction of life, and there expresses itself here in Kierkegaard an outright life-hostile tendency. In his later years, accordingly, he read Schopenhauer's writings with great interest.

It was under the influence of this line of thought that he caught sight of the great problem (already stressed by Schopenhauer and Feuerbach) of the relation between primitive Christianity and modern Christianity, not in a dogmatic but in an ethical

respect. His violent polemic against the established Church in his last days came as a natural consequence of his whole course of development. In "the Christianity of the New Testament" he found his criterion of the highest view of life (the highest "stage") fulfilled. That the established Church now considered itself in continuity with the Christianity of the apostles was in Kierkegaard's eyes an insolent presumption, and in his *Øjeblikke* he carried through with great pathos and cutting mockery the thesis: the Christianity of the New Testament does not exist! — Thus the harmony which the thought of Romanticism had thought to find between knowledge and faith, between culture and religion, was burst. It was a work for which Kierkegaard by his whole nature was suited. He was not *[290]* one of those who were able to open ways to a solution by posing the problems in new forms or widening the horizon by way of experience. But he has passionately asserted anew the qualitative differences and decisions which speculation was inclined to efface, and he has thereby worked for the power and fullness of life and enriched far more than those who stand near him in view of life and standpoint[70]).

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Ninth Book

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Positivism.

For the spiritual life of the nineteenth century two intellectual tendencies are characteristic: Romanticism and positivism. The first proceeds from the needs of the heart and from the ideals of thought, the second seeks its support in what is factually given; for by the word *positive* is meant first and foremost the same as *real.* The two tendencies thus seem to be as opposed as can well be. They proceed from opposite starting points. And yet there is an inner kinship between them. They are borne by the same interest and rest on the same presupposition. Both are animated by a need of full spiritual content and of absorption in great realities. Romanticism wishes to grasp reality just as much as positivism does; only the former believes it can grasp it by the subjective way, while the latter will build on objective facts. The common presupposition is that an ideal which stands wholly outside reality is a false one. Therefore men turn away from the eighteenth century's criticism by the understanding and immerse themselves with enthusiasm in the great course of development of nature and of history. The concept of development reigns both in Romanticism and in positivism. Men seek especially to find the continuous connection within history which the age of Enlightenment and Revolution had broken apart. A more thorough understanding of the past and of the conditions for the development of the spiritual life forms an essential part of the significance of both tendencies in the history of thought. Even though it must be said that the spiritual oppositions — the oppositions in the domain of faith and of views of life — have steadily grown sharper in *[292]* the course of the nineteenth century, it is nevertheless the great debt we owe to these two intellectual tendencies that they have taught us to enter into standpoints which are essentially different from our own. By means of the historical sense, which is a form of human sympathy in general, a bridge has been thrown over oppositions which could not be overcome by logical discussion. This marks a turning point in the history of the human life of the spirit. A new organ has been acquired. What Immanuel Kant set up as

the program of the human sciences: to find the moving forces which have produced the works of the past, to investigate their laws and their range, and thereby to set them free to do the work of the future, — this Romanticism and positivism have each in its own way sought to fulfill.

It would therefore be wrong to regard positivism merely as a reaction against the philosophy of Romanticism. It would be wrong even chronologically; for the origin of positivism lies in the time before the full development of Romanticism. It was not surfeit with the one tendency that led to the other, even though positivism found more favorable conditions for spreading in wider circles once the dominion of Romanticism had been broken.

Yet there is one very essential point in which the two tendencies from the very outset diverge from each other. In its enthusiasm for the unity of thought Romanticism overlooked the multiplicity of reality, and in its assurance of the truth of the ideal it overlooked the strict mechanical conformity to law to which everything must submit that is to attain subsistence in the world of reality. Since positivism takes its starting point in what is factually given, it opens itself to the differences and oppositions of reality and strives to find the laws according to which phenomena arise and develop in the real world. Its difficulties and problems therefore come to lie at other points than those of Romanticism. Positivism seeks to go up from what is factually given toward the unity of thought and the validity of the ideal, while Romanticism went the opposite way. The great thinkers of positivism assume just as confidently that it must be possible
[293] to climb from below upward as the Romantics assumed that one must be able to climb from above downward. Comparative consideration must decide which way is the better.

While the philosophy of Romanticism had its home in Germany, positivism has its home in France and England. National differences show themselves here in a characteristic manner. The English school, which in Hume had reached a certain conclusion, now revives and appears in new forms, which still preserve the realistic stamp that distinguished English thought right from the Middle Ages onward. Only when the word positivism is taken in a wider sense does it also include the main tendency in the English philosophy of the nineteenth century. Its most important spokesmen have protested against being called positivists — but they were then thinking of the narrower sense of the word, in which it designates Auguste Comte's philosophy. It does not diminish the significance of modern English philosophy that it is here presented as that which has completed and extended the way of looking at things which Auguste Comte — the second great figure of French philosophy beside Descartes — had shaped in its broad

outlines. And all the less so, as Comte himself stood in considerable debt to the older English school.

Both France and England have in our century made valuable contributions to the development of philosophical thought which cannot be brought under positivism, even where this word is taken in the wider sense. But in the general history of philosophy we must lay the chief weight on the most significant tendency, and therefore also let it furnish the title for the Book that treats French and English philosophy in the nineteenth century. *[294]*

A. Comte and French Philosophy

1. Philosophy in France in the First Decades of the Century

a. The Renewal of the Principle of Authority

The French Revolution had wished to throw off the bridge to the past. The age of the Church and of the old faith was to be over. But even before the Revolution's opponents had triumphed outwardly, it became evident that as a spiritual power the Church was not defeated. Eminent writers praised religion for its significance in the service of humanity and for the poetry it shed over life, and amid the mighty upheavals there were those who sought an absolutely fixed point in the general swell of the waves. With the outward shipwreck of the Revolution, moreover, the ideas of the eighteenth century, as the French philosophers and their English predecessors had maintained them, also seemed to have played out their part. They had — so history now seemed to testify — been unable to lead to any fixed social order, while the Church had maintained and renewed itself "in spite of the syllogism, the scaffold and the epigram", to use an utterance of the principle of authority's most important representative, Joseph De Maistre *(Le pape.* 14. éd. p. 477). In defiance of everything the eighteenth century had thought and fought for, a theological school now sought to lead all ordering of human life and all understanding of existence back to supernatural principles. This school has more interest for the history of literature than for that of philosophy, and I therefore confine myself here to some characteristic features from de Maistre's chief works, while, as regards the tendency as a whole, I refer to the account in Georg Brandes: *Reaktionen i Frankrig* (2nd ed. 1892).

When modern philosophy, as the preceding sections of this work have shown, supports itself on natural science, partly as a further application of its results, partly as a discussion of its presuppositions, de Maistre seeks to deprive it of this support. He

[295] disputes that there can be purely physical causal explanations. What is material cannot — he asserts — be a cause. A physical cause is a self-contradiction[71]). All material motion is derived from original impulses, which can only stem from spiritual beings. In the consciousness of the influence of our own will we have a proof that motion begins with an act of willing. Scientific explanations, such as, e.g., the origin of water by the combination of oxygen and hydrogen, the origin of ebb and flood by the influence of sun and moon, the influence of chemical processes on the formation of the strata of the earth, de Maistre therefore declares to be "dogmas" which may rightly be doubted. Yet he is quite willing to let the learned amuse themselves in the domain of natural science, provided only they do not apply consequences from it to social and religious affairs. It is not science but faith that is to govern men. The truths that sustain life God proclaims not through the academies but through the authorities in Church and State. It is prelates, noblemen and high officers of state who are to teach the nations what is true and false in the moral and the spiritual domain. History has now furnished proof of the impotence of human reason to guide men. How few are able to think correctly! And no one is able to think correctly about all things. Then it is best to begin with authority. The ground of all misfortunes is that speech has been set free. It began with the Reformation — one of the greatest crimes men have committed against God! And it was continued in the philosophy of the eighteenth century — one of the most shameful episodes in the history of the human mind! The philosophers of the eighteenth century preached error as religion. They were not in good faith, but had formed a conspiracy (la cabale) against what is holy. Voltaire — this blasphemous buffoon (bouffon sacrilège) — gave himself the appearance of pleading the cause of innocence, although the innocence of Calas is not proven. By setting Locke on a pedestal he has only shown his fanaticism and at the same time his lack of patriotism. — An end to the misfortunes will come only when the absolute infallibility of the Pope is acknowledged. Without it the universal
[296] character of the Church cannot be maintained, social peace cannot be preserved, and the sovereignty of princely power cannot be upheld. It is a mysterious principle that is here appealed to: but everything is mystery in the social and in the physical world. Reason condemns war — and yet war reigns throughout all nature as a mysterious means to the subsistence of life. The executioner is the horror of society — and yet at the same time a force that holds society up: take away this incomprehensible social factor, and chaos will take the place of order! Only tradition and authority, not human reason, can guide us in this world, which everywhere presents terrible mysteries. Nothing therefore is more ridiculous than the assumption that man should gradually have raised himself

from barbarism to science and civilization. —

These ideas, which de Maistre develops especially in his *Soirées de St. Petersbourg*, a series of dialogues which are assigned to 1809 but appeared only after his death, in the year 1821, turn upside down everything that thought had sought to build up ever since the days of the Renaissance. As so often, there is here a kinship between the opposites. Just as unhistorically and uncomprehendingly as the philosophers of the eighteenth century had stood toward the Church and the Middle Ages, just as unhistorically and uncomprehendingly did de Maistre and the men of his tendency stand toward the philosophy of the eighteenth century, — only with the difference that the wicked philosophers now took the place of the wicked priests as the source of all evil. The philosophy of the eighteenth century was regarded as a deliberate conspiracy against authority, as a purely arbitrary process. And this was in good agreement with the fact that authority itself was conceived as what is purely arbitrary, commanding from without or from above. Hereby there arose again a new likeness with the philosophy of the eighteenth century: for when all knowledge is to be built on authority, its self-activity must be subdued, its free unfolding of its inner possibilities denied. De Maistre therefore consistently comes — just as Condillac did — to assume the complete passivity of man. It is then no wonder that the possibility that man should have arrived at civilization by the way of natural development appears to him as utterly absurd. *[297]*

b. The Psychological School

The significance of the authoritarian school lay in its passionate bringing forward of the value of the historical powers, in opposition to the psychology of the Enlightenment and the Revolution, which saw in the heart and understanding of the single individual an independent force. But as the principle of authority was conceived here, it was an external, brutal principle; its inner connection with mental life was not investigated — and could not be investigated from this standpoint, since every independent value that was ascribed to the individual's own inner life would entail a limitation of the absolute authority. Philosophically, therefore, as already remarked, the authoritarian school stood (when certain mystical stirrings, of which it itself becomes afraid, are discounted) on Condillac's standpoint. All the greater is the interest of seeing how, out of Condillac's own school, through the use of psychological observation and reflection, a deeper psychology develops, which places itself in definite opposition both to the French philosophy of the Enlightenment and to the new authoritarian school.

When the Revolution broke out, Condillac had triumphed. His philosophy became the

official one, and his adherents occupied the philosophical section in the National Institute established by the Convention. Before this philosophical academy the physician PIERRE JEAN GEORGES CABANIS in the winter of 1797—98 read a series of memoirs on the relation between soul and body, which were printed in the Academy's transactions and later in expanded form appeared as a book (1802) under the title: *Rapports du physique et du moral de l'homme.* Cabanis speaks of Condillac with great appreciation, but modifies his doctrine on essential points. Condillac lays the weight exclusively on the external senses; according to him man receives from the surrounding world in a passive manner the whole content of his consciousness and all its forms. Cabanis reproaches him for not having taken account of that in consciousness which corresponds to the organism's own inner states. Impressions constantly come to the brain from the various inner organs, and besides, the state of the brain and nerves themselves always plays a part
[298] as well. There is an obscure feeling or sense perception which is immediately bound up with the subsistence of life, independently of external sense impressions, and which has therefore already been present before the individual is at birth brought into relation to the great surrounding world, from whose influences Condillac derives everything in consciousness. Cabanis introduced the feeling of life into modern psychology. With this the passivity toward the surrounding world was already limited; for in the feeling of life the individual has an original fund which gains influence on everything it later takes into itself, and gives it color and stamp. And Cabanis brings instinct into close connection with the feeling of life. In instinctive actions too he finds facts that are incompatible with Condillac's theory. Instinct presupposes a fund of original force, which is released by impressions from the inner functions of life, as is shown especially in the reproductive and parental instincts. Cabanis lays so great weight on the concept of instinct that he even hints at the thought that a universal instinct might be at work in all processes of nature, a thought in which the germ of Schopenhauer's philosophy of nature has rightly been found. It was not, however, Cabanis's intention to put forward a philosophical system. He wishes to proceed in a purely psychological and physiological way and not to answer the ultimate questions. Although in a single place he expresses himself in a strongly materialistic way (in that he lets thought be secreted in the brain, as bile is in the liver), it would nevertheless be wrong to assign his work to the materialistic literature proper. His standpoint is characterized by the following statement (Rapports, XI, 8. éd. Peisse p. 597): "It took a long time before men came to acknowledge only a single force in nature; perhaps an even longer time will be needed to acknowledge rightly that, since we cannot compare this force with anything else, we cannot form any true notion of its

properties." In a later work *(Lettre sur les causes premières)* he gave a fuller development of his thoughts, which departs still further from materialism than the famous chief work had done.

Napoleon did not regard Condillac's school with favorable eyes. It did not, after all, content itself with investigating our sensations and representations; beside its *ideology* (so DESTUTT DE TRACY, an ardent Condillacien, called the doctrine of the origin of *[299]* representations, cf. his work *Eléments d'Idéologie* 1801) it also busied itself with theories of morality and law in the spirit of the eighteenth century. Napoleon derived all the misfortunes of France from the "ideologists", and in any case he would rather think for France, just as he acted for it. He suppressed the Académie des sciences morales et politiques established by the Convention, and writings of a freer philosophical content could no longer appear in France. Destutt de Tracy had to have his commentary on Montesquieu printed anonymously in America in English translation. The ideologists withdrew into narrower circles. In particular there gathered around Cabanis and Destutt de Tracy in Auteuil a band of younger men who were interested in philosophical studies. From this circle came forth France's first psychologist of this century, MAINE DE BIRAN (b. 1766, d. 1824).

Biran held administrative posts under the Revolution, the Empire and the Restoration and was a member of legislative assemblies. It was not, however, the great outward events of the time that filled his mind. He was a good patriot, but lacked the inclination and the capacity for outward action. The great historical drama worked upon him by strengthening the emotions in his own mind, emotions which he early observed both with the interest of the theorist and with an inward longing for peace and harmony within. His constitution must have been of such a nature that the relation between the passive and the active, the involuntary and the voluntary, or, as he himself aptly designates it, between temperament and character, became for him a theoretical and practical problem all his life long. Of this his *Journal intime* bears witness, the most remarkable of the works he left behind (published by E. Naville in his book: *Maine de Biran. Sa vie et ses pensées.* 1857), which contains entries from the beginning, the middle and the end of his life as a thinker. How early and how definitely his peculiar problem presented itself to him can be seen from a statement from the period when he still adhered to Condillac's doctrine (27 May 1794). "What is properly the so-called mental activity? I always feel the state of the soul determined by this or that bodily state *[300]* . . . If I could some day undertake a connected piece of work, I might have a wish to investigate in what degree the soul is active, in what degree it can alter the external

impressions, increase or diminish their strength by the attention it directs upon them, and try how far it is master of this attention . . . It would be highly to be wished that a man practiced in observation would analyze the will, as Condillac has analyzed the understanding." At the same time he complains of the constant shifting and streaming in his inner states (cette révolution perpétuelle, cette roue toujours mobile de l'existence), which hinders fixed moods and gives rise to doubt and unrest even toward what he seeks with honest will to hold fast.

In his first works Biran kept to the doctrine of Condillac and Cabanis. But gradually he laid ever more weight on the activity which in his view comes immediately to our consciousness in the effort of will. "The self makes itself known to itself through the inner sense in the effort or the movement of will, which the soul perceives (aperçoit) within itself as a product of its activity, as an effect that has its cause in its will." *(Rapports du physique et du moral de l'homme,* Oeuvres philos. 1841. IV, p. 75). Here there is for Biran an original fact which puts an end to Condillac's doctrine of passivity. But he does not forget — constantly repeated inner experiences saw to that — the involuntarily arising sensations and moods. He reproached the moralists who write on the problem of happiness for giving themselves up to general reflections and postulates and believing that one can immediately command one's feelings and inclinations. Wholly independently of our conscious willing a multitude of changing phenomena stir within us, which the self finds before it when it becomes conscious of itself, and which must stem from another inner cause than the self. There is also, outside the self or consciousness and independently of the relation to the surrounding world, a series of inner phenomena which self-observation discovers, but which can subsist without it. (Journal intime. 24
[301] Oct. 1814). This theory of sensations that were to subsist outside the self aroused great attention and much opposition in the small philosophical society which Biran gathered about him in Paris, and in whose meetings men such as the physicist Ampère, the historian Guizot and the philosopher Royer Collard took part. Only Ampère joined Biran definitely. They were both agreed in setting the immediate consciousness (apperception immédiate) of the energy of will as the central psychological fact. But outward from it mental life fades off into the unconscious, and when the act of self-consciousness sets in, the self finds a whole series of elements given — in any case it hears their echo. So also on awakening from torpor or on reflection upon habitual states. In a hitherto unprinted *Mémoire sur les perceptions obscures* (of 1807), which has been brought to light by Alexis Bertrand (La psychologie de l'effort et les doctrines contemporaines. Chap. 2), Biran had sought to explain this opposition between the passive and the

active in mental life physiologically by assuming different cooperating nerve centers. He also uses the phenomena of somnambulism to illuminate the duality within us. In his zeal for exact self-observation and for supplementing self-observation by other sources of psychological knowledge Biran is a forerunner of modern psychology. From his psychological standpoint he declares himself definitely against the authoritarian school, which attributed no significance to man's free self-activity. Only through absorption in the nature and laws of mental life can one, Biran maintains, find the point where the origin of the moral and religious relation lies. The men of the authoritarian school, who wholly disregarded the individual nature of the soul and "ex abrupto" set up authority as the foundation of all science, lacked in his eyes love of truth and were led merely by the need to act, to work upon men. And they fell, he continues, into skepticism and materialism, in that they made the soul purely passive and let its whole direction be determined from without, by authority! (Journal intime. 28 July 1823).

In the immediate consciousness of self-activity Biran sees the primitive and fundamental principle of all our knowledge. In it not only does our own self proclaim itself to us; but since self-activity meets resistance, we obtain consciousness of the existence of *[302]* the material world. The self-activity we are aware of is in fact the soul's acting upon the organism, where there is always a certain resistance to overcome. Thus Biran gives a spiritualistic interpretation of the consciousness of our acting and of the checking of our acting that may take place; and thereby there is gained at the same time an explanation of the opposition between our willing proper and what is involuntary in our inner states. Yet he does not adhere to the Cartesian doctrine of the soul as a substance: we know the self only as active force; there could be no immediate consciousness of the substance. — The force or acting which we thus perceive in ourselves serves us as a type to which we refer all external phenomena. We come to know in our self-consciousness the concepts of force, cause, unity and identity, and only because we can draw these concepts from the primitive inner fact, concepts which are very different from concepts of kinds, can we apply them to the phenomena of experience. — Neither external sense perception nor authority, but the immediate consciousness of our self-activity is the foundation of all our knowledge. Concerning the reality of this foundation Biran harbors no doubt; modern psychology will hardly be able to grant, however, that he has overcome Hume's doubt about the possibility of an immediate apprehension of causality[72]). —

Biran's most important writings were published only long after his death. He worked for a time on a larger work which was to comprise all his psychological studies *(Essai sur les fondements de la psychologie);* but political circumstances prevented him from

finishing it for publication, and later he laid his draft aside in order to work out a new exposition *(Nouveaux essais d'anthropologie).* Both works were published in 1859 by Naville (Oeuvres inédites de M. de Biran). The reason why in his last years he rejected his older exposition was that he had caught sight of a deeper layer in the soul than those he had hitherto taken into account in his reflections. Hitherto he had described the currents which spread from the feeling of life over the whole inner life, and, in opposition to
[303] them, the feeling of energy in which our self seeks to work its way out of the passive and involuntary. In the feeling of energy he had in stoic fashion found his refuge amid the outer and inner emotions. But this refuge was not enough for him in the long run. It did not suffice against the unrest in the outer and in the inner world. The need awakens in him for an absolutely fixed point of support, for a devotion of another and higher kind than that to the moods of the feeling of life, a devotion to something that is independent of the soul's own imperfect activity, and which yet comes not from without but from within. He uses Kant's distinction between noumena and phenomena and seeks his firm standing ground in the inner relation to something that lies beyond all phenomena. It is the fundamental thoughts of religious mysticism in which he now lands by the way of his own independent experience, at the same time as the study of the Gospel of John, De imitatione and Fénelon replaces the study of Descartes and Leibniz, which in its time had in turn replaced the study of Condillac and Cabanis. Biran's religiosity takes on no confessional character. He holds fast that religious feeling cannot be produced from without; it must arise involuntarily, and just as involuntarily under its influence the images take shape in which the soul finds the notion of the eternal and infinite expressed. Religion is feeling rather than faith; faith is subordinate to feeling. What our spiritual activity can do here is to bring about receptivity for the higher life, the spiritual life proper, *la vie de l'esprit* in opposition to the active life in will and reason, *la vie humaine,* which again is opposed to the currents of the feeling of life, *la vie animale.* This higher life Biran also designates as "the mystical life of enthusiasm", "the highest stage the human soul can reach, in that, so far as it lies with it, it makes itself one with its highest object and thus returns to the source from which it has sprung". (Oeuvres inédites. III. p. 541. 521).

But Biran's diary shows that his passion for psychological observation and reflection did not fall away with his passing over to mysticism. Again and again he discusses the question how far the psychological causal explanation can reach with respect to the most inward and highest states. Now it is said that those who seek to explain everything
[304] by natural causes cannot be prevented from asking whether even the highest state of

blessed rest and contemplation, where the soul is in ecstasy and feels itself under divine influence, — whether even that might not be explained by the working of organic moods, so that heavenly bliss would be replaced by unrest and tossing as soon as the organic state changed. Now it is declared as settled that the idea or feeling which the soul has of the perfect, the great, the beautiful and the eternal cannot spring from the soul itself; moral and religious truths have another source than psychological truths. (Journal intime. 26 August 1818 and 19 September 1818). Right up to the last leaves of the diary there is discussed in particular the question how our own mental activity in thinking and in concentration of the mind can prepare the way for that higher devotion by virtue of the constant reciprocal action between the active and the passive in us: and how, it is then asked, can one distinguish between what as a consequence of such concentration wells up from the soul's own ground and what is due to the influence of divine forces? Beyond a "whether — or" (soit—soit) Biran does not get, however great his need to hold fast that when we seek to penetrate into truth, truth also seeks to penetrate into us. (Journal intime. October 1823).

Only few had knowledge of Maine de Biran's thought before his writings — for the most part long after his death — were made public. To these few belonged first and foremost the famous physicist André-Marie Ampére (b. 1775, d. 1836), who stood in lively oral and written communication with him and together with him found the starting point of philosophy in the immediate consciousness of the energy of the self. Ampère acknowledged his friend's priority on this point, but while Biran all his life continued to circle about this one point, Ampère sought to give a detailed psychological and epistemological theory. Philosophical studies occupied him to a high degree; beside his scientific and mathematical lectures he also lectured on philosophy. Ørsted's discovery of electromagnetism led him into the brilliant series of investigations which have especially made his name famous; but he did not forsake his old love, and in his later years he was *[305]* occupied with studies on the encyclopedia and classification of the sciences, which led to the publication of his *Essai sur la philosophie des sciences* (first volume 1834, second volume with introductions by Ste. Beuve and Littré 1843), in which several of his interesting psychological and philosophical ideas were expressed. Later his correspondence with Biran and some fragments of a treatise were published with an introduction by his son, under the title: *Philosophie des deux Ampère,* publiée par Barthélemy St. Hilaire (Paris 1866). (Of the son's philosophy, however, there is not much in the book).

Ampère stands as a philosopher to Biran roughly as he stands as a physicist to Ørsted. In Ørsted's discovery he divined a single example of a general law, which he succeeded

in developing mathematically and demonstrating experimentally. Electromagnetism was extended by him into the electrodynamic theory. The influence of the electric current on the magnetic needle led him to discover the influence of electric currents on one another and the influence of the earth on electric currents. Biran's observation of the relation between the active and the passive in mental life led Ampère to two series of investigations, partly to an investigation of how our sensations and representations involuntarily combine prior to and independently of conscious activity, partly to an investigation of how the scientific knowledge of existence becomes possible on the basis of the mind's conscious use of its faculties. By the first investigation he has given interesting contributions to the psychological doctrine of association, by the second to epistemology.

In psychological respects Ampère applies the same method as the English association psychologists. He does not — like Biran and most other French psychologists — stop at description, but seeks to explain the origin of compound phenomena of consciousness by the fusion and combination of simpler elements. A fusion (concrétion) takes place, e.g., between the sensation of color and the sensation of resistance, which enter into so close a combination that one is inclined to take such compound combinations for
[306] simple and primitive ones. It is only analysis that discovers that a fusion has taken place, and that our own involuntary activity and the limits it meets have thereby played an essential part. Another example of concrétion is had in recognition, where the present sensation of an object and the reproduction of an earlier sensation fuse together. By the same process Ampère further explains that when during the performance of an opera one has the text before one, one can immediately *hear* the words, while without the help of the text one would not be able to hear them. The words one reads and the sounds one hears fuse immediately together[73]). It was this kind of explanation which, according to Ste. Beuve, made Ampère's psychological lectures so attractive.

In the very fundamental fact on which he was agreed with Biran, the immediate consciousness of self-activity, Ampère makes a distinction which Biran did not grant. He distinguishes between self-consciousness (autopsie [i.e. heautopsie], later: émesthèse) and the muscular sensation, showing that this latter sensation we have also when another person moves our arm or our leg. In one's own movement self-consciousness is the phenomenon that contains the cause, the muscular sensation the phenomenon in which the effect appears. Precisely through the fact that the muscular sensation can also be produced by another, I discover that I myself am in some cases the cause of it. In this experience, then, the causal relation dawns upon me. Immediately I have it given only in

my own activity, and all other acting I therefore apprehend by analogy with this. Together with the immediate consciousness, at which Biran stopped, there works according to Ampère the faculty of apprehending relations, the cognitive faculty proper. It is by it that we apprehend the relation between cause and effect, between places in space and time, etc. And it is by it alone that we can know more than mere phenomena. However true it is, according to Ampère, that everything that appears to us is only phenomenon, he is nevertheless convinced that the relations (rapports, relations) which obtain between the phenomena among themselves without depending on the phenomena's qualitative character have absolute (noumenal) validity. The mere concepts of kinds, which are *[307]* obtained by comparison and abstraction, have no higher validity than the sense qualities which make up their content. But the concepts which express pure relations and modes of coordination, such as causality, number, time and space, hold absolutely and form the bridge by which our knowledge reaches from the world of phenomena to that of noumena. On this point Ampère declares himself definitely against Kant, whom he otherwise highly admires. No less definitely does he declare himself against Reid, who assumes an immediate observation of external reality, whereby according to Ampére he confuses a "concrétion" with an elementary observation. Noumenal reality (matter in itself as the cause of sense impressions, the soul in itself as the cause of our own acting, God as the cause of everything) we discover only by way of inference and in the form of hypothesis. What we immediately apprehend is only phenomena, but the relations, independent of the qualities, which we intuit between the phenomena hold for the noumena too. By relations Ampère means the same as what Boyle and Locke called the primary qualities.

On this question Biran at first inclined to Reid's view. When his philosophical friend had eagerly urged him to study Kant, he became more of a Kantian than Ampère cared for. Against the latter's theory of relations he maintains (in a conversation which is mentioned in Journal intime 30 Oct. 1816) that a very great difficulty is bound up with making the transition from the consciousness of our own activity (the fundamental fact common to Ampère and Biran) to the assumption of external causes. Here, according to Biran, is an abyss over which analysis and induction cannot carry us. From the fact that I am not the cause of a passive state in me it cannot be inferred that there is necessarily a cause which produces everything that I do not produce. We are ourselves causes, and therefore it is natural for us to apprehend other things as causes: but that is no proof! — When Ampère held that the absolute validity of relations was proved by the fact that all inferences we draw from our theories of relations are confirmed by experience, Biran

answers (in a hitherto unprinted letter, used by A. Bertrand: La psychol. de l'effort p. 188):
[308] "What experience can teach us whether the forms of the coordination of phenomena are absolutely in things or only in the mind that apprehends? Can this doubt of reflection ever be cleared up by an external experience? Do not both possibilities agree equally well with the phenomena?" — To this Ampère has hardly been able to give any satisfactory answer from his theory of relations[74]).

In his world-view Ampère is a Cartesian. He divides the sciences into cosmological and noological sciences, of which the first concern the material, the latter the mental phenomena. The noological sciences are to be studied after the cosmological, partly because these show us the use of the human faculties for the knowledge of the world and thereby throw light on these faculties' nature, partly because the knowledge of the world, especially of man's physical nature, which they give, is of importance for understanding man's intellectual and moral faculties. As regards Ampère's classification of the sciences in detail, it is very intricate and cannot in simplicity and clarity measure up to the one Auguste Comte set up at about the same time. But there are in his "Essai sur la philosophie des sciences" many interesting remarks, and if he had been able to carry out his wish to give not only a classification but also an exposition of the most important truths, methods, problems and hypotheses in the various domains, there is no doubt that so excellent a thinker and investigator would have produced an excellent work and an interesting corrective to Comte's system. — At about the same time as Ampère and Comte the woman mathematician Sophie Germain (b. 1776, d. 1831) was also busied with ideas on the course of development of the sciences in its connection with the standard of truth which lies in the very nature of human knowledge. Under the evident influence of Kant she shows that by reason of the nature of our consciousness we have a need of unity, order and connection, a need which guides us not only in scientific inquiry but also in the moral and aesthetic domain. Her philosophical work *Considérations générales sur l'état des sciences et des lettres aux différentes époques de leur*
[309] *culture* (published 1833, new ed. 1879 in "Oeuvres philosophiques de Sophie Germain" by H. Stupuy) aims to show that the same type guides us in science, morality and art. "There is only one single type of the true. In morality, in science, in literature, in art we always seek the unity, order and proportion of existence among the parts of one and the same whole." The causal principle is only a special form of this general principle. We find application for the causal principle when we do not see the object we are investigating in its whole. It shows itself to us as a fragment, and we then ask what unity encloses it. We see it as a part and seek the whole to which it belongs. This need of unity and wholeness

has led to bold systems, to arbitrary use of analogies and to the assumption of mystical causes. But it has also led to many happy ideas, and the progress of science depends on its being alive. The coherence in the history of science depends on it. But men gradually learn to put methods in the place of systems and to ask *how and how much* in the place of *why.* Instead of creating the world according to the whims of his imagination man learns to trace out the real connection of the world, and when it thus gradually succeeds in finding a great unity in everything, the imagination will regain in surer form what it has lost by its bold images having been displaced by critical inquiry. —

While in the philosophical closets of Biran and Ampère a thinking was developing which led beyond the reigning Condillacism, this was defeated in philosophical teaching by the appearance of Royer Collard and Victor Cousin. ROYER COLLARD'S lectures at the Sorbonne (1811—14) introduced Reid as a philosophical classic in France. He laid great weight on psychological inquiry in the direction of the Scottish school and emphasized the significance of instinctive apprehension and of immediate moral conviction as against the limited point of view and abstract analyses of ideology. This was at the expense of the sharpness of the problems; an appeal to sound common sense all too often took the place of an argument. But a new tendency had been inaugurated, and the eye was opened to sides of the spiritual life which the hitherto reigning school had left in shadow. Royer Collard's considerable personality gave his appearance as a philosopher a peculiar *[310]* weight. His successor Victor Cousin (b. 1792, d. 1867) was an enthusiastic speaker with great talent for awakening the interest of the young. Young himself, he shared the young's expectation that with the free constitution and the fall of military despotism there would come a brilliant time for French spiritual life. His teaching had a decidedly historical stamp. He brought the history of philosophy into academic teaching in France. His own philosophical points of view he got partly from the Scots and Royer Collard, partly from Biran and Ampère, in whose philosophical gatherings he took part. Biran saw well enough that Cousin "was hunting on his preserves"; but he expected to get a share of the game, since Cousin's teaching would produce receptivity for his own great psychological work when it came (which unfortunately it did only some thirty-odd years after Biran's death). Cousin combined Reid's theory of the immediate apprehension of absolute reality with Biran's doctrine of the consciousness of our own self-activity and with Ampère's doctrine of absolute relations. And with this he later also united Schelling's and Hegel's doctrine of absolute reason. On journeys in Germany he had come to know German speculative philosophy, after Md. de Staël's book *De l'Allemagne* (which was written in 1810, but could appear only after Napoleon's fall) had given a

general impression of German philosophy from Kant's time. Cousin's philosophy became an eclecticism which from the various systems drew out what supposedly had lasting value: the systems were each by itself incomplete, but none of them wholly false! The standard for the choice was at first psychological observation after the pattern of Reid and Biran, later the universal reason of Schelling and Hegel, and in the last part of Cousin's career — when after the Revolution of 1830 he had become official director of all philosophical teaching in France — the chief doctrines of natural religion and of Cartesian spiritualism. The boldness and the inspiring quality of his first appearance[75]) as a philosophical teacher were damped by his historical studies and his official position. Aptly has Renan, who moreover long after Cousin's death expressed in warm words
[311] what he owed him, cited him as an example of the fact that it is not good for philosophy to win an all too complete victory. It must be added that the victory was only outward. Napoleon had looked with pleasure on Royer Collard's appearance, and it was the July Monarchy that supported Cousin.

Cousin's most important work is, Du *vrai, du beau, du bien,* his lectures of 1818, which were published by Garnier twenty years later in their original form, but in later editions were mutilated by Cousin himself, whom the heresies of his youth now embarrassed. It was this work that made so great an impression on Renan. Besides this, one must seek his ideas in the prefaces to five volumes of *Fragments philosophiques,* in which he treats subjects from the history of philosophy. In the preface to the first of these volumes (1826) he presents his characteristic doctrine of "impersonal reason", in which he has combined Reid, Ampère and Schelling. It is by means of the psychological method, by deepening self-observation, that he believes he has reached a point to which not even Kant had penetrated, a point where the apparent subjectivity and relativity of the necessary principles vanishes, in that there is revealed an involuntary grasping of truth as the foundation of all logical reflection and of all setting up of necessary concepts. Only by becoming an object of reflection does reason become subjective; in itself it is a light which, independently of all personal differences, shines within every human being as a universal revelation.

While Cousin did indeed believe that he was supporting himself on psychological observation, but very soon rose above it on the wings of rhetoric and fancy, Théodore Jouffroy (1796 —1842) had a greater and more faithful interest in psychological inquiry. He did, to be sure, keep one-sidedly to self-observation and disdain other sources of psychological knowledge, in that in a strictly spiritualistic fashion he distinguished between psychology and physiology as two subjects which had nothing to do with each

other. But he was in earnest — a whole series of investigations shows it — about getting clear on the relation between self-observation on the one side and natural science and philosophical speculation on the other. He does not have Cousin's romantic virtuosity in throwing bridges over abysses. His philosophy sprang from his purely personal need of *[312]* clarity in knowledge, after he had through a hard struggle seen the theological notions lose their value for him. But although there is a strong personal interest in his thinking, it does not lead him to compromise or to content himself with solutions that do not fully satisfy his thought. Rather, he then stops at the recognition that the solution of the problem is unattainable. "There are", he says in one of his lectures *(Mélanges philosophiques.* 3 éd. p. 350 f.), "two ways in which the man who thinks can obtain peace in his soul and rest for his mind; the first is to possess, or believe one possesses, the truth concerning the questions that interest humanity; the second is to see clearly that this truth is inaccessible, and to know why it is so. . . Since the facts we can observe are limited, the inferences we can draw from these facts are so likewise. Science therefore has its horizon, beyond which it cannot see; it is its task gradually to determine this horizon. At this outermost limit of its domain it must part from poetry, which alone can go further. This separation is its duty toward humanity, toward which it has the task of discovering the truth, and which has only all too often suffered from having hoped for and sought the truth where it was, and for ever will be, inaccessible to it." — It is far more by the manly way in which Jouffroy faced the problems than by his treatment of them in detail that he stands as a fine and attractive figure in the history of French philosophy. In so far as he points to definite results, these go in the direction of the eclecticism founded by Cousin.

c. The Social School

When Cousin's disciple Damiron in the year 1828 published a work on French philosophy in the nineteenth century, he set up eclecticism as the higher unity or the right mean as against the Condillacian school on the one side and the theological school on the other. The peculiar intellectual work of Biran and Ampère, which went deeper than *[313]* eclecticism, had not come before the public. But in the second edition (which came out the same year) he found himself induced to speak of the ideas of Saint-Simon and Auguste Comte, although he admitted that he could not clearly see their significance. At that time only Comte's first writings were available. Saint-Simon's career, on the other hand, was concluded. The germ had been laid from which modern socialism and positivism were to develop.

Saint-Simon (whose full name was Claude Henri de Rouvroy, comte de Saint-Simon) was more a journalist and social reformer than a philosopher. But he gains significance for the history of philosophy through his living conviction that a new social order becomes possible only if a new world-view makes its way. And this new world-view can, of this he is no less convinced, only be built up on the foundation of the positive sciences. Saint-Simon himself had not received any scientific education. As a young man (he was born in 1760) he took part with bravery in the North American War of Independence. Thereafter he threw himself into industrial enterprises, renounced his noble rank during the Revolution, took to speculating in national lands and appeared as a "grand seigneur sansculotte". He was in prison during the Reign of Terror, but was freed at Robespierre's fall. His wealth he used (as long as he had it) not only to live in high style, but also to prepare the scientific and social reform of which he had early conceived the thought. He surrounded himself with men of science from the Polytechnic and the Medical School in order to train himself to write an encyclopedia of the sciences. His great plans he did not give up because he became poor; on the contrary, he worked with ever greater zeal. He had a rare capacity for drawing eminent men to himself and getting them to work with him and for him. Augustin Thierry and Auguste Comte were for a time his secretaries and collaborators and called themselves with pride his pupils, until their ways parted. While Saint-Simon in his earlier period had put the production of a new encyclopedia in the first line, because only on the foundation of such a one
[314] could a new catechism be worked out, after the upheavals of 1814—15 he believed he could proceed directly to the new organization of society, whose main feature was that industrial and economic life should stand in the first line and the government system proper take a subordinate place. In place of noblemen and jurists, great industrialists and men of science were now to stand as the heads of society. The final purpose was to be that the working class, the class that suffers most, should be helped forward in intellectual and economic respects. While in France after the Restoration men were absorbed in constitutional and parliamentary struggles, Saint-Simon declared that it is the social renewal that matters, that the external form of government is indifferent, and that a government is on the whole only a necessary evil. As the leading motive to carry the new order he at first thought of insight into the harmony of interests, much as Helvetius and later Bentham conceived it. But in his last years he appealed to the love of man (philanthropy) and wished to found a new Christianity, which consisted essentially in the commandment of love, while at the same time it differed from the old Christianity by letting earthly life come into its full right and not making it a means to a

supernatural existence. The individual right of property can in his view be founded only on the universal utility which this institution offers, not on the single person's claim in and for itself. What matters now is association in order to exploit the earth. It is things, not men, that are to be governed. As his disciples later expressed it: in place of the mutual exploitation of men by one another the exploitation of the globe is to come. The state is to gather the human forces for great works, such as the building of canals and roads, drainage, cultivation. Labor and brotherhood was Saint-Simon's watchword. Surrounded by a small circle of adherents he died in 1825.

As Saint-Simon by his political program stood in opposition to both of the contending parties of his day, the Liberals as well as the Legitimists, so in the domain of ideas he stood in opposition both to the theological school and to the philosophy of the eighteenth century. Significant here is the way in which he conceives and evaluates the Middle Ages already in writings from 1807 and 1813. The Middle Ages are for him a great period *[315]* of organization. The civilized world was held together by brotherhood and common faith. The priests were not, as Voltaire thought, charlatans, but the most advanced part of the nation. Since the medieval system has fallen before criticism and revolution, we live in a spiritual and social chaos. Negation and egoism flourish. What matters is to reach a new organic period, and the spiritual power that is to lead to it can be none other than positive science, which it is a matter of putting into a system. Saint-Simon's friend the physician BURDIN put forward, in a conversation with him, ideas on the development of the sciences which Saint Simon took up, and which contain interesting intimations of positivism. All sciences began on the foundation of a small number of experiences and therefore had in their first period the character of guesses, often fantastic guesses. Astronomy began as astrology, chemistry as alchemy. As experience advances, the sciences pass from a conjectural to a positive form. Mathematics, astronomy, physics and chemistry have already reached the positive form; physiology and psychology are on the point of doing so. Finally philosophy too will become a positive science; it is due to the imperfect state of the particular sciences that there is still so great a distance between general science and the special sciences. On this conception of the history of the sciences Saint-Simon built his hope of a new world-view on a purely scientific foundation. The expression *positive philosophy* he was the first to use. Further than the general thought he did not get. He was not the man to carry through a great, connected work. And yet he marks a decisive turning point. By his recognition of the Middle Ages as a distinct period of culture he betrays a historical sense which was rare at the time when he appeared. And he thereby succeeds in being able to conceive history as a continuous process of

development. While in the period of Enlightenment the Middle Ages had come to stand as an arbitrary interruption of development, they now stood as a period of spiritual and
[316] social organization after a preceding period of dissolution. In return the period after the Middle Ages came to stand as an age of dissolution. Criticism and liberalism are for him only means of getting the obsolete system out of the way, and are not themselves a new system. Such a system had first to be developed, and in Saint-Simon's conviction it could not be reached except on the foundation of empirical science. The time must come when humanity would build its faith and its morality on experience and science. Only by this way can there arise a new total conception of existence about which all can gather, so that life can again be carried on with united forces.

This doctrine of Saint-Simon's his school generalized, in that throughout history it distinguishes between critical and organic periods. In Kant and Fichte, indeed already in Rousseau, a similar opposition occurs. Such a conception was bound naturally to come forward in an age that had seen criticism overthrow an old order of things in the domain of faith and of society without being able to produce a new order and without being able to root out the need of one. However much there was of charlatanry in Saint-Simon, he nevertheless had on a decisive point a correct historical eye and felt the spiritual and social needs of the time. It thereby becomes understandable that he had a great historian and a great philosopher among his disciples.

In Saint-Simon there is found only the germ of socialism. The germ was developed in his school[76]), which reshaped his doctrine in such a way that he would not have acknowledged it. They wished to restrict the right of inheritance and to entrust to the state the distribution of the product of labor according to each person's capacity and work. The school took on more and more a fantastic and utopian character. A new hierarchy was set up with "Father" ENFANTIN at its head; but the school was dissolved when Enfantin came out with ordinances concerning the sexual relations of the faithful, which brought conviction and imprisonment for the leaders. Before these last extravagances the sober and critical elements had withdrawn. The enthusiasm that had bound the Saint-Simonians together — the enthusiasm for the goal of completing man's dominion over nature by the cooperation of human forces — was nevertheless not without fruit.
[317] A great part of France's railways, canals, factories and banks owe their coming into being to former Saint-Simonians. The cutting through of the isthmuses of Suez and Panama was a Saint-Simonian thought. It was pupils from the Polytechnic School who in particular had joined the new doctrine with enthusiasm. This school, which had been established by the Convention, was the home of the sciences which according to

Burdin and Saint-Simon had already become positive, and which had the calling to form the foundation of the faith of the future. Positive philosophy is the philosophy of the Polytechnic School, while eclecticism reigned at the École normale (where teachers were trained for the higher educational institutions). The real founder of positivism, Auguste Comte, himself a pupil of the Polytechnic School, speaks in a letter (Lettres à Valat. p. 229) of the hidden and involuntary strife between *Normaliens* and *Polytechniciens,* which in his eyes was a special form of the strife between the metaphysical and the positive school. It was the philosophy of the Polytechnicians that was to become France's most significant contribution to the history of philosophy in the nineteenth century.

2. Auguste Comte

a. Biography and Characterization

At the root of the rise of modern positivism lies the development of the empirical sciences in the last centuries, and in particular the magnificent advance in the domains of chemistry and physiology which the French naturalists began at the end of the eighteenth and the beginning of the nineteenth century. One by one all the domains of nature had been drawn under the scientific principles and methods which Kepler, Galilei and Newton had established. LAVOISIER and BICHAT had continued in chemistry and physiology what the founders of modern natural science had accomplished in the domains of astronomy and physics. The question now is: what significance has this whole development for our view of life and our world-view? Positivism's answer is: Should it be the case that the progress of science is a joy only to the specialists and to the *[318]* industrialists? Should it not be a change in the whole spiritual life of man, in human faith and conduct of life, that results from all the domains of nature being subject to the method of the empirical sciences? There is, after all, in the human mind an involuntary need to apply the same method and the same theory everywhere. Now when we understand nature by explaining its phenomena according to the laws that experience points out, are we then, in our faith and in our way of conducting and ordering life, to be able to build on an entirely different foundation? In fact we find that empirical science constantly exerts a determining influence on men's religious and speculative representations, however much these are asserted to have sprung from entirely different sources. A certain degree of positivism is found in every world-view. But now all the domains of nature have been drawn under *positive science,* i.e. under the knowledge that builds on facts and on the laws which experience establishes for these facts. The moment then seems to have come when that influence is no longer to be hidden and limited, but when men can consciously

and resolutely use in their view of life and world-view only such representations as positive science can recognize and confirm. There is still lacking, to be sure, a positive science of the spiritual life of man; it must then be the task of positivism to found one. If that succeeds, what then stands in the way of carrying out the program of positivism? — The task Auguste Comte set himself was a double one: to make the human sciences into a positive science, and then to give a systematic exposition of the chief facts, chief laws and chief methods of all the positive sciences. The foundation for the world-view of the future would thereby be given. The time of theology and speculation was over; the positive philosophy now stood as the only possible tendency. The questions that man cannot get answered with the help of positive science would then gradually fall away.

There nevertheless always remains a difference between positive science and positive
[319] philosophy. For it is not the single isolated facts, nor the special laws, that can condition a world-view. The human mind seeks, as Comte expressly emphasizes, unity in method and in doctrine. The positive philosophy first comes into being when the single positivities are worked together into a whole. The point, then, is to unite positivity and generality or totality. Positivism therefore cannot use the given as it immediately lies before us, but must submit it to a working-up, unite it into a whole. Here the difficulty of positivism comes to lie; for in this working-together there is necessarily need of hypotheses and presuppositions, and the question then is on what their validity rests.

Auguste Comte finds his philosophical predecessors in the eighteenth century. He refers especially to Diderot and Hume, to Kant and the Scottish school. But he admires de Maistre and regards the Middle Ages as the last *organic* period we have had. It was his conviction that only by carrying out positivism in faith and conduct of life would we regain the unity and harmony in our spiritual and social life which was lost through modern criticism and through the Revolution. Side by side with the enthusiasm for positive thinking there goes in Comte the fervent, at last mystical love of humanity, a need of feeling that demanded its satisfaction with such energy that it several times threatened to burst his consciousness. The founder of positivism felt himself spiritually akin to the mystics of the Middle Ages. And at last he even wanted to found a new religion, which was to be positive in a sense other than the one in which people are accustomed to speak of positive religion. —

Auguste Comte's life shows us how the various motives that led to his work developed in him. He was born on 19 January 1798 in Montpellier. His family was strictly Catholic. By his own account (Cours de philosophie positive. Tome VI. Préface personelle) he was scarcely fourteen years old when he "ran through all the essential degrees

of the revolutionary intellectual tendency and felt the need of a general political and religious regeneration." Perhaps it was precisely the rapid development that led him away from the representations of his childhood which brought it about that the need of settled dogmas and definite symbols became so vivid in his later years, after his best *[320]* strength had been spent on his main philosophical work. The period in which he was seeking was only short, and he did not reap sufficient experience of what value such a period can have for the freedom, inwardness and fullness of personal life. He came to share the age's aversion to the *critical* periods. Already as a boy he was refractory toward all compulsion and all regulation; but full of reverence before intellectual and moral superiority. (Cf. the characterization he gives of himself as a young man in a letter printed in Littré: *Auguste Comte et la philosophie positive,* 2. éd. p. 493). In the year 1814 he entered the Polytechnic School in Paris. This school for the exact sciences, founded by the Convention, he later regarded (cf. a letter to Stuart Mill of 22 July 1842) as the first germ of a true scientific corporation. It comprised the sciences that had decidedly reached the *positive* stage, and it was for him the foundation on which all higher education should build. There was, moreover, among the school's pupils a republican temper and a brotherhood which united them in a common enthusiasm over the significance their subjects could have for the development of culture. It was this that won Saint-Simon's ideas so great a following among the school's pupils. The thought of positivity and the thought of humanity sprouted forth together. The reactionary government used, in the year 1816, a demonstration against a teacher as the occasion for closing the school and sending the pupils home. But Auguste Comte could not bear to live away from the center of the movement. In spite of his parents' prohibition he came back to Paris the same year to continue his education. He studied biology and history to supplement his polytechnical training, and acquired an excellent encyclopedic foundation. He earned his living by teaching mathematics. Of great significance for his development was his coming into close relation with Saint-Simon. He writes of this in a letter from that time (1818): "Through collaboration and friendship with one of the men who see farthest in the domain of philosophical politics I have learned a host of things which I would have sought in vain in books, and my mind has, in the half year *[321]* I have been in contact with him, made greater progress than it would have made in three years had I been alone. This work has developed my conception of the political sciences and has besides indirectly raised my representations of the other sciences, so that I have noticed that I have got more philosophy in my head and a more correct, more elevated view of things." *(Lettres à Valat.* p. 37). What Comte owed to Saint-Simon was

partly the notion of the necessity of a new spiritual authority in place of the hierarchy of the Middle Ages, which in the opinion of both had been an admirable institution for its time, partly the interest in the social question, with which the sharp emphasis on the opposition between militarism and industrialism is connected. In a treatise of 1820 *(Sommaire appréciation de l'ensemble du passé moderne)* Comte dates the dissolution of the old social system back to the twelfth century, finding the first germs of this dissolution in the emancipation of the communes and in the introduction of the positive sciences into Europe by the Arabs. Thereby the ground was laid for the industrial system founded on labor gradually taking the place of the territorial system founded on conquest, and for positive science replacing theology. Even if Comte had already, by his own path, arrived at similar ideas, his relation to Saint-Simon nevertheless hastened his development, and it was with great injustice that Comte later regarded it as a great misfortune for himself that he had come to know Saint-Simon[77]). The two men were, however, too different in their nature and line of thought to be able to go on working together. The break came when Comte felt his independence. In principle they came to disagree about the place science was to take in relation to labor and to the new "priesthood" Saint-Simon had in mind in his later years. As we have seen, the latter, in the last part of his career, pushed the theoretical reformation aside in order to think directly of the practical one. In this Comte disagreed with him. The last work in which Comte still calls himself a pupil of Saint-Simon occasioned the break; Saint-Simon dissociated himself from it, and
[322] Comte later published it in his own name. This work, whose title was: *Plan des travaux scientifiques nécessaires pour réorganiser la société* (1822, published again in 1824 under the title: Politique positive), marks Comte's breakthrough as an independent thinker. The most important obstacle to the development of civilization he sees in the constant preponderance of the revolutionary tendency, especially in freedom of conscience and the sovereignty of the people, which are critical, not organic, principles. There is lacking the capacity to enter into a coherent circle of representations and to recognize a rationally grounded authority. A new social system cannot be constructed all at once, but must develop successively. That is sufficiently proved by the many failed constitutions in the most recent history of France. The fundamental condition for working together toward a common goal is a common way of thinking and temper. The old society had such a common foundation in theology. In order that the new society may get a corresponding foundation, there must first be brought about a comprehensive edifice of thought that can come forward with the same authority as that with which the results of the single sciences come forward in the special domains. Most important of all is that politics be

made a positive science. While in the mathematical, physical and biological domains one recognizes definite laws for all phenomena, one still believes that in the social domain one can intervene at one's own discretion, e.g. introduce a thought-out social system without further ado. This belief will fall away when the conformity to law of human relations is recognized. And such a conformity to law Comte seeks to point out in the close connection that exists at every point in history between the social organization and the whole of civilization, that is to say, between the order of society and spiritual development as this appears in science, art and industry. Civilization in its turn springs from man's instinctive need of perfection[78]). The various stages that civilization runs through Comte, relying on the history of the sciences, reduces to three: the theological, the metaphysical and the positive, and he seeks to show how definite stages in social development correspond to them. Since *the law of the three stages* is a chief point in *[323]* Comte's philosophy, the detailed account of it must be deferred to the exposition of that philosophy. — This writing of Comte's aroused great attention among politicians, historians and scientists alike. Guizot, the mathematician Poinsot, Alexander von Humboldt, the Duke of Broglie and others expressed their approval. It went so deep that it led beyond the contending parties, while at the same time it did justice to what was of value in their tendencies. But Comte's real plan had still not been carried out. Only now did he proceed to the chief work of his life: to give an encyclopedia of the content and methods of the positive sciences. He writes to his friend Valat (1824): "I will work all my life and with all my powers at the founding of the positive philosophy." And at about the same time he writes to another friend (Gustave d'Eichthal) that his real task is "an encyclopedic recasting of all our positive knowledge, which in reality ought to be conceived as a single mass." A few years later he set out his "positive philosophy" for the first time before a small but select audience (to which there belonged, among others, A. von Humboldt, Poinsot, the mathematician Fourier, the political economist Dunoyer, the physicians Broussais and Esquirol, the engineer Hippolyte Carnot).

Comte supported himself by literary work and by teaching mathematics. He had married a young, gifted Parisian woman. Very early the relationship seems to have become less happy, partly because Comte's parents were against the union (which in their eyes was already abominable through having been contracted in civil form), partly also, as is seen from a characteristic letter to Valat (16 Nov. 1825), because Comte had not found in his wife the qualities he valued most in a woman: "devotion of heart and gentleness of character, together with the kind of subordination which the feeling of her husband's intellectual superiority can inspire in her," qualities which "it is certainly very

difficult nowadays to find united in a woman of outstanding intellectual gifts." Comte wishes these qualities to his friend's bride in such a way that it shines through clearly
[324] that he misses them in his own wife. She had been too independent for him. She showed, however, great faithfulness and energy during an attack of mental illness which — no doubt chiefly as a result of overwork — struck Comte. He was placed in an asylum, and his parents sought to use the opportunity to bring him under their influence and wanted to place him in a monastery. Madame Comte, whose civil marriage the parents had wished to pass over in silence, asserted her right and demanded that her husband be placed in his own home, and her nursing then succeeded in bringing back his mental health, so that he could resume his interrupted lectures. (Comte's parents nevertheless had him — while he was still insane — married in church, and thereby relieved their hearts of the mortal sin of having given their consent to the civil marriage).

Now there followed for Comte a series of vigorous and happy years, in which he worked out and published his main work, *Cours de philosophie positive* (in 6 volumes, 1830—42). The content of each single volume he first thought through at length on solitary walks. As far as the material was concerned, he could rely on his great memory. When the meditations were concluded, he wrote them down in the course of a comparatively short time. Little weight was thereby laid on the form of the exposition. His style is heavy, unaesthetic, and suffers from repetitions and a heavy use of technical expressions; but it is distinguished by clarity, thoroughness and weight, and bears the stamp of energy and earnestness. Often (especially in the last three volumes of the work, which deal with the social sciences) the stream of thought becomes so mighty that it carries the reader along. Comte sought to work for the positive philosophy in a popular way through public lectures. From the twenties on, former pupils of the Polytechnic School had held popular lectures on natural science round about the country. In the year 1830 Comte and some of his comrades founded the Association polytechnique, which was to work in this direction. For a long series of years (until the coup d'état of 1851 put an end to it) Comte gave free public popular lectures, first on astronomy, later (after 1848) on positive
[325] philosophy as a whole. In a little writing, *Discours sur l'esprit positif* (1844), he has set out the general considerations with which he introduced his astronomical course; it gives the best introduction to Comte's philosophy. —

A fixed official position Comte never obtained. After 1830 he had hoped for a professorship in the history of the positive sciences, for which he was in the highest degree qualified; but an approach to Guizot, now so powerful, was without result. In his Mémoires Guizot even, by an incomprehensible lapse of memory, presents it as

though it were the first time he saw this man, with whom he had nevertheless, in the years of opposition in the twenties, had so many conversations and felt a community of thought. Nor did he obtain a professorship of mathematics at the Polytechnic School, although he had for a year temporarily filled this post in a manner that won general recognition. He had to be content with the more modest position of répétiteur and of examiner at the entrance examination, in which last capacity he made annual tours of the provinces. This last post too, to which one had to be elected annually by the teaching council of the Polytechnic School, he lost when in the preface to the last volume of the "Cours" he had made a sharp attack on mathematical specialism and its arrogance. It was now, he held, biologists and sociologists who were to stand in the front rank in intellectual respects: the supremacy of the mathematicians was over, since the more concrete domains had been drawn under positive scientific treatment. The provocative manner in which he expressed this caused him not to be re-elected examiner, and he was now again reduced to living by private teaching. English and French admirers and friends (among the English are to be mentioned Stuart Mill and the historian Grote, among the French the learned Littré) nevertheless made his economic circumstances easier for him by giving him support. — In his domestic circumstances too it came to a crisis, his life together with his wife coming to an end after a long period of growing discord. Yet his wife continued in the finest manner to show her interest both in his ideas and in his personal circumstances.

Amid all these conflicts and the emotions they caused, and after the great exertion *[326]* of strength he had spent in the twelve years he worked on his main work, it came to a new nervous crisis, not indeed so severe as the earlier one, but yet of such a kind that, in his own expression in a letter to Stuart Mill, there was "danger cérébral"[79]). Other circumstances worked together in this crisis and gave it its peculiar character. He had come to know a woman who became for him what Beatrice was for Dante: the one toward whom his need of feeling, which he had carried all his life without getting satisfaction for it in any intimate relationship, could find a complete outlet. For the first time he came to know the whole inwardness of mood and of devotion, and this young woman, whose name was Clotilde de Vaux, continued, when she died after the lapse of a year, to stand for him as the representative of humanity (just as Beatrice for Dante was that of theology), to whom he addressed his daily thoughts and the "solemn outpourings of noble feeling" (solennelle effusion des sentiments généreux), as he called his prayers. She became the genius who inspired him to his second main work, which was to systematize the feelings, just as the first had systematized the ideas. This work

appeared under the title *Politique positive ou traité de sociologie, instituant la religion de l'humanité* (4 volumes, 1851 — 54). In a letter to Stuart Mill (of 14 July 1845) Comte states that the new work, the plan for which he had already formed earlier, had received its decisive character during that crisis and the "méditation exceptionelle" it brought with it. For it dawned on him that the second part of his philosophical activity ought to differ from the first in this, that feeling should get as great a place in it as the understanding had in the former. Now that the purely theoretical work has been done, the point is its social application, and this consists essentially in a systematization of the human feelings, a consequence of the systematization of ideas and an indispensable basis for the systematization of institutions. — Had the new work aimed only at this, it would not have marked any decisive opposition to the earlier one, but a continuation and extension
[327] or deepening of it. In reality it went, as the title already shows, far further. It was to found a new religion. And whereas the earlier work started from the world or nature and on the basis of the knowledge of nature sought to arrive at an understanding of man, this objective method was now to be replaced by the subjective one, which started from man, regarded the whole of nature from the human point of view, and humanity itself as the highest being (le grand être). In subjective absorption in the thought of humanity, in fervent devotion to it, consisted the new religion, whose high priest Comte soon felt himself to be. He would not give up the positive philosophy; its content was to be the dogmas of the new religion, the religion of humanity; but he added cult and practice. In his *Catechisme positiviste ou sommaire exposition de la religion universelle* (1852) he has given a shorter exposition of what the "Politique positive" develops at great length.

This last turn in his line of thought had been prepared by what he called his "hygiene of the brain" (hygiène cérébrale), which consisted in his abstaining on principle from all study and absorbing himself solely in the working out of his works. Thereby he would keep disturbance away and see to it that the unity of the plan was held fast. Earlier he had pursued far-reaching studies, and his solid memory enabled him now to use what he had gathered at that time. Together with this isolation from everything new in science and literature, an isolation which brought with it the cessation of all real discussion and of critical testing of his own ideas, there went absorption in music, Italian and Spanish poetry, and the reading of De imitatione Christi. He demanded of every positivist that he read every day in a poetic masterpiece at least as much as corresponds to a canto of Dante. The writing De imitatione he regarded as a great poem on human nature; in his reading he everywhere put humanity in place of God, and so found in the old mystical book an aid to a fervent entering into the life of humanity. According to an

eyewitness's description of Comte in his last years, his whole being was pervaded by gentleness and goodness. He had, however, had the disappointment that his most famous disciple Littré, and several with him, withdrew from him when he passed from appearing *[328]* as a philosopher to feeling himself the high priest of the religion of humanity, just as he himself had withdrawn from Saint-Simon when the latter voiced a similar pretension. After the long and severe labor of thought, the mythological need of childhood had now, favored by the cerebral hygiene, broken through again, and on this last part of his way he was followed by only a few[80]). Yet the new religion of humanity has its congregations and its places of worship round about in France, England, Sweden and America. This religion without theology is a characteristic sign of the times. For Comte himself it was a resting place where his thought gathered the memory of the greatest and best it had found in humanity's knowledge and action, and from which it looked with hope toward the future that could be expected for the constantly progressing species. Love as principle, order as foundation and progress as goal! that was the motto of the religion of humanity. — On 5 September 1857 Auguste Comte died.

b. The Law of the Three Stages

Our knowledge runs through, according to Comte, *three stages of development,* which can be pointed out in the case of each single science. The more intricate and composite the subject matter of a science, the longer it will be before it has run through this scale. The abstract sciences come first to the definitive stage, the concrete ones later. In Comte's opinion the point of time has now come when the most concrete of all the sciences, the science of society (sociology), is about to enter the third stage. With that the circle is closed, and a retrospect becomes possible.

The first stage is *the theological.* Here a very limited number of observations lies at the basis, and imagination therefore plays the greatest part. The explanation of natural phenomena, the connecting bond between the given facts which the human mind by its constitution cannot do without, is found at this stage in the intervention of personal *[329]* beings. It is only with the help of the representations of gods and spirits that man can from the beginning make the world comprehensible to himself. Herein lies the great benefit these representations have rendered to the development of human knowledge. Had these representations not been available, and had the wish to find again the working of those beings in what happened round about in nature not asserted itself, the activity of knowing would not have got under way; the original torpor would have persisted. The understanding man believes he gains at this stage is, moreover, absolute; for beyond

those divine beings he thinks of nothing, and if he can explain a natural phenomenon by their intervention, he has nothing more to search for. Therefore man at first has no doubt at all about the possibility of reaching an absolute knowledge. — In practical respects the theological representations have exerted an influence no less important. They have furnished a firm and common ground for moral and social life. This stage of development is the time of authority. Men live in a common trust in unshakable powers. In politics the age of kingship corresponds to this.

The theological stage comprises a series of steps. In fetishism a mental life like the human one is ascribed directly to the objects of nature. In polytheism, the step most characteristic of the theological stage, material objects are deprived of their immediate life, and the source of their motions and changes is sought in various beings, mostly invisible, who form a higher world. The transition from fetishism to polytheism is the most important step taken within the theological stage. The god is removed from the natural phenomenon to which he corresponds, and thereby the notion of matter as living can be replaced by the notion of matter as subject to the law of inertia. It is owing to the beginning capacity for observation and induction that this transition is made. In monotheism the distance and the opposition between the principle from which the explanation is taken and the phenomena that are to be explained are still greater. It is
[330] therefore vague and brings with it less spiritual community than polytheism. It forms the transition to the second great stage: *the metaphysical.* Here the explanation is found not in personal beings but in abstract ideas, principles and forces. There prevails here a striving to reduce various phenomena to one principle, a striving which already within the theological stage leads over from fetishism to polytheism and from polytheism to monotheism. The metaphysical stage continues this striving by assuming as many forces as there are peculiar groups of phenomena, thus a chemical force, a vital force, etc. At last the striving is then to reduce all these various forces to one fundamental force, one fundamental being, Nature, an equivalent of the concluding unity that monotheism made possible at the theological stage. Common to these first two stages is the tendency to seek absolute solutions. Metaphysics, just as much as theology, seeks to explain the inmost nature of things, the origin and destiny of all things, and the manner in which the phenomena are produced. The difference is only that the abstract is put in place of the concrete, argumentation in place of fantasy. As great a preponderance as fantasy had over observation at the first stage, so great a preponderance has argumentation here.

Comte conceives the metaphysical stage essentially as a stage of transition and as a process of dissolution. Argumentation bores its way into the theological circle

of representations, points out the contradictions there, puts constant ideas or forces in place of the incalculable wills, but thereby weakens the living impression and the authoritative influence of the powers that are assumed to rule in nature and in human life. Something new it cannot build up; real equivalents it cannot give. In practical respects the dissolution shows itself in the rule of doubt and egoism. The individual is torn out of the living connection with society, and the understanding is cultivated at the expense of feeling; Comte even speaks in his later years of the long revolt of the understanding against the heart. In politics it is the age of the peoples, as the first stage was that of the kings; the jurists are the leading men; society is conceived as having arisen by contract, and the state is built on the principle of the sovereignty of the people. *[331]*

At the *positive* stage both fantasy and argumentation are subordinated to observation. Every proposition that is put forward concerns a fact, either a special or a universal fact. Agreement with facts is the only criterion. Not, however, to be understood as though it were the single, isolated facts that are stopped at. Positivism is as far from empiricism as from mysticism: just as little as it departs from the facts in order to lose itself in supernatural beings or in abstract principles, just as little does it scatter itself in incoherent observations. But instead of seeking absolute *causes and* tracing the production of things out of them, it seeks the *laws* of phenomena, that is to say, the constant relations that subsist between the observed phenomena. Whether it is thoughts and feelings or impact and gravity that are to be understood, the relations under which they appear are the only thing our knowledge can ascertain. Science builds on the invariability of the laws of nature, which first dawned on thought with the founding of mathematical astronomy among the Greeks, but in modern times has been found again in one domain after another. Bacon, Galilei and Descartes stated this principle and thereby stand as the founders of the positive philosophy. Even if the laws that hold in the particular have not yet been found everywhere, an irresistible analogy now leads us to apply that great philosophical principle of the invariability of the laws of nature to all phenomena and events.

Whereas the first two stages showed us a striving to reduce everything in existence to an absolute principle, whether this was thought of in theological or in metaphysical form — as God or as Nature —, it follows from the nature of the positive philosophy that such an absolute and objective conclusion is not possible. The strict demand for confirmation by experience makes it impossible to reduce everything to one principle; experience always shows us only a limited connection, and there will always be many phenomena and events which it will not be possible to bring into connection with the

[332] others. There are groups of phenomena (at least as many as there are different sciences that cannot be reduced). The many laws cannot be reduced to a single one. Only a subjective, not an objective unity can our knowledge reach. The subjective unity consists in the same method being applied everywhere, which brings with it uniformity and convergence of the various theories. (Quant à *la source intérieure* des théories humaines . . . nos connaissances tendent vers une entière systématisation. Discours p. 24). Subjectively regarded, then, we get only a single science. And it can become common to all human subjects. The positive method brings about unity not only within the consciousness of the single subject, but also between the various subjects. Thereby the positive philosophy becomes the intellectual foundation of society among men. Even now it is only on subjects that have been drawn into the domain of positive science that agreement prevails among men, while otherwise opinions go in the most various directions. Catholicism, as long as it ruled without intellectual competition from the side of scientific thought, offered a spiritual community which stands as a model of what the positive philosophy will one day bring. The positive philosophy finds in the subjective unity that lies in the concept of humanity the only possible equivalent of what the concept of God was for the theological philosophy and the concept of nature for the metaphysical philosophy.

The connection of theory and practice is far closer at this stage than at the first two. For knowledge of the laws of phenomena makes it possible for us to intervene in the future development of phenomena, and the wish to be able to do this is a contributing motive in the transition from the first two stages to the third. See in order to be able to foresee (voir pour prévoir!) is the motto of positive science. To this third stage therefore corresponds industry, in the sense of man's working-up of nature. Yet the consideration must not here be restricted to external nature. If there are also laws for man's states and actions, individual and social development, no less than the purely physical, can come to
[333] be determined by knowledge.

Comte points out (see especially "Discours sur l'esprit positif" p. 41—44) that all the various meanings in which the word *"positive"* can be used fit the philosophy he calls the positive. Positive can mean the same as *real,* and the positive philosophy will, after all, first and foremost build on facts. It also means *useful* as opposed to idle, and the positive philosophy, after all, aims at the improvement of our individual and social existence, seeks to be more than the mere satisfaction of curiosity. By the positive we also often mean *the certain and indubitable, and* it is, after all, the task of the positive philosophy to lead us beyond the constant doubt and debate of the earlier philosophies. The positive

can further mean *the precisely determined, and* the positive philosophy will, after all, put the laws, the constant and determined relations, in place of the indeterminate and shifting representations of earlier stages. Finally, "positive" is also used as the *opposite of negative, and* that fits the third stage, whose task it is to organize, whereas the second stage essentially signified a process of dissolution. The positive philosophy cannot, indeed, use the explanations with which the theological philosophy rests content; but it wages no direct battle with them. It even concedes that it is as impossible to bring a proof for the rejection of the beings in which one believes in the theological philosophy as to bring a proof for their acceptance. No one has ever brought a proof that Apollo or Minerva did not exist; the belief in them fell away when it no longer fitted together with the whole spiritual condition. Not only does the positive philosophy not polemize against the theological; it also seeks to do it full justice, by investigating historically the conditions of its rise and evaluating its influence on human development. Such an understanding was not possible at the earlier stages, where one absolute theory stood over against another, and where each single system had to regard it as its task to annihilate the others.

Only one peculiarity of the positive stage does not lie in the word "positive" itself. It is that it everywhere puts *the relative* in place of the absolute. But this relative character *[334]* follows for Comte necessarily from the other marks of the positive philosophy. When it puts the law in place of the cause, it stops at a mere relationship, at a relation. It asks: how? not: why? The inner production and the first grounds it does not seek. The single laws, moreover, cannot be reduced to a single law, and in any case there remains the relativity that arises from our always regarding the world from man's point of view. All knowledge presupposes an opposition between the individual and the surrounding world. Kant therefore rightly distinguished between the subjective and the objective, and deserves an eternal admiration because by this distinction he put an end to the absolute philosophy, although he was not sufficiently positive in his thinking to be able to prevent his successors from returning to it[81]). Although the world presupposes man if it is to be known, it can nevertheless exist without him. And although man depends on the world, he is nevertheless not a result of it. (L'homme dépend du monde, mais il n'en résulte pas. Catéchisme positiviste. *2.* éd. p. 146). In vain has materialism sought to abolish the independence and self-activity (spontanéité) of organic life by exaggerating the significance of the influence of the inorganic surrounding world. The duality remains.

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The law of the three stages Comte derives from the history of the sciences, thus by

the way of experience. But after he has set it up, he seeks to show that it can be derived from what we know of the nature of the human mind; the induction is confirmed by a deduction. For the human mind cannot do without modes of view and concepts that can bring connection among the single facts. Before the bond between the phenomena can be found by investigating the phenomena themselves, as at the positive stage, it must be found in the mythological representations that arise involuntarily, or in the metaphysical ideas that are formed by giving these representations a more abstract and constant form. At the same time the human mind has an inclination to conceive everything after its
[335] own likeness, to carry its own inner feeling over into things. That is an easy way of gaining an explanation. But if men did not trust that the explanation could easily be found, they would not engage in any investigation. —

It would be interesting to make a comparison between Comte's law of the three stages and similar thoughts in Kant, Fichte, Hegel and Saint-Simon, indeed already in Rousseau and Lessing. Beside characteristic divergences there would turn out to be agreement at least on the one point, that the spiritual life of man, after the period when authority ruled in theory and in practice, has had a period when criticism, reflection and doubt performed a dissolving work, after which the point now is to reach a standpoint with a positive and common foundation for faith and conduct of life. It is a great historical experience that has been formulated by the thinkers named in the law of the three stages. As far as Comte is concerned, it is really only the first and the third stage that stand out with a definite characterization. One often gets the impression that Comte really assigns to the metaphysical stage all such endeavors in thought and life as he harbors an antipathy toward. He sympathizes with the old Catholic system, in which he sees one of the most marvelous productions of the human spirit, and he sympathizes with that system of the future which, on the ground of empirical science, was to lead to a new spiritual community. The intervening critical and revolutionary period he looked on with aversion, although he had to concede that it had performed an indispensable work. The various marks he adduces of the metaphysical stage do not necessarily hang together. In particular it will not be possible to find a connection between the tendency to explain the phenomena by assuming specific forces or faculties and the overvaluation of the intelligence or the predominance of egoism. That tendency appears at a certain point within the history of each single science and cannot be used to characterize the common human development; in any case it can very well be united with the recognition of the central significance of the life of feeling and of the reality of the sympathetic feelings. An overvaluation of the intelligence is found perhaps quite as much among cultivators

of the special sciences as among speculative philosophers. At the conclusion of his main *[336]*
work Comte did, after all, feel vividly the opposition in which he stood to scientific specialism, and he expressed it so strongly that it had drastic personal consequences for him. But for this opposition between the positive philosophy and positive special research he finds no place in his doctrine of the three stages. And yet it is a very essential characteristic of modern times that the division of labor in the scientific domain has advanced so far that an intellectual unity, a common world-view, becomes harder and harder to reach. Here there surely lies a greater problem of the history of culture than in any of the defects Comte finds in the metaphysical stage. It was Comte's enthusiasm and sanguineness that kept him from seeing this clearly. The problem of the relation between positivity and generality — of the possibility of reaching a general world-view on "positive" ground — did not stand sharply enough before him. That is connected, as will appear later, with his shortcomings in respect of epistemology. The positive stage itself is therefore unclearly drawn in several respects. Thus it remains unclear how narrowly the positive philosophy binds us to the facts before us. How far may we depart from these by the way of hypothesis? What justification has a hypothesis that is a logical consequence of given experiences but cannot be verified? What justification has a faith that cannot indeed be derived from experience, but yet does not conflict with experience? Such questions, extraordinarily important for the relation between science and view of life, Comte passes by.

Absolute boundaries between the various stages Comte does not really assume, although he has not sufficiently emphasized the unity of human thought through all the stages. It was not his opinion that the theological stage was wholly without positivity. The influence of experience has always asserted itself, e.g. in the transition from fetishism to polytheism and from polytheism to monotheism; but as long as the circle of experience was small, its influence could not become one of principle. It is nevertheless the forward-driving power of experience that expresses itself in the metaphysical principles. These *[337]*
represent in reality groups of experience which are demanded to be withdrawn from mythological arbitrariness. Metaphysics systematized, as Comte says in one place (Discours sur l'esprit positif, p. 36), the opposition of nascent science to the old theology. As long as the positive spirit was still only little developed, it could not formulate its own tendencies, and it then used metaphysics as its organ. Thus knowledge moves forward through continuous transitions. A complete breakthrough and a general dominion of *esprit positif* Comte by no means expected at the beginning of his career. He writes to Valat on 30 March 1825: "Although I hope for some use from my endeavors, I do not hide

from myself that in my lifetime they will not be able to produce any remarkable result, even if they should be able to give an impulse to all minds that are capable of taking part with effect in the great work, something which only the future will be able to show." Later, when under the influence of his cerebral hygiene and of the mystical absorption in his ideas he had lost all understanding of what was going on around him in the scientific and the political world, he cherished a greater and ever increasing sanguineness in this respect.

Whether Comte laid most weight on the negative character of the second stage or on its significance as a necessary intermediate link, it was his conviction that the decisive battle in the spiritual world would come to be fought between Catholicism and positivism. Repeatedly in his letters to Stuart Mill he expresses the wish that a direct debate between these two tendencies — with the exclusion of Protestantism, deism and other inconsistent intermediate forms — might come about. It was to him a significant sign of the times that the Catholics now demanded freedom of teaching, while the "metaphysical" school (Cousin, Guizot, Thiers, Villemain) demanded that the state monopoly be maintained. Both schools had thereby denied their principles, the former the principle of authority, the latter the principle of freedom. But if discussion and teaching became free, he hoped
[338] that the metaphysical school's fear would be fulfilled, while Catholicism would then stand over against a new adversary it had not known before. He only regretted that the representatives of Catholicism had become inferior since de Maistre's time. That the oppositions sharpened, and that Catholic orthodoxy became ever stricter and more systematic, was for him fully understandable, after the revolutionary movement had betrayed the tendency to build spiritual and social development on an entirely different foundation from that of authority. In the face of these radical prospects de Maistre had inculcated ultramontanism as the only consistent religious system, and it had succeeded in shattering Gallicanism, whose fate other intermediate standpoints too would come to share. Comte therefore rejoices that his friend Valat, in a breakthrough of religious faith, took the step fully back to pure Catholicism, without stopping at any intermediate standpoint, although in the beautiful letter he wrote to Valat on that occasion (25 August 1843) he made no secret of his doubt that Catholicism could now bestow the peace and harmony which in its classical time it could give to those who had the highest intellectual and moral development of that day. In particular he refers in this respect to the misgivings about the dogma of the eternal damnation of those who practice the highest morality even if they reject every article of faith, a dogma which is nevertheless indispensable to the whole Catholic organism.

c. The Classification of the Sciences

By philosophy Comte understands the whole system of human concepts. This systematizing can, as we have seen, take place in three different ways. Hitherto the theological and the metaphysical method have ruled; Comte himself believes he is the first to carry out the systematization by the positive method, that is to say, by the same method as that by which the concepts or laws are won within the single sciences. It is the task of the positive philosophy to gather and order the laws that have been extracted from the facts. By thus connecting what occurs scattered in the various sciences, it seeks to *[339]* remedy the drawbacks the division of labor brings with it in the domain of knowledge. Comte regards it as hopeless to attempt to reduce all special laws to a single one, as the theological and the metaphysical philosophy had each in its own way attempted. The positive philosophy would certainly become more perfect if that were possible; but experience speaks against it, since it presents diversities that cannot be reduced. While the unity of the positive philosophy does not consist in all laws being reduced to a single one, it comes out clearly, on the other hand, in the common method, which everywhere proves applicable, and which, in spite of the irreducible diversities, points to a uniformity in the various classes of phenomena. To each of these classes there corresponds a peculiar science, and the classification of the sciences therefore becomes the most important task of the positive philosophy.

Comte's classification of the sciences orders them in *the sequence in which each of them historically entered the positive stage.* Here comes first mathematics, then astronomy, physics, chemistry, biology and sociology. But this sequence shows us at the same time a progressive transition from *simplicity* in the phenomena considered to *compositeness* (complication). The simpler a science's content is, the more quickly will it be able to run through the various stages; the more complex the phenomena that are to be investigated, the longer it will be before the childhood diseases are overcome. Biology and sociology therefore come last in the series. The simpler the relations a science concerns, the more *universal validity* will it at the same time have; for the simpler relations recur in the more composite ones. Mathematics holds for all phenomena, and the most concrete sciences, biology and sociology, have the *narrowest extension.* (This relation between généralité décroissante and complication croissante Comte could have found demonstrated in logic's doctrine of the inverse relation between the content and the extension of concepts). Finally, the methods of the various sciences also present a corresponding sequence. The simpler and more universal a science's foundation is, the greater the preponderance the *deductive* method gains over the *inductive.* Thus mathematics is the most deductive *[340]*

science, sociology the least deductive. In sociology the demonstration of the historical course of development plays the greatest part; after this has been established by the inductive way, the task then becomes to deduce it from human nature, as Comte himself has attempted by his setting up of the law of the three stages. The inductive foundation of mathematics is so simple that one often overlooks it and regards mathematics as a pure science of reason. But a pure science of reason there is not at all. Even mathematics is a natural science; its concepts, just as much as those of all other sciences, stem from experience. But the facts mathematics concerns are so simple that one can consider them as easily in fantasy as in reality, and that one can abstract from the physical and chemical properties of bodies. Number, extension and motion are properties we can represent to ourselves just as well in an indeterminate medium thought of as including all the bodies of the universe as in the real bodies themselves. This explains the apparent independence of mathematics from experience. On the foundation of fantasies of space and number the mathematical deductions can unfold unconcerned about experience. If mathematics had no foundation in experience at all, it would be altogether impossible to understand how the mathematical deductions could be applied to the study of real nature. — Between the as good as purely deductive method of mathematics and the as good as purely inductive method of sociology lie the other chief methods, each corresponding to its chief science. Through astronomy's use of deductively set up hypotheses that are confirmed by observation, we come to the experimental method of physics, to the rational method of division of chemistry, and to the comparative method of biology; this last forms the transition to the historical method of sociology.

The six groups of concepts or laws that Comte thus sets up in the "Cours de philosophie positive" he regards as irreducible. In the transition from one sphere to the other a leap takes place; a new principle comes into force which cannot be derived from the preceding one. It is in Comte's eyes materialism to seek to derive the higher (i.e. more
[341] complex, more limited and more inductive) science from the lower. Just as little can any of the single groups be made the all-embracing one. And the same discontinuity that thus for Comte prevails between the six groups of fundamental concepts prevails also within each single one of them. The various branches of physics will thus — he asserts — always remain separate from one another. It is with the organic species as with the physical forces. The concept of species would, says Comte, entirely lose its scientific significance if one were to admit transformations of one species into another under the influence of the external conditions of life. Between the plant and the animal world in particular he maintains "a real and deep discontinuity, which cannot be effaced by any

transition whatsoever."

This point in Comte is an instructive example of how tendencies of thought can turn over into their opposites. The difference of the positive philosophy from the metaphysical was to consist, after all, in laws being put in place of ideas or forces, and relative explanations being put in place of absolute ones. But Comte in reality himself sets up a Platonic world of ideas through his irreducible groups of concepts. He takes the discontinuity too dogmatically, instead of taking it as a mere fact, a fact which is perhaps due only to the imperfection of science. By his emphasis on the discontinuity in existence he sets his positivism in the sharpest opposition to German Romanticism, which sought to dissolve all differences in ideal continuity. It is, as Comte also concedes, the constant task of science to reduce the diversities, the breaks in the continuity of the phenomena, to the least possible. Every discontinuity can therefore signify only a provisional limit. And the later development of science, precisely in the period that begins with the conclusion of Comte's main work (c. 1840), has indeed led to demonstrating or conjecturing continuity where Comte had considered it impossible. Thus the law of the conservation of energy, in particular the doctrine of heat as a form of motion, and the demonstration of the identity of light and electricity have given an insight into the unity of nature for which Comte had no room in his system. In the organic domain the hypothesis of evolution *[342]* has led to the well-founded assumption that species signify only stages in, or offshoots of, connected processes; in particular it has become harder and harder to maintain sharp boundaries between the animal and the plant world. Comte already pronounced against the hypothesis of evolution as it existed in his time (in Lamarck's form); and his disciples (Charles Robin, Littré) have since been eager opponents of it. This much can be said without anticipating anything: that the relation between continuity and discontinuity has proved to be more intricate than Comte had supposed in his philosophy.

The connection between the various domains rests, according to Comte, on this, that the preceding domain, as the simpler and more universal, lies at the basis of the following, more complex and special one (even if the latter cannot be derived from the former). Mathematics is the most abstract and universal science, and within mathematics, again, the theory of number is simpler and more universal than geometry and the theory of motion; it therefore becomes the rational basis of the whole system of our positive knowledge. Comte has here overlooked that there are fundamental concepts of our knowledge that are still more simple and universal than the concept of number, namely the concepts of identity and difference, as these are conceived in logic: not only as identity and difference between magnitudes, but also between qualities. In his series

of branches of knowledge Comte has thus not got the first member in. Moreover, a difficulty comes up for him when he teaches that geometry and the theory of motion — which in abstraction and universality come right after the theory of number — hold for all phenomena, the former in so far as these are conceived in equilibrium (from the static point of view), the latter in so far as they are conceived in motion (from the dynamic point of view). Extension and motion are forms only of material phenomena, not of mental phenomena, and since Comte himself rejects materialism because it introduces the points of view of the lower science into the higher and seeks to *derive* man from the world, not merely to point out his dependence on the world, the science of mental phenomena does not fit into Comte's sequence of the sciences, in which each preceding
[343] one is to hold for the following ones, although these are more special. Logic and the theory of number hold directly for the science of mental phenomena as well as for the science of material phenomena; but this cannot be said of geometry and the theory of motion, as long as one does not, with consistent materialism, regard the phenomena of consciousness without further ado as extended in space. The classification of the sciences is not so simple a matter as Comte presupposed. His system presents here a difficulty similar to the one we have already pointed out in Hobbes (in the first part of this work, p. 264 f.). Comte himself would not concede this. He does not regard psychology as a special science and therefore gives it no independent place in the series either. He denies the possibility of self-observation. One observes — he says[82]) — all phenomena with one's mind (esprit); but with what, then, is one to observe one's mind? One cannot, after all, divide one's mind into two parts, of which the one acts while the other investigates how it goes about it! Yet Comte concedes that the feelings can be observed more easily than thinking, because, as he believes, they have a different organ from it. Our intellectual activities must be studied in their products and results or in the organs to which they are tied. Instead of setting up psychology as an independent science, Comte distributes the psychological investigations between biology and sociology. Biology alone would not let the fact that the individual's mental life is determined by the influence of history and of society come into its own. It is thus by a purely objective way that Comte wants the life of consciousness studied. It was meritorious that the objective method was emphasized so strongly in opposition to the purely subjective and spiritualist psychology of Cousin and Jouffroy. But Comte has not seen that the subjective method in reality always lies at the basis of the objective one, and he has on the whole not let the peculiarity of the phenomena of consciousness come into its own. Whereas elsewhere he emphasizes discontinuity more strongly than is defensible, here he emphasizes it too little. There

would, as already indicated, also have arisen a difficulty for his classificatory system if the qualitative peculiarity of the phenomena of consciousness were to be stressed; *[344]* they would break the sequence, since, even if they are in and for themselves simple and elementary, they are in fact tied to the most complicated material phenomena. Should they then be placed first or last? — Comte sets himself in definite opposition not only to spiritualism but also to the psychology of Condillac, which he reproaches for conceiving the single individual as purely isolated, and also for taking account only of the external influences, not of the inner conditions. The single individual is a scholastic abstraction; only the species is a reality. (Cours[2] VI, p. 590). In biological respects Comte found the most important work done in Gall's physiology of the brain, if one disregarded his unfortunate doctrine of localization. Gall's list of mental faculties would indeed, Comte concedes, have to be considerably reduced; but no fewer than ten could be assumed; between ten and fifteen faculties was what he was most inclined to assume (Lettres à Stuart Mill. p. 51—55). Gall, Comte believes, first brought the study of the intellectual and moral functions into the positive stage and maintained the significance of the inner dispositions. This admiration of Comte's for Gall conflicts with his own conception of the relation between the metaphysical and the positive stage; for Gall's doctrine of faculties presents all the marks of "metaphysics." — If one asks how Comte unites the assumption of 10—15 peculiar faculties with the unity of the life of consciousness, the answer is that this unity is, according to Comte, not original but a result, resting on the harmony between man's various tendencies and forces (Cours[2] III, p. 545). — The problem of the relation between discontinuity and continuity extends into the life of consciousness and appears here in a way that demands its own peculiar investigation, an investigation, however, that Comte does not come to take up, because he carves psychology up into a biological and a sociological part, without the central psychological questions getting room to come forward. On this point Comte is supplemented by the new English school, which lays a peculiar weight precisely on psychology. *[345]*

d. Sociology and Ethics

More than half of Comte's chief work — the last three and largest volumes — treats of the science of society (sociology — as he calls it with a word of his own coining, which in spite of all philological misgivings has now made its way through). It comprises an essential part of psychology, the whole of political economy and ethics, together with the philosophy of history. Just as Comte protests against treating the psychology of the single individual apart from the spiritual development of the whole species, so he also

protests against political economy and ethics being isolated from the general theory of society, and the later development of these sciences has proved him right in this protest[83]). Neither psychology, political economy nor ethics can be treated without regard to human development through history. In all the different scientific domains Comte points out the relation between statics and dynamics. The world is regarded statically in geometry, dynamically in the doctrine of motion. Within physics and chemistry forces are regarded partly in equilibrium, partly in activity. In the organic domain statics is represented by anatomy, which investigates organization, dynamics by physiology, which investigates functions. To sociology there belong partly a social statics, which studies the constant conditions of existence of society, partly a social dynamics, which studies the laws of the steady development of society: the fundamental idea of the former is order, of the latter progress. Statics and dynamics belong closely together, for order and progress mutually condition each other — something which neither the reactionary nor the revolutionary school has seen.

a) Social statics. Society forms a whole, within which all elements stand in the most intimate interaction, so that one of them cannot be altered without more or fewer of the others undergoing a corresponding alteration. Thus, e.g., the political and social constitution is closely connected with the whole of civilization. There is a close connection between ideas, manners and institutions, and no external authority — whether of revolu-
[346] tionary or reactionary kind — is able to accomplish anything by forcing upon society institutions which do not correspond to the prevailing ideas and manners. Naturally the institutions in turn react upon ideas and manners; but this reaction presupposes a longer undisturbed subsistence and is most strongly prominent in the childhood of the human race. The ideas and the manners, too, stand in mutual interaction. It is the task of political institutions to regulate the common life which has developed involuntarily during the progress of humanity in intellectual, moral and physical respects. The elements which play the greatest part in this development will in the end become the ruling ones in society. Authority rests on free co-operation, not the reverse.

Within social statics *ethics* finds, on an essential side, its place. The ethical laws express the solidarity in all human life. This solidarity is discovered involuntarily, as men follow their need of fellowship. It is self-contradictory to explain the origin of social life out of calculation of the utility man can have from living together with others. Utility can only show itself after a sufficiently long common life has taken place, and can therefore not be the original motive for the common life's taking place. Comte here sets himself in opposition to the explanation frequent in the eighteenth century, which stopped at the

shrewd calculation of isolated individuals. In his view an instinctive impulse to social life, independent of personal calculation, asserted itself originally; feeling here, as on the whole, precedes knowledge. In this view of his Comte is, by his own account, a disciple partly of Hume and Adam Smith[84]), who led him beyond the current theory of egoism, partly of Gall, who assumed definite organs of the brain for the social instincts. Comte finds the first germ of sociability at the stage in the animal world where the sexes are separate and where care for offspring appears. Yet even in man the egoistic inclinations at first have the upper hand over the social inclinations, which latter Comte, in order to designate them in their marked opposition to egoism, designates as altruism (from alter, one's neighbor). Nor is personal interest to be rooted out. Altruism would become a vague and unfruitful love if it did not acknowledge the need of personal satisfaction *[347]* both in others and in the individual himself. What matters is that the egoistic instinct be subordinated, and this happens through the successive development of intelligence and sympathy. Through sympathy intelligence is freed from working in the service of egoism alone, and through intelligence the eye for social solidarity is sharpened. The social state favors the sympathetic feelings. As the individual extends his feeling to embrace the whole species, he obtains through this social expansion satisfaction for his need to make himself eternal, seeing the continued life of the species as a continuation of his own life.

As isolated and single the individual is only an abstraction. The social unit is the family, where the first germs appear of the dispositions which characterize the social organism. Here the individual learns to go beyond himself, to live in others, at the same time as it obeys its most energetic instincts. It is the most intimate society. It is a unity, not a combination (union, not association). Co-operation no doubt also plays a part in the family, but not the chief part, as in the larger societies, which rest on common labor and on the supplementing made necessary by the division of labor. By performing a definite labor each individual and each family exercises a social function. The authority which is to direct this function must proceed from society itself and found its power on the confidence it inspires and the free adherence it finds. Every society will inevitably come to have a government. While in the individual life it is the personal instincts, and in the family sympathy, that rule, in the larger society it will essentially be the intellectual faculties that act to order it. A certain intellectual community is necessary for a society; the co-operation of interests and immediate sympathy are not sufficient. Positive science has a very great significance for ethics, for it will be a matter of determining as exactly as possible the real, direct or indirect influence which every action, inclination or feeling has on human existence, both on the single individuals and on society as a whole.

[348] It was, according to Comte, the great merit of Catholicism that it emancipated morality from politics, to which in the polytheistic period it had been wholly subordinated. This emancipation found its expression in the doctrine of the two powers and of the independence of the spiritual power from the temporal. But Catholicism showed itself unable to satisfy the growing intellectual needs of men, and metaphysical criticism began its dissolving work. Moreover Catholicism furthered egoism by letting the individual be occupied with his own salvation, whereby it no doubt gained a mighty motive, but hindered the free and pure unfolding of the sympathetic and high-minded feelings. Only the positive intellectual tendency directly favors the unfolding of these feelings by impressing upon us that our whole development takes place *within* society, so that it is the individual and not society that is an abstraction. In social life the egoistic inclinations must in many ways be checked; only the social drives can unfold freely, and the heightened activity which the unlimited horizon thus opened brings with it contains a source of happiness, of inner satisfaction, which is independent of external rewards. The concept of duty springs from the view of the whole (esprit d'ensemble) which positive philosophy asserts, and which shows the single individual as a member of the whole species, so that the rules for its action must follow from a universal order of things, not merely from purely individual interests. The highest idea in the ethical domain is the idea of humanity as that whose development is uninterruptedly determined by the co-operation of all individual and social organs. Comte polemicizes against the sharp separation between private and public functions. This distinction is purely empirical and comes forward only in times of transition, when a new civilization is developing, and when the new elements have difficulty in finding their place. It did not exist among the Greeks and Romans, nor in the theocracy of the Middle Ages; it first came forward at the end of the Middle Ages, especially after the industrial upswing which followed the abolition of serfdom. The masses of the proletariat have since that time not truly been incorporated into the social system. And only with such an incorporation comes the
[349] right personal self-respect, which consists in feeling oneself a fellow worker within a great whole. One day such a feeling will ennoble even the humblest occupation, namely when through *positive* education the consciousness has become alive of the significance of every single endeavor for society as a whole.

It was Comte's conviction that what was needed for the time being was political quiet, the avoidance of great political upheavals, so that ideas, feelings and manners could undergo the involuntary change which the positive stage will bring with it. The most important social difficulties were in his eyes not political, but chiefly moral, and

could be overcome only through the influence of ideas and manners. In him the concept of duty decidedly preponderates over the concept of right. His philosophy contains no elements for a philosophy of law and political theory proper. For he holds the belief that institutions will easily develop once ideas and manners are changed. Here, already in his first and proper chief work, there asserts itself a utopian tendency, arisen as a reaction against the preceding period's strong emphasis on individual rights as well as against the many constitutional struggles with which the age was occupied. He thought that a comparatively short-lived "progressive dictatorship" would suffice to bring about the quiet ripening of ideas and feelings which was a necessary condition for the Occident to arrive at its lasting social institutions, and for the great Revolution to obtain its positive counterpart.

β) Social dynamics. One important law which belongs to social dynamics we already know: the law of the three stages. In the intellectual domain, as Comte emphasizes, progress can most easily be demonstrated. But to the different stages of intellectual development there correspond, as we indicated in the exposition of the three stages, stages of development of a social and political kind. To the theological stage corresponds the *military.* Its significance consists in this, that regularity and discipline, necessary conditions of political organization, are implanted and developed. Forces are gathered about pressing common purposes. Just as the first spiritual authority necessarily had to be theological, so the first governments necessarily had to be military. Only external *[350]* force could gather and hold together. And only through war does society grow at the earliest stages of development. War in turn brings slavery: in order that the warriors might have free disposal of their force, the economic labor had to be done by slaves. To the transitional period which in intellectual respects is designated as the metaphysical there corresponds in the social and political domain a period in which the *jurists* ("the legists") rule. A defensive military organization takes the place of the earlier offensive one. The warlike spirit gradually gives way to the productive spirit. The middle classes push forward and demand political rights. It is the task of the ruling jurists to weigh the different claims against one another. It is a vague and restless transitional stage, the social phase in which we still find ourselves. To the positive stage corresponds the *industrial* phase, in which the producing forces determine the arrangement of institutions and the distribution of power. Social questions now take the place of political ones. The proletarians gradually discover that political revolutions cannot solve the great social problems whose sting they feel most of all. They will naturally sympathize with the endeavor of positive philosophy to set duties above rights, so that the attention of all

may be directed to solving the task which is the properly social one: to procure for all access to spiritual development and access to work. There will therefore be a natural understanding between proletarians and positive philosophers.

This demonstration of humanity's solidary course of development becomes of great significance for ethics, which thus belongs not only to social statics but also to social dynamics. Not only is the social feeling strengthened by the demonstration of the solidary development, but an essential part of the content of ethics can be determined only on a dynamic basis. For the task of ethics is to contribute to the development of the specifically human qualities — in contrast to the animal and vegetative ones. And here ethics has its presupposition not only in what social dynamics teaches about the different
[351] stages, but already in the results of comparative biology, which show how the animal functions, the higher one comes up in the series of animals, play an ever greater part in relation to the vegetative ones. In man the specifically human qualities, intelligence and sociability, in turn develop in relation to the animal ones, which Comte — following Gall's interpretation of the brain — finds expressed in the stronger development of the frontal region of the brain in relation to the part of the brain which lies nearest the spinal cord. Within the human species Comte assumes an ever further-reaching development of intelligence and sympathy (altruism), the condition for the single individual's being able to make himself one with the species. Although Comte is an opponent of Lamarck's doctrine of evolution, he nevertheless acknowledges the great influence which uniform and steady exercise has on faculties and qualities. The nobler tendencies in our nature will through social life be ever further developed; the baser instincts will gradually be weakened, partly by self-control, partly by lack of exercise, or they will gradually be compelled to serve the social order itself. — A more definite form Comte is unable to give his ideas on this point, partly because he lacks the doctrine of the displacement of motives founded by Spinoza and Hartley, which English psychology later carried further, partly also because he could not make use of the close connection which evolutionism assumes between the development of the individual and of the whole species.

In concluding his exposition of human ethics, Comte declares that when one wishes to compare the morality of positive philosophy with religious morality, one must not forget that while the former is scarcely thought out and cannot work through regular institutions, the latter is fully developed through centuries of spiritual labor and has for long ages found support in a great social apparatus. In his later years — after the remarkable nervous crisis in the middle of the forties — he became more sanguine and no longer held fast to the conviction that we stand only at the beginning of the development.

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e. Theory of Knowledge

Although Comte has not made the theory of knowledge the object of special investigation and exposition, the whole of positive philosophy nevertheless rests clearly enough on definite epistemological presuppositions, and in some places (especially in the last volume of "Cours" and then in "Discours sur l'esprit positif") he expresses himself on it at greater length. It will be of great interest to attempt a determination of Comte's epistemological standpoint.

Positive philosophy is, according to Comte's express and repeated declarations, not empiricism. Already in 1825 he shows in an essay *(Considérations philosophiques sur les sciences et les savants),* that an absolute empiricism is impossible. Science does not consist in a mere accumulation of facts. Its essential task is the connection of facts; it consists of laws, not of mere facts. No isolated fact is incorporated in science; it becomes so only when the isolation is removed by its being connected — at least by means of a hypothesis — with another fact. Comte even makes the interesting remark (Cours[2] IV, p. 300) that no isolated and purely empirical observation can be certain. It was indeed also, as we have seen, through the necessity of a bond between phenomena that Comte made it intelligible that our knowledge must pass through the theological and the metaphysical stage before it can reach the positive. At the positive stage the laws of phenomena are sought, that is to say a factual connection between them. The connection can be brought about in two ways. Either it is a question of simultaneously given phenomena, and then the explanation is gained by demonstrating similarity in the relations and laws which hold for different groups of phenomena. Or else it is a question of successively appearing phenomena, and then the task is to demonstrate a continuous connection between them. In the first case we have a static, in the second case a dynamic explanation. Whether we explain *par similitude* or *par filiation,* we connect phenomena in such a way that we can foresee their occurrence. And in both cases we satisfy our mind's need of unity and find again the constant in the midst of the variations. (Discours sur l'esprit positif, p. 20—21). *[353]*

Comte's theory of knowledge does not go beyond these developments, interesting in themselves, on this point. He does not pursue the connecting activity of the mind more closely, although he regards it as fundamental: *tout se réduit toujours à lier!* A closer investigation would here have led him further into subjective psychology than he wished and considered possible. Nor does he enter into the question what validity can be ascribed to those laws of similarity and of succession. He expresses himself somewhat

waveringly on this point. Now he says that science has nothing to do with the first fundamental propositions; they spring up involuntarily in the human mind and cannot be discussed. (Discours sur l'esprit positif. p. 46). Here he approaches Reid's theory of knowledge built on common sense. Now, on the other hand, he says that the principle of the invariability of natural laws, on which positive science rests, cannot be grounded a priori, but will always in the end prove to rest on simple observation and induction (Discours. p. 17). This last consideration would evidently, if carried through consistently, drive him over into pure empiricism and into the difficulties in which it finds itself with regard to the causal principle, difficulties which first came clearly to light through Stuart Mill's significant resumption of Hume's problem. — When Comte expects that the principle of the invariability of natural laws will gradually make its way in all domains, it is, as already remarked, because he thinks that an irresistible analogy will compel us to carry it through. It will, as he declares in one of his older essays, in the long run be impossible for the human mind to think positively in some domains, metaphysically or theologically in others. It will strive after unity in method and theory. Here, then, the power of habit is at work, to which Hume already pointed. But habit can explain only the principle's psychological power over the mind, not its real validity. The sting of the problem of knowledge proper Comte has not felt. He sought to systematize positive
[354] knowledge; but he did not set himself the task of discussing the ultimate foundation of this knowledge.

From another side, however, Comte approaches the problem of knowledge, namely in his emphasis on the relativity of knowledge. He finds it, as we have seen, to be an essential characteristic of positive philosophy that it everywhere puts the relative in place of the absolute which was sought in the earlier forms of philosophy. The relativity of knowledge he grounds in two ways. In the first place, positive science can teach us only relations of similarity and of succession between things, not the absolute causes of things, nor the very inmost nature of the things between which those relations obtain. The relations themselves, too, stand as mere facts which cannot be fathomed further. In the second place, all our knowledge is determined by the relation between our organism and its environment — Comte even speaks of "the great elementary dualism between the understanding and the milieu" — and in such a way that the environment does affect and determine the understanding and gives it its material, but the working up takes place, as in the process of nutrition, according to the laws of our own organization and in its forms. There is therefore a relation partly to the subject, partly to the object, which asserts itself in all knowledge. In this biological epistemology[85]) Comte feels himself a

successor of Aristotle, Leibniz and Kant (Cours[2] VI, p. 620 ff. Catéchisme positiviste, p. 150 f.). It leads him to see that our knowledge can reach only an approximation to reality; but on account of his practical standpoint he does not come to discuss the question how far our knowledge can rightly be said to be a reflection of reality. It is enough for him that the knowledge we have can be used practically for orientation. On the other hand he lays great weight on this, that the very subject to which knowledge stands in relation undergoes change and development. All knowledge is determined by the stage of development which the individual and the species have reached. All our knowledge therefore has a historical character. Sociology leads, as already the doctrine of the three stages has shown, no less than biology to an emphasis on the relativity of knowledge. As long as biology and sociology were not yet developed in positive form, the relativity *[355]* could be overlooked, as happened in the whole period when mathematics stood as the chief science. Now the scepter has passed to sociology. It is sociology to which it belongs to state the concluding scientific points of view.

That Comte thus came to stress the biological and sociological point of view for our knowledge, or in other words came to see knowledge as determined by man's nature, needs and degree of development, brought with it an important change in his philosophical standpoint. From the first he had laid the chief weight on "the necessary and rational subordination of the concept man to the concept world" (Cours[2] III, p. 188); now he lays more and more weight on the subjective side of knowledge. Knowledge comes to stand for him essentially as the satisfaction of a subjective need, and he regards every means of reaching such satisfaction as justified. The need of knowledge becomes for him an artistic need. He asserts the justification of following the simplest hypotheses and disregards the necessity of a confirmation by experience. In what relation the satisfaction of the need of unity and simplicity in the conception of the world stands to positive reality he does not discuss, occupied as he more and more became with the mystical world to which he had gained access after the nervous crisis and the "méditation exceptionelle" it brought with it. His plan now became to set a subjective system in place of the objective system which had been set forth in "Cours". Hitherto he had explained man by the world; now he would explain the world from man.

f. Comte as Mystic

Comte's great work as a thinker was done with his exposition of positive philosophy. But later this stood for him only as an introduction to the highest mysteries. Of the psychological causes of this change in his views we have already spoken. We shall now

merely mention briefly what the ideas were which occupied him in his last years. If
[356] they have no significance as scientific thoughts, they nevertheless have significance as symptoms.

In his positive philosophy Comte had gone from the world to man; he had advanced from the abstract and universal relations to the more complicated — and man is the being that lives under the most complicated relations. Now he will go from man to the world, will with full consciousness take up the subjective standpoint which men in the childhood of the species took as their basis without being conscious of it. The world is now regarded as the supporting foundation of human life. From the most complicated — and therefore the highest — standpoint in existence the lower stages are now to be looked back upon. Within human nature it was in the exposition of positive philosophy necessarily intelligence that had come to stand in the first line. It is through intelligence that we know the laws of existence, and it is its influence on feeling that leads to new manners and institutions. Now feeling is set in the first line, and it becomes the task to enlighten intelligence by means of the heart. In place of analysis and the specialties synthesis, the unified conception, is now set. But only a subjective unity in our knowledge is possible. The forces of existence cannot, as positive philosophy has established, be reduced to absolute unity. But a unity in conception can be gained by regarding the whole of existence (so far as we know it) in relation to humanity. But Comte now feels the necessity of making a change in his earlier classification of the sciences, namely that of separating ethics from sociology and setting it up as the seventh fundamental science. For when humanity is to be the point of unity about which all thoughts gather and from which they go out, then the task is to immerse oneself in human nature in its whole fullness. But in sociology it is properly only thought and action that become of significance; the feelings play a part there only as motives, and since the different motives can on the whole balance one another, there will from a purely sociological point of view be no reason to lay great weight on the feelings. Ethics now becomes precisely the most complicated science of all by bringing forward an element
[357] which in sociology still stood as subordinate. The addition of ethics as an independent member in the system of the sciences thus takes place in accordance with the law of advance from the more abstract to the more concrete[86]). All sciences are now to be regarded as parts of ethics and to be cultivated with it in view, so that each single one is treated as preparation for the following, more complex one, until we land in ethics, the concluding science. Care must be taken that analysis does not again gain the upper hand, and all studies should be dropped which do not further our knowledge of the order of

nature and our ability to adapt ourselves to it, or which cannot serve as guidance for our activity. New chief works should be composed which can reduce the fundamental sciences to their necessary extent. — Characteristically Comte now (in *Politique positive* and in *Catéchisme positif)* calls the chief doctrines of positive philosophy "the positive dogma". His thought seeks rest, and he becomes impatient with every investigation that leads away from immediate absorption in the religion of humanity. Feeling stands for him high above knowledge and action, which have value only through their results, which moreover are dependent on external circumstances, while feeling causes direct and inner satisfaction. And from this follows in turn the precedence of art over science. The source of art is in feeling. It gently leads the abstract considerations of the theorists back to reality, while it instills in the practical men a noble interest in great ideas. Its physiological origin lies in the involuntary movements which are bound up with the feelings and express our inner states, while they also react upon these. It forms ideal types which by steady contemplation perfect our thoughts and instincts. In the priests of the religion of humanity philosophy and poetry are to melt together.

The religion of humanity, of which Comte regarded himself as the founder, is a worship of humanity as *le grand être,* in which we all have a share, the sum of all deceased, now living and future beings who in a free way have worked for human progress and happiness in small or great circles. It is an ideal concept; for only those will live in the continued memory of the species who have devoted themselves to it *[358]* and worked for it. In drawing up a positivist calendar, in which every month and every day is named after men who have had significance for the development of humanity, Comte himself has given directions for a public cult, which consists in periodically recurring, enthusiastic and grateful remembrance of the benefactors of the species. In the private cult persons who have stood nearer to the individual will represent for him the ideal of the species. Those who have put themselves in the way of progress — as, e.g., the Emperor Julian and Napoleon, "ces deux principaux rétrogradeurs que nous offre l'ensemble de l'histoire" — will be remembered in the positivist cult only in order to receive "the deserved periodical flogging". The single individual's dignity rests on its absorption in "the great being" through the smaller and larger circles (family, fatherland, etc.) within it. Toward the preservation and perfecting of this being all thoughts and actions are to be directed. Altruism (vivre pour autrui) contains at once the highest duty and the highest happiness.

The direction of the cult as well as of education is, according to Comte, to be entrusted to a priesthood consisting of encyclopedically trained philosophers, who are also to be

poets and physicians. Through their authority, resting on free acknowledgment, they will gain influence on public opinion. Comte seeks on the whole to carry over into the religion of positivism analogies to "everything great and heartfelt which the Catholic system of the Middle Ages has realized or at least sketched". Sociocracy, the social order he has in mind, is far more akin to theocracy than to the intervening individualist period with its "endless agitation". Indeed, Comte even seeks further back. Precisely because positivism has completely worked itself free of the old prejudices, it will be able without misgiving to resume the fetishistic view of nature and ascribe life and soul to everything in nature. That will give language force and favor the artistic imagination, and it will quicken the feeling for everything that can serve the subsistence and development of "the great being". Every animal species is regarded as a human being that has been arrested
[359] in its growth (un grand être plus ou moins avorté). It is the time of images that now comes again after the long critical period. The new religion then regards cosmic space as "the great medium" in which the earth, "the great fetish", has formed itself. The great fetish has in turn let its mighty elementary forces rest and has sacrificed itself out of longing that "the great being", in which the highest perfection appears within a narrow frame, should develop. (The highest beings are, as Comte already taught in "Cours", also the most dependent). The great medium, the great fetish and the great being make up the positivist trinity. —

Comte also gives hints concerning the constitution of the future sociocracy. The idea of right is to disappear completely. No one possesses any other right than that of doing his duty. Individuals are regarded not as separate beings but as so many organs of the great being. The directing power in external affairs belongs to the industrial chiefs (bankers, manufacturers and landowners). These *patricians* are by their wealth freed from covetousness and will be able to ennoble labor by taking it on by free choice, motivated by the highest personal feelings. Their task is to direct industrial labor in such a way that all can come to share in family life, which is the foundation of human happiness. It is in the interest of the proletarians that capital be concentrated in the hands of a few patricians; thereby a common and intelligent direction becomes possible. The time of the middle class is past. The patricians correspond to the nutritive organs in the organism. Over against them stand the cerebral organs, the organs of reason *(the philosophers),* of heartfelt feeling *(the women)* and of energy *(the proletarians).* While the patricians chiefly represent order, the proletarians represent progress. It is the business of the philosophers (or priests) to find out what the welfare of "the great being" demands; the business of the women is to awaken the right feeling to drive the work. Philosophers

and women will sympathize with the proletariat and find support in it against possible encroachments on the part of the patricians, and the proletarians have in public opinion and in refusal of co-operation a means of preventing abuse on the part of the spiritual or temporal authorities. — *[360]*

Comte's utopia has the interest which such descriptions can have when they come from eminent minds with an eye for the tendencies of the age. One traces his historical presuppositions, just as in Plato's Republic a multitude of features from Greek conditions have been found again. His sketch of a religion of the future has significance partly as the testimony of an earnest and heartfelt man that all is not done with criticism and negation of the views of life of the past, partly in that it is the idea of humanity and of love that he makes the center. What he would not admit was that, provided only this idea can come to dominion, the religious representations would be able to develop with individual freedom and in great multiplicity among those who feel a need of them. Hardly many will share the taste Comte displays in his arrangement of a cult of the future, apart from some brilliant thoughts, e.g. about a new calendar with the names of historical personalities in place of the unknown names of saints. Many will think that the real religious problem only begins where Comte's religion ends, namely with the question how the development of the world is related to the development of the human species and its ideals. Comte's new trinity gives no answer to this. — Just as little as Comte will give individuality freer play in religious respects, just as little will he do so in political respects, so that he even wholly excludes the concept of a legal order and of a representative constitution. Public opinion and the strike will hardly be sufficient correctives when there are no fixed forms of public and private life to which those in power are bound. — Most characteristic is the suppression of free inquiry in the face of the closed "positive dogma". It can be excused only as an effect of the cerebral hygiene which the tired thinker, after his energetic labor, had had to impose on himself. But it would conflict with the spirit of his own most excellent work if his proposals were followed. *[361]*

B. John Stuart Mill and the Renewal of English Philosophy in the Nineteenth Century

In England the clash between the thought of the eighteenth and of the nineteenth century was not so violent as in France and in Germany. The English people's wonderful ability to carry out radical upheavals without breaking the continuity of its development displays

itself also in the philosophical domain. There does indeed appear (as already remarked in the first part), after the time of Hume and Adam Smith, an ebb in English philosophy. In the domain of epistemology Hume had drawn such radical consequences from the preceding English philosophy of experience that there was hardly anything to be done for the time being from that standpoint. The practical, political and religious interests, moreover, came into the first line. The men who held fast to the philosophical interest at the transition to the new century, Jeremy Bentham and James Mill, are accordingly guided essentially by practical interest. The ethics and psychology of the older English school were taken into the service of the ideas of reform. And here it is remarkable to see in how different a way essentially the same circle of thoughts is made fruitful in England and in France. There is not the heaven-storming quality in Bentham and James Mill that there is in the French philosophers of the Enlightenment; but there is a concentration of thought on definite purposes and a sense for practical application of general principles, which forms a salutary contrast to the pointed phrases and declamations of the revolutionary Frenchmen. The English do not like to fire off their artillery into the air; they care about hitting their target, even if the thunder of the cannon sounds less resounding. Through their practical sense and their steadfast upholding of the principles of the older English school those two men mark significant extensions of the best in the eighteenth century into the new time. They brought up a young generation which could receive the thoughts of the new time without forfeiting the yield of the preceding time's intellectual
[362] labor. Stuart Mill's personality as a thinker obtains a central position in the history of English philosophy by the fullness and understanding with which he is able to take up into himself the thoughts of the old and of the new time and to work them over in such a way that with him a whole renewal of English philosophy sets in. He resumes Hume's problems on a broader basis, at once with greater consistency and with greater many-sidedness. With conscientious energy he holds fast to the points of view of the older school as long as it is possible for him, without concealing the difficulties and without lacking an eye for the significance of the new experiences. He has, as regards both theoretical and practical problems, carried out a work of survey and of reckoning of the greatest significance for the spiritual and social development of the nineteenth century, and in the forties and fifties he was undoubtedly the greatest philosophical thinker of the age. Then came the doctrine of evolution with its new points of view.

Critical philosophy and Romantic speculation, however, also made their way into the English life of the spirit and gave it mighty impulses. A deepening and an enlargement became possible for which the older English school, left to itself, could hardly have

prepared the way. Against empiricism the tendency which Schiller had called "the philosophy of the ideal" asserted that there were spiritual values which the philosophy of experience had not been able to explain. In England this tendency was represented by Coleridge and especially by Carlyle. In the latter Goethe's humanism and Fichte's doctrine of personality are transformed into a peculiar historical view of life, which has intervened powerfully in the development of many individuals and given forceful guidance for preserving, in the struggle with the problems of the age and amid the century's sharp oppositions, the inmost kernel of the personal life. There was one question, however, which Carlyle always pushed aside, in so far as he did not answer it by cutting the knot. It was this: how this assertion of the value of the personal life is related to the attempts to find a thoroughgoing conformity to law both in the spiritual domain and in the rest of existence. Here lies the epistemological problem which in the spirit of critical philosophy is taken up by inquirers such as Whewell and Hamilton, and *[363]* which Stuart Mill then treats in the direction of absolute empiricism.

1. Philosophy in England before 1840

a. The Philosophy of Reform

Jeremy Bentham (born in London on February 15, 1748, died June 8, 1832) belongs rather to the history of jurisprudence and of philanthropy than to that of philosophy. But he intervened powerfully in the development of philosophical ethics by the energy with which he declared and applied the principle that every action and every institution is to be judged by the tendency it has to further happiness and keep unhappiness away. He would found all ethics and all jurisprudence on the one fundamental proposition that well-being is to be preferred to ill-being. Only prejudices, especially religious prejudices, and the lust for domination could, he thought, hinder the open acknowledgment of this principle, and we therefore also see men follow it everywhere where they use their reason without external and internal hindrances. The principle of the greatest possible happiness for the greatest possible number, or, as it is also called, the principle of utility (by utility is understood the tendency to produce happiness), is according to Bentham a self-evident principle: it grounds all practical valuation, but itself needs no grounding. So he sets it up already in his first writing, which appeared in the year of Hume's death. (*A Fragment on Government.* 1776. Chap. 1. § 48. Here *The principle of utility* is said to be "the principle which contains the only reason [reason] that does not in turn depend on any higher reason, but is itself the one, fully sufficient reason [the sole and all-sufficient reason] for every kind of action".)

This principle, however, Bentham did not discover himself. It had been set up in a similar formulation by Hutcheson and by several after him. Bentham himself declares (in *Deontology),* that he has it from Priestley. But it was, he says, in an interesting statement
[364] about his course of development (Fragm. on Gov. Chap. 1. § 36 Note), the reading of the third part of Hume's "Treatise," that made the scales fall from his eyes. He had been brought up in an orthodox Tory family, in strictly conservative views. He had regarded Charles I as a martyr and rebellion as an impiety. His legal studies had later led him to the natural-law doctrine of the original contract, from which it was deduced that when the prince broke his obligations, the subjects were no longer obliged to obey him. This doctrine he now broke with, influenced by Hume, partly because no conclusion of any such contract could be demonstrated in history, partly because, even if one assumed an original contract, there still remained the question to be answered *why* one should keep contracts or promises at all. The only possible answer to this question is: the keeping of contracts is to the advantage of society, and therefore each single individual ought to keep his promises; if he does not keep them, he ought to be punished, for the suffering the punishment brings him is amply outweighed by the good that the keeping of promises brings society (Fragm. Chap. 1. § 42). Thereby the theory of natural law is replaced by the theory of utility, the original contract by the principle of utility. And this transition has, as Bentham himself emphasizes, the great significance that we now pass from the world of fictions over into the world of facts. For whether an action or an institution is useful or not must be demonstrable in experience. Discussion will therefore now consist essentially in the ascertaining of facts. The right of free criticism of actions and institutions is therefore of extraordinary significance. There could — Bentham thinks — be no danger in such freedom of expression. For every institution that is really useful will have those as its defenders who draw utility from it, and so there will be no lack of answers to charges. — In his later writings Bentham preferred to speak of *the principle of happiness* rather than of the principle of utility.

Through his principle of utility Bentham came to make front on two sides. He came forward as a severe critic of the traditional institutions and especially of the chaotic English legislation. His proper field is the censure of the existing legal and social order
[365] (censorial jurisprudence, see Fragm. on Gov. Preface). He brought forward and chastised existing abuses with such effect that it has been said of him that he worked against his own fame: the evils he attacked disappeared to such a degree that later one could not sufficiently appreciate the significance of the labor that made them disappear! — In the Italian philosopher of law Beccaria he saw in this domain his most important

predecessor. Beccaria had already set up the principle of the greatest happiness for the greatest number as the purpose of laws and applied it in his criticism of criminal law. Bentham gave it a far more comprehensive application. While he thereby came into sharp conflict with the conservatives, he opposed on the other hand the French revolutionaries' appeal to natural law and their setting up of general rights of man. Rights the individual has, in Bentham's view, only in so far as the welfare of the whole society brings this with it. And he thought that the proclamation of the rights of man led to favoring egoism, which was strong enough already, while what mattered was precisely to get the individual to submit to the sacrifices the good of the majority demanded. When nevertheless in revolutionary declamations people spoke of natural and inalienable rights, that was for Bentham only "nonsense upon stilts".

In his chief philosophical work *Principles of Morals and Legislation,* which appeared in 1789, Bentham goes more into detail with regard to the application of the principle of utility. He investigates which feelings of happiness are to be preferred to others, and in this demands regard for the certainty of the feeling of pleasure, for its intensity, duration, propinquity, purity and capacity to propagate itself. He investigates further by what rewards and punishments men can be led to perform actions that bring happiness, and in close connection with this, what different motives determine men's actions in general, and what moral value they have. Everywhere he shows himself strong in distinction, definition and classification. In his domain he is something of a scholastic. Of particular interest is his investigation of motives. For it is by no means the case that he overlooks their significance. On the contrary, the principle of utility itself gives him a standard *[366]* for the valuation of the inner sources of actions. Those motives are to be called good which prove to work in the direction of harmony between the individual's own interests and those of others, bad those which lead to separating the interests. The motive that will most surely of all lead in the direction of what the principle of utility demands is benevolence (good-will, benevolence): "the demands of utility are neither more nor less than what the most comprehensive and enlightened benevolence requires". Then comes the need of others' esteem, the wish to win their love, religion, and finally the individual impulse of self-preservation and the wish for enjoyment, advantage and power. (Princ. of Mor. and Legisl. Chap. X. § 29—42).

Bentham's chief interest was the reform of legislation in a humane direction: codification of the laws (the very word "codification" he coined), so that everyone can know and understand them, and so as to keep abuses and costs away, improvement of the prison system, and the development of the constitution into a democracy through the

introduction of universal suffrage. In his struggle for the reforms he uses the principle of utility as his starting point. And since he constantly proceeds from it, he takes it as a dogmatic principle, as settled, and feels no need to investigate on what its acknowledgment is grounded. He does not see that a similar question can be put to him to the one he himself put to the adherents of natural law. As he asked them why one ought to keep one's promise, so one can ask him why one is to work for the happiness of the many. Logically self-evident it is not. If one investigates what possibilities of an answer to this question are to be found in Bentham, one gets no clear result out of it. As we have seen, he declares the demands of the principle of utility to be the same as the demands of comprehensive and enlightened benevolence. So far one might believe that the acknowledgment of the principle of utility as a standard springs, according to Bentham too, from sympathetic feelings. In Bentham personally it probably was so. He was a man easily moved to pity and sympathy, just as he was also, for his own part, very susceptible to pain. But he lacked the ability to enter into the moods and states of others, and moreover the many prejudices and hindrances springing from
[367] egoism which he met on his way did not give him any great notion of the power of disinterested sympathy in the world. While his own sympathy led him not only to work his whole life long, theoretically and practically, to further human happiness, a labor which brought him himself neither advantage nor external honor, and while it moreover inspired in him a great hope that the demands of the principle of utility would make their way, he nevertheless built, both in his activity and in his hopes, rather on there proving to be among men's egoistic interests a harmony which made it possible for the general happiness to be reached, provided only each individual energetically and shrewdly worked for his own happiness. It is the harmony of well-understood interests on which he properly builds. In a writing which was composed on the basis of his notes by one of his disciples and appeared after his death *(Deontology)* this side of his view comes out especially strongly. He here takes a standpoint which recalls that of Helvetius; he also always rated this French author very highly. Although disciples of Bentham have protested against the views developed in the said writing being without further ado ascribed to Bentham himself, it is nevertheless not improbable that it contains a part of Bentham's views of whose relation to other parts he himself had not made himself clear. All virtue is here traced back to individual prudence, which sees that one must serve others in order to receive services from them in return, and the hope for the future is built on public opinion gaining greater and greater significance, and on its becoming more and more insight into the harmony of well-understood interests that determines

the judgments of public opinion. The notion of this great harmony inspired Bentham, just as Adam Smith was inspired by the harmony of economic interests. Bentham has properly extended what in Smith is political economy to make up the whole of ethics. He builds on a conception of the species as consisting of isolated individuals, each striving eagerly to procure for himself as many goods as possible at as little cost as possible. Even so great an admirer of Bentham as Stuart Mill has nevertheless declared that he knew only "the business side of human life" (the business part of human affairs)[87]). Only it *[368]* must be held fast that Bentham himself did not stand in any "businesslike" relation to his views and endeavors[88]).

Early on Bentham's ideas gained influence on men who were active in public life, and when he learned what resistance his ideas of reform met with on the part of state and church, he himself became more politically interested and more anticlerical than he had previously been. But no school proper or closed party formed itself around him. He lived very retired and worked properly only through his writings. Toward the end of his life his tendency got its own organ (Westminster Review), which entered into sharp opposition to the journals of the old parties (Quarterly Review and Edinburgh Review). Bentham's friend James Mill and the latter's son Stuart Mill were the most active contributors. The adherents of Bentham's principle of utility were commonly called Utilitarians, a word which Stuart Mill (Levned. Danish trans. p. 81) believes he brought into currency, although Bentham himself had already used it. Like the word positivism, the word utilitarianism, too, designates rather different views; for an ethical view is not completely characterized by the standard it applies in valuation. The ethical problem contains more questions than this. A utilitarian (in a wider sense) need not, in particular, conceive the psychological foundation of ethics in the same way as Bentham.

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Among Bentham's collaborators James Mill stands in the first rank. His philosophical significance consists especially in this, that he seeks to supply the psychological foundation which Bentham's ethics lacked. In this respect his education at the University of Edinburg, where he heard lectures by Dugald Stewart, a disciple of Reid, had given him a preparation. Later he is influenced by Thomas Brown, whose philosophical standpoint forms a transition between the older Scottish school's appeal to induction, instinct or common sense and pure association psychology. But on James Mill's definitive standpoint the study of Hartley had the greatest positive influence. He was born in the south of Scotland on April 6, 1773, and grew up in humble circumstances. His father was a *[369]* village shoemaker. His mother was of a more well-to-do family, and it was owing to her

self-respect and ambition that the son's good intellectual abilities received sufficient development. Later James Mill received support from a landowner, Sir John Stuart, in whose family he became tutor. In Edinburgh he studied theology, but it seems that he early gave up the thought of becoming a clergyman, even if his theological views only much later (and still later than his son relates in his Autobiography) developed in a radical direction. In his thirtieth year he went to London, where he made his living by literary work. These became straitened years for him, when he married and had a numerous family which he had to support with his pen. In this hard struggle for existence he nevertheless did not give up the struggle for humane purposes, any more than he abated his gradually developed radical views. The chief work from this period of his life is *History of British India,* in which he gives a sharp criticism, but one built on thorough knowledge, of the East India Company's government in India. It is characteristic of English conditions that Mill was nonetheless appointed in the Company's service when he applied for a vacant post; one held to the knowledge and was not afraid of the criticism. Soon he rose to a very high position and gained great influence on the Indian administration. Of great significance for him was his acquaintance with Bentham, which became a firm friendship that lasted in spite of passing ill feelings and clashes between the two sharply marked and self-assured men. Mill worked for the application of the principle of utility in a whole series of philanthropic and political domains. His endeavors in these directions are still interesting and instructive to follow; Alexander Bain gives in his *work James Mill. A Biography* (London 1882) a detailed survey of his labors from year to year. Mill worked not only through his writings, but quite as much through his conversations with a circle of younger men who gathered around him, members of the radical political party and contributors to the Westminster Review. He is the spiritual father of the first parliamentary reform. With a zeal and a sharpness which at times became too strong
[370] even for Bentham, who otherwise did not mince matters, he attacked the aristocracy and the hierarchy. From his study he directed the great class struggle which brought English politics onto new paths. He showed himself a more practical politician than Bentham by not proclaiming universal suffrage without further ado, but restricting the program to the emancipation of the middle class. It was his conviction that the suffrage could be extended to larger circles only as enlightenment and instruction spread. From a politics which rested on general popular enlightenment and suffrage and was guided by the principle of utility, he expected an unlimited progress for the human race. For him, as for Bentham, the decisive thing was that men should be enlightened about their interests and be allowed to follow their reason. He proceeded, like Bentham, predominantly

deductively. The principle of utility was for him too an eternal truth from which it was only a matter of drawing the consequences. Neither he nor Bentham had a sense for the manifoldly nuanced concrete circumstances under which the general ethical and legal-philosophical principles were to be applied. To give laws was for Bentham far from so difficult a matter as Montesquieu had thought. Bentham did indeed write a special essay "on the influence of time and place on legislation"; but he nevertheless thought that — especially under the influence of prejudices and of zeal to defend traditional abuses — the significance of historical circumstances was exaggerated. What in James Mill marks an advance in relation to Bentham is the endeavor to give a more definite foundation for the general principle of valuation from which the ethical and political propositions were to be deduced. To this he was led by the psychological studies which he steadily continued during his official activity and during his philanthropic and political endeavors, and which found their conclusion in his famous work *Analysis of the Human Mind* (1829).

This work is of great significance in the history of psychology as the most thoroughly carried-through attempt to trace all mental phenomena back to association between representations. It resumes the attempt which Hartley had in his time made in the same direction. There is far greater clarity and fullness in James Mill's exposition than *[371]* in Hartley's. He not only explains all phenomena in consciousness as arisen through association, but all association he in turn traces — in a somewhat artificial way — back to association between representations which have frequently occurred together (what has later been called association by contiguity). It is the principle of simplicity that is at work here. James Mill sets up as a rule that as few original facts as possible are to be acknowledged. Just as Bentham would build the whole of ethics on the one proposition that pleasure is to be preferred to pain, so James Mill would build the whole of psychology on the one proposition that what has once been experienced can be recalled again when experiences which occurred together with it in time or space are repeated. — If this proposition could bear the whole of psychology, psychological theory would undeniably gain a quite extraordinary simplicity and perspicuity. And at the same time the prospect would be opened of so consistent a carrying through of the philosophy of experience as the older English school had not yet known. Not only all single representations each by itself, but also all combinations of representations would then be wholly determined by what had appeared to consciousness from without. An unlimited prospect of influencing the human life of the spirit and determining its direction opened here; for through legislation, education and in general the ordering of external circumstances one determines the combinations of representations which come

to rule in men. Association psychology thus gives access not only to understanding men's representations as they have once and for all combined and formed themselves, but also to deciding which representations and combinations of representations are to rule in the future. It could be the foundation both for criticism and for new formation, a mighty weapon against prejudices by demonstrating their origin, and a means of progress by putting new, justified combinations of representations in the place of the prejudices. — Beside Hartley's influence that of Helvetius shows itself here. The book "De l'esprit" was in Mill's circle a favorite work, just like "Observations on Man".

When now association psychology, in the form in which it here appears, is to serve
[372] as the foundation for the acknowledgment of the principle of utility, James Mill — following Hartley — lays strong weight on this, that association can lead not only to one representation calling up another representation or awakening a feeling of pleasure or pain, but also to several representations and feelings entering into so close a combination that they cannot be separated, and that the new whole they together make up has qualities which the parts each by itself do not have. The new whole formed by association can, as James Mill expresses himself in the clearest exposition he has given of this (Appendix B to the polemical writing *Fragment on Mackintosh,* an essay of classic significance in psychology), itself become a supporting principle in human nature (a substantive principle of human nature). One can therefore not infer from the nature and value of a feeling to its origin. It has given occasion to many disputes and misunderstandings that these two things, value and origin, have been confused. It has been believed that a mental faculty must either be absolutely original, or else that the factors to which it owes its coming into being must be homogeneous with it. Mental processes have been compared with mechanical combinations, and it has been overlooked that they must often be conceived by analogy with chemical processes. This mistake has led to either (as in "the selfish system") tracing, for the sake of clarity and simplicity, all feelings back to egoism, or assuming several different original kinds of feelings (some selfish, others disinterested). By means of association psychology, on the other hand, the view can be grounded that the disinterested feelings have developed out of elementary feelings just as much as the egoistic ones. As Butler already demonstrated, egoism proper cannot be original, since it presupposes conscious calculation; but just as little is disinterested benevolence original: it develops through pleasure and pain being very early transferred from the nearest and more elementary causes to more remote ones, which stand in connection with them as conditions and means of their working. Secondary feelings form themselves, which are attached to something that was at first a means but gradually

comes to stand as a purpose. Thus motives which at first have value only as conditions of action can acquire independent value; and the unconditional value we ascribe to *[373]* the moral feeling or conscience thereby obtains its explanation. The welfare of others, which at first was only a means to the individual's welfare, can secondarily become a purpose for him. James Mill expected, by virtue of this possibility, great things of the political education a people undergoes by living under a good constitution. He says of this in an essay ("Education", written 1818): "When the political machine is such that the great objects of desire are seen to be the natural rewards of great and noble conduct, of considerable services to humanity and of the high-minded and benevolent cast of mind from which great endeavors in the service of humanity proceed, then one will also naturally find zeal to acquire the qualities which lead to admirable actions spread abroad in humanity: such qualities as great intelligence, perfect self-control and all-conquering goodness!" — And secondary feelings and qualities can be just as firm and immediate as the primary ones. Nor will — Mill asserts — the analytic investigation of the origin of such secondary feelings and qualities through association weaken the value they can have for us: "Gratitude remains gratitude, resentment remains resentment and generosity generosity in the consciousness of those who feel them, — after the analysis as before it. He who can trace such feelings back to their elements does not cease to have them, any more than he who never thought of investigating them . . . . They are constituent parts of human nature. How we are affected by having them is a matter of experience which everyone knows from himself. Their working is what it is, whether they are simple or compound. Does a compound motive cease to be a motive as soon as it is discovered that it is compound?" (Fragm. on Mackintosh. p. 51 f.). "The disinterested nature of a feeling is not impaired by the consideration that it has come into being through one of the great psychical processes by which we with great probability explain the coming into being of the other habitual states of human consciousness (habitual states)" (ib. p. 421). Mill here answers an objection which both then and later was often made against the psychological explanation of the origin of the moral feeling — not only against the *[374]* explanation the older association psychology gave, but also against the one the doctrine of evolution has given.

But in the very statement of Mill's just quoted it is implied that the deductive application of association psychology has its definite difficulties and limits. First the concrete states must be studied in their *historical* coming into being before their elements can be discovered. By the reference to immediate experience it is admitted that the matter is not so simple as it might seem according to the principles of association psychology.

The new qualities of the product cannot, after all, be deduced from the factors! And the historical element which thus asserts itself in the individual's life of consciousness as a — at least provisional — hindrance to analysis appears in far higher degree in the social and political domain as a barrier to the deductive application of the principle of utility. The Utilitarians of the older school satisfied only in a very imperfect way the principle of experience which it was their task and their pride to assert.

It was now precisely the irreducible, that which resists analysis, on which the philosophy of Romanticism laid weight. With great passion the most important representative of the Romantic school in England in particular asserts that in life which does not seem intelligible by mechanical explanation, the original and unique which all personal and historical development presents.

b. The Romantic Philosophy of Personality

Bentham and James Mill are two mighty rocks that jut into the new century, and against which the waves of Romanticism break. It was of extraordinarily great importance for the life of the spirit in England that the new currents met such superior representatives of the old time, so firm of character. The interaction between the old and the new thereby became as fruitful as it seldom is. The clash between the two tendencies was sharp and cut deep into many a personal development, but it gave birth to forms and endeavors in
[375] the view of life and in thought that have epoch-making importance for the life of the spirit in Europe as a whole.

Through Samuel Taylor Coleridge (1772—1834) the new German literature and philosophy of the age of Romanticism gained influence in England. In his youth he had (as he relates in his *Biographia litteraria*) been an adherent of Hume and Hartley. But he was repelled by the eighteenth century's attempt to lead all mental phenomena back by analysis to elementary functions, and by the endeavor to demonstrate mechanical laws for the life of the mind: the unity and activity of the mind were lost! The study of German philosophy led him decidedly away from the English school. As a good Romantic he reveled in absolute ideas, in which the differences and oppositions of the finite world faded and vanished. He set the whole, intuited by presentiment, against the piecemeal work of empiricism, synthesis against analysis. In his own opinion he anticipated some of Schelling's ideas; so thoroughly had he lived himself into the speculative line of thought that he could continue it on his own in the same direction as its original founders. Occasionally, in editing his writings, he confused pieces he had translated from Schelling with his own compositions, which gave occasion to charges of plagiarism. This poetic,

enthusiastic spirit, who was quite as much poet and preacher as thinker, could not properly distinguish between what he had himself conceived and what he had found in others. What he took from German philosophy was above all Kant's distinction between reason (in the narrower sense), as the faculty of ideas of the unconditioned, and understanding, as the faculty of categories that make limited knowledge possible. This distinction he used above all to find a reconciliation between thought and religion. All the objections the preceding age had held to be decisive were now — not exactly rejected, but — set down to the account of the "understanding" and pushed aside by an appeal to the deeper conception of "reason." Behind this appeal there lay hidden the awakening sense for what in history and in life escaped, or had hitherto escaped, scientific explanation. It was a protest against the sufficiency of the science developed hitherto, a *[376]* protest that was given and taken as a new solution, a new foundation. Coleridge did not get beyond presentiment and annunciation. He worked mostly through conversations (or rather through his monologues before listeners). His great work on the agreement between philosophy and Christianity, to which he referred his listeners at the difficult points, he never got written. He placed himself in opposition both to the philosophy of the eighteenth century (Hume and Voltaire) and to the ecclesiastical theology that had stiffened into outward forms and formulas. Like the German systematizers of Romanticism, he wished to bring about a higher unity of reflection and dogmatic faith. In his fundamental thoughts he recalls the so-called right wing of Hegel's school. He found an explanation of the Trinity, e.g., by applying the schema thesis, antithesis and synthesis, which from Fichte's time on played so great a role in German speculation. (Under the evident influence of Schelling's later doctrine, however, he is in danger of deducing a quaternity, since he posits God as "the absolute will or identity" = prothesis, as the foundation for the interplay of thesis and antithesis that is harmonized in synthesis! See *Table Talk*, 8 July 1827.) But coherent thinking was not Coleridge's affair. It is surely apt when he has been called a religious epicurean. He reveled in religious thoughts and feelings and in ideas that could furnish symbols for them. But he was a weak character. He could master neither his conduct of life nor his line of thought. He fled, as Carlyle (who in *Life of Sterling*, chap. VIII, gives an interesting characterization of him) has said, from life into the world of theosophic dreams; what he gave was "transcendental moonshine," which showed many a young doubting and seeking nature the old things in a new light, but did not contain any really new life. A man who was himself for a time strongly influenced by Coleridge, the friend of Carlyle and of Stuart Mill, John Sterling, has said of him: "His misfortune was that he appeared at a time when there was a man's

work to be done — and he did not do it. He had not sufficient strength of character; doctrines he had ceased to believe in he professed, in order to avoid the unrest of strife."
[377] What this charge, which Sterling repeats in a letter to Carlyle, is aimed at is not clear. But when it is speculation and fantasy that become the chief organs of religious thought, it easily happens that more doctrines are acknowledged than one can really turn into personal property.

Meanwhile, Coleridge had awakened a new sense and new ways of seeing in many of the younger generation. No less a man than Stuart Mill, who was otherwise not favorably disposed toward German speculative philosophy, has placed him beside Bentham and regarded these two men as the two great fertilizing minds (seminal minds) of their time. Mill's own great and honest attempt to learn from both of them we shall later have occasion to bring forward. Most of those whom Coleridge influenced for the time being would not learn from Bentham as well, but were led through the "moonshine" to orthodoxy, and soon to a more pronounced orthodoxy than the old. Perhaps Carlyle is right when he says that had there been no Coleridge, there would have been no Anglo-Catholic movement (Puseyism). Several of the men of this movement at any rate began with poetic and speculative ideas before the positively ecclesiastical line of thought gathered speed[89]). That the oppositions in the domain of the view of life sharpen in the course of the nineteenth century — at any rate as regards the outward formulas — is something to which the life of the spirit of every country bears witness.

Thomas Carlyle (1795—1881) found nourishment for his spirit in a wholly different side of German philosophy. Not a higher speculative knowledge, but a more vigorous assertion of the value of personality, a new faith, was what he gained through the study of German poetry and thought. Through the German poets he was led to the German thinkers; through Schiller and Goethe he was led to Kant and Fichte. He saw clearly that the Scottish school had not overcome Hume's doubt by its appeal to instinct *(Signs of the Times* 1829), whereas Kant and his successors went to work more thoroughly and attacked Hume, and the mechanical philosophy in general, at the center, not in the outworks. Like Coleridge, Carlyle lays particular weight on the distinction between
[378] understanding and reason; it was for him from the first the most characteristic thing in Kant's philosophy *(State of German literature.* 1827). But Goethe was and remained the great prophet of his youth. However different his heavy Scottish nature, in which the mists did not wholly vanish even on clear days, was from Goethe's bright humanism, Goethe was nevertheless his model, and his confession of faith was the same as Faust's. From Kant he took into his definitive philosophy above all the distinction between

the thing in itself and phenomena. All that nature shows us is only phenomena. The "philosophy of clothes," which he sets forth in his profound and humorous work *Sartor Resartus* (1833), rests on the thought that just as little as the clothes make the man, so little has one fathomed existence because one knows the phenomena. Existence is and remains an unfathomable wonder, which must fill us with reverence. The world is the garment of the Deity. Natural science merely demonstrates the outward mechanism, but the kernel of existence is not thereby reached. Nature is a great symbol, a revelation of ideas that cannot be grasped by means of any scientific method. The world is no dead machine, as science would make it. Even the purely outward connection — the clothes — is inexhaustible and beyond survey from our little corner of existence. And however far our experience and our thought might be able to reach, beyond the forms of space and time we do not get, and they are after all only forms of knowledge; they too belong to the garments of the Deity, which are forever woven anew. The wonder of existence hides itself from us because from dull habit we take the forms for realities, and because our first fundamental propositions are often only assumptions we are accustomed to and therefore do not call into doubt. But what is philosophy other than a constant struggle against habit? It is "transcendental" precisely because it oversteps (transcends) the sphere of blind habit. It is then the task of philosophy to awaken anew the feeling for the wonder of existence when the mind is dulled by mechanical representations. "The man who cannot wonder, and who does not constantly wonder and worship — were he President of innumerable scientific societies and did he carry in his single head the whole Mécanique céleste and Hegel's philosophy and the sum of the work of all observatories and laboratories — is but a pair of spectacles behind which there is no eye. Let those *[379]* who have eyes look through him, then he may be of some use." (Sartor Resartus. I, 10).

It is easy to see that Carlyle applies the distinction between the thing in itself and phenomena not in the spirit of Kant but in the spirit of Romanticism. For Kant it was the task of science to find the lawful connection of phenomena, and the concept of the thing in itself marked the limit of this endeavor. But Romanticism despises and cancels the scientific endeavor to link phenomenon to phenomenon according to definite laws. One never comes to an end that way, after all, and one does not reach the kernel of things! Carlyle himself, like his hero in Sartor Resartus (Book II. Chap. 3 and 7), had a time in his youth when the world was to him a dead machine. (Froude: *Thomas Carlyle.* I. p. 101)[90]). The study of Hume, Gibbon and d'Alembert had robbed him of the religious conception of his childhood. For a time he was wholly absorbed in mathematical studies, and his first literary work was a translation of Legendre's Geometry. It was then that he was led

through Mme. de Staël to German literature, and Goethe's great world opened itself to him. "Four years ago," he writes to Goethe in 1824, "I read Faust on the mountains of my Scottish homeland, and then it was my dream that I might reveal to you, as to a father, all the pains and wild wanderings that my heart, whose innermost secrets you seem to know so completely, has gone through." Like Faust he grew impatient with science. What is the use of being able to dig a few feet deeper than one has reached hitherto, when there is still an infinitely long way left? Or, as he later said in a conversation with Charles Darwin, how ridiculous is it not to care whether a glacier moves a little faster or a little slower, or moves at all! As a good Romantic he believed in Goethe's optics and would not grant the physicists the right to criticize it. His hints of a view of nature recall Schelling's, especially the eleventh lecture on "Methode des akademischen Studiums," where it is stated, among other things, that "the ideas *symbolize themselves* in things,"
[380] and that "empiricism apprehends reality quite independently of its meaning, since it is the nature of the symbol to have a life of its own in itself": natural science thus takes as absolute reality what is only symbol.

But it is not outward nature that interests Carlyle. Man is for him the real revelation, the true Shekinah, the highest symbol. No analysis can exhaust his essence. Locke and his successors sought to analyze and mechanize the mind through and through. And in particular men sought to make the whole conduct of life mechanical, to lead it back to the mechanics of the feelings of pleasure and pain, to make the principle of utility the rule, and to exclude every original, spontaneous, independent action. Bentham and utilitarianism are objects of Carlyle's sharp criticism and bitter scorn. He ascribes to the principle of utility only a negative, dissolving significance. It may be serviceable for criticizing the old order of things, but not for producing a new one. It puts a dead mechanism of interests in place of the living, personal morality that springs from the heart's yearning to fulfill its inner ideals. As eagerly as Carlyle polemicizes against what he calls the "philosophy of cause and effect," just as eagerly does he polemicize against an ethics that builds on "profit and loss" (virtue by profit and loss), and he sees in Bentham's theory a solution of the problem: "Given a world full of knaves, to show how virtue becomes the result of their united endeavors." *(On Heroes and Heroworship.* Lect. VI). What Carlyle holds against the analytical psychology and the doctrine of utility is not least that they lay too great weight on conscious reflection. Everything great comes into being and ripens in silence. Only a man who can keep silent becomes a great man; and a great deed that is done with full consciousness of its greatness is in reality a small deed. Rightly do the poets sing hymns to the night. Full, clear consciousness makes

everything small and mechanical. The highest truth can exist for man only in the form of the symbol: the symbol speaks and is silent, reveals and conceals at once.

Toward theological dogma Carlyle is no more favorably disposed than toward natural science and empirical philosophy. Only in the form of the symbol can man have the truth. But symbols grow old and wear out like clothes, and new ones must be formed in place *[381]* of the old. Philosophy, which is a struggle with habit, has in particular to struggle with the habitual and mechanized religious representations. Carlyle describes his standpoint as *natural* supernaturalism. It is divine forces that work in us and outside us. But they work in an inward, natural way. As the living garment of the Deity is constantly woven anew in outward nature, so the inner sense constantly fashions new forms for the life of the spirit. Each individual has to find his symbol and his religion, just as he has his deed to cast into the ever-flowing stream of time. "By *religion*," says Carlyle *(On Heroes and Hero-Worship.* Lect. I), "I do not understand the church-creed a man professes, the articles of faith he will sign . . ., but that which a man practically believes (often *without* declaring it to himself, and still less to others) — that which a man practically lays to heart and holds for certain concerning his inner relation to this mysterious universe." To speculate further on the essence of the Deity is useless. We have each of us enough to do in performing the deed that lies in our power. Carlyle's religious conception is nevertheless clearly enough the same as Goethe's, and as Fichte's in his later writings ("Grundzüge des gegenwärtigen Zeitalters" and "Wesen des Gelehrten"), to which he even refers explicitly. Fichte was the one among the German philosophers with whom Carlyle most sympathized. The thought of the Deity pervades everything, the spiritual and the physical world; every spiritual being is a spark from it. Whether his concept of God is theistic or pantheistic, he will not decide. John Sterling, in a letter that contains a criticism of Sartor Resartus, had objected among other things that the God spoken of in the book is no personal God; Carlyle declares it an abstract question that he cannot or will not discuss. (Life of Sterling. Part II, Chap. 2). — In his later years he declared that it was best for the time being to keep silent about religion. Its time will come again; but the dishonesty must first be done away with (Froude: Thomas Carlyle. II. 2. p. 394).

Carlyle's protest both against empirical and critical science and against theology was bound to make him lonely. He felt himself also a stranger in his time. It was a time *[382]* dissolved by skepticism and analysis, a decrepit (decadent) time, the age of mechanism. Spirit was lacking and faith was lacking. Faith is the expression of the truth of life, mysterious and indescribable like all the activities of life. But next after the end of the Roman world there has been no time so skeptical, untrue and decadent as the eighteenth

century, whose after-effects reach into the nineteenth century. It is a time in which no ideal can grow or flower. (On Heroes. Lect. V. — History of the French Revol. Vol. I, Chap. 2). The law of the three stages appears plainly here in Carlyle, who very likely has it from Fichte and the Saint-Simonians. But more strongly than any of the other writers in whom this law appears, he speaks out about the wretchedness of the time. Both in speech and in writing he complains of having come into being in so miserable a time. His mighty fantasy, his intense and easily stirred mood, and his deep melancholy, which was constantly reinforced by bodily infirmity (dyspepsia), made him dwell in particular on the dark sides of the time, which he exaggerated in comparison with those of earlier times. He found it hard to imagine that this time could be a time of transition to a better, positive time, as according to the law of the three stages it should be. He himself really assumes no gaps in development: it is a misunderstanding, he says, to believe that the phoenix burns itself up completely and then lies as a heap of ashes until a new bird comes into being in a miraculous way; no, destruction and creation go hand in hand, and the new garment is woven in the very midst of the process of the old one's dissolution (Sartor Resartus Chap. 7). But Carlyle was not himself able to find the new threads. He is convinced that none of the spiritual works of the past is lost: "Whatever mechanical science and spiritual torpor may say, man and his universe are eternally divine, and what the past has possessed of nobleness and of divine revelation can and shall not be lost." (Life of Sterling. Part I. Chap. 8). This conviction of the subsistence of values he will not, like Coleridge and Hegel, ground by a symbolic interpretation of dogmas and customs. But he wants the same reverence, the same wonder, the same
[383] energy in renunciation and in work, to which the faith of the past partly bore witness and which it partly conditioned, preserved for the conduct of life of the future. By this endeavor his standpoint is understood, his opposition at once to scientific criticism and to ecclesiastical faith. He seeks the foundation for the preservation of values much farther within personality than analysis and tradition can reach. The living garment of the divine force that never dies must be woven from within. But will these inner forces now also be capable of being awakened after the time of dissolution in which we are placed? Carlyle is thrown back and forth between faith in the subsistence of values, or of the inner forces, and the dark picture the age presented. The picture became, as it seems, even darker and darker to him. In his youth he was indignant at Bentham's philanthropy, in his old age at Gladstone's zeal for England's political and material development. In a conversation in March 1867 (see Caroline Fox's diaries) he declares that the whole thing is going downhill and means nothing to him. Like "a mourning Jeremiah" he strikes

up the old lament he had begun to chant forty years earlier, and still can find no bright spots. The more vigorous faith of his younger days weakened in his later years.

The cause of Carlyle's dark view of the time cannot be sought merely in his own melancholy and easily wounded heart. It lies also in the very unrest and doubt of the time, in the ever sharper opposition between the contending powers in the domain of the spiritual life, the ever advancing settlement between the old and the new, a settlement in which debit and credit are so extraordinarily difficult to apportion. We have seen a whole series of thinkers, right from Rousseau, Lessing and Kant, stand before this problem. Few could by their whole nature feel the sting as sharply as the Scottish peasant's son who, starting from the strict Presbyterian faith, had lived through the spiritual powers of the time and tested them in his personal life. The problem will follow the race on its way until "the third kingdom" comes — if it comes. That Carlyle saw nothing but darkness around him came from the strict demands he made on the solution of the problem. It was his indwelling ideality, his own inner light, that made the darkness around him so deep. He says somewhere that to every young and fiery mind that enters this world with *[384]* a great goal before its eyes and with a clear view of life, the world will appear as bad: for what else should its strength and heroism be used for? Were the world good, he would be of no use! It is thus human strength and ideality that make the world appear as evil: evil is the shadow of ourselves, a shadow that does not stay outside us but stretches into our own mind. Man's misfortune comes from his greatness, from the infinite that stirs in him and to which he cannot give vent in the forms of finitude. (Sartor Resartus. II, 9. — Life of Sterling. I, 5).

But the cause lay also in Carlyle's being an incorrigible Romantic. He had all Romanticism's lack of ability to discover the ideality in the constant, faithful work that adds link to link and through the small is led to the great. He has only scorn for science and for all coherent endeavors in the practical domain. And yet precisely he, who has spoken such beautiful, and often such loud, words on the significance of silence and of quiet preparation, should naturally have been able to come to understand that proud buildings can rise from stones each of which by itself is insignificant. That is the faith that lies behind the detailed investigations of science. But inquiring natures Carlyle did not understand. For men like Stuart Mill, Faraday, Charles Darwin and George Grote he has only superficial mockery to spare. Froude (II, 2. p. 259) says of him: "He fought against Darwin's doctrine of the transformation of species; but I could see that he feared it might be true!" His idealism was shy, because he lacked understanding of the significance of intellectual and practical work. His faith in the personal life was not so

firmly established that he could trust that it would surely subsist, however much analysis and criticism were applied to it. The kernel of life can surely bear a purgatory. Carlyle is right that we cannot live on criticism, but he did not see the ideal forces that could lie behind the critical work. On this point the Romantic became a philistine.

When Carlyle speaks out on what is to bear the spiritual and social life in the future,
[385] he points to what in his conviction has also borne it in the past: reverence for heroes, for the great men. The infinite force that works and symbolizes itself in everything appears above all in man, and most especially in the great man. As in Fichte, faith in the rise and significance of the chosen spirits stands in close connection with faith in the working of the Deity in all finite beings. By heroes or great men Carlyle understands the leaders and models of men, the founders of everything that the multitude of men strive to accomplish and attain. All the works we see realized around us in the human world have had their first germ in the mind of the great man. There the invisible forces work that, after the time of quiet conception and pregnancy, at last set the new into the visible world. The history of great men is therefore the soul of world history. The hero can appear as prophet, poet, statesman, but in all his shapes he represents the great, collected force of life, as opposed to all externality, partiality and limitation. Before them the deep need of reverence, the divine spark in all, breaks forth. (On this point too we are reminded in Carlyle of Fichte, especially of the latter's "Reden an die deutsche Nation"). It is the hero who finds the hidden thoughts of existence and of the times and proclaims them to others in word and deed, and thereby leads the race forward. More than ever, such heroes are needed in our time, an enlightened aristocracy to lead men. The great social gulf that the whole of history proclaims grows broader and broader in our day. Already in "Sartor Resartus," before his social writings proper, Carlyle points to it (in the chapter "Helotage," III, 4), lamenting the bodily and spiritual hunger of the many. Popular education and emigration on a great scale are the means he points to for relieving the distress. He misses new prophets and new Hengists and Horsas to set these means to work. But to go more closely into the social question was not Carlyle's bent; for that he despised philanthropists, political economists and politicians far too much. In its place he turned to history, in order to bring forward great figures and models, in
[386] case they might have an awakening and concentrating effect on the slack and divided race. It is through his historical writings above all that Carlyle has worked on wider circles. It is through these writings that his thoughts work on more and more people in our day. For many he has himself been one of the heroes he wrote about, one of those who have the ability to feel, see and express the needs of the time and the thoughts of

the time. Of his significance as a historian we shall not speak here; it belongs before another forum. Only we find Romanticism again in him in the excessive weight he lays on what is original, spontaneous, inexplicable in the great figures. He makes no real attempt to show the dependence of the great on the race — although greatness does after all show itself also in the great capacity to run into debt, because the spiritual sense and consumption of the great are greater than those of others. Only a mystical explanation of "the hero" does Carlyle give, by regarding him as an incarnation of the infinite force that works in everything. Here too the Romantic shyness of analysis and of the "philosophy of cause and effect" shows itself.

Carlyle does not, however, go as far as a later conception that finds "the goal of history" in the great men. He regards them as the great causes, and thus as the great servants. He proclaims the worship of heroes — but he sees clearly that to "be worshipped by" or to rule over a stupid mass is no heroic deed. "Hero-worship," he says *(Past and Present.* I, 6), "yes, my friends, but consisting first and foremost in our having a heroic mind ourselves. A whole world of heroes — not a world of bunglers, where no hero-king can rule — that is what we strive for."

c. Critical Philosophy

In the same year in which James Mill's "Analysis," which marked a vigorous resumption of the work of English philosophy of experience, appeared (1829), the "Edinburgh Review" brought out an essay by William Hamilton, *Philosophy of the Unconditioned,* by which the critical philosophy was transplanted onto English ground. Hamilton was born in Glasgow in 1788 and had studied there and at Oxford. When his epoch-making essay appeared, he was professor of history at the University of Edinburgh, after having lived *[387]* for a number of years as an advocate. His starting point as a philosopher lies in the Scottish school founded by Reid, which marked a reaction against Hume's skeptical and negative results. Reid and his successors sought to correct these results by more exact psychological observation, but were led in their zeal to set up a whole series of original faculties and instincts. All too often they dismissed the results of analysis and criticism with an appeal to "common sense" — an appeal that called forth Kant's apt remark that one proves one's common sense by using it, not by invoking it, and that Hume's common sense might be just as sound as Reid's. Kant appears here (in the introduction to the "Prolegomena") as advocate for Hume, to whom he owed so much. But he sought in his own way to refute Hume, and thereby came to have the same opponent as Reid and his disciples. William Hamilton now seeks to combine the doctrines of Reid and

Kant. In the essay mentioned and in a series of essays following upon it, which were likewise published in the "Edinburgh Review," and among which *Philosophy of Perception* (1830) is especially to be noted, he developed his peculiar views, which were of no small interest in epistemological and religious-philosophical respects. These essays he later published collected under the title *Discussions on Philosophy.* In 1836 he became professor of philosophy at Edinburgh, and there, until his death (1856), he gave well-attended lectures on psychology and logic, which became of great importance for philosophical development in Scotland. The lectures were later published under the titles *Lectures on Metaphysics* (2 vols.) and *Lectures on Logic* (2 vols.). Hamilton was an acute thinker, an eager seeker after truth, akin to Lessing in spite of his Scottish gravity; he was also in possession of comprehensive learning in the history of philosophy.

In his essay on the philosophy of the unconditioned, which is a criticism of Schelling and Cousin, and in part also of Kant, Hamilton seeks to show that only the conditioned and limited can be an object of knowledge, and that the attempts to set up a philosophy
[388] of the unconditioned cannot be carried through. But he gives not only criticism; he gives the outline of a whole epistemology, which on particular points he worked out further in his later writings. His main proposition is: *to think is to set conditions* (to think is to condition). By this he means that everything we are to be able to apprehend and comprehend we determine by its relation to something else that conditions and limits it. An absolute whole we cannot apprehend; every whole is for us always part of a greater whole. An absolute part we can just as little apprehend; every part can in turn be thought of as divided and is thus a whole. An infinite whole we cannot apprehend either; for an infinite time would be required to go through all its parts. Thus neither the absolutely limited nor the absolutely unlimited can be an object of knowledge. Our knowledge concerns the conditionally limited. The philosophy of the unconditioned, which credits itself with the ability to construct a scientific concept of the unconditioned as that which is to be at once absolute totality and infinity, sins against the nature of our knowledge. In a later appendix to the essay (Discussions p. 577 ff: *Conditions of the Thinkable*) it is developed how all knowledge consists in a judgment that connects two terms: therewith the relativity of all thought is given. All knowledge rests, moreover, on a relation between subject and object; beyond this opposition we cannot get, unless with Schelling we are willing to sink into the absolute unity in a mystical way, whereby all knowledge is excluded. Another relation that plays an essential role in our knowledge is the relation between thing and property: the thing and the property are known only in relation to each other. In the closer determination of things we determine them in time

(protensively) or in space (extensively) or with respect to the degree of their properties (intensively); but all these determinations are relative: the one determination of time, space or degree presupposes the other. The concept of cause too Hamilton brings under what one might call the epistemological law of relation *(law of relativity).* He takes it as a limitation of our knowledge that we understand things only by finding their causes. We are not able to think an unconditioned beginning. When a phenomenon begins, this *[389]* can for us be only apparent. Just as little can we think an unconditioned cessation. The disappearance of a phenomenon too can be only apparent for us. When we seek the conditions of the phenomenon, this means that we connect it with other phenomena, regard it as a term in a relation — because of our impotence to think an unconditioned! Every existence is for our knowledge relative, is bound to a preceding existence. As soon as something new is added to what exists that is more than a transformation of the earlier into a new form, and as soon as something is taken from what exists that does not come again under a new form, we stand before the incomprehensible. Cause and effect cover each other and can be found again in each other. The concept of cause is thus not, as Reid and Kant each in his own way assumed, an independent, isolated concept. It is a particular form of our consciousness's general way of working, a particular example of the fundamental law that our knowledge by its nature can grasp only the conditionally limited.

By this epistemological law of relation Hamilton has cast light on an essential side of our knowledge. In place of Reid's appeal to the instincts of common sense and in place of Kant's scholastic table of categories he has put an analysis of the fundamental forms of the knowing consciousness and has demonstrated a common characteristic mark for them. Psychology is for him the fundamental philosophical science. Consciousness is for the philosopher what the Bible is for the theologian, and the original testimonies of consciousness must be taken as true, since they lie at the foundation of all our knowledge. On this point Hamilton sometimes speaks just like Reid. "The root of our nature cannot be a lie" — therefore the assumption of an opposition between consciousness and its object, which immediate consciousness according to Hamilton entertains, must be true. This is Hamilton's simple solution of the problem of the existence of a world independent of consciousness. The problem is thereby hardly allowed to come forward. "Natural realism" has, for Hamilton, right on its side.

On the particular psychological and logical questions that Hamilton has treated with great acuteness and learning we cannot enter here[91]). We must, on the other hand, dwell *[390]* on the consequences Hamilton draws from the limitation of knowledge. The value of

philosophy cannot for him, naturally, depend on its bringing us absolute truth. Its value depends altogether not on the finished results, but on the constant release of mental energy into activity and striving. By speculation our mental forces are exercised. Lessing is right when he places the eternal search above the possession of truth. A waking error is better than a sleeping truth. Philosophy does not lose its value because it ends with an acknowledgment of our ignorance grounded on insight into the conditions of our knowledge (docta ignorantia, as Hamilton calls it with Cusanus). Before it reaches this its conclusion, it has developed and exercised the noblest mental forces.

This ignorance concerning the absolute essence of existence rests on the fact that our thought ends in a dilemma whose horns are equally inconceivable. Were we to know the unconditioned, it would have to stand before us either as an unconditionally limited, an absolute totality, or else as an unconditionally unlimited, an infinite in time, space and degree; but neither of the two can be grasped by us, because both conflict with the law of relativity. Hamilton does not think, however, that we must or can remain stuck in this dilemma. For by virtue of the logical principle of the excluded middle between two mutually contradictory possibilities, one of the two assumptions *must* be true. (The logical principles thus have for Hamilton not — like the law of relation — validity only for our knowledge, but are laws of things, not merely laws of thought. See on this Discussions p. 577 f.). There is then here not merely room for a faith, but necessity for a faith. We have learned that our knowledge is no measure of existence, and we have learned that a choice must be made. The idea of an unconditioned is not, as Kant thought, a positive idea that has its natural root in the nature of our knowledge; it signifies on the contrary the negation of all knowledge. While Coleridge held to Kant's doctrine of ideas, Hamilton parts from Kant precisely on this point. He finds the possibility and the necessity of a faith given in this, that, as regards the idea of the unconditioned, we
[391] stand before two possibilities that are both inconceivable, and between which we must nevertheless choose with logical necessity.

The choice is determined, according to Hamilton, by practical, moral motives. We need an unconditioned being that can uphold and secure the subsistence of our mind. And in the representation of this being we lean on the analogy with our own being. Hamilton does not agree with Kant in his criticism of spiritualistic psychology. Consciousness, he says (Lectures on Metaphysics. I. p. 158), is indeed the condition of all inner phenomena; but it is itself only a phenomenon; there must lie something behind it whose property it is, and this must be something different from what lies behind material phenomena. The same relation as there is between our mind and our body we assume — by an analogy —

between the unconditioned spiritual being we believe in and the world of relativities that knowledge presents to us. Through this analogy Hamilton makes the transition from philosophy to theology. The end of philosophy is the beginning of theology. But he says at the same time that no difficulty arises in theology that does not already appear in philosophy. And his own doctrine confirms it: for that unconditioned being which is to be the object of faith is set up by an inference from analogy and is thus to *stand in relation to* the world as the soul to the body; but then it becomes, after all, conditioned, relative! Hamilton has not shown the possibility of escaping, by passing over to faith, the epistemological difficulties he himself has so energetically demonstrated. He agrees with Fichte that a God who could be comprehended would be a conditioned being; but a God who is placed in relation to a world would, even if he could not be comprehended, be a conditioned being, and if he is to be maintained as an unconditioned being, his concept would contain a self-contradiction. — As regards the content of faith, there is confirmed here in Hamilton what we have been able to establish in a whole series of thinkers from Leibniz's time on: that every attempt at a speculative or religious world-view must, consciously or unconsciously, lean on an inference from analogy. — Hamilton's whole standpoint, which is characterized by his psychological starting point, his critical analysis of the conditions and limits of knowledge, at once with an adoption and a correction *[392]* of Kant's line of thought, and finally his philosophy of religion grounded on analogy, recalls such German inquirers as Fries and Beneke.

Two years after Hamilton's death his disciple Henry Mansel gave a series of lectures at Oxford, which are published under the title *Limits of Religious Thought* (Danish translation by N. Teisen: "Grænserne for den religiøse Tænkning." Odense 1888). Here he draws from Hamilton's doctrine the consequence that a scientifically grounded theology is impossible, because our knowledge cannot reach the unconditioned. But at the same time he maintains that no objections can be raised from scientific thought against revelation; only if we had an absolute knowledge, a philosophy of the infinite, could revelation be refuted. And in agreement with Joseph Butler's "Analogy" — a work that has altogether played a great role in religious discussion in England in our century — he maintains that the difficulties and contradictions that can be demonstrated in the theological assumptions occur within every attempt at a conclusive world-view, even if it moves on the foundation of nature and reason. From this he concludes that the difficulties arise — not from revelation, but from the limitation of reason. We must then believe what we cannot comprehend. It is our duty to assume the personality of God, although it is for us self-contradictory that an unconditioned being should be a personality, since

personality presupposes opposition and limitation. We must believe in the dogma of the atonement and in the dogma of eternal punishment, even if these dogmas conflict with our notions of love and justice. What is love and justice in our eyes is perhaps not so in God's! We see only the part, not the whole. If our horizon were widened, we would apprehend everything differently! Just as Hamilton had already taught that it is owing to an imperfection of our nature that we apprehend things as causes and effects, so Mansel teaches that human ethics hangs together with limited human nature, for which reason no inference can be drawn from the ethics of man to the ethics of the
[393] Deity. For man it is, e.g., a duty to forgive, because he needs a bridle on his selfishness; but this grounding of the necessity of forgiveness cannot hold for God!! — Both human thought and human conscience are thus here cut off from all influence on the valuation of religious representations. Without rudder and without compass, according to Mansel, man is to sail the ocean of religion. Whether the theological voyage will go better in this way is perhaps a question — until such time as the superhuman thought and the superhuman conscience that are to escape the logical and ethical difficulties have been demonstrated. Mansel's philosophy of religion was at first received with great approval as a good weapon against rationalism and speculation. But people seem to have come back somewhat from the first enthusiasm[92]), having discovered that it was a weapon that easily wounded the one who used it. It is also clear, after all, that however much one may wish to inculcate the necessity and the possibility of leaping from thought and conscience into faith, man cannot, even by the boldest salto mortale, once and for all leap over his own shadow. —

Besides Hamilton and his disciples there must still be named an inquirer who in a fruitful way brought Kant's epistemology to bear in England. William Whewell (1795—1866), who worked at the University of Cambridge first as a natural scientist (mineralogist), later as a philosopher, published in 1837 his work *History of the Inductive Sciences,* which some years later was followed by *Philosophy of the Inductive Sciences, founded upon their History* (1840). In these works he sought by the historical road to establish the correctness of Kant's fundamental thoughts, which, to be sure, he did not always apprehend with sharpness and clarity. He stands on the ground of the English school in maintaining that all knowledge develops on the foundation of experience. Both the human sciences and natural science are inductive sciences. But induction does not mean merely the collecting and juxtaposing of facts; it means the grouping of the facts under a guiding thought, their reduction under a general law. And the history of the inductive sciences now proves that this grouping and reduction become possible only

when there are already at work in the inquirer ideas and points of view that can make *[394]* possible the discovery of the law of the facts' connection. The facts must be connected by a mental act whose possibility must lie in the nature of the mind. Every inquirer therefore starts from presuppositions that cannot be derived from the particular facts each taken by itself. By a closer investigation of the way in which inquirers such as Kepler and Newton came to their discoveries, this conception of the method of induction is further illuminated. Kepler had the concept of an ellipse beforehand, just as Newton had the concept of attraction, and only because these concepts were present in advance did their discoveries become possible. It is not Whewell's opinion that such concepts lie fully finished in the mind, but that no discovery can be made without the cooperation of a mental activity that follows its own laws. Even if the special ideas (as, e.g., of ellipse and of attraction) have themselves developed under the influence of experience, we nevertheless come back at last to fundamental concepts that are the expression of nothing other than the laws of the cognitive faculty itself, which assert themselves in all experience, from the simplest sense perception to the most comprehensive induction. By attending to the way in which our knowledge works, we find a series of forms or fundamental concepts — the concepts of time and space as the foundation of mathematics, the concept of cause as the foundation of the mechanical sciences, the concept of purpose as the foundation of the organic sciences, and the concept of duty as the foundation of ethics — concepts that according to Whewell cannot be reduced to simpler forms or derived from observation. He contents himself with setting them up in a series without subjecting them to a closer examination, and his conception thereby acquires a likeness to Reid's, whereas Hamilton, as we have seen, made an interesting attempt to demonstrate the various fundamental concepts as various expressions of the epistemological law of relation. Nor does he apprehend the relation between philosophy and theology with the critical sharpness that distinguishes Hamilton. His significance lies in the preparatory work he has contributed toward a theory of induction. Here he has in a significant way illuminated the method of induction as a *method of discovery,* whereas — as his great *[395]* opponent Stuart Mill showed — the method of induction as a *method of proof* did not come into its own with him. For one who occupied himself particularly with the history of the inductive sciences, the point of view so energetically stressed by Whewell was bound, quite naturally, to come especially into the foreground.

2. John Stuart Mill

a. Biography and Characterization

John Stuart Mill, James Mill's son, was born on 20 May 1806 in London. He is an example of early intellectual development, but he also developed under a high pressure of intellectual influence that a less vigorous and original nature could hardly have endured, and that left lasting consequences in him too, both in his intellectual tendency and for his bodily health. His father taught him himself. At three years old he began Greek and soon after arithmetic; at the same time he naturally learned English reading and grammar. Latin he began when he was eight. History he learned on his own from detailed works, and then gave his father an account, on walks, of what he had read. After a great part of Greek and Roman literature had been gone through, he began logic when he was twelve years old, and here too what he had read on his own was gone through and discussed thoroughly on walks. Then followed reading in political economy and a careful study of Demosthenes and Plato, with particular regard to argument and method. Very early the boy had to teach younger brothers and sisters, and thereby got practice in mastering and applying what he had learned. There was no cramming. The father endeavored that understanding should keep pace step by step with instruction, indeed preferably go ahead of it: "What could be found out by thinking," says Stuart Mill in his autobiography *(Levned.* Danish trans. by Vilh. Arntzen, p. 34), "my father never told me until I had
[396] exhausted my strength in finding it out myself." — Mill recommends the way in which he himself was taught for imitation. He thinks that what he achieved here any boy or girl of fairly good abilities and a healthy nature could achieve. Even if what was demanded often lay beyond the child's ability, Mill nevertheless holds it pedagogically right to make such demands: "A pupil from whom nothing is ever demanded that he cannot do never does what he can do!" He would never admit that this upbringing, which, as he says, had given him a quarter of a century's start over his contemporaries[93]), had exercised any questionable influence on his development and on his health. And yet there is hardly any doubt that a mental crisis he later went through, as well as a nervous debility from which he suffered all his life from his thirtieth year on, were in great part consequences of the one-sided and forced development of his childhood years. That at least was the view of his friends (see A. BAIN: *John Stuart Mill.* A Criticism. London 1882). Mill would never admit that work could be harmful. Although an eager philosopher of experience, he was nevertheless so spiritualistic that he denied all essential influence of physiological dispositions and states. Therefore he not only believed that mental development could be hastened without harm to health, but he also did not acknowledge

any original difference between individuals. It was always a foundation of the hope he had for the future of humanity that all qualities of character were effects of upbringing and of outward, social conditions, so that by pedagogical and social reforms one would be able to alter human characters to indeterminable degrees and in the same direction. In this respect he constantly paid homage to the doctrine of Helvetius, of which his father and Bentham were such eager adherents. — Apart from the physical consequences of the forced upbringing, it suffered also from the defect that it was only the understanding, not feeling and fantasy, that was developed. The whole spontaneous unfolding of moods and images was checked under this strictly rationalistic upbringing. Whereas in his old age, in his autobiography, he recommended the way in which he had been brought up, he had at an earlier stage himself felt its drawbacks. In a conversation of 1840, which the *[397]* Quaker Caroline Fox records in her diaries, he says expressly that he will not recommend the early intensive strain in the upbringing of all children, since it easily comes at the expense of the liveliness of childhood and favors reflection at the cost of activity. "I have never been a boy," he said, "have never played cricket; it is better after all to let nature go its own ways."

Another point on which Mill's autobiography must also be corrected by what has come to light by other roads about his development is what he relates about his religious notions in childhood. He relates in the autobiography that he received no religious upbringing at all, since his father had long since given up all religious faith. Supported by definite recollections on the part of members of the family, Bain (in his biography of James Mill) has shown that Stuart Mill remembered wrongly as regards the definitive change in his father's religious standpoint. In Stuart Mill's childhood his father still went to church, and the son with him, and the rest of the family continued to do so later too. Definite utterances of Stuart Mill as a boy are preserved that show that he read the Bible with enthusiasm. But the father began very early to discuss religious questions with the boy. He taught him not to break off a line of thought or let go of a problem out of a feeling of religious devotion before he had found out whether it could be solved or not. He impressed upon him what was for him the decisive difficulty not only in the faith of the church but also in faith in a good and almighty Creator: the impossibility of reconciling the experience of the physical and moral evil in the world with such a faith. It was Butler's "Analogy" that had led James Mill to the purely negative standpoint he took in religious respects in his later years. Not for the first and only time has a work of defense here had the opposite effect from that intended — here, however, by its depth and consistency. The line of thought into which Stuart Mill was thus early led

determined his later position on the religious problem, even when, after a mental crisis, he took a more positive standpoint, and even in his posthumous essays in the philosophy
[398] of religion the after-effects of the father's conversations with the little boy are to be felt.

A year of his early youth (1820—21) Stuart Mill spent in the south of France with a brother of Bentham's, who had an estate there. He ascribes to this stay a great influence on his development. Above all there was awakened in him a love of France and an interest in French literature and politics to which he remained faithful all his life. The French historical writers later, at an important point in his development, gained great influence on him, and he found in the French a capacity for enthusiasm and for self-renewal that he often missed in his own countrymen. After his return home he pursued legal studies and studied Bentham's writings. In Bentham he found an idea that could bring coherence into all notions and endeavors. He felt himself a new man when he had made a penetrating acquaintance with Bentham's "principle of utility." He says of this in the autobiography (Levned. p. 68 f.): "This principle gave unity to my notions of things. I now had a conviction, a faith, a doctrine, a philosophy, indeed a religion in the best sense of the word, the spreading and inculcating of which was to be the chief task of my life. I pictured to myself a great number of changes in the condition of men that were to be carried through by this doctrine . . . . . The prospect of progress that Bentham opened for me was sufficiently wide and brilliant to make my life bright and to give my endeavors a definite form."

With zeal Mill joined the band of young men who strove to work for the philosophical and political ideas of Bentham and James Mill. Hartley's psychology, Malthus's political economy and Bentham's moral philosophy were the foundation on which they worked. The "Westminster Review" was their organ, in which Stuart Mill's first larger essays were printed, and what they fought for first and foremost was freedom of expression and the extension of the suffrage. They hoped that if only full access were given to enlightenment and knowledge, all social problems would be solved. By prudent self-control the workers would be able to prevent the strong increase of population, which is
[399] a chief factor in pressing wages down. Class rule would be broken through the extension of the suffrage. Human notions would be altered, and through them in turn human character, by virtue of Hartley's doctrine of inseparable associations. Despite all the zeal with which these ideals and prospects for the future were held fast, Mill nevertheless states that the expression "thinking-machine," which has often wrongly been applied to Bentham's adherents, really fitted him in this period. As he expected the regeneration of the human race not through disinterested love of justice but from "the enlightening

influence of the cultivated understanding upon the selfish feelings," so in himself it was essentially the understanding that had the floor. "My zeal," he says in his autobiography, "was in this period of my life not much else than zeal for speculative opinions. It did not have its root in any genuine benevolence or sympathy with the human race, though these qualities had their due place in my ethical conception. Nor did it stand in connection with any ideal elevation. And yet my fantasy was very receptive to this feeling; but there was at that time a stoppage in its natural nourishment, poetic culture, while there was a superabundance of what stood in conflict with it, logic and analysis. Add to this that my father's teaching had a tendency to undervalue feeling." (Levned p. 112). It was the rationalism of the eighteenth century that here, through Stuart Mill's upbringing, continued its effect far into the new century, nourished and goaded by the constant struggle against prejudices and sentimental declamations.

There came upon Mill, when he was twenty, a mental crisis that on essential points gave him another view of life. The foundation of his whole mental life suddenly collapsed for him. He fell into a state of dejection in which he asked himself whether he would now really feel happy if everything he expected of intellectual and political progress were realized — and he had to answer No. Life now became heavy and desolate to him. He performed his work purely mechanically, but it seemed to him as if the source of mental life in him had dried up. Constant analysis had made everything cold for him. *[400]* The psychology by which he had been brought up had indeed seen that the feelings are influenced and altered by the development of representations, but had not seen that deep and strong feelings can be produced and preserved in this way only when they get sufficient nourishment through spontaneous experiences and through constantly fresh and immediate influence. Now he felt like one who is stranded at the beginning of his voyage, with a well-equipped ship and a rudder — but without a sail! — It was a reaction against the violent strain of childhood and early youth and against the one-sided development of the understanding. And it was at the same time a reaction against the line of thought of the eighteenth century, a reaction that was bound quite naturally to come forth in a young man who had been brought up in the thoughts of the old time and stood in the midst of those of the new, even if he had not yet let them come in over his threshold. — The mental crisis was resolved for Mill through new experiences, new ideas and a new human relationship. — He was deeply moved by a touching passage in a biography and so learned that the source of feeling was not exhausted in him: "I was no longer hopeless; I felt that I was neither a stick nor a stone." Thereby he gained new courage. He immersed himself in the poets, of whom Wordsworth above all came to have

great significance for him. He learned what indispensable spiritual nourishment poetry can give; hitherto he had been inclined, with Bentham, to fear that it would impart false notions to men. He learned to give the unconscious and the spontaneous their due in life, and found out how important it is not to stare at one's own happiness as the final purpose, but to set oneself a higher purpose: then happiness will be found in the endeavor to realize it. "Only those are happy who have their gaze directed on something other than their own happiness . . . . Ask yourself whether you are happy — and you will cease to be so!" — To this were added new studies, especially of the French historians (Guizot, Michelet, Tocqueville), who taught him that institutions and laws develop according to natural laws; they correspond to the stage of development of society and of views, and their perfection must be regarded in relation to this stage, and only to a certain degree
[401] can the order that progress follows be altered by arbitrary intervention. In particular the writings of Comte and the Saint-Simonians made a strong impression on him. He came to know the difference between critical and organic periods, and Carlyle's works acted on him as a mighty poetry of personality. He saw that one could not, as he had hitherto believed with his father and Bentham, treat human affairs purely deductively, or ascribe to outward conditions a wholly determining influence. He came — which was after all also the most consistent course for an adherent of the philosophy of experience — to give experience its full due. In his religious views too this crisis and its resolution intervened, although he does not speak of this point more closely in the autobiography, where he says only that he now saw the significance of the development of the inner man. The foundation of the religious views he developed in his posthumous essays was laid in these years, as will later be shown more closely. By Bentham's friends he was regarded as an apostate, while in other circles he was received as one who had broken with sectarian prejudices in order to grasp the truth. In a remark recorded by Caroline Fox, Mill's friend John Sterling says of him that he has sacrificed the position of undisputed leader of a mighty party for the higher honor of being a private in the army of truth, and Sterling hopes that Mill's books will work to give the feeling of reverence the place in life that so many have denied it. — But what a stirring of feeling and self-forgetfulness had begun, and what a widened historical view and religious sense had developed, gained its proper fullness in his life only through his acquaintance with the woman who later became his wife, and to whom he himself ascribes a wholly overwhelming influence on his mental development, and whom he regards as a supremely superior mind. By his overflowing eulogies of her he set his friends, and — especially by his autobiography — his readers, a riddle, since he seems to ascribe to her everything he has accomplished. The secret is no

doubt that Eros, here as so often, has been the great teacher, who has given an eye for ideals and a sense for intimate, personal qualities, and has set feeling in a motion that does not stop at the object that has awakened it. This explains why those nearest to Mill *[402]* and his friends could see nothing extraordinary in the woman who for him was a genius of the highest rank. One of his brothers, who knew her well, used to say that she was an able and remarkable woman, "but nothing like what John took her for." Where Mill gives a closer account of what he owes her, he exempts his purely scientific works (his Logic and the theoretical parts of the Political Economy) from her influence, whereas he characterizes her as the one who had the freest and boldest eye for the ideal and for what the future might bring, while at the same time she possessed a great sense for concrete, factual conditions and a lively ability to make them vivid, just as she also had a sure apprehension of the relative significance of different points of view[94]). One understands that Mill here found something he had missed all his life, and to which his upbringing had made him less receptive, until it came to meet him in personal reality. He found his Beatrice, as Comte found his.

Stuart Mill entered practical activity early, being employed already at seventeen in the service of the East India Company, where he gradually rose to fill the high post his father had held at the last. For a time he directed the education department, later the Indian government's correspondence with the Indian princes and with foreign states, and at last, as Chief Examiner of the Indian Correspondance, had the superintendence of the whole administration. His activity in these positions has won great recognition from competent quarters, and on the dissolution of the India Company he was offered a seat in the new "Indian Council," which, however, he declined for reasons of health. In his leisure he continued his studies. In his younger days he was an eager member of a debating club in which Bentham's disciples discussed philosophical, literary and political questions with young men from other circles, partly Tory lawyers, partly Coleridgians. As an author he worked for a long time only in reviews and newspapers. His most important essays he collected in a series of volumes under the name *Dissertations and* *[403]* *Discussions.* His definitive views on philosophical questions were founded around 1830. Of influences from a later time must especially be named those he received from Comte, with whom he stood for a long time in lively correspondence, and from Carlyle, whom he admired as a prophet but could not follow as a philosopher. — In those essays one finds the astonishing balance and many-sidedness that distinguish Mill's line of thought. After the strict school of clear thinking he had gone through in his childhood had been supplemented by richer experiences, he was able to show appreciation and to cast light

in many directions.

The reaction against the line of thought of the eighteenth century, and in particular against the ways of seeing that derived from Bentham and James Mill, acted more strongly on Mill's social, literary and religious conception than on his general philosophical (epistemological and moral-philosophical) standpoint. In Caroline Fox's diaries we meet him in the circle of eminent men who frequented the house of the gifted Quaker family in Falmouth[95]). Mill had gone out to Cornwall to nurse a brother who was ill with consumption. The utterances and conversations of his that are reported bear an intimate, personal stamp, and bear witness to a deep seriousness about life and its tasks. He speaks much in a Carlylean spirit, and in that circle one hardly had the feeling of being in the company of one of the most radical men of the century. Later, when the book "On Liberty" appeared, the eager listener felt deeply indignant (as, for that matter, she had also felt at Carlyle's "Life of Sterling"). For Mill there was no contradiction between the philosophy of personality into which he had been led and the strictly rational consideration of knowledge and life that he applied in his inquiry. His significance rests in an essential respect precisely on his having freed the philosophy of personality from the Romantic character, shy of knowledge, that it had in Carlyle. For him personal inwardness and enthusiasm were not incompatible with an untiring search for grounds and causal explanation. He learned — after his crisis — that one can have a great view of life without therefore fearing or despising thought. That Carlyle could never get into his
[404] head. "Poor fellow," he said of Mill, " he has had to work his way out of Benthamism, and the emotions and sufferings he has gone through have helped him to thoughts that never came into Bentham's head. But he is all the same far too eager to prove everything. If John Mill got to heaven, he would hardly be content until he had found out how everything was arranged. For my part I should not care much about the machinery of the place, but should quite give up the question whether there was to be an artisan class or a working class among the angels." *(Memories of old Friends from the Journals of Caroline Fox.* 6 June 1842). In humorous form Carlyle has here aptly indicated the difference between himself and Stuart Mill. Yet it should perhaps be added that if heaven is not reserved especially for Romantics, insight into and work with the mechanism of life would be necessary to prepare the way to heaven. And besides, heaven must be assumed to furnish satisfaction for the intellectual need as well as for every other personal need. What Carlyle reproaches Mill with, he could just as well reproach Socrates with, who imagined, after all, that in the other world he would spend the time as here in testing and inquiring. — A comparison between the way in which Stuart Mill speaks of Carlyle

in his autobiography and the way in which Carlyle in conversation and letters (see e.g. Froude II, 1. p. 74; II, 2. p. 419) spoke of Mill, even just after the latter's death, establishes which of the two men had the better ability and the more honest will to understand and to appreciate. Carlyle, who himself always had the valve open and loudly preached the gospel of silence, could not reconcile himself to the subdued quality in Mill's personality. And when he had had to give up the hope of getting an unconditional adherent in Mill, he had almost nothing but scorn and mockery to spare for him. —

Stuart Mill attached an extraordinarily great practical significance to the purely theoretical debate carried on in modern philosophy between empiricism and speculation. All false assumptions and tendencies in the moral, religious and social domain would — he thinks — be invincible if one did not refute the claim that truths can be won by immediate intuition, by the road of pure thought, independently of experience and observation. *[405]* This claim in reality makes an assumption its own proof! Never has a better means been devised for securing the preservation of all ingrown prejudices! In opposition to this Mill sets the derivation of all knowledge from experience and the explanation of all intellectual and moral qualities by means of the laws of the association of representations. Here there are tasks enough to solve, whereas the intuitive philosophy, which marks the reaction of the nineteenth century against the eighteenth, favors human indolence while at the same time being a bulwark for every kind of conservative prejudice. It was this line of thought (see Levned. p. 226 f. 277 f) that lay at the foundation of the working out of his two chief philosophical works *System of Logic* (1843) and *Examination of Sir William Hamilton's Philosophy* (1865). — In his Logic he has for the first time given a theory of induction, systematized the methods of experience, just as Aristotle systematized the methods in which deductive reasoning moves. Aristotle had as his predecessors the Greek philosophers and sophists, and as foundation the eager discussions that were held in Athens, especially in the Socratic schools. Stuart Mill built on the history of modern natural science through three centuries and on an analysis of the forms in which its work has moved. His work marks the most thoroughgoing presentation of empiricism in epistemology that has been given, just as his father in his "Analysis" had carried empiricism through in the psychological domain. There appears in the foundation of Mill's Logic a clear and fateful influence of his father's psychological theory, a theory that he later (partly in the notes to a new edition of the "Analysis," partly in the work on Hamilton's philosophy) gives up, without becoming aware of what effect this giving up really has on epistemology. His great love of truth, which shows itself in his untiring readiness to go into objections and let them have their full effect, has also opened his

eyes to truths for which at the beginning of his career he did not yet have an eye. He began writing his Logic in 1830, and still in the last edition of the work on Hamilton's
[406] philosophy, at the end of his life, he makes modifications in his conception that arose from his uninterrupted work with the problems. Whereas he starts from James Mill's conception of consciousness as a series or heap of independent elements, which are brought into connection with one another in a purely external and really inexplicable way according to the laws of association, he ends by acknowledging the unity and connection in consciousness as a fundamental psychological fact. "The uniting principle," which Hume wished to push aside, Mill has now acknowledged as a cornerstone. This marks a break with the older English school whose range Mill was not aware of.

Besides the purely philosophical writings named (to which must still be added an interesting book on Auguste Comte), there is a whole series of significant writings of ethical, social and political content. To these belong his *Principles of Political Economy* (1848), in which work he conceives political economy as a part of the whole theory of society and carries through a distinction between the laws of production and the laws of distribution, a distinction that makes it possible for him to acknowledge what is justified in the socialist systems. (It was, however, only in the second edition of the book, 1849, that this acknowledgment properly came forward.) Whereas he had hitherto, in the footsteps of his father and Bentham, fought against class rule and for personal freedom, it was now clear to him that behind political emancipation there stood a far more serious question, the social one. The Saint-Simonians had already opened his eyes to it. That he did not lose sight of the significance of personal freedom is shown by the book *Om Friheden* (1859). In the essay *Utilitarianismen* (translated by G. Brandes under the title: Moral, grundet paa Nytte- eller Lykkeprincipet) he defends the ethical doctrine of utility against its opponents, and in the work *Om repræsentativ Regering* (1860) he gives his political theory.

Mill had a seat in Parliament 1865—1868 and won recognition for his clarity and expert knowledge, although he spoke above all on the most unpopular questions. Gladstone has in a letter (of 19 Dec. 1888, cited in GOMPERZ: *John Stuart Mill.* Ein Nachruf. Wien 1889. p. 46) expressed himself on Mill's parliamentary activity in the following way: "Mr.
[407] Mill's eminent gifts of mind were well known to us all before he entered Parliament. What his conduct there revealed to us, to me at least, was the uncommon nobility of his character. I used at that time in conversation to call him the saint of rationalism . . . . On the other hand he was indeed a philosopher, but not a man of fixed ideas. He united the sound sense and practical tact of the statesman with the lofty independence of a solitary

thinker. I need not say that I greeted his appearance with joy and deeply regretted his disappearance, and that in the interest of the whole House of Commons. He was a boon to us all. In every party, within every political tendency, such men are, I must confess with pain, rare." — The reason Mill was not re-elected was partly that his religious views were used against him, partly that his radical political ideas went too far for the class of voters who had elected him the first time.

During a stay in Avignon Stuart Mill died on 5 May 1873. In him passed away one of the greatest, most honest and noblest minds of the nineteenth century. The course of his life, as he himself has depicted it, is a source of instruction for everyone who strives, just as his writings cast light on some of the most important subjects of human thought[96]).

b. Inductive Logic

Mill's chief strength as a thinker lies in his untiring discussion, his unwearied turning and twisting of problems in order to reach their ultimate presuppositions. In the later editions of the *System of Logic,* in which account is taken of the objections raised against his theories, his exposition takes on a dialogical character. Just as the author of a dialogue, when it truly deserves its name, endeavors to let the different standpoints come forward in as clear and characteristic a form as possible, so it is also Mill's endeavor to do full justice to the views and objections of his opponents; he does not really regard them as opponents, but as fellow workers. That his understanding of other standpoints *[408]* had its limit goes without saying; it was the same limit that his personality and his historical starting point as a thinker set to his inquiry altogether. This limit stands out to a critical examination all the more clearly as the psychological foundation of his whole epistemology shifts under him, without his seeming to have become aware of it. Whoever would assess Mill must first and foremost see him as the great seeker. His merit in philosophy does not stand or fall with the answer to the question whether he succeeded in carrying through empiricism in so absolute a form as even Hume had not yet conceived it.

It is the logic of experience that Mill sets over against the logic of pure thinking, partly in opposition to it, partly as an extension of it. Pure thinking cannot extend our knowledge, but only lead us to preserve agreement among our representations. New truths, however, are obtained only by observation and experience, and the question then arises by what paths, on the basis of observation, we can prove the new truths. It is on the proof, not on discovery itself, that Mill lays the chief weight. His guiding interest is to let all assumptions pass through a purgatory before they enter the heaven of truth. How

assumptions arise in the first place interests him less; it is the justification that matters. The significance and practical worth of philosophy rest for Mill, as for Carlyle, on its being a struggle with habit. The assumptions which have formed in the individual by involuntary association of representations, or which he has received by tradition, must be sifted and tested, so that one can exclude prejudices that hinder progress. For his great hope of the advance of the human race the consistent tracing back of all assumptions to experience stood as an essential support.

Apart from such cases in which a general proposition is laid down by an authority, as with theological and juridical propositions, every general proposition, Mill maintains, is only a summing up of a series of particular observations. Deductive logic, which begins
[409] with general propositions, therefore presupposes an inductive logic, which shows how we arrive at these general propositions from the particular observations. Pure thinking always presupposes experience. If, e.g., I reason thus: "All men are mortal — the Duke of Wellington [who was still living when Mill wrote his Logic] is a man — therefore the Duke of Wellington is mortal", — then it is clear that, in order to have the right to set up the first proposition, about the mortality of *all* men, I must really be certain that the Duke of Wellington too will die. In reality, says Mill, I do not infer from the death of *all* men to Wellington's death, but I infer from a great series of experiences of the death of particular men to the death of this particular man, Wellington. If I really knew that *all* men were mortal, I should not need to infer at all, since Wellington's mortality would be included in the general mortality. My inference is then really this: John is mortal, Thomas is mortal, etc., etc. — so Wellington is presumably mortal too. At the bottom of every syllogism whose major premise is not established by authority there lies, then, in reality an *inference from particular to particular.* This is the form of inference from which both induction and deduction spring, — first induction, and then deduction. The beginning of the whole process of knowledge consists in two phenomena (man and death, e.g.) occurring together for me. The next time the first of them occurs, the expectation of the second will then occur. If this expectation is confirmed, I sum up all these experiences in a *general* proposition, that is to say, in a proposition which gives me *all my experiences* in brief compendium. More it cannot give me, if I am not to make an unjustified generalization. All inference is therefore from particular (or particulars) to particular (from particulars to particulars). This inference occurs already in children. When the child sees the flame on which it has burned itself before, it draws its hand back, not because it has set up any general proposition, but because the sight of the fire immediately calls up the representation of pain. Animals too can infer in this way, for

not only the burned child, but the burned dog too shuns the fire. (Logic. Book II. Chap. 3 § 3). *[410]*

Mill of course sees clearly that this immediate transition from an observation to a representation or an expectation is merely an association, whose justification has first to be tested. It is an association of the kind to which James Mill sought to reduce all association, association "by contiguity": we have often seen A and B together, and now expect B because A occurs again. But can *proof* now be given of the validity of such an expectation? For what is at issue in logic is not what we acknowledge as valid, but what we *ought* to acknowledge as valid (Evidence is not that which the mind does or must yield to, but that which it ought to yield to. Logic. Book III. Chap. 21. § 1). By what right, then, do we infer from the occurrence of one phenomenon (A) to the occurrence of another, different from it (B)?

On the basis of the history of the empirical sciences, a basis in whose use Mill acknowledges having had predecessors in Comte, Whewell and John Herschel (On the Study of Natural Science. London 1830), he now sets up four chief methods by whose help one can distinguish between valid and invalid connections of representations. The very detailed exposition of these methods (whose indefinite main features Herschel had given) is one of the most important parts of Mill's work. Here we can keep only to one main point, the weight Mill lays on the negative cases, the experiences in which a phenomenon does not occur, although the circumstances otherwise, save for a few particular ones, are the same as in the experiences in which it does occur. In such cases the *method of difference,* the chief inductive method, finds application. (It is suggested by Bacon as one of the most important "instantiæ solitariæ"). In the particular circumstances by which an experience that presents the phenomenon differs from an experience that does not present the phenomenon, we must then see something that stands in causal connection with it (either as its cause or its effect or in such a way that both are effects of the same cause). Only then, therefore, can we rightly expect B after A, when it has been shown that with the absence of A follows the absence of B. And we then need only two *[411]* cases, one positive and one negative, to establish a law concerning the relation between A and B.

Only in very simple cases, however, is the inference so simple. In complicated conditions, where many different elements act together, we must first resolve the phenomenon into its particular elements, then by simple inductions investigate how each of these acts, then again by deduction find out what the result of their combined action will be, and finally by observation show that what we have inferred or calculated also really

agrees with experience. The whole process of knowledge thus consists of three members, induction, deduction and verification. — So far is Mill from failing to recognize the significance of the deductive method that he even sees a proof of a science's perfection in the role deduction plays within it. What he insists on is only that all deduction rests in the end on inductions and must be confirmed by the agreement of its results with experience. He rejects pure thinking only when it would begin without a foundation in experience and end without confirmation by experience.

The method of difference itself, which is to bear the whole edifice of knowledge, is, as Mill clearly sees, demonstrative only if we proceed from the assumption that there is such a connection in nature that what has once happened will happen again when exactly the same conditions occur. The uniformity of nature, or the law of causality, is thus the presupposition of the validity of all our inferences concerning real phenomena. In discussing how the causal principle is proved, Mill takes up the great problem which Hume and Kant had discussed. He answers it in Hume's spirit, but in such a way that he would illuminate it from the logical side, whereas Hume declared it logically insoluble and went straight over to giving a psychological explanation of how the assumption of a causal connection could have arisen.

Mill denies that the causal principle is founded on immediate belief, intuition or instinct. In the first place, belief or instinct are not proofs. A strong and firm association
[412] of representations can produce a conviction that cannot be shaken by any counter-proof; but its strength is not itself a proof. On the other hand, Mill holds, every assumption of cause can be suspended and "instinct" pushed aside. Anyone who is accustomed to abstraction and analysis, and whose imagination can move freely, will not find it impossible to conceive an absolute chaos in which phenomena follow one another in ever-changing ways, without any definite law. (Logic III, 21, 1. 4). And often we are compelled to stop at mere "collocations", a sum of factually given conditions which cannot in turn be shown to be caused. Nor, indeed, have men at all times believed in the law of causality; they have assumed "chance" and ascribed reality to "free will". Nor, in order to understand the validity of empirical science, need we assume that the law of causality holds for all phenomena, if only we assume it valid for the domains in which our investigation moves; the motion of the planets may, e.g., be subject to definite laws, even if wind and weather are not. And we lack all right to extend the validity of the law of causality further than to the part of the universe known to us. Experiences can be grounded only by experiences; the very foundation of empirical science must be established by way of experience. Experience must tell us how far we can rely on

experience; we have no higher authority to appeal to. We must make experience its own measure. *(We make experience its own test.* Logic. III, 4, 2).

The causal principle itself, then, is to be grounded on experience and proved by induction. Our conviction that a phenomenon does not occur unless a certain definite other phenomenon goes before it is grounded on our having experienced such a connection between phenomena countless times. The situation with the conviction of the validity of the law of causality for the particular phenomenon is thus the same as with the conviction that the Duke of Wellington will die. Here too it is an inference from particular to particular. But Mill holds that the law of causality rests on so many experiences that it may rightly be said to be the most comprehensive generalization we possess. By giving our more limited pieces of knowledge a support in it, we raise their certainty to a higher level. *[413]*

Yet it was precisely the justification of connecting two representations (e.g. A and B) in the future, because they had previously occurred together, that inductive logic was to establish; the "inference" from particular to particular is, after all, only an association. But we have in reality come no nearer to seeing that justification by being told that it is grounded on the causal principle, when this principle itself in turn proves to be grounded on inference from particular to particular — that is, on association, however countless the repetitions that strengthen this association. (In the causal principle A is "the occurrence of a phenomenon", B "the occurrence of a certain definite other phenomenon"). Mill moves in a circle[97]), or rather: he does not really get off the spot. What he has really attained by his reference to the causal principle as the ultimate basis of the particular inductions is only to establish that the most indissoluble associations will be able to hold up the looser, more unstable ones; a new habit can be strengthened by an old habit of kindred nature. But that is no proof.

Mill has surely let himself be carried too far by his zeal to root out all a priori principles. He has — like so many of his opponents — confused the assumption that a principle springs from the very nature of consciousness with the proof of its real validity. He is right that, however much a principle has its ground in the nature of our consciousness, its real validity is not thereby proved. But it might be that there were hypotheses which we were driven by the nature of our consciousness to set up, and which it was the task of empirical science to test in detail. The task of scientific knowledge will then be partly the making precise, partly the verifying of involuntarily posited presuppositions. Mill harbored suspicion every time it was maintained that involuntary assumptions could become of significance for knowledge; he feared that one would at once smuggle in a

dogma. A march on the spot was the nemesis his suspiciousness brought down on his own epistemology.

An inference from particular to particular does not in reality exist, if by inference
[414] one understands a process that leads *with necessity* from the one assumption to the other, and if one will not abolish the difference between inference and mere association of representations. If the transition from the one particular to the other is to be justified, this can only be done by the likeness between the first particular and an earlier one: I infer that, since B_1 corresponded to A_1, B_2 will correspond to A_2. It is an analogy that here guides the transition, and on a test of the analogy's validity the decision as to the justification of the transition depends. Mill too comes to admit the significance of the relation of likeness for all inferring (Logic. II, 3, 3 and III, 3, 1); but he does not see that this abolishes the possibility of an inferring from particular to particular such as he has described it, and that the principle of identity becomes the ultimate presupposition of all inference, whether it appears in inductive or deductive form. A blind coupling together of representations gives no logical connection: only where the relation of identity can be shown is the blindness removed[98]).

But the logical fundamental propositions themselves Mill regards, in consequence of his pure empiricism, as grounded on experience. I experience early that light and darkness, motion and rest, past and future are different and incompatible predicates. I experience that I cannot at the same time believe something and not believe it. Belief and non-belief are incompatible mental states. From this I then derive by generalization the proposition that what is self-contradictory cannot be true. But to this proposition there can belong no necessity that goes beyond what experience teaches. Even if we stop at certain assumptions which we cannot reject without feeling self-contradiction, we may not therefore infer that they are in themselves, in reality, impossible. There may be associations of representations that are so firm that we cannot dissolve them. But the history of science shows us examples enough that what at one time was regarded as unthinkable was nevertheless true. When we see that inconceivable things in the course of time become conceivable, the inconceivability of the opposite of an assumption cannot be any proof of the correctness of this assumption. What is possible and real we learn only by way of experience, not through subjective criteria. — Of the two
[415] questions, whether there are self-contradictions that are more than factually indissoluble associations of representations, and whether we have the right to assume that the self-contradictory cannot exist, it is evidently the latter that has the greater interest for Mill. He even expresses himself in his later writings (Examination of Sir William Hamilton's

Philosophy. 2. ed. p. 67) doubtfully as to whether the incompatibility of two mutually contradictory assumptions is grounded on the original nature of our consciousness or springs from experience. To change the logical principles, he remarks (ib. chap. 21), experiences would be required which the conditions of our life do not make possible. But in any case he holds fast that a subjective necessity cannot prove an objective reality. What Mill overlooks here is the stimulating, forward-driving force of self-contradiction. It is the self-contradiction in some of the inferences we draw from experience that sets us problems and spurs us to work to procure the self-contradiction away — by extending experience. If we did not constantly use self-contradiction as a criterion, our thinking would soon fall asleep.

The mathematical fundamental propositions are for Mill, like the logical fundamental propositions and the causal principle, mere generalizations from experience. Mathematics is indeed a science of reason which by way of thinking leads to necessary results. But it rests on fundamental propositions that can only have been taken from experience. That two magnitudes which are equal to one and the same third are equal to each other is a proposition of experience and nothing more. Likewise it is a proposition of experience that two straight lines that have intersected will always diverge. The geometrical definitions contain elements from experience, although these elements of experience are represented as more perfect and more exact than experience presents them. The definition of the circle thus implies that all the radii are equal, which they cannot be shown to be in any real circle. All real circles nevertheless contain greater or lesser approximations to the ideal circle about which geometry proves its propositions, and the greater the approximation, the greater the validity the application of these propositions acquires in reality. We make a leap from the approximately equal to the absolutely equal, because we can thereby derive consequences; and in applying these *[416]* consequences we then take the degree of approximation into account. Geometry thus rests on hypotheses, or, if one will, on fictions. If geometry is to have real validity, the presupposition is that the nature of space corresponds to the observations we have used to form our ideal hypotheses or fictions. But whether space is so constituted everywhere in the world we cannot know a priori. — Had Mill gone more closely into the idealization which leads to the setting up of the hypotheses of geometry, his whole epistemology would perhaps have taken on another character, since he would have seen that one may very well acknowledge an a priori element in our knowledge without therefore assuming that assumptions about the nature of existence can be founded on it without further ado.

Mill's attempt to carry through empiricism and to make earnest of the doctrine of

the blank tablet cannot, then, be said to have succeeded. But by venturing the attempt he has, through his acute and ramified investigations, thrown light on the nature of our knowledge. The attempt itself may be regarded as a counterpart to Hegel's dialectical method: whereas Hegel would develop truth by the self-unfolding of pure thought, Mill would develop truth by piecing together independent representations taken up from without. —

A peculiarity of Mill's epistemology is the dependence in which it stands on his psychological presuppositions. He builds his logic on his father's psychology. Association by contiguity as the fundamental form of all connection of representations, and indissoluble association as the explanation of what stands as inconceivable: these are the means he operates with. *Psychology* is for him the fundamental science which bears all other sciences. The laws of association are the fundamental laws of all our knowledge.

Mill has not himself given any exposition of psychology, but has in his Logic treated its method and in other writings ("Examination", 1865, and in the notes to James Mill's
[417] "Analysis", 1869) treated particular psychological questions in an interesting way. The exposition of psychology with which in his later years he most nearly sympathized was the one his friend and disciple Alexander Bain gave in his two chief works *The Senses and the Intellect* (1856) and *The Emotions and the Will* (1859). But in these works the strict association psychology, as James Mill taught it, is abandoned, and the relation of likeness is — under the influence of William Hamilton — acknowledged as the condition of all association of representations, association by contiguity included. Mill associates himself with this in his notes to the "Analysis". The relation of likeness should thus, to be consistent, now have gained a greater influence in epistemology than Mill had granted it in his Logic, where, as we have seen, he grounded all inference on association by contiguity. He did indeed already in the Logic (I, 3, 11; 5, 6) acknowledge the relations of likeness and of difference as distinct, irreducible relations; but he did not investigate more closely the role these relations play in all connection of representations and all thinking.

Even in his later writings Mill would hold fast to Hume's and James Mill's conception of consciousness as a series of states, only he adds: — real or *possible* states. But he finds a difficulty at which his predecessors had not stopped. In memory and expectation the presupposition is contained that *I myself and no other* have had at another time, or will have, such a state of consciousness as I now have the representation of. There must then be more in consciousness than the particular members that make up the series. How can a series know that it is a series, — that it has past members and will have new

members? The presence of *a bond* is here made evident between the particular members of consciousness, the particular sensations and representations, a bond that is just as real as the particular members themselves and is not a mere product of thought. If we are to have a name for this original connecting element in our consciousness, we must call it our ego or our self. (Examination. Chap. 12. The most characteristic and decisive statements were first added in the later editions). In the notes to the "Analysis" (II. p. 175) Mill expresses himself thus: "There is a certain bond between all the members of the series which makes me say that these were sensations of a person who was the *[418]* same person throughout . . . . and this bond constitutes for me my self". Further than this — Mill adds — we can at least not get by way of psychological analysis. With this, association psychology is definitively abandoned. Mill has, as has been aptly said, opened a trapdoor in the middle of his own philosophy. For Hume "the uniting principle" consistently stood as a riddle, because he proceeded from the particular sensations and representations as the only reality. When Mill now acknowledges the uniting bond as just as real as the particular elements, he corrects the whole concept of consciousness from which Hume, and after him James Mill, proceeded. The laws of association can now only be special forms of "the uniting principle", and the English school has thus by its untiring analysis corrected its own principle[99]). It brings contradictions into Mill that he shows the relation of likeness and the bond to be realities, although he builds his logic on purely external and accidental association. But it bears witness to the honesty and acuteness of his search that he thus arrives at results which lead beyond his own original presuppositions. —

A peculiar application that Mill makes of the laws of association must still be mentioned. He seeks to show how belief in an external world can be explained from them. — What I know of the world, I know through my sensations. But does there then exist in reality a world in the sense of something different from my sensations? — What is really given are the sensations I have at this moment. But besides, in memory and expectation I have representations of possible sensations, representations which by repetition and association can come to form fixed, coherent groups. And these groups continue to stand before us in the same way in our representation, whether they are given as sensations or not. We thereby get the representation of something that stands fast, whether we perceive it or not. What is the so-called "external" object other than the regular possibility that certain sensations recur in the same way as I have had them before? — In the assumption that there exists something independent of my consciousness I am *[419]* confirmed when I experience that other sentient beings have a similar regular series of

sensations to my own. — What I mean by "matter" is then nothing but a permanent possibility of sensations. It is the fixed and common associations of representations that lead us to set up the concept of matter. And since we find that the concept of cause holds within the series of our sensations and representations, we involuntarily apply it also to the whole series, the whole sum of our sensations and representations, and regard matter as their cause. — The practical belief, the strong inclination to assume an external world, is for Mill no proof. By virtue of the principle of fixed association he holds that such an assumption is not necessary. One need not assume anything existing other than consciousness — only that besides the consciousness really given one must also assume the *possibility* of new states of consciousness. It is clear that in this "possibility" Kant's "thing in itself" lies hidden. When Mill says (Examination. 2. ed. p. 189): "The not-self might be merely a form in which consciousness represents to itself the possible modifications of the self", then it must be asked whether the self can itself turn these possibilities into realities, so that its changes of state could be understood out of itself. If the answer is yes, the self will stand as creating its inner world out of nothing; if the answer is no, one must either deny the causal principle or assume a reality besides the self. Mill does not go more closely into this side of the matter and in general brings out too little the connection in which the problem of the reality of an external world stands with the problem of causality.

c. The Principles of Ethics

Nowhere does Stuart Mill's peculiar position within the history of thought show itself more clearly than in his ethics. Brought up in the spirit of the eighteenth century, he seeks with honest will to make the points of view of the new age his own; but since the presuppositions are taken from the old school, while the goal he would reach and the
[420] points of view he acknowledges are due to the influence of new motives, there appears at decisive points in his line of thought a duality which he himself does not always notice. In the ethical domain this duality consists in his holding fast to the principle of utility, which he had learned to acknowledge in Bentham's school, but wishing to supplement it by bringing it into agreement with the philosophy of personality, whose ethics is of a more subjective kind. He saw clearly that this task is the most important one for the future of ethics, although he did not succeed in giving a theoretical solution of it. The spiritual crisis he went through at a certain point of his development was conditioned precisely by this problem. That in practice, in the art of life, in his great and significant activity, he united in a rare degree regard for the inner development of the personal life

with regard for the outer effects of particular actions, his biography bears witness. But the theory of this art he was not able to give.

The reason why he did not succeed in this must doubtless be sought chiefly in his great piety toward Bentham and his father. He had at one time, in the subjectively strongly agitated period after that crisis, spoken out strongly against Benthamite utilitarianism, which he reproached for taking a purely external business standpoint. Now he believed he had got beyond the reaction against the view into which he had been initiated in his early youth, and he now came forward as its defender — without being clear how far he himself had altered the theory whose defender he would be. He did not distance himself sufficiently from the older utilitarianism, and at particular points his theory is also determined more by it than, to be consistent, it should be. That is why his treatise "Utilitarianism" has become one of his least clear writings.

A clear and natural measure in ethical valuation we obtain — of this Mill is constantly convinced — only by keeping to the effects of actions. If we ask *why* an action is good, the decision will in the end always depend on whether it produces feeling of pleasure at some point or other and in some extent or other. Human nature is once and for all so constituted that it desires nothing which is not either the whole of happiness, a part of *[421]* happiness or a means to happiness. Every action is judged and must be judged according to the degree in which it leads toward what is thus always the ultimate object of human desire. In this consists the application of the principle of utility.

To the question *whose* happiness it is that is to furnish the measure of valuation, Stuart Mill answers: "Not the agent's own greatest happiness, but the greatest total sum of happiness". And this assertion he grounds not, as Bentham in the "Deontology", by pointing to the harmony of well-understood egoistic interests, but by giving a psychological explanation of the origin of the moral feeling. It is the moral feeling that makes it possible for us to strive to produce happiness, even if it is not our own happiness that we produce.

Stuart Mill does not take the moral feeling to be innate, but to be a highly composite product. Its most important elements are sympathy, fear, religious feeling of various kinds, experiences of the effects of actions, self-esteem, desire for the esteem of others. In its extraordinarily composite character must be sought the cause of the mystical character with which the notion of moral obligation appears. There is a great quantity of feelings that must be rooted out before one can do anything that conflicts with what one regards as right. The association between those various elements is besides so strong that the feeling appears with an indissoluble unity. And since the laws of association

are laws of nature, the moral feeling is a natural feeling, notwithstanding that it has a history of its genesis. It is, after all, natural for men to speak, to draw inferences of reason, to till the soil and to build cities, and yet these skills are not innate, but acquired. If any single element in the moral feeling were innate, it would have to be sympathy. But the greatest weight must be laid on the fact that social life accustoms all individuals to having a common interest, to working with united forces, to constantly taking account of one another. There forms a kind of instinct that leads individuals to solidarity in feeling and in action. And the more social life develops by the barriers between the
[422] different classes being broken down, the more solidarity grows. The feeling of unity can become a religion when it is constantly favored by education and by the ordering of institutions and is enjoined by public opinion. There takes place through social life an education of egoism, so that even if regard for others was at first only a means, it later becomes itself an end. Whereas Bentham — at any rate in practice — proceeded from self-interest as the universal motive, Stuart Mill was convinced of the reality of disinterested feelings. Thus he remarks, in his interesting essay on Plato, apropos of the doctrine of the dialogue Gorgias that it is better to suffer than to do wrong: "The advance which 'Gorgias' marks is one of the greatest that has ever been made in moral culture, namely the development of a disinterested choice of duty for its own sake, such that this choice signifies a higher state than that in which selfish inclinations are sacrificed to a more remote selfish interest". (Diss. and Disc. III, p. 340).

Stuart Mill thus embraces an individual doctrine of development, like Spinoza, Hartley and James Mill. The principle of indissoluble association serves him — like his predecessors — to ground and explain the inner foundation of ethics, while the principle of utility serves to ground the judgment of particular actions. In clarity and simplicity, however, his exposition cannot measure itself with Spinoza's and James Mill's. There is a certain wavering in him as to what is original and what is acquired, and he also entangles himself in a difficulty his predecessors did not know, by assuming differences of quality between feelings. One feeling of pleasure is to be able not only to be stronger, more lasting, more unmixed with pain, more fruitful in happiness-bringing effects, more common to many than another feeling of pleasure; it is also to be able to be of a qualitatively higher kind, which shows itself in that whoever has experienced it seeks it, even if it can be obtained only by going through great suffering, and prefers it to the greatest possible quantity of another feeling of pleasure. The question is whether the "difference of quality" Mill here posits does not find its explanation precisely in certain feelings of pleasure being
[423] sanctioned and favored by the moral feeling whose origin Stuart Mill himself has sought

to explain. In any case the question whether there are qualitative differences between feelings is far too complicated a question to be solved by the simple proof Mill adduces. —

While Stuart Mill, by laying so great a weight on the subjective foundation and by assuming a difference of quality between feelings, moves far away from Bentham's standpoint, he is still far too much a Benthamite in distinguishing between the value of the action and the value of the acting personality and in holding that ethics has to do only with the former. Precisely from a utilitarian standpoint this distinction must be untenable. For since every motive contains possibilities of far more actions than the particular one being judged at the moment, valuation must extend to it as well. Even if the particular action does not give rise to effects that conflict with the principle of welfare, actions may nevertheless spring from the same motive which must be condemned, and the motive itself becomes in such cases ethically questionable. It is consistent that valuation goes back to the source. In the book "Om Friheden" the distinction mentioned plays, as will appear, a large role too.

d. Social Ethics

Of the whole series of ethical and economic questions which Mill has treated in his various writings and essays, only a few will be brought forward here to illustrate his ethical standpoint.

α. *Individual and Society.* (Om Friheden. 1859). — Mill fought for emancipation and self-development. But he experienced that political freedom does not bring with it real, spiritual freedom and independence. In place of the physical compulsion that is gradually displaced, public opinion, the moral police, easily steps in. And this tyranny is more dangerous than the political one, since it leaves fewer ways of escape open, penetrates into daily life and makes the soul itself a slave. One follows the current, even when it concerns only which pleasures one is to enjoy, instead of following one's own drive *[424]* and one's own judgment. In England especially, says Mill, the yoke of public opinion is heavy, while the yoke of the law is lighter than in other countries. A great danger threatens here: the rule of the masses, the rule of collective mediocrity. The mass must certainly always be led by individuals — but it prefers to take its opinions from people who do not stand too high above it. And yet the impulse to everything that is noble and wise always comes from individual outstanding people. The few original people are the salt of the earth! — Mill will, however, have no cult of great men. All they can demand is full freedom to utter the new ideas. Power to compel others would first and

foremost corrupt the great themselves. The means of avoiding that danger, which lies in the tyranny of public opinion and in the lowering of the level by the rule of the masses, is *freedom,* the only constant, never-failing source of progress, because it creates as many independent centers of improvement as there are personalities.

As a general fundamental proposition for the interference of others in the affairs of the individual, Mill lays down that an individual's freedom of action may be restricted only when an interference is necessary to prevent his conduct from bringing suffering on others. Care for the individual's own good is not sufficient ground for such interference. The only part of his conduct for which the individual owes society an account is the part of it that concerns others. — By interference Mill understands partly physical compulsion, partly the spiritual compulsion that lies in the judgments of others. In the latter respect he distinguishes between "moral disapproval or condemnation" and "expression of displeasure or withdrawal of esteem". (Om Friheden. Danish trans. p. 142). This, however, is a very fine distinction, since to withdraw my esteem from an individual is to inflict a pain on him — if, that is, he sets store by my esteem; and if he does not, my "disapproval" will surely not pain him either. The distinction is just as difficult as that between actions which have effects only for ourselves and actions which have effects for others as well. Whoever does not develop his personality, or who shows
[425] a lack of prudence and personal dignity, can thereby deprive others of a power to which they rightly had a claim. Precisely when one maintains, as Mill does, that what matters is to get as many independent starting points as possible, it is impossible to mark off any domain that could not acquire significance beyond the limits of individuality. This distinction between what concerns the individual himself and what concerns others is as little tenable as the previously mentioned distinction between the value of the agent and the value of the action. And in both places Mill would precisely by virtue of the principle of utility have been able to arrive at a more correct conception. —

The importance of as great a freedom as possible Mill insists on as regards both opinions and actions. — If an opinion is true, what harm is there then in its being freely discussed? Should it contain only a part of the truth, then only free discussion can decide how great a part of the truth it makes up. And only by being kept alive through discussion does it gain influence on characters and actions. Even if, finally, an opinion could be useful without being true, its utility must nevertheless be discussed just as well as its truth; in our days — Mill holds — the utility of opinions plays quite as large a role as their truth. — As for actions, they cannot of course have as great a freedom as opinions. But Mill maintains that the value of the different ways of living must be tested

in experience, and that this can only be done by giving differences of character as much scope as is possible without causing others suffering. It is a condition of the individual's own happiness as well as of individual and social progress that its own character, and not custom and usage, determine its manner of acting. Strong drives and desires are a good; they are the stuff of which heroes are made, and men do not act badly because their desires are strong, but because their conscience is weak. —

Even if the boundary between individual and society is not so easy to draw as Mill thinks, he has nevertheless by his investigation (which, moreover, takes up Kant's fundamental thought in his "Rechtslehre") shown that the burden of proof always lies on whoever would have freedom restricted, just as he has impressed the responsibility *[426]* that is attached to every moral judgment.

β. The Woman Question. (Kvindernes Underkuelse. 1869). — The question of woman's right is for Mill only a single side of the great question of emancipation. Here too what essentially matters is the struggle against the abuse of power, the removal of the hindrances to the free unfolding of personality, the abolition of the right of the stronger and of all external authority where this is not necessary as protecting and educating. Besides, in his investigation of this question he finds application for his favorite theory of indissoluble association, to explain the opinions that hinder development. The notions people have of the female nature rest, he maintains, only on habit and tradition, not on real experience. What we call the "female nature" is only an artificial product. Only by opening the barriers that have hitherto been set to the development and use of female capacities can one come to know the real female nature. It is the same line of thought as in the book on liberty: compulsion must go, so that experience can be gained. It is a practical application of the method of difference. But Mill believes he knows the result beforehand; he maintains that the mental capacities of women and men are equal — an assumption which is of course not necessary to ground equality of rights.

γ. The Question of Representation. (Om den repræsentative Regering. 1861). — In discussing this question too Mill pursues his main thought: how freedom can be secured not only against the despotism of individuals and estates, but against the despotism of the majority. He discusses the dangers and advantages of democracy. The constitutions between which the struggle now stands are in reality only two: democracy and bureaucracy. The only possible solution is that democracy take bureaucracy into its service and merely reserve to itself superintendence and control. Parliament should represent only the decision of will in the state; for drafting laws it is not suited; for that commissions of experts should be appointed. To secure the right of the minority in

[427] decisions Mill recommends, among other things, proportional representation. Clear as his eye is, as the book on liberty also shows, for the defects of democracy, he nevertheless sees clearly how shortsighted are those who sigh for a despotism in order to get their plans of reform carried through. They forget that the development of the people itself is the most important condition of continued progress, and they do not see that a good despotism in a fairly civilized country is even more harmful than a bad one, because it is far more enervating to the spirit and strength of the people.

δ. The Social Question. (Principles of Political Economy. 1848). — In the course of Mill's development it became more and more clear to him that the social question takes precedence over the political. His democratic conviction he never gave up; but partly, as we have already seen, he came to have an open eye for the defects of democracy, partly there formed for him an ideal of the future which lay far beyond the democratic program. Without great social changes individual and political freedom would in any case not really be able to benefit all.

Already in a letter from the year 1842 (to Robert Barcley Fox, printed in Caroline Fox's Diaries. II. p. 272) Mill speaks of the significance of the social question: "I believe that ever since the changes that took place in the constitution through Catholic Emancipation and the Reform Act, a great part of the ruling class, especially younger men, have gradually had their eyes opened, and the progress of Chartism is producing, I believe, the feeling that those who govern must strive, both from duty and from prudence, to a far higher degree than they have long been accustomed to, to look after the temporal and spiritual interests of the poor". But he must add: "As to the means, on the other hand, of curing or even merely alleviating great social evils, people are just as far from harbor as they were before". To a definite solution Mill did not come later either. He acknowledges openly that here there are two fundamental thoughts, both of which he must hold fast, although in their consequences — at any rate apparently — they conflict with each other. They are the ideas of individualism and of socialism, of which Mill (Levned. p. 259) declares that they are diametrically opposed to each other. How it is then possible to unite them,
[428] he declares he does not know; that he leaves to the future. It was his conviction that we do not yet know either of the two systems in its best possible form. We know neither yet what free self-activity in its highest degree and form can accomplish, nor what possibilities there are for a thoroughgoing social ordering of the external conditions of life. Nowhere perhaps do Mill's great qualities show themselves so clearly as here. Together with a great hope for the future, indeed perhaps precisely supporting himself on it, he takes up a critically discussing position toward the different possibilities that

experience presents. What is merely handed down and customary does not impress him; he calmly investigates the conditions of its subsistence and looks to see whether these conditions might not be replaced by others.

In his investigation of the socialist systems he took up a far more unprejudiced standpoint than the economists of his time were wont to. That followed from the spirit of his philosophy. He could not, like the usual opponents of socialism, see in the right of inheritance and the right of private property dogmas that did not need to be grounded. The way in which the yield of production is distributed depends, in his conception, to so high a degree on the actual social order and on the habit and will of men that it was a great piece of sleight of hand when the economists were wont to regard the mode of distribution now prevailing as grounded on eternal natural necessity. As regards private property in particular, it was not introduced for any of the reasons for which it is now upheld. Originally society, for the sake of peace, maintained the individual in the possession of what he had once and for all seized. Whether this arrangement is to be retained for the future or replaced by an arrangement such as socialist systems conceived, that is now an open question, and a question which is the object of discussion through all classes in the civilized countries. Mill finds precisely what is peculiar to our time in this, that all principles are discussed, and that even those who suffer most under the existing institutions take part in the discussion of the value and justification of these principles. It is the first time in history that this has been the case. What will the result be? Mill answers (Principles of Polit. Economy. II, 1, 3): "If I may venture a conjecture, *[429]* the decision will probably depend chiefly on one consideration, namely which of the two systems is consistent with the greatest degree of human freedom and self-activity. Once the means of sustaining life are secured, the need of freedom is the strongest of all the personal wants of men, and it grows in strength, instead of diminishing, the more the intellectual and moral faculties are developed . . . An education which taught men, or social institutions which commanded them, to give up control over their own actions in order to obtain comfort and abundance, be it in ever so high a degree, or to renounce freedom for the sake of equality, would rob them of one of the most exalted characteristics of human nature". Now Mill does think that this objection to socialism is often exaggerated, and he maintains that the compulsion this system would entail would deserve to be called freedom in comparison with the compulsion under which the great working class now lives. But he sees no reason to give up the system of private property, if only the laws will do as much to prevent its drawbacks as they now do to aggravate them. The present order favors the development of selfishness to a degree

which is not in itself necessary, even if the system of private property is retained. The socialists wrongly blame competition for all social evils. They forget that where there is no competition, there is monopoly and with it taxation of the industrious in favor of the idle. Competition among the workers themselves certainly presses wages down; but all other competition is to the good of the workers by making the necessaries of life cheaper. Not competition, but the subjection of labor to capital is the cause of the drawbacks. (IV, 7, 7).

What according to Mill essentially matters is, by vigorous intervention on the part of the state, to raise the whole level of the working class, its demands on life and on itself. What matters is to reach by peaceful means such an advance as the French working class gained by the French Revolution. By improved education, by the parceling out of land
[430] and by an emigration carried through on a large scale the working class will be able to attain such a social position and to have acquired such ideal and material needs that, in order not to lower the level and diminish wages, it will show the necessary self-control and avoid improvident increase of population, even if this effect only occurs in a later generation that has grown up under the more favorable conditions. (II, 11, 2). But Mill also expected much of the free associations (trade unions and especially productive associations) and followed their development in France and England with great interest. Their chief significance he saw in their favoring the virtues of independence: justice and self-control. In such smaller circles experiments in a socialist direction could moreover be made which might become of significance for the continued discussion of the social question.

e. The Religious Problem

In the writings Stuart Mill himself published he touches only in passing on religious questions. Most fully in the work on Hamilton's philosophy. He here disputes two conceptions. The one (which Hamilton too, from his standpoint, had disputed) was the one Schelling and Hegel had asserted, that by way of pure thinking a scientific concept of God could be set up, a concept of an absolute and infinite being, the author and end of all things. Mill maintains that in the religious question too we must proceed from experience, and that it must be the contemplation of nature that leads us to the assumption of a God. He seems also to hold that it is possible to ground such an assumption, and in accordance with this he maintains (in his work on Comte and positivism) that even if one assumes with Comte the law of the three stages, there will still remain on the positive stage *open questions,* and that one may very well hold fast to theological assumptions, provided only

these are framed so that they do not come into conflict with what experience teaches. The other conception Mill disputes is the one Hamilton and especially Mansel embraced, that even if the scientific investigation of the concept of God as the absolute and infinite, *[431]* almighty and all-good being should lead to self-contradictions and to consequences that conflict with what the human conscience must maintain, we must nevertheless believe in such a being, since neither our logic nor our ethics can find application to the Deity. Against this doctrine Mill expresses himself with great indignation. "If", he says (Examination. 2. ed. p. 103), "instead of the glad tidings that there exists a being in whom all the excellences which the highest human thought can conceive are found in a degree inconceivable to us, I am told that the world is governed by a being whose attributes are infinite, without our knowing what they are, any more than we can understand what principles it follows in its government, except that 'the highest human morality which we are able to conceive' does not sanction them, — convince me of it, and I will bear my fate as well as I can. But when I am told that I am to believe this and at the same time call this being by names which express the highest human morality, then I say in plain words that I will not. Whatever power such a being may have over me, there is one thing it shall not be able to do: it shall not compel me to worship it. I will call no being good who is not what I mean by it when I apply this word to my fellow creatures; and if such a being can sentence me to hell, well then, to hell I will go".

From this forceful statement, which aroused great attention and offense and was used against Mill by political opponents, one will see what for Mill was the kernel of the religious problem, namely the moral impossibility of acknowledging a higher measure of moral valuation than the human one. Already Bentham had (in Deontology. Part. I. Chap. 8) spoken out sharply against calling an attribute in God love which in man would be the opposite of love; that is to call a dagger thrust a kiss! And according to what Stuart Mill relates in his Levned (p. 42), it was the impossibility of reconciling the evil in the world with belief in an almighty and all-good creator that determined James Mill's rejection of all religious assumptions and made him feel a certain sympathy with the Manichaean assumption of a good and an evil principle that struggle with each *[432]* other for dominion in the world. For his own part Stuart Mill did not express himself definitely in any of the writings that appeared in his lifetime. It was certainly not from fear of men that he refrained from uttering his views. What he did utter was enough to brand him as a man who departed to a high degree from the usual religious notions. His utterances were indeed used as a means of agitation against him, and a church organ declared, when he was dead: "His death is no loss to anyone, for he was a gross

unbeliever, however nicely he behaved, and a very dangerous person. The sooner the 'lights of thought' who hold the same opinions as he go to the same place where he has gone, the better it will be both for church and state". What opinions Mill held, even the church organ, which was hardly in doubt about Mill's fate after death, would have had trouble saying. The explanation must be sought in his not himself finding his ideas on the religious problem fully developed; the question was for him in reality open, and he would not speak publicly about it before he had satisfied his own demands for clarity and thoroughness. Even his friends knew no more of his religious standpoint than has been set out above, until his *Essays on Religion* appeared some time after his death. Of the three essays this work contains, two (on the utility of religion and on theism) were not finished; only the third (on nature) did he himself consider worthy of publication. — A contributing reason for his reticence may perhaps also have been that he would not mix several problems together, but have them discussed each by itself. He felt no need to fire off his artillery in all directions at once. The secret of the great influence which, in spite of his radical standpoint, he gained in his time has partly lain in his taking one point at a time. As he himself said to a friend: "I spare no prejudice, but I attack one at a time".

The fundamental thought of the essay *on nature* is that one cannot set up nature, as it
[433] is without human intervention, as a model for man, any more than one can infer from it an all-wise, all-good and almighty author. Nature can impress us by its might and power; but its conduct bears the stamp of terrorism and injustice; it gives to those who have, and is merciless to those who have not. The only possibility of uniting belief in a God with experience of the real world consists in the assumption that the Deity is indeed good, but not almighty. Omnipotence must be sacrificed to goodness. In the essay *on theism* this is developed more closely. The author of the world order has been obliged to conform to conditions that were independent of his will. The matter and the force in the world are not created; their properties and laws are independent of the will of the orderer of the world. It is then no wonder that there are so many imperfections! Everything that checks the tendency, traceable in nature, to produce something purposive and good must be put down to the account of the material hindrances with which the Deity has to struggle. The mood that will be peculiar to the religion of the future must be the inspiring feeling of being the Deity's fellow worker, a feeling that cannot without self-contradiction be united with belief in an almighty God. What the Deity's will is can only be discovered by attending to everything in nature that tends toward general welfare and a higher development of life. In this struggle between the good and the evil forces in the world

even the slightest contribution can be of significance. To do something, however little it may be, to further the victory of the good is the most living and strengthening thought that can animate a man. This thought will be a leading thought in the religion of the future! — Already in a letter of 1841 (to Robert Barclay Fox, see Caroline Fox's Diaries. II, p. 260) Mill mentions this thought, referring to the Saint-Simonians' doctrine of continuing the work of creation.

One point in Mill's polemic against Hamilton first becomes properly intelligible when one knows that he conceived the Deity as a limited being. For he cannot understand all the difficulties Hamilton finds theoretically in the concept of God, namely not how *[434]* the law of relativity can be a hindrance to holding fast this concept in thought. God is thought in relation to the world — what contradiction is there in that? asks Mill. He surely presupposes here his own concept of a limited God; for the self-contradiction is of course present only so long as the Deity is to be thought as an absolute and infinite being — and yet in relation to something that is not itself, so that it is thus conditioned and limited, i.e. becomes relative! — It was not logical but ethical reasons that led Mill to his standpoint in the philosophy of religion, which recalls the one Voltaire and Rousseau took.

When Mill, from the tendency to purposiveness and heightened life which can be traced in the world alongside wholly opposite tendencies, infers the assumption of a Deity, this inference is shaken if a purely scientific explanation of purposiveness and the development of life were possible. Mill rejects the dogma of creation because every phenomenon must be explained by another phenomenon; but must not for the same reason the inference too which he believes he can draw from purposiveness and the development of life be rejected, since here too definite natural causes must be sought in each particular case? It is these causes that the hypothesis of evolution seeks to show, and Mill does indeed declare that if the hypothesis of evolution is adopted, it will not indeed make impossible the assumption of a creative intervention, but will nevertheless greatly weaken the proof of such an assumption. "Let us leave this remarkable hypothesis to the fate which the progress of science reserves for it".

Even if religion could not be proved, it will — in Mill's conception — not fall away so long as it is of benefit to man. Mill investigates in a separate essay *the utility of religion.* Religion has this in common with poetry, that it springs from a need of grander and more beautiful images than those prosaic life brings with it. Human life will, in spite of all progress, continue to be so wretched and narrow that man will always need greater breadth and height for his destiny. Imagination must be freed from the limitation that

[435] positive experience brings with it. Even if religion differs from poetry by the weight it lays on the real significance of ideals, the essential thing in all religion is nevertheless that feeling and longing are directed in a forceful and earnest way toward an ideal object. This condition is fulfilled by Comte's religion of humanity better than by any of the religions that embrace belief in God as the author of the world. The feeling of unity with humanity, of deep sympathy with its welfare and progress, is a disinterested feeling which does not contain anything that can become questionable in a logical or ethical respect. Only one form of supernatural belief is there that is neither illogical nor immoral, namely the one described above, which regards nature as the product of a struggle between a good and wise being on the one side and on the other either matter (as Plato assumed) or an evil principle (as the Manichaeans assumed). Whoever embraces this belief can be sure that the evil in the world is not a work of the being he worships. Even if this belief could not be proved, even if it is rather a hope than a belief, true wisdom of life nevertheless demands that it be held fast with all one's might. It brings with it a bright and cheerful way of taking life, and it does not exclude reason and criticism. By one-sidedly and needlessly turning one's gaze toward what is dark, narrow and unhappy in life one only hampers one's capacity for action. The hope that beyond the limit set to our experience there are good forces and an immortal life has its essential significance in giving the scale of the feelings a greater extent. Mill ascribed significance to belief in immortality especially for the sympathetic feelings; he even at times placed the whole significance of religion at this point, as when he said to a friend who had lost one of those nearest to him: "In my view the only lasting value of religion consists in alleviating the feeling of complete separation which is so dreadful in a great sorrow". But in his work on the philosophy of religion he dwells also especially on the significance Christianity has had and has in setting up an exalted model for the race; it is far more by its image of Christ than by its idea of God that Christianity has worked. And
[436] the influence of this model will — Mill maintains — not fall away because one conceives it in a purely historical, human manner. —

Mill's religion has this in common with Kant's, that it is rather a hope than a belief. He is quite clear that the decisive question is this: "Is it irrational to let oneself be determined by the imagination to a hope for whose realization one may perhaps never be able to give any probable ground? Must one combat this hope as a departure from the principle of reason which bids us regulate our feelings as well as our opinions according to strict proofs?" Mill holds that on this point there will long be dispute among thinkers, and that each of them will decide it according to his temperament. But he holds as

well that the question itself has not received the serious discussion its great significance demands.

Mill's own solution leaves one side of the problem standing. If the good principle struggles with material chaos, there must nevertheless be a world order that embraces them both and makes the struggle possible. Struggle is interaction, and interaction presupposes an order of things that makes it possible for the different acting forces to meet. Thus the old problem rises again behind Mill's solution. Mill's critics in the philosophy of religion in England[100]), who for the most part operate with Hegelian weapons, have especially brought forward this point, while they have not felt the ethical sting of the problem as strongly as Mill. — Psychologically regarded, Mill has passed over a factor that plays a large role in religion, namely the need of absolute absorption and rest in a being that is not itself placed in the struggle of life. This need is in the religions curiously at odds with the need of having in the Deity a living and struggling model. The religious problem is sharpened when both spiritual needs demand satisfaction at once.

But Mill's greatness does not lie in his results. His pure empiricism he succeeded in grounding and carrying through as little as Plato succeeded in carrying through his pure idealism. This comparison lies all the closer as Mill was all his life a great admirer of Plato. One can apply to him what he himself says of Plato (Levned p. 24): "I have always *[437]* felt that the name Platonist belongs with far greater right to those who have been trained in and have endeavored to apply Plato's mode of investigation than to those who have only adopted certain of his dogmatic propositions". Mill's significance in the history of philosophy consists precisely in his mode of investigation, in the way in which he has set experience and critical thinking to work in a whole series of theoretical and practical domains. He has awakened a philosophical spirit which is of greater significance than any single one of his results. — And with this we leave the thinker at once all of a piece and ever striving, logically clear and strongly moved by feeling, who in his development at once reflects the problems of his time and gives weighty help in their treatment.

C. The Philosophy of Evolution

The general fundamental thought on which the positivist philosophy rests is that our views are to be built on observations. In explaining the phenomena that experience presents, we are then to rely on causes that can themselves be demonstrated in experience. Positivism is really only a carrying-through of the demand for "true causes" that Kepler

and Newton made. As Auguste Comte has shown, the difference between the various main forms of philosophy depends on which causes one assumes. But even if one adheres to this principle, there may still be room for significant oppositions. Comte and Stuart Mill stopped at the conception of the causal relation as a relation between two different phenomena which for us simply were, as a matter of fact, connected. They laid no weight on that side of the causal relation by which, the more it is deepened and seen through, it shows us a continuous connection between the phenomena. Therefore they did not come to see the close connection there is between the concept of cause and the concept
[438] of evolution. In the social domain this latter concept does, to be sure, play an essential role with them; but it does not become determinative for their whole view of nature. They laid too great weight on the differences that experience presents, and did not see that all knowledge aims at reducing them as much as possible. They stopped at the problem of how new forms arise. Within the individual they assumed a development, and in the species likewise, through the influence of traditions and institutions. But that the concept of evolution could become a leading thought for the whole conception of the world they did not see, and the limits of their thinking were in great part determined by this.

The change in the view of nature that Charles Darwin's discoveries and hypotheses brought about may with good reason be compared with the changes caused by Copernicus and Bruno, Galilei and Newton. Copernicanism enlarged the world and made it infinite; the earth and human life no longer stood as that about which everything turned. Newton showed that a fixed lawfulness holds in the whole great world, since the distant globes follow the same laws that hold here on earth. Darwin's view of nature signifies a similar widening of the gaze as regards organic life. In the domain of living beings no great connection, no general law of origin and development, had yet been demonstrated. It had been shown by Caspar Friedrich Wolff and Karl Ernst Von Baer that the single organism develops through a series of stages from a germ that bears no likeness to the fully developed individual. It has been shown by Spinoza, Hartley and James Mill that a psychological development can take place in the single individual, through which, by virtue of the laws of association, mental forms take shape that resemble the original foundation as little as the forms of the grown organism resemble those of the germ. And the historical school, founded by Montesquieu and flourishing after the Revolution especially in France and Germany, had asserted the concept of development with regard to the origin of social and political forms. In the philosophy of Romanticism the concept of evolution likewise played a great role. The inward connection of the individual

thing with the whole was held fast, and existence was conceived as a series of steps *[439]*
on which the content of the world appeared in ever higher forms. But the use that was made there of the concept of evolution was a purely ideal one. Development was not conceived as a process by which one phenomenal form arose out of another; the continuity was thought to lie in the ideal kernel of existence, not between the single phenomena. And no one asked after the efficient causes that could lead over from one step to the next. A great example of how new forms and states can take shape by virtue of laws that constantly hold was, on the other hand, given in KANT's and LAPLACE's hypothesis of the development of the solar system out of a primeval nebula through the influence of definite physical and chemical relations, and in LYELL's theory of the origin of the present state of the earth's surface through the constant action of the same physical and chemical causes that are at work to this day. Darwin's doctrine holds that the organic species arise because constant variations of the organic forms take place, but that only those variations can survive which under the given conditions of life (physical conditions and relations to other living beings) offer certain purposive properties. It is this that he figuratively designates by the expression "the struggle for existence." When the conditions of life then continue to be the same through long periods, the purposive properties will be fixed and heightened. Darwin has widened the view of nature and let us discover an immeasurable course of development out of which the species now living have come forth. And on the other hand he has sharpened the eye for small things, in that he has taught us to see, in the often inconspicuous causes that are at work around us, the forces by whose quiet but uninterrupted working the living species have arrived at the forms under which they now appear to us.

What was new in Darwin's doctrine was not the general thought that organic species and forms have arisen through natural causes. This thought had come forward in many shapes before him. His significance lies in his having supported it by demonstrating definite causes that work in the direction of such a new formation. The concept of evolution itself is, as already indicated, so closely bound up with the concept of cause and the concept of continuity that it could come forward as a leading idea before Darwin gave *[440]*
it the mighty confirmation of experience through his investigations. Thus HERBERT SPENCER had, several years before Darwin's epoch-making work appeared, already adopted the hypothesis of evolution, and in particular he had (in the first edition of his Psychology) shown what significance this hypothesis had through the possibility that the properties and faculties that cannot find their explanation in the single individual's experience could be explained by the experience of the whole species. It was an extension

of the philosophy of experience, by which it became possible for it to let points of view that hitherto only the speculative and critical philosophy had asserted come into their own more fully than had happened before. Later Spencer has, in a detailed, systematic exposition, sought to show the concept of evolution to be a principal concept to which our experience and our thinking lead us back from all sides.

1. Charles Darwin

a. Biography and Course of Development

The justification for drawing this great naturalist into the history of philosophy lies — in a similar way as with Copernicus, Kepler, Galilei and Newton — first and foremost in this, that his method and his results have significance far beyond his special field and mark a turning point for scientific research and for the view of nature as a whole. Quite shortly after the idea of his hypothesis had arisen in him (1837), twenty years before his famous work appeared, he wrote in his notebook: "My theory will lead to a whole philosophy." But it was not only through its consequences, not only by raising the question of how the new theory bore on the whole world-view, that Darwin acquired significance for philosophy. He has also himself treated psychological and ethical questions and expressed himself on the limits of knowledge. And as we know him through his autobiography
[441] and his letters, he stands before us as a peculiar Socratic figure in the research of our time, venerable by his energy, his love of truth and his feeling of humanity.

Charles Darwin was born at Shrewsbury on 12 Feb. 1809. After having tried medical studies in Edinburgh and then theological studies in Cambridge, without having got anything out of it, his early awakened interest in natural history led him to go along on a voyage round the world that the ship "Beagle" made from 1831—1836. The observations he made on this voyage form the first foundation of his theory. He had compared the present fauna of South America with the extinct species and wondered at the likeness in structure alongside the differences, especially in size. He had compared the animal species in the northern part of South America with those in the southern part, and here too found likeness and difference combined in a peculiar way. But above all he had wondered to find on the Galopagos Islands, some 150 miles from South America, an animal and plant world that quite recalled South America's, although it consisted of species that were found nowhere else in the world: the species occurred only on these islands, but they belonged to genera that were found on the nearest mainland. It was as if one and the same fundamental form had been transformed partly to live on the mainland, partly on the islands. And besides, each one of the islands had its peculiar

species, which were not found on the other islands. How is this kinship on the one side, this difference on the other, to be explained? With this problem Darwin returned home from his great voyage, and he spent his whole life answering it. He lived in the country near London under the happiest circumstances, uninterruptedly occupied with collecting facts that could throw light on his problem. The question to be solved was why precisely such forms and properties are preserved and developed as are of benefit to animals and plants under the conditions in which they live. What had struck Darwin was precisely the variation in forms and properties in accordance with the conditions of life. He started from the assumption that there must be a natural cause of this — but which? The reading (in the year 1838) of MALTHUS's work on the population question set his thoughts going. Malthus showed that living beings have a tendency to multiply *[442]* to a higher degree than the quantity of the means of nourishment can increase. But then, Darwin concluded, living beings must surely be one another's opponents, must fight or compete with one another for access to what they need in order to live. Life is and must be a "struggle for existence," and the individual or the group of individuals that for some reason or other has acquired a faculty or an organ that the others lack, and that suits the circumstances, will more easily hold its own in the struggle than the others. The multiplication of the species will take place chiefly through them, and forms that lack this faculty or this organ will gradually die out. And when a group of living beings has to live in a limited space, this will be all the more easily done, the more different forms there are within the group, since their wants will thereby more easily be able to be satisfied. When all have to live on one and the same thing, not so many can live as when there is difference in food. It is therefore an advantage in the struggle for existence that a capacity for variation is present in a species.

Already at the beginning of the forties Darwin worked out a treatise in which he set forth his theory, but he laid it aside in order to undertake further investigations. In the middle of the fifties he began the final redaction of his work, and this appeared in 1859 under the name of *Origin of Species.* (Danish translation by J. P. Jakobsen. 1872). To it was joined a long series of other writings, of which the most important in philosophical and psychological respects are *Variation of Animals and Plants under Domestication* (1868), *Descent of Man* (1871, Danish translation by J. P. Jacobsen 1874) and *Expression of the Emotions in Men and in Animals* (1872). — Darwin's long, faithful work in the service of science ended only with his death on 19 April 1882[101]). Three years before, he had written in his autobiography: "I believe that I have acted rightly in devoting my life to science. I feel no remorse for having committed any sin, but I have again and again

[443] regretted that I have not done more direct good to my fellow men." —

b. Theory and Method

The result of Darwin's research was a victory for the principle of natural causes and for the assertion that nature makes no leaps. It had indeed earlier been assumed that the varieties are due to definite natural causes; but the species themselves were regarded as supernaturally produced or at any rate as inexplicable. Continuity was now demonstrated in a domain where hitherto one had either assumed supernatural intervention and interruption or had stopped at original, inexplicable differences, or at the most had appealed to an inner impulse of development that was to lead from one step to the next. By the explanation Darwin gave of the origin of species, through the struggle for existence, he moreover threw a light on the conditions of life, not only physical but also mental life, which is quite as important as the solution of the special task he had set himself. It was an old English thought that he, without knowing it, took up again. For his "struggle for existence" recalls Hobbes's "war of all against all." The English have always had an eye for the conditions of life and for the necessity of putting force into the effort to master them. By this is conditioned the peculiar mixture of empiricism and idealism that is often regarded as a contradiction. Their sense for practical circumstances as they simply are does not, however, dull their conviction of the value of the inner forces, even if they do not express it in the mystical or speculative way into which the thinkers of the Continent often lapse.

The explanation Darwin gives of the origin of species is entirely in the spirit of positivism, and it was therefore inconsistent when Comte's disciples took up a dismissive attitude toward the new hypothesis. The cause to which Darwin appeals is given, *positive*. He declares in a letter that to say that the species were created thus or thus is no scientific explanation, but only a pious way of saying that it is as it is. And according to other statements, when at the close of the "Origin of Species" he spoke of the first forms of life
[444] as having been created, he meant only that we know nothing of how life arises; later he expressly regretted that out of regard for common opinion he had used the word "created."

The expression *the struggle for existence* is figurative. It is meant to designate the dependence of one being on another and on the conditions of life, and in such a way that not only the single individual is thought of, but also its offspring. A plant that grows on the edge of a desert must struggle for life, that is to say, its subsistence depends on how little moisture it can make do with. An even greater role than the relation to physical

conditions is played in the organic world by the relation to other living beings. Mistletoes on the same branch "struggle" with one another for room and nourishment; but they also struggle with other fruit-bearing plants, since their multiplication depends on whether the birds swallow their seeds rather than those of the others, so that sowing can take place to a higher degree. While the expression the struggle for existence designates the organism's adaptation to the living or lifeless environment, Darwin designates by the expression *natural selection* the other side of the relation, namely the way in which the conditions of life favor certain properties — "the principle according to which every variation, however slight, that is useful to the individual is preserved." There is struggle everywhere in nature. It is always a matter of: to be or not. Every being subsists by virtue of a victory it has won at some time or other of its life. And the struggle for existence is in turn a consequence of the strong multiplication of organic beings. The whole economy of nature becomes unintelligible to us if we for a single moment forget that each single species seeks to increase extraordinarily strongly, and that there is always some obstacle or other that checks or stops its spread, even if we cannot always catch sight of it. It is because of this collision between spread and obstacle that a selection becomes necessary. The individuals that offer purposive variations survive, the others die out; thus the rise of differences is favored, and there take shape at last the species that are now in the eyes of so many essentially different. We have here a key to understanding how forms of different kinds can be related to one another, since with increasing divergence of character they develop out of a common fundamental form. What C. F. Wolff had shown *[445]* for the different organs of the same individual, that in spite of their unlikeness they develop out of certain simple rudiments, Darwin showed for the different species within the organic world as a whole.

Darwin's method is an interesting model for inductive research. His course of development as a researcher presents with unusual clarity the three main stages that a truth built on experience runs through. The first foundation of his theory was, as we saw, the observations made on the voyage. They called forth the idea, the provisional hypothesis, and — after the reading of Malthus — the explanation by deduction from the strong impulse of multiplication in its relation to the conditions of life. The third link in the series of labors was the verification, the confirmation by experience. It consisted essentially in four groups of facts. The English animal breeders had for many years produced new races by "artificial selection." They noticed small variations, suited to their purposes, in the different individuals, and then, by seeing to it that only individuals which offered such variations propagated the species, succeeded in breeding new forms.

What happens here consciously and methodically happens through "natural selection" unconsciously and in a more irregular way. The second fact that confirms the theory is the kinship of extinct species with species now living. They can as a rule be ranged as links in groups now existing and be conceived as lower forms of development of these. The third fact lies in the geographical distribution of the species, which is best understood by assuming that different forms and properties in related beings have arisen through the spread of the stock over tracts of different character. That the lowest beings are the most widely distributed is explained by their seeds and eggs being small and well fitted to be carried far and wide by water and current. Finally the agreement there can be in the embryonic state between animal forms that, when fully grown, differ greatly from one another, testifies to a distant kinship, and rudimentary organs (which are found especially in the embryonic state) likewise become intelligible only on the assumption of such a kinship.

[446] There was a remarkable kinship between Darwin's character and the direction and method of his research. What distinguished him as a researcher was his open and childlike mind, for which nothing was mean and insignificant. Everywhere he divined a connection. Nature was to him not a dead collection of objects to which the naturalists were to give name and number, but a living reality, where the subsistence and growth of the one depends on that of the other. The insect and the flower, the bird and the plant, the mold and the worm, the life of animals, their structure, their ornaments, their love and their fights, all this stood for him in mutual connection and could only in an unnatural way be separated. The word natural history he has made a truth. With the childlike openness toward nature there was joined in him the manly understanding that fixed, definite laws hold everywhere, and that even the highest in nature is not withdrawn from such laws, but precisely shows its highness by developing by virtue of these laws. He used to say that one could not be a good observer if one could not speculate. It was not possible for him to see anything new in nature without at once forming a hypothesis about how it could have come to be. But on the other hand he had a peculiar capacity for seizing an objection and holding it fast. He followed what he called the golden rule, of noting down every fact and every thought he came across that conflicted with the results he had reached. Very few objections were therefore made against his views that he had not himself noticed and sought to answer. He was in this respect more critical than several of his adherents; these sometimes thought that he laid too much weight on the objections that could be made against his doctrine. He was fully aware that his hypothesis really cannot be proved directly. Of one of his critics he

writes in a letter: "He is one of the very few who see that the change of species cannot be directly proved, and that my doctrine will sink or swim according as it is able to group and explain the phenomena. It is remarkable how few judge it in this way, which is the one right way." The proof of his doctrine he found in the "thread of thought" (the intelligible thread) with which it connects a whole series of facts. His doctrine was for *[447]* him not a dogma but a means of work, a light over nature that could produce more light. The significance of scientific hypotheses is, after all, not least that of giving occasion for continued research: we try whether it is possible to connect *all* our experiences in the way we have succeeded in connecting *some* of them; we learn what we are to ask nature about. In this respect Darwin's doctrine of the origin of species by natural selection has been extraordinarily fruitful. It teaches us everywhere in natural science to trace the significance that a property, a faculty or a form has in the struggle for existence. It starts from the assumption that nothing can subsist or develop that does not have a definite significance in the whole connection of life.

c. The Limits of the Theory

Darwin acknowledges a limit to his research in the question of the first origin of the individual differences between which both artificial and natural selection take place. A selection presupposes something that can be selected, differences and variations. Darwin takes it as a fact that such variations arise, and the more frequently, the more favorable the conditions of life are, and the more developed a species already is. He admits that "we are very ignorant on all points concerning the causes of variability," and the enigma of the origin of the original variations became for him for a time all the greater, as he was inclined to ascribe to external circumstances a very slight direct influence. In a letter to Huxley shortly after the "Origin of Species" had appeared, he writes (25 Nov. 1859): "You have very acutely found a point that has greatly perplexed me: if — as I believe — external conditions produce little *direct* effect, what the devil determines each particular variation?" (See already the letter to Hooker of 23 Nov. 1856). Later he came to ascribe to the conditions of life a greater direct influence on the variations (see his letters to Carus 4 May 1869, to Wagner 13 Oct. 1876, to Neumayr 9 March 1877). In " Variation of Animals and Plants under Domestication" (Chap. 22—26) Darwin *[448]* discusses the question more closely and emphasizes not only the direct action of the conditions of life but also the effect of the use or non-use of organs and faculties. (See also the "Descent of Man" ch. 4). — There is all the more reason in studying Darwin to be attentive to this point, as his critics have often confused the original origin of the

variations with the natural selection of properties that have come to be through such varying. "Natural selection," says Darwin against this (Variation of Animals and Plants. London 1868. II p. 272), "depends on the best endowed individuals remaining alive under different and complex circumstances, but has nothing whatever to do with the original cause of any change whatsoever in structure." Darwin's hypothesis concerns chiefly the effects of selection among the variations, not the origin of the variations. Naturally he assumed that there were causes of this too; but it was not the main thing for him to find them. Every hypothesis must build on a certain foundation that it cannot draw into the proof along with it. There is then no contradiction in Darwin's taking the variations essentially merely as factually given, and it is a strange objection that has sometimes been directed against him that he builds on "chance." Darwin does indeed use a few times the expression "chance-variations," thus in the letter to Hooker mentioned above, but by it he means variations whose cause is not known, and he himself declares the expression inexact. Variability depends more on the nature of the organism than on the character of the surroundings (More Letters from Ch. Darwin I. p. 377); this explains its mysterious character. Darwin declares (in a letter of 1881: More Letters I. p. 391) that the discovery of the causes of variability will be one of the most important tasks of the research of the future. — Just as Darwin regarded the first origin of the variations as enigmatic in several respects, so he also declared the origin of life as a whole to be an unsolved enigma.

On the other hand Darwin found no reason to assume that in the origin of the human
[449] species from lower forms quite peculiar forces were at work. As soon as the conviction had dawned on him that the species had arisen by natural development, it became clear to him that the human species could form no exception to this. But in the "Origin of Species" he contented himself with hinting that the new doctrine would also be able to throw light on man and his history; there was no room here for special investigation of a single species of living beings. When Darwin later took up this question in detail (in the work "On the Descent of Man"), it was, as he declares in a letter, partly because he was accused of not daring to come forward with his views on this point. He stood here even more alone than on the general question of the origin of species. Even men like Lyell and Wallace, who otherwise adhered to him, became doubtful here. Darwin himself felt no conflict between feeling and the impulse of research. Man's real worth and highness was for him not diminished by his having developed from lower forms. Against the theological and romantic conception that regards man as a fallen angel he set the realistic conception of man as an animal that has developed into a mental being.

Neither in mental nor in bodily respects did he assume more than differences of degree between man and animal. There is, he asserts, a far greater leap between the mental faculties of one of the lowest vertebrates (the lamprey or the lancelet) and those of one of the highest apes than there is between the apes' and men's mental faculties. And he shows how difficult it is to draw the boundary between mere instinct and reason proper. The very fact that animals can learn by harm shows that one dare not deny them reason. And the case is similar with memory, the sense of beauty and sympathetic instincts.

It has sometimes been concluded from Darwin's doctrine that there must really take place an uninterrupted progress in perfection in all living beings, and one has then found a great objection to the doctrine in the fact that the lowest organic forms continually go on existing. Here too Darwin himself has already answered the objection. Natural selection does not necessarily bring progress with it. What use would an intestinal worm *[450]* or an earthworm have of more perfect organs than it has? An organ that does not help in the struggle for existence takes up room and force to no use. Where circumstances for some reason or other exclude strong rivalry, it is perfectly intelligible that a form of life remains as it is through immeasurably long periods of time. (Origin of Species ch. IV. p. 154—155. Cf. ch. X. p. 422 f). Natural selection can only make every organic being as perfect as it needs to be in order to rival or "struggle for existence" with other organic beings, and the remarkable thing, according to Darwin, is that experience shows us so few cases where such perfection is not present. But perfection must always be measured by the relation to the conditions of life. (The natural selection of each species implies improvement in that species *in relation to its conditions of life.* Letter to Lyell. 25 Oct. 1859). In this it is then also implied that natural selection can sometimes bring about a regression to simpler and more elementary forms of life, namely when the conditions of life for some reason or other are simplified, so that various organs become superfluous, if they do not become harmful. There is no inherent or necessary tendency in organic beings to rise upward on the ladder of organization. (Variation etc. I. p. 8).

d. Ethical-Religious Consequences

Apart from the objections that point to supposed self-contradiction or disagreement with experience, objections of an ethical-religious kind have also been raised against Darwin's theory, and that not only from the theological side; even so radical a thinker as Eugen Dühring has here taken offense on ethical grounds and expressed himself in the most violent way against the theory. The very idea of the struggle for existence seems to many incompatible with an ethical conception of life: how are love of mankind and

conscience to be reconciled with this idea?

[451] As regards the ethical objection, it is so far from being the case that Darwin has overlooked it, that he has with full consciousness discussed the ethical problem in connection with his theory. As a moral philosopher he stands at a standpoint similar to the one that Shaftesbury and Hutcheson founded, and that in the nineteenth century Comte and Spencer carried further, only that Darwin's natural-historical point of view and his doctrine of natural selection give a broader basis for the whole standpoint.

Darwin finds in the fact that man is the only being that can with certainty be designated a moral being the greatest difference between him and the animal. But from this it does not follow that the moral feeling should not have developed in a natural way, and there is nothing in it either that conflicts with the struggle for existence or with natural selection. One must only hold fast that the properties and faculties that natural selection fosters are not only such as are for the good of the single individual, but also such as are for the good of the whole group or species. In all such animals for which it is most fortunate to live in close association, those individuals that found the greatest pleasure in the company of others would most easily escape dangers. And since the continued subsistence of the species depends on the preservation of the very often helpless offspring, it will be easy to understand how the love of parents for their offspring can have been developed by natural selection. Experience shows that animals sometimes expose themselves to danger to save other animals. A tribe or group of animals or men in which mutual compassion and the need to help one another prevail will be especially fortunately placed in the struggle for existence, more fortunately than other groups where each cares only for himself, and where the forces are not united toward a common end. Thus in the single individual, by virtue of the law of natural selection, properties can be preserved and developed that are beneficial to the subsistence of society, not especially to that of the single individual. Not only the selfish but also the unselfish feelings have their natural history. However great the distance may be between what stirs in animals when they show love and self-sacrifice for one another, and the highest
[452] human morality, there are between the two countless degrees, and one has at no point the right to assume that natural development is interrupted. Within the human world itself there are, after all, very great distances in moral respects. There are states and forms of human life that stand even far below what animal life can show us. Darwin declares that he would rather be descended from the ape that risked its life to save its keeper than from the savage who delights in torturing his enemies, kills his children without remorse, treats his wives as slaves and is himself a slave of the most dreadful

superstition[102]).

The moral feeling presupposes, according to Darwin, besides the desire to live together with others (sociability) and the capacity for compassion with others (sympathy), also the capacity to remember and compare. When these presuppositions are present, past actions will be able to be drawn forth and judged by what the strongest (in the moment of remembrance, if not at all moments the strongest) feeling demands. When the faculty of speech is developed, mutual praise and blame will act on the single individuals. A public opinion will take shape. Further, constant habit and practice in working for common interests will root and strengthen the social motives and instincts. Perhaps an inheritance of dispositions in this direction will also take place. — A confirmation by experience of his moral-philosophical theory Darwin finds in the investigation of the properties that at different times and among different peoples have been acknowledged as virtues, and of the different extent of the circles of individuals whose weal and woe the moral feeling has at different times taken into account.

The very theory of natural selection thus shows us that not every adaptation, every selection, every form of the struggle for existence is approved. From lower forms of this struggle we are led over to higher forms, first and foremost to a stage where a valuation of the forms of the struggle can take place[103]). —

It was found not only that Darwinism was an immoral doctrine, but also that it was a materialistic and godless doctrine.

Darwin has nowhere expressed himself on the relation between the mental and the material. He keeps to the fact that mental life in men and animals is, as a matter of fact, *[453]* bound up with the activity of material organs and so far can be studied as natural history like other organic phenomena. He therefore investigates the development of mental life through the struggle for life from lower to higher steps, and the definite laws of this development. In this there is nothing materialistic. Just as he declares the origin of life an unsolved problem, so he would surely also have declared the origin of mental life unexplained; perhaps the two problems coincided for him. Darwin has made interesting special psychological investigations, namely, besides those on the moral feeling, on the sexual feelings and instincts, on instincts in general, on the expression of the emotions, on the development of the child in its first year. One finds all through his writings, often where one does not expect it (e.g. in the little book on the influence of earthworms on the formation of mold), interesting psychological traits. But to discuss the principles of psychology itself was not his business.

If by materialism one merely means the reduction of phenomena to definite natural

laws that exclude supernatural intervention, then Darwin is certainly a materialist. The organic forms have, according to him, not stood as fully finished ideals before the Creator's imagination, to be thereupon set into material reality, but they are results of long processes of development from the most inconspicuous beginnings, under the constant action of the conditions of life. Darwin has widened the domain of natural connection; he has brought it about that the habit of thinking positively and doing without theological causes has increased among naturalists and thereafter also in wider circles. In principle nothing new has in a certain sense been added by this; for the principle of natural causes had long since been laid down — but it is not frequent in the history of science that this principle finds so brilliant an application as here. When in recent times several English theologians have accepted Darwin's doctrine, they help themselves by placing the creative act at the beginning of life, when the primeval form or forms arose, and natural selection then becomes for them the way, fixed by the Deity,
[454] in which the single species arise. To this expedient Darwin himself, however, could not resort. In no less a degree than Stuart Mill he found it impossible to reconcile the suffering and the disharmonies in the world with the providential government of an almighty being. He was no pessimist; he had the conviction, which to be sure he found very difficult to prove, that happiness in the world decidedly outweighed unhappiness. But natural selection, which brings suffering and unhappiness with it and breeds hatred and cruelty no less than love and gentleness, could not in his eyes be regarded as a means in the hand of a providence. Should the ichneumon wasp have been created on purpose to devour living caterpillars, the cat to play with the mouse? Already in his writings (especially "Variation." II. p. 432) he expressed himself in this direction, still more strongly in his letters and in his autobiography. It was the fact of evil that he again and again came back to when he was invited to express himself in religious respects. When he came home from his great voyage he still believed in revelation, and when he published the "Origin of Species" he was still a theist. But little by little, without any painful break, his views changed, and he at last declared himself (in a piece of the autobiography written in the year 1876) an *agnostic*, that is to say, one who is conscious that our knowledge cannot solve the problem. The expression had first (1859) been used by Darwin's friend and disciple Huxley (who on this point was influenced by William Hamilton). As little as he could reconcile himself to the thought that the world as it is was due to conscious intention (design), so little could he reconcile himself to the thought that it should be a result of chance (chance or brute force). "The safest conclusion," he wrote to a young man who wanted to know his opinion, "seems to me to be that the

whole question lies beyond the reach of the human understanding. But man can do his duty."

Agnosticism really means that what stands for us as a dilemma need not be one in itself, since there could be more possibilities besides those that are for us the only ones: design—chance. Darwin ends with the same result as Kant in the "Kritik der Urtheilskraft," who declared that the distinction between mechanism and teleology perhaps belonged *[455]* to the oppositions that we must set up by virtue of our knowledge, but that we had no right to take as valid for existence itself.

2. Herbert Spencer

a. Biography and Characterization

The year before Darwin's famous work appeared, Herbert Spencer drew up an outline of a systematic exposition that was to show the significance of the concept of evolution in the various domains. His earlier works had led him to conceive this concept as a principal concept in our knowledge, and he now wished to devote his life to carrying out this conception in detail. As he was without means, it was his wish to be enabled with the government's help to carry out his plan. He was warmly recommended by men who knew his earlier works, men such as the philosophers Stuart Mill and Fraser (Hamilton's successor in Edinburgh), the historian Grote, the physiologist Huxley, the botanist Hooker and the physicist Tyndall. The work was reckoned first at seven, later at ten volumes; in the actual execution it came to comprise nine volumes. Although Spencer obtained no government support, he has nevertheless with great perseverance, and in spite of his often failing strength, carried his plan through: the first part of the work *(First Principles)* began to appear in the autumn of 1860, and the last part (the conclusion of the *Principles of Ethics)* appeared in the spring of 1893. The intervening parts treat biology, psychology and sociology. It will be of interest to see how the thought of this comprehensive work arose and matured[104]).

Herbert Spencer was born at Derby on 27 April 1820. His father was an excellent schoolmaster, who laid great weight on the view that sound mental development comes only through self-education, and therefore sought to get his pupils to think and observe on their own. Spencer showed an early interest in natural science and history. It was his delight to follow the development of insects; in his father's library he made himself *[456]* acquainted with books of various content, and together with his brothers he eagerly discussed scientific, political and religious questions. His parents were Methodists; but the father gradually became dissatisfied with the clerical organization of this communion and

joined the Quakers, without, however, adopting their special doctrines; what appealed to him was their "unpriestly system." The son Herbert then went on Sundays with his father to the morning service of the Quakers and with his mother to the evening service of the Methodists. The result was that in the end he could not bear to hear Bible texts. Later Spencer came to live in the house of an uncle who was a clergyman (of the Broad Church tendency), an excellent man who took an eager part in the agitation for the repeal of the Corn Laws and had merits in the regulation of poor relief. Spencer's critical position toward all far-reaching attempts to order social relations by the intervention of the power of the state, and his great faith in free development, were founded and nourished by the religious and political views he met with in his early youth. It was intended that he should be a schoolmaster, like his father; but for a number of years he worked as a civil engineer. Mathematical studies and mechanical inventions together with political agitation filled a great part of his time. The first works mentioned from his hand are articles on the proper sphere of governmental authority and a treatise on the nature of sympathy, in which he set forth a theory similar to the one Adam Smith had in his time given. Already earlier he had, through his natural-historical studies, become an adherent of the theory of the natural development of species. He has himself declared that the study of Lyell's geology, which in the older editions combated Lamarck's theory, was precisely what led him (in the year 1839) to see the correctness of the theory of natural development. This also had an influence on his religious notions, which changed without his being able to trace the decisive change back to any definite point of time.

In his first considerable work, *Social Statics* (1850), he conceived social development
[457] by analogy with organic development, a conception that also plays a great role in his later writings. The perfect unfolding of life is here for him a divine idea that is to be realized, and of which nature offers intimations and approximations. Spencer has himself declared that in this work he was influenced by Coleridge and through him by Schelling. This theological mode of consideration he later abandoned for a purely factual demonstration of development under its various forms. The greatest influence on his theory of development he himself ascribes, already for this period, to "the truth that Harvey's embryological investigations first dimly indicated, that was later grasped with greater clarity by [the anatomist] Wolff, and that received its definite form from von Baer, the truth that all organic development is a change from a state of homogeneity to a state of heterogeneity." This truth, which had been found valid for the development of the single organism, stood for Spencer as valid for development in all domains, and it dawned on him at the same time that in all domains development takes place. In a little

essay on the development hypothesis from the year 1852 he makes a comparison between the theory of development and the theory of creation, and assumes — relying on the variations of domestic animals and cultivated plants, on the difficulty of distinguishing between species and variety, and on the likeness of different forms in the embryonic state — that the species have acquired their present forms by development under the influence of external circumstances. Decisive for his whole point of view was the working-out of his *Principles of Psychology* (first edition 1855). What is significant about this work is that — although it takes its stand on the ground of the philosophy of experience — it insists on the impossibility of explaining the individual consciousness out of the single individual's experiences alone. All earlier empiricism had trusted that it was only a matter of finding the experiences the single individual has, in order to understand its conscious life. It was a conviction that had not only theoretical but also practical significance. For if man's conscious life is determined merely by the individual's own experience, there opens up a prospect of giving it, through purposive arrangement of education and social relations, the stamp one wishes. Stuart Mill and Comte built to a great extent their hopes for the *[458]* future development of the species on this conviction, although both these thinkers' deep understanding of historical development stood in a remarkable conflict with it. Spencer now arrived at the conviction that development in the mental as well as in the material domain proceeds very slowly, through countless intermediate links and stages, of which none can be skipped. It is the constant influence of the conditions of life that carries life forward to new forms, among others also to the forms with which it now appears. We cannot therefore understand the conscious life of the single individual from the experiences to which it has itself been subject, but must go back to the experiences of the species. And it is not here only the traditions that are at work; in the innermost structure of our mind, in the way in which representations are combined, and in the direction in which feelings and drives unfold, hereditary tendencies are at work that become intelligible only by assuming an after-effect of the experiences of preceding generations. Heredity had hitherto stood as a curiosity to which no significance was ascribed outside single striking examples; now it came to stand as a constantly co-operating factor even in the very highest forms of life. It was this widening of the horizon in the psychological domain that led Spencer to search out the general laws of development, and at the same time to investigate whether these laws might not be derivable from the fundamental laws of our knowledge itself. With this the idea of his philosophy of evolution was given. The first outline of it appeared in an essay *Progress, its Law and Cause* (1857). (Danish translation in "Mindre Afhandlinger af Herbert Spencer." Copenhagen 1877). Here he

carried through the thought that all development is a transition from homogeneity to heterogeneity, and sought to derive it from the law of the conservation of energy. In the first edition of "First Principles" he still sets up only this one form of the law of evolution; later he supplemented it with two others: evolution as a transition from dispersion to coherence and as a transition from an indefinite to a definite state. And thereafter the publication of the great work went forward.

There were several preliminary works besides those mentioned. In a series of essays
[459] that were published in various journals, Spencer had already, before the working-out of his great work, treated various subjects of a philosophical, natural-scientific, sociological, aesthetic and ethical kind. His systematic works are only further, broad (often far too broad) elaborations of what is laid down in these short, clear and often also in form masterly essays, which are published collected in three volumes of *Essays*, and which are perhaps the part of Spencer's production that will last longest.

During the working-out of the first edition of his Psychology, Spencer through overexertion brought on himself a nervous debility that he has never since got over. He came to suffer from chronic sleeplessness. At last it threatened that he would have to give up all work. But in later years (from 1890) he has in his advanced age not only been able to complete his work, but also with life and vigor to take part in the newest discussions (called forth by Weissmann's hypothesis) in the domain of the biological doctrine of evolution. The last thing he has published is *Facts and Comments* (1902), a collection of short essays. He declares that it will be his last book.

The most decisive question in evaluating Spencer's theory becomes whether, by the extension he gives to empiricism and positivism with the help of the concept of evolution, he has succeeded in reaching the reconciliation of hitherto mutually conflicting views at which he aims. But even if he should not have succeeded in this, — even if old problems, after having passed through the purgatory of the philosophy of evolution, rose up in a new form — his widening of the psychological horizon and his demonstration of the concept of evolution as the principal concept of special research do not lose their significance. It must, in evaluating his work, be held fast that it was in this widening and this demonstration that his real new thought consisted, not in the setting-up of any definite epistemological or psychological theory. Moreover it cannot be otherwise than that during the working-out of a work that extends through some thirty-odd years unevennesses and inconsistencies in the treatment must arise, especially when his views
[460] on single points change before the completion of the work.

Spencer's works do not give us much access to knowledge of his inner life. The

impression one gets is that he is not so subjective a nature as Mill and Comte, in spite of all their positivism, were. Yet one finds between his long inductions and deductions, between his formulas of evolution and his statements of fact, utterances of warm feeling and quiet enthusiasm. He does not have, like Mill, the need to influence men directly. The doctrine of evolution has opened his eyes to the hard struggle in which everything living is placed, and to the difficulties with which especially man's spiritual and social life has to contend. It has taught him that ideals that to the preceding generation still seemed able to find their realization in the nearest future presuppose a long series of intermediate links that must all be gone through, and therefore come to lie as an immeasurably distant goal for the wandering of the species. Therefore resignation comes to play a great role in the view of life that arises on the basis of his doctrine. Not as if it were his opinion that the present should altogether fade or be sacrificed for the distant goal. But in such a way that the present is not understood merely by being seen in its connection with the past out of which it has developed, but only when one at the same time holds fast that it is a transition to a more developed state. Spencer surely belongs himself among those of whom he speaks in the words with which his great work closes (Princ. of Eth. II, p. 433): "The highest ambition of the good man will be to have a share — even if it is an extremely small and hidden share — in the development of what is human ('the making of Man'). Experience shows us examples of interest arising from the pursuit of altogether unselfish ends; and in the course of time there will be more and more of those whose unselfish end is the further development of humanity. When from the heights of thought they behold the distant form of the life of the species, which they themselves will never come to enjoy, but which will first fall to the lot of a distant posterity, they will feel a quiet joy in helping toward its coming."

The year after Spencer had conceived the idea of his philosophy of evolution, Darwin's book on the origin of species appeared, through which the doctrine of evolution acquired a wholly new and firmer basis. Darwin names Spencer in the preface among *[461]* his predecessors and later calls him (in a letter to Ray Lankester 15 March 1870) "the greatest living philosopher in England." Darwin's whole theory fitted excellently into Spencer's system, only that Spencer — with all acknowledgment of the significance of natural selection — laid more weight on direct development through the use of the faculties under the influence of circumstances than Darwin did in his earlier years. — Of no small interest is it that Stuart Mill, who at first was doubtful about Spencer's evolutionist psychology, later declared himself convinced that not only in the individual but in the race, through inherited dispositions, a mental development takes place. This

changed conviction he expressed the year before his death in a letter to the physiologist Carpenter (printed in the latter's Mental Physiology. p. 486).

b. Religion and Science

Spencer begins his systematic exposition *(System of Synthetic Philosophy)* with a section on the Unknowable[105]). It contains a doctrine of the limits of knowledge and at the same time examines the relation between religion and science. It is to show what validity can be ascribed to the scientific world-view that the remaining parts of the system seek to build up. There is here something inexpedient in the manner of exposition of which Spencer has not been aware. Neither epistemology nor the philosophy of religion can attain full clarity and grounding when they are not connected with psychology — but psychology comes only far along in the system!

The line of thought in the first section is as follows. In recent times religion and science stand as opposites. This comes from religion's wanting to solve problems that only science can solve, and from science's wanting to intrude upon domains where only religion belongs. Every religion has hitherto sought to give a theoretical explanation of existence, presenting itself as an unveiling of the riddle of the world. Different as the
[462] religions are, they nevertheless agree in this, that they would be revelations of something that would otherwise be unknown. The difference between the lower and the higher religions consists in this, that the lower — fetishism, and polytheism as well — think that it is very easy to form a notion of what is at work in existence. It is above all the belief in the spirits of ancestors that forms the foundation of primitive theology. One has, no doubt, a presentiment that the world is at bottom a riddle; but this presentiment is very weak, and the riddle is held to be easily solved. In the higher religions ever more weight is laid on the mystery, and it comes at last to this, that every image, every thought becomes inadequate to express the innermost of all things, and it is declared a blasphemy to think the Deity in the way in which alone we *can* think it. The religious development will be complete when it is recognized that the mystery is greater than the religions have hitherto assumed. The religions assumed a relative mystery, a mystery that could be unveiled. But the mystery is absolute when no image, no thought any longer suffices. The contradictions to which the religious notions constantly lead arise from the wish to assume an absolute cause. Whether one will assume that the world has produced itself, or that an external cause has produced it, one ends in the contradiction that something should be able to be the cause of itself; in the first case the world would have to be the cause of itself, in the second case the being that has produced the world. The more

special contradictions in the religious notions — between omnipotence on the one side, goodness and justice on the other, between justice and grace etc. — are, as Mansel has already shown, only special consequences of that fundamental contradiction.

But it now fares, Spencer continues, with the underlying scientific concepts just as it fares with the religious attempts to express the innermost nature of existence. Concepts such as time and space, motion and force, consciousness and personality are clear and serviceable so long as we keep within the limited and relative world of experience, but lead to contradiction when we would use them as expressions of the nature of an absolute being. Our scientific knowledge moves — as regards both the outer and the inner world — in the midst of a multiplicity of constant changes, without our being able *[463]* to see either beginning or end, either the first cause or the final end. It follows, moreover, as Hamilton and Mansel had already shown, from the very nature of our knowledge that we comprehend only the finite and limited. All knowledge presupposes a distinguishing; but the Absolute cannot be distinguished from anything else, since there can be nothing besides it. Spencer adds that all knowledge also presupposes an apprehension of likeness, a referring of what is to be known to something kindred; but the Absolute cannot be kindred with anything else, since there can be nothing besides it.

Yet Spencer does not grant Hamilton and Mansel that the Absolute is a purely negative concept. Knowledge must assume that there is other and more than it can comprehend. Its activity consists in distinguishing, apprehension of likeness, determining and limiting; but there must be something that is determined and limited, something that is formed, and that has subsistence independently of the determinate form it receives for our consciousness. What thus forms the constant foundation, for us indeterminate, of the whole content of our knowledge, we represent by analogy with what we sense as our own *force* in muscular exertion. There is a force to which it is owing that anything at all can become an object of our consciousness. It is this force which not only produces the individual changes that we distinguish and apprehend, but which also underlies that in us which persists through all changes. A concept of this force we cannot form; we can only approach it by successively thinking away the limits under which the unknowable force appears in each individual case in our experience. It is, then, the primitive sensation of force that in the end gives our knowledge its content, both in the shifting changes and in what remains constant while these take place (something constant under all modes).

Religion and science will in the end be able to meet in the conviction that the innermost essence of existence is unknown and incomprehensible to us, whereas we can have a scientific knowledge of the way in which it manifests itself in the world of *[464]*

experience. Our consciousness of existence is, seen from the one side, religion, seen from the other side, science. The strife will cease when the limit of knowledge is fixed in the right way; then religion will not draw anything that can be comprehended under *its* authority, and science will not draw anything that cannot be comprehended under *its* authority. There can be no strife between our intellectual and our moral faculties: it therefore cannot be our duty to believe something that contains a self-contradiction. It cannot, as Mansel assumed, be our duty to think God as personal and yet infinite, when personality and infinity exclude each other. Our duty can only be to submit to the limits of our knowledge and to acknowledge the mystery that really is there. When the concept of personality cannot serve as an expression of the Absolute, it is not because it is too high a concept, but precisely because it is too narrow a concept. A divine psychology we cannot give, and the strife between religion and science continues so long as the adherents of religion claim for themselves an intimate acquaintance with what is an eternal mystery, and therefore set up concepts that can be attacked by criticism.

Spencer is fully aware that his attempt to mediate between religion and science will meet with resistance. Men need living and concrete notions, need to represent what underlies all things as a being akin to themselves, and only when this need can find expression in their faith can their faith gain influence on their actions. So long as there has not developed an *organic morality,* which lets man act ethically from inner need, so long is it of importance that the influence of religious opinions should not be lost. Every faith stands in a definite relation to the stage of development on which man stands, and a hasty rejection of religious faith will therefore easily let the imperfections of our nature come forward more than they did while the old faith ruled. In any case so profound a spiritual revolution cannot take place without pain. He who is impatient because he finds that the process of transition lasts too long must remember that in every religion,
[465] even the lowest, there is "a soul of truth", — that even the lowest religions give their adherents something they could not get in any other way, — and that all forms of faith are links in a great process of development that is still far from its end. If he considers this, he will more readily put up with the sophistical reasons by which traditional opinions are defended, with the unworthy flattery with which men worship their deity, with the arrogance of ignorance, which far surpasses the arrogance of science, and with the condemnation of actions that proceed from unselfish compassion and pure love of the good. —

One misgiving about this doctrine of the relation between religion and science, which Spencer does not touch upon, is that the Absolute and the Relative are distinguished

in an all too external way[106]). Through this relation of opposition the Absolute itself becomes relative. That the opposition between the Absolute and the Relative cannot consistently be so external on Spencer's own conception is seen from his statement that a completed concept of the Absolute could be reached only by the removal of *all* limits and relations: such a removal is, after all, an infinite process, and we cannot, therefore, in reality set the Absolute and the Relative over against each other as two *finished* concepts. When, moreover, Spencer states in some places that the Absolute, which for him is one with the Unknowable, works through all phenomena, the inner as well as the outer (thus First Principles § 34, 46 and 93), then its essence must manifest itself in the forms and laws under which the phenomena appear; it cannot in reality be wholly unknowable, and Spencer is wrong to declare the chief concept of experience, the concept of evolution, wholly invalid as regards "the Absolute". It would be a self-contradiction if the general laws and forms that experience presents should have no significance whatever for what lies behind or beyond experience. No proof can be given that evolution should concern only the shell of existence and not its kernel at all. A dualism has here remained in Spencer. *[466]*

c. Philosophy as Unified Knowledge

The problem of the relation between religion and science arises not only every time a special science draws a new phenomenon within its domain, but also when an attempt is made to unite the various truths that the special sciences have found into universal truths. It is, according to Spencer, the task of philosophy to carry further the work of demonstrating unity in the multiplicity of phenomena, which the special sciences each carry on in its own domain, by seeking laws that are common to all our various domains of experience. The point is to find a highest truth from which the fundamental propositions of mechanics, physical and psychological principles and social laws can be derived. We should then have founded a philosophy as completely unified knowledge (completely unified knowledge).

Philosophy begins with a provisional assumption of the validity of the fundamental propositions on which all thinking rests. This provisional assumption is justified later by the consequences drawn from it agreeing with experience. The validity of an assumption can only be established by its agreement with all other assumptions. Truth can for us consist only in complete agreement between our representations (representations of things) and our observations (presentations of things). When our expectations and our observations do not agree, the assumption from which we started has been invalid.

The complete establishment of an assumption's validity by its agreement with all other assumptions leads us again to that working together of all knowledge into unity in which philosophy consists. But this also implies that every demonstration of the validity of individual assumptions presupposes the validity of the activity of thought by which we find things like or different. This activity of thought underlies all knowledge, both observation and inference, whatever it concerns. Every proof presupposes this *primordial act,* and its validity therefore cannot be disproved; a disproof is, after all, also a proof.

[467] We think always in relations. Next after the relation of likeness and difference, the relation of succession and that of coexistence are of importance. The relation of coexistence is derived from the relation of succession: we represent as simultaneous such different phenomena as can be represented just as well in the one order as in the other. The fundamental experience that underlies every apprehension of likeness or difference, succession or coexistence, we already know; it is the sensation of force: of something that offers resistance or produces change, and that we represent to ourselves by analogy with our own sensation of effort. Matter and motion are only manifestations of force, and time and space are forms of manifestations of force. Both in the domain of inner and in that of outer experience the concept of force is the ultimate concept to which we come back. We rightly distinguish between the self and the not-self; but both are only different ways in which the concept of force manifests itself. The matter and content of our thinking are all the different forms of force.

The concept of force is thus of a symbolic kind. It is the ultimate symbol (the ultimate symbol). It points back to our subjective experience, by analogy with which we apprehend everything else. Spencer here recalls the metaphysical idealism that uses this analogy to solve the problem of existence. He concedes that if we had the choice between reducing the mental elements to the material or the material to the mental, we should have to choose the latter. It is absurd to reduce the known to the unknown; an intelligible hypothesis we get only when we reduce the unknown to the known, and thus regard the objective, material elements as being in their essence of the same nature as the subjective elements of our consciousness. In his "Principles of Psychology" § 63 Spencer says: "Let me say here, once for all, that if we were obliged to choose between the two possibilities, of reducing mental phenomena to physical phenomena and of reducing physical phenomena to mental phenomena, this latter possibility would stand as the more reasonable (the most acceptable) of the two." But such a reduction Spencer considers impossible, since we cannot avoid making the mental elements clear
[468] to ourselves by means of forms and relations that are taken from external nature. A clear

intuition of the nature of consciousness we get only by using images from the material world. Moreover, we cannot derive and explain the physical phenomena from the mental phenomena. We must therefore stop at the difference between psychical and material phenomena and content ourselves with showing that the same laws of experience hold for both kinds of manifestation of force.

Now it is, according to Spencer, a fundamental presupposition of all science that force neither arises nor passes away in existence. All thinking consists in setting something in relation to something else. If, now, force were to perish, or if force arose out of nothing — which in outer experience would show itself in matter's being annihilated or arising, or in motion's being interrupted or beginning, — then we should have a relation between something and nothing or between nothing and something; but a relation one of whose terms has vanished from consciousness or has not arisen in it is a self-contradiction. The force that must thus always be assumed to persist is not any relative, phenomenal form of force, but the absolute force that stirs in all things, and so we here come back again to the thought that an Absolute must underlie all the phenomena of experience, the fundamental thought common to religion and science. It is the business of empirical science to demonstrate the special transitions that take place between the various forms of force. But the principle of the conservation of force itself cannot be proved by way of experiment, since every experiment, on closer reflection, turns out to presuppose its validity. All weighing and measuring presupposes, after all, that the unit we take as a basis does not change while the weighing or measuring goes on. If the force with which the weight tends toward the earth varies while the chemical experiment that is to determine the atomic weight is being performed, then the conclusion drawn from the experiment is invalid. The conservation of force is therefore a principle or a postulate on which all investigation of the real world rests.

For Spencer, as for Hamilton and for Kant, the principle of the conservation of force coincides with the causal principle. We shall also find that several of the natural scientists who set up the principle of the conservation of force on experimental grounds assumed *[469]* that it was really a proposition of reason, so that what mattered was to show how it manifests itself in experience. So Robert Mayer, Joule and Colding. There is, however, surely a distinction to be made here. For there could, after all, be lawfulness in nature even if force arose and force passed away, provided only that this arising and passing away were bound to definite conditions. One could imagine that when the condition A was present, B always arose, although B in quantitative respect might be put = A + x, and that when the condition C was present, D always arose, although D was = C—y. In that

case the causal principle would hold without the principle of the conservation of force holding. This shows that this latter principle is a more special principle than the causal principle, when the latter merely expresses a rule for the occurrence of phenomena. All regularity in nature would not be gone because the law of the conservation of force did not hold. It is another matter that we wish, as far as possible, to establish that just as in the logical inference there is no more than was already in the premises, so in the real effect, too, there is no more than was already in the cause; for thereby the principle of causality and the principle of grounding will be brought into the greatest possible agreement with each other. Even without this complete analogy with logical grounding, however, the causal relation could have validity. Therefore Spencer's deduction does not suffice, even if he is right that every attempt to prove the conservation of force experimentally in a certain sense presupposes it.

From the principle of the conservation of force Spencer derives that motion goes in the direction of greatest attraction or least resistance, — that all motion is rhythmical, — and that all phenomena are subject to evolution and dissolution. By this road he passes from the consideration of philosophy as unified knowledge to his consideration of philosophy as the doctrine of evolution. But before we follow him to this consideration, we must first ask what consequences Spencer draws from the law of the conservation of
[470] force with regard to the relation between the mental and the material phenomena. It has not been Spencer's intention to discuss this problem in particular, and one finds in his writings, as they now stand, a certain wavering on this point, which is explained by Spencer's having changed his conception in the interval between the various editions of his books, without this change having been carefully carried through at every point in the new editions. Hence arise the contradictory utterances that are found in him[107]). At first (in the first edition of "Principles of Psychology" and "First Principles") he conceived the relation between the mental and the material after the likeness of the relation between the various forces of nature, and assumed a transition from the material to the mental just as from motion to heat. The arising of sensations is thus explained by the law of the conservation of force. Insofar as one should wish to find (as some critics did find) materialism in this conception, Spencer reminded them that matter and motion are for him in the end only symbolic expressions for the unknowable force that stirs in all things. But later he became aware that the arising of the phenomena of consciousness could not be derived from the principle of the conservation of physical energy. In particular, the arising of sensation out of motion, as an equivalent for the latter, is incompatible with the doctrine of the continuity of motion. In later editions, therefore, Spencer conceived

the relation between the mental and the material according to the identity hypothesis, in such a way that they stand as two phenomenal forms of the unknowable force, mutually irreducible. With regard to the doctrine of evolution, which is the main thing for Spencer, it comes to the same which hypothesis one holds to. For whether one assumes a relation of interaction or a relation of identity between the mental and the bodily, evolution in both domains can present the same fundamental features, and that is what is decisive for the philosophy of evolution. According to the identity hypothesis, to be sure, mental evolution cannot be explained by deduction from the conservation of physical force; but the material process corresponding (obverse) to mental life can nevertheless be explained in this way[108]). When Spencer in his doctrine of evolution explains all evolution by the laws of matter and motion, and yet uses examples from psychology and sociology no *[471]* less than from astronomy and physiology to illustrate the forms of evolution, this is justified by the fact that the mental phenomena, whichever hypothesis one starts from, are assumed to be bound in a lawful way to material phenomena.

d. Philosophy as the Doctrine of Evolution

The unified knowledge that philosophy seeks is won, according to Spencer, not only by a demonstration that there is a common presupposition on which all inquiry rests, but also by the demonstration of a law that is common to all phenomena that experience presents. Spencer systematizes positivism partly by reducing all positive knowledge (knowledge of facts) to a common presupposition, partly by setting up a common law or form for everything positive, all phenomena. Every phenomenon has its history: it comes into being for us, and it vanishes again for us. Each single science describes the history of *its* phenomena; the point now is to examine whether these various historical processes do not present common features, so that *a general law of evolution* can be set up. It then turns out that all evolution more or less clearly presents three different characteristic features, which together give us the complete concept of evolution. — As already remarked, Spencer had at first set up only one characteristic feature of evolution, namely the transition from homogeneity to heterogeneity. Later he saw that there were quite simple forms of evolution in which this feature was wholly subordinate, and that, moreover, in order to be able to distinguish sharply between processes of evolution and of dissolution, it was necessary to set up yet a third characteristic feature.

1) *Evolution as concentration* (or integration). When a phenomenon comes into being, there takes place a gathering, combination and concentration of elements that previously occurred in a dispersed state. When a cloud forms in the sky or a heap of sand on the

beach, an evolution of the simplest kind has taken place, the process consisting almost exclusively in a gathering and heaping up. Such a process of concentration took place,
[472] according to the hypothesis of Kant and Laplace, at the first coming into being of our solar system out of the primeval nebula, whose parts were previously in a more dispersed state. All organic growth takes place through the taking up into the organic tissue of elements that were hitherto found dispersed in the surrounding world. A psychological example is had in every generalization, every formation of general concepts and laws: we thereby concentrate in a single thought a whole multitude of different observations and representations. Social evolution consists first and foremost in a joining together of individuals or groups of individuals who formerly lived in a more dispersed state.

2) *Evolution as differentiation.* Only in the very simplest cases can evolution be described as a mere process of concentration. There will not only take place a separation of the whole mass from its surroundings; but within the mass thus set apart special concentrations will in turn take place, so that the evolution becomes compound. And in the course of evolution these special concentrations will come forward more and more, so that on comparing the earlier and later stages we find a transition from homogeneity to heterogeneity. In the evolution of the solar system there occurs a separation between different globes, each with its peculiarity. Organic evolution advances from the homogeneous germ to the organism furnished with different kinds of tissue and with differently built and differently acting organs. The whole of organic life has, according to the hypothesis of Lamarck and Darwin, been more homogeneous at earlier stages, the differences of species now existing being due to evolution from common ancestral forms. The senses develop, when we compare earlier and later stages, from a less distinct and less exact faculty of apprehension to ever greater distinctness and exactness, so that more differences can be apprehended. Mental life is measured not only by its concentration but also by its richness. In the evolution of society different estates and classes are formed through the division of labor.

3) *Evolution as determination.* In a process of dissolution, too, differences arise in a hitherto homogeneous mass. To distinguish evolution from dissolution it must then
[473] be added that in evolution there occurs a transition from a more indeterminate and disordered state to a more determinate and ordered state. Evolution advances from a chaos whose parts are dispersed and homogeneous to a gathered whole whose parts are heterogeneous and at the same time stand in a determinate connection with one another. Thus the solar system, the organism, consciousness and society are more or less ordered wholes. This third point of view really arises from the combination of the first

two: an ordered whole is one in which differentiation of the parts and concentration of the whole go hand in hand with each other.

Everywhere in existence — in the great and in the small, in the mental and in the material world — processes of evolution of the kind now described take place. The philosophy of evolution formulates, on the basis of a comparative consideration of these processes, the fundamental features of the general history of every phenomenon. But what has thus been found inductively must now be grounded deductively by being derived from the law of the conservation of force.

In the first place, a concentration of homogeneous parts will arise when they are acted upon in a homogeneous way by the same force. This happens when heaps of withered leaves or heaps of sand or clouds form under the influence of the wind. Natural selection works in this way by requiring quite definite variations of the living beings that are to be able to subsist under definite conditions: thereby these are set apart as new species from those not fit to live. In the second place, when a homogeneous whole has formed, differences will arise when the homogeneous parts are acted upon by heterogeneous forces. And when difference in the state of the parts has thus been produced, homogeneous forces will act differently upon them. An organic species will vary under different physical conditions, and even if the varieties are subjected to a homogeneous influence, this will nevertheless not have the same effect on them all. — This follows from the conservation of force, which would be abolished if the same causes produced heterogeneous effects on homogeneous objects, or heterogeneous causes produced homogeneous effects on homogeneous objects, or the same causes homogeneous effects on heterogeneous objects. *[474]*

In order to understand Spencer rightly one must remember that his doctrine of evolution does not apply to the world as a whole, of which, owing to the relativity of knowledge, we can form no notion, but that it applies to the individual wholes that arise within the circle of our experience. It has sometimes been objected against him that his "evolution" becomes incomprehensible if it is to begin as an absolute homogeneity; whence — it is asked — are the differences then to come? Spencer himself declares expressly (First Principles. § 116, 149, 165) that he speaks only of limited phenomena, and that he assumes only a relative homogeneity and heterogeneity, just as the concepts of concentration and differentiation, simplicity and complexity, must also be taken in a relative sense. "Since we know," says Spencer, "no absolute dispersion or absolute concentration, change can never be anything but a transition from a more dispersed state to a less dispersed state, from looser coherence to firmer coherence. Likewise,

since no concrete phenomena present absolute simplicity, — since nothing is absolutely uniform, — since we nowhere find complete homogeneity, the transformation in reality always takes place in the direction of greater compositeness, or increased multiplicity, or a higher degree of heterogeneity. This qualification the reader must constantly keep hold of"[108 b]). *If* one could imagine the whole universe in a state of complete equilibrium, where wholly homogeneous centers of force were distributed with absolute homogeneity over the whole of space, then everything would remain in equilibrium for all eternity. But such a thought is excluded already by the fact that no limit can be set to space. Of all homogeneous masses that we know it holds that they will necessarily become differentiated in a lower or higher degree. —

Evolution must (naturally, when it is not interrupted from without) necessarily lead to a state of equilibrium in which both concentration and differentiation have reached their culmination. As regards human evolution this state means the highest perfection and bliss, consisting in the greatest possible harmony between men and nature, and
[475] among men themselves. But since the external influences constantly continue to exert their effect, this state of equilibrium will not be able to maintain itself in the course of time. Evolution will be followed by dissolution when energy is lacking to preserve, under the continued influences, the harmony between concentration and differentiation. And through the various stages of dissolution one ends up in a new chaos. — Just as processes of evolution go on uninterruptedly within the circle of our experience, so processes of dissolution of greater and smaller wholes also go on uninterruptedly. Even if our solar system too — and all solar systems —, as some natural scientists think, should have the germ of dissolution within it, there will always be a possibility of new formations of systems, since there will always be external forces to set the processes of evolution going. All motion is rhythmical, and evolution and dissolution will therefore alternate endlessly.

While Hegel in his idealist doctrine of development guaranteed an eternal progress, the "higher unities" (which correspond to Spencer's harmony between concentration and differentiation) always becoming the foundation for new evolutions, Spencer, as a good positivist and by virtue of the doctrine of the relativity of knowledge, cannot say anything about whether evolution or dissolution is the stronger process in existence. By virtue of his dualistic conception of the relation between the Absolute and the Relative he will not even ascribe any absolute significance to evolution (nor, therefore, to the whole rhythm of evolution and dissolution). Here, as already shown, a contradiction comes out in him, since he acknowledges a connection between the phenomena and the unknowable order of things that underlies them (the connection between the phenomenal order and the

ontological order). Only this he holds firmly, that just as little as we can set limits to time and space, just as little can we set limits to the force that works in the world of phenomena. But his proper task he has accomplished by his characterization of the history that every phenomenon runs through. *[476]*

e. The Concept of Evolution in the Domain of Biology and Psychology

It is in the biological and the sociological domain that Spencer moves with the greatest expert knowledge. A biological idea gave, as we have seen, the impulse to his whole system, and this biological idea (of evolution as differentiation) he in turn connected directly with a sociological idea (of the division of labor). Social life is for him a form of life that obeys the general laws of life, and the same holds for mental life and the ethical life. Biology is thus the dominant concrete science in his system. It is the evolution of life at all stages and under all forms that his inquiry will follow.

Life consists in a constant adaptation of inner relations to outer. In a living body an impression from without produces not only a direct effect but also an indirect effect, a state being called forth whereby the living being becomes able to meet a following change in the surrounding world. An inner activity is set going which comes, or can come, to be of use on a later occasion. The organic tissue has two apparently opposite properties which in combination make adaptation possible. It has a plasticity that lets the impressions from without send a tremor through the whole organic mass, or at least through large parts of it; and it has a polarity, a definite manner of arrangement of the ultimate organic particles, which Spencer calls the physiological units and regards as far simpler than the cells but more compound than the chemical molecules, a polarity which brings it about that the effect of the impressions on the organic mass is determined predominantly by the mass's own constitution. The formation of the organic mass in the first place is an obscure question. Spencer rejects the assumption of the arising of organic *forms* out of inorganic matter. He is, on the other hand, inclined to assume that organic mass of a wholly *structureless* kind was formed at a certain definite point in the cooling process of the earth's surface. Since this original organic mass was without definite structure, this assumption is thus not one with the assumption of a first organism. There has been organic life without organization. (See especially the Appendix to the first volume of *[477]* "Principles of Biology".) The organic forms have arisen successively under the action of outer conditions on inner. By virtue of the general laws of evolution, constant external influences produce differences in the mass, first and foremost a difference between the surface and the interior; and through the different ways in which the different parts

react to the impressions, they gradually take on a different nature. Spencer even sets up the proposition that function precedes structure, and that it is the constant functioning in a definite way that produces the definite structure that the individual organs possess. This direct influence of the surroundings on the organic tissue is really presupposed by the doctrine of natural selection; this selection takes place precisely among the incipient structures called forth by the direct influence, among the various variations that the organic mass undergoes under the external influence. Even where the variations arise from other, unknown causes, it is nevertheless only when favorable modes of action are bound up with these so-called "spontaneous variations" that natural selection can preserve them. (Princ. of Biol. § 61). The change that can take place in the structure through a certain constant functioning will be able to be inherited when it comes to produce an alteration in the organic polarity, that is, in the manner peculiar to this particular organism in which the physiological units are arranged.

Darwin's doctrine of natural selection (which Spencer prefers to call the doctrine of the subsistence of the most purposive forms of life: the survival of the fittest) Spencer acknowledges in the fullest measure and with great admiration. But just as little as Darwin himself does he think that selection can explain everything. Without the assumption of the inheritance of acquired characters the origin of species is for him incomprehensible, and he has therefore in recent years carried on a polemic against "the natural scientists who would be more Darwinian than Darwin himself", especially against the Weissmannian hypothesis, which denies all influence of practiced functions on the cells that contain the germ of the offspring, and must therefore derive all evolution from natural
[478] selection. Especially in the case of the higher organic beings, which have a developed nervous and muscular system, he thinks that the influence that the use of the powers has on the organization plays a considerable part. He ascribes to the question not only biological but also ethical and social significance. "If", he says in his essay *Factors of Organic Evolution* (1886), "a nation is altered as a whole by the inheritance of the effects that the nature of its members undergoes through the forms of daily activity that its institutions and conditions of life determine, then these institutions and these conditions of life must mold the citizens far more quickly and completely than they could if the only cause of adaptation is the more frequent survival of individuals that have varied in a favorable direction". The ethical and social responsibility thus becomes greater when one assumes the inheritance of acquired characters than when everything rests on selection. It is this that gives the problem so great an interest in Spencer's eyes. —

Consciousness too is an activity that arises in the adaptation of living beings to

their surroundings. When the impressions grow in number, the adaptation to them can only become exact if they are ordered in a series, as happens in consciousness. Psychology is to that extent a part of biology. But Spencer here distinguishes between objective and subjective psychology. It is really only objective psychology, the one that studies the life of consciousness through the material functions to which it is bound, that belongs in the series of the concrete sciences. Subjective psychology, which builds on immediate self-observation and wins its results by analysis of it, stands in the same relation to all other sciences as the subject in general stands to the object; it is the science of subjective existence, while all other sciences together concern objective existence. Therefore subjective psychology is a science wholly unique of its kind. (Princ. of Psychol. § 56. — Classif. of the Sciences. 3. ed. p. 26 b).

Like Stuart Mill, then, Spencer maintains the independence of psychology in opposition to Comte's overlooking of it. And he differs at the same time from the older English school in maintaining that one cannot regard consciousness as a mere sum of sensations *[479]* and representations, since there is always something that connects these and maintains the unity of the circle of representations against attempts to divide it. There must — Spencer teaches — be something substantial in consciousness as opposed to all shifting forms, but this substantial something we cannot know, since it does not come forth in any single state, although every state is a particular form of it.

For all the peculiarity with which the life of consciousness appears in subjective experience, it nevertheless presents the same fundamental features as life and as evolution in general. Its evolution consists in progressive concentration, differentiation and determination. All the transitions take place gradually, and it is therefore wrongly that faculties differing in essence have been distinguished in the soul. From reflex movement through instinct and memory to reason, from the simplest distinguishing and recognizing to the highest scientific act of thought, there is a continuous series of stages. The qualitative and quantitative richness of consciousness corresponds to the richness of the living being's relations to the surrounding world. There is always a correspondence between the life of consciousness and the external relations the individual has to do with. There is a continuous series of stages from the intestinal worm in an organic tissue to a Newton or Shakespeare whose thought embraces the world.

With regard to the question of the origin of knowledge Spencer makes a stand against Leibniz and Kant on the one side, Locke and Mill on the other side. He combats empiricism, which does not see that the matter of experience is always taken up and worked over in a manner determined by the original nature of the individual, and which

moreover comes to lack a criterion of truth. An original organization is needed to explain the influence that impressions have on the different individuals; and there is needed, every time one would ground an assertion, a criterion consisting in this, that the opposite would contain a self-contradiction. In the innate nature of the individual and in the logical principle on which we rely in every inference we thus have something a priori, something that cannot be derived from experience. To that extent Spencer grants Leibniz
[480] and Kant that they are right as against Locke and Mill. But only so long as it is a question of the experience of the single individual. For the single individual there exists an a priori, but not for the species. For the conditions and forms of knowledge and feeling that are found originally in the single individual, and therefore cannot be derived from its experience, are due to inheritance from earlier generations. The forms of thought are one with the accumulated and inherited changes of structure that are latently present in every newborn individual and are gradually unfolded by its experiences. And their first origin, then, is empirical after all: the fixed and universal relations between things must in the course of evolution form fixed and universal connections within the organism; by the incessant repetition of absolute external uniformities there are bred in the species necessary forms of knowledge, indissoluble connections of thought, which express the net yield of the experience of perhaps millions of generations right down to the present time. The single individual cannot take apart what has thus become an ingrown connection in the organization of the species, and therefore it brings with it in its innate nature the foundation on which the necessary truths are built. (See Princ. of Psychol. § 208, 216 cf. First Princ. § 53: Absolute uniformities of experience generate absolute uniformities of thought). Although the inductive school has, in Spencer's opinion, gone too far when it would derive everything by way of induction, whereby induction itself comes to hang in the air, he would nevertheless, if he must choose, rather count himself of Locke's school than of Kant's. For *in the end*, according to him too, all knowledge and all forms of thought nevertheless stem from experience. Max Müller has held that when Spencer conceded that there is something in consciousness that is not a product of one's own experience, he is a pure Kantian (a thorough Kantian). But Spencer has answered this: "The evolutionist point of view is wholly empirical. It distinguishes itself from the older view of the empiricists by the extension it gives this view . . . . But Kant's view is evidently absolutely unempirical". (Essays. III. p. 274. 277).

Spencer is, however, struck here by his own criticism of empiricism when he assumes
[481] that the species at any stage whatever of its evolution has been able to receive influences from without without there being, for receiving these influences and determining their

results, any previously given organization. There is here an unclearness in his psychology that goes back to his biology. For his proposition that function determines structure seems unable to be understood otherwise than as saying that the organic tissue at the lowest stage behaves absolutely passively toward the influences that produce activity in it. Such a relation of absolute passivity is, however, excluded by his own definition of life as adaptation — and does not in reality take place even where it is inorganic beings that are acted upon, since surely the stone is affected otherwise by heat than the wax. — Moreover, Spencer has overlooked that the *validity* of our knowledge is not absolutely guaranteed by the first presuppositions' being derived from the experiences of countless generations. This establishes at most that these presuppositions have hitherto proved practically serviceable in the struggle for existence. A proof of their absolute truth cannot be built on it[109]). —

Even more than knowledge, feeling, according to Spencer, presupposes a foundation that is original for every individual and received from the species. The constant interaction with the surrounding world, the constant struggle with the conditions of life, engenders a tendency to moods for which one does not always succeed in finding clear expression in representations. So it is with the sympathetic feelings, e.g. the little child's pleasure at a smiling face, its pain at a sorrowful or threatening face. The feeling of justice, the immediate indignation at encroachment, can arise without there being any distinct concept of the limits of the individuals' freedom of action. Spencer objects to the earlier utilitarians that they let the conscious insight into the effects of utility precede the appearance of the actions and motives that are valued. In order to experience that sympathy is beneficial, one must first act sympathetically without deliberation. It is thus feeling that makes it possible to discover utility, not the discovery of utility that calls forth the feeling in the first place. It is from the constant education to which man is subjected through the interaction of the conditions of life that Spencer expects continued changes and progress in human emotional life, not from the mere communication of *[482]* finished knowledge. He opposes the proposition from which Comte, especially in his earlier writings, started, that ideas rule the world. It is feeling and character that drive the work, and which ideas become dominant will depend on the character of the people, which in turn rests on a foundation inherited from earlier generations.

f. The Concept of Evolution in Sociology and Ethics

For Spencer's sociology the relation between individual and species that his biology and psychology have found becomes of decisive significance. When every individual has an

original foundation for its character that goes back to the earlier history of the species, the evolution of social life will not be able to be changed suddenly by the influences that one may be able to exert on the single individuals through reforms in education and constitution. Spencer has not so immediate a hope for the future as that which animated Bentham, Mill and Comte. Evolution advances slowly, because it is the very natural foundation in the individuals, not merely their thoughts and knowledge, that must be changed. During his stay in America Spencer, in a conversation, strongly censured the abuses in the public life of the United States, which he derived from the fact that the country had got its constitution, good in itself, by a happy accident, but also had only *got* it; it had not grown forth spontaneously, and therefore it was no wonder that its consequences turned out other than had been imagined. When an American friend asked whether instruction and the spread of political knowledge would not remedy this, Spencer answered: "No. It is essentially a question of character, and only in a subordinate sense a question of knowledge. It is a common illusion that instruction can be a universal remedy for political evils". Changes of character take place, according to Spencer, only through a race's standing for generations in direct interaction with the real conditions of life. Only through practical adaptation, through the development and exercise of the powers that are used in the struggle of life, do firm characters and healthy feelings
[483] arise. In his work on education Spencer therefore impresses the necessity of the child's gaining experiences on its own account, coming itself to know natural relations and to feel the effects of its own actions. Screens are to be used as little as possible, since they involve double work: first adaptation to the screen, and then to what lies behind the screen, the real relations. The child had best burn itself; otherwise it does not rightly learn to shun the fire. And what holds of the character of the individual holds of that of the race, and of the whole species. In his largely and broadly planned work *Principles of Sociology* Spencer seeks to illustrate this by means of an extensive material, drawn especially from the races standing lowest. Together with a circle of collaborators he has published, under the title *Descriptive Sociology*, a series of tabular surveys (supported by references to sources) of the conditions of cultural history among various peoples. In a lively written book, *The Study of Sociology*, he has discussed the difficulties with which a purely scientific sociology has to contend.

It is a favorite proposition of Spencer's that societies and constitutions grow, but are not made. He demonstrates in detail the analogies between a society and an organism. The single individuals correspond to the cells, or rather perhaps to the physiological units. Both within society and within the organism the common life is to a certain

degree independent of the fate of the single units each taken by itself. But there is a great and significant difference between society and the organism: while in the latter consciousness (where it is present) is bound to the central organs, in relation to which the other organs and units have only subordinate significance, in society it is precisely the single units that have consciousness, while the central organization as such has no consciousness of its own. In the organism the parts exist for the sake of the whole; in society the whole exists for the sake of the parts. On this difference between society and organism, according to Spencer, no less weight must be laid than on the analogies between them. (Cf. Princ. of Sociol. § 222. cf. Princ. of Ethics. II. § 102, 137). For on it rests the fundamental proposition, whose correctness the history of social life establishes, that centralization always has significance only as a necessary means, while the decisive *[484]* thing always remains what goes on in the single individuals, where human life proper is led. In social life as in individual education the point is to reduce screens and tutelage to the least possible, since they always complicate and retard evolution by necessitating a double adaptation, first to the artificial, then to the natural relations. Experience shows that wherever demands that have their root in nature are enjoined by external authorities, obedience to the authorities comes to take the first place at the expense of the practical relation to natural reality. And experience shows as well that artificially set up and appointed authorities do not work as ably and purposively as the free, individual forces themselves. The social machinery uses too much force to reach its end — and easily reaches other ends than those intended. The single individuals experience far more urgently than the authorities of the state a direct connection between labor and yield. Although Spencer theoretically holds fast (what he had already taught in "Social Statics") that the land is the property of the whole people, he declares himself, from aversion to "officialism", against the state's taking over the cultivation of the land, as the "Nationalists" propose. Quite apart from the financial difficulties of the state's taking over the land, difficulties that in Spencer's view would in and of themselves make the proposal impossible to carry out, the notorious imperfection of state administration in comparison with private administration is in his eyes sufficient to reject it. (See Appendix B to Princ. of Ethics. Part. IV).

Experience shows us societies at the most different stages of evolution; the degree of evolution is measured here as everywhere by concentration, differentiation and determination. But there is an opposition that in sociological — and no less in ethical — respect stands in the first line. It is that between militarism and industrialism. Two types of society appear here, of which the first prevails especially at lower stages and

gradually, but very slowly and with many fluctuations, gives way to the second. The military society arises from the necessity of defending the social group against external
[485] enemies, often also from the drive to acquire livelihood and power at the expense of other groups. Characteristic of this type is the absolute subordination of the individuals to the whole. They are means, not ends. Obedience becomes the highest duty. Peaceful labor at procuring the means of subsistence is left to women and slaves. In industrial society, on the other hand, labor stands in the first line. The main thing here is the free, personal intercourse between the individuals, their working together in the service of common interests. Instead of the mixture of savagery and submission that the military type fosters, the industrial type sets the individuals freely over against one another and lets them learn in daily intercourse to attain their ends in such a way that they acknowledge the equal right of others to attain theirs. It is a discipline that gains influence on characters, manners and constitutional conditions. While the regulating apparatus under the military type was all in all, it now becomes more and more restricted to the office of maintaining peace and right among the members of society. For carrying out the functions that the state formerly attended to as a kind of providence, voluntary associations are formed, insofar as the spontaneous working together of individuals is not sufficient. While formerly it was society that stamped its impress on the single individuals, it is now more the single individuals who order society according to their needs. — The struggle between militarism and industrialism is still undecided. After long years of peace militarism got a new upswing on the Continent, in which for the second time the Bonaparte family, the greatest of all modern curses (that greatest of all modern curses, the Bonaparte family), intervened in the course of things, and now it flourishes in all countries more or less and introduces its spirit and type again in other domains too than the purely military. Compulsory regulation now takes the place of free self-development in a series of domains. In the ideal of social democracy, of an army of workers, each with his appointed occupation and his appointed wage, the same type expresses itself, and it is then no wonder that socialism, like militarism, thrives best in one and the same country, Germany. (Princ. of Ethics. II § 26, 72). And the prospects of
[486] improvement are for the present not bright: " So long as the European nations divide among themselves the parts of the earth that are inhabited by lower peoples, with cynical indifference to the rights of these peoples, it is foolish to expect that within each single one of those nations a tender regard for the right of individuals should be shown on the part of the government". (ib. § 119). — Yet Spencer descries the possibility of a third type, which stands as high above industrialism as the latter stands above militarism. The

industrial type is easily carried into the one-sidedness of everything's being directed to labor for the sake of gain. The third and higher type of human life will be one in which free devotion to activities that are immediately satisfying, and do not merely serve as a means of existence, comes to occupy a far greater place than now. Within our present social order it is especially efforts and institutions for the furtherance of intellectual and aesthetic ends that contain indications in the direction of the third type[110]).

The closer working out of what this highest type of human life contains within it belongs to ethics, not to sociology. For this type of the future is, according to Spencer, the goal toward which all ethical striving moves. So long as this goal has not been reached, no strict ethics is possible: the realization of absolute ethics presupposes a perfect human life in a perfect society. According to Spencer, scientific ethics is a system of deductions from the highest laws of biology and of the general doctrine of evolution. But he concedes that such deductions cannot hold without further ado. At the imperfect stages the conditions are so intricate that only approximations to the absolutely right and compromises between the absolutely right and what the subsistence of life under the actual conditions requires become possible. But relative ethics must constantly be checked and regulated by keeping the ideal principles of absolute ethics in view. Ethics must construct a picture of the conditions on which the perfect life rests, without at once directing itself by whether these conditions are already actually present now. If we compare ethical development at various stages, we find that it bears the general characteristic marks of all evolution. Ethical action presents greater concentration and coherence than unethical *[487]* action: compare e.g. self-control with dissoluteness, truthfulness with mendacity! It likewise presents greater richness in differences, greater differentiation: for he who merely looks after his own egoistic interest gets a narrower range of vision and range of action than he who works for others as well; all of a man's faculties and possibilities do not come to development when he will work only for selfish ends, and the highest degree of development can therefore be present only where the individual's activity furthers the welfare of others. Finally, the more perfect ethical conduct also bears the stamp of greater determinateness than the less perfect: it takes definite considerations into account and sets definite limits to the individual's own impulses and to those of others, which in themselves might spread into the indeterminate; conscientiousness, justice and moderation are examples of this.

In the perfect type of life the development of the individual will be limited only by the equal right of others to development; the individual will, moreover, spontaneously, from his own inner need, refrain from encroachment on the healthy development of

others, and for his part help it forward as much as lies in his power; finally, it will not be necessary, for the attainment of more remote ends, to undertake labor that does not bring immediate satisfaction.

The construction of these conditions for the perfect form of life rests on the principle of welfare. Spencer does indeed criticize the utilitarianism of Bentham and Mill; but only because it is too empirical, too inclined to keep to the nearest effects of actions without catching sight of the more remote ones, which can be established only by way of deduction. Here, as in his theory of knowledge, he seeks to show that the empirical and the a priori conception can be united by means of the philosophy of evolution. The significance of a priori ("intuitive") ethics he finds partly in its maintaining the significance of deduction and setting up ideal principles that are not directly built on experience, partly in its finding a deeper psychological foundation for ethical feeling than the individual's own experience can afford. In his construction of the doctrine of right he comes to the same result as Kant: that the fundamental right is freedom for the
[488] individual, insofar as he does not come to commit encroachment on the equal freedom of others; this is, after all, as stated, the first characteristic mark of the perfect type of life. This agreement of Spencer's with Kant, which surprises him himself (see Appendix A to Princ. of Eth. Part. IV), cannot surprise us, since we have seen Kant's ethics and doctrine of right proceed from an evolutionist line of thought. The great significance of a priori ethics consists in not letting itself be entangled in regard for the nearest effects of actions; but it is mistaken in thinking that the ethical principles are not in the end determined by regard for whether the actions cause happiness or unhappiness. And even if there is an a priori foundation for ethical feeling, independent of individual experience of happiness, this very foundation is in turn to be explained as the result of the doing and suffering of earlier generations. In his closer conception of ethical feeling Spencer parts from Kant by assuming that the feeling of duty belongs only to a certain stage of evolution. The feeling of duty consists in an inner control that one feeling exerts over another; but with advanced evolution such control will be unnecessary; there will have been formed an "organic morality" which makes the performance of the actions required by the ethical principles as spontaneous and as immediately satisfying as the mother's care for the child and the artist's absorption in his work already are now. Man will then be completely adapted to the social state — and the latter to him.

Until perfect evolution is reached, we must make do with compromises. As a strict science ethics is not possible before the highest stage of life. Spencer thus really assumes that no ethics can be given before it is — superfluous! Paradoxical as this may seem,

there might nevertheless lie in it a correct recognition of the difficulties of principle with which every attempt at a scientific ethics has to contend under the intricate conditions under which we have to lead our life. Perhaps one must even go still further than Spencer, in his great confidence in the victorious course of evolution, has thought necessary; in particular, he takes too little account of individual differences and their influence on ethical determinations[111]). — We shall bring forward some examples to illustrate what *[489]* Spencer means by the opposition between absolute and relative ethics.

Sociology has shown us that the present stage of development of the human race is characterized by a conflict between militarism and industrialism. In this period of conflict the freedom of the individual will in many ways be more limited than absolute ethics permits. Slavery is an institution that belongs to militarism. With the development of industrialism personal freedom comes into force in ever wider circles. But through the worker's dependence on the employer something constantly maintains itself of the relation of obedience in which the slave stood to the master, although the relation now rests on contract, and so on consent and obligation on the part of both parties. Whether this can ever be wholly changed we cannot know; the task of relative ethics, on the other hand, is to impress the necessity of as great an approximation to the ideal relation of equality as the conditions in any way permit. — While slavery existed, the poor enjoyed the protection of the masters, which was often paternal. With the abolition of unfreedom has followed the abolition of protection, and with it in turn the sufferings that the struggle for existence brings. By state compulsion, then, men have sought to remedy this through the arrangement of poor relief, since it was clear that the principle of the subsistence of the best adapted could not be allowed to prevail in all its severity. But the intervention of the state has here caused greater evils than were to be remedied: it has protected the weak and incapable, made it possible for them to bring offspring into the world, and maintained them at the expense of the able and industrious! Men have wanted to soothe their feeling of the suffering that is in the human world by applying a system that on closer consideration makes evil worse. The present system of state aid is a kind of social opium-eating. The sufferings that evolution once and for all brings with it cannot be escaped. And they are not escaped by letting feelings that have their proper domain in more personal relations and within the family gain influence on the arrangement of political and social relations. Family ethics and state ethics must not be confused; the former concerns the rearing of helpless offspring, the latter the ordering of the mutual relations of adults. When adult age is reached, no state aid is to hinder *[490]* the conditions of life from taking effect. It is another matter with free charity, which

uses its own means, while the state *extorts* its means. But it too will only be able to hinder superfluous suffering and soften the necessary; if it goes further, it weakens the vital power of the race. — Spencer must himself concede, however, that just as there are many transitional forms between the child and the adult, so there is room for many compromises between family ethics and state ethics. These systems do not form so thoroughgoing an opposition in principle to each other as one would think from many of Spencer's utterances. Spencer's correct fundamental thought is that compassion with suffering must not exclude the discipline through interaction with the real conditions of life that is necessary for the healthy development of the individual and of the whole race.

In its crudest form the struggle for existence favors egoism. But altruism, which has its root in sympathy, slowly works its way forward, first in the form of care for the helpless, later also in other and higher forms. There is no absolute opposition between them: evolution will lead to such a change of human nature that the individual finds his highest happiness in devotion, without, however, hampering by his intervention the independent development of the other, and that, on the other side, no one is so egoistic as to wish to be the object of an absolute self-sacrifice by another individual. What now characterizes the exceptionally highly placed may be expected one day to characterize all: for what the best human nature is capable of lies within the compass of human nature as a whole, and evolution is not ended so long as there is still a possibility of a fuller and richer life through the unfolding of faculties that at once bring the individual himself immediate satisfaction and do others good. Free activity and unfolding of life which is more than a means to a future end is, as we have seen, what is characteristic of the higher type of life to which Spencer looks forward. Already now there are, as we have also seen, actions and efforts that bear this stamp, so that "the third kingdom", to
[491] which Spencer with so many other thinkers in the nineteenth century looks forward, does not lie only in the future.

Appendix

In English literature there remain to be mentioned some works of a more special nature that are of interest for philosophical thought in general. These are the logical works of Boole and Jevons and the ethical work of Sidgwick.

After Stuart Mill had treated inductive logic in such a way that the limits of the probative force of the purely inductive method came clearly to light, George Boole in his work *An Investigation of the Laws of Thought* (London 1854) gave a new exposition

of deductive logic, in which he assigned to deduction the significance of deriving all the combinations of certain concepts that are logically possible when a definite connection between these concepts is given. The task of deduction is thus to give exhaustive information about the logical value of a given judgment. Boole's method is acute, but very artificial in its form. His fundamental thought was carried out in a far simpler form by Stanley Jevons in a series of works that began with *Pure Logic or The Logic of Quality apart from Quantity* (1864) and ended with *Principles of Science* (1874). Jevons takes the content of concepts, not their extension, as his basis, and thereby brings out more sharply the difference of pure logic from pure mathematics. He also carries through (what Hamilton and several before him had already begun by quantifying the predicate) the principle of identity far more sharply, already in the formulation of judgments, than had hitherto been done in logic — if some long unprinted treatises of Leibniz are excepted. His logical work goes far more into particulars in its exposition of the methods of research than Mill's logic. Jevons's logic has a considerable epistemological interest through the clear way in which it establishes that a deductive inference underlies every induction, the proof of the correctness of an induction always consisting in this, that deduction from the proposition tentatively set up leads back to the given experiences, *[492]* neither to more nor to less than them. Since deduction in turn presupposes the validity of the logical principles, the incorrectness of pure empiricism is thereby established. In the last years of his life (1877—79) Jevons published in "Contemporary Review" a very sharp criticism of Stuart Mill's philosophy, in which he sets pure empiricism in the sharpest opposition to evolutionism and attaches himself decidedly to Spencer's philosophy.

Henry Sidgwick by his work *Methods of Ethics* (1877) gave ethical discussion new life and new clarity. In particular he drew attention to the fact that behind the name utilitarianism there were hidden two different ethical systems, of which the one took egoism, the other altruism as its basis, while they both saw the ethical standard in the principle of welfare. He undertakes a careful investigation of how far the two systems can each reach by itself. He also seeks to show that the current moral propositions (the morality of Common Sense) all lead back to utilitarianism, and that the gaps or contradictions found in them can be made to vanish when one carries through this unconscious utilitarianism[112]).

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Tenth Book

Philosophy in Germany 1850–1880.

Of the two great tendencies in nineteenth-century philosophy it was positivism that *[493]* had best maintained its connection both with the thought of the eighteenth century and with empirical science. The philosophy of Romanticism, by contrast, was a definite and conscious attempt at reaction against both these tendencies; it wanted even to do over again what had been founded with the rise of natural science in the seventeenth century. In Germany, the hearth of Romanticism and of its philosophy, that tendency was the dominant one at the middle of the century. The adherents of the critical undercurrent were the only ones who held fast to continuity with the other sciences — leaving aside Schopenhauer and Feuerbach, who at that time still stood as solitary thinkers for whom the hour of recognition had not come.

As little as positivism was to be explained as a reaction against Romanticism, so little is the philosophical movement which takes place in Germany after the middle of the century to be explained as a continuation of Franco-English positivism. It has its presuppositions in Germany itself. It receives its problems above all from the renewed rise of natural science toward the middle of the century. Not only do natural-scientific studies and results now become the object of more general interest than in the earlier part of the century, when poetry, religion and speculative philosophy had laid claim to men's minds; natural science itself now returned, with clearer consciousness, to the great *[494]* principles its founders had in their day set up. The demand for an understanding of outer nature as a series of demonstrable causes and effects — in other words, for a mechanical explanation of nature — was now asserted again, as in the seventeenth century. And at the same time a new and comprehensive law of nature was discovered and demonstrated: the law that in physical nature no energy arises or perishes, but that what happens in every apparent arising or perishing is a conversion of energy into another form, and in such a way that the different forms of energy stand in a determinate quantitative relation to one another. This law — which, together with Darwin's doctrine of the origin of species

through the struggle for existence, is the most significant result of natural-scientific investigation in the nineteenth century — could not but set philosophical thinking in motion, just as in their day the Copernican theory and Galilei's founding of mechanics had led philosophy into new paths. For German philosophy in particular it had to become a principal question: how far can the thoughts developed by the philosophy of Romanticism be held fast, if one accepts the new natural-scientific way of regarding things? This question was answered in three different ways. Modern materialism rejected those thoughts as pure illusions and proclaimed the doctrine that only the material exists, as a necessary consequence of natural science. Lotze and Fechner, by contrast, attempted to exhibit the fundamental thought of speculative philosophy of religion as the last and concluding presupposition of the picture of existence that can be built up by the help of the natural-scientific method. Finally Albert Lange and Eugen Dühring approached critical philosophy and positivism, by bringing out the significance of the problem of knowledge and by maintaining practical idealism's independence of empirical science, while at the same time insisting upon experience's right to determine the actual content of the world-picture.

1. Robert Mayer and the Principle of the Conservation of Energy

[495] Not only in philosophy, but in the other sciences too, knowledge moves forward in rhythmic fashion, through actions and reactions. At the close of the eighteenth century natural science had a great and fruitful period, in which some of the most important truths in the domains of chemistry and physiology were discovered. Lavoisier introduced the quantitative method into chemistry, and by its help the truth of the old thought was established that no matter arises or perishes, but that the same mass of matter subsists through all changes, under different forms. Chemistry was founded as an exact science. Priestley, Ingenhouss, Senebier and Saussure found the principal laws of the metabolism of plants and animals, and thereby founded the great doctrine of the circulation of matter in nature, whereby the organic and the inorganic world appear in their close mutual connection. It was discovered that it is under the influence of sunlight that organic matter is formed in the green plant cells by the taking up and dissolution of the carbonic acid of the air. The carbon stored in the plant cells then serves animals for nutriment; by the animal functions a combustion takes place, and the carbonic acid thereby formed passes out into the air in respiration, whence the same circulation can

begin anew. A great cosmic connection had here been exhibited.[6] The picture of the world sketched by Copernicus, Bruno, Kepler and Galilei, which Newton had enriched by exhibiting universal attraction as a binding and ordering tie, hereby received a new and significant enrichment.

But the age was too much taken up with revolution and world war, with Romanticism and speculation, and soon with orthodoxy and mysticism as well, for these great ideas to gain the influence upon general opinion that they deserved. Even within natural science their victory was hindered by other tendencies. It was within the science of organic life that resistance was chiefly offered to the new points of view, because men were still governed by the reaction against the Cartesian doctrine of the organism as a machine. The *[496]* expressions peculiar to the organism were derived from a peculiar vital force, supposed to be wholly different from the other forces of nature. This mode of explanation, so-called vitalism, could not but lead to a rejection of the attempts to draw the organic phenomena into the universal circulation of matter.[7] Besides, the prevailing interest among botanists and zoologists was to describe the forms and determine their systematic place, not to find the developmental process through which the forms come to be, and the causes that determine that process. Romantic philosophy of nature favored this aesthetic and formal conception — and had itself proceeded from it. When "philosophy of nature" is often given the whole blame for the fact that the points of view of the close of the eighteenth century did not sooner prevail in botany and zoology, it must not be forgotten that it was quite as much an effect of the condition of natural science as a cause of it.

From about 1840 a change came about in this respect. Life began to be conceived as something other than a play of forms or a revelation of ideas. Dumas and Liebig insisted upon the significance of the chemical processes for plant and animal life. And from the medical side too a strict mechanical explanation began to be demanded of the processes that go on in the organism. In some of his earliest writings Hermann Lotze worked in this direction, especially in *Allgemeine Pathologie und Therapie als mechanische Naturwissenschaften* (1842) and in the treatise *Leben, Lebenskraft* (1843, in R. Wagner's Handwörterbuch der Physiologie). The task was set of making physiology a purely mechanical science. Most epoch-making, however, was the fundamental proposition set up by the physician and physicist Robert Mayer (b. 1814, d. 1878) in his work *Die*

[6] On "the period of the great discoveries" at the close of the eighteenth century see *Rasmus Pedersen*: *Planternes Næringsstoffer.* Historisk Indledning. København 1883, p. 44 ff. *[596]*

[7] On vitalism in its various forms see *Louis Peisse*: *La Médicine et les Médicins.* Paris 1857. I, p. 226–297. — It must be firmly held that by vitalism is here meant a tendency which finds the *explanation* of vital phenomena in the assumption of a "vital force" — not a tendency which maintains that there is much *[597]* in the vital phenomena that has hitherto not been explained by physical and chemical causes. See my treatise *Om Vitalisme* (in "Mindre Arbejder").

organische Bewegung in ihrem Zusammenhange mit dem Stoffwechsel (1845): that in the process of life there takes place only a transformation of force as well as of matter, never a creation of them. Mayer here points back to the great law he had discovered a few years earlier, when he set the law of the conservation of force beside the law of the conservation of matter deriving from Lavoisier.

[497] Robert Mayer's life as an investigator is borne by a single thought, which he came upon early; but it was the tragedy of his life that he was unable to carry it through in detail by experimental means. As a child he had tried to construct a *perpetuum mobile*, and it made a strong impression upon him that he did not succeed. From that time he brooded constantly upon the relation between cause and effect — first and foremost within physiology, which as a medical man lay nearest to him, but thereafter within chemistry and physics as well. On a voyage to the East Indies as ship's doctor he arrived at the new idea of the imperishability of force in nature, under the influence partly of investigations into the origin of organic heat, partly of the fact that heat arises in the wave-motion of the sea. His point of departure is thus definitely observation. But that observation led him to ask whether a law similar to the one in chemistry might not hold in physics too, so that not only matter but force also was imperishable, and every change a mere conversion. And he then came back to his earlier broodings upon cause and effect. It was for him a self-evident proposition that there could not be more in the effect than in the cause: *causa æqvat effectum*! It follows from "the law of the logical ground"! Mayer thus does not distinguish, any more than the dogmatic philosophers do, between ground and cause; thereby he comes, like William Hamilton and Herbert Spencer, to regard it as a matter of course that the cause cannot become nothing when the effect arises, but that the effect must furnish an equivalent for what seemed to vanish with the cause. Hitherto men had rested content with the fact that a motion ceased when it met a sufficiently strong resistance, and that heat arose from frictional resistance. But — Mayer asked — does the motion then become nothing, and does heat then arise out of nothing? If that is so, then "the red thread of science" is cut through, just as it would be if in chemistry we were to assume that oxygen and hydrogen became nothing when they combined, and that water then came to be out of nothing. Just as we assume that oxygen and hydrogen are transformed into water, so we must also assume that the motion does not become
[498] nothing, but is transformed into heat.[8] And by the expression *transformed* Mayer meant

[8]Cf. *Mayer's* letter to Griesinger of 20 July 1844: "*Wenn ich sage, Wärme lässt sich in Bewegung verwandeln, und umgekehrt, so will dies nichts meinen, als dass zwischen Wärme und Bewegung finden hin und her dieselben qvantitativen Beziehungen statt.*" (*Kleine Schriften und Briefe* von *Robert Mayer.* Herausg. von *Weyrauch.* Stuttgart 1893, p. 225.) — In his paper *Robert Mayers Entdeckung und Beweis des Energieprincipes* (in the Festschrift for Sigwart. Tübingen 1900) A. *Riehl* illuminates Mayer's doctrine from

to designate only that a constant quantitative relation subsists between the cause, which disappears, and the effect, which comes in its place. If the train of thought put forward is correct, experience must establish such a relation of equivalence between the different forms of force that succeed one another. On the basis of experiments already available, Mayer sought already in his first paper (*Bemerkungen über die Kräfte der unbelebten Natur*, printed in Liebig's "Annalen der Chemie und Pharmacie", 1842) to determine the quantitative relation between heat and motion. It was this idea of an invariable relation of magnitude that was the main thing for Mayer, and that in the following period proved so fruitful for natural science. It was a significant extension of the abstract law of the conservation of force, already maintained by Huyghens and Leibniz, so as to hold for the relation of the different forces of nature to one another. Mayer derived from his train of thought that there really exists only a single force, which appears under different forms, and these stand to one another in a determinate quantitative relation.

Although Mayer's attempt to give a philosophical derivation of the law of the conservation of energy does not hold, his train of thought has nevertheless an epistemological interest. For it was in itself fully legitimate to raise the question whether between the event we call the cause and the event we call the effect there might not obtain a relation similar to that between logical ground and logical consequent. And Mayer's train of thought leads naturally to raising that question. Experience must then decide what the answer is to be. — It is interesting that the Danish physicist Colding, who in the year after Mayer's first paper was printed set up the proposition of the conservation of energy by a road of his own and confirmed it by experiment, proceeded from the view that it was really a proposition of reason; and likewise that Helmholtz, in his paper *Ueber die Erhaltung der Kraft* (1847), proceeded from an epistemological postulate. The Englishman Joule, who like Colding and Helmholtz had come by a road of his own to the same result as Mayer — a remarkable example of how several may simultaneously be on the track of one and the same discovery — proceeds predominantly experimentally, *[499]* but nevertheless indicates the a priori improbability that force should be annihilated without a corresponding effect.[9]

an epistemological point of view.

[9] "If the resistance to electrolysis which is over and above that due to chemical change were not accounted for elsewhere, *it would prove the annihilation of a part of the power of the circuit, without any corresponding effect.* We shall see that this is not the case, but that in the evolution of heat, where the excess of resistance takes place, an exact equivalent is restored" (*Joule*: *On the Heat evolved during the Electrolysis of Water.* 1843. Scientific Papers. I. London 1884, p. 115). "*We might reason, a priori, that absolute destruction of living force cannot possibly take place,* because it is manifestly absurd to suppose that the powers with which God has endowed matter can be destroyed ... but we are not left with this argument alone, decisive as it must be to every unprejudiced mind" (*Joule*: *Matter, Living Force and Heat.* 1847. Scientif. Pap. I, p. 269). — Joule's standpoint is therefore not in principle different from Mayer's and

The new proposition had difficulty in winning recognition, but gradually made its way in the course of eighteen years (1842–1860), especially as it proved fruitful in leading to new investigations and discoveries. Its great importance consists precisely in the determinate questions it leads the investigator of nature to ask, every time a peculiar manifestation of force appears or ceases. Just as it is the importance of the general causal proposition that at every change which occurs it leads us to look about for a preceding change of which the new one might be the effect, so it is the importance of the proposition of the conservation of force — or, as one now prefers to say, following the proposal of English investigators, of energy — that one is at once brought to set oneself the task of finding out in what relation the manifestations of force that succeed one another stand to one another.

Philosophically regarded, the question had above all to become how the spiritual phenomena stood to the new law. It is characteristic in this respect that almost all its discoverers proceeded from decidedly spiritualistic and teleological views. Mayer repeatedly declared himself a definite opponent of materialism, and expressed as his conviction that the scientific truths stand to the Christian religion as brooks and rivers to the ocean. At the congress of natural scientists in Innsbruck in 1869 he made a declaration to this effect, to the offense of Carl Vogt and his friends. Colding's standpoint is seen from the following utterance: "My first thought that the forces of nature must be imperishable I drew from the view that the forces in nature must be akin to the spiritual in nature, to the eternal reason as well as to the human mind. It was thus the religious conception of life that led me to the thought of the imperishability of the forces of nature." Colding and Joule conceive that God at the creation laid a certain sum of force into nature, and that this sum can neither be increased nor diminished as a whole, but only
[500] distributed in different ways — as Descartes in his day assumed of motion. And both assumed that with the conservation of energy the conservation of what is valuable in the world was secured, since they no more than Leibniz distinguished between energy itself and its employment in favor of life and development.[10]

Colding's. For the latter compare an utterance in his *Undersøgelse om de almindelige Naturkræfter og deres gensidige Afhængighed.* (Videnskabernes Selskabs Skrifter. Femte Række. Naturv. mathem. Afd. II, p. 129): "The thought that an activity could disappear in the corporeal without appearing again as operative cause seems to me contrary to reason." — It is of interest in this connection to recall that *Lavoisier* too proceeded in his day from a fundamental proposition he regarded as a matter of course, namely that the weight of a compound body is equal to the weight of the substances of which it consists. Self-evident as that presupposition was, it nevertheless had to be proved to those who took heat for a substance; for if that assumption were correct, the release of heat which accompanies a chemical combination would have to entail a diminution of weight. Compare *Ernst von Meyer*: *Geschichte der Chemie.* Leipzig 1889, p. 137.

[10] *Mayer*: *Kleinere Schriften und Briefe*, p. 339 f., 460. *Die Mechanik der Wärme.* 3. Ausg., p. 356 f. — *Colding*: *Naturvidenskabelige Betragtninger over Slægtskabet mellem det aandelige Livs Virksomheder og de*

But a closer investigation and grounding of these presuppositions the natural scientists named do not undertake; they proceeded, like the philosophers of the seventeenth century, from given religious representations. A closer investigation of the relation of the law exhibited — and of the knowledge of nature so far won generally — to the phenomena of spiritual life had therefore naturally to be called forth. And it is then no wonder that the ways parted, and that different tendencies stepped forth against one another. All that was given was this: that when one kind of physical energy ceases, another kind of physical energy enters, in a certain determinate quantity, in its place. The question was what significance this knowledge must consistently come to have for the conception of the world.

2. Materialism

The strongest reaction against the philosophy of Romanticism — apart from crass orthodoxy — is marked by the materialist literature which flourished in Germany at the middle of the century. If there the Idea had been everything, now matter became everything. Enthusiasm for natural science, and for the great points of view it had succeeded in carrying through in the doctrine of the conservation of matter and of force, could naturally lead to regarding that doctrine itself as a sufficient philosophy, and to explaining every side of existence by its help. The simplest way of solving the question what significance the law of the conservation of matter and force has for our world-view would undeniably be to say that this law contains our whole world-view. The philosophy which modern materialism sets up will be nothing other than the simple consequence of natural science. Thereby a firm basis seemed at the same time to have been won for all our representations and for our conduct of life. In place of a mystical or spiritualistic *[501]* foundation we now get a wholly tangible foundation to build upon in theory and in practice. And it is a doctrine easy to popularize. It is childlike, intuitable, easily accessible, and its presentation gives good occasion to bring forward a mass of natural-scientific facts of general interest. Several of the spokesmen of German materialism were able and independent investigators of nature, such as Carl Vogt and Jacob Moleschott. Others, like Louis Büchner, worked especially by clear, tasteful and enthusiastic presentation. It is in its popularizing tendency that this literature has its greatest merit. A

almindelige Naturkræfter. (Oversigt over det kgl. danske Videnskabernes Selskabs Forhandlinger. 1856), p. 155, 166. — *Joule*: *Scientific Papers.* I, p. 269, 273. — As regards Mayer, it appears from his address at Innsbruck (*Die Mech. d. Wärme*, p. 357) that he did not assume any break in the series of material *[598]* phenomena, but assumed a parallelism between the spiritual processes and the brain processes.

wealth of knowledge has been spread abroad by it in wide circles. Büchner's *Kraft und Stoff* is one of the most read popular-scientific books of our century; from 1855 to 1899 nineteen German editions appeared, and it has besides been translated into several other languages. Even if there is something dogmatic about materialism, it has nevertheless done good service by setting its dogmatism against ecclesiastical dogmatism, and thereby awakening reflection upon problems which had been pushed aside at the dissolution of Romantic philosophy. The whole materialist movement in Germany was moreover borne by an idealistic interest in humanity and progress, and Büchner was fully justified in protesting against the confusion of materialism as method and theory with materialism in the sense of a practical direction of life. Materialism can very well acknowledge the value of the highest and noblest ideas and feelings — even if it holds that they, like all spiritual phenomena, are only products of, or forms of, material processes.

When materialism claims to be the simple consequence of the results of natural science, it is of interest to observe — not only that the discoverers of the law of the conservation of energy proceeded throughout from spiritualistic presuppositions, but that the most important collisions in the materialist controversy took place between parties who all stood upon the ground of natural science. This does not necessarily show that materialism is wrong; but it shows how difficult it is to draw the true consequences, and how many different motives come to influence the world-view of individuals.

[502] Moleschott's famous book *Der Kreislauf des Lebens* (1852) is directed against Liebig's theologizing utterances in his "Chemische Briefe", in which he had particularly attacked an earlier proposition of Moleschott's: "Ohne Phosphor kein Gedanke!" At the same time a controversy broke out between the physiologist Rudolph Wagner in Göttingen and the zoologist Carl Vogt in Geneva. This controversy reached its height when Wagner, at a congress of natural scientists in Göttingen in 1854, declared himself in favor of the assumption of an ethereal soul-substance which moves the brain's fibres as the musician moves the pianoforte, and which is propagated from parents to children by division [!] — an assumption he supported upon the Bible's teaching, but which he admitted he could maintain only by a sharp distinction between faith and knowledge. For when one goes beyond the domain of exact knowledge one has, according to Wagner, only an unprovable faith to hold to — and he adds: "In Sachen des Glaubens liebe ich den schlichten, einfachen Köhlerglauben am meisten." Against this Vogt came forward with a robust polemic: *Köhlerglaube und Wissenschaft* (1855). In a later work, *Vorlesungen über den Menschen* (1863), Vogt gives a more scientific presentation of his standpoint. He lays weight above all upon the brain's being the organ of consciousness, and upon

consciousness standing to the brain as every function stands to the corresponding organ. Although he admits that it cannot be explained *how* consciousness arises in the brain's cells, he holds fast that it is indissolubly bound to them; and he does not give up the teasing proposition he had already set up earlier, that thought stands in the same relation to the brain as bile to the liver or urine to the kidneys — notwithstanding that a function stands in a different relation to its organ than a product does to the place where it is produced.

Moleschott relies, as the title of his work shows, especially upon the doctrine of the conservation of matter. It is for him the great thought which the Encyclopedists of the eighteenth century had already brought forward, which subsequent natural science has confirmed, and upon which the faith of the future must be built. He feels reverence for the great circulation in nature. The miner fetches phosphate of lime up out of the earth in the sweat of his brow — and perhaps thereby the matter destined for the best head and the highest thoughts passes through his hands: the farmer manures his soil with the *[503]* phosphate of lime; this becomes a constituent of the wheat that nourishes men's bodies and brains! With matter, life circulates through all parts of the world; with life, thought — and from thought there springs in turn the will to make life better and happier. If we can but supply the organism and the brain with the best substances, thought and will too will attain their highest development. The investigator of nature is our age's Prometheus, and chemistry is the highest science. The social question finds its solution if one can but [sic] find the right distribution of the substances to which the life of thought and of will is bound. — Only the first edition of Moleschott's book has been taken into account here; in later editions it was altered and expanded. Moleschott, who was born in Holland in 1822, was working as a lecturer in Heidelberg when the book appeared; but when his freedom of teaching was interfered with there, he moved to Zürich; later he worked as professor of physiology in Turin and in Rome, where he died in 1893. After his death his autobiography appeared (*Für meine Freunde.* Lebenserinnerungen von Jacob Moleschott. Giessen 1895). It gives a fine picture of the idealistically minded investigator of nature, striving after an all-round culture. For a right understanding of his standpoint the following utterance is of service, which comes after the train of thought of "Kreislauf des Lebens" has been developed: "One-sidedly 'materialistic' this is only for those who can imagine a matter without force or a force without material bearer. I was clearly conscious that one could turn the whole way of representing it round: since all matter is a bearer of force, is endowed with force or penetrated by spirit (*geistesdurchdrungen*), one may just as well call it an idealistic view" (p. 221). Moleschott's standpoint is rather to be

called monism than materialism: "It is a question of a true, indivisible two-in-oneness, and the dispute is not between materialism and spiritualism, but between two-in-oneness (*Zweieinigkeit*) and divided duality (*zwiespältige Anschauung*)" (p. 222).

Louis Büchner (1824–1899) does indeed declare that he will not explain what the relation is between spirit and matter, force and substance; he will only maintain that
[504] there is a necessary and indissoluble relation between them. But even this is more than can be scientifically proved; and Büchner moreover harbors no doubt that spirit is only a property of matter, force only a property of substance. The impulse to his famous book "Kraft und Stoff" he had received from reading Moleschott's "Kreislauf des Lebens", and it is the eternity of matter that is the ultimate foundation in him too. He holds that the conservation of force is really a self-evident consequence of the conservation of matter, and could therefore already have been drawn from Lavoisier's chemistry. It was only in the fifth edition of his book, however, that he inserted a special chapter on "the immortality of force", in which the circulation of force is set as a necessary correlate beside the circulation of matter, so that "the two together form, from eternity and to eternity, the sum of phenomena which we call the world". Yet he constantly holds fast to the assumption that "in matter dwell all the forces of nature and of spirit; matter is the primal ground of all being". And as regards the spiritual, he regards it — notwithstanding that he has disclaimed all insight into its relation to matter — as a mere product. "As the steam engine produces motion, so the complicated organic composition of force-endowed substances in the animal body produces a total sum of certain effects which, when they are bound together into a unity, are called by us spirit, soul, thought" (Kraft und Stoff. 7. Aufl., p. 130). In a later utterance Büchner goes — without appearing to notice it — a step further. Thus it runs, in an utterance directed polemically against the author of the present work: "The opposition between physical and psychical energy is tenable only so long as one conceives the concepts of body and spirit, or more generally of matter and force, in a dualistic way, whereas for the materialistic-monistic conception they coincide into one and perhaps present only *different sides or forms of expression of one and the same primal ground of all things*. It is precisely the law of the conservation of force that in the problem of the soul leads necessarily to materialistic consequences, as I, supported by physiological-psychological facts and considerations, believe I have established to evidence in several of my writings, and recently again in the fifth of my
[505] letters on *Das künftige Leben und die moderne Wissenschaft* (Leipzig 1889). I believe I have there shown that psychical activity *is not, and cannot be, anything other than the radiation, taking place between the cells of the gray cerebral cortex, of a movement initiated*

by outer impressions" (from an article in the weekly *Menschthum.* Gotha 1889, No. 46).

It is clear that Büchner in this utterance confuses two different theories: the one, which conceives spirit and matter as forms of expression of one and the same substance, and the other, for which matter is "the primal ground" and psychical activities are therefore consistently *nothing other than* motion (or "radiation of motion"). In one and the same breath he professes both. The tireless champion of the influence of natural-scientific results upon our world-view has not seen this inconsistency, which perhaps in his eyes did not much matter, since his real aim was not to inculcate any one determinate philosophical theory. In his great zeal he has also believed that he had "established to evidence" more than can be established.

With sharper criticism Heinrich Czolbe (1819–1873), a physician like Büchner, sought to determine the consequences of the teaching of natural science. In his first works he is a consistent materialist. In *Neue Darstellung des Sensualismus* (1855) and *Die Entstehung des Selbstbewusstseins* (1856), as well as in the paper *Die Elemente der Psychologie vom Standpunkte des Materialismus* (Zeitschrift für Philosophie. XXVI), he develops the view that materialism conflicts with natural science so long as the latter assumes the doctrine of the specific energy of the sensory nerves, and thus assumes a difference between the sensations and the outer processes corresponding to them. In opposition to that doctrine Czolbe sets up the assumption that it is essentially one and the same motion which propagates itself from the surrounding world, through sense-organs and nerves, to the brain. At no point does any qualitative change take place; the difference between the different senses is to be explained by the different intensity with which the motion proceeds in the different organs. The unity of consciousness is to be explained by the movements in the brain acquiring a direction that runs back into itself; *[506]* by making such circular movements possible, the brain is the organ of consciousness. Thus Czolbe would conceive both sensation and self-consciousness, plainly and simply, as movements in space. But — clear and consistent thinker that he is — he becomes aware that if he identifies sensation and self-consciousness with motion without more ado, then one must also be able to reverse the proposition and say that wherever motion of a certain intensity and a certain direction is present, there consciousness is present too, so that the consequence is a universal ensoulment of nature. Thus the consistent carrying through of materialism led beyond it.[11]

[11]The consequence Czolbe draws from materialism was already drawn by *Henry More*, who (*De anima.* Roterodami 1677, p. 64) sets up the following fundamental proposition: "If sensation can be ascribed to matter, then the motion of one material part, or its reaction upon another material part, or at any rate the continuation of that motion, is one and the same as sensation and apprehension. This proposition ... is confirmed by the voice of my chief opponent, Hobbes." He then refers to Hobbes's doctrine of sensation as

Later — in the work *Die Grenzen und der Ursprung der menschlichen Erkenntniss* (1865), and in the exceedingly interesting paper *Die Mathematik als Ideal für alle andere Erkenntniss und das Verhältniss der empirischen Wissenschaften zur Philosophie* (Zeitschrift für exakte Philosophie, 1866) — Czolbe saw that it was impossible to explain existence from a single principle, whether one finds that principle with Büchner in matter, or with the speculative philosophers in spirit, or with the theologians in God. We reach an explanation only by proceeding from several elements, calling that an element which we are unable to analyze. Such elements, not reducible to one another, are the material atoms, the organic forces, and the mental elements (which last make up in sum the world-soul). Between these three kinds of element a harmonious co-operation takes place, by which a purposive connection in nature is realized. Not in its origin, but in its direction and tendency, does existence establish itself as a unity. — Thus it appeared that the problem of existence was more intricate than materialism supposed. In his later investigations, which are only partly published, Czolbe approached Spinoza's fundamental thoughts, of which he wished to give an empirical execution. — On the different stages of development of this energetic thinker's philosophizing, reference must here be made to Vaihinger's paper *Die drei Phasen des Czolbeschen Naturalismus* (Philosophische Monatshefte XII).

[507] What lies at the ground in Czolbe from first to last is the demand for clarity and intuitability in all fundamental concepts. He wanted everything mystical and supersensible away, and sought so far as possible to think all ideas with geometrical clarity. It was more with respect to intuitability than with respect to stringency of proof that he proclaimed mathematics the ideal of science. In his youth he was enthusiastic for Hölderlin's poetry, and had set himself the task of maintaining Greek clarity and joy in life against all mysticism and dualism. Intuitability in the domain of thought was for him closely akin to joy in the natural world. Just as it may require labor to carry through the principle of intuitability in thinking, so it may cost resignation to carry through joy in actual life in practice; but in the theoretical domain as in the practical, Czolbe would combat what he calls "der Blödsinn der Transcendenz". He admitted already in his materialist period that the innermost motive of his philosophy was the subjective need for intuitability and for holding fast to the actual; materialism was to him a postulate — which was later replaced by other postulates. —

Ernst Haeckel (b. 1834, from 1865 professor of zoology in Jena) is often reckoned to materialism, but has himself determined his standpoint as a monism which leads beyond the opposition between spiritualism and materialism, and which takes as its

a motion. Hobbes, however, denied that consequence, saying expressly that the proposition that sensation is motion cannot be reversed. (*De corpore* XXV, 5).

ground the great pantheistic thought of the unity of nature. For monism, he says, there exists neither spirit nor matter in the usual sense, but only one thing which is both at once. It was in his *Generelle Morphologie* (1862–1866) that he first uttered his general philosophical views. The mental is for him — as for Czolbe — an original element in existence, though it exists in exceedingly different degrees, from the atom-soul and the cell-soul to the souls of the highest organisms. This theory of ensoulment becomes questionable in Haeckel through his often letting the soul's action serve as an explanation of organic movements, instead of seeking a purely natural-scientific explanation. On the other hand he sometimes declares mental phenomena to be mere effects of the most complicated and most unstable carbon compounds, the albumen molecules in nerve *[508]* tissue. He believes himself to be beyond spiritualism and materialism, and yet one finds in him both propositions that sound spiritualistic and propositions that sound materialistic. He has not arrived at his principles through a sharp thinking-through of the laws of the conservation of matter and of force. It is Darwin's doctrine that set his thinking in motion, and he has sought a highest principle that could be united with the great unity in nature to which the new doctrine pointed. Monism leads, according to Haeckel (Gen. Morphol. II, p. 445–451), to the most exalted idea of God. In the great, all-embracing law of causality the Godhead reveals itself as embracing the whole of nature and operating in every natural phenomenon. The usual conception of God, by contrast, is belief in two gods — amphitheism, not monotheism — because it sets the natural causes beside the Godhead.

Haeckel was one of the first German investigators of nature to attach himself to Darwin, and he fought with great ardor for his theory, to which he ascribed a certainty, and gave an extension, that could not always be approved by the theory's critical and level-headed founder. He was untiring in constructing genealogical trees for the species now living, and he had no scruple about assuming a constant arising of organic matter. He has no clear eye for the limits of hypotheses and for the necessity of verification; only thereby is it explained that he can without more ado set Darwin's hypothesis alongside Newton's. He accordingly also reproached Darwin with laying too much weight upon the objections to the theory. The great investigator shook his head at his young adherent's zeal: "Your boldness sometimes makes me tremble", he wrote to him (19 November 1868). — In Haeckel's most recent "philosophical" work (*Die Welträthsel.* Gemeinverständliche Studien über monistische Philosophie. Bonn 1899) the qualities mentioned appear in a still more glaring form than in his earlier utterances. —

The doctrine of the conservation of matter and of force, and the doctrine of the

development of species through natural selection, were — as already remarked — the natural-scientific results which in the period after the middle of the century had above all
[509] to set reflection in motion. Having now considered the attempts made by investigators of nature to found a philosophy upon these results, we shall go on to examine the way in which professional philosophers took up their position with respect to them.

3. Idealist Constructions on a Realist Basis

a. Rudolph Hermann Lotze

Idealist philosophy has in Lotze its most important representative in the last half of the century. In his personality and in his development he stands as one who has taken up into himself, in an interesting way, both the ideal motives upon which the philosophy of Romanticism built and that strict carrying through of the mechanical view of nature which toward the middle of the century was working its way forward in science. Lotze is a master of conceptual development — of discussing a thought so that all its nuances come out, of taking up a problem again and again and treating it from different points of view. His ideal was in the last instance the same as the one that floated before the philosophy of Romanticism: to derive all development and all connection in the world from an eternal Idea, which should contain the ultimate ground of all that happens, as well as of the value of what happens. But he saw clearly that such a derivation exceeds what human thought can accomplish, and that when the thinkers of Romanticism nevertheless believed they had given one, it was because in them the poetic-religious tendency involuntarily merged with the philosophical. Lotze therefore separates the poetic-religious element from the speculative. His own fine poetic sense let him understand that their Romantic confusion was not a happy one. And this poetic sense is one with a sense for individual nuances and relations, which speculative philosophy all too often dissipated by its abstractions. Through this sense Lotze's speculative interest hangs together with his
[510] realist tendency. What matters for him is first and foremost to apprehend the phenomena in their concrete nature and in their determinate, law-governed connection; it is then the task of philosophical thinking to find out the presuppositions upon which that real connection rests. The poetic, the natural-scientific and the philosophical element in Lotze thus hang closely together, and he was equipped as few have been to treat the task he had set himself, and which the intellectual conditions of the age suggested: to attempt a reconstruction of an idealist philosophy upon a realist basis. It was his conviction that the philosophy of Romanticism had overlooked the real conditions and the mechanical

connection of nature, without which even the most significant ideas are helpless idols; just as it was his conviction that materialism made *that* into the be-all and end-all which was only a — quite certainly necessary — form and frame for the valuable content of existence. His philosophy is essentially an analysis of the very concept of natural mechanism, designed to show that this concept leads with necessity to the assumption of an ideal principle of existence, and that it at any rate does not exclude the assumption that such a principle is the eternal source of all that is good and valuable in the world.

In Lotze's development the different motives and interests that animated him appear clearly. He was born in Bautzen in Saxony on 21 May 1817, in the same region from which Lessing and Fichte come. At the university of Leipzig he studied philosophy, medicine and physics. He was here initiated into the two circles of thought which it was later to be his task to unite, after having worked through them separately. It was the aesthetician and philosopher of religion Chr. Hermann Weisse who was his teacher in philosophy, and Lotze later said that he owed to Weisse not merely many particular influences, but that he was in particular introduced by him into a circle of thoughts which he has since felt no need to give up. Weisse was the most eminent representative of philosophical theism. Through him Lotze's religious-philosophical conception stands in historical connection with Schelling's — and through that with old Jacob Böhme's. Lotze emphasizes, in a retrospect upon his development (*Streitschriften.* 1857, p. 5 ff.), that *[511]* he appropriated the ideas of speculative philosophy not as a closed doctrinal edifice but as a peculiar form of intellectual culture. The presupposition that the ultimate ground of things could be thought only as spiritual was early laid down in him and never given up. Medicine and physics he studied under E. H. Weber, Volkmann and G. Th. Fechner. Here he came to know the natural-scientific method and conception at first hand. In one and the same year he took the doctorate in philosophy and in medicine, and now came forward as lecturer in both subjects — also after a professorship in philosophy had been conferred upon him in 1842. After working some years in Leipzig he became professor in Göttingen, as Herbart's successor, and here fall his most important works. The year before his death he received a call to Berlin, but there soon succumbed to an illness from which he had long suffered (1881). His life was quiet, devoted to study, thought and academic teaching. With a rare union of many-sidedness and thoroughness he knew how to orient himself in highly different fields — something to which not only his medical and philosophical works but also a whole series of smaller papers and reviews (published after his death in four volumes under the title *Kleine Schriften*) bear witness. From his more rigorous scientific labors he rested in occupation with art and literature: thus he

occasionally translated "Antigone" into Latin verse.

As a medical writer Lotze set himself, as already touched upon, the task of maintaining physiology's character as a mechanical natural science. He would not have the peculiarity of the organic phenomena explained by appeal to a mysterious vital force, but by exhibiting the determinate, law-governed manner in which the general forces of nature operate within the organism. As everywhere in nature, so here too we must find the explanation in the interaction of real elements. Not by being withdrawn from the mechanical connection of nature, but by the peculiar manner in which connected series of effects arise, does organic life distinguish itself from the inorganic world. In speculative philosophy of nature the idea of the organism had been made the type for the
[512] whole conception of the world; that this idea was corrected had therefore to be of general philosophical significance, and one understands that not only Lotze's medical but also his philosophical interest was satisfied by his labors in this direction (*Allgemeine Pathologie und Therapie als mechanische Naturwissenschaften.* 1842. — *Allgemeine Physiologie des körperlichen Lebens.* 1851). These works even gave some materialists occasion to see a fellow-thinker in Lotze, notwithstanding that he stated definitely that mechanism was for him only a part of his conception of nature, not the whole. It is characteristic of Lotze that he thus divides the problems. His general philosophical views he had developed in his *Metaphysik* (1841). But he by no means felt himself finished when he began his teaching activity, and especially as regards the concluding ideas he constantly referred to future works. In *Medicinische Psychologie oder Physiologie der Seele* (1852) he expressed himself in detail upon the relation between the spiritual and the material, and at the same time gave psychological investigations of lasting interest. A long-cherished plan he carried out in the following period (1856–1864) with the publication of his *Mikrokosmus* (3 vols.). This work, conceived as a counterpart to Humboldt's "Kosmos" and Herder's "Ideen", gives a psychology in close connection with physiology and cultural history, and ends with a development of Lotze's cosmological and religious-philosophical ideas. For all his interests, for all the different roads his thought and feeling had traveled, he had here found expression. The work is laid out as a popular presentation and has won a considerable circulation. After publishing a spirited *Geschichte der Aesthetik in Deutschland* (1868), Lotze proceeded to give a concluding systematic presentation of his philosophy. But only two parts of it did he succeed in finishing (*Drei Bücher der Logik.* 1874. — *Drei Bücher der Metaphysik.* 1879). The third volume, which was to comprise aesthetics, ethics and philosophy of religion, was not worked out. His philosophy is thus unfinished; it was not granted him to set the crown upon the work and show how

the many threads of thought he had spun could be woven together into unity. — Short presentations of Lotze's teaching in the different philosophical fields are to be had in the *[513]* dictated notes (*Grundzüge*) from his lectures, published after his death. Among these must especially be singled out *Grundzüge der Logik und Encyclopädie der Philosophie*, which (in its second part) gives a short and clear survey of his whole system. Here we give a characterization of the three most important points in his philosophy.

α. The Mechanical View of Nature

Lotze's thinking has two points of departure: a living feeling for the value of spiritual life, for the fact that the highest thing for us is bound up with spiritual development and its ideals — but at the same time a firm conviction that a system of mechanical causes and laws is necessary to the realization even of the highest ideals. He can, as he says somewhere, believe in ideas that operate, but not in ideas that bewitch. The two points of departure do not, however, stand wholly isolated in him. He seeks, by proceeding from the one, to show the possibility of acknowledging the other. The most characteristic and the most scientific feature of his philosophy is his attempt to establish, through an analysis of the very concept of mechanism, that its ultimate presupposition is the concept of a principle which — as soon as we conceive it in a way intelligible to us — shows itself as the bearer of, and the source of, the highest ideas. Lotze means to take the bull by the horns: by drawing out realism's consequences he will found idealism. Only investigations conducted in realism's spirit can, in his view, bring us nearer to the goal idealism sets itself: the knowledge of the world as the expression of a valuable idea. Romanticism's philosophy had believed it could derive the forms of reality from the highest Idea. That has shown itself to be an insoluble task. We can, by contrast, attempt by way of inference to go back from what is given to its presuppositions. Not a *deduction* but a *reduction* is possible. Thinking must always apply its forms to a given material. Both general culture and the several sciences here operate with a mass of concepts whose origin, meaning and validity they do not investigate more closely. Such concepts are cause and effect, matter and force, end and means, freedom and necessity, *[514]* matter and spirit. It is philosophy's task to bring unity and connection into the circle of representations by making these concepts — presupposed in practical life and in the several sciences — the object of a peculiar investigation, and by determining the limits of their validity (*Grundzüge der Logik und Encyclopädie der Philosophie*, § 88).

Of these concepts the most important is the one we presuppose in every investigation of reality — every time we make an experiment, every time we seek an explanation: the

concept of a pervasive, all-embracing causal connection. This concept is not grounded upon experience, but is presupposed in all experience. But since the understanding of nature so far won rests upon its reality, it can itself be described as the expression of a fact; and that fact — that the single element in our experience is bound by a law-governed connection to other elements, the fact of mechanical connection — is what philosophy has to think through and to draw all the consequences of. Deduce it, philosophy cannot; but it may perhaps discover what lies in it. A *multiplicity of real elements in mutual interaction* is the foundation upon which the mechanical view of nature builds; and Lotze was, as we have seen, so convinced of that foundation's inevitability that he himself worked with zeal to gain it recognition within physiology, where hitherto vitalism had reigned with its appeal to a single forming and directing "vital force".

But because mechanical connection is an inevitable feature of our conception of the world, it need not be the only, all-determining feature. On the contrary: even if its validity is unlimited, it may perfectly well have a subordinate significance in the world as a whole. (*Die unbeschränkte Gültigkeit des Mechanismus, aber zugleich seine durchaus untergeordnete Bedeutung im Ganzen der Welt! Drei Bücher der Metaphysik*, p. 462.) And a closer analysis of the concept of mechanism leads us to see that this is precisely how the matter stands.

[515] The mechanical view of nature stops — when it is made into a concluding world-view — at a multiplicity of elements (atoms) in mutual interaction. It proclaims a pluralism. But how do the elements stand to the connection in which they stand? Can they have independent subsistence apart from that connection, so that it is a relation indifferent to their essence — or are they not determined right through by the combination in which they stand with the world as a whole? Interaction and connection cannot, after all, float freely above or between the elements, but presuppose an inner unity of them. For if I assume that the two elements *A* and *B* are wholly independent, interaction between them becomes unintelligible. An effect cannot be transferred as a fully finished state from *A* to *B*. What goes on in *A* can acquire significance for what goes on in *B* only if *A* and *B* are not in reality independent, absolutely separated beings, but if their states are states in one and the same, more comprehensive being. A multiplicity of independent beings would make mechanical interaction unintelligible; intelligibility comes only with the assumption of an infinite, all-embracing being, of which the single elements are moments, or points of action. The concept of a force's or an influence's "passing over" from one independent element to another is untenable. Not a *transient* cause (*causa transiens*), only an *immanent* cause (*causa immanens*) is intelligible. The states of one

and the same being can stand to one another as ground and consequent; the states of two mutually independent beings cannot. — Among the numerous presentations Lotze has given of this train of thought, *Logik und Encyclopädie* §§ 99–100 and *Drei Bücher der Metaphysik* §§ 50–81 must be singled out.

By analysis of the concepts of the causal relation and of interaction — the fundamental concepts of the mechanical view of nature — Lotze has thus arrived at the idea of a primal substance, an all-embracing principle. His thinking here moves along the same paths as those which in their day led Spinoza to his concept of substance, and which Kant walked especially in his youthful writings, without however abandoning them in his later works. He is reminiscent of Leibniz too at several points, without there seeming here, any more than in Spinoza's case, to have been any influence through study. When he heard that *[516]* he was being placed alongside Spinoza and Leibniz, he said (see Falckenberg: *Hermann Lotze*. Stuttgart 1901, p. 85) that he did not feel himself to stand in any inner relation to them, though he admitted that there was occasion enough to institute a comparison.

The concept which Romanticism's philosophy would lay at the ground, and from which it would derive everything, becomes for Lotze the last postulate — or, as he also expresses it, the last fact for our thinking. A closer, intuitable development of the universal principle presupposed in the simplest relation of interaction is impossible. We stand here at a *limiting concept* which we can neither do without nor develop. (*Drei Bücher der Metaphysik*, §§ 73 and 246.) It is nevertheless this concept that makes it possible for Lotze to hold fast to what was for him the essential thing in the idealist train of thought into which his teacher Weisse had introduced him. By it he found a connection between the opposed currents of the world of thought. Neither materialism's absolute atoms, nor Leibniz's monads, nor Herbart's reals could for him mark the conclusion of thinking: pluralism must necessarily give place to *monism.* As regards the atoms in particular, Lotze maintains that natural science's interest leads only to assuming elements which are in fact indissoluble for our experience, but that the assumption of a multiplicity of extended elements — even if these are thought of as vanishingly small — cannot be any conclusion for thought. Either one must give up the atom's unity or its extension: within an extended atom every effect will take time and pass from part to part — and these parts will then be more fundamental unities than the whole "atom". In the concluding concept of the atom we must abstract from all extension and think of the atoms as centers of force which — according to the analysis of the concept of mechanism given above — are each of them points of departure for the primal substance's operation. (*Drei Bücher der Metaphysik*, §§ 190–191 and 245; cf. the interesting review of Fechner's

"Atomenlehre" in *Kleine Schriften* III, p. 215–238.)

[517] Lotze's assertion of the mechanical view of nature, and the consequences he draws from it, make up the most significant part of his philosophy — notwithstanding that it is as a rule not brought forward when Lotze is praised. It is the spiritualistic and theologizing tendency in him that is most frequently emphasized, by admirers and opponents alike. And yet he has earned his greatest merit as a thinker by his analysis of the fundamental concept of the scientific view of nature. It is a defect in him that he does not undertake an epistemological investigation of the limiting concept he arrives at, but lets it stand in the form in which Romanticism's philosophy and the earlier dogmatism had conceived it. Lotze had in general no understanding of the importance of epistemology.

β. Metaphysical Idealism

In Lotze's train of thought, so far as we have presented it hitherto, there is still a certain indeterminacy. It might still be asked whether we then have no way at all of forming a closer representation of the elements and of the primal substance. In now bringing forward Lotze's answer to this question, it must first of all be emphasized that he is entirely clear that any answer to it rests upon analogies, and is determined by motives other than the purely theoretical. The law of analogy, and the need to find our own spiritual being again in existence — which have shown themselves to lie at the ground in all metaphysical idealists, and especially in Leibniz, Berkeley, Schelling, Schopenhauer and Beneke — appear with full consciousness in Lotze.

Lotze is an atomist — but he does not conceive the atoms as material, because extension, as well as all other sensible qualities, must be explained by the interaction of atoms, which therefore cannot themselves have that property. Like life and all qualities of experience, the sensible fact of extension is due to an interplay of force-points, which are in turn to be conceived as initial points of the infinite primal being's inner operation. Lotze does not regard it as impossible that wholly new beginnings may occur in the *[518]* course of the world: this does not conflict with universal lawfulness, for the law always expresses only the order in which the states arise, not an outer fate. The number of the elements need not be constant, but may be thought to rise or fall according to the demands of the highest world-purpose. (*Drei Bücher d. M.* §§ 33 and 235.) Nor need the single elements be wholly homogeneous with one another. A certain agreement or commensurability must indeed be present if one world-order is to embrace them all; but a complete likeness is not necessary. A law of nature may very well bind together qualitatively different elements for which we are unable to find any common denominator.

It is in the end not a *logical* but an *aesthetic* necessity that makes existence intelligible to us (*Drei Bücher d. M.* § 59): not a formal consistency in the primal being's operation, but the richness and fullness with which that operation proceeds, is decisive for the constitution of the single elements. At this point Lotze plainly enough passes over from purely theoretical motives to motives of feeling — without either the one or the other enabling him to determine more definitely how great *differences of quality* are compatible with the *commensurability* which the fact of interaction presupposes.[12]

If we would now form a representation at all of the inner nature of the elements, we must conceive them on the analogy of our own spiritual being. The mechanical view of nature really instructs us only about the mutual relation between the elements, not about their inner nature. It concerns the outer circumstances, moves the whole time around the outside of things — is a *cognitio circa rem.* It is perhaps even inclined — like the popular conception of the world — to hold that it is wholly indifferent to things whether we apprehend them or not. In opposition to this Lotze maintains (already in his youthful work *Metaphysik*, 1841, p. 313) that subjectivity is just as much a part of reality as the outer objects are: *Nicht nur was zwischen den Wesen, sondern auch was in ihnen sich zuträgt, ist ein wahres und wirkliches Geschehen.* In his later writings — first in *Medicinische Psychologie* — he goes a step further, maintaining that our own *[519]* subjective being is the only case in which we know a thing's interior and have a *cognitio rei*, not merely a *cognitio circa rem*: the only way of forming a representation of the interior of other things is then to conceive them on the analogy of ourselves — that is, as feeling beings (if not as representing beings, since feeling is a more original expression of consciousness than representation). The use of this analogy is the only means of thinking things as real beings subsisting for themselves, which are more than mere images for us. We thereby merely make use of the general method that leads the unknown back to the known. (*Drei Bücher d. M.* §§ 96–98.) And the "aesthetic necessity" demands this too, since it cannot be reconciled with an existence in which a great part — perhaps the greatest — should be merely a dark foundation for a mental life extended to only a few regions (see already *Allg. Physiologie* § 129). The elements of existence must therefore be assumed to be ensouled in manifold degrees.

As it stands with the elements of existence, so it stands also with the all-embracing world-ground that makes their interaction intelligible. If we follow the law of analogy and attempt to lead the unknown back to the known, then according to Lotze we must conceive the world-ground as a spiritual being. When inner states are to be bound into

[12] On the indeterminacy in Lotze at this point see *Max Wentscher*: *Lotzes Gottesbegriff und dessen metaphysische Begründung.* Halle 1893, p. 19–29.

unity and connection, our own inner, spiritual life is the only example we know in which the possibility of such a preservation of unity through the different states is realized — at least approximately realized. Following his teacher C. H. Weisse, Lotze conceives the world-ground as an absolute personality, and he defends this transference of the concept of personality to the absolute being on the ground that only an absolute being can be a personality, because only it has inner independence and originality, while the concept of personality finds only imperfect application to finite beings, which are dependent upon outer conditions. Lotze admits, to be sure, that to a personal life there belong resistance to be overcome, and a capacity to suffer and to receive as well as to act. But
[520] when one then asks how an absolute being, subject to no limits, can suffer, Lotze answers (*Grundzüge der Religionsphilosophie* § 34, cf. § 52) that what sets the Godhead's feeling in motion must be "inner products of its own creative imagination"! — It is surely a great question, however, whether such a self-created resistance can mean anything serious — especially as it can be annihilated at whatever moment one pleases. The personal life we know has in any case precisely not self-created and not so easily removable limits to struggle with, and the analogy upon which Lotze relies therefore seems to fail at just the decisive point. And to this it must be added that Lotze — on the most probable interpretation of his obscure and shifting utterances[13] — differs from Weisse in that he does not hold the form of time applicable to the absolute being. A personal being that does not develop in time — a timeless life, and a timeless suffering and acting: these are concepts that make far too great demands upon our capacity for analogy! — It need only further be remarked that Lotze — as regards both the elements and the world-ground — presupposes that the opposition between spirit and matter is contradictory, so that no

[13] *Reinhold Geijer*, in his paper *Hermann Lotzes tankar om tid och timlighet i kritisk belysning* (Lund 1886), drew attention to the shifting standpoints Lotze seems to have taken on the question of time's absolute validity. In my paper *Lotze og den svenske Filosofi* (Letterstedtske Tidsskrift 1888) I associated myself with Geijer's interpretation, though I had to conceive the problem itself otherwise than my Swedish colleague. Yet the matter seems to stand otherwise than Geijer, and I with him, had reason to assume. According to what *Richard Falckenberg* has shown in his paper *Die Entwickelung der Lotzeschen Zeitlehre* (Zeitschrift für Philosophie und philosophische Kritik, vol. 105), the *Grundzüge der Metaphysik* published after Lotze's death date from as early as 1865, and with that an essential support falls away for the assumption that Lotze first denied the reality of time toward the close of his life. When in earlier writings he speaks without qualification of the time-relation as absolutely valid, this is according to Falckenberg a legitimate use of popular representations where the context did not require a more stringent usage. Lotze's definitive conception was the one presented in *Drei Bücher der Metaphysik* (1879) and in *Grundzüge der Religionsfilosofi* (from 1878–79), according to which succession holds for finite beings, while the Godhead is raised above every difference of time. *This* view departs from his earliest presentation (in *Metaphysik*, 1841) in asserting this distinction, which admittedly does not make the whole question clearer — though the *psychological* distinction between succession and abstract time-form may well be defensible. — As little as my Swedish colleague am I able to connect any meaning with the expression "ein zeitloses Geschehen und Wirken" (Grundzüge der Metaphysik § 58). My German colleague, by contrast, finds a significant thought in it.

further possibilities could be thought than the two: only on that presupposition is there compelling necessity to conceive the elements and the world-ground as spiritual beings, when one cannot or will not conceive them as material. But there is no logical, only a factual necessity for choosing between spirit and matter in determining the concluding concepts. We in fact know only these two forms of existence; but there would be nothing *unthinkable* in one or more other forms existing. Therefore metaphysical idealism cannot be proved.[14]

Lotze is now also fully conscious that the conclusion of his whole world-view rests upon motives of feeling, not upon strict thinking. The necessity of going back from the multiplicity of the elements to the unity of the world-ground he asserts, indeed, on purely theoretical grounds: "*Gar kein Weltlauf, weder ein harmonischer noch ein unharmonischer, ist mir ohne die Voraussetzung jener Einheit begreiflich, die alles gegenseitige Wirken* [521] *des Einzelnen erst möglich macht; die Störungen der Dinge durch einander bezeugen die beständige Gegenwart dieses Einen eben so eindringlich, wie das Zusammenstimmen der Kräfte zum Zweck*" (*Drei Bücher d. M.* § 233). But when he leads everything back to a single principle, he knows very well that we cannot construct the world-connection and the elements out of it: we always lack the second premiss! (*Drei Bücher d. M.* § 93.) Only in practical conviction, then, can the thought be held fast that all beings and all events have their ultimate ground in the highest purpose maintained by the world-ground. What that purpose is we do not know — any more than we know how the conflict of forces in nature and the reality of evil in the moral world are compatible with the validity of the world-plan. (*Mikrokosmus*, Buch IX, Kap. 5. — *Drei Bücher d. M.* § 233.) But in an ethical idea Lotze found, at the beginning as at the close of his career, the final conclusion for his thinking: metaphysics goes back for him to ethics; in what *ought* to be lies the ultimate explanation of what *is*. —

If by pantheism one understands the doctrine which maintains an inner relation between God and the world — whether the Godhead is conceived as a personal being or not — then Lotze's religious-philosophical standpoint is to be described as ethical pantheism. He himself felt closely akin to Fichte; the kinship with Spinoza he would not properly acknowledge. In the use of theological words and turns of phrase he went — especially in *Mikrokosmus* and in the *Grundzüge* of philosophy of religion and of moral philosophy published after his death — further than was fitting and warranted. By his

[14]Compare my *Psykologi*[4], p. 80; *Psykologiske Undersøgelser*, p. 91 f.; *Filosofiske Problemer* (Universitetsprogram 1902), p. 62–65. — It is interesting that the founder of metaphysical idealism, Leibniz, in his [599] earlier period, like Spinoza, laid great weight upon the duality spirit–matter being purely factual, not logically necessary. *Philos. Schr.*, ed. Gerhardt I, p. 237, 242, 268. Compare *Ludw. Stein: Leibniz und Spinoza*, p. 93 f.

usage (for it is nothing more) it came to look as though he stood nearer to the popular religious circle of ideas than in fact he did. He does, however, occasionally emphasize (especially *Mikrokosmus*, Buch VIII, Kap. 4) the difference between free religiosity and the positive religions.

γ. Spiritualistic Psychology

[522] The elements whose interaction produces the phenomena that experience presents need not — as we have seen — according to Lotze be absolutely homogeneous. There need be only such homogeneity as the interaction itself presupposes. By this very indeterminate rule Lotze makes it possible for himself to hold fast to the popular assumption of an interaction of soul and body. The elements which the concept of mechanism presupposes are for him partly psychical, partly physical. He disputes the assumption of a continuous series of physical phenomena. At certain points the physical motion is broken off, to be "absorbed", that is, to pass over into psychical states. It is indeed doubtful whether Lotze, when he first put forward this claim (in the year 1851; *Allg. Physiol.* § 424), had any living conception of the significance of the newly discovered law of the conservation of energy — though neither did he later find in it any reason to depart from the standpoint he had once seized. There were for him compelling grounds which made the assumption of a peculiar soul-substance necessary. Only two possibilities presented themselves to him: either to derive the psychical phenomena from a soul as a principle peculiar to them, or to explain them by the co-operation of the physical forces. (See *Selbstanzeige der medicinischen Psychologie*, Kleine Schriften III, p. 4.) Since the latter is impossible — inasmuch as the interaction of physical forces cannot explain the unity that stamps even the simplest expression of mental life — only the first way out remains for Lotze. A great part of Lotze's psychological investigations is therefore devoted to finding out the manner in which the psychical elements stand in interaction with the physical ones. His psychology here recalls Descartes's. There are, on his view, certain regions in which the physical elements act upon the soul; thereafter the mental processes may for a time run their course exclusively according to their own laws, and they may then in turn release a mechanical force for the production of new physical changes. (Kleine Schriften III, p. 7.)

[523] Notwithstanding that the present condition of nerve physiology and of psychology would seem to make a separation of these different stages hopeless, Lotze nevertheless proceeds confidently (especially in *Medicin. Psychol.* and *Mikrokosm.*) to a closer investigation. The significance which the material processes have for mental life cannot — according to the conception Lotze has urged of the nature of interaction — consist in finished

sensations or representations being conveyed from them to the soul; they can only give signals, which the soul must translate into its own language, just as, conversely, the soul's inner states are not as such transferred to the material organs, but only become the occasion of these entering into function. The material organs work in the service of the higher spiritual activities by delivering and preparing the material upon which the soul is to exercise its powers, and in that their whole significance consists. The higher spiritual activities themselves — memory, thinking, aesthetic and moral feeling — take place, once the material is given, in the soul's interior, and it is wholly unnecessary to assume peculiar material processes corresponding to them. Lotze would in any case sooner interpret such material processes as effects of the mental states — brought about by these vibrations propagating themselves from soul to brain — than as their causes. So especially when we see a state of feeling persist long after the original occasion has fallen away.

Upon Lotze's special psychological theories we cannot here enter. Particular mention must be made of his brilliant doctrine of local signs, by which the development of the apprehension of space becomes more intelligible,[15] and of his assertion of feeling's importance as a fundamental element in mental life, over against Hegelian and Herbartian psychology. On a mass of particular points Lotze throws new light by his fine, spirited conception and his often happy manner of expression. It conflicts with good psychological method, on the other hand, when Lotze, instead of seeking the laws that connect the mental phenomena with one another, conceives the relation in the arising of the different mental phenomena in such a way that the representations react upon "the soul itself" and release the feelings out of it, which must in turn act upon "the soul itself" in order to release from it the expressions of will. Likewise when he lets a peculiar faculty of *[524]* thinking appear in opposition to the association of representations. Here his spiritualistic metaphysics has forced upon psychology an all too artificial and scholastic mode of explanation. It takes its revenge here that Lotze does not acknowledge psychology as an independent science of experience, but regards it as applied metaphysics. (*Logik und Encyclopädie* § 93.)

When Lotze's psychology recalls Descartes's on essential points, it must nevertheless be held fast that this likeness is present only so long as Lotze holds to the popular usage which he employs, in his psychology no less than in his philosophy of religion, to a great extent. The soul a thing for itself and the body a thing (or group of things)

[15] See on this *Reinhold Geijer*: *Hermann Lotzes lära om rummet* (Nyt svensk tidskrift 1880) and *Darstellung und Kritik der Lotzeschen Lehre von den Localzeichen* (Philosophische Monatshefte 1885). Cf. my paper *Lotze og den svenske Filosofi* (Letterstedtske Tidsskrift 1890) and my *Psykologi* V C, 7 c.

for itself — from many of Lotze's utterances that might seem to be his doctrine, as it was Descartes's and still is that of popular metaphysics. But it is only a provisional conception for Lotze. From Descartes he differs first and foremost by his doctrine of the subjectivity of extension and by his analysis of the concept of matter, which leads him to the assumption that matter is only a phenomenal form of the interaction of psychical elements. Thereby the dualistic conception of the relation between soul and body is abolished. The qualitative difference between the interacting elements is thereby considerably reduced: instead of an interaction of psychical and physical elements we get a mutual interaction of psychical elements. And this reduction Lotze makes — as must be well noted — not because he finds any difficulty in popular or in Cartesian metaphysics. The problem of the relation between soul and body really does not exist for him at all; the interaction between them presents to him no greater riddle than the relation of interaction that takes place between two bodies striking one another.[16] It is general philosophical considerations that guide him here — considerations that would retain their validity even if he had, instead of laying Cartesian dualism at the ground of his psychology, built upon Spinoza's identity hypothesis. Metaphysical idealism is independent of every more special psychological hypothesis. Since Lotze began as a
[525] Cartesian, those considerations lead him to a conception that may be described as a *monistic spiritualism*, in that it regards soul and body as two *different substances* which are nevertheless of *homogeneous* constitution. Herbart's theory, and a theory that appeared within the Wolffian school (see Note 1 to this volume), form predecessors of this peculiar theory.

On another point, and one more essential for his whole philosophy, Lotze also separates himself from Descartes — and here from Leibniz and Herbart as well. As we have seen, he regards the single elements in existence as moments or expressions (actions) of the primal substance, which is the one thing that is in truth real. The soul can then, no more than the physical atom, be substance in the stricter sense of the word. The independence of finite things is only apparent. "Our monistic conception" — says Lotze — "has *wholly and without reserve* laid the ordering of the world, and the existence and efficacy of every single thing, in the hands of the one infinite being upon which alone the possibility of all interactions rested." (*Drei Bücher d. M.* § 245, cf. §§ 72–73.) The words substance and being he therefore takes, when he uses them of the soul,

[16] *Mikrokosmus* Buch III, Kap. 1: "*Warum sollte nicht ein Atom des Nervensystems ebenso auf die Seele oder sie auf jenes stossen und drücken können, da doch jeder gemeine Stoss und Druck sich für die nähere Betrachtung nicht als ein Mittel zur Wirkung, sondern nur als die anschauliche Form eines viel zarteren Ereignissen zwischen den Elementen ausweist?*"

only in the sense of that which can act and suffer (ib. § 243) — and he admits at last (ib. § 307) that it might perhaps be best to refrain from using these words, which can easily mislead. Already in his paper *Seele und Seelenleben* (in Wagner's "Physiologisches Handwörterbuch", 1846; "Kleine Schriften" II, p. 198) Lotze said that every soul possesses only the reality that belongs to it on account of its significance in the whole of the world, and that its immortality depends not upon its nature but upon its place within the ethical world-order. In "Mikrokosmus" and in his principal work (*Drei Bücher der Metaphysik*) Lotze returns to the same thought. Just as it was not an interest in the soul's immortality that led him to his spiritualistic psychology, so he sees clearly that this question cannot be answered by scientific means at all. "No other fundamental proposition stands here at our disposal than the general conviction: all that has come to be will subsist, if its subsistence belongs to the world-plan, and so long as it belongs to it; all will perish *[526]* whose reality had its warranted place only in a passing phase of the world's course. That this fundamental proposition cannot be applied in detail by human hands scarcely needs saying; we certainly do not know the merits that could give one being a right to eternal subsistence, any more than the defects that could deny another that right." (*Drei Bücher d. M.* § 245.)

When one notices the agreement between Lotze's earlier (1846 and 1864) and later (1879) utterances upon this point, one will scarcely be led to assume, with some recent critics of Lotze,[17] that Lotze made essential alterations in his doctrine of the soul's substantiality and thereby approached Spinozism more nearly than before. But it would surely have been to the advantage of clarity if the excellent thinker — whose disposition assuredly permitted no unworthy accommodation — had already in his earlier presentations brought out his divergence from the current assumptions as definitely as in the latest utterances.

[17] *Kresto Krestoff*: *Lotzes metaphysischer Seelenbegriff.* Halle 1890, p. 46 f.; 73. — *Max Wentscher*: *Lotzes Gottesbegriff und dessen metaphysische Begründung.* Halle 1893, p. 11 ff. — The opposition in which Lotze's conception stands from the very first to Herbart's, despite all the points of contact it has with it, also testifies that no change of principle has taken place in Lotze's thinking in the course of the years. Compare on this *Max Nath*: *Die Psychologie Hermann Lotzes in ihrem Verhältniss zu Herbart.* Halle 1892. — With Lotze's ethical monism his optimism hangs together. Although he himself repeatedly emphasizes the reality of physical and of moral evil as an obstacle to the carrying through of his world-view, an otherwise well-disposed critic has nevertheless complained that he lacks interest in "the fundamental dogma of all religions, the atonement", because he does not take sufficient account of evil. G. *Vorbrodt*: *Principen der Ethik und Religionsphilosophie Lotzes.* Dessau und Leipzig 1891, p. 97.

b. Gustav Theodor Fechner

Fechner and Lotze stand as two Dioscuri in German philosophy in the last part of the century. They are kindred spirits in their ideal disposition, their great natural-scientific culture, their poetic sense and their need for unity in the conception of the world. They pursue kindred ends, but in part by different roads. A comparison between them is therefore interesting, especially as on decisive points one must choose between them. The choice will — since the fundamental proposition, an idealist world-view upon a realist foundation, is the same in both — depend upon which of them lets the realist foundation come most fully into its own. That is a measure Lotze has himself acknowledged, in the remark that idealism can be maintained only by investigations conducted in realism's spirit. Even if Lotze takes the first place as regards the working-through of the general philosophical principles, Fechner may well have the precedence as regards the
[527] consistent and resolute holding fast to the natural-scientific foundation. This is all the more remarkable in that Fechner's imagination has from the first a far greater influence upon his thinking than is possible in his critical, deliberate companion. Fechner is — like Kepler, of whom he strongly reminds one — an interesting example of how one may be led through bold and fantastic speculations to positive and exact results, provided only one holds fast to a definite fundamental thought and is able to extract it from its mystical wrapping. Just as Kepler was led successively from mystical speculations to the discovery of his famous laws, which satisfied his need to find determinate mathematical relations expressed in the universe, so Fechner was led through bold analogies to the conviction of a determinate quantitative relation between the spiritual and the material, and, in working that thought out more closely, came to found psychophysics, or experimental psychology.

Fechner was born in Lausitz in 1801 and in his youth pursued medical and physical studies. In 1835 he became professor of physics in Leipzig and made himself known by able work in his subject. The philosopher of religion C. H. Weisse belonged to his closest friends and had great influence upon his religious conception — as upon Lotze's. It is nevertheless doubtful whether Fechner would have come to play so great a part in the history of philosophy as he has, had he not, by studying subjective phenomena of light and color (in the winter of 1839–40), ruined his eyes, so that after long suffering he had to give up his professorship of physics. In philosophical thinking, in free fancy, in absorption in the inner world of the mind, he sought compensation for the bright and colourful world in which his weakened sight no longer allowed him to dwell as persistently as before, but which he did not forget. On the contrary, it became his life's task to work the two

worlds together and to find the law of their connection. In opposition to Romanticism's philosophy he would go from below upwards, not from above downwards. And where empirical inquiry gives out, he would rather let imagination range in the boldest analogies than let abstract thinking spin its slender threads. His bent for paradox he gave vent to in *[528]* a series of humorous writings, under the pseudonym Dr. Mises. His remaining writings divide naturally into two groups. The one — within which *Zendavesta* (1851), *Ueber die Seelenfrage* (1861) and *Die drei Motive des Glaubens* (1863) are especially to be named — develops his poetic-speculative world-view; to the other belong his most epoch-making works in the fields of natural philosophy and psychology: *Ueber die physikalische und die philosophische Atomenlehre* (1855), *Elemente der Psychophysik* (1860) and *Vorschule der Aesthetik* (1876).

Right into his advanced old age Fechner was intellectually active and productive. Even in his eighty-sixth year he made an interesting contribution to the discussion of his psychophysical theory (*Ueber die psychischen Massprincipien und das Weber'sche Gesetz*, Wundt's Philosophische Studien IV). Shortly afterwards he died, in 1887.[18]

α. Poetic-Speculative World-View

In the field of the world-view Fechner combats every conception which takes the rich and bright world of mental life to be borne by, or to be the product of, dark, intuitable things or beings. Materialism's "matter", spiritualism's "soul-substance" and Kant's "Ding an sich" are again and again objects of his polemic. No less does he combat those who separate God and world, spirit and nature. He rejects in equal measure belief in a natureless God and belief in a spiritless matter — orthodoxy as much as materialism. He reproaches the usual conception of the world with dualistically separating the infinite and the finite: "One sets the infinite over against, above, beyond, outside, behind the finite, perhaps even places an immense partition between them, as though they had nothing whatever to do with one another. And yet they by no means lie outside one another. On the contrary, the finite is the content of the infinite .... The infinite is therefore not incomprehensible either; it can be grasped by countless handles within the finite; only it cannot be encompassed." (Ueber die Seelenfrage, p. 111.)

[18] Interesting biographical notes in A. *Elsas*: *Zum Andenken Gustav Theodor Fechners* (Grensboten 1888). A fine characterization of his great predecessor was given by *Wundt* in his memorial words at the graveside (printed in "Philosophische Studien" IV) and later, more fully, in his address on the centenary of Fechner's birth (*Gustav Theodor Fechner.* Rede. Leipzig 1901). A rich biography has been given by J. E. *Kuntze* (*G. Th. Fechner.* Ein deutsches Gelehrtenleben. Leipzig 1892), and *Kurt Lasswitz* has given a critically searching presentation of Fechner's scientific standpoint (*G. Th. Fechner.* Stuttgart 1896, the first volume of "Frommanns Klassiker der Philosophie").

[529] Closely connected with the prejudices mentioned is, according to Fechner, the opinion that only human beings and animals are ensouled, and nothing else — at least upon our earth. It is thought that experience forbids extending ensoulment further. But only of my own soul have I direct experience; to the existence of other souls I must infer by way of analogy. And what then can prevent me from extending the analogy further, from human beings and animals to plants and to the heavenly bodies, when good grounds show themselves for it? That plants have no nerves is no counter-proof; for the lowest animal species have no nerves either. If it is said that the plant is no closed individual, the answer is that individuality is always relative, since no living being stands absolutely separated from its surroundings. There is so continuous a transition between animal and plant life that it is unwarranted to lay down an opposition between them as between the ensouled and the soulless. The plant's mental life might stand as far below the animal's as the animal's below man's. — And why should the heavenly bodies not be ensouled? Human beings and animals belong to the earth, and the earth-soul might stand to the individual human and animal souls as the earth's body does to the individual human and animal bodies. It is only an artificial abstraction when one sets human and animal life as ensouled in opposition to the life of the whole globe. The living does not proceed from the lifeless; rather the lifeless is a product of organic metabolism and of the perishing of organisms — a conception which according to Fechner is easier to defend than the opposite. In his work *Ideen zur Schöpfungs- und Entwickelungsgeschichte der Organismen* (1873) he develops especially the thought that variation and differentiation proceeded with greater rapidity in earlier periods of the earth than in the present one. — The lower souls must be supposed to stand to the higher as representations and motives stand to the individual soul. In the end all souls are parts of the highest, all-embracing soul, whose life and whose reality make themselves known in the law of causality — the principle of all particular laws of nature, and thereby of all connection and order in existence.

[530] As for Lotze, then, the law-governed connection in the world is the foundation upon which the religious-philosophical view is built. What for many makes the idea of God impossible or superfluous makes it, for Fechner and Lotze, precisely necessary. In that universal law they find the highest unity expressed — the eternal and unchangeable which embraces all. The concept of law is the last thing in all our knowledge; upon it, therefore, according to Fechner, our highest idea must be built. (See especially his paper *Ueber das Causalgesetz* in "Berichte der sächs. Societät der Wissenschaften" 1849. Cf. Seelenfrage p. 205 f.; Atomenlehre, 2. Aufl., p. 125.)

The concept of the world goes back for Fechner, as for Lotze, into the concept of

God. The widest conception of the concept of God Fechner regards as the best grounded; the narrower is in his eyes an abstraction. And since the world's life is God's life, this cannot be complete in itself, but develops and unfolds with and through the world's development. God's perfection consists not in finished completeness but in unlimited advance. The divine world-plan is not finished once for all, but is shaped in accordance with the world's events, which are at the same time divine experiences. — In this doctrine — in which Fechner follows his friend Weisse, while Lotze did not think he could ascribe validity to time for the Godhead — Fechner sees a carrying through of Christianity's declaration that God is spirit and shall be worshipped in spirit and in truth, and that in him we are, live and move: declarations which for the most part stand as mere words! He admits that his doctrine is certainly not Christian, if one reckons as an essential constituent of Christianity "belief in the bite of the apple in paradise and its mystical consequences, in the eternal damnation of the non-elect, in miracles against the laws of nature, in God's remoteness from the world, and in all the unedifying stuff out of which the theologians build up Christianity" (Seelenfrage p. 194). Fechner's bold theology makes it easier for him to understand how man can be God's fellow-worker, and how evil can be foreign to God's essence and yet a necessary member in the world's development.

β. Psychophysics

Early on, in accordance with his whole fundamental view, Fechner had arrived at the *[531]* conviction that the difference between the spiritual and the material could not be a difference between two beings, of which the one was immaterial and the other spiritless. The material world is the Godhead's outer, the spiritual world the Godhead's inner side. The difference is phenomenal, resting upon a difference in the spectator's standpoint, not upon a difference in substance. (Zendavesta II, p. 341.) Or, as he later expressed it (Elemente der Psychophysik I, p. 2), the difference between the spiritual and the material is to be likened to the difference between the concave and the convex side of one and the same circle: he who stands within the circle sees only the concave side, he who stands outside it only the convex — and he who can change his standpoint may perhaps believe that he has two different things before him. It is Spinoza's identity hypothesis that Fechner here arrives at, led by the physicist's interest in maintaining the continuity of the physical processes, and by the philosopher of religion's interest in thinking the Godhead, the all-embracing spiritual principle, in such a way that it expresses itself at every point in material phenomena, so that these are not loosened from it and do not fall outside it. Fechner arrives, so to speak, at a theophysics before he develops a

psychophysics.

A confirmation of this fundamental conception he found in the law of the conservation of energy, and he is the first to use the newly discovered law in this way. (Elemente der Psychophysik I, p. 21–45.) He admits that no proof has been given that this law holds for the material processes bound up with spiritual activities; but he holds that all experience points in this direction, and that we must therefore, until a counter-proof is produced, assume the law to be valid. Experience establishes that the material processes to which conscious life is bound stand in interaction with the other material processes within and outside the body, so that the total physical energy at our disposal may now be consumed predominantly by the former, now by the latter. Physical energy is drawn upon both
[532] when we think and when we chop wood — and therefore we cannot do both at once as well as we can do each of them separately.

From Spinoza he differs in this doctrine only by laying strong weight upon finding out a mathematical functional relation[19] between the two sides of existence. He first proceeded from the view that there must be a directly proportional relation between them. Later — he has noted the date and the situation: 22 October 1850, in the morning, in bed — he hit upon the thought that the spiritual did not rise and fall directly with the material, but that the changes of the former corresponded to the *proportional* changes of the latter, so that the change ($d\gamma$) in the intensity of a spiritual state is to be set equal to the ratio between the change ($d\beta$) in the energy of the corresponding material state and the energy (β) already present — that is, $d\gamma = d\beta/\beta$. With this he had come down from the region of indeterminate speculations to an assumption that could be checked experimentally. He found it at once confirmed by the fact that the strength of the sensation of light does not grow as fast as the physical intensity of the light. His own experiments and his researches in the literature gave him a material on the basis of which he set up the general law that a sensation's increase does not correspond directly to the increase of the physical impression, but to the ratio between that increase and the whole impression present — so that with one and the same increase of impression the sensation

[19] In my earlier presentation (*Filosofien i Tyskland.* 1872, p. 264 f.) I allowed myself to be misled by some utterances in *Zendavesta* (II, p. 352 ff.) into saying that Fechner assumes an interaction between spirit and
[600] matter. What Fechner there develops is, however, only the importance of changing standpoint and of finding out the formula for the mutual relation of the two sides. As little there as later does he assume a passage from the one side to the other. In *Elemente der Psychophysik* I, p. 8 it runs: "*Wir nennen das Psychische Function des Physischen, davon abhängig und umgekehrt, insofern eine derartige constante oder gesetzliche Beziehung zwischen beiden besteht, dass von dem Dasein und den Veränderungen des Einen auf die des Anderen geschlossen werden kann.*" For this kind of dependence Fechner often uses (for example Elemente I, p. 18; II, p. 393) the expressions *Simultanbedingungen, simultane oder Wechselabhängigkeit.* On his conception the brain processes corresponding to the activities of consciousness cannot — as Lotze and the spiritualists hold — be regarded as releasing impressions (*Reize*) upon the soul (Elemente I, p. 18).

grows the less, the stronger the impression already is. This law Fechner calls the Weber law, in memory of his teacher, the physiologist E. H. Weber, from whom some of the most significant experiments upon which it was built derived. — Of the law's validity, its limitation and its interpretation nothing further shall be said here; for that, reference must be made to more recent psychological works. Here it need only be remarked that by the step Fechner thus took, exact experiment became possible in the psychological field: he made — to use an expression of Galilei's — something measurable that had not hitherto been so. Through its relation to the outer impression the sensation, which in and for itself, as a purely subjective element, cannot be measured, can be determined quantitatively. Therewith the road was pointed out which *experimental psychology* had to *[533]* take: it had to seek points of connection between psychical processes and such processes as can be directly counted, weighed or measured.

What Fechner had first been seeking was certainly not what the new science came to occupy itself with. He had wanted to find the relation between the mental activities and the corresponding brain processes. But that relation is inaccessible to us. What we immediately apprehend is, from the subjective side, the mental state, and from the objective side, the physical impression. For the relation between these the law Fechner found holds, within certain limits; and it is most probable that brain process and mental process are directly proportional — which also agrees best with the identity hypothesis Fechner espoused. Psychophysics has so far become something other than Fechner himself had conceived. But this does not lessen the importance of his brilliant work *Elemente der Psychophysik*: between the human sciences and natural science there has been inserted — in consequence of the great law of the division of labor — a new discipline. And it was in Fechner's free spirit, rich in possibilities and in ideas, that this budding was accomplished. The investigator who after Fechner has done most in the new science's field said at his graveside: "Psychophysics, which he founded, was only the first conquest in a domain whose further appropriation could no longer present significant difficulties, once this beginning had been made" (Wilhelm Wundt: *Zur Erinnerung an Fechner.* Philos. Studien IV, p. 477).

In defense of the psychophysical law he had set up, Fechner composed a series of writings which are models of truth-loving and careful discussion and of chivalrous polemic; they are marked too by the author's freshness and humor. —

Akin to Fechner's endeavors toward the founding of an experimental psychology is his endeavor toward an aesthetics grounded upon experience. His *Vorschule der Aesthetik* (1876) is an epoch-making work, into whose special content we cannot here enter more

closely.

γ. Philosophy of Nature

[534] His opposition to Romanticism's philosophy Fechner has expressed above all in his "Atomenlehre". Speculation, he says, descends into nature as the bear into a beehive. It enjoys the coherent honey, but does not notice that it has been brought together by the activity of a multitude of small individual beings. Natural science shows — in physics and in chemistry alike — that the material processes can be understood only by being regarded as processes of interaction of small parts which for us are indivisible. Every part of the material world, however small, natural science conceives as a little world with globes and systems of globes. What seems to us a continuous mass is, like so many nebulous masses in the sky, a group of different parts. Natural science does not assume absolute atoms, but only that divisibility goes further than the eye and the microscope discover. It assumes only relative atoms — that is, atoms indissoluble for us, or by the forces of nature known to us — led by the need to refer every effect to a determinate place in space as its point of departure. Philosophical atomism goes to the end of the road along which physical atomism had reason to travel only part of the way. We obtain a philosophical conclusion to the conception of nature when we assume that the smaller one thinks the atoms, the more exact will be the results one can derive by thinking of matter as consisting of atoms. It was in order to have points of departure for the effects that natural science conceived atoms. The share with which every such point of departure contributes to the fulfilment of a law of nature we call the force proceeding from that point. What we know about the atoms is now precisely only that these contributions proceed from them: therefore philosophical atomism designates them as centers of force. There is thus no reason to ascribe extension to the atoms — the absolute, philosophical atoms — and the concept of matter is no longer materialistic. Philosophy stands here for Fechner as a hypothesis in which we think as absolute what in reality is always given only relatively.

[535] The foundation of our whole knowledge of existence is the law of the connection of phenomena; from the law we come to determine the forces in the co-operating elements. Fechner will not, with Leibniz, Herbart and Lotze, see psychical elements in the atoms. These are for him merely the last points at which spiritual beings arrive when they analyze the content of their consciousness.

A closer explanation Fechner regards as superfluous. The atom is the lower limit of our knowledge, just as the universal world-law, which testifies to the reality of an

all-embracing being, is the higher limit. Between these two limiting concepts lies all our knowledge. (Seelenfrage, p. 215–216.)

c. Eduard von Hartmann

In the year 1869 there appeared a book that caused a great stir and enjoyed a circulation rare for voluminous philosophical works. It was Eduard von Hartmann's *Philosophie des Unbewussten*. Its motto — "Speculative Resultate nach inductiv-wissenschaftlicher Methode" — shows that the author aimed at a grounding and development of philosophical ideas similar to Lotze's and Fechner's. By his clear and broad presentation, by his detailed use of natural-scientific examples, by meeting the reader at once with a realist foreground and a romantic-mystical background, and finally by taking up pessimism — which, under the influence partly of Schopenhauer's now much-read writings, partly of prevailing currents of the time, was a widespread train of thought — as a member of his system, Hartmann could find readers in wide circles. The attention he aroused is seen from the fact that in the course of twenty years ten editions of his principal work appeared, and that in the years 1870–1875 alone there appeared fifty-eight writings dealing with him.

Eduard von Hartmann was born in Berlin in 1842, was the son of a general, and himself entered upon a military career. In his free time he cultivated music, painting and philosophy. A knee injury compelled him in 1865 to resign his commission. And when he became clear that his real talent did not lie in the field of art, he gave himself — "bankrupt *[536]* in everything but thought" — to philosophy. He had philosophized and worked out philosophical essays earlier. He now proceeded to the working out of "Philosophie des Unbewussten", which must still be regarded as his principal work, notwithstanding that it has been followed by some thirty larger and smaller works. The most considerable of these is certainly *Phänomenologie des sittlichen Bewusstseins* (1879), which is on the whole Hartmann's best work. For the characterization of his philosophical standpoint I shall here keep to two principal points, in full consciousness that Hartmann's untiring activity in developing his ideas in all directions, and illuminating them from all points of view, does not thereby quite come into its own. Hartmann has complained that I have "halved" him by mentioning only the part of his activity that lies before 1880. That, however, lay in this book's plan; and I have besides, it must be admitted, the view that the two works named are and remain Hartmann's most considerable contributions to philosophy. Those who wish to supplement the short presentation of Hartmann's philosophy that I can give here I must refer to the detailed work, sprung from excessive admiration, of

Arthur Drews: *Hartmanns philosophisches System im Grundriss* (Heidelberg 1902).

α. Philosophy of Nature and Psychology

While Lotze and Fechner in principle laid the natural-scientific mode of explanation at the ground, and would ground their philosophy by drawing the consequences of its presuppositions and results, Hartmann's philosophy bears far more the stamp of a neo-Romantic reaction against natural science's realism. He sets out to show that the natural-scientific mode of explanation is not sufficient, but that together with the causes the mechanical view of nature assumes there must operate a spiritual principle which he — in order not to make it too anthropomorphic — calls *the Unconscious*. He appeals to
[537] this principle wherever he thinks the causes empirical science can exhibit do not suffice. It is thus that he thinks to arrive by "inductive" means at "speculative" results.

Hartmann agrees with Fechner that matter must be conceived as a system of atomic forces. No single atom can have mass, any more than one can speak of the magnitude of a unit. If, then, we keep to atomic force as the last element to which the analysis of material reality leads, there can — Hartmann holds — be determinate connection and consistency in the atomic force's striving only when that striving is conceived as a willing, with an unconscious representation of its goal. We understand force only by thinking of it as a willing. Matter itself is therefore representation and will — and the difference between matter and spirit falls away. In organic growth there likewise expresses itself an unconscious willing and representing: in the organism's coming to be a purpose is realized, a purpose in relation to which the single substances and processes are to be regarded as means, whose provision and combination become intelligible only through the purpose, notwithstanding that this is not an object of consciousness. There is only a difference of degree between organic formation and instinctive action. Instinct cannot, Hartmann claims, be explained by the material organization or rest upon a nerve-mechanism implanted once for all; for then it could not vary in its expressions when the organization does not change. As little does it rest upon conscious deliberation. There remains, then, only to conceive it as unconscious willing and representing. In the human mind the Unconscious expresses itself in sense-perception, in that the intuition of the outer object comes to be by an unconscious working-together of the sensations. The association of representations proceeds only through representations being brought forward, without conscious search, which stand in connection with the present ones. Feelings and motives arise on account of unconscious processes and are therefore often unintelligible to ourselves. Even where the will is bound up with consciousness, an

unconscious will is required if it is to lead to a willed movement: consciousness does not know which nerve-center in the brain must be affected for the movement to come about. In history the Unconscious is able to get individuals to work in the service of great world-purposes while they believe they are working only for their own limited ends. *[538]* What is fate or providence other than the Unconscious's dominion in men's actions, so long as their conscious understanding is not mature enough to make history's purposes its own? — The Unconscious is everywhere that which makes co-operation between individual beings possible. There are individuals of every possible degree — from the atom to universal nature — and it is one and the same Unconscious that stirs in them all. Plurality has only phenomenal, not metaphysical significance. Hartmann declares himself in principle in agreement with speculative theism (as it appears in Schelling, Weisse, Lotze and Fechner), his "Unconscious" being rather to be called "superconscious". But he prefers the negative expression in order to keep anthropomorphic representations at a distance. He thereby overlooks that even if the name he has chosen removes such representations, they are in high degree favored by the manner in which he lets "the Unconscious" operate.

It is an unscientific procedure that Hartmann employs when, wherever he thinks he finds a hole in the scientific explanation, he puts his "Unconscious" in as a magical means that works equally well everywhere. Of it holds what Galilei said of the appeal to an almighty will: it explains nothing, because it explains everything equally well. The real interest in the concept of the Unconscious has in Hartmann scarcely arisen from a striving after understanding of the phenomena either. It is due rather to a need for rest from reflection, analysis and criticism, and to a faith in the value of what emerges involuntarily and unprepared. Thus it runs in one place in Hartmann (Philosophie des Unbewussten, 3. Aufl., p. 369): "Conscious reason is only negating, criticizing, controlling, correcting, measuring, comparing, combining, subordinating, inducing and generalizing. Never is it creatively productive, never inventive; in this respect man is entirely dependent upon the Unconscious, and when he loses the capacity to receive the Unconscious's promptings he loses his source of life, without which he will drag out his life in monotonous fashion *[539]* under dry schematizing. Therefore the Unconscious is indispensable, and woe to the age that violently suppresses it and, in one-sided overvaluation of the consciously rational, will let that alone count!" Consciousness cannot be the purpose of development, but can only be a means, necessary at a certain stage of development. (Phil. d. Unbew., 3. Aufl., p. 739.) — It is this psychological-historical experience of the significance of unconscious life that is the warranted foundation of Hartmann's thinking. But he then

makes the Unconscious into a mythological being, a demon that everywhere intervenes in the natural connection, directs the atoms, displaces the brain molecules, adjusts the circumstances. Hartmann's philosophy takes on a mythological and dualistic character, which is certainly most prominent in the first editions of "Philosophie des Unbewussten", but is too closely bound up with his principle for it to be able, despite all correctives, to disappear entirely. Characteristic above all is his position on Darwinism, in that he seeks to show that the selection of qualities does not explain the arising of new forms, but is only a mechanical means which "the Unconscious" makes use of when it wants to prepare the way for such forms. Darwin keeps — Hartmann holds — to the conditions, but overlooks the properly producing force.

It was then no wonder that Hartmann was often attacked from the natural-scientific side. Now, in order to show that he himself had a very good command of the natural-scientific points of view, he wrote an anonymous critique of himself: *Das Unbewusste vom Standpunkt der Physiologie und der Descendenzlehre* (1872). The critique was so good and so telling that his opponents praised the anonymous work and used it against him. In the second edition Hartmann named himself and added notes in which he sought to refute the objections he had himself raised. It is doubtful, however, how far he has succeeded; he has conjured up spirits that will not vanish again. He admits that he had undervalued the mechanical causes and overlooked many intermediate links. But his procedure continues to be that of seeking out the holes which scientific inquiry always leaves, and filling them by the help of his mystical *deus ex machina*. Since there is
[540] always the possibility that the holes may be filled scientifically, it is a dubious method to employ. Lotze employed this method in his spiritualistic psychology; Hartmann extends it to his whole philosophy, without attending to Lotze's clear and energetic emphasis upon the significance of mechanism. His "speculative results" are not won by "inductive method". He accordingly admits (Das Unbewusste etc., 2. Aufl., p. 359 f.) the insufficiency of induction, and ascribes to it only the significance of serving to confirm what has been thought out by another way.

β. Pessimism and Ethics

Hartmann declares it a great misunderstanding that he should himself, in his personal temper, be a pessimist; he teaches pessimism as a *theoretical view*, after disappointments and sufferings in his youth opened his eyes to it. "With the chapter on pessimism [in "Philosophie des Unbewussten"] I have", he says, "written myself free once for all from world-sorrow as such, and purified it into an objective, calm *knowledge* of existence's

wretchedness — but thereby also won back for myself that undisturbed cheerfulness which may be found in the philosopher who floats in the ether of pure thought and from there regards the world, and his own pain, as an alien object of investigation." (*Mein Entwickelungsgang.* Gesammelte Studien und Aufsätze, p. 38.) He emphasizes as well that pessimism does not even theoretically constitute his whole philosophy, and does not give it its principal stamp. (*Die Stellung des Pessimismus in meinem philosophischen System.* Zur Geschichte und Begründung des Pessimismus, 2. Aufl., p. 18–28.) He describes his task as that of uniting pessimism — as the result of valuing the world by the standard of happiness — with the philosophy of development. He would thus reconcile Schopenhauer's philosophy with Hegel's, just as in his very concept of the Unconscious he has sought to combine Schopenhauer's absolute will with Hegel's absolute Idea. In a peculiar way Hartmann here finds a point of attachment in Kant. When Kant declared every attempt to solve the problem of evil hopeless, that was consistent from his standpoint, since he proceeded from the Christian concept of God, with which the *[541]* reality of evil stands in insuperable contradiction. But the problem must be posed thus: how must the world's absolute ground be thought, if it is, without contradicting itself, to have come to posit such a world? And when the question is so posed, Kant has prepared the way for its treatment, since he has given indications in a pessimistic and in an evolutionist direction alike. Schopenhauer pursued the one indication, Hegel the other: now the time has come to gather up what lies in both. (*Kant als Vater des modernen Pessimismus.* Zur Gesch. u. Begr. des Pess., 2. Aufl., p. 136 f.) (Compare Note 17 in the present volume.)

Since now — so Hartmann reasons — an unprejudiced consideration shows more unhappiness than happiness in the world, the world's coming to be cannot be derived from reason. It cannot be due to a rational but must be due to an irrational principle. The explanation must then be sought in the will-element within the Unconscious having separated itself from the representation-element. Hartmann takes up Böhme's and Schelling's mystical doctrine of a dark element in the Godhead whose breaking loose leads to the world's development. In a groundless manner the in itself blind willing has torn itself out of connection with representation, or the Idea. But this seeing element in the Unconscious continues to work in the world's development and seeks to lead that blind and refractory element to harmony and reconciliation. Hence the double-sided impression we get of the world. Both pessimism and evolutionism are right. There are two principles at work in the world. Hartmann's philosophy of religion comes here to recall Voltaire's, Rousseau's and Stuart Mill's — save that development is, on his

belief, a progressive redemption of the suffering God, since what matters is, through the abolition of the will, to undo what was done when the world became actual. Of great importance in this respect is that the knowledge of life's wretchedness should awaken, and Hartmann therefore holds that with Schopenhauer a new world-period has begun. (Phänomenologie des sittlichen Bewusstseins, p. 782.) Only slowly does this
[542] knowledge dawn in mankind. It happens through a successive abolition of illusions. First one expects happiness in the present life. When one discovers that the goods it can bring are only apparent, one sets one's hope upon another life. That is the illusion's second stage. Belief in the beyond cannot maintain itself; one gains fresh confidence in earthly life, but, taught by experience, one does not expect an immediate enjoyment of happiness, and instead hopes for a state of bliss for mankind at some time in the future, straining every nerve to prepare the way for it. That is the illusion's third and last stage, at which mankind will probably remain for long ages. But in the very fact that a small minority has seen through the illusion in all three of its forms, Hartmann sees sufficient warrant for the pessimistic theory; its correctness cannot be settled by a vote.

Hartmann's ethics hangs closely together with his pessimistic theory. There is for him an irreconcilable conflict between culture and happiness. Culture's advances are bound up with a decline in happiness. The more complicated life's apparatus becomes, the more occasion for displeasure is given. Susceptibility to pain grows greater, and the increasing reflection sees through the illusions more easily. Culture lets needs grow faster than the means of satisfying them. One must then choose between culture and happiness, between the doctrine of development and the doctrine of bliss. Happiness presupposes peace and rest — and will therefore bring stagnation and dissolution with it. Development leads ever further, until all possibilities are exhausted.[20] Through development dissatisfaction is increased and the illusions are abolished, until life at last comes to a stop. Then redemption is reached, and the blind will's breaking loose is annulled. The highest form of ethics is according to Hartmann that which builds not merely upon love of mankind and the urge to development, but makes compassion with God — the need to release the bound and suffering world-principle — the foundation of morality. One then makes the Unconscious's purposes one's own.

Hartmann's philosophy is theology inverted. He operates confidently with the thought of an absolute world-purpose, without seriously engaging the difficulties that
[543] thought contains. In his ethical work (Phän. d. sittl. Bew.) there are, however, excellent particulars, especially in the characterization and criticism of the different standpoints.

[20] On the relation between culture and happiness compare my *Etik*, Kap. 7, and my *Etiske Undersøgelser*, Kap. 2.

And in detail his train of thought is independent of pessimism's metaphysics. He has overcome that metaphysics to a still higher degree than he says. In practice he is an evolutionist and an optimist. For provisionally the affirmation of the will to life is the one right thing, since something can be accomplished for the world-process only by full devotion to life and its pains, not by cowardly renunciation and self-surrender. Pessimism's time really comes only after long ages — ages so long that one wonders doubly at Hartmann's confident knowledge of what will happen when they are past. By this long postponement pessimism loses the personal, practical significance it had in Schopenhauer. Hartmann stands to Schopenhauer as the modern Christians, for whom the day of judgment has been pushed out into the indefinite, stand to the first Christians, whose whole life was determined by the expectation of its speedy coming. Distant possibilities cannot be determining for our conduct of life, and cannot shake life's value. Only the life we know and live can we value, and within that life itself we must find what we are to be able to acknowledge as valuable.

4. Criticism and Positivism

a. Friedrich Albert Lange

It was natural that the dissolution of Romanticism's philosophy, the rise of natural science, and materialism's claim to be science's last word, should bring to greater influence the tendency which in the century's first half had been only an undercurrent. Already the greater attention and adherence which Schopenhauer's and Herbart's ideas now became the object of are signs in this direction. There is a need for a critical revision of the very foundation of our knowledge, in order to be able to take up the right attitude to the new material of knowledge. In Lotze and Fechner, and in part in Hartmann too, investigations *[544]* in this direction are indeed to be found, but the problem of knowledge proper is not taken up. They take realism as given, and then seek, more or less consistently, to find a foundation for their idealist constructions either in realism's presuppositions or in its holes. Great as the interest of Lotze's attempt in this direction is, it was nevertheless the problem of existence rather than the problem of knowledge that he treated. One of the first to draw attention to the necessity of taking up Kant's work upon the problem of knowledge again was the famous historian of philosophy Eduard Zeller, in a small work *Ueber die Bedeutung und Aufgabe der Erkenntnisstheorie* (1862), which caused all the more stir in that its author had come from Hegel's school. A few years later Friedrich Albert Lange, in his brilliant historical-philosophical work *Geschichte des Materialismus*

(1866), pointed to epistemology as a court of appeal overlooked not only by Romantic but also by materialist philosophy, and sought upon its basis to carry on the struggle for an idealist world-view.

Lange is one of the most attractive figures in the history of recent German philosophy. Ideal enthusiasm, sharp criticism, great fullness of knowledge and a vigorous realist eye have entered in him upon a rare union. He commands besides a great talent as a writer. Unfortunately his career was short. He was born in 1828 near Solingen in Rhenish Prussia, but lived his early youth in Zürich, whither his father, the well-known theologian J. P. Lange, was called as professor of theology in 1841. He thus received the foundation of his intellectual development in the free air of Switzerland, and it was to be traced through his whole life and striving. His university studies he pursued in Bonn. Educational work stood before him as his principal aim, and he studied philology as a means to it, without however neglecting natural-scientific and philosophical studies. His philosophical direction was early laid down. In a letter of 1851 (in *Ellissen*: *Friedrich Albert Lange.* Eine Lebensbeschreibung. Leipzig 1891, p. 69–73) he develops the view
[545] that all representations man can form of existence — of God and the world, of good and evil and so on — are developed according to the laws of the human mind: those laws are then the last foundation of our knowledge, beyond which we cannot get. He was here led to the psychological-epistemological standpoint which Ludwig Feuerbach had reached through criticism of the religious and speculative ideas, though Feuerbach does not seem to have influenced him. What influenced him philosophically in his youth was above all Hegel's "Phenomenology", Herbart's psychology and Schleiermacher's writings — of the last he speaks with enthusiasm in a letter of 1849.

After some time as lecturer and schoolmaster, Lange worked as editor and as a political and social agitator. He took part in the opposition to Bismarck's domestic policy and in the German workers' associations' efforts to win a firm organization. The work *Die Arbeiterfrage* (1865) derives from this period. It is one of the first writings — next after Mill's political economy — in which the social question is conceived from a more general point of view and in an impartial manner. It has interest besides in bringing the Darwinian ideas into connection with the workers' question, in that this is seen to spring from the struggle for existence in the social field. Lange conceives the matter thus, that what is required is to wage a struggle against the struggle for existence — a struggle which precisely constitutes man's higher spiritual vocation. (Since at this time only "The Origin of Species", and not "The Descent of Man", had appeared, Lange could not make use of the points of view Darwin has given for conceiving the ethical endeavors too as

a struggle for existence.) He saw the political significance of the workers' question in its exerting a constant pressure upon the conservative institutions, whose dykes will be broken through if canals are not dug in time. The liberal-minded men in the higher classes now have the choice between strengthening those dykes and helping to dig the canals. Lange's own choice was easily made. But in Germany men were busiest with the dykes, and Lange therefore emigrated to Switzerland. In the midst of his journalistic and agitational activity he had nevertheless found time to put the last hand to his principal work *Geschichte des Materialismus* (which according to Ellissen, p. 145, appeared in the *[546]* autumn of 1865, though the title page bears 1866). It is not a purely historical work. Through a spirited depiction of materialism's principal forms and of the development of the natural sciences, Lange seeks to illuminate the problem of the relation of the spiritual and the ideal to the material, and he sets the theoretical question at the same time in connection with the practical and social one. Although many parts of the work are now of course out of date, no one will read it without enrichment and stimulus. — Shortly afterwards there appeared a new social-philosophical work of Lange's: *Mill's Ansichten über die soziale Frage* (1866).

In Switzerland Lange worked as bookseller, teacher and editor in Winterthur, and unfolded an astonishingly many-sided activity. He took eager part in the agitation for a revision of the constitution of the canton of Zürich. Later he became professor in Zürich and could now live for his studies again. He was soon noticed, however, in his old fatherland, back to which he also longed, and in 1872 he became professor in Marburg. But by then he had already suffered for several years from cancer of the abdomen, and his last years were painful. His ideal disposition and his great enthusiasm for research and for freedom sustained him; his spiritual nourishment he found above all in Schiller's philosophical poems. He died in 1875. — An unfinished but interesting work on logic (*Logische Studien.* 1877) was published shortly after his death. Later (1897) Ellissen published *Einleitung und Kommentar zu Schillers Philosophischen Gedichten*, a work Lange had prepared during his stay in Zürich but laid aside for other labors. Schiller was, as Lange said in the preface to his principal work, the Kantian with whom he felt himself most akin. Schiller's poems contained Lange's religion.

As the title of Lange's book indicates, his problem is how one is to take up a position with respect to materialism. His standpoint here is the same as Fechner's. He would overcome materialism precisely by carrying it through. What he praises in materialism, and what for him constitutes its great historical significance, is its consistent demand for the exhibition of material causes for all natural phenomena. In idealism, by contrast, *[547]*

Lange sees a great danger in the theoretical field, since it easily leads to relaxing the strict mechanical explanation and to inserting subjectively constructed members into the series of objective causes. The functions of the nervous system and of the brain too must be explained according to the general physical and chemical laws. One must — relying upon the law of the conservation of matter and force — follow the influences from without through sense-organs, afferent nerves, brain, motor nerves and muscles, and seek a complete and unbroken connection in this whole process. It is unwarranted to assume the connection broken at any point. Consciousness itself is not a member of this connection. It is the subjective state of the individual in whom the process takes place — another side of the process, but not itself a part of it. Here materialism has its limit: precisely because the states of consciousness are not members or parts of the material processes, and cannot be explained by the law of the conservation of matter and force, precisely therefore do they fall outside materialism, and materialism comes into contradiction with itself when it thinks it has explained everything. It can at most explain the objective, material process, which is the way in which the conscious individual's subjective state appears to outward consideration. Lange regards it as conceivable that one will some day be able to determine that part of the material processes which coincides in time with a certain state of consciousness in the individual; but he does not think one will ever be able to give a closer determination of the relation between the subjective state of consciousness and the objective nerve process. It is a warranted conjecture that behind the two corresponding worlds — the world of matter and the world of consciousness — there lies an unknown third as their common cause.

It is remarkable to see how the great hypotheses of the seventeenth century are renewed after the rise of natural science at the middle of the nineteenth. As Vogt, Moleschott and Büchner renew Hobbes's standpoint, so Lotze renews Descartes's, and
[548] Fechner and Lange Spinoza's. It is a testimony that these hypotheses indicate the possibilities between which our thinking must here make its choice. On closer investigation one will find that it is not a simple resumption but a renewal of the older hypotheses that has taken place. Not only the natural-scientific foundation but also the philosophical elaboration is other than it was in the great dogmatists of the seventeenth century. There is greater critical circumspection, a clearer consciousness that what is at issue is finding the best and most consistent points of view in a discussion that can scarcely be brought to a close — not the setting up of definitive systems. One notices that the critical philosophy, whose breakthrough lies between the two epochs, has not worked in vain.

Lange's standpoint, which from one side points back to Spinoza, now precisely from

another side points back to Kant. He does not stop at the double form of existence — as motion and as consciousness — but calls attention to the fact that the whole of matter, the world outside us as well as our own organism, brain and sense-organs included, is only there for our consciousness, is a product of consciousness. It is in virtue of our organization that we apprehend the world as we do apprehend it. That Kant showed, and the later physiology of the senses has confirmed it. Here again materialism is overcome by being consistently carried through. For even if it be conceded to materialism that our whole conception of the world is a product of our material organization, that material organization is itself only an object of consciousness, and exists immediately only in and for consciousness. Matter and organization are our representations — even if they are representations we necessarily form, and form according to determinate laws. The dispute between matter and spirit is thus ended in the latter's favor. The naive belief in matter must vanish upon closer investigation, and it can vanish without natural research's campaign of conquest being in the least degree hindered thereby. — In the way Lange here takes up his position against materialism he has a predecessor in Schopenhauer. At about the same time as Lange, the famous Viennese pathologist *Rokitansky* expressed a similar view.[21]

There is in Lange's presentation in "Geschichte des Materialismus" a certain wavering *[549]* as to how far he will extend subjectivism.[22] Occasionally he says that the very opposition between the thing in itself and the phenomenon is a product of our organization, and that we therefore do not know whether that opposition has any significance whatever outside our experience. In other places — especially in the second edition — he speaks of an order of things corresponding to our knowledge. In a letter to the natural scientist Dohrn (in Ellissen, p. 258–262) Lange develops the view that with respect to this whole question four different stages must be distinguished: 1) the naive belief in sensible appearance; 2) the recognition that the senses do not give us things as they are in themselves, but vibrations of air as sound, vibrations of ether as color and so on — that is natural science's standpoint; 3) the recognition that our understanding as well as our senses belongs to our organization, and that therefore even the scientifically corrected

[21] *Karl Rokitansky's* spirited paper *Der selbständige Werth des Wissens* (Sitzungsberichte der Wiener-akademie 1867) did indeed appear more than a year after Gesch. d. Mat., but as early as 1862 he had in an address pointed to Kant's idealism as the true consequence of natural-scientific thinking. Compare Th. *Meynert*: *Karl Rokitansky.* Ein Nachruf. (Sammlung von populär-wissenschaftlichen Vorträgen. Wien und Leipzig 1892, p. 76 f.) — *Helmholtz* too (in his *Physiologische Optik.* 1867, p. 443 ff.) maintains that it is in the last instance unwarranted to ascribe to our representations any validity other than a practical and symbolic one. The physiologist A. *Fick* arrives at a similar result in his lecture *Die Welt als Vorstellung.* Würzburg 1870.

[22] *Max Heinze* has shown this in his interesting paper *Der Idealismus Friedrich Albert Langes* (Vierteljahrschrift für wissensch. Philos. I, p. 183–185).

apprehension of sense does not give us the thing in itself — that is Kant's standpoint; 4) finally the recognition that the very opposition between thing in itself and phenomenon is a product of our organization, so that its validity may be doubted. This fundamental doubt cannot, according to Lange, be removed by way of knowledge. It marks science's limit, beyond which only poetry, not thinking, can lead. Lange reproaches Kant for stopping at the third stage; Kant's Romantic successors he reproaches with having "tumbled out beyond the fourth stage into poetry, and mixed criticism and poetry in their systems instead of bringing them into a clearly delimited relation to one another".

In this conception Lange has, however, overlooked that when our knowledge applies at its very limit the same principles as within the limit, it must necessarily hold fast to the opposition between subject and object, knowledge and thing: in the contrary case it would in one form or another come to assume a creation out of nothing, a causality proceeding from an absolute unity. It is a law for all knowledge that what we understand is understood only by the help of *several* premisses or conditions.[23] If
[550] one does not acknowledge that law, one becomes entangled in difficulties which from Nicolaus Cusanus onwards emerge again and again in the history of thought.

Lange, like Schopenhauer, uses idealism — in the sense of subjectivism — to unseat materialism. But he also espouses idealism in the word's practical sense. It is not enough for him to maintain reality's subjective character; reality would not thereby become other than it is, bad or good. But he maintains that just as deeply grounded in human nature as the need to apprehend and explain reality according to natural-scientific laws is the need to form ideal images and to regard them as the expression of the highest reality. The producing power which expresses itself in sensation and in thinking does not rest content with what is given, but strives beyond it in order to form an ideal world. Man cannot do without a supplementing of reality. All religions and all speculative systems are products of this need. They differ from the scientific picture of reality in that it is not the organization common to all, the race's nature, that operates in their arising, but individual and personal tendencies and requirements — and also in that a complete confirmation by experience is not possible. It is of great importance to hold fast to this difference, so that one does not lay down as literal truth what is only a symbol. Every significance for the knowledge of reality must be denied to the religious and speculative ideas. But on the other hand a distinction must also be drawn between such ideas and mere figments. Although Lange in single places lets poetry begin where knowledge ends, he maintains in other places (1st ed. p. 539, 541; 2nd ed. II, p. 540) that speculation stands

[23] Compare my *Psykologi* V D, and *Filosofiske Problemer* (Universitetsprogram 1902), p. 48–52.

between empirical science and poetry: in the one the material rules, in the other free form; speculation seeks to unite the two. Clearer is the difference Lange finds in this, that the idea is to be judged by its value for the development of our spiritual life, not by its grounding and origin (1st ed. p. 346 f.; 2nd ed. II, p. 545). We do after all regard Cologne cathedral as something other and more than a heap of stones. Indeed, Lange even speaks of practical idealism as possibly hanging together with the unknown truth, though in *[551]* another way than materialism does: the idea is image and symbol for an unknowable Absolute. (1st ed. p. 346, 539, 541, 545. — 2nd ed. II, p. 494 and elsewhere.)

Here again it shows itself how difficult it is to place an absolute limit. Lange becomes here involuntarily a Platonist. How does he know that the ideas which for us express the highest value are images or symbols? And on the other hand — if his sharp distinction between ideas and scientific knowledge is held fast — who will then work at the development of the ideas, if they have no real significance at all? The speculative ideas can be maintained in their significance only by being seen as hypotheses, in which human thought draws the last consequences of what stands fast for it at the given time. It must then be left to closer testing what theoretical and practical significance they may have; but practical significance they can scarcely have if they have no theoretical significance whatever. As regards the practical significance itself, finally, Lange gives no definite criterion, no measure. Where he indicates his ethical conception, he says that the natural sympathy which for Auguste Comte was the foundation of ethics is not enough for an idealist ethics, and he regards the idea of a totality to which we belong as the a priori foundation of ethics, by which alone the moral law's inexorable strictness becomes intelligible. But at once he becomes doubtful again: does the idea not often lead astray, and may it not be asked, especially with respect to the religious systems, whether it is not better simply to give oneself over to the ennobling effect of natural sympathy than to listen to prophetic voices which only too often have led to the most terrible fanaticism? — A closer investigation of the relation between formalism and realism in ethics Lange has not given. In his ethical ideas he was a disciple of Kant, or rather of Schiller, though he saw the limitation of their conception.

His doctrine of the symbolic character and practical value of the religious ideas Lange applies not only to philosophical speculation but also to the positive religions. When materialism attacks Christianity as charcoal-burner's faith, it overlooks — Lange *[552]* maintains — the possibility of a freer, ideal conception of the content of the ecclesiastical traditions. One should look to the ideas' value and not to their literal formulation. In a certain sense religion's ideas are imperishable. Who will refute a mass of Palestrina's, or

establish that Raphael's Madonna is an error? In the inwardly moved spiritual life of the genuinely devout something stirs which the ideally minded freethinker must understand and must sympathize with. Lange desires a reinterpretation of the ecclesiastical doctrines, in the spirit of Kant, Fichte and Schleiermacher, so that Christianity's ethical and ideal content may be drawn out of its literal husk; Hegel's philosophy of religion was for him too conservative. Direct polemic against positive religion he does not want, lest the people's unity and the strength of its spiritual life should suffer. Of the two, he would rather sacrifice enlightenment for a time than strength, provided only that freedom of teaching is secured and priestly rule kept away. He has an open eye for the great oppositions of our time, and expects from philosophy, as a contribution to softening the struggle, above all a more penetrating understanding of the nature and the laws of human development. He closes his brilliant work by expressing the hope that serious work in this direction will bear its fruit. —

The great interest in Kant's philosophy, to which Lange's work partly testified and which it partly awakened, led to an eager study of the great philosopher whom Romanticism's philosophy believed it had overcome. On several points disagreement showed itself as to how Kant was to be understood. A rich Kant literature blossomed, which in several respects led to a more exact conception of Kant, his task, his method and his position in the history of philosophy. Even if one often lost oneself in investigations of more philological than philosophical significance, it was nevertheless of great importance that Kant should be understood more correctly than before. This is due especially to works by Cohen, Fr. Paulsen, Laas and Vaihinger, to which were later added the contributions to the illumination of Kant's development and thought which B. Erdmann, R. Reicke and others brought forward from Kant's unpublished notes and letters. In
[553] the great edition of Kant's works begun by the Berlin Academy, the interest in critical philosophy will set itself a fine memorial.

In a spirited paper *Was uns Kant sein kann* (Vierteljahrsschr. für wissensch. Philos. 1881) Friedrich Paulsen has sought to determine the significance Kant can have for our time. That significance rests not upon a direct resumption of his doctrine, but upon a treatment of the problems in his spirit. So-called *neo-Kantianism* therefore designates no closed school, but an endeavor toward epistemological testing of the concepts we operate with. Wilhelm Wundt, in his paper *Was soll uns Kant nicht sein?* (Philosophische Studien VII), has spoken against some neo-Kantians' unduly dogmatic adherence to the old master, and has shown why Kant's doctrine cannot be set up as an epistemology valid for the science of our own time.

b. Eugen Dühring

The year before Lange's "Geschichte des Materialismus" there appeared one of the most excellent epistemological works of the century's last half, namely Eugen Dühring's *Natürliche Dialektik*. It takes its point of departure in critical philosophy in so far as it means, in a more radical way, to take up again Kant's problem, which Romanticism's "deluge" had washed away. Among later German works it mentions only Trendelenburg's "Logische Untersuchungen" with great recognition as a predecessor; among foreign thinkers Comte and Stuart Mill are named, who have not, however, specially treated the book's subject. The book is already now a literary rarity, since the author has not republished it — perhaps because his standpoint has since altered in a more positivist direction. And yet it is surely the best of all Dühring's books, in form and in content alike. It is an interesting contribution to the illumination of the relation between critical philosophy and positivism, two tendencies which contend not only in Dühring but in the whole of the most recent philosophy. By two fundamental thoughts Dühring here already approaches positivism. Still more sharply than Kant and Schopenhauer he maintains that *[554]* the law of grounding — the principle of sufficient reason — is only a law for our thinking, not for reality, which is more comprehensive than our thinking. And he sets up, on the other hand, the principle of the lacking ground, whereby the burden of proof is laid upon whoever puts forward a new claim, what has hitherto been given being retained so long as ground is lacking for abandoning it. By the first fundamental thought he rejects rationalist dogmatism, by the second idealism and dualism; through both he works his way forward to building upon the ground of the given reality, without the admixture of representations not drawn from it. He continues Ludwig Feuerbach's work and is positivism's most important representative in Germany. He himself calls his philosophy the philosophy of reality. The great fundamental features of reality, which experience teaches us to know, are to form the basis both of a view of life and of the conduct of life. Philosophy is for him not merely theory, but the expression of a personal disposition.

In Dühring's view the nineteenth century has a decidedly reactionary character — the technical field alone forms an exception. The seventeenth century stood for him as the classical age in scientific respects. Serious science begins only with it, and our century has brought forth nothing that can eclipse it. Dühring is thinking here above all of Galilei, Huyghens and Newton in natural science, and of Bruno, Hobbes and Spinoza in philosophy. It is the fundamental thoughts of these great investigators that we still live upon. It is characteristic of Dühring that the great, vigorous and fruitful initiating thoughts have for him far greater significance than the special working-out and the

confirmation by experience. This hangs together with his historical eye and with his sense for what is inspired; but it leads him to misjudge the importance of the great thoughts finding their confirmation in particular experiences — an importance which a positivist of all people ought least to misjudge. This tendency in Dühring, which was surely nourished by his blindness preventing him from seeing and orienting himself for
[555] himself, increased in the course of his development and became of fateful significance not only for his philosophy but also for his outward circumstances.

While he praises the seventeenth century as the age of great thoughts, he praises the eighteenth for the reforming application of those thoughts. When his own time seems to him poor, he turns with all the greater hope toward the future. He feels himself the herald of a tendency in thought and in life which will be able to win currency only in a distant future. His great energy and fullness of thought are united with the reformer's self-feeling — unfortunately also with a bitterness and a suspiciousness which his fate has nourished, but which make his later writings in many parts unpleasant through coarse attacks upon those he calls his enemies. His writings have a strongly personal stamp, in the word's good and in its less good sense alike. He has himself depicted his personality and his life in the remarkable book *Sache, Leben und Feinde. Als Hauptwerk und Schlüssel zu seinen sämmtlichen Schriften* (Karlsruhe und Leipzig 1882). He is right that his cause and his life hang closely together, and to his honor it must be said that with him the cause stands before the life; less right is he perhaps in supposing that his enemies hang together with his cause and his life as closely as he suggests in the title.

Eugen Dühring was born in Berlin in 1833. His family came — a thing he set great store by — from Sweden. By his father he was brought up in Rousseau's spirit, under the influence of a liberal religious way of thinking. The upbringing developed independence and firmness in his thinking. Mathematics and astronomy early became principal interests. After his father's death he entered a Gymnasium which was also a boarding house, but where he was as little satisfied with the spiritual as with the bodily fare. He then studied law and worked for a time as a practicing lawyer. An eye disease which brought blindness with it made this activity impossible, and gave occasion for his resolving to use his great store of knowledge and thought to enter upon a scientific writer's career. His wife, and later his son, was his secretary. A series of philosophical,
[556] economic and scientific-historical works testify that the energetic thinker did not put his trust in vain in his inner resources when the outer ones were closed to him. Of his writings there are to be named, besides the one already mentioned, *Der Werth des Lebens* (1865), *Cursus der Philosophie* (1875) — reworked as *Wirklichkeitsphilosophie* (1895)

— *Logik und Wissenschaftstheorie* (1878) and *Ersatz der Religion durch Vollkommeneres* (1883). He worked as a Privatdocent at the University of Berlin, where his lectures were among the best attended. His indignation at having been passed over in the filling of professorships, and his criticism of conditions at the university, brought him into sharp opposition to the leading circles there; and when he came to believe that Helmholtz had usurped the credit for the discovery of the law of the conservation of energy, which belonged to Robert Mayer — in whom Dühring saw the Galilei of the nineteenth century — his indignation found particular vent in repeated violent attacks upon the famous physicist. The ministry, on the recommendation of the philosophical faculty, punished him by depriving him of the right to lecture. That this was an unfair harshness toward the blind philosopher seems, from everything available, to be clear, even if his polemic was improper in content and in tone alike. The charge Dühring brought against Helmholtz becomes intelligible only through his not having followed the literature carefully, on account of his blindness; for Helmholtz had in fact, as soon as he discovered that he had a predecessor in Mayer, neglected no opportunity of giving him the honor that was his due. But there operated here in Dühring as well that overvaluation of first ideas as against their exact carrying through which we have mentioned above; what let him place Robert Mayer so high above all his competitors was the same fundamental consideration that let him set the seventeenth century so far above the nineteenth. And finally his suspiciousness contributed, probably nourished by the isolation blindness brought with it, and which let him find "enemies" on every side. Next after the professors, the Social Democrats and the Jews were his "enemies". "Opponents" concerned with truth and not with their own advantage he does not think he has met upon his way. (Leben, Sache und Feinde, Kap. 14.) He is so certain of having all good powers on his side that he says at *[557]* the close of his autobiography: "*Ueberhaupt sind meine Feinde wesentlich keine andern Elemente gewesen, als diejenigen, welche auch Feinde der besseren Menscheit sind!*"

After working for some years by giving free lectures, Dühring has now withdrawn to a small town near Berlin, occupied with natural-scientific and literary-historical labors. A great and noble force has here been pushed aside, though not broken, by misfortune and by his own fault and others'.

α. Epistemology

In his investigation of knowledge Dühring works (especially in "Natürliche Dialektik") entirely in the spirit of critical philosophy. He seeks to exhibit our knowledge's peculiarity in form and in procedure, in order to be able to decide how far we may regard its forms

and its results as expressions of existence. He thus attempts a critical testing of the relation between thinking and reality.

A principal point for him here is that thinking always seeks *continuous connection* and always seeks to *continue* its course. In mathematics the concept of infinity arises thereby, since no limits can be set to a magnitude's decrease or increase. In logic we enter, by virtue of the principle of sufficient reason, into an infinite series, one Why constantly succeeding another. Against this tendency to continual connection and continuation Dühring sets what he calls *the law of the determinate number*. Every *actual* given, every *factual* result is limited, and exhibits a determinate number of parts in time and in space. Infinity and unlimitedness mean only the possibility of continuation — a possibility not always really warranted. And where the continuation actually comes, it happens through successive additions of single, determinate elements. — There appears here, then, according to Dühring, a characteristic difference between thought and reality.

Thought must be supple and untiring in order to be able to follow reality in all its
[558] oscillations, in order to be able to rise and sink with the phenomena and to discover all relations of dependence between them. But the law it must follow in order to fulfill its task must not — Dühring insists — be transferred to nature, to actual existence itself. This is seen clearly by comparing the real science with the formal. In astronomy the determinate given magnitudes stand in contrast to pure mathematics' unlimited possibility of production. In chemistry the atomic numbers are examples of the law of the determinate number. Indeed, all really given numbers and magnitudes are finite: there is a determinate number of globes in the heavens, a determinate number of atoms in matter; at every moment a wholly determinate number of parts of time has elapsed; the earth has moved round the sun a certain determinate number of times; the causal series consists of a finite number of members.

With great resoluteness Dühring draws from the law of the determinate number the conclusion that the process of nature — the sum of all changes in nature — must have had a beginning. An infinite regress cannot be thought. Prior to the time filled with changes there lies, then, an eternal being, in which the differences, the inner distinctions that bring succession and change with them, had not yet appeared *as successive*. Simultaneous multiplicity may very well have been present, and must have been present, for it contains the germs — or, as Dühring says, the roots — of all that we call process and apprehend as determinate existence (Wirklichkeitsphilosophie, p. 15). Time and the causal series first came to be with the transition from that homogeneous and unchangeable condition to the unfolding of differences and changes. — This regressive conclusion of the causal series

in an unchangeable condition is for Dühring satisfying not only from the standpoint of thought but also from that of feeling.

But Dühring has emphasized the opposition between thought's ideal continuity and the reality of experience — always given in separation and parceling — so strongly that he ends in an insoluble riddle. For how is a transition from absolute equilibrium and homogeneity to succession and change possible? He has here, in his positivist thinking, come to face the same problem that Böhme and Schelling had put forward in mythologizing and speculative form. He would soften the paradox by pointing out *[559]* that every time a new phenomenon [a new quality] arises within our experience, we really have an absolute beginning. (Cursus p. 79. Logik p. 191.) But the difference is that what is new and beginning in our experience awakens wonder and gives thought a problem that can occupy it, whereas that sharp opposition between absolute identity and change makes thought's work hopeless once for all. Not merely a difference but a conflict between thinking and reality is what Dühring here comes to lay down. — When one has the choice between assuming such a conflict and assuming that existence is as incapable of being concluded as thinking is, this latter possibility seems preferable. There is a constant interplay between thought and reality: as inquiry advances there shows itself new continuity behind the apparent discontinuity, and so far reality confirms thought's ideal; but new discontinuity is also discovered where there seemed to be continuity in what is given, and the problem of how far reality can be understood is then posed anew. In Dühring the dogmatist has got the upper hand of the critical philosopher, on account of his great zeal to let reality come into its own.

On the other hand Dühring lays very great weight upon the inner connection between thinking and reality. Existence itself continues on into our thinking, where this unfolds according to its true essence. To the relation between premisses and conclusion (ground and consequent) there corresponds, in the real processes, the relation between cause and effect. To logic's principle of identity there corresponds the single thing's identity with itself, notwithstanding that it occurs in different connections. To our thinking's combinations and deductions there corresponds nature's unity and the interplay of forces in the arising of phenomena. We do not thereby conceive nature on the analogy of human consciousness, but as operating in such a way that human knowledge gets a material to investigate that corresponds to its own constitution. And even if our analysis will never be able to exhaust the great connection in nature, the necessity does not cease *[560]* of maintaining the kinship between what operates in things and what operates in the understanding.

What appears in our scientific principles is according to Dühring the fruit of forces that have long worked in a determinate direction before coming to clear consciousness. Consciousness and thought rest upon something that is as little consciousness and thought as the moving mechanical force is itself a phenomenon of movement. But it is a warranted hypothesis to represent to ourselves the ground of our thinking activity — a ground not appearing in consciousness — as agreeing with the ground of nature's connection showing itself to us to be ordered in accordance with the understanding. (Nat. Dial. p. 155, 225. Logik p. 173. Leben, Sache etc. p. 303.) What makes us think is thus the same as what makes nature operate.

This last train of thought now gradually gains such preponderance over Dühring that in his later writings he conducts a violent polemic against the critical philosophy, which maintains that we do not know things in themselves, because our knowledge is determined not merely by the nature of things but also by our own nature. He sees in that conception a bogging-down of science, a hypocritical accommodation of thought's results to what the need to assume a world beyond demands. Here again it is the law of the determinate number that underlies the matter: number is an example of our commanding concepts which express what is, in such a way that between reality and the concept no insertion conditioned by the constitution of the I takes place at all. (Logik p. 165.)

As sharply as Dühring elsewhere emphasizes the opposition between thinking and being, so definitely does he here emphasize the possibility of their completely covering one another. There he has formal knowledge chiefly in view, here real knowledge. But they cannot be kept wholly apart, as is shown by the fact that the law of the determinate numbers precisely sets inquiry new tasks — namely, to find out why changes occur just when those determinate numbers are reached.

It is Dühring's merit with respect to recent epistemology that he sets the problem
[561] of the relation between quality and continuity, between the determinate turning points and the connected processes, in so clear and distinct a light. His whole standpoint is determined by a contention between criticism and positivism, and contains an invitation to a closer investigation of the relation between those two tendencies.

β. Conception of the World

Dühring lays great weight upon the development of a *world-schematism*, which is to give a fully formed picture of actual existence's general features. In this the task of the philosophy of reality consists. It is to lead to a system of experience, whereas the earlier

metaphysics was a kind of alchemy which sought by mystical ways to fathom existence's essence. — At this point he recalls Comte and Spencer. He is in his "schematism" more abstract than Spencer and less original than Comte.

My system, he says, knows not two realities but only a single one — nature and its parts. He approaches materialism in emphasizing that everything in the world has a material side, and that by following the material phenomena in their intuitable connection one must be able to find everything that has reality. But he will not stop at the mere phenomena. Thereby he distances himself from "the usual, rather coarse materialism". He often uses the word *matter* in the sense of *being, world-medium, the bearer of all that is actual.* But he expressly emphasizes that the concept of matter must be widened and deepened so that it becomes clear that the dispositions to the phenomena of consciousness are found in nature just as much as the possibilities of all other phenomena. *The philosophical concept of matter* contains more than the physicists' and the chemists', since it is to be the concept of what underlies all bodily and spiritual states and comprehends the whole and full reality. (Cursus p. 62, 75. Leben, Sache etc. p. 302 f.) Materialism is for Dühring only a "pedestal", a real foundation, an outer, mechanical frame for the conception of the world and the view of life — not the exhaustive, ample explanation of existence itself.

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A doctrine one does not expect to find in Dühring is the assumption of *purposes* in nature. To understand him rightly on this point one must notice that he distinguishes between purpose (*Zweck*) and intention (*Absicht*). (Logik p. 203.) When he ascribes validity to the concept of purpose for our conception of nature, it is because the forces and dispositions operating in nature always exhibit certain tendencies, work in certain directions, and above all because they *operate together* and thereby make nature appear as a systematic unity. Those tendencies do not always lead — as little in outer nature as in conscious human life — to completed results. But whether the goal is reached or not, *the combined operation of manifold elements* appears as a characteristic principal feature of all processes. In apprehending nature one must attend not merely to the general laws but also to the directions and tendencies of the processes that go on, and to the types and forms in which nature's activity appears. Nature forms a scale of stages, the lower forms of existence being the foundation of the higher. The measure by which we distinguish between lower and higher forms can be sought only in the results of the different tendencies, in the degree to which they lead to their natural goal. And the last goal must be the arising of beings which not only exist and operate but also become conscious of their existence and their operating: only as the foundation of the history

of conscious beings has the history of the mechanical whole of nature a meaning; a thoroughly unconscious world would — says Dühring (Cursus p. 104) — be a foolish half-measure, a theater without actors and without spectators. And that the phenomena of consciousness are, on the other hand, essentially to be regarded as purposes is seen from the fact that unconscious and conscious operation can accomplish one and the same thing; it would be mere luxury for men and animals to become conscious of their impulses and motives, if it were only a question of attaining outer results, and if we were not here facing something that was in itself purpose. Our feelings in particular — the feeling of love, the social feeling — do not have their value merely by being means or
[563] motives leading to actions necessary for life's subsistence; they have immediate value as expressions of a higher stage of life than the purely elementary one. (Cursus p. 158 f. Der Werth des Lebens, 2. Aufl., p. 241.)

If one were to object to this last thought that conscious life is nevertheless not always bound up with pleasure, but that pain occupies so great a place that conscious life does not seem to be a real "purpose", Dühring would point to what he calls *the law of difference* — a law which under different forms and names plays a great part in recent psychology. (The most suitable name would be: the law of ratio.) For Dühring it is not merely a psychological law but a world-law. The antagonism of forces is a principal feature in the world-schematism. All expression of force, all movement and all development is conditioned by distance, difference, opposition. All consciousness too presupposes difference. Without difference, no sensation. Within consciousness the same holds. Life's play would lose its attraction if there were no resistance and hindrance to overcome, and if need and satisfaction did not succeed one another. It is existence's rhythms and differences that make it valuable for us: our feeling of life is set in motion by the transitions between mutually different states. Without what is harsh, bitter and painful, the deep satisfaction in life would not be felt. — The same emphasis upon feeling's being determined by oppositions which made Schopenhauer a pessimist makes Dühring an optimist. Schopenhauer is, as regards the "law of difference", more consistent than Dühring, in so far as he assumes that unrest and movement lie as an original tendency in the world-ground's nature; whereas for Dühring's philosophy it becomes a great riddle how successive differences and rhythmically changing states could have succeeded the resting being which that philosophy assumes in virtue of the law of the determinate number.

The *combination* of the different forces is what makes development possible. Whether this combination leads to a state of equilibrium or to a new movement, the tendencies of

the several forces are preserved in the development's result. A constant strife between the forces would be an absurdity — but it is not found in nature. The avoidance of the *[564]* absurd is a natural-logical principle. Only in exceptional cases does this principle lead to untenable forms being annihilated and dissolved. As a rule nature takes the positive way. Therefore, on Dühring's view, the idea of the struggle for existence is false: it one-sidedly emphasizes the negative point of view and lays the weight upon antagonism instead of upon combination. Dühring gives Lamarck's theory the preference over Darwin's.

γ. Ethics

The ethical has according to Dühring its foundation in the human impulses. Only its foundation — for nature can err, and therefore a closer development and correction is necessary. The higher a being stands, the greater the possibility that it goes wrong. Besides, what matters ethically is to combine elements which from nature's hand are not yet connected, but are perhaps even in mutual strife.

The germs of the good Dühring finds, like Comte, in the sympathetic instincts. Nature has itself seen to it that another's suffering determines our own feeling in a painful way, and there is perhaps no feeling that has undergone so conspicuous a development in the course of culture as this one. It does not lead to subduing and abolishing our own individuality. On the contrary, ethical development consists both in an individualization and in a socialization; the two necessarily belong together, since the full development of the single individuality first becomes possible in a more highly developed society. Under more imperfect social conditions the free, distinctive development of individuals is hindered. But since our present states are essentially built upon compulsion, they constantly hamper that development. A free society — the germs of which are given in the organizations formed by free association — will first let both individuality and community come into their own. In such a society all relations of production and of consumption would be brought under social direction, so that the individual's whole interest can be attached to the work and not to the work's yield. Only thereby does a thoroughgoing *[565]* ennoblement of life become possible. From socialism Dühring's conception differs partly in that he thinks of free labor communes and not of an absolute centralization of production,[24] partly in that he does not regard it as necessary that development should be interrupted by a historical miracle: the building of the future will on his view arise, not — as Karl Marx holds — through evil increasing, but through the good growing quietly. On this point too Dühring is decidedly an optimist, however much baseness he

[24] On Dühring's position on the social question, and on the different standpoints he has taken here, I refer to B. *Friedländer*: *Die vier Hauptrichtungen der modernen socialen Bewegung.* Berlin 1901.

may think he can establish in the world. He is a definite opponent of pessimism. Only the pessimism of indignation (*Entrüstungspessimismus*), as represented by Byron and called forth by determinate social conditions, has his approval; whereas he despises the Romantic "pessimism of the beyond" and the "pessimism of rottenness" frequent in the higher classes.

Beyond the feeling for the good forces that work their way forward within the human world, there arises a *universal affect* when the gaze is raised to the great unity of which the human world is only a single ramification — when the thinker feels himself as belonging to that unity and as moved by its forces — and when the thought of his own fate is taken up into the thought of the great order of things, within which so many good germs have been able to unfold. —

Not only by the problems it poses, but also by the noble, antique style in which it is carried through, and by the connection it establishes between thinking and personality, Dühring's philosophy stands as one of the most characteristic attempts of thought in the generation that followed Romanticism's fall and natural science's new rise at the middle of the nineteenth century. The year 1880 forms here a characteristic turning point, in that a circle of the more considerable investigators conclude their activity about that time — namely, besides Dühring, Lotze, Fechner, Darwin and Spencer; shortly before, Lange had succumbed. In a separate work some principal forms of the history of philosophy in the following quarter-century are to be presented.

Notes to the Second Volume

[1]) P. 4. Only a single point shall be touched on here, which could not well find a place in the text. Wolff had transformed the Spinoza-Leibnizian identity hypothesis into an external parallel theory, which could rightly bear the name duplicism. (Cf. note 76 to the first volume). It was this theory that had especially aroused offense among his theological opponents; but for other reasons too it could not but be found difficult, and it became the object of much criticism and was soon abandoned by several Wolffians. An acute and independent thinker, Kant's teacher, *Martin Knutzen* in Königsberg, who could as little put up with the usual doctrine of interaction as with Wolff's external parallelism, sought a solution of the difficulty by taking up again Leibniz's thought of the ultimate elements of matter as "representing" (psychical) beings. With this the heterogeneity of the elements which, on the assumption of an interaction between soul and body, were to act on one another, fell away. The soul-monads were taken to differ from the body-monads only in degree, not in essence. The Cartesian dualism, which Wolff had renewed in his own way, thereby fell away, and the problem of the relation between soul and body was widened into a problem of the interaction of things in general. Knutzen thereby set up a theory that was later developed by Herbart and especially, in a characteristic way, by Lotze. — Cf. *Benno Erdmann*: *Martin Knutzen und seine Zeit.* Leipzig 1876. p. 64—97. — A similar theory was, moreover, set up by the eclectic Berlin academician *Prémontval.* See *Dessoir*: *Geschichte der neueren deutschen Psychologie* I. p. 45. *[566]*

[2]) P. 6. "It is usual to divide the faculties of the soul into the cognitive faculty and the faculty of desire, and to count the feeling of pleasure and pain as belonging to the faculty of desire. But it seems to me that the soul's approval, approbation and satisfaction, which are nevertheless properly very different from desire, lie between cognition and desire. We regard the beauty of nature and of art with pleasure and satisfaction, without the least stirring of desire . . . . It seems to me most expedient to give the soul's satisfaction or dissatisfaction, which is indeed a germ of desire but not desire itself, a name of its own . . . I will call it the faculty of valuation (Billigungsvermögen), in order thereby to distinguish it both from the knowledge of truth and from striving after the good," *Mendelssohn*: *Morgenstunden.* Berlin 1786. p. 118—119. *[567]*

[3]) P. 20. That Lessing uses the concepts of education and revelation in a figurative sense there can be no doubt, given his own declaration in the preface that the positive religions represent the course of development of the human spirit. To speak popularly he had to speak

anthropomorphically. His standpoint appears clearly in some philosophical fragments from his younger years (e.g. *Das Christenthum der Vernunft*) and in the conversation with Jacobi. — The account of Lessing's standpoint that is given in the text has, with a few changes, been printed as part of my essay *Apologi for Lessing* in Nordisk tidsskrift, published by the Letterstedt Society. 1889.

[4]) P. 24. As regards Jacobi's rendering of the famous conversation, both the thoughts and the often highly characteristic expressions testify that we have Lessing himself speaking before us. Herder declared: "Lessing ist so dargestellt, dass ich ihn reden sehe und höre", Cf. *Erich Schmidt*: *Lessing.* II. p. 667. — As remarked in the text, one must naturally not take Lessing's "Spinozism" quite literally and externally. See on this the interesting investigation in C. *Hebler*: *Lessing-Studien.* Bern 1862. Ch. 5. — Jacobi sent Herder a copy of the account of the conversation with Lessing. In his reply Herder (who was at that time Superintendent in Weimar) declared that, by unexpectedly finding a fellow believer in Lessing, he had been strengthened in his adherence to the divine Spinoza, whose philosophy, however, he does not embrace on every point. He moreover points out to his "dear extramundane personalist" that his salto mortale is impossible, "denn wir sind in der Schöpfung auf ebnem Boden". (R. Haym: *Herder nach seinem Leben und seinen Werken.* II. Berlin 1885. p. 275 f.). — *Goethe* joined Herder, although his artistic standpoint led him to lay greater weight on the incomprehensibility of that which cannot take on a definite, individual form. In Weimar Spinoza was eagerly studied in the winter of 1784—85. An interesting document from this study is a recently found little essay, written in Frau v. Stein's hand, probably from Goethe's dictation. It shows clearly how Goethe sought to arrange Spinoza's ideas for himself in his own way. Cf. on this *Dilthey's* essay: *Aus der Zeit der Spinoza-Studien Goethe's* (Archiv für Geschichte der Philosophie. VII). — Jacobi was dismayed at Herder's adherence to Lessing and showed Herder's letter to *Wizenmann* (whose relation to Jacobi is described by the words: "I love his spirit and marvel at his unbelief"), but must here again have been dismayed when the young man, a positive believer, answered Jacobi's question as to what he
[568] thought of Herder's confession of faith: " I have the same faith". What appealed to Wizenmann in Spinoza was the inward relation in which God in his doctrine stands to nature. (Goltz: *Thomas Wizenmann.* Gotha 1859. I. p. 342 f.). On the other hand Wizenmann agreed with Jacobi in finding Goethe's poem "Prometheus" blasphemous (ib. p. 311). In the anonymous work *Die Resultate der Jacobischen und Mendelsohnschen Philosophie,* kritisch untersucht von einem Freiwilligen (Leipzig 1786) Wizenmann afterwards took Jacobi's side in the dispute against Mendelssohn, since he too contested the possibility of a religion of reason. In a note to "Kritik der practischen Vernunft" (I, 2, 2) Kant speaks of Wizenmann as "a fine and clear head, whose early death is to be lamented".

[5]) P. 33. As regards the more detailed grounding of my conception of Kant's development, I must refer to my monograph: *Kontinuiteten i Kants filosofiske Udviklingsgang* (Det kgl. danske Videnskabernes Selskabs Skrifter. 6. Række. 1893), where the references to the sources are found. — When Kant (see my monograph mentioned above, p. 6) says in several places in his letters

that Kritik der reinen Vernunft is the fruit of twelve years' work, *Emil Arnoldt (Kritische Excurse auf dem Gebiete der Kant-Forschung.* Königsberg 1894. p. 182) would count these twelve years back from the time when Kant wrote the book down, which in his opinion happened in the year 1778. This might in itself be quite reasonable; but when one also takes into account other passages (cited on the same page of the monograph) which point directly to the year 1769 as the decisive turning point, it becomes more natural to count the twelve years from the publication of the work. In favor of the year 1766 it could be said that Kant, in his announcement *(Nachricht* etc.) of lectures for 1765—66, for the first time uses the expression "Kritik der Vernunft" of the epistemological task he has set himself. But then a distinction must be made between the beginning of reflection and its first result. This latter, and with it the decisive, turning point must in any case be put at 1769. — I take the opportunity to remark that, when I worked out the fourth chapter of the monograph and showed in it that Kant's theory of the intuition of space arose through a subjectivizing of Newton's world-space, a transposition from sensorium dei to sensorium hominis, — I did not know the second part of *Vaihinger's Kommentar zu Kants Kritik der reinen Vernunft,* in which a similar explanation is given. My essay was worked out in the summer of 1892, the same year in which Vaihinger's work appeared.

[6]) P. 36. The book has a prehistory which is not without interest and which has only recently been fully illuminated by *Dilthey (Der Streit Kants mit der Censur über das Recht freier Religionsforschung.* Archiv für Gesch. d. Philos. III. p. 418 ff.) and *Emil Fromm (Immanuel Kant und die preussische Censur.* Hamburg und Leipzig 1894). The book consisted of several essays, which began to appear separately in the Berliner Monatsschrift. But when the second essay was to be printed, the board of censors imposed a ban. The editor of the journal complained to the King, and the matter came before the council of ministers. But just at that same time the ministers received a very ungracious royal letter, because they had not shown themselves willing to go *[569]* along with further steps against the press. They were told that they "were pleading the cause of the so-called men of enlightenment", and the King expressed the expectation that, by watching over literature, they would protect positive religion and thereby in turn the order of the state. (A characteristic causal series!) Under these circumstances the council of ministers resolved that the decision of the board of censors must stand. It was then that Kant published the essays, collected as a book. — The expressions I have used of Wöllner in the text are too mild, if Frederick the Great was right when, in his day, on a proposal to raise him to the nobility, he answered with the following marginal note: "Der Wölner ist ein betriegerischer und intriguanter Pfaff, weiter Nichts" (E. *Fromm* p. 19). Cf. also *Emil Arnoldt*: *Beiträge zu dem Material der Geschichte von Kant's Leben und Schriftstellerthätigkeit in Bezug auf seine "Religionslehre" und seinen Conflict mit der preussischen Regierung.* Königsberg 1898. (It is shown here, among other things, that Kant — for what reason is uncertain — got the philosophical faculty in Jena, not in Königsberg, to give the imprimatur).

[7]) P. 50. Already S. *Maimon* and *Herbart* criticized Kant's division of judgments and the

use he made of it in his theory of knowledge. — Cf. the investigation of the relation between primary and secondary judgments in my *Formel Logik.* — On Kant's division in relation to the divisions current before him see *Adickes*: *Kants Systematik als systembildender Factor.* Berlin 1887. p. 30—41.

[8]) P. 50. Kant, who on the whole is very loose in the use of his terminology, often lets the wider and the narrower meaning of the word reason pass into one another. This holds also with regard to the meaning in which he takes it in the title of his main work, since by the pure reason he means to investigate he thinks sometimes of the capacity for knowledge independent of *experience*, sometimes particularly of the capacity for *knowledge* of *something* that cannot become an *object* of experience. Cf. on this H. *Vaihinger*: *Commentar zu Kants Kritik der reinen Vernunft.* I. Stuttgart 1881. p. 453 f.

[9]) P. 51. Cf. on this *Schopenhauer* in his *Kritik der Kantischen Philosophie,* which forms an appendix to *Die Welt als Wille und Vorstellung,* first part. (6. Aufl. Leipzig 1887. I. p. 572—577). — A. *Riehl Der philosophische Kriticismus.* I. Leipzig 1876. p. 441. 446.

[10]) P. 53. Kritik der reinen Vernunft. I. Aufl. p. 229. — Cf. my essay on Kontinuiteten i Kants filosofiske Udviklingsgang. p. 16 f.

[11]) P. 56. Kant's criticism of Leibniz's doctrine of monads is found in Kritik der reinen Vernunft. I. Aufl. p. 265 f. — From Kant's notes it seems to follow that in his lectures he did not, theoretically either, hold fast to the unknowability of the Ding an sich as sharply and definitely as
[570] the consequence of his epistemological investigation properly demanded. Cf. *Reflexionen Kants zur kritischen Philosophie.* Aus Kants handschriftlichen Aufzeichnungen herausgegeben von *Benno Erdmann.* II. Leipzig 1884. Nr. 1151; 1156—58. *Lose Blätter aus Kants Nachlass.* Mitgetheilt von R. *Reicke.* I. Königsberg 1889. — Cf. on Kant's pedagogical and didactic accommodations as a lecturer the interesting remarks in *Emil Arnoldt*: *Kritische Excurse auf dem Gebiete der Kant-Forschung.* p. 387—399; 402—403; 454. — *Max Heinze (Vorlesungen Kants über Metaphysik aus drei Semestern.* Abhandlungen der phil. hist. Classe der kgl. sächs. Gesellsch. der Wissensch. 1894. p. 658) grants Arnoldt that Kant — according to what is available — must have expressed himself more cautiously in his lectures than in his writings, but adds: "Manches klingt da allerdings recht dogmatisch, da er die kritische Einschränkung nicht stets beifügt; aber dann neigt *er innerlich auch diesen dogmatischen Sätzen zu.*" This statement aptly expresses a result that can be drawn from the manuscripts of Kant's lectures (partly by himself, partly by his hearers) which have been brought to light in recent years by B. Erdmann, Reicke, Arnoldt and Heinze. — One more remark shall be taken from Heinze's essay (ib. p. 518). Whereas Kant, as is seen from his "Nachricht" of 1766, earlier began his philosophical course with empirical psychology, he later, as notes taken at his lectures show, went back to the order followed by the Wolffians, in which psychology came after ontology and cosmology. — *Benno Erdmann (Einleitung zu Kants Prolegomena.* Leipzig 1878. p. LXV) and O. Riedel *(Die monadologischen Bestimmungen in Kants Lehre vom Ding an sich.* Hamburg und Leipzig 1884) have shown how the doctrine of monads persisted in the

background of Kant's consciousness. In his excellent book on Kant *(Immanuel Kant. Sein Leben und seine Lehre.* Stuttgart 1898) *Friedrich Paulsen* very strongly emphasizes that Kant, in spite of his critique of reason, always held fast to a certain vague metaphysical idealism as his final conviction. It was not Kant's intention to root out metaphysics; he only wanted to give it other grounds than those on which it had hitherto relied. Yet Paulsen aptly remarks of the lecture notebooks through which alone we know Kant's metaphysics: "Freilich, sollten wir Kant allein aus diesen Heften kennen lernen, dann würden wir wohl niemals zu einer deutlichen Vorstellung von seiner Denkart kommen, wahrscheinlich es auch nicht der Mühe wert halten" (p. 237).

[12]) P. 57. The first reason: Kritik der reinen Vernunft. I. Aufl. p. 26 f. 34 f. 286. Prolegomena. Riga 1783. p. 163 f. — The second reason: Kritik der reinen Vernunft, I. Aufl. p. 19. 288. 494. 637. Prolegomena p. 104 f. Grundlegung zur Metaphysik der Sitten. 3. Aufl. p. 106. Ueber eine Entdeckung, nach der alle neue Critik der reinen Vernunft durch eine ältere entbehrlich gemacht werden soll. Königsberg 1790. p. 56. It is this last passage that is translated in the text. Since the translation is free, the German text is given here: Die Gegenstände, als Dinge an sich, geben den *[571]* Stoff zu empirischen Anschauungen (sie enthalten den Grund, das Vorstellungsvermögen seiner Sinnlichkeit gemäss zu bestimmen).

[13]) P. 57. It is of no use, with A. *Riehl* (Der philosophische Kriticismus. I. p. 434), to distinguish between ground and cause. For the matter is given only through the faculty of representation being affected; the relation of time cannot be kept out. The real contradiction in Kant surely lies rather precisely in this, that he will apply the concept of cause to the Ding an sich without daring also to apply the relation of time. For Kant himself sought to disarm the objection by distinguishing between thinking and knowledge (Kritik der practischen Vernunft. I. Aufl. p. 94 f. and the preface to Kritik der reinen Vernunft. 2. Aufl.): in mere thinking we operate with the pure category; to real knowledge belongs sense intuition. But the merely "thinking" application of the concept of cause itself nevertheless becomes meaningless when the relation of time may not also be applied. — This whole problem presents itself differently, partly when Kant's concept of time is subjected to a revision, partly when Kant's proof of the validity of the causal principle is restricted to the extent within which alone it really has significance. — The whole doctrine of the "Ding an sich" appears with a more realist stamp in the 2nd edition of "Kritik der reinen Vernunft" than in the 1st edition, something to which *Jacobi* and *Schopenhauer* already drew attention. But in the main point and in the main difficulty both editions agree, and the most important difference between them is to be found on another point (namely, as remarked in "Kontinuiteten i Kants filosofiske Udviklingsgang" p. 28. 55 f.: in that the psychological analysis, the subjective deduction, recedes more in the later edition, as well as in the "Prolegomena", and the whole weight is laid on the objective or transcendental deduction). — Cf. on the significance of the "Ding an sich" in the critical philosophy, once Kant's errors have been corrected, my essay *Filosofiske Problemer* (Universitetsprogram 1902), p. 50 f. I am unable to see that any epistemology which will not derive the "matter" of knowledge from its "form" can avoid positing a "Ding an

sich" at some point or other. It seems to me that this side of the problem has not come into its own in the otherwise interesting and acute essay by my young friend Mag. art. *Anton Thomsen*: *Bemerkungen zur Kritik des Kantischen Begriffes des Dinges an sich.* (Kantstudien. 1903).

[14]) P. 58. To the quotations cited in the monograph must be added the letter to Reinhold of 12 May 1789: "Das Realwesen von Raum und Zeit und der erste Grund, warum jenem drei, dieser nur eine Abmessung zukommt, ist uns unerforschlich". Further, Reflexionen II. Nr. 1187—88, where Kant distinguishes between the succession and the ground of the succession. —

[15]) P. 59. Kant's statements are not entirely clear, but nevertheless seem to tend rather in the direction of the identity hypothesis. *Benno Erdmann* (Einleitung in die Prolegomena. p. LXV) and
[572] O. *Riedel* (Die monadol. Bestimmungen. p. 7; 27 f.) interpret Kant's statements in the direction of a theory like the "monistic-spiritualistic" hypothesis set up by Martin Knutzen (see note 1) and later by Herbart and Lotze. Against this speaks: 1) Kant expressly rejects the assumption of a soul-substance, which Knutzen, Herbart and Lotze alike embrace. 2) The possibility on which he dwells is: "Eben dasselbe, was in einer Beziehung körperlich heisst, würde in einer andern zugleich ein denkend Wesen sein". (Kritik. 1. Aufl. p. 359). 3) Kant rejects three hypotheses, namely dualism, materialism and pneumatism (Kr. 1. Aufl. p. 379), the last of which is surely a "monistic spiritualism". 4) In any case Kant differs from the monistic spiritualists in definitely denying the possibility of an interaction between matter (as phenomenon) and the soul (as phenomenon): "Das commercium der Seele mit der Materie als Phänomen kann gar nicht gedacht werden; denn das müsste im Raum sein. Die Seele ist aber kein Gegenstand äusserer Anschauung". Reflexionen II. Nr. 1197 (cf. Nr. 1131). Cf. also Lose Blätter I. p. 160. Tugendlehre. 2. Aufl. p. 66. In a notebook of Kant's lectures of 1793—94, recently brought to light (by *Emil Arnoldt*: Kritische Excurse p. 502), it is likewise said: "Die Körper als Körper können auf die Seele nicht wirken und umgekehrt, weil Körper gar nicht Relationen auf ein denkendes Wesen haben können. Die äussere Relation, in der ein Körper mit einer Substanz steht, ist nur im Raum, also muss diese Substanz auch im Raum, mithin ein Körper sein."

[16]) P. 61. In Kritik der reinen Vernunft 1. Aufl. p. 549 note Kant does indeed seek to foist a dogmatic line of thought on the antitheses; but he does not succeed. It is indeed also clear that the contradictory opposite of absolute completion is not absolute (given) infinity, but non-completion (continual progress). Kant's doctrine of the antinomies appears very early in him, and he seems originally to have wanted to use it as the foundation for the critique of the whole of dogmatic metaphysics. It is an artificial schematism, devised later, that leads him to distinguish between three ideas, to which three different critical investigations must then also correspond, and the doctrine of the antinomies was then restricted to the investigation of the concept of the world. Cf. on this question Fr. *Paulsen*: *Immanuel Kant.* p. 209—213.

[17]) P. 73. In his essay *Kant als Vater des modernen Pessimismus Eduard* von *Hartmann* (Zur Geschichte und Begründung des Pessimismus. 2. Aufl. Leipzig 1891) has in a tendentious way exaggerated the pessimistic side of Kant's conception. In particular he makes Kant depict the

disproportion between culture and individual happiness in far stronger terms than he actually uses. When Kant *(Anthropologie.* 2. Aufl. p. 314) speaks of "Perfectionirung des Menschen durch fortschreitende Cultur, wenn gleich mit mancher *Aufopferung* der *Lebensfreuden* desselben", Hartmann (p. 104) renders the last words without further ado by: " auf *Kosten* ihrer *Lebensfreuden*", *[573]* which sounds far stronger and can be understood absolutely. When Kant *(Kritik der Urtheilskraft.* 2. Aufl. p. 392) speaks of culture's increasing inequality and torments, Hartmann (p. 104) uses this statement as if it were Kant's last word on this matter, which, as the context shows, it is not.

[18]) P. 74. In the work *Zum ewigen Frieden* (1795) Kant develops the line of thought of "Idee" and "Muthmasslicher Anfang" with particular regard to the possibility of the cessation of wars through the formation of a world-embracing union. As in the two earlier essays, Kant here abstracts entirely from moral motives and sympathetic feelings and merely investigates which egoistic interests can be assumed to work in the direction of the ideal end: a state of right that embraces all men. Yet he declares that it is a *duty* to use the mechanism of the interests in this direction. — Kant has taken up, on a more deep-going psychological foundation, Lessing's idea of the education of the human race. He expressly defends this idea against *Mendelssohn*, who (in his *Jerusalem)* held that there could be talk of education and progress only for the single individual, not for the species. (See Kant's essay: *Ueber den Gemeinspruch: Das mag in der Theorie richtig sein, taugt aber nicht für die Praxis.* 1793). — On the significance that *Rousseau* probably had for Kant in the transition to the definitive form of his ethics I have expressed myself in *Jean Jacques Rousseau og hans Filosofi* (1896). p. 106— 108 and in the essay *Rousseaus Indflydelse paa den definitive Form for Kants Etik* (Oversigt over det kgl. danske Videnskabernes Selskabs Forhandlinger. 1896).

[19]) P. 75. Cf. what I remarked in this respect about Kant's ethics in my work *Om Grundlaget* for den humane *Etik.* (1876). p. 52.

[20]) P. 78. It was Kant's formalism that brought him close to mysticism. *Hamann* rightly spoke of "a mystical love of form" and "a gnostic hatred of matter" as traits that could characterize Kant's philosophy *(Metakritik.* Schriften ed. Roth. VII. p. 7, cf. his oral statements to Kant VI. p. 227, see also p. 212. 213). — One of Kant's disciples, C. A. *Willmans*, wrote a dissertation *De similitudine inter Mysticismum purum et Kantianam religionis doctrinam* (Halis Sax. 1797), of which Kant — with some reservation — reprinted a piece as an appendix to *Streit der Facultäten.* Later Kant found himself prompted to make a still more definite reservation, so that his doctrine should not be confused with a doctrine of the possibility of supersensible experiences, a possibility he definitely rejected as self-contradictory. See the statement of Kant reprinted in R. *Reicke*: *Kantiana.* Königsberg 1860. p. 81 f., and *Jachmann*: *Immanuel Kant geschildert in Briefen an einen Freund,* Königsberg 1804. p. 116 ff.

[21]) P. 81. Cases of conflict as examples (the method of reagents): *Kritik der practischen Vernunft* *[574]* (Kehrbach's edition). p. 36. 112. 186 ff. *Anthropologie.* 2. Aufl. p. 243. *Tugendlehre.* Einleitung § IX. (In the last-named place it is developed that virtue is strength, and that strength is known only by resistance!). — See also the first chapter of the essay: *Ueber den Gemeinspruch: Das mag*

in der Theorie richtig sein, taugt aber nicht für die Praxis (1793). The passage cited in the text immediately following is found in this essay.

[22]) P. 84. Cf. Reflexionen Kants. II. Nr. 1534: "Das Vermögen die Motive des Willens schlechthin selbst hervorzubringen ist die Freiheit. Dieser Actus beruht nicht selbst auf dem Willen, sondern ist die Spontaneität der Causalität des Wollens". — Freedom in this sense is not a capacity to choose: "Die Freiheit ist ein positives Vermögen, nicht etwa zu wählen, denn hier ist keine Wahl, sondern das Subject zu bestimmen". (Lose Blätter. I. p. 570).

[23]) P. 85. *Herbart* has already *(Gespräche über das Böse.* Königsberg 1817. p. 145—147) pointed out two meanings of the word "freedom" in Kant. I agree with *Jodl (Geschichte der Ethik in der neueren Philosophie.* II. Stuttgart 1889. p. 30—37) that three different meanings of this word must be distinguished in Kant. — As I have shown in my *Etik* (ch. 5), the meanings of the vague word are by no means exhausted with this.

[24]) P. 95. Kritik der reinen Vernunft. 1. Aufl. p. 695 ff. Prolegomena. 1. Aufl. p. 174. 179. Kritik der Urtheilskraft. §§ 59. 88. 91 Anm.

[25]) P. 96. This appears clearly from the information that *Emil Arnoldt* has recently given about Kant's lectures in his *Kritische Excurse auf dem Gebiete der Kant-Forschung.* See, besides what is cited in note 11, also Arnoldt's work cited, p. 465 f., where Kant's lectures on the philosophy of religion are especially discussed, in which he used the current religious usage without drawing attention to the modifications in the meaning of the concepts which criticism and symbolism bring with them.

[26]) P. 102. It is quite characteristic of the history of the development of taste that, whereas Kant in his *Beobachtungen über das Gefühl des Schönen und Erhabenen* (1763) mentions "in Figuren geschnittene Bäume" (p. 4) as an example of beautiful phenomena, it is said in *Kritik der Urtheilskraft* (1790) that "the English taste in gardening", precisely by its approach to the grotesque and by its avoidance of all constraint, gives "taste in the designs of the imagination" occasion to show its greatest perfection. (§ 22 Anm.).

[27]) P. 103. In the period immediately preceding (the Sturm und Drang period) the word genius had been in fashion. It expressed a craving into the boundless, a wish to free oneself from all rules and laws. Every unclarity and unruliness found it easy to adorn itself with this word. "The word genius", says *Goethe* (Aus *meinem Leben.* Book 19), "was so misused that it was even
[575] considered necessary to banish it from the German language. And so the Germans might perhaps have deprived themselves of this word, seemingly foreign but belonging to all peoples in equal degree, had not the sense for the highest and best, founded anew by a deeper philosophy, happily gained power." That Goethe here alludes to Kant's definition of genius in Kr. d. Urth. is seen from what precedes, where it is said of the genius period's misuse of the word genius: "It was still long until the time when it could be declared that genius is the power in man that gives law and rule through doing and acting". — There is a certain connection between Kant's concept of genius and his conception of the consciousness of synthesis. In every consciousness a hidden

art is exercised in the synthesizing activity which is the essence of all consciousness, but which need not itself be an object of consciousness. And the laws and rules for thinking and acting that can be set up consciously are in the end derived from that hidden art. The quality of genius then denotes only the culmination of a form of activity that is present in all mental life.

[28]) P. 116. In *Eduard Allwills Papiere* (Vermischte Schriften. 1780) Jacobi depicted a "beautiful soul" who insists on the right of the exception and uses it for excesses. But although Allwill's conduct is evidently disapproved of by Jacobi, he nevertheless later, in his own name (Jacobi an Fichte. p. 32), uses the example of Desdemona, which is already used by the hero in "Allwills Papire" (p. 236) to defend the right of individual feeling to make exceptions. — The expression "beautiful soul" was perhaps taken up by Schiller and Goethe from "Allwills Papiere" (p. 229). Yet the expression goes back to Rousseau. — When I have somewhere called Jacobi a modern Herbert of Cherbury, it may be added that the similarity also shows itself in that Jacobi, like Herbert, wished for a sign in confirmation — only it did not come. *Wizenmann* relates in a letter from Jacobi's estate Pempelfort near Düsseldorf that Jacobi, as they sat in the garden one evening, said: "Here I often sit and look at the setting sun and then imagine the joy that would flame up in me if a miracle of God's grace assured me of his existence. Then I stand up, inspired by the thought: God! and would like to go out, call together the whole world and proclaim him to it." His enthusiasm thus did not fall away because the sign did not come. When he heard the evening wind rustle in the trees of the garden, and nature stirred for him in its depth and sublimity, it was his conviction: "Das ist doch nicht die Macht eines Chaos, die mich so harmonisch durchregt!" (*Goltz*: Thomas Wizenmann. I. p. 310 f).

[29]) P. 123. In Maimon there is a certain wavering with regard to the concept of the thing in itself. The comparison with an imaginary number($\sqrt{}$-a) is found in *Kritische Untersuchungen* (Leipzig 1797) p. 158. 191. But according to other passages the thing in itself (like matter and form) is an idea that can be represented approximately, like the irrational number $\sqrt{2}$. Thus Maimon expresses himself when he leaves it open where the thing in itself is to be sought: in an absolute object or an absolute subject (e.g. *Versuch einer neuen Logik.* Berlin 1794. p. 142). *[576]* The solution may perhaps be sought in this, that Maimon, where he altogether rejects the Ding an sich, is thinking of Kant's conception of it, whereas he himself has another concept of it, according to which the contradictions fall away. In *Philosophisches Wörterbuch* (Berlin 1791) — with regard to form and clarity Maimon's best work — it is said: "I differ from Herr Kant merely in the following way: according to him things in themselves are the foundation, existing outside us, of their phenomena in us . . . According to me, on the other hand, the knowledge of things in themselves is nothing other than *the complete knowledge of phenomena* . . . . We continually approach the knowledge of them, the more complete our knowledge of phenomena becomes" (p. 276 f.). And with this agrees a note in his autobiography *(Salomon Maimons Lebensgeschichte. Von ihm selbst geschrieben.* Berlin 1792. II. p. 43), where it is said: "The nature of irrational numbers shows us that, without having any concept of *a thing as object in itself*, one can nevertheless

determine its relation to other things".

[30]) P. 124. Some years after Maimon's first writings there appeared *Jakob Sigismund Beck's* abstract of Kant's writings, in which he came to a result similar to Maimon's, namely that epistemologically there could be no question at all of a thing in itself, and in which an attempt was made to show how forms of intuition and categories denote different stages of development of the activity of the understanding through which alone there are things for us. It is of interest to see how different thinkers strike out on the same path, but it is not necessary to go further into Beck's expositions. Interesting information about his personal relation to Kant is obtained from the letters from Beck to Kant published by R.*Reicke* (Reicke: *Aus Kants Briefwechsel.* Königsberg 1885) and from the letters from Kant to Beck published by *Dilthey* (Archiv für die Geschichte der Philosophie. II). (These letters have since been included in the new edition of Kant's writings undertaken by the Berlin Academy, in the 11th and 12th volumes).

[31]) P. 129. With regard to the relation between the two essays cf. F. *Ueberweg*: *Schiller als Historiker und Philosoph.* Leipzig 1884. p. 242—248 and F. *Jodl*: *Geschichte der Ethik in der neueren Philosophie.* II. Stuttgart 1889. p. 56 f. cf. p. 507 f. I cannot, however, agree with the result of these two authors, that Schiller constantly holds fast to a relation of opposition between grace and dignity. Not only does he already indicate in the older essay that grace, harmony, is what is highest in value, but he does so still more definitely in "Briefe üb aesth. Erz.". Thus it is said in the 27th letter: "In the midst of the fearful realm of forces and in the midst of the holy realm of laws the aesthetic formative drive builds at a third realm, the realm of joy, of play and of seeming, in which it takes from man the fetters of all relations and frees him from everything that is called constraint, in the physical as in the moral domain." And Schiller finds this third realm realized where "man with bold simplicity and calm innocence goes through the most intricate relations
[577] and has neither need to injure another's freedom in order to assert his own, nor to throw away his dignity in order to show grace". Thus a union of grace and dignity (which were earlier set up as opposites) in the aesthetic state, in which the opposition between the physical and the moral has also fallen away.

[32]) P. 133. This comparison comes out in an interesting way in one of *Friedrich Schlegel's* fragments in *Athenäum:* "Die Französische Revolution, Fichtes Wissenschaftslehre und Goethes Meister sind die grössten Tendenzen des Zeitalters. Wer an dieser Zusammenstellung Anstoss nimt, wem keine Revolution wichtig scheinen kann, die nicht laut und materiell ist, der hat sich noch nicht auf den hohen weiten Standpunkt der Geschichte der Menschheit erhoben." (Athen. 1, 2. p. 56) — Yet *Baggesen* had often hinted at a similar comparison, first admiringly, later indignantly. Cf. in the latter respect his statement in a letter to Erhard of 17 May 1797 about "die neuesten Ichtischen und Fichtischen Sanskulotterien a priori, die jetzt in Deutschland auf die politischen Sanskulotterien a posteriori in Frankreich gefolgt sind."

[32 b]) P. 138. A more special demonstration of the way in which Fichte's independent philosophy developed has been given — on the basis of hitherto unprinted manuscripts of Fichte — by *Willy*

Kabitz in his work *Studien zur Entwickelungsgeschichte der Fichteschen Wissenschaftslehre aus der Kantischen Philosophie.* (Berlin 1902). Of particular interest is — besides the decisive influence of Kant's writings — the preceding influence of Rousseau and Pestalozzi (work cited p. 8) and the later influence of "Ænesidemus" (work cited p. 56).

[33]) P. 145. On the influence that Fichte's later writings have had on the spiritual development in Denmark, through *Grundtvig*'s being so strongly influenced by them in his youth, I have made some remarks in my account of philosophy in Denmark in the nineteenth century in P. Hansen's danske Literaturhistorie.

[34]) P. 147. Fichte thinks of the relation between the ego and the body thus: the body is the outer material form under which the ego must appear in order to be able to struggle with the material limits. Since the non-ego appears in the form of space, the ego must do so too: "Matter can be displaced from its space only by other matter; therefore the mind, as a force that is to act in a material world, must itself be matter, thus an immediately given body, determinate and limited in space . . . The resistance lies in matter; therefore the force must also be taken up into the medium of matter." *Die Thatsachen des Bewusstseins.* 1817. p. 81. 83. Cf. already *Naturrecht* § 5, and *Sittenlehre* § 9 (Nr. V). — Compare this view of Fichte's with Kant's view mentioned in note 15.

[35]) P. 148. Fichte often declared that the special content of experience and of life cannot be deduced from general principles. Most clearly and finely perhaps in his lectures *Ueber das Wesen des Gelehrten* (Berlin 1806) p. 33 f. "Life in time can be comprehended only in general, according to *[578]* its essence, as the presentation of the one, original and divine life; but in the particular, according to its real content, it must be immediately lived and experienced; and only in and through this experience can it be reproduced in representation and in consciousness . . . . In each single part of human life something remains over which does not go wholly into the concept and therefore cannot be anticipated or replaced by any concept either, but which must be immediately lived if it is to come into consciousness at all. This is called the domain of bare and pure empiricism or experience." If one overlooks this, one will "forfeit life itself, in striving to explain it out completely". — Among what must thus be taken as given, Fichte counts first and foremost each single ego's definite, individual peculiarity, "the point at which we find ourselves when we begin to use our freedom", and which conditions the path we afterwards describe. (*Wissenschaftslehre.* 2. ed. p. 267). For even if an ethical explanation can be given of the limitation of the pure ego to finite egos, no explanation is thereby given of the definite individual peculiarity in the single ego. In his later expositions Fichte calls this splitting into peculiar individualities a "projectio irrationalis".

[36]) P. 152. Cf. on this my *Etik* VIII, 4. — J. H. *Löwe*: *Die Philosophie Fichtes.* Stuttgart 1862 p. 155—158 has clearly illuminated the opposition in which Fichte's doctrine of individuality stands to other elements in his system.

[37]) P. 153. In his youthful work *Kritik aller Offenbarung* (Königsberg 1793. p. 225) Fichte even

declares that the living wish to exercise spiritual influence and the sure conviction that this can happen only by means of the representation of a revelation can produce an at any rate momentary faith through the influence of enthusiasm on the imagination. A wish that the revelation may be true can be justified, he thinks, when this wish is itself ethically motivated (ibid. p. 216).

[38]) P. 155. Fr. Schlegel's and his wife's letters to Schleiermacher are found in *Aus Schleiermachers Leben.* In Briefen. 3. Bd. Berlin 1861. p. 129. 132—134. — On the peculiar way in which Fr. Schlegel transposed Fichte's Wissenschaftslehre into romantic "irony" reference must be made here to S. *Kierkegaard*: *Om Begrebet Ironi.* Copenhagen 1841. p. 287—320. Yet it must be well noted with regard to Kierkegaard's account of Schlegel that he takes account only of the novel "Lucinde", which marks an extreme point of oscillation in Schlegel's movement of thought. In the fragments in the journal *Athenäum* and elsewhere Schlegel developed the concept from a far more significant side, as an expression of the capacity to immerse oneself in single domains or individuals as though they were everything, and thereby to see them in their peculiarity, a capacity that presupposes an inner infinity in the one who possesses it; one must oneself be a
[579] world in order to be able to understand others as peculiar parts of the world. (Cf. Athenäum I. 2. p. 31 f.). It is a concept that expresses the historical method from an essential side, but which can indeed be turned so that it swings out into pure arbitrariness and sensuality. The Romantic school was in constant oscillation. Romanticism swings between unbridled self-feeling (the pure ego being made one with the empirical ego) and mystical self-surrender (the empirical ego vanishing in the pure, infinite ego), the interval between the two poles is filled by a living need to revel in all the forms of the life of the spirit (poetry, philosophy, religion) — preferably all at once. No poetry and no science could in the end satisfy this need. No wonder that in the end it led so many over to the old Catholic Church, which in doctrine and cult is still untouched by the modern distinction between the different domains of the spirit. Thus Dorothea Schlegel wrote to Schleiermacher later on (1806 or 7): "Trost, Mittheilung, Liebe, Musik, Malerei und wohlthätige Thränen finde ich in der Kirche". (Aus Schleiermachers Leben. III. p. 416). In Schelling there corresponds to this transition the transition to the period of philosophy of religion in his speculation.

[39]) P. 157. When Schelling says: "Sensibilität als Phänomen steht an der Grenze aller empirischen Erscheinungen, und an ihre Ursache als das höchste ist in der Natur alles geknüpft" (Erster Entwurf p. 174), this statement recalls Hobbes's proposition that of all phenomena in nature the most remarkable is this, that anything at all can become a phenomenon, so that, while the phenomena are the condition for knowing everything else, sense perception is in turn the condition for the knowledge of this condition. De corpore XXV, I. (Cf. the account of Hobbes's philosophy in the first volume of this work. p. 266).

[40]) P. 166. A detailed account of Weisse's system of philosophy of religion I gave in my day in my book *Filosofien i Tyskland efter Hegel* (Copenhagen 1872) p. 181—219. On philosophical theism in general I have had occasion to make some remarks at the end of my essay *Lotze og den svenske Filosofi* (Nordisk Tidsskrift, published by the Letterstedt Society. 1890).

[41]) P. 167. In "Filosofien i Tyskland efter Hegel" p. 121—149 I have given a survey of Schelling's later doctrine.

[41 b]) P. 169. The characterization of Hegel's conception of Christianity in the notes from his youth I owe to Mr. Mag. art. *Anton Thomsen.*

[41 c]) P. 177. *Ellis* Mc. *Taggart*: *Hegel's Treatment of the subjective notion.* (Mind. 1897). p. 164. Cf. the same author's essay *The Changes of method in Hegel's Dialectic* (Mind. 1892). — Cf. also F. C. *Sibbern's* acute remarks on the different kinds of triadic relation ("trilogies") that appear in Hegel: *Hegels Philosophie betragtet i Forhold til vor Tid.* Copenhagen 1838. p. 125—132. Sibbern aptly remarks that the Hegelian logic in its spiral course has no consistent room for real *[580]* synthetic unities, i.e. such unities as are formed when two elements that stand on the same line *meet* one another (as when two feelings, each of which has arisen by ways of its own, meet, or when thought and feeling meet after each has developed with a certain independence). In Hegel, consistently, the second member should always be derived from the first.

[42]) P. 191. Cf. note 38 above. — On Schleiermacher's conception of Fr. Schlegel's personality, which he had to defend not only against opponents but also against those closest to him, see *Aus Schleiermachers Leben.* In Briefen. I. p. 320. 349 ff. The last statement in particular testifies both to Schleiermacher's sense for other personalities (particularly for that in them which connects their shell with their kernel) and to his faithful and chivalrous disposition. It will stand as one of the most important documents when a comparative ethics of friendship comes some day to be written.

[43]) P. 194. The lectures that Steffens gave in Copenhagen in 1802, and which became of such great significance for the life of the spirit in Denmark through the awakening influence they had on Grundtvig and Oehlenschläger, are in large part an account of the fundamental thoughts of the "Beiträge", only with the addition of aesthetic and historical considerations, whereby the line of thought is extended from the domain of the philosophy of nature to that of the philosophy of spirit. Cf. *Indledning til filosofiske Forelæsninger.* Copenhagen 1803. p. 91. 107 f., where the at once individualizing and universalizing tendency in nature is emphasized. See moreover on these lectures my essays on Steffens in "Tilskueren" (1902) and in "Nordisk Universitetstidsskrift" (1903).

[44]) P. 195. *Dilthey* (*Leben Schleiermachers.* p. 351) remarks that especially the first draft of Schleiermacher's ethics, which dates from 1804, recalls in its content the writings of Steffens mentioned here. — When the University of Berlin was to be opened, Schleiermacher declared that it was desirable to appoint Steffens in order to avoid one-sidedness both in philosophy and in natural science, and that for his own part he particularly wished for this appointment, since his lectures on ethics would thereby find a foundation in general philosophy; to achieve this he was willing in the first years to forgo a considerable part of his salary. (Aus Schleiermachers Leben. IV. p. 175). It did not succeed, however. Steffens went instead to Breslau, and only several years after Schleiermacher's death to Berlin.

[45]) P. 197. Compare with this Fichte's view mentioned in note 37.

[46]) P. 201. When it has been said that for Schleiermacher religion was a variety of the sense for art (see *Albrecht Ritschl*: *Schleiermachers Reden über die Religion und ihre Nachwirkungen auf die evangelische Kirche Deutschlands.* Bonn 1874. p. 53 cf. 91 f.), it is rather the reverse that is
[581] correct: it is the sense for art that for Schleiermacher is a kind or variety of the religious need to express the relation of feeling to the unity of life. Cf. W. *Bender*: *Schleiermachers Theologie mit ihrem philosophischen Grundlegen.* Nördlingen 1876—78. I. p. 163. The correctness of Bender's conception is seen especially from "Philosophische Sittenlehre" § 290.

[47]) P. 202. *Philos. Sittenlehre* § 343: "Where the individual is what is properly ethically productive in actions, there only the agent himself can be his judge". — Cf. *Die christliche Sitte* p. 65: "Insofar as an action has its ground in the individuality of the man, no one other than he himself can judge it. But in this respect each is only his own judge, not his own teacher". — Perhaps it is characteristic that this last remark occurs only in Schleiermacher's "Christian" ethics, not in the "philosophical" one; but from a philosophical point of view too it is correct, and it finds its grounding in his whole doctrine.

[48]) P. 205. J. E. *Erdmann*: *Grundriss der Geschichte der Philosophie.* Berlin 1866. II. p. 465. 477. Cf. also *Weissenborn*: *Darstellung und Kritik der Schleiermacherschen Dogmatik.* Leipzig 1849. p. 40—65, and *Alb. Ritschl*: *Schleiermachers Reden über die Religion* etc. p. 59. — A more correct conception is put forward by W. *Bender*: *Schleiermachers Theologie.* I. p. 173—175. — *Delbrück* found the "Doctrine of Faith" pantheistic and incompatible with the 2nd edition of the Speeches (Aus Schleiermachers Leben. II. p. 366 f.), — thus took the reverse view of it from that which Erdmann and Ritschl took. — When Schleiermacher in the "Doctrine of Faith", instead of the expression "feeling", also uses the expression "immediate self-consciousness", he himself explains this in "Philos. Sittenlehre" thus: "The expression 'immediate self-consciousness' has its advantages, since many assign feeling only to a lower region; but the word self-consciousness cannot be used without the addition 'immediate', since otherwise one thinks of reflected self-consciousness. Therefore the expression 'feeling' is the best." *Philos. Sittenlehre* § 253. — A difference between the conception of religion in the "Speeches" and in the later writings, on which great weight has often been laid, is this, that religion is there depicted as a purely passive feeling, whereas later (and, as *Dilthey* has shown, already in sermons contemporary with the "Speeches") the practical character of religion was more emphasized. I believe, however, that here too the standpoint of the "Speeches" is often characterized somewhat one-sidedly; this holds e.g. of *Jodl's* otherwise excellent account of Schleiermacher's standpoint in his "Geschichte der Ethik". For it is an express doctrine in the "Speeches" that "in order to take up into himself the life of the world-spirit and to have religion, man must first have found humanity, and he finds it only in and through love"; — and the development of humanity is not concluded; it goes on uninterruptedly anew from manifold individual starting points (2. Rede). Since religion is conceived as that which provides the highest point of view for, or rather the fundamental mood

in, participation in human life and its development, the religious feeling is by no means merely *[582]* passive. When Schleiermacher calls it the music of life, it is a music that serves not merely for mental entertainment but also to deepen the interest in work, even if it does not itself directly set definite tasks, any more than it sets up definite theoretical assumptions. Here too there is only a difference of degree between the "Speeches" (I. ed.) and the later writings. In the "Doctrine of Faith" (§ 9) Schleiermacher distinguishes definitely between aesthetic religiousness, where the highest is the beauty of the soul, and teleological religiousness, where the highest state is not one of rest, but at the same time works for the advance of the "Kingdom of God". This distinction is unknown (any more than the "Dialectic's" distinction between God and the world) in the "Speeches", where the whole relation is left more indefinite. — In the "Dialectic" too a closer determination of the relation between feeling and the other sides of mental life is given than in the "Speeches". See especially Appendix C § LI: "Es [das Gefühl] scheint bisweilen allein hervorzutreten und darin Gedanke und That unterzugehen; aber dies scheint nur, es sind immer Spuren des Wollens und Keime des Denkens oder umgekehrt beides, wenn auch wieder scheinbar verschwindend, darin mitgesetzt."

[49]) P. 215. The consideration that leads Schopenhauer to call philosophy an art does not, as will easily be seen, coincide with the one I have put forward in my essay *Filosofi som Kunst* in "Mindre Arbejder".

[50]) P. 202. Cf. also *Helmholtz's Handbuch der physiologischen Optik. 2.* Aufl. p. 248 f., where Fichte's work "Die Thatsachen des Bewusstseins" is mentioned with recognition, and where it is then added that what is correct in Schopenhauer's corresponding expositions must for the most part be traced back to this source. — Possibly, however, Schopenhauer was also influenced on this point, in his theory of the principle of causality, by hints in his teacher G. E. *Schulze.* Cf. *Ernst Fischer*: *Von G. E. Schulze zu A. Schopenhauer.* Ein Beitrag zur Geschichte der Kantischen Erkenntnisstheorie. Aarau 1901. p. 68 and 102. Yet Fischer is decidedly wrong when he declares that Schopenhauer could learn nothing from the lectures he heard from Fichte in Berlin (among others on "Die Thatsachen der Bewusstseins"). It should moreover be noted that Fichte was strongly influenced by Schulze's "Ænesidemus", so that both influences may have combined here.

[51]) P. 222. The difficulty mentioned was already emphasized by *Herbart* in his criticism of Schopenhauer's main work immediately on its appearance (reprinted in Herbart's *Sämttl. Werke.* XII). It is not impossible that Herbart's objection led Schopenhauer — despite his great contempt for this and other critics — to the closer explanations in the second volume; it is striking that he here repeatedly returns to it. That these explanations do not remove the difficulty is at bottom granted also by P. *Deussen* (Archiv für Gesch. d. Phil. III. p. 164), who seeks to defend them, but *[583]* in fact goes back from Schopenhauer to Kant — In a conversation with Karl Bähr in the year 1856 Schopenhauer declared that he had read Herbart's review only once in his life, namely when it appeared in 1820. *(Gespräche und Briefwechsel mit Arthur Schopenhauer.* Aus dem Nachlasse von Karl Bähr. Leipzig 1894. p. 19). It may nevertheless well have had its effect for all that.

[52]) P. 227. See on this psychological theory my *Psykologi.* VI E, 6.

[53]) P. 228. Schopenhauer in his philosophy of nature, as regards the conception of organic life, is strongly influenced by the French investigators Cabanis, *Bichat* and *Lamarck*. Cf. an interesting essay by *Paul Janet*: *Schopenhauer et la physiologie française.* (Revue des deux mondes. 1880).

[54]) P. 232. Cf. *Lindner* und *Frauenstädt*: *Arthur Schopenhauer. Von ihm. Ueber ihn.* Berlin 1863. p. 592 ff. — Frauenstädt is surely right when he (work cited p. 434 f.) argues that Schopenhauer in the first edition of his main work has a more subjectivist conception than later, in that he was at first inclined to regard the individual differences as merely phenomenal, but later (especially in "Parerga und Paralipomena") assumes that they have their ground in the thing in itself. — Cf. *Schopenhauers Briefe,* herausg. von Griesebach p. 211.

[55]) P. 244. See thus W. *Windelband*: *Geschichte der Philosophie.* Freiburg 1892. p. 492. — The disparaging way in which Th. *Graae* in his book on Uwe Jens Lornsen occasionally speaks of Fries is, on the other hand, due rather to a lack of acquaintance.

[56]) P. 244. Against this utterance (which is found in the preface to his Allgemeine Metaphysik) protest was made from the side of strict Kantianism. In a letter (of 26 March 1833) Herbart then declares that he has called himself a Kantian because the changes he has made in the interior of the science, especially by his criticism of the metaphysical starting points and his transformation of psychology, are of subordinate significance in relation to the great fundamental features of philosophizing set up by Kant. Moreover he has the same enemies as Kant: speculative theology and the remnants of the old scholasticism. (See *Ungedruckte Briefe von und an Herbart.* Herausg. von R. Zimmermann. Wien 1877. p. 101—103).

[56 b]) P. 250. Cf. already G. *Hartenstein*: *Die Probleme und Grundlehren der allgemeinen Metaphysik.* Leipzig 1836. p. 532. "Was aus dem Gegensatze der Vorstellungen folgt (die, als Selbsterhaltungen der Seele, das *einzige* Beispiel darbieten, wo das wirkliche Geschehen uns erkenntbar wird), muss überall, wo entgegensetzte Selbsterhaltungen in Einem Realen eintreten, ein Analogon finden." — Cf. with this my essay *Filosofiske Problemer* (Universitetsprogram 1902), p. 64, and note 52.

[584] [57]) P. 251. Herbart's theory of the relation between soul and body thus stands closest to the one set up by Martin Knutzen and Prémontval (see note 1). Within Herbart's school this theory has been developed by, among others, *Ludwig Strümpell*: *Grundriss der Psychologie.* Lepzig 1884. Ch. 15—16.

[58]) P. 252. Cf. on this my *Psykologi.* V B, 6. — Moreover Herbart's psychology and metaphysics here again come into conflict with one another: for if he really succeeds in explaining the unity of consciousness by the interaction of the elements, he does not need the metaphysical explanation of the unity and connection in consciousness by a soul-substance; and if he starts from the soul-substance from the outset, he cannot take the unity for a mere product. — The "apperceiving masses of representations" correspond roughly to what in my Psykologi (V B, 5) I call the real self.

[59]) P. 254. Among Herbart's merits in the domain of theoretical philosophy his treatment of logic should not be forgotten (in *Hauptpunkte der Logik* 1808 and in *Einleitung in die Philosophie).* Of interest is especially his reduction of the doctrine of judgments, in that he rejects all divisions except that into affirmative and negative, and his hint at the doctrine of the quantification of the predicate concept. See especially "Einleitung" § 53. In the attempt to reduce the artificial classification of logical judgments, which played so great and fateful a role in Kant, Herbart moreover has Maimon as a predecessor.

[60]) P. 256. *Lotze* remarks in his excellent account of Herbart's philosophy *(Geschichte der deutschen Philosophie seit Kant.* Leipzig 1882) that Herbart consistently had to conceive God as a real which, without relation to other reals, could unfold no activity. (§ 72). This criticism also hits Leibniz and — Lotze himself. — With regard to immortality Herbart's philosophy is, as Lotze also remarks, really all too favorably placed, since by virtue of the doctrine of the eternal reals it leads not only to an individual immortality but also to "the inconvenient assumption of the eternal pre-existence of the soul!"

[61]) P. 259. In an interesting way *Fr. Jodl (Geschichte der Ethik in der neueren Philosophie.* II. p. 251—266) has characterized Beneke's place in the history of ethical thought by emphasizing his opposition to Kant and to speculation, his being influenced by the English school (particularly Bentham, one of whose works he translated) and his position as Feuerbach's predecessor. — Mag. art. *Anton Thomsen* has in his essay *Ueber die Entwickelung der ethischen Theorie Benekes* (Archiv für die Geschichte der Philosophie. XVI, 2. 1903) shown the interesting difference between Beneke's earlier, predominantly subjective standpoint (in "Phys. d. Sitten") and the later, more objective standpoint influenced by Bentham (in "Sittenlehre"). The text of my account has been changed in the 2nd edition in accordance with Thomsen's result. — Interesting biographical information on Beneke is given by O. *Gramzow*: *Fr. E. Benekes Leben und Philosophie* (Bern 1899). (Cf. my review of this work in Göttinger gelehrte Anzeigen 1901. Nr. 2). — *Kuno Fischer* *[585]* has (see his work *Hegels Leben, Werke und Lehre.* Heidelberg 1901. I. p. 157 f.) examined the documents concerning Beneke's dismissal and found no trace of Hegel's participation. He derives the dismissal from "the privy councillors' fear of demagogues".

[62]) P. 262. Cf. the interesting remarks on Beneke's position on the problem of soul and body in *Ueberweg's Grundriss der Geschichte der Philosophie der neunzehnten Jahrhundert.* 9. Aufl. von *Max Heinze.* 1902. p. 133 note.

[63]) P. 263. Beneke is of interest also as a logician, in that he (already in his *Lehrbuch der Logik.* 1832) showed that it is the content of concepts, not in the first instance their extension, that matters in logical thinking (§ 21. 57), and in that he led the doctrine of inference back to the principle of substitution, whereby inference becomes a simple "taking up of one judgment into another", the concepts being "divided" (or, as it has later been called, quantified) (§ 170). — Beneke and his disciples hold that William Hamilton, in his doctrine of the quantification of the predicate, made himself guilty of plagiarism of Beneke. See *Dressler*: *Characteristik der Werke Benekes.*

(Appendix to the 3rd printing of Beneke's "Lehrbuch der Psychologie". p. 299). — Beneke's logic aroused attention in England. *Stuart Mill* writes in a letter to Bain (1844): "I am reading a book on logic by a German professor named Beneke, which he sent me after reading mine, and which had previously been recommended to me by Austin and Herschel as agreeing with the spirit of my theories. This agreement is present to a certain degree, although his book is more psychological than my plan entailed. Although I consider much of his psychology unsound, because he has not properly grasped the principle of association (he now and then comes very near to it), there is much in it of a thought-provoking kind." (A. *Bain*: *John Stuart Mill.* A Criticism. London 1882. p. 79).

[64]) P. 264. On the dispute concerning the belief in immortality see *Poul* Møller: *Om Muligheden af Beviser for Menneskets Udødelighed.* (Efterladte Skrifter. V) and my book *Filosofien i Tyskland efter Hegel,* p. 17—29. — The last-named work also gives a detailed account of Schelling's later doctrine and of the philosophy of Weisse and of the younger Fichte.

[65]) P. 267. A detailed account of Trendelenburg's philosophical views is given in my book on Filosofien i Tyskland efter Hegel (p. 222—246).

[66]) P. 271. The self-characterization that Feuerbach gave in the statements cited only quite recently became known through *Wilhelm Bolin's* book: *Ludwig Feuerbach, sein Wirken und seine Zeitgenossen.* Mit Benutzung ungedruckten Materials. Stuttgart 1891. — A new, critical edition of Feuerbach's writings has been begun by W. Bolin and Fr. Jodl.

[586] [67]) P. 274. This line of thought of Feuerbach's will receive illumination and confirmation from what is developed in my *Psykologi* V B. 4 and VI F. — Feuerbach has in part a predecessor for his psychological theory in Hume. Cf. the first part of this work.

[68]) P. 285. With regard to the course of development of Swedish philosophy reference must be made to *Axel Nyblæus's* great work *Den filosofiska forskningen i Sverige från slutet af adertonde århundradet,* framställd i sitt sammenhang med filosofiens allmänna utveckling. Lund 1873 ff. Cf. also the same author's essay *Om den Boströmska Filosofien* (Lund 1883) (offprint from Ny svensk tidsskrift). — In my essay *Filosofien i Sverig* (Nordisk Tidsskrift, published by the Letterstedt Society 1879) I have given a more detailed characterization and criticism of Swedish philosophy.

[69]) P. 286. A more detailed characterization of Sibbern and his philosophy I have given in "Tilskueren" 1885. (Reprinted in "Mindre Arbejder". 1899). — On Danish philosophy in the nineteenth century cf. my article in P. Hansen's Literaturhistorie vol. 2. In a somewhat more detailed form this survey is reproduced in *Archiv für die Geschichte der Philosophie.* 2. Bd.

[70]) P. 290. In a philosophical respect the most important of Kierkegaard's writings are: *Begrebet Angest* (1844). *Stadier paa Livets Vej* (1845). *Uvidenskabelig Efterskrift* (1846). — With regard to the more detailed grounding of the characterization of Kierkegaard given in the text I must refer to my book *Søren Kierkegaard som Filosof* (Copenhagen 1892). — On Kierkegaard's dogmatic standpoint interesting demonstrations are found in Chr. *Schrempf*'s essay: *Kierkegaards Stellung zu Bibel und Dogma.* (Zeitschrift für Theologie und Kirche. Freiburg in Br. I. p. 179—229).

[71]) P. 295. While de *Maistre* by his principle of authority and by his emphasis on war recalls Hobbes, by his doctrine of causation he recalls Malebranche, whom he also mentions with great admiration, just as he also follows him in his mystical-pantheistic intellectual tendency, as when he speaks of the divine ocean that shall one day take up all things and all men. He feels himself that he stands here at the border of heresy, for he adds at once: Je me garde cependant de vouloir toucher à la personnalité, sans laquelle l'immortalité n'est rien. (*Les* Soirées de St. Petersbourgh. 7. éd. II. p. 203).

[72]) P. 302. Cf. on this question my *Psykologiske Undersøgelser.* (Det kgl. danske Videnskabernes Selskabs Skrifter. 6. Række) p. 68—81. — When Biran speaks of sensations that fall outside the self, he means by the self more or less what in my *Psykologi* (V B, 5) is called the real self, the central and more active part of the content of consciousness. Outside the formal self, which expresses the connection and the unity in consciousness, no sensations can fall. — In his criticism of Condillac Biran had a predecessor in Charles *Bonnet (Essai analytique sur les facultés de l'âme.* Copenhague 1760).

[73]) P. 306. See, on the phenomena of concretion, *Philosophie des deux Ampère.* p. 316 f. *Essai sur la philosophie des sciences* (1834) p. LVIII ff. — A. *Bertrand (La psychol. de l'effort* p. 62) uses *[587]* the expression *concrétion* wrongly, in that he uses it of free association between two different representations; for that kind of combination Ampére uses the expression *commémoration.* When I recognize the tree, it is concrétion; when I see the tree again and then come to think of the dog that lay at the root of the tree last time, it is commémoration. — I regret that I did not know Ampère's account when I wrote the chapter on immediate recognition in my "Psykologiske Undersøgelser". But it is a great joy to me to see how close an agreement there is between his conception and the one I have developed.

[74]) P. 308. Ampère, moreover, makes the difference between his theory and Kant's greater than it is, by ascribing to Kant the conception that the relations, the forms of coordination, stand in no connection whatever with things in themselves.

[75]) P. 310. Of Cousin's first appearance as a philosophical teacher *Jouffroy* gives an interesting characterization in his *Nouveaux Mélanges philosophiques* (2. éd. p. 85—95). — *Renan (Feuilles detachées,* 4. éd. 1892. p. 298 f.) has expressed himself about Cousin, and about what he owes him, in the following way: A travers une foule de défauts, quel haut sentiment de l'infini! quelle vue juste du spontané et de l'inconscient! quel accent religieux, inoui depuis Malebranche, quand il parle de la raison! Que l'on comprend bien les traces que gardèrent de ce premier enseignement des hommes tel que Jouffroy! Je connus le cours de 1818 sous les ombrages d'Issy vers 1842. L'impression fut sur moi on ne peut pas plus profonde. J'ai la conscience que plusieurs des cadres de mon esprit viennent de là . . . M. Cousin a été non un de mes pères, mais un des excitateurs de ma pensée.

[76]) P. 316. On the relation between Saint-Simon and his school see *Paul Janet*: *Saint-Simon et le Saint-Simonisme.* Paris 1878. — In the characterization of Saint-Simon I rely especially

on *George Weill*: *Saint-Simon et son oeuvre.* Paris 1894 and on Chr. *Adam*: *La philosophie en France* (Première moitié du 19e siècle). Paris 1894. — On the relation of the Polytechnic School to Saint-Simonism cf. G. *Pinet*: *L'Ecole polytechnique et les Saint-Simonisme.* (La Revue de Paris. 15 May 1894). — While Saint-Simon's writings are incoherent and make only constant fresh starts, his school produced a coherent development of doctrine, namely the one edited by *Hippolyte Carnot*, the *Doctrine de Saint-Simon* (1829), which gives the content of *Bazard's* lectures, and which in time lies before the fantastically extravagant period that ended with the dissolution of the school. The further development that the doctrine of critical and organic periods received in "Doctrine de Saint-Simon" surely took place, however, under the influence of *Auguste Comte's* first important work (Plan des travaux scientifiques nécessaires pour réorganiser la société. 1822); although he still called himself Saint-Simon's disciple, he was nonetheless fully emancipated.
[588] — In his Autobiography (Danish trans. p. 164—169) *Stuart Mill* speaks in an interesting way of the significance that the Saint-Simonians' doctrine of critical and organic periods had for him. — The whole doctrine had, as was shown earlier, first been developed by Kant and Fichte, the first thinkers who saw both that criticism and the Revolution had triumphed and that they could nevertheless not be the last word of history. In Rousseau and Lessing the same thought is dawning. — Moreover, Lessing's "Erziehung des Menschengeschlechts" was one of the Saint-Simonians' favorite books.

[77]) P. 321. Comte's disciples go a step further than the master, in that they maintain that it is rather Saint-Simon who learned from Comte than the latter from the former. *Littré*: *Auguste Comte et la philosophie positive.* 2 éd. p. 90 f. *Robinet*: *Notice sur l'oeuvre et sur la vie d'Auguste Comte.* 2. éd. p. 139. Even if Saint-Simon — as is natural — did not remain uninfluenced by the gifted younger men who worked together with him, it nevertheless stands firm that he had developed his main thoughts before he came into contact with Comte. The priority with respect to the idea of a positive philosophy as the new spiritual authority that was necessary after the age of criticism and revolution cannot be denied to Saint-Simon. But — as *Georges Weill* (Saint-Simon et son oeuvre. p. 208—210), who has clearly demonstrated this, rightly remarks — Comte's great significance is not in reality diminished by it.

[78]) P. 322. Instead of perfecting, *Comte* later (Cours de philosophie positive. 2. éd. IV. p. 262. 264) prefers to use the expression development (developpement), because a moral valuation might lie in the word perfecting.

[79]) P. 326. In a letter to Stuart Mill of 27 June 1845 he says: Le trouble a consisté en insomnies opiniâtres, avec mélancolie douce, mais intense, et oppression profonde, longtemps mêlée d'une extrême faiblesse. He connects the attack with the fact that a few days before he had begun to work out his new work, and adds: *L'ensemble de ma composition aura beaucoup gagné à cette période exceptionelle,* où ma méditation était loin d'éprouver l'atonie de ma motilité. — The year after, when Chlotilde de Vaux had died, he confessed that the feeling that had arisen in him toward her had contributed to that nervous crisis. (Lettres d'Auguste Comte à Stuart Mill, p. 413

f.). — Madame Chlotilde de Vaux's husband had been sentenced to penal servitude for life for a degrading crime, but she regarded her marriage as indissoluble, and the relation between her and Comte therefore did not go beyond an intimate friendship.

[80]) P. 328. Of the two most important writings on Comte, Littré's maintains that Comte's real significance ends with the publication of "Cours de philosophie positive", while *Robinet's* regards "Politique positive" as the real conclusion and polemicizes against Littré as an ungrateful and uncomprehending apostate. — It follows of itself that Stuart Mill's recognition of Comte *[589]* applies only to the first period. The correspondence between them ceased when Comte stood at the entrance to the second. In his work *Auguste Comte and Positivism* (1865) Mill has given an interesting valuation of Comte's doctrine in both of its stages. — The schism that Littré caused has, moreover, later repeated itself within the circle of the "orthodox" positivists. Some of these (with *Lafitte* at their head) hold that the positivist priesthood is to work upon the heart through the intelligence, while others (with the Englishman *Congreve* at their head) hold that positivism must triumph, as Christianity did in its time, through direct appeal to proletarians and women, so that the heart must be worked upon without its happening through the intelligence. See on this *Caird*: *The social philosophy and religion of Comte.* Glasgow 1885. p. 171 ff.

[81]) P. 334. *Cours de philosophie positive.* 2. éd. VI. p. 612: Kant a réellement mérité une éternelle admiration en tendant, le premier, d'échapper directement à l'absolu philosophique par sa célèbre conception de la double réalité, à la fois objective et subjective, qui indique un si juste sentiment de la saine philosophie. Cf. *Discours sur l'esprit positif.* p. 24. *Catéchisme positiviste.* 2. éd. p. 8 and 150. — According to what Comte himself states in one place, he did not read Kant's chief work. Yet he early came to know his ideas at second hand. In a letter to Valat of 3 Nov. 1824 he states that in Kantianism there are "very good things alongside a mass of extravagances", and he protests against Cousin's account of Kant being regarded as valid: Cousin est bien loin de comprendre la portée des Idées-mères du philosophe de Koenigsberg. Yet he here counts Kant as belonging to the metaphysical stage, and draws a parallel between his own relation to Kant and Galilei's relation to the Peripatetic physics. — The only writing of Kant's that he knew is the little treatise "Idee zu einer allgemeinen Weltgeschichte", which his friend Gustave d'Eichthal translated for him (1824), and which he admired greatly. Had he known it six or seven years earlier, he could, he says, have spared himself the trouble of working out his treatises [of 1820 and 22]. Letter to Eichthal of 10 Dec. 1824. — On Comte's relation to "metaphysics" cf. the interesting discussion in *Bulletin de la société française de philosophie.* January 1903. (Report of the meeting of 27 Nov. 1902).

[82]) P. 343. The first time Comte touches on the question of the possibility of self-observation is in *Lettres à Valat* p. 89 (letter of 24 Sept. 1819). In his treatise *Examen du traité de Broussais sur l'irritation* (1828) (reprinted in the interesting collection *Opuscules de philosophie sociale,* Paris 1883, which contains the most important of Comte's works before the publication of Cours de philosophie positive) he praises Broussais for his polemic against self-observation and remarks:

One can observe one's feelings when they are not too strong, because they depend on an organ other than that of observation; but one's own intellectual activity one cannot observe, for then
[590] observer and object would be identical, and who would then make the observation? What has won subjective psychology a large following in the last ten years [before 1828] is, he thinks, the justified criticism of the ideology of Condillac and Helvetius, which regarded our knowledge as the mere product of external influences, without taking into account the necessity of inner dispositions; but this criticism had been better given by Bonnet and Cabanis, and above all by Gall and Spurzheim. (Opuscules p. 293—296). In *Cours de philosophie positive* (I p. 32 f.; III p. 564; VI p. 605. 2. éd.) he repeats his polemic against self-observation, remarking as well that an elementary analysis cannot be applied to the most complicated phenomena of all. — Since he assumed that self-observation was nevertheless possible as regards the feelings, he has given a practical proof of it in one of his letters to Valat, in which he analyzes the motives that underlie his activity as an author (28 Sept. 1819). — On the whole question I refer to my Psykologi I, 8.

[83]) P. 345. Cf., as regards political economy, the interesting treatise by *Ingram*: *Nationaløkonomiens nuværende Stilling og Udsigter.* (Translated from the English). p. 16 ff. — To Mill, who had told him of his plan to write a political economy, Comte stated that although political economy properly belongs to sociology as a whole, it can nevertheless do provisional service by being treated in isolation, especially when the working out is undertaken by so good a head as Mill. *Lettres à Stuart Mill.* p. 231 f. 254 f. Cf., on the other hand, *Cours de la philosophie positive.* (2. éd.) IV. p. 255: Toute étude isolée des divers éléments sociaux est donc, par la nature de la science, essentiellement stérile, à l'exemple de notre économie politique, fût-elle même mieux cultivée.

[84]) P. 346. Cf. *Cours de philosophie positive.* 2. éd. IV p. 392. Also *Lettres à Stuart Mill.* p. 121: — cette noble école, qui . . . fut certainement la plus avancée de toutes celles du dernier siècle. — p. 275: C'est à l'école écossaise et non, comme beaucoup d'autres, à l'école germanique, que j'ai dû la première rectification des graves aberrations, à la fois morales et intellectuelles, propres à ce qu'on appelle l'école française: je n'oublierai jamais combien ma propre évolution a été d'abord redevable surtout à quelques lumineuses inspirations de Hume et d'Adam Smith. — (By the Scottish school Comte evidently means the whole tendency that began with Shaftesbury and Hutscheson, not what is called the Scottish school in a narrower sense: Reid and his successors). — I may perhaps add the remark here that the fundamental conception on which I build in my various ethical writings developed in me especially in the study of Comte, as my treatise *Om Grundlaget for den humane Etik* (Copenhagen 1876) will also bear witness.

[85]) P. 354. It is therefore not correct to say with *Fiorentino*: *Manuale di storia della filosofia,* 2. ed. Napoli 1887. p. 581, that Comte takes account only of the relations between objects among
[591] themselves, not of the relations between object and subject. To be sure, he has not drawn the consequences of this latter relation.

[86]) P. 357. See on this *Politique positive.* IV p. 187; 233 f. *Catéchisme positiviste.* p. 165 f. 222. — In his later period Comte also makes another characteristic change in his classification. He sums

up in a series (15) of propositions the most general principles of the positive philosophy, and gives this "systematization of the positive dogma" the name of *philosophie première.* The name he takes from Bacon. Then comes *philosophie seconde:* the series of the seven fundamental sciences, which are, however, again divided into two groups: *philosophie naturelle* (the first five sciences) and *philosophie morale* (the last two sciences). Politique positive IV p. 226. — In the Catéch. pos. (p. 167 f.) he speaks a little differently, mathematics, astronomy, physics and chemistry making up *cosmology,* and biology, sociology (in the narrower sense) and ethics *sociology.*

[87]) P. 368. In his essay on Bentham (Dissertations and Discussions. I) — where the utterance quoted is found — *Stuart Mill* says: "Bentham conceives the world as a collection of persons, each pursuing his separate interest or pleasure, and who can be kept from colliding more than is necessary by hope and fear under the influence of the law, of religion and of public opinion". (p. 362). — *Leslie Stephen* (The *Utilitarians.* I. London 1900. p. 314 and 325) remarks that the manuscript of the "Deontology", which is now at University College in London, seems to show that Bentham was essentially the author of this work, and that the editor, who was probably not very intelligent, is responsible only for some rhetorical exaggerations, especially in the second part.

[88]) P. 368. By my distinction between the motive of valuation and the motive of action I have sought to avoid the difficulty that arises for Bentham when the question of the justification of the principle of utility comes up. Cf. my *Etik* ch. III, 1—2; 18 and the first treatise in my *Etiske Undersøgelser.*

[89]) P. 377. Cf. J. H. *Newmann*: *Apologia pro vita sua.* London 1879. p. 10. 189 f. — See also L. E. *Fløystrup*: *Den anglokatholske Bevægelse i det nittende Aarhundrede.* Copenhagen 1891. p. 28. 30 and *George Worley*: *The Catholic Revival of the 19th Century.* London 1894. p. 45—51. — Coleridge's philosophical ideas are perhaps best got to know in brief from his work *Church and State,* where Appendix B sets out his doctrine of reason and understanding, and where Appendix E develops his relation to earlier philosophers. In *Biographia literaria* I chap. 12 he develops his ideas in Schelling's terminology and with a *very* extensive use of Schelling's own sentences. — Sterling's utterances about Coleridge are found in *Journals of Caroline Fox* (28. january 1842) and in a letter to Carlyle (in the latter's *Life of Sterling* p. 273).

[90]) P. 379. See, besides *Froude*: *Thomas Carlyle* (London 1882— 84), also *Dilthey's* characterization of Carlyle in Archiv für Geschichte der Philosophie IV, and *Leslie Stephen's* in *The Utilitarians* III. p. 462—477. *[592]*

[91]) P. 390. In my *Psykologiske Undersøgelser* p. 58 I have touched on Hamilton's merit in regard to the doctrine of association in setting up the law of the whole (law of redintegration). Of older writers there should have been named here, besides Kant and Fries, also *Beneke*: *Psychol. Skizzen.* I. p. 382 ff., and already Chr. *Wolff*: *Vernünftige Gedanken von Gott, der Welt und der Seele des Menschen.* § 238. Hamilton may have known both these writings. — In my *Formel Logik* § 18 I have touched on Hamilton's standpoint in logic. — In my work *Den engelske Filosofi i vor*

Tid (Copenhagen 1874) there is a fuller discussion of Hamilton. — Among Hamilton's disciples and critics it has been much discussed how he conceives the knowledge of the so-called primary qualities of things (extension and form): whether it too is determined by the subject, or whether we are here to know things as they are in themselves. *Stuart* Mill found (in his *Examination of Sir William Hamiltons* Philosophy*)* here a contradiction of Hamilton's doctrine that all knowledge is relative, conditioned by the relation between object and subject. *Mansel (Philosophy of the Conditioned.* London and New-York 1866. p. 84) and *Veitch* (*Hamilton.* Edinburgh and London 1882. p. 179) interpret their teacher as assuming a relative knowledge of the primary qualities (which are thus primary only in relation to the secondary ones, the qualities of sense usually so called). *George Grote* (*Minor Works.* p. 293) thinks that Hamilton changed his standpoint with respect to this question, so that in the dissertations found in his edition of Reid's works he assumes a knowledge of the primary qualities in themselves, whereas earlier he assumed only a relative knowledge of them. *Leslie Stephen* expresses himself in a similar direction on this point in Hamilton (*The Utilitarians.* III. p. 385—401).

[92]) P. 393. Cf. *Masson*: *Recent British Philosophy.* 3. ed. London 1877. p. 252 f. — *Edinburgh Review.* July 1884. p. 218.

[93]) P. 396. Yet there was one among these whose early development and early learning is quite as remarkable as Mill's, namely *William Rowan* Hamilton (b. 1805, d. 1865), famous for his optical and mathematical discoveries (not to be confused with the philosopher William Hamilton). At three years old he could read and do sums; at four he was proficient in geography; at five he could translate from Latin, Greek and Hebrew and was familiar with Homer, Milton and Dryden; at eight he read Italian and French and improvised Latin poems; before he was ten he studied Arabic and Sanskrit, and when he was twelve he wrote a Syriac grammar. And all this with full preservation of bodily and mental health. He appropriated everything without effort and with understanding. He had a strong feeling for poetry, was a friend of Wordsworth, and was himself the author of fine poems. From his ninth year on he was already a good swimmer, so that his
[593] bodily development was not neglected. *Nature.* 3. May 1883. — If one compares his development with Mill's, and both with that of other children, one gets an impression of how very different the original foundation is on which the work of education builds.

[94]) P. 402. Cf. *Stuart Mill*: *Levned.* Danish trans. p. 190 f. 243. 245. 256. — *Bain* (*John Stuart Mill.* A Criticism. London 1882. p. 163—174) gives an interesting characterization, grounded on various testimonies and on psychological investigation, of the relation between Mill and his wife. — *Theodor Gomperz* (*John Stuart Mill.* Ein Nachruf. Wien 1889. p. 18) describes a visit to Stuart Mill's house. Mill's wife takes part in the conversation, especially by interjecting "brilliant and telling words", and even when the discussion passes over into the purely philosophical domain, "her husband, listening almost with devotion, asked her opinion, which she put forward in clear and well-ordered form".

[95]) P. 403. See *Memories of old Friends from the Journals of Caroline Fox.* Edited by Horace

S. Pym. (Tauchniz Edition. Leipzig 1881). I. p. 150 ff. II. p. 218 f. (cf. 77. 143). — These diaries give a most interesting glimpse into the spiritual life of a whole series of distinguished English thinkers, poets and men of science. For a long series of years Caroline Fox wrote down every evening what she remembered of the utterances of the eminent guests in her parents' house.

[96]) P. 407. In my book *Den engelske Filosofi i vor Tid.* Copenhagen 1874. p. 47 I was led by my criticism of some of Mill's views, — a criticism I still maintain, — to use disparaging expressions about Mill which I would not use now, which I already had struck out in the German translation (Einleitung in die englische Philosophie unserer Zeit. Uebers. von Dr. H. Kurella. Leipzig 1889), and which Mill's posthumous treatises on the philosophy of religion, which were not available at the time, moreover refute. Repeated occupation with Mill and his writings has increased my admiration for him. — An excellent characterization of Stuart Mill's personality and significance has recently been given by *Leslie Stephen* in the third volume of his work *The Utilitarians.* (London 1900).

[97]) P. 413. This circular reasoning has been demonstrated with great force and sharpness by *Stanley Jevons* in the fragments of the "Examination of J. S. Mills Philosophy", which his early death prevented him from completing. See *Pure Logic and other Minor Works.* London 1890. p. 254 ff. From his correct criticism of Mill on this point, however, Jevons draws all too far-reaching consequences with regard to Mill's significance as a thinker as a whole. — See also *Arne Löchen*: *Om Stuart Mills Logik.* Kristiania and Copenhagen 1885. p. 165 ff. Jevons's criticism had already been published in "Contemporary Review" 1877—79, so that Löchen was able to profit by it. But Löchen's work is rich in interesting and instructive remarks and investigations, and gives a good characterization of Mill as logician and epistemologist.

[98]) P. 414. On this I already built my criticism of Mill's epistemology in my earlier account. *[594]* See *Den engelske Filosofi i vor Tid* p. 37 f. (On p. 37, instead of: "comparison is the condition of synthesis" there should stand: "synthesis is the condition of comparison"). — In his essay *Criteria of truth and error* (Mind 1900. p. 15) *Henry Sidgwick* remarks: I think it impossible to establish the general truths of the accepted science by processes of cogent inference on the basis of merely particular premisses; and I think the chief service that J. S. Mill rendered to philosophy, by his elaborate attempt to perform this task, was to make this impossibility as clear as day.

[99]) P. 418. In my *Psykologi i Omrids paa Grundlag af Erfaring* (Copenhagen 1882. 4. ed. 1898) I have tried to make "the unifying principle" the guiding point of view and to show how both the laws of association and the laws of sensations lead back to this principle, which Kant (under the name of synthesis) had already made the basis of the conception of consciousness. It has therefore surprised me more than once to be counted without further ado among the adherents of the English school. *Paul Carus* (*Primer of Philosophy.* Chicago 1893. p. 175) thus names me among "the psychologists who hold that the doctrine of association gives a key to all the problems of the soul". *Wundt* (*Physiologische Psychologie.* 4. Aufl. Leipzig 1893. II. p. 482) has named me among "the representatives of association psychology", who hold that "all mental

processes can be derived from the associations". Already in my book on English philosophy (1874) I criticized association psychology, the conception that makes the sensations and representations in their independence the real in consciousness and regards association as an external connection between them, inessential to themselves. And through the whole of my Psykologi there runs a criticism of this conception, combined, to be sure, with attempts to preserve the truths it has led to discovering. I also believe that association psychology can be corrected only by seeking out its own hidden presuppositions, not by assuming a special faculty of thought or "apperception" which should cancel or limit the associations. Cf. my *Psykologiske Undersøgelser* p. 5 — 6 and 59 — 67.

[100]) P. 436. A survey of the attempts in the philosophy of religion in England before and after Mill's time is given by O. *Pfleiderer*: *Die Entwickelung der protestantischen Theologie in Deutschland seit Kant und in Grossbritannien seit 1825.* Freiburg 1891. (The account of Mill's views in this work is, however, not accurate. Pfleiderer commits a bad historical error on p. 407 by having Hamilton, who had died 9 years before Mill's "Examination" appeared, reply to Mill's criticism!!). — Cf. also N. H. *Marshall*: *Die gegenwärtigen Richtungen der Religionsphilosophie in England und ihre erkenntnisstheoretische Grundlage.* Berlin 1902. — The somewhat naive and often not very profound character of Mill's treatises on the philosophy of religion speaks for
[595] the correctness of *Leslie Stephen's* conjecture (*The Utilitarians.* III. p. 86) that Mill did not know Hume's writings on the philosophy of religion, any more than he seems to have known the "Treatise".

[101]) P. 442. A somewhat fuller biography and characterization of Darwin I have given in my popular sketch: *Charles Darwin.* Copenhagen 1889. (Studentersamfundets Smaaskrifter Nr. 91—92). Some passages of this little book will be found again in the account. — The chief source is *The Life and Letters of Charles Darwin, including an autobiographical chapter.* Edited by Francis Darwin. London 1887. (Translated in "Bibliotek for de tusen hjem"). To this must be added the later published *More Letters of Charles Darwin.* A record of his work in a series of hitherto unpublished letters. Edited by Francis Darwin and A. C. Seward. London 1903.

[102]) P. 452. It is not impossible that this utterance is owed to the recollection of the well-known scene at the meeting of naturalists in Oxford in 1860 between Huxley and the Bishop of Oxford (see Life and Letters. II. p. 320 —22), — except that Huxley preferred the ape as ancestor not to a savage but to — the Bishop of Oxford.

[103]) P. 452. Cf. my remarks on the relation between Darwinism and ethics in *Grundlaget for den humane Etik* (1876). p. 30—39, and in *Etiske Undersøgelser* (1891). p. 22 f.; 32—38.

[104]) P. 455. For Spencer's biography and course of development I use, besides utterances in his writings (especially the interesting remarks in the essay *The Classification of the Sciences, with Reasons for Dissenting from the Philosophy of M. Comte.* 3. ed. p. 31, 34, 46), above all a short biography by Prof. *Youmans*, based on information from Spencer, in the American periodical *The Popular Science Monthly* (March 1876), which I translated at the time in my translation of "Mindre

Afhandlinger af Herbert Spencer" (Copenhagen 1877), and a draft of a speech by Youmans, found in the report of a celebration for Spencer in New York in the autumn of 1882. (*Herbert Spencer on the Americans, and the Americans on Herbert Spencer.* New York 1882. p. 68—76).

[105]) P. 461. This first part of "First Principles" has been translated into Danish under the title *Religion og Filosofi* by S. *Hansen* (Copenhagen 1887).

[106]) P. 463. Already in my book *Den engelske Filosofi i vor Tid* (1874) p. 123 f., 132, 145 f. I put forward this objection.

[107]) P. 470. In the year 1876 I drew Spencer's attention to these contradictions in a private letter, and in his answer he explained them in the way stated, remarking that he had not until then noticed them.

[108]) P. 470. Cf. *First Principles.* 3. ed. p. 318 f. "Although the evolution of the various products of human activity cannot be said directly to furnish examples of the accumulation of matter and the dissipation of motion [which Spencer assumes in every evolution, the parts of the whole that is forming losing their independent motion], they are nevertheless *indirect* examples of *[596]* it." p. 391: "The phenomena that are subjectively apprehended as changes in consciousness are objectively apprehended as nervous processes, which science now interprets as processes of motion." — *Principles of Psychology.* 2. ed. I. p. 508: "Although the evolution of mind itself cannot be explained by a series of deductions from the conservation of force, it is nevertheless possible that its correlate (its obverse). The evolution of physical changes in a physical organ, can be so explained."

[108 b]) P. 474. Strangely enough, even the most acute of Spencer's English critics, *James Ward*, has overlooked this in his work *Naturalism and Agnosticism* (London 1899), interesting in epistemological respects. He puts, e.g., "the universe" without further ado in place of "a universe". Cf. the work named, I. p. 192 with 187.

[109]) P. 481. Already in *Den engelske Filosofi i vor Tid* (1874). p. 171—173 I put forward this criticism of Spencer's theory of knowledge. In my *Psykologi* VII C, 4 I have discussed this subject again.

[110]) P. 486. An interesting criticism of Spencer's sociology is found in *Émile Durckheim*: *De la division du travail social.* Paris 1893. p. 218—247.

[111]) P. 488. Cf., concerning the difficulties for a scientific ethics, my work *Etiske Undersøgelser.* Copenhagen 1891. — An instructive criticism of Spencer's attempt to ground ethics immediately on biology (on the organic force and breadth of life instead of on its value as experienced in pleasure and pain) — and thereby of Spencer's inclination to apply a deductive method in ethics, is given in *Henry Sidgwick's Lectures on the Ethics of Green, Spencer and Martineau.* (London 1902). (Thus Sidgwick states: So long as life is partially painful, it may still be true in any particular case that conduct conducive to maximum quantity of life is not conducive to maximum of pleasure. p. 148. — Sidgwick also emphasizes the possibility of a conflict between what the development of the individual organism and what the development of the social organism requires. p. 143 f).

[112]) P. 492. In the concluding section of the German translation of my book on modern English philosophy I have already named the works of Boole, Jevons and Sidgwick, besides several others that I must pass over here. (*Einleitung in die englische Philosophie unserer Zeit.* Leipzig 1889. p. 243—249).

[Notes 113–130, to the Tenth Book, are given as footnotes in that Book.]

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